



MATHEMATICS

TREND ANALYSIS

(2016-2014)

S.No.	Chapter Name	2016 (I)	2015 (II)	2015 (I)	2014 (II)
1	Number System	11	10	12	13
2	HCF and LCM of Numbers	4	3	4	4
3	Decimal Fractions	-	1	1	-
4	Square Roots and Cube Roots	2	1	2	1
5	Time and Distance	7	10	7	5
6	Time and Work	1	6	1	4
7	Percentage	2	4	2	2
8	Simple Interest	1	-	2	1
9	Compound Interest	-	1	2	1
10	Profit and Loss	1	1	-	1
11	Ratio and Proportion	2	3	2	1
12	Logarithm	1	-	-	-
13	Basic Operations and Factorisation	11	5	5	1
14	HCF and LCM of Polynomials	4	3	4	4
15	Rational Expressions	-	-	-	1
16	Linear Equations	1	1	1	1
17	Quadratic Equations and Inequations	7	4	1	9
18	Set Theory	-	1	1	-
19	Measurement of Angles and Trigonometric Ratio	8	11	12	11
20	Height and Distances	2	2	4	1
21	Lines Angles	3	-	1	-
22	Triangles	12	2	3	7
23	Quadrilateral and Polygon	3	5	4	4
24	Circle	7	4	5	6
25	Locus	-	-	-	-
26	Area and Perimeter of Plane Figures	4	6	11	4
27	Surface Area and Volume of Solids	-	11	8	11
28	Statistics	6	5	5	7
Total		100	100	100	100

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NUMBER SYSTEM

Generally (10-12) questions have been asked from this chapter. Questions, from this section usually test your basic knowledge of numbers and are mostly based on various properties of multiplication and division. A good number of statement based questions have been asked from this chapter.



NUMBER SYSTEM

Numbers are collection of certain symbols or figures called digits. The common number system in use is decimal number system. In this system, we use ten symbols each representing a digit. These are 0, 1, 2, 3, 4, 5, 6, 7, 8, and 9. A combination of these figures representing a number is called a numeral.

Types of Numbers

1. **Natural numbers** Numbers which are used for counting i.e. 1, 2, 3, 4, ... are called natural numbers. The set of natural numbers is denoted by ' N '. Smallest natural number is 1 but we cannot find the largest natural number as successor of every natural number is again a natural number.
2. **Whole numbers** Natural numbers including zero are known as whole numbers. The set of whole numbers is denoted by W .
 - Every natural number is a whole number.
 - Zero (0) is the only whole number which is not a natural number.
3. **Even numbers** The numbers which are divisible by 2 are called as even numbers. e.g 2, 4, 6, 8, 10, In general these are represented by $2m$, where $m \in N$.
4. **Odd numbers** The number which are not divisible by 2 are called as odd numbers. e.g. 1, 3, 5, 7, 9, In general, these are represented by $(2m - 1)$, where $m \in N$.
5. **Prime numbers** Those numbers which are divisible by 1 and the number itself are known as prime numbers. e.g. 2, 3, 5, 7, ..., etc. are prime numbers.
 - If a number is not divisible by any of the prime numbers upto square root of that number, then it is a prime number.
 - 2 is the only even number which is prime.
 - The prime numbers upto 100 are : 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89 and 97 i.e. there are 25 prime numbers upto 100.

6. **Coprime numbers** Two natural numbers x and y are said to be coprime, if they do not have any common divisor other than 1.

e.g. (9, 2), (5, 6), (11, 15) are the pairs of coprime numbers.

- If x and y are any two coprimes, a number p is divisible by x as well as by y , then the number is also divisible by xy .
- Coprime are also called as relatively prime numbers.
- **Twin primes** Twin primes are pair of primes which differ by 2. e.g. (3, 5), (7, 9), (11, 13) etc.

7. **Composite numbers** A composite number is any number greater than one that is not a prime number. e.g. 4, 6, 8, 9, ... all are composite numbers.

- '1' is neither prime nor composite.

INTEGERS

The collection of positive numbers, negative numbers and zero are called integers. The set of integers is denoted by Z or I .

Thus, Z or $I = \{\dots, -4, -3, -2, -1, 0, 1, 2, 3, 4, \dots\}$ is the set of integers. Every natural number and whole number is a part of integer. So, $N \subset W \subset I$.

Types of Integers

Integers are of three types

- Positive Integers** It is a set of all positive numbers. It is denoted by I^+ , $I^+ = \{1, 2, 3, 4, \dots\}$
- Negative Integers** It is a set of all negative numbers. It is denoted by I^- , $I^- = \{\dots, -3, -2, -1\}$
- Non-negative Integers** An integer that is either 0 or positive is called non-negative integer. $\{0, 1, 2, 3, \dots\}$

Note '0' is neither positive nor negative.

EXAMPLE 1. The smallest 3 digit prime number is

- a. 101 b. 103 c. 109 d. 113

Sol. a. The smallest 3 digit number is 100, which is divisible by 2.

$\therefore$ 100 is not a prime numbers.

$\sqrt{101} < 11$ and 101 is not divisible by 2, 3, 5 and 7.

$\therefore$ 101 is a prime number.

Hence, 101 is the smallest 3 digit prime number.

RATIONAL NUMBERS

The numbers which are expressed in the form of p/q where p and q are integers and are coprimes, $q \neq 0$ are called rational numbers.

The set of rational numbers is represented by 'Q'.

e.g. $\frac{3}{5}, \frac{-3}{7}, 7, -6$ are rational numbers.

- The decimal expansion of every rational number is either terminating or non-terminating repeating.

e.g. $\frac{1}{5} = 0.2$, $\frac{1}{3} = 0.333\dots$, $\frac{8}{44} = 0.181818\dots$ etc.

- The recurring decimal have been given a short notation as $0.3333 = 0.\bar{3}$, $0.181818 = 0.\bar{18}$

Note • Zero is a rational number, since we can write $0 = 0/1$.
• Every natural number, whole number and integer is a rational number.

IRRATIONAL NUMBERS

The number which cannot be expressed in the form p/q , where p and q both are integers and $q \neq 0$ are known as irrational numbers. The irrational numbers when expressed in decimal form are in non-terminating and non-repeating form. e.g. $\sqrt{2}, \sqrt{5}, \sqrt{7}$, 0.101005001, etc.

- Here, it is notable that exact value of π is not $\frac{22}{7}$ or 3.14, as $\frac{22}{7}$ is a rational number while π is irrational.

IMPORTANT FACTS

1. If $a + \sqrt{b} = x + \sqrt{y}$, where a and x are rational and $\sqrt{b}$ and $\sqrt{y}$ are irrational, then $a = x$ and $b = y$.
2. The sum or difference of a rational and an irrational number is irrational number.
3. The product of rational and irrational number is also an irrational number.
4. If we add, subtract, multiply or divide two irrational numbers, we may get an irrational number or rational number.

EXAMPLE 2. The rational number lying between $\sqrt{2}$ and $\sqrt{3}$ is

- a. $\frac{49}{28}$ b. $\frac{56}{35}$
c. $\frac{63}{45}$ d. $\frac{85}{68}$

Sol. b. We have, $\sqrt{2} = 1.414\dots$ and $\sqrt{3} = 1.732\dots$

$$\frac{49}{28} = \frac{7}{4} = 1.75, \quad \frac{63}{45} = \frac{7}{5} = 1.4$$

$$\frac{56}{35} = \frac{8}{5} = 1.6, \quad \frac{85}{68} = \frac{5}{4} = 1.25$$

Clearly, 1.6 lies between $\sqrt{2}$ and $\sqrt{3}$.

Hence, $\frac{56}{35}$ lies between $\sqrt{2}$ and $\sqrt{3}$.

REAL NUMBERS

The collection of all rational and all irrational numbers together forms the set of real numbers and is denoted by 'R'. Thus, all natural numbers, whole numbers, integers, rational and irrational numbers are real numbers.

Properties of Real Numbers

Properties of real numbers are as follows

General Properties of R

1. If x and y are two real numbers, then either $x > y$, $y > x$ or $x = y$.
2. If x and y are two real numbers, then
 - (i) $x > y \Rightarrow \frac{1}{x} < \frac{1}{y}$
 - (ii) $x > y \Rightarrow -x < -y$
 - (iii) $x > y \Rightarrow x + a > y + a$
 - (iv) $x > y \Rightarrow xa > ya$, when $a > 0$
3. If $xy = 0 \Rightarrow x = 0$ or $y = 0$

Properties of operations on R

Let '*' be any operation defined on R.

1. **Closure Property** If $a \in R$ and $b \in R$, then $a * b \in R$.
 2. **Associative Property** If $a, b, c \in R$, then $a * (b * c) = (a * b) * c$.
 3. **Commutative Property** If $a, b \in R$, then $a * b = b * a$.
 4. **Identity** Let 'I' be the identity, then $I * a = a * I = a \forall a \in R$.
 5. **Inverse** Let a' be the inverse of a , then $a' * a = a * a' = I \forall a \in R$.
- The above properties hold for addition and multiplication on R.
 - The additive identity of R is 0 and additive inverse of $a \in R$ is $-a$.
 - The multiplicative identity of R is 1 and multiplicative inverse of $a \in R$ is $1/a$.
 - R is closed for subtraction, if $a, b \in R$, then $a - b \in R$.
 - If $a, b, c \in R$, then $a \times (b + c) = (a \times b) + (a \times c)$, is the distributive property on R.

Absolute Value of a Real Number

The absolute value of a real number x is denoted by $|x|$. Thus, $|3| = 3$ and $|-4| = 4$.

If x is any real number, then $|x| = \begin{cases} x, & \text{when } x > 0 \\ 0, & \text{when } x = 0 \\ -x, & \text{when } x < 0 \end{cases}$

e.g.

1. If $x = 5$, $|5| = 5$, if $x = 0$, $|0| = 0$
If $x = -5$, $|-5| = -(-5) = 5$

2. $|x - 2| = x - 2$, if $x > 2$, $|x - 2| = 0$, if $x = 2$
 $|x - 2| = 2 - x$, if $x < 2$

Some Properties of Absolute Values

1. $|x| \geq 0$ for all real x .
2. $|x| = a$ means $x = a$ or $x = -a$
3. $|x| > a$ means $x > a$ or $x < -a$
4. If $n = x^2$; $\sqrt{n} = \sqrt{x^2} = |x| = x$, if $x > 0 = -x$, if $x < 0$

EXAMPLE 3. Find the value of x which satisfy the inequalities $|x| \geq x$ and $2x - 1 > 3$.

- a. All positive number
- b. All positive number greater than 2
- c. All negative number less than -2
- d. All negative number

Sol. b. $|x| \geq x$ is true for all real values of x . Now, consider $2x - 1 > 3$ or $2x > 4$ or $x > 2$.
So, the solution set is all positive number greater than 2.

FACTORS IN SET OF INTEGERS

Let $a, b \in I$, we say that a is a factor of b , if there exists an integer p such that $b = ap$; in short we write a / b read it ' a ' divides ' b '.

If a is factor of b , then b is called a multiple of a .

Properties of factor and multiples

For real numbers a and b , if $b = ac$ then ' a ' is a factor ' b ' and we write a / b .

- (i) $a / b, b / c \Rightarrow a / c$ (This law is known as transitivity)
- (ii) $a / a, \forall a \in R$ (This law is known as reflexivity)
- (iii) a / b and $a / c \Rightarrow a / b + c$ and $a / b - c$
- (iv) If p is prime number and p divides ab where a, b are integers, then p divides ' a ' or p divides ' b '. Thus $p / ab \Rightarrow p / a$ or p / b .

EXAMPLE 4. How many factors of $2^5 \times 3^6$ are perfect squares?

- a. 9
- b. 12
- c. 18
- d. 4

Sol. b. Any factor of this number should be of the form $2^a \times 3^b$.

For the factor to be a perfect square a, b have to be even a can take values 0, 2, 4 and b can take values 0, 2, 4, 6.

$\therefore$ Total number of perfect squares = $3 \times 4 = 12$

To find the Unit's Place Digit of a given Expression

When Number is in the form of Product

To find the unit digit in the product of two or more number we take unit digit of every number and then multiply them. Then, the unit digit of the resultant product is the unit digit of the product of original numbers.

When Number is in the form of Index

Let the exponential number of the form be a^n and $n \in I$.

1. In case, if a is any of (0, 1, 5, 6), then the unit's place digit is 0, 1, 5 and 6, respectively.
2. In case, if a is any of (4 and 9)
 - (i) and if power is odd, then the unit's place digit is 4 and 9, respectively.
 - (ii) and if power is even, then the unit's place digit is 6 and 1, respectively.
3. In case, if a is any of (2, 3, 7, 8), then see the following steps

Step I First, divide the exponent of a by 4.

Step II If any remainder comes on division. Put it as the power of a and get the result.

Step III If any remainder does not come on division. Put 4 as the power of a and get the result.

EXAMPLE 5. Find the unit digit of $207 \times 781 \times 39 \times 94$.

- a. 4 b. 2 c. 1 d. 5

Sol. b. Taking unit digit of every number and then multiplying them $= 7 \times 1 \times 9 \times 4 = 7 \times 36$

Again taking unit digit and then multiplying $= 7 \times 6 = 42$
 $\therefore$ Required unit digit = 2

EXAMPLE 6. What is the last digit in $7^{402} + 3^{402}$?

- a. 0 b. 4 c. 8 d. None of these

Sol. c. On division of 402 by 4, we get 2 as remainder.

$\therefore$ Last digit of $7^{402} = \text{Last digit of } 7^2 = \text{Last digit of } 49 = 9$
 Last digit of $3^{402} = \text{Last digit of } 3^2$
 $= \text{Last digit of } 9 = 9$

$\therefore$ Last digit of $7^{402} + 3^{402} = \text{Last digit of } (9 + 9) = 8$

DIVISION ON NUMBERS (DIVISION ALGORITHM)

Let ' a ' and ' b ' be two integers such that $b \neq 0$. On dividing ' a ' by ' b ', ' q ' will be the quotient and ' r ' will be the remainder, then the relationship between a , b , q and r is $a = bq + r$, where $0 \leq r < b$. Or in general, we have

$$\text{Dividend} = \text{Divisor} \times \text{Quotient} + \text{Remainder}$$

EXAMPLE 7. When a positive integer n is divided by 5, the remainder is 2. What is the remainder when the number $3n$ is divided by 5?

- a. 1 b. 2 c. 3 d. 4

Sol. a. Let $n = 5q + 2$ and $3n = 3(5q + 2)$

$\Rightarrow 3n = 15q + 6 = 15q + (5 + 1) = 5(3q + 1) + 1$
 when $3n$ is divided by 5, then remainder is 1.

Divisibility Test

Divisor	Condition with Example
2	If the unit place of a number is '0' or divisible by 2 i.e. unit place is even. e.g. 17980, 314782, 6148, 316 etc.
3	If the sum of the digits of the given number is divisible by 3. e.g. 24375, here $2 + 4 + 3 + 7 + 5 = 21 \div 3 = 7$ Hence, 24375 is divisible by 3.
4	If last two digits of number is divisible by 4. e.g. 589372, 72 is divisible by 4. Hence, 589372 is also divisible by 4.
5	If digit at unit place is 5 or 0. e.g. 895, 700 etc.
6	If given number is divisibly by both 2 and 3. e.g. 759312. Here the last digit is divisible by 2. And $(7 + 5 + 9 + 3 + 1 + 2) = 27$, is divisible by 3. Hence, 759312 is divisible by 6.
7	If twice the number at units place is subtracted from rest of the digits and the remainder is divisible by 7. e.g. (a) $875 = 87 - 2(5) = 87 - 10 = 77 \div 7 = 11$ Hence, 875 is divisible by 7. (b) $5103 = 510 - 2(3) = 510 - 6 = 504 \div 7 = 72$ Hence, 5103 is divisible by 7. Note Trick is applicable for number greater than 99.
8	If last three digits are divisible by 8. e.g. (a) $96432 \rightarrow 432 \div 8 = 54$ (b) $16000 \rightarrow 000 \div 8 = 0$
9	If sum of all the digits are divisible by 9. e.g. $317349 \Rightarrow (3 + 1 + 7 + 3 + 4 + 9) = 27 \div 9 = 3$ Hence, 317349 is divisible by 9.
10	If last digit of a number is '0'. e.g. 130, 36980, etc.
11	If the difference between sum of digits at even places and sum of digits at odd places is divisible by 11. e.g. 10615. Sum of digits at odd place $= 1 + 6 + 5 = 12$ Sum of digits at even place $= 0 + 1 = 1$ Difference $= 12 - 1 = 11 \div 11 = 1$ So, 10615 is divisible by 11.

EXAMPLE 8. What is the remainder when 4^{1000} is divided by 7?  2014 I

- a. 1 b. 2 c. 4 d. 5

Sol. c. On division of $4^1, 4^2, 4^3, 4^4, 4^5, 4^6$, and 4^7 by 7,

we get remainders 4, 2, 1, 4, 2, 1 and 4 respectively.

Now, 4^4 gives us same remainder as 4^1 , so the cyclicity is of 3.

So, any power of 3 or a multiple of 3 will give a remainder of 1.

So, 4^{999} will give a remainder of 1.

$\therefore$ Remainder of $\frac{4^{1000}}{7} = 4$

Theorem of Divisibility

1. If N is a composite number of the form $N = a^p \cdot b^q \cdot c^r \dots$, where a, b and c are primes, then the number of divisors of N , represented by m is given by $m = (p+1)(q+1)(r+1)\dots$
2. The sum of the divisors of N , represented by S is given by

$$S = \frac{(a^{p+1} - 1)}{(a - 1)} \cdot \frac{(b^{q+1} - 1)}{(b - 1)} \cdot \frac{(c^{r+1} - 1)}{(c - 1)}$$

SOME IMPORTANT RESULTS ON DIVISION

1. If p divides q and r , then p also divides their sum and difference also.
2. For any natural number n , $(n^3 - n)$ is divisible by 6.
3. The product of three consecutive natural numbers is always divisible by 6.
4. $(x^m - a^m)$ is divisible by $(x + a)$ for even values of m .
5. $(x^m + a^m)$ is divisible by $(x + a)$ for odd values of m .
6. $(x^m - a^m)$ is divisible by $(x - a)$ for all values of m .

EXAMPLE 9. $19^5 + 21^5$ is divisible by

- a. Only 10 b. Only 30
c. Both 10 and 20 d. Neither 10 nor 20

Sol. c. We can check divisibility of $19^5 + 21^5$ by 10 by adding the unit digits of 9^5 and 1^5 which is equal to $9 + 1 = 10$. So, it must be divisible by 10.

Now, for divisibility by 20 we add 19 and 21 which is equal to 40.

So, it is clear that it is also divisible by 20.

So, $19^5 + 21^5$ is divisible by both 10 and 20.

Shortcut Methods for Multiplication

1. Multiplication of a Given Number by 9, 99 etc.

Method Place as many zeros at the right of the multiplicand as is the number of nines in the multiplier and subtract from the number, so formed from the multiplicand, to get the result. e.g.

- (i) Multiply 8886 by 9999

$$\text{So, } 8886 \times 9999 = 88860000 - 8886 = 88851114$$

- (ii) Multiply 56985 by 999

$$\text{So, } 56985 \times 999 = 56985000 - 56985 = 56928015$$

2. Multiplication of a Given Number by a Power of 5

Method Put as many zeros to the right of the multiplicand as is the power of 5 in the multiplier. Divide the number so formed by 2 raised to the same power as is the power of 5.

e.g. Multiply 6798 by 125

$$\begin{aligned} \text{Here, } 6798 \times 125 &= 6798 \times 5^3 = \frac{6798000}{2^3} \\ &= \frac{6798000}{8} = 849750 \end{aligned}$$

Important Identities

The identities given below are very useful for quick multiplication.

Here, if a, b and c are real numbers, then

1. (i) $(a + b)^2 = (a + b)(a + b) = a^2 + b^2 + 2ab$
(ii) $(a + b)^3 = a^3 + b^3 + 3ab(a + b)$
2. (i) $(a - b)^2 = (a - b)(a - b) = a^2 + b^2 - 2ab$
(ii) $(a - b)^3 = a^3 - b^3 - 3ab(a - b)$
3. (i) $(a + b) = \sqrt{(a - b)^2 + 4ab}$
(ii) $(a - b) = \sqrt{(a + b)^2 - 4ab}$
4. (i) $(a + b)^2 + (a - b)^2 = 2(a^2 + b^2)$
(ii) $(a + b)^2 - (a - b)^2 = 4ab$
5. $ab = \left(\frac{a+b}{2}\right)^2 - \left(\frac{a-b}{2}\right)^2$
6. $a^2 - b^2 = (a - b)(a + b)$

EXAMPLE 10. The value of

$$\frac{1.073 \times 1.073 - 0.927 \times 0.927}{1.073 - 0.927} + \frac{(3^4)^4 \times 9^6}{(27)^7 \times 3^9} \text{ is}$$

- a. $2\frac{1}{3}$ b. $2\frac{1}{5}$
c. $2\frac{1}{9}$ d. 3

$$\begin{aligned} \text{Sol. c. } & \frac{1.073 \times 1.073 - 0.927 \times 0.927}{1.073 - 0.927} + \frac{(3^4)^4 \times (9)^6}{(27)^7 \times (3)^9} \\ &= \frac{(1.073)^2 - (0.927)^2}{1.073 - 0.927} + \frac{(3^4)^4 \times (3^2)^6}{(3^3)^7 \times (3)^9} \\ &= \frac{(1.073 + 0.927)(1.073 - 0.927)}{1.073 - 0.927} + \frac{3^{28}}{3^{30}} \\ &= 2 + \frac{1}{3^2} = 2 + \frac{1}{9} = 2\frac{1}{9} \end{aligned}$$

SIMPLIFICATION

The word simplification refers to a procedure of converting a complex arithmetical expression into a simple expression.

For simplifying an expression, follow the rule 'VBODMAS'. The letter of word means

V	—	Under line portion i.e. bar
B	—	Brackets
O	—	Orders or powers
D	—	Division
M	—	Multiplication
A	—	Addition
S	—	Subtraction

- The operations have to be carried out in the order, in which they appear in the word VBODMAS.
- The order of brackets to be simplified is $()$, $\{\}$, $[\]$.

EXAMPLE 11. Simply the following expression $52 - 4$ of $(17 - 12) + 4 \times 7$.

- a. 38 b. 85 c. 60 d. 72

Sol. c. $52 - 4$ of $(17 - 12) + 4 \times 7$
 $= 52 - 4$ of $5 + 4 \times 7$ (simplifying parenthesis)
 $= 52 - 4 \times 5 + 4 \times 7$ (simplifying of)
 $= 52 - 20 + 4 \times 7$ (simplifying multiplication)
 $= 52 - 20 + 28$ (simplifying multiplication)
 $= 32 + 28$ (simplifying subtraction)
 $= 60$ (simplifying addition)

PRACTICE EXERCISE

- Let 'a' and 'b' be natural number, not necessarily distinct. For all values of 'a' and 'b' the natural number would be
 (a) $(a + b)$ (b) a/b (c) $a - b$ (d) $\log(ab)$
- x and y are two natural numbers such that x is less than y , q is the quotient and r is the remainder when y is divided by x . Therefore,
 (a) $r = 0$ (b) $r < 0$ (c) $r > x$ (d) $0 \leq r < x$
- If p is a prime number and p divides ab i.e. $p|ab$, where 'a' and 'b' are integers, then
 (a) $p|a$ or $p|b$ (b) $p|a$ and $p|b$
 (c) $p|a - b$ (d) None of these
- Which one of the following is a prime number?
 (a) 161 (b) 171
 (c) 173 (d) 221
- If n is a natural number, then $\sqrt{n}$ is
 (a) always a whole number
 (b) always a natural number
 (c) sometimes a natural number and sometimes an irrational number
 (d) always an irrational number
- The product of a rational number and an irrational number is
 (a) natural number (b) an irrational number
 (c) a composite number (d) a rational number
- What is the value of x for which x , $x + 1$, $x + 3$ are all prime numbers?
 (a) 0 (b) 1
 (c) 2 (d) 101
- In a division operation, the divisor is 5 times the quotient and twice the remainder. If the remainder is 15, then what is the dividend?
 (a) 175 (b) 185 (c) 195 (d) 250
- If $42 * 8$ is a multiple of 9, then the digit represented by $*$ is
 (a) 0 (b) 1 (c) 2 (d) 4
- The value at $*$ in $\frac{16}{7} \times \frac{16}{7} - \frac{*}{7} \times \frac{9}{7} + \frac{9}{7} \times \frac{9}{7} = 1$ is
 (a) 33 (b) -33 (c) -11 (d) 32
- The value of $10 \div 4 + 6 \times 4$ is
 (a) 4 (b) $1/4$ (c) 5 (d) None of these
- If $\frac{a}{4} = \frac{b}{5} = \frac{c}{6}$, then the value of $\frac{a+b+c}{b}$ is
 (a) 3 (b) 2 (c) 6 (d) 4
- The value of $[a - b + (b - a)] - [2a - 2b + (b - 2a)]$ is
 (a) $b - 2a$ (b) $a - 2b$ (c) $b + 2a$ (d) b
- If $3.325 \times 10^k = 0.0003325$, the value of 'k' is
 (a) 4 (b) -4 (c) -3 (d) -2
- If $\sqrt{[0.04 \times 0.4 \times x]} = 0.4 \times 0.04 \times \sqrt{y}$, then the value of x/y is
 (a) 0.0016 (b) 0.16 (c) 0.016 (d) 0.160
- The unit digit in the product $(127)^{170}$ is
 (a) 3 (b) 9 (c) 7 (d) 3
- The unit digit in the product $(7^{71} \times 6^{59} \times 3^{65})$ is
 (a) 6 (b) 2 (c) 4 (d) 1

- 18.** When n is divided by 4, the remainder is 3. What is the remainder when $2n$ is divided by 4?
 (a) 0 (b) 2 (c) 6 (d) 3
- 19.** If $\frac{\frac{1}{4} - \frac{1}{6} - \frac{1}{48}}{\frac{1}{4} - \left(\frac{1}{6} - \frac{1}{48}\right)} \div \frac{\frac{1}{4} \times \frac{1}{6} - \frac{1}{48}}{\frac{1}{4} \times \left(\frac{1}{6} - \frac{1}{48}\right)} = x$, the value of x is
 (a) $\frac{20}{21}$ (b) $-\frac{21}{20}$ (c) $\frac{21}{20}$ (d) $-\frac{20}{21}$
- 20.** If x is negative real number, then
 (a) $|x| = x$ (b) $|x| = -x$ (c) $|x| = \frac{1}{x}$ (d) $|x| = -\frac{1}{x}$
- 21.** If $\frac{x}{y} = \frac{3}{5}$, then the value of $\frac{x-y}{x+y}$ is
 (a) $-\frac{1}{4}$ (b) $\frac{1}{4}$ (c) $\frac{1}{6}$ (d) -6
- 22.** If x is real, then $\left|\frac{7-x}{3}\right| < 2$, if and only if
 (a) $1 < x < 13$ (b) $-1 < x < 13$ (c) $x < 13$ (d) $x > 13$
- 23.** A number is divisible by 25 only, if
 (a) the last digit of the number is zero
 (b) the last digit of the number is 5
 (c) the last two digit of the number is divisible by 15
 (d) the last two digit of the number is divisible by 25
- 24.** The value of $\frac{2.48 \times 2.48 - 1.52 \times 1.52}{0.96}$ is
 (a) 4 (b) 0.96 (c) 16 (d) 15.04
- 25.** Which one of the following is correct regarding the number 222222?
 (a) It is divisible by 3 but not divisible by 7
 (b) It is divisible by 3 and 7 but not divisible by 11
 (c) It is divisible by 2 and 7 but not divisible by 11
 (d) It is divisible by 3, 7 and 11
- 26.** What is the sum of positive integers less than 100 which leave a remainder 1 when divided by 3 and leave a remainder 2 when divided by 4?
 (a) 416 (b) 620 (c) 1250 (d) 1314
- 27.** What is the last digit in the expansion of $(2457)^{754}$?
 (a) 3 (b) 7 (c) 8 (d) 9
- 28.** A three-digit number is divisible by 11 and has its digit in the unit's place equal to 1. The number is 297 more than the number obtained by reversing the digits. What is the number?
 (a) 121 (b) 231 (c) 561 (d) 451
- 29.** The remainder on dividing given integers a and b by 7 are, respectively 5 and 4. What is the remainder when ab is divided by 7?
 (a) 3 (b) 4 (c) 5 (d) 6
- 30.** If k is any even positive integer, then $(k^2 + 2k)$ is
 (a) divisible by 24
 (b) divisible by 8 but may not be divisible by 24
 (c) divisible by 4 but may not be divisible by 8
 (d) divisible by 2 but may not be divisible by 4
- 31.** The number 2784936 is divisible by which one of the following numbers?
 (a) 86 (b) 87 (c) 88 (d) 89
- 32.** Which among the following is the largest four-digit number that is divisible by 88?
 (a) 9988 (b) 9966 (c) 9944 (d) 8888
- 33.** The pair of rational numbers that lies between $\frac{1}{4}$ and $\frac{3}{4}$ is
 (a) $\frac{262}{1000}, \frac{752}{1000}$ (b) $\frac{24}{100}, \frac{74}{100}$
 (c) $\frac{9}{40}, \frac{31}{40}$ (d) $\frac{252}{1000}, \frac{748}{1000}$
- 34.** By adding x to 1254934, the resulting number becomes divisible by 11, while adding y to 1254934 makes the resulting number divisible by 3. Which one of the following is the set of values for x and y ?
 (a) $x = 1, y = 1$ (b) $x = 1, y = -1$
 (c) $x = -1, y = 1$ (d) $x = -1, y = -1$
- 35.** What is the last digit in the expansion of 3^{4798} ?
 (a) 1 (b) 3 (c) 7 (d) 9
- 36.** What least value must be given to $\otimes$, so that the number $84705 \otimes 2$ is divisible by 9?
 (a) 0 (b) 1 (c) 2 (d) 3
- 37.** What is the total number of three digit numbers with unit digit 7 and divisible by 11?
 (a) 6 (b) 7 (c) 8 (d) 9
- 38.** Find the value of 'a' and 'b' $3\frac{7}{a} \times b\frac{3}{15} = 8$.
 (a) 2, 11 (b) 11, 2 (c) 1, 1 (d) 2, 1
- 39.** If a, b and c are real numbers such that $a < b$ and $c < 0$, then which of the statements is true?
 (a) $(a/c) < (b/c)$ (b) $ac < bc$
 (c) $(c/a) > (c/b)$ (d) $ac > bc$
- 40.** If we divide a positive integer by another positive integer, what is the resulting number?
 (a) It is always a natural number
 (b) It is always an integer
 (c) It is a rational number
 (d) It is an irrational number
- 41.** What can be said about the expansion of $2^{12n} - 6^{4n}$, where n is a positive integer?
 (a) Last digit is 4 (b) Last digit is 8
 (c) Last digit is 2 (d) Last two digits are zero

- 42.** If p is an integer, then every square integer is of the form
 (a) $2p$ or $(4p-1)$ (b) $4p$ or $(4p-1)$
 (c) $3p$ or $(3p+1)$ (d) $4p$ or $(4p+1)$
- 43.** If r and s are any real numbers such that $0 \leq s \leq 1$ and $r + s = 1$, then what is the maximum value of the product?
 (a) 1 (b) $\frac{3}{4}$ (c) $\frac{1}{2}$ (d) $\frac{1}{4}$
- 44.** A number when divided by 2, 3 or 5 gives remainder 1. The number is
 (a) 31 (b) 47 (c) 49 (d) 53
- 45.** The largest integer that divides product of any four consecutive integers is
 (a) 4 (b) 6 (c) 12 (d) 24
- 46.** Which one of the following three digit numbers divides 9238 and 7091 with the same remainder in each case?
 (a) 113 (b) 209 (c) 317 (d) 191
- 47.** For a positive integer n , define $d(n)$ = The number of positive divisors of n . What is the value of $d(d(d(12)))$?
 (a) 1 (b) 2 (c) 4 (d) None of these
- 48.** If three sides of a right angled triangle are integers in their lowest form, then one of its sides is always divisible by
 (a) 6 (b) 5 (c) 7 (d) None of these
- 49.** If $-1 \leq x \leq 3$ and $1 \leq y \leq 3$, then the maximum value of $(3y - 4x)$ is
 (a) 18 (b) 13 (c) 5 (d) -6
- 50.** If ' p ' is an integer greater than 3, then on dividing $p^{11} + 1$ by $p - 1$, we would get the remainder as
 (a) 2 (b) 0 (c) -2 (d) -1
- 51.** If A is the set of squares of natural numbers and x and y are any two element of A , then the correct statement is
 (a) $x + y$ belongs to A (b) $x - y$ belongs to A
 (c) $\frac{x}{y}$ belongs to A (d) xy belongs to A
- 52.** A ten-digit number is divisible by 4 as well as by 5. What could be the possible digit at the ten's place in the given number?
 (a) 0, 1, 2, 4, or 6 (b) 1, 2, 4, 6 or 8
 (c) 2, 3, 4, 6 or 8 (d) 0, 2, 4, 6 or 8
- 53.** If $x < 0 < y$, then which one of the following relations is correct?
 (a) $\frac{1}{x^2} < \frac{1}{xy} < \frac{1}{y^2}$ (b) $\frac{1}{x^2} > \frac{1}{xy} < \frac{1}{y^2}$
 (c) $\frac{1}{x} < \frac{1}{y}$ (d) $\frac{1}{x} > \frac{1}{y}$
- 54.** What will be the remainder when 19^{100} is divided by 20?
 (a) 19 (b) 20 (c) 3 (d) 1
- 55.** $7^{12} - 4^{12}$ is exactly divisible by which of the following number?
 (a) 34 (b) 33 (c) 36 (d) 35
- 56.** If $2^p + 3^q = 17$ and $2^{p+2} - 3^{q+1} = 5$, then find the value of p and q .
 (a) -2, 3 (b) 2, -3 (c) 3, 2 (d) 2, 3
- 57.** If $\frac{49}{15} = 3 + \frac{1}{x + \frac{1}{y + \frac{1}{z}}}$, where x, y and z are natural numbers, then what is z equal to?
 (a) 1 (b) 2 (c) 3
 (d) Cannot be determined due to insufficient data
- 58.** The least number which is a perfect square and has 540 as a factor is
 (a) 8100 (b) 6400 (c) 4900 (d) 3600
- 59.** If A is real and $1 + A + A^2 + A^3 = 40$, then A is equal to
 (a) -3 (b) -1 (c) 1 (d) 3
- 60.** How many factors of 1080 are perfect squares?
 (a) 4 (b) 6 (c) 8 (d) 5
- 61.** Consider the following statements:
 I. In a given whole number, if the sum of the odd numbered digit is equal to the sum of even numbered digits, then the number is divisible by 11.
 II. In a given whole number, if the difference of sum of odd numbered digits and even numbered digits is divisible by 11, then the number is divisible by 11.
 Which of the statement(s) given above is/are correct?
 (a) Only I (b) Both I and II
 (c) Only II (d) None of these
- 62.** Consider the following statements:
 A number $a_1a_2a_3a_4a_5$ is divisible by 9, if
 I. $a_1 + a_2 + a_3 + a_4 + a_5$ is divisible by 9.
 II. $a_1 - a_2 + a_3 - a_4 + a_5$ is divisible by 9.
 Which of the statement(s) given above is/are correct?
 (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II
- 63.** Consider the following statements:
 If p is a prime such that $p + 2$ is also a prime, then
 I. $p(p + 2) + 1$ is a perfect square.
 II. 12 is a divisor of $p + (p + 2)$, if $p > 3$

Which of the statement(s) given above is/are correct?

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

64. Consider the following statements:

- I. If x and y are composite integers, so also is $x + y$.
II. If x and y are composite integers and $x > y$, then $x - y$ is also a composite integer.

III. If x and y are composite integers, so also in xy .

Which of the statement(s) given above is/are correct?

- (a) I, II and III (b) I and II
(c) Only III (d) None of these

65. Consider the following statements:

- I. The product of any three consecutive integers is divisible by 6.
II. Any integer can be expressed in one of the three forms $3k$, $3k + 1$, $3k + 2$, where k is an integer.

Which of the statement(s) given above is/are correct?

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

66. Consider the following statements:

- I. A natural number is divisible by 2, if its last digit is divisible by 2.
II. A natural number is divisible by 2, if its last digit is either zero or 2.
III. A natural number is divisible by 2, if its last digit is even.

Which of the statements given above are correct?

- (a) I and II (b) I and III (c) II and III (d) All of these

67. Consider the following statements for natural numbers a , b and c :

- I. If ' a ' is divisible by ' b ' and ' b ' is divisible by ' c ', then ' a ' must be divisible by ' c '.
II. If ' a ' is a factor of both ' b ' and ' c ', then ' a ' must be a factor of ' $b + c$ '.
III. If ' a ' is a factor of both ' b ' and ' c ', then ' a ' must be a factor of ' $b - c$ '.

Which of the statements given above are correct?

- (a) I, II and III (b) I and II
(c) II and III (d) None of these

68. Consider the following statements:

- I. Set of positive powers of 2 is closed under multiplication.
II. The set $\{1, 0, -1\}$ is closed under multiplication.
III. The number 35 has exactly four divisors.
IV. The set $\{1, 0, -1\}$ is closed under addition.

Which of the statements given above are correct?

- (a) I, II and III are true (b) Only III
(c) Only IV (d) All of these

PREVIOUS YEARS QUESTIONS

69. What number should be added to 231228 to make it exactly divisible by 33? **2012 I**
(a) 1 (b) 2 (c) 3 (d) 4

70. If a positive integer leaves remainder 28 when divided by 143, then what is the remainder obtained on dividing the same number by 13? **2012 I**
(a) 0 (b) 2 (c) 9 (d) 10

71. Which one of the following is neither prime number nor composite number? **2012 II**
(a) 1 (b) 2 (c) 3 (d) None of these

72. Which one of the following has least number of divisors? **2012 II**
(a) 88 (b) 91 (c) 96 (d) 99

73. How many numbers between -11 and 11 are multiples of 2 or 3? **2012 II**
(a) 11 (b) 14 (c) 15 (d) None of these

74. Consider the following statements
I. If n is a prime number greater than 5, then $n^4 - 1$ is divisible by 2400.
II. Every square number is of the form $5n$ or $(5n - 1)$ or $(5n + 1)$, where n is a whole number.

Which of the statement(s) given above is/are correct? **2012 II**

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

75. If N , $(N + 2)$ and $(N + 4)$ are prime numbers, then the number of possible solutions for N are **2013 I**
(a) 1 (b) 2 (c) 3 (d) None of these

76. The two-digit number, which when divided by sum of the digits and product of the digits, respectively leaves the same remainder the number is and the difference of quotients is one. **2013 I**
(a) 14 (b) 23 (c) 32 (d) 41

77. If x is positive even integer and y is negative odd integer, then x^y is **2013 I**
(a) odd integer (b) even integer
(c) rational number (d) None of these

78. Consider the following statements
I. There is a finite number of rational numbers between any two rational numbers.
II. There is an infinite number of rational numbers between any two rational numbers.
III. There is a finite number of irrational number between any two rational numbers.

Which of the statement(s) given above is/are correct? **2013 I**

- (a) Only I (b) Only II
(c) Only III (d) Both II and III

- 79.** If k is a positive integer, then every square integer is of the form **2013 II**
 (a) $4k$ (b) $4k$ or $4k + 3$
 (c) $4k + 1$ or $4k + 3$ (d) $4k$ or $4k + 1$
- 80.** If b is the largest square divisor of c and a^2 divides c , then which one of the following is correct (where, a , b and c are integers)? **2013 II**
 (a) b divides a (b) a does not divide b
 (c) a divides b (d) a and b are coprime
- 81.** Every prime number of the form $3k + 1$ can be represented in the form $6m + 1$ (where, k and m are integers), when **2013 II**
 (a) k is odd
 (b) k is even
 (c) k can be both odd and even
 (d) No such form is possible
- 82.** Consider the following statements
 I. 7710312401 is divisible by 11.
 II. 173 is a prime number.
 Which of the statement(s) given above is/are correct? **2013 II**
 (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II
- 83.** Consider the following statements
 I. To obtain prime numbers less than 121, we are to reject all the multiples of 2, 3, 5 and 7.
 II. Every composite number less than 121 is divisible by a prime number less than 11.
 Which of the statement(s) given above is/are correct? **2013 II**
 (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II
- 84.** Consider the following statements
 I. No integer of the form $4k + 3$, where k an integer, can be expressed as the sum of two squares.
 II. Square of an odd integer can expressed in the form $8k + 1$, where k is an integer.
 Which of the statement(s) given above is/are correct? **2014 I**
 (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II
- 85.** What is $26^2 + 97^2$ equal to? **2014 I**
 (a) $27^2 + 93^2$ (b) $34^2 + 93^2$ (c) $82^2 + 41^2$ (d) $79^2 + 62^2$
- 86.** If n is a whole number greater than 1, then $n^2(n^2 - 1)$ is always divisible by **2014 I**
 (a) 12 (b) 24 (c) 48 (d) 60
- 87.** What is the remainder when $(17^{23} + 23^{23} + 29^{23})$ is divided by 23? **2014 II**
 (a) 0 (b) 1 (c) 2 (d) 3
- 88.** What is the remainder when $(1235 \times 4523 \times 2451)$ is divided by 12? **2014 II**
 (a) 1 (b) 3 (c) 5 (d) 7
- 89.** What is the number of divisors of 360? **2014 II**
 (a) 12 (b) 18
 (c) 24 (d) None of these
- 90.** The multiplication of a three-digit number $XY5$ with digit Z yields $X215$. What is $X + Y + Z$ equal to? **2014 II**
 (a) 13 (b) 15 (c) 17 (d) 18
- 91.** How many pairs of X and Y are possible in the number $763X4Y2$, if the number is divisible by 9? **2014 II**
 (a) 8 (b) 9 (c) 10 (d) 11
- 92.** If $a_n = 3 - 4n$, then what is $a_1 + a_2 + a_3 + \dots + a_n$ equal to? **2014 II**
 (a) $-n(4n - 3)$ (b) $-n(2n - 1)$ (c) $-n^2$ (d) $-n(2n + 1)$
- 93.** Consider all those two-digit positive integers less than 50, which when divided by 4 yield unity as remainder. What is their sum? **2014 II**
 (a) 310 (b) 314 (c) 218 (d) 323
- 94.** p, q and r are prime numbers such that $p < q < r < 13$. In how many cases would $(p + q + r)$ also be a prime number? **2014 II**
 (a) 1 (b) 2 (c) 3 (d) None of these
- 95.** The digit in the units place of the product $81 \times 82 \times 83 \times 84 \times \dots \times 99$ is **2015 I**
 (a) 0 (b) 4 (c) 6 (d) 8
- 96.** What is the remainder obtained when $1421 \times 1423 \times 1425$ is divided by 12? **2015 I**
 (a) 1 (b) 2 (c) 3 (d) 4
- 97.** What is the remainder when 4^{96} is divided by 6? **2015 I**
 (a) 4 (b) 3
 (c) 2 (d) 1
- 98.** What is the maximum value of m , if the number $N = 35 \times 45 \times 55 \times 60 \times 124 \times 75$ is divisible by 5^m ? **2015 I**
 (a) 4 (b) 5
 (c) 6 (d) 7
- 99.** If $a - b = 4$ and $a^2 + b^2 = 40$, where a and b are positive integers, then $a^3 + b^6$ is equal to **2015 I**
 (a) 264 (b) 280
 (c) 300 (d) 324
- 100.** If n is a natural number and $n = p_1^{x_1} p_2^{x_2} p_3^{x_3}$, where p_1, p_2, p_3 are distinct prime factors, then the number of prime factors for n is **2015 I**
 (a) $x_1 + x_2 + x_3$ (b) $x_1 x_2 x_3$
 (c) $(x_1 + 1)(x_2 + 1)(x_3 + 1)$ (d) None of these

- 101.** If $\frac{37}{13} = 2 + \frac{1}{x + \frac{1}{y + \frac{1}{z}}}$, where x, y and z are natural numbers, then what is z equal to? **2015 I**
 (a) 1 (b) 2 (c) 3
 (d) Cannot be determined due to insufficient data
- 102.** A student was asked to multiply a number by 25. He instead multiplied the number by 52 and got the answer 324 more than the correct answer. The number to be multiplied was **2015 I**
 (a) 12 (b) 15 (c) 25 (d) 32
- 103.** The number of pairs (x, y) , where x, y are integers satisfying the equation $21x + 48y = 5$, is **2015 II**
 (a) Zero (b) One (c) Two (d) Infinity
- 104.** The largest natural number which divides every natural number of the form $(n^3 - n)(n - 2)$, where n is a natural number greater than 2, is **2015 II**
 (a) 6 (b) 12 (c) 24 (d) 48
- 105.** The digit in the units place of the resulting number of the expression $(234)^{100} + (234)^{101}$, is **2015 II**
 (a) 6 (b) 4 (c) 2 (d) 0
- 106.** A number when divided by 7 leaves a remainder 3 and the resulting quotient, when divided by 11 leaves a remainder 6. If the same number when divided by 11 leaves a remainder m and the resulting quotient when divided by 7 leaves a remainder n . What are the values of m and n , respectively? **2015 II**
 (a) 1 and 4 (b) 4 and 1 (c) 3 and 6 (d) 6 and 3
- 107.** The seven digit number $876p37q$ is divisible by 225. The values of p and q can be respectively **2015 II**
 (a) 9, 0 (b) 0, 0 (c) 0, 5 (d) 5, 9
- 108.** Let x and y be positive integers such that $x > y$. The expressions $3x + 2y$ and $2x + 3y$, when divided by 5 leave remainders 2 and 3, respectively. What is the remainder when $(x - y)$, is divided by 5? **2015 II**
 (a) 4 (b) 2 (c) 1 (d) 0
- 109.** The sum of first 47 terms of the series $\frac{1}{4} + \frac{1}{5} - \frac{1}{6} - \frac{1}{4} - \frac{1}{5} + \frac{1}{6} + \frac{1}{4} + \frac{1}{5} - \frac{1}{6} - \dots$, is **2015 II**
 (a) 0 (b) $-\frac{1}{6}$ (c) $\frac{1}{6}$ (d) $\frac{9}{20}$
- 110.** The value of the expression $\frac{(243 + 647)^2 + (243 - 647)^2}{243 \times 243 + 647 \times 647}$ is equal to **2016 I**
 (a) 0 (b) 1 (c) 2 (d) 3
- 111.** What is the maximum value of m , if the number $N = 90 \times 42 \times 324 \times 55$ is divisible by 3^m ? **2016 I**
 (a) 8 (b) 7 (c) 6 (d) 5
- 112.** $7^{10} - 5^{10}$ is divisible by **2016 I**
 (a) 5 (b) 7 (c) 10 (d) 11
- 113.** If $a^3 = 117 + b^3$ and $a = 3 + b$, then the value of $a + b$ is (given that $a > 0$ and $b > 0$) **2016 I**
 (a) 7 (b) 9 (c) 11 (d) 13
- 114.** Let m be a non-zero integer and n be a positive integer. Let R be the remainder obtained on dividing the polynomial $x^n + m^n$ by $(x - m)$. Then, **2016 I**
 (a) R is a non-zero even integer
 (b) R is odd, if m is odd
 (c) $R = s^2$ for some integer s , if n is even
 (d) $R = t^3$ for some integer t , if 3 divides n
- 115.** If $\frac{61}{19} = 3 + \frac{1}{x + \frac{1}{y + \frac{1}{z}}}$ where x, y and z are natural numbers, then what z is equal to? **2016 I**
 (a) 1 (b) 2 (c) 3 (d) 4
- 116.** Let x and y be positive integers such that x is prime and y is composite. Which of the following statements are correct?
 I. $(y - x)$ can be an even integer.
 II. xy can be an even integer.
 III. $0.5(x + y)$ can be an even integer.
 Select the correct answer using the code given below **2016 I**
 (a) I and II (b) II and III
 (c) I and III (d) All of these
- 117.** Consider the following statements:
 I. Every natural number is a real number.
 II. Every real number is a rational number.
 III. Every integer is real number.
 IV. Every rational number is a real number.
 Which of the above statements are correct?
 (a) I, II and III (b) I, III and IV **2016 I**
 (c) II and III (d) III and IV
- 118.** Consider the following statements in respect of two different non-zero integers p and q .
 I. For $(p + q)$ to be less than $(p - q)$, q must be negative.
 II. For $(p + q)$ to be greater than $(p - q)$, both p and q must be positive.
 Which of the above statements is/are correct?
 (a) Only I (b) Only II **2016 I**
 (c) Both I and II (d) Neither I nor II

119. If a and b are negative real numbers and c is a positive real number, then which of the following is/are correct?

- I. $a - b < a - c$
 II. If $a < b$, then $\frac{a}{c} < \frac{b}{c}$
 III. $\frac{1}{b} < \frac{1}{c}$

Select the correct answer using the code given below

☑ 2016 I

- (a) Only I (b) Only II
 (c) Only III (d) II and III

120. If m and n are distinct natural numbers, then which of the following is/are integer/integers?

- I. $\frac{m}{n} + \frac{n}{m}$ II. $mn\left(\frac{m}{n} + \frac{n}{m}\right)(m^2 + n^2)^{-1}$
 III. $\frac{mn}{m^2 + n^2}$

Select the correct answer using the code given below

☑ 2016 (I)

- (a) I and II (b) Only II
 (c) II and III (d) Only III

ANSWERS

1		2		3		4		5		6		7		8		9		10	
11		12		13		14		15		16		17		18		19		20	
21		22		23		24		25		26		27		28		29		30	
31		32		33		34		35		36		37		38		39		40	
41		42		43		44		45		46		47		48		49		50	
51		52		53		54		55		56		57		58		59		60	
61		62		63		64		65		66		67		68		69		70	
71		72		73		74		75		76		77		78		79		80	
81		82		83		84		85		86		87		88		89		90	
91		92		93		94		95		96		97		98		99		100	
101		102		103		104		105		106		107		108		109		110	
111		112		113		114		115		116		117		118		119		120	

HINTS AND SOLUTIONS

- (a) $(a + b)$ always represent a natural number $\forall a, b \in \mathbb{N}$.
- (d) As, $y = qx + r$, so $0 \leq r < x$.
- (a) As p is prime, so p/a or p/b .
- (c) Consider the given options
 - 161 is divisible by 7. Hence, 161 is not a prime number.
 - Since, $1 + 7 + 1 = 9$. Therefore, 171 is divisible by 3. Hence, 171 is not a prime number.
 - $173 < (14)^2$
173 is not divisible by any of the numbers 2, 3, 5, 7, 11, 13. Hence, 173 is a prime number.
 - 221 is divisible by 13. Hence, 221 is not a prime number.
- (c) Consider, $2, 4 \in \mathbb{N}$
So, $\sqrt{4} = 2$, a natural number
and $\sqrt{2}$ = irrational number.

- (b) We know that, the product of a rational number and an irrational number is an irrational number.
- (c) If $x = 2$, then, $x, x + 1$ and $x + 3$ are all prime numbers.
- (c) $\because \text{Dividend} = D \times Q + R$
Given, $D = 5Q$ and $D = 2R$
When $R = 15, D = 2 \times 15 = 30$
 $\therefore Q = \frac{D}{5} = \frac{30}{5} = 6$
 $\therefore \text{Dividend} = 30 \times 6 + 15 = 195$
- (d) As, $4 + 2 + * + 8 = 14 + *$; $42 * 8$ is divisible by 9, if $14 + *$ divisible by 9.
So, $14 + * = 18$ (nearest multiple of 9)
 $\Rightarrow * = 18 - 14 = 4$
- (d) Here, $\frac{16}{7} \times \frac{16}{7} - \frac{*}{7} \times \frac{9}{7} + \frac{9}{7} \times \frac{9}{7} = 1$
Put $* = x$

$$\Rightarrow \frac{256}{49} - \frac{9x}{49} + \frac{81}{49} = 1$$

$$\Rightarrow 256 + 81 - 9x = 49$$

$$\Rightarrow 9x = 288 \Rightarrow x = 32$$

$$\Rightarrow 9x = 288 \Rightarrow x = 32$$

$$\begin{aligned} \text{11. (a) By BODMAS, } 10 \div 4 + 6 \times 4 \\ = 10 \div (4 + 6) \times 4 \\ = 10 \div 10 \times 4 = 1 \times 4 = 4 \end{aligned}$$

$$\begin{aligned} \text{12. (a) } \frac{a}{4} = \frac{b}{5} = \frac{c}{6} = k \text{ (say)} \\ \Rightarrow a = 4k, b = 5k \text{ and } c = 6k \\ \text{So, } \frac{a+b+c}{b} = \frac{4k+5k+6k}{5k} = \frac{15k}{5k} = 3 \end{aligned}$$

$$\begin{aligned} \text{13. (d) } [a - b + (b - a)] \\ - [2a - 2b + (b - 2a)] \\ \Rightarrow [a - b + b - a] - [2a - 2b + b - 2a] \\ \Rightarrow 0 - [2a - 2a - b] = 0 + b = b \end{aligned}$$

14. (b) $10^k = \frac{0.0003325}{3.325}$ (given)

$$= \frac{3.325 \times 10^{-4}}{3.325} = 10^{-4}$$

$$\Rightarrow 10^k = 10^{-4}$$

$$\therefore k = -4$$

15. (c) $\sqrt{0.04 \times 0.4 \times x} = 0.4 \times 0.04 \times \sqrt{y}$ (given)

$$\therefore \frac{\sqrt{x}}{\sqrt{y}} = \frac{0.4 \times 0.04}{\sqrt{0.04 \times 0.4}}$$

$$\frac{x}{y} = \frac{0.4 \times 0.4 \times 0.04 \times 0.04}{0.04 \times 0.4} = 0.016$$

(squaring both sides)

16. (b) Since, unit digit in 7^4 is 1.

$$\therefore 7^{168} = (7^4)^{42} \text{ give unit digit 1.}$$

$$\therefore 7^{170} = 7^{168} \times 7^2 \text{ gives the unit digit as } 1 \times 7 \times 7 = 9$$

17. (c) Unit place in $7^4 = 1$, so unit place in 7^{68} is 1.

$$\therefore \text{Unit place in } 7^{68} \times 7^3 = 3. \text{ Similarly, unit place in } 6^{59} \text{ is 6 and unit place in } 3^4 \text{ is 1, so unit place in } 3^{65} = 3^{64} \times 3 = 3.$$

$$\therefore \text{Unit place in } 7^{71} \times 6^{59} \times 3^{65} \text{ is the unit place of } 3 \times 6 \times 3 = 4$$

18. (b) As, n is divided by 4 the remainder is 3, so

$$n = 4q + 3, \text{ where } q \text{ is quotient.}$$

$$\Rightarrow 2n = 8q + 6$$

$$\Rightarrow 2n = (8k + 4) + 2 = 4(2k + 1) + 2$$

$$\text{So, if } 2n \text{ is divided by 4 the quotient is } 2k + 1 \text{ and remainder is 2.}$$

19. (c) Given, $\frac{1 - \frac{1}{6} - \frac{1}{48}}{\frac{1}{4} - \left(\frac{1}{6} - \frac{1}{48}\right)} \div \frac{1 \times \frac{1}{6} - \frac{1}{48}}{\frac{1}{4} \times \left(\frac{1}{6} - \frac{1}{48}\right)}$

$$= x$$

$$\Rightarrow x = \frac{12 - 8 - 1}{48} \div \frac{1 - \frac{1}{24}}{\frac{1}{4} \times \left(\frac{8 - 1}{48}\right)}$$

$$\Rightarrow x = \frac{\frac{3}{5}}{\frac{48}{5} \div \frac{48}{7 \times 48}}$$

$$\Rightarrow x = \frac{3}{5} \div \frac{4}{7} \Rightarrow x = \frac{21}{20}$$

20. (b) Clearly, absolute value is defined by $|x| = -x$.

21. (a) $\frac{x-y}{x+y} = \frac{x/y-1}{x/y+1} = \frac{3/5-1}{3/5+1}$

$$= \frac{(3-5)/5}{(3+5)/5} = \frac{-2}{8} = -\frac{1}{4}$$

22. (a) As, $\left|\frac{7-x}{3}\right| < 2$

$$\Rightarrow \frac{7-x}{3} < 2 \text{ or } -\left(\frac{7-x}{3}\right) < 2$$

$$[\because |x| < a \Rightarrow x < a \text{ or } -x < a]$$

$$\Rightarrow 7-x < 6 \text{ or } \frac{x-7}{3} < 2$$

$$\Rightarrow -x < -1 \text{ or } x-7 < 6$$

$$\Rightarrow x > 1 \text{ or } x < 13$$

$$\Rightarrow 1 < x < 13$$

23. (d) A number is divisible by 25 when its last 2 digits are either zero or divisible by 25.

24. (a) $\frac{(2.48)^2 - (1.52)^2}{0.96}$

$$= \frac{(2.48 - 1.52)(2.48 + 1.52)}{0.96}$$

$$[\because a^2 - b^2 = (a-b)(a+b)]$$

$$= \frac{0.96 \times 4}{0.96} = 4$$

25. (d) Given, number is 222222.

$$\text{Here, sum of digits}$$

$$= 2 + 2 + 2 + 2 + 2 + 2 = 12,$$

$$\text{which is divisible by 3. So, given number is divisible by 3.}$$

$$\text{Now, sum of odd terms of digits - Sum of even terms of digits} = 6 - 6 = 0,$$

$$\text{it is divisible by 11.}$$

$$\text{Also, in a number if a digit is repeated six times, then the number is divisible by 7, 11 and 13.}$$

$$\text{Hence, the given number is divisible by 3, 7 and 11.}$$

26. (a) Required numbers are of the form of $12q - 2$

$$\text{i.e. } 10, 22, 34, 46, 58, 70, 82, 94$$

$$\therefore \text{Total sum} = 10 + 22 + 34 + 46 + 58 + 70 + 82 + 94 = 416$$

27. (d) The last digit in the expansion of $(2457)^{754}$ is equal to last digit of $(7)^{754}$.

$$(7)^{754} = (7^4)^{188} \times 7^2 = 1 \times 7^2$$

$$= (7)^2 = 49$$

$$\text{Hence, last digit is 9.}$$

28. (d) On taking option (d),

$$\text{The reverse digit of 451 is 154.}$$

$$\text{Now, } 154 + 297 = 451 \text{ is equal to the original number.}$$

29. (d) Let $a = 7p + 5$ and $b = 7q + 4$

$$\text{where, } p \text{ and } q \text{ are natural numbers.}$$

$$\therefore ab = (7p + 5)(7q + 4)$$

$$ab = 49pq + (4p + 5q)7 + 20$$

$$= 7(7pq + 4p + 5q) + 7 \times 2 + 6$$

$$\text{when } ab \text{ is divided by 7, we get the remainder 6.}$$

30. (b) If k is any even positive integer, then $k^2 + 2k$ is divisible by 8 but may not be divisible by 24.

$$\text{Let } k = 2m, m \in \mathbb{N}, \text{ then}$$

$$k^2 + k \cdot 2 = 4m^2 + 4m = 4m(m + 1)$$

$$\text{which is divisible by 8.}$$

31. (c) Given number is 2784936.

$$\text{Sum of digits at odd places} = 25 \text{ and sum of digits at even places} = 14$$

$$\therefore \text{Difference} = 25 - 14 = 11$$

$$\text{So, number is divisible by 11.}$$

$$\text{Also last three digits of } 2784936 \text{ i.e. } 936 \text{ is divisible by 8.}$$

$$\text{Hence, } 2784936 \text{ is divisible by both 8 and 11 i.e. } 88.$$

32. (c) A number divisible by 88, if it is divisible by 8 and 11.

$$\text{In the given options } 9944 \text{ and } 8888 \text{ are divisible by 88. Hence, maximum number is } 9944.$$

33. (d) $\because \frac{1}{4} = 0.25$ and $\frac{3}{4} = 0.75$

$$\text{Only option (d) lies between } 0.25 \text{ and } 0.75. \text{ Since } \frac{252}{1000} = 0.252$$

$$\text{and } \frac{748}{1000} = 0.748$$

34. (b) Difference of sums of even and odd places digit of 1254934

$$= (1 + 5 + 9 + 4) - (2 + 4 + 3)$$

$$= 19 - 9 = 10$$

$$\text{This number will be divisible by 11, after adding } x, \text{ if } x = 1.$$

$$\text{Also, the sum of digits of } 1254934$$

$$= 1 + 2 + 5 + 4 + 9 + 3 + 4 = 28$$

$$1254934 \text{ will be divisible by 3, after adding } y, \text{ if } y = -1$$

35. (d) Last digit in the expansion of 3^{4798}

$$= \text{Last digit in the expansion of } (3^4)^{1199} \cdot 3^2$$

$$= \text{Last digit in the expansion of } 3^2 = 9$$

36. (b) The given number is divisible by 9, if sum of the digits is divisible by 9.

$$\text{Here, sum of digits}$$

$$= 8 + 4 + 7 + 0 + 5 + 2 + \otimes = 26 + \otimes$$

$$\text{If } \otimes = 1, \text{ then } 26 + 1 = 27 \text{ is divisible by 9.}$$

37. (c) The total number of three-digit numbers with unit digit 7 and divisible by 11 are 187, 297, 407, 517, 627, 737, 847, 957.

$$\therefore \text{Total numbers} = 8$$

38. (b) Apply hit and trial method from the given option. As, here when $a = 11$, $b = 2$, then

$$3 \frac{7}{a} \times b \frac{3}{15} = 3 \frac{7}{11} \times 2 \frac{3}{15} \\ = \frac{40}{11} \times \frac{33}{15} = 8$$

39. (d) Since, $a < b \Rightarrow a - b < 0$. Also, $c < 0$
 $\therefore (a - b)c > 0 \Rightarrow ac - bc > 0 \Rightarrow ac > bc$

40. (c) When we divide a positive integer by another positive integer, the resultant will be a rational number i.e. in the form of p/q , where p and q are positive integers and $q \neq 0$.

41. (d) $2^{12n} - 6^{4n} = (2^{12})^n - (6^4)^n$
 $= (4096)^n - (1296)^n$
 $\therefore 2^{12n} - 6^{4n} = (4096 - 1296)k$
 As $x^n - y^m$ is always divisible by $(x - m)$
 $= 2800(k)$

Hence, last two digits are always be zero.

42. (c) We know that
 $4 = 3p + 1$, for $p = 1$, $9 = 3p$, for $p = 3$
 $16 = 3p + 1$, for $p = 5$,
 $25 = 3p + 1$, for $p = 8$
 $36 = 3p$, for $p = 12$
 Hence, every square integer is of form either $3p$ or $3p + 1$.

43. (d) Given, $r + s = 1$
 For maximum product, $r = s = \frac{1}{2}$
 $\therefore rs = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

44. (a) Required number
 $= [\text{LCM of } 2, 3 \text{ and } 5] + 1$
 $= 30 + 1 = 31$

45. (d) The largest integer that divides product of any four consecutive integers is 24.
 e.g. 1, 2, 3, 4 are four consecutive integers.
 Multiplication $= 1 \times 2 \times 3 \times 4 = 24$
 which divided by 24.

46. (a) When we divide the number 9238 and 7091 by 113, we get the same remainder 85.

47. (d) $d(d(d(12))) = d(d(6))$
 $[\because \text{positive integer divisor of } 12 = 1, 2, 3, 4, 6, 12]$
 $d(d(6)) = d(4)$
 $[\because \text{positive integer divisor of } 6 = 1, 2, 3, 6]$
 and $d(4) = 3$
 $[\because \text{positive integer divisor of } 4 = 1, 2, 4]$

48. (b) Given, by pythagoras theorem,
 $(3)^2 + (4)^2 = (5)^2$
 Let the sides of a right triangle be 3, 4, 5.

Hence, one of its sides is always divisible by 5.

49. (b) $\therefore 1 \leq y \leq 3 \Rightarrow 3 \leq 3y \leq 9 \dots(i)$
 and $-1 \leq x \leq 3 \Rightarrow -4 \leq 4x \leq 12$
 $\Rightarrow -12 \leq -4x \leq 4 \dots(ii)$
 From Eqs. (i) and (ii), we get
 $-9 \leq 3y - 4x \leq 13$

Maximum value is 13.

50. (a) When $p^{11} + 1$ is divided by $(p - 1)$, then the remainder is
 $(1)^{11} + 1 = 1 + 1 = 2$

51. (d) Let $x = a^2$ and $y = b^2$ for some $a, b \in N$, then $xy = a^2b^2 = (ab)^2$, where $ab \in N$.

So, $xy \in A$.

But $\frac{x}{y} = \frac{a^2}{b^2} = \left(\frac{a}{b}\right)^2 \notin A$ as $\frac{a}{b} \notin N$.

52. (d) Since, a ten-digit number is divisible by 4 as well as by 5, then this number must be divisible by 20.

We known that any number is divisible by 20, if last two digits is divisible by 20. It means unit place will be zero and ten's place may be 0, 2, 4, 6 or 8.

53. (c) As, $x < 0 < y$ [given]
 $\Rightarrow x < 0$ and $y > 0$
 $\therefore \frac{1}{x} < 0$ and $\frac{1}{y} > 0 \therefore \frac{1}{x} < \frac{1}{y}$

54. (d) On division of $(19)^n$ by 20, we get remainder either 19 or 1.
 Since, last digit of $(19)^{100}$ is 1.
 $\therefore$ Remainder of $\frac{(19)^{100}}{20}$ is 1.

55. (b) We know that $(x^n - y^n)$ is divisible by $(x - y)$ for all 'n' and is divisible by $(x + y)$ for even 'n'.
 $\therefore (7^{12} - 4^{12})$ is divisible by $(7 + 4) = 11$ and $(7 - 4) = 3$
 Thus, $(7^{12} - 4^{12})$ is divisible by 33.

56. (c) Here, $2^p + 3^q = 17 \dots(i)$
 $2^{p+2} - 3^{q+1} = 5$
 or $4 \cdot 2^p - 3 \cdot 3^q = 5 \dots(ii)$
 On multiplying Eq. (i), by 3 and adding it with Eq. (ii), we get
 $7 \cdot 2^p = 56$, $2^p = 8 = 2^3 \Rightarrow p = 3$
 Put $p = 3$ in Eq. (i), we get
 $2^3 + 3^q = 17$, $3^q = 17 - 8 = 9 = 3^2$
 $\therefore q = 2$

$$57. (c) \frac{49}{15} = 3 + \frac{1}{x + \frac{1}{y + \frac{1}{z}}}$$

$$\Rightarrow \frac{1}{x + \frac{1}{yz + 1}} = \frac{49}{15} - 3 = \frac{4}{15}$$

$$\Rightarrow \frac{1}{x + \frac{1}{yz + 1}} = \frac{4}{15}$$

$$\Rightarrow \frac{yz + 1}{xyz + x + z} = \frac{4}{15}$$

$$\Rightarrow 15(yz + 1) = 4(xyz + x + z)$$

$$\Rightarrow 15yz + 15 = 4xyz + 4x + 4z$$

$$\Rightarrow 15 - 4x = (4xy + 4 - 15y)z$$

$$z = \frac{15 - 4x}{(4xy + 4 - 15y)}$$

As x, y and z are natural numbers.

Again, let $x = 3$ and $y = 1$

$$\text{Then, } z = \frac{15 - 4(3)}{4(3)(1) + 4 - 15(1)}$$

$$= \frac{15 - 12}{12 + 4 - 15} = \frac{3}{1}$$

$\Rightarrow z = 3$, which is a natural number.

58. (a) Required number is
 $x^2 = 540 \times q = 3 \times 3 \times 3 \times 2 \times 2 \times 5 \times q$
 In order to make x^2 a perfect square, the least number we must have to put is
 $q = 3 \times 5 = 15$
 $\therefore x^2 = 540 \times 15 = 8100$

59. (d) We have, $1 + A + A^2 + A^3 = 40$
 $\Rightarrow (1 + A) + A^2(1 + A) = 40$
 $\Rightarrow (1 + A)(1 + A^2) = 40$
 Only $A = 3$ satisfies the above equation,
 so, $A = 3$.

60. (a) We have, $1080 = 2^3 \times 3^3 \times 5$
 For any perfect square, all the powers of the primes have to be even numbers.
 So, if the factor is of the form

$$2^a \times 3^b \times 5^c$$

The values a can be 0 and 2, b can be 0 and 2 and c can take the value 0.

Totally, there are 4 possibilities.

61. (b) Clearly, both statements satisfies divisibility rule of 11.

62. (a) As, we know that a number $a_1 a_2 a_3 a_4 a_5$ is divisible by 9, if sum of the digits, i.e. $a_1 + a_2 + a_3 + a_4 + a_5$ is divisible by 9. Hence, only statement I is true.

63. (c) Taking $p = 11$
 $p + 2 = 13$ [a prime number]
 I. $11 \times 13 + 1 = 144$ [a square number]
 II. $11 + 13 = 24$ [12 is a divisor of 24]
 Hence, both statements I and II are correct.
64. (c) I. If $x = 15$ and $y = 14$, then $x + y = 15 + 14 = 29$, which is a prime number. So, if x and y are composite, then $x + y$ is not always composite.
 II. If $x = 15$ and $y = 14$, then $x - y = 15 - 14 = 1$ which is neither prime nor composite, hence again $x - y$ is not always composite.
 III. Third condition is satisfied for all measure.
 Hence, only III is correct.
65. (c) I. The product of any three consecutive integers is divisible by 6.
 II. Here, $3k = \{\dots - 6, -3, 0, 3, 6, \dots\}$
 $3k + 1 = \{\dots - 5, -2, 1, 4, 7, \dots\}$
 and $3k + 2 = \{\dots - 4, -1, 2, 5, 8, \dots\}$
 $\therefore \{3k, 3k + 1, 3k + 2\}$
 $= \{\dots - 6, -5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5, 6, \dots\}$
 Hence, it is true.
66. (d) All statements are true.
67. (a) (i) If b / a and c / b , then $a = bx$ and $b = cy$ for $x, y \in N$
 $\Rightarrow a = bx = cy(x) = cxy \Rightarrow c / a$
 (ii) If a / b and a / c , then $b = ax$ and $c = ay$ for $xy \in N$,
 $\Rightarrow (b + c) = ax + ay = a(x + y)$
 So, $a / (b + c)$
 (iii) From (ii), we have $(b - c) = a(x - y)$
 $\Rightarrow a / (b - c)$
68. (d) All are true.
69. (c) Given, $33 \mid 231228$ (7006)

$$\begin{array}{r} 231 \\ 228 \\ \hline 198 \\ 198 \\ \hline 30 \end{array}$$

 Now, $33 - 30 = 3$
 So, on adding 3 to 231228, it will be completely divisible by 33.
70. (b) Given, $N = 143k + 28$
 $\Rightarrow N = 143k + 26 + 2$
 $\Rightarrow N = 13(11k + 2) + 2$
 $\therefore$ When the number is divided by 13 the remainder is 2.
71. (a) 1 is neither prime number nor composite number.

72. (b) Here, $88 = 2 \times 2 \times 2 \times 11 = (2)^3 \times (11)^1$
 $91 = (7)^1 \times (13)^1$
 $96 = 2 \times 2 \times 2 \times 2 \times 3$
 $= (2)^5 \times (3)^1$
 and $99 = 3 \times 3 \times 11 = (3)^2 \times (11)^1$
 So, 91 has least number of divisors.
73. (c) Following are the numbers between -11 and 11 which are multiples of 2 or 3.
 $-10, -9, -8, -6, -4, -3, -2,$
 $0, 2, 3, 4, 6, 8, 9, 10$
 So, the numbers of multiples 2 or 3, between -11 and 11 are 15.
74. (b) I. Given, n is a prime number greater than 5.
 Now, $n^4 - 1 = (n^2 - 1)(n^2 + 1)$
 $= (n - 1)(n + 1)(n^2 + 1)$
 Put $n = 11$,
 $n^4 - 1 = (11 - 1)(11 + 1)(121 + 1)$
 [prime number greater than 5]
 $= 10 \times 12 \times 122$
 $= 14640$ which is not divisible by 2400.
 So, statement I is not true.
 II. Every square number can be of the form $5n$ or $(5n \pm 1)$ or $(5n \pm 4)$.
 So, statement II is true.
75. (a) When N is a natural number, then there is only one possible case that N , $(N + 2)$, $(N + 4)$ are prime numbers.
 When $N = 3$, then N , $(N + 2)$, $(N + 4)$
 $= 3, 5, 7$ all are primes numbers.
76. (c) From options,
 (c) $\frac{32}{(3 + 2)} = \frac{32}{5} = 2$ (remainder)
 and $\frac{32}{(3 \times 2)} = \frac{32}{6} = 2$ (remainder)
 $\therefore$ Remainder are same.
 and difference of quotients $= 6 - 5 = 1$
77. (c) If x is a positive even integer and y is negative odd integer, then x^y is a rational number.
78. (b) We know that, between any two rational numbers, there are an infinite number of rational and irrational numbers.
 Hence, only statement II is correct.
79. (d) If k is a positive integer, then every square integer is of the form $4k$ or $4k + 1$, as every square number is either a multiple of 4 or exceeds multiple of 4 by unity.
80. (c) Since, b is largest square divisor of c .
 So, $c = bx$
 [where, x is not a whole square number]
 Also, a^2 divides c .

So, a^2 will divide bx or a will divide b .
 [since, it cannot divide x as it is not a whole square]

81. (b) Every prime number of the form $3k + 1$ can be represented in the form $6m + 1$ only, when k is even.
82. (c) I. In 7710312401, difference between sum of even place digits and the sum of odd place digits 0. So, it is divisible by 11.
 II. Since, $173 < (14)^2$ and it is not divisible by 2, 3, 5, 7, 11 and 13. So, it is a prime number.
 Hence, both statements I and II are correct.
83. (c) Both the statements given are correct. As 121 is the square of 11. So, to obtain prime numbers less than 121, we reject all the multiples of prime numbers less than 11 i.e. 2, 3, 5 and 7. Similarly, every composite number less than 121 is divisible by a prime number less than 11 i.e. 2, 3, 5 or 7.
84. (a) I. $f(k) = 4k + 3$
 For $k = 1$, $f(1) = 4 \times 1 + 3 = 7$
 For $k = 2$, $f(2) = 4 \times 2 + 3 = 11$
 For $k = 3$, $f(3) = 4 \times 3 + 3 = 15$
 Values of $f(k)$ for $k = 1, 2, \dots$ cannot be expressed as sum of two squares, since $1^2 + 2^2 = 5$, $1^2 + 3^2 = 10$,
 $2^2 + 3^2 = 13$.
 II. $f(k) = 8k + 1$
 For $k = 1$, $f(1) = (8 \times 1) + 1 = 9$
 For $k = 2$, $f(2) = (8 \times 2) + 1 = 17$
 For $k = 3$, $f(3) = (8 \times 3) + 1 = 25$
 For $k = 4$, $f(4) = (8 \times 4) + 1 = 33$
 For $k = 5$, $f(5) = (8 \times 5) + 1 = 41$
 $f(k) = 8k + 1$ is square of an odd integer only for some values of k .
 So, only statement I is correct.
85. (d) To check which option is equal to $26^2 + 97^2$, we take the sum of unit digit's square of both number of the question as well as the answer options. Whichever answer option shows the same result will be the answer.
 Here, in $26^2 + 97^2$, $6^2 = 36$
 and $7^2 = 49$
 So, $36 + 49 = 85$
 For option (a), $7^2 + 3^2 = 49 + 9 = 58$
 For option (b), $4^2 + 3^2 = 16 + 9 = 25$
 For option (c), $2^2 + 1^2 = 4 + 1 = 5$
 For option (d), $9^2 + 2^2 = 81 + 4 = 85$
 Only option (d) satisfies the condition.
86. (a) If n is greater than 1, then $n^2(n^2 - 1)$ is always divisible by 12.
Illustration 1 Put $n = 2$, then

$$n^2(n^2 - 1) = (2)^2(2^2 - 1) = 4 \times 3 = 12$$

Illustration 2 Put $n = 3$, then

$$n^2(n^2 - 1) = (3)^2(3^2 - 1) = 9 \times 8 = 72$$

- 87. (a)** We have, $\frac{17^{23} + 29^{23}}{23} + \frac{23^{23}}{23}$
 $(17^{23} + 29^{23})$ is divisible by
 $(17 + 29)$ i.e. 46

So, it is divisible by 23.

Also, 23^{23} is always divisible by 23.

$\therefore$ Remainder = 0

- 88. (b)** Let $E = (1235 \times 4523 \times 2451)$
 $= (12 \times 102 + 11)(12 \times 376 + 11)$
 $\times (12 \times 204 + 3)$

When we divide E by 12, then

Remainder = Remainder when

$11 \times 11 \times 3$ or 363 is divided by $12 = 3$

- 89. (c)** $\because 360 = 2^3 \times 3^2 \times 5^1$
 $\therefore$ Number of divisors
 $= (3 + 1)(2 + 1)(1 + 1)$
 $= 4 \times 3 \times 2 = 24$

- 90. (a)** Given, three-digit number = $XY5$
and $\frac{XY5}{X215} = Z$

Here, Z can take values 1, 3, 5, 7 and 9.

But only 9 satisfies it,

$$\therefore \begin{array}{r} 135 \\ \times 9 \\ \hline 1215 \end{array}$$

then $X = 1, Y = 3$ and $Z = 9$

Now, $X + Y + Z = 1 + 3 + 9 = 13$

- 91. (d)** Given number is $763X4Y2$.
Since, given number is divisible by 9.
 $\therefore 7 + 6 + 3 + X + 4 + Y + 2 = 9k$
 $\Rightarrow 22 + X + Y = 9k$
It is clear that LHS is divisible by 9, if
 $X + Y = 5, 14$

When sum of X and Y is 5, then possible pairs are (1, 4), (4, 1), (2, 3), (3, 2), (0, 5), (5, 0). When sum of X and Y is 14, then possible pairs are (5, 9), (9, 5), (6, 8), (8, 6) and (7, 7).

Hence, the possible pairs are 11.

- 92. (b)** Given, $a_n = 3 - 4n$
 $\therefore \Sigma a_n = \Sigma(3 - 4n)$
 $= 3n - 4 \frac{n \times (n + 1)}{2}$
 $= 3n - 2n^2 - 2n$
 $= n - 2n^2 = -n(2n - 1)$

- 93. (a)** Let the two-digit numbers less than, 50 which when divided by 4 yield unity as remainder be 13, 17, ..., 49.

Here, first term, $a = 13$, common difference, $d = 4$ and $n = 10$

$\therefore$ Required sum = $\frac{n}{2}[2a + (n - 1)d]$

$$= \frac{10}{2}[2 \times 13 + (10 - 1)4]$$

$$= \frac{10}{2}[26 + 36] = \frac{10 \times 62}{2} = 310$$

- 94. (b)** The prime numbers less than 13 are 2, 3, 5, 7, 11.

Also, $p < q < r < 13$

and $p + q + r$ is a prime number.

Hence, only two possible pairs exist i.e. (3, 5, 11) and (5, 7, 11).

- 95. (a)** Product of unit digits
 $= 1 \times 2 \times 3 \times 4 \times 5 \times 6 \times 7 \times 8 \times 9$
 $\times 1 \times 2 \times 3 \times \dots \times 8 \times 9$
 $= 0$

$\therefore$ Required digit in the unit place is 0.

- 96. (c)** See question 88.

- 97. (d)** See example 8.

- 98. (c)** $N = 35 \times 45 \times 55 \times 60 \times 124 \times 75$
 $= 5 \times 7 \times 5 \times 9 \times 5 \times 11 \times 5 \times 12 \times 124$
 $\times 5^2 \times 3$
 $= 5^6 \times 7 \times 9 \times 11 \times 12 \times 124 \times 3$

Hence, the maximum value of m is 6.

- 99. (b)** Given $a - b = 4$... (i)

On squaring both sides, we get

$$(a - b)^2 = (4)^2 \Rightarrow a^2 + b^2 - 2ab = 16$$

$$\Rightarrow 40 - 2ab = 16 \quad [\because a^2 + b^2 = 40]$$

$$\Rightarrow 2ab = 24 \Rightarrow ab = 12$$

$$\therefore a + b = \sqrt{a^2 + b^2 + 2ab}$$

$$= \sqrt{40 + 2 \times 12} = 8$$

$$\therefore a + b = 8 \quad \dots (ii)$$

On solving Eq. (i) and Eq. (ii), we get
 $a = 6$ and $b = 2$

$$\text{Now, } a^3 + b^6 = 6^3 + 2^6 = 2^3 \times 3^3 + 2^6$$

$$= 2^3(3^3 + 2^3)$$

$$= 8(27 + 8) = 8 \times 35 = 280$$

Hence, the value of $a^3 + b^6$ is 280.

- 100. (c)** If factor of given number is of the form of $p_1 \alpha_1, p_2 \alpha_2, \dots, p_n \alpha_n$, then number of prime factors are

$$(\alpha_1 + 1)(\alpha_2 + 1) \dots (\alpha_n + 1)$$

Hence, the prime factor of n are

$$(x_1 + 1)(x_2 + 1) \text{ and } (x_3 + 1)$$

- 101. (b)** See question 57.

- 102. (a)** Let x be the required number.

$$\therefore 52x - 25x = 324 \Rightarrow 27x = 324$$

$$\Rightarrow x = \frac{324}{27} = 12$$

Hence, the required number is 12.

- 103. (a)** $21x + 48y = 5 \Rightarrow 3(7x + 16y) = 5$

If x, y are integer, then LHS of the above equation is multiple of 3, but the RHS of above equation is not multiple of 3.

$\therefore$ There is no any integral values of x and y exist.

- 104. (c)** Let $x = (n^3 - n)(n - 2)$, where $n > 2$
Take $n = 3$, we get

$$x = (3^3 - 3)(3 - 2) = (27 - 3)(1) = 24$$

[which is divisible by 6, 12 and 24]

$$\text{Take } n = 4, \text{ we get } x = (4^3 - 4)(4 - 2)$$

$$= (64 - 4) \times 2 = 120$$

[which is again divisible by 6, 12 and 24]

Now, take $n = 5$, we get

$$x = (5^3 - 5)(5 - 2) = (125 - 5) \times 3$$

$$= 120 \times 3 = 360$$

[which is again divisible by 6, 12 and 24]

Hence, 24 is the largest natural number.

- 105. (d)** We have, $(234)^{100} + (234)^{101}$
 $\Rightarrow (234)^{100}(1 + 234) = 235(234)^{100}$

We know that square of any number having 4 at unit place is a number in which 6 at unit place.

Any exponent of a number 6 at unit place is always 6 at unit place.

$$\therefore (235)(\dots 6) = \dots 30$$

Resulting number have 0 at unit place.

- 106. (a)** Let the number be y .

$$\therefore y = 7q + 3 \text{ and } q = 11p + 6$$

$$\therefore y = 7(11p + 6) + 3$$

$$\Rightarrow y = 77p + 45$$

When divided by 11 remainder is 1 and quotient is $7p + 4$ and when the quotient is divided by 7 remainder is 4.

- 107. (c)** Seven digits number $876p37q$ is divisible by 225, if this number is divisible by 9 and 5.

If this number is divisible by 9.

Then, sum of its digits is divisible by 9.

Now, sum of digits

$$= 8 + 7 + 6 + p + 3 + 7 + q$$

$$= 31 + p + q$$

$$\therefore p + q = 5 \text{ or } p + q = 14$$

$\therefore$ Given number is divisible by 225 when $q = 5$

if $q = 5, p = 0$ or 9.

- 108. (a)** We have, when $3x + 2y$ is divided by 5, remainder is 2.

$$\therefore 3x + 2y = 5q + 2 \quad \dots (i)$$

and when $2x + 3y$ is divided by 5, the remainder is 3.

$$\therefore 2x + 3y = 5m + 3 \quad \dots(ii)$$

Subtract Eq. (ii) from Eq.(i), we get

$$x - y = 5(q - m) - 1$$

$$x - y = 5(q - m) - 5 + 4$$

$$x - y = 5(q - m - 1) + 4$$

$\therefore x - y$ is divided by 5 remainder is 4.

- 109. (b)** The sum of first 47 terms of the series

$$\frac{1}{4} + \frac{1}{5} - \frac{1}{6} - \frac{1}{4} - \frac{1}{5} + \frac{1}{6} + \frac{1}{4} + \frac{1}{5} - \frac{1}{6} - \frac{1}{4} - \frac{1}{5} + \frac{1}{6} \dots 47 \text{ term}$$

It is clear that sum of first 6 term is zero.

Similarly, sum of first 42 terms is zero.

Sum of last 5 terms

$$= \frac{1}{4} + \frac{1}{5} - \frac{1}{6} - \frac{1}{4} - \frac{1}{5} = \frac{-1}{6}$$

$$\begin{aligned} \text{110. (c)} \quad & \frac{(243 + 647)^2 + (243 - 647)^2}{243 \times 243 + 647 \times 647} \\ &= \frac{2[(243)^2 + (647)^2]}{(243)^2 + (647)^2} = 2 \\ & [\therefore (a + b)^2 + (a - b)^2 = 2(a^2 + b^2)] \end{aligned}$$

$$\begin{aligned} \text{111. (b)} \quad & \text{We have, } N = 90 \times 42 \times 324 \times 55 \\ &= 3^2 \times 10 \times 3 \times 14 \times 3^4 \times 4 \times 55 \\ &= 3^7 \times 10 \times 14 \times 4 \times 55 \end{aligned}$$

Hence, N is divisible by 3^7 .

So, the maximum value of m is 7 when N is divisible by 3^m .

$$\begin{aligned} \text{112. (d)} \quad & 7^{10} - 5^{10} = (7^5)^2 - (5^5)^2 \\ &= (7^5 + 5^5)(7^5 - 5^5) \\ &= (16807 + 3125)(7^5 - 5^5) \\ &= 19932 \times (7^5 - 5^5) \end{aligned}$$

Since, 19932 is divisible by 11.

Hence, $7^{10} - 5^{10}$ is divisible by 11.

$$\begin{aligned} \text{113. (a)} \quad & \text{We have, } a - b = 3 \\ & \text{Squaring both sides, we get} \\ & a^2 + b^2 - 2ab = 9 \quad \dots(i) \end{aligned}$$

$$\begin{aligned} \text{Also,} \quad & a^3 - b^3 = 117 \\ \Rightarrow & (a - b)(a^2 + b^2 + ab) = 117 \\ \Rightarrow & a^2 + b^2 + ab = \frac{117}{3} = 39 \quad \dots(ii) \end{aligned}$$

On subtracting Eq. (i) from Eq. (ii)

$$3ab = 30 \Rightarrow ab = 10$$

$$\begin{aligned} \text{Now, } a + b &= \sqrt{(a - b)^2 + 4ab} \\ &= \sqrt{9 + 40} = \sqrt{49} = 7 \end{aligned}$$

- 114. (a)** If R is the remainder obtained by dividing the polynomial $x^n + m^n$ by $(x - m)$, then $(x^n + m^n - R)$ is divisible by $(x - m)$.

$$\begin{aligned} \text{Let } f(x) &= x^n + m^n - R \\ \therefore f(x) &\text{ is divisible by } (x - m). \end{aligned}$$

$$\begin{aligned} \therefore f(m) &= 0 \\ \Rightarrow (m)^n + (m)^n - R &= 0 \\ R &= 2(m)^n \end{aligned}$$

As, m is a non-zero integer and n is a positive integer, then R is a non-zero even integer.

- 115. (c)** See question 57.

- 116. (d)** x is prime and y is composite number.

Since, x is prime number.

$\therefore x$ may be $\{2, 3, 5, 7, 11, \dots\}$ and y may be $\{4, 6, 8, 9, 10, 12, 14, 15, \dots\}$

Here, y may be even or odd number and x is odd number except 2.

$\therefore (y - x)$ can be even number.

Also, xy can be even number.

If y is odd integer and $x \geq 3$.

Then, $0.5(x + y)$ is an even integer.

Hence, statements I, II and III are correct.

- 117. (b)** A real number is collection of all rational and irrational numbers.

So, statement II is false.

- 118. (a)** Given, p and q are non-zero integers.

$$\text{I. } \therefore p + q < p - q$$

$$\Rightarrow q + q < 0$$

$$\Rightarrow 2q < 0$$

$$\Rightarrow q < 0$$

$\therefore q$ must be negative.

Hence, statement I is correct.

$$\text{II. } \therefore p + q > p - q$$

$$\Rightarrow q + q > 0$$

$$\therefore 2q > 0$$

$\therefore q$ must be positive irrespective of p .

Hence, statement II is incorrect.

$$\begin{aligned} \text{119. (d) I. } a - b &< a - c \\ &\Rightarrow b - c > 0 \quad \dots(i) \end{aligned}$$

As, b is negative real number and c is positive real number, then Eq. (i) is not true.

II. If $a < b$, when a and b are negative real numbers and c is a positive real number, then $\frac{a}{c} < \frac{b}{c}$ is always true for $b > a$.

III. $\frac{1}{b} < \frac{1}{c}$ is always true, as c is a positive real number and b is a negative real number.

- 120. (d)** I. If m and n are distinct natural numbers, then $\frac{m}{n} + \frac{n}{m}$ is integer if and only if $m = n$. Hence statement I is incorrect.

$$\begin{aligned} \text{II. } mn \left(\frac{m}{n} + \frac{n}{m} \right) (m^2 + n^2)^{-1} \\ = mn \left(\frac{m^2 + n^2}{mn} \right) \left(\frac{1}{m^2 + n^2} \right) = 1 \end{aligned}$$

Hence, statement II is correct for all values of m and n .

III. Now, $\frac{mn}{m^2 + n^2}$ is a fraction.

So, statement III is incorrect.

02

SEQUENCE AND SERIES

Usually (1-2) questions have been asked from this chapter. Questions are mostly based on relation between arithmetic geometric and harmonic mean.



SEQUENCE

A set of numbers arranged in a definite order according to some definite rule is called a sequence.

e.g. 2, 4, 6, 8, ... is a sequence.

SERIES

If $a_1, a_2, a_3, \dots, a_n$ is a sequence, then the expression $a_1 + a_2 + a_3 + \dots + a_n$ is called the series.

The series is said to be finite or infinite depending upon the last term is given or not. e.g.

(i) $a_1, a_2, a_3, \dots, a_n$ is a finite sequence and is denoted as $\{a_k\}_{k=1}^n$.

(ii) $a_1, a_2, a_3, a_4, \dots$ is an infinite sequence and is denoted by $\{a_n\}_{n=1}^{\infty}$ or simply $\{a_n\}$.

Here, a_1 is called the first term and in case of finite sequence a_n is called the last term.

e.g. 2, 4, 6, 8, ..., 100 is a finite sequence.

1, 2, 3, 4, ... is an infinite sequence.

PROGRESSION

Sequence following certain patterns are called progressions.

e.g. 2, 3, 4, 5, ... is a progression, here each term is increasing by 1.

Arithmetical Progression (AP)

An arithmetic progression is a sequence in which the difference between any term and its preceding term is constant throughout. The constant ' d ' is called the common difference. The first term of an AP is represented by ' a '.

If an AP has first term $= a$ and common difference $= d$, then the general form of an AP is

$$a, a + d, a + 2d, a + 3d, \dots, a + (n-1)d$$

- n th term of an AP is $T_n = a + (n-1)d$.
- Sum of the first n terms of an AP is

$$S_n = \frac{n}{2}[2a + (n-1)d] \text{ or } S_n = \frac{n}{2}[a + l]$$

where, l = last term

EXAMPLE 1. Find the sum of 11 terms of $-7, -2, 3, 8 \dots$

- a. 200 b. 198 c. 326 d. 137

Sol. d. Here, $a = -7$, $l = -7 + (10 \times 5) = 43$

$$\begin{aligned} \therefore S_{11} &= n \left(\frac{a+l}{2} \right) = 11 \times \left(\frac{-7+43}{2} \right) \\ &= 11 \times \frac{36}{2} = 198 \end{aligned}$$

Arithmetic Mean (AM)

When three terms are in AP, then the middle term is called arithmetic mean of the other two.

If a, b and c are in AP, then b is AM of a and c .

Let a and b be two numbers and M be the AM between them. Then, a, M, b are in AP, $M = \frac{a+b}{2}$.

EXAMPLE 2. Find the arithmetic mean (AM) between 3 and 9.

- a. 4 b. 6
c. 8 d. None of these

Sol. b. The arithmetic mean is $\frac{3+9}{2} = \frac{12}{2} = 6$.

So, 3, 6, 9 are in AP.

Geometric Progression

A geometric progression is a progression of numbers, whose first term is non-zero and each of the term is obtained by multiplying its preceding term by a constant quantity. This constant quantity is called the common ratio of the GP.

Thus, if t_1, t_2 and t_3 are in GP, then common ratio

$$r = \frac{\text{Second term}}{\text{First term}} = \frac{t_2}{t_1}$$

If 'a' is the first term and 'r' is the common ratio, then GP can be written as $a, ar, ar^2, ar^3, \dots, ar^{n-1} = (a \neq 0)$

- n th term of a GP is $T_n = ar^{n-1} = l$.
[where, l = last term]

- Sum of the n terms of a GP is $S_n = \frac{a(r^n - 1)}{r - 1}$,

$$\text{if } r > 1 \text{ and } S_n = \frac{a(1-r^n)}{1-r}, \text{ if } r < 1.$$

- Sum of infinite terms of a GP is $S_n = \frac{a}{1-r}$.

Geometric Mean (GM)

If three terms are in GP, then the middle term is called the geometric mean of the other two. If a, b and c are in GP, then b is the GM of a and c .

Let a and b be two numbers and G be the GM between them. Then, a, G, b are in GP, $G = \sqrt{ab}$, $a > 0$, $b > 0$.

EXAMPLE 3. If the 4th, 10th and 16th terms of a GP are x, y and z respectively, then x, y, z are in

- a. AP b. GP
c. AGP d. HP

Sol. b. Let the first term and common ratio be a and r , respectively.

$$\text{Given, } T_4 = x \Rightarrow ar^{4-1} = x \Rightarrow ar^3 = x \quad \dots(i)$$

$$T_{10} = y \Rightarrow ar^{10-1} = y \Rightarrow ar^9 = y \quad \dots(ii)$$

$$T_{16} = z \Rightarrow ar^{16-1} = z \Rightarrow ar^{15} = z \quad \dots(iii)$$

On multiplying Eq. (i), by Eq. (iii), we get

$$ar^3 \times ar^{15} = xz \Rightarrow a^2 r^{18} = xz$$

$$\Rightarrow (ar^9)^2 = xz \Rightarrow y^2 = xz \quad [\text{from Eq. (ii)}]$$

Hence, x, y and z are in GP.

Harmonic Progression (HP)

A sequence is said to be harmonic progression (HP). If the reciprocals of its terms are in arithmetic progression (AP).

e.g. The sequence, $1, \frac{1}{3}, \frac{1}{5}, \frac{1}{7} \dots$ is an HP because the sequence $1, 3, 5, 7 \dots$ is an AP.

If $a_1, a_2, a_3, \dots, a_n$ are in HP, then $\frac{1}{a_1}, \frac{1}{a_2}, \frac{1}{a_3}, \dots, \frac{1}{a_n}$ are in AP.

- n th term of an HP is $T_n = \frac{1}{\frac{1}{a_1} + (n-1)\left(\frac{1}{a_2} - \frac{1}{a_1}\right)}$

Harmonic Mean (HM)

If three terms are in HP, then the middle term is called the harmonic mean of the other two. If a, b and c are in HM, then b is the HM of a and c .

Let a and b be two numbers and H be the HM between them.

Then, a , b and c are in HP

$$\therefore H = \frac{2ab}{a+b}$$

Relation between Arithmetic, Geometric and Harmonic Mean

Let A , G and H be the arithmetic, geometric and harmonic means between a and b , then

$$(i) A \geq G \geq H \quad (ii) G^2 = AH$$

EXAMPLE 4. If a, b, c are in AP, p, q, r are in HP and ap, bq, cr are in GP, then $\frac{p}{r} + \frac{r}{p}$ is equal to

a. $\frac{a}{c} - \frac{c}{a}$
d. $\frac{b}{q} + \frac{q}{b}$

b. $\frac{a}{c} + \frac{c}{a}$
c. $\frac{b}{q} - \frac{q}{b}$

Sol. b. Since, a, b, c are in AP $\Rightarrow b = \frac{a+c}{2}$... (i)

$$p, q, r \text{ are in HP} \Rightarrow q = \frac{2pr}{p+r} \quad \dots (ii)$$

and ap, bq, cr are in GP, $b^2q^2 = apcr$

$$\Rightarrow \left(\frac{a+c}{2}\right)^2 \left(\frac{4p^2r^2}{(p+r)^2}\right) = apcr \quad [\text{from Eqs. (i) and (ii)}]$$

$$\Rightarrow \frac{(a+c)^2 pr}{(p+r)^2} = ac$$

$$\Rightarrow \frac{(a+c)^2}{ac} = \frac{(p+r)^2}{pr}$$

$$\Rightarrow \frac{p^2}{pr} + \frac{r^2}{pr} + 2 = \frac{a^2}{ac} + \frac{c^2}{ac} + 2 \Rightarrow \frac{p}{r} + \frac{r}{p} = \frac{a}{c} + \frac{c}{a}$$

Sum to n Terms of Special Series

The sum of first n terms of some special series is given below

1. The sum of first n natural numbers

$$= \sum n = 1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$$

2. The sum of square of the first n natural numbers

$$= \sum n^2 = 1^2 + 2^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

3. The sum of cubes of the first n natural numbers

$$= \sum n^3 = 1^3 + 2^3 + \dots + n^3 = \left[\frac{n(n+1)}{2}\right]^2$$

> PRACTICE EXERCISE

1. If $\frac{3+5+7+\dots+n}{5+8+11+\dots+10 \text{ terms}} = 7$, then the value of n

is

- (a) 35 (b) 36 (c) 37 (d) 40

2. If sum of n terms of an AP is $3n^2 + 5n$ and $T_m = 164$, then m is equal to

- (a) 26 (b) 27 (c) 28 (d) None of these

3. In a GP, if the $(m+n)$ th term be p and $(m-n)$ th term be q , then its m th term is

- (a) $\sqrt{pq}$ (b) $\sqrt{p/q}$ (c) $\sqrt{q/p}$ (d) $\sqrt{p+q}$

4. The sum of the first ' n ' terms of the series $\frac{1}{2} + \frac{3}{4} + \frac{7}{8} + \frac{15}{16} + \dots$ is

- (a) $2^n - n - 1$ (b) $1 - 2^{-n}$
(c) $n + 2^{-n} - 1$ (d) $2^n - 1$

5. An AP consists of n (odd terms) and its middle term is m . Then, the sum of the AP is

- (a) $2mn$ (b) $\frac{1}{2}mn$ (c) mn (d) mn^2

6. The sum of $1 + \frac{2}{5} + \frac{3}{5^2} + \frac{4}{5^3} + \dots \infty$ upto n terms is

- (a) $\frac{25}{16}$ (b) $\frac{15}{16}$
(c) $\frac{5}{16}$ (d) $\frac{3}{2}$

7. The sum of n terms of an AP is $an(n-1)$. The sum of the squares of these terms is equal to

- (a) $a^2n^2(n-1)^2$ (b) $\frac{a^2}{6}n(n-1)(2n-1)$
(c) $\frac{2a^2}{3}n(n-1)(2n-1)$ (d) $\frac{2a^2}{3}n(n+1)(2n+1)$

8. If S be the sum to infinity of a GP, whose first term is a , then the sum of first n terms is
- (a) $S \left(1 - \frac{a}{S}\right)^n$
 (b) $S \left[1 - \left(1 - \frac{a}{S}\right)^n\right]$
 (c) $a \left[1 - \left(1 - \frac{a}{S}\right)^n\right]$
 (d) None of the above
9. If the non-zero numbers a, b, c are in AP and $\tan^{-1} a, \tan^{-1} b, \tan^{-1} c$ are also in AP, then
- (a) $a = b = c$ (b) $b^2 = 2ac$
 (c) $a^2 = bc$ (d) $c^2 = ab$
10. $\frac{1}{b-a} + \frac{1}{b-c} = \frac{1}{a} + \frac{1}{c}$, then a, b, c are in
- (a) AP (b) GP
 (c) HP (d) None of these
11. The value of $x + y + z$ is 15, if a, x, y, z, b are in AP while the value of $\frac{1}{x} + \frac{1}{y} + \frac{1}{z}$ is $\frac{5}{3}$, if a, x, y, z, b are in HP. Then, a and b are
- (a) 1, 9 (b) 3, 7
 (c) 7, 3 (d) None of these
12. If the m th and n th term of a HP are n and m respectively, then the mn th term is
- (a) 0 (b) 1 (c) 2 (d) $\frac{1}{2}$
13. If $a, 2a + 2, 3a + 3$ are in GP, then what is the fourth term of the GP?
- (a) -13.5 (b) 13.5 (c) -27 (d) 27
14. If the sum of first ' n ' natural numbers is $\frac{n(n+1)}{2}$. Then, what will be the sum of first ' n ' terms of the series of alternate positive and negative numbers when ' n ' is even?
- $$1^2 - 2^2 + 3^2 - 4^2 + 5^2 - \dots$$
- I. $\frac{n(n+1)}{2}$ II. $\frac{n^2(n+1)}{2}$ III. $\frac{-n(n+1)}{2}$
- Which of the above statement(s) is/are correct?
- (a) Only I (b) Only III (c) Only II (d) None of these

> PREVIOUS YEARS QUESTIONS

15. If A, G and H are the arithmetic, geometric and harmonic means between a and b respectively, then which one of the following relations is correct?
- (a) G is the geometric mean between A and H **2015 I**
 (b) A is the arithmetic mean between G and H
 (c) H is the harmonic mean between A and G
 (d) None of the above
16. Consider the following statements in respect of the expression $S_n = \frac{n(n+1)}{2}$, where ' n ' is an integer.
- I. There are exactly two values of n for which $S_n = 861$.
 II. $S_n = S_{-(n+1)}$ and hence for any integer m we have two values of n for which $S_n = m$.
- Which of these statement(s) is/are correct? **2016 (I)**
- (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II

> ANSWERS

1		2		3		4		5		6		7		8		9		10	
11		12		13		14		15		16									

> HINTS AND SOLUTIONS

1. (a) $\therefore \frac{\frac{n}{2}[2 \times 3 + (n-1)2]}{\frac{10}{2}[2 \times 5 + (10-1) \times 3]} = 7$

$$\Rightarrow \frac{n(n+2)}{5 \times 37} = 7$$

$$\Rightarrow n^2 + 2n - 1295 = 0$$

$$\Rightarrow n^2 + 37n - 35n - 1295 = 0$$

$$\Rightarrow (n+37)(n-35) = 0$$

$$\therefore n = 35$$

2. (b) $\therefore T_m = S_m - S_{m-1}$

$$\Rightarrow 164 = 3(2m-1) + 5 \cdot 1$$

$$\Rightarrow 6m = 162$$

$$\therefore m = 27$$

3. (a) $T_{m+n} = ar^{m+n-1} = p$,
 $T_{m-n} = ar^{m-n-1} = q$
 On multiplying, we get $a^2 r^{2m-2} = pq$
 $\therefore T_m = ar^{m-1} = \sqrt{pq}$

4. (c) $\frac{1}{2} + \frac{3}{4} + \frac{7}{8} + \frac{15}{16} + \dots + n$ terms

$$= \left(1 - \frac{1}{2}\right) + \left(1 - \frac{1}{4}\right) + \left(1 - \frac{1}{8}\right) + \dots + \left(1 - \frac{1}{2^n}\right)$$

$$= n - \left(\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots + \frac{1}{2^n}\right)$$

$$= n - \frac{1}{2} \left(\frac{1 - (1/2)^n}{1 - 1/2}\right) = n + 2^{-n} - 1$$

5. (c) Middle term = $T_{\frac{n+1}{2}}$
 $\therefore a + \left(\frac{n+1}{2} - 1\right)d = m$ [given]
 $2a + (n-1)d = 2m \quad \dots(i)$
 Now, $S_n = \frac{n}{2}[2a + (n-1)d] = nm$

6. (a) The given sequence is arithmetic geometric series, where $r = \frac{1}{5}$ and $d = 1$

$$\begin{aligned} S_{\infty} &= \frac{a}{1-r} + \frac{dr}{(1-r)^2} \\ &= \frac{1}{1-\frac{1}{5}} + \frac{1 \times \frac{1}{5}}{\left(1-\frac{1}{5}\right)^2} \\ &= \frac{5}{4} + \frac{5}{16} = \frac{25}{16} \end{aligned}$$

7. (c) Let $S_n = an(n-1)$, then
 $S_{n-1} = a(n-1)(n-2)$
 $\therefore T_n = S_n - S_{n-1} = 2a(n-1)$
 $T_{n-2} = 4a^2(n-1)^2$
 $\therefore \text{Sum} = \Sigma T_n^2 = 4a^2 \frac{(n-1)(n)(2n-1)}{6}$
 $= \frac{2a^2 n(n-1)(2n-1)}{3}$

8. (b) Let r be the common ratio of GP, then

$$\begin{aligned} S &= \frac{a}{1-r}, \quad r = 1 - \frac{a}{S} \\ \therefore S_n &= \frac{a(1-r^n)}{1-r} = \frac{a}{1-r} (1-r^n) \\ &= S \left[1 - \left(1 - \frac{a}{S}\right)^n \right] \end{aligned}$$

9. (a) Since, $2b = a + c \quad \dots(i)$
 and $2 \tan^{-1} b = \tan^{-1} a + \tan^{-1} c$
 $\Rightarrow \frac{2b}{1-b^2} = \frac{a+c}{1-ac}$
 $\Rightarrow b^2 = ac$ [from Eq. (i)]
 $\Rightarrow 4b^2 = 4ac$
 $\Rightarrow (a+c)^2 - 4ac = 0$ [from Eq. (i)]
 $\Rightarrow (a-c)^2 = 0$
 $\Rightarrow a = c = b$

10. (c) Given, $\frac{1}{b-a} + \frac{1}{b-c} = \frac{1}{a} + \frac{1}{c}$
 $\Rightarrow \frac{1}{b-a} - \frac{1}{c} = \frac{1}{a} - \frac{1}{b-c}$

$$\begin{aligned} \Rightarrow \frac{(c-b+a)}{d(b-a)} &= \frac{(b-c-a)}{a(b-c)} \\ \Rightarrow \frac{1}{d(b-a)} &= -\frac{1}{a(b-c)} \\ \Rightarrow ba - ca &= -cb + ac \\ \Rightarrow ab + bc &= 2ac \\ \therefore b &= \frac{2ac}{a+c} \end{aligned}$$

Hence, a, b, c are in HP.

11. (a) $\therefore a, x, y, z, b$ are in AP.

$$\begin{aligned} \therefore x + y + z &= 3\left(\frac{a+b}{2}\right) \\ \Rightarrow 15 &= 3\left(\frac{a+b}{2}\right) \\ \Rightarrow a + b &= 10 \quad \dots(i) \\ \text{Also, } a, x, y, z, b &\text{ are in HP.} \\ \Rightarrow \frac{1}{a}, \frac{1}{x}, \frac{1}{y}, \frac{1}{z}, \frac{1}{b} &\text{ are in AP.} \\ \Rightarrow \frac{1}{x} + \frac{1}{y} + \frac{1}{z} &= 3\left(\frac{a+b}{2ab}\right) \\ \Rightarrow \frac{5}{3} &= \frac{3 \times 10}{2ab} \quad [\because a+b=10] \\ \Rightarrow ab &= 9 \end{aligned}$$

On solving Eqs. (i) and (ii), we get
 $a = 1, b = 9$ or $b = 1, a = 9$

12. (b) Let a be the first term and d the common difference of corresponding AP.

So, m th and n th term of AP are $\frac{1}{n}$ and $\frac{1}{m}$.

$$\therefore \frac{1}{n} = a + (m-1)d \quad \dots(i)$$

$$\text{and } \frac{1}{m} = a + (n-1)d \quad \dots(ii)$$

On solving Eqs. (i) and (ii), we get,

$$d = \frac{1}{mn} \text{ and } a = \frac{1}{mn}$$

$\therefore m$ th term of AP

$$\begin{aligned} &= \frac{1}{mn} + (mn-1) \times \frac{1}{mn} \\ &= \frac{1+mn-1}{mn} = 1 \end{aligned}$$

$\therefore m$ th term of HP = 1

13. (c) $a, 2a+2, 3a+3$ are in GP.
 $\Rightarrow (2a+2)^2 = (3a+3)a$
 $\Rightarrow 4a^2 + 4 + 8a = 3a^2 + 3a$
 $\Rightarrow a^2 + 5a + 4 = 0$
 $\Rightarrow a = -1, -4$

Now, $a = -1$ does not satisfy the given series.

$\therefore -4, -6, -9$ are in GP.

$$\therefore t_4 = -4 \left(\frac{3}{2}\right)^3 = -13.5$$

14. (b) When ' n ' is even.

$$\begin{aligned} \text{Let } n &= 2m, \text{ then} \\ &= 1^2 - 2^2 + 3^2 - 4^2 + 5^2 - \dots \\ &= (1^2 - 2^2) + (3^2 - 4^2) + (5^2 - 6^2) \\ &\quad + \dots + (2m-1)^2 - (2m)^2 \\ &= (1+2)(1-1) + (3+4)(-1) \\ &\quad + (5+6)(-1) + \dots \\ &\quad + (2m-1+2m)(-1) \\ &= -(1+2+3+4+\dots+2m) \\ &= \frac{-2m(2m+1)}{2} = \frac{-n(n+1)}{2} \end{aligned}$$

15. (a) Given, A, G and H are the arithmetic, geometric and harmonic means between a and b , respectively.

$$\begin{aligned} \therefore A &= \frac{a+b}{2}, \quad G = \sqrt{ab} \text{ and } H = \frac{2ab}{a+b} \\ (i) \quad (ii) \quad (iii) \end{aligned}$$

On multiplying Eq. (i) and (iii), we get

$$\therefore AH = \frac{a+b}{2} \times \frac{2ab}{a+b} = ab = (\sqrt{ab})^2$$

[from Eq. (ii)]

$$AH = G^2$$

Hence, the option (a) is correct.

16. (a) I. $S_n = \frac{n(n+1)}{2} = 861$

$$\Rightarrow n^2 + n - 861 \times 2 = 0$$

$$\Rightarrow (n+42)(n-41) = 0$$

$$\Rightarrow n = -42, 41$$

Hence, statement I is correct.

- II. Given, $S_n = S_{-(n+1)}$
 If $S_n = m$, then we have two values of n if and only if m is positive integer.

Hence, statement II is incorrect.

03

HCF AND LCM OF NUMBERS

Usually (2-3) questions have been asked from this chapter. Generally questions based on LCM and HCF are related to traffic lights, racetracks, largest size of tile etc. This concept is also useful in the chapters of time and distance, time and work, pipes and cisterns etc.



FACTORS

If number ' a ' divides the number b without leaving a remainder, then ' a ' is said to be the factor of ' b '.

e.g. (i) 4 is a factor of 16 as $16 = 4 \times 4$ (ii) 9 is a factor of 729 as $729 = 9 \times 9 \times 9$

Prime Factors

The factors that cannot be again factorized i.e. the factors which are prime numbers are called prime factors.

e.g. (i) 9 is a factor of 729, but 3 is a prime factor of 729.

(ii) 8 and 9 are factors of 72, but 2 and 3 are prime factors of 72.

Least Common Multiple (LCM)

1. **Common Multiple** A common multiple of two or more numbers is a number which is exactly divisible (without leaving remainder) by each of them.

e.g. 30 is a common multiple of 2, 3, 5, 6, 10 and 15 because 30 is exactly divisible by each number.

Similarly, 72 is a common multiple of 2, 3, 4, 6, 9, 12 and 18.

2. **Least Common Multiple** The least common multiple of two or more given numbers is the least number which is exactly divisible by all the given numbers.

e.g. 84, 162, 252 are the common multiples of 2, 3, 4, 7. But, 84 is the LCM of 2, 3, 4, 7.

Methods of Finding LCM

1. **Prime Factorization Method** Write down the given numbers as the product of prime factors. Then, the LCM is the product of the highest powers of all the prime factors.

EXAMPLE 1. The LCM of 30, 250, 490 is

- a. 46750 b. 36750
c. 26750 d. None of these

Sol. b. Here, $30 = 2 \times 3 \times 5$
 $250 = 5 \times 5 \times 2 \times 5 = 2 \times 5^3$
 and $490 = 7 \times 7 \times 2 \times 5 = 2 \times 5 \times 7^2$
 $\therefore$ LCM of 30, 250 and 490 $= 2 \times 5^3 \times 7^2 \times 3 = 36750$

2. **Division Method** Write the given number in a row and divide them with the common prime divisor. On division. Write the quotient in each case below the number. If any number is not divisible by the respective divisor, then write it as such in the next row. Keep on dividing the quotients until you get 1. Multiply all the divisors to get the required LCM.

EXAMPLE 2. What is the LCM of 120, 144, 160 and 180.

- a. 1450 b. 1620 c. 1440 d. 1380

Sol. c.

2	120,	144,	160,	180
2	60,	72,	80,	90
2	30,	36,	40,	45
3	15,	18,	20,	45
3	5,	3,	10,	15
5	5,	1,	10,	5
2	1,	1,	2,	1
	1,	1,	1	1

$$\therefore \text{LCM of } 120, 144, 160 \text{ and } 180 = 2 \times 2 \times 2 \times 3 \times 3 \times 5 \times 2 \\ = 1440$$

IMPORTANT POINTS

- The least number which when divided by x, y and z leaving the remainders a, b and c , respectively is given by $[\text{LCM of } (x, y, z) - p]$, where $p = (x - a) = (y - b) = (z - c)$.
- The least number which when divided by x, y and z leaving the same remainder R in each case is given by $[\text{LCM of } (x, y, z) + R]$.

EXAMPLE 3. What is the least number which when divided by 42, 72 and 84 leaves the remainders 25, 55 and 67, respectively?

- a. 521 b. 512 c. 504 d. 487

Sol. d. Here, difference $= (42 - 25) = (72 - 55) = (84 - 67) = 17$
 Now, $\text{LCM}(42, 72, 84) = 504$
 $\therefore$ Required number $= 504 - 17 = 487$

EXAMPLE 4. Find the least number which when divided by 5, 6, 7 and 8 leaves a remainder 3 but when divided by 9, leaves no remainder.

- a. 1620 b. 1683
c. 1635 d. 1672

Sol. b. $\text{LCM}(5, 6, 7, 8) = 840$.

Here, $R = 3 \Rightarrow$ Number is of the form $840k + 3$.

Least value of k for which $(840k + 3)$ when divided by 9 leaves no remainder is 2.

$$\therefore \text{Required number} = 840 \times 2 + 3 = 1683$$

Highest Common Factor (HCF)

- Common Factor** A common factor of two or more numbers is a number which divides each of them exactly. e.g. 2 is common factor of 2, 10, 20.
- Highest Common Factor** The Highest Common Factor (HCF) of two or more numbers is the largest number that divides all the given numbers exactly. It is also known as Greatest Common Divisor (GCD). HCF is always a factor of LCM. e.g. HCF of the numbers 18 and 24 is 6.

Methods of Finding HCF

- HCF by Prime Factorization** Write the given number as product of prime factors and then find the product of least powers of common prime factors. This product is the required HCF of given numbers.

EXAMPLE 5. The HCF of 65, 75 and 105 is

- a. 4 b. 5 c. 6 d. 8

Sol. b. Here, $65 = 13 \times 5$, $75 = 5 \times 5 \times 3$
 and $105 = 7 \times 3 \times 5$
 $\therefore$ HCF of 65, 75 and 105 $= 5$

- HCF by Division Method** Suppose we have to find the HCF of two given numbers, divide the large number by the smaller one. Now, you will get the remainder. Divide the divisor by the remainder. Repeat this process until no remainder is left, the last divisor used in this process is the desired greatest common divisor i.e. HCF. In order to find the HCF of three numbers, then, HCF of $[(\text{HCF of any two}) \text{ and } (\text{third number})]$ gives the HCF of given three numbers.

EXAMPLE 6. The HCF of 204, 1190 and 1445 is

- a. 85 b. 15 c. 17 d. 75

Sol. c. Here, $1190 \div 1445$ (1

$$\begin{array}{r} 1190 \\ 255 \overline{) 1190} \quad (4 \\ \underline{1020} \\ 170 \quad 255 \quad (1 \\ \underline{170} \\ 85 \overline{) 170} \quad (2 \\ \underline{170} \\ \hline \times \end{array}$$

So, HCF of 1190 and 1445 is 85.

$$\begin{array}{r}
 \text{Now,} \quad 85) 204 (2 \\
 \underline{170} \\
 34) 85 (2 \\
 \underline{68} \\
 17) 34 (2 \\
 \underline{34} \\
 \times
 \end{array}$$

∴ HCF of 85 and 204 is 17.
Hence, HCF of 204, 1190 and 1145 is 17.

IMPORTANT POINTS

1. For integers x , y and z , if $\text{HCF}(x, y) = 1$ and $\text{HCF}(x, z) = 1$, then $\text{HCF}(x, y, z)$ is always 1.
2. The greatest number that will divide x , y and z leaving remainders a , b and c , respectively is given by HCF of $(x - a)$, $(y - b)$, $(z - c)$.
3. The greatest number that will divide x , y and z leaving the same remainder in each case is given by $[\text{HCF of } |x - y|, |y - z|, |z - x|]$

EXAMPLE 7. Find the greatest number which will divide 400, 435 and 541 leaving 9, 10 and 14 as remainders respectively.

- a. 19 b. 17 c. 13 d. 9 **2014 I**

Sol. b Required number = $\text{HCF of } (400 - 9, 435 - 10, 541 - 14)$
= $\text{HCF of } (391, 425, 527) = 17$

EXAMPLE 8. For any integers ' a ' and ' b ' with $\text{HCF}(a, b) = 1$, what is $\text{HCF}(a + b, a - b)$ equal to?

- a. It is always 1 b. It is always 2 **2014 I**
c. Either 1 or 2 d. None of these

Sol. c. Put arbitrary values of a and b .

Illustration 1 Let $a = 9$ and $b = 8$.

$$\therefore \text{HCF}(8 + 9, 9 - 8) \Rightarrow \text{HCF}(17, 1) = 1$$

Illustration 2 Let $a = 23$ and $b = 17$.

$$\therefore \text{HCF}(17 + 23, 23 - 17) \Rightarrow \text{HCF}(40, 6) = 2$$

Hence, $\text{HCF}(a + b, a - b)$ can either be 1 or 2.

How to Calculate LCM and HCF of Fractions

The LCM and HCF of fractions can be obtained from the following formula

1. $\text{HCF of fractions} = \frac{\text{HCF of numerators}}{\text{LCM of denominators}}$
2. $\text{LCM of fractions} = \frac{\text{LCM of numerators}}{\text{HCF of denominators}}$

EXAMPLE 9. The HCF of $\frac{14}{3}, \frac{21}{9}, \frac{7}{15}$ is

- a. $\frac{7}{45}$ b. $3\frac{2}{5}$ c. $\frac{7}{30}$ d. None of these

Sol. a. HCF of numerators i.e. 14, 21 and 7 is 7 and LCM of denominators i.e. 3, 9 and 15 is 45. So, HCF of given fractions = $\frac{7}{45}$.

Relation between LCM and HCF of Two Numbers

Product of two numbers = (Their HCF) $\times$ (Their LCM)

EXAMPLE 10. The LCM of two numbers is 90 times their HCF. The sum of LCM and HCF is 1456. If one of the numbers is 160, then what is the other number?

2014 II

- a. 120 b. 136 c. 144 d. 184

Sol. c. Let the HCF of two numbers be x .
LCM of two numbers be $90x$.

According to the question,

$$\text{LCM} + \text{HCF} = 1456$$

$$\Rightarrow 90x + x = 1456$$

$$\Rightarrow 91x = 1456$$

$$\Rightarrow x = 16$$

$$\therefore \text{HCF of two numbers} = 16$$

$$\text{and LCM of two numbers} = 90 \times 16 = 1440$$

We know that,

$$\text{LCM} \times \text{HCF} = \text{Product of two numbers}$$

$$\Rightarrow 1440 \times 16 = 160 \times \text{Second number}$$

$$\therefore \text{Second number} = \frac{1440 \times 16}{160} = 144$$

EXAMPLE 11. What is the greatest number that divides 13850 and 17030 and leaves a remainder 17?

- a. 477 b. 159 c. 107 d. 87 **2012 II**

Sol. b. Required number = $\text{HCF of } (13850 - 17, 17030 - 17)$
= $\text{HCF of } (13833, 17013) = 159$

EXAMPLE 12. There are three drums with 1653 litre 2261 litre and 2527 litre of petrol. The greatest possible size of the measuring vessel with which we can measure the petrol of any drum while every time the vessel must be completely filled is

- a. 31 b. 27 c. 19 d. 41

Sol. c. The maximum capacity of the vessel = $\text{HCF of } 1653, 2261 \text{ and } 2527 = 19$

EXAMPLE 13. John, Kate and Smith at same time, same point and in same direction to run around a circular in 150 seconds. Find after what time will they meet again?

- a. 30 min b. 25 min c. 20 min d. 15 min

Sol. b. LCM of 250, 300 and 150 = 1500 sec
= 25 min.

Hence, John, Kate and Smith meet after 25 min.

> PRACTICE EXERCISE

- If $x = 2^3 \times 3^2 \times 5^4$ and $y = 2^2 \times 3^2 \times 5 \times 7$, then HCF of x and y is
(a) 180 (b) 360 (c) 540 (d) 35
- LCM of $2^3 \times 3 \times 5$ and $2^4 \times 5 \times 7$ is
(a) $2^{12} \times 3 \times 5^2 \times 7$ (b) $2^4 \times 5 \times 7 \times 9$
(c) $2^4 \times 3 \times 5 \times 7$ (d) $2^3 \times 3 \times 5 \times 7$
- LCM of $\frac{4}{5}$, $\frac{3}{10}$ and $\frac{7}{15}$ is
(a) $8\frac{2}{3}$ (b) $\frac{8}{15}$ (c) 20 (d) $16\frac{4}{5}$
- What is the HCF of $\frac{3}{2}$, $\frac{9}{7}$ and $\frac{15}{14}$?
(a) $\frac{3}{7}$ (b) $\frac{3}{14}$ (c) $\frac{3}{2}$ (d) 3
- The least number divisible by 12, 15, 20 and is a perfect square is
(a) 900 (b) 400 (c) 36 (d) 256
- The least number which when divided by 5, 6, 7 and 8 leaves a remainder 3 is
(a) 423 (b) 843 (c) 1683 (d) 2523
- The HCF of two numbers is $\frac{1}{5}$ th of their LCM. If the product of the two numbers is 720, then the HCF of the numbers is
(a) 13 (b) 12 (c) 14 (d) 18
- The LCM of two numbers is 39780 and their ratio is 13 : 15. Then, the numbers are
(a) 273, 315 (b) 2652, 3060
(c) 516, 685 (d) None of these
- If the highest common factor of two positive integers is 24, then their least common multiple cannot be
(a) 72 (b) 216 (c) 372 (d) 600
- If the HCF of three numbers 144, x and 192 is 12, then the number x cannot be
(a) 180 (b) 84 (c) 60 (d) 48
- Consider those numbers between 300 and 400 such that when each number is divided by 6, 9 and 12, it leaves 4 as remainder in each case. What is the sum of the numbers?
(a) 692 (b) 764 (c) 1080 (d) 1092
- What is the smallest positive integer which when divided by 4, 5, 8 and 9 leaves remainder 3, 4, 7 and 8, respectively?
(a) 119 (b) 319 (c) 359 (d) 719
- What is the HCF of $a^2b^4 + 2a^2b^2$ and $(ab)^7 - 4a^2b^9$?
(a) ab (b) a^2b^2 (c) a^2b^3 (d) a^3b^2
- If a number is exactly divisible by 11 and 13, which of the following types the number must be?
(a) Divisible by (11+13) (b) Divisible by (13-11)
(c) Divisible by (11×13) (d) Divisible by (13÷11)
- What is the sum of the digits of the least number which when divided by 52, leaves 33 as remainder, when divided by 78 leaves 59 and when divided by 117, leaves 98 as remainder?
(a) 17 (b) 18 (c) 19 (d) 21
- For any integer n , what is the HCF of integers $m = 2n + 1$ and $k = 9n + 4$?
(a) 3 (b) 1 (c) 2 (d) 4
- For any three natural numbers a, b and c , if HCF $\left(\frac{a}{c}, \frac{b}{c}\right)$ is
(a) a/c (b) b/c (c) c (d) Always 1
- Raj, Rachit and Asha begin to tag around a circular stadium. They complete their revolutions in 42 s, 56 s and 63 s, respectively. After how many seconds will they be together at the starting point?
(a) 366 (b) 252 (c) 504 (d) 605
- Find the side of the largest possible square slabs which can be paved on the floor of a room 2m 50cm long and 1m 50cm broad. Also, find the number of such slabs to pave the floor.
(a) 25,20 (b) 30,15 (c) 50,15 (d) 55,10
- Four bells begin to toll together and toll, respectively at intervals of 5, 6, 8 and 12 s. How many times will they toll together in an hour excluding the one at the start?
(a) 10 (b) 19 (c) 13 (d) 9
- 21 mango trees, 42 apple trees and 56 orange trees have to be planted in rows such that each row contains the same number of trees of one variety only. What is the minimum number of rows in which the above trees may be planted?
(a) 3 (b) 15 (c) 17 (d) 20
- A person has four iron bars whose lengths are 24 m, 36 m, 48 m and 72 m, respectively. This person wants to cut pieces of same length from each of four bars. What is the least number of total pieces, if he is to cut without any wastage?
(a) 10 (b) 15 (c) 20 (d) 25

- 23.** For two natural numbers m and n , let g_{mn} denote the greatest common factor of m and n . Consider the following in respect of three natural numbers k, m and n .

I. $g_{m(nk)} = g_{(mn)k}$ II. $g_{mn}g_{nk} = g_{mk}$

Which of the above statement(s) is/are correct?

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II
- 24.** Consider the following in respect of integers a and b
- I. $\text{HCF}(a, b) = \text{HCF}(a + b, b)$
II. $\text{HCF}(a, b) = \text{HCF}(a, b - a)$ for $b > a$
- Which of the above statement(s) is/are correct?
- (a) Only I (b) Only II
(c) Both I and II (d) None of these

PREVIOUS YEARS' QUESTIONS

- 25.** The HCF and LCM of two natural numbers are 12 and 72, respectively. What is the difference between the two numbers, if one of the numbers is 24? ✎ 2012 I
- (a) 12 (b) 18 (c) 21 (d) 24
- 26.** The sum of two numbers is 232 and their HCF is 29. What is the number of such pairs of numbers satisfying the above condition? ✎ 2012 I
- (a) One (b) Two (c) Four (d) None of these
- 27.** The product of HCF and LCM of 18 and 15 is ✎ 2012 II
- (a) 120 (b) 150 (c) 175 (d) 270
- 28.** Three planets revolve round the Sun once in 200, 250 and 300 days, respectively in their own orbits. When do they all come relatively to the same position at a certain point of time in their orbits? ✎ 2012 II
- (a) After 3000 days (b) After 2000 days
(c) After 1500 days (d) After 1200 days
- 29.** The LCM of two numbers is 2376 while their HCF is 33. If one of the numbers is 297, then the other number is ✎ 2013 I
- (a) 216 (b) 264 (c) 642 (d) 792
- 30.** The HCF of two numbers is 98 and their LCM is 2352. The sum of the numbers may be ✎ 2013 II
- (a) 1372 (b) 1398 (c) 1426 (d) 1484
- 31.** If for integers a, b and c , $\text{HCF}(a, b) = 1$ and $\text{HCF}(a, c) = 1$, then which one of the following is correct? ✎ 2013 II
- (a) $\text{HCF}(a, bc) = 1$ (b) $\text{HCF}(a, bc) = a$
(c) $\text{HCF}(a, bc) = b$ (d) None of these
- 32.** What is the number of integral solutions of the equations $\text{HCF}(a, b) = 5$ and $a + b = 65$? ✎ 2014 I
- (a) Less than 65 (b) Infinitely many
(c) Exactly one (d) None of these
- 33.** For any integer n , HCF of $(22n + 7, 33n + 10)$ is equal to ✎ 2014 I
- (a) n (b) 1 (c) 11 (d) None of these
- 34.** In a fire range, 4 shooters are firing at their respective targets. The first, the second, the third and the fourth shooter hit the target once in every 5 s, 6 s, 7 s and 8 s, respectively. If all of them hit their target at 9 : 00 am, when will they hit their target together again? ✎ 2014 I
- (a) 9 : 04 am (b) 9 : 08 am
(c) 9 : 14 am (d) None of these
- 35.** If a and b be positive integers, then HCF of $\left(\frac{a}{\text{HCF}(a, b)}, \frac{b}{\text{HCF}(a, b)}\right)$ equal to? ✎ 2014 I
- (a) a (b) b (c) 1 (d) $\frac{a}{\text{HCF}(a, b)}$
- 36.** The HCF of two natural numbers m and n is 24 and their product is 552. How many sets of values of m and n are possible? ✎ 2014 II
- (a) 1 (b) 2 (c) 4
(d) No set of m and n is possible satisfying the given conditions
- 37.** The LCM of two integers is 1237. What is their HCF? ✎ 2014 II
- (a) 37 (b) 19
(c) 1 (d) Cannot be determined
- 38.** There are 48 cricket balls, 72 hockey balls and 84 tennis balls and they have to be arranged in several rows in such a way that every row contains the same number of balls of one type. What is the minimum number of rows required for this to happen? ✎ 2014 II
- (a) 12 (b) 16 (c) 17 (d) 19
- 39.** Consider all positive two digit numbers each of which when divided by 7 leaves a remainder 3. What is their sum? ✎ 2015 II
- (a) 661 (b) 666 (c) 676 (d) 777
- 40.** What is the sum of digits of the least multiple of 13, which when divided by 6, 8 and 12 leaves 5, 7 and 11, respectively, as the remainders? ✎ 2015 II
- (a) 5 (b) 6 (c) 7 (d) 8
- 41.** The LCM of two numbers is 12 times their HCF. The sum of HCF and LCM is 403. If one of the numbers is 93, then the other number is ✎ 2015 II
- (a) 124 (b) 128 (c) 134 (d) 138
- 42.** Consider the following in respect of natural numbers a, b and c ✎ 2016 I
- I. $\text{LCM}(ab, ac) = a \text{ LCM}(b, c)$
II. $\text{HCF}(ab, ac) = a \text{ HCF}(b, c)$
III. $\text{HCF}(a, b) < \text{LCM}(a, b)$
IV. $\text{HCF}(a, b)$ divides $\text{LCM}(a, b)$.
- Which of the above statement(s) is/are correct?
- (a) III and II (b) III and IV
(c) I, II and IV (d) All of these

ANSWERS

1		2		3		4		5		6		7		8		9		10	
11		12		13		14		15		16		17		18		19		20	
21		22		23		24		25		26		27		28		29		30	
31		32		33		34		35		36		37		38		39		40	
41		42																	

HINTS AND SOLUTIONS

1. (a) Given, $x = 2^3 \times 3^2 \times 5^4$
and $y = 2^2 \times 3^2 \times 5 \times 7$
 $\therefore \text{HCF} = 2^2 \times 3^2 \times 5 = 4 \times 9 \times 5 = 180$

2. (c) Here, say $a = 2^3 \times 3 \times 5$ and

$$b = 2^4 \times 5 \times 7, \text{ then}$$

$$\text{LCM} = 2^4 \times 3 \times 5 \times 7$$

3. (d) Here, LCM

$$= \frac{\text{LCM of } 4, 3, 7}{\text{HCF of } 5, 10, 15} = \frac{84}{5} = 16\frac{4}{5}$$

4. (b) $\text{HCF} \left(\frac{3}{2}, \frac{9}{7}, \frac{15}{14} \right) = \frac{\text{HCF}(3, 9, 15)}{\text{LCM}(2, 7, 14)}$

$$= \frac{3}{14}$$

5. (a)
- | | | | |
|---|----|----|----|
| 2 | 12 | 15 | 20 |
| 2 | 6 | 15 | 10 |
| 3 | 3 | 15 | 5 |
| 5 | 1 | 5 | 5 |
| | 1 | 1 | 1 |

LCM of 12, 15 and 20 = $2 \times 2 \times 3 \times 5$

$\therefore$ Required perfect square

$$= 2 \times 2 \times 3 \times 3 \times 5 \times 5 = 900$$

6. (b) To find out the least number, firstly we find out the LCM of given numbers.

$$\therefore \text{LCM}(5, 6, 7, 8)$$

$$= \text{LCM}(5, 2 \times 3, 7, 2^3) = 840$$

$$\therefore \text{Required number} = 840 + 3 = 843$$

7. (b) Let $\text{LCM} = 5x$, then $\text{HCF} = x$
Now, product of numbers = 720

$$\text{So, } 5x \times x = 720$$

$$\Rightarrow 5x^2 = 720 \Rightarrow x = 12$$

8. (b) The numbers are $13x$ and $15x$.

So, x is the HCF. Now,

$$\text{HCF} \times \text{LCM} = \text{Product of numbers}$$

$$x \times 39780 = 13x \times 15x$$

$$\Rightarrow x \times 39780 = 13 \times 15 \times x^2$$

$$\Rightarrow x = \frac{39780}{13 \times 15} = 204.$$

$\therefore$ Numbers are $13 \times 204 = 2652$ and $15 \times 204 = 3060$

9. (c) In the given options, only 372 is not divisible by 24. Therefore, LCM of numbers cannot be 372.

10. (d) Here, we know that

$$144 = 12 \times 2 \times 2 \times 3$$

$$\text{and } 192 = 12 \times 2 \times 2 \times 2 \times 2$$

By taking option (d), $48 = 12 \times 2 \times 2$

Hence, the value of x will not be 48 otherwise the HCF of given numbers becomes 48.

11. (a) LCM of 6, 9 and 12 = 36
So, number is the form of $36p + 4$.

Since, the required numbers are between 300 and 400.

$$\therefore p = 9 \text{ and } 10$$

$$\therefore \text{Required sum} = 328 + 364 = 692$$

12. (c) Here,

$$4 - 3 = 5 - 4 = 8 - 7 = 9 - 8 = 1$$

$$\text{Now, } 4 = 2 \times 2, 5 = 5$$

$$8 = 2 \times 2 \times 2, 9 = 3 \times 3$$

$$\therefore \text{LCM} = 5 \times 2 \times 2 \times 2 \times 3 \times 3 = 360$$

$$\text{Required number} = 360 - 1 = 359$$

13. (b) $a^2 b^4 + 2a^2 b^2 = a^2 b^2 (b^2 + 2) \dots (i)$

$$\text{and } (ab)^7 - 4a^2 b^9 = a^7 b^7 - 4a^2 b^9$$

$$= a^2 b^2 (a^5 b^5 - 4b^7) \dots (ii)$$

From Eqs. (i) and (ii), we get

$$\text{HCF} = a^2 b^2$$

14. (c) LCM of 11 and 13 will be (11×13) . Hence, if a number is exactly divisible by 11 and 13, then the same number must be exactly divisible by their LCM i.e. (11×13) .

15. (a) Here, $52 - 33 = 78 - 59$

$$= 117 - 98 = 19$$

$$\text{Now, } 52 = 13 \times 2 \times 2$$

$$\Rightarrow 78 = 13 \times 2 \times 3$$

$$\Rightarrow 117 = 13 \times 3 \times 3$$

$$\therefore \text{LCM} = 13 \times 2 \times 2 \times 3 \times 3 = 468$$

$$\therefore \text{Required number} = 468 - 19 = 449$$

Hence, the sum of the digits is 17.

16. (b) Since, $m = 2n + 1$ is an odd integer, so its factors may be 1 or 3 and $k = 9n + 4$ its factors may be 1, 2 and 4.

Hence, HCF of (m, k) is 1.

17. (d) It is always 1

Illustrations Let $a = 21$ and $b = 35$

Then, $\text{HCF}(21, 35) = 7$

$$\therefore \text{HCF} \left(\frac{21}{7}, \frac{35}{7} \right) = \text{HCF}(3, 5) = 1$$

18. (c) Required time = LCM of 42, 56 and 63 s

2	42, 56, 63
3	21, 28, 63
7	7, 28, 21
	1, 4, 3

$\therefore$ Required time

$$= 2 \times 3 \times 7 \times 4 \times 3 = 504 \text{ s.}$$

19. (c) Side of largest possible square slab is the HCF of 250 cm and 150 cm

$$250 = 5 \times 5 \times 5 \times 2$$

$$150 = 5 \times 5 \times 3 \times 2$$

$\therefore$ HCF is 50. Then, number of slabs

$$= \frac{\text{Area of floor}}{\text{Area of slab}} = \frac{250 \times 150}{50 \times 50} = 15$$

20. (a) Here, LCM of 5, 6, 8 and 12 is 360, so the bells will toll after 360 s.

So, in an hour they will toll together

$$= \frac{60 \times 60}{360} = 10 \text{ times}$$

21. (c) The HCF of (21, 42, 56) is 7.

$\therefore$ The minimum number of rows

$$= \frac{21}{7} + \frac{42}{7} + \frac{56}{7}$$

$$= 3 + 6 + 8 = 17$$

22. (b) Here, $24 = 12 \times 2$, $36 = 12 \times 3$

$$48 = 12 \times 4, \quad 72 = 12 \times 6$$

$$\therefore \text{HCF}(24, 36, 48, 72) = 12$$

$$\text{Total piece} = \frac{24}{12} + \frac{36}{12} + \frac{48}{12} + \frac{72}{12}$$

$$= 2 + 3 + 4 + 6 = 15$$

23. (d) I. Let three natural numbers are
 $m = 16, n = 15, k = 20$

$$\therefore g_{m(nk)} = g_{16(300)} = 4$$

$$\text{and } g_{(mn)k} = g_{240(20)} = 20$$

$$\therefore g_{m(nk)} \neq g_{(mn)k}$$

II. $g_{mn} g_{nk} = g_{16,15} g_{15,20} = 1 \times 5 = 5$
 and $g_{mk} = g_{16,20} = 4$
 $\therefore g_{mn} g_{nk} \neq g_{mk}$

So, neither I nor II is correct.

24. (c) I. Let $a = 4$, and $b = 10$

$$\therefore a + b = 14$$

$$\text{HCF } (4, 10) = 2$$

$$\text{and HCF } (14, 10) = 2$$

$$\therefore \text{HCF } (a, b) = \text{HCF } (a + b, b)$$

- II. Let $a = 6$ and $b = 15$

$$\therefore b - a = 15 - 6 = 9$$

$$\text{HCF } (6, 15) = 3$$

$$\text{HCF } (6, 9) = 3$$

$$\therefore \text{HCF } (a, b) = \text{HCF } (a, b - a)$$

25. (a) Second number = $\frac{\text{LCM} \times \text{HCF}}{\text{First number}}$
 $= \frac{72 \times 12}{24} = 36$

$$\therefore \text{Difference between two numbers} = 36 - 24 = 12$$

26. (b) Let two numbers be $29x$ and $29y$.

$$29x + 29y = 232 \Rightarrow x + y = 8$$

$$\therefore (x, y) = (1, 7), (3, 5)$$

Hence, one such pair is 87 and 145.
 and the other pair is 203 and 29.

27. (d) Here, $18 = 2 \times 3 \times 3$ and $15 = 3 \times 5$

$$\text{HCF of } 18 \text{ and } 15 = 3$$

$$\text{LCM of } 18 \text{ and } 15 = 2 \times 3 \times 3 \times 5 = 90$$

$$\therefore \text{Product of HCF and LCM of both numbers} = 3 \times 90 = 270$$

28. (a) Given that, three planets revolves the Sun once in 200, 250 and 300 days, respectively in their own orbits.

$$\therefore \text{Required time} = \text{LCM of } 200, 250 \text{ and } 300 = 3000 \text{ days}$$

Hence, after 3000 days they all come relatively to the same position at a certain point of time in their orbits.

29. (b) Given, LCM of two numbers = 2376

$$\text{HCF of two numbers} = 33$$

$$\text{One of the number} = 297$$

$$\therefore (\text{HCF of two numbers}) \times (\text{LCM of two numbers})$$

$$= (\text{First number}) \times (\text{Second number})$$

$$\therefore \text{Second number} = \frac{33 \times 2376}{297} = 264$$

30. (a) It is given that, the HCF of two numbers is 98. This means that both the numbers are multiples of 98. Therefore, the sum of these two numbers must also be a multiple of 98. Among all the four options given, only option (a) satisfies this condition.

31. (a) For integers a, b and c , if $\text{HCF } (a, b) = 1$ and $\text{HCF } (a, c) = 1$ then, $\text{HCF } (a, b, c) = 1$

32. (a) $\therefore \text{HCF } (a, b) = 5$

$$\text{Let } a = 5x \text{ and } b = 5y$$

$$\therefore 5x + 5y = 65 \Rightarrow x + y = 13$$

$$\therefore \text{Number of pairs of } (x, y) = (1, 12), (2, 11), (3, 10), (4, 9), (5, 8), (6, 7)$$

Hence, number of solutions is less than 65.

33. (b) HCF of $(22n + 7, 33n + 10)$ is always 1.

$$\text{Illustration For } n = 1, \text{HCF } (29, 43) = 1$$

$$\text{For } n = 2, \text{HCF } (51, 76) = 1$$

$$\text{For } n = 3, \text{HCF } (73, 109) = 1$$

34. (c) Time after which they will hit the target together again

$$= \text{LCM of } 5, 6, 7 \text{ and } 8$$

$$= 5 \times 3 \times 7 \times 2 \times 2 \times 2 = 840 \text{ s}$$

Duration after which they will hit target together

$$= \frac{840}{60} = 14 \text{ min.}$$

So, they will hit the target together after 14 min.

Hence, they will hit together again at 9:14 am.

35. (c) $\text{HCF} \left(\frac{a}{\text{HCF}(a, b)}, \frac{b}{\text{HCF}(a, b)} \right)$ is always equal to 1.

Illustration Let the two positive integers be $a = 24$ and $b = 36$.

$$\therefore \text{HCF} \left(\frac{24}{\text{HCF}(24, 36)}, \frac{36}{\text{HCF}(24, 36)} \right)$$

$$\text{HCF} \left(\frac{24}{12}, \frac{36}{12} \right) \Rightarrow \text{HCF}(2, 3) = 1$$

36. (d) LCM of two natural numbers

$$= \frac{\text{Product of } m \text{ and } n}{\text{HCF of } m \text{ and } n} = \frac{552}{24} = 23$$

Here, no set of m and n is possible satisfying the given conditions as LCM is always a multiple of HCF.

37. (c) Given, LCM of two integers is 1237, which is a prime number.

So, their HCF is 1.

38. (c) Given, number of cricket balls

$$= 48 = 2^4 \times 3$$

$$\text{Number of hockey balls} = 72 = 2^3 \times 3^2$$

$$\text{and number of tennis balls}$$

$$= 84 = 2^2 \times 3 \times 7$$

$$\therefore \text{HCF of } 48, 72 \text{ and } 84 = 2^2 \times 3 = 12$$

Now, minimum number of rows

$$= \frac{48}{12} + \frac{72}{12} + \frac{84}{12}$$

$$= 4 + 6 + 7 = 17$$

39. (c) The required numbers are 10, 17, 24, ..., 94.

Total number of numbers is 13.

$\therefore$ Sum of these numbers

$$= \frac{13}{2} [10 + 94] = \frac{13}{2} \times 104$$

$$= 13 \times 52 = 676$$

40. (d) Here, $6 - 5 = 1, 8 - 7 = 1$

$$12 - 11 = 1$$

$$\text{LCM of } 6, 8 \text{ and } 12 = 24$$

Required number

$$= 24k - 1, k \text{ is any natural number}$$

$$\text{For } k = 6, \text{the number} = 144 - 1 = 143$$

which is multiple of 13

$$\text{So, sum of digits} = 1 + 4 + 3 = 8$$

41. (a) Let other number be b and HCF be x .

$$\Rightarrow \text{LCM} = 12x$$

$$\text{We have, } x + 12x = 403 \Rightarrow 13x = 403$$

$$\therefore x = 31$$

$\therefore$ Product of two numbers

$$= \text{LCM} \times \text{HCF}$$

$$\therefore 93 \times b = x \times 12x$$

$$\Rightarrow 93 \times b = 12 \times 31 \times 31$$

$$\therefore b = 124$$

42. (d) a, b and c are natural numbers.

I. $\text{LCM of } (ab, ac) = abc$

$$a \times \text{LCM of } (b, c) = abc$$

Hence, statement I is correct.

II. $\text{HCF}(ab, ac) = a \text{HCF}(b, c)$

$$\text{HCF of } (ab, ac) = \text{Common factor of } (ab, ac)$$

$$\text{and } a \times \text{HCF}(b, c) = a \times \text{common factor of } (b, c)$$

Hence, statement II is correct.

III. We know that HCF is always less than LCM.

Hence, statement III is correct.

IV. $\text{HCF}(a, b)$ divides $\text{LCM}(a, b)$

because a common factor between a, b always divides $(a \times b)$.

Hence, statement IV is correct.

04

DECIMAL FRACTIONS

Regularly (1-2) questions have been asked from this chapter. It is one of the most common chapters which we are studying from the very starting age.



FRACTION

If any unit is divided into some parts, then each of these parts is called fraction of that unit. A fraction is represented as p/q , where $q \neq 0$ and here q is called as denominator and p is called as numerator.

e.g. $1/2, 3/4, 6/7$ etc. are fractions.

Simple fraction

The fraction which has denominator other than power of 10, is called simple fraction. e.g. $\frac{6}{5}, \frac{2}{9}$ etc.

Note Simple fraction is also known as vulgar fraction.

DECIMAL FRACTION

Fraction that has powers of 10 in the denominator are called decimal fraction. e.g.

- (i) $\frac{1}{10}$ is the tenth part of 1 is written as 0.1. (ii) $\frac{1}{100}$ is the hundredth part of 1 is written as 0.01.

Types of Decimals

1. **Recurring Decimals** A decimal number in which a digit or set of digits repeats regularly is called non-terminating repeating decimals or recurring decimals. To represent these fractions, a line is drawn on repeating digits.

e.g. $\frac{1}{7} = 0.1428571428571 \dots = 0.\overline{142857}$

$$\frac{2}{3} = 0.66666 \dots = 0.\overline{6}$$

$$\frac{4}{3} = 1.33333 \dots = 1.\overline{3}$$

2. **Pure recurring decimal** A decimal fraction in which all the digits after the decimal points are repeated, is called a pure recurring decimal.

e.g. $0.786786786... = 0.\overline{786}$
 $0.77777... = 0.\overline{7}$

EXAMPLE 1. Which one of the following is a non-terminating repeating decimal?

- a. $\frac{13}{8}$ b. $\frac{3}{16}$ c. $\frac{3}{11}$ d. $\frac{137}{25}$

Sol. c. $\because \frac{3}{11} = 0.272727...$

$= 0.\overline{27}$

$\frac{13}{8} = \frac{13 \times 125}{8 \times 125} = \frac{1625}{1000} = 1.625$

$\frac{3}{16} = \frac{3 \times 625}{16 \times 625} = \frac{1875}{10000} = 0.1875$

$\frac{137}{25} = \frac{137 \times 4}{25 \times 4} = \frac{548}{100} = 5.48$

From above it is clear that all of these are terminating decimals. Hence, $\frac{3}{11}$ is a non-terminating repeating decimal.

3. **Mixed recurring decimal** A decimal fraction in which some digits after the decimal point are not repeated while some are repeated, is called a mixed recurring decimal.

e.g. $0.179999... = 0.17\overline{9}$
 $0.053939... = 0.05\overline{39}$

Note The decimal expansion of a rational number is either terminating or non-terminating recurring. In other words, a number whose decimal expansion is terminating or non-terminating recurring is a rational number.

Operations on Decimal Fractions, Addition and Subtraction

While making addition or subtraction of decimal fractions the numbers are placed in such a way that the decimal point lie in one column. Then, the numbers can be added or subtracted as usual.

EXAMPLE 2. $7.093 - 3.57 = ?$

- a. 3.523 b. 3.513
c. 3.143 d. 3.532

Sol. a. Here,

$$\begin{array}{r} 7.093 \\ - 3.570 \\ \hline 3.523 \end{array}$$

EXAMPLE 3. $51.3 + 7.078 + 1.38 + 0.9 = ?$

- a. 61.668 b. 59.238
c. 60.658 d. None of these

Sol. c. Here,

$$\begin{array}{r} 51.300 \\ 7.078 \\ 1.380 \\ + 0.900 \\ \hline 60.658 \end{array}$$

Note Adding/Annexing zeroes to the extreme right of a decimal fraction does not change its value.

Multiplication

While multiplying two or more decimal fractions, consider them without decimal point and multiply them as usual. In the product so obtained the decimal point is marked off as many places from the right as in the sum of the decimal places in the given numbers.

EXAMPLE 4. $5 \times 0.25 \times 6.301 \times 0.00394 = ?$

- a. 0.0310324220 b. 0.0310324210
c. 0.0310324250 d. None of these

Sol. c. For the product of $5 \times 0.25 \times 6.301 \times 0.00394$

Here, $5 \times 25 \times 6301 \times 394 = 310324250$

Total decimal places $(2 + 3 + 5) = 10$

$\therefore 5 \times 0.25 \times 6.301 \times 0.00394 = 0.031032425$

Division

While dividing the given decimal fraction by a natural number, divide it without the decimal point to get quotient. Here, in the quotient so obtained, place the decimal point after as many places from the right as are there in the dividend.

For dividing a decimal fraction by a decimal fraction, multiply the dividend and the divisor by a suitable power of 10 to make the divisor a whole number and then proceed as in the previous rule.

EXAMPLE 5. $0.01834 \div 13 = ?$

- a. 141.077 b. 1.41077
c. 0.00141077 d. None of these

Sol. c. Here, for $0.01834 \div 13$

$\Rightarrow \frac{1834}{13} = 141.077$

So, $0.01834 \div 13 = 0.00141077$

EXAMPLE 6. $0.0066 \div 0.22 = ?$

- a. 0.03 b. 0.3 c. 3 d. None of these

Sol. a. $0.0066 \div 0.22$

So, $\frac{0.0066 \times 10000}{0.22 \times 10000} = \frac{66}{22 \times 100} = \frac{3}{100} = 0.03$

To Convert a Pure Recurring Decimal into a Simple Fraction

In order to convert a pure recurring decimal fraction into simple (vulgar) fraction, we write the repeated figures only once in the numerator without decimal point and write as many nines in the denominator as the number of repeating figure.

e.g. $0.\overline{5} = \frac{5}{9}$ or $0.\overline{45} = \frac{45}{99}$

EXAMPLE 7. $1.\overline{27}$ in the form p/q is equal to

- a. $\frac{127}{100}$ b. $\frac{73}{100}$ c. $\frac{14}{11}$ d. $\frac{11}{14}$

Sol. c. $1.\overline{27} = 1 + 0.\overline{27} = 1 + \frac{27}{99} = \frac{99 + 27}{99} = \frac{126}{99} = \frac{14}{11}$

To Convert Mixed Recurring Decimal into a Simple Fraction

The numerator is obtained by taking the difference between the number formed by all the digits after decimal point. (Here, repeated digits are taken only once) and the number formed by non-repeating digits. The denominator is obtained by placing, as many nines as in repeating digits followed by as many zeroes as the number of non-repeating digits.

EXAMPLE 8. The vulgar fraction of $0.12\overline{36}$ is

- a. $\frac{120}{823}$ b. $\frac{102}{825}$ c. $\frac{93}{825}$ d. None of these

Sol. b. $0.12\overline{36} = \frac{1236 - 12}{9900} = \frac{1224}{9900} = \frac{102}{825}$

EXAMPLE 9. What is the value of $1.\overline{34} + 4.1\overline{2}$?

- a. $\frac{133}{90}$ b. $\frac{371}{90}$ c. $5\frac{219}{990}$ d. $5\frac{461}{990}$

Sol. d. $\therefore 1.\overline{34} = \frac{134 - 1}{99} = \frac{133}{99}$ and $4.1\overline{2} = \frac{412 - 41}{90} = \frac{371}{90}$
 $\therefore 1.\overline{34} + 4.1\overline{2} = \frac{133}{99} + \frac{371}{90} = \frac{1330 + 4081}{990} = \frac{5411}{990} = 5\frac{461}{990}$

Comparison of Fractions By Converting in Decimal Form

To compare the fraction using decimal, convert each one of the given fractions in decimal form and now arrange them in ascending or descending order as per the requirement.

By Equating Denominators

For comparison of fractions, take LCM of the denominator of all fractions. So, that their denominators are same. Now, the fractions having largest numerator is the largest fraction.

EXAMPLE 10. Arrange $\frac{2}{3}, \frac{5}{6}, \frac{16}{25}, \frac{3}{7}$ in ascending order.

- a. $\frac{2}{3}, \frac{5}{6}, \frac{16}{25}$ and $\frac{3}{7}$ b. $\frac{16}{25}, \frac{5}{6}, \frac{3}{7}$ and $\frac{2}{3}$
 c. $\frac{3}{7}, \frac{16}{25}, \frac{2}{3}$ and $\frac{5}{6}$ d. None of these

Sol. c. Firstly, we convert each of the fraction in decimal form.

$$\frac{2}{3} = 0.\overline{666}, \frac{5}{6} = 0.8\overline{33}, \frac{16}{25} = 0.64, \frac{3}{7} = 0.\overline{428571}$$

Here, $0.\overline{428571} < 0.64 < 0.\overline{666} < 0.8\overline{33}$

Hence, $\frac{3}{7} < \frac{16}{25} < \frac{2}{3} < \frac{5}{6}$ and so $\frac{3}{7}, \frac{16}{25}, \frac{2}{3}$ and $\frac{5}{6}$ are in ascending order.

EXAMPLE 11. Arrange $\frac{7}{8}, \frac{5}{6}, \frac{6}{7}$ in descending order

- a. $6/7, 5/6, 7/8$ b. $7/8, 5/6, 6/7$
 c. $7/8, 6/7, 5/6$ d. $6/7, 7/8, 5/6$

Sol. c. LCM of 8, 6, 7 = 168

$$\therefore \frac{7}{8} = \frac{7 \times 21}{8 \times 21} = \frac{147}{168} \Rightarrow \frac{5}{6} = \frac{5 \times 28}{6 \times 28} = \frac{140}{168}$$

$$\text{and } \frac{6}{7} = \frac{6 \times 24}{7 \times 24} = \frac{144}{168}$$

$$\therefore \frac{140}{168} < \frac{144}{168} < \frac{147}{168}$$

So, descending order of fractions is $\frac{7}{8}, \frac{6}{7}, \frac{5}{6}$.

HCF and LCM of Decimal Fractions

For finding the HCF or LCM of decimal fractions, first make the given fractions to have same number of decimal places by annexing zeroes, if needed. Now, find the HCF or LCM of the numbers without considering decimal. Finally, in the result mark as many decimal places as are there in each of the given numbers.

EXAMPLE 12. The HCF and LCM of 1.600, 8 and 2.4 is

- a. 0.800 and 24 b. 0.6000 and 38
 c. 0.900 and 42 d. None of these

Sol. a. We can write the given numbers as 1.600, 8.000, 2.400.

So, without decimal, the numbers are 1600, 8000 and 2400.

Now, HCF of 1600, 8000 and 2400 is 800.

$\therefore$ HCF of 1.600, 8.000 and 2.400 is 0.800.

And LCM of 1600, 8000, 2400 is 24000.

$\therefore$ LCM of 1.600, 8.000 and 2.400 is 24.000.

> PRACTICE EXERCISE

1. If $(15.9273 - x) = 11.0049$, then the value of x is
(a) 4.9224 (b) 0.4922 (c) 0.4294 (d) 6.932
2. If $(15.39 + 0.236 + 5.290 + 0.0002) = x$, then the value of x is
(a) 0.20916 (b) 2.0916 (c) 209.16 (d) 20.9162
3. If $175 \times 1.24 = 2.17$, then the value of 1.75×124 is
(a) 217 (b) 0.0217
(c) 2.17 (d) None of these
4. If $111.744 \div 28.8 = 3.88$, then the value of $1117.44 \div 288$ is
(a) 3.88 (b) 0.388 (c) 388.0 (d) 38.8
5. The value of $\frac{(0.5)^4 - (0.4)^4}{(0.5)^2 + (0.4)^2}$ is equal to
(a) 0.9 (b) 0.09 (c) 9 (d) 0.009
6. The value of $\frac{0.004 \times 0.0008}{0.02}$ equals to
(a) 0.000016 (b) 0.00016
(c) 0.0016 (d) None of these
7. Which of the following sets of the fractions is in ascending order?
(a) $\frac{6}{8}, \frac{7}{9}, \frac{5}{6}, \frac{11}{13}$ (b) $\frac{5}{6}, \frac{6}{8}, \frac{7}{9}, \frac{11}{13}$
(c) $\frac{11}{13}, \frac{5}{6}, \frac{7}{9}, \frac{6}{8}$ (d) $\frac{11}{13}, \frac{7}{9}, \frac{6}{8}, \frac{5}{6}$
8. When $0.232323\dots$ is converted into a fraction, then the result is
(a) $\frac{1}{5}$ (b) $\frac{2}{9}$ (c) $\frac{23}{99}$ (d) $\frac{23}{100}$
9. If $3.245 \times 10^k = 0.0003245$, then the value of k is
(a) 4 (b) -4 (c) 3 (d) 5
10. The value of $(0.\overline{6} + 0.\overline{8} + 0.\overline{7})$ is
(a) $2\frac{1}{8}$ (b) $2\frac{1}{9}$ (c) $2\frac{1}{3}$ (d) $1\frac{1}{9}$
11. The value of $(6.\overline{88} - 2.\overline{58})$ is
(a) $4.\overline{30}$ (b) $4.\overline{29}$ (c) $3.\overline{22}$ (d) $4.\overline{38}$
12. The expression $(11.98 \times 11.98 + 11.98 \times x + 0.02 \times 0.02)$ will be a perfect square for x equal to
(a) 0.02 (b) 0.2 (c) 0.04 (d) 0.4
13. If $2.5x = 0.5y$, then the value of $\frac{x+y}{x-y}$ is
(a) -1.3 (b) -1.5 (c) 1.5 (d) 1.3
14. 7.2 exceeds its one-tenth by
(a) 8.48 (b) 5.48 (c) 6.48 (d) 5.28
15. If $2.5252525\dots = \frac{p}{q}$ (in the lowest form), then what is the value of $\frac{q}{p}$?
(a) 0.4 (b) 0.42525 (c) 0.0396 (d) 0.396
16. What decimal of an hour is a second?
(a) 0.25 (b) 0.0256 (c) 0.00027 (d) 0.0125
17. If $\sqrt{5} = 2.24$, then the value of $\frac{3\sqrt{5}}{2\sqrt{5} - 0.48}$ is
(a) 1.68 (b) 16.8 (c) 168 (d) 0.168
18. The value of $\left\{ \frac{(0.1)^2 - (0.01)^2}{0.0001} + 1 \right\}$ is equal to
(a) 1001 (b) 11 (c) 101 (d) 100
19. The value of $\frac{3}{3 + \frac{0.3 - 3.03}{3 \times 0.91}}$ is equal to
(a) 0.75 (b) 1.5 (c) 15 (d) 0.15
20. What should be subtracted from the multiplication of 0.527 and 2.013 to get 1?
(a) 0.939085 (b) 0.060851 (c) 1.91984 (d) 2.16085
21. The value of $0.34\overline{67} + 0.13\overline{33}$ is equal to
(a) 0.48 (b) $0.48\overline{01}$
(c) $0.4\overline{8}$ (d) 0.48
22. The greatest fraction out of $\frac{2}{5}, \frac{5}{6}, \frac{11}{12}$ and $\frac{7}{8}$ is
(a) $\frac{7}{8}$ (b) $\frac{5}{6}$ (c) $\frac{11}{12}$ (d) $\frac{2}{5}$
23. Which of the following fractions is greater than $\frac{3}{4}$ and less than $\frac{5}{6}$.
(a) $\frac{1}{2}$ (b) $\frac{2}{3}$ (c) $\frac{4}{5}$ (d) $\frac{9}{10}$
24. Consider the following statements:
I. $\frac{1}{22}$ cannot be written as a terminating decimal.
II. $\frac{2}{15}$ can be written as a terminating decimal.
III. $\frac{1}{16}$ can be written as a terminating decimal.
Which of the statement(s) given above is/are correct?
(a) Only I (b) Only II
(c) I and III (d) II and III

25. Consider the following decimal numbers:

- I. 1.1666666... II. 1.181181118...
 III. 2.010010001... IV. 1.454545...

Which of the above numbers represent(s) rational number(s)?

- (a) Only IV (b) II and III
 (c) I and IV (d) None of these

PREVIOUS YEARS' QUESTIONS

26. What is the value of $0.00\overline{7} + 17.\overline{83} + 310.020\overline{2}$?

- (a) $327.86\overline{638}$ (b) $327.86\overline{638}$ **✓ 2012 I**
 (c) $327.86\overline{683}$ (d) $327.86\overline{68}$

27. What is the value of $0.242424...$? **✓ 2012 II**

- (a) $\frac{23}{99}$ (b) $\frac{8}{33}$ (c) $\frac{7}{33}$ (d) $\frac{47}{198}$

28. Representation of $0.\overline{2341}$ in the form $\frac{p}{q}$, where p and q are integers, $q \neq 0$, is **✓ 2013 I**

- (a) $\frac{781}{3330}$ (b) $\frac{1171}{4995}$ (c) $\frac{2341}{9990}$ (d) $\frac{2339}{9990}$

29. Let p be a prime number other than 2 or 5. One would like to express the vulgar fraction $1/p$ in the form of a recurring decimal. Then, the decimal will be **✓ 2015 I**

- (a) a pure recurring decimal and its period will be necessarily $(p-1)$
 (b) a mixed recurring decimal and its period will be necessarily $(p-1)$
 (c) a pure recurring decimal and its period will be some factor of $(p-1)$
 (d) a mixed recurring decimal and its period will be some factor of $(p-1)$

30. The value of $(0.\overline{63} + 0.\overline{37})$ is **✓ 2015 II**

- (a) 1 (b) $\frac{100}{91}$ (c) $\frac{100}{99}$ (d) $\frac{1000}{999}$

ANSWERS

1		2		3		4		5		6		7		8		9		10	
11		12		13		14		15		16		17		18		19		20	
21		22		23		24		25		26		27		28		29		30	

HINTS AND SOLUTIONS

1. (a) $x = 15.9273 - 11.0049 = 4.9224$

2. (d)
$$\begin{array}{r} 15.3900 \\ 0.2360 \\ 5.2900 \\ + 0.0002 \\ \hline 20.9162 \\ x = 20.9162 \end{array}$$

3. (a) $1.75 \times 1.24 = 2.17$
 $\Rightarrow 1.75 \times 124 = 217$

4. (a) Since, $\frac{111.744}{28.8} = 3.88$
 $\Rightarrow \frac{111744}{28800} = 3.88$ [Rule 6]

Now, $\frac{1117.44}{288} = \frac{11744}{28800} = 3.88$ [Rule 4]

5. (b)
$$\frac{[(0.5)^2]^2 - [(0.4)^2]^2}{(0.5)^2 + (0.4)^2}$$

$$[(a^2 - b^2) = (a + b)(a - b)]$$

$$\begin{aligned} &= \frac{[(0.5)^2 - (0.4)^2][(0.5)^2 + (0.4)^2]}{[(0.5)^2 + (0.4)^2]} \\ &= (0.5)^2 - (0.4)^2 = (0.5 - 0.4)(0.5 + 0.4) \quad [\text{Rule 1}] \\ &= 0.1 \times 0.9 = 0.09 \quad [\text{Rule 2}] \end{aligned}$$

6. (b)
$$\frac{0.004 \times 0.0008}{0.002} = \frac{0.0000032}{0.02} = 0.00016$$

7. (a) Here, $\frac{6}{8} = 0.75$, $\frac{7}{9} = 0.\overline{7}$, $\frac{5}{6}$
 $= 0.8\overline{3}$, $\frac{11}{13} = 0.846$
 So, $0.75 < 0.\overline{7} < 0.8\overline{3} < 0.846$

8. (c) Given, $0.232323... = 0.\overline{23}$
 (which is a recurring decimal) $= \frac{23}{99}$

9. (b) Here, $10^k = \frac{0.0003245}{3.245}$

$$= \frac{3.245 \times 10^{-4}}{3.245}$$

$10^k = 10^{-4}$
 So, $k = -4$

10. (c) $0.\overline{6} + 0.\overline{8} + 0.\overline{7} = \frac{6}{9} + \frac{8}{9} + \frac{7}{9}$
 $= \frac{21}{9} = \frac{7}{3} = 2\frac{1}{3}$

11. (a) Here, $6.\overline{88} - 2.\overline{58}$
 $= \left(6 + \frac{88}{99}\right) - \left(2 + \frac{58}{99}\right)$
 $= (6 - 2) + \left(\frac{88}{99} - \frac{58}{99}\right)$
 $= 4 + \frac{88 - 58}{99} = 4 + \frac{30}{99} = 4.\overline{30}$

12. (c) Given expression
 $= (11.98)^2 + (0.02)^2 + 11.98 \times x$
 For the given expression to be a perfect square, we must have
 $11.98 \times x = 2 \times 11.98 \times 0.02$
 $\Rightarrow x = 0.04$
 [by using $(a + b)^2 = a^2 + b^2 + 2ab$]

$$13. (b) 2.5x = 0.5y \\ \Rightarrow \frac{x}{y} = \frac{0.5}{2.5} = 0.20$$

Now, the expression is [dividing numerator and denominator by y]

$$\frac{x+y}{x-y} = \frac{x/y+1}{x/y-1} = \frac{0.20+1}{0.20-1} = \frac{1.20}{-0.80} = -15$$

Shortcut Method

$$\frac{x}{y} = \frac{0.5}{2.5} \Rightarrow \frac{x+y}{x-y} = \frac{0.5+2.5}{0.5-2.5}$$

[using componendo and dividendo]

$$\Rightarrow \frac{x+y}{x-y} = \frac{3}{-2} = -1.5$$

$$14. (c) 7.2 - \frac{7.2}{10} \\ \Rightarrow 7.2 - 0.72 = 6.48$$

$$15. (d) \text{ Given, } \frac{p}{q} = 2.5\overline{2} \quad \dots(i)$$

Now, 100 multiply both sides, we get

$$\frac{100p}{q} = 252.\overline{2} \quad \dots(ii)$$

On subtracting Eq. (i) from Eq. (ii), we get

$$\frac{99p}{q} = 250 \Rightarrow \frac{p}{q} = \frac{99}{250} = 0.396$$

$$16. (c) \text{ Required decimal fraction} = \frac{1}{60 \times 60} \\ = \frac{1}{3600} = 0.00027$$

$$17. (a) \frac{3\sqrt{5}}{2\sqrt{5} - 0.48} = \frac{3 \times 2.24}{2 \times 2.24 - 0.48} \\ [\because \sqrt{5} = 2.24] \\ = \frac{6.72}{4.48 - 0.48} = \frac{6.72}{4} = 1.68$$

$$18. (d) \text{ Here, } \left[\frac{(0.1)^2 - (0.01)^2}{0.0001} + 1 \right] \\ = \frac{0.01 - 0.0001}{0.0001} + 1 = \left(\frac{0.0099}{0.0001} + 1 \right) \\ = (99 + 1) = 100$$

$$19. (b) \frac{3}{3 + 0.3 - 3.03} = \frac{3}{3 - 2.73} \\ = \frac{3}{3 \times 0.91} = \frac{3}{3 \times 91}$$

$$= \frac{3}{3-1} = \frac{3}{2} = 1.5$$

$$20. (b) \text{ Let } x \text{ should be subtracted} \\ (0.527 \times 2.013) - x > 1 \\ 1.060851 - x > 1 \\ x = 0.060851$$

$$21. (b) 0.34\overline{67} + 0.13\overline{33} \\ = \frac{3467 - 34}{9900} + \frac{1333 - 13}{9900} \\ = \frac{3433 + 1320}{9900} = \frac{4753}{9900} \\ = \frac{4801 - 48}{9900} = 0.48\overline{01}$$

$$22. (c) \frac{2}{5} = 0.4, \frac{5}{6} = 0.8\overline{3}, \frac{11}{12} = 0.91\overline{6} \\ \text{and } \frac{7}{8} = 0.875$$

Clearly, the greatest fraction is $0.91\overline{6}$ i.e. $\frac{11}{12}$.

$$23. (c) \frac{3}{4} = 0.75, \frac{5}{6} = 0.83\overline{3} \\ \frac{1}{2} = 0.5, \frac{2}{3} = 0.6\overline{6}, \frac{4}{5} = 0.8 \text{ and } \frac{9}{10} = 0.9 \\ \text{Clearly, } 0.8 \text{ lies between } 0.75 \text{ and } 0.83\overline{3}. \\ \therefore \frac{4}{5} \text{ lies between } \frac{3}{4} \text{ and } \frac{5}{6}.$$

$$24. (c) \text{ In } \frac{1}{22} \text{ and } \frac{2}{15}, 22 \text{ and } 15 \text{ are not in} \\ \text{the form of } 2^m \times 5^n \text{ but in } \frac{1}{16}. \\ 16 \text{ in the form of } 2^4 \times 5^0. \text{ So, } \frac{1}{16} \text{ can be} \\ \text{written as a terminating decimal.} \\ \text{Hence, statements I and III are correct.}$$

25. (c) Since, $1.16666\dots$ and $1.454545\dots$ are recurring numbers and we know that, recurring numbers represent rational numbers.

Hence, statements I and IV are rational numbers.

$$26. (b) 0.00\overline{7} = 0.007777777 \\ \Rightarrow 17.\overline{83} = 17.83838383 \\ 310.020\overline{2} = 310.02022222 \\ = 327.8663838 \text{ on adding} \\ = 327.866\overline{38}$$

$$27. (b) \text{ Given, } 0.242424\dots = 0.2\overline{4} \\ [\text{which is a recurring decimal number}] \\ = \frac{24}{99} = \frac{8}{33}$$

$$28. (d) 0.23\overline{41} = \frac{2341 - 2}{9990} = \frac{2339}{9990}$$

$$29. (c) \text{ Value } p \text{ may be } 3, 7, 11, 13. \\ \frac{1}{3} = 0.\overline{3}, \text{ Period} = 1$$

Here, $p - 1 = 2$ and 1 is a factor of 2.

$$\frac{1}{7} = 0.1428\overline{57}, \text{ Period} = 6$$

Here, $p - 1 = 6$ and 6 is a factor of 6.

$$\frac{1}{11} = 0.09\overline{09}, \text{ Period} = 2$$

Here, $p - 1 = 10$ and 2 is a factor of 10.

$$30. (c) \text{ We have, } 0.6\overline{3} + 0.3\overline{7}$$

Let $x = 0.636363\dots$

$$100x = 63.636363\dots, 99x = 63$$

$$\text{and } x = \frac{63}{99}$$

$$\text{Similarly, } y = 0.3\overline{7} = 0.373737$$

$$\therefore y = \frac{37}{99}$$

$$\therefore x + y = \frac{63}{99} + \frac{37}{99} = \frac{100}{99}$$

05

SQUARE ROOTS AND CUBE ROOTS

Generally (1-3) questions have been asked from this chapter. This section will be useful in solving simplification questions and will save lots of time while doing fuzzy calculations.



SQUARE

The square of a number is obtained by multiplying the number by itself.

e.g. Square of $5 = 5 \times 5 = 25$, square of $6 = 6 \times 6 = 36$

Perfect square A natural number n is called a perfect square or a square number, if there exists a natural number m such that $n = m^2$.

e.g. The numbers 1, 4, 9, 16, 25, ... are perfect square.

Properties of Perfect Square

The properties of perfect squares are discussed below.

- (i) A number having 2, 3, 7 or 8 at unit's place is never a perfect square.
e.g. 172, 2783
- (ii) The squares of odd numbers are odd and the squares of even numbers are even.
- (iii) The difference of squares of two consecutive natural numbers is equal to their sum.
e.g. $7^2 - 6^2 = 7 + 6 = 13$
- (iv) The product of four consecutive natural numbers plus one is always a square.
- (v) A number ending in an odd number of zeroes is never a perfect square.

Square Root

The square root of a number x is that number which when multiplied by itself gives x as the product. It is denoted by $\sqrt{\quad}$.

In general, $y^2 = x \Rightarrow y = \sqrt{x}$

Here, y is the square root of x , if and only if x is the square of y .

e.g. $\sqrt{4} = \sqrt{2 \times 2} = 2$

Thus, 2 is the square root of 4.

EXAMPLE 1. If $\sqrt{\frac{x}{64}} = \frac{4}{8}$, then the value of x is

- a. 16 b. 12 c. 8 d. 4

Sol. a. We have, $\sqrt{\frac{x}{64}} = \frac{4}{8}$,

On squaring both sides, we get

$$\left(\sqrt{\frac{x}{64}}\right)^2 = \left(\frac{4}{8}\right)^2 \Rightarrow \frac{x}{64} = \frac{16}{64} \Rightarrow x = 16$$

Methods to Find Square Root

The square root of a given number can be determined by using any of the following two methods

1. Prime Factorization Method
2. General Method/Division Method

Prime Factorization

In this method, we express the given number as the product of prime factors. Now, for finding square root, we take the product of these prime factors choosing one out of every pair of same prime factors.

EXAMPLE 2. The square root of 213444 is

- a. 332 b. 368 c. 432 d. 462

Sol. d. The prime factorisation of 213444 is

$$213444 = 2 \times 2 \times 3 \times 3 \times 7 \times 7 \times 11 \times 11$$

$$\therefore \sqrt{213444} = \sqrt{2 \times 2 \times 3 \times 3 \times 7 \times 7 \times 11 \times 11}$$

Now, taking one number from each pair and multiplying them, we get $\sqrt{213444} = 2 \times 3 \times 7 \times 11 = 462$

EXAMPLE 3. What is the square root of $\frac{0.324 \times 0.64 \times 129.6}{0.729 \times 1.024 \times 36}$?

- a. 4 b. 3
c. 2 d. 1

Sol. d.
$$\sqrt{\frac{0.324 \times 0.64 \times 129.6}{0.729 \times 1.024 \times 36}} = \sqrt{\frac{324 \times 64 \times 1296}{729 \times 1024 \times 36}}$$

$$= \sqrt{\frac{(2^2 \times 3^4) \times (2^6) \times (2^4 \times 3^4)}{(3^6) \times (2^{10}) \times (2^2 \times 3^2)}} = \frac{(2 \times 3^2) \times 2^3 \times (2^2 \times 3^2)}{3^3 \times 2^5 \times (2 \times 3)}$$

$$= \frac{18 \times 8 \times 36}{27 \times 32 \times 6} = 1$$

Division Method

To find the square root of a given number using this method, the following steps are to be followed.

Step I In the given number, place bars over every pair of digits starting with the unit's digit. Each pair and the remaining one digit (if any) on the extreme left is called a period.

Step II Think of the largest number whose square is less than or equal to the first period.

Step III Put the quotient above the period and write the product of divisor and quotient just below the first period.

Step IV Subtract the product of divisor and quotient from the first period and bring down the next period to the right of the remainder. This becomes the new dividend.

Step V Double the quotient and enter it with a blank on its right.

Step VI Guess a largest possible digit to fill the blank which will also become the next digit in the quotient, such that when the new divisor is multiplied to the new quotient the product is less than or equal to the dividend.

Step VII Repeat the above steps till all the periods have been taken up.

Now, the quotient so obtained is the required square root of the given number.

EXAMPLE 4. The square root of 1522756 is

- a. 1182 b. 1222
c. 1234 d. 1334

Sol. c

1	1 52 27 56	1234
1		
22	52	
x2	44	
243	0827	
x3	729	
2464	9856	
x4	9856	
	x	

Hence, the square root of 1522756 is 1234.

EXAMPLE 5. The least number of four digits which is a perfect square is 2012 II

- a. 1204 b. 1024
c. 1402 d. 1420

Sol. b. Here, the greatest three-digit number = 999

31	9 99
3	9 99
x3	9
61	99
	61
	38

Here, the greatest number of three digits which is perfect square = $999 - 38 = 961 = (31)^2$

So, the smallest four-digit number which is perfect square = $(31 + 1)^2 = (32)^2 = 1024$

CUBE

If a number is multiplied three times with itself, then the result of this multiplication is called the cube of that number.
e.g. $8 = 2 \times 2 \times 2 = 2^3$

Properties of Cube

1. Cubes of all even natural numbers are even and all odd natural numbers are odd.
2. Cubes of the numbers ending in digit 0, 1, 4, 5, 6 and 9 are the numbers ending in the same digit.
3. Cubes of negative integers are negative.
4. Cubes of numbers ending in digits 3 and 7 ends in digit 7 and 3, respectively.

CUBE ROOT

The cube root of a given number is the number whose cube is the given number.

In general, we can say that $a^3 = b$, then cube root of 'b' is 'a'. Cube root of a number 'a' is denoted as $\sqrt[3]{a}$.

e.g. $\sqrt[3]{64} = \sqrt{4 \times 4 \times 4} = 4$. Thus, 4 is the cube root of 64.

EXAMPLE 6. If $\sqrt[3]{\frac{x}{27}} = \frac{5}{3}$, then the value of x is

- a. 125 b. 25 c. 27 d. 9

Sol. a. We have $\sqrt[3]{\frac{x}{27}} = \frac{5}{3}$

On cubing both sides, we get

$$\left(\sqrt[3]{\frac{x}{27}}\right)^3 = \left(\frac{5}{3}\right)^3 \Rightarrow \frac{x}{27} = \frac{125}{27}$$

$$\therefore x = 125$$

Method to Find Cube Root

Method to calculate the cube root of a number is as follow

Prime Factorization Method

In this method, we express the given number as the product of prime factors.

Now, for finding cube root, we take the product of these prime factors choosing one out of every three, from the factors.

EXAMPLE 7. Find the cube root of 373248.

- a. 42 b. 52 c. 62 d. 72

Sol. d. Here, $373248 = 8 \times 8 \times 8 \times 9 \times 9 \times 9$

$$373248 = \overbrace{2 \times 2 \times 2} \times \overbrace{2 \times 2 \times 2} \times \overbrace{2 \times 2 \times 2} \times \overbrace{3 \times 3 \times 3} \times \overbrace{3 \times 3 \times 3} \times \overbrace{3 \times 3 \times 3}$$

$$\Rightarrow \sqrt[3]{373248} = 2 \times 2 \times 2 \times 3 \times 3 = 72$$

Hence, the cube root of 373248 is 72.

POWERS OR EXPONENTS

An expression that represents repeated multiplication of the same factor is called a power. It is written as ' x^n ' and is read as 'x' raised to the power n.

Here, x is called the 'base'.

and n is called the 'exponent'.

e.g. $6^5 = 6 \times 6 \times 6 \times 6 \times 6 = 7776$.

Laws of Exponents

Let $\frac{x}{y}$ be any rational number and m, n be any integers.

Then,

$$(i) \left(\frac{x}{y}\right)^m \times \left(\frac{x}{y}\right)^n = \left(\frac{x}{y}\right)^{m+n}$$

$$(ii) \left(\frac{x}{y}\right)^m \div \left(\frac{x}{y}\right)^n = \left(\frac{x}{y}\right)^{m-n} \quad (iii) \left[\left(\frac{x}{y}\right)^m\right]^n = \left(\frac{x}{y}\right)^{mn}$$

$$(iv) \left(\frac{x}{y}\right)^{-n} = \left(\frac{y}{x}\right)^n \quad (v) \left(\frac{x}{y}\right)^0 = 1$$

EXAMPLE 8. If $2^m + 2^{1+m} = 24$, then the value of m is

- a. 1 b. 2 c. 3 d. 5

Sol. c. Given, $2^m + 2^{1+m} = 24 \Rightarrow 2^m + 2^1 \times 2^m = 24$

$$\Rightarrow 2^m(1 + 2) = 24 \Rightarrow 2^m \times 3 = 24 \Rightarrow 2^m = 8 = 2^3$$

Now, on comparing both sides, we get

$$\therefore m = 3$$

Hence, the value of m is 3.

SURDS OR RADICALS

Let x be a rational number and n be a positive integer, such that $\sqrt[n]{x}$ is irrational, then $\sqrt[n]{x}$ is called a radical or surd of order n and here x is called the radicand.

- A surd of order 2 is called a quadratic surd i.e. $\sqrt{2}, \sqrt{3}$.
- A surd of order 3 is called a cubic surd. i.e. $\sqrt[3]{2}, \sqrt[3]{3}$.
- A surd of order 4 is called a biquadratic surd i.e. $\sqrt[4]{5}, \sqrt[4]{7}$.

Laws of Radicals

The laws of exponents which are applicable to the surds are

$$(i) (\sqrt[n]{x})^n = x$$

$$(ii) \sqrt[n]{xy} = \sqrt[n]{x} \cdot \sqrt[n]{y}$$

$$(iii) \sqrt[n]{\frac{x}{y}} = \frac{\sqrt[n]{x}}{\sqrt[n]{y}}$$

$$(iv) (\sqrt[n]{x})^m = (\sqrt[n]{x^m})$$

$$(v) \sqrt[m]{\sqrt[n]{x}} = \sqrt[mn]{x} = \sqrt[n]{\sqrt[m]{x}}$$

Types of Surds

- Pure surds** A surd which has only 1 as a rational factor is called a pure surd.
e.g. $\sqrt{2}$, $\sqrt[3]{2}$, $\sqrt[4]{3}$ all are pure surds.
- Mixed surds** A surd which is not a pure surd or has factor other than unity is called a mixed surd.
e.g. $2\sqrt{3}$, $5\sqrt[3]{12}$, $\frac{4}{3}\sqrt[3]{35}$ are mixed surds.
- Like and Unlike surds** When the radicands of two surds are same, then those are known as like surds.
e.g. $6\sqrt{3}$ and $2^2 \cdot \sqrt{3}$.
When radicands of two surds are different, then they are called unlike surds. e.g. $4\sqrt{7}$ and $6\sqrt{5}$.

Comparison of Two Surds

- If two surds are of the same order, then the one whose radicand is larger, is the larger of the two.
e.g. $\sqrt{17} > \sqrt{13}$, $\sqrt[3]{21} > \sqrt[3]{16}$
- If any two surds have different orders are to be compared, then we will first reduce them to the same but smallest order and then compare them.

Note Let $\sqrt[n]{x}$ be a surd of order n .
Then, $\sqrt[n]{x} = \sqrt[m]{x^{m/n}}$ i.e. $x^{1/n} = (x^{m/n})^{1/m}$ is a surd of order m .

EXAMPLE 9. Convert $\sqrt{2}$ into a surd of order 4.

- a. $\sqrt[4]{2}$ b. $\sqrt[4]{8}$ c. $\sqrt[4]{4}$ d. None of these

Sol. c. $\sqrt{2} = 2^{1/2} = (2^{4/2})^{1/4} = (2^2)^{1/4} = \sqrt[4]{4}$

EXAMPLE 10. Consider the following in respect of the numbers $\sqrt{2}$, $\sqrt[3]{3}$, and $\sqrt[6]{6}$. ☞ 2014 (I)

- $\sqrt[6]{6}$ is the greatest number.
- $\sqrt{2}$ is the smallest number.

Which of these statements is/are correct?

- a. Only I b. Only II c. Both I and II d. Neither I nor II

Sol. d. Taking LCM of 2, 3 and 6 = 12

$$\text{Now, } \sqrt{2} = 2^{1/2} = 2^{6/12} = \sqrt[12]{2^6} = \sqrt[12]{64}$$

$$\begin{aligned} \sqrt[6]{6} &= (6)^{1/6} = (6)^{2/12} = \sqrt[12]{6^2} = \sqrt[12]{36} \\ \sqrt[3]{3} &= 3^{1/3} = 3^{4/12} = \sqrt[12]{81} \end{aligned}$$

So, neither I nor II is correct.

Operations on Surds

Addition and Subtraction of Surds

Addition or subtraction takes place only between like surds, using laws of numbers. e.g.

- $2\sqrt{3} + 4\sqrt{3} + \sqrt{5} = 6\sqrt{3} + \sqrt{5}$
- $6\sqrt{5} - 3\sqrt{2} + 7\sqrt{5} + 4\sqrt{2} = (6\sqrt{5} + 7\sqrt{5}) + (4\sqrt{2} - 3\sqrt{2})$
 $= 13\sqrt{5} + \sqrt{2}$

Multiplication and Division of Surds

Suppose given surds are $\sqrt[p]{a}$, $\sqrt[q]{b}$ and $\sqrt[r]{c}$, then first find the LCM of p , q and r to make the radical powers of the surds same and then their multiplication and division can be done using following rules.

$$(i) \sqrt[p]{x} \times \sqrt[q]{y} = \sqrt[pq]{xy} \quad (ii) \frac{\sqrt[p]{x}}{\sqrt[q]{y}} = \sqrt[pq]{\frac{x}{y}}$$

EXAMPLE 11. Simplify $\sqrt[6]{12} \div \sqrt{3} \cdot \sqrt[3]{2}$.

- a. $\sqrt[3]{3}$ b. $\sqrt{3}$
c. $\sqrt[3]{2}$ d. $\sqrt[3]{\frac{1}{3}}$

Sol. d. Here, the order of $\sqrt{3}$, $\sqrt[3]{2}$ is 2 and 3, respectively.

So, LCM 2, 3 and 6 = 6

$$\sqrt{3} = 3^{1/2} = (3^{6/2})^{1/6} = (3^3)^{1/6} = (27)^{1/6} = \sqrt[6]{27}$$

$$\sqrt[3]{2} = (2)^{1/3} = (2^{6/3})^{1/6} = (2^2)^{1/6} = \sqrt[6]{4}$$

$$\begin{aligned} \text{So, } \frac{\sqrt[6]{12}}{\sqrt{3} \cdot \sqrt[3]{2}} &= \frac{\sqrt[6]{12}}{\sqrt[6]{27} \cdot \sqrt[6]{4}} = \frac{\sqrt[6]{12}}{\sqrt[6]{27 \times 4}} = \sqrt[6]{\frac{12}{27 \times 4}} = \sqrt[6]{\frac{1}{9}} \\ &= \left[\left(\frac{1}{3} \right)^2 \right]^{1/6} = \left[\frac{1}{3} \right]^{2/6} = \left[\frac{1}{3} \right]^{1/3} = \sqrt[3]{\frac{1}{3}} \end{aligned}$$

Rationalisation of Surds

Process of converting surd into rational number is called as rationalisation of surds.

When the product of two surds is a rational number, then each surd is called a rationalising factor of each other.

$$\text{e.g. } \sqrt[3]{25} \times \sqrt[3]{5} = \sqrt[3]{25 \times 5} = \sqrt[3]{125} = \sqrt[3]{5^3} = 5$$

So, $\sqrt[3]{25}$ and $\sqrt[3]{5}$ are rationalising factor of each other.

Rationalising factors of a given surd

- Rationalising factor of $\frac{1}{\sqrt{a}} = \sqrt{a}$
- Rationalising factor of $\frac{1}{\sqrt{a} \pm \sqrt{b}} = \sqrt{a} \mp \sqrt{b}$

EXAMPLE 12. $\frac{\sqrt{5} + \sqrt{3}}{\sqrt{5} - \sqrt{3}} + \frac{\sqrt{5} - \sqrt{3}}{\sqrt{5} + \sqrt{3}}$ is equal to

- a. 16 b. 8 c. 4 d. $\sqrt{15}$

Sol. b.
$$\begin{aligned} &\frac{\sqrt{5} + \sqrt{3}}{\sqrt{5} - \sqrt{3}} + \frac{\sqrt{5} - \sqrt{3}}{\sqrt{5} + \sqrt{3}} \\ &= \frac{(\sqrt{5} + \sqrt{3})(\sqrt{5} + \sqrt{3})}{(\sqrt{5} - \sqrt{3})(\sqrt{5} + \sqrt{3})} + \frac{(\sqrt{5} - \sqrt{3})(\sqrt{5} - \sqrt{3})}{(\sqrt{5} + \sqrt{3})(\sqrt{5} - \sqrt{3})} \\ &= \frac{(\sqrt{5} + \sqrt{3})^2}{\sqrt{5}^2 - \sqrt{3}^2} + \frac{(\sqrt{5} - \sqrt{3})^2}{\sqrt{5}^2 - \sqrt{3}^2} = \frac{(\sqrt{5} + \sqrt{3})^2 + (\sqrt{5} - \sqrt{3})^2}{(\sqrt{5})^2 - (\sqrt{3})^2} \\ &= \frac{2[(\sqrt{5})^2 + (\sqrt{3})^2]}{5 - 3} = \frac{2(5 + 3)}{2} = 8 \end{aligned}$$

PRACTICE EXERCISE

- If $\sqrt{\frac{x}{49}} = \frac{4}{7}$, then the value of x is
(a) 9 (b) 25 (c) 16 (d) 8
- Which of the following cannot be a digit in the unit place of a perfect square?
(a) 1 (b) 5 (c) 7 (d) 0
- The number $\sqrt{0.0001}$ is
(a) a rational number less than 0.01
(b) a rational number
(c) an irrational number
(d) Neither a rational number nor an irrational number
- If m and n are natural numbers, then $\sqrt[n]{m}$ is
(a) always irrational
(b) irrational unless n is the m th power of an integer
(c) irrational unless m is the n th power of an integer
(d) irrational unless m and n are coprime
- What is that fraction which when multiplied by itself gives 227.798649?
(a) 15.093 (b) 15.099 (c) 14.093 (d) 9.0019
- If $x = \sqrt{3018 + \sqrt{36 + \sqrt{169}}}$, then the value of x is
(a) 55 (b) 44 (c) 63 (d) 42
- If $\sqrt[3]{\sqrt{x}} = x/5$, then x is equal to
(a) $\sqrt[5]{5}$ (b) $\sqrt[6]{5^5}$ (c) $\sqrt[5]{5^6}$ (d) $\sqrt{5}$
- What is the value of
 $\sqrt{29.16} + \sqrt{0.2916} + \sqrt{0.002916} + \sqrt{0.00002916}$?
(a) 5.9949 (b) 5.9894 (c) 5.9984 (d) 5.9994
- $(\sqrt{5})^{-5/2} \times (\sqrt{5})^{-3/2}$
(a) $\frac{1}{625}$ (b) $\frac{1}{25}$ (c) $\frac{1}{125}$ (d) None of these
- If $a = 3$, $b = 9$ and $c = 10$, then the value of
 $\sqrt{13+a} + \sqrt{112+b} + \sqrt{c-1}$ is
(a) 15 (b) 18 (c) 16 (d) 10
- If $\sqrt{0.9 \times 0.09 \times x} = 0.9 \times 0.09 \times \sqrt{z}$, then the value of $\frac{x}{z}$ is
(a) 0.081 (b) 0.810 (c) 0.81 (d) 8.09
- $\sqrt{2\sqrt{2\sqrt{2\sqrt{2\sqrt{2}}}}}$ is equal to
(a) 0 (b) 2 (c) 1 (d) $2^{31/32}$
- If $\sqrt{35} = 5.9160$, then the value of $\frac{\sqrt{7} + \sqrt{5}}{\sqrt{7} - \sqrt{5}}$ is
(a) 9.1060 (b) 10.9160 (c) 11.9160 (d) 12
- Find the value of
 $\sqrt{343 + \sqrt{307 + \sqrt{273 + \sqrt{241 + \sqrt{225}}}}}$
(a) 18 (b) 19 (c) $\sqrt{19}$ (d) $18 + \sqrt{3}$
- What is $\frac{1}{\sqrt{9} - \sqrt{8}} - \frac{1}{\sqrt{8} - \sqrt{7}} + \frac{1}{\sqrt{7} - \sqrt{6}} - \frac{1}{\sqrt{6} - \sqrt{5}}$
 $+ \frac{1}{\sqrt{5} - \sqrt{4}}$ equal to?
(a) 0 (b) 1 (c) 5 (d) $\frac{1}{3}$
- What is one of the square roots of $9 - 2\sqrt{14}$?
(a) $\sqrt{7} - \sqrt{3}$ (b) $\sqrt{6} - \sqrt{3}$ (c) $\sqrt{7} - \sqrt{5}$ (d) $\sqrt{7} - \sqrt{2}$
- If $p^x = r^y = m$ and $r^w = p^z = n$, then which one of the following is correct?
(a) $xw = yz$ (b) $xz = yw$
(c) $x + y = w + z$ (d) $x - y = w - z$
- If $a^x = b^y = c^z$ and $abc = 1$, then what is $xy + yz + zx$ equal to
(a) xyz (b) $x + y + z$ (c) 0 (d) 1
- If $a = \frac{\sqrt{5} + 1}{\sqrt{5} - 1}$ and $b = \frac{\sqrt{5} - 1}{\sqrt{5} + 1}$, then the value of
 $\left(\frac{a^2 + ab + b^2}{a^2 - ab + b^2}\right)$ is
(a) $3/4$ (b) $4/3$ (c) $3/5$ (d) $5/3$
- If $3\sqrt{5} + \sqrt{125} = 17.88$, then what will be the value of $\sqrt{80} + 6\sqrt{5}$
(a) 13.41 (b) 20.46 (c) 21.66 (d) 22.35
- If $N = 2^{0.15}$ and $N^b = 16$, then b is equal to
(a) $80/3$ (b) $5/3$ (c) 4 (d) $3/5$
- If $6^A = 2$, $6^B = 5$ and $6^Q = 15$, then Q equals to
(a) $B + 3$ (b) $5A + B$ (c) $B^2 - 2B$ (d) $B - A + 1$
- Arrange $\sqrt[4]{3}$, $\sqrt[6]{10}$, $\sqrt[12]{25}$ in descending order.
(a) $\sqrt[6]{10} > \sqrt[4]{3} > \sqrt[12]{25}$ (b) $\sqrt[12]{25} > \sqrt[4]{3} > \sqrt[6]{10}$
(c) $\sqrt[6]{10} > \sqrt[12]{25} > \sqrt[4]{3}$ (d) $\sqrt[4]{3} > \sqrt[12]{25} > \sqrt[6]{10}$
- If $(3.7)^x = (0.037)^y = 10000$, then what is the value of $\frac{1}{x} - \frac{1}{y}$?
(a) 1 (b) 2 (c) $\frac{1}{2}$ (d) $\frac{1}{4}$
- The greatest six digit number which is a perfect square is
(a) 998004 (b) 998006 (c) 998049 (d) 998001

- 26.** What is the smallest number by which 26244 must be divided to get a perfect cube?
(a) 4 (b) 6 (c) 36 (d) 16
- 27.** What is the smallest number that must be added to 1780 to make it a perfect square?
(a) 39 (b) 49 (c) 59 (d) 69
- 28.** A gardener plants 17956 trees in such a way that there are as many rows as there are trees in a row. The number of trees in a row are
(a) 136 (b) 164 (c) 134 (d) 166
- 29.** A group of student decided to collect as many paise from each member of the group as is the number of members. If the total collection amounts to ₹ 32.49 the number of members in the group, is
(a) 37 (b) 47 (c) 57 (d) 27
- 30.** A general arranges his soldiers in rows to form a perfect square. He find that in doing, so 60 soldiers are left out. If the total number of soldiers be 8160. Then, the number of soldiers in each row is
(a) 90 (b) 91 (c) 92 (d) 80
- 31.** A ball is dropped from a height 64 m above the ground and every time it hits the ground it rises to a height equal to half of the previous. What is the height attained after it hits the ground for the 16th time?
(a) 2^{-12} m (b) 2^{-11} m (c) 2^{-10} m (d) 2^{-9} m
- 32.** If $x = \frac{\sqrt{a+3b} + \sqrt{a-3b}}{\sqrt{a+3b} - \sqrt{a-3b}}$, then $3bx^2 - 2ax + 3b$ is equal to (given that, $b \neq 0$).
(a) 1 (b) 0 (c) ab (d) $2ab$
- 37.** If $16 \times 8^{n+2} = 2^m$, then m is equal to **2013 I**
(a) $n + 8$ (b) $2n + 10$
(c) $3n + 2$ (d) $3n + 10$
- 38.** When a ball bounces, it rises to $\frac{2}{3}$ of the height from which it fell. If the ball is dropped from a height of 36 m, how high will it rise at the third bounce? **2013 I**
(a) $10\frac{1}{3}$ m (b) $10\frac{2}{3}$ m (c) $10\frac{1}{3}$ m (d) $12\frac{2}{3}$ m
- 39.** The product of four consecutive natural numbers plus one is **2014 I**
(a) a non-square
(b) always sum of two square numbers
(c) a square
(d) None of the above
- 40.** The difference of two consecutive cubes **2014 I**
(a) is odd or even (b) is never divisible by 2
(c) is always even (d) None of these
- 41.** The square root of $\frac{(0.75)^3}{1 - 0.75} + [0.75 + (0.75)^2 + 1]$ is **2015 I**
(a) 1 (b) 2 (c) 3 (d) 4
- 42.** What is $\frac{5 + \sqrt{10}}{5\sqrt{5} - 2\sqrt{20} - \sqrt{32} + \sqrt{50}}$ equal to? **2015 I**
(a) 5 (b) $5\sqrt{2}$ (c) $5\sqrt{5}$ (d) $\sqrt{5}$
- 43.** If $x = \frac{91}{216}$, then the value of $3 - \frac{1}{(1-x)^{1/3}}$ is **2015 II**
(a) $\frac{9}{5}$ (b) $\frac{5}{9}$ (c) $\frac{4}{9}$ (d) $\frac{4}{5}$

> PREVIOUS YEARS' QUESTIONS

- 33.** If $a^x = b$, $b^y = c$ and $xyz = 1$, then what is the value of c^z ? **2012 I**
(a) a (b) b (c) ab (d) a/b
- 34.** If $196x^4 = x^6$, then x^3 is equal to which one of the following? **2012 I**
(a) $\frac{x^6}{14}$ (b) $14x^4$ (c) $\frac{x^2}{14}$ (d) $14x^2$
- 35.** If $\sqrt{10 + \sqrt[3]{x}} = 4$, then what is the value of x ? **2012 I**
(a) 150 (b) 216 (c) 316 (d) 450
- 36.** The expression $[(\sqrt{2})^{\sqrt{2}}]^{\sqrt{2}}$ gives **2013 I**
(a) a natural number
(b) an integer and not a natural number
(c) a rational number but not an integer
(d) a real number but not a rational number
- 44.** Which one of the following is correct? **2015 II**
(a) $\sqrt{2} < \sqrt[4]{6} < \sqrt[3]{4}$ (b) $\sqrt{2} > \sqrt[4]{6} > \sqrt[3]{4}$
(c) $\sqrt[4]{6} < \sqrt{2} < \sqrt[3]{4}$ (d) $\sqrt[4]{6} > \sqrt{2} > \sqrt[3]{4}$
- 45.** If $x = \sqrt{3} + \sqrt{2}$, then the value of $x^3 + x + \frac{1}{x} + \frac{1}{x^3}$, is **2015 II**
(a) $10\sqrt{3}$ (b) $20\sqrt{3}$ (c) $10\sqrt{2}$ (d) $20\sqrt{2}$
- 46.** If $x = 2^{1/3} + 2^{-1/3}$, then the value of $2x^3 - 6x - 5$ is equal to **2016 I**
(a) 0 (b) 1 (c) 2 (d) 3
- 47.** If $4^x 2^y = 128$ and $3^{3x} 3^{2y} - 9^{xy} = 0$, then the value of $x + y$ can be equal to **2016 I**
(a) 7 (b) 5 (c) 3 (d) 1
- 48.** If $x = \frac{\sqrt{a+2b} + \sqrt{a-2b}}{\sqrt{a+2b} - \sqrt{a-2b}}$, then $bx^2 - ax + b$ is equal to (given that $b \neq 0$) **2016 I**
(a) 0 (b) 1 (c) ab (d) $2ab$

ANSWERS

1		2		3		4		5		6		7		8		9		10	
11		12		13		14		15		16		17		18		19		20	
21		22		23		24		25		26		27		28		29		30	
31		32		33		34		35		36		37		38		39		40	
41		42		43		44		45		46		47		48					

HINTS AND SOLUTIONS

1. (c) Given, $\sqrt{\frac{x}{49}} = \frac{4}{7}$ or $\frac{\sqrt{x}}{7} = \frac{4}{7}$

Now, on squaring both sides, we get

$$(\sqrt{x})^2 = (4)^2 \Rightarrow x = 16$$

Hence, the value of x is 16.

2. (c) The digits 2, 3, 7 and 8 cannot be at the unit place of a perfect square number.

3. (b) $\sqrt{0.0001} = 0.01$ which is a rational number.

4. (b) If m and n are natural numbers, then $\frac{m}{n}$ is irrational unless n is m th power of an integer.

5. (a) Let fraction be x , then

$$x^2 = 227.798649$$

$$\Rightarrow x = \sqrt{227.798649} = 15.093$$

6. (a) Given, $x = \sqrt{3018 + \sqrt{36 + \sqrt{169}}}$
 $= \sqrt{3018 + \sqrt{36 + 13}}$
 $= \sqrt{3018 + \sqrt{49}} = \sqrt{3018 + 7}$
 $= \sqrt{3025} = 55$

Hence, the value of x is 55.

7. (c) $\sqrt[3]{\sqrt{x}} = x/5$

On cubing both sides, we get

$$\sqrt{x} = x^3/5^3$$

On squaring both sides, we get

$$x = x^6/5^6 \Rightarrow 5^6 = \frac{x^6}{x} = x^5$$

$$\Rightarrow x^5 = 5^6 \Rightarrow x = \sqrt[5]{5^6}$$

8. (d) $\sqrt{29.16} + \sqrt{0.2916} + \sqrt{0.002916}$
 $\quad \quad \quad + \sqrt{0.00002916}$
 $= 5.4 + 0.54 + 0.054 + 0.0054$
 $= 5.9994$

9. (b) $(\sqrt{5})^{-5/2} \times (\sqrt{5})^{-3/2}$
 $= (\sqrt{5})^{-5/2 - 3/2} = (\sqrt{5})^{-8/2}$

$$= (\sqrt{5})^{-4} = \left(\frac{1}{\sqrt{5}}\right)^4$$

$$= \left(\frac{1}{5}\right)^{(1/2) \times 4} = \left(\frac{1}{5}\right)^2 = \frac{1}{25}$$

10. (b) Given, $a = 3, b = 9$ and $c = 10$

$$\therefore \sqrt{13+a} + \sqrt{112+b} + \sqrt{c-1}$$

$$= \sqrt{13+3} + \sqrt{112+9} + \sqrt{10-1}$$

$$= \sqrt{16} + \sqrt{121} + \sqrt{9} = 4 + 11 + 3 = 18$$

11. (a) Here, $\sqrt{\frac{x}{z}} = \frac{0.9 \times 0.09}{\sqrt{0.9 \times 0.09}}$
 $= \frac{\sqrt{0.9 \times 0.09} \times \sqrt{0.9 \times 0.09}}{\sqrt{0.9 \times 0.09}}$
 $= \sqrt{0.9 \times 0.09}$

$$\Rightarrow \left(\sqrt{\frac{x}{z}}\right)^2 = (\sqrt{0.9 \times 0.09})^2$$

$$\Rightarrow \frac{x}{z} = 0.081$$

12. (d) Here, $\sqrt{2\sqrt{2\sqrt{2\sqrt{2}}}}$
 $= \sqrt{2\sqrt{2\sqrt{2 \times 2^{1/2}}}} = \sqrt{2\sqrt{2\sqrt{2 \times 2^{3/4}}}}$
 $= \sqrt{2\sqrt{2 \times 2^{7/8}}} = \sqrt{2 \times 2^{15/16}} = 2^{31/32}$

13. (c) Here, $\frac{\sqrt{7} + \sqrt{5}}{\sqrt{7} - \sqrt{5}}$
 $= \frac{\sqrt{7} + \sqrt{5}}{\sqrt{7} - \sqrt{5}} \times \frac{\sqrt{7} + \sqrt{5}}{\sqrt{7} + \sqrt{5}} = \frac{(\sqrt{7} + \sqrt{5})^2}{7 - 5}$
 $= \frac{7 + 5 + 2\sqrt{35}}{2} = \frac{12 + 2\sqrt{35}}{2}$
 $\quad \quad \quad [\because (a+b)^2 = a^2 + b^2 + 2ab]$
 $= 6 + \sqrt{35} = 6 + 5.9160 = 11.9160$

14. (b) $\sqrt{343 + \sqrt{307 + \sqrt{273 + \sqrt{241 + \sqrt{225}}}}}$
 $= \sqrt{343 + \sqrt{307 + \sqrt{273 + \sqrt{241 + 15}}}}$
 $= \sqrt{343 + \sqrt{307 + \sqrt{273 + 16}}}$

$$= \sqrt{343 + \sqrt{307 + 17}}$$

$$= \sqrt{343 + 18} = \sqrt{361} = 19$$

15. (c) $\frac{1}{\sqrt{9-\sqrt{8}}} - \frac{1}{\sqrt{8-\sqrt{7}}} + \frac{1}{\sqrt{7-\sqrt{6}}}$
 $\quad \quad \quad - \frac{1}{\sqrt{6-\sqrt{5}}} + \frac{1}{\sqrt{5-\sqrt{4}}}$
 $= (\sqrt{9} + \sqrt{8}) - (\sqrt{8} + \sqrt{7})$
 $\quad \quad \quad + (\sqrt{7} + \sqrt{6}) - (\sqrt{6} + \sqrt{5})$
 $\quad \quad \quad + (\sqrt{5} + \sqrt{4})$
 $\quad \quad \quad [\text{on rationalisation}]$
 $= \sqrt{9} + \sqrt{4} = 3 + 2 = 5$

16. (d) $\sqrt{9-2\sqrt{14}} = \sqrt{7+2-2 \times \sqrt{7} \times \sqrt{2}}$
 $= \sqrt{(\sqrt{7}-\sqrt{2})^2} = \sqrt{7}-\sqrt{2}$

17. (a) Given, $p^x = r^y = m$ and
 $r^w = p^z = n$
Now, $p^x = r^y$

On multiply power with w on both sides, we get

$$(p^x)^w = (r^y)^w \Rightarrow p^{xw} = r^{yw}$$

$$\Rightarrow p^{xw} = (r^w)^y \quad \dots (i)$$

On putting $r^w = p^z$ in Eq. (i), we get

$$p^{xw} = (p^z)^y \Rightarrow p^{xw} = p^{zy}$$

On comparing both sides, we get

$$\therefore xw = zy$$

18. (c) Given, $a^x = b^y = c^z = k$ [Let]

$$\Rightarrow a = k^{1/x}, b = k^{1/y}$$

and

$$c = k^{1/z}$$

$$\therefore abc = k^{\frac{1}{x} + \frac{1}{y} + \frac{1}{z}}$$

$$\Rightarrow 1 = k^{\frac{1}{x} + \frac{1}{y} + \frac{1}{z}} = k^0$$

$$[\because abc = 1, \text{ given}]$$

On comparing both sides, we get

$$\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 0$$

$$\Rightarrow xy + yz + zx = 0$$

$$\begin{aligned}
 19. (b) \quad a &= \frac{\sqrt{5}+1}{\sqrt{5}-1} \times \frac{\sqrt{5}+1}{\sqrt{5}+1} \\
 &= \frac{(\sqrt{5}+1)^2}{5-1} = \frac{3+\sqrt{5}}{2} \\
 b &= \frac{\sqrt{5}-1}{\sqrt{5}+1} \times \frac{\sqrt{5}-1}{\sqrt{5}-1} \\
 &= \frac{(\sqrt{5}-1)^2}{5-1} = \frac{3-\sqrt{5}}{2} \\
 \therefore a^2 + b^2 &= \left(\frac{3+\sqrt{5}}{2}\right)^2 + \left(\frac{3-\sqrt{5}}{2}\right)^2 \\
 &= \frac{2(9+5)}{4} = 7
 \end{aligned}$$

$$\begin{aligned}
 \text{Also, } ab &= \frac{\sqrt{5}+1}{\sqrt{5}-1} \times \frac{\sqrt{5}-1}{\sqrt{5}+1} = 1 \\
 \therefore \frac{a^2 + ab + b^2}{a^2 - ab + b^2} &= \frac{(a^2 + b^2) + ab}{(a^2 + b^2) - ab} \\
 &= \frac{7+1}{7-1} = \frac{4}{3}
 \end{aligned}$$

$$\begin{aligned}
 20. (d) \quad 3\sqrt{5} + \sqrt{125} &= 17.88 \\
 \Rightarrow 3\sqrt{5} + 5\sqrt{5} &= 17.88 \\
 \Rightarrow 8\sqrt{5} &= 17.88 \\
 \Rightarrow \sqrt{5} &= \frac{17.88}{8} = 2.235 \\
 \therefore \sqrt{80} + 6\sqrt{5} &= 4\sqrt{5} + 6\sqrt{5} \\
 &= 10\sqrt{5} = 10 \times 2.235 = 22.35
 \end{aligned}$$

$$\begin{aligned}
 21. (a) \quad N &= 2^{0.15} \\
 \Rightarrow N &= (2)^{3/20} \Rightarrow (N)^b = (2)^{3b/20} \\
 \text{But, } N^b &= 16 \\
 \therefore 16 &= (2)^{3b/20} \Rightarrow 2^4 = (2)^{3b/20} \\
 \Rightarrow 4 &= \frac{3b}{20} \Rightarrow b = \frac{80}{3}
 \end{aligned}$$

$$\begin{aligned}
 22. (d) \quad \text{Since } 15 &= 3 \times 5 = 3 \times 6^B \\
 \text{As, } 3 &= \frac{6}{2} = \frac{6^1}{6^A} = 6^{1-A} \\
 \therefore 15 &= 3 \times 6^B = 6^{1-A} \times 6^B \\
 \Rightarrow 6^Q &= 6^{1-A} \times 6^B \Rightarrow 6^Q = 6^{1-A+B} \\
 \Rightarrow Q &= 1-A+B
 \end{aligned}$$

$$\begin{aligned}
 23. (a) \quad \text{LCM of } 4, 6, 12 &= 12 \\
 \sqrt[4]{3} &= \sqrt[12]{3^3} = \sqrt[12]{27} \\
 \sqrt[6]{10} &= \sqrt[12]{10^2} = \sqrt[12]{100} \\
 \sqrt[12]{25} &= \sqrt[12]{25}
 \end{aligned}$$

$$\begin{aligned}
 \text{Clearly, } \sqrt[12]{100} &> \sqrt[12]{27} > \sqrt[12]{25} \\
 \text{Thus, } \sqrt[6]{10} &> \sqrt[4]{3} > \sqrt[12]{25}
 \end{aligned}$$

$$\begin{aligned}
 24. (c) \quad \text{Given, } (3.7)^x &= (0.037)^y = 10000 \\
 \Rightarrow (3.7)^x &= 10^4 \quad \text{and} \quad (0.037)^y = 10^4 \\
 \Rightarrow 37 &= 10^{4/x+1} \quad \text{and} \quad 37 = 10^{4/y+3} \\
 \Rightarrow 10^{4/x+1} &= 10^{4/y+3}
 \end{aligned}$$

$$\begin{aligned}
 \text{On comparing both sides, we get} \\
 \frac{4}{x} + 1 &= \frac{4}{y} + 3
 \end{aligned}$$

$$\begin{aligned}
 \Rightarrow \frac{4}{x} - \frac{4}{y} &= 3 - 1 \Rightarrow \frac{1}{x} - \frac{1}{y} = \frac{1}{2} \\
 \text{Hence, the value of } \frac{1}{x} - \frac{1}{y} &\text{ is } \frac{1}{2}.
 \end{aligned}$$

$$\begin{aligned}
 25. (d) \quad \text{Here, the greatest six-digit number} \\
 &= 999999
 \end{aligned}$$

$$\begin{array}{r}
 999 \\
 9 \overline{) 99 \, 99 \, 99} \\
 \underline{81} \\
 189 \\
 \times 9 \\
 \hline
 1989 \\
 \times 9 \\
 \hline
 19899 \\
 \times 9 \\
 \hline
 17901 \\
 \hline
 1998
 \end{array}$$

$$\begin{aligned}
 \text{Here, the greatest number of six digit} \\
 \text{which is perfect square} \\
 &= 999999 - 1998 = 998001
 \end{aligned}$$

$$\begin{aligned}
 26. (c) \quad \text{We have,} \\
 26244 &= 2 \times 2 \times 3 \times 3 \times 3 \times 3 \times 3 \times 3 \times 3 \times 3 \\
 \text{So, to get a perfect cube,} \\
 26244 &\text{ must be divided by } 2 \times 2 \times 3 \times 3 \\
 \text{i.e., } 36.
 \end{aligned}$$

$$\begin{aligned}
 27. (d) \quad \text{We know that, } (42)^2 &= 1764 \quad \text{and} \\
 (43)^2 &= 1849 \\
 \text{Since, } 1764 < 1780 < 1849 \\
 \text{Hence, the smallest number that must be} \\
 \text{added to 1780 is } (1849 - 1780), \text{ i.e. } 69.
 \end{aligned}$$

$$\begin{aligned}
 28. (c) \quad \text{Given, total number of trees} &= 17956 \\
 \therefore \text{Number of trees in each row} \\
 &= \sqrt{17956} = 134
 \end{aligned}$$

$$\begin{aligned}
 29. (c) \quad \text{Total rupees collected} &= ₹ 32.49 \\
 &= 32.49 \times 100 \text{ paise} = 3249 \text{ paise} \\
 \therefore \text{Number of member in the group} \\
 &= \sqrt{3249} = 57
 \end{aligned}$$

$$\begin{aligned}
 30. (a) \quad \text{Here, number of soldier arranged} \\
 &= 8160 - 60 = 8100 \\
 \text{As number of soldier in each row is} \\
 \text{equal to number of row.} \\
 \text{So, number of soldier in each row} \\
 &= \sqrt{8100} = 90
 \end{aligned}$$

$$\begin{aligned}
 31. (c) \quad \text{After Ist hit, height of the ball} \\
 &= \frac{1}{2} (64) \\
 \text{After IInd hit, height of the ball} \\
 &= \left(\frac{1}{2}\right)^2 (64)
 \end{aligned}$$

$$\begin{aligned}
 \text{After 16th hit, height of the ball} \\
 &= \left(\frac{1}{2}\right)^{16} (64) \\
 &= \frac{1}{2^{16}} (2^6) = 2^{-10} \text{ m}
 \end{aligned}$$

$$\begin{aligned}
 32. (b) \quad \text{Given, } x &= \frac{\sqrt{a+3b} + \sqrt{a-3b}}{\sqrt{a+3b} - \sqrt{a-3b}} \\
 \text{By rationalising, we get} \\
 x &= \frac{(\sqrt{a+3b})^2 + (\sqrt{a-3b})^2}{(\sqrt{a+3b})^2 - (\sqrt{a-3b})^2} \\
 &\quad + \frac{2\sqrt{(a+3b)(a-3b)}}{(\sqrt{a+3b})^2 - (\sqrt{a-3b})^2} \\
 x &= \frac{a+3b + a-3b + 2\sqrt{a^2 - 9b^2}}{a+3b - a+3b} \\
 x &= \frac{2a + 2\sqrt{a^2 - 9b^2}}{a+3b - a+3b} \\
 &= \frac{(a + \sqrt{a^2 - 9b^2})}{(3b)}
 \end{aligned}$$

$$3bx^2 - 2ax + 3b = 0$$

$$\begin{aligned}
 33. (a) \quad \text{Given, } a^x &= b, \quad b^y = c \quad \text{and} \quad xyz = 1 \\
 \text{Now, } a^x &= b \\
 \text{On multiplying both sides by } y \text{ in} \\
 \text{power, we get} \\
 (a^x)^y &= (b)^y \Rightarrow a^{xy} = b^y \\
 \text{Here, } b^y &= c \Rightarrow a^{xy} = c \\
 \text{Again, multiply with } z \text{ in power, we get} \\
 (a^{xy})^z &= c^z \Rightarrow a^{xyz} = c^z \\
 \text{Given, } xyz &= 1 \Rightarrow a = c^z \\
 \therefore c^z &= a \\
 \text{Hence, the value of } c^z &\text{ is } a.
 \end{aligned}$$

$$34. (d) 196x^4 = x^6, 196 = \frac{x^6}{x^4} = x^2$$

$$\begin{aligned}
 \Rightarrow x &= \sqrt{196} = 14 \\
 \Rightarrow x^3 &= 14x^2
 \end{aligned}$$

$$\begin{aligned}
 35. (b) \quad \text{Given, } \sqrt{10 + \sqrt[3]{x}} &= 4 \\
 \text{On squaring both sides, we get} \\
 10 + \sqrt[3]{x} &= 16 \\
 \Rightarrow \sqrt[3]{x} &= 16 - 10 = 6 \\
 \text{Now, on cubing both sides, we get} \\
 x &= (6)^3 = 216 \\
 \text{Hence, the value of } x &\text{ is } 216.
 \end{aligned}$$

$$\begin{aligned}
 36. (d) \quad \text{Expression} &= [(\sqrt{2})^{\sqrt{2}}]^{\sqrt{2}} \\
 &= (\sqrt{2})^{(\sqrt{2})^{\sqrt{2}}} = (\sqrt{2})^{(2)^{\frac{1}{\sqrt{2}}}} \\
 &= (2)^{\frac{1}{2} \times 2^{\frac{1}{\sqrt{2}}}} = (2)^{(2)^{\left(\frac{1}{\sqrt{2}} - 1\right)}}
 \end{aligned}$$

which denotes a real number but not a rational number.

$$\begin{aligned}
 37. (d) \quad \text{Given, } 16 \times 8^{n+2} &= 2^m \\
 \Rightarrow (2^4) \times 2^{3(n+2)} &= 2^m \\
 \Rightarrow 2^{(4+3n+6)} &= 2^m \Rightarrow 2^{(3n+10)} = 2^m \\
 \text{On comparing, we get} \\
 3n + 10 &= m \\
 \Rightarrow m &= 3n + 10
 \end{aligned}$$

38. (b) After first bounce, height of ball

$$= \left(\frac{2}{3}\right)^1 \times 36$$

and after third bounce, height of ball

$$= \left(\frac{2}{3}\right)^3 \times 36$$

$$= \frac{8}{27} \times 36 = \frac{8 \times 4}{3} = \frac{32}{3} \text{ m} = 10\frac{2}{3} \text{ m}$$

Hence, the required height at third bounce is $10\frac{2}{3}$ m.

39. (c) Product of four consecutive numbers plus one is always a square.

Illustration 1 Let four consecutive numbers be 3, 4, 5 and 6.

$$\therefore (3 \times 4 \times 5 \times 6) + 1 = 361 = (19)^2$$

Illustration 2 Let four consecutive numbers be 9, 10, 11 and 12.

$$\therefore (9 \times 10 \times 11 \times 12) + 1 = 11881 = (109)^2$$

40. (b) The difference of two consecutive cubes is never divisible by 2.

Illustration 1 Let the two consecutive numbers be 4 and 5.

$$\therefore (5)^3 - (4)^3 = 125 - 64 = 61$$

Illustration 2 Let the two consecutive numbers be 9 and 10.

$$\therefore (10)^3 - (9)^3 = 1000 - 729 = 271$$

41. (b) $\sqrt{\frac{(0.75)^3}{1-0.75} + [0.75 + (0.75)^2 + 1]}$

$$= \sqrt{\frac{(0.75)^3}{0.25} + [1.75 + (0.75)^2]}$$

$$= \sqrt{3 \times (0.75)^2 + 1.75 + (0.75)^2}$$

$$= \sqrt{4 \times (0.75)^2 + 1.75} = \sqrt{4} = 2$$

Hence, the required square root is 2.

42. (d) $\frac{5 + \sqrt{10}}{5\sqrt{5} - 2\sqrt{20} - \sqrt{32} + \sqrt{50}}$

$$= \frac{5 + \sqrt{10}}{5\sqrt{5} - 4\sqrt{5} - 4\sqrt{2} + 5\sqrt{2}}$$

$$= \frac{(5 + \sqrt{10})}{(\sqrt{5} + \sqrt{2})} \times \frac{(\sqrt{5} - \sqrt{2})}{(\sqrt{5} - \sqrt{2})}$$

$$= \frac{5\sqrt{5} - 5\sqrt{2} + \sqrt{50} - \sqrt{20}}{\sqrt{5^2 - 2^2}} \quad [\text{by rationalisation}]$$

$$[\because (a-b)(a+b) = a^2 - b^2]$$

$$= \frac{5\sqrt{5} - 5\sqrt{2} + 5\sqrt{2} - 2\sqrt{5}}{5 - 2} = \frac{3\sqrt{5}}{3} = \sqrt{5}$$

43. (a) We have, $x = \frac{91}{216}$

$$\Rightarrow 1 - x = 1 - \frac{91}{216} = \frac{125}{216} = \left(\frac{5}{6}\right)^3$$

$$\Rightarrow (1-x)^{\frac{1}{3}} = \frac{5}{6} \Rightarrow \frac{1}{(1-x)^{\frac{1}{3}}} = \frac{6}{5}$$

$$\Rightarrow 3 - \frac{1}{(1-x)^{\frac{1}{3}}} = 3 - \frac{6}{5} = \frac{9}{5}$$

44. (b) We have, $\sqrt{2}, 6^{1/4}, 4^{1/3}$

The LCM of 2, 4, 3 are 12.

$$(2^6)^{\frac{1}{12}}; (6^3)^{\frac{1}{12}}; (4^4)^{\frac{1}{12}}$$

$$(64)^{\frac{1}{12}}; (216)^{\frac{1}{12}}; (256)^{\frac{1}{12}}$$

$$\sqrt{2} < 6^{\frac{1}{4}} < 4^{\frac{1}{3}} \text{ or } \sqrt{2} > \sqrt[4]{6} > \sqrt[3]{4}$$

45. (b) We have, $x = \sqrt{3} + \sqrt{2}$

$$\therefore \frac{1}{x} = \frac{1}{\sqrt{3} + \sqrt{2}} \times \frac{\sqrt{3} - \sqrt{2}}{\sqrt{3} - \sqrt{2}}$$

$$= \frac{\sqrt{3} - \sqrt{2}}{3 - 2} = \sqrt{3} - \sqrt{2}$$

$$\therefore x + \frac{1}{x} = 2\sqrt{3}$$

On cubing both sides, we get

$$x^3 + \frac{1}{x^3} + 3\left(x + \frac{1}{x}\right) = 24\sqrt{3}$$

$$\therefore x^3 + \frac{1}{x^3} + x + \frac{1}{x} = 24\sqrt{3} - 2\left(x + \frac{1}{x}\right)$$

$$= 24\sqrt{3} - 2 \times 2\sqrt{3}$$

$$= 24\sqrt{3} - 4\sqrt{3} = 20\sqrt{3}$$

46. (a) Given, $x = 2^{1/3} + 2^{-1/3}$

$$\therefore 2x^3 - 6x - 5 = 2(2^{1/3} + 2^{-1/3})^3 - 6(2^{1/3} + 2^{-1/3}) - 5$$

$$= 2[2 + 2^{-1} + 3(2^{1/3} + 2^{-1/3})] - 6(2^{1/3} + 2^{-1/3}) - 5$$

$$= 4 + 2 \times \frac{1}{2} + 6(2^{1/3} + 2^{-1/3}) - 6(2^{1/3} + 2^{-1/3}) - 5$$

$$= 4 + 1 - 5 = 0$$

47. (b) We have, $4^x 2^y = 128$

$$\Rightarrow 2^{2x} 2^y = (2)^7$$

$$\Rightarrow 2^{2x+y} = (2)^7$$

$$\Rightarrow 2x + y = 7$$

$$\Rightarrow y = 7 - 2x \quad \dots(i)$$

Now, $3^{3x} \cdot 3^{2y} = 9^{xy}$

$$\Rightarrow 3^{3x+2y} = 3^{2xy}$$

$$\Rightarrow 3x + 2y = 2xy \quad \dots(ii)$$

$$\Rightarrow 3x + 2(7 - 2x) = 2x(7 - 2x)$$

[from Eq. (i)]

$$\Rightarrow 3x + 14 - 4x = 14x - 4x^2$$

$$\Rightarrow 4x^2 - 15x + 14 = 0$$

$$\Rightarrow 4x^2 - 8x - 7x + 14 = 0$$

$$\Rightarrow 4x(x - 2) - 7(x - 2) = 0$$

$$\Rightarrow (4x - 7)(x - 2) = 0$$

$$\Rightarrow x = \frac{7}{4} \text{ or } x = 2$$

$$\therefore y = \frac{7}{2} \text{ or } y = 3$$

$$\therefore x + y = \frac{21}{4} \text{ or } x + y = 5$$

48. (a) Refer to question 32.

06

TIME AND DISTANCE

Regularly (4-7) questions have been asked from this chapter. Generally the type of questions vary from simple to quite complex. But if you know the basics, then this section becomes easy to score area.



Distance

The length of the path travelled by any object or a person between two places is known as distance. The unit of distance is m or km.

Time

The duration in hours, minutes or seconds spent to cover a certain distance is called time.

Speed

The distance travelled by any object or a person per unit time is known as speed of that object or a person. The unit of speed is m/s or km/h.

So,
$$\text{speed} = \frac{\text{distance}}{\text{time}}$$

EXAMPLE 1. There are 20 poles with a constant distance between each pole. A car takes 24 s to reach the 12th pole. How much time will it take to reach the last pole?

- a. 25.25 s b. 17.45 s c. 35.75 s d. 41.45 s

Sol. d. Suppose distance between each pole is 1m.

∴ Total distance = 19 m

It takes 24 s to cover 11 m.

∴ To cover 19 m, it will take $\frac{24}{11} \times 19 = 41.45$.

Conversion of Unit

- (i) To convert speed of an object from km/h to m/s, multiply it by $\frac{5}{18}$.

e.g. $36 \text{ km/h} = 36 \times \frac{5}{18} \text{ m/s} = 10 \text{ m/s}$

- (ii) To convert speed of an object from m/s to km/h, multiply it by $\frac{18}{5}$.

e.g. $5 \text{ m/s} = 5 \times \frac{18}{5} \text{ km/h} = 18 \text{ km/h}$

Average Speed

The average speed of an object is defined as total distance travelled divided by total time taken.

i.e. Average speed = $\frac{\text{Total distance}}{\text{Total time}}$

EXAMPLE 2. A man completes 30 km of a journey at 6 km/h and the remaining 40 km of the journey in 5 h. Then, his average speed for the whole journey is

- a. 5 km/h b. 7 km/h
c. 7.5 km/h d. None of these

Sol. b. Average speed = $\frac{\text{Total distance travelled}}{\text{Total time taken}} = \frac{70}{10} = 7 \text{ km/h}$

Relative Speed

The speed of an object with respect to other is called relative speed. Suppose two bodies are moving with speeds of x km/h and y km/h respectively, then their relative speed will be

- (i) $(x + y)$ km/h, if both the bodies are moving in opposite directions.
(ii) $(x - y)$ km/h, if both the bodies are moving in same direction.

EXAMPLE 3. Two persons 27 km apart setting out at the same time are together in 9 h, if they walk in the same direction but in 3 h, if they walk in opposite directions. Then, their rates of walking (speeds) are

- a. 2 km/h and 4 km/h b. 3 km/h and 5 km/h
c. 4 km/h and 8 km/h d. None of these

Sol. d. Let the first person be walking faster with speed x km/h and second walking with speed y km/h.

Case I Both walking in same directions.

$\therefore$ Distance travelled by first person in 9 h = $9x$ km
and distance travelled by second person in 9 h = $9y$ km
As both are 27 km apart

$\therefore 9x - 9y = 27 \Rightarrow x - y = 3 \quad \dots(i)$

Case II Both walking in opposite directions.

$\therefore$ Distance travelled by first person in 3 h = $3x$
and distance travelled by second person in 3 h = $3y$
So, by condition, $3x + 3y = 27 \Rightarrow x + y = 9 \quad \dots(ii)$

On adding Eqs. (i) and (ii), we get

$\Rightarrow 2x = 12 \Rightarrow x = 6 \text{ km/h}$

Put the value of x in Eq. (ii), we get

$6 + y = 9 \Rightarrow y = 3 \text{ km/h}$

So, their speeds are 6 km/h and 3 km/h.

Important Rules and Formulae

Rule 1 If A travels with speed x km/h for t_1 h and with speed y km/h for t_2 h, then average speed = $\frac{xt_1 + yt_2}{t_1 + t_2}$

EXAMPLE 4. A man walks at the rate of 5 km/h for 6 h and at 4 km/h for 12 h. Find out the average speed (in km/h) of the man.

- a. $2\frac{1}{3}$ b. $5\frac{1}{3}$ c. $7\frac{1}{3}$ d. None of these

Sol. d. Here, $x = 5$ km/h, $y = 4$ km/h, $t_1 = 6$ h and $t_2 = 12$ h

$\therefore$ Average speed = $\frac{xt_1 + yt_2}{t_1 + t_2} = \frac{5 \times 6 + 4 \times 12}{6 + 12} = \frac{30 + 48}{18}$
 $= \frac{78}{18} = 4\frac{1}{3} \text{ km/h}$

Rule 2 If a man/vehicle covers two equal distances with the speed of x km/h and y km/h respectively, then the average speed of the man/vehicle for complete journey will be $\frac{2xy}{x + y}$.

EXAMPLE 5. A man covers half of his journey at 6 km/h and the remaining half at 3 km/h. Then, his average speed is

- a. 1 km/h b. 2 km/h c. 3 km/h d. 4 km/h

Sol. d. Here, $x = 6$ km/h and $y = 3$ km/h

$\therefore$ Average speed = $\frac{2xy}{x + y} = \frac{2 \times 6 \times 3}{6 + 3} = \frac{36}{9} = 4 \text{ km/h}$

Rule 3 If a man changes his speed to $\left(\frac{x}{y}\right)$ of his usual

speed and gets late by t min, then the usual time taken by him = $\frac{t \times x}{(y - x)}$.

Note If a person changes his speed to $\left(\frac{x}{y}\right)$ of his usual and reaches early by t min, then usual time taken by him = $\frac{t \times x}{(x - y)}$.

EXAMPLE 6. If a man travels with a speed of $\frac{2}{5}$ times of his original speed and he reached his office 15 min late to fixed time, then the time taken with his original speed, is

- a. 10 min b. 15 min c. 20 min d. 25 min

Sol. a. Here, $x = 2$, $y = 5$ and $t = 15$ min

$\therefore$ Required time = $\frac{x \times t}{y - x} = \frac{2 \times 15}{5 - 2}$
 $= \frac{2 \times 15}{3} = 2 \times 5 = 10 \text{ min}$

Rule 4 When a man travels from A to B with a speed x km/h and reaches t_1 h later after the fixed time and when he travels with a speed y km/h from A to B , he reaches his destination t_2 h before the fixed time, then distance between A and $B = \frac{xy(t_1 + t_2)}{y - x}$ km.

EXAMPLE 7. Walking at 3 km/h, Rajeev reaches his school 5 min late, if he walks at 4 km/h, he will be 5 min early. The distance of Rajeev's school from his house is

- a. $1\frac{1}{2}$ km b. 2 km c. $2\frac{1}{2}$ km d. 5 km

Sol. b. Here, $x = 3$ km/h, $y = 4$ km/h, $t_1 = \frac{5}{60}$ h and $t_2 = \frac{5}{60}$ h

$$\therefore \text{Required distance} = \frac{xy(t_1 + t_2)}{y - x} = \frac{3 \times 4 \left(\frac{5}{60} + \frac{5}{60} \right)}{4 - 3} = 3 \times 4 \times \frac{10}{60} = 2 \text{ km}$$

Rule 5 A person travels from A to B with a speed x km/h and reaches t_1 h late. Later he increases his speed by y km/h to cover the same distance and he still gets late by t_2 h, then the distance between A and $B = (t_1 - t_2)(x + y) \frac{x}{y}$.

EXAMPLE 8. A boy walking at a speed of 20 km/h reaches his school 30 min late. Next time he increased his speed by 5 km/h but still he is late by 10 min. What is the distance of the school from his home?

- a. $\frac{100}{3}$ km b. $\frac{50}{3}$ km c. $\frac{100}{7}$ km d. $\frac{2}{5}$ km

Sol. a. Given, $x = 20$ km/h, $y = 5$ km/h, $t_1 = 30 \text{ min} = \frac{1}{2}$ h

$$\text{and } t_2 = 10 \text{ min} = \frac{1}{6} \text{ h}$$

$$\therefore \text{Required distance} = (t_1 - t_2)(x + y) \frac{x}{y} = \left(\frac{1}{2} - \frac{1}{6} \right) (20 + 5) \times \frac{20}{5} = \frac{100}{3} \text{ km}$$

Problem Based on Trains

Problems based on trains are same as the problems related to speed, Time and Distance. The only difference is that the length of the moving object (train) is taken into consideration in these types of problems.

Rule 6 (i) Time taken by a train x m long in passing a single post or pole or standing man = Time taken by the train to cover x m.

(ii) Time taken by a train x m long in passing an object of length y m = Time taken by the train to cover $(x + y)$ m.

EXAMPLE 9. A 100 m long train is moving at a speed of 60 km/h. In what time will it cross a signal pole?

- a. 6 s b. 12 s c. 15 s d. None of these

Sol. a. Here, speed of train = 60 km/h = $60 \times \frac{5}{18} = \frac{50}{3}$ m/s

and distance covered by train in passing a pole = Length of the train

$\therefore$ Length of train = 100 m

$$\therefore \text{Time taken to pass the pole} = \frac{\text{Distance travelled by train}}{\text{Speed of train}} = 6 \text{ s}$$

Rule 7 Time taken by a train of length l_1 metre moving at a speed of s_1 m/s to cross another train of length l_2 moving at a speed of s_2 m/s in same/opposite direction is $\frac{l_1 + l_2}{s_1 \pm s_2}$

Use '+' for opposite direction.

Use '-' for same direction.

EXAMPLE 10. Two trains of lengths 110 m and 130 m travel on parallel track. If they move in the same direction, the first one which is faster takes one minute to pass the other one completely. If they move in opposite directions then they pass each other in 3s, then the speed of the trains is

- a. 41 m/s and 39 m/s b. 32 m/s and 43 m/s
c. 42 m/s and 38 m/s d. None of these

Sol. c. Let v_1 be the velocity of faster train and v_2 be the velocity of slower train.

Case I If they move in the same direction, then

$$\frac{110 + 130}{v_1 - v_2} = 60 \text{ s} \Rightarrow v_1 - v_2 = 4 \quad \dots(i)$$

Case II If they move in opposite directions,

$$\frac{110 + 130}{v_1 + v_2} = 3 \Rightarrow v_1 + v_2 = 80 \quad \dots(ii)$$

On adding Eqs. (i) and (ii), we get $2v_1 = 84 \Rightarrow v_1 = 42$ m/s

Now, putting the value of v_1 in Eq. (ii), we get

$$42 + v_2 = 80, \quad v_2 = 38 \text{ m/s} = 3 \times 4 \times \frac{10}{60} = 2 \text{ km}$$

Rule 8 If two trains start at the same time from points A and B towards each other and after crossing each other, they take t_1 and t_2 time in reaching points B and A respectively, then ratio of their speed = $\sqrt{\frac{t_2}{t_1}}$

or
$$\frac{S_1}{S_2} = \sqrt{\frac{t_2}{t_1}}$$

EXAMPLE 11. Two trains, one from Howrah to Patna and the other from Patna to Howrah, start simultaneously. After they meet, the trains reach their destinations after 9 h and 16 h, respectively. What is the ratio of their speeds?

- a. 4:3 b. 3:1
c. 4:5 d. 3:2

Sol. a. Let us name the trains as A and B . Then, A 's speed : B 's speed = $\sqrt{t_2} : \sqrt{t_1} = \sqrt{16} : \sqrt{9} = 4 : 3$.

Problems Based on Boat and Stream

Boats and streams is an application of speed, time and distance. Some of the important terms are explained below

- **Still water** If the speed of water is zero, then water is considered to be still water.
- **Stream water** If the water of a river is moving at a certain speed, then it is called stream water.
- **Downstream motion** In water, the direction along the stream is called downstream.
- **Upstream motion** In water, the direction against the stream is called upstream.

Some rules and problems related to boat and stream are given below

Rule 9 If the speed of a boat in still water be x km/h and speed of stream be y km/h, then

- (i) Speed of boat downstream = $(x + y)$ km/h
(ii) Speed of boat upstream = $(x - y)$ km/h

EXAMPLE 12. A sailor goes 8 km downstream in 40 min and returns back in 1 h. Then, the speed of the sailor in still water and the speed of the current is

- a. 5 km/h and 3 km/h b. 10 km/h and 2 km/h
c. 7 km/h and 10 km/h d. None of these

Sol. b. Let the speed of the sailor in still water = x km/h and speed of the current (stream) = y km/h
Then, speed of the sailor downstream = $(x + y)$ km/h and speed of the sailor upstream = $(x - y)$ km/h

$$\therefore \text{Time to travel 8 km downstream} = 40 \text{ min} = \frac{40}{60} \text{ h} = \frac{2}{3} \text{ h}$$

$$\Rightarrow \frac{8}{x + y} = \frac{2}{3} \Rightarrow x + y = \frac{24}{2} \quad \left[\because \frac{\text{distance}}{\text{speed}} = \text{time} \right]$$

$$\Rightarrow x + y = 12 \quad \dots(i)$$

$$\text{and time to return} = 1 \text{ h} \Rightarrow \frac{8}{x - y} = 1$$

$$\Rightarrow x - y = 8 \quad \dots(ii)$$

On adding Eqs. (i) and (ii), we get $x = 10$ and $y = 2$

Hence, speed of sailor in still water = 10 km/h

Speed of current (stream) = 2 km/h

Rule 10 Let the speed of boat in downstream = u km/h and speed of boat in upstream = v km/h, then

- (i) Speed of boat in still water = $\frac{1}{2}(u + v)$ km/h
(ii) Speed of stream (current) = $\frac{1}{2}(u - v)$ km/h

EXAMPLE 13. The speed of boat upstream and speed of boat downstream are 7 km/h and 13 km/h, respectively. Then, the speed of stream and speed of boat in still water is

- a. 10 km/h and 3 km/h b. 15 km/h and 9 km/h
c. 20 km/h and 6 km/h d. 40 km/h and 12 km/h

Sol. a. Let the speed of boat in downstream and upstream be u km/h and v km/h, respectively.

Given, $v = 7$ km/h and $u = 13$ km/h

$$\text{Speed of boat in still water} = \frac{1}{2}(7 + 13) = \frac{1}{2}(20) = 10 \text{ km/h}$$

$$\text{Speed of the stream} = \frac{1}{2}(13 - 7) = \frac{1}{2}(6) = 3 \text{ km/h}$$

RACES

A contest of speed in running, driving, riding, sailing or rowing over a specified distance is called a race.

Suppose A and B are two contestants in a race. If A beats B by x m and length of the track is d m.

Then, Distance travelled by B when A finishes the race

$$= d - x \text{ and } \frac{\text{Speed of } A}{\text{Speed of } B} = \frac{d}{d - x}$$

Note (i) A gives B a start of x m means when A starts at starting point, B starts x m ahead from starting point.

(ii) A gives B a start of t s means A starts t s after B starts from the same point.

EXAMPLE 14. In a 200 m race, A can beat B by 50 m and B can beat C by 8 m. In the same race, A can beat C by what distance?

- a. 60 m b. 72 m c. 56 m d. 66 m

Sol. c. A can beat B by 50 m.

$$\therefore \frac{\text{speed of } A}{\text{speed of } B} = \frac{200}{200 - 50} = \frac{200}{150}$$

Also, B can beat C by 8 m,

$$\therefore \frac{\text{speed of } B}{\text{speed of } C} = \frac{200}{200 - 8} = \frac{200}{192}$$

$$\begin{aligned} \text{Now, } \frac{\text{speed of } A}{\text{speed of } C} &= \frac{\text{speed of } B}{\text{speed of } C} \times \frac{\text{speed of } A}{\text{speed of } B} \\ &= \frac{200}{150} \times \frac{200}{192} = \frac{200}{144} \end{aligned}$$

So, A beats C by $(200 - 144) = 56$ m

> PRACTICE EXERCISE

1. A car completes a journey in 6 h with a speed of 50 km/h. At what speed must it travel to complete the journey in 5 h?
(a) 60 km/h (b) 55 km/h (c) 45 km/h (d) 61 km/h
2. Kiran covers a certain distance at 80 km/h and returns back to the same point at 20 km/h. Then, the average speed during the whole journey is
(a) 35 km/h (b) 32 km/h (c) 30 km/h (d) 28 km/h
3. Normally Sarita takes 3 h to travel between two stations with a constant speed. One day her speed was reduced by 12 km/h and she took 45 min more to complete the journey. Then, the distance between the two stations is
(a) 60 km (b) 120 km (c) 180 km (d) 95 km
4. Rani goes to school at 10 km/h and reaches the school 6 min late. Next day, she covers this distance at 12 km/h and reaches the school 9 min earlier than the scheduled time. What is the distance of her school from her house?
(a) 16 km (b) 12 km (c) 10 km (d) 15 km
5. A man travels first 50 km at 25 km/h next 40 km at 20 km/h and then 90 km at 15 km/h. His average speed (in m/s) for the whole journey is
(a) 18 (b) 5 (c) 10 (d) 36
6. A boy is running at a speed of p km/h to cover a distance of 1 km. But due to the slippery ground, his speed is reduced by q km/h ($p > q$). If he takes r h to cover the distance, then
(a) $\frac{1}{r} = \frac{pq}{p+q}$ (b) $\frac{1}{r} = p+q$ (c) $r = p-q$ (d) $\frac{1}{r} = p-q$
7. A train passes telegraph post in 40 s moving at a rate of 36 km/h. Then, the length of the train is
(a) 400 m (b) 500 m (c) 450 m (d) 395 m
8. A person can run around a circular path of radius 21 m in 44 s. In what time will the same person run a distance of 3 km?
(a) 18 min 40 s (b) 16 min 30 s
(c) 18 min 30 s (d) 16 min 40 s
9. A car is ahead of a scooter by 30 km. Car goes at the rate of 50 km/h and the scooter goes at the rate of 60 km/h. The scooter overtakes the car after
(a) 3 h (b) 3.5 h (c) 4 h (d) $3\frac{1}{4}$ h
10. Two towns A and B are 250 km apart. A bus starts from A to B at 6 : 00 am at a speed of 40 km/h. At the same time another bus starts from B to A at a speed of 60 km/h. The time of their meeting is
(a) 8 : 30 am (b) 8 : 00 am (c) 9 : 00 am (d) 9 : 15 am
11. A train T_1 leaves a place P at 5:00 am and reaches another place Q at 9 : 00 am another train T_2 leaves the place Q at 7 : 00 am and reaches the place P at 10 : 30 am. The time at which the two trains cross each other is
(a) 8 : 26 am (b) 7 : 56 am (c) 8 : 15 am (d) 8 : 00 am
12. A certain distance is covered at a certain speed. If half of the distance is covered in double time, then the ratio of the two speeds is
(a) 4 : 1 (b) 1 : 4 (c) 2 : 1 (d) 1 : 2
13. Two trains start running at the same time from two stations which are 210 km apart and going in opposite directions cross each other at a distance of 100 km from one of the station. The ratio of their speed is
(a) 11 : 9 (b) 10 : 11 (c) 11 : 10 (d) 9 : 11
14. A police car is ordered to chase a speeding car that is 5 km ahead. The thief car is travelling at an average speed of 80 km/h and the police car pursues it at an average speed of 100 km/h. How long does it take for the police car to overtake the other car?
(a) 17 min (b) 19 min (c) 13 min (d) 15 min
15. Points A and B are 70 km apart on a highway. A car starts from A and another car starts from B at the same time. If they travel in the same direction they meet in 7 h, but they travel towards each other they meet in 1 h. What are the speeds of the cars?
(a) 30 km/h, 40 km/h (b) 36 km/h, 40 km/h
(c) 19 km/h, 20 km/h (d) 40 km/h, 50 km/h
16. Assume that the distance that a car runs on 1 L of petrol varies inversely as the square of the speed at which it is driven. It gives a run of 25 km/L at a speed of 30 km/h. At what speed should it be driven to get a run of 36 km/L?
(a) 12.5 km/h (b) 25 km/h (c) 30 km/h (d) 40 km/h
17. A man standing on a railway platform observes that a train going in one direction takes 4 s to pass him. Another train of same length going in the opposite direction takes 5 s to pass him. The time taken (in seconds) by the two trains to cross each other will be
(a) $\frac{32}{9}$ (b) $\frac{33}{7}$ (c) $\frac{40}{9}$ (d) $\frac{49}{9}$

- 18.** A bullock cart has to cover a distance of 80 km in 10 h. If it covers half of the journey in $\frac{3}{5}$ th time, what should be its speed to cover the remaining distance in the time left?
(a) 5 km/h (b) 10 km/h (c) 15 km/h (d) 18 km/h
- 19.** By walking at $\frac{3}{4}$ of his usual speed, a man reaches his office 25 min later than usual. His usual time is
(a) 60 min (b) 70 min (c) 75 min (d) 80 min
- 20.** A train covers a distance in 50 min, if it runs at a speed of 48 km/h on an average. The speed at which the train must run to reduce the time of journey to 40 min, will be
(a) 10 km/h (b) 20 km/h (c) 40 km/h (d) 60 km/h
- 21.** Two trains travel in the same direction at 50 km/h and 32 km/h, respectively. A man in the slower train observes that 15 s elapse before the faster train completely passes him. What is the length of the faster train?
(a) 75 m (b) 125 m (c) 150 m (d) $\frac{625}{3}$ m
- 22.** A father and his son start at a point A with speeds of 12 km/h and 18 km/h, respectively and reach another point B. If his son starts 60 min after his father at A and reaches B, 60 min before his father, what is the distance between A and B?
(a) 90 km (b) 72 km (c) 36 km (d) None of these
- 23.** Two trains of lengths 100 m and 150 m are travelling in opposite directions at speeds of 75 km/h and 50 km/h, respectively. What is the time taken by them to cross each other?
(a) 7.4 s (b) 7.2 s (c) 7 s (d) 6.8 s
- 24.** A man can walk uphill at the rate of 2.5 km/h and downhill at the rate of 3.25 km/h. If the total time required to walk a certain distance up the hill and return to the starting position is 4 h 36 min, what is the distance he walked up the hill?
(a) 3.5 km (b) 4.5 km (c) 5.5 km (d) 6.5 km
- 25.** A boat goes 30 km upstream and 44 km downstream in 10 h. In 13 h it can go 40 km upstream and 55 km downstream. The speed of the boat in still water is
(a) 9 km/h (b) 8 km/h (c) 4 km/h (d) 3 km/h
- 26.** A motorboat takes 2 h to travel a distance of 9 km down the current and it takes 6 h to travel the same distance against the current. What is the speed of the boat in still water (in km/h)?
(a) 3 (b) 2 (c) 1.5 (d) 10
- 27.** Ram travels from P to Q at 10 km/h and return at 15 km/h. Shyam travels from P to Q and return at 12.5 km/h. If he takes 12 min less than Ram, then what is the distance between P and Q?
(a) 60 km (b) 45 km (c) 36 km (d) 30 km
- 28.** A person x started 5 min earlier at 60 km/h from a place P, then another person y followed him at 48 km/h, started his journey at 3:05 pm. Which of the following is/are correct.
I. At 3 : 15 pm, x and y are 7 km apart.
II. At 3 : 25 pm, y will overtake x
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II
- 29.** A boy walking at a speed of 20 km/h reaches his school 30 min late. Next time he increases his speed by 4 km/h but still he is late by 10 min.
Which of the following statement is/are correct?
I. The distance of the school from his home is 50 km/h.
II. If he increases his speed by 10 km/h but still late by 10 min, then distance of the school will be 20 km.
(a) Both I and II are correct (b) Only I is correct
(c) Only II is correct (d) Neither I nor II are correct.
- Directions** (Q. Nos.30-31) Ramesh and Prateek start running at same time in opposite directions from two points and after passing each other they complete their journeys in 'x' and 'y' h, respectively.
Then, Speed of Ramesh : Speed of Prateek = $\sqrt{y} : \sqrt{x}$
- 30.** Find the ratio of the speeds of Ramesh and Prateek, if Ramesh and Prateek completed their journeys in 16 h and 25 h after passing each other.
(a) 5 : 4 (b) 5 : 3 (c) 4 : 5 (d) 3 : 5
- 31.** What if two persons completed their journey with speeds in the ratio 7 : 9, then how much time they should have taken to complete their journey after they meet each other.
(a) 81 : 49 (b) $\sqrt{7} : \sqrt{9}$ (c) 49 : 81 (d) 14 : 18

PREVIOUS YEARS' QUESTIONS

- 32.** A train 280 m long is moving at a speed of 60 km/h. What is the time taken by the train to cross a platform 220 m long? ☑ 2012 I
(a) 45 s (b) 40 s (c) 35 s (d) 30 s
- 33.** A car travels along the four sides of a square at speeds v , $2v$, $3v$ and $4v$, respectively. If u is the average speed of the car in its travel around the square, then which one of the following is correct? ☑ 2012 I
(a) $u = 2.25v$ (b) $u = 3v$
(c) $v < u < 2v$ (d) $3v < u < 4v$
- 34.** A car is travelling at a constant rate of 45 km/h. The distance travelled by car from 10 : 40 am to 1 : 00 pm is ☑ 2012 II
(a) 165 km (b) 150 km (c) 120 km (d) 105 km

- 35.** A person travels a certain distance at 3 km/h and reaches 15 min late. If he travels at 4 km/h, he reaches 15 min earlier. The distance he has to travel is **✎ 2013 I**
 (a) 4.5 km (b) 6 km (c) 7.2 km (d) 12 km
- 36.** If a body covers a distance at the rate of x km/h and another equal distance at the rate of y km/h, then the average speed (in km/h) is **✎ 2013 I**
 (a) $\frac{x+y}{2}$ (b) $\sqrt{xy}$ (c) $\frac{2xy}{x+y}$ (d) $\frac{x+y}{xy}$
- 37.** A sailor sails a distance of 48 km along the flow of a river in 8 h. If it takes 12 h to return the same distance, then the speed of the flow of the river is **✎ 2013 I**
 (a) 0.5 km/h (b) 1 km/h (c) 1.5 km/h (d) 2 km/h
- 38.** A train running at the speed of 72 km/h goes past a pole in 15 s. What is the length of the train? **✎ 2013 II**
 (a) 150 m (b) 200 m (c) 300 m (d) 350 m
- 39.** Two cars A and B start simultaneously from a certain place at the speed of 30 km/h and 45 km/h, respectively. The car B reaches the destination 2 h earlier than A . What is the distance between the starting point and destination? **✎ 2013 II**
 (a) 90 km (b) 180 km (c) 270 km (d) 360 km
- 40.** A man cycles with a speed of 10 km/h and reaches his office at 1 : 00 pm. However, when he cycles with a speed of 15 km/h, he reaches his office at 11 : 00 am. At what speed should he cycle, so that he reaches his office at 12 noon? **✎ 2013 II**
 (a) 12.5 km/h (b) 12 km/h (c) 13 km/h (d) 13.5 km/h
- 41.** A train takes 9 s to cross a pole. If the speed of the train is 48 km/h, the length of the train is **✎ 2014 I**
 (a) 150 m (b) 120 m (c) 90 m (d) 80 m
- 42.** A train takes 10 s to cross a pole and 20 s to cross a platform of length 200 m. What is the length of the train? **✎ 2014 II**
 (a) 50 m (b) 100 m (c) 150 m (d) 200 m
- 43.** A train travels at a speed of 40 km/h and another train at a speed of 20 m/s. What is the ratio of speed of the first train to that of the second train? **✎ 2014 II**
 (a) 2 : 1 (b) 5 : 9 (c) 5 : 3 (d) 9 : 5
- 44.** The distance between two points (A and B) is 110 km. X starts running from point A at a speed of 60 km/h and Y starts running from point B at a speed of 40 km/h at the same time. They meet at a point C , somewhere on the line AB . What is the ratio of AC to BC ? **✎ 2014 II**
 (a) 3 : 2 (b) 2 : 3 (c) 3 : 4 (d) 4 : 3
- 45.** A man rides one-third of the distance from A to B at the rate of x km/h and the remainder at the rate of $2y$ km/h. If he had travelled at a uniform rate of $6z$ km/h, then he could have ridden from A to B and back again in the same time. Which one of the following is correct? **✎ 2014 II**
 (a) $z = x + y$ (b) $3z = x + y$
 (c) $\frac{1}{z} = \frac{1}{x} + \frac{1}{y}$ (d) $\frac{1}{2z} = \frac{1}{x} + \frac{1}{y}$
- 46.** In a flight of 600 km, an aircraft was slowed down due to bad weather. Its average speed for the trip was reduced by 200 km/h and the time of flight increased by 30 min. The duration of the flight is **✎ 2015 I**
 (a) 1 h (b) 2 h (c) 3 h (d) 4 h
- 47.** With a uniform speed, a car covers a distance in 8 h. Had the speed been increased by 4 km/h, the same distance could have been covered in 7 h and 30 min. What is the distance covered? **✎ 2015 I**
 (a) 420 km (b) 480 km (c) 520 km (d) 640 km
- 48.** A car travels the first one-third of a certain distance with a speed of 10 km/h, the next one-third distance with a speed of 20 km/h and the last one-third distance with a speed of 60 km/h. The average speed of the car for the whole journey is **✎ 2015 I**
 (a) 18 km/h (b) 24 km/h (c) 30 km/h (d) 36 km/h
- 49.** A man rows 32 km downstream and 14 km upstream, and he takes 6 h to cover each distance. What is the speed of the current? **✎ 2015 I**
 (a) 0.5 km/h (b) 1 km/h (c) 1.5 km/h (d) 2 km/h
- 50.** A runs $1\frac{2}{3}$ times as fast as B . If A gives B a start of 80 m, how far must the winning post from the starting point be so that A and B might reach it at the same time? **✎ 2015 I**
 (a) 200 m (b) 300 m (c) 270 m (d) 160 m
- 51.** A thief is noticed by a policeman from a distance of 200 m. The thief starts running and the policeman chases him. The thief and the policeman run at the speed of 10 km/h and 11 km/h, respectively. What is the distance between them after 6 min? **✎ 2015 I**
 (a) 100 m (b) 120 m (c) 150 m (d) 160 m
- 52.** Two persons A and B start simultaneously from two places c km apart and walk in the same direction. If A travels at the rate of p km/h and B travels at the rate of q km/h, then A has travelled before he overtakes B a distance of **✎ 2015 I**
 (a) $\frac{qc}{p+q}$ km (b) $\frac{pc}{p-q}$ km (c) $\frac{qc}{p-q}$ km (d) $\frac{pc}{p+q}$ km

- 53.** By increasing the speed of his car by 15 km/h, a person covers 300 km distance by taking an hour less than before. The original speed of the car was **2015 II**
(a) 45 km/h (b) 50 km/h (c) 60 km/h (d) 75 km/h
- 54.** Two trains, one is of 121 m in length at the speed of 40 km/h and the other is of 99 m in length at the speed of 32 km/h are running in opposite directions. In how much time will they be completely clear from each other from the moment they meet? **2015 II**
(a) 10 s (b) 11 s (c) 16 s (d) 21 s
- 55.** The speeds of three buses are in the ratio 2 : 3 : 4. The time taken by these buses to travel the same distance will be in the ratio. **2015 II**
(a) 2 : 3 : 4 (b) 4 : 3 : 2
(c) 4 : 3 : 6 (d) 6 : 4 : 3
- 56.** Two trains are moving in the same direction at 1.5 km/min and 60 km/h, respectively. A man in the faster train observes that it takes 27 s to cross the slower train. The length of the slower train is **2015 II**
(a) 225 m (b) 230 m (c) 240 m (d) 250 m
- 57.** In a race of 100 m, A beats B by 4 m and A beats C by 2 m. By how many metres (approximately) would C beat B in another 100 m race assuming C and B run with their respective speeds as in the earlier race? **2015 II**
(a) 2 (b) 2.04 (c) 2.08 (d) 3.2
- 58.** Three athletes run a 4 km race. Their speeds are in the ratio 16 : 15 : 11. When the winner wins the race, then the distance between the athlete in the second position to the athlete in the third position is **2015 II**
(a) 1000 m (b) 800 m (c) 750 m (d) 600 m
- 59.** A motorboat, whose speed is 15 km/h in still water goes 30 km downstream and comes back in a total of 4h and 30 min. The speed of the stream is **2015 II**
(a) 4 km/h (b) 5 km/h
(c) 6 km/h (d) 10 km/h
- 60.** A bike consumes 20 mL of petrol per kilometre, if it is driven at a speed in the range of 25-50 km/h and consumes 40 mL of petrol per kilometre at any other speed. How much petrol is consumed by the bike in travelling a distance of 50 km, if the bike is driven at a speed of 40 km/h for the first 10 km, at a speed of 60 km/h for the next 30 km and at a speed of 30 km/h for the last 10 km? **2015 II**
(a) 1 L (b) 1.2 L (c) 1.4 L (d) 1.6 L
- 61.** A passenger train takes 1 h less for a journey of 120 km, if its speed is increased by 10 km/h from its usual speed. What is its usual speed? **2016 I**
(a) 50 km/h (b) 40 km/h (c) 35 km/h (d) 30 km/h
- 62.** A man walking at 5 km/h noticed that a 225 m long train coming in the opposite direction crossed him in 9 s. The speed of the train is **2016 I**
(a) 75 km/h (b) 80 km/h (c) 85 km/h (d) 90 km/h
- 63.** A cyclist moves non-stop from A to B, a distance of 14 km, at a certain average speed. If his average speed reduces by 1 km/h, then he takes 20 min more to cover the same distance. The original average speed of the cyclist is **2016 I**
(a) 5 km/h (b) 6 km/h
(c) 7 km/h (d) None of these
- 64.** In a race of 1000 m, A beats B by 100 m or 10 s. If they start a race of 1000 m simultaneously from the same point and if B gets injured after running 50 m less than half the race length and due to which his speed gets halved, then by how much time will A beat B? **2016 I**
(a) 65 s (b) 60 s (c) 50 s (d) 45 s
- 65.** In a race A, B and C take part. A beats B by 30 m, B beats C by 20 m and A beats C by 48 m.
Which of the following is/are correct?
I. The length of the race is 300 m.
II. The speeds of A, B and C are in the ratio 50 : 45 : 42.
Select the correct answer using codes given below.
(a) Only I (b) Only II **2016 I**
(c) Both I and II (d) Neither I nor II

ANSWERS

1	a	2	b	3	c	4	d	5	b	6	d	7	a	8	d	9	a	10	a
11	b	12	a	13	c	14	d	15	a	16	b	17	c	18	b	19	c	20	d
21	a	22	b	23	b	24	d	25	b	26	a	27	d	28	a	29	c	30	a
31	a	32	d	33	c	34	d	35	b	36	c	37	b	38	c	39	b	40	b
41	b	42	d	43	b	44	a	45	c	46	a	47	b	48	a	49	c	50	a
51	a	52	b	53	c	54	b	55	d	56	a	57	b	58	a	59	b	60	d

HINTS AND SOLUTIONS

1. (a) Let speed be v km/h.

We know that,

Distance = Speed $\times$ Time

$$6 \times 50 = 5 \times v$$

$$\Rightarrow v = \frac{6 \times 50}{5} = 60 \text{ km/h}$$

2. (b) Average speed = $\frac{2xy}{x+y}$ [Rule 2]

$$= \frac{2 \times 80 \times 20}{80 + 20} = 32 \text{ km/h}$$

3. (c) Let Sarita's speed be u and distance is constant.

$$\text{Then, } u \times 3 = (u - 12) \frac{15}{4}$$

$$\left[\because 3 \text{ h} + 45 \text{ min} = \frac{15}{4} \text{ h} \right]$$

$$\Rightarrow u = \frac{45 \times 4}{3} = 60 \text{ km/h}$$

$\therefore$ Distance = Speed $\times$ Time

$$= 60 \times 3 = 180 \text{ km}$$

4. (d) Here, $x = 10$, $y = 12$, $t_1 = \frac{6}{60}$ h and

$$t_2 = \frac{9}{60} \text{ h}$$

$$\therefore \text{ Required distance} = \frac{xy(t_1 + t_2)}{y - x}$$

[Rule 4]

$$= \frac{10 \times 12 \times 15}{60 \times 2}$$

$$= 15 \text{ km}$$

5. (b) Total distance covered

$$= 50 + 40 + 90 = 180 \text{ km}$$

Total time taken

$$= \left(\frac{50}{25} + \frac{40}{20} + \frac{90}{15} \right) = 10 \text{ h}$$

$\therefore$ Average speed for the whole journey

$$= \frac{\text{Total distance travelled}}{\text{Total time taken}}$$

$$= \frac{180}{10} = 18 \text{ km/h}$$

$$\therefore 18 \text{ km/h} = \frac{18 \times 5}{18} \text{ m/s} = 5 \text{ m/s}$$

6. (d) Actual speed of boy = $(p - q)$ km/h

$$\text{Time taken to cover 1 km} = \frac{1}{p - q}$$

$$\therefore \frac{1}{p - q} = r \Rightarrow \frac{1}{r} = p - q$$

7. (a) Length of train = Distance covered in 40 s at the rate of 36 km/h.

[Rule 6 (i)]

$\therefore$ Length of train

$$= 40 \times 36 \times \frac{5}{18} = 400 \text{ m}$$

8. (d) Distance travelled in 44 s = $2\pi r$

$$= 2 \times \frac{22}{7} \times 21 = 132 \text{ m}$$

$$\therefore \text{ Speed} = \frac{132}{44} = 3 \text{ m/s}$$

$$\left[\because \text{ speed} = \frac{\text{distance}}{\text{time}} \right]$$

$$\text{Time taken to travel 3 km} = \frac{3000}{3}$$

$$= 1000 \text{ s} = \frac{1000}{60} \text{ min} = 16 \text{ min } 40 \text{ s}$$

9. (a) Distance between car and scooter = 30 km

Relative speed

$$y = 60 - 50 = 10 \text{ km/h}$$

So, the time taken by scooter to overtake

$$\text{the car} = \frac{30}{10} = 3 \text{ h}$$

10. (a) Let the buses meet x h after 6:00 am.

Then, the distance covered by the two buses is 250 km.

$$\therefore 40x + 60x = 250$$

$$\Rightarrow x = \frac{250}{100} = 2.5 \text{ h} = 2 \text{ h and } 30 \text{ min}$$

So, they will meet at 8 : 30 am.

11. (b) Let the distance between P and Q be x km and let the two trains meet y h after 7 am.

Then, T_1 covers x km in 4h and T_2 covers

x km in $3\frac{1}{2}$ h.

$$\therefore \text{ Speed of train } T_1 = \frac{x}{4} \text{ km/h}$$

$$\text{and speed of train } T_2 = \frac{2x}{7} \text{ km/h}$$

According to the question,

$$\frac{x(y+2)}{4} + \frac{2xy}{7} = x$$

$$\Rightarrow \frac{(y+2)}{4} + \frac{2y}{7} = 1$$

$$\Rightarrow \frac{7(y+2) + 8y}{28} = 1$$

$$\Rightarrow 7y + 14 + 8y = 28$$

$$\therefore y = \frac{14}{15} \text{ h} = \frac{14}{15} \times 60 \text{ min} = 56 \text{ min}$$

So, trains will meet at 7 : 56 am.

12. (a) Let x km distance be covered in y h.

Then, speed of object in first case

$$= \frac{x}{y} \text{ km/h}$$

As, half of this distance is covered in double time.

Then, speed of object in second case

$$= \frac{x}{2} \div 2y = \frac{x}{2} \times \frac{1}{2y} = \frac{x}{4y} \text{ km/h}$$

$\therefore$ Ratio of first and second speeds

$$= \frac{x}{y} : \frac{x}{4y} = 1 : \frac{1}{4} = 4 : 1$$

13. (c) Let the speed of two trains be x km/h and y km/h, respectively. Then, the time taken by first train to cover 110 km = Time taken by second train to cover 100 km.

$$\text{Thus, } \frac{110}{x} = \frac{100}{y} \Rightarrow \frac{x}{y} = \frac{110}{100}$$

$$\therefore x : y = 11 : 10$$

14. (d) Distance travelled by thief car in one hour = 80 km

Distance travelled in one hour by police car = 100 km

So, police travels extra 20 km in 1 h.

So, to overtake thief, police car has to travel 5 km extra.

$$\therefore \text{ Time} = \frac{5}{20} = \frac{1}{4} \text{ h} = \frac{60}{4} \text{ min} = 15 \text{ min}$$

15. (a) Let the speed of car A = x km/h

Let the speed of car B = y km/h

Total distance covered by both the cars in 1h = 70 km

$$x(1) + y(1) = 70$$

$$x + y = 70 \quad \dots(i)$$

[in opposite direction]

Distance covered by both the cars in

$$7 \text{ h} = 70 \text{ km}$$

But in same direction

$$7x - 7y = 70 \Rightarrow x - y = 10 \quad \dots(ii)$$

added Eq. (i) and (ii), we get

$$2x = 80, x = 40 \text{ km/h}, y = 30 \text{ km/h}$$

16. (b) Let the speed of car be v and distance covered by car in one litre be A .

$$\therefore A \propto \frac{1}{v^2} \Rightarrow A = \frac{K}{v^2}$$

Then, $A = 25 \text{ km/h}$ and $v = 30 \text{ km/h}$

$$\text{So, } 25 = \frac{K}{(30)^2} \Rightarrow K = 900 \times 25$$

$$\Rightarrow K = 22500 \Rightarrow A = \frac{22500}{v^2}$$

When $A = 36 \text{ km/L}$

$$v^2 = \frac{22500}{36} \Rightarrow v = \sqrt{\frac{22500}{36}} = \frac{150}{6}$$

Hence, the speed of a car is 25 km/h .

17. (c) Let the length of each train be $I \text{ m}$.

$$\Rightarrow \text{Speed of first train} = \left(\frac{I}{4}\right) \text{ m/s}$$

$$\text{and speed of second train} = \left(\frac{I}{5}\right) \text{ m/s}$$

As, both trains are moving in opposite direction.

Time taken to cross each other

$$= \frac{I + I}{\frac{I}{4} + \frac{I}{5}} \quad [\text{Rule 7}]$$

$$= \left(\frac{2I}{\frac{9I}{20}}\right) \text{ s} = \left(\frac{20 \times 2}{9}\right) = \frac{40}{9} \text{ s}$$

18. (b) Given, total distance to cover in 10 h
 $= 80 \text{ km}$

If it covers 40 km in $\frac{3}{5}$ th of time i.e.
 40 km in 6 h .

$$\therefore \text{Remaining time} = 10 - 6 = 4 \text{ h}$$

$$\text{Remaining distance} = 40 \text{ km}$$

$$\therefore \text{Required speed} = (40 \div 4) \text{ km/h}$$

$$= \frac{40}{4} = 10 \text{ km/h}$$

19. (c) Here, $x = 3 \text{ m/s}$, $y = 4 \text{ m/s}$
 and $t = 25 \text{ min}$

$$\therefore \text{Required time} = \frac{x \times t}{y - x} \quad [\text{Rule 3}]$$

$$= \frac{3 \times 25}{4 - 3} = 75 \text{ min}$$

20. (d) Distance travelled in $1 \text{ h} = 48 \text{ km}$

$$\therefore \text{Distance travelled in } 50 \text{ min}$$

$$= \frac{48}{60} \times 50 = 40 \text{ km}$$

$$\text{Time to be reduced} = \frac{40}{60} \text{ h}$$

$$\therefore \text{Required speed}$$

$$= \frac{40}{40/60} = \frac{40 \times 60}{40} = 60 \text{ km/h}$$

21. (a) Let the length of faster train be $x \text{ km}$.

$\therefore$ Trains travel in the same direction.

$$\therefore \text{Relative speed} = (50 - 32) = 18 \text{ km/h}$$

$$\text{Elapsed time} = 15 \text{ s} = \frac{15}{3600} \text{ h}$$

$$\text{Now, time} = \frac{\text{distance}}{\text{speed}}$$

$$\Rightarrow \frac{15}{3600} = \frac{x}{18} \Rightarrow x \times 3600 = 18 \times 15$$

$$\therefore x = \frac{18 \times 15}{3600} = 0.075 \text{ km} = 75 \text{ m}$$

22. (b) Let distance between A and B be $x \text{ km}$.

By given condition,

$$\frac{x}{12} - \frac{x}{18} = 2 \Rightarrow 6x = 2 \times 18 \times 12$$

$$\therefore x = \frac{2 \times 18 \times 12}{6} = 72 \text{ km}$$

Hence, the required distance is 72 km .

23. (b) Relative speed when trains are in opposite direction

$$= V_1 + V_2$$

$$= 75 + 50 = 125 \text{ km/h}$$

$$= \frac{125 \times 5}{18} \text{ m/s}$$

and total distance covered

$$= (100 + 150) = 250 \text{ m}$$

$\therefore$ Time taken to cross each other

$$= \frac{\text{Total covered distance}}{\text{Relative speed}}$$

$$= \frac{250 \times 18}{125 \times 5} = 7.2 \text{ s}$$

24. (d) Let he walked up the hill at a distance of $x \text{ km}$.

By given condition,

$$\frac{x}{2.5} + \frac{x}{3.25} = 4 \frac{36}{60}$$

$$\Rightarrow x \left(\frac{1}{2.5} + \frac{1}{3.25} \right) = \frac{276}{60}$$

$$\Rightarrow x \left(\frac{2}{5} + \frac{4}{13} \right) = \frac{276}{60}$$

$$\therefore x = \frac{276}{60} \times \frac{65}{46} = 6.5 \text{ km}$$

25. (b) Let speed of boat in still water

$$= x \text{ km/h}$$

and speed of boat in current water

$$= y \text{ km/h}$$

$$\therefore \text{Downstream speed} = (x + y) \text{ km/h}$$

$$\text{Upstream speed} = (x - y) \text{ km/h} \quad [\text{Rule 9}]$$

According to the question,

$$\frac{30}{x - y} + \frac{44}{x + y} = 10 \quad \dots(i)$$

$$\text{and } \frac{40}{x - y} + \frac{55}{x + y} = 13 \quad \dots(ii)$$

$$\text{Let } \frac{1}{x - y} = u \text{ and } \frac{1}{x + y} = v \quad \dots(iii)$$

$$\text{Then, } 30u + 44v = 10$$

$$40u + 55v = 13$$

On solving the above equations, we get

$$u = \frac{1}{5}, v = \frac{1}{11}$$

From Eq. (iii), we get

$$x - y = 5 \text{ and } x + y = 11$$

On solving the above equations, we get

$$x = 8, y = 3$$

Thus, the speed of boat in still water

$$= 8 \text{ km/h}$$

26. (a) Let the speed of motorboat be $x \text{ km/h}$ and the speed of water be $y \text{ km/h}$.

Now, speed of boat downstream

$$= (x + y) \text{ km/h} \quad [\text{Rule 9}]$$

and speed of boat upstream

$$= (x - y) \text{ km/h}$$

By given condition,

$$\frac{9}{x + y} = 2 \Rightarrow 2x + 2y = 9 \dots(i)$$

$$\text{and } \frac{9}{x - y} = 6 \Rightarrow 6x - 6y = 9$$

$$\Rightarrow 2x - 2y = 3 \quad \dots(ii)$$

On solving Eqs. (i) and (ii), we get

$$x = 3 \text{ km/h and } y = 1.5 \text{ km/h}$$

$\therefore$ Speed of boat = 3 km/h

27. (d) Let the distance between P and $Q = d \text{ km}$. Total time taken by Ram

$$= \frac{d}{10} + \frac{d}{15} = \frac{25d}{150}$$

$$\text{Total time taken by Shyam} = \frac{2d}{12.5} = \frac{4d}{25}$$

According to question,

$$\frac{25d}{150} - \frac{4d}{25} = \frac{12}{60}$$

$$\Rightarrow d \left[\frac{25 - 24}{150} \right] = \frac{1}{5}$$

$$\Rightarrow \frac{d}{150} = \frac{1}{5} \Rightarrow d = 30 \text{ km}$$

28. (a) Distance travelled by x in 15 min

$$= 60 \times \frac{15}{60} = 15 \text{ km}$$

Distance travelled by y in 10 min

$$= 48 \times \frac{10}{60} = 8 \text{ km}$$

$$\text{Difference} = (15 - 8) \text{ km} = 7 \text{ km}$$

Hence, at 3:15 pm they are 7 km apart.

So, statement I is true. As speed of x is greater than y . So, y will never overtake x .

Thus, statement II is false.

29. (c) I. Here, $x = 20 \text{ km/h}$, $y = 4 \text{ km/h}$,

$$t_1 = 30 \text{ min}, t_2 = 10 \text{ min}$$

According to formula,

$$\therefore \text{Required distance} = (t_1 - t_2)(x + y) \frac{x}{y} \quad [\text{Rule 5}]$$

$$= \frac{(30 - 10)}{60} (20 + 4) \left(\frac{20}{4} \right)$$

$$= \frac{20}{60} \times 24 \times \frac{20}{4} = 5 \times 8 = 40 \text{ km}$$

So, I is incorrect.

II. Here, $x = 20$ km/h, $y = 10$ km/h
 $t_1 = 30$ min, $t_2 = 10$ min

According to formula,

∴ Required distance

$$= \left(\frac{30 - 10}{60} \right) (20 + 10) \left(\frac{20}{10} \right)$$

$$= \frac{20}{60} \times 30 \times \frac{20}{10} = 20 \text{ km}$$

So, II is correct.

30. (a) $x = 16$, $y = 25$

According to formula,

$$\therefore \frac{\text{Ramesh's Speed}}{\text{Prateek's Speed}} = \sqrt{\frac{y}{x}} = \sqrt{\frac{25}{16}} = \frac{5}{4}$$

or 5:4

31. (a) Here, we know the relation

$$\frac{\text{Speed of Ramesh}}{\text{Speed of Prateek}} = \sqrt{\frac{y}{x}} \Rightarrow \frac{7}{9} = \sqrt{\frac{y}{x}}$$

On squaring both sides, $\frac{49}{81} = \frac{y}{x} = 81:49$

32. (d) To cover a distance by train

$$= 280 + 220 = 500 \text{ m}$$

Speed of train = 60 km/h

$$= 60 \times \frac{5}{18} \text{ m/s}$$

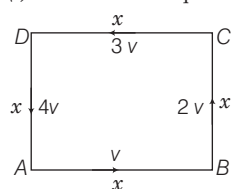
$$= \frac{50}{3} \text{ m/s}$$

∴ Time taken by train

$$= \frac{\text{total distance}}{\text{speed}}$$

$$= \frac{500}{50/3} = 30 \text{ s}$$

33. (c) Let side of a square be x .



∴ Average speed (u) = $\frac{\text{total distance}}{\text{total time}}$

$$= \frac{(x + x + x + x)}{\frac{x}{v} + \frac{x}{2v} + \frac{x}{3v} + \frac{x}{4v}}$$

$$= \frac{4x}{x \left(\frac{1}{v} + \frac{1}{2v} + \frac{1}{3v} + \frac{1}{4v} \right)}$$

$$= \frac{4 \times v}{1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4}} = \frac{48v}{25} = 1.92 v$$

which lies in the interval $v < u < 2v$.

34. (d) Given, speed of a car = 45 km/h
 Time taken by the car

$$= 140 \text{ min} = \frac{140}{60} \text{ h}$$

So, required distance travelled by car

$$= \text{speed} \times \text{time}$$

$$= 45 \times \frac{140}{60} = 105 \text{ km}$$

35. (b) Given, $x = 3$ km/h, $y = 4$ km/h,

$$t_1 = 15 \text{ min} = \frac{15}{60} \text{ h}$$

$$\text{and } t_2 = 15 \text{ min} = \frac{15}{60} \text{ h}$$

$$\therefore \text{Required distance} = \frac{xy(t_1 + t_2)}{y - x}$$

[Rule 4]

$$= \frac{3 \times 4 \left(\frac{15}{60} + \frac{15}{60} \right)}{4 - 3}$$

$$= 12 \times \frac{30}{60} = 12 \times \frac{1}{2} = 6 \text{ km}$$

36. (c) If a body covers a distance at the rate of x km/h and another equal distance at the rate of y km/h, then = $\frac{2xy}{x + y}$

[Rule 2]

37. (b) Let speed of the flow of water be v km/h and rate of sailing of sailor be u km/h.

Speed of sailor downstream

$$= (u + v) \text{ km/h}$$

[Rule 9]

Speed of sailor upstream = $(u - v)$ km/h

$$\text{Condition I } u + v = \frac{48}{8}$$

$$\Rightarrow u + v = 6 \quad \dots(i)$$

$$\text{Condition II } u - v = \frac{48}{12}$$

$$\Rightarrow u - v = 4 \quad \dots(ii)$$

On subtracting Eqs. (ii) from (i), we get

$$v = 1 \text{ km/h}$$

38. (c) Speed of train = 72 km/h

$$= 72 \times \frac{5}{18}$$

$$= 20 \text{ m/s}$$

Length of train = distance covered

$$x = 20 \times 15$$

$$x = 300 \text{ m}$$

Length of the train is 300 m.

39. (b) Let the distance between two points be x km

Time taken by A - Time taken by B = 2

$$\Rightarrow \frac{x}{30} - \frac{x}{45} = 2$$

$$\Rightarrow x \left[\frac{3 - 2}{90} \right] = 2$$

$$\Rightarrow x = 90 \times 2 = 180 \text{ km}$$

40. (b) Let t h be the time taken by the man to reach his office at speed of 15 km/h.

Then, time taken to reach office at the speed of 10 km/h = $(t + 2)$ h

$$\text{Now, } s_1 \times t_1 = s_2 \times t_2$$

$$\Rightarrow 10 \times (t + 2) = 15 \times t$$

$$\Rightarrow 10t + 20 = 15t \Rightarrow 5t = 20$$

$$\therefore t = 4 \text{ h}$$

Now, distance covered to reach office

$$= s_1 \times t_1$$

$$= 10 \times (4 + 2) = 10 \times 6 = 60 \text{ km}$$

Now, speed required to reach office at 12 noon in 5 h.

$$\therefore \text{Speed} = \frac{60}{5} = 12 \text{ km/h}$$

41. (b) Speed of train = 48 km/h

$$= \left(48 \times \frac{5}{18} \right) \text{ m/s}$$

Let the length of train be x m

$$x = 48 \times \frac{5}{18} \times 9$$

$$x = 120 \text{ m}$$

Length of the train is 120 m.

42. (d) Let the speed of a train be x m/s and length be y m.

Condition I When $t = 10$ s

$$\text{time} = \frac{\text{distance}}{\text{speed}}$$

$$\Rightarrow 10 = \frac{y}{x} \therefore y = 10x \quad \dots(i)$$

Condition II When $t = 20$ s

and total distance = $y + 200$ m

$$\therefore \text{Time} = \frac{\text{Total distance}}{\text{Speed}}$$

$$20 = \frac{y + 200}{x} \Rightarrow 20x = y + 200$$

$$20 \times \frac{y}{10} = y + 200$$

$$\Rightarrow 2y - y = 200 \Rightarrow y = 200$$

Hence, the length of train is 200 m.

43. (b) Given, speed of a train = 40 km/h

$$= 40 \times \frac{5}{18} \text{ m/s}$$

Speed of another train = 20 m/s

∴ Required ratio

$$= \frac{\text{Speed of first train}}{\text{Speed of second train}}$$

$$= \frac{40 \times \frac{5}{18}}{20} = \frac{40 \times 5}{20 \times 18} = \frac{2 \times 5}{18}$$

$$= \frac{10}{18} = \frac{5}{9} \text{ or } 5:9$$

44. (a) Given, distance between two points (A and B) = 110 km

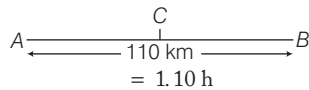
Their relative speed

$$= 60 + 40 = 100 \text{ km/h}$$

Time after which they meet

$$= \frac{\text{Total distance}}{\text{Relative speed}}$$

$$= \frac{110}{100}$$



Distance covered by A in 1.1 h

$$= AC = 60 \times 1.1 = 66 \text{ km}$$

Remaining distance

$$= BC = 110 - 66 = 44 \text{ km}$$

∴ Required ratio = AC : BC

$$= 66 : 44 = 3 : 2$$

45. (c) Let the total distance be d .

Time taken to cover $\frac{1}{3}$ rd distance,

$$t_1 = \frac{\frac{1}{3}d}{x} = \frac{d}{3x} \left[\because \text{time} = \frac{\text{distance}}{\text{speed}} \right]$$

Remaining distance = $d - \frac{1}{3}d = \frac{2}{3}d$

Time taken to cover $\frac{2}{3}$ rd distance,

$$t_2 = \frac{\frac{2}{3}d}{2y} = \frac{2d}{6y} = \frac{d}{3y}$$

Time taken to cover distance from A to B and B to A,

$$\therefore t = \frac{2d}{6z} = \frac{d}{3z}$$

According to the question,

$$t_1 + t_2 = t \Rightarrow \frac{d}{3x} + \frac{d}{3y} = \frac{d}{3z}$$

$$\Rightarrow \frac{1}{3x} + \frac{1}{3y} = \frac{1}{3z} \therefore \frac{1}{x} + \frac{1}{y} = \frac{1}{z}$$

46. (a) Let the original speed of an aircraft be x km/h and its reduced speed = $(x - 200)$ km/h.

Condition I

Time taken by aircraft to cover 600 km = $\frac{600}{x}$ h.

Condition II Time taken by aircraft to cover

$$600 \text{ km} = \frac{600}{(x - 200)} \text{ h}$$

According to the question,

$$\frac{600}{x - 200} - \frac{600}{x} = \frac{1}{2} \left[\because 30 \text{ min} = \frac{1}{2} \text{ h} \right]$$

On dividing both sides by 600, we get

$$\frac{1}{x - 200} - \frac{1}{x} = \frac{1}{600 \times 2}$$

$$\Rightarrow \frac{x - (x - 200)}{x(x - 200)} = \frac{1}{1200}$$

$$\Rightarrow x^2 - 200x - 240000 = 0$$

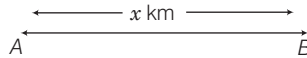
$$\Rightarrow (x - 600)(x + 400) = 0$$

$$\Rightarrow x = 600 \text{ km/h}$$

$$\text{or } x \neq -400 \text{ km/h}$$

$$\therefore \text{Required time} = \frac{600}{600} = 1 \text{ h.}$$

47. (b) Let the distance between A and B be x km and speed be V km/h.



Case I Given, distance = x km, speed = V km/h and time = 8 h.

$$\therefore \text{Speed} = \frac{\text{Distance}}{\text{Time}} \Rightarrow V = \frac{x}{8} \dots (i)$$

Case II If speed = $(V + 4)$ km/h

$$\text{and time} = 7\frac{1}{2} \text{ h} = \frac{15}{2} \text{ h}$$

$$\text{then, } V + 4 = \frac{x}{15/2} \Rightarrow V + 4 = \frac{2x}{15}$$

$$\Rightarrow \frac{x}{8} + 4 = \frac{2x}{15} \quad [\text{from Eq. (i)}]$$

$$\Rightarrow \frac{2x}{15} - \frac{x}{8} = 4 \Rightarrow \frac{x}{120} = 4$$

$$\Rightarrow x = 480 \text{ km}$$

48. (a)

Let total distance of AB be x km.

$$\text{For distance AC, } t_1 = \frac{x/3}{10} = \frac{x}{30}$$

$$\text{For distance CD, } t_2 = \frac{x/3}{20} = \frac{x}{60}$$

$$\text{For distance BD, } t_3 = \frac{x/3}{60} = \frac{x}{180}$$

∴ Total time taken

$$= t_1 + t_2 + t_3 = \frac{x}{30} + \frac{x}{60} + \frac{x}{180}$$

$$= \frac{10x}{180} = \frac{x}{18}$$

$$\therefore \text{Average speed} = \frac{\text{Total distance}}{\text{Total time taken}} = \frac{x}{\frac{x}{18}} = 18 \text{ km/h}$$

49. (c) Let speed of a boat in still water be x km/h and speed of the stream be y km/h.

Then, speed of the boat downstream

$$= (x + y) \text{ km/h} \quad [\text{Rule 9}]$$

and speed of the boat upstream

$$= (x - y) \text{ km/h}$$

According to the question,

$$\frac{32}{x + y} = 6 \quad \left[\because \text{speed} = \frac{\text{distance}}{\text{time}} \right]$$

$$\text{and } \frac{14}{x - y} = 6$$

$$\Rightarrow 6x + 6y = 32 \quad \dots (i)$$

$$\text{and } 6x - 6y = 14 \quad \dots (ii)$$

On subtracting Eq. (ii) from Eq. (i), we get

$$12y = 18$$

$$y = \frac{18}{12} = 1.5 \text{ km/h}$$

50. (a) Let the speed of B be x m/s.

$$\therefore \text{Speed of A} = 1\frac{2}{3}x = \frac{5x}{3} \text{ m/s}$$

Ratio of speed of rates of A and B

$$= \frac{5x}{3} : x = 5 : 3$$

∴ 2 m are gained in a race of 5 m.

∴ 1 m are gained in a race of $\frac{5}{2}$ m.

So, 80 m are gained in a race of

$$\frac{5}{2} \times 80 \text{ m} = 200 \text{ m}$$

51. (a) Given, speed of thief = 10 km/h

$$= \frac{10 \times 1000}{60} \text{ m/min} = \frac{500}{3} \text{ m/min}$$

and speed of policeman = 11 km/h

$$= \frac{11 \times 1000}{60} \text{ m/min} = \frac{550}{3} \text{ m/min}$$

Now, distance travelled by thief in 6 min

$$= \frac{500}{3} \times 6 = 1000 \text{ m}$$

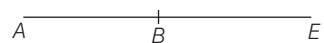
and distance travelled by policeman in

$$6 \text{ min} = \frac{550}{3} \times 6 = 1100 \text{ m}$$

∴ Difference = $(1100 - 1000) = 100 \text{ m}$

Hence, the distance between them after 6 min is 100 m.

52. (b) Let A and B will meet after t h at point E.



Distance travelled by A = pt h

and distance travelled by B = qt h

According to the question,

$$pt = qt + c \Rightarrow pt - qt = c$$

$$\Rightarrow t(p - q) = c \Rightarrow t = \frac{c}{p - q} \quad \dots (i)$$

∴ Distance travelled by A = pt

$$= \frac{pc}{p - q} \quad [\text{from Eq. (i)}]$$

53. (c) Let the original speed of car be x km/h.

$$\text{Time taken to cover 300 km} = \frac{300}{x} \text{ h}$$

If the speed of car is increased by 15 km/h.

$$\text{Then, } \frac{300}{x+15} = \frac{300}{x} - 1$$

$$\Rightarrow 1 = \frac{300}{x} - \frac{300}{x+15}$$

$$= 300 \left[\frac{x+15-x}{x(x+15)} \right]$$

$$\Rightarrow 1 = \frac{4500}{x^2 + 15x}$$

$$\Rightarrow x^2 + 15x - 4500 = 0$$

$$\Rightarrow (x+75)(x-60) = 0$$

$$\therefore x = 60 \text{ km/h}$$

54. (b) Total length of trains

$$= 121 + 99 = 220 \text{ m}$$

Relative speed of trains $= (40 + 32)$ km/h

$$= 72 \text{ km/h}$$

$$= 72 \times \frac{5}{18} \text{ m/s} = 20 \text{ m/s}$$

$$\therefore \text{Time} = \frac{\text{distance}}{\text{speed}} = \frac{220}{20} = 11 \text{ s}$$

55. (d) Let the distance be x .

Now, ratio of time taken to travel the distance by each bus is

$$\frac{x}{2} : \frac{x}{3} : \frac{x}{4} \Rightarrow \frac{12}{2} : \frac{12}{3} : \frac{12}{4}$$

$$= 6 : 4 : 3$$

56. (a) Speed of faster train is 1.5 km/min.

i.e. 90 km/h or 25 m/s

and speed of slower train $= 60$ km/h

$$= \frac{50}{3} \text{ m/s}$$

Since, these trains are moving in same direction.

So, relative speed of train

$$= \left(25 - \frac{50}{3} \right) \text{ m/s} = \frac{25}{3} \text{ m/s}$$

Time taken by crossing slower from

$$= 27 \text{ s}$$

$\therefore$ Distance $= \text{speed} \times \text{time}$

$$= \frac{25}{3} \times 27 = 225 \text{ m}$$

57. (b) When A covers 100 m, then B covers 96 m and C covers 98 m.

i.e. when C covers 98 m, then B covers 96 m.

$\therefore$ when C covers 100 m, then B covers 100×96

$$\frac{98}{98} \text{ m, i.e. } 97.96$$

So, C beats B by approximately 2.04 m.

58. (a) Ratio of speeds are 16 : 15 : 11.

Let speed of winner athletic be $16x$ km/h.

Similarly, speed of athletic in second and third position be $15x$ and $11x$, respectively.

Total distance travelled by winner is 4 km.

$$\therefore \text{Time} = \frac{4}{16x} \text{ h}$$

Distance travelled by second athletic in

$$\frac{4}{16x} \text{ h} = \frac{4}{16x} \times 15x = \frac{15}{4} \text{ km}$$

Similarly, distance travelled by third athletic in

$$\frac{4}{16x} \text{ h} = \frac{4}{16x} \times 11x = \frac{11}{4} \text{ km}$$

$\therefore$ Difference between their distances

$$= \left(\frac{15}{4} - \frac{11}{4} \right) \text{ km}$$

$$= 1 \text{ km} = 1000 \text{ m}$$

59. (b) Refer to example 12.

60. (d) Petrol consumed by the bike

$$= (10 \times 20) + (30 \times 40)$$

$$+ (10 \times 20) \text{ mL}$$

$$= (200 + 1200 + 200) \text{ mL}$$

$$= 1600 \text{ mL} = 1.6 \text{ L}$$

61. (d) Refer to question 53.

62. (c) Let the speed of the train $= x$ km/h

Then, relative speed of train

$$= (x + 5) \text{ km/h}$$

and length of the train

$$= 225 \text{ m} = 0.225 \text{ km} \quad [\text{given}]$$

Time taken by train to cross the man

$$= \frac{0.225}{x+5} \text{ h}$$

According to the question,

$$\frac{0.225}{x+5} = \frac{9}{3600}$$

$$\Rightarrow x + 5 = 90$$

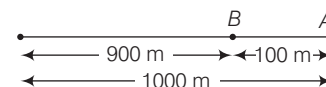
$$\Rightarrow x = 85 \text{ km/h}$$

Hence, the speed of the train is 85 km/h.

63. (c) Refer to question 46.

64. (a) Since, either A beats B by 100 m or 10 s. It means that B runs 100 m in 10 s.

$$\therefore \text{Speed of } B = \frac{100}{10} = 10 \text{ m/s}$$



$\therefore B$ gets injured at a distance of 450 m and his speed gets halved.

So, time taken by B to cover 1000 m

$$= \frac{450}{10} + \frac{550}{5} = 155$$

$\therefore$ Ratio of speed of A and B is equal to ratio of distance covered by A and B .

$$A : B = 1000 : 900 = 10 : 9$$

$$\text{Now, speed of } A = \frac{10}{9} \times 10 = \frac{100}{9} \text{ m/s}$$

Time taken by A to cover

$$1000 \text{ m} = \frac{1000}{\frac{100}{9}} \times 9 = 90 \text{ s}$$

Hence, A beat B by length of time

$$= (155 - 90) = 65 \text{ s}$$

65. (c) I. Let the length of race be x m.

Then, distance covered by $A = x$ m.

Distance covered by B when A reaches the destination $= x - 30$

Distance covered by C when A reaches the destination $= x - 48$

$$\text{Ratio of speed of } B \text{ to } A = \frac{x-30}{x}$$

$$\text{Ratio of speed of } C \text{ to } A = \frac{x-48}{x}$$

and distance covered by C when B reaches the destination $= x - 20$

$$\text{Ratio of speed } B \text{ to } C = \frac{x}{x-20}$$

$$\frac{x-30}{x} = \frac{x-48}{x-20} \left[\frac{B}{A} = \frac{C}{A} \times \frac{B}{C} \right]$$

$$\Rightarrow x^2 - 50x + 600 = x^2 - 48x$$

$$\therefore x = 300 \text{ m}$$

II. The speeds of A, B and C are in the ratio

$$300 : 270 : 252 = 50 : 45 : 42$$

07

TIME AND WORK

Usually (2-4) questions have been asked from this chapter. Generally questions are asked on Man-hours formulae, total wages paid for a work, number of days to complete a certain work and pipes and cisterns.



In this chapter, we will study the relationship among the quantity of work given, wages given, stipulated time, number of persons etc., and after it, we will be able to calculate the work in stipulated time by arranging some persons according to the work but before that let's discuss some basic rules.

IMPORTANT RULES AND FORMULAE

Rule 1 If a person can do a piece of work in ' n ' days, then he will do $\frac{1}{n}$ of the work in one day and if a person can do $\frac{1}{n}$ th of work in one day, then he will complete the work in n days.

e.g. If Raj can do a piece of work in 20 days, then he will do 1/20th of the work in one day.

Note In the problems related to time and work it is always considered that a man/woman works at uniform rate.

Rule 2 A and B can do a piece of work in x days and y days, respectively. Then, time taken by $(A + B)$ to complete the work is equal to reciprocal of $(A + B)$'s one day's work, i.e. $\frac{xy}{x + y}$.

Rule 3 If A and B can complete a work in x days and A alone can finish that work in y days, then number of days required to complete the work by $B = \frac{xy}{y - x}$ days

Rule 4 If A and B can do a piece of work in x days. B and C can do same work in y days, C and A can do same work in z days. Then, they will complete the same work in $\frac{2xyz}{xy + yz + zx}$ days by working together.

EXAMPLE 1. Raj can do a piece of work in 20 days and Rohan can do it in 12 days. How long will they take if both work together?

- a. $5\frac{1}{2}$ days b. $7\frac{1}{2}$ days c. $3\frac{1}{2}$ days d. $9\frac{1}{2}$ days

Sol. b. Raj's one day's work = $\frac{1}{20}$

Rohan's one day's work = $\frac{1}{12}$

$$\therefore (\text{Raj} + \text{Rohan})\text{'s one day's work} = \frac{1}{20} + \frac{1}{12} = \frac{3+5}{60} = \frac{8}{60}$$

$\therefore$ Number of day's taken by Raj and Rohan together to complete the work = $\frac{60}{8}$ days or $7\frac{1}{2}$ days

Here, $x = 20$ days and $y = 12$ days

$$\begin{aligned} \therefore \text{Required time} &= \frac{xy}{x+y} \quad [\text{Rule 2}] \\ &= \frac{20 \times 12}{20 + 12} = \frac{20 \times 12}{32} = \frac{15}{2} = 7\frac{1}{2} \text{ days} \end{aligned}$$

EXAMPLE 2. A and B together can do a piece of work in 12 days and A alone can do it 18 days. In how many days can B alone do it?

- a. 14 days b. 24 days c. 36 days d. 28 days

Sol. c. Here, $x = 12$ and $y = 18$

$$\therefore \text{Time taken by B} = \frac{xy}{y-x} = \frac{12 \times 18}{18-12} = 36 \text{ days} \quad [\text{Rule 3}]$$

EXAMPLE 3. A and B can do a piece of work in 3 days. B and C in 9 days and A and C in 12 days. Find the time in which A, B and C can finish the work, working together.

- a. 3 days b. $3\frac{1}{4}$ days
c. $3\frac{15}{19}$ days d. None of these

Sol. c. Here, $x = 3$, $y = 9$ and $z = 12$

According to rule,

$$\begin{aligned} \text{Required time taken by A, B and C} &= \frac{2xyz}{xy + yz + zx} \quad [\text{Rule 4}] \\ &= \frac{2 \times 3 \times 9 \times 12}{3 \times 9 + 9 \times 12 + 3 \times 12} = \frac{6 \times 12}{19} = \frac{72}{19} = 3\frac{15}{19} \text{ days} \end{aligned}$$

(Rule 5) If two groups, M_1 persons of the first group can do ' W_1 ', work in ' D_1 ' days working T_1 h in a day earning a sum of ₹ R_1 and M_2 persons of the second group can do ' W_2 ' work in ' D_2 ' days working T_2 h in a day earning a sum of ₹ R_2 . If each person of both group has the same efficiency of work, then

$$\frac{M_1 D_1 T_1}{W_1 R_1} = \frac{M_2 D_2 T_2}{W_2 R_2}$$

EXAMPLE 4. 15 men complete a work in 16 days. If 24 men are employed, then the time required to complete that work will be

☞ 2014 I

- a. 7 days b. 8 days c. 10 days d. 12 days

Sol. c. Let the work done be 1.

Here, $M_1 = 15$, $D_1 = 16$, $W_1 = W_2 = 1$, $M_2 = 24$ and $D_2 = ?$

Now, according to the formula,

$$\begin{aligned} M_1 D_1 W_2 &= M_2 D_2 W_1 \Rightarrow 15 \times 16 \times 1 = 24 \times D_2 \times 1 \\ \Rightarrow D_2 &= \frac{15 \times 16}{24} = \frac{240}{24} = 10 \text{ days} \end{aligned}$$

Therefore, 10 days are required to complete the work.

(Rule 6) If ' m ' men or ' n ' women can do a piece of work in ' a ' days, then x men and y women can do the same work in $\frac{1}{\frac{x}{m \times a} + \frac{y}{n \times a}}$ days.

EXAMPLE 5. If 3 men or 4 women can reap a field in 43 days. How long will 7 men and 5 women take to reap it?

- a. 3 days b. 7 days c. 12 days d. 15 days

Sol. c. Here, $m = 3$, $n = 4$, $a = 43$, $x = 7$ and $y = 5$

$$\begin{aligned} \therefore \text{Required days} &= \frac{1}{\frac{x}{m \times a} + \frac{y}{n \times a}} = \frac{1}{\frac{7}{3 \times 43} + \frac{5}{4 \times 43}} \quad [\text{Rule 6}] \\ &= \frac{1}{\frac{7+5}{43}} = \frac{1}{\frac{12}{43}} = \frac{43}{12} = 12 \text{ days} \end{aligned}$$

(Rule 7) If A can do a work in x days and B can do $y\%$ fast than A, then B will complete the work in $\frac{100x}{(100+y)}$ days.

EXAMPLE 6. x can do a work in 16 days. In how many days will the work be completed by y , if the efficiency of y is 60% more than that of x ? ☞ 2013 II

- a. 10 days b. 12 days c. 25 days d. 30 days

Sol. a. Here, $x = 16$ days and $y = 60\%$ faster

$$\therefore \text{Required days} = \frac{100x}{100+y} = \frac{100 \times 16}{100+60} = \frac{1600}{160} = 10 \text{ days}$$

[Rule 7]

(Rule 8) If A, B and C can do a piece of work in x , y and z days, respectively and they received ₹ k as wages by working together, then

$$\text{share of A} = ₹ \frac{yz}{xy + yz + zx} \times k$$

$$\text{share of B} = ₹ \frac{xz}{xy + yz + zx} \times k$$

and

$$\text{share of C} = ₹ \frac{xy}{xy + yz + zx} \times k$$

Note If A and B can do a piece of work in x days and y days, respectively and they received ₹ k as wages by working together, then share of $A = ₹ \frac{yk}{x+y}$ and share of $B = ₹ \frac{xk}{x+y}$.

Wages are directly proportional to the work done and indirectly proportional to the time taken by the individual.

EXAMPLE 7. X completes a job in 2 days and Y completes it in 3 days and Z takes 4 days to complete it. If they work together and get ₹ 3900 for the job, then how much amount does Y get?

- a. ₹ 1800 b. ₹ 1200 c. ₹ 900 d. ₹ 800

Sol. b. Here, $a = 2$ days, $b = 3$ days and $c = 4$ days and $k = ₹ 3900$

$$\begin{aligned} \text{Now, amount of } Y &= \frac{ac}{ab+bc+ca} \times k && [\text{Rule 8}] \\ &= \frac{2 \times 4}{2 \times 3 + 3 \times 4 + 4 \times 2} \times 3900 \\ &= \frac{8}{6 + 12 + 8} \times 3900 = \frac{8 \times 3900}{26} = ₹ 1200 \end{aligned}$$

SOME OTHER FORMULAS

1. If A and B can do a piece of work in x and y days, respectively. A and B started working together but A left the work t days before completing the work, then time taken to complete the work will be $\frac{(x+t)y}{(x+y)}$ days.
2. A and B do a piece of work in ' a ' and ' b ' days, respectively. Both begin together but after some days, A leaves the work and the remaining work is completed by B in x days. Then, the time after which A left is given by $T = \frac{(b-x)a}{a+b}$
3. If A and B can do a piece of work in x and y days, respectively. They start working together and after t days B leaves the work, then time taken to finish the whole work will be $\frac{x}{y} \times (y-t)$ days.

EXAMPLE 8. Akshu can do a piece of work in 10 days and Harshal can do same work in 12 days. They started working together but Akshu left the work 2 days before completion of work, then time taken to complete the work?

- a. $6\frac{6}{11}$ days b. $5\frac{3}{10}$ days c. $4\frac{3}{2}$ days d. $7\frac{2}{5}$ days

Sol. a. $6\frac{6}{11}$ days $= \frac{12 \times (10+2)}{10+12} = \frac{12 \times 12}{22} = 6\frac{6}{11}$ days

EXAMPLE 9. A can do a piece of work in 10 days and B can do same work in 15 days. They started working together but after 2 days, A left work. The remaining work was completed by B alone, then time taken to complete the work?

- a. $8\frac{3}{11}$ days b. $8\frac{2}{3}$ days c. $5\frac{3}{4}$ days d. $2\frac{5}{3}$ days

Sol. b. $8\frac{2}{3}$ days $= \frac{10}{15} (15-2) = \frac{26}{3} = 8\frac{2}{3}$ days

PIPES AND CISTERNS

Problems on pipes and cisterns are based on the basic concept of time and work. Pipes are connected to a tank or cistern and are used to fill or empty the tank or cistern.

Important Rules and Formulae

(Rule 9) If there are two pipes A and B takes ' a ' and ' b ' h respectively to fill a tanker, then the two pipes together fill $\left(\frac{1}{a} + \frac{1}{b}\right)$ part of the tank in 1h and time taken to fill the tank will be $\left(\frac{ab}{a+b}\right)$ h. If a pipe fills a tank in ' a ' h, then in 1 h only $\frac{1}{a}$ of the tank is filled.

EXAMPLE 10. Pipe A can fill a tank in 45 h and pipe B can fill it in 36 h. If both the pipes are opened in the empty tank. In how many hours will it be full?

- a. 10 h b. 15 h c. 20 h d. 28 h

Sol. c. Here, $a = 45$ h and $b = 36$ h.

$$\therefore \text{Required time} = \frac{ab}{a+b} = \frac{45 \times 36}{45+36} = \frac{1620}{81} = 20 \text{ h}$$

- (Rule 10)** (i) If a pipe can fill a tank in ' a ' h and another can fill the tank in ' b ' h but a third pipe empties the filled tank in ' c ' h, then in one hour $\left(\frac{1}{a} + \frac{1}{b} - \frac{1}{c}\right)$ part of the tank will be filled (when all the three pipes are open) and time taken to fill the tank will be $\frac{abc}{bc+ac-ab}$ h.
- (ii) If a pipe can fill a tank in a_1 h and another pipe can empty the filled tank in a_2 h, then in one hour $\left(\frac{1}{a_1} - \frac{1}{a_2}\right)$ part of the tank will be filled. (when both pipes are open) and time taken to fill the tank will be $\left(\frac{a_1 a_2}{a_2 - a_1}\right)$ h.

EXAMPLE 11. Two pipes ' A ' and ' B ' can fill a tank in 36 min and 45 min, respectively. A waste pipe ' C ' can empty the tank in 30 min. In how much time the tank is full if all three pipes are opened?

- a. 60 min b. 90 min c. 115 min d. None of these

Sol. a. Here, $a = 36$, $b = 45$ and $c = 30$.

So, time taken to fill the tank will be

$$\begin{aligned} &= \frac{abc}{ac+bc-ab} && [\text{Rule 8}] \\ &= \frac{36 \times 45 \times 30}{36 \times 30 + 45 \times 30 - 36 \times 45} = 60 \text{ min} \end{aligned}$$

> PRACTICE EXERCISE

1. X can do $\frac{3}{4}$ of a work in 12 days. In how many days X can finish the $\frac{1}{2}$ work?
(a) 8 days (b) 16 days (c) 12 days (d) 24 days
2. A can do a piece of work in 10 days and B can do the same work in 12 days. How long will they take to finish the work, if both work together?
(a) $5\frac{5}{11}$ days (b) $6\frac{5}{11}$ days
(c) $5\frac{1}{5}$ days (d) None of these
3. A , B and C working together take 30 min to address a pile of envelopes. A and B together would take 40 min, A and C together would take 45 min. How long would each take working alone?
(a) A : 72 min, B : 90 min, C : 120 min
(b) A : 42 min, B : 90 min, C : 120 min
(c) A : 72 min, B : 90 min, C : 100 min
(d) A : 72 min, B : 80 min, C : 120 min
4. A and B together can do a piece of work in 12 days and A alone can do it in 36 days. In how many days can B alone do it?
(a) 18 days (b) 12 days
(c) 15 days (d) 20 days
5. Ram can do a piece of work in 6 days and Shyam can finish the same work in 12 days. How much work will be finished, if both work together for 2 days?
(a) One-fourth of the work (b) One-third of the work
(c) Half of the work (d) Whole of the work
6. A and B can do given work in 8 days; B and C can do the same work in 12 days and A , B , C complete it in 6 days. In how many days can A and C finish it?
(a) 12 (b) 8 (c) 14 (d) 16
7. A can finish a work in 8 days and B can do it in 12 days. After A had worked for 3 days, B also joins A to finish the remaining work. In how many days will the remaining work be finished?
(a) 2 days (b) 3 days
(c) 4 days (d) 5 days
8. Two taps can fill a tub in 5 min and 7 min, respectively. A pipe can empty it in 3 min. If all the three are kept open simultaneously, when will the tub be full?
(a) 60 min (b) 85 min (c) 90 min (d) 105 min
9. A and B can do a piece of work in 40 days and 50 days, respectively. Both begin together but after a certain time, A leaves off. In this case B finishes the remaining work in 20 days. After how many days did A leave?
(a) 14 days (b) $13\frac{1}{3}$ days (c) 13 days (d) 15 days
10. P and Q can do a job in 2 days, Q and R can do it in 4 days and P and R in $\frac{12}{5}$ days. What is the number of days required for P alone to do the job?
(a) $\frac{5}{2}$ (b) 3 (c) $\frac{14}{5}$ (d) 6
11. $\frac{1}{48}$ of a work is completed in half a day by 5 persons. Then, $\frac{1}{40}$ of the work can be completed by 6 persons in how many days?
(a) 1 (b) 2 (c) 3 (d) $\frac{1}{2}$
12. A garrison of ' n ' men had enough food to last for 30 days. After 10 days, 50 more men joined them. If the food now lasted for 16 days, what is the value of n ?
(a) 200 (b) 240 (c) 280 (d) 320
13. Two pipes A and B can fill a tank in 12 and 16 min, respectively. If both the pipes are opened simultaneously, after how much time should B be closed so that the tank is full in 9 min?
(a) 3 min (b) 5 min (c) 4 min (d) 2 min
14. 2 men undertake to do a job for ₹ 1400. One can do it alone in 7 days and the other in 8 days. With the assistance of a boy they finish the work in 3 days. How should the money be divided?
(a) ₹ 600, ₹ 525, ₹ 275 (b) ₹ 550, ₹ 500, ₹ 350
(c) ₹ 650, ₹ 470, ₹ 280 (d) None of these
15. A , B and C can do a piece of work individually in 8, 10 and 15 days, respectively. A and B start working but A quits after working for 2 days. After this, C joins B till the completion of work. In how many days will the work be completed?
(a) $\frac{53}{9}$ days (b) $\frac{34}{7}$ days
(c) $\frac{85}{13}$ days (d) $\frac{53}{10}$ days
16. A can do a piece of work in ' x ' days and B can do the same work $3x$ days. To finish the work together they take 12 days. What is the value of ' x '?
(a) 8 (b) 10 (c) 12 (d) 16

- 17.** If 6 men and 8 boys can do a piece of work in 10 days while 26 men and 48 boys can do the same in 2 days, what is the time taken by 15 men and 20 boys in doing the same type of work?
(a) 4 days (b) 5 days (c) 6 days (d) 7 days
- 18.** Four taps can individually fill a cistern of water in 1h, 2h, 3h and 6h, respectively. If all the four taps are opened simultaneously, the cistern can be filled in how many minutes?
(a) 20 (b) 30 (c) 35 (d) 40
- 19.** 76 ladies complete a job in 33 days. Due to some reason some ladies did not join the work and therefore, it was completed in 44 days. The number of ladies who did not report for the work is
(a) 17 (b) 18 (c) 19 (d) 20
- 20.** 9 men finish one-third work in 10 days. The number of additional men required for finishing the remaining work in 2 more days will be
(a) 78 (b) 81 (c) 55 (d) 30
- 21.** Ravi and Sneha working separately can finish a job in 8 and 12 h, respectively. If they work for an hour alternately, Ravi beginning at 9:00 am. When will the job be finished?
(a) 7 : 30 pm (b) 7 : 00 pm (c) 6 : 30 pm (d) 6 : 00 pm
- 22.** Consider the following statements:
I. If 18 men can earn ₹ 1440 in 5 days, then 10 men can earn ₹ 1280 in 6 days.
II. If 16 men can earn ₹ 1120 in 7 days, then 21 men can earn ₹ 800 in 4 days.
Which of the statement(s) given above is/are correct?
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

Directions (Q. Nos. 23-24) If a_1 men and b_1 boys can complete a work in x days, while a_2 men and b_2 boys can complete the same work in y days, then

$$\frac{\text{One day work of 1 man}}{\text{One day work of 1 boy}} = \frac{(yb_2 - xb_1)}{(xa_1 - ya_2)}$$

- 23.** If 14 men and 12 boys can finish a work in 4 days, while 8 men and 16 boys can finish the same work in 5 days. Compare the 1 day work of 1 man and 1 boy.
(a) 2 (b) $1\frac{1}{2}$ (c) $\frac{1}{2}$ (d) 3
- 24.** 28 men and ' m ' boys can finish a work in 4 days, while 20 men and 6 boys can finish the same work in 5 days. Now, if the ratio of 1 day work of 1 man and 1 boy is 1 : 2, what will be the value of ' m '?
(a) 7 (b) 6 (c) 8 (d) 12
- 25.** 45 people take 18 days to dig a pond. If the pond would have to be dig in 15 days, then the number of people to be employed will be **2012 I**
(a) 50 (b) 54 (c) 60 (d) 72
- 26.** A and B can do a piece of work in 10 h. B and C can do it in 15 h, while A and C take 12 h to complete the work. B independently can complete the work in **2012 I**
(a) 12 h (b) 16 h (c) 20 h (d) 24 h
- 27.** X can do a piece of work in 25 days. Y is 25% more efficient than X. The number of days taken by Y is **2012 II**
(a) 15 days (b) 20 days (c) 21 days (d) 30 days
- 28.** A mason can build a tank in 12 h. After working for 6 h, he took the help of a boy and finished the work in another 5 h. The time that the boy will take alone to complete the work is **2013 I**
(a) 30 h (b) 45 h (c) 60 h (d) 64 h
- 29.** X can complete a job in 12 days. If X and Y work together, they can complete the job in $6\frac{2}{3}$ days. Y alone can complete the job in **2013 I**
(a) 10 days (b) 12 days (c) 15 days (d) 18 days
- 30.** Pipe A can fill a tank in 10 min and pipe B can empty it in 15 min. If both the pipes are opened in an empty tank, the time taken to make it full is **2013 I**
(a) 20 min (b) 25 min
(c) 30 min (d) None of these
- 31.** 4 goats and 6 sheeps can graze a field in 50 days. 2 goats and 9 sheeps can graze the field in **2013 II**
(a) 100 days (b) 75 days (c) 50 days (d) 25 days
- 32.** A can finish a work in 15 days, B in 20 days and C in 25 days. All these three worked together and earned ₹ 4700. The share of C is **2013 II**
(a) ₹ 1200 (b) ₹ 1500 (c) ₹ 1800 (d) ₹ 2000
- 33.** 20 workers working for 5 h per day complete a work in 10 days. If 25 workers are employed to work 10 h per day, what is the time required to complete the work? **2013 II**
(a) 4 days (b) 5 days (c) 6 days (d) 8 days
- 34.** 18 men can earn ₹ 360 in 5 days. How much money will 15 men earn in 9 days? **2013 II**
(a) ₹ 600 (b) ₹ 540 (c) ₹ 480 (d) ₹ 360
- 35.** 2 men and 1 woman can complete a piece of work in 14 days, while 4 women and 2 men can do the same work in 8 days. If a man gets ₹ 90 per day, what should be the wages per day of a woman? **2013 II**
(a) ₹ 48 (b) ₹ 60 (c) ₹ 72 (d) ₹ 135

PREVIOUS YEARS' QUESTIONS

- 36.** A can do a piece of work in 4 days and B can complete the same work in 12 days. What is the number of days required to do the same work together? **2013 II**
(a) 2 days (b) 3 days (c) 4 days (d) 5 days
- 37.** A , B and C can do a piece of work individually in 8, 12 and 15 days, respectively. A and B start working but A quits after working for 2 days. After this, C joins B till the completion of work. In how many days will the work be completed? **2014 II**
(a) $5\frac{8}{9}$ days (b) $4\frac{6}{7}$ days (c) $6\frac{7}{13}$ days (d) $3\frac{3}{4}$ days
- 38.** A is thrice as efficient as B and hence completes a work in 40 days less than the number of days taken by B . What will be the number of days taken by both of them when working together? **2014 II**
(a) 22.5 days (b) 15 days (c) 20 days (d) 18 days
- 39.** The efficiency of P is twice that of Q , whereas the efficiency of P and Q together is three times that of R . If P , Q and R work together on a job, in what ratio should they share their earnings? **2015 I**
(a) 2 : 1 : 1 (b) 4 : 2 : 1 (c) 4 : 3 : 2 (d) 4 : 2 : 3
- 40.** A and B are two taps which can fill a tank individually in 10 min and 20 min, respectively. However, there is a leakage at the bottom, which can empty a filled tank in 40 min. If the tank is empty initially, then how much time will both the taps take to fill the tank with leakage? **2015 II**
(a) 2 min (b) 4 min (c) 5 min (d) 8 min
- 41.** If 4 men working 4 h per day for 4 days complete 4 units of work, then how many units of work will be completed by 2 men working for 2 h per day in 2 days? **2015 II**
(a) 2 (b) 1 (c) $\frac{1}{2}$ (d) $\frac{1}{8}$
- 42.** If m persons can paint a house in d days, then how many days will it take for $(m + 2)$ persons to paint the same house? **2015 II**
(a) $md + 2$ (b) $md - 2$ (c) $\frac{m+2}{md}$ (d) $\frac{md}{m+2}$
- 43.** Two pipes A and B can fill a tank in 60 min and 75 min, respectively. There is also an outlet C . If A , B and C are opened together, then the tank is full in 50 min. How much time will be taken by C to empty the full tank? **2016 I**
(a) 100 min (b) 110 min (c) 120 min (d) 125 min

ANSWERS

1	a	2	a	3	a	4	a	5	c	6	b	7	b	8	d	9	b	10	b
11	d	12	a	13	c	14	a	15	d	16	d	17	a	18	b	19	c	20	b
21	c	22	d	23	a	24	b	25	b	26	d	27	b	28	c	29	c	30	c
31	d	32	a	33	a	34	b	35	b	36	b	37	a	38	b	39	a	40	d
41	c	42	d	43	a														

HINTS AND SOLUTIONS

- 1.** (a) Since, X can do $\frac{3}{4}$ of work in 12 days.
[Rule 1]
So, X can do 1 work in $\frac{12 \times 4}{3}$ days.
 $\therefore X$ can do $\frac{1}{2}$ work in $\frac{12 \times 4 \times 1}{3 \times 2} = 8$ days
- 2.** (a) Here, $x = 10$ and $y = 12$
 $\therefore$ Number of days taken by A and B
 $= \frac{xy}{x+y} = \frac{12 \times 10}{12+10}$ [Rule 2]

$$= 5\frac{5}{11} \text{ days}$$

- 3.** (a) B 's one min's work
 $= (A + B + C)$'s one min's work
 $-(A + C)$'s one min's work
 $= \frac{1}{30} - \frac{1}{45} = \frac{6-4}{180} = \frac{2}{180} = \frac{1}{90}$
 C 's one min's work $= (A + B + C)$'s one min work $-(A + B)$'s one min work

$$= \frac{1}{30} - \frac{1}{40} = \frac{4-3}{120} = \frac{1}{120}$$

A 's one min's work $= (A + B)$'s one min work $- B$'s one min work

$$= \frac{1}{40} - \frac{1}{90} = \frac{9-4}{360} = \frac{5}{360} = \frac{1}{72}$$

Hence, A , B and C alone can finish the work in 72 min, 90 min and 120 min, respectively.

4. (a) Here, $x = 12$ and $y = 36$
 $\therefore$ Time taken by $B = \frac{xy}{y-x} = \frac{12 \times 36}{36-12}$
 $= \frac{12 \times 36}{24} = 18$ days [Rule 3]

5. (c) $\therefore$ One day work of Ram $= \frac{1}{6}$
 $\therefore$ One day work of Shyam $= \frac{1}{12}$
Hence, one day work of, Ram and Shyam
 $= \frac{1}{6} + \frac{1}{12} = \frac{2+1}{12} = \frac{3}{12} = \frac{1}{4}$
 $\therefore$ Two day's work $= \frac{1}{2}$

Thus, if they work together for 2 day's, then half of the work will be complete.

6. (b) Here, $x = 8, y = 12, z = 12$
Now, time taken by $(A + B + C)$
 $= \frac{2xyz}{xy + yz + zx}$ [Rule 4]
 $\Rightarrow 6 = \frac{2 \times 8 \times 12 \times 12}{8 \times 12 + 12 \times 12 + 12 \times 8}$

$$\Rightarrow 96 + 20z = 32z$$

$$\Rightarrow 96 = 12z$$

$$\Rightarrow z = 8$$

So, A and C can do the work in 8 days.

7. (b) One day's work of $A = \frac{1}{8}$
One day's work of $B = \frac{1}{12}$
3 day's work of $A = \frac{3}{8}$
Remaining work of $A = 1 - \frac{3}{8} = \frac{5}{8}$

One day's work of A and B together
 $= \frac{1}{8} + \frac{1}{12} = \frac{3+2}{24} = \frac{5}{24}$

Number of days to finish the work
 $= \frac{5}{\frac{5}{24}} = 24$ days

8. (d) 1 min work of first tap $= \frac{1}{5}$
1 min work of second tap $= \frac{1}{7}$
and 1 min work of third $= \frac{1}{3}$

$$\therefore 1 \text{ min work of all taps} = \frac{1}{5} + \frac{1}{7} + \frac{1}{3} = \frac{21+15+35}{105} = \frac{1}{105}$$

Hence, tap will be filled in 105 min, if they work together.

9. (b) Here, $a = 40$ days, $b = 50$ days, $x = 20$ and $T = ?$
 $\therefore$ Required time $= \frac{(b-x)a}{a+b} = \frac{(50-20) \times 40}{(40+50)} = \frac{30 \times 40}{90} = \frac{40}{3} = 13\frac{1}{3}$ days

10. (b) Here, $x = 2, y = 4$ and $z = \frac{12}{5}$
 $\therefore$ Time taken by $(P + Q + R)$
 $= \frac{2xyz}{xy + yz + zx}$ [Rule 4]

$$= \frac{2 \times 2 \times 4 \times \frac{12}{5}}{2 \times 4 + 4 \times \frac{12}{5} + \frac{12}{5} \times 2} = \frac{192}{40 + 48 + 24} = \frac{192}{112} = \frac{12}{7} \text{ days}$$

$$= \frac{192}{40 + 48 + 24} = \frac{192}{112} = \frac{12}{7} \text{ days}$$

Now, P one days work

$$= \frac{7}{12} - \frac{1}{4} = \frac{7-3}{12} = \frac{4}{12} = \frac{1}{3}$$

Hence, P alone can do in 3 days.

11. (d) Using the formula $\frac{M_1 D_1}{W_1} = \frac{M_2 D_2}{W_2}$

12. (a) By given condition,
 $n \times 30 = n \times 10 + (n + 50) \times 16$
 $\Rightarrow 20n = 16n + 800$
 $\therefore n = \frac{800}{4} = 200$

13. (c) Here, $x = 12, y = 16$ and $t = 9$
Two pipes A and B can fill a tank in x h and y h, respectively. If both the pipes ax opened simultaneously, then the time after which B should be closed, so that the tank is filled in t h
 t h $= [y(1 - t/x)]^n$
Required time after which B should be closed $= y \left(1 - \frac{t}{x}\right) = 16 \left(1 - \frac{9}{12}\right) = 16 \times \frac{3}{12} = 4$ min

14. (a) Let the boy completes the work in x days.

According to the condition,

$$\frac{1}{7} + \frac{1}{8} + \frac{1}{x} = \frac{1}{3} \Rightarrow \frac{1}{x} = \frac{1}{3} - \frac{1}{8} - \frac{1}{7} \Rightarrow \frac{1}{x} = \frac{56-21-24}{168} = \frac{11}{168}$$

$$\therefore x = \frac{168}{11} \text{ days}$$

So, money is to be shared in the ratio

$$\frac{1}{7} : \frac{1}{8} : \frac{11}{168} \text{ or } 24 : 21 : 11$$

Thus,

$$A's \text{ amount} = \frac{24}{56} \times 1400 = ₹ 600$$

$$B's \text{ amount} = \frac{21}{56} \times 1400 = ₹ 525$$

$$\text{Boy's amount} = \frac{11}{56} \times 1400 = ₹ 275$$

15. (d) A 's work in one day $= \frac{1}{8}$, B 's work in one day $= \frac{1}{10}$

$$C's \text{ work in one day} = \frac{1}{15}$$

$$(A + B)'s \text{ work in one day} = \frac{1}{8} + \frac{1}{10} = \frac{5+4}{40} = \frac{9}{40}$$

$$(A + B)'s \text{ work in two days} = \frac{2 \times 9}{40} = \frac{9}{20}$$

$$\text{Remaining work} = 1 - \frac{9}{20} = \frac{11}{20}$$

$$(B + C)'s \text{ work in one day} = \frac{1}{10} + \frac{1}{15} = \frac{3+2}{30} = \frac{5}{30} = \frac{1}{6}$$

Since, $(B + C)$ complete the work in 6 days.

$$\therefore \frac{11}{20} \text{ work will be completed in}$$

$$6 \times \frac{11}{20} = \frac{11 \times 3}{10} = \frac{33}{10} \text{ days}$$

$$\therefore \text{Total number of days} = 2 + \frac{33}{10} = \frac{20+33}{10} = \frac{53}{10} \text{ days}$$

16. (d) 1 day work of $A = \frac{1}{x}$

$$1 \text{ day work of } B = \frac{1}{3x}$$

$$\therefore 1 \text{ day work of both } A \text{ and } B$$

$$= \frac{1}{x} + \frac{1}{3x} = \frac{4}{3x}$$

$$\text{given, one day work of both } A \text{ and } B = \frac{1}{12}$$

$$\Rightarrow \frac{4}{3x} = \frac{1}{12} \Rightarrow 3x = 48 \Rightarrow x = 16$$

Hence, the value of x is 16.

- 17.** (a) Given, $6M + 8B = 10$ days ... (i)
and $26M + 48B = 2$ days ... (ii)
 $15M + 20B = ?$

Here, $M_1 D_1 = M_2 D_2$
 $\Rightarrow (6M + 8B) \times 10 = (26M + 48B) \times 2$
 $\Rightarrow 60M + 80B = 52M + 96B$
 $\Rightarrow 8M = 16B$
 $\therefore M = 2B$
 Then, $15M + 20B = 15 \times 2B + 20B$
 $= 30B + 20B = 50B$

On putting the value of M in Eq. (i), we get

$$\begin{aligned} \Rightarrow 6 \times 2B + 8B &= 10 \text{ days} \\ \Rightarrow 12B + 8B &= 10 \text{ days} \\ \therefore 20 \text{ boys finish the work in } &10 \text{ days} \\ \text{One boy finish the work in} &10 \times 20 \text{ days} = 200 \text{ days} \\ \text{and 50 boys finish the work in } &\frac{200}{50} \text{ days} \\ &= 4 \text{ days} \end{aligned}$$

- 18.** (b) Part filled by first tap in 1h = 1
 Part filled by second tap in 1h = $\frac{1}{2}$
 Part filled by third tap in 1h = $\frac{1}{3}$
 Part filled by fourth tap in 1h = $\frac{1}{6}$
 Total tank filled by all taps in 1h
 $= 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{6}$
 $\therefore$ Required time = $\frac{1}{1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{6}}$
 $= \frac{6}{6 + 3 + 2 + 1} = \frac{6}{12} \text{ h} = \frac{6}{12} \times 60 \text{ min}$
 $= 30 \text{ min}$

- 19.** (c) Given, $M_1 = 76, D_1 = 33$
 Let number of ladies who did not report for the work = x
 By given condition,

$$M_2 = 76 - x \text{ and } D_2 = 44$$

We know that,

$$\begin{aligned} M_1 D_1 &= M_2 D_2 \quad [\text{Rule 5}] \\ \therefore 76 \times 33 &= (76 - x) \times 44 \\ \Rightarrow 76 - x &= \frac{76 \times 3}{4} = 19 \times 3 \Rightarrow x = 19 \end{aligned}$$

Hence, the number of ladies is 19.

- 20.** (b) Refer to Question No 11
 [Using Rule 5]

- 21.** (c) Work done by Ravi in one hour = $\frac{1}{8}$

$$\text{Work done by Sneha in one hour} = \frac{1}{12}$$

Total work done by them in 2 h

$$= \frac{1}{8} + \frac{1}{12} = \frac{5}{24}$$

Work done is 4 pairs of hours

$$= \frac{5}{24} \times 4 = \frac{5}{6}$$

Remaining work after 8 hours

$$= 1 - \frac{5}{6} = \frac{1}{6}$$

Now, it's Ravi's turn

$$\text{So, work left after 9 hours} = \frac{1}{6} - \frac{1}{8} = \frac{1}{24}$$

$$\frac{1}{24} \text{ work will be done by sneha in}$$

$$12 \times \frac{1}{24} \text{ h} = 30 \text{ min}$$

Total time taken = 9 h and 30 min

Required time = 6 : 30 pm

- 22.** (d) I. 18 men can earn in 5 days = ₹ 1440

$$1 \text{ man can earn in 1 day} = \frac{1440}{18 \times 5}$$

$\therefore$ 10 men can earn in 6 days

$$= \frac{1440}{18 \times 5} \times 6 \times 10$$

$$= ₹ 960 \neq ₹ 1280$$

II. 16 men can earn in 7 days = ₹ 1120

$$1 \text{ man can earn in 1 day} = \frac{1120}{16 \times 7}$$

$\therefore$ 21 men can earn in 4 days

$$= \frac{1120}{16 \times 7} \times 21 \times 4 = ₹ 840 \neq ₹ 800$$

So, neither statement I nor II is correct.

- 23.** (a) Here, $a_1 = 14, b_1 = 12, x = 4,$

$$a_2 = 8, b_2 = 16 \text{ and } y = 5$$

$$\frac{\text{One day work of 1 man}}{\text{One day work of 1 boy}} = \frac{(yb_2 - xb_1)}{(xa_1 - ya_2)}$$

$$= \frac{5 \times 16 - 4 \times 12}{4 \times 14 - 5 \times 8} = 2$$

- 24.** (b) Refer to Question No 23.

- 25.** (b) Using the formula, $M_1 D_1 = M_2 D_2$

- 26.** (d) Here, $x = 10, y = 15, z = 12.$

$\therefore$ Time taken by $(A + B + C)$

$$= \frac{2xyz}{xy + yz + zx} \quad [\text{Rule 4}]$$

$$= \frac{2 \times 10 \times 15 \times 12}{10 \times 15 + 15 \times 12 + 12 \times 10}$$

$$= \frac{3600}{450} = 8 \text{ h}$$

$$\begin{aligned} \text{So, 1 h work of } B &= \frac{1}{8} - \frac{1}{12} \\ &= \frac{3-2}{24} = \frac{1}{24} \end{aligned}$$

Hence, B alone can do in 24 h.

- 27.** (b) Here, $x = 25$ days and $y = 25$

$$\begin{aligned} \therefore \text{Required days} &= \frac{100x}{100 + y} = \frac{100 \times 25}{100 + 25} \\ &= \frac{2500}{125} = 20 \text{ days} \end{aligned}$$

- 28.** (c) Mason work for 1 h = $\frac{1}{12}$

$$\text{Mason Work for 6 h} = \frac{6}{12} = \frac{1}{2}$$

$$\text{Work left} = 1 - \frac{1}{2} = \frac{1}{2}$$

Now, let the boy can finish the work in x h.

$$\text{Then, their 1 h work} = \frac{1}{12} + \frac{1}{x} = \frac{x + 12}{12x}$$

$$\therefore \frac{x + 12}{12x} \times 5 = \frac{1}{2} \Rightarrow \frac{5x + 60}{12x} = \frac{1}{2}$$

$$\Rightarrow 10x + 120 = 12x \Rightarrow 120 = 2x$$

$$\therefore x = 60 \text{ h}$$

- 29.** (c) X 's one day work = $\frac{1}{12}$ and $(X + Y)$'s

$$\text{one day work} = \frac{3}{20}$$

$\therefore Y$'s one day work

$$= \frac{3}{20} - \frac{1}{12} = \frac{4}{60} = \frac{1}{15}$$

$\therefore$ Number of day's taken by Y to complete the work = 15 days

- 30.** (c) Part filled by pipe A in 1 min = $\frac{1}{10}$

$$\text{and part emptied by pipe B in 1 min} = \frac{1}{15}$$

$\therefore$ Total tank filled in minutes

$$= \frac{1}{10} - \frac{1}{15} = \frac{3-2}{30} = \frac{1}{30}$$

Hence, the tank will be filled in 30 min.

- 31.** (d) Here, $m = 4, n = 6, a = 50$ and $x = 2, y = 9$

$$\therefore \text{Required days} = \frac{1}{\frac{x}{m \times a} + \frac{y}{n \times a}}$$

$$= \frac{1}{\frac{2}{4 \times 50} + \frac{9}{6 \times 50}} \quad [\text{Rule 6}]$$

$$= \frac{1}{\frac{1}{100} + \frac{3}{100}} = \frac{1}{\frac{4}{100}} = \frac{100}{4} = 25 \text{ days}$$

32. (a) Here, $x = 15$ days, $y = 20$ days and $z = 25$ days and $K = ₹4700$

$$\begin{aligned}\text{Share of } C &= ₹ \left(\frac{kxy}{xy + yz + zx} \right) \\ &= ₹ \frac{4700 \times 15 \times 20}{15 \times 20 + 20 \times 25 + 25 \times 15} \\ &= ₹ \frac{4700 \times 15 \times 20}{1175} = ₹ 1200\end{aligned}$$

33. (a) Refer to Question No 11

[Using Rule 5]

34. (b) Refer to Question No 22

35. (b) $\therefore$ 14 days work of 2 men and 1 women
 $= 8$ days work of 4 women and 2 men
 $\Rightarrow$ 1 day work of 28 men and 14 women
 $=$ 1 day work of 32 women and 16 men
 $\Rightarrow$ 28 men + 14 women
 $=$ 32 women + 16 men
 $\Rightarrow 28m - 16m = 32w - 14w$
 $\Rightarrow 12m = 18w \Rightarrow \frac{m}{w} = \frac{18}{12} = \frac{3}{2}$

So, efficiency of 1 man and 1 woman is 3 : 2.

So, their wages must be in the same ratio

$$\therefore \frac{90}{x} = \frac{3}{2}$$

[where, x = wages per day of a woman]

$$\therefore x = \frac{90 \times 2}{3} = ₹ 60$$

36. (b) A's one day work = $\frac{1}{4}$

$$\text{and } B\text{'s one day work} = \frac{1}{12}$$

One day work of A and B together

$$= \frac{1}{4} + \frac{1}{12} = \frac{3+1}{12} = \frac{4}{12} = \frac{1}{3}$$

Days required by A and B together to do the work

$$= \frac{1}{\text{One day work of A and B together}} = 3 \text{ days}$$

37. (a) Work done by A in one day = $\frac{1}{8}$

$$\text{Work done by B in one day} = \frac{1}{12}$$

$$\text{and work done by C in one day} = \frac{1}{15}$$

Let the work will be completed in x days.

$$\text{Then, } \frac{2}{8} + \frac{x}{12} + \frac{x-2}{15} = 1$$

$$\Rightarrow \frac{1}{4} + \frac{x}{12} + \frac{x-2}{15} = 1$$

$$\Rightarrow \frac{15 + 5x + 4(x-2)}{60} = 1$$

$$\Rightarrow 15 + 5x + 4x - 8 = 60 \Rightarrow 9x + 7 = 60$$

$$\Rightarrow 9x = 53 \Rightarrow x = \frac{53}{9} = 5\frac{8}{9}$$

Hence, the work will completed in $5\frac{8}{9}$ days.

38. (b) Let efficiency of A be x days.

Then, efficiency of B = $3x$ days.

$$\therefore B\text{'s one day work} = \frac{1}{3x}$$

$$\text{and } A\text{'s one day work} = \frac{1}{x}$$

Again, let time taken by B = t days

Then, time taken by A = $(t - 40)$ days

$$\text{Now, } \frac{t}{3x} = \frac{t-40}{x} \Rightarrow t = 3(t-40)$$

$$\Rightarrow 3t - t = 120, t = \frac{120}{2} = 60 \text{ days}$$

So, B complete the work in 60 days and A complete the work in 20 days.

(A + B)'s 1 day work

$$= \frac{1}{60} + \frac{1}{20} = \frac{4}{60} = \frac{1}{15}$$

Hence, they will complete the work in 15 days when they work together.

39. (a) Let efficiency of P be x days.

Then, efficiency of Q = $2x$ days

$$\therefore Q\text{'s one day work} = \frac{1}{2x}$$

$$\text{and } P\text{'s one day work} = \frac{1}{x}$$

Now, (P + Q)'s one day work

$$= \frac{1}{x} + \frac{1}{2x} = \frac{3}{2x}$$

$\therefore$ (P + Q) will complete the whole work in $\frac{2x}{3}$ days.

According to the question,

R will complete this work = $2x$ days

$$\therefore R\text{'s one day work} = \frac{1}{2x}$$

$$\begin{aligned}\text{Required ratio} &= \frac{1}{x} : \frac{1}{2x} : \frac{1}{2x} = 1 : \frac{1}{2} : \frac{1}{2} \\ &= 2 : 1 : 1\end{aligned}$$

40. (d) Part of tank filled by A in 1 min = $\frac{1}{10}$

$$\text{Part of tank filled by B in 1 min} = \frac{1}{20}$$

$$\begin{aligned}\text{Part of tank emptied by leakage in 1 min} \\ &= \frac{1}{40}\end{aligned}$$

$$\text{Part of tank filled by both taps and leakage is } \left(\frac{1}{10} + \frac{1}{20} - \frac{1}{40} \right) = \frac{1}{8}$$

Hence, total time taken to fill the tank is 8 min.

41. (c) Here, $M_1 = 4, T_1 = 4, W_1 = 4, D_1 = 4$ and $M_2 = 2, T_2 = 2, D_2 = 2, W_2 = ?$

Using the formula,

$$\frac{M_1 \times T_1 \times D_1}{W_1} = \frac{M_2 \times T_2 \times D_2}{W_2} \quad [\text{Rule 5}]$$

$$\Rightarrow \frac{4 \times 4 \times 4}{4} = \frac{2 \times 2 \times 2}{W_2}$$

$$\Rightarrow W_2 = \frac{8}{16} = \frac{1}{2}$$

42. (d) m persons paint a house in d days.

$\therefore$ 1 person paints a house in $(m \times d)$ days.

and $m + 2$ persons paint a house in

$$\left(\frac{md}{m+2} \right) \text{ days.}$$

43. (a) Since, two pipes A and B fill a tank in 60 min and 75 min, respectively.

$\therefore$ Part of tank filled by pipe A in 1 min

$$= \frac{1}{60}$$

and part of tank filled by pipe B in 1 min

$$= \frac{1}{75}$$

Now, part of tank filled by A and B together in 1 min

$$= \frac{1}{60} + \frac{1}{75} = \frac{9}{300} = \frac{3}{100}$$

Let part of tank emptied by pipe C in 1 min = $\frac{1}{C}$.

So, net part of tank filled by pipes A, B and C together in 1 min

$$= \frac{3}{100} - \frac{1}{C} = \frac{1}{50} \quad [\text{Given}]$$

$$\Rightarrow \frac{1}{C} = \frac{3}{100} - \frac{1}{50} = \frac{1}{100}$$

$\therefore C = 100$

Hence, the time taken by pipe C to empty the tank is 100 min.

08

PERCENTAGE

Generally (2-3) questions have been asked from this chapter. Generally questions which are asked from this chapter are tricky but you can easily solve them by using the short-tricks formulae in very less time.



PER CENT

Per cent is a fraction whose denominator is 100 and numerator of the fraction is called the rate per cent. Per cent is denoted by the symbol '%’.

Important Rules and Formulae

Rule 1 For expressing $a\%$ as a fraction, we can write $a\% = \frac{a}{100}$, i.e. divide 'a' by 100 and reduce it to the lowest form, e.g. $25\% = \frac{25}{100} = \frac{1}{4}$

Rule 2 For expressing a fraction $\frac{a}{b}$ as a per cent, we can write $\frac{a}{b} = \left(\frac{a}{b} \times 100\right)\%$

e.g. $\frac{2}{3} = \frac{2}{3} \times 100\% = \frac{200}{3}\%$

Rule 3 To find how much per cent one quantity is of another quantity, we can write

Required percentage = $\frac{\text{The quantity to be expressed in per cent}}{\text{2nd quantity (in respect of which the per cent has to be obtained)}} \times 100\%$

EXAMPLE 1. 20g is what per cent of 1000g?

a. 2%

b. 5%

c. 7%

d. 9%

Sol. a. Required percentage = $\frac{20}{1000} \times 100\% = 2\%$

Rule 4 Percentage increase or decrease in a quantity when it increases or decreases by some value.

Percentage = $\frac{\text{Increase or decrease in the value}}{\text{Initial value}} \times 100\%$

EXAMPLE 2. A person's salary has increased from ₹ 7200 to ₹ 8100. What is the percentage increase in his salary? ✎ 2013 II

- a. 25% b. 18% c. $16\frac{2}{3}\%$ d. $12\frac{1}{2}\%$

Sol. d. Percentage increase in salary = $\frac{8100-7200}{7200} \times 100\%$
 $= \frac{900}{7200} \times 100\% = 12.5\% = 12\frac{1}{2}\%$

(Rule 5) If A is $x\%$ more than ' B ', then B would be $\left[\frac{x}{(100+x)} \times 100 \right]\%$ less than ' A '.

(Rule 6) If A is $x\%$ less than ' B ', then B would be $\left[\frac{x}{(100-x)} \times 100 \right]\%$ more than A .

EXAMPLE 3. Raj get 10% less marks than Rohit in an examination. What percentage of marks does Rohit gets more than Raj?

- a. $11\frac{1}{6}\%$ b. $11\frac{1}{9}\%$ c. $12\frac{1}{9}\%$ d. None of these

Sol. b. Given, $x = 10\%$
 $\therefore$ Rohit more percentage than Raj = $\frac{10}{100-10} \times 100$
 $= \frac{10 \times 100}{90} = 11\frac{1}{9}\%$

(Rule 7) If the price of a commodity is increased (or decreased) by $x\%$, then the decrease (or increase) in consumption, so as not to increase or (decrease) the expenditure is $\left(\frac{x}{100 \pm x} \times 100 \right)\%$.

EXAMPLE 4. If the price of the cooking gas rises by 15%, by what per cent should a family reduce its consumption so as not to exceed the budget on cooking gas?

- a. $12\frac{1}{23}\%$ b. $13\frac{1}{23}\%$ c. $14\frac{1}{23}\%$ d. None of these

Sol. b. Let initial price of cooking gas be ₹ 100.

Price after increase = ₹ 115

On ₹ 115 he should reduce ₹ 15 on ₹ 100, he should reduce

$$= \frac{15}{115} \times 100 = 13\frac{1}{23}\%$$

Shortcut Method

Here, $x = 15\%$

$$\therefore \text{Reduction in consumption} = \left(\frac{x}{100+x} \times 100 \right)\% \\ = \frac{15}{115} \times 100\% = 13\frac{1}{23}\%$$

(Rule 8) If the value of an object is first changed (increased or decreased) by $x\%$ and then changed (increased or decreased) by $y\%$, then

$$\text{Net effect} = \left(\pm x \pm y + \frac{(\pm x)(\pm y)}{100} \right)\%$$

Here, + sign is used in case of increment and - sign is used in case of decrement.

EXAMPLE 5 The price of an article is first increased by 20% and later on the price were decreased by 25% due to reduction in sales. Find the net percentage change in final price of article.

- a. 20% b. 10% c. 30% d. 15%

Sol. b. Here, $a = 20\%$, $b = 25\%$

$$\text{So, required percentage} = \frac{10}{100} \times 100 = 10\%$$

$$\begin{aligned} \text{Required change} &= \left[(\pm a) + (\pm b) + \frac{(\pm a)(\pm b)}{100} \right]\% \\ &= \left[20 - 25 + \frac{20 \times (-25)}{100} \right]\% \begin{cases} \text{positive sign for increase} \\ \text{negative sign for decrease} \end{cases} \\ &= [-5 - 5]\% \\ &= -10\% \end{aligned}$$

$\therefore$ Net percentage change is a decrease of 10 % because final result is negative.

(Rule 9) If the population of a town is P and it increases (or decreases) at the rate of $R\%$ per annum, then

$$(i) \text{ Population after } n \text{ yr} = P \left(1 \pm \frac{R}{100} \right)^n$$

$$(ii) \text{ Population, } n \text{ yr ago} = \frac{P}{\left(1 \pm \frac{R}{100} \right)^n}$$

EXAMPLE 6 The population of a town is 352800. If it increases at the rate of 5% per annum, then what will be its population 2 yr hence, also find the population 2 yr ago.

- a. 315000 b. 316500 c. 200045 d. 320000

Sol. d. Given that, $p = 352800$, $R = 5\%$ and $n = 2$

According to the formula,

$$\begin{aligned} \text{Population after 2 yr} &= P \left(1 + \frac{R}{100} \right)^n = 352800 \times \left(1 + \frac{5}{100} \right)^2 \\ &= 352800 \times \left(\frac{100+5}{100} \right)^2 = 352800 \times \left(\frac{21}{20} \times \frac{21}{20} \right) = 388962 \end{aligned}$$

$$\begin{aligned} \text{Population 2 yr ago} &= \frac{P}{\left(1 + \frac{R}{100} \right)^n} = \frac{352800}{\left(1 + \frac{5}{100} \right)^2} = 352800 \times \frac{20}{21} \times \frac{20}{21} = 320000 \end{aligned}$$

Rule 10 If the present population of a city is P and there is an increment or decrement of $R_1\%$, $R_2\%$ and $R_3\%$ in first, second and third year respectively, then

Population of city after 3 yr

$$= P \left(1 \pm \frac{R_1}{100} \right) \left(1 \pm \frac{R_2}{100} \right) \left(1 \pm \frac{R_3}{100} \right)$$

Note Use '+' sign for increases and '-' sign for decreases.

EXAMPLE 7 Population of a city in 2004 was 1000000. If in 2005 there is an increment of 15%, in 2006 there is a decrement of 35% and in 2007 there is

an increment of 45%, then find the population of city at the end of year 2007.

a. 1083875 b. 1083000 c. 1089000 d. 1135000

Sol a. Given that, $P = 1000000$, $R_1 = 15\%$, $R_2 = 35\%$ (decrease) and $R_3 = 45\%$
Population of city at the end of year 2007.

$$\begin{aligned} &= P \left(1 + \frac{R_1}{100} \right) \left(1 - \frac{R_2}{100} \right) \left(1 + \frac{R_3}{100} \right) \\ &= 1000000 \left(1 + \frac{15}{100} \right) \left(1 - \frac{35}{100} \right) \left(1 + \frac{45}{100} \right) \\ &= 1000000 \times \frac{115}{100} \times \frac{65}{100} \times \frac{145}{100} \\ &= 1083875 \end{aligned}$$

> PRACTICE EXERCISE

- If 90% of $A = 30\%$ of B and $B = x\%$ of A , the value of x is
(a) 700 (b) 600 (c) 300 (d) 1100
- When 40% of a number is added to 42, the result is the number itself. The number is
(a) 70 (b) 90 (c) 82 (d) 72
- If $x\%$ of y is $13x$, then the value of y is
(a) 880 (b) 1300 (c) 1200 (d) 700
- What is the number when 20% of number is 30% of 40?
(a) 90 (b) 80 (c) 60 (d) 50
- What is the number when increased by 20% becomes 300?
(a) 250 (b) 200 (c) 180 (d) 280
- If 50% of $(x - y) = 40\%$ of $(x + y)$, then what percent of x is y ?
(a) $10\frac{1}{9}\%$ (b) $11\frac{1}{9}\%$ (c) $13\frac{1}{9}\%$ (d) $21\frac{1}{9}\%$
- Two candidates fought an election one get 65% of the votes and won by 300 votes. The total number of votes polled in the election is
(a) 700 (b) 950 (c) 1000 (d) 900
- A man spends ₹ 3500 and saves $12\frac{1}{2}\%$ of his income. His monthly income (in ₹) is
(a) 4000 (b) 3800 (c) 4200 (d) 4500
- In an examination 52% of the candidates failed in English, 42% in Mathematics and 17% in both. The number of those who passed in both the subjects is
(a) 23% (b) 40% (c) 53% (d) 33%
- To pass an examination, a candidate needs 40% marks. All questions carry equal marks. A candidate just passed by getting 10 answers correct by attempting 15 of the total questions. How many questions are there in the examination?
(a) 25 (b) 30
(c) 40 (d) 45
- If salary of X is 20% more than salary of Y , then by how much percentage is salary of Y less than X ?
(a) 25 (b) 20
(c) $\frac{50}{3}$ (d) $\frac{65}{4}$
- Water contains $14\frac{2}{7}\%$ of hydrogen and the rest is oxygen. In 350 g of water, oxygen will be
(a) 300 g (b) 250 g
(c) 200 g (d) None of these
- The income of 'A' is 20% higher than that of 'B'. The income of 'B' is 25% less than of 'C'. What per cent less is A's income from 'C's income?
(a) 7% (b) 8%
(c) 10% (d) 12.5%
- 38 L of milk was poured into a tub and the tub was found to be 5% empty. To completely fill the tub, what amount of additional milk must be poured?
(a) 1 L (b) 2 L (c) 3 L (d) 4 L
- 10% of the inhabitants of a certain city left that city. Later on 10% of the remaining inhabitants of that city again left the city. What is the remaining percentage of population of that city?
(a) 80% (b) 80.4% (c) 80.4% (d) 81%

- 16.** If after 24% of wastage the net output of a coal-mine is 68400 quintals. Then, the total output of the coal-mine in quintals is
(a) 70000 (b) 90000 (c) 80000 (d) 89000
- 17.** A rise of 25% in the price of grapes compels a person to buy 1.5 kg of grapes less for ₹ 240. Then, the original price of grapes per kg is
(a) ₹ 40 (b) ₹ 32 (c) ₹ 30 (d) ₹ 28
- 18.** The price of an item is increased by 20% and then decreased by 20% the final price as compared to original price is
(a) 4% less (b) 4% more (c) 20% less (d) 20% more
- 19.** Sohan saves 14% of his salary while George saves 22%. If both gets the same salary and George saves ₹ 1540. Then, the salary of each of them is
(a) ₹ 9500 (b) ₹ 17000 (c) ₹ 7000 (d) ₹ 7500
- 20.** A's salary is half that of B. If A got a 50% rise in his salary and B got a 25% rise in his salary, then the percentage increase in combined salaries of both is
(a) 13% (b) $33\frac{1}{3}\%$ (c) 33% (d) 45%
- 21.** A man donated 4% of his salary to a charity and deposited 10% of the rest in the bank. If now he has ₹ 10800, then his income was
(a) ₹ 13500 (b) ₹ 14500
(c) ₹ 40000 (d) ₹ 12500
- 22.** Rohit saves 30% of his salary. When his expenses increased by 30%, he is able to save ₹ 1215 per month. His monthly salary is
(a) ₹ 13500 (b) ₹ 14500
(c) ₹ 30000 (d) ₹ 12500
- 23.** A man spends 75% of his income. If his income is increased by 20% and he increased his expenditure by 10%. His savings percentage is increased by
(a) 25% (b) 50% (c) 75% (d) 10%
- 24.** The price of wheat has increased by 60%. In order to restore to the original price, the new price must be reduced by
(a) 37.5% (b) 33% (c) 34% (d) 40%
- 25.** If the numerator of a fraction increased by 20% and its denominator be diminished by 10%. The value of the fraction is $\frac{16}{27}$, then the fraction is
(a) $\frac{4}{9}$ (b) $\frac{3}{2}$ (c) $\frac{3}{8}$ (d) $\frac{9}{4}$
- 26.** 1 L of water is evaporated from 6 L of a solution having 4% of sugar. The percentage of sugar in the remaining solution is
(a) 4% (b) 5% (c) $4\frac{4}{5}\%$ (d) $\frac{4}{5}\%$
- 27.** Two numbers are less than a third number by 30% and 40%, respectively. How much per cent is the second number less than the first?
(a) 35% (b) 36% (c) 14% (d) $14\frac{2}{7}\%$
- 28.** A sample of 5 L of glycerine is formed to be adulterated to the extent of 20%. Find how glycerine should be added to bring down percentage of impurity to 5%?
(a) 10 L (b) 25 L (c) 15 L (d) 20 L
- 29.** The daily wages of a worker increase by 20% but the number of hours worked by him also dropped by 20%. If originally he was getting ₹ 200 per week, his wages per week now is
(a) ₹ 160 (b) ₹ 192 (c) ₹ 210 (d) ₹ 198
- 30.** 140 L of a liquid contains 90% of acid and the rest water. How much water must be added to make the water 12.5% of the resulting mixture?
(a) 4 L (b) 10 L (c) 12 L (d) 3 L
- 31.** A person spends 30% of monthly salary on rent, 25% on food, 20% on children's education and 12% on electricity and the balance of ₹ 1040 on the remaining items. What is the monthly salary of the person?
(a) ₹ 8000 (b) ₹ 9000 (c) ₹ 9600 (d) ₹ 10600
- 32.** In an office, 40% of the staff is female and rest is male. 60% of the male and 40% of female voted for Ramesh. The percentage of votes Ramesh got was
(a) 24% (b) 42% (c) 50% (d) 52%
- 33.** In a class-X of 30 students, 24 passed in first class; in another class-Y of 35 students, 28 passed in first class. In which class was the percentage of students passed first class more?
(a) Class-X (b) Class-Y
(c) Both X and Y (d) None of these
- Directions** (Q. Nos. 34-36) *The population of the town is 126800. It increases by 15% in the 1st year due to increase crime in the city.*
- 34.** What is the population of the town at the end of 2nd year if the population decreases by 20% in the second year?
(a) 174984 (b) 135996 (c) 116656 (d) 145820
- 35.** What is the population of the town, if the population decrease by 15% at the end of 2nd year?
(a) 123749 (b) 123479
(c) 123947 (d) None of these
- 36.** What is the population of the town, if the population increases by 20% at the end of 2nd year?
(a) 194784 (b) 174984 (c) 179484 (d) 178494

> PREVIOUS YEARS' QUESTIONS

- 37.** A man losses 20% of his money. After spending 25% of the remainder, he has ₹ 480 left. What is the amount of money he originally had? **☑ 2012 I**
(a) ₹ 600 (b) ₹ 720 (c) ₹ 800 (d) 840
- 38.** The price of an article is ₹ 25. After two successive cuts by the same percentage, the price becomes ₹ 20.25. If each time the cut was $x\%$, then **☑ 2012 II**
(a) $x = 9$ (b) $x = 10$ (c) $x = 11$ (d) $x = 11.5$
- 39.** What is 5% of 50% of 500? **☑ 2012 II**
(a) 12.5 (b) 25 (c) 1.25 (d) 6.25
- 40.** X, Y and Z had taken a dinner together. The cost of the meal of Z was 20% more than that of Y and the cost of the meal of X was $\frac{5}{6}$ as much as the cost of the meal of Z. If Y paid ₹ 100, then what was the total amount that all the three of them had paid? **☑ 2013 II**
(a) ₹ 285 (b) ₹ 300
(c) ₹ 355 (d) None of these
- 41.** A water pipe is cut into two pieces. The longer piece is 70% of the length of the pipe. By how much percentage is the longer piece longer than the shorter piece? **☑ 2013 II**
(a) 140% (b) $\frac{400}{3}\%$
(c) 40% (d) None of these
- 42.** If $m\%$ of $m + n\%$ of $n = 2\%$ of $(m \times n)$, then what percentage of m is n ? **☑ 2014 II**
(a) 50% (b) 75%
(c) 100% (d) Cannot be determined
- 43.** The price of a commodity increased by 5% from 2010 to 2011, 8% from 2011 to 2012 and 77% from 2012 to 2013. What is the average price increase (approximate) from 2010 to 2013? **☑ 2014 II**
(a) 26% (b) 32% (c) 24% (d) 30%
- 44.** A person could save 10% of his income. But 2 yr later, when his income increased by 20%, he could save the same amount only as before. By how much percentage has his expenditure increased? **☑ 2015 I**
(a) $22\frac{2}{9}\%$ (b) $23\frac{1}{3}\%$
(c) $24\frac{2}{9}\%$ (d) $25\frac{2}{9}\%$
- 45.** 20% of a number when added to 20 becomes the number itself, then the number is **☑ 2015 II**
(a) 20 (b) 25 (c) 50 (d) 80
- 46.** A's salary was increased by 40% and then decreased by 20%. On the whole A's salary is increased by **☑ 2015 II**
(a) 60% (b) 40% (c) 20% (d) 12%
- 47.** In an election 10% of the voters on the voter list did not cast their vote and 60 voters cast their ballot papers blank. There were only two candidates. The winner was supported by 47% of total voters in the voter list and he got 308 votes more than his rival. The number of voters on the voter list is **☑ 2015 II**
(a) 3600 (b) 6200 (c) 6028 (d) 6400
- 48.** The salary of a person is increased by 10% of his original salary. But he received the same amount even after increment. What is the percentage of his salary he did not receive? **☑ 2016 I**
(a) 11% (b) 10% (c) $\frac{100}{11}\%$ (d) $\frac{90}{11}\%$
- 49.** The expenditure of a household for a certain month is ₹ 20000, out of which ₹ 8000 is spent on education, ₹ 5900 on food, ₹ 2800 on shopping and the rest on personal care. What percentage of expenditure is spent on personal care? **☑ 2016 I**
(a) 12% (b) 16.5% (c) 18% (d) 21.8%

> ANSWERS

1	c	2	a	3	b	4	c	5	a	6	b	7	c	8	a	9	a	10	a
11	c	12	a	13	c	14	b	15	d	16	b	17	b	18	a	19	c	20	b
21	d	22	a	23	b	24	a	25	a	26	c	27	d	28	c	29	b	30	a
31	a	32	d	33	c	34	c	35	c	36	b	37	c	38	b	39	a	40	d
41	b	42	c	43	d	44	a	45	b	46	d	47	b	48	c	49	b		

HINTS AND SOLUTIONS

1. (c) Given, 90% of $A = 30\%$ of B

$$\frac{90A}{100} = \frac{30B}{100} \quad [\text{Rule 1}]$$

$$\Rightarrow \frac{A}{B} = \frac{3}{9} \Rightarrow B = 3A$$

Now, $B = x\%$ of A , $3A = \frac{x \cdot A}{100}$ [Rule 1]

$$\therefore x = 300$$

Hence, the value of x is 300.

2. (a) Let the number be x ,
 then, $42 + 40\%$ of $x = x$

$$\Rightarrow x = 42 + \frac{40}{100}x \quad [\text{Rule 1}]$$

$$\Rightarrow \frac{3}{5}x = 42$$

$$\therefore x = \frac{42 \times 5}{3} = 70$$

Hence, the required number is 70.

3. (b) Given, $x\%$ of $y = 13x$

$$\Rightarrow \frac{x}{100}y = 13x \quad [\text{Rule 1}]$$

$$\therefore y = 13 \times 100 = 1300$$

4. (c) Let the number be x .
 By given condition, 20% of x

$$= 30\% \text{ of } 40$$

$$\Rightarrow \frac{20x}{100} = \frac{30 \times 40}{100} \Rightarrow 20x = 30 \times 40$$

$$\therefore x = 60$$

Hence, the required number is 60.

5. (a) Let the number be x .

$$\therefore x + \frac{20x}{100} = 300$$

$$\Rightarrow x = \frac{300 \times 100}{120} = 250$$

6. (b) Given, 50% of $(x - y) = 40\%$ of $(x + y)$

$$\frac{50}{100} \times (x - y) = \frac{40}{100} \times (x + y)$$

$$\Rightarrow 5x - 5y = 4x + 4y \quad [\text{Rule 1}]$$

$$\Rightarrow x = 9y \quad \dots (i)$$

 Let $r\%$ of $x = y \Rightarrow \frac{r}{100} \times x = y$

$$\Rightarrow \frac{r}{100} \times 9y = y \quad [\text{from Eq. (i)}]$$

$$\therefore r = \frac{100}{9} = 11\frac{1}{9}\%$$

7. (c) Let total number of votes polled be x .

$$\therefore 65\% \text{ of } x - 35\% \text{ of } x = 300$$

$$\frac{65}{100}x - \frac{35}{100}x = 300$$

$$\Rightarrow \frac{30}{100}x = 300 \Rightarrow x = \frac{300 \times 100}{30}$$

$$\therefore x = 1000 \text{ votes}$$

Hence, the total number of votes is 1000.

8. (a) Let monthly income be ₹ x .

$$\Rightarrow 87\frac{1}{2}\% \text{ of } x = ₹ 3500$$

$$\Rightarrow \frac{175}{2 \times 100}x = ₹ 3500$$

$$\therefore x = \frac{3500 \times 2 \times 100}{175} = ₹ 4000$$

9. (a) Number of candidates who passed in both the subject = $100 - (x + y - z)\%$

$$= 100 - (52 + 42 - 17)\%$$

$$= 23\%$$

10. (a) Let the total number of questions in examination be x .

By given condition, 40% of $x = 10$

$$\Rightarrow \frac{x \times 40}{100} = 10 \Rightarrow x = \frac{1000}{40} = 25$$

11. (c) Let the salary of y be ₹ x .

$$\therefore \text{Salary of } x = ₹ \frac{120x}{100} = ₹ \frac{6x}{5}$$

Difference of their salaries

$$= ₹ \left(\frac{6x}{5} - x \right) = ₹ \frac{x}{5}$$

$$\therefore \text{Required percentage} = \frac{x/5}{6x/5} \times 100\% = \frac{50}{3}\%$$

[Rule 3]

Shortcut Method

Here, $x = 20\%$

$$\therefore \text{Required percentage} = \left(\frac{x}{100 + x} \times 100 \right)\% \quad [\text{Rule 5}]$$

$$= \left(\frac{20}{100 + 20} \times 100 \right)\%$$

$$= \frac{2000}{120}\% = \frac{50}{3}\%$$

12. (a) Percentage of oxygen in water

$$= 100 - 14\frac{2}{7} = 100 - \frac{100}{7} = 85\frac{5}{7}\%$$

Percentage of oxygen in 350 g of water

$$= 85\frac{5}{7}\% \text{ of } 350 = \frac{600}{7} \times \frac{350}{100} = 300 \text{ g}$$

13. (c) Let the income of C be ₹ x .

$$\therefore \text{Income of } B = x - x \times 25\% = x - \frac{25x}{100}$$

$$= x - \frac{x}{4} = \frac{3x}{4}$$

and, the income of $A = \frac{3x}{4} + \frac{3x}{4} \times 20\%$

$$= \frac{3x}{4} + \frac{3x}{4} \times \frac{1}{5} = \frac{9x}{10}$$

Now, difference between C 's salary and A 's salary

$$= x - \frac{9x}{10} = \frac{x}{10}$$

$\therefore$ The required percentage

$$= \frac{x}{10 \times x} \times 100\% = 10\%$$

14. (b) Amount of milk in tub = 38 L

$\therefore$ Tub found to be 5% empty

$\therefore$ Total quantity of milk in per cent = 95%

Now, completely fill the tub, total amount of additional milk

$$= \frac{38 \times 5}{95} = 2 \text{ L}$$

15. (d) Remaining percentage

$$= \left(1 - \frac{10}{100} \right) \left(1 - \frac{10}{100} \right)$$

$$= \frac{90}{100} \times \frac{90}{100} = 81\%$$

16. (b) Let total output be x quintal.

$\therefore$ Useful output = $100 - 24 = 76\%$

$$\Rightarrow \frac{76}{100}x = 68400$$

$$\Rightarrow x = \frac{68400 \times 100}{76}$$

$$\therefore x = 90000 \text{ quintal}$$

17. (b) Let original price of grapes be ₹ x .

$$\therefore \text{Increased price} = x + \frac{25}{100}x$$

$$= ₹ \frac{125}{100}x \quad \dots (i)$$

Increased price of 1.5 kg of grapes

$$= \frac{25}{100} \times 240 = ₹ 60$$

$$\therefore \text{Increased price of 1 kg} = \frac{60}{1.5}$$

$$= ₹ 40 \quad \dots (ii)$$

So, from Eqs. (i) and (ii), we have

$$\frac{125}{100}x = ₹ 40$$

$$\therefore x = \frac{40 \times 100}{125} = ₹ 32$$

18. (a) The price of item first increased by 20% and then decreased by 20%.

$$\therefore \text{Net effect} = \left(20 - 20 + \frac{20 \times (-20)}{100} \right)$$

$$= \frac{-400}{100} = -4\% \quad [\text{Rule 8}]$$

19. (c) Let salary of each of them be ₹ x .
George saves 22% of x and his saving amount is ₹ 1540.

$$\Rightarrow \frac{22}{100}x = 1540$$

$$\Rightarrow x = \frac{1540 \times 100}{22} = ₹ 7000$$

20. (b) Let A 's salary = x . Then,

$$B\text{'s salary} = 2x$$

$$\text{New salary of } A = 150\% \text{ of } x = \frac{3}{2}x$$

$$\text{Total salary of } B = 125\% \text{ of } 2x = \frac{5}{2}x$$

$$\text{Total combined salary} \\ = \left(\frac{3}{2}x + \frac{5}{2}x \right) = 4x$$

$$\therefore \text{Required increment in salary} \\ = \frac{4x - 3x}{3x} \times 100 = \frac{100}{3} = 33\frac{1}{3}\%$$

[Rule 4]

21. (d) Let total income be ₹ x .

$$\text{Income deposited} = 10\% \text{ of } \left[x - \frac{4}{100}x \right]$$

$$= \frac{10}{100} \left(x - \frac{4}{100}x \right) = \frac{96}{1000}x$$

$$\text{Remaining income} = ₹ 10800$$

$$\therefore \frac{4}{100}x + \frac{96x}{1000} + 10800 = x$$

$$\Rightarrow 10800 = x - \frac{136x}{1000}$$

$$\Rightarrow 10800 = \frac{864x}{1000} \Rightarrow x = \frac{10800 \times 1000}{864}$$

$$\therefore x = ₹ 12500$$

22. (a) Let the salary of Rohit be ₹ 100, then saving = ₹ 30

$$\text{Expenses} = ₹ 70$$

$$\text{New expenses} = (100 + 30)\% \text{ of } ₹ 70 \\ = ₹ 91$$

$$\text{New saving} = ₹ (100 - 91) = ₹ 9$$

$$\text{He saves ₹ 9, his salary} = ₹ 100$$

$$\text{If he saves ₹ 1215.}$$

$$\text{Then, his salary} = ₹ \left(\frac{100}{9} \times 1215 \right) \\ = ₹ 13500$$

23. (b) Let income of man be ₹ 100.

$$\text{Then, his expenditure} = ₹ 75$$

$$\text{and savings} = ₹ 25$$

$$\text{New income} = ₹ (100 + 20) = ₹ 120$$

$$\text{New expenditure} = ₹ (75 + 7.5) \\ = ₹ 82.50$$

$$\text{New saving} = ₹ (120 - 82.5) = ₹ 37.50$$

$$\text{Saving difference} = 37.50 - 25.0 = 12.5$$

$$\therefore \text{Percentage increase saving} \\ = \frac{12.5}{25} \times 100 = 50\%$$

24. (a) New price reduced by

$$= \left(\frac{x}{100 + x} \right) \times 100\% \quad [\text{Rule 7}]$$

$$= \left(\frac{60}{100 + 60} \right) \times 100\% = 37.5\%$$

25. (a) Let fraction be $\frac{x}{y}$.

$$\text{New fraction} = \frac{120\% \text{ of } x}{90\% \text{ of } y} = \frac{4x}{3y}$$

According to question,

[by given condition]

$$\frac{4x}{3y} = \frac{16}{27} \Rightarrow \frac{x}{y} = \frac{16}{27} \times \frac{3}{4} = \frac{4}{9}$$

26. (c) Amount of sugar in 6 L of solution

$$= \frac{4}{100} \times 6 = 0.24 \text{ L}$$

$$\text{After evaporation, sugar in 5 L} = 0.24 \text{ L}$$

$\therefore$ Percentage of sugar

$$= \left(\frac{0.24}{5} \times 100 \right) = 4\frac{4}{5}\%$$

27. (d) Let the third number be z .

$$\therefore \text{First number } x = \frac{(100 - 30)}{100} \times z = \frac{7z}{10}$$

Second number

$$y = \frac{(100 - 40)}{100} \times z = \frac{6z}{10}$$

Difference between first and second number

$$= (x - y) = \frac{7z}{10} - \frac{6z}{10} = \frac{z}{10}$$

Hence, the required percent [Rule 4]

$$= \frac{\frac{z}{10}}{\frac{7z}{10}} \times 100\% = \frac{100}{7}\% = 14\frac{2}{7}\%$$

28. (c) Glycerine in the given sample = 80% of 5 L

$$= \frac{80}{100} \times 5 = 4 \text{ L}$$

Let x L of glycerine be added, then

$$\frac{4 + x}{(5 + x)} \times 100 = 95$$

$$\Rightarrow 80 + 20x = 95 + 19x$$

$$\therefore x = 15 \text{ L}$$

29. (b) Increased wages of the worker = 200 + 20% of 200 = ₹ 240

Also, let he worked for x h.

$\therefore$ Reduced working hours

$$= x - \frac{20}{100}x = 0.80x$$

$$\text{Required wages} = \frac{240}{x} \times 0.80x = ₹ 192$$

30. (a) Water in the mixture = 10% of 140 L
 $= \frac{10}{100} \times 140 = 14 \text{ L}$

Let x L of water added in the mixture,
then $\left(\frac{14 + x}{140 + x} \right) \times 100 = 12.5$

$$\Rightarrow 1400 + 100x = 1750 + 12.5x$$

$$\Rightarrow 87.5x = 350$$

$$\Rightarrow x = \frac{350}{87.5} = 4 \text{ L}$$

31. (a) Let the monthly salary of the person be ₹ x .

By given condition,

$$\frac{(100 - 30 - 25 - 20 - 12) \times x}{100} = 1040$$

$$\Rightarrow \frac{13x}{100} = ₹ 1040 \Rightarrow x = \frac{1040 \times 100}{13}$$

$$\Rightarrow x = ₹ 8000$$

Hence, the monthly salary of the person is ₹ 8000.

32. (d) Let total number of staff be 100.

$$\text{Female staff} = 40$$

$$\text{Male staff} = (100 - 40) = 60$$

Votes casted by females

$$= \frac{40}{100} \times 40 = 16$$

$$\text{Votes casted by males} = \frac{60}{100} \times 60 = 36$$

Votes casted by both males and females

$$= 16 + 36 = 52$$

$\therefore$ Percentage votes obtained = 52%.

33. (c) For class-X, let the student passed in first class = $a\%$.

Then, by condition given in question,

$$a\% \text{ of } 30 = 24 \Rightarrow \frac{a \times 30}{100} = 24$$

$$\therefore a = 80\%$$

Now, for class-Y let the student passed in first class = $b\%$.

According to the question,

$$b\% \text{ of } 35 = 28$$

$$\Rightarrow \frac{b}{100} \times 35 = 28$$

$$\therefore b = 80\%$$

Hence, both classes have equal percentage of students getting first class.

34. (c) Given, $R_1 = 15\%$ and $R_2 = 20\%$.

$\therefore$ Required population

$$= P \left(1 + \frac{R_1}{100} \right) \left(1 - \frac{R_2}{100} \right) \quad [\text{Rule 10}]$$

$$= 126800 \left(1 + \frac{15}{100} \right) \left(1 - \frac{20}{100} \right)$$

$$= 126800 \left(1 + \frac{3}{20} \right) \left(1 - \frac{1}{5} \right)$$

$$= 126800 \left(\frac{23}{20} \right) \left(\frac{4}{5} \right) = 116656$$

35. (c) Given, $R_1 = 15\%$, $R_2 = 15\%$

$$\therefore \text{Population} = P \left(1 + \frac{R_1}{100} \right) \left(1 - \frac{R_2}{100} \right)$$

$$= 126800 \left(1 + \frac{15}{100} \right) \left(1 - \frac{15}{100} \right)$$

$$= 126800 \times \frac{115}{100} \times \frac{85}{100} = 123947$$

36. (b) Given, $R_1 = 15\%$ and $R_2 = 20\%$

$\therefore$ Required population

$$= P \left(1 + \frac{15}{100} \right) \left(1 + \frac{20}{100} \right)$$

$$= 126800 \left(\frac{23}{20} \right) \left(\frac{6}{5} \right) = 174984$$

37. (c) Refer to question 21.

38. (b) Refer to question 34.

39. (a) 5% of 50% of 500
- $$= \frac{5}{100} \times \frac{50}{100} \times 500 = 12.5 \text{ [Rule 1]}$$

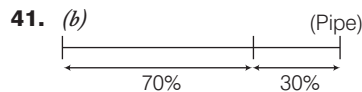
40. (d) Given, the cost of meal of Y = ₹ 100
Now, according to the question,
Cost of the meal of Z = 20% more than that of Y
- $$= \left(100 + \frac{20}{100} \times 100 \right)$$

$$= (100 + 20) = ₹ 120$$

and, the cost of the meal of X = $\frac{5}{6}$ as much as the cost of the meal of Z = $\frac{5}{6} \times 120 = ₹ 100$

$\therefore$ Total amount that all the three of them has paid

$$= 100 + 100 + 120 = ₹ 320$$



$\therefore$ Increase in percentage of longer piece compared to shorter piece

$$= \frac{70 - 30}{30} \times 100\% \quad [\text{Rule 4}]$$

$$= \frac{40}{30} \times 100\% = \frac{400}{3}\%$$

42. (c) Refer to question 6.

43. (d) Average price increase
- $$= \left(\frac{5 + 8 + 77}{3} \right) \% = \frac{90}{3} \% = 30\%$$

44. (a) Refer to question 23.

45. (b) Refer to question 2.

46. (d) Refer to question 18.

47. (b) Let the number of voters on the voter list be x .

Total cast vote = 90% of $x - 60$

Winner was supported by 47% of total voter. i.e. 47% of x .

Hence, rival got vote = (90% of $x - 60$) - 47% of x .

$$= 43\% \text{ of } x - 60$$

It is given that difference between their vote is 308.

Then, 47% of $x - 43\% \text{ of } x - 60 = 308$

$$\Rightarrow 4\% \text{ of } x = 308 - 60$$

$$\Rightarrow \frac{4}{100} x = 248$$

$$\therefore x = \frac{248 \times 100}{4} = 6200$$

48. (c) Let the original salary be x .

Then, increased salary

$$= \left(\frac{110}{100} \right) x = ₹ \frac{11x}{10}$$

$\therefore$ He received the same salary even after increment.

Amount of salary he did not receive

$$= \frac{11x}{10} - x = ₹ \frac{x}{10}$$

$\therefore$ Amount of salary in percentage

$$= \left(\frac{\frac{x}{10}}{\frac{11x}{10}} \right) \times 100\%$$

$$= \frac{x}{11x} \times 100\% = \frac{100}{11}\%$$

49. (b) Refer to question 31.

09

SIMPLE INTEREST

Regularly (1-2) questions have been asked from this chapter. Generally direct formula based questions are asked from this chapter and hence making it easy to score area.



INTEREST

When money is borrowed by a person, then customarily the money lender used to charge some extra money in lieu of the money lent by him. This extra money earned by the money lender is called interest.

Some terms related to interest are given below

Principal (P) The money which is borrowed from a money lender, is called Principal.

Amount (A) The sum of the principal and the interest is called Amount i.e. Amount (A) = Principal (P) + Interest (I).

Rate (R) It is the rate at which the interest is charged on Principal.

Time (T) The time period, for which the money is lent or deposited or borrowed, is called Time.

SIMPLE INTEREST

If the interest is calculated on the original Principal for any length of time, then it is called simple interest.

$$\text{Simple Interest (SI)} = \frac{\text{Principal} \times \text{Rate} \times \text{Time}}{100} \quad \text{or} \quad \text{SI} = \frac{P \times R \times T}{100}$$

$$\text{Amount (A)} = \text{Principal (P)} + \text{Interest (I)} \quad \text{and} \quad P = \frac{100 \times A}{100 + RT}$$

Where, $A \rightarrow$ Amount

$R \rightarrow$ Rate of simple Interest

$T \rightarrow$ Time

EXAMPLE 1. The sum required to earn a monthly interest of ₹400 at 10% per annum at simple interest is

a. ₹ 2000

b. ₹ 12000

c. ₹ 24000

d. ₹ 48000

Sol. d. Given, $SI = ₹ 400$, $R = 10\%$ and $T = 1 \text{ month} = \frac{1}{12} \text{ yr}$

$$SI = \frac{P \times R \times T}{100} \quad \text{or} \quad P = \frac{100 \times SI}{R \times T} = \frac{100 \times 400 \times 12}{10 \times 1} = ₹ 48000$$

EXAMPLE 2. In what time will the simple interest on ₹ 400 at 10% per annum be the same as the simple interest on ₹ 1000 for 4 yr at 4% per annum?

- a. 2 yr b. 3 yr c. 4 yr d. 6 yr

Sol. c. Here, $P = ₹ 1000$, $T = 4 \text{ yr}$, and $R = 4\%$

$$\therefore \text{Simple interest} = \frac{P \times R \times T}{100} = \frac{1000 \times 4 \times 4}{100} = ₹ 160$$

Now, simple interest = ₹ 160, $P = ₹ 400$, $R = 10\%$,

$$\text{then, } T = \frac{100 \times SI}{P \times R} = \frac{100 \times 160}{400 \times 10} = 4 \text{ yr}$$

EXAMPLE 3. A sum at simple interest of 4% per annum amounts to ₹ 3120 in 5 yr. Find the sum.

- a. ₹ 2500 b. ₹ 1300 c. ₹ 4000 d. ₹ 2600

Sol. d. Here, $T = 5 \text{ yr}$, $R = 4\%$, $A = ₹ 3120$

$$\text{We know that, } P = \frac{100 \times A}{100 + RT} = \frac{100 \times 3120}{100 + 4 \times 5} = \frac{100 \times 3120}{120} = ₹ 2600$$

Important Rules and Formulae

(Rule 1) If a sum of money becomes n times in ' T ' yr at simple interest, then rate of interest will be,
 $R = \left(\frac{100(n-1)}{T} \right) \%$

EXAMPLE 4. At what rate per cent per annum will a sum of money double in 8 yr?

- a. 12% b. $12\frac{1}{2}\%$
 c. 13% d. 15%

Sol. b. Here, $n = 2$ and $T = 8 \text{ yr}$

$$\therefore \text{Rate} = \frac{100(n-1)}{T} = \frac{100 \times (2-1)}{8} = \frac{100}{8} = 12\frac{1}{2} \%$$

(Rule 2) If a certain Principal Amounts to ₹ A_1 in t_1 yr and to ₹ A_2 in t_2 yr, then the sum (P) is given by
 $₹ \left(\frac{A_2 t_1 - A_1 t_2}{t_1 - t_2} \right)$ and the rate per cent (R) per annum is
 given by $\left[\frac{100 (A_2 - A_1)}{A_1 t_2 - A_2 t_1} \right] \%$.

EXAMPLE 5. A certain sum amounts to ₹ 1586 in 2 yr and ₹ 1729 in 3 yr. Find the rate and the sum.

- a. 8%, ₹ 1200
 b. 9%, ₹ 1300
 c. 10%, ₹ 1000
 d. 11%, ₹ 1300

Sol. d. Here, $A_1 = ₹ 1586$, $t_1 = 2 \text{ yr}$, $A_2 = 1729$ and $t_2 = 3 \text{ yr}$

$$\therefore \text{Required principle} = \frac{A_2 t_1 - A_1 t_2}{t_1 - t_2} = \frac{1729 \times 2 - 1586 \times 3}{2 - 3} = ₹ 1300$$

$$\text{and required rate } R = \left[\frac{(A_2 - A_1) \times 100}{A_1 t_2 - A_2 t_1} \right] \%$$

$$= \frac{(1729 - 1586) \times 100}{1586 \times 3 - 1729 \times 2} = \frac{143 \times 100}{1300} = 11\%$$

(Rule 3) At the same rate of simple interest, if a sum of money becomes n_1 times of itself in t_1 yr and n_2 times in t_2 years, then $t_2 = \frac{(n_2 - 1)}{(n_1 - 1)} t_1 \text{ yr}$.

EXAMPLE 6. A sum of money becomes 3 times in 5 yr at simple interest. In how many years, will the same sum become 6 times at the same rate of simple interest?

- a. 10 yr
 b. 12 yr
 c. 12.5 yr
 d. 10.5 yr

Sol. c. Here, $n_1 = 3$, $t_1 = 5 \text{ yr}$ and $n_2 = 6$, $t_2 = ?$

$$\therefore \text{Required time } (t_2) = \frac{(n_2 - 1)}{(n_1 - 1)} t_1 = \frac{(6 - 1)}{(3 - 1)} \times 5 = \frac{25}{2} = 12.5 \text{ yr}$$

> PRACTICE EXERCISE

- Find the amount on a sum of ₹ 400 for 3 yr at simple interest at 5% per annum.
 (a) ₹ 460 (b) ₹ 415 (c) ₹ 435 (d) ₹ 412
- Find what sum of money will amount to ₹ 900 in 4 yr at 5% per annum on simple interest?
 (a) ₹ 750 (b) ₹ 650
 (c) ₹ 500 (d) ₹ 550
- If a certain sum is doubled in 8 yr on simple interest, in how many years will it be four times?
 (a) 24 yr (b) 16 yr (c) 32 yr (d) 12 yr
- A sum of money at simple interest amount to ₹ 1260 in 2 yr and ₹ 1350 in 5 yr, then the rate per cent per annum is
 (a) 30% (b) 10% (c) 2.5% (d) 5%

5. The difference of 13% per annum and 12% of a sum in 1 yr is ₹ 110. Then, the sum is
(a) ₹ 12000 (b) ₹ 13000 (c) ₹ 11000 (d) ₹ 16000
6. The simple interest on a sum of money at 10% per annum for 6 yr is half the sum. Then, the sum is
(a) ₹ 5000 (b) not possible (c) ₹ 4000 (d) ₹ 6000
7. The sum which amounts to ₹ 840 in 5 yr at the rate of 8% per annum simple interest is
(a) ₹ $\frac{100 \times 8 \times 5 + 100}{840}$ (b) ₹ $\frac{100 + 840}{100 + 5 \times 8}$
(c) ₹ $\frac{100 \times 840}{100 + 5 \times 8}$ (d) ₹ $\frac{840 \times 5 \times 8}{100}$
8. A certain sum at simple interest amounts to ₹ 1040 in 3 yr and to ₹ 1360 in 7 yr. Then, the sum is
(a) ₹ 750 (b) ₹ 800 (c) ₹ 900 (d) ₹ 1000
9. A sum becomes 6 fold at 5% per annum. At what rate, the sum becomes 12 fold?
(a) 10% (b) 12% (c) 9% (d) 11%
10. A man borrowed ₹ 40000 at 8% simple interest per year. At the end of second year he paid back certain amount and at the end of fifth year he paid back ₹ 35960 and cleared the debt. What is the amount did he pay back after the second year?
(a) ₹ 16200 (b) ₹ 17400
(c) ₹ 18600 (d) None of these
11. A sum of ₹ 1550 was lent partly at 5% and partly at 8% simple interest. The total interest received after 3 yr was ₹ 300. The ratio of money lent at 5% to 8% is
(a) 11 : 12 (b) 16 : 15 (c) 12 : 21 (d) 11 : 13
12. Rahim buys a house and pays ₹ 8000 cash and ₹ 9600 at 5 yr credit at 4% per annum simple interest. Then, the cash price of the house
(a) ₹ 10000 (b) ₹ 9600 (c) ₹ 17000 (d) ₹ 16000
13. At what rate per cent per annum simple interest, will a sum of money triple itself in 25 yr?
(a) 8% (b) 9% (c) 10% (d) 12%
14. A man invested ₹ 1000 on a simple interest at a certain rate and ₹ 1500 at 2% higher rate. The total interest in 3 yr is ₹ 390. What is the rate of interest for ₹ 1000?
(a) 4% (b) 5% (c) 6% (d) 8%
15. In what time the simple interest on a sum of money be $\frac{3}{8}$ of the principal with rate of interest $3\frac{1}{8}\%$?
(a) 9 yr (b) 6 yr (c) 12 yr (d) 15 yr
16. If the rate of simple interest is 12% per annum the amount that would fetch interest of ₹ 6000 per annum is
(a) ₹ 7200 (b) ₹ 72000 (c) ₹ 50000 (d) ₹ 48543.69
17. A lends a sum of money for 10 yr at 5% simple interest, B lends double that amount for 5 yr at the same rate of interest. Which of the following statement is true in this regard?
(a) A will get twice the amount of interest that B would get
(b) B will get twice the amount of interest that A would get.
(c) A and B will get the same amount as interest.
(d) B will get four times the amount of interest that A would get
18. Out of a sum of ₹ 625 a part was lent at 5% and the other at 10% simple interest. If the interest on the first part after 2 yr is equal to the interest on the second part after 4 yr, then the second sum (in ₹) is
(a) ₹ 125 (b) ₹ 200 (c) ₹ 250 (d) ₹ 300
19. A man invests an amount of ₹ 15860 in the names of his three sons A, B and C in such a way that they get the same interest after 2, 3 and 4 yr, respectively. If the rate of simple interest is 5%, then the ratio of the amounts invested among A, B and C will be
(a) 10 : 15 : 20 (b) $\frac{1}{10} : \frac{1}{15} : \frac{1}{20}$
(c) 110 : 115 : 120 (d) $\frac{1}{110} : \frac{1}{115} : \frac{1}{120}$
20. A sum was put at simple interest at a certain rate for 2 yr. Had it been put at 3% higher rate, it would have fetched ₹ 72 more. The sum is
(a) ₹ 1200 (b) ₹ 1600 (c) ₹ 1900 (d) ₹ 1400
21. Harsha makes a fixed deposit of ₹ 20000 in Bank of India for a period of 3 yr. If the rate of interest be 13% SI per annum charged half yearly, what amount will he get after 42 months?
(a) ₹ 27800 (b) ₹ 28100 (c) ₹ 29100 (d) ₹ 30000
22. A sum of money lent on simple interest triples itself in 15 yr and 6 months. In how many year still it be doubled?
(a) 6 yr and 3 months (b) 7 yr and 9 months
(c) 8 yr and 3 months (d) 9 yr and 6 months
23. Mr Pawan invests an amount of ₹ 24200 at the rate of 4% per annum for 6 yr to obtain a simple interest, later he invests the principal amount as well as the amount obtained as simple interest for another 4 yr at the same rate of interest. What amount of simple interest will be obtained at the end of the last 4 yr?
(a) ₹ 4800 (b) ₹ 4850.32 (c) ₹ 4801.28 (d) ₹ 4700
24. A person invested some amount at the rate of 12% simple interest and the remaining at 10%. He received yearly an interest of ₹ 130. Had he interchanged the amounts invested, he would have received an interest of ₹ 134. How much money did he invest at different rates?
(a) ₹ 500 at the rate of 10%, ₹ 800 at the rate of 12%
(b) ₹ 700 at the rate of 10%, ₹ 600 at the rate of 12%
(c) ₹ 800 at the rate of 10%, ₹ 400 at the rate of 12%
(d) ₹ 700 at the rate of 10%, ₹ 500 at the rate of 12%

Directions (Q. Nos. 25-27) If SI for a certain sum P_1 for time T_1 and rate of interest R_1 is I_1 and SI for another sum P_2 for time T_2 and rate of interest R_2 is I_2 , then difference of SI = $I_2 - I_1 = \frac{P_2 R_2 T_2 - P_1 R_1 T_1}{100}$.

- 25.** Simple interest for the sum of ₹ 1500 is ₹ 50 in 4 yr and ₹ 80 in 8 yr, the rate of SI is
(a) 0.6% (b) 5% (c) 0.05% (d) 0.5%
- 26.** Simple interest for the sum of ₹ 1230 for 2 yr is ₹ 10 more than the simple interest for ₹ 1130 for the same duration. Find the rate of interest.
(a) 5% (b) 6% (c) 8% (d) 2%
- 27.** If the annual payment of ₹ A will discharge a debt of ₹ 1092 due in 2 yr at 12% simple interest, then
I. A will be ₹ 515 (approx.).
II. A will be ₹ 530 (approx.), if interest rate is 6%.
(a) Only I (b) Both I and II
(c) Only II (d) Neither I nor II

PREVIOUS YEARS' QUESTIONS

- 28.** The principal on which a simple interest of ₹ 55 will be obtained after 9 months at the rate of $3\frac{2}{3}\%$ per annum is **2013 I**
(a) ₹ 1000 (b) ₹ 1500
(c) ₹ 2000 (d) ₹ 2500
- 29.** In how much time would the simple interest on a principal amount be 0.125 times the principal amount at 10% per annum? **2015 I**
(a) $1\frac{1}{4}$ yr (b) $1\frac{3}{4}$ yr (c) $2\frac{1}{4}$ yr (d) $2\frac{3}{4}$ yr
- 30.** If a sum of money at a certain rate of simple interest per year doubles in 5 yr and at a different rate of simple interest per year becomes three times in 12 yr, then the difference in the two rates of simple interest per year is **2016 I**
(a) 2% (b) 3% (c) $3\frac{1}{3}\%$ (d) $4\frac{1}{3}\%$

ANSWERS

1	a	2	a	3	a	4	c	5	c	6	b	7	c	8	b	9	d	10	b
11	b	12	d	13	a	14	a	15	c	16	c	17	c	18	a	19	b	20	a
21	c	22	b	23	c	24	d	25	d	26	a	27	b	28	c	29	a	30	c

HINTS AND SOLUTIONS

- 1.** (a) Given, $P = ₹ 400$, $R = 5\%$ and $T = 3$ yr
Simple interest = $\frac{P \times R \times T}{100}$
 $SI = \frac{400 \times 5 \times 3}{100} = ₹ 60$
 $\therefore \text{Amount} = P + SI = 400 + 60 = ₹ 460$
- 2.** (a) Let the sum of money be ₹ x .
 $\therefore \text{Amount} = \text{Sum} + SI$
 $\therefore \text{Amount} = x + \frac{x \times 5 \times 4}{100}$
 $\left[\because SI = \frac{P \times R \times T}{100} \right]$
But amount = ₹ 900
 $\therefore 900 = x + \frac{20x}{100}$
 $\Rightarrow 900 = \frac{6x}{5} \Rightarrow x = \frac{900 \times 5}{6} = ₹ 750$
Shortcut Method
Here, $A = ₹ 900$, $R = 5\%$ and $T = 4$ yr

- $\therefore \text{Sum} = \frac{100 \times A}{100 + RT} = \frac{100 \times 900}{100 + 5 \times 4} = ₹ 750$
- 3.** (a) Let the sum be ₹ x , so amount = ₹ $2x$
 $\therefore SI = ₹ x$
Let R be rate of interest.
 $\therefore R = \frac{100 \times SI}{P \times T} = \frac{100 \times x}{x \times 8} = 12.5\%$
Now, the needed amount = ₹ $4x$
 $\therefore SI = ₹ (4x - x) = ₹ 3x$
 $\therefore T = \frac{100 \times SI}{P \times R} = \frac{100 \times 3x}{x \times 12.5} = 24$ yr
Shortcut Method
Here, $n_1 = 2$, $n_2 = 4$ and $t_1 = 8$ yr
 $\therefore \text{Required time} = \frac{(n_2 - 1)}{(n_1 - 1)} \times t_1$
 $= \frac{(4 - 1)}{(2 - 1)} \times 8 = 3 \times 8 = 24$ yr [Rule 3]

- 4.** (c) Simple interest in 3 yr
 $= ₹ (1350 - 1260) = ₹ 90$
 $\therefore \text{Simple interest for 2 yr} = \frac{2}{3} \times 90 = ₹ 60$
 $\therefore \text{Principal} = ₹ (1260 - 60) = ₹ 1200$
 $\therefore \text{Rate, } R = \frac{100 \times SI}{P \times T} = \frac{100 \times 60}{1200 \times 2} = \frac{60}{24} = \frac{5}{2}$
 $R = 2.5\%$
Shortcut Method
Here, $A_1 = ₹ 1260$, $t_1 = 2$ yr,
 $A_2 = ₹ 1350$ and $t_2 = 5$ yr
 $\therefore \text{Required rate} = \left[\frac{(A_2 - A_1)}{A_1 t_2 - A_2 t_1} \right] \times 100\%$ [Rule 2]
 $= \left[\frac{(1350 - 1260)}{1260 \times 5 - 1350 \times 2} \right] \times 100\% = 2.5\%$

5. (c) Let the sum be ₹ x .

$$\text{Then, } \frac{x \times 13 \times 1}{100} - \frac{x \times 12 \times 1}{100} = 110$$

$$\Rightarrow \frac{x}{100} = 110$$

$$\Rightarrow x = 110 \times 100 = ₹ 11000$$

Shortcut Method

Here, $P = ₹ 110$, $r_1 = 13\%$, $r_2 = 12\%$
and $t_1 = t_2 = 1\text{yr}$

∴ Short trick

$$= \frac{110 \times 100}{13 - 12}$$

$$= ₹ 11000$$

6. (b) Let the sum (principal) be ₹ x .

$$\therefore \text{Simple interest} = ₹ \frac{x}{2}$$

and $T = 6\text{ yr}$, $R = 10\%$ per annum

$$\therefore \text{SI} = \frac{P \times R \times T}{100}$$

$$\Rightarrow \frac{x}{2} = \frac{x \times 10 \times 6}{100} \Rightarrow \frac{1}{2} = \frac{6}{10}$$

Which is not true, so it is not a possible case.

7. (c) Let the sum be ₹ 100.

Then, amount = (Sum + SI)

$$= \left(100 + \frac{100 \times 8 \times 5}{100} \right) = (100 + 8 \times 5)$$

So, when the amount is $(100 + 8 \times 5)$, then sum = 100

When the amount is ₹ 840, then sum

$$= ₹ \left(\frac{100 \times 840}{100 + 8 \times 5} \right)$$

8. (b) Simple interest for 4 yr
= ₹ $(1360 - 1040) = ₹ 320$

Simple interest for 3 yr

$$= ₹ \left(\frac{320}{4} \times 3 \right) = ₹ 240$$

∴ Sum or principle

$$= ₹ (1040 - 240) = ₹ 800$$

Shortcut Method

Here, $A_1 = ₹ 1040$, $t_1 = 3\text{ yr}$,

$A_2 = ₹ 1360$ and $t_2 = 7\text{ yr}$

∴ Required principle (sum)

$$= ₹ \left(\frac{A_2 t_1 - A_1 t_2}{t_1 - t_2} \right) \quad [\text{Rule 2}]$$

$$= \frac{1360 \times 3 - 1040 \times 7}{3 - 7} = ₹ 800$$

9. (d) SI at 5% = $6P - P = 5P$

$$\therefore 5P = \frac{P \times 5 \times T}{100} \Rightarrow T = 100\text{ yr}$$

Now, for new rate (R),

$$11P = \frac{P \times R \times 100}{100}$$

$$\therefore R = 11\%$$

Shortcut Method

As, $R_1 = 5\%$, $n_1 = 6$, $n_2 = 12$.

According to the formula,

$$R_2 = \frac{n_2 - 1}{n_1 - 1} \times R_1 = \frac{12 - 1}{6 - 1} \times 5 \quad [\text{Rule 3}]$$

$$= \frac{11}{5} \times 5 = 11\%$$

10. (b) Total borrowed money = ₹ 40000
and rate of interest = 8%

$$\text{The interest for 2 yr} = \frac{40000 \times 8 \times 2}{100}$$

$$= ₹ 6400$$

Let he paid ₹ x at the end of second year.

∴ Interest will be calculated on

$$₹ (40000 - x + 6400)$$

$$\text{Interest for 3 yr} = \frac{(46400 - x) \times 3 \times 8}{100}$$

$$= ₹ \frac{6}{25} (46400 - x)$$

$$\therefore \frac{6}{25} (46400 - x) + 46400 - x = 35960$$

$$\Rightarrow x = \frac{21576 \times 25}{31} = ₹ 17400$$

11. (b) Let sum lent at rate 5% be ₹ x .

Then, sum lent at rate 8%

$$= ₹ (1550 - x)$$

$$\therefore \text{Simple interest at rate 5\%} = \frac{x \times 5 \times 3}{100}$$

$$\text{Simple interest at rate 8\%}$$

$$= \frac{(1550 - x) \times 8 \times 3}{100}$$

$$\therefore \frac{x \times 15}{100} + \frac{(1550 - x) \times 24}{100} = 300$$

$$\Rightarrow \frac{15x}{100} + \frac{37200}{100} - \frac{24x}{100} = 300$$

$$\Rightarrow x = ₹ 800$$

Amount lent at rate 8%

$$= ₹ (1550 - 800) = ₹ 750$$

$$\therefore \text{Required ratio} = \frac{800}{750} = \frac{16}{15} = 16 : 15$$

12. (d) Let the amount remaining to pay be ₹ x .

∴ Price of house = ₹ $(x + 8000)$

$$\Rightarrow 9600 - \frac{x \times 4 \times 5}{100} = x$$

$$\Rightarrow 9600 - \frac{x}{5} = x$$

$$\Rightarrow 9600 = \frac{6x}{5} \Rightarrow \frac{9600 \times 5}{6} = x$$

$$\Rightarrow x = ₹ 8000$$

∴ Cash price of the house

$$= ₹ (8000 + 8000) = ₹ 16000$$

13. (a) Let the principal be ₹ P

As, amount = $3P$ and $T = 25\text{ yr}$

$$\therefore \text{SI} = 3P - P = 2P$$

$$\therefore \text{Rate} = \frac{100 \times \text{SI}}{\text{principal} \times T}$$

$$= \frac{100 \times 2P}{P \times 25} = 8\%$$

14. (a) Let a man invest ₹ 1000 at a $R\%$.

Now, rate is increased by 2%.

$$\therefore \text{New rate} = (R + 2)\%$$

By given condition,

$$\frac{1000 \times R \times 3}{100} + \frac{1500 \times (R + 2) \times 3}{100} = 390$$

$$\Rightarrow 30R + 45R + 90 = 390$$

$$\Rightarrow 75R = 300 \Rightarrow R = 4\%$$

15. (c) Here, rate of interest = $3\frac{1}{8}\% = \frac{25}{8}\%$

Let principal be ₹ x .

$$\text{and simple interest} = ₹ \frac{3}{8} x$$

$$\therefore \frac{3}{8} x = \frac{x \times \frac{25}{8} \times T}{100}$$

$$\Rightarrow \frac{300}{25} = T \Rightarrow T = 12\text{ yr}$$

16. (c) Given, rate of interest = 12% per annum

Simple interest = ₹ 6000 per annum

Let principal is ₹ P .

$$\therefore 6000 = \frac{P \times 1 \times 12}{100}$$

$$\therefore P = \frac{6000 \times 100}{12} = ₹ 50000$$

Hence, the required Principal amount is ₹ 50000.

17. (c) For A Let the amount be ₹ x .

Rate of interest = 5% and Time = 10 yr

$$\therefore \text{Simple interest} = \frac{x \times 5 \times 10}{100} = \frac{x}{2}$$

For B The amount be ₹ $2x$.

Rate of interest = 5% and Time = 5 yr

$$\text{SI} = \frac{2x \times 5 \times 5}{100} = \frac{x}{2}$$

So, A and B both will get the same amount as interest.

18. (a) Let the first part and the second part be ₹ x and ₹ $(625 - x)$, respectively.

∴ The simple interest on both part in same.

$$\text{So, } \frac{x \times 5 \times 2}{100} = \frac{(625 - x) \times 10 \times 4}{100}$$

$$\Rightarrow 10x = (625 - x)40$$

$$\Rightarrow x = (625 - x)4$$

$$\therefore x = ₹ 500$$

$$\therefore \text{Second part} = ₹ (625 - 500) = ₹ 125$$

19. (b) Let the amount of $A = ₹ a$,
time = 2 yr and rate = 5%

∴ Simple Interest of

$$A = \frac{a \times 2 \times 5}{100} = \frac{10a}{100}$$

Let the amount of $B = ₹ b$, rate = 5%
and time = 3 yr.

∴ Simple interest of

$$B = \frac{b \times 3 \times 5}{100} = \frac{15b}{100}$$

Let the amount of $C = ₹ c$, time = 4 yr
and rate = 5%

∴ Simple interest of $C = \frac{c \times 4 \times 5}{100} = \frac{20c}{100}$

$$\text{But } \frac{a \times 10}{100} = \frac{b \times 15}{100} = \frac{c \times 20}{100}$$

$$\Rightarrow 10a = 15b = 20c = k$$

$$\text{So, } a = \frac{k}{10}, b = \frac{k}{15}, c = \frac{k}{20}$$

$$\therefore a : b : c = \frac{1}{10} : \frac{1}{15} : \frac{1}{20}$$

20. (a) Let the sum be $₹ x$ and the original
rate $r\%$, then

$$\text{Simple interest} = \frac{x \times r \times 2}{100}$$

Now, rate is increased by 3%.

∴ New rate = $(r + 3)\%$

$$\therefore \text{Simple interest} = \frac{x \times (r + 3) \times 2}{100}$$

$$\therefore \frac{x \times (r + 3) \times 2}{100} - \frac{x \times r \times 2}{100} = 72$$

$$\Rightarrow \frac{(xr + 3x)2}{100} - \frac{2xr}{100} = 72$$

$$\Rightarrow \frac{2xr + 6x - 2xr}{100} = 72$$

$$\therefore x = ₹ 1200$$

21. (c) As rate of interest is charged half
yearly,

$$\text{So, rate} = \frac{13}{2}\% \text{ half yearly}$$

$$\text{time} \left(\frac{42}{12} \times 2 \right) \text{ half yearly} = 7 \text{ half yearly}$$

$$\text{SI} = \frac{20000 \times 13 \times 7}{100 \times 2} = ₹ 9100$$

$$\therefore \text{Amount (A)} = 20000 + 9100 = ₹ 29100$$

22. (b) Let initial amount be $₹ P$, then
 $A = ₹ 3P$ and

$$T = 15 \text{ yr and } 6 \text{ months} = \frac{31}{2} \text{ yr}$$

$$\therefore \text{SI} = A - P = 3P - P = ₹ 2P$$

$$\Rightarrow P \times \frac{31}{2} \times \frac{R}{100} = 2P$$

$$\Rightarrow R = \frac{2 \times 2 \times 100}{31}\% = \frac{400}{31}\%$$

Let amount doubled be t_1 yr.

Now, $\text{SI} = 2P - P = ₹ P$

$$\therefore t_1 = \frac{\text{SI} \times 100}{P \times R}$$

$$\Rightarrow t_1 = \frac{P \times 100 \times 31}{P \times 400} = \frac{31}{4} = 7 \text{ yr and } 9 \text{ months}$$

23. (c) Case I $\text{SI} = \frac{P \times R \times T}{100}$

$$R = \frac{24200 \times 4 \times 6}{100} = ₹ 5808$$

∴ Amount = Principal + SI

$$\text{SI} = 24200 + 5808 = 30008$$

$$\text{Case II } \text{SI} = \frac{30008 \times 4 \times 4}{100} = ₹ 4801.28$$

24. (d) Let the person invest amount x and y
into two different rates of interest.

$$\therefore \frac{x \times 12 \times 1}{100} + \frac{y \times 10 \times 1}{100} = 130$$

$$\left[\because \text{SI} = \frac{\text{PRT}}{100} \right]$$

$$\Rightarrow 12x + 10y = 13000 \quad \dots(i)$$

$$\text{and } \frac{y \times 12 \times 1}{100} + \frac{x \times 10 \times 1}{100} = 134$$

$$\Rightarrow 12y + 10x = 13400 \quad \dots(ii)$$

On solving Eqs. (i) and (ii), we get

$$x = ₹ 500 \text{ and } y = ₹ 700$$

25. (d) Here, $I_1 = 50$, $I_2 = 80$, $T_1 = 4$ yr,
 $T_2 = 8$ yr, and $P = ₹ 1500$

According to formula,

$$I_2 - I_1 = \frac{P \times R \times (T_2 - T_1)}{100}$$

$$\Rightarrow 80 - 50 = \frac{1500 \times R \times (8 - 4)}{100}$$

$$\Rightarrow 30 = 15 \times R \times 4$$

$$\Rightarrow 2R = 1$$

$$\Rightarrow R = \frac{1}{2} = 0.5\%$$

26. (a) Here, $I_1 - I_2 = ₹ 10$, $P_1 = ₹ 1230$,

$$P_2 = ₹ 1130 \text{ and } T = 2 \text{ yr}$$

According to formula,

$$I_2 - I_1 = \frac{T \times R \times (P_2 - P_1)}{100}$$

$$-10 = \frac{R \times 2(1130 - 1230)}{100}$$

$$-10 = \frac{-200R}{100}$$

$$\therefore R = 5\%$$

27. (b) Case I Annual payment

$$= \frac{100P}{100T + \frac{RT(T-1)}{2}}$$

$$= \frac{1092 \times 100}{100 \times 2 + \frac{24(2-1)}{2}}$$

$$= \frac{1092 \times 100}{212} = ₹ 515.09 \approx ₹ 515$$

Case II Annual payment

$$= \frac{1092 \times 100}{100 \times 2 + \frac{12(2-1)}{2}}$$

$$= \frac{1092 \times 100}{206} = ₹ 530.09 \approx ₹ 530$$

28. (c) Let P be the principal amount.

$$\text{SI} = ₹ 55, \text{ time, } t = 9 \text{ months} = \frac{9}{12} \text{ yr}$$

$$\text{and rate, } r = 3\frac{2}{3}\% = \frac{11}{3}\%$$

$$\therefore \text{SI} = \frac{P \times R \times T}{100}$$

$$\Rightarrow P = \frac{\text{SI} \times 100}{R \times T} = \frac{55 \times 100}{\frac{11}{3} \times 9} \times 3 \times 12$$

$$= 5 \times 100 \times 4 = ₹ 2000$$

$$\therefore \text{Principal (P)} = ₹ 2000$$

29. (a) Let the principal be $₹ x$ and the time
be t yr.

Rate = 10%

$$\therefore \text{Simple interest} = \frac{P \times R \times T}{100}$$

According to the question,

$$0.125 \times \text{Principal} = \text{Simple interest}$$

$$\therefore 0.125 P = \frac{P \times 10 \times T}{100}$$

$$\Rightarrow \frac{125P}{1000} = \frac{10 \times P \times T}{100}$$

$$\Rightarrow \frac{125}{100} = T$$

$$\Rightarrow T = \frac{5}{4} = 1\frac{1}{4} \text{ yr}$$

30. (c) Let principal = $₹ P$, then amount of
money = $₹ 2P$

$$\therefore \text{SI} = 2P - P = ₹ P$$

$$\text{Now, } P = \frac{P \times r \times 5}{100} \Rightarrow r = 20\%$$

Amount of money after 12 yr = $₹ 3P$

$$\therefore \text{SI} = 3P - P = 2P$$

$$\text{Now, } 2P = \frac{P \times R \times 12}{100} \Rightarrow R = \frac{50}{3}\%$$

∴ Difference between two interest rates

$$= \left(20 - \frac{50}{3} \right)\% = \frac{10}{3}\% = 3\frac{1}{3}\%$$

10

COMPOUND INTEREST

Usually (1-2) questions have been asked from this chapter. Questions from this chapter are based on the direct application of compound interest formula.



COMPOUND INTEREST

When the interest is calculated on the amount of previous year, then it is known as compound interest. It will make a deposit or loan grow at a faster rate than simple interest.

Important Rules and Formulae

Consider P = Principal (amount borrowed), A = Amount (Principal + Interest)
 R = Rate of interest per annum and n = Number of years

Rule 1 When interest is compounded annually, then

$$(i) \text{ Amount} = P \left(1 + \frac{R}{100}\right)^n \quad (ii) \text{ CI} = A - P = P \left[\left(1 + \frac{R}{100}\right)^n - 1\right]$$

EXAMPLE 1. The amount and the compound interest on ₹ 24000 compounded annually for 2 yr at the rate of 10% per annum is

- a. ₹ 39040 and ₹ 4040 b. ₹ 29040 and ₹ 5040 c. ₹ 19040 and ₹ 3040 d. None of these

Sol. b. Here, $P = ₹ 24000$, $R = 10\%$ per annum and $n = 2$ yr

$$\therefore A = P \left(1 + \frac{R}{100}\right)^n = 24000 \left(1 + \frac{10}{100}\right)^2 = 24000 \left(\frac{110}{100}\right)^2 = 24000 \times \frac{110}{100} \times \frac{110}{100} = ₹ 29040$$

$$\therefore \text{Compound interest} = \text{Amount} - \text{Principal} = ₹ 29040 - ₹ 24000 = ₹ 5040$$

So, amount = ₹ 29040 and compound interest = ₹ 5040

Rule 2 When interest is compounded half-yearly (every 6 months), then

$$(i) \text{ Amount} = P \left(1 + \frac{r}{100 \times 2}\right)^{2n} \quad (ii) \text{ CI} = \left[\left(1 + \frac{R}{100 \times 2}\right)^{2n} - 1\right]$$

EXAMPLE 2. The compound interest on ₹ 24000 compounded semi-annually for $1\frac{1}{2}$ yr at the rate of 10% per annum is

- a. ₹ 3783 b. ₹ 2783 c. ₹ 2763 d. ₹ 3763

Sol. a. Given, $P = ₹ 24000$

Rate of interest = 10% per annum

and time = $1\frac{1}{2}$ yr = $\frac{3}{2}$ yr

Since, the interest is compounded half-yearly.

Then, rate $(R) = \frac{10}{2}\%$ half-yearly

and time $(n) = 2 \times \frac{3}{2} = 3$ half-yearly

$$\begin{aligned} \therefore A &= 24000 \left(1 + \frac{5}{100}\right)^3 \quad \left[\because A = P \left(1 + \frac{R}{100}\right)^n\right] \\ &= 24000 \left(1 + \frac{1}{20}\right)^3 = 24000 \left(\frac{21}{20}\right)^3 \quad [\text{Rule 1}] \\ &= 24000 \times \frac{21}{20} \times \frac{21}{20} \times \frac{21}{20} = ₹ 27783 \end{aligned}$$

Now, compound interest = $A - P$

$$= ₹ 27783 - ₹ 24000 = ₹ 3783$$

(Rule 3) When interest is compounded quarterly (every 3 months), then

$$(i) \text{ Amount} = P \left(1 + \frac{R}{100 \times 4}\right)^{4n}$$

$$(ii) \text{ CI} = P \left[\left(1 + \frac{R}{100 \times 4}\right)^{4n} - 1 \right]$$

EXAMPLE 3. The amount and the compound interest on ₹ 100000 compounded quarterly for 9 months at the rate of 4% per annum is

- a. ₹ 106060 and ₹ 6060.10 b. ₹ 103030 and ₹ 3030.10
c. ₹ 103030.10 and ₹ 3030.10 d. ₹ 106060.10 and ₹ 6060.10

Sol. c. Given, $P = ₹ 100000$, rate of interest = 4% per annum and time = 9 months

Since, the interest is compounded quarterly.

Then, rate $(R) = \frac{4}{4} = 1\%$ quarterly

and time $(n) = 9 \text{ months} = \frac{9}{12} \times 4 = 3$ quarter yearly

$$\therefore \text{Amount, } A = P \left(1 + \frac{R}{100}\right)^n$$

$$\therefore A = 100000 \left(1 + \frac{1}{100}\right)^3 = 100000 \left(\frac{101}{100}\right)^3$$

$$= 100000 \times \frac{101}{100} \times \frac{101}{100} \times \frac{101}{100} = ₹ 103030.10$$

$$\therefore \text{Compound interest} = A - P = ₹ 103030.10 - ₹ 100000 = ₹ 3030.10$$

(Rule 4) When interest is compounded annually but time is given in fraction (say, $t\frac{a}{b}$ yr), then

$$\text{Amount} = P \left(1 + \frac{R}{100}\right)^t \times \left(\frac{1 + (a/b)R}{100}\right)$$

EXAMPLE 4. The amount and compound interest on ₹ 5000 compounded annually for 2 yr 6 months at the rate of 10% per annum is

- a. ₹ 6252.5 and ₹ 1342.5 b. ₹ 6000 and ₹ 1300
c. ₹ 6250 and ₹ 1340 d. ₹ 6352.5 and ₹ 1352.5

Sol. d. Here, time = 2 yr and 6 months

$$\therefore t = 2 \text{ yr and } \frac{a}{b} = \frac{1}{2} \text{ yr}$$

$$\therefore \text{Amount} = P \left(1 + \frac{R}{100}\right)^t \times \left(\frac{1 + \frac{a}{b}R}{100}\right)$$

$$\therefore A = 5000 \left(1 + \frac{10}{100}\right)^2 \left(1 + \frac{\frac{1}{2} \cdot 10}{100}\right)$$

$$= 5000 \left(\frac{110}{100}\right)^2 \left(\frac{105}{100}\right)$$

$$= 5000 \times \frac{11}{10} \times \frac{11}{10} \times \frac{21}{20} = ₹ 6352.50$$

$$\therefore \text{Compound interest} = A - P$$

$$\text{CI} = 6352.50 - 5000 = ₹ 1352.50$$

(Rule 5) When rate of interest for n_1 , n_2 and n_3 yr are R_1 , R_2 and R_3 respectively, then

$$\text{Amount} = P \left(1 + \frac{R_1}{100}\right)^{n_1} \left(1 + \frac{R_2}{100}\right)^{n_2} \left(1 + \frac{R_3}{100}\right)^{n_3}$$

EXAMPLE 5. The compound interest on ₹ 5000 for 4 yr if the rate of interest is 10% per annum for the first two years and 15% for the next two years is

- a. ₹ 3000 b. ₹ 3001 c. ₹ 3002 d. None of these

Sol. b. Here, $R_1 = 10\%$, $n = n_1 + n_2 = 4$ yr

$$n_1 = 2 \text{ yr, } R_2 = 15\%, n_2 = 2 \text{ yr}$$

and principal = ₹ 5000

$$\therefore \text{Amount} = P \left(1 + \frac{R_1}{100}\right)^{n_1} \left(1 + \frac{R_2}{100}\right)^{n_2}$$

$$\therefore A = 5000 \left(1 + \frac{10}{100}\right)^2 \left(1 + \frac{15}{100}\right)^2$$

$$= 5000 \times \left(\frac{110}{100}\right)^2 \left(\frac{115}{100}\right)^2$$

$$= 5000 \times \frac{11}{10} \times \frac{11}{10} \times \frac{23}{20} \times \frac{23}{20} = ₹ 8001.125$$

$$\therefore \text{Compound interest}$$

$$= A - P = ₹ 8001.125 - ₹ 5000 \approx ₹ 3001$$

Rule 6 The difference D between simple and compound interest accrued on ₹ P at the rate of interest of $R\%$ is given by

$$(i) \text{ For 2 yr, } D = \frac{PR^2}{(100)^2}$$

$$(ii) \text{ For 3 yr, } D = \frac{PR^2(300+r)}{(100)^3}$$

EXAMPLE 6. The difference between the compound interest and the simple interest on a certain sum at 12% per annum for 2 yr is ₹ 90. Then, sum is

- a. ₹ 6250 b. ₹ 6350 c. ₹ 6520 d. ₹ 6950

Sol. a. Let the sum be ₹ 100.

Case I Then, simple interest on ₹ 100 for 2 yr

$$= \left(\frac{100 \times 12 \times 2}{100} \right) = ₹ 24$$

Case II Amount when ₹ 100 is borrowed for 2 yr

$$= 100 \times \left(1 + \frac{12}{100} \right)^2 \left[A = P \left(1 + \frac{R}{100} \right)^n, \text{Rule 1 (i)} \right]$$

$$= 100 \times \frac{28}{25} \times \frac{28}{25} = ₹ \left(\frac{3136}{25} \right)$$

$$\therefore \text{Compound interest for 2yr} = \left(\frac{3136}{25} - 100 \right) = ₹ \left(\frac{636}{25} \right)$$

$$\text{Difference between CI and SI} = \left(\frac{636}{25} - 24 \right) = ₹ \frac{36}{25}$$

If the difference between CI and SI is ₹ $\frac{36}{25}$, then the sum = ₹ 100

If the difference between CI and SI is ₹ 1,

$$\text{then the sum} = ₹ \left(\frac{100 \times 25}{36} \right)$$

If the difference between CI and SI is ₹ 90, then the sum

$$= \left(100 \times \frac{25}{36} \times 90 \right) = ₹ 6250$$

Here, the sum is ₹ 6250.

Shortcut Method

Let the required sum be ₹ P .

Here, difference = ₹ 90 and $R = 12\%$

$\therefore$ Difference between CI and SI

$$90 = \frac{PR^2}{(100)^2} = \frac{(P)(12)^2}{(100)^2} \Rightarrow 90 = \frac{P \times 144}{100 \times 100}$$

$$\therefore P = ₹ 6250$$

Required sum = ₹ 6250

Rule 7 If a certain sum at compound interest becomes A_1 , in n yr and A_2 in $(n+1)$ yr, then

$$(i) \text{ Rate of compound interest} = \frac{(A_2 - A_1)}{A_1} \times 100\%$$

$$(ii) \text{ Sum} = A_1 \left(\frac{A_2}{A_1} \right)^n$$

EXAMPLE 7. A sum of money on compound interest amount to ₹ 9680 in 2 yr and to ₹ 10648 in 3 yr. What is the rate of interest per annum?

- a. 5% b. 10% c. 15% d. 20%

Sol. b. Given, $A_1 = ₹ 9680$ and $A_2 = ₹ 10648$

$$\therefore \text{Rate of compound interest} = \frac{(A_2 - A_1)}{A_1} \times 100\%$$

$$= \frac{10648 - 9680}{9680} \times 100\% = 10\%$$

DEPRECIATION

The value of certain things like the machine, vehicle etc., decreases over a period of time. The decrement in the value things is called depreciation. The depreciation per unit time is called the rate of depreciation.

Rule 8 If the present value of a article is P and it depreciate at the rate of $R\%$ per annum. Then,

$$(i) \text{ Value of article after } n \text{ yr} = P \times \left(1 - \frac{R}{100} \right)^n$$

$$(ii) \text{ Value of article } n \text{ yr ago} = \frac{P}{\left(1 - \frac{R}{100} \right)^n}$$

EXAMPLE 8. Given that carbon - 14 (C_{14}) decays at a constant rate in such a way that it reduces to 50% in 5568 yr. Then, the age of an old wooden piece in which the carbon is only 12.5% of the original is

- a. 16000 yr b. 16244 yr c. 16702 yr d. None of these

Sol. d. Let the rate of decay be $R\%$ and the age of the wooden piece be n yr.

$$\text{Then, } \left(\frac{50}{100} \right) = \left(1 - \frac{R}{100} \right)^{5568} \Rightarrow \left(1 - \frac{R}{100} \right) = (0.5)^{1/5568} \dots (i)$$

$$\Rightarrow \frac{12.5}{100} = \left(1 - \frac{R}{100} \right)^n \Rightarrow (0.125) = \left(1 - \frac{R}{100} \right)^n$$

$$\Rightarrow (0.125)^{1/n} = \left(1 - \frac{R}{100} \right) \Rightarrow (0.5)^{3/n} = \left(1 - \frac{R}{100} \right) \dots (ii)$$

$$\text{From Eqs. (i) and (ii), } (0.5)^{3/n} = (0.5)^{\frac{1}{5568}}$$

On comparing both sides, we get

$$\frac{3}{n} = \frac{1}{5568} \Rightarrow n = 3 \times 5568 = 16704 \text{ yr}$$

INSTALMENTS

When a borrower pays the sum in parts, then we say that he/she is paying in instalments.

$$\therefore P = \frac{x}{\left(1 + \frac{R}{100} \right)} + \frac{x}{\left(1 + \frac{R}{100} \right)^2} + \dots + \frac{x}{\left(1 + \frac{R}{100} \right)^n}$$

x = value of each instalment

Total amount paid in instalments, $A = P \left(1 + \frac{R}{100} \right)^n$

n = number of instalments

EXAMPLE 9. Sapna borrowed some money on compound interest and returned it in 3 yr in equal annual instalments. If rate of interest is 15% per annum and annual instalment is ₹ 486680, then the sum borrowed is.

- a. 111300 b. 122400
c. 1111200 d. 154320

Sol. c. Given that, rate of interest, $R = 15\%$ pa

Annual instalment, $x = ₹ 486680$

Total number of instalments, $n = 3$

$$\therefore P = \left[\frac{x}{\left(1 + \frac{R}{100}\right)} + \frac{x}{\left(1 + \frac{R}{100}\right)^2} + \frac{x}{\left(1 + \frac{R}{100}\right)^3} \right]$$

$$\begin{aligned} \Rightarrow P &= x \left[\frac{100}{(100 + R)} + \frac{(100)^2}{(100 + R)^2} + \frac{(100)^3}{(100 + R)^3} \right] \\ &= 486680 \left[\left(\frac{100}{100 + 15} \right) + \left(\frac{100}{100 + 15} \right)^2 + \left(\frac{100}{100 + 15} \right)^3 \right] \\ &= 486680 \times \left[\frac{20}{23} + \left(\frac{20}{23} \right)^2 + \left(\frac{20}{23} \right)^3 \right] \\ &= 486680 \times \frac{20}{23} \left(1 + \frac{20}{23} + \frac{400}{529} \right) \\ &= 1111200 \\ \therefore \text{Principal borrowed} &= ₹ 1111200 \end{aligned}$$



SOME IMPORTANT RESULTS

1. Simple interest and compound interest are equal for first year period on the same sum and at the same rate.
2. If A and B are the amounts of a certain sum for two consecutive years, then simple interest for 1 yr = $B - A$
3. If certain sum at compound interest becomes x times in n_1 yr and y times in n_2 yr, then $x^{1/n_1} = y^{1/n_2}$

> PRACTICE EXERCISE

- Kiran purchased a scooter for ₹ 24000. The value of scooter is depreciating at the rate of 5% per annum. Then, its value after 3 yr is
(a) ₹ 20577 (b) ₹ 20977 (c) ₹ 20677 (d) ₹ 20877
- If P be the principal amount and the rate of interest be $r\%$ per annum and the compound interest is calculated k times in a year, then what is the amount at the end of n yr?
(a) $P \left(1 + \frac{r}{100k}\right)^{nk}$ (b) $P \left(1 + \frac{kr}{100}\right)^{nk}$
(c) $P \left(1 + \frac{kr}{100}\right)^{n/k}$ (d) $P \left(1 + \frac{kr}{100k}\right)^{n/k}$
- The amount of a certain sum at compound interest for 2 yr at 5% is ₹ 4410. The sum is
(a) ₹ 4000 (b) ₹ 4200 (c) ₹ 3900 (d) ₹ 3800
- A person borrowed ₹ 7500 at 16% compound interest. How much does he have to pay at the end of 2 yr to clear the loan?
(a) ₹ 9900 (b) ₹ 10092 (c) ₹ 11000 (d) ₹ 11052
- If the rate of interest is 10% per annum and is compounded half-yearly, then the principal of ₹ 400 in $3/2$ yr will amount to
(a) ₹ 463.00 (b) ₹ 463.05
(c) ₹ 463.15 (d) ₹ 463.20
- At compound interest, if a certain sum of money doubles in n yr, then the amount will be four fold in
(a) $2n^2$ yr (b) n^2 yr (c) $2n$ yr (d) $4n$ yr
- The simple interest on a certain sum of money for 3 yr at 8% per annum is half the compound interest on ₹ 4000 for 2 yr at 10% per annum. What is the sum placed on simple interest?
(a) ₹ 1550 (b) ₹ 1650 (c) ₹ 1750 (d) ₹ 2000
- What is the least number of complete years in which a sum of money at 20% compound interest will be more than doubled?
(a) 7 (b) 6 (c) 5 (d) 4
- The difference between the simple interest and the compound interest (compounded annually) on ₹ 1250 for 2 yr at 8% per annum will be
(a) ₹ 18 (b) ₹ 2 (c) ₹ 8 (d) ₹ 4
- The compound interest on a sum for 2 yr is ₹ 832 and the simple interest on the same sum at the same rate for the same period is ₹ 800. What is the rate of interest?
(a) 6% (b) 8% (c) 10% (d) 12%
- A saving bank gives interest which compounds annually. Raju deposited ₹ 100 and received ₹ 121 at the end of second year. Rate of compound interest per annum is
(a) 10% (b) 15% (c) 11.5% (d) 20.5%
- The amount of a certain sum at compound interest for 4 yr at 10% is ₹ 4410. The sum is
(a) 3012.08 (b) 3015 (c) 3020.16 (d) 3016.9
- The compound interest on ₹ 5000 for 3 yr at 8% for first year, 10% for second year and 12% for third year will be
(a) ₹ 1560.40 (b) ₹ 1500 (c) ₹ 1565.60 (d) ₹ 1652.80
- An amount of ₹ x at compound interest at 20% per annum for 3 yr becomes y . What is $y : x$?
(a) 3 : 1 (b) 36 : 25
(c) 216 : 125 (d) 125 : 216
- The compound interest on ₹ 2000 for 1 yr at the rate of 8% per annum, when the interest is compounded semi-annually
(a) ₹ 163.20 (b) ₹ 2163.20 (c) ₹ 3153.20 (d) ₹ 1163
- ₹ 16000 invested at 10% per annum compounded semi-annually amounts to ₹ 18522. Then, the period of investment is
(a) $1\frac{1}{2}$ yr (b) 3 yr (c) 2 yr (d) $\frac{5}{2}$ yr
- A sum compounded annually becomes $\frac{25}{16}$ times of itself in 2 yr. Then, the rate of interest per annum is
(a) 25% (b) 20% (c) 15% (d) $7\frac{1}{2}\%$
- A sum of ₹ 3200 invested at 10% per annum compounded quarterly amounts to ₹ 3362, then the time period is
(a) $1\frac{1}{2}$ yr (b) $\frac{1}{2}$ yr (c) 2 yr (d) 1 yr
- A sum amount to ₹ 9680 in 2 yr and to ₹ 10648 in 3 yr compounded annually. Then, the sum and rate of interest, respectively are
(a) ₹ 8000, 10% (b) ₹ 8500, 10%
(c) ₹ 8500, 9% (d) ₹ 8000, 9%
- If the value of a machine depreciates by 10% of its value at the beginning of the year and its present value is estimated as ₹ 10935, then what was its value three years back?
(a) ₹ 15000 (b) ₹ 7000
(c) ₹ 8050 (d) None of these

HINTS AND SOLUTIONS

1. (a) Given, $P = ₹ 24000$, $R = 5\%$ per annum and $n = 3$ yr

$$\begin{aligned}\therefore A &= P \left(1 - \frac{R}{100}\right)^n \quad [\text{Rule 8 (i)}] \\ &= 24000 \left(1 - \frac{5}{100}\right)^3 \\ &= 24000 \times \left(\frac{95}{100}\right)^3 \\ &= ₹ 20577\end{aligned}$$

2. (a) Given, principal amount = ₹ P
Rate of interest, $R = \frac{r}{k}\%$ per annum and
Time, $T = nk$

$$\therefore A = P \left(1 + \frac{R}{100}\right)^T \quad [\text{Rule 1. (i)}]$$

$$\therefore A = P \left(1 + \frac{r}{100k}\right)^{nk}$$

3. (a) Let the principal be = ₹ x .
Given, $n = 2$ yr, $R = 5\%$, $A = ₹ 4410$

$$\therefore A = P \left(1 + \frac{R}{100}\right)^n \quad [\text{Rule 1. (i)}]$$

$$\therefore 4410 = x \left(1 + \frac{5}{100}\right)^2$$

$$\Rightarrow 4410 = x \left(\frac{21}{20}\right)^2$$

$$\Rightarrow x = \frac{4410 \times 400}{441} = ₹ 4000$$

4. (b) Given, $P = ₹ 7500$, $R = 16\%$ and $n = 2$ yr

$$\therefore A = P \left(1 + \frac{R}{100}\right)^n \quad [\text{Rule 1. (i)}]$$

$$= 7500 \left(1 + \frac{16}{100}\right)^2$$

$$= 7500 \left(\frac{116}{100}\right)^2$$

$$= 7500 \times \frac{29}{25} \times \frac{29}{25}$$

$$= 12 \times 841 = ₹ 10092$$

5. (b) Refer to example 2.

6. (c) Let the principal be ₹ x .

$\therefore$ Rate = ₹ R , Amount = ₹ $2x$
and Time = n yr

$$\therefore A = P \left(1 + \frac{R}{100}\right)^n \quad [\text{Rule 1. (i)}]$$

$$\therefore x \left(1 + \frac{R}{100}\right)^n = 2x$$

$$\Rightarrow \left(1 + \frac{R}{100}\right)^n = 2 \quad \dots(i)$$

Let it becomes four fold in N yr.

$$\text{Then, } x \left(1 + \frac{R}{100}\right)^N = 4x$$

$$\Rightarrow \left(1 + \frac{R}{100}\right)^N = 4$$

$$\Rightarrow 2^2 = \left(1 + \frac{R}{100}\right)^N$$

$$\Rightarrow \left(1 + \frac{R}{100}\right)^{2n} = \left(1 + \frac{R}{100}\right)^N \quad [\text{from Eq. (i)}]$$

$$\therefore N = 2n \text{ yr}$$

7. (c) Let the principal amount be ₹ P .

By given condition, $SI = \frac{1}{2} CI$

$$\Rightarrow \frac{P \times 8 \times 3}{100} = \frac{1}{2} \left[4000 \left(1 + \frac{10}{100}\right)^2 - 4000 \right]$$

[Rule 1 (ii)]

$$\Rightarrow \frac{24P}{100} = \frac{1}{2} \left[4000 \times \frac{121}{100} - 4000 \right]$$

$$= \frac{1}{2} [4840 - 4000]$$

$$\Rightarrow \frac{24P}{100} = 420$$

$$\therefore P = \frac{420 \times 100}{24} = ₹ 1750$$

8. (d) Let the sum of money be ₹ P

$\therefore$ Amount = ₹ $2P$

$$\Rightarrow A = P \left(1 + \frac{R}{100}\right)^n \quad [\text{Rule 1. (i)}]$$

$$\Rightarrow 2P = P \left(1 + \frac{20}{100}\right)^n$$

$$\Rightarrow \frac{2P}{P} = \left(\frac{6}{5}\right)^n$$

$$\Rightarrow 2 = \left(\frac{6}{5}\right)^n$$

On putting $n = 4$, we get

$$\left(\frac{6}{5}\right)^4 = \frac{1296}{625} = 2 \text{ (approx)}$$

$\therefore$ Least number of years = 4

9. (c) Given, $P = ₹ 1250$, $R = 8\%$ per annum and $n = 2$ yr

$$\therefore \text{Simple interest} = \frac{P \times R \times T}{100}$$

$$\therefore \text{Simple interest} = \frac{1250 \times 8 \times 2}{100}$$

$$= ₹ 200$$

$\therefore$ Compound interest

$$= P \left[\left(1 + \frac{R}{100}\right)^n - 1 \right] \quad [\text{Rule 1. (ii)}]$$

$$\therefore CI = 1250 \left(1 + \frac{8}{100}\right)^2 - 1250$$

$$= 1250 \times \left(\frac{108}{100}\right)^2 - 1250$$

$$= 1458 - 1250 = ₹ 208$$

$\therefore$ Difference in SI and CI

$$= 208 - 200 = ₹ 8$$

Shortcut Method

Here, $p = ₹ 1250$ and $R = 8\%$

$$\therefore \text{Required difference} = \frac{PR^2}{(100)^2} \quad [\text{Rule 6. (i)}]$$

$$= \frac{1250 \times 64}{100 \times 100} = ₹ 8$$

10. (b) Given, $CI = ₹ 832$, $SI = ₹ 800$,
 $n = 2$ yr

$$CI = P \left\{ \left(1 + \frac{R}{100}\right)^n - 1 \right\}$$

[Rule 1. (ii)]

$$\therefore 832 = P \left\{ \left(1 + \frac{R}{100}\right)^2 - 1 \right\} \quad \dots(i)$$

$$\text{Also, } SI = \frac{P \times R \times T}{100}$$

$$\Rightarrow 800 = \frac{P \times R \times 2}{100}$$

$$\Rightarrow P = \frac{40000}{R}$$

From Eq. (i),

$$832 = \frac{40000}{R} \left(\frac{R^2}{10000} + \frac{2R}{100} \right)$$

$$\Rightarrow 832 = \frac{40000}{100} \left(\frac{R}{100} + 2 \right)$$

$$\Rightarrow 832 = 4R + 800$$

$$\therefore R = \frac{32}{4} = 8\%$$

- 11. (a)** Given, principal (P) = ₹ 100
Amount (A) received after 2 yr = ₹ 121
Let rate of interest = $R\%$ per annum

$$\therefore A = P \left(1 + \frac{R}{100} \right)^n \quad [\text{Rule 1. (i)}]$$

$$\therefore 121 = 100 \left(1 + \frac{R}{100} \right)^2$$

$$\Rightarrow \frac{121}{100} = \left(\frac{100 + R}{100} \right)^2$$

$$\Rightarrow \left(\frac{11}{10} \right)^2 = \left(\frac{100 + R}{100} \right)^2$$

$$\Rightarrow \frac{11}{10} = \frac{100 + R}{100}$$

$$\Rightarrow 100 + R = \frac{11 \times 100}{10}$$

$$\Rightarrow 100 + R = 110$$

$$\therefore R = 110 - 100 = 10$$

$$\therefore \text{Rate of interest} = 10\%$$

- 12. (a)** Refer to example 3.

- 13. (d)** Given, $P = ₹ 5000$,

$$R_1 = 8\%, R_2 = 10\%, R_3 = 12\%$$

$$\text{and } n_1 = n_2 = n_3 = 1 \text{ yr}$$

$$\therefore \text{Amount}$$

$$= P \left(1 + \frac{R_1}{100} \right)^{n_1} \left(1 + \frac{R_2}{100} \right)^{n_2} \left(1 + \frac{R_3}{100} \right)^{n_3}$$

$$[\text{Rule 5}]$$

$$= \left[5000 \times \left(1 + \frac{8}{100} \right) \left(1 + \frac{10}{100} \right) \left(1 + \frac{12}{100} \right) \right]$$

$$= \left(5000 \times \frac{27}{25} \times \frac{11}{10} \times \frac{28}{25} \right) = ₹ 6652.80$$

$$\therefore \text{Compound interest} = 6652.80 - 5000$$

$$= ₹ 1652.80$$

- 14. (c)** Given, $P = ₹ x$, $R = 20\%$ per annum,
 $n = 3$ yr

$$\text{and } A = ₹ y$$

$$\therefore A = P \left(1 + \frac{R}{100} \right)^n \quad [\text{Rule 1. (i)}]$$

$$\therefore y = x \left(1 + \frac{20}{100} \right)^3$$

$$\Rightarrow y = x \left(\frac{6}{5} \right)^3 \Rightarrow \frac{y}{x} = \frac{216}{125}$$

$$\Rightarrow y : x = 216 : 125$$

- 15. (a)** Given, principal (P) = ₹ 2000

$$\text{Time} = 1 \text{ yr}$$

$$\text{Rate of interest} = 8\% \text{ per annum}$$

$$\text{Since, the interest is compounded semi-annually.}$$

$$\text{Then, rate } (R) = \frac{8}{2}\% = 4 \text{ half-yearly}$$

$$\text{and time } (n) = 1 \times 2 = 2 \text{ half-yearly}$$

$$\therefore \text{CI} = P \left[\left(1 + \frac{R}{100} \right)^n - 1 \right]$$

$$[\text{Rule 1. (ii)}]$$

$$\text{CI} = 2000 \left(1 + \frac{4}{100} \right)^2 - 2000$$

$$= 2000 \times \left(\frac{26}{25} \right)^2 - 2000$$

$$= 2163.20 - 2000 = ₹ 163.20$$

- 16. (a)** Given, principal (P) = ₹ 16000

$$\text{Amount received at the end of period } (A) = ₹ 18522$$

$$\text{Let time} = t \text{ yr}$$

$$\therefore n = 2t$$

$$\text{Rate} = 10\% \text{ per annum i.e } 5\% \text{ half-yearly}$$

$$\therefore A = P \left(1 + \frac{R}{100} \right)^n \quad [\text{Rule 1. (i)}]$$

$$\therefore 18522 = 16000 \left(1 + \frac{5}{100} \right)^n$$

$$\Rightarrow \left(\frac{18522}{16000} \right) = \left(\frac{21}{20} \right)^n$$

$$\Rightarrow \left(\frac{9261}{8000} \right) = \left(\frac{21}{20} \right)^n \Rightarrow \left(\frac{21}{20} \right)^3 = \left(\frac{21}{20} \right)^n$$

$$\Rightarrow n = 3$$

$$\therefore n = 2t \Rightarrow 2t = 3$$

$$\therefore t = \frac{3}{2} \text{ yr} = 1 \frac{1}{2} \text{ yr}$$

- 17. (a)** Let sum/ principal be ₹ x and the rate be $R\%$ per annum.

$$\therefore \text{Amount, } A = \frac{25}{16} x \text{ and } n = 2 \text{ yr}$$

$$\therefore A = P \left(1 + \frac{R}{100} \right)^n \quad [\text{Rule 1. (i)}]$$

$$\therefore \frac{25}{16} x = x \left(1 + \frac{R}{100} \right)^2$$

$$\Rightarrow \frac{25}{16} = \left(1 + \frac{R}{100} \right)^2$$

$$\Rightarrow \left(\frac{5}{4} \right)^2 = \left(1 + \frac{R}{100} \right)^2$$

$$\Rightarrow \frac{5}{4} = \left(1 + \frac{R}{100} \right) \Rightarrow \frac{R}{100} = \frac{1}{4}$$

$$\therefore R = 25\%$$

- 18. (b)** Here, $P = ₹ 3200$,

$$A = ₹ 3362$$

$$[\text{since, amount is payable quarterly}]$$

$$\therefore R = 10\% \text{ per annum} = \frac{10}{4}\% \text{ quarterly}$$

$$\text{and let time} = t$$

$$\Rightarrow n = 4t$$

$$\therefore A = P \left(1 + \frac{R}{100} \right)^n \quad [\text{Rule 1. (i)}]$$

$$\Rightarrow 3362 = 3200 \left(1 + \frac{10}{4 \times 100} \right)^n$$

$$\Rightarrow \frac{3362}{3200} = \left(\frac{410}{400} \right)^n \Rightarrow \left(\frac{41}{40} \right)^2 = \left(\frac{41}{40} \right)^n$$

$$\Rightarrow n = 2 \Rightarrow 4t = 2 \text{ yr} \quad [\text{as } n = 4t]$$

$$\therefore t = \frac{1}{2} \text{ yr}$$

- 19. (a)** Given, $A_1 = ₹ 9680$, $A_2 = ₹ 10648$
and $n = 2$ yr

$$\therefore \text{Rate of compound interest}$$

$$= \frac{A_2 - A_1}{A_1} \times 100\% \quad [\text{Rule 7 (i)}]$$

$$= \frac{10648 - 9680}{9680} \times 100\% = 10\%$$

$$\therefore \text{Sum} = A_1 \left(\frac{A_2}{A_1} \right)^n \quad [\text{Rule 7 (ii)}]$$

$$= 9680 \left(\frac{10648}{9680} \right)^2$$

$$= 9680 \left(\frac{110}{121} \right)^2 = 9680 \left(\frac{10}{11} \right)^2$$

$$= \frac{9680 \times 100}{121} = ₹ 8000$$

- 20. (a)** Let the value of machine 3 yr ago be ₹ x .

$$\text{and given, } P = ₹ 10935, R = 10\% \text{ and } n = 3 \text{ yr}$$

$$\therefore x = P \left(1 - \frac{R}{100} \right)^n \quad [\text{Rule 8 (i)}]$$

$$\therefore x \left(1 - \frac{10}{100} \right)^3 = 10935$$

$$\Rightarrow x \left(\frac{90}{100} \right)^3 = 10935$$

$$\therefore x = \frac{10935 \times 10 \times 10 \times 10}{9 \times 9 \times 9} = ₹ 15000$$

- 21. (d)** Let the sum be ₹ x and rate be $R\%$ per annum.

$$\therefore \text{Amount, } A = ₹ 2x$$

$$\text{and time, } n = 15 \text{ yr}$$

$$\therefore A = P \left(1 + \frac{R}{100} \right)^n \quad [\text{Rule 1. (i)}]$$

$$\text{Then, } x \left(1 + \frac{R}{100} \right)^{15} = 2x$$

$$\Rightarrow \left(1 + \frac{R}{100} \right)^{15} = 2 \quad \dots(i)$$

$$\text{Suppose the sum becomes eight times in 'n' yr, then}$$

$$x \left(1 + \frac{R}{100} \right)^n = 8x \quad [\text{by given condition}]$$

$$\Rightarrow \left(1 + \frac{R}{100} \right)^n = 8 = 2^3 \quad \dots(ii)$$

$$\begin{aligned} \text{[but } 2^3 &= \left[\left(1 + \frac{R}{100} \right)^{15} \right]^3 \text{ from Eq. (i)]} \\ \Rightarrow \left(1 + \frac{R}{100} \right)^n &= 2^3 = \left(1 + \frac{R}{100} \right)^{45} \\ &\text{[on comparing]} \\ \therefore n &= 45 \text{ yr} \end{aligned}$$

Shortcut Method

Since, a sum of money at compound interest doubles itself in 15 yr.

Hence, it will become eight times (2^3 times) in $= 15 \times 3 = 45$ yr

22. (c) Amount due at the end of 1 yr
 $= 4000 \left(1 + \frac{8}{100} \right) = ₹ 4320$

$\therefore$ Amount due after 1 yr
 $= (4320 - 1500) = ₹ 2820$

Amount due at the end of second year
 $= 2820 \left(1 + \frac{8}{100} \right) = ₹ 3045.60$

$\therefore$ Amount due after second year
 $= (3045.60 - 1500) = ₹ 1545.60$

Amount due at the end of third year
 $= 1545.60 \left(1 + \frac{8}{100} \right) = ₹ 1669.25$

Amount due after the payment of third instalment
 $= (1669.25 - 1500) = ₹ 169.25$

23. (a) Let the sum be ₹ x , then
 $x \left(1 + \frac{R}{100} \right)^5 = 2x \Rightarrow \left(1 + \frac{R}{100} \right)^5 = 2$... (i)

The amount after 20 yr,
 $x \left(1 + \frac{R}{100} \right)^{20} = x \left[\left(1 + \frac{R}{100} \right)^5 \right]^4$
 $= 2^4 x = 16x$ [from Eq. (i)]
 $= 16 \times 10000 = ₹ 160000$
 [put $x = ₹ 10000$]

24. (b) Data is insufficient as the rate of interest is not given to calculate further.

25. (d) Let amount be ₹ x and rate of interest is R % annually.

According to the questions,

Amount after 1st yr = ₹ 1200

$$x \left(1 + \frac{R}{100} \right) = 1200 \quad \dots (i)$$

Amount after 3rd yr = ₹ 1587

$$x \left(1 + \frac{R}{100} \right)^3 = ₹ 1587 \dots (ii)$$

On dividing Eq. (ii) by Eq. (i), we get

$$\begin{aligned} \left(1 + \frac{R}{100} \right)^2 &= \frac{1587}{1200} = \frac{529}{400} \\ 1 + \frac{R}{100} &= \frac{23}{20} \Rightarrow \frac{R}{100} = \frac{3}{20} \end{aligned}$$

$\therefore R = 15\%$

Put $R = 15\%$ in Eq. (i),

$$x \left(1 + \frac{15}{100} \right) = ₹ 1200 \Rightarrow x = \frac{1200 \times 100}{115}$$

$\therefore x = ₹ 1043.478$

26. (d) I. Given, $R = 4\%$, $n = 2$ yr and

$A = ₹ 169$, $P = ?$

$$A = P \left(1 + \frac{R}{100} \right)^n \quad \text{[Rule 1. (i)]}$$

$$169 = P \left(1 + \frac{4}{100} \right)^2$$

$$\Rightarrow 169 = P \left(\frac{26}{25} \right)^2$$

$$P = \frac{169 \times 25 \times 25}{26 \times 26} = ₹ 156.25$$

II. Given, $SI = ₹ 120$, $n = 2$ yr

and $CI = ₹ 129$.

$$SI = \frac{P \times R \times T}{100}$$

$$120 = \frac{P \times R \times 2}{100} \Rightarrow PR = ₹ 6000$$

$\therefore P = \frac{6000}{R} \quad \dots (i)$

$$CI = P \left[\left(1 + \frac{R}{100} \right)^n - 1 \right] \quad \text{[Rule 1. (ii)]}$$

$$129 = P \left[\left(1 + \frac{R}{100} \right)^2 - 1 \right]$$

$$1290000 = P [(100 + R)^2 - 100^2]$$

$$= P [R^2 + R \times 200]$$

$$= \frac{6000}{R} [R^2 + R \times 200]$$

[from Eq. (i)]

$$1290000 = 6000R + 1200000$$

$$R = \frac{90000}{6000} = 15\%$$

Hence, both statements are correct.

27. (d) $P = ₹ 1600$, $r = 25\%$ and $n = 2$ yr

$$\therefore A = P \left[1 + \frac{R}{100} \right]^n \quad \text{[Rule 1. (i)]}$$

$$= 1600 \left[1 + \frac{25}{100} \right]^2$$

$$= 1600 \times \frac{5}{4} \times \frac{5}{4} = ₹ 2500$$

$\therefore$ Compound interest

$$= ₹ 2500 - ₹ 1600 = ₹ 900$$

28. (b) Refer to example 6.

29. (a) Given, principal = ₹ 15000

Rate, $R = ?$

Difference, $D = ₹ 96$

For 2 yr Difference, $D = \frac{PR^2}{(100)^2}$ [Rule 6 (i)]

$$\Rightarrow 96 = \frac{15000 \times R^2}{100 \times 100}$$

$$\Rightarrow R^2 = 64$$

$$\therefore R = 8\%$$

30. (c) $\therefore$ Let the principal be ₹ x .

Then, $SI = ₹ \frac{60x}{100}$

$$\therefore SI = \frac{\text{Principal} \times \text{Rate} \times \text{Time}}{100}$$

$$\Rightarrow \frac{60x}{100} = \frac{x \times \text{Rate} \times 6}{100} \quad [\because \text{Time} = 6 \text{ yr}]$$

$$60 = \text{Rate} \times 6$$

$$\therefore \text{Rate} = 10\%$$

Again principal = ₹ 12000, Time = 3 yr

$$\text{Amount} = \text{Principal} \left(1 + \frac{\text{Rate}}{100} \right)^{\text{Time}}$$

$$= 12000 \left(1 + \frac{10}{100} \right)^3$$

$$= 12000 \left(\frac{11}{10} \right)^3$$

$$= 12000 \times \frac{11 \times 11 \times 11}{1000}$$

$$= 12 \times 121 \times 11 = ₹ 15972$$

$$\therefore CI = \text{Amount} - \text{Principal}$$

$$= 15972 - 12000 = ₹ 3972$$

31. (b) Refer to example 2.

11

PROFIT AND LOSS

Generally (2-4) questions have been asked from this chapter. Generally questions are based on basic formulae and tricks explained in this chapter which are helpful to solve the problems more easily and quickly.



Cost Price (CP)

The price, at which an article is bought is called its cost price. All the overhead expenses in the transaction like freight, damage etc., are added to the cost price.

Selling Price (SP)

The price at which an article sold is called its selling price.

Profit ($SP > CP$) When an article is sold at a price more than its cost price, then profit is earned.

- Profit = Selling price – Cost price
- Loss = Cost price – Selling price

Loss ($CP > SP$) When an article is sold at a price lower than its cost price, then loss is incurred.

- Profit % = $\frac{\text{Profit}}{CP} \times 100\% = \frac{(SP - CP)}{CP} \times 100\%$
- Loss % = $\frac{\text{Loss}}{CP} \times 100\% = \frac{(CP - SP)}{CP} \times 100\%$

Note Profit and loss percentage is always calculated as the percentage of CP unless otherwise specified.

List Price/Marked Price

The price of an article which is displayed on the price tag is known as marked price. It is the normal price of the thing/products without a discount. Sometimes the shopkeeper increases or decreases the cost price, then this price is taken as the list price of the article.

Important Rules and Formulae

Rule 1 If there is a profit of $r\%$, then

$$(i) SP = \frac{(100 + r) \times CP}{100} \quad (ii) CP = \frac{100 \times SP}{(100 + r)}$$

EXAMPLE 1. If cost price of a fan is ₹ 720. If there is a profit of $16\frac{2}{3}\%$. Then, selling price is

- a. ₹ 840 b. ₹ 940
c. ₹ 1050 d. None of these

Sol. a. Given, $r = 16\frac{2}{3}\%$ and cost price = ₹ 720

$$\begin{aligned} \therefore \text{Selling price (SP)} &= \frac{CP \times (100 + r)}{100} = \frac{720 \times \left(100 + 16\frac{2}{3}\right)}{100} \\ &= \frac{720 \times \left(100 + \frac{50}{3}\right)}{100} = \frac{720 \times 350}{300} = ₹ 840 \end{aligned}$$

Rule 2 If there is a loss of $r\%$, then

$$(i) SP = \frac{(100 - r)}{100} \times CP \quad (ii) CP = \frac{100 \times SP}{(100 - r)}$$

EXAMPLE 2. When Selling price is ₹ 75 and there is a loss of 12%. Then, cost price is

- a. ₹ 22 b. ₹ 44
c. ₹ 55 d. ₹ 85.22

Sol. d. Given, $r = 12\%$ and selling price = ₹ 75

$$\therefore \text{Cost price} = \frac{100 \times SP}{(100 - r)} = \frac{100 \times 75}{(100 - 12)} = \frac{100 \times 75}{88} = 85.22$$

Rule 3 (i) When there are two successive profits or losses of $x\%$ and $y\%$, then the resultant profit or loss percent is given by

$$\left(\pm x \pm y + \frac{(\pm x)(\pm y)}{100} \right)\%$$

- Note** • Take positive for profit and negative for loss.
• If the sign of result is positive, then there is a total gain.
• If the sign of result is negative, then there is a total loss.

EXAMPLE 3. By selling a watch for ₹ 132 a tradesman got two successive profits of 10% and 20%, then the resultant profit percentage is

- a. 22% b. 30%
c. 32% d. 34%

Sol. c. Given, the successive profits are 10%, 20%.

$$\begin{aligned} \text{So, resultant profit percentage} &= \left(10 + 20 + \frac{20 \times 10}{100} \right)\% \\ &= 32\% \end{aligned}$$

Rule 4 When two different articles are sold at the same selling price, getting gain/loss of $x\%$ on the first and gain/loss of $y\%$ on the second, then the overall % gain or % loss in the transaction is given by

$\left[\frac{100(x + y) + 2xy}{(100 + x) + (100 + y)} \right]\%$. The above expression represent overall gain or loss accordingly as its sign is positive or negative.

When two different articles are sold at the same selling price getting a gain of $x\%$ on the first and loss of $x\%$ on the second, then the overall % loss in the transaction is given by $\left(\frac{x}{10} \right)^2\%$.

Note That in such questions there is always a loss.

EXAMPLE 4. A trader sells two cycles at ₹ 1188 each and gains 10% on the first and loses 12% on the second. What is the profit or loss percent on the whole?

- a. 1% loss b. 1% gain c. No loss no gain d. 3.2% loss

Sol. d. When there is a profit of $x\%$ and loss of $y\%$, then the resultant

$$\text{Profit/Loss percentage} = \left(x - y - \frac{xy}{100} \right)\%$$

Here, $x = 10\%$ and $y = 12$

$$\therefore \text{Profit/Loss percentage} = \left(10 - 12 - \frac{10 \times 12}{100} \right)\% = -3.2\%$$

EXAMPLE 5. Harish sold two scooters, each for ₹ 24000. If he makes 20% profit on the first and 15% loss on the second. What is his gain or loss percent in the transaction?

- a. loss $\frac{20}{41}\%$ b. gain $\frac{20}{41}\%$ c. loss $\frac{32}{73}\%$ d. gain $\frac{32}{73}\%$

Sol. a. Here, $x = 20$ and $y = -15$

$$\begin{aligned} \therefore \text{Overall gain/loss \%} &= \left[\frac{100(x + y) + 2xy}{(100 + x) + (100 + y)} \right]\% \\ &= \left[\frac{100(20 - 15) + 2 \times 20 \times -15}{(100 + 20) + (100 - 15)} \right]\% \\ &= \frac{-100}{205}\% = \frac{-20}{41}\% \end{aligned}$$

Which represents loss.

EXAMPLE 6. Vikram sold two horses for ₹ 990 each gaining 10% on the one and losing 10% on the other. Find his gain or loss percent.

- a. 1% gain b. 1% loss c. 4% gain d. 4% loss

Sol. b. Here, $x = 10$

$$\therefore \text{Overall loss \%} = \left(\frac{x}{10}\right)^2 \% \\ = \left(\frac{10}{10}\right)^2 \% = 1\%$$

(Rule 5) A person buys two items for ₹ S . One is sold at a loss of $r\%$ and the other at a gain of $R\%$, if each item was sold at the same price, then

$$\text{Cost price of item sold at loss} = \frac{S \times (100 + R)}{(100 - r) + (100 + R)}$$

$$\text{Cost price of item sold at gain} = \frac{S \times (100 - r)}{(100 - r) + (100 + R)}$$

EXAMPLE 7. Sudhir bought two boxes for ₹ 1300. He sold one box at a profit of 20% and the other at a loss of 12%. If the selling price of both boxes is the same, then the cost price of each box is

- a. ₹ 700 and ₹ 500 b. ₹ 750 and ₹ 550
c. ₹ 800 and ₹ 600 d. None of these

Sol. b. Given, cost of two boxes $S = ₹ 1300$, profit on first box $R = 20\%$

and loss on second box $r = 12\%$

$$\text{Cost price of box sold at loss} \\ = \frac{1300 \times (100 + 20)}{(100 - 12) + (100 + 20)} = \frac{1300 \times 120}{88 + 120} = ₹ 750$$

$$\text{Cost price of box sold at gain} \\ = \frac{1300 \times (100 - 12)}{(100 - 12) + (100 + 20)} = \frac{1300 \times 88}{88 + 120} = ₹ 550$$

(Rule 6) If a trader professes to sell his goods at cost price, but uses false weights, then gain %

$$= \frac{\text{true weight} - \text{false weight}}{\text{false weight}} \times 100$$

EXAMPLE 8. A dishonest dealer sells his goods at cost price, but he uses a weight of 960 gm for the kg weight, then the percentage of gain is

- a. $2\frac{1}{3}\%$ b. $4\frac{1}{6}\%$ c. $6\frac{1}{9}\%$ d. $3\frac{1}{2}\%$

Sol. b. Gain % = $\frac{\text{true weight} - \text{false weight}}{\text{false weight}} \times 100$

$$= \frac{1000 - 960}{960} \times 100 = \frac{40}{960} \times 100 = 4\frac{1}{6}\%$$

(Rule 7) If cost price of ' n ' articles is equal to the selling price of ' m ' articles, then

$$\text{Profit percentage} = \left(\frac{n - m}{m}\right) \times 100\% \quad (n > m)$$

$$\text{and loss percentage} = \left(\frac{m - n}{m}\right) \times 100\% \quad (m > n)$$

EXAMPLE 9. If the cost price of 18 chairs be equal to selling price of 16 chairs. The gain percent is

- a. 12% b. 12.5%
c. 14% d. 15.5%

Sol. b. Given, $n = 18$, $m = 16$ [$\because n > m$]

$$\therefore \text{Profit percentage} = \left(\frac{18 - 16}{16}\right) \times 100 \quad \left[\because \frac{n - m}{m} \times 100\right] \\ = \frac{2}{16} \times 100 = \frac{100}{8} = 12.5\%$$

DISCOUNT

The reduction allowed on the marked price of an article is called as discount. Discount is always reckoned on the marked price.

Selling price = Marked price – Discount

(Rule 8) If discount allowed is $r\%$, then

$$\text{Selling price} = \frac{(100 - r)}{100} \times \text{Marked price}$$

EXAMPLE 10. If the marked price of a fan is ₹ 700 and a discount of 10% is given on it, then what is the selling price of the fan?

- a. ₹ 500 b. ₹ 575
c. ₹ 610 d. ₹ 630

Sol. d. Given, $r = 10\%$

$$\text{Selling price} = \frac{(100 - 10)}{100} \times 700 \\ = \frac{90 \times 700}{100} = ₹ 630$$

(Rule 9) If a shopkeeper marks his items at $x\%$ above the cost price and allows customers a discount of $y\%$ then there is $\left(x - y - \frac{xy}{100}\right)\%$ profit or loss according to positive or negative sign, respectively.

EXAMPLE 11. A dealer marked his goods 20% above the cost price and allows a discount of 10%. Then, his gain percent is

- a. 2% b. 4%
c. 6% d. 8%

Sol. d. Given, $x = 20\%$ and $y = 10\%$

$$\text{Profit/Loss percentage} = \left(x - y - \frac{xy}{100}\right)\% \\ = \left(20 - 10 - \frac{20 \times 10}{100}\right)\% = 8\%$$

Hence, positive sign shows profit i.e. profit of 8%.

Successive Discounts (Discount Series)

Rule 10 Single discount equivalent to three successive discounts $r_1\%$, $r_2\%$ and $r_3\%$

$$= \left[1 - \left(1 - \frac{r_1}{100} \right) \left(1 - \frac{r_2}{100} \right) \left(1 - \frac{r_3}{100} \right) \right] \times 100\%$$

Note This formula also can be apply for more than three successive discounts.

EXAMPLE 12. A single discount which is equivalent to two successive discounts of 20% and 5% is

- a. 20% b. 22% c. 24% d. 25%

Sol. c. Given, $r_1 = 20\%$, $r_2 = 5\%$

$\therefore$ Single discount equivalent to two successive discount 20% and 5%

$$\begin{aligned} &= \left[1 - \left(1 - \frac{20}{100} \right) \left(1 - \frac{5}{100} \right) \right] \times 100\% \\ &= \left(1 - \frac{4}{5} \times \frac{19}{20} \right) \times 100\% = \left(1 - \frac{19}{25} \right) \times 100\% \\ &= \frac{6}{25} \times 100\% = 24\% \end{aligned}$$

Rule 11 If marked price of an item is ₹ x and the successive discount rates are $r_1\%$, $r_2\%$, $r_3\%$ and so on, then selling price of the item

$$SP = x \times \left(1 - \frac{r_1}{100} \right) \left(1 - \frac{r_2}{100} \right) \left(1 - \frac{r_3}{100} \right)$$

EXAMPLE 13. The marked price of a watch is ₹ 1600. The shopkeeper gives successive discount of 10%, $r\%$ to the customer. If the customer pays ₹ 1224 for the watch, then the value of r is

- a. 20% b. 35% c. 37% d. 15%

Sol. d. Given, cost price $x = ₹ 1600$, $r_1 = 10\%$

and $r_2 = r\%$

$$\therefore \text{Selling price} = \text{Cost price} \left(1 - \frac{r_1}{100} \right) \left(1 - \frac{r_2}{100} \right)$$

$$1224 = 1600 \left(1 - \frac{10}{100} \right) \left(1 - \frac{r}{100} \right)$$

$$\Rightarrow 1 - \frac{r}{100} = \frac{1224}{1600 \times 9} \Rightarrow \frac{r}{100} = 1 - \frac{1224}{1440}$$

$$\therefore r = 15\%$$

> PRACTICE EXERCISE

- By selling an article for ₹ 247.50, Sonu get a profit of 12.5%. The cost of the article is
(a) ₹ 220 (b) ₹ 205 (c) ₹ 210 (d) ₹ 200
- If cost price of a fan is ₹ 720 and its SP is ₹ 840. Find the gain percent.
(a) 16% (b) $16\frac{2}{3}\%$ (c) $16\frac{1}{3}\%$ (d) $16\frac{7}{3}\%$
- By selling an article for ₹ 110, a man losses 12%. For how much should he sell it to gain 8%?
(a) ₹ 120 (b) ₹ 125 (c) ₹ 135 (d) ₹ 140
- A man buys 4 tables and 5 chairs for ₹ 1000. If he sells the tables at 10% profit and chairs 20% profit, he earns a profit of ₹ 120. Then, what is the cost of one table?
(a) ₹ 200 (b) ₹ 220 (c) ₹ 240 (d) ₹ 260
- If selling price of 8 articles is equal to the cost price of 10 articles, then per cent gain or loss is
(a) 20% (b) 25% (c) 30% (d) 35%
- A man sells fans at the same price on one he gain 20% and losses 20% on the other. His gain or loss is
(a) 4% loss (b) 4% gain
(c) Neither gain nor loss (d) 1% loss
- Sneha gains 10% on selling a pen. If she sells it at double the price, the profit per cent is
(a) 120% (b) 60% (c) 100% (d) 200%
- On selling an article for ₹ 240, a trader loses 4%. In order to gain 10%, he must sell the article for
(a) ₹ 275 (b) ₹ 280 (c) ₹ 285 (d) ₹ 300
- By selling 8 dozen pencils, a shopkeeper gains the selling price of 1 dozen pencils. What is the gain?
(a) $12\frac{1}{2}\%$ (b) $13\frac{1}{7}\%$ (c) $14\frac{2}{7}\%$ (d) $87\frac{1}{2}\%$
- A man purchased a watch for ₹ 400 and sold it at a gain of 20% of the selling price. The selling price of the watch is
(a) ₹ 300 (b) ₹ 320 (c) ₹ 440 (d) ₹ 500
- A person A sells a table costing ₹ 2000 to a person B and earns a profit of 6%. The person B sells it to another person C at a loss of 5%. At what price did B sell the table?
(a) ₹ 2054 (b) ₹ 2050 (c) ₹ 2024 (d) ₹ 2014
- What price did the seller mark at the printed price of a watch purchased at ₹ 380, so that after giving 5% discount, there is 25% profit?
(a) ₹ 400 (b) ₹ 450 (c) ₹ 500 (d) ₹ 600

- 13.** A discount series of 10%, 20% and 40% is equal to a single discount of
(a) 50% (b) 60% (c) 56.8% (d) 70.28%
- 14.** Successive discounts of $12\frac{1}{2}\%$ and $7\frac{1}{2}\%$ are given on the marked price of a cupboard. If the customer pays ₹ 2590, then what is the marked price?
(a) ₹ 3108 (b) ₹ 3148 (c) ₹ 3200 (d) ₹ 3600
- 15.** The difference between a discount of 40% on ₹ 1000 and two successive discounts of 35% and 5% on the same amount is
(a) ₹ 15.50 (b) ₹ 16.50 (c) ₹ 17.50 (d) ₹ 18.00
- 16.** A dealer buys an article listed at ₹ 100 and gets two successive discounts of 10% and 20%. He spends 10% of the cost price on transport etc. At what price should he sell the article to earn a profit of 15%?
(a) ₹ 90 (b) ₹ 91 (c) ₹ 91.08 (d) ₹ 91.10
- 17.** An item costing ₹ 200 is being sold at 10% loss. If the price is further reduced by 5%, then the selling price will be
(a) ₹ 170 (b) ₹ 171 (c) ₹ 175 (d) ₹ 179
- 18.** A man bought a number of oranges at 3 for a rupee and an equal number at 2 for a rupee. At what price per dozen should he sell them to make a profit of 20%?
(a) ₹ 4 (b) ₹ 5 (c) ₹ 6 (d) ₹ 7
- 19.** A milk vendor bought 28 L of milk at the cost of ₹ 8.50 per L. After adding some water, he sold the mixture at the same price. If he gains 12.5%, then how much water did he add?
(a) 5.5 L (b) 4.5 L
(c) 3.5 L (d) 2.5 L
- 20.** A man sold two watches, each for ₹ 495. If he gained 10% on one watch and suffered a loss of 10% on the other, then what is the loss or gain percentage in the transaction?
(a) 1% gain (b) 1% loss
(c) 100/99% loss (d) No gain no loss
- 21.** Jyoti bought a computer system for ₹ 40000. She sold it to Brajesh at a loss of 4%. If Brajesh sells it for ₹ 40320 to Yash, then the profit per cent earned by Brajesh is
(a) 3% (b) 5%
(c) 7% (d) 10%
- 22.** List price of a video cassette is ₹ 100. A dealer sells three video cassettes for ₹ 274.50 after allowing discount at certain rate. The rate of discount allowed is
(a) 7% (b) 7.5%
(c) 8% (d) 8.5%
- 23.** A fruit-seller buys lemons at 2 for a rupee and sells them at 5 for three rupees. What is his gain per cent?
(a) 10% (b) 15% (c) 20% (d) 25%
- 24.** A trader marks 10% higher than the cost price. He gives a discount of 10% on the marked price. In this kind of sales how much per cent does the trader gain or loss?
(a) 5% gain (b) 2% gain (c) 1% loss (d) 3% loss
- 25.** One saree was purchased for ₹ 564 after getting a discount of 6% and another saree was purchased for ₹ 396 after getting a discount of 1%. Taking both the items as a single transaction, what is the percentage of discount?
(a) 3.5% (b) 4% (c) 7% (d) 7.5%
- 26.** A dishonest dealer professes to sell his good at cost price but uses a false weight and thus gains 25%. For a kilogram he uses a weight of
(a) 700 g (b) 750 g (c) 800 g (d) 850 g
- 27.** A person bought two old scooters for ₹ 9000. By selling one at a profit of 25% and the other at a loss of 20%, to neither gain nor loses. The cost of each scooter is
(a) ₹ 3500, ₹ 500 (b) ₹ 4500, ₹ 4500
(c) ₹ 4000, ₹ 5000 (d) ₹ 5300, ₹ 3700
- 28.** The manufacturer of a certain item can sell all he can produce at the selling price of ₹ 60 each. It costs him ₹ 40 in materials and labour to produce each item and he has overhead expenses of ₹ 3000 per week in order to operate the plant. The number of items he should produce and sell in order to make a profit of atleast ₹ 1000 per week is
(a) 400 (b) 300 (c) 250 (d) 200
- 29.** A person sold a table at a profit of $6\frac{1}{2}\%$. If he had sold it for ₹ 1250 more, he would have gained 19%
I. The CP of the table is ₹ 10000.
II. CP will ₹ 19000, if he had sold it for ₹ 2400 more and gained same profit
Which one is correct?
(a) Only II (b) Only I
(c) Neither I nor II (d) Both I and II
- 30.** A bookseller sells a book at a gain of 10%. If he had bought it at 4% less and sold it for ₹ 6 more, he would have gained $18\frac{3}{4}\%$
I. The SP of the book is ₹ 165.
II. The CP of the book is ₹ 150.
Which one is correct?
(a) Neither I nor II (b) Both I and II
(c) Only I (d) Only II

> PREVIOUS YEARS' QUESTIONS

- 31.** A person bought 8 quintal of rice for certain rupees. After a week, he sold 3 quintal of rice at 10% profit, 3 quintal of rice with neither profit nor loss and 2 quintal at 5% loss. In this transaction, what is the profit? **☑ 2012 I**
(a) 10% (b) 20% (c) 25% (d) None of these
- 32.** The cost of two articles are in the ratio 3 : 5. If there is 30% loss on the first article and 20% gain on the second article, then what is overall percentage of loss or gain? **☑ 2012 I**
(a) 2.25% gain (b) 5.25% loss
(c) 2% loss (d) None of these
- 33.** A cloth store is offering Buy 3, get 1 free. What is the net percentage discount being offered by the store? **☑ 2012 II**
(a) 20% (b) 25% (c) 30% (d) $33\frac{1}{3}\%$
- 34.** A person sold an article for ₹ 136 and got 15% loss. Had he sold it for ₹ x , he would have got a profit of 15%. Which one of the following is correct? **☑ 2012 II**
(a) $190 < x < 200$ (b) $180 < x < 190$
(c) $170 < x < 180$ (d) $160 < x < 170$
- 35.** A man buys a television set which lists for ₹ 5000 at 10% discount. He gets an additional 2% discount (after the first discount) for paying cash. What does he actually pay for the set? **☑ 2012 II**
(a) ₹ 4410 (b) ₹ 4400 (c) ₹ 4000 (d) ₹ 4500
- 36.** A merchant earns a profit of 20% by selling a basket containing 80 apples which costs ₹ 240 but he gives one-fourth of it to his friend at cost price and sells the remaining apples. In order to earn the same profit, at what price must he sell each apple? **☑ 2012 II**
(a) ₹ 3.00 (b) ₹ 3.60 (c) ₹ 3.80 (d) ₹ 4.80
- 37.** A person sold an article for ₹ 3600 and got a profit of 20%. Had he sold the article for ₹ 3150, how much profit would he have got? **☑ 2013 II**
(a) 4% (b) 5% (c) 6% (d) 10%
- 38.** Two lots of onions with equal quantity, one costing ₹ 10 per kg and the other costing ₹ 15 per kg, are mixed together and whole lot is sold at ₹ 15 per kg. What is the profit or loss?

☑ 2013 II

- (a) 10% loss (b) 10% profit
(c) 20% profit (d) 20% loss

- 39.** On a 20% discount sale, an article costs ₹ 596. What was the original price of the article?

☑ 2014 I

- (a) ₹ 720 (b) ₹ 735 (c) ₹ 745 (d) ₹ 775

- 40.** A man buys 200 oranges for ₹ 1000. How many oranges for ₹ 100 can be sold, so that his profit percentage is 25%? **☑ 2014 II**

- (a) 10 (b) 14 (c) 16 (d) 20

- 41.** When an article is sold at 20% discount, the selling price is ₹ 24. What will be the selling price when the discount is 30%. **☑ 2014 II**

- (a) ₹ 25 (b) ₹ 23 (c) ₹ 21 (d) ₹ 20

- 42.** A shopkeeper sells his articles at their cost price but uses a faulty balance which reads 1000 g for 800 g. What is his actual profit percentage?

- (a) 25% (b) 20% **☑ 2014 II**
(c) 40% (d) 30%

- 43.** A person selling an article for ₹ 96 finds that his loss per cent is one-fourth of the amount of rupees that he had paid for the article. What can be the cost price? **☑ 2014 II**

- (a) Only ₹ 160 (b) Only ₹ 240
(c) Either ₹ 160 or ₹ 240 (d) Neither ₹ 160 nor ₹ 240

- 44.** A milkman claims to sell milk at its cost price but he is making a profit of 20% since, he has mixed some amount of water in the milk. What is the percentage of milk in the mixture? **☑ 2015 I**

- (a) 80% (b) $\frac{250}{3}\%$ (c) 75% (d) $\frac{200}{3}\%$

- 45.** The value of a single discount on some amount which is equivalent to a series of discounts of 10%, 20% and 40% on the same amount, is equal to **☑ 2015 II**

- (a) 43.2% (b) 50% (c) 56.8% (d) 70%

- 46.** A cloth merchant buys cloth from a weaver and cheats him by using a scale which is 10 cm longer than a normal metre scale. He claims to sell cloth at the cost price to his customers, but while selling uses a scale which is 10 cm shorter than a normal metre scale. What is his gain? **☑ 2016 I**

- (a) 20% (b) 21% (c) $22\frac{2}{9}\%$ (d) $23\frac{1}{3}\%$

> ANSWERS

1	a	2	b	3	c	4	a	5	b	6	a	7	a	8	a	9	c	10	d
11	d	12	c	13	c	14	c	15	c	16	c	17	b	18	c	19	c	20	b
21	b	22	d	23	c	24	c	25	b	26	c	27	c	28	d	29	b	30	b
31	d	32	d	33	b	34	b	35	a	36	c	37	b	38	c	39	c	40	c
41	c	42	a	43	c	44	b	45	c	46	a								

HINTS AND SOLUTIONS

1. (a) Given, selling price of article
 $= ₹ 247.50$ and gain $= \frac{25}{2}\%$
 $\therefore$ Cost price
 $= ₹ \left\{ \frac{100}{\left(100 + \frac{25}{2}\right)} \times 247.50 \right\}$ [Rule 1]
 $= ₹ \frac{100 \times 2 \times 247.50}{225} = ₹ 220$
2. (b) Total gain = SP - CP
 $= (840 - 720) = ₹ 120$
 $\therefore$ Gain percent $= \frac{120}{720} \times 100 = 16\frac{2}{3}\%$
3. (c) Given, SP = ₹ 110 and loss = 12%
 $\therefore$ Cost price $= ₹ \left(\frac{100}{88} \times 110 \right) = ₹ 125$
 Now, CP = ₹ 125, gain required = 8%
 $\therefore$ SP $= ₹ \left(\frac{(100 + 8)}{100} \times 125 \right) = ₹ 135$
4. (a) Let cost of one table and one chair be ₹ x and ₹ y , respectively.
 $\therefore$ Cost of 4 tables and 5 chairs
 $= ₹ 4x + 5y$
 $\therefore 4x + 5y = 1000$... (i)
 $\therefore$ Now, profit on 4 tables
 $= 4x \times 10\% = ₹ \frac{4x}{10}$
 Profit on 5 chairs $= 5y \times 20\% = ₹ y$
 $\therefore \frac{4x}{10} + y = 120$... (ii)
 Now, on solving Eqs. (i) and (ii), we get
 $x = ₹ 200$
5. (b) Refer to example 9.
6. (a) Refer to example 4.
7. (a) Let the selling price be ₹ 100.
 $\therefore$ Cost price $= \frac{SP \times 100}{100 + 10}$ [Rule 1]
 $= \frac{100 \times 100}{110} = ₹ \frac{1000}{11}$
 Now, if SP is ₹ 200.
 $\therefore$ Gain $= ₹ \left(200 - \frac{1000}{11} \right) = ₹ \frac{1200}{11}$
 Gain percent $= \frac{1200 / 11}{1000 / 11} \times 100 = 120\%$
8. (a) Given, selling price of an article
 $= ₹ 240$
 Loss = 4%
 $\therefore$ Cost price of an article for the loss of 4%
 $= ₹ \frac{240 \times 100}{96}$ [Rule 2]

Selling price of an article for a profit of 10%
 $= \frac{240 \times 100}{96} \times \frac{110}{100} = \frac{240 \times 110}{96}$
 $= ₹ 275$

9. (c) Let selling price of 1 dozen pencil be ₹ x .
 $\therefore$ Selling price of 8 dozen pencils = ₹ $8x$ and profit = ₹ x
 $\therefore$ Cost price of 8 dozen pencils
 $= 8x - x = ₹ 7x$
 $\therefore$ Gain, percent $= \frac{x}{7x} \times 100\% = 14\frac{2}{7}\%$
10. (d) Given, CP = ₹ 400, let selling price be ₹ x .
 Then, $400 + 20\%$ of $x = x$
 $\Rightarrow 400 + \frac{x}{5} = x \Rightarrow \frac{4x}{5} = 400$
 $\Rightarrow x = \frac{400 \times 5}{4} \Rightarrow x = ₹ 500$
 $\therefore$ Selling price = ₹ 500
11. (d) The cost price of table for person B
 $= 2000 + 6 \times \frac{2000}{100}$
 $= 2000 + 120 = ₹ 2120$
 Selling price for person B
 $= 2120 - \frac{2120 \times 5}{100}$
 $= 2120 - 106 = ₹ 2014$
12. (c) Let marked price be ₹ x .
 Selling price after 5% discount
 $= x - \frac{5}{100}x = \frac{19}{20}x$
 Profit = SP - CP $= \frac{19}{20}x - 380$
 Profit % $= \frac{\frac{19}{20}x - 380}{380} \times 100$
 $= \frac{\frac{19}{20}x - 380}{380} \times 100$
 $25 = \frac{\frac{19}{20}x - 380}{380} \times 100$
 $x = 475 \times \frac{20}{19} = ₹ 500$
13. (c) Refer to example 12.
14. (c) Refer to example 13.
15. (c) Case I Discount = 40 %
 $\Rightarrow$ Selling price = 60% of 1000
 $= \frac{60}{100} \times 1000 = ₹ 600$
 Case II Two successive discounts are of 35% and 5%.

- $\Rightarrow$ Selling price
 $= 1000 \times \left(1 - \frac{35}{100}\right) \left(1 - \frac{5}{100}\right)$ [Rule 11]
 $= \frac{65}{100} \times \frac{95}{100} \times 1000 = ₹ 617.50$
 $\therefore$ Difference $= (617.50 - 600) = ₹ 17.50$
16. (c) List price of article be ₹ 100.
 Cost price for dealer
 $= \frac{(100 - 20)}{100} \times \frac{(100 - 10)}{100} \times 100$
 $= \frac{80 \times 90 \times 100}{100 \times 100} = ₹ 72$
 Money spent on transport
 $= \frac{10}{100} \times 72 = ₹ 7.20$
 $\therefore$ Total cost price $= 72 + 7.20 = ₹ 79.20$
 $\therefore$ Selling price
 $= \frac{(100 + 15)}{100} \times 79.20$
 $= \frac{115 \times 79.20}{100} = ₹ 91.08$
 17. (b) Given, cost of article = ₹ 200
 $\therefore$ Selling price of article = 95% of (90% of 200)
 $= \frac{95}{100} \times \frac{90}{100} \times 200 = ₹ 171$
 18. (c) Given, CP of 3 oranges of Ist variety = ₹ 1
 CP of 1 orange of Ist variety $= ₹ \frac{1}{3}$
 CP of 2 oranges of IInd variety = ₹ 1
 CP of 1 orange of IInd variety $= ₹ \frac{1}{2}$
 Total CP of 2 oranges of different variety $= \frac{1}{3} + \frac{1}{2} = ₹ \frac{5}{6}$
 Profit on 2 oranges of different variety
 $= 20\%$ of $\frac{5}{6} = \frac{20}{100} \times \frac{5}{6} = ₹ \frac{1}{6}$
 $\therefore$ SP of 2 oranges of different variety
 $= \frac{5}{6} + \frac{1}{6} = ₹ 1$
 Hence, SP of 12 oranges is ₹ 6.
 19. (c) Given, cost price of 1L = ₹ 8.50
 $\therefore$ Total CP of milk $= 28 \times 8.50 = ₹ 238$
 $\therefore$ Profit = 12.5% of 238
 $= \frac{12.5}{100} \times 238 = ₹ 29.75$

Let he added x L of water.

$$\therefore \text{Profit} = x \times 8.5 \Rightarrow 29.75 = x \times 8.5$$

$$\therefore x = 3.5 \text{ L}$$

20. (b) Refer to example 6.

$$\begin{aligned} \text{21. (b) Cost price of computer for Brajesh} \\ = 40000 - \frac{4}{100}(40000) = ₹ 38400 \end{aligned}$$

$$\begin{aligned} \text{Selling price of computer for Brajesh} \\ = ₹ 40320 \end{aligned}$$

$$\therefore \text{Profit amount} = (40320 - 38400) = ₹ 1920$$

$$\begin{aligned} \therefore \text{Profit percentage earned by Brajesh} \\ = \frac{1920}{38400} \times 100 = 5\% \end{aligned}$$

22. (d) Given, list price of a video cassette = ₹ 100

Let the rate of discount be $r\%$.

$$\begin{aligned} \text{Selling price of 3 video cassette} \\ = ₹ 274.50 \end{aligned}$$

$$\therefore \text{Selling price of 1 video cassette} = ₹ \frac{274.50}{3} = ₹ 91.50$$

$$\therefore 100 - \frac{r}{100} \times 100 = ₹ 91.50$$

$$\Rightarrow 100 - 91.50 = r \Rightarrow 8.50 = r$$

$$\therefore \text{Rate of discount} = 8.5\%$$

23. (c) CP of 1 lemon = $\frac{1}{2}$

$$\text{SP of 1 lemon} = \frac{3}{5}$$

$$\therefore \text{Gain \%}$$

$$= \frac{\frac{3}{5} - \frac{1}{2}}{\frac{1}{2}} \times 100 = \frac{6-5}{10} \times 2 \times 100 = 20\%$$

24. (c) Let the cost price be ₹ x .

$$\text{Marked price} = \frac{x \times 110}{100} = ₹ \frac{11x}{10}$$

$$\therefore \text{SP} = \frac{11x}{10} \times \frac{90}{100} = \frac{99x}{100}$$

$$\therefore \text{Required gain/loss per cent}$$

$$= \frac{\frac{99x}{100} - x}{x} \times 100 = -1\% \quad [\text{Rule 8}]$$

[negative sign i.e. loss]

25. (b) Let marked price of two sarees be ₹ x and ₹ y , respectively.

$$\therefore x - \frac{6x}{100} = 564 \Rightarrow \frac{94x}{100} = 564$$

$$\Rightarrow x = ₹ 600 \text{ and } y - \frac{y}{100} = 396$$

$$\Rightarrow \frac{99y}{100} = 396 \Rightarrow y = ₹ 400$$

$$\therefore \text{Total MP amount} = 600 + 400 = ₹ 1000$$

$$\begin{aligned} \text{Total amount after discount} \\ = 564 + 396 = ₹ 960 \end{aligned}$$

$$\therefore \text{Discount per cent}$$

$$= \frac{1000 - 960}{1000} \times 100 = \frac{40}{10}\% = 4\%$$

26. (c) Here, true weight = 1000

and gain = 25%

$$\Rightarrow 25 = \frac{1000 - \text{false weight}}{\text{false weight}} \times 100$$

$$\Rightarrow \frac{\text{false weight}}{4} = 1000 - \text{false weight}$$

$$\Rightarrow \text{false weight} = 1000 \times \frac{4}{5} = 800$$

Hence, he uses the weight 800 g.

27. (c) Let cost price of one scooter be ₹ x .

$$\begin{aligned} \therefore \text{Cost price of other scooter} \\ = ₹ (9000 - x) \end{aligned}$$

$$\begin{aligned} \therefore \text{Selling price of 1st scooter} \\ = x + \frac{25x}{100} = \frac{125x}{100} \end{aligned}$$

Also, selling price of 2nd scooter

$$\begin{aligned} &= (9000 - x) \left(1 - \frac{20}{100}\right) \\ &= (9000 - x) \left(\frac{80}{100}\right) \end{aligned}$$

Total selling price of both scooters

$$= \frac{125x}{100} + (9000 - x) \frac{80}{100}$$

$$\therefore \frac{125x}{100} + (9000 - x) \frac{80}{100} = 9000 \quad [\text{given}]$$

$$\Rightarrow \frac{45x}{100} + 7200 = 9000$$

$$\therefore x = 4000$$

$$\therefore \text{Cost price of 1st scooter} = ₹ 4000$$

$$\begin{aligned} \text{Hence, cost price of 2nd scooter} \\ = (9000 - x) = ₹ 5000 \end{aligned}$$

28. (d) Let the number of items be x .

Then, selling price of items = $60x$

Cost of material of items = $40x$

Overhead expenses = ₹ 3000

$$\therefore 60x - (40x + 3000) = 1000$$

$$\Rightarrow 20x = 4000$$

$$\therefore x = \frac{4000}{20} = 200$$

29. (b) I. Let CP = x

$$\text{Then, SP} = \frac{213x}{200}$$

$$\text{If SP} = \frac{213x}{200} + 1250, \text{ then gain} = 19\%$$

$$\Rightarrow 19 = \frac{\frac{213x}{200} + 1250 - x}{x} \times 100$$

$$\Rightarrow \frac{19x}{100} - \frac{213x}{200} = 1250$$

$$\Rightarrow x = 1250 \times 8 = 10000$$

$$\text{II. If SP} = \frac{213x}{200} + 2400 \text{ and gain}$$

= 19%, then

$$19 = \frac{\frac{213x}{200} + 2400 - x}{x} \times 100$$

$$\Rightarrow \frac{19x}{100} - \frac{13x}{200} = 2400$$

$$\Rightarrow x = 2400 \times 8 = 19200$$

Hence, I is correct but II is incorrect.

30. (b) Let CP = x

$$\text{then SP} = \frac{110x}{100} = \frac{11x}{10}$$

$$\text{Now, CP} = 96\% \text{ of } x = \frac{96x}{100} = ₹ \frac{24x}{25}$$

According to the questions

$$\text{SP} = ₹ \left(\frac{11x}{10} + 6 \right)$$

$$\therefore \left(\frac{11x}{10} + 6 \right) = 118 \frac{3}{4} \% \text{ of } \frac{24x}{25}$$

$$\Rightarrow \frac{11x + 60}{10} = \frac{475}{400} \times \frac{24x}{25} = \frac{57x}{50}$$

$$\Rightarrow 570x = 550x + 3000$$

$$\Rightarrow x = \frac{3000}{20} = 150$$

$$\therefore \text{CP} = ₹ 150$$

$$\therefore \text{SP} = \frac{11}{10} \times 150 = ₹ 165$$

Hence, I and II both are correct.

31. (d) Let CP of 8 quintal be ₹ x .

$$\therefore \text{CP of 1 quintal} = ₹ \frac{x}{8}$$

$$\therefore \text{SP of 3 quintal rice at 10\% profit}$$

$$= \frac{3x}{8} + \frac{3x}{8} \times \frac{1}{10} = \frac{3x}{8} + \frac{3x}{80} = \frac{33x}{80}$$

SP of 3 quintal rice without profit or

$$\text{loss} = ₹ \frac{3x}{8}$$

SP of 2 quintal rice at 5% loss

$$= \frac{2x}{8} - \frac{2x}{8} \times \frac{5}{100} = \frac{x}{4} - \frac{x}{4 \times 20}$$

$$= \frac{19x}{4 \times 20} = \frac{19x}{80}$$

$$\therefore \text{Total SP} = \frac{33x}{80} + \frac{3x}{8} + \frac{19x}{80}$$

$$= \frac{33x + 30x + 19x}{80} = \frac{82x}{80}$$

$$\therefore \text{Profit percentage} = \frac{\text{SP} - \text{CP}}{\text{CP}} \times 100\%$$

$$= \frac{\frac{82x}{80} - x}{x} \times 100\% = \frac{(82-80)x}{80x} \times 100\%$$

$$= \frac{2}{80} \times 100 = 2.5\%$$

32. (d) Let the CP of two articles be $3x$ and $5x$, respectively.

$$\therefore \text{SP of first article} = \frac{3x \times 70}{100} = \frac{21x}{10}$$

[Rule 2]

$$\text{and SP of second article} \\ = \frac{5x \times 120}{100} = 6x$$

$$\therefore \text{Total SP} \\ = 6x + \frac{21x}{10} = \frac{60x + 21x}{10} = \frac{81x}{10}$$

$$\therefore \text{Total CP} = 3x + 5x = 8x$$

$$\therefore \text{Profit} = \frac{81x}{10} - 8x = \frac{81x - 80x}{10} = \frac{x}{10}$$

$$\therefore \text{Overall percentage of gain percentage} \\ = \frac{\frac{x}{10} \times 100}{8x} = \frac{x \times 100}{10 \times 8x} = 1.25\%$$

33. (b) We know that,

$$\text{Net percentage discount} \\ = \frac{\text{Discount}}{\text{Cost price}} \times 100\% = \frac{1}{4} \times 100\% = 25\%$$

34. (b) We know that,

$$\text{Cost price} = \left(\frac{100}{100 - \text{Loss}\%} \right) \times \text{SP}$$

[Rule 2]

$$= \left(\frac{100}{100 - 15} \right) \times 136$$

$$= \frac{136 \times 100}{85} = ₹ 160$$

$\therefore$ Given, profit percentage = 15%

$$\therefore \text{Selling price (x)} \\ = \frac{160 \times (100 + 15)}{100} = \frac{160 \times 115}{100} = ₹ 184$$

So, option (b) is correct because $180 < x < 190$.

35. (a) Given, list price of television = ₹ 5000 and discount = 10%

$$\therefore \text{After discount television cost} \\ = 5000 - 5000 \times 10\% \\ = 5000 - \frac{5000 \times 10}{100} = ₹ 4500$$

Additional discount = 2%

$$\therefore \text{Actually price of television} \\ = 4500 - \frac{4500 \times 2}{100} = ₹ 4410$$

36. (c) $\therefore$ CP of 80 apples = ₹ 240

$$\therefore \text{CP of 1 apple} = \frac{240}{80} = ₹ 3$$

$\therefore$ CP of 20 apples = ₹ 3 $\times$ 20 = ₹ 60
To earn a profit of 20%, SP = 120% of 240 = ₹ 288

But he sold $\frac{1}{4}$ of his apples i.e. 20 apples for ₹ 60.

$$\therefore \text{SP of remaining 60 apples} \\ = (288 - 60) = ₹ 228$$

$$\therefore \text{SP of 1 apple} = \frac{228}{60} = ₹ 3.80$$

37. (b) Let the cost price of the article be ₹ x .

Given, profit percentage = 20%

$$\text{Now, } x + \frac{20x}{100} = 3600 \Rightarrow \frac{120x}{100} = 3600$$

$$\therefore x = ₹ 3000$$

Now, profit percentage when the article is sold for ₹ 3150

$$\frac{3150 - 3000}{3000} \times 100 = \frac{150}{3000} \times 100 = 5\%$$

38. (c) Let each lot of onion contains x kg. Then, total cost price of these two lots together = $10x + 15x = 25x$

Selling price of whole lot

$$= 15 \times (x + x) = 15 \times 2x = 30x$$

$$\therefore \text{Profit percentage} = \frac{30x - 25x}{25x} \times 100 \\ = \frac{5x}{25x} \times 100 = 20\%$$

Hence, the profit is 20%.

39. (c) Let the original price be ₹ x .

Since, at discount of 20% article costs ₹ 596.

$$\text{Then, } 596 = \frac{80}{100} \times x \Rightarrow x = \frac{596 \times 100}{80} \\ = ₹ 745$$

Hence, the original price of an article is ₹ 745.

40. (c) Given, cost price of 200 oranges = ₹ 1000

$$\therefore \text{Cost price of 1 orange} = \frac{1000}{200} = ₹ 5$$

$$\text{Selling price of 1 orange} = 5 \left(\frac{100 + 25}{100} \right) \\ = 5 \times \frac{125}{100} = ₹ 6.25$$

Now, in ₹ 6.25 number of oranges can be sold = 1

$$\text{In ₹ 100, number of oranges can be sold} \\ = \frac{1}{6.25} \times 100 = 16$$

Hence, 16 oranges can be sold in ₹ 100 for profit 25%.

41. (c) Let the cost price of article be ₹ x .

Discount = 20% and selling price = 24

[given]

$$\therefore x = 24 + 20\% \text{ of } x$$

$$\Rightarrow x = 24 + \frac{20}{100}x \Rightarrow x = 24 + \frac{x}{5}$$

$$\Rightarrow x - \frac{x}{5} = 24 \Rightarrow \frac{4x}{5} = 24$$

$$\Rightarrow x = \frac{24 \times 5}{4} = 30$$

$\therefore$ Cost price of an article = ₹ 30

Now, selling price after 30% discount

$$= 30 - 30\% \text{ of } 30 = 30 - \frac{30}{100} \times 30$$

$$= 30 - 9 = ₹ 21$$

42. (a) $\therefore$ Actual profit percentage $= \frac{\text{Fair weight} - \text{Unfair weight}}{\text{Unfair weight}} \times 100\%$

$$= \frac{1000 - 800}{800} \times 100\% = 25\%$$

43. (c) Let the cost price of an article be ₹ x . and selling price of an article = ₹ 96

[given]

$$\frac{x - 96}{x} \times 100 = \frac{1}{4}x$$

$$\Rightarrow x^2 - 400x + 38400 = 0$$

$$\Rightarrow x^2 - 160x - 240x + 38400 = 0$$

$$\Rightarrow (x - 160)(x - 240) = 0$$

$$\therefore x = 160 \text{ or } 240$$

Hence, the cost price of the article is either

₹ 160 or ₹ 240.

44. (b) Let CP of 1 L of milk be ₹ x .

$$\therefore \text{SP of 1 L of milk} \\ = x \times 120\% = ₹ 1.2x$$

Now, as in ₹ 1.2x, the quantity of milk sold = 1 L

$$\therefore \text{In ₹ } x, \text{ quantity of milk sold} \\ = \frac{1}{1.2x} \times x = \frac{5}{6} \text{ L}$$

According to the question,

CP of milk and SP of mixture are same, therefore in mixture, quantity of milk must be $\frac{5}{6}$ L.

Hence, the required percentage

$$= \frac{5}{6} \times 100\% = \frac{250}{3}\%$$

45. (c) Refer to example 12.

46. (a) Let the amount of cloth = 100 cm

Then, amount of cloth purchased = 110 cm

and amount of cloth sold = 90 cm

$$\therefore \text{Gain \%} = \frac{110 - 90}{100} \times 100\%$$

$$= \frac{20}{100} \times 100\% = 20\%$$

12

RATIO AND PROPORTION

Regularly (1-2) questions have been asked from this chapter. Generally both the topics i.e. ratio and proportion have same importance but more emphasis is given on proportion. The concepts of this chapter are extremely used in other chapters like-age, average. etc.



RATIO

A ratio is the comparison of two or more quantities of the same type (or kind) by division. i.e. if a and b are two quantities of same kind (or same unit), then the fraction a/b is called the ratio a to b and we write it as $a : b$. Here, a is called antecedent and b is called the consequent.

e.g. If a fruit box contains 8 oranges and 7 lemons then the ratio of oranges to lemons is 8 to 7 or 8:7.

Note If both terms a and b of a ratio are multiplied or divided by the same quantity, then ratio remains unchanged.

i.e. $a:b$ is same as $na:nb \Rightarrow \frac{a}{b} = \frac{na}{nb}$ and $a:b$ is same as $\frac{a}{n}:\frac{b}{n} \Rightarrow \frac{a}{b} = \frac{a/n}{b/n}$.

EXAMPLE 1. If the ratio of 90 cm to 1.5 m is same as 3 : x , then x is

a. 4

b. 5

c. 2

d. 1

Sol. b. 1.5 m = 150 cm (units must be same)

$$\text{Ratio of 90 cm to 1.5 m} = \frac{90}{150} = \frac{90/30}{150/30} = \frac{3}{5}.$$

As this ratio is same as 3 : x

$$\therefore \frac{3}{x} = \frac{3}{5}$$

$$\Rightarrow 3x = 5 \times 3$$

$$\Rightarrow x = 5$$

Types of Ratios

1. **Compound Ratio** When two or more ratios are multiplied together, they are said to be compound ratio.

If $\frac{a}{b}$, $\frac{c}{d}$, $\frac{e}{f}$ and $\frac{g}{h}$ are all ratios, then their compound ratio is $aceg : bdfh$.

e.g. Compound ratio of $\frac{2}{3}$, $\frac{4}{7}$ and $\frac{1}{3} = \frac{2 \times 4 \times 1}{3 \times 7 \times 3} = \frac{8}{63}$.

2. **Duplicate Ratio** When a ratio is compounded with itself, the resulting ratio is called the duplicate ratio.

So, $a^2 : b^2$ is duplicate ratio of $a : b$.

EXAMPLE 2. The duplicate ratio of the ratio $2\sqrt{2} : 3\sqrt{5}$ is

- a. 4 : 9 b. 8 : 45 c. 2 : 3 d. 6 : 45

Sol. b. The duplicate ratio of $2\sqrt{2} : 3\sqrt{5}$ is

$$\frac{2\sqrt{2}}{3\sqrt{5}} \times \frac{2\sqrt{2}}{3\sqrt{5}} = \frac{4\sqrt{4}}{9\sqrt{25}} = \frac{8}{45} = 8 : 45$$

3. **TriPLICATE Ratio** If a ratio is compounded three times with itself, then resulting ratio is called triplicate ratio.

So, $a^3 : b^3$ is the triplicate ratio of $a : b$.

EXAMPLE 3. If $4x + 3 : 9x + 10$ is the triplicate ratio of $3 : 4$. Then, the value of x is

- a. 2 b. 6 c. 8 d. 12

Sol. b. Since, $4x + 3 : 9x + 10$ is the triplicate ratio of $3 : 4$.

$$\text{Therefore, } \frac{4x+3}{9x+10} = \frac{3}{4} \times \frac{3}{4} \times \frac{3}{4} \Rightarrow \frac{4x+3}{9x+10} = \frac{27}{64}$$

$$\Rightarrow 64(4x+3) = 27(9x+10)$$

$$\Rightarrow 256x + 192 = 243x + 270$$

$$\Rightarrow 256x - 243x = 270 - 192 \Rightarrow 13x = 78$$

$$\therefore x = \frac{78}{13} = 6$$

Hence, the value of x is 6.

4. **Subduplicate Ratio** If two numbers are in ratio, then the ratio of their square roots is called subduplicate ratio.

If $a : b$ is ratio, then the subduplicate ratio is $\sqrt{a} : \sqrt{b}$.

e.g. Subduplicate ratio of $9 : 16$ is $\sqrt{9} : \sqrt{16} = 3 : 4$.

5. **Subtriplicate Ratio** If two numbers are in ratio, then the ratio of their cube roots is called subtriplicate ratio.

If $a : b$ is ratio, then the subtriplicate ratio is $\sqrt[3]{a} : \sqrt[3]{b}$.

e.g. Subtriplicate ratio of $64 : 27$ is $\sqrt[3]{64} : \sqrt[3]{27} = 4 : 3$

6. **Reciprocal Ratio** If $a : b$ is a ratio, then $\frac{1}{a} : \frac{1}{b}$ is its reciprocal ratio, i.e. $b : a$.

e.g. Reciprocal ratio of $3 : 7$ is $\frac{1}{3} : \frac{1}{7}$ i.e. $7 : 3$.

EXAMPLE 4. Compound ratio of the duplicate ratio of $5 : 6$, the reciprocal ratio of $25 : 42$ and the subduplicate ratio of $36 : 49$ is

- a. 1 : 2 b. 2 : 3 c. 1 : 1 d. 4 : 9

Sol. c. The duplicate ratio of $5 : 6$ is $\frac{5}{6} \times \frac{5}{6} = \frac{25}{36}$

The reciprocal ratio of $25 : 42$ is $\frac{1}{25} : \frac{1}{42}$ i.e. $\frac{42}{25}$

and subduplicate ratio of $\frac{36}{49}$ is $\frac{\sqrt{36}}{\sqrt{49}} = \frac{6}{7}$

$\therefore$ Compound ratio is $\frac{25}{36} \times \frac{42}{25} \times \frac{6}{7} = \frac{1}{1} = 1 : 1$

IMPORTANT POINTS

- Usually, the ratio is expressed in its lowest terms.
- Ratio exists only between quantities of the same kind.
- Ratio is a fraction, so it has no units.
- Ratio is taken only between positive quantities.

PROPORTION

The equality of two ratios is called proportion.

' $:$ ' is the sign of proportion.

So, if $a : b = c : d$, we write it as

$a : b :: c : d$ and say that a, b, c and d are in proportion.

The quantities a, b, c and d are called the first, second, third and fourth terms of proportion respectively. First and fourth terms are called **extreme terms** and second and third terms are called **means** or **middle terms**.

If quantities a, b, c and d are in proportion, then

$$\frac{a}{b} = \frac{c}{d} \Rightarrow ad = bc$$

i.e. product of extreme terms = product of middle terms.

This is also called cross-product rule.

Types of Proportion

1. **Continued Proportion** The non-zero quantities of same kind, $a, b, c, d, e, f, \dots$ are said to be continued proportion, if $\frac{a}{b} = \frac{b}{c} = \frac{c}{d} = \frac{d}{e} = \dots$

EXAMPLE 5. If a, b, c, d and e are in continued proportion, then a/e is equal to

- a. a^3/b^3 b. a^4/b^4 c. b^3/a^3 d. b^4/a^4

Sol. b. Since, a, b, c, d and e are in continued proportion.

$$\begin{aligned} \therefore \frac{a}{b} &= \frac{b}{c} = \frac{c}{d} = \frac{d}{e} \Rightarrow \frac{e}{d} = \frac{d}{c} = \frac{c}{b} = \frac{b}{a} \Rightarrow c = \frac{b^2}{a} \\ d &= \frac{c^2}{b} = \frac{b^4}{a^2} \cdot \frac{1}{b} = \frac{b^3}{a^2}, \quad e = \frac{d^2}{c} = \frac{b^6}{a^4} \cdot \frac{a}{b^2} = \frac{b^4}{a^3} \\ \therefore \frac{a}{e} &= \frac{a}{(b^4/a^3)} = \frac{a^4}{b^4} \end{aligned}$$

2. **Mean Proportional** If a, b and c are in continued proportion, then b is called the mean proportional of a and c .

i.e. $\frac{a}{b} = \frac{b}{c} \Rightarrow b^2 = ac \Rightarrow b = \sqrt{ac}$

EXAMPLE 6. What is the mean proportional between $(15 + \sqrt{200})$ and $(27 - \sqrt{648})$? ✎ 2012 I

- a. 4 b. $14\sqrt{7}$ c. $3\sqrt{5}$ d. $5\sqrt{3}$

Sol. c. Let mean proportional between $(15 + \sqrt{200})$ and $(27 - \sqrt{648})$ is x

$$\begin{aligned} \text{Then, } x &= \sqrt{(15 + \sqrt{200})(27 - \sqrt{648})} \\ &= \sqrt{(15 + 10\sqrt{2})(27 - 18\sqrt{2})} \\ &= \sqrt{5(3 + 2\sqrt{2}) \cdot 9(3 - 2\sqrt{2})} = \sqrt{45[(3)^2 - (2\sqrt{2})^2]} \\ &= \sqrt{45(9 - 8)} = \sqrt{45} = 3\sqrt{5} \end{aligned}$$

3. **Third Proportional** If $a : b :: b : c$, then c is called the 3rd proportional to a and b . Now, c will be calculated as below:

$$a : b :: b : c \Rightarrow a : b = b : c \Rightarrow c = \frac{b^2}{a}$$

EXAMPLE 7. The two numbers such that their mean proportional is 24 and the third proportional is 1536, are

- a. 3 and 98 b. 6 and 96 c. 4 and 92 d. None of these

Sol. b. Let a and b be the required numbers, then mean proportional between a and b is 24.

$$\therefore a : 24 = 24 : b \Rightarrow ab = 24^2, \quad ab = 576 \quad \dots(i)$$

and third proportional to a and b is 1536.

$$\Rightarrow a : b = b : 1536$$

$$\Rightarrow b^2 = 1536a \Rightarrow a = \frac{b^2}{1536} \quad \dots(ii)$$

On putting the value of a in Eq. (i), we get

$$\Rightarrow \frac{b^2}{1536} \cdot b = 576 \Rightarrow b^3 = 576 \times 1536$$

$$\Rightarrow b^3 = (96 \times 6) \times (96 \times 16) = 96^3 \Rightarrow b = 96$$

$$\therefore a = \frac{576}{b} = \frac{576}{96} = 6 \quad [\text{from Eq. (i)}]$$

Hence, the required numbers are 6 and 96.

4. **Fourth Proportional** If four numbers or quantities a, b, c and d are in proportion, then d is known as fourth proportional.

i.e. $a : b :: c : d$

EXAMPLE 8. The fourth proportional to

$$p^2 - pq + q^2, p^3 + q^3, p - q \text{ is}$$

- a. $p + q$ b. $p - q$ c. $p^2 + q^2$ d. $p^2 - q^2$

Sol. d. Let the fourth proportional be x .

$$\begin{aligned} \text{Then, } p^2 - pq + q^2 : p^3 + q^3 &= p - q : x \\ \Rightarrow \frac{p^2 - pq + q^2}{p^3 + q^3} &= \frac{p - q}{x} \Rightarrow x = \frac{(p - q)(p^3 + q^3)}{p^2 - pq + q^2} \end{aligned}$$

$$\therefore x = \frac{(p - q)(p^3 + q^3)}{p^2 - pq + q^2} = (p - q)(p + q) = (p^2 - q^2)$$

Properties of Proportion

If four non zero quantities a, b, c and d are in proportion, then properties of proportion are as follows.

1. **Invertendo** If $a : b :: c : d$, then $b : a :: d : c$

i.e. $\frac{a}{b} = \frac{c}{d} \Rightarrow \frac{b}{a} = \frac{d}{c}$

2. **Alternendo** If $a : b :: c : d$, then $a : c :: b : d$

i.e. $\frac{a}{b} = \frac{c}{d} \Rightarrow \frac{a}{c} = \frac{b}{d}$

3. **Componendo** (Adding the Denominator)

If $a : b :: c : d$, then $(a + b) : b :: (c + d) : d$

i.e. $\frac{a}{b} = \frac{c}{d} \Rightarrow \frac{a + b}{b} = \frac{c + d}{d}$

4. **Dividendo** (Subtracting the Denominator)

If $a : b :: c : d$, then $a : (a - b) :: c : (c - d)$

i.e. $\frac{a}{b} = \frac{c}{d} \Rightarrow \frac{a}{a - b} = \frac{c}{c - d}$

5. **Componendo and Dividendo**

If $a : b :: c : d$, then $(a + b) : (a - b) :: (c + d) : (c - d)$

i.e. $\frac{a}{b} = \frac{c}{d} \Rightarrow \frac{a + b}{a - b} = \frac{c + d}{c - d}$

EXAMPLE 9. If $(a - b) : (a + b) = 1 : 5$, then what is $(a^2 - b^2) : (a^2 + b^2)$ equal to?

- a. $\frac{13}{5}$ b. $\frac{4}{13}$ c. $\frac{5}{13}$ d. None of these

Sol. c. Given, $\frac{a - b}{a + b} = \frac{1}{5} \Rightarrow \frac{a + b}{a - b} = \frac{5}{1} \Rightarrow \frac{a + b + a - b}{a + b - a + b} = \frac{5 + 1}{5 - 1}$
(using componendo and dividendo rule)

$$\Rightarrow \frac{2a}{2b} = \frac{6}{4} = \frac{3}{2} \Rightarrow \frac{a}{b} = \frac{3}{2} \Rightarrow \frac{a^2}{b^2} = \frac{9}{4}$$

Again, using componendo and dividendo rule

$$\Rightarrow \frac{a^2 + b^2}{a^2 - b^2} = \frac{9 + 4}{9 - 4} = \frac{13}{5}$$

$$\Rightarrow \frac{a^2 - b^2}{a^2 + b^2} = \frac{5}{13} \quad \therefore (a^2 - b^2) : (a^2 + b^2) = 5 : 13$$

MIXTURE OR ALLIGATION

If two or more quantities are mixed together in a certain ratio, then the product is called mixture.

Mean Price

Cost price of a unit quantity of the mixture is called the mean price.

Rule of Mixture or Alligation

It is the rule to find the ratio in which two or more ingredients are mixed or to be mixed.

If two quantities are mixed in the ratio $n_1:n_2$, then

$$\frac{n_1}{n_2} = \frac{P_2 - P_m}{P_m - P_1}$$

where P_1 , P_2 and P_m are cheaper price, dearer price and mean price respectively.

It can also be expressed as

$$\begin{array}{ccc} \text{Cheaper price} & & \text{Dearer price} \\ P_1 & & P_2 \\ & \swarrow \quad \searrow & \\ & \text{Mean price} & \\ & P_m & \\ & \swarrow \quad \searrow & \\ P_2 - P_m & & P_m - P_1 \\ \therefore \frac{\text{Cheaper quantity}}{\text{Dearer quantity}} = \frac{n_1}{n_2} = \frac{P_2 - P_m}{P_m - P_1} \end{array}$$

Note $P_1 < P_m < P_2$

EXAMPLE 10. In what proportion, must wheat at ₹ 6.20 per kg be mixed with wheat at ₹ 7.20 per kg, so that the mixture be worth ₹ 6.50 per kg?

a. 6 : 4 b. 7 : 4 c. 7 : 3 d. 5 : 2

Sol. c. Given, cost price of cheaper quantity = ₹ 6.20 per kg

Cost price of dearer quantity = ₹ 7.20 per kg

Mean price = ₹ 6.50 per kg

According to the rule of alligation,

$$\begin{array}{ccc} \text{CP of cheaper} & & \text{CP of dearer} \\ \text{(620 paise)} & & \text{(720 paise)} \\ & \swarrow \quad \searrow & \\ & \text{Mean price} & \\ & \text{(650 paise)} & \\ & \swarrow \quad \searrow & \\ (720 - 650) & & (650 - 620) \\ = 70 \text{ paise} & & = 30 \text{ paise} \end{array}$$

$\therefore$ Required ratio = 70 : 30 = 7 : 3

VARIATIONS

When a change in a certain quantity leads to a certain change in another quantity then it is called variations.

Direct Variation

If x varies directly with y , then as x increases (decrease) y also increases (decreases)

i.e. $x \propto y$

$\Rightarrow x = ky$, where, k is the constant of variation.

Inverse Variation

If x varies inversely with y , then an increase is the value of x causes a decrease in the value of y .

i.e. $x \propto \frac{1}{y}$

$\Rightarrow x = \frac{k'}{y}$

where, k' is the constant of variation.

EXAMPLE 11. If a quantity y varies as the sum of three quantities of which the first varies as x , the second varies as $-x + x^2$, the third varies as $x^3 - x^2$, then what is y equal to?

- a. kx^3 , where k is a constant
- b. $kx + lx^2 + mx^3$, where k , l and m are constants
- c. kx^2 , where k is a constant
- d. kx , where k is a constant

Sol. b. Since, first term $\propto x$

$\Rightarrow$ First term = c_1x ,

$\Rightarrow$ Second term $\propto (-x + x^2)$

$\Rightarrow$ Second term = $c_2(-x + x^2)$

and third term $\propto (x^3 - x^2) \Rightarrow$ Third term = $c_3(x^3 - x^2)$

Also, $y \propto [c_1x + c_2(-x + x^2) + c_3(x^3 - x^2)]$ (given)

$\Rightarrow y = c_4[(c_1 - c_2)x + (c_2 - c_3)x^2 + c_3x^3]$

$$\begin{aligned} &= c_4(c_1 - c_2)x + (c_2 - c_3)c_4x^2 + c_3c_4x^3 \\ &= kx + lx^2 + mx^3 \end{aligned}$$

where, $k = c_4(c_1 - c_2)$,

$$l = (c_2 - c_3)c_4,$$

and $m = c_3c_4$

> PRACTICE EXERCISE

- The compounded ratio of $2a:6b$, $7a:49b$ and $3a:12b$ is
(a) $a:84b$ (b) $a^3:84b^3$ (c) $a:24b$ (d) $2a:3b$
- The subduplicate ratio of $16x^4:625y^6$ is
(a) $4x^2:25y^3$ (b) $4x^2:25y^2$ (c) $4x:25y$ (d) $8x^2:125y^3$
- If $x+5:3x+4$ is the duplicate ratio of $5:8$, then the value of x is
(a) 16 (b) 18 (c) 20 (d) 22
- If $x:6::5:3$, then the value of x is
(a) 8 (b) 10 (c) 12 (d) 13
- The third proportional to 9 and 12 is
(a) 12 (b) 14 (c) 16 (d) 18
- The mean proportional between $(2+\sqrt{3})$ and $(8-\sqrt{48})$ is
(a) 2 (b) 3 (c) 4 (d) 5
- The fourth proportional to 7, 11, 14 is
(a) 16 (b) 18 (c) 20 (d) 22
- In a ratio which is equal to $7:8$, if the antecedent is 35, then the consequent is
(a) 30 (b) 32 (c) 36 (d) 40
- If $x:y=1:3$, $y:z=5:k$, $z:t=2:5$ and $t:x=3:4$, then what is the value of k ?
(a) $1/2$ (b) $1/3$ (c) 2 (d) 3
- If the ratio of x to y is 25 times the ratio of y to x , then what is the ratio of x to y ?
(a) $1:5$ (b) $5:1$ (c) $25:1$ (d) $1:25$
- If $p\%$ of ₹ x is equal to t times $q\%$ of ₹ y , then what is the ratio of x to y ?
(a) $pt:q$ (b) $p:qt$ (c) $qt:p$ (d) $q:pt$
- If $a:b=2:3$ and $x:y=3:4$, then the value of $\frac{4ay-3bx}{5ax-2by}$ is
(a) $\frac{5}{3}$ (b) $\frac{5}{6}$ (c) $\frac{4}{5}$ (d) $\frac{5}{4}$
- If $\frac{x^3+3x}{3x^2+1}=\frac{341}{91}$, then the value of x is
(a) 16 (b) 14 (c) 13 (d) 11
- What is the number which has to be added to each term of the ratio $49:68$, so that it becomes $3:4$?
(a) 3 (b) 5 (c) 8 (d) 9
- Divide 1870 into three parts in such a way that half of the first part, one-third of the second part and one-sixth of the third part are equal. Then, the third part is
(a) 340 (b) 510 (c) 1020 (d) 1320
- Out of the ratios $7:20$, $13:25$, $17:30$ and $11:15$, the smallest one is
(a) $11:15$ (b) $7:20$ (c) $11:16$ (d) $17:30$
- What is the ratio of the numbers of 0.5 of a number is equal to 0.07 of another?
(a) $50:7$ (b) $5:7$ (c) $1:14$ (d) $7:50$
- What is the ratio whose terms differ by 40 and the measure of which is $\frac{2}{7}$?
(a) $14:56$ (b) $15:56$ (c) $16:56$ (d) $16:72$
- If $P:Q=\frac{3}{5}:\frac{5}{7}$ and $Q:R=\frac{3}{4}:\frac{2}{5}$, then what is $P:Q:R$ equal to?
(a) $\frac{3}{5}:\frac{5}{7}:\frac{2}{5}$ (b) $\frac{9}{20}:\frac{15}{28}:\frac{2}{7}$ (c) $\frac{3}{5}:\frac{3}{4}:\frac{2}{5}$ (d) $\frac{3}{5}:\frac{5}{7}:\frac{3}{4}$
- Two numbers are in the ratio $3:5$. If 9 is subtracted from each number, then they are in the ratio of $12:23$. What is the second number?
(a) 44 (b) 55 (c) 66 (d) 77
- The mean proportional between two numbers is 28 and their third proportional to them is 224. The two numbers are
(a) 7 and 112 (b) 14 and 56 (c) 28 and 28 (d) 21 and 36
- ₹ 770 have been divided among A, B, C in such a way that A receives $\frac{2}{9}$ th of what B and C together receive. What is A 's share?
(a) ₹ 154 (b) ₹ 140 (c) ₹ 250 (d) ₹ 254
- x varies directly as y and inversely as square of z . When $y=4$ and $z=14$, then $x=10$. If $y=16$ and $z=7$, what is x ?
(a) 180 (b) 160 (c) 154 (d) 140
- If $x=\frac{\sqrt{m+3n}+\sqrt{m-3n}}{\sqrt{m+3n}-\sqrt{m-3n}}$, then
(a) $3nx^2-2mx+3n=0$ (b) $2nx^2-2mx+3n=0$
(c) $3nx^2-2mx-3n=0$ (d) $3nx^2+3mx+3n=0$
- If $p+r=2q$ and $\frac{1}{q}+\frac{1}{s}=\frac{2}{r}$, then
(a) $p:s=r:q$ (b) $p^2:r^2=q:s$
(c) $p:q=r:s$ (d) $p^2=2r+s$

- 26.** If q is the mean proportional between p and r , then $\frac{p^2 - q^2 + r^2}{p^{-2} - q^{-2} + r^{-2}}$ is equal to
 (a) p^2q^3 (b) q^3 (c) q^4 (d) $p^2r^2q^4$
- 27.** The monthly incomes of A and B are in the ratio 4 : 3. Each of them saves ₹ 600. If the ratio of the expenditure is 3 : 2, then what is the monthly income of A ?
 (a) ₹ 2400 (b) ₹ 1800 (c) ₹ 2000 (d) ₹ 3600
- 28.** Three numbers are in the ratio 3 : 2 : 5 and the sum of their squares is 1862. What are the three numbers?
 (a) 18, 12, 30 (b) 24, 16, 40
 (c) 15, 10, 25 (d) 21, 14, 35
- 29.** The speeds of three cars are in the ratio 4 : 3 : 2. What is the ratio between the times taken by the cars to cover the same distance?
 (a) 2 : 3 : 4 (b) 3 : 4 : 6
 (c) 1 : 2 : 3 (d) 4 : 3 : 2
- 30.** The ratio between the ages of A and B is 2 : 5. After 8 yr, their ages will be in the ratio 1 : 2. What is the difference between their present ages?
 (a) 20 yr (b) 22 yr (c) 24 yr (d) 25 yr
- 31.** Let y is equal to the sum of two quantities of which one varies directly as x and the other inversely as x . If $y = 6$ when $x = 4$ and $y = 10/3$, when $x = 3$, then what is the relation between x and y ?
 (a) $y = x + (4/x)$ (b) $y = -2x + (4/x)$
 (c) $y = 2x + (8/x)$ (d) $y = 2x - (8/x)$
- 32.** If ₹ 8400 is divided among A, B and C in the ratio $\frac{1}{5} : \frac{1}{6} : \frac{1}{10}$, what is the share of A ?
 (a) ₹ 1800 (b) ₹ 3000 (c) ₹ 3600 (d) ₹ 4000
- 33.** A certain amount of money has to be divided between two persons P and Q in the ratio 3 : 5. But it was divided in the ratio of 2 : 3 and there by Q loses ₹ 10. What was the amount?
 (a) ₹ 250 (b) ₹ 300 (c) ₹ 350 (d) ₹ 400
- 34.** 20 L of a mixture contains milk and water in the ratio 4 : 3. If 6 L of this mixture are replaced by 6 L of milk, the ratio of milk to water in the new mixture will become
 (a) 7 : 3 (b) 8 : 3 (c) 9 : 7 (d) 4 : 6
- 35.** Two vessels are full with milk and water in the ratios 1 : 3 and 3 : 5, respectively. If both are mixed in the ratio 3 : 2, what is the ratio of milk and water in the new mixture?
 (a) 4 : 15 (b) 3 : 7
 (c) 6 : 7 (d) None of these
- 36.** A mixture contains milk and water in the ratio 5 : 1. On adding 5 L of water, the ratio of milk and water becomes 5 : 2. What is the quantity of milk in the original mixture?
 (a) 5 L (b) 25 L
 (c) 27.5 L (d) 32.5 L
- 37.** If x varies as the m th power of y , y varies as the n th power of z and x varies as the p th power of z , then which one of the following is correct?
 (a) $p = m + n$ (b) $p = m - n$
 (c) $p = mn$ (d) None of these
- 38.** The wages of labourers in a factory has increased in the ratio 22 : 25 and their number decreased in the ratio 3 : 2. What was the original wage bill of the factory, if the present bill is ₹ 5000?
 (a) ₹ 4000 (b) ₹ 6000
 (c) ₹ 8000 (d) None of these
- 39.** The ratio of A to B is $x : 8$ and the ratio of B to C is $12 : z$. If the ratio of A to C is $2 : 1$, then what is the ratio of $x : z$?
 (a) 2 : 3 (b) 3 : 2 (c) 4 : 3 (d) 3 : 4
- 40.** A bag contains ₹ 112 in the form of ₹ 1, 50 paise and 10 paise coins in the ratio 3 : 8 : 10. What is the number of 50 paise coins?
 (a) 112 (b) 108 (c) 96 (d) 84
- 41.** In a class, the number of boys is more than the number of girls by 12% of the total students. What is the ratio of number of boys to that of girls?
 (a) 11 : 14 (b) 14 : 11
 (c) 28 : 25 (d) 25 : 28
- 42.** A sum of ₹ 53 is divided among A, B, C in such a way that A gets ₹ 7 more than what B gets and B gets ₹ 8 more than what C gets. The ratio of their shares is
 (a) 25 : 18 : 10 (b) 6 : 7 : 8
 (c) 12 : 14 : 9 (d) 15 : 8 : 30
- 43.** A person P started a business with a capital of ₹ 2525 and another person Q joined P after some months with a capital of ₹ 1200. Out of the total annual profit of ₹ 1644, P 's share was ₹ 1212. When did Q join as partner?
 (a) After 2 months (b) After 3 months
 (c) After 4 months (d) After 5 months
- 44.** Seats for Mathematics, Physics and Biology in a school are in the ratio 5 : 7 : 8. There is a proposal to increase these seats by 40%, 50% and 75%, respectively. What will be the ratio of increased seats?
 (a) 2 : 3 : 4 (b) 6 : 7 : 8
 (c) 6 : 8 : 9 (d) 5 : 7 : 12

- 45.** The ratio between the number of passengers travelling by I and II class between the two railway stations is 1:50, whereas the ratio of I and II class fares between the same station is 3:1. If on a particular day, ₹ 1325 were collected from the passengers travelling between these stations, then what was the amount collected from the II class passengers?

(a) ₹ 750 (b) ₹ 1000
(c) ₹ 850 (d) ₹ 1250

- 46.** If x varies inversely as the square of y in such a way that, if $x = 1$, then $y = 6$.

I. If $y = 3$, then $x = 4$ II. If $y = 6$, then $x = 1$

Which of the following options is correct?

(a) Only I (b) Both I and II
(c) Only I (d) Neither I nor II

- 47.** If $a : b = c : d = e : f = 1 : 2$, then

$$\text{I. } \frac{3a+5c+7e}{3b+5d+7f} = \frac{1}{3} \quad \text{II. } \frac{a^2+c^2+e^2}{b^2+d^2+f^2} = \frac{1}{2}$$

Which of the following option is correct?

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

- 48.** A cat takes 5 leaps for every 4 leaps of a dog but 3 leaps of the dog are equal to 4 leaps of the cat. Now, the

I. Ratio of the speeds of the cat to that of the dog is 15 : 16.

II. Ratio of the distance of the cat to that of the dog is 15:16, covered in 30 min.

Which of the following option is correct?

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

- 49.** Determine the ratio of the number of people having characteristic X to the number of people having characteristic Y in a population of 100 subjects from the following table:

Having X and Y	10
Having X but not Y	30
Having Y but not X	20
Having neither X nor Y	40

(a) 4:3 (b) 3:2 (c) 1:2 (d) 2:3

- 50.** Fresh grapes contain 90 percent water by weight while dried grapes contain 20 percent water by weight. What is the weight of dry grapes contain 20 percent water by weight. What is the weight of dry grapes available from 20 kg of fresh grapes?

(a) 2 kg (b) 2.4 kg
(c) 2.5 kg (d) None of these

PREVIOUS YEARS' QUESTIONS

- 51.** If $x : y = 7 : 5$, then what is the value of $(5x - 2y) : (3x + 2y)$? ☑ 2012 I

(a) $5/4$ (b) $6/5$ (c) $25/31$ (d) $31/42$

- 52.** Two numbers are in the ratio 2 : 3. If 9 is added to each number, they will be in the ratio 3 : 4. What is the product of the two numbers? ☑ 2012 I

(a) 360 (b) 480 (c) 486 (d) 512

- 53.** A milkman bought 15 kg of milk and mixed 3 kg of water in it. If the price per kg of the mixture becomes ₹ 22, what is cost price of the milk per kg? ☑ 2012 II

(a) ₹ 28.00 (b) ₹ 26.40 (c) ₹ 24.00 (d) ₹ 22.60

- 54.** Sex ratio is defined as the number of females per 1000 males. In a place, the total inhabitants are 1935000, out of which 935000 are females. What is the sex ratio for the place? ☑ 2012 II

(a) 935 (b) 1000
(c) 1935 (d) 9350

- 55.** In a certain school, the ratio of boys to girls is 7 : 5. If there are 2400 students in the school, then how many girls are there? ☑ 2013 I

(a) 500 (b) 700 (c) 800 (d) 1000

- 56.** If $A : B = 2 : 3$, $B : C = 5 : 7$ and $C : D = 3 : 10$, then what is $A : D$ equal to? ☑ 2014 I

(a) 1 : 7 (b) 2 : 7 (c) 1 : 5 (d) 5 : 1

- 57.** $(x + y) : (x - y) = 3 : 5$ and $xy = \text{positive}$ imply that

(a) x and y are both positive ☑ 2014 II
(b) x and y are both negative
(c) one of them is positive and one of them is negative
(d) no real solution for x and y exists

- 58.** The ratio of ages of A and B is 2 : 5 and the ratio of ages of B and C is 3 : 4. What is the ratio of ages of A, B and C ? ☑ 2014 II

(a) 6 : 15 : 20 (b) 8 : 5 : 3
(c) 6 : 5 : 4 (d) 2 : 15 : 4

- 59.** The height of a tree varies as the square root of its age (between 5 to 17 yr). When the age of the tree is 9 yr, its height is 4 ft. What will be the height of the tree at the age of 16 yr? ☑ 2014 II

(a) 5 ft 4 inch (b) 5 ft 5 inch
(c) 4 ft 4 inch (d) 4 ft 5 inch

- 60.** 16 L of a mixture contains milk and water in the ratio 5 : 3. If 4 L of milk is added to this mixture, the ratio of milk to water in the new mixture would be ☑ 2015 I

(a) 2 : 1 (b) 7 : 3 (c) 4 : 3 (d) 8 : 3

- 61.** If $a:b=3:5$ and $b:c=7:8$, then $2a:3b:7c$ is equal to **2015 II**
 (a) 42 : 105 : 320 (b) 15 : 21 : 40
 (c) 6 : 15 : 40 (d) 30 : 21 : 350
- 62.** The speeds of three buses are in the ratio 2 : 3 : 4. The time taken by these buses to travel the same distance will be in the ratio. **2015 II**
 (a) 2 : 3 : 4 (b) 4 : 3 : 2
 (c) 4 : 3 : 6 (d) 6 : 4 : 3
- 63.** In a mixture of milk and water of volume 30 L, the ratio of milk and water is 7 : 3. The quantity of water to be added to the mixture to make the ratio of milk and water 1 : 2 is **2015 II**
 (a) 30 (b) 32
 (c) 33 (d) 35
- 64.** The annual incomes of two persons are in the ratio 9 : 7 and their expenses are in the ratio 4 : 3. If each of them saves ₹ 2000 per year, what is the difference in their annual incomes? **2016 I**
 (a) ₹ 4000 (b) ₹ 4500 (c) ₹ 5000 (d) ₹ 5500
- 65.** If $\frac{a}{b} = \frac{b}{c} = \frac{c}{d}$, then which of the following is/are correct?
 I. $\frac{b^3 + c^3 + d^3}{a^3 + b^3 + c^3} = \frac{d}{a}$ II. $\frac{a^2 + b^2 + c^2}{b^2 + c^2 + d^2} = \frac{a}{d}$
 Select the correct answer using the code given below. **2016 I**
 (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II

ANSWERS

1	b	2	a	3	c	4	b	5	c	6	a	7	d	8	d	9	a	10	b
11	c	12	b	13	d	14	c	15	c	16	b	17	d	18	c	19	b	20	b
21	b	22	b	23	b	24	a	25	c	26	c	27	a	28	d	29	b	30	c
31	d	32	c	33	d	34	a	35	b	36	b	37	c	38	d	39	c	40	a
41	b	42	a	43	b	44	a	45	d	46	b	47	d	48	c	49	a	50	c
51	c	52	c	53	b	54	a	55	d	56	a	57	d	58	a	59	a	60	b
61	c	62	d	63	c	64	a	65	a										

HINTS AND SOLUTIONS

- 1.** (b) The required compounded ratio is
 $\frac{2a}{6b} \times \frac{7a}{49b} \times \frac{3a}{12b} = \frac{a^3}{84b^3} = a^3 : 84b^3$
- 2.** (a) The subduplicate ratio of
 $16x^4 : 625y^6$ is
 $\frac{\sqrt{16x^4}}{\sqrt{625y^6}} = \frac{4x^2}{25y^3} = 4x^2 : 25y^3$
- 3.** (c) Duplicate ratio of $5 : 8 = \frac{5^2}{8^2}$
 $\Rightarrow \frac{5^2}{8^2} = \frac{x+5}{3x+4} \Rightarrow \frac{25}{64} = \frac{x+5}{3x+4}$
 $\Rightarrow 25(3x+4) = 64(x+5)$
 $\Rightarrow 75x + 100 = 64x + 320$
 $\Rightarrow 75x - 64x = 320 - 100$
 $\Rightarrow 11x = 220$
 $\therefore x = \frac{220}{11} = 20$
- 4.** (b) Given, $x : 6 :: 5 : 3$
 $\therefore \frac{x}{6} = \frac{5}{3} \Rightarrow 3x = 30 \Rightarrow x = \frac{30}{3} = 10$
- 5.** (c) Let the third proportional to 9 and 12 be x .
 $\therefore 9 : 12 :: 12 : x$
 $\Rightarrow \frac{9}{12} = \frac{12}{x} \Rightarrow 9x = 144$
 $\therefore x = \frac{144}{9} = 16$
- 6.** (a) Mean proportional between $(2 + \sqrt{3})$ and $(8 - \sqrt{48})$
 $= \sqrt{(2 + \sqrt{3})(8 - \sqrt{48})}$
 $= \sqrt{(2 + \sqrt{3})4(2 - \sqrt{3})}$
 $= 2\sqrt{(2 + \sqrt{3})(2 - \sqrt{3})}$
 $= 2\sqrt{4 - 3} = 2 \times 1 = 2$
 $[\because (a-b)(a+b) = a^2 - b^2]$
- 7.** (d) Let fourth proportional be x , then
 $7 : 11 :: 14 : x$
 $\Rightarrow \frac{7}{11} = \frac{14}{x} \Rightarrow 7x = 11 \times 14$
 $\Rightarrow x = 2 \times 11$
 $\therefore x = 22$
- 8.** (d) Let the consequent be x .
 $\therefore 35 : x = 7 : 8$
 Then, $\frac{35}{x} = \frac{7}{8}$ [by given condition]
 $\Rightarrow 7x = 35 \times 8 \therefore x = \frac{35 \times 8}{7} = 40$
- 9.** (a) Given, $x : y = 1 : 3$, $y : z = 5 : k$,
 $z : t = 2 : 5$ and $t : x = 3 : 4$
 $\frac{x}{y} \times \frac{y}{z} \times \frac{z}{t} \times \frac{t}{x} = 1$
 $\Rightarrow \frac{1}{3} \times \frac{5}{k} \times \frac{2}{5} \times \frac{3}{4} = 1 \Rightarrow k = \frac{1}{2}$
- 10.** (b) Given, $\frac{x}{y} = 25 \left(\frac{y}{x}\right)$
 $\Rightarrow \frac{x^2}{y^2} = \frac{25}{1} \Rightarrow \frac{x}{y} = \frac{5}{1}$ or, $5 : 1$
- 11.** (c) According to the question,
 $p\% \text{ of } x = t \text{ (} q\% \text{ of } y)$
 $\Rightarrow \frac{xp}{100} = \frac{yq}{100} \times t$
 $\therefore x : y = qt : p$

12. (b) $\frac{4ay - 3bx}{5ax - 2by}$ [dividing numerator and denominator by by]

$$\begin{aligned} &= \frac{\frac{4ay}{by} - \frac{3bx}{by}}{\frac{5ax}{by} - \frac{2by}{by}} = \frac{4\left(\frac{a}{b}\right) - 3\left(\frac{x}{y}\right)}{5\left(\frac{a}{b}\right)\left(\frac{x}{y}\right) - 2} \\ &= \frac{4\left(\frac{2}{3}\right) - 3\left(\frac{3}{4}\right)}{5\left(\frac{2}{3}\right)\left(\frac{3}{4}\right) - 2} = \frac{\frac{8}{3} - \frac{9}{4}}{\frac{5}{2} - 2} \\ &= \frac{\frac{32 - 27}{12}}{\frac{5 - 4}{2}} = \frac{12}{1} = \frac{5}{12} \times \frac{2}{1} = \frac{5}{6} \end{aligned}$$

13. (d) Given, $\frac{x^3 + 3x}{3x^2 + 1} = \frac{341}{91}$
By componendo and dividendo rule,
 $\frac{x^3 + 3x + 3x^2 + 1}{x^3 + 3x - 3x^2 - 1} = \frac{341 + 91}{341 - 91} = \frac{432}{250}$

$$\Rightarrow \frac{(x+1)^3}{(x-1)^3} = \frac{216}{125}$$

[$\because (a+b)^3 = a^3 + b^3 + 3ab(a+b)$]

On cube roots both sides, we get
 $\Rightarrow \frac{x+1}{x-1} = \frac{6}{5} \Rightarrow 5(x+1) = 6(x-1)$

$$\therefore x = 11$$

14. (c) Let the number be x

$$\begin{aligned} \therefore \frac{49+x}{68+x} &= \frac{3}{4} \\ \Rightarrow 196 + 4x &= 204 + 3x \\ \therefore x &= 8 \end{aligned}$$

15. (c) Let the three parts be A, B and C .
By given condition,

$$\begin{aligned} \frac{1}{2}A &= \frac{1}{3}B = \frac{1}{6}C = x \quad [\text{say}] \\ \therefore \frac{1}{2}A &= x \Rightarrow A = 2x \\ \frac{1}{3}B &= x \Rightarrow B = 3x \\ \frac{1}{6}C &= x \Rightarrow C = 6x \end{aligned}$$

As, $A + B + C = 1870$
[by given condition]

$$\begin{aligned} \therefore 2x + 3x + 6x &= 1870 \\ \Rightarrow 11x &= 1870 \\ \therefore x &= \frac{1870}{11} = 170 \end{aligned}$$

$\therefore$ Third part i.e. $C = 6 \times 170 = 1020$

16. (b) LCM of 20, 25, 30, 15 is 300.

$$\begin{aligned} \text{So, } \frac{7}{20} &= \frac{7}{20} \times \frac{15}{15} = \frac{105}{300} \\ \Rightarrow \frac{13}{25} &= \frac{13}{25} \times \frac{12}{12} = \frac{156}{300} \end{aligned}$$

$$\Rightarrow \frac{17}{30} = \frac{17}{30} \times \frac{10}{10} = \frac{170}{300}$$

$$\text{and } \frac{11}{15} = \frac{11}{15} \times \frac{20}{20} = \frac{220}{300}$$

As, $\frac{105}{300}$ is the smallest, so $\frac{7}{20}$ or, 7 : 20 is the smallest ratio.

17. (d) Let the numbers be A and B .

$$\therefore 0.5A = 0.07B \text{ [by given condition]}$$

$$\Rightarrow \frac{A}{B} = \frac{0.07}{0.5} = \frac{7}{50} \text{ i.e. } 7 : 50$$

18. (c) Let the ratio be x and $(x + 40)$.

Then, by given condition, $\frac{x}{x+40} = \frac{2}{7}$

$$\Rightarrow 7x = 2x + 80 \Rightarrow x = 16$$

So, the required ratio is 16 : 56.

19. (b) Given, $P : Q = \frac{3}{5} : \frac{5}{7}$... (i)

and $Q : R = \frac{3}{4} : \frac{2}{5}$... (ii)

From Eq. (i), $P : Q = \frac{3}{5} \times \frac{3}{4} : \frac{5}{7} \times \frac{3}{4}$
 $= \frac{9}{20} : \frac{15}{28}$... (iii)

From Eq. (ii), $Q : R = \frac{3}{4} \times \frac{5}{7} : \frac{2}{5} \times \frac{5}{7}$
 $= \frac{15}{28} : \frac{2}{7}$... (iv)

From Eqs. (iii) and (iv),

$$P : Q : R = \frac{9}{20} : \frac{15}{28} : \frac{2}{7}$$

20. (b) Let two numbers be $3x$ and $5x$, respectively.

According to the question,

$$(3x - 9) : (5x - 9) :: 12 : 23$$

$$\Rightarrow \frac{3x-9}{5x-9} = \frac{12}{23}$$

$$\Rightarrow 69x - 207 = 60x - 108$$

$$69x - 60x = 207 - 108 \Rightarrow 9x = 99$$

$$\therefore x = 11$$

$$\therefore \text{Second number} = 5x = 5 \times 11 = 55$$

21. (b) Let the numbers are a and b and its mean proportional is x

$$\therefore a : x :: x : b$$

$$a : 28 :: 28 : b (\because x = 28)$$

$$\Rightarrow 28^2 = ab$$
 ... (i)

Again, $a : b :: b : 224$

$$\Rightarrow b^2 = a \times 224 \Rightarrow a = \frac{b^2}{224}$$
 ... (ii)

From Eq. (i) say,

$$28^2 = \frac{b^2}{224} \times b$$

$$\Rightarrow b^3 = 28 \times 28 \times 224$$

$$\Rightarrow b = 56$$

Again from Eq. (i) say,

$$a = \frac{28 \times 28}{56} = 14$$

22. (b) As, $A : (B + C) = 2 : 9$ [given]

$$\text{Sum of ratios} = 2 + 9 = 11$$

$$\text{So, } A's \text{ part} = 770 \times \frac{2}{11} = ₹ 140$$

23. (b) Given, $x \propto y$ and $x \propto \frac{1}{z^2}$

$$\text{Now, } x \propto \frac{y}{z^2} \Rightarrow x = \frac{ky}{z^2}$$

$$\therefore x = 10, y = 4 \text{ and } z = 14$$

$$\therefore 10 = \frac{k \cdot 4}{196} \Rightarrow k = \frac{1960}{4} = 490$$

Now, $z = 7$ and $y = 16$, then

$$x = \frac{490 \times 16}{7 \times 7} = 160$$

24. (a) $\frac{x}{1} = \frac{\sqrt{m+3n} + \sqrt{m-3n}}{\sqrt{m+3n} - \sqrt{m-3n}}$

By componendo and dividendo rule,

$$\begin{aligned} \frac{x+1}{x-1} &= \frac{\sqrt{m+3n} + \sqrt{m-3n}}{\sqrt{m+3n} - \sqrt{m-3n}} \\ &\quad + \frac{\sqrt{m+3n} - \sqrt{m-3n}}{-\sqrt{m+3n} + \sqrt{m-3n}} \\ \Rightarrow \frac{x+1}{x-1} &= \frac{\sqrt{m+3n}}{\sqrt{m-3n}} \end{aligned}$$

On squaring both sides, we get

$$\frac{(x+1)^2}{(x-1)^2} = \frac{m+3n}{m-3n}$$

Again applying componendo and dividendo rule,

$$\begin{aligned} \frac{(x+1)^2 + (x-1)^2}{(x+1)^2 - (x-1)^2} &= \frac{m+3n+m-3n}{m+3n-m+3n} \\ \Rightarrow \frac{2(x^2+1)}{2(2x)} &= \frac{2m}{2(3n)} \end{aligned}$$

$$\Rightarrow \frac{x^2+1}{2x} = \frac{m}{3n} \Rightarrow 3n(x^2+1) = 2mx$$

$$\therefore 3nx^2 - 2mx + 3n = 0$$

25. (c) Given, $p + r = 2q \Rightarrow 2 = \frac{p+r}{q}$

$$\text{and } \frac{1}{q} + \frac{1}{s} = \frac{2}{r}$$

$$\Rightarrow \frac{1}{q} + \frac{1}{s} = \frac{p+r}{qr} \Rightarrow \frac{s+q}{sq} = \frac{p+r}{qr}$$

$$\Rightarrow r(s+q) = s(p+r)$$

$$\Rightarrow rq = sp \Rightarrow \frac{p}{q} = \frac{r}{s}$$

$$\therefore p : q = r : s$$

26. (c) As, given $q^2 = pr$

$$\frac{p^2 - q^2 + r^2}{p^{-2} - q^{-2} + r^{-2}} = \frac{p^2 - q^2 + r^2}{\frac{1}{p^2} - \frac{1}{pr} + \frac{1}{r^2}}$$

$$= \frac{p^2 - q^2 + r^2}{\frac{(p^2 r^2)(p^2 - q^2 + r^2)}{r^2 - pr + p^2}} = \frac{(p^2 r^2)(p^2 - q^2 + r^2)}{(r^2 - pr + p^2) p^2 r^2}$$

$$= \frac{(p^2 r^2)(p^2 - pr + r^2)}{(r^2 - pr + p^2)}$$

$$= p^2 r^2 = (pr)^2 = (q^2)^2 = q^4$$
27. (a) Let monthly incomes of A's and B's be $4x$ and $3x$ respectively
 Each saving = ₹ 600
 $\therefore$ Income - saving = Expenditure
 $\therefore \frac{4x - 600}{3x - 600} = \frac{3}{2}$
 $\Rightarrow 2(4x - 600) = 3(3x - 600)$
 $8x - 1200 = 9x - 1800$
 $\Rightarrow 1800 - 1200 = 9x - 8x \Rightarrow x = 600$
 $\therefore$ A's income = $4x = 4 \times 600 = ₹ 2400$
28. (d) Let three numbers be $3x$, $2x$ and $5x$, respectively.
 According to the question,
 $(3x)^2 + (2x)^2 + (5x)^2 = 1862$
 $\Rightarrow 9x^2 + 4x^2 + 25x^2 = 1862$
 $\Rightarrow x^2 = \frac{1862}{38} = 49 \quad \therefore x = 7$
 Hence, the numbers are 3×7 , 2×7 and 5×7 i.e. 21, 14 and 35.
29. (b) Speed $\propto \frac{1}{\text{Time}}$
 $\therefore$ Required ratio = $\frac{1}{4} : \frac{1}{3} : \frac{1}{2} = 3 : 4 : 6$
30. (c) Let the ages of A and B be $2x$ yr and $5x$ yr, respectively.
 After 8 yr age of A = $(2x + 8)$ yr
 and age of B = $(5x + 8)$ yr
 According to the question, $\frac{2x + 8}{5x + 8} = \frac{1}{2}$
 $\Rightarrow 4x + 16 = 5x + 8$
 $\Rightarrow x = 8$
 $\therefore$ Difference between their present ages
 $= 5x - 2x = 3x = 3 \times 8 = 24$ yr
31. (d) By given condition,
 $y = lx + \frac{m}{x} \quad \dots(i)$
 where, l and m are proportionality constants.
 when $y = 6$ and $x = 4$,
 then $6 = 4l + \frac{m}{4}$
 $\Rightarrow 16l + m = 24 \quad \dots(ii)$

when $y = \frac{10}{3}$ and $x = 3$,
 then $\frac{10}{3} = 3l + \frac{m}{3} \Rightarrow 9l + m = 10 \dots(iii)$
 On solving Eqs. (ii) and (iii),
 $l = 2$ and $m = -8$
 From Eq. (i), $y = 2x - \frac{8}{x}$

32. (c) Given, $A : B : C = \frac{1}{5} : \frac{1}{6} : \frac{1}{10}$
 $= 6 : 5 : 3$
 $\therefore$ Share of A = $\frac{6}{6 + 5 + 3} \times 8400$
 $= \frac{6}{14} \times 8400 = ₹ 3600$
33. (d) Let the amount be ₹ x
 In first condition, Q's part = $\frac{5x}{5 + 3} = \frac{5}{8}x$
 In second condition, Q's part
 $= \frac{3x}{2 + 3} = \frac{3}{5}x$
 By given condition, $\frac{5}{8}x - \frac{3}{5}x = 10$
 $\Rightarrow \frac{x}{40} = 10 \Rightarrow x = ₹ 400$
34. (a) Remaining mixture = $20 - 6 = 14$ L
 Quantity of milk = $14 \times \frac{4}{7} = 8$ L
 $\therefore$ Water contents = $14 - 8 = 6$ L
 $\therefore$ Milk in new mixture
 $= 8 + 6 = 14$ L and water = 6 L
 $\therefore$ Ratio of milk and water in new mixture = $\frac{14}{6} = \frac{7}{3}$
 Hence, the required ratio 7 : 3.
35. (b) Let x be the part of milk in new mixture.
 By law of mixture,
 $\therefore \frac{\frac{3}{8} - x}{x - \frac{1}{4}} = \frac{3}{2} \Rightarrow \frac{3}{4} - 2x = 3x - \frac{3}{4}$
 $\Rightarrow 5x = \frac{6}{4} = \frac{3}{2} \quad \therefore x = \frac{3}{10}$
 Required ratio = $\frac{3}{10 - 3} = \frac{3}{7}$ or, 3 : 7
36. (b) Let quantities of milk and water be $5x$ and x L, respectively.
 According to the question, $\frac{5x}{x + 5} = \frac{5}{2}$
 $\Rightarrow 10x = 5x + 25$
 $\therefore x = 5$
 Hence, the quantity of milk in the original mixture = $5 \times 5 = 25$ L

37. (c) Given, $x \propto y^m \quad \dots(i)$
 $y \propto z^n \quad \dots(ii)$
 $x \propto z^p \quad \dots(iii)$
 On putting the values of x and y from Eqs. (ii) and (iii) in Eq. (i), we get
 $z^p \propto (z^n)^m \Rightarrow z^p \propto z^{nm}$
 $\therefore p = mn$
38. (d) Let the initial and final salary be ₹ $22x$ and ₹ $25x$, respectively.
 and let initial and final number of employees be $3y$ and $2y$, respectively.
 $\therefore$ Present bill = Final salary $\times$ Final number of employees
 $5000 = 25x \times 2y$
 $\Rightarrow \frac{5000}{50} = xy \Rightarrow xy = 100$
 $\therefore$ Original bill = Initial salary $\times$ Initial number of employees
 $= 22x \times 3y$
 $= 66xy = 66 \times 100 = ₹ 6600$
39. (c) Given,
 $A : B = x : 8$ and $B : C = 12 : z$
 $\Rightarrow A : C = \frac{A}{C} = \frac{A}{B} \times \frac{B}{C} = \frac{x}{8} \times \frac{12}{z} = \frac{3x}{2z}$
 But $A : C = 2 : 1$
 $\Rightarrow \frac{3x}{2z} = \frac{2}{1} \Rightarrow x : z = 4 : 3$
40. (a) Let the number of coins of ₹ 1, 50 paise and 10 paise be $3x$, $8x$ and $10x$, respectively
 According to the question,
 $\frac{3x}{1} + \frac{8x}{2} + \frac{10x}{10} = 112$
 $\Rightarrow 3x + 4x + x = 112$
 $\Rightarrow x = \frac{112}{8} = 14$
 $\therefore$ Number of 50 paise coins
 $= 14 \times 8 = 112$
41. (b) Let the number of boys and girls be x and y , respectively.
 $\therefore$ Total number of students = $(x + y)$
 According to the question,
 Number of boys = Number of girls
 $+ \text{Total Number of students} \times 12\%$
 $x = y + (x + y) \times 12\%$
 $x - y = \frac{(x + y) \times 12}{100}$
 $\Rightarrow 25x - 25y = 3x + 3y$
 $\Rightarrow 22x = 28y$
 $\therefore x : y = 14 : 11$
42. (a) Let C gets ₹ x .
 Now, by given conditions,
 Amount of B = ₹ $(x + 8)$
 Amount of A = ₹ $(x + 15)$

$$\begin{aligned}\therefore x + (x + 8) + (x + 15) &= 53 \\ \Rightarrow x &= 10 \\ \therefore A : B : C &= (10 + 15) : (10 + 8) : 10 \\ &= 25 : 18 : 10\end{aligned}$$

43. (b) Let Q join for x months.

$$\begin{aligned}\therefore \text{Ratio of capital} &= 2525 \times 12 : 1200 \times x \\ &= 2525 : 100x = 101 : 4x\end{aligned}$$

$$\therefore P\text{'s profit} = \frac{101}{101 + 4x} \times 1644$$

$$\Rightarrow 1212 = \frac{101 \times 1644}{101 + 4x}$$

$$\Rightarrow \frac{1}{137} = \frac{1}{101 + 4x}$$

$$\Rightarrow 4x = 36 \Rightarrow x = 9$$

Hence, Q joined for 9 months i.e. he joined after 3 months.

44. (a) Originally, let the numbers of seats for Mathematics, Physics and Biology be $5x$, $7x$ and $8x$, respectively.

Number of increased seats are (140% of $5x$) (150% of $7x$) and (175% of $8x$).

Required ratio can be obtained as

$$\begin{aligned}\Rightarrow \left(\frac{140}{100} \times 5x\right) : \left(\frac{150}{100} \times 7x\right) : \left(\frac{175}{100} \times 8x\right) \\ \Rightarrow \left(7x : \frac{21}{2}x : 14x\right) \times 2 \\ \Rightarrow (14x : 21x : 28x) \div 7x \\ \therefore 2 : 3 : 4\end{aligned}$$

45. (d) Let the number of passengers travelling by class I and class II be x and $50x$ respectively.

Then, amount collected from class I and II will be ₹ $3 \times x$ and ₹ $50x$ respectively.

$$\text{Given, } 3x + 50x = 1325$$

$$53x = 1325 \Rightarrow x = 25$$

$$\therefore \text{Amount collected from class II}$$

$$= 50 \times 25 = ₹ 1250$$

46. (b) As $x \propto \frac{1}{y^2} \Rightarrow x = \frac{k}{y^2}$... (i)

$$\text{If } x = 1$$

$$\text{and } y = 6$$

$$1 = \frac{k}{6^2} \Rightarrow k = 36$$

On putting the value of k in Eq. (i), we get

$$x = \frac{36}{y^2} \quad \dots (ii)$$

$$\text{I. On putting } y = 3 \text{ in Eq. (ii), } x = \frac{36}{9}$$

$$x = 4$$

II. On putting $y = 6$ in Eq. (ii), we get

$$x = \frac{36}{36} = 1$$

$$x = 1$$

Both statements I and II are correct.

47. (d) Given that, $\frac{a}{b} = \frac{c}{d} = \frac{e}{f} = \frac{1}{2}$

$$\Rightarrow a = \frac{b}{2}, c = \frac{d}{2}, e = \frac{f}{2}$$

$$\begin{aligned}\text{I. } \frac{3a + 5c + 7e}{3b + 5d + 7f} &= \left(\frac{\frac{3b}{2} + \frac{5d}{2} + \frac{7f}{2}}{3b + 5d + 7f} \right) \\ &= \frac{1}{2} \frac{(3b + 5d + 7f)}{3b + 5d + 7f} = \frac{1}{2}\end{aligned}$$

$$\text{II. } \frac{a^2 + c^2 + e^2}{b^2 + d^2 + f^2} = \frac{\frac{b^2}{4} + \frac{d^2}{4} + \frac{f^2}{4}}{b^2 + d^2 + f^2} = \frac{1}{4}$$

$\therefore$ Neither I nor II correct.

48. (c) I. 4 leaps of cat = 3 leaps of dog

$$\Rightarrow 1 \text{ leap of cat} = \frac{3}{4} \text{ leap of dog}$$

Cat takes 5 leap for every 4 leaps of dog.

$\therefore$ Required Ratio

$$= (5 \times \text{cat's leap}) : (4 \times \text{dog's leap})$$

$$= \left(5 \times \frac{3}{4} \text{ dog's leap}\right) : (4 \times \text{dog's leap})$$

$$= 15 : 16$$

$$\text{Thus, } \frac{\text{Speed of cat}}{\text{Speed of dog}} = \frac{15}{16}$$

$$\text{II. } \frac{\text{Distance (cat)}}{\text{Distance (dog)}} = \frac{s(\text{cat}) \times t}{s(\text{dog}) \times t}$$

$$= \frac{s(\text{cat}) \times 30}{s(\text{dog}) \times 30} = \left(\frac{s(\text{cat})}{s(\text{dog})} = \frac{15}{16} \right)$$

Thus, both statements I and II are correct.

49. (a) Number of people having characteristic X

$$= 10 + 30 = 40$$

$$\text{Number of people having characteristic } Y$$

$$= 10 + 20 = 30$$

$$\text{Required ratio} = 40 : 30 = 4 : 3.$$

50. (c) Fresh grapes contain 10% pulp.

$$\therefore 20 \text{ kg fresh grapes contain 2 kg pulp.}$$

$$\text{Dry grapes contain 80\% pulp.}$$

$$2 \text{ kg pulp would contain}$$

$$\frac{2}{0.8} = \frac{20}{8} = 2.5 \text{ kg dry grapes.}$$

51. (c) Refer to question 12

52. (c) Refer to question 20

$$\therefore \text{Their product} = 18 \times 27 = 486$$

53. (b) Let cost price of milk be ₹ x per kg.

Then, according to the rule of mixture,

$$22 : (x - 22) = 15 : 3$$

$$\Rightarrow \frac{22}{x - 22} = \frac{15}{3}$$

$$\Rightarrow \frac{22}{x - 22} = 5$$

$$\therefore x = ₹ 26.40$$

54. (a) Total number of inhabitants = 1935000

$$\text{Total number of females} = 935000$$

$$\therefore \text{Total number of males}$$

$$= 1935000 - 935000 = 1000000$$

$$\therefore \text{sex ratio} = \frac{935000}{1000000} \times 1000 = 935$$

55. (d) Given, ratio of boys to girls = 7 : 5

$$\text{and total number of students} = 2400$$

$$\text{Sum of ratio} = 7 + 5 = 12$$

$$\therefore \text{Number of girls} = \frac{5}{12} \times 2400 = 1000$$

56. (a) Given, $A : B = 2 : 3$, $B : C = 5 : 7$

$$\text{and } C : D = 3 : 10$$

$$\begin{aligned}\therefore \frac{A}{D} &= \frac{A}{B} \times \frac{B}{C} \times \frac{C}{D} \\ &= \frac{2}{3} \times \frac{5}{7} \times \frac{3}{10} = \frac{1}{7}\end{aligned}$$

$$\therefore A : D = 1 : 7$$

57. (d) Given that, $(x + y) : (x - y) = 3 : 5$

$$\therefore \frac{x + y}{x - y} = \frac{3}{5}$$

On applying componendo and dividendo rule, we get

$$\frac{(x + y) + (x - y)}{(x + y) - (x - y)} = \frac{3 + 5}{3 - 5}$$

$$\Rightarrow \frac{2x}{2y} = \frac{8}{-2} \Rightarrow \frac{x}{y} = -4$$

$$\therefore x = -4y$$

But it is given, xy is positive

$$\therefore -4y \times y = \text{positive}$$

$$\Rightarrow -4y^2 = \text{positive, which is not possible.}$$

Hence, no real solution for x and y exists.

58. (a) Given, ratio of ages of A and B i.e.

$$A : B = 2 : 5$$

$$\text{Ratio of ages of } B \text{ and } C \text{ i.e.}$$

$$B : C = 3 : 4$$

$$\therefore \text{Ratio of ages of } A, B \text{ and } C = \frac{2}{3} : \frac{5}{4} : 1$$

$$= 2 \times 3 : 3 \times 5 : 5 \times 4 = 6 : 15 : 20$$

59. (a) Let height of tree be h ft and age be a yr.

Then, according to the question,

$$h \propto \sqrt{a} \Rightarrow h = k\sqrt{a} \quad \dots (i)$$

where, k is a constant.

If age = 9 yr, then height = 4 ft

From Eq. (i), we get $4 = k\sqrt{9}$

$$\Rightarrow 4 = k \times 3 \therefore k = \frac{4}{3}$$

If age = 16 yr, then $h = k\sqrt{16} = \frac{4}{3} \times 4$

$$\left[\therefore k = \frac{4}{3} \right]$$

$$\therefore h = \frac{16}{3} \text{ ft} = 5\frac{1}{3} = 5 \text{ ft } 4 \text{ inch}$$

60. (b) Refer to question 34.

61. (c) We have, $\frac{a}{b} = \frac{3}{5}$

$$\therefore b = \frac{5a}{3} \text{ and } \frac{b}{c} = \frac{7}{8}$$

$$\Rightarrow c = \frac{8}{7}b = \frac{8}{7} \times \frac{5a}{3} = \frac{40a}{21}$$

$$\therefore a : b : c = a : \frac{5a}{3} : \frac{40a}{21}$$

$$\text{Now, } 2a : 3b : 7c = 2a : 5a : \frac{40a}{3}$$

$$= 2 : 5 : \frac{40}{3} = 6 : 15 : 40$$

62. (d) Let the distance be x .

Now, ratio of time taken to travel the distance by each bus is

$$\frac{x}{2} : \frac{x}{3} : \frac{x}{4} \Rightarrow \frac{12}{2} : \frac{12}{3} : \frac{12}{4}$$

$$\Rightarrow 6 : 4 : 3$$

63. (c) Since, milk = $7x$ and water = $3x$

$$\text{Then, } 7x + 3x = 30 \Rightarrow 10x = 30$$

$$\therefore x = 3$$

So, milk = 21 L and water = 9 L

Let y L of water to be added.

$$\text{Then, } \frac{21}{9+y} = \frac{1}{2}$$

$$\Rightarrow 9 + y = 42$$

$$\therefore y = 33 \text{ L}$$

64. (a) Refer to question 27.

65. (a) $\frac{a}{b} = \frac{b}{c} = \frac{c}{d} = k$ (say)

$$\Rightarrow a = bk, b = ck, c = dk$$

$$\therefore a = dk^3, b = dk^2, c = dk$$

$$\text{I. } \frac{b^3 + c^3 + d^3}{a^3 + b^3 + c^3}$$

$$= \frac{b^3 + c^3 + d^3}{(bk)^3 + (ck)^3 + (dk)^3}$$

$$= \frac{1(b^3 + c^3 + d^3)}{k^3(b^3 + c^3 + d^3)} = \frac{1}{k^3} = \frac{d}{a}$$

Hence, statement I is correct.

$$\text{II. Similarly } \frac{a^2 + b^2 + c^2}{b^2 + c^2 + d^2} = k^2 = \frac{b}{d}$$

Hence, statement II is incorrect.

13

LOGARITHM

Usually (2-3) questions have been asked from this chapter. Generally questions are based on fundamental rules of logarithm. It is especially used to solve complicated mathematical calculations. Applications of square, cube, surds and indices are extremely used here.



DEFINITION

If 'a' is a positive real number, other than 1 and 'b' is a rational number such that $a^b = N$, then we say that logarithm of N to base 'a' is b or 'b' is the logarithm of N to base 'a', written as $\log_a N = b$

So, $a^b = N \Leftrightarrow \log_a N = b$, e.g. $7^0 = 1 \Leftrightarrow \log_7 1 = 0$, $81^{1/4} = 3 \Leftrightarrow \log_{81} 3 = 1/4$

EXAMPLE 1. Which one of the following has the value of $x = 4$?

a. $\log_2 x = 3$

b. $\log_5 x = 3$

c. $\log_{81} x = \frac{3}{2}$

d. $\log_{\sqrt{2}} x = 4$

Sol. d. a. $\log_2 x = 3 \Leftrightarrow x = 2^3 = 8$ b. $\log_5 x = 3 \Leftrightarrow x = 5^3 = 125$ [by definition]

c. $\log_{81} x = 3/2 \Leftrightarrow x = (81)^{3/2} = (3^4)^{3/2}, \Leftrightarrow x = 3^6 \Leftrightarrow x = 729$

d. $\log_{\sqrt{2}} x = 4 \Leftrightarrow x = (\sqrt{2})^4, x = (2^{1/2})^4 = 2^2 = 4$

EXAMPLE 2. If $\log_5 y = x$ and $\log_2 z = x$, then the value of 20^x in terms of y and z is

a. zy

b. z^2y

c. xy^2

d. z/y

Sol. b. Given, $\log_5 y = x$ and $\log_2 z = x \Rightarrow y = 5^x$ and $z = 2^x$... (i)

Now, $20^x = (2^2 \times 5)^x = 2^{2x} \times 5^x = (2^x)^2 \cdot 5^x = z^2 \cdot y$ [from Eq. (i)]

$\therefore 20^x = z^2 \cdot y$

Fundamental Rules of Logarithms

Rule 1 If m and n are positive rational number, then $\log_a (mn) = \log_a m + \log_a n$

If $m_1, m_2, m_3, \dots, m_n$ are positive rational numbers, then

$$\log_a (m_1 m_2 m_3 \dots m_n) = \log_a m_1 + \log_a m_2 + \dots + \log_a m_n$$

EXAMPLE 3. The value of $\log_{10} (15)$ is

- a. $\log_{10} 2 - \log_{10} 5$ b. $\log_{10} 5$
 c. $\log_{10} 3 + \log_{10} 5$ d. $\log_{10} 2 + \log_{10} 5$

Sol. c. $\log_{10} 15 = \log_{10} (3 \times 5) = \log_{10} 3 + \log_{10} 5$

(Rule 2) If m and n are positive rational numbers, then

$$\log_a \left(\frac{m}{n} \right) = \log_a m - \log_a n$$

EXAMPLE 4. The value of x in

$$\log x - \log (x - 1) = \log 3 \text{ is}$$

- a. $2/3$ b. $3/2$ c. $1/2$ d. $1/4$

Sol. b. $\log x - \log (x - 1) = \log 3$

$$\Rightarrow \log \frac{x}{x-1} = \log 3 \Rightarrow \frac{x}{x-1} = 3$$

$$\Rightarrow x = 3(x-1) \Rightarrow x = 3x - 3 \Rightarrow -2x = -3$$

$$\therefore x = 3/2$$

(Rule 3) If m and n are positive rational numbers, then,

$$\log_a (m^n) = n \log_a m$$

EXAMPLE 5. The value of x in

$$\log_x 4 + \log_x 16 + \log_x 64 = 12 \text{ is}$$

- a. 1 b. 2 c. 3 d. 4

Sol. b. $\log_x 4 + \log_x 16 + \log_x 64 = 12$

$$\Rightarrow \log_x 2^2 + \log_x 2^4 + \log_x 2^6 = 12$$

$$\Rightarrow 2 \log_x 2 + 4 \log_x 2 + 6 \log_x 2 = 12$$

$$\Rightarrow 12 \log_x 2 = 12 \Rightarrow \log_x 2 = 1$$

$$\therefore x = 2$$

EXAMPLE 6. The value of x in $\frac{\log 256}{\log 16} = \log x$ is

- a. 1 b. 10 c. 100 d. 0

Sol. c. $\frac{\log 256}{\log 16} = \frac{\log 16^2}{\log 16} = \frac{2 \log 16}{\log 16} = 2$

$$\Rightarrow \log x = 2 \Rightarrow \log_{10} x = 2 \quad [\because \log x = \log_{10} x]$$

$$\therefore x = 10^2 = 100$$

(Rule 4) If m is a positive rational number and a, b are positive real numbers i.e. $a \neq 1$ and $b \neq 1$, then

$$\log_a m = \frac{\log_b m}{\log_b a}$$

EXAMPLE 7. The value of

$$\log_3 4 \cdot \log_4 5 \cdot \log_5 6 \cdot \log_6 7 \cdot \log_7 8 \cdot \log_8 9 \text{ is}$$

- a. 1 b. 2 c. -1 d. None of these

Sol. b. $\log_3 4 \cdot \log_4 5 \cdot \log_5 6 \cdot \log_6 7 \cdot \log_7 8 \cdot \log_8 9$

$$= \frac{\log 4}{\log 3} \times \frac{\log 5}{\log 4} \times \frac{\log 6}{\log 5} \times \frac{\log 7}{\log 6} \times \frac{\log 8}{\log 7} \times \frac{\log 9}{\log 8}$$

$$= \frac{\log 9}{\log 3} = \frac{\log 3^2}{\log 3} = \frac{2 \log 3}{\log 3} = 2$$

SOME USEFUL RESULTS

- logarithm of 1 with any base is always zero i.e. $\log_a 1 = 0$
- logarithm of base to itself is always 1 i.e. $\log_a a = 1$
- $x > y \Rightarrow \log_a x > \log_a y$ for $a > 1$
- $x > y \Rightarrow \log_a x < \log_a y$ for $0 < a < 1$
- for $x > 0, 1 \neq a > 0, \log_a n(x) = \frac{1}{n} \log_a x$
- for $1 \neq a, b > 0, x > 0, a \log_b x = x \log_b a$
- If ' a ' is a positive real number and n is a positive rational number, then
 - (i) $\log_a n^x = \frac{x}{y} \log_a n$.
 - (ii) $a^{\log_a n} = n$
 - (iii) $\log_a n = \log_{a^2} n^2 = \log_{a^3} n^3 = \log_{a^m} n^m$

Common Logarithm and Natural Logarithm

Logarithm can have any positive base other than 0. But there are two most important base which are generally used.

1. **Common Logarithm** Logarithm to the base '10' is called common logarithm. It is also called Brigg's logarithm.

$$\text{e.g. } \log 100 = \log_{10} 100 = 2$$

$$\log 1000 = \log_{10} 1000 = 3$$

2. **Natural Logarithm** Logarithm to the base e is called natural logarithm. It is also called Napier logarithm. $\log_e x$ is usually denoted by $\ln x$. e is irrational number between 2 and 3.

$$\ln x = y \Rightarrow e^y = x$$

ANTILOGARITHM

The positive number ' a ' is called the antilogarithm of a number b , if $\log a = b$. If ' a ' is antilogarithm of b , it is written as $a = \text{antilog } b$.

$$\text{So, } a = \text{antilog } b \Leftrightarrow \log a = b$$

Characteristic and Mantissa of a Logarithm

The logarithm of positive real number ' n ' consists of two parts

1. The integral part is known as the characteristic. It is always an integer (positive, negative or zero).
2. The decimal part is called as the mantissa. The mantissa is never negative and is always less than one.

To Find the Characteristic

Case I The characteristic of the log of a number greater than 1 is positive and numerically one less than the number of digits to the left of the decimal part.

e.g.

Value	Characteristic
$\log 3.257$	$1 - 1 = 0$
$\log 32.57$	$2 - 1 = 1$
$\log 325.7$	$3 - 1 = 2$

Case II The characteristic of log of a number less than 1 is negative and numerically one more than the number of zeroes immediately after the decimal point. It is represented by bar over the digit.

The bar over the characteristic indicates that it is negative.

e.g.

Value	Characteristic
$\log 0.3257$	$-(0 + 1) = -1$ i.e. $\bar{1}$
$\log 0.03257$	$-(1 + 1) = -2$ i.e. $\bar{2}$
$\log 0.003257$	$-(2 + 1) = -3$ i.e. $\bar{3}$

Rules for Inserting Decimal Point

Two rules are used for inserting a decimal point.

Rule 5 When the characteristic of the logarithm is positive, we insert the decimal point after the $(n + 1)$ th digit, where n is the characteristic.

Rule 6 When the characteristic of the logarithm is negative, we insert the decimal point such that the

first significant figures is at n th place, where $\bar{n}$ is the characteristic.

EXAMPLE 8. The characteristic of the logarithm of the number 0.00000014 is

- a. 1 b. 7
c. -7 d. None of these

Sol. c. $0.00000014 = 1.4 \times \frac{1}{10000000} = 1.4 \times 10^{-7}$

$\therefore$ The characteristic of $\log 0.00000014 = -7$

EXAMPLE 9. The characteristic of the logarithm of the number 8.188 is

- a. 2 b. 1
c. -1 d. 0

Sol. d. $8.188 = 8.188 \times 1 = 8.188 \times 10^0$

$\therefore$ The characteristic of $\log 8.188 = 0$

EXAMPLE 10. The characteristic of the logarithm of the number 566.37 is

- a. 2 b. -2
c. 4 d. None of these

Sol. a. $566.37 = 56637 \times 10^{-2}$

$\therefore$ The characteristic of $\log 566.37 = 2$

EXAMPLE 11. The characteristic of the logarithm of the number 313 is

- a. 3 b. 2
c. 1 d. 0

Sol. b. $313 = 3.13 \times 10^2$

$\therefore$ The characteristic of $\log 313 = 2$

SOME IMPORTANT POINTS

1. The base of a logarithm is never taken as zero and negative.
2. Log of negative integers are not defined and also $\log_e 0$ is not defined.
3. Logarithmic function is positive as well as negative but exponential function is always positive.

PRACTICE EXERCISE

- What is the value of $\log_{100} 0.1$?
(a) $1/2$ (b) $-1/2$ (c) -2 (d) 2
- The value of $3 \log 3 + 2 \log 2$ is
(a) $\log 108$ (b) $\log 106$ (c) $\log 109$ (d) None of these
- If $\log_a \sqrt{2} = \frac{1}{6}$, then the value of a is
(a) $(\sqrt{2})^6$ (b) $(6)^{1/2}$ (c) 3 (d) -6
- If $\log_3 x = -2$, then the value of x is
(a) $\frac{1}{9}$ (b) $-\frac{1}{9}$ (c) $\frac{1}{8}$ (d) $-\frac{1}{8}$
- Find the logarithm of 1728 to the base $2\sqrt{3}$.
(a) 3.124 (b) 3.1732 (c) 6 (d) 5
- What is the value of
 $(\log_{1/2} 2)(\log_{1/3} 3)(\log_{1/4} 4) \dots (\log_{1/1000} 1000)$?
(a) 1 (b) -1 (c) 1 or -1 (d) 0
- What is the value of
 $\frac{1}{2} \log_{10} 25 - 2 \log_{10} 3 + \log_{10} 18$?
(a) 2 (b) 3 (c) 1 (d) 0
- What is the value of $[\log_{10} (5 \log_{10} 100)]^2$?
(a) 4 (b) 3 (c) 2 (d) 1
- The value of $\log_y x \cdot \log_z y \cdot \log_x z$ is
(a) $\log xyz$ (b) xyz (c) 1 (d) 0
- The value of $\log_3 (27 \times \sqrt[4]{9} \times \sqrt[3]{9})$ is
(a) 4 (b) $4\frac{1}{3}$ (c) $8\frac{1}{3}$ (d) $4\frac{1}{6}$
- The value of $\log_2 [\log_2 \log_2 \log_2 (65536)]$ is
(a) 8 (b) 16 (c) 4 (d) 1
- What is the value of $[\log_{13}(10)]/[\log_{169}(10)]$?
(a) $\frac{1}{2}$ (b) 2 (c) 1 (d) $\log_{10} 13$
- What is the value of
 $\left(\frac{1}{3} \log_{10} 125 - 2 \log_{10} 4 + \log_{10} 32 + \log_{10} 1\right)$?
(a) 0 (b) $\frac{1}{5}$ (c) 1 (d) $\frac{2}{5}$
- If $\log_r 6 = m$ and $\log_r 3 = n$, then what is $\log_r (r/2)$ equal to?
(a) $m - n + 1$ (b) $m + n - 1$ (c) $1 - m - n$ (d) $1 - m + n$
- What is $\log_{10} \left(\frac{3}{2}\right) + \log_{10} \left(\frac{4}{3}\right) + \log_{10} \left(\frac{5}{4}\right) + \dots$ upto 8 terms equal to?
(a) 0 (b) 1 (c) $\log_{10} 5$ (d) None of these
- The value of $\frac{1}{\log_{xy}(xyz)} + \frac{1}{\log_{yz}(xyz)} + \frac{1}{\log_{zx}(xyz)}$ is
(a) xyz (b) 2 (c) 0 (d) 1
- The value of
 $\frac{1}{1 + \log_x(yz)} + \frac{1}{1 + \log_y(xz)} + \frac{1}{1 + \log_z(xy)}$ is
(a) 1 (b) $\frac{1}{xy^2}$ (c) $x = yz$ (d) 0
- If $\log_4 (x^2 + x) - \log_4 (x + 1) = 2$, then the value of x is
(a) 4 (b) 8 (c) 16 (d) 1
- If $\log_4 x + \log_2 x = 6$, then the value of x is
(a) 16 (b) 4 (c) 2 (d) 1
- Given $\log_{10} 2 = 0.3010$, the value of $\log_{10} 5$ is
(a) 0.6990 (b) 0.6919 (c) 0.6119 (d) 0.7525
- If $\log \frac{x}{y} + \log \frac{y}{x} = \log (x + y)$, then
(a) $x + y = 1$ (b) $x - y = 0$ (c) $x - y = 1$ (d) $x = y$
- The characteristic in $\log 6.7482 \times 10^{-5}$ is
(a) 6 (b) -4 (c) 5 (d) -5
- If $10^x = 1.73$ and $\log_{10} 1730 = 3.2380$, then x equals to
(a) 2.380 (b) 0.2380 (c) 2.2380 (d) 1.380
- If $2^{2x+3} = 6^{x-1}$, then x equals
(a) $\frac{4 \log 2 + \log 3}{\log 3 - \log 2}$ (b) $\frac{3 \log 2 + 2 \log 3}{\log 3 - 2 \log 2}$
(c) $\frac{\log 48}{\log 7}$ (d) None of these
- The value of $10^{\log_{10} m + 2 \log_{10} n + 3 \log_{10} p}$ is
(a) $m^2 np^3$ (b) $mn^2 p^3$ (c) $m^3 np^2$ (d) None of these
- Given that $\log_{10} 2 = 0.3010$, $\log_{10} 3 = 0.4771$ and $\log_{10} 7 = 0.8491$, then $\log_{10} \frac{108}{\sqrt{7}}$ is equal to
(a) 2.6123 (b) 1.6088 (c) 1.6320 (d) 2.4558
- If a, b and c are three consecutive integers, then $\log(ac + 1)$ is equal to
(a) $\log(2b)$ (b) $(\log b)^2$ (c) $2 \log b$ (d) None of these
- If $\log_r p = 2$, $\log_r q = 3$, then the value of $\log_p q$ is
(a) $\frac{1}{3}$ (b) $\frac{2}{3}$ (c) $\frac{3}{2}$ (d) 6
- If $\log x^2 y^2 = a$ and $\log \frac{x}{y} = b$, then $\frac{\log x}{\log y}$ is equal to
(a) $\frac{a - 3b}{a + 2b}$ (b) $\frac{a + 3b}{a - 2b}$ (c) $\frac{a + 2b}{a - 2b}$ (d) $\frac{a - 2b}{a + 3b}$

- 30.** If $\log_{10} 5 = 0.70$, then $\log_5 10$ is equal to
(a) 1.35 (b) 1.40 (c) 1.43143 (d) 1.56

- 31.** The value of $\log_3 \left(1 + \frac{1}{3}\right) + \log_3 \left(1 + \frac{1}{4}\right) + \log_3 \left(1 + \frac{1}{5}\right) + \dots + \log_3 \left(1 + \frac{1}{24}\right)$ is
(a) $-1 + 2 \log_3 5$ (b) 2
(c) 3 (d) 4

- 32.** If $\log(x + y) = \log x + \log y$ and $x = 1.1568$, then y is equal to
(a) 7.3776 (b) $\bar{7}.3776$ (c) 5.3776 (d) 5.3116

- 33.** If $\log_8 x + \log_4 x + \log_2 x = 11$. Then, the value of x is
(a) 128 (b) 16 (c) 32 (d) 64

- 34.** What is the value of $\log_y x^5 \log_x y^2 \log_z z^3$?
(a) 10 (b) 30 (c) 20 (d) 60

- 35.** If $(\log_3 x)(\log x^{2x})(\log_{2x} y) = \log_x x^2$, then what is the value of y ?
(a) $\frac{9}{2}$ (b) 9 (c) 18 (d) 27

- 36.** If $\log_{10} 2, \log_{10}(2^x - 1), \log_{10}(2^x + 3)$ are three consecutive terms of AP, then which one of the following is correct?
I. $x = 1$ II. $x = \log_2 5$
(a) Both I and II (b) Only II
(c) Only I (d) None of these

- 37.** Consider the following statements
I. $(\log_{10} 0.1)^2 + \log_{10} 10 \cdot \log_{10} 100 = 3$
II. $\log_{10} \log_{10} 10 = 1$ III. $\log_{10} \sqrt{10} + \log_{10} \sqrt{10} = 1$
Which of the statements given above are correct?
(a) I and III (b) II and III
(c) I and II (d) All are correct

- 38.** If $y = (a^x)^{(a^x)} \dots \infty$, then which one of the following is correct?
(a) $\log y = xy \log a$ (b) $\log y = x + y \log a$
(c) $\log y = y + x \log a$ (d) $\log y = (y + x) \log a$

> PREVIOUS YEARS' QUESTIONS

- 39.** What is the logarithm of 0.0001 with respect to base 10? ☑ 2012 II
(a) 4 (b) 3 (c) -4 (d) -3
- 40.** If $\log_{10} a = p$ and $\log_{10} b = q$, then what is the value of $\log_{10}(a^p b^q)$? ☑ 2012 II
(a) $p^2 + q^2$ (b) $p^2 - q^2$ (c) $p^2 q^2$ (d) $\frac{p^2}{q^2}$
- 41.** What are the possible solutions for x of the equation $x^{\sqrt{x}} = \sqrt[n]{x^x}$, where x and n are positive integers? ☑ 2015 I
(a) 0, n (b) 1, n (c) n, n^2 (d) 1, n^2
- 42.** The value of $\frac{1}{5} \log_{10} 3125 - 4 \log_{10} 2 + \log_{10} 32$ is ☑ 2016 I
(a) 0 (b) 1 (c) 2 (d) 3

> ANSWERS

1	b	2	a	3	a	4	a	5	c	6	b	7	c	8	d	9	c	10	d
11	d	12	b	13	c	14	d	15	c	16	b	17	a	18	c	19	a	20	a
21	a	22	d	23	b	24	a	25	b	26	b	27	c	28	c	29	c	30	c
31	a	32	a	33	d	34	b	35	b	36	b	37	a	38	a	39	c	40	a
41	a	42	b																

> HINTS AND SOLUTIONS

1. (b) $\log_{100} 0.1 = \log_{10^2} \left(\frac{1}{10}\right)$
 $= \frac{1}{2} \log_{10} \left(\frac{1}{10}\right) = \frac{1}{2} \log_{10} (10)^{-1}$
 $= -\frac{1}{2} \log_{10} 10 = -\frac{1}{2}$ [Rule 3]

2. (a) $3 \log 3 + 2 \log 2 = \log 3^3 + \log 2^2$
 $= \log 27 + \log 4$ [Rule 3]
 $= \log (27 \times 4) = \log 108$ [Rule 1]

3. (a) $\because \log_a \sqrt{2} = \frac{1}{6} \Rightarrow a^{1/6} = \sqrt{2}$

$\therefore a = (\sqrt{2})^6$

4. (a) $\log_3 x = -2 \Rightarrow x = 3^{-2} = \frac{1}{3^2} = \frac{1}{9}$

5. (c) Let $\log_{2\sqrt{3}} 1728 = x$
 $\Rightarrow (2\sqrt{3})^x = 1728$
 $\therefore 1728 = 2^6 (\sqrt{3})^6 = (2\sqrt{3})^6$
 $(2\sqrt{3})^x = (2\sqrt{3})^6$

On comparing both sides, we get $x = 6$

6. (b) $(\log_{1/2} 2)(\log_{1/3} 3)(\log_{1/4} 4) \dots (\log_{1/1000} 1000)$
 $= \left(\frac{\log 2}{\log 1/2}\right) \left(\frac{\log 3}{\log 1/3}\right) \left(\frac{\log 4}{\log 1/4}\right) \dots \left(\frac{\log 1000}{\log 1/1000}\right)$
 $\left[\because \log_b a = \frac{\log a}{\log b} \right]$ [Rule 4]

- $$= \left(\frac{\log 2}{-\log 2} \right) \left(\frac{\log 3}{-\log 3} \right) \left(\frac{\log 4}{-\log 4} \right) \dots \left(\frac{\log 1000}{-\log 1000} \right)$$
- $$= (-1) \times (-1) \times (-1) \times \dots \times (-1)$$
- $$[\because \text{number of terms is odd}]$$
- $$= -1$$
7. (c) $\frac{1}{2} \log_{10} 25 - 2 \log_{10} 3 + \log_{10} 18$
- $$= \log_{10} 25^{1/2} - \log_{10} 3^2 + \log_{10} 18$$
- $$[\text{Rule 3}]$$
- $$= \log_{10} 5 - \log_{10} 9 + \log_{10} 18$$
- $$= \log_{10} \frac{5 \times 18}{9} = \log_{10} \frac{90}{9}$$
- $$= \log_{10} 10 = 1$$
8. (d) $[\log_{10} (5 \log_{10} 100)]^2$
- $$= [\log_{10} (5 \log_{10} 10^2)]^2$$
- $$= [\log_{10} (10 \log_{10} 10)]^2$$
- $$= [\log_{10} 10]^2 \quad [\because \log_{10} 10 = 1]$$
- $$= 1^2 = 1$$
9. (c) $\log_y x \cdot \log_z y \cdot \log_x z$
- $$= \frac{\log x}{\log y} \times \frac{\log y}{\log z} \times \frac{\log z}{\log x} = 1$$
- $$\left[\because \log_a b = \frac{\log b}{\log a} \right]$$
10. (d) Let $\log_3 (27 \times \sqrt[4]{9} \times \sqrt[3]{9}) = x$
- $$\Rightarrow 3^x = 27 \times \sqrt[4]{9} \times \sqrt[3]{9}$$
- $$\Rightarrow 3^x = 3^3 \times 3^{2/4} \times 3^{2/3}$$
- $$3^x = 3^{25/6}$$
- On comparing both sides, we get
- $$\Rightarrow x = \frac{25}{6} = 4 \frac{1}{6}$$
11. (d) $\log_2 \log_2 \log_2 \log_2 2^{16}$
- $$= \log_2 \log_2 \log_2 (16) \quad [\because \log_2 2 = 1]$$
- $$= \log_2 \log_2 \log_2 (2^4) = \log_2 \log_2 (4)$$
- $$= \log_2 \log_2 (2^2) = \log_2 (2) = 1$$
12. (b) $\frac{\log_{13} (10)}{\log_{169} (10)} = \frac{\log_{13} (10)}{\log_{13^2} (10)}$
- $$= \frac{\log_{13} 10}{\frac{1}{2} \log_{13} 10} \left[\because \log_{a^b} c = \frac{1}{b} \log_a c \right]$$
- $$= \frac{1}{1/2} = 2$$
13. (c) We have, $\frac{1}{3} \log_{10} 125 - 2 \log_{10} 4$
- $$+ \log_{10} 32 + \log_{10} 1$$
- $$= \log_{10} (125)^{1/3} - 2 \log_{10} (2)^2$$
- $$+ \log_{10} (2)^5 + 0 \quad [\because \log_{10} 1 = 0]$$

- $$= \log_{10} 5 - 4 \log_{10} 2 + 5 \log_{10} 2$$
- $$= \log_{10} 5 + \log_{10} 2 = \log_{10} 5 \times 2$$
- $$= \log_{10} 10 = 1 \quad [\text{Rule 1}]$$
14. (d) Given, $\log_r 6 = m$ and $\log_r 3 = n$
- $$\therefore \log_r 6 = \log_r (2 \times 3) = \log_r 2 + \log_r 3$$
- $$\therefore \log_r 3 + \log_r 2 = m$$
- $$\Rightarrow n + \log_r 2 = m$$
- $$\Rightarrow \log_r 2 = m - n$$
- $$\therefore \log_r \left(\frac{r}{2} \right) = \log_r r - \log_r 2 \quad [\text{Rule 2}]$$
- $$= 1 - m + n$$
15. (c) $\log_{10} \left(\frac{3}{2} \right) + \log_{10} \left(\frac{4}{3} \right) + \log_{10} \left(\frac{5}{4} \right)$
- $$+ \dots + 8\text{th term}$$
- $$\therefore T_n = \log_{10} \left(\frac{n+2}{n+1} \right)$$
- $$\Rightarrow T_8 = \log_{10} \left(\frac{10}{9} \right)$$
- $$\therefore \log_{10} \left(\frac{3}{2} \right) + \log_{10} \left(\frac{4}{3} \right) + \log_{10} \left(\frac{5}{4} \right)$$
- $$+ \dots + \log_{10} \left(\frac{10}{9} \right)$$
- $$= \log_{10} \left(\frac{3}{2} \times \frac{4}{3} \times \frac{5}{4} \times \dots \times \frac{10}{9} \right) \quad [\text{Rule 1}]$$
- $$= \log_{10} \left(\frac{10}{2} \right) = \log_{10} 5$$
16. (b) The given expression is
- $$= \log_{xyz} (xy) + \log_{xyz} (yz) + \log_{xyz} (zx)$$
- $$= \log_{xyz} (xy \cdot yz \cdot zx) \quad [\text{Rule 1}]$$
- $$= \log_{xyz} (xyz)^2 = 2 \log_{xyz} xyz = 2$$
17. (a) The given expression is
- $$= \frac{1}{\log_x (yz) + \log_x x}$$
- $$+ \frac{1}{\log_y (xz) + \log_y y}$$
- $$+ \frac{1}{\log_z (xy) + \log_z z}$$
- $$= \frac{1}{\log_x xyz} + \frac{1}{\log_y xyz} + \frac{1}{\log_z xyz}$$
- $$= \log_{xyz} x + \log_{xyz} y + \log_{xyz} z$$
- $$= \log_{xyz} xyz = 1 \quad [\text{Rule 1}]$$
18. (c) $\log_4 (x^2 + x) - \log_4 (x + 1) = 2$
- $$\Rightarrow \log_4 \left(\frac{x^2 + x}{x + 1} \right) = 2 \Rightarrow 4^2 = \frac{x^2 + x}{x + 1}$$
- $$\Rightarrow 16x + 16 = x^2 + x$$
- $$\Rightarrow x^2 - 15x - 16 = 0$$
- $$\therefore x = 16 \text{ or } x = -1 \quad [\text{not possible}]$$

19. (a) Given, $\log_4 x + \log_2 x = 6$
- $$\Rightarrow \frac{\log x}{\log 4} + \frac{\log x}{\log 2} = 6$$
- $$\Rightarrow \frac{\log x}{2 \log 2} + \frac{\log x}{\log 2} = 6$$
- $$\Rightarrow 3 \log x = 12 \log 2$$
- $$\Rightarrow \log x = 4 \log 2 \Rightarrow \log x = \log 2^4$$
- $$\Rightarrow \log x = \log 16$$
- On comparing both sides, we get
- $$\therefore x = 16$$
20. (a) $\log_{10} 5 = \log_{10} \frac{10}{2}$
- $$= \log_{10} 10 - \log_{10} 2 = 1 - 0.3010$$
- $$= 0.6990$$
21. (a) Given, $\log \frac{x}{y} + \log \frac{y}{x} = \log (x + y)$
- $$\Rightarrow \log \frac{x}{y} \cdot \frac{y}{x} = \log (x + y)$$
- $$\Rightarrow \log 1 = \log (x + y)$$
- $$\therefore x + y = 1$$
22. (d) The characteristic in $\log 6.7482 \times 10^{-5}$ is -5.
23. (b) Given, $10^x = 1.73$, $x = \log_{10} 1.73$
- $$= \log_{10} 1730 - \log_{10} 1000$$
- $$= \log_{10} 1730 - \log_{10} 10^3$$
- $$= 3.2380 - 3 = 0.2380$$
24. (a) $2^{2x+3} = 6^{x-1}$
- Taking log on both sides, we get
- $$(2x + 3) \log 2 = (x - 1) \log 6$$
- $$\Rightarrow 2x \log 2 + 3 \log 2$$
- $$= (x - 1) (\log 2 + \log 3)$$
- $$\Rightarrow 2x \log 2 + 3 \log 2 = x (\log 2 + \log 3)$$
- $$- \log 2 - \log 3$$
- $$\Rightarrow x (\log 2 - \log 3) = -4 \log 2 - \log 3$$
- $$\therefore x = \frac{4 \log 2 + \log 3}{\log 3 - \log 2}$$
25. (b) $10^{\log_{10} m} + 2 \log_{10} n + 3 \log_{10} p$
- $$= 10^{\log_{10} m + \log_{10} n^2 + \log_{10} p^3}$$
- $$\Rightarrow 10^{\log_{10} mn^2 p^3} = mn^2 p^3 \quad [\because a^{\log_a p} = p]$$
26. (b) $\log_{10} \frac{108}{\sqrt{7}} = \log_{10} 108 - \log_{10} \sqrt{7}$
- $$= \log_{10} 2^2 \times 3^3 - \log_{10} 7^{1/2}$$
- $$= 2 \log_{10} 2 + 3 \log_{10} 3 - \frac{1}{2} \log_{10} 7$$
- $$= 2 \times (0.3010) + 3(0.4771)$$
- $$- \frac{1}{2} (0.8491)$$
- $$= 0.6020 + 1.4313 - 0.4245 = 1.6088$$

27. (c) Let first integer be a .

Then, $b = a + 1$ and $c = a + 2$

$$\begin{aligned}\therefore ac + 1 &= a(a + 2) + 1 \\ &= a^2 + 2a + 1 = (a + 1)^2 \\ ac + 1 &= b^2\end{aligned}$$

So, $\log(ac + 1) = \log b^2 = 2 \log b$

28. (c) Given, $\log_r p = 2$ and $\log_r q = 3$

$$\text{By relation, } \log_p q = \frac{\log_r q}{\log_r p} = \frac{3}{2}$$

29. (c) Given, $\log x^2 y^2 = a$

$$[\because \log(mn) = \log m + \log n]$$

$$\therefore \log x^2 + \log y^2 = a$$

$$\Rightarrow 2 \log x + 2 \log y = a \quad \dots(i)$$

$$\text{Also, } \log x - \log y = b \quad \dots(ii)$$

On solving Eqs. (i) and (ii), we get

$$\log x = \frac{a + 2b}{4} \text{ and } \log y = \frac{a - 2b}{4}$$

$$\therefore \frac{\log x}{\log y} = \frac{a + 2b}{a - 2b}$$

30. (c) $\log_5 10 = \frac{\log_{10} 10}{\log_{10} 5} = \frac{1}{0.70} = 1.43143$

$$[\because \log_{10} 10 = 1]$$

31. (a) $\log_3 \left(1 + \frac{1}{3}\right) + \log_3 \left(1 + \frac{1}{4}\right)$
 $+ \log_3 \left(1 + \frac{1}{5}\right) + \dots + \log_3 \left(1 + \frac{1}{24}\right)$
 $= \log_3 \frac{4}{3} + \log_3 \frac{5}{4} + \log_3 \frac{6}{5} +$
 $\dots + \log_3 \frac{25}{24}$
 $= \log_3 4 - \log_3 3 + \log_3 5 - \log_3 4$
 $+ \log_3 6 - \log_3 5$
 $+ \dots + \log_3 25 - \log_3 24$
 $= -\log_3 3 + \log_3 25 = -1 + 2 \log_3 5$

32. (a) Given, $\log(x + y) = \log x + \log y$

$$\Rightarrow \log(x + y) = \log xy$$

$$\Rightarrow x + y = xy$$

$$\begin{aligned}\Rightarrow y &= \frac{x}{x - 1} = \frac{11568}{11568 - 1} = \frac{11568}{0.1568} \\ &= 7.37755 = 7.3776\end{aligned}$$

33. (d) Given, $\log_8 x + \log_4 x + \log_2 x = 11$

$$\Rightarrow \frac{\log x}{\log 8} + \frac{\log x}{\log 4} + \frac{\log x}{\log 2} = 11$$

$$\Rightarrow \frac{\log x}{\log 2^3} + \frac{\log x}{\log 2^2} + \frac{\log x}{\log 2} = 11$$

$$\Rightarrow \frac{\log x}{3 \log 2} + \frac{\log x}{2 \log 2} + \frac{\log x}{\log 2} = 11$$

$$\Rightarrow \frac{11 \log x}{6 \log 2} = 11 \Leftrightarrow \log x = 6 \log 2$$

$$\Rightarrow x = 2^6, \therefore x = 64$$

34. (b) $\therefore \log_y x^5 \log_x y^2 \log_z z^3$
 $= 5 \log_y x \cdot 2 \log_x y \cdot 3 \log_z z$
 $[\because \log_a b^n = n \log_a b]$
 $= 5 \log_y x \cdot 2 \log_x y \cdot 3 \times 1$
 $= \frac{5 \log x}{\log y} \times 2 \cdot \frac{\log y}{\log x} \times 3 \left[\because \log_a b = \frac{\log b}{\log a} \right]$
 $= 5 \times 2 \times 3 = 30$

35. (b) $(\log_3 x)(\log_x 2x)(\log_{2x} y) = \log_x x^2$

$$\begin{aligned}\Rightarrow \frac{\log x}{\log 3} \times \frac{\log 2x}{\log x} \times \frac{\log y}{\log 2x} &= \frac{\log x^2}{\log x} \\ \left[\because \log_b a &= \frac{\log a}{\log b} \right]\end{aligned}$$

$$\Rightarrow \frac{\log y}{\log 3} = \frac{2 \log x}{\log x} [\because \log a^b = b \log a]$$

$$\Rightarrow \log y = 2 \log 3 \Rightarrow \log y = \log 3^2$$

$$[\because \log m = \log n \Rightarrow m = n]$$

$$\Rightarrow \log y = \log 9$$

$$\therefore y = 9$$

36. (b) Given that, $\log_{10} 2, \log_{10} (2^x - 1)$ and $\log_{10} (2^x + 3)$ are in AP.

$$\begin{aligned}\text{then, } 2 \log_{10} (2^x - 1) &= \log_{10} 2 + \log_{10} (2^x + 3) \\ \Rightarrow \log_{10} (2^x - 1)^2 &= \log_{10} (2^{x+1} + 6) \\ \Rightarrow (2^x - 1)^2 &= (2^{x+1} + 6) \\ \Rightarrow 2^{2x} + 1 - 2 \cdot 2^x &= 2 \cdot 2^x + 6 \\ \Rightarrow 2^{2x} - 4 \cdot 2^x - 5 &= 0 \\ \Rightarrow (2^x)^2 - 4(2^x) - 5 &= 0 \\ \Rightarrow y^2 - 4y - 5 &= 0\end{aligned}$$

$$\therefore y = 5, -1$$

$$\text{Hence, } 2^x = -1 \text{ or } 2^x = 5$$

$$\therefore x = \log_2 (-1) \text{ is not possible}$$

$$\text{or } x = \log_2 (5)$$

Then, $x = \log_2 (5)$ is answer.

Hence, II is correct.

37. (a) I. $[\log_{10} (0.1)]^2 + \log_{10} 10 \cdot \log_{10} 100$
 $= [-\log_{10} 10]^2 + \log_{10} 10 \cdot \log_{10} 10^2$
 $= (-1)^2 + (1) \cdot 2 \log_{10} 10$
 $= 1 + (1) \cdot 2 = 1 + 2 = 3$

Hence, statement I is correct.

$$\text{II. } \log_{10} \log_{10} 10 = \log_{10} (1) = 0 \neq 1$$

Hence, statement II is incorrect.

$$\begin{aligned}\text{III. } \log_{10} \sqrt{10} + \log_{10} \sqrt{10} \\ &= \frac{1}{2} \log_{10} 10 + \frac{1}{2} \log_{10} 10 \\ &= \frac{1}{2} + \frac{1}{2} = 1\end{aligned}$$

Hence, statement III is correct.

38. (a) Given, $y = (a^x)^{(a^x)^{\dots \infty}}$

$$\therefore y = (a^x)^y$$

On taking log both sides, we get

$$\log y = y \log a^x$$

$$\therefore \log y = xy \log a$$

39. (c) Let, $\log_{10} 0.0001 = x$

$$\text{Then, } x = \log_{10} \frac{1}{(10)^4}$$

$$= \log_{10} 1 - \log_{10} (10)^4$$

$$= 0 - 4 = -4$$

40. (a) Given, $\log_{10} a = p$ and $\log_{10} b = q$

$$\log_{10} (a^p b^q) = \log_{10} a^p + \log_{10} b^q$$

$$\Rightarrow \log_{10} (a^p b^q) = p \log_{10} a + q \log_{10} b$$

$$\begin{aligned}\therefore \log_{10} (a^p b^q) &= p \times p + q \times q \\ &= p^2 + q^2\end{aligned}$$

41. (a) Given, $x^{\sqrt{x}} = \sqrt[n]{x^x}$, where x and n are positive integers.

On taking log both sides, we get

$$\log x^{\sqrt{x}} = \log [\sqrt[n]{x^x}]$$

$$\Rightarrow \sqrt{x} \log x = \log (x^x)^{1/n} = \log x^{x/n}$$

$$\Rightarrow \sqrt{x} \log x = \frac{x}{n} \log x$$

$$\Rightarrow \sqrt{x} \log x - \frac{x}{n} \log x = 0$$

$$\Rightarrow \log x \left[\sqrt{x} - \frac{x}{n} \right] = 0$$

$$\therefore \log x \neq 0$$

$$\therefore \sqrt{x} - \frac{x}{n} = 0$$

$$\Rightarrow \sqrt{x} = \frac{x}{n}$$

$$\therefore \frac{x}{\sqrt{x}} = n$$

On squaring both sides, we get

$$\frac{x^2}{x} = n$$

$$\Rightarrow x^2 - nx = 0$$

$$\Rightarrow x(x - n) = 0$$

$$\therefore x = 0, x = n$$

Hence, the possible solution of x is 0, n .

42. (b) $\frac{1}{5} \log_{10} 3125 - 4 \log_{10} 2 + \log_{10} 32$
 $= \log_{10} [(5)^5]^{1/5} - \log_{10} (2)^4 + \log_{10} 32$
 $= \log_{10} 5 - \log_{10} 16 + \log_{10} 32$
 $= \log_{10} \left(\frac{5 \times 32}{16} \right) = \log_{10} 10 = 1$

14

ALGEBRAIC OPERATIONS

Generally (3-4) questions have been asked from this chapter. Questions which are asked from this chapter are mostly based on direct identities and factor theorem.



Algebraic Expressions

A combination of constants and variables connected by the four fundamental operations $+$, $-$, $\times$ and $\div$ is called an algebraic expression.

POLYNOMIALS

A polynomial is an algebraic expression consisting of variables and coefficients, having non-negative integral powers. e.g. $x^3 + 5x^2 - 1$, $3x^3 + 4x^2y + 17$ etc.

- (i) $3x^2 + 9x - 1 + 7/x$ is not a polynomial as it contains a term, namely $7/x$, having negative integral power of variable x .
- (ii) $5x^2 - 7x^{7/2} + x - 1$ is not a polynomial as the term $-7x^{7/2}$ contains rational power of variable x .

Degree of a Polynomial

The exponent of the highest degree term in a polynomial, is known as degree of polynomial.

- e.g. (i) $4x^3 - 9x^2 + 7x + 9$ is a polynomial with variable x of degree 3.
(ii) $4x^3y - 3x^2 + 2xy + 1$ is a polynomial of degree 4.

Various Types of Polynomials

Polynomials are classified on the basis of degree of polynomial and number of terms which are given below.

Based on Degree of Polynomial

- (i) **Constant polynomial** A polynomial of degree zero, is called constant polynomial. e.g. $f(x) = 8$.

Note The polynomial $f(x) = 0$ is called the zero polynomial. The degree of zero polynomial is not defined.

- (ii) **Linear polynomial** A polynomial of degree 1, is called linear polynomial. e.g. $9x + 7$, $x - 9$, $x + 2$ etc.
- (iii) **Quadratic polynomial** A polynomial of degree 2, is called quadratic polynomial.
The general form of quadratic polynomial in x is $ax^2 + bx + c$, where a , b and c are real and $a \neq 0$.
e.g. $x^2 - 7x + 12$, $5y^2 - 7$, $z^2 - 4$ etc., are all quadratic polynomials.
- (iv) **Cubic polynomial** A polynomial of degree 3, is called cubic polynomial.
The general form of a cubic polynomial in x is $ax^3 + bx^2 + cx + d$, where a , b , c and d are real and $a \neq 0$.
e.g. $3x^3 + 2x^2 + 7x - 1$, $8y^3 + 5y + 2$, $x^3 - 8$ etc., are all cubic polynomials.
- (v) **Biquadratic polynomial** A polynomial of degree 4, is called a biquadratic polynomial.
The general form of a biquadratic polynomial in x is $ax^4 + bx^3 + cx^2 + dx + e$, where a , b , c , d and e are real numbers and $a \neq 0$.
e.g. $6x^4 + 3x^3 + 2x^2 + x + 2$, $x^4 - 8$ etc., are all biquadratic polynomials.

Based on Number of Terms

- (i) **Monomial** A polynomial containing only one term is called a monomial. e.g. $6x^2$
- (ii) **Binomial** A polynomial containing two terms is called a binomial. e.g. $5x^3 + 7xy$
- (iii) **Trinomial** A polynomial containing three terms is called a trinomial. e.g. $7x^2y + 5x + 6$

FACTORISATION

To express a polynomial as the product of other polynomials of degree less than that of the given polynomial is called as factorisation.

e.g. $x^2 - 49 = x^2 - 7^2 = (x - 7)(x + 7)$

Important Identities

- $(a^2 - b^2) = (a + b)(a - b)$
- $(a + b)^2 = a^2 + b^2 + 2ab$ and $(a - b)^2 = a^2 + b^2 - 2ab$
- $(a + b)^2 - (a - b)^2 = 4ab$
- $(a + b)^2 + (a - b)^2 = 2(a^2 + b^2)$
- $(a + b)^3 = a^3 + b^3 + 3ab(a + b)$
- $(a - b)^3 = a^3 - b^3 - 3ab(a - b)$
- $(a^3 + b^3) = (a + b)(a^2 + b^2 - ab)$

- $(a^3 - b^3) = (a - b)(a^2 + b^2 + ab)$
- $(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ac)$
- $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ac)$
- If $a + b + c = 0$, then $a^3 + b^3 + c^3 = 3abc$
- $a^4 + a^2b^2 + b^4 = (a^2 + ab + b^2)(a^2 - ab + b^2)$

Methods of Factorisation

These methods are given below

Factorisation by Taking Out the Common Factor

If each term of an expression has a common factor, take out the common factors.

EXAMPLE 1. Factorise value of $4(3a - 2b)^2 - 5(3a - 2b)$ is

- a. $(3a - 2b)(12a - 8b - 5)$ b. $(2a - 3b)(12a - 8b)$
c. $(2a - 3b)(8a - 12b - 5)$ d. None of these

Sol. a. $4(3a - 2b)^2 - 5(3a - 2b) = (3a - 2b)[4(3a - 2b) - 5]$
 $= (3a - 2b)[12a - 8b - 5]$

Factorisation by Grouping

Sometimes in a given polynomial, it is not possible to take out a common factor directly. However, on rearranging the terms of the polynomial and grouping them such that all the terms have a common factor, the polynomial can be easily factorised.

EXAMPLE 2. Factorise $x^2 + y - xy - x$

- a. $(x - y)(x + 1)$ b. $(x - y)(x^2 - 1)$
c. $(x + y)(x^2 + 1)$ d. $(x - y)(x - 1)$

Sol. d. **Method I** $x^2 + y - xy - x = x^2 - x + y - xy$
 $= x(x - 1) + y(1 - x) = x(x - 1) - y(x - 1) = (x - 1)(x - y)$

Method II $x^2 + y - xy - x = x^2 - xy + y - x$
 $= x(x - y) - 1(x - y) = (x - y)(x - 1)$

EXAMPLE 3. Factorise $x^2 + \frac{4}{x^2} + 4 - 2x - \frac{4}{x}$

- a. $\left(x - \frac{2}{x}\right)\left(x + \frac{2}{x} + 2\right)$ b. $\left(x - \frac{2}{x}\right)\left(x + \frac{2}{x} - 2\right)$
c. $\left(x + \frac{2}{x}\right)\left(x + \frac{2}{x} - 2\right)$ d. None of these

Sol. c. $\left(x^2 + \frac{4}{x^2} + 4 - 2x - \frac{4}{x}\right) = \left(x^2 + \frac{4}{x^2} + 4\right) - 2\left(x + \frac{2}{x}\right)$
 $= \left(x + \frac{2}{x}\right)^2 - 2\left(x + \frac{2}{x}\right) = \left(x + \frac{2}{x}\right)\left(x + \frac{2}{x} - 2\right)$

Factorisation of Perfect Square Polynomials

If the polynomial is given as the perfect square quadratic polynomial, then use the following Identities to factorise it.

$$\bullet a^2 + 2ab + b^2 = (a + b)^2 \quad \bullet a^2 - 2ab + b^2 = (a - b)^2$$

EXAMPLE 4. Factorise $16x^2 - 8x + 1$

a. $(2x-1)^2$ b. $(x-4)^2$ c. $(x-2)^2$ d. $(4x-1)^2$

Sol. d. $16x^2 - 8x + 1 = (4x)^2 - 2(4x) + 1^2 = (4x - 1)^2$
 $= (4x - 1)(4x - 1)$

EXAMPLE 5. Factorise $x^2 - 2\sqrt{5}x + 5$

a. $(x\sqrt{5}-1)^2$ b. $(x-5)^2$ c. $(x+\sqrt{5})(x-\sqrt{5})$ d. $(x-\sqrt{5})^2$

Sol. d. $x^2 - 2\sqrt{5}x + 5 = x^2 - 2\sqrt{5}x + (\sqrt{5})^2 = (x - \sqrt{5})^2$
 $= (x - \sqrt{5})(x - \sqrt{5})$

Factorising the Difference of Two Squares

In case, the polynomial is given in the form of $a^2 - b^2$, to evaluate it, following identity is applied.

$$\bullet a^2 - b^2 = (a - b)(a + b)$$

EXAMPLE 6. Factorise $2a^5 - 32a$

a. $(a^2 + 4)(a - 2)(a + 2)$ b. $2a(a^2 + 4)(a - 2)(a + 2)$
 c. $2a(a^2 + 4)$ d. None of these

Sol. b. $2a^5 - 32a = 2a(a^4 - 16)$
 $= 2a[(a^2)^2 - (4)^2] = 2a(a^2 + 4)(a^2 - 4)$
 $= 2a(a^2 + 4)(a^2 - 2^2) = 2a(a^2 + 4)(a - 2)(a + 2)$

Factorisation of Quadratic Polynomials

Quadratic polynomials of the type $ax^2 + bx + c$, where $a \neq 0$, a and b are coefficients of x^2 and x respectively and c is constant, can be factorised by splitting the middle term. We find two numbers p and q such that

$$p + q = b \text{ and } pq = ac, \text{ then}$$

$$ax^2 + bx + c = ax^2 + (p + q)x + c = ax^2 + px + qx + c$$

EXAMPLE 7. Factorise $x^2 + 9x + 14$

a. $(x + 2)(x + 7)$ b. $(x - 2)(x - 7)$
 c. $(x + 2)(x - 7)$ d. $(x - 2)(x + 7)$

Sol. a. $x^2 + 9x + 14 = x^2 + (7 + 2)x + 14 = x^2 + 7x + 2x + 14$
 $= x(x + 7) + 2(x + 7) = (x + 7)(x + 2)$

EXAMPLE 8. Factorise $4\sqrt{3}x^2 + 5x - 2\sqrt{3}$

a. $(2x + \sqrt{3})(4x - \sqrt{3})$ b. $(2x - \sqrt{3})(4x + \sqrt{3})$
 c. $(\sqrt{3}x + 2)(4x - \sqrt{3})$ d. None of these

Sol. c. $4\sqrt{3}x^2 + 5x - 2\sqrt{3} = 4\sqrt{3}x^2 + 8x - 3x - 2\sqrt{3}$
 $[\because 4\sqrt{3} \times (-2\sqrt{3}) = -24 = 8 \times (-3)]$
 $= 4x(\sqrt{3}x + 2) - \sqrt{3}(\sqrt{3}x + 2)$
 $= (\sqrt{3}x + 2)(4x - \sqrt{3})$

Factorisation of Sum and Difference of Cubes

In case, the polynomial is given in the form of $a^3 + b^3$ or $a^3 - b^3$. To factorise it, following identities can be applied.

$$\bullet a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

$$\bullet a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

EXAMPLE 9. Factorise $(2x + 3y)^3 - (2x - 3y)^3$

a. $18(4x^2 + 3y^2)$ b. $18y(3x^2 + 4y^2)$
 c. $xy(4x^2 + 3y^2)$ d. $18y(4x^2 + 3y^2)$

Sol. d. $(2x + 3y)^3 - (2x - 3y)^3$

Put $2x + 3y = a$ and $2x - 3y = b$

$$\Rightarrow a^3 - b^3 = (a - b)(a^2 + ab + b^2) = (2x + 3y - 2x + 3y)$$

$$[(2x + 3y)^2 + (2x + 3y)(2x - 3y) + (2x - 3y)^2]$$

$$= 6y[4x^2 + 9y^2 + 12xy + 4x^2 - 9y^2 + 4x^2 + 9y^2 - 12xy]$$

$$= 6y[12x^2 + 9y^2] = 18y[4x^2 + 3y^2]$$

Hence, $(2x + 3y)^3 - (2x - 3y)^3 = 18y[4x^2 + 3y^2]$

Factorisation of Polynomial of form $a^3 + b^3 + c^3 - 3abc$

Here, it is easy to use.

$$\bullet a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ac)$$

$$\bullet \text{ If } a + b + c = 0, \text{ then } a^3 + b^3 + c^3 = 3abc$$

EXAMPLE 10. Factorise $x^3 + 27y^3 + 8x^3 - 18xyz$

a. $(x + 3y + 2z)(x^2 + 9y^2 + 4z^2 + 3xy + 6yz + 2xz)$
 b. $(x + 3y + 2z)(x^2 + 9y^2 + 4z^2 - 3xy - 6yz - 2xz)$
 c. $(x - 3y - 2z)(x^2 + 9y^2 - 4z^2 - 3xy + 6yz + 2xz)$
 d. None of the above

Sol. b. $x^3 + 27y^3 + 8z^3 - 18xyz$

$$= x^3 + (3y)^3 + (2z)^3 - 3(x)(3y)(2z)$$

$$= (x + 3y + 2z)[x^2 + 9y^2 + 4z^2 - x(3y) - 3y(2z) - x(2z)]$$

$$= (x + 3y + 2z)[x^2 + 9y^2 + 4z^2 - 3xy - 6yz - 2xz]$$

EXAMPLE 11. Factorise

$$(2x - 3y)^3 + (3y - 5z)^3 + (5z - 2x)^3$$

a. $3(2x - 3y)(3y - 5z)(5z - 2x)$
 b. $(2x - 3y)(3y - 5z)(5z - 2x)$
 c. $3(2x - 3y)(3y + 5z)(5z - 2x)$
 d. None of the above

Sol. a. Here, $2x - 3y = a$, $3y - 5z = b$ and $5z - 2x = c$

$$\begin{aligned}\therefore a + b + c &= 0 \Rightarrow a^3 + b^3 + c^3 = 3abc \\ \therefore (2x - 3y)^3 + (3y - 5z)^3 + (5z - 2x)^3 &= 3(2x - 3y)(3y - 5z)(5z - 2x)\end{aligned}$$

Remainder Theorem

Let $p(x)$ be a polynomial of degree n greater than or equal to 1 (i.e. $n \geq 1$) and a be any real number. If $p(x)$ is divided by the linear polynomial $(x - a)$, then the remainder is $p(a)$. Remainder can be evaluated by substituting $x - a = 0$, i.e. $x = a$ in $p(x)$.

EXAMPLE 12. The remainder when $12x^3 - 13x^2 - 5x + 9$ is divided by $(3x + 2)$ is

- a. 1 b. 2 c. 3 d. 0

Sol. c. Let $p(x) = 12x^3 - 13x^2 - 5x + 9$ and $q(x) = 3x + 2$
When $p(x)$ is divided by $(3x + 2)$, then remainder is $p\left(\frac{-2}{3}\right)$.

$$\begin{aligned}\text{Now, } p\left(\frac{-2}{3}\right) &= 12\left(\frac{-2}{3}\right)^3 - 13\left(\frac{-2}{3}\right)^2 - 5\left(\frac{-2}{3}\right) + 9 \\ &= 12 \times \left(\frac{-8}{27}\right) - 13 \times \frac{4}{9} + \frac{10}{3} + 9 = 3\end{aligned}$$

Hence, the required remainder is 3.

Factor Theorem

Let $p(x)$ be a polynomial, if degree $n \geq 1$ and a be any real number. Then,

- If $p(a) = 0$, then $(x - a)$ is a factor of $p(x)$.
- If $(x - a)$ is a factor of $p(x)$, then $p(a) = 0$.

EXAMPLE 13. For what values of k will $4x^5 + 9x^4 - 7x^3 - 5x^2 - 4kx + 3k^2$ contain $x - 1$ as a factor?

- a. 3, $-\frac{1}{2}$ b. 3, -1 c. 0, $\frac{1}{3}$ d. 1, $\frac{1}{3}$

Sol. d. Given, $P(x) = 4x^5 + 9x^4 - 7x^3 - 5x^2 - 4kx + 3k^2$

$$\begin{aligned}\text{Since, } (x - 1) \text{ is a factor of } P(x). \therefore P(1) &= 0 \\ \Rightarrow 4 \times (1)^5 + 9 \times (1)^4 - 7 \times (1)^3 - 5 \times (1)^2 - 4 \times k \times (1) + 3 \times k^2 &= 0 \\ \Rightarrow 3k^2 - 4k + 1 &= 0 \Rightarrow 3k^2 - 3k - k + 1 = 0 \\ \Rightarrow (3k - 1)(k - 1) &= 0 \Rightarrow k = \frac{1}{3}, 1\end{aligned}$$

Division Synthetic Algorithm

If $p(x)$ and $g(x)$ any two polynomials with $g(x) \neq 0$, then we can find polynomials $q(x)$ and $r(x)$ such that

$$p(x) = g(x) \times q(x) + r(x)$$

i.e. Dividend = (Divisor $\times$ Quotient) + Remainder

where, $r(x) = 0$ or degree of $r(x) <$ degree of $g(x)$, then we say that $p(x)$ divided by $g(x)$, gives $q(x)$ as quotient and $r(x)$ as remainder.

If the remainder $r(x)$ is zero, we say that divisor $g(x)$ is a factor of $p(x)$.

EXAMPLE 14. If two factors of $a^4 - 2a^3 - 9a^2 + 2a + 8$ are $(a + 1)$ and $(a - 1)$, then what are the other two factors?

- a. $(a - 2)$ and $(a + 4)$ b. $(a + 2)$ and $(a + 4)$
c. $(a + 2)$ and $(a - 4)$ d. $(a - 2)$ and $(a - 4)$

Sol. c. Let $f(a) = a^4 - 2a^3 - 9a^2 + 2a + 8$
and $(a + 1)$ and $(a - 1)$ are two factors of $f(a)$.
Now, by division method,

$$\begin{array}{r} a^2 - 1 \quad a^4 - 2a^3 - 9a^2 + 2a + 8 \quad (a^2 - 2a - 8) \\ \underline{a^4 - 2a^3} \\ - 9a^2 + 2a + 8 \\ \underline{- 9a^2 + 18a + 8} \\ 2a - 8 \\ \underline{2a - 8} \\ 0 \end{array}$$

So, the required factor of $f(a)$ is $(a^2 - 2a - 8)$.
Further factorise it by splitting the middle term, i.e.

$$\begin{aligned}a^2 - 2a - 8 &= a^2 - 4a + 2a - 8 \\ &= a(a - 4) + 2(a - 4) = (a + 2)(a - 4)\end{aligned}$$

Hence, $(a + 2)$ and $(a - 4)$ are other two factors of $f(a)$.

Factorisation of Polynomials Using Factor Theorem

To factorise the polynomial, follow the following steps

- Let $p(x)$ be a given polynomial.
- find all the possible factors of constant term in $p(x)$.
- Take any one of the factors, say (α) and find $p(\alpha)$. If $p(\alpha) = 0$ the $(x - \alpha)$ is a factor of $p(x)$.
- Divide the polynomial $p(x)$ by $(x - \alpha)$ to find its all other factors.

EXAMPLE 15. Factorise $2x^3 - 5x^2 - 19x + 42$.

- a. $(x - 2)(2x - 7)(x + 3)$ b. $(x + 2)(2x + 7)(x - 3)$
c. $(x - 1)(3x - 7)(x - 4)$ d. $(2x + 3)(4x - 7)(x - 1)$

Sol. a. Let $p(x) = 2x^3 - 5x^2 - 19x + 42$

Hence, constant term is 42 and its all factors are $\pm 1, \pm 2, \pm 3, \pm 6, \pm 7, \pm 14, \pm 21$ and ± 42 .

$$\begin{aligned}\text{At } x = 1, \quad p(1) &= 2(1)^3 - 5(1)^2 - 19 + 42 \\ &= 2 - 5 - 19 + 42 = 20 \neq 0\end{aligned}$$

So, $(x - 1)$ is not a factor of $p(x)$.

$$\text{At } x = 2, \quad p(2) = 2(2)^3 - 5(2)^2 - 19(2) + 42 = 58 - 58 = 0$$

So, $(x - 2)$ is a factor of $p(x)$.

Now, on dividing $p(x)$ by $(x - 2)$, we get $(2x^2 - x - 21)$ as quotient i.e. other factor of $p(x)$.

$$\begin{aligned}\text{So, } p(x) &= (x - 2)[2x^2 - x - 21] \\ &= (x - 2)(2x^2 - 7x + 6x - 21) \\ &= (x - 2)[x(2x - 7) + 3(2x - 7)]\end{aligned}$$

$$\therefore p(x) = (x - 2)(2x - 7)(x + 3)$$

PRACTICE EXERCISE

- If $x = 5, y = 3$ and $z = 2$, then the value of $x^2 + y^2 + z^2 - 2xy + 2yz - 2zx$ is
(a) 125 (b) 0 (c) -25 (d) 10
- The factors of $x^2 - 2\sqrt{3}x + 3$ are
(a) $(x + \sqrt{3})^2$ (b) $(x - \sqrt{3})^2$
(c) $(x + \sqrt{3})(x - \sqrt{3})$ (d) $(x + 2)(x + \sqrt{3})$
- The factors of $(a^4b^4 - 16c^4)$ are
(a) $4(a^2b^2 + c^2)(ab - 2c)(ab + 2c)$
(b) $(a^2b^2 - 4c^2)(ab + 2c)^2$
(c) $(a^2b^2 + 4c^2)(ab + 2c)(ab - 2c)$
(d) $(a^2b^2 - 4c^2)^2(ab + 2c)(ab + 4c)$
- The factors of $(x^8 - y^8)$ are
(a) $(x^4 + y^4)(x^2 + y^2)(x + y)(x - y)$
(b) $(x^2 + y^2)^2(x + y)(x - y)$
(c) $(x^4 + y^4)(x^2 + y^2)^2$
(d) $(x^2 + y^2)(x - y)^2$
- The factors of $\left(a^2 + a + \frac{1}{4}\right)$ are
(a) $\left(a + \frac{1}{2}\right)^2$ (b) $\left(a + \frac{1}{2}\right)(a + 2)$
(c) $\left(a + \frac{1}{2}\right)\left(a - \frac{1}{2}\right)$ (d) $\left(a - \frac{1}{2}\right)^2$
- The factors of $8 - 4x - 2x^3 + x^4$ are
(a) $(2 - x)(4 - x^3)$ (b) $(2 + x)(4 - x^3)$
(c) $(2 + x)(3 - x^3)$ (d) $(2 - x)(x^3 - 4)$
- The factors of $(a^2 - b^2 - 4ac + 4c^2)$ are
(a) $(a + 2c + b)(a - 2c - b)$ (b) $(a - 2c + b)(a - 2c - b)$
(c) $(a - 2b + c)(a + b + 2c)$ (d) $(a - 2b)(a + 2b + 2c)$
- What are the factors of $x^2 + 4y^2 + 4y - 4xy - 2x - 8$?
(a) $(x - 2y - 4)$ and $(x - 2y + 2)$
(b) $(x - y + 2)$ and $(x - 4y + 4)$
(c) $(x - y + 2)$ and $(x - 4y - 4)$
(d) $(x + 2y - 4)$ and $(x + 2y + 2)$
- Factorise $a^3 - \frac{1}{a^3} - 2a + \frac{2}{a}$
(a) $\left(a + \frac{1}{a}\right)\left(a^2 + \frac{1}{a^2}\right) - 2\left(a - \frac{1}{a}\right)$
(b) $\left(a - \frac{1}{a}\right)\left(a^2 + \frac{1}{a^2} + 1\right) + 2\left(a - \frac{1}{a}\right)$
(c) $\left(a + \frac{1}{a}\right)\left(a - \frac{1}{a}\right)$
(d) None of the above
- What are the factors of $\left(\frac{1}{3}x^2 - 2x - 9\right)$?
(a) $\frac{1}{3}(x - 9)(x + 3)$ (b) $\frac{1}{3}(x - 9)(x - 3)$
(c) $\frac{1}{3}(x + 9)(x + 3)$ (d) $\frac{1}{3}(x + 9)(x - 3)$
- What are the factors of $(6\sqrt{3}x^2 - 47x + 5\sqrt{3})$?
(a) $(3\sqrt{3}x + 5\sqrt{3})(2x - 5\sqrt{3})$
(b) $(3\sqrt{3}x - 5\sqrt{3})(x\sqrt{3}x + 1)$
(c) $(2x + 5\sqrt{3})(3\sqrt{3}x + 1)$
(d) $(2x - 5\sqrt{3})(3\sqrt{3}x - 1)$
- The factors of $(8a^3 + 125b^3 - 64c^3 + 120abc)$ are
(a) $(2a + 5b - 4c)(2a + 5b + 4c)$
(b) $(2a - 5b - 4c)(2a + 5b + 4c)$
(c) $(2a + 5b - 4c)(4a^2 + 25b^2 + 16c^2 - 10ab + 20bc + 8ac)$
(d) $(2a + 5b + 4c)(4a^2 + 25b^2 + 16c^2 - 10ab + 20bc + 8ac)$
- If $x^{1/3} + y^{1/3} + z^{1/3} = 0$, then
(a) $x + y + z = 0$ (b) $(x + y + z)^3 = 27xyz$
(c) $x + y + z = 3xyz$ (d) $x^3 + y^3 + z^3 = 0$
- If $x + \frac{1}{x} = \sqrt{5}$, then the value of $x^3 + \frac{1}{x^3}$ is
(a) $8\sqrt{5}$ (b) $2\sqrt{5}$ (c) $5\sqrt{5}$ (d) $7\sqrt{5}$
- If $\left(x + \frac{1}{x}\right) = 6$, then $\left(x^2 + \frac{1}{x^2}\right)$ is equal to
(a) 32 (b) 38 (c) 34 (d) 44
- If $x - \frac{1}{x} = \frac{1}{3}$, then what is $9x^2 + \frac{9}{x^2}$ equal to?
(a) 18 (b) 19 (c) 20 (d) 21
- If $a + b + c = 6$ and $a^2 + b^2 + c^2 = 26$, then what is $ab + bc + ca$ equal to?
(a) 0 (b) 2 (c) 4 (d) 5
- If $(x^4 + x^{-4}) = 322$, then what is one of the value of $(x - x^{-1})$?
(a) 18 (b) 16 (c) 8 (d) 4
- If $a + b + c = 0$, then what is the value of $\frac{a^2}{bc} + \frac{b^2}{ca} + \frac{c^2}{ab}$?
(a) -3 (b) 0 (c) 1 (d) 3
- If $a + b + c = 10$ and $ab + bc + ca = 31$, then the value of $a^2 + b^2 + c^2$ is
(a) 48 (b) 38 (c) 28 (d) 18

- 21.** The factors of $x^2 + \frac{1}{4x^2} + 1 - 2x - \frac{1}{x}$ are
 (a) $\left(x - \frac{1}{2x}\right)\left(x - \frac{1}{2x} + 2\right)$ (b) $\left(x + \frac{1}{2x}\right)\left(x + \frac{1}{2x} - 2\right)$
 (c) $\left(x - \frac{1}{2x}\right)\left(x - \frac{1}{2x} - 2\right)$ (d) $\left(x + \frac{1}{2x}\right)\left(x - \frac{1}{2x} - 2\right)$
- 22.** The factors of $(a^2 - b^2)(c^2 - d^2) - 4abcd$ are
 (a) $(ac + bd + bc + ad)(ac - bd - bc - ad)$
 (b) $(ac - bd - bc + ad)(ac + bd + dc + ab)$
 (c) $(ac - bd + bc + ad)(ac - bd - bc - ad)$
 (d) $(ac - bd - bc - ad)(ac + bd + dc + ab)$
- 23.** If 'a' is an integer such that $a + \frac{1}{a} = \frac{17}{4}$, then the value of $\left(a - \frac{1}{a}\right)$ is
 (a) 4 (b) $\frac{13}{4}$ (c) $\frac{17}{4}$ (d) $\frac{15}{4}$
- 24.** $x^4 + 4y^4$ is divisible by which one of the following?
 (a) $(x^2 + 2xy + 2y^2)$ (b) $(x^2 + 2y^2)$
 (c) $(x^2 - 2y^2)$ (d) None of these
- 25.** If $x(x + y + z) = 9$, $y(x + y + z) = 16$ and $z(x + y + z) = 144$, then x equal to
 (a) $\frac{9}{5}$ (b) $\frac{9}{7}$ (c) $\frac{9}{13}$ (d) $\frac{16}{13}$
- 26.** If $x = (b - c)(a - d)$, $y = (c - a)(b - d)$ and $z = (a - b)(c - d)$, then what is $x^3 + y^3 + z^3$ equal to?
 (a) xyz (b) $2xyz$ (c) $3xyz$ (d) $-3xyz$
- 27.** If the expression $(px^3 + x^2 - 2x - q)$ is divisible by $(x - 1)$ and $(x + 1)$, what are the values of p and q respectively?
 (a) 2, -1 (b) -2, 1 (c) -2, -1 (d) 2, 1
- 28.** If the expression $px^3 + 3x^2 - 3$ and $2x^3 - 5x + p$ when divided by $x - 4$ leave the same remainder, then what is the value of p ?
 (a) -1 (b) 1 (c) -2 (d) 2
- 29.** Let $f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \dots + a_{n-1}x + a_n$, where $a_0, a_1, a_2, \dots, a_n$ are constants. If $f(x)$ is divided by $ax - b$, the remainder is
 (a) $f\left(\frac{b}{a}\right)$ (b) $f\left(-\frac{b}{a}\right)$ (c) $f\left(\frac{a}{b}\right)$ (d) $f\left(-\frac{a}{b}\right)$
- 30.** Which one of the following statements is correct?
 (a) Remainder theorem is a special case of factor theorem
 (b) Factor theorem is a special case of remainder theorem
 (c) Factor theorem and remainder theorem are two independent results
 (d) None of the above
- 31.** What is $x(y - z)(y + z) + y(z - x)(z + x) + z(x - y)(x + y)$ equal to?
 (a) $(x + y)(y + z)(z + x)$ (b) $(x - y)(x - z)(z - y)$
 (c) $(x + y)(z - y)(x - z)$ (d) $(y - x)(z - y)(x - z)$
- 32.** If the remainder of the polynomial $a_0 + a_1x + a_2x^2 + \dots + a_nx^n$ when divided by $(x - 1)$ is 1, then which one of the following is correct?
 (a) $a_0 + a_2 + \dots = a_1 + a_3 + \dots$
 (b) $a_0 + a_2 + \dots = 1 + a_1 + a_3 + \dots$
 (c) $1 + a_0 + a_2 + \dots = -(a_1 + a_3 + \dots)$
 (d) $1 - a_0 - a_2 - \dots = a_1 + a_3 + \dots$
- 33.** If u, v and w are real numbers such that $u^3 - 8v^3 - 27w^3 = 18uvw$, then which one of the following is correct?
 (a) $u - v + w = 0$ (b) $u = -v = -w$
 (c) $u - 2v = 3w$ (d) $u + 2v = -3w$
- 34.** The value of $(a^{1/8} + a^{-1/8})(a^{1/8} - a^{-1/8})(a^{1/4} + a^{-1/4})(a^{1/2} + a^{-1/2})$ is
 (a) $(a + a^{-1})$ (b) $(a - a^{-1})$ (c) $(a^2 + a^{-2})$ (d) $(a^2 - a^{-2})$
- 35.** The polynomial $f(x) = x^4 - 2x^3 + 3x^2 - ax + b$, when divided by $(x - 1)$ and $(x + 1)$ leaves the remainders 5 and 19, respectively. The values of a and b are
 (a) $a = 4, b = 3$ (b) $a = 3, b = 4$
 (c) $a = 5, b = 8$ (d) $a = 9, b = 7$
- 36.** If the expression $ax^2 + bx + c$ is equal to 4 when $x = 0$, leaves a remainder 4 when divided by $x + 1$ and a remainder 6 when divided by $x + 2$, then the values of a, b and c are respectively:
 (a) 1, 1, 4 (b) 2, 2, 4 (c) 3, 3, 4 (d) 4, 4, 4
- 37.** If $(x^{3/2} - xy^{1/2} + x^{1/2}y - y^{3/2})$ is divided by $x^{1/2} - y^{1/2}$ the quotient is
 (a) $x - y$ (b) $x + y$ (c) $x^{1/2} + y^{1/2}$ (d) $x^2 - y^2$
- 38.** If $9x^2 + 3px + 6q$ when divided by $3x + 1$ leaves a remainder $-\frac{3}{4}$ and $qx^2 + 4px + 7$ is exactly divisible by $x + 1$, then the values of p and q respectively are
 (a) $0, 7/4$ (b) $-7/4, 0$
 (c) $\frac{7}{4}, 0$ (d) None of these
- 39.** If $a + b + c = 1$, then which of the following statements are true?
 I. $(a + b)(b + c)(c + a) = bc + ac + ab - abc$
 II. $a^2 + b^2 - c^2 + 2ab = a + b - c$
 (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II
- 40.** If $x + \frac{1}{x} = 4$, then the value of expression
 I. $x^3 + \frac{1}{x^3} = 52$ II. $x = 2 + \sqrt{3}$
 Which of the following is correct?
 (a) Only I (b) Only II
 (c) Neither I nor II (d) Both I and II

PREVIOUS YEARS' QUESTIONS

- 41.** What is the value of k that $(2x - 1)$ may be a factor of $4x^4 - (k - 1)x^3 + kx^2 - 6x + 1$? **2012 I**
 (a) 8 (b) 9 (c) 12 (d) 13
- 42.** $x^4 + xy^3 + x^3y + xz^3 + y^4 + yz^3$ is divisible by **2012 I**
 (a) Only $(x - y)$
 (b) Only $(x^3 + y^3 + z^3)$
 (c) Both $(x + y)$ and $(x^3 + y^3 + z^3)$
 (d) None of the above
- 43.** If $x^3 + 5x^2 + 10k$ leaves remainder $-2x$ when divided by $x^2 + 2$, then what is the value of k ? **2012 I**
 (a) -2 (b) -1 (c) 1 (d) 2
- 44.** If $x + \frac{1}{x} = a$, then what is the value of $x^3 + x^2 + \frac{1}{x^3} + \frac{1}{x^2}$? **2012 I**
 (a) $a^3 + a^2$ (b) $a^3 + a^2 - 5a$
 (c) $a^3 + a^2 - 3a - 2$ (d) $a^3 + a^2 - 4a - 2$
- 45.** If $p(x)$ is a common multiple of degree 6 of the polynomials $f(x) = x^3 + x^2 - x - 1$ and $g(x) = x^3 - x^2 + x - 1$, then which one of the following is correct? **2012 I**
 (a) $\rho(x) = (x - 1)^2 (x + 1)^2 (x^2 + 1)$
 (b) $\rho(x) = (x - 1) (x + 1) (x^2 + 1)^2$
 (c) $\rho(x) = (x - 1)^3 (x + 1) (x^2 + 1)$
 (d) $\rho(x) = (x - 1)^2 (x^4 + 1)$
- 46.** What is the value of k which will make the expression $4k^2 + 12k + k$ a perfect square? **2012 II**
 (a) 5 (b) 7 (c) 8 (d) 9
- 47.** The factors of $5px - 10qy + 2rpx - 4qry$ is/are **2013 I**
 (a) Only $(5 + 2r)$
 (b) Only $(px - 2qy)$
 (c) Both $(5 + 2r)$ and $(px - 2qy)$
 (d) Neither $(5 + 2r)$ nor $(px - 2qy)$
- 48.** $(a + 1)^4 - a^4$ is divisible by **2013 I**
 (a) $-2a^2 + 2a - 1$ (b) $2a^3 - 2a - 1$
 (c) $2a^3 - 2a + 1$ (d) $2a^2 + 2a + 1$
- 49.** One of the factors of the polynomial $x^4 - 7x^3 + 5x^2 - 6x + 81$ is **2013 I**
 (a) $x + 2$ (b) $x - 2$ (c) $x + 3$ (d) $x - 3$
- 50.** If the expression $x^3 + 3x^2 + 4x + k$ has a factor $x + 5$, then what is the value of k ? **2013 II**
 (a) -70 (b) 70 (c) 48 (d) -48
- 51.** The quantity which must be added to $(1 - x)(1 + x^2)$ to obtain x^3 is **2013 II**
 (a) $2x^3 + 3x^2 + x + 1$ (b) $2x^3 + x^2 + x - 1$
 (c) $2x^3 - x^2 + x - 1$ (d) $-x^3 + x - 1$
- 52.** $x(y^2 - z^2) + y(z^2 - x^2) + z(x^2 - y^2)$ is divisible by **2013 II**
 (a) Only $(y - z)$
 (b) Only $(z - x)$
 (c) Both $(y - z)$ and $(z - x)$
 (d) Neither $(y - z)$ nor $(z - x)$
- 53.** Consider the following statements
 I. $x + 3$ is the factor of $x^3 + 2x^2 + 3x + 8$.
 II. $x - 2$ is the factor of $x^3 + 2x^2 + 3x + 8$.
 Which of the statement(s) given above is/are correct? **2013 II**
 (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II
- 54.** $x^3 + 6x^2 + 11x + 6$ is divisible by **2014 I**
 (a) Only $(x + 1)$ (b) Only $(x + 2)$
 (c) Only $(x + 3)$ (d) All of these
- 55.** What should be added to the $x(x + a)(x + 2a)(x + 3a)$, so that the sum be a perfect square?
 (a) $9a^2$ (b) $4a^2$ **2014 I**
 (c) a^4 (d) None of these
- 56.** If $\left(x^2 + \frac{1}{x^2}\right) = \frac{17}{4}$, then what is $\left(x^3 - \frac{1}{x^3}\right)$ equal to? **2014 I**
 (a) $\frac{75}{16}$ (b) $\frac{63}{8}$ (c) $\frac{95}{8}$ (d) None of these
- 57.** What is the remainder when $x^5 - 5x^2 + 125$ is divided by $x + 5$? **2014 II**
 (a) 0 (b) 125 (c) -3125 (d) 3125
- 58.** If $ax + by - 2 = 0$ and $axby = 1$, where $a \neq 0, b \neq 0$, then what is $(a^2x + b^2y)$ equal to? **2014 II**
 (a) $a + b$ (b) $2ab$ (c) $a^3 + b^3$ (d) $a^4 + b^4$
- 59.** If $(x + k)$ is the common factor of $x^2 + ax + b$ and $x^2 + cx + d$, then what is k equal to? **2014 II**
 (a) $(d - b)/(c - a)$ (b) $(d - b)/(a - c)$
 (c) $(d + b)/(c + a)$ (d) $(d - b)/(c + a)$
- 60.** Consider the following statements:
 I. $(a - b - c)$ is one of the factors of $3abc + b^3 + c^3 - a^3$.
 II. $(b + c - 1)$ is one of the factors of $3bc + b^3 + c^3 - 1$.
 Which of the statement(s) given above is/are correct? **2014 II**
 (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II
- 61.** For what value of k is $(x - 5)$ a factor of $x^3 - 3x^2 + kx - 10$? **2015 I**
 (a) -8 (b) 4 (c) 2 (d) 1
- 62.** If $x + y + z = 0$, then $x^3 + y^3 + z^3 + 3xyz$ is equal to **2015 I**
 (a) 0 (b) $6xyz$ (c) $12xyz$ (d) xyz

- 63.** The expression $x^3q^2 - x^3pt + 4x^2pt - 4x^2q^2 + 3xq^2 - 3xpt$ is divisible by ☑ 2015 I
 (a) Only $(x-1)$ (b) Only $(x-3)$
 (c) Both $(x-1)$ and $(x-3)$ (d) Neither $(x-1)$ nor $(x-3)$
- 64.** Which one of the following is correct? ☑ 2015 II
 (a) $(x+2)$ is a factor of $x^4 - 6x^3 + 12x^2 - 24x + 32$
 (b) $(x+2)$ is a factor of $x^4 + 6x^3 - 12x^2 + 24x - 32$
 (c) $(x-2)$ is a factor of $x^4 - 6x^3 + 12x^2 - 24x + 32$
 (d) $(x-2)$ is a factor of $x^4 + 6x^3 - 12x^2 + 24x - 32$
- 65.** For what value of k $(x+1)$ is a factor of $x^3 + kx^2 - x + 2$? ☑ 2016 I
 (a) 4 (b) 3 (c) 1 (d) -2

- 66.** If the polynomial $x^6 + px^5 + qx^4 - x^2 - x - 3$ is divisible by $(x^4 - 1)$, then the value of $p^2 + q^2$ is ☑ 2016 I
 (a) 1 (b) 9 (c) 10 (d) 13
- 67.** If the linear factors of $ax^2 - (a^2 + 1)x + a$ are p and q , then $p + q$ is equal to ☑ 2016 I
 (a) $(x-1)(a+1)$ (b) $(x+1)(a+1)$
 (c) $(x-1)(a-1)$ (d) $(x+1)(a-1)$
- 68.** If $\sqrt{\frac{x}{y}} = \frac{10}{3} - \sqrt{\frac{y}{x}}$ and $x - y = 8$, then the value of xy is equal to ☑ 2016 I
 (a) 36 (b) 24 (c) 16 (d) 9

ANSWERS

1	b	2	b	3	c	4	a	5	a	6	a	7	b	8	a	9	d	10	a
11	d	12	c	13	b	14	b	15	c	16	b	17	d	18	d	19	d	20	b
21	b	22	c	23	d	24	a	25	c	26	c	27	d	28	b	29	a	30	b
31	b	32	d	33	c	34	b	35	c	36	a	37	b	38	c	39	c	40	d
41	d	42	c	43	c	44	c	45	a	46	d	47	c	48	d	49	d	50	b
51	c	52	c	53	d	54	d	55	c	56	b	57	c	58	a	59	a	60	c
61	a	62	b	63	c	64	c	65	d	66	c	67	a	68	d				

HINTS AND SOLUTIONS

1. (b) $x^2 + y^2 + z^2 - 2xy + 2yz - 2zx$
 $= (x - y - z)^2 = (5 - 3 - 2)^2 = 0$
 $[\because x = 5, y = 3 \text{ and } z = 2]$

2. (b) $x^2 - 2\sqrt{3}x + 3$
 $= x^2 - \sqrt{3}x - \sqrt{3}x + 3$
 $= x(x - \sqrt{3}) - \sqrt{3}(x - \sqrt{3})$
 $= (x - \sqrt{3})(x - \sqrt{3}) = (x - \sqrt{3})^2$

3. (c) $(a^4b^4 - 16c^4) = [(a^2b^2)^2 - (4c^2)^2]$
 $= [a^2b^2 + 4c^2][a^2b^2 - 4c^2]$
 $[\because a^2 - b^2 = (a - b)(a + b)]$
 $= [a^2b^2 + 4c^2][(ab)^2 - (2c)^2]$
 $= [a^2b^2 + 4c^2][(ab + 2c)(ab - 2c)]$

4. (a) $x^8 - y^8 = [(x^4)^2 - (y^4)^2]$
 $= [x^4 + y^4][x^4 - y^4]$
 $= [x^4 + y^4][(x^2)^2 - (y^2)^2]$
 $= (x^4 + y^4)(x^2 + y^2)(x^2 - y^2)$
 $= (x^4 + y^4)(x^2 + y^2)(x + y)(x - y)$

5. (a) $\left(a^2 + a + \frac{1}{4}\right) = a^2 + \frac{1}{2}a + \frac{1}{2}a + \frac{1}{4}$
 $= a\left(a + \frac{1}{2}\right) + \frac{1}{2}\left(a + \frac{1}{2}\right)$
 $= \left(a + \frac{1}{2}\right)\left(a + \frac{1}{2}\right) = \left(a + \frac{1}{2}\right)^2$

6. (a) $8 - 4x - 2x^3 + x^4$
 $= 4(2 - x) - x^3(2 - x)$
 $= (2 - x)(4 - x^3)$

7. (b) $(a^2 - b^2 - 4ac + 4c^2)$
 $= (a^2 - 4ac + 4c^2) - b^2$
 $[(a)^2 - 2(2c)(a) + (2c)^2] - b^2$
 $= (a - 2c)^2 - b^2$
 $= (a - 2c - b)(a - 2c + b)$

8. (a) $x^2 + 4y^2 + 4y - 4xy - 2x - 8$
 $= x^2 - 4xy + 4y^2 - 2x + 4y - 8$
 $= (x - 2y)^2 - 2(x - 2y) - 8$
 Let $x - 2y = z$

$\therefore z^2 - 2z - 8 = z^2 - 4z + 2z - 8$
 $= z(z - 4) + 2(z - 4)$
 $= (z + 2)(z - 4)$

Now, put $z = x - 2y$
 $\therefore (x - 2y + 2)(x - 2y - 4)$
 and $(x - 2y - 4)$ are required factors.

9. (d) $a^3 - \frac{1}{a^3} - 2a + \frac{2}{a}$
 $= a^3 - \left(\frac{1}{a}\right)^3 - 2\left(a - \frac{1}{a}\right)$
 $= \left(a - \frac{1}{a}\right)\left(a^2 + \frac{1}{a^2} + 1\right) - 2\left(a - \frac{1}{a}\right)$
 $= \left(a - \frac{1}{a}\right)\left(a^2 + \frac{1}{a^2} - 1\right)$

10. (a) $\left(\frac{1}{3}x^2 - 2x - 9\right) = \frac{(x^2 - 6x - 27)}{3}$
 $= \frac{1}{3}[x^2 - 9x + 3x - 27]$
 $= \frac{1}{3}[x(x - 9) + 3(x - 9)]$
 $= \frac{1}{3}(x + 3)(x - 9)$

$$\begin{aligned}
 11. (d) & 6\sqrt{3}x^2 - 47x + 5\sqrt{3} \\
 &= 6\sqrt{3}x^2 - 45x - 2x + 5\sqrt{3} \\
 &= 3\sqrt{3}x(2x - 5\sqrt{3}) - 1(2x - 5\sqrt{3}) \\
 &= (2x - 5\sqrt{3})(3\sqrt{3}x - 1)
 \end{aligned}$$

$$\begin{aligned}
 12. (c) & 8a^3 + 125b^3 - 64c^3 + 120abc \\
 &= (2a)^3 + (5b)^3 - (4c)^3 + 120abc \\
 &= (2a)^3 + (5b)^3 + (-4c)^3 - 3(2a)(5b)(-4c) \\
 &= (2a + 5b - 4c)(4a^2 + 25b^2 + 16c^2 - 10ab + 20bc + 8ac) \\
 &[\because a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ac)]
 \end{aligned}$$

$$\begin{aligned}
 13. (b) & \text{Since, } x^{1/3} + y^{1/3} + z^{1/3} = 0 \\
 \therefore & (x^{1/3})^3 + (y^{1/3})^3 + (z^{1/3})^3 - 3x^{1/3}y^{1/3}z^{1/3} = 0 \\
 \Rightarrow & x + y + z - 3(xyz)^{1/3} = 0 \\
 \Rightarrow & x + y + z = 3(xyz)^{1/3} \\
 \Rightarrow & (x + y + z)^3 = 27xyz
 \end{aligned}$$

$$\begin{aligned}
 14. (b) & \because \left(x + \frac{1}{x}\right)^3 = x^3 + \frac{1}{x^3} + 3\left(x + \frac{1}{x}\right) \\
 & (\sqrt{5})^3 = x^3 + \frac{1}{x^3} + 3(\sqrt{5}) \\
 & \left[\because x + \frac{1}{x} = \sqrt{5}\right] \\
 \Rightarrow & x^3 + \frac{1}{x^3} = 5\sqrt{5} - 3\sqrt{5} = 2\sqrt{5}
 \end{aligned}$$

$$\begin{aligned}
 15. (c) & \text{Given, } \left(x + \frac{1}{x}\right) = 6 \\
 \text{On squaring both sides, we get} \\
 & \left(x + \frac{1}{x}\right)^2 = 6^2 \\
 \Rightarrow & x^2 + \frac{1}{x^2} + 2 = 36 \\
 \Rightarrow & x^2 + \frac{1}{x^2} = 34
 \end{aligned}$$

$$\begin{aligned}
 16. (b) & \text{Given, } x - \frac{1}{x} = \frac{1}{3} \\
 \Rightarrow & 3x - \frac{3}{x} = 1 \\
 \text{On squaring both sides, we get} \\
 & \left(3x - \frac{3}{x}\right)^2 = (1)^2 \\
 & 9x^2 + \frac{9}{x^2} - 2 \times 9 = 1 \\
 \Rightarrow & 9x^2 + \frac{9}{x^2} = 19
 \end{aligned}$$

$$\begin{aligned}
 17. (d) & \because (a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca) \\
 \therefore & (6)^2 = 26 + 2(ab + bc + ca) \\
 \Rightarrow & 2(ab + bc + ca) = 10 \\
 \Rightarrow & ab + bc + ca = 5
 \end{aligned}$$

$$\begin{aligned}
 18. (d) & \text{Given, } x^4 + \frac{1}{x^4} = 322 \\
 \Rightarrow & \left(x^2 + \frac{1}{x^2}\right)^2 - 2 = 322 \\
 \Rightarrow & \left(x^2 + \frac{1}{x^2}\right)^2 = 324 = 18^2 \\
 \Rightarrow & x^2 + \frac{1}{x^2} = 18 \\
 \Rightarrow & \left(x - \frac{1}{x}\right)^2 + 2 = 18 \Rightarrow x - \frac{1}{x} = 4
 \end{aligned}$$

$$\begin{aligned}
 19. (d) & \text{Given, } a + b + c = 0 \\
 \Rightarrow & a^3 + b^3 + c^3 = 3abc \\
 \text{On dividing both sides by } abc, \text{ we get} \\
 \Rightarrow & \frac{a^3}{abc} + \frac{b^3}{abc} + \frac{c^3}{abc} = 3 \\
 \Rightarrow & \frac{a^2}{bc} + \frac{b^2}{ac} + \frac{c^2}{ab} = 3
 \end{aligned}$$

$$\begin{aligned}
 20. (b) & \because (a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ac \\
 & (10)^2 = (a^2 + b^2 + c^2) + 2(31) \\
 & a^2 + b^2 + c^2 = 100 - 62 \\
 \Rightarrow & a^2 + b^2 + c^2 = 38
 \end{aligned}$$

$$\begin{aligned}
 21. (b) & x^2 + \frac{1}{4x^2} + 1 - 2x - \frac{1}{x} \\
 &= \left(x^2 + \frac{1}{4x^2} + 1\right) - 2\left(x + \frac{1}{2x}\right) \\
 &= \left(x + \frac{1}{2x}\right)^2 - 2\left(x + \frac{1}{2x}\right) \\
 &= \left(x + \frac{1}{2x}\right)\left(x + \frac{1}{2x} - 2\right)
 \end{aligned}$$

$$\begin{aligned}
 22. (c) & (a^2 - b^2)(c^2 - d^2) - 4abcd \\
 &= a^2c^2 - a^2d^2 - b^2c^2 + b^2d^2 - 4abcd \\
 &= (a^2c^2 + b^2d^2 - 2abcd) - (b^2c^2 + a^2d^2 + 2abcd) \\
 &= (ac - bd)^2 - (bc + ad)^2 \\
 &= (ac - bd + bc + ad)(ac - bd - bc - ad)
 \end{aligned}$$

$$\begin{aligned}
 23. (d) & \because \left(a - \frac{1}{a}\right) = \sqrt{\left(a + \frac{1}{a}\right)^2 - 4} \\
 &= \sqrt{\left(\frac{17}{4}\right)^2 - 4} \\
 & \left[\because \text{given } \left(a + \frac{1}{a}\right) = \frac{17}{4}\right] \\
 &= \sqrt{\frac{289 - 64}{16}} = \sqrt{\frac{225}{16}} = \frac{15}{4}
 \end{aligned}$$

$$\begin{aligned}
 24. (a) & x^4 + 4y^4 \\
 &= x^4 + 4y^4 + 4x^2y^2 - 4x^2y^2 \\
 &= (x^2 + 2y^2)^2 - (2xy)^2 \\
 &= (x^2 + 2y^2 - 2xy)(x^2 + 2y^2 + 2xy)
 \end{aligned}$$

From above it is clear that $x^4 + 4y^4$ is divisible by $x^2 + 2y^2 + 2xy$

$$\begin{aligned}
 25. (c) & \text{Given, } x(x + y + z) = 9, \\
 & y(x + y + z) = 16 \\
 & \text{and } z(x + y + z) = 144 \\
 \text{On adding all three equations, we get} \\
 & x(x + y + z) + y(x + y + z) + z(x + y + z) \\
 &= 9 + 16 + 144 \\
 \Rightarrow & (x + y + z)(x + y + z) = 9 + 16 + 144 \\
 \Rightarrow & (x + y + z)^2 = 169 \\
 \Rightarrow & x + y + z = 13 \\
 \therefore & x(x + y + z) = 9 \\
 \Rightarrow & x(13) = 9 \Rightarrow x = \frac{9}{13}
 \end{aligned}$$

$$\begin{aligned}
 26. (c) & \text{Given, } x = (b - c)(a - d), \\
 & y = (c - a)(b - d) \\
 \text{and } & z = (a - b)(c - d) \\
 \therefore & x + y + z = (b - c)(a - d) + (c - a)(b - d) + (a - b)(c - d) \\
 &= 0 \\
 \text{Hence, } & x^3 + y^3 + z^3 = 3xyz
 \end{aligned}$$

$$\begin{aligned}
 27. (d) & \text{Here, } px^3 + x^2 - 2x - q \text{ is divisible by } (x - 1) \text{ and } (x + 1). \\
 \therefore & p(1)^3 + (1)^2 - 2(1) - q = 0 \quad \dots(i) \\
 \Rightarrow & p - q = 1 \\
 \text{and } & p(-1)^3 + (-1)^2 - 2(-1) - q = 0 \quad \dots(ii) \\
 \Rightarrow & -p + q = 3
 \end{aligned}$$

On solving Eqs. (i) and (ii), we get

$$\begin{aligned}
 & p = 2 \text{ and } q = 1 \\
 28. (b) & \text{Let } f(x) = px^3 + 3x^2 - 3 \\
 \text{and } & g(x) = 2x^3 - 5x + p. \\
 \text{On dividing } f(x) \text{ by } (x - 4), \text{ remainder is} \\
 & f(4) = p(4)^3 + 3(4)^2 - 3 \\
 &= 64p + 48 - 3 = 64p + 45 \\
 \text{Similarly } & g(4) = 2(4)^3 - 5(4) + p \\
 &= 128 - 20 + p = 108 + p \\
 \text{But } & f(4) = g(4) \quad [\text{given}] \\
 \therefore & 64p + 45 = 108 + p \\
 \Rightarrow & 63p = 63 \Rightarrow p = 1
 \end{aligned}$$

$$\begin{aligned}
 29. (a) & ax - b = 0 \Rightarrow x = b/a \\
 \text{By remainder theorem, if } f(x) \text{ is divisible by } ax - b, \text{ the remainder is } f(b/a).
 \end{aligned}$$

30. (b) Factor theorem is a special case of remainder theorem.

$$\begin{aligned}
 31. (b) & x(y - z)(y + z) + y(z - x)(z + x) \\
 &= x(y^2 - z^2) + y(z^2 - x^2) + z(x^2 - y^2) \\
 &= x(y^2 - z^2) + yz^2 - yx^2 + zx^2 - zy^2 \\
 &= (y - z)(xy + xz - x^2 - yz) \\
 &= (y - z)[y(x - z) + x(z - x)] \\
 &= (y - z)(z - x)(x - y) \\
 &= (x - y)(x - z)(z - y)
 \end{aligned}$$

32. (d) Let $f(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$

Since, $(x-1)$ is a factor of $f(x)$.

Put $x = 1$ in $f(x)$, then

$$f(1) = a_0 + a_1 + a_2 + \dots + a_n$$

$$\Rightarrow 1 = a_0 + a_1 + a_2 + \dots + a_n$$

$$\therefore 1 - a_0 - a_2 - \dots = a_1 + a_3 + \dots$$

33. (c) Given, $(u)^3 + (-2v)^3 + (-3w)^3 = 3 \times (-2)(-3)uvw$

$$\therefore u + (-2v) + (-3w) = 0$$

$$[\because a^3 + b^3 + c^3 = 3abc, \text{ then } a + b + c = 0]$$

$$\Rightarrow u - 2v - 3w = 0$$

$$\therefore u - 2v = 3w$$

34. (b) We know that, $(a+b)(a-b) = a^2 - b^2$

$$\begin{aligned} \text{So, } & [(a^{1/8} + a^{-1/8})(a^{1/8} - a^{-1/8}) \\ & (a^{1/4} + a^{-1/4})(a^{1/2} + a^{-1/2})] \\ & = [(a^{1/8})^2 - (a^{-1/8})^2(a^{1/4} + a^{-1/4})] \\ & = (a^{1/4} - a^{-1/4})(a^{1/4} + a^{-1/4}) \\ & \quad (a^{1/2} + a^{-1/2}) \\ & = [a^{1/2} - a^{-1/2}][a^{1/2} + a^{-1/2}] \\ & = (a - a^{-1}) \end{aligned}$$

35. (c) $f(x) = x^4 - 2x^3 + 3x^2 - ax + b$

Put $x = 1$ as $f(1) = 5$, $f(-1) = 19$

$$f(1) = 1^4 - 2(1)^3 + 3(1)^2 - a + b$$

$$\Rightarrow 1 - 2 + 3 - a + b = 5$$

$$\Rightarrow -a + b = 3 \quad \dots(i)$$

and $f(-1) = (-1)^4 - 2(-1)^3 + 3(-1)^2 - a(-1) + b$

$$\Rightarrow 1 + 2 + 3 + a + b = 19$$

$$\Rightarrow a + b = 13 \quad \dots(ii)$$

On solving Eqs. (i) and (ii), we get

$$a = 5, b = 8$$

36. (a) Let $f(x) = ax^2 + bx + c$

Given $f(0) = 4$

$$\Rightarrow 0 + 0 + c = 4 \Rightarrow a - b + 4 = 4 \Rightarrow c = 4$$

On dividing $f(x)$ by $x + 1$, remainder is 4

$$\therefore f(-1) = 4$$

$$\Rightarrow a - b + c = 4 \Rightarrow a - b + 4 = 4$$

$$\Rightarrow a - b = 0 \quad \dots(i)$$

On dividing $f(x)$ by $x + 2$, remainder is 6

$$\therefore f(-2) = 6$$

$$\Rightarrow 4a - 2b + c = 6$$

$$\Rightarrow 4a - 2b = 2 \quad \dots(ii)$$

$$\Rightarrow a - b + 4 = 4$$

On solving Eq. (i) and eq. (ii), we get

$$a = 1, b = 1$$

Hence,

$$a = 1, b = 1, c = 4.$$

37. (b) $x^{3/2} - xy^{1/2} + x^{1/2}y - y^{3/2}$
 $= x(x^{1/2} - y^{1/2}) + y(x^{1/2} - y^{1/2})$
 $= (x^{1/2} - y^{1/2})(x + y)$

$\Rightarrow$ Quotient

$$= \frac{(x^{1/2} - y^{1/2})(x + y)}{(x^{1/2} - y^{1/2})} = x + y$$

38. (c) Let $f(x) = 9x^2 + 3px + 6q$

Given, $f(-1/3) = -3/4$

$$\Rightarrow 9\left(-\frac{1}{3}\right)^2 + 3p\left(-\frac{1}{3}\right) + 6q = -3/4$$

$$\Rightarrow 1 - p + 6q = -3/4$$

$$\Rightarrow 24q - 4p + 7 = 0 \quad \dots(i)$$

$$\text{Let } g(x) = qx^2 + 4px + 7$$

Since, $(x + 1)$ is a factor of $g(x)$

$$\therefore g(-1) = 0 \Rightarrow q - 4p + 7 = 0 \dots(ii)$$

On solving Eq. (i) and Eq. (ii), we get

$$q = 0 \text{ and } p = 7/4$$

39. (c) I. $(a+b)(b+c)(c+a)$

$$= (1-c)(1-a)(1-b)$$

$$[\because a + b + c = 1]$$

$$= (1-c)[1 - b - a + ab]$$

$$= 1 - b - a + ab - c + bc + ac - abc$$

$$= 1 - (a + b + c) + ab + bc + ac - abc$$

$$= ab + bc + ac - abc$$

II. $a^2 + b^2 - c^2 + 2ab$

$$= a^2 + b^2 + 2ab - c^2$$

$$= (a + b)^2 - c^2$$

$$= (a + b + c)(a + b - c) = a + b - c$$

$$[\because a + b + c = 1]$$

So, both statements are correct.

40. (d) We have, $x + \frac{1}{x} = 4$

I. $\left(x + \frac{1}{x}\right)^3 = x^3 + \frac{1}{x^3} + 3(x)$

$$\times \left(\frac{1}{x}\right) \left(x + \frac{1}{x}\right)$$

$$(4)^3 = x^3 + \frac{1}{x^3} + 3(4)$$

$$64 - 12 = x^3 + \frac{1}{x^3} \Rightarrow x^3 + \frac{1}{x^3} = 52$$

II. $x + \frac{1}{x} = 4 \Rightarrow x^2 - 4x + 1 = 0$

$$x = \frac{4 \pm \sqrt{16 - 4 \times 1 \times 1}}{2 \times 1}$$

$$= \frac{4 \pm 2\sqrt{3}}{2} = 2 \pm \sqrt{3}$$

Hence, both statements are correct.

41. (d) Let $f(x) = 4x^4 - (k-1)x^3 + kx^2 - 6x + 1 \quad \dots(i)$

Since, $(2x - 1)$ is a factor of $f(x)$

$$\therefore f\left(\frac{1}{2}\right) = 0$$

$$\Rightarrow 4 \times \left(\frac{1}{2}\right)^4 - (k-1) \times \left(\frac{1}{2}\right)^3 + k \times \left(\frac{1}{2}\right)^2 - 6 \times \frac{1}{2} + 1 = 0$$

$$\Rightarrow \frac{1}{4} - \frac{(k-1)}{8} + \frac{k}{4} - 2 = 0$$

$$\Rightarrow \frac{k}{4} - \frac{(k-1)}{8} = 2 - \frac{1}{4}$$

$$\Rightarrow \frac{2k - (k-1)}{8} = \frac{8-1}{4} = \frac{7}{4}$$

$$\Rightarrow \frac{8k - 4k + 4}{8} = \frac{7}{4}$$

$$\Rightarrow 4k = 52 \Rightarrow k = 13$$

Hence, the value of k is 13.

42. (c) Given, $x^4 + xy^3 + x^3y + xz^3 + y^4 + yz^3$

$$= (x^4 + xy^3 + xz^3) + (x^3y + y^4 + yz^3)$$

$$= x(x^3 + y^3 + z^3) + y(x^3 + y^3 + z^3)$$

$$= (x + y)(x^3 + y^3 + z^3)$$

Hence, given polynomial is divisible by both

$$(x + y) \text{ and } (x^3 + y^3 + z^3).$$

43. (c) $x^2 + 2 \mid x^3 + 5x^2 + 10k(x + 5)$

$$\begin{array}{r} x^3 + 2x \\ \hline 5x^2 - 2x + 10k \end{array}$$

$$\begin{array}{r} 5x^2 + 10 \\ \hline -2x - 10 + 10k = \text{Remainder} \end{array}$$

But given that, remainder = $-2x$

$$\therefore -2x - 10 + 10k = -2x$$

$$\Rightarrow -10 + 10k = 0$$

$$\Rightarrow 10k = 10 \Rightarrow k = 1$$

44. (c) Given that, $x + \frac{1}{x} = a \quad \dots(i)$

$$\text{Then, } x^3 + x^2 + \frac{1}{x^3} + \frac{1}{x^2}$$

$$= \left(x^3 + \frac{1}{x^3}\right) + \left(x^2 + \frac{1}{x^2}\right)$$

$$= \left(x + \frac{1}{x}\right)^3 - 3\left(x + \frac{1}{x}\right) + \left(x + \frac{1}{x}\right)^2 - 2$$

$$= a^3 - 3a + a^2 - 2 = a^3 + a^2 - 3a - 2$$

45. (a) Given, $f(x) = x^3 + x^2 - x - 1$

$$\text{and } g(x) = x^3 - x^2 + x - 1$$

$$\begin{aligned} p(x) &= f(x) \cdot g(x) \\ &= (x^3 + x^2 - x - 1)(x^3 - x^2 + x - 1) \\ &= [x^2(x+1) - 1(x+1)] \end{aligned}$$

$$[x^2(x-1) + 1(x-1)]$$

$$= [(x+1)(x^2-1)][(x-1)(x^2+1)]$$

$$= (x+1)(x+1)(x-1)(x-1)(x^2+1)$$

$$= (x+1)^2(x-1)^2(x^2+1)$$

46. (d) Put the value of all options in
 $4k^2 + 12k + k$
 when, $k = 9$,
 then, $4(9)^2 + 12(9) + 9 = 441 = (21)^2$

47. (c) Given expression
 $= 5px - 10qy + 2rpx - 4qry$
 $= (5px + 2rpx) - (10qy + 4qry)$
 $= px(5 + 2r) - 2qy(5 + 2r)$
 $= (5 + 2r)(px - 2qy)$
 So, factors of given expression are both
 $(5 + 2r)$ and $(px - 2qy)$.

48. (d) Given, $(a + 1)^4 - a^4$
 $= \{(a + 1)^2 - a^2\} \{(a + 1)^2 + a^2\}$
 $= \{(a + 1) + a\} \{(a + 1) - a\}$
 $\quad \quad \quad \{a^2 + 1 + 2a + a^2\}$
 $= (2a + 1)(1)(2a^2 + 2a + 1)$
 $= (2a + 1)(2a^2 + 2a + 1)$
 So, $(a + 1)^4 - a^4$ is divisible by
 $(2a^2 + 2a + 1)$

49. (d) Let $f(x) = x^4 - 7x^3 + 5x^2 - 6x + 81$
 $f(3) = (3)^4 - 7(3)^3 + 5(3)^2 - 6(3) + 81$
 $= 81 - 189 + 45 - 18 + 81 = 0$
 $\therefore (x - 3)$ is a factor of $f(x)$

50. (b) Let $f(x) = x^3 + 3x^2 + 4x + k$
 $\therefore (x + 5)$ is a factor of $f(x)$
 $\therefore f(-5) = 0$
 $\Rightarrow (-5)^3 + 3(-5)^2 + 4(-5) + k = 0$
 $\Rightarrow -125 + 75 - 20 + k = 0$
 $\Rightarrow -70 + k = 0 \Rightarrow k = 70$
 Hence, the value of k is 70.

51. (c) Let $p(x)$ be added to obtain x^3 .
 Then, $(1 - x)(1 + x^2) + p(x) = x^3$
 $\Rightarrow 1 + x^2 - x - x^3 + p(x) = x^3$
 $\Rightarrow p(x) = x^3 - 1 - x^2 + x + x^3$
 $\therefore p(x) = 2x^3 - x^2 + x - 1$

So, $2x^3 - x^2 + x - 1$ is added to
 $(1 - x)(1 + x^2)$ to obtain x^3 .

52. (c) Let $f(x) = x(y^2 - z^2) + y(z^2 - x^2)$
 $\quad \quad \quad + z(x^2 - y^2)$

If it is divisible by $(y - z)$, then $y - z$ is
 a factor of $f(x)$.

$$\therefore y - z = 0 \Rightarrow y = z$$

On putting $y = z$, we get

$$x(z^2 - z^2) + z(z^2 - x^2) + z(x^2 - z^2)$$

$$= z^3 - zx^2 + zx^2 - z^3 = 0$$

Hence, $y - z$ is a factor, so it is divisible
 by $(y - z)$.

On putting $z = x$, we get

$$x(y^2 - x^2) + y(x^2 - x^2)$$

$$\quad \quad \quad + x(x^2 - y^2)$$

$$= xy^2 - x^3 + x^3 - xy^2 = 0$$

So, $(z - x)$ is also a factor, so it is also
 divisible by $(z - x)$.

53. (d) Put $x = -3$ in $p(x)$, we get

$$\text{Let } p(x) = x^3 + 2x^2 + 3x + 8$$

$$p(-3) = (-3)^3 + 2(-3)^2 + 3(-3) + 8$$

$$= -27 + 18 - 9 + 8$$

$$= -10 \neq 0$$

So, $(x + 3)$ is not the factor of
 $x^3 + 2x^2 + 3x + 8$.

Similarly, put $x = 2$ in $p(x)$, we get

$$p(2) = (2)^3 + 2(2)^2 + 3(2) + 8$$

$$= 30 \neq 0 \text{ So, } x - 2 \text{ is also not the factor}$$

$$\text{of } x^3 + 2x^2 + 3x + 8.$$

54. (d) Let $f(x) = x^3 + 6x^2 + 11x + 6$

Put $x = -1$ in $f(x)$, we get

$$f(-1) = (-1)^3 + 6(-1)^2 + 11(-1) + 6$$

$$= -1 + 6 - 11 + 6$$

$$= -12 + 12 = 0$$

Hence, $(x + 1)$ is a factor of $f(x)$.

$$\therefore f(x) = (x + 1)(x^2 + 5x + 6)$$

$$= (x + 1)(x^2 + 3x + 2x + 6)$$

$$= (x + 1)(x + 2)(x + 3)$$

Hence, $(x + 1)$, $(x + 2)$ and $(x + 3)$ are
 the factors of $f(x)$.

55. (c) $x(x + a)(x + 2a)(x + 3a)$

$$= (x + a)(x + 2a)x(x + 3a)$$

$$= (x^2 + 3ax + 2a^2)(x^2 + 3ax)$$

$$= (x^2 + 3ax + a^2 + a^2)$$

$$\quad \quad \quad (x^2 + 3ax + a^2 - a^2)$$

$$= (x^2 + 3ax + a^2)^2 - a^4$$

So, a^4 must be added to
 $x(x + a)(x + 2a)(x + 3a)$ to make it a
 perfect square.

56. (b) Given, $\left(x^2 + \frac{1}{x^2}\right) = \frac{17}{4}$

$$\Rightarrow x^2 + \frac{1}{x^2} - 2 + 2 = \frac{17}{4}$$

$$\Rightarrow \left(x - \frac{1}{x}\right)^2 + 2 = \frac{17}{4}$$

$$\Rightarrow \left(x - \frac{1}{x}\right)^2 = \frac{17}{4} - 2$$

$$\Rightarrow \left(x - \frac{1}{x}\right)^2 = \frac{9}{4}$$

$$\Rightarrow \left(x - \frac{1}{x}\right) = \frac{3}{2} \quad \dots(i)$$

On cubing both sides, we get

$$\left(x - \frac{1}{x}\right)^3 = \left(\frac{3}{2}\right)^3$$

$$\Rightarrow x^3 - \frac{1}{x^3} - 3 \times \frac{1}{x} \cdot x \left(x - \frac{1}{x}\right) = \frac{27}{8}$$

$$\Rightarrow x^3 - \frac{1}{x^3} = \frac{27}{8} + 3 \times \left(\frac{3}{2}\right)$$

[From Eq. (i)]

$$\Rightarrow x^3 - \frac{1}{x^3} = \frac{27}{8} + \frac{9}{2}$$

$$\therefore \left(x^3 - \frac{1}{x^3}\right) = \frac{63}{8}$$

57. (c) Let $f(x) = x^5 - 5x^2 + 125$

Now, put $x = -5$,

$$\therefore \text{Required remainder} = f(-5)$$

$$= (-5)^5 - 5(-5)^2 + 125$$

$$= -3125 - 125 + 125 = -3125$$

58. (a) Given, $ax + by - 2 = 0$

$$\Rightarrow ax + by = 2$$

Squaring on both sides, we get

$$(ax + by)^2 = (2)^2$$

$$\Rightarrow a^2x^2 + b^2y^2 + 2axy = 4$$

$$\Rightarrow a^2x^2 + b^2y^2 + 2 = 4 \quad [\because axy = 1]$$

$$\Rightarrow a^2x^2 + b^2y^2 = 2$$

$$\Rightarrow a^2x^2 = 1 \text{ and } b^2y^2 = 1$$

$$\Rightarrow ax = 1 \text{ and } by = 1$$

$$\Rightarrow x = \frac{1}{a} \text{ and } y = \frac{1}{b}$$

$$\therefore a^2x + b^2y = a^2 \cdot \frac{1}{a} + b^2 \cdot \frac{1}{b} = a + b$$

59. (a) Given, $x + k$ is the common factor of
 $x^2 + ax + b$ and $x^2 + cx + d$. Then, put
 $x = -k$ in given polynomial.

$$\therefore k^2 - ka + b = 0$$

$$\text{and } k^2 - kc + d = 0$$

$$\text{Then, } k^2 - ka + b = k^2 - kc + d$$

$$\Rightarrow k(c - a) = d - b \Rightarrow k = \frac{d - b}{c - a}$$

60. (c) I. We have, $3abc + b^3 + c^3 - a^3$

$$= -(a^3 - b^3 - c^3 - 3abc)$$

$$= -[a^3 + (-b)^3 + (-c)^3 - 3(a)(-b)(-c)]$$

$$= -(a - b - c)(a^2 + b^2 + c^2 + ab$$

$$\quad \quad \quad -bc + ac)$$

So, $(a - b - c)$ is a factor of

$$3abc + b^3 + c^3 - a^3.$$

Hence, statement I is correct.

II. Now, $3bc + b^3 + c^3 - 1$

$$= b^3 + c^3 - (1)^3 - 3b(-1)(-1)$$

$$= (b + c - 1)[b^2 + c^2 + 1^2 - bc + c + b]$$

So, $(b + c - 1)$ is a factor of

$$3bc + b^3 + c^3 - 1.$$

Hence, statement II is also correct.

61. (a) Let $p(x) = x^3 - 3x^2 + kx - 10$

Then, $(x-5)$ is a factor of $p(x)$.

$$\begin{aligned}\therefore p(5) &= 0 \\ \Rightarrow 5^3 - 3(5)^2 + k \times 5 - 10 &= 0 \\ \Rightarrow 125 - 75 + 5k - 10 &= 0 \\ \Rightarrow 40 + 5k &= 0 \Rightarrow k = -8.\end{aligned}$$

62. (b) $\because x^3 + y^3 + z^3 - 3xyz = (x+y+z)(x^2 + y^2 + z^2 - xy - yz - zx)$

Given, $x + y + z = 0$

$$\begin{aligned}\therefore x^3 + y^3 + z^3 - 3xyz &= 0 \\ \Rightarrow x^3 + y^3 + z^3 + 3xyz &= 6xyz\end{aligned}$$

63. (c) $x^3q^2 - x^3pt + 4x^2p^2t - 4x^2q^2$
 $+ 3xq^2 - 3xpt$
 $= x^3q^2 - x^3pt - 4x^2q^2$
 $+ 4x^2pt + 3xq^2 - 3xpt$
 $= x^3(q^2 - pt) - 4x^2(q^2 - pt)$
 $+ 3x(q^2 - pt)$
 $= (x^3 - 4x^2 + 3x)(q^2 - pt)$
 $= x(x^2 - 4x + 3)(q^2 - pt)$
 $= x(x^2 - 3x - x + 3)(q^2 - pt)$
 $= x[x(x-3) - 1(x-3)](q^2 - pt)$
 $= x(x-3)(x-1)(q^2 - pt)$

64. (c) In option (c), we have

$$\begin{aligned}(2)^4 - 6(2)^3 + 12(2)^2 - 24 \times 2 + 32 \\ = 16 - 6(8) + 12(4) - 48 + 32 \\ = 16 - 48 + 48 - 48 + 32 = 0\end{aligned}$$

Hence, $(x-2)$ is a factor of

$$x^4 - 6x^3 + 12x^2 - 24x + 32$$

65. (d) Let $f(x) = x^3 + kx^2 - x + 2$

If $(x+1)$ is a factor of $f(x)$.

Then, $f(-1) = 0$

$$\begin{aligned}\Rightarrow (-1)^3 + k(-1)^2 - (-1) + 2 &= 0 \\ \Rightarrow -1 + k + 1 + 2 &= 0 \Rightarrow k = -2\end{aligned}$$

66. (c) Let $f(x) = x^6 + px^5 + qx^4 - x^2 - x - 3$

$\because f(x)$ is divisible by $(x^4 - 1)$.

Then, $f(1) = f(-1) = 0$

Now, $f(1) = 1 + p + q - 1 - 1 - 3 = 0$

$$\Rightarrow p + q = 4 \quad \dots(i)$$

and $f(-1) = 1 - p + q - 1 + 1 - 3 = 0$

$$\Rightarrow q - p = 2 \quad \dots(ii)$$

On solving Eqs. (i) and (ii), we get $q = 3$ and $p = 1$

$$\therefore p^2 + q^2 = 3^2 + 1^2 = 10$$

67. (a) Let $f(x) = ax^2 - (a^2 + 1)x + a$
 $= a x^2 - x - a^2 x + a$

$$= x(ax - 1) - a(ax - 1)$$

$$= (x - a)(ax - 1)$$

If p and q are linear factors of $f(x)$, then

$$p = (x - a) \text{ and } q = (ax - 1)$$

$$\begin{aligned}\therefore p + q &= (x - a) + (ax - 1) \\ &= (x - 1)(a + 1)\end{aligned}$$

68. (d) We have, $x - y = 8 \quad \dots(i)$

$$\text{and } \sqrt{\frac{x}{y}} = \frac{10}{3} - \sqrt{\frac{y}{x}}$$

$$\Rightarrow \sqrt{\frac{x}{y}} + \sqrt{\frac{y}{x}} = \frac{10}{3}$$

$$\Rightarrow \frac{(\sqrt{x})^2 + (\sqrt{y})^2}{\sqrt{xy}} = \frac{10}{3}$$

$$\Rightarrow x + y = \frac{10}{3}\sqrt{xy}$$

On squaring both sides, we have

$$(x + y)^2 = \frac{100}{9}xy$$

$$\Rightarrow (x - y)^2 + 4xy = \frac{100}{9}xy$$

$$\begin{aligned}[\because (x + y)^2 &= (x - y)^2 + 4xy] \\ \Rightarrow (8)^2 &= \left(\frac{100}{9} - 4\right)xy \quad [\text{from Eq. (i)}]\end{aligned}$$

$$\Rightarrow 64 = \frac{64}{9}xy \Rightarrow xy = 9$$

15

HCF AND LCM OF POLYNOMIALS

Regularly (1-2) questions have been asked from this chapter. Generally questions are based on your prior knowledge of GCD and LCM of numbers and expressions.



DIVISOR

A polynomial $d(x)$ is said to be a divisor of polynomial $p(x)$ if $d(x)$ is a factor of $p(x)$ i.e., $p(x)$ can be written as $p(x) = d(x) \cdot q(x)$, where $q(x)$ is a quotient polynomial e.g. $(x-3)$ is a divisor of $(x-3)^2(x+1)$.

HCF (GCD) of Polynomials

A polynomial $h(x)$ is called the HCF or GCD of two or more given polynomials, if $h(x)$ is a polynomial of highest degree dividing each one of the given polynomial without leaving any remainder.

Note The coefficient of highest degree term is always taken as positive.

EXAMPLE 1. The HCF of polynomials $(x+2)^2(x-3)^2$ and $(x+1)(x+2)^2(x-3)$ is

a. $(x+2)(x-3)$

b. $(x+2)^2(x-3)$

c. $(x+1)(x+2)(x-3)$

d. $(x+2)^2(x-3)^2$

Sol. b. Let $p(x) = (x+2)^2(x-3)^2$ and $q(x) = (x+1)(x+2)^2(x-3)$

The common factor between two polynomials is $(x+2)^2(x-3)$, $\therefore$ HCF = $(x+2)^2(x-3)$

HCF by Factorisation Method

Following are the steps for calculating HCF through factorisation method

Step I Resolve the given polynomials in the complete factored form.

Step II Find the HCF of the numerical factors (if any) of given polynomial.

Step III Find the factors of highest degree which is common to all given polynomials.

Step IV The product of all such common factors and HCF of the numerical factors is the HCF of given polynomials.

EXAMPLE 2. The HCF of $p(x) = 24(6x^4 - x^3 - 2x^2)$ and $q(x) = 20(2x^6 + 3x^5 + x^4)$ is

- a. $4(2x+1)$ b. $x^2(2x+1)$ c. $4x^2$ d. None of these

Sol. d. Here, $P(x) = 24(6x^4 - x^3 - 2x^2) = 2^3 \cdot 3 \cdot x^2 \cdot (6x^2 - x - 2)$
 $= 2^3 \cdot 3 \cdot x^2 \cdot (6x^2 - 4x + 3x - 2) = 2^3 \cdot 3 \cdot x^2(2x+1) \cdot (3x-2)$
 and $q(x) = 20(2x^6 + 3x^5 + x^4) = 2^2 \cdot 5 \cdot x^4(2x^2 + 3x + 1)$
 $= 2^2 \cdot 5 \cdot x^4 \cdot (2x^2 + 2x + x + 1) = 2^2 \cdot 5 \cdot x^4(2x+1)(x+1)$
 HCF of numerical factor $= 2^2 = 4$, Highest degree common factor $= x^2(2x+1)$
 $\therefore$ Required HCF $= 4x^2(2x+1)$

HCF by Division Method

To find the HCF of polynomials which cannot be factorised easily, we use successive division method.

- Step I** Arrange the given polynomials in descending order of powers of its variables.
Step II If any common factor is present in the terms of each polynomial, it should be taken out.
Step III Divide the polynomial of highest degree by the polynomial of lowest degree.
 or If both the polynomials are of the same degree then any one of them can be taken as divisor or dividend.
Step IV After the first division take the remainder as the new divisor and first divisor as new dividend.
Step V Continue this process of dividing the last divisor by the last remainder until the remainder becomes zero.
Step VI The product of common factors obtained from step II and last divisor is the HCF of given polynomials.

- Note**
- If the first term of a remainder is negative at any stage, the sign of all of its term must be changed.
 - If at any stage, the remainder contains common factor it should be taken out.

EXAMPLE 3. The HCF of $22x^6 - 78x^5 - 16x^2$ and $2x^5 - 78x^2 - 44x$ is

- a. $(x^2 - 3x - 2)$ b. $2x(x^2 - 3x - 2)$
 c. $22x(x^2 - 3x - 2)$ d. None of these

Sol. b. Let $p(x) = 22x^6 - 78x^5 - 16x^2 = 2x^2(11x^4 - 39x^3 - 8)$
 $q(x) = 2x^5 - 78x^2 - 44x = 2x(x^4 - 39x - 22)$
 Let us divide $(11x^4 - 39x^3 - 8)$ by $(x^4 - 39x - 22)$

$$\begin{array}{r} x^4 - 39x - 22 \overline{) 11x^4 - 39x^3 - 8} \\ \underline{11x^4 - 429x - 242} \\ -39x^3 + 429x + 234 \end{array}$$

-39 is taken out as common factor from remainder.

$$\begin{array}{r} x^3 - 11x - 6 \overline{) x^4 - 11x^2 - 6x} \\ \underline{-x^3 + 11x + 6} \\ 11x^2 - 33x - 22 \end{array}$$

Again 11 is taken out as common factor from remainder

$$\begin{array}{r} x^2 - 3x - 2 \overline{) x^3 - 11x - 6} \\ \underline{x^3 - 3x^2 - 2x} \\ 3x^2 - 9x - 6 \\ \underline{3x^2 - 9x - 6} \\ 0 \end{array}$$

$\therefore$ HCF of $11x^4 - 39x^3 - 8$

and $x^4 - 39x - 22$ is $x^2 - 3x - 2$.

Also, HCF of $2x^2$ and $2x$ is $2x$.

$\therefore$ Required HCF $= 2x(x^2 - 3x - 2)$

LCM of Polynomials

A polynomial $b(x)$ is called the LCM of two or more polynomials, if it is a polynomial of smallest degree which is divided by each one of the given polynomials without leaving any remainder.

EXAMPLE 4. The LCM of $12x^2y^3z^2$ and $18x^4y^2z^3$ is

- a. $x^4y^3z^3$
 b. $9x^2y^2z^3$
 c. $36x^3y^3z^4$
 d. $36x^4y^3z^3$

Sol. d. Here, $12x^2y^3z^2 = 2^2 \times 3^1 \times x^2 \times y^3 \times z^2$
 $18x^4y^2z^3 = 2^1 \times 3^2 \times x^4 \times y^2 \times z^3$
 $\therefore$ Required LCM $= 2^2 \times 3^2 \times x^4 \times y^3 \times z^3$
 $= 36x^4y^3z^3$

LCM by Factorisation Method

Following are the steps for calculating LCM through factorisation method

- Step I** Resolve the given polynomials in the complete factored form
Step II Find the LCM of the numerical factors (if any) of given polynomials
Step III The required LCM is the product of LCM of numerical factors and each factor raised to the highest power.

EXAMPLE 5. The LCM of $x^3 - 2x^2 - x + 2$ and $x^3 - x^2 - 4x + 4$ is

- a. $(x^2 - 1)(x^2 - 4)$ b. $(x^2 + 2)(x^2 - 4)$
 c. $(x^2 + 1)(x^2 + 2)$ d. None of these

Sol. a. Here, $x^3 - 2x^2 - x + 2 = x^2(x - 2) - 1(x - 2)$

$$\Rightarrow (x - 2)(x^2 - 1) = (x - 2)(x - 1)(x + 1)$$

$$= (x - 1)(x + 1)(x - 2)$$

$$\text{and } x^3 - x^2 - 4x + 4 = x^2(x - 1) - 4(x - 1)$$

$$\Rightarrow (x - 1)(x^2 - 4) = (x - 1)(x - 2)(x + 2)$$

$$\therefore \text{ Required LCM} = (x - 1)(x + 1)(x - 2)(x + 2)$$

$$= (x^2 - 1)(x^2 - 2^2)$$

$$[\because (a - b)(a + b) = a^2 - b^2]$$

$$= (x^2 - 1)(x^2 - 4)$$

For any two polynomials $p(x)$ and $q(x)$

$$p(x) \times q(x) = (\text{Their HCF}) \times (\text{Their LCM})$$

i.e. Product of two polynomials

= Product of their HCF and LCM

EXAMPLE 6. The HCF of two polynomials is $x + 3$ and their LCM is $x^3 - 7x + 6$. If one of the polynomials is $x^2 + 2x - 3$. Then, the other is

- a. $x^2 - x - 6$ b. $x^2 - x + 6$
 c. $x^2 + x + 6$ d. None of these

Sol. d. Here, HCF = $x + 3$ and LCM = $x^3 - 7x + 6$

Also, one polynomial = $x^2 + 2x - 3$

As, LCM $\times$ HCF = Product of two polynomials

$$\text{Other polynomial} = \frac{\text{LCM} \times \text{HCF}}{\text{One polynomial}}$$

$$= \frac{(x + 3) \times (x^3 - 7x + 6)}{x^2 + 2x - 3}$$

$$= \frac{(x + 3)(x^3 - 7x + 6)}{(x + 3)(x - 1)}$$

$$= \frac{(x + 3)(x - 1)(x^2 + x - 6)}{(x + 3)(x - 1)} = x^2 + x - 6$$

So, other polynomial is $x^2 + x - 6$.

PRACTICE EXERCISE

- The LCM of $(x - 1)(x - 2)$ and $x^2(x - 2)(x + 3)$ is
 (a) $(x - 1)$ (b) $(x - 1)(x - 2)(x + 3)$
 (c) $x^2(x - 1)(x - 2)(x + 3)$ (d) None of these
- The LCM of $2(a^2 - b^2)$, $3(a^3 - b^3)$, $4(a^4 - b^4)$ is
 (a) $6(a - b)(a + b)(a^2 + b^2)$ (b) $12(a^4 - b^4)(a^2 + ab + b^2)$
 (c) $a^3 - b^3$ (d) $12(a^4 - b^4)$
- The HCF of two expressions a and b is 1. Their LCM is
 (a) $(a + b)$ (b) $a - b$ (c) ab (d) $\frac{1}{ab}$
- The LCM of the polynomials $(x + 3)^2(x - 2)(x + 1)^2$; $(x + 1)^3(x + 3)(x + 4)$ is
 (a) $(x - 2)(x + 1)^3(x + 3)^2(x + 4)$
 (b) $(x - 2)(x + 1)^3(x + 3)(x + 4)$
 (c) $(x - 2)(x + 3)(x + 4)$
 (d) $(x - 2)^2(x + 1)(x + 3)^2(x + 4)$
- HCF of $4y^4x - 9y^2x^3$ and $4y^2x^2 + 6yx^3$ is
 (a) $y^2x(2y + 3x)$ (b) $yx(3x + 2y)$
 (c) $yx^2(x + 3)$ (d) None of these
- What is the HCF of $(x^4 - x^2 - 6)$ and $(x^4 - 4x^2 + 3)$?
 (a) $x^2 - 3$ (b) $x + 2$ (c) $x + 3$ (d) $x^2 + 3$
- The HCF of the polynomial A and B where $A = (x + 3)^2(x - 2)(x + 1)^2$ and $B = (x + 1)^2(x + 3)(x + 4)$ is given by
 (a) $(x + 1)^2(x + 3)$ (b) $(x + 1)(x + 3)^2$
 (c) $(x + 1)(x + 3)$ (d) $(x + 3)^2(x + 1)^2$
- The HCF of $22x(x + 1)^2$ and $36x^2(2x^2 + 3x + 1)$ is
 (a) $2x(x + 1)$ (b) $x(x + 1)$ (c) $2(x + 1)$ (d) $2(x + 1)^2$
- The LCM of $a^2 - b^2 - c^2 - 2bc$, $b^2 - c^2 - a^2 - 2ac$ and $c^2 - a^2 - b^2 - 2ab$ is
 (a) $(a + b + c)$
 (b) $(a - b - c)(a + b + c)$
 (c) $(a + b + c)(c - a - b)$
 (d) $(a + b + c)(a - b - c)(b - c - a)(c - a - b)$
- LCM of $[(x + 3)(x - 2)]^2$ and $[(x - 2)(x - 6)]$ is
 (a) $(x + 3)(x - 2)^3(x - 6)$ (b) $(x + 3)(x - 2)^2(x - 6)$
 (c) $(x + 3)(x - 2)(x - 6)$ (d) $(x + 3)(x - 6)$

- 11.** If $(z - 1)$ is the HCF of $(z^2 - 1)$ and $pz^2 - q(z + 1)$, then
 (a) $2p = q$ (b) $p = 2q$ (c) $3p = 2q$ (d) $3p = 2p$
- 12.** The HCF of two expressions is $3x^2 + 4x - 4$ and their LCM is $3x^4 + 4x^3 - 7x^2 - 4x + 4$. The expressions are
 (a) $(x - 1)(3x^2 + 4x - 4)$ and $(3x^2 + 4x - 4)$
 (b) $(x + 1)(3x^2 + 4x - 4)$ and $(x + 2)(3x^2 + 4x - 4)$
 (c) $(x + 2)(3x^2 + 4x + 4)$ and $(x - 1)$
 (d) $(x + 1)(3x^2 + 4x - 4)$ and $(x - 1)(3x^2 + 4x - 4)$
- 13.** The LCM and HCF of two polynomials $p(x)$ and $q(x)$ are $36x^2(x + a)(x^3 - a^3)$ and $x^2(x - a)$, respectively. If $p(x) = (4x^2)(x^2 - a^2)$, then $q(x)$
 (a) $4x^3(x^3 - a^3)$ (b) $12x^3(x^2 - a^2)$
 (c) $9x^3(x^3 - a^3)$ (d) $36x^3(x^3 - a^3)$
- 14.** The LCM of two polynomials $p(x)$ and $q(x)$ is $x^3 - 7x + 6$. If $p(x) = x^2 + 2x - 3$ and $q(x) = x^2 + x - 6$, then the HCF is
 (a) $(x + 3)$ (b) $(x - 3)$
 (c) $(x + 3)(x - 2)$ (d) $(x - 1)$
- 15.** If $(x + k)$ is the HCF of $(x^2 + ax + b)$ and $(x^2 + px + q)$, then the value of k is
 (a) $\left(\frac{b+q}{a+p}\right)$ (b) $\left(\frac{b-q}{a-p}\right)$ (c) $\left(\frac{a+b}{p+q}\right)$ (d) $\left(\frac{a-b}{p-q}\right)$
- 16.** What is the value of k for which the HCF of $2x^2 + kx - 12$ and $x^2 + x - 2k - 2$ is $(x + 4)$?
 (a) 5 (b) 7 (c) 10 (d) -4
- 17.** If the HCF of $(x^2 + x - 12)$ and $(2x^2 - kx - 9)$ is $(x - k)$, then what is the value of k ?
 (a) -3 (b) 3 (c) -4 (d) 4
- 18.** If GCD of the polynomials $(x^3 - 2x^2 + px + 6)$ and $(x^2 - 5x + q)$ is $(x - 3)$. Then, the value of $5q + 6p$ is
 (a) -1 (b) 1
 (c) 0 (d) None of these
- 19.** The sum and the difference of two expressions is $5x^2 - x - 4$ and $x^2 + 9x - 10$ respectively, then their LCM would be equal to
 (a) $(x - 1)$ (b) $(2x + 3)(3x + 7)$
 (c) $(2x - 3)(3x + 7)$ (d) $(x - 1)(2x - 3)(3x + 7)$
- 20.** Find the values of a and b so that the polynomials $p(x)$ and $q(x)$ have $(x + 1)(x + 3)$ as their HCF
 $p(x) = (x^2 + 3x + 2)(x^2 + 2x + a)$
 and $q(x) = (x^2 + 7x + 12)(x^2 + 7x + b)$
 (a) -3, 6 (b) 3, -6
 (c) 6, -3 (d) None of these
- 21.** If $f(x)$ and $g(x)$ are two polynomials with integral coefficients which vanish at $x = 1/2$, then what is the factor of HCF of $f(x)$ and $g(x)$?
 (a) $x - 1$ (b) $x - 2$ (c) $2x - 1$ (d) $2x + 1$
- 22.** The HCF of two polynomials $p(x)$ and $q(x)$ is $2x(x + 2)$ and LCM is $24x(x + 2)^2(x - 2)$. If $p(x) = 8x^3 + 32x^2 + 32x$, then what is $q(x)$ equal to?
 (a) $4x^3 - 16x$ (b) $6x^3 - 24x$ (c) $12x^3 + 24x$ (d) $12x^3 - 24x$
- 23.** If $(x - 2)$ is the HCF of $(ax^2 + bx + c)$ and $(bx^2 + ax + c)$, then value of c is
 (a) $2(a + b)$ (b) $(a + b)$ (c) $-3(a + b)$ (d) $-(a + b)$
- 24.** We have three polynomials $A = 8p + p^2 + 12$, $B = p^2 + 2p - 24$ and $C = p^2 + 15p + 54$
 I. Their LCM is $(p + 6)(p - 4)(p + 2)(p + 9)$
 II. Their HCF is $(p + 6)(p - 2)$
 Then, which of the following codes is/are correct
 (a) Only I (b) Only II
 (c) Neither I nor II (d) Both I and II
- 25.** Which of the following statements are true?
 I. HCF of $x^2 - 6x + 9$ and $x^3 - 27$ is $(x - 3)$.
 II. LCM of $10x^2yz$, $15xyz$, $20xy^2z^2$ is $120x^2y^2z^2$.
 III. HCF of $(6x^2 - 7x - 3)$ and $(2x^2 + 11x - 21)$ is $(2x - 3)$.
 Select the correct answer using the codes given below
 (a) I and III (b) I, II and III
 (c) II and III (d) None of these
- 26.** Consider the following statements :
 I. The HCF of $x + y$ and $x^{10} - y^{10}$ is $x + y$.
 II. The HCF of $x + y$ and $x^{10} + y^{10}$ is $x + y$.
 III. The HCF of $x - y$ and $x^{10} + y^{10}$ is $x - y$.
 IV. The HCF of $x - y$ and $x^{10} - y^{10}$ is $x - y$.
 Which of the statement(s) given above is/are correct?
 (a) I and II (b) II and III (c) I and IV (d) II and IV
- Directions** (Q. Nos. 27-29) A student wrote five polynomials such as $A = pq - np$, $B = pq - mq$, $C = q^2 - 3nq + 2n^2$, $D = pq - 2pn - mq + 2mn$, $E = pq - np - mq + mn$. Now, he divide the polynomials into groups and calculate the HCF and LCM.
- 27.** Calculate the HCF of A , C and E
 (a) $(q - 2n)$ (b) $(p - n)$ (c) $(q - n)$ (d) $(q - n)(q - 2n)$
- 28.** Calculate the LCM of D and E
 (a) $(p - m)(q - 2n)$ (b) $(p - m)(q - n)(q - 2n)$
 (c) $(q - n)(q - 2n)(m - p)$ (d) $(q - n)(p - m)$
- 29.** The HCF of all five polynomials together is
 (a) $(q - n)$ (b) $(q - 2n)$ (c) 1 (d) $(p - m)$

PREVIOUS YEARS' QUESTIONS

- 30.** What is the LCM of $a^3b - ab^3$, $a^3b^2 + a^2b^3$ and $ab(a+b)$? ☑ 2012 I
 (a) $a^2b^2(a^2 - b^2)$ (b) $ab(a^2 - b^2)$
 (c) $a^2b^2 + ab^3$ (d) $a^3b^3(a^2 - b^2)$
- 31.** What is the HCF of $36(3x^4 + 5x^3 - 2x^2)$, $9(6x^3 + 4x^2 - 2x)$ and $54(27x^4 - x)$? ☑ 2012 I
 (a) $9x(x+1)$ (b) $9x(3x-1)$ (c) $18x(3x-1)$ (d) $18x(x+1)$
- 32.** What is the HCF of the polynomials $x^3 + 8$, $x^2 + 5x + 6$ and $x^3 + 2x^2 + 4x + 8$? ☑ 2013 II
 (a) $x+2$ (b) $x+3$ (c) $(x+2)^2$ (d) None of these
- 33.** The LCM of $(x^3 - x^2 - 2x)$ and $(x^3 + x^2)$ is ☑ 2013 II
 (a) $x^3 - x^2 - 2x$ (b) $x^2 + x$ (c) $x^4 - x^3 - 2x^2$ (d) $x - 2$
- 34.** The HCF of $(x^4 - y^4)$ and $(x^6 - y^6)$ is ☑ 2013 II
 (a) $x^2 - y^2$ (b) $x - y$ (c) $x^3 - y^3$ (d) $x^4 - y^4$
- 35.** What is the LCM of $x^2 + 2x - 8$, $x^3 - 4x^2 + 4x$ and $x^2 + 4x$? ☑ 2013 II
 (a) $x(x+4)(x-2)^2$ (b) $x(x+4)(x-2)$
 (c) $x(x+4)(x+2)^2$ (d) $x(x+4)^2(x-2)$
- 36.** What is the HCF of $a^2b^4 + 2a^2b^2$ and $(ab)^7 - 4a^2b^9$? ☑ 2013 II
 (a) ab (b) a^2b^3 (c) a^2b^2 (d) a^3b^2
- 37.** What is the HCF of $8(x^5 - x^3 + x)$ and $28(x^6 + 1)$?
 (a) $4(x^4 - x^2 + 1)$ (b) $2(x^4 - x^2 + 1)$ ☑ 2014 I
 (c) $(x^4 - x^2 + 1)$ (d) None of these
- 38.** What is the highest common factor of $2x^3 + x^2 - x - 2$ and $3x^3 - 2x^2 + x - 2$? ☑ 2014 II
 (a) $x-1$ (b) $x+1$ (c) $2x+1$ (d) $2x-1$
- 39.** The HCF and LCM of two polynomials are $(x+y)$ and $(3x^5 + 5x^4y + 2x^3y^2 - 3x^2y^3 - 5xy^4 - 2y^5)$, respectively. If one of the polynomials is $(x^2 - y^2)$, then the other polynomial is ☑ 2015 I
 (a) $3x^4 - 8x^3y + 10x^2y^2 + 7xy^3 - 2y^4$
 (b) $3x^4 - 8x^3y - 10x^2y^2 + 7xy^3 + 2y^4$
 (c) $3x^4 + 8x^3y + 10x^2y^2 + 7xy^3 + 2y^4$
 (d) $3x^4 + 8x^3y - 10x^2y^2 + 7xy^3 + 2y^4$
- 40.** If $(x+1)$ is the HCF of $Ax^2 + Bx + C$ and $Bx^2 + Ax + C$ where $A \neq B$, then the value of C is ☑ 2015 II
 (a) A (b) B (c) $A - B$ (d) 0
- 41.** The sum and difference of two expressions are $5x^2 - x - 4$ and $x^2 + 9x - 10$ respectively. The HCF of the two expressions will be ☑ 2016 I
 (a) $x+1$ (b) $(x-1)$ (c) $(3x+7)$ (d) $(2x-3)$

ANSWERS

1	c	2	b	3	c	4	a	5	b	6	a	7	a	8	a	9	d	10	b
11	b	12	d	13	c	14	a	15	b	16	a	17	b	18	c	19	d	20	a
21	c	22	b	23	c	24	a	25	a	26	c	27	c	28	b	29	c	30	a
31	c	32	a	33	c	34	a	35	a	36	c	37	a	38	a	39	c	40	d
41	b																		

HINTS AND SOLUTIONS

- 1. (c)** LCM = product of the largest power of each factor

$$= x^2(x-1)(x-2)(x+3)$$

- 2. (b)** Here, $2(a^2 - b^2) = 2(a+b)(a-b)$

$$3(a^3 - b^3) = 3(a-b)(a^2 + ab + b^2)$$

$$\text{and } 4(a^4 - b^4) = 4(a+b)(a-b)(a^2 + b^2)$$

LCM of numerical coefficients = 12
 and LCM of algebraic expressions

$$= (a-b)(a+b)(a^2 + b^2)$$

$$(a^2 + ab + b^2)$$

$$= (a^4 - b^4)(a^2 + ab + b^2)$$

$\therefore$ LCM of polynomials

$$= 12(a^4 - b^4)(a^2 + ab + b^2)$$

- 3. (c)** $\text{LCM} = \frac{\text{Product of expressions}}{\text{HCF}}$

$$= \frac{a \times b}{1} = ab$$

- 4. (a)** Given, $(x+3)^2(x-2)(x+1)^2$
 and $(x+1)^3(x+3)(x+4)$

LCM

$$= (x-2)(x+1)^3(x+3)^2(x+4)$$

- 5. (b)** $4y^4x - 9y^2x^3 = y^2x(4y^2 - 9x^2)$

$$= y^2x(2y-3x)(2y+3x)$$

$$4y^2x^2 + 6yx^3 = 2yx^2(2y+3x)$$

$$\therefore \text{Required HCF} = xy(2y+3x)$$

- 6. (a)** Let $p(x) = x^4 - x^2 - 6$

$$= x^4 - 3x^2 + 2x^2 - 6$$

$$= x^2(x^2 - 3) + 2(x^2 - 3)$$

$$= (x^2 + 2)(x^2 - 3)$$

$$q(x) = x^4 - 4x^2 + 3$$

$$= x^4 - 3x^2 - x^2 + 3$$

$$= x^2(x^2 - 3) - 1(x^2 - 3)$$

$$= (x^2 - 3)(x^2 - 1)$$

$$\text{HCF of } p(x), q(x) = x^2 - 3$$

- 7. (a)** $A = (x+3)^2(x-2)(x+1)^2$ and

$$B = (x+1)^2(x+3)(x+4)$$

$\therefore$ HCF of polynomials

$$= (x+3)(x+1)^2$$

8. (a) Here, HCF of 22 and 36 is 2.

Now, $x(x+1)^2 = x(x+1)(x+1)$
 $x^2(2x^2+3x+1) = x^2(2x+1)(x+1)$
 Common factors of $x(x+1)^2$
 and $x^2(2x^2+3x+1)$ are $x(x+1)$.
 Hence, required HCF = $2x(x+1)$

9. (d) $a^2 - b^2 - c^2 - 2bc = a^2 - (b^2 + c^2 + 2bc)$
 $= a^2 - (b+c)^2 = (a+b+c)(a-b-c)$
 $b^2 - c^2 - a^2 - 2ac = b^2 - (c^2 + a^2 + 2ac)$
 $= b^2 - (c+a)^2$
 $= (b-c-a)(b+c+a)$
 $= (a+b+c)(b-c-a)$
 and $c^2 - a^2 - b^2 - 2ab$
 $= c^2 - (a^2 + b^2 + 2ab) = c^2 - (a+b)^2$
 $= (c-a-b)(c+a+b)$
 $= (a+b+c)(c-a-b)$
 $\therefore$ Required LCM
 $= (a+b+c)(a-b-c)$
 $(b-c-a)(c-a-b)$

10. (b) Given, $[(x+3)(x-2)^2]$
 and $[(x-2)(x-6)]$

$\therefore$ LCM = $(x+3)(x-2)^2(x-6)$

11. (b) Since, $(z-1)$ is the HCF, so it will divide each one of the given polynomials.
 So, $z=1$ will make each one zero.

$$\therefore p(1)^2 - q(1+1) = 0 \Rightarrow p = 2q$$

12. (d) Let $p(x)$ and $q(x)$ be two polynomials
 and $p(x) \div \text{HCF} = a$

and $q(x) \div \text{HCF} = b$

$$\therefore p(x) \times q(x) = \text{LCM} \times \text{HCF}$$

$$\Rightarrow a \times \text{HCF} \times b \times \text{HCF} = \text{LCM} \times \text{HCF}$$

$$\Rightarrow a \times b \times \text{HCF} = \text{LCM}$$

$$ab = \frac{\text{LCM}}{\text{HCF}}$$

$$= \frac{3x^4 + 4x^3 - 7x^2 - 4x + 4}{3x^2 + 4x - 4}$$

$$= (x+1)(x-1)$$

Let $a = (x+1)$ and $b = (x-1)$,
 then the required expression are
 $(x+1)(3x^2+4x-4)$
 and $(x-1)(3x^2+4x-4)$.

13. (c) Here, LCM = $36x^3(x+a)(x^3-a^3)$
 and HCF = $x^2(x-a)$

$$p(x) = 4x^2(x^2 - a^2)$$

But $p(x) \times q(x) = \text{HCF} \times \text{LCM}$

$$q(x) = \frac{\text{HCF} \times \text{LCM}}{p(x)}$$

$$= \frac{x^2(x-a)36x^3(x+a)(x^3-a^3)}{4x^2(x^2-a^2)}$$

$$= 9x^3(x^3-a^3)$$

14. (a) Given,

$$p(x) = x^2 + 2x - 3 = (x+3)(x-1)$$

$$q(x) = x^2 + x - 6 = (x+3)(x-2)$$

and LCM

$$= x^3 - 7x + 6 = (x-1)(x^2 + x - 6)$$

$$= (x-1)(x+3)(x-2)$$

$$\therefore \text{HCF} = \frac{p(x) \times q(x)}{\text{LCM}}$$

$$= \frac{(x+3)(x-1) \times (x+3)(x-2)}{(x-1)(x+3)(x-2)}$$

$$= (x+3)$$

15. (b) Since, $(x+k)$ is the HCF, it will divide both the polynomials without leaving any remainder, thus $x = -k$ will make both of them zero.

$$\therefore k^2 - pk + q = k^2 - ak + b$$

$$\text{or } -ak + b = -pk + q$$

$$\Rightarrow ak - pk = b - q$$

$$\therefore k = \frac{b-q}{a-p}$$

16. (a) Since, $(x+4)$ is HCF, so it will divide both the expressions i.e. $x = -4$ will make each one zero.

$$\therefore 2(-4)^2 + k(-4) - 12 = 0$$

$$\Rightarrow 32 - 12 = 4k$$

$$\therefore 20 = 4k \Rightarrow k = 5$$

17. (b) Since, HCF of $x^2 + x - 12$ and $2x^2 - kx - 9$ is $(x-k)$, then $(x-k)$ will be the factor of $2x^2 - kx - 9$.

$$\therefore 2k^2 - k^2 - 9 = 0$$

$$\Rightarrow k^2 - 9 = 0$$

$$\Rightarrow k = \pm 3$$

and factor of $x^2 + x - 12$ are
 $(x+4)(x-3)$.

Hence, value of k is 3.

18. (c) Here, $(x-3)$ is GCD, so is a factor of both of them.

$\therefore$ Putting $x=3$, in both makes each polynomial zero.

$$3^3 - 2(3)^2 + p(3) + 6 = 0 \Rightarrow p = -5$$

$$3^2 - 5(3) + q = 0 \Rightarrow q = 6$$

$$\therefore 5q + 6p = 5(6) + (-5)(6)$$

$$= 30 - 30 = 0$$

19. (d) Let the expressions be $p(x)$ and $q(x)$, then

$$p(x) + q(x) = 5x^2 - x - 4 \quad \dots(i)$$

$$p(x) - q(x) = x^2 + 9x - 10 \quad \dots(ii)$$

On adding Eqs. (i) and (ii), we have

$$2p(x) = 6x^2 + 8x - 14$$

$$\Rightarrow p(x) = 3x^2 + 4x - 7$$

$$= 3x^2 + 7x - 3x - 7$$

$$= x(3x+7) - 1(3x+7)$$

$$\therefore p(x) = (3x+7)(x-1)$$

Subtracting Eqs. (i) and (ii), we have

$$2q(x) = 4x^2 - 10x + 6$$

$$\Rightarrow q(x) = 2x^2 - 5x + 3$$

$$= 2x^2 - 3x - 2x + 3$$

$$= x(2x-3) - 1(2x-3)$$

$$= (x-1)(2x-3)$$

$\therefore$ LCM of $p(x)$ and $q(x)$

$$= (x-1)(2x-3)(3x+7)$$

20. (a) $p(x) = (x+2)(x+1)(x^2+2x+a)$
 $q(x) = (x+3)(x+4)(x^2+7x+b)$

As HCF is $(x+1)(x+3)$, then both $(x+1)$ and $(x+3)$ must be factors of $p(x)$ and $q(x)$.

For $p(x)$, $(x+1)$ is already a factor, so $(x+3)$ must be a factor of x^2+2x+a .

$$\text{So, } (-3)^2 + 2(-3) + a = 0$$

$$\Rightarrow 9 - 6 + a = 0$$

$$\therefore a = -3$$

For $q(x)$, $(x+3)$ is already factor.

$\therefore (x+1)$ must be a factor of

$$x^2 + 7x + b.$$

$$\therefore (-1)^2 + 7(-1) + b = 0$$

$$\therefore b = 6$$

So, $a = -3$ and $b = 6$ is solution.

21. (c) Given, $f(x)$ and $g(x)$ vanish at $x = \frac{1}{2}$

So, $(2x-1)$ is a factor of $f(x)$ and $g(x)$ both.

Hence, HCF of $f(x)$ and

$$g(x) = 2x - 1.$$

22. (b) Refer to question 13.

23. (c) As $(x-2)$ is the HCF of (ax^2+bx+c) and (bx^2+ax+c)

So, it will divide both the expressions,

$$\therefore a(2)^2 + b(2) + c = 0$$

$$\Rightarrow 4a + 2b + c = 0 \quad \dots(i)$$

$$\text{and } b(2)^2 + a(2) + c = 0$$

$$\Rightarrow 4b + 2a + c = 0 \quad \dots(ii)$$

adding Eqs. (i) and (ii), we get

$$\Rightarrow 6a + 6b + 2c = 0$$

$$\Rightarrow 2c = -6a - 6b$$

$$\Rightarrow c = -3(a+b)$$

24. (a) $A = p^2 + 8p + 12 = (p+2)(p+6)$

$$B = p^2 + 2p - 24 = (p-4)(p+6)$$

$$C = p^2 + 15p + 54 = (p+9)(p+6)$$

I. LCM of A, B

$$\text{and } C = (p+2)(p+6)(p-4)(p+9)$$

Thus, I is correct

II. HCF of A, B and $C = (p+6)$

Hence, II is incorrect.

25. (a) I. $x^2 - 6x + 9 = (x-3)(x-3)$
 and $x^3 - 27 = x^3 - (3)^3$
 $= (x-3)(x^2 + 3x + 9)$
 $\therefore$ HCF = $x-3$.
 Hence, it is true.
- II. LCM of $10x^2yz$, $15xyz$
 and $20xy^2z^2$ is $60x^2y^2z^2$.
 Hence, it is false.
- III. $6x^2 - 7x - 3 = (2x-3)(3x+1)$
 and $2x^2 + 11x - 21$
 $= (x+7)(2x-3)$
 Hence, HCF = $(2x-3)$, it is also true.
 Hence, the statement I and III are correct.

26. (c) We know that, $(x+y)$ and $(x-y)$ are the factors of $(x^{10} - y^{10})$.

Hence, statements I and IV are true.

27. (c) We have, $A = pq - np = p(q-n) \dots(i)$
 $C = q^2 - 3nq + 2n^2$
 $= q^2 - 2nq - nq + 2n^2$
 $= q(q-2n) - n(q-2n)$
 $= (q-n)(q-2n) \dots(ii)$
 $E = pq - np - mq + mn$
 $= p(q-n) - m(q-n)$
 $= (p-m)(q-n) \dots(iii)$
 $\therefore$ HCF of A, C and $E = (q-n)$
28. (b) We have, $E = (p-m)(q-n) \dots(i)$
 $D = pq - 2np - mq + 2mn$
 $= p(q-2n) - m(q-2n)$
 $= (p-m)(q-2n) \dots(ii)$
 $\therefore$ LCM of D and E
 $= (p-m)(q-n)(q-2n)$

29. (c) The factors of given polynomials are as follows

$$A = pq - np = p(q-n) \dots(i)$$

$$B = pq - mq = q(p-m) \dots(ii)$$

$$C = (q-n)(q-2n) \dots(iii)$$

$$D = (p-m)(q-2n) \dots(iv)$$

$$E = (p-m)(q-n) \dots(v)$$

There is no such factor which is common to all given five polynomials.
 Thus, HCF $(A, B, C, D, E) = 1$

30. (a) Here, $a^3b - ab^3 = ab(a^2 - b^2)$
 $= ab(a-b)(a+b)$
 $a^3b^2 + a^2b^3 = a^2b^2(a+b)$
 $\therefore$ LCM $[(a^3b - ab^3), (a^3b^2 + a^2b^3),$
 $ab(a+b)]$
 $= a^2b^2(a+b)(a-b)$
 $= a^2b^2(a^2 - b^2)$

31. (c) Let $P(x) = 36(3x^4 + 5x^3 - 2x^2)$
 $= 36x^2(3x^2 + 5x - 2)$
 $= 36x^2(x+2)(3x-1)$
 $Q(x) = 9(6x^3 + 4x^2 - 2x)$
 $= 18x(3x^2 + 2x - 1)$
 $= 18x(3x-1)(x+1)$
 $R(x) = 54(27x^4 - x) = 54x(27x^3 - 1)$
 $= 54x(3x-1)(9x^2 + 1 + 3x)$
 HCF of $[36, 18, 54] = 18$
 $\therefore$ HCF of $[P(x), Q(x), R(x)]$
 $= 18x(3x-1)$

32. (a) Let $f(x) = x^3 + 8 = x^3 + 2^3$
 $= (x+2)(x^2 - 2x + 4)$
 $= (x+2)(x-2)(x+2)$
 $g(x) = x^2 + 5x + 6$
 $= x^2 + 3x + 2x + 6$
 $= (x+3)(x+2)$
 and $h(x) = x^3 + 2x^2 + 4x + 8$
 $= (x+2)(x^2 + 4)$
 $\therefore$ HCF of $\{f(x), g(x), h(x)\} = (x+2)$

33. (c) Let $f(x) = x^3 - x^2 - 2x$
 $= x(x^2 - x - 2)$
 $= x(x+1)(x-2)$
 and $g(x) = x^3 + x^2$
 $= x^2(x+1) = x \cdot x(x+1)$
 $\therefore$ LCM of $[f(x), g(x)]$
 $= x^2(x+1)(x-2)$
 $= x^2(x^2 - x - 2)$
 $= x^4 - x^3 - 2x^2$

34. (a) Let $f(x) = (x^4 - y^4)$
 $= (x^2 - y^2)(x^2 + y^2)$
 $= (x-y)(x+y)(x^2 + y^2)$
 and $g(x) = (x^6 - y^6)$
 $= (x^3 + y^3)(x^3 - y^3)$
 $= (x+y)(x^2 - xy + y^2)(x-y)$
 $= (x-y)(x+y)(x^2 - xy + y^2)$
 $= (x-y)(x+y)(x^2 + xy + y^2)$
 $\therefore$ HCF of

$$[f(x), g(x)] = (x-y)(x+y)$$

$$= x^2 - y^2$$

35. (a) $x^2 + 2x - 8 = (x-2)(x+4)$
 $x^3 - 4x^2 + 4x = x[x^2 - 4x + 4]$
 $= x(x-2)^2$
 $x^2 + 4x = x(x+4)$

Now, LCM of $(x^2 + 2x - 8)$,
 $(x^3 - 4x^2 + 4x)$ and $(x^2 + 4x)$
 $= x(x-2)^2(x+4)$

36. (c) $a^2b^4 + 2a^2b^2 = a^2b^2(b^2 + 2) \dots(i)$
 and $(ab)^7 - 4a^2b^9 = a^7b^7 - 4a^2b^9$
 $= a^2b^2(a^5b^5 - 4b^7) \dots(ii)$

From Eqs. (i) and (ii), HCF = a^2b^2

37. (a) Let $p(x) = 8(x^5 - x^3 + x)$
 $= 4 \times 2 \times x(x^4 - x^2 + 1)$
 and $q(x) = 28(x^6 + 1)$
 $= 7 \times 4[(x^2)^3 + (1)^3]$
 $= 4 \times 7 \times (x^2 + 1)(x^4 - x^2 + 1)$
 $\therefore$ HCF of $p(x)$ and $q(x)$
 $= 4(x^4 - x^2 + 1)$

38. (a) Let $f(x) = 2x^3 + x^2 - x - 2$
 $= (x-1)(2x^2 + 3x + 2)$
 and $g(x) = 3x^3 - 2x^2 + x - 2$
 $= (x-1)(3x^2 + x + 2)$

Hence, the highest factor of $f(x)$ and $g(x)$ is $(x-1)$.

39. (c) Given, HCF = $(x+y)$ and
 LCM = $3x^5 + 5x^4y + 2x^3y^2$
 $- 3x^2y^3 - 5xy^4 - 2y^5$
 $= x^3(3x^2 + 5xy + 2y^2) - y^3$
 $(3x^2 + 5xy + 2y^2)$
 $= (3x^2 + 5xy + 2y^2)(x^3 - y^3)$

We know that,

Product of two polynomials

$$= \text{HCF} \times \text{LCM}$$

$\therefore$ Required polynomial

$$= \frac{(x+y)(x^3 - y^3)(3x^2 + 5xy + 2y^2)}{(x-y)(x+y)}$$

$$= \frac{(x-y)(x^2 + y^2 + xy)(3x^2 + 5xy + 2y^2)}{(x-y)}$$

$$= (x^2 + y^2 + xy)(3x^2 + 5xy + 2y^2)$$

$$= 3x^4 + 8x^3y + 10x^2y^2 + 7xy^3 + 2y^4$$

40. (d) $(x+1)$ is the HCF of
 $Ax^2 + Bx + C$ and $Bx^2 + Ax + C$
 $\therefore A(-1)^2 + B(-1) + C = 0$
 $\Rightarrow A - B + C = 0$
 $\Rightarrow C = B - A$
 and $B(-1)^2 + A(-1) + C = 0$
 $\Rightarrow B - A + C = 0$
 $\Rightarrow C = A - B$
 $\therefore C = 0$

41. (b) Refer to question 19.

16

RATIONAL EXPRESSIONS

Usually (1-2) questions have been asked from this chapter. Generally questions are asked from this chapter are based on simplification of rational expressions.



RATIONAL EXPRESSIONS

An expression in the form of $\frac{p(x)}{q(x)}$, where $p(x)$ and $q(x)$ are polynomials and $q(x) \neq 0$ is called a rational expression.

- In the rational expression $\frac{p(x)}{q(x)}$, $p(x)$ is called the numerator and $q(x)$ is called the denominator of the rational expression.
- Every polynomial is a rational expression. Since, $p(x)$ can always be written as $\frac{p(x)}{1}$ and a constant function 1 is a polynomial of degree 0.
- Every rational expression need not be a polynomial.

Working Rule to Reduce the Given Rational Expression in its Lowest Term

1. Firstly, factorize both the polynomials $p(x)$ and $q(x)$.
2. Find the HCF of $p(x)$ and $q(x)$. If HCF of $p(x)$ and $q(x)$ is one, then rational expression $\frac{p(x)}{q(x)}$ is in its lowest terms.
3. If HCF is not equal to 1. Then, divide both $p(x)$ and $q(x)$ by their HCF and the rational expression is obtained in the lowest terms.

EXAMPLE 1. The lowest term of an expression

$$\frac{a^3 - b^3}{a^2 + ab + b^2} \text{ is}$$

- a. $a + b$ b. $a - b$ c. ab d. $\frac{a}{b}$

Sol. b. Rational expression

$$\begin{aligned} &= \frac{a^3 - b^3}{a^2 + ab + b^2} = \frac{(a - b)(a^2 + ab + b^2)}{(a^2 + ab + b^2)} = a - b \\ &[\because (a^3 - b^3) = (a - b)(a^2 + ab + b^2)] \end{aligned}$$

EXAMPLE 2. The lowest term of an expression

$$\frac{3x^2 - 11x - 4}{6x^2 - 7x - 3} \text{ is}$$

- a. $\frac{x+4}{2x+3}$ b. $\frac{x+4}{2x-3}$ c. $\frac{x-4}{2x+3}$ d. $\frac{x-4}{2x-3}$

Sol. d.
$$\begin{aligned} \frac{3x^2 - 11x - 4}{6x^2 - 7x - 3} &= \frac{3x^2 - 12x + x - 4}{6x^2 - 9x + 2x - 3} \\ &= \frac{3x(x-4) + 1(x-4)}{3x(2x-3) + 1(2x-3)} = \frac{(3x+1)(x-4)}{(3x+1)(2x-3)} = \frac{x-4}{2x-3} \end{aligned}$$

EXAMPLE 3. The lowest term of an expression

$$\frac{12x^3y^5z^4}{18x^2y^6z^5} \text{ is}$$

- a. $\frac{2x}{yz}$ b. $\frac{xy}{z}$ c. $\frac{2x}{3yz}$ d. $\frac{3x}{2yz}$

Sol. c.
$$\frac{12x^3y^5z^4}{18x^2y^6z^5} = \frac{6 \times 2 \times x^2 \times x \times y^5 \times z^4}{6 \times 3 \times x^2 \times y \times y^5 \times z \times z^4} = \frac{2x}{3yz}$$

Operations on Rational Expressions

1. Addition or Subtraction of Rational Expressions with like denominators

It $\frac{p(x)}{q(x)}$ and $\frac{b(x)}{q(x)}$ are two rational expressions then

$$\frac{p(x)}{q(x)} \pm \frac{b(x)}{q(x)} = \frac{p(x) \pm b(x)}{q(x)}$$

2. Addition or subtraction of Rational Expressions with unlike denominators

To add or subtract rational expressions with unlike denominators, follow the steps given below

Step I Write each denominator in the factor form

Step II Find the LCM of the denominators

Step III Rewrite each rational expression with LCM as the denominator

Step IV Add or subtract the numerators.

EXAMPLE 4. The sum of $\frac{x+1}{x-1}$ and $\frac{x-1}{x+1}$ is

a. $\frac{x^2+1}{x^2-1}$ b. $\frac{1}{x^2+1}$ c. $\frac{2x^2+2}{x^2-1}$ d. $\frac{x^2+2}{x^2-1}$

Sol. c.
$$\begin{aligned} \frac{x+1}{x-1} + \frac{x-1}{x+1} &= \frac{(x+1)}{(x-1)} \times \frac{(x+1)}{(x+1)} + \frac{(x-1)}{(x+1)} \times \frac{(x-1)}{(x-1)} \\ &= \frac{(x+1)^2 + (x-1)^2}{(x^2-1)} \\ &= \frac{x^2+1+2x+x^2+1-2x}{x^2-1} = \frac{2x^2+2}{x^2-1} \end{aligned}$$

$$[\because (a+b)^2 = a^2 + b^2 + 2ab \text{ and } (a-b)^2 = a^2 + b^2 - 2ab]$$

EXAMPLE 5. What should be subtracted from

$$\frac{7x}{x^2+x-12} \text{ to get } \frac{4}{x+4}?$$

a. $\frac{1}{x-3}$ b. $\frac{3}{x-3}$ c. $\frac{1}{x+3}$ d. $\frac{3}{x+3}$

Sol. b. Let $p(x)$ is to be subtracted, then

$$\begin{aligned} \frac{7x}{x^2+x-12} - p(x) &= \frac{4}{x+4} \\ \Rightarrow p(x) &= \frac{7x}{x^2+x-12} - \frac{4}{x+4} = \frac{7x}{(x+4)(x-3)} - \frac{4}{(x+4)} \\ &= \frac{7x}{(x+4)(x-3)} - \frac{4(x-3)}{(x+4)(x-3)} = \frac{7x-4(x-3)}{(x+4)(x-3)} \\ &= \frac{3x+12}{(x+4)(x-3)} = \frac{3(x+4)}{(x+4)(x-3)} = \frac{3}{x-3} \end{aligned}$$

3. Multiplication of Rational Expressions If $\frac{p(x)}{q(x)}$ and

$\frac{g(x)}{h(x)}$ are two rational expressions, then their product

$$\text{is given by } \frac{p(x)}{q(x)} \times \frac{g(x)}{h(x)} = \frac{p(x) \cdot g(x)}{q(x) \cdot h(x)}$$

• Multiplicative inverse of $\frac{p(x)}{q(x)}$ is $\frac{q(x)}{p(x)}$.

• 1 is the multiplicative identity.

EXAMPLE 6. The product of $\frac{x^2-1}{x^2+1}$ and $\frac{x+2}{x+1}$ is

a. $\frac{(x-1)}{(x^2+1)}$ b. $\frac{(x-2)}{(x^2+1)}$
c. $\frac{(x-1)(x+2)}{(x^2+1)}$ d. $\frac{x+2}{(x^2-1)}$

Sol. c. Here, product = $\frac{x^2-1}{x^2+1} \times \frac{(x+2)}{(x+1)} = \frac{(x-1)(x+1)(x+2)}{(x^2+1)(x+1)}$

$$= \frac{(x-1)(x+2)}{(x^2+1)} \quad [\because a^2-b^2 = (a-b)(a+b)]$$

4. Division of Rational Expressions If $\frac{p(x)}{q(x)}$ and $\frac{g(x)}{h(x)}$

are two rational expressions, then their division is

$$\text{given by } \frac{p(x)}{q(x)} \div \frac{g(x)}{h(x)} = \frac{p(x)}{q(x)} \times \frac{h(x)}{g(x)}$$

= Product of $\frac{p(x)}{q(x)}$ and the reciprocal of $\frac{g(x)}{h(x)}$

i.e. $\frac{b(x)}{g(x)}$.

EXAMPLE 7. The lowest term of an expression

$$\frac{x^2+8x+12}{x^2-7x+12} \div \frac{x^2+4x-12}{x-4} \text{ is}$$

a. $\frac{(x-2)}{(x-3)(x+2)}$ b. $\frac{x-2}{x+2}$
c. $\frac{(x+2)}{(x-3)(x-2)}$ d. $\frac{x+3}{x-2}$

Sol. c. Here,
$$\begin{aligned} \frac{x^2+8x+12}{x^2-7x+12} \div \frac{x^2+4x-12}{x-4} &= \frac{x^2+8x+12}{x^2-7x+12} \times \frac{x-4}{x^2+6x-2x-12} \\ &= \frac{(x+6)(x+2)}{(x-4)(x-3)} \times \frac{x-4}{(x+6)(x-2)} = \frac{x+2}{(x-3)(x-2)} \end{aligned}$$

PRACTICE EXERCISE

1. Which of the following are rational expressions?

I. $\frac{x^3 - 3x^2 + 2}{x^2 + 1}$ II. $\frac{z^3 - 3z^2}{2z + 3}$
 III. $\frac{x^2 - x + 2}{x + 3}$ IV. $\frac{x^3 + 3x^2 - 1}{x^2 + \sqrt{x - 1}}$

Select the correct answer using the codes given below

- (a) I, II and III (b) II, III and IV
 (c) All of these (d) None of these

2. The simplified form of $\frac{a+2}{a+3} - \frac{(a+1)}{(a+2)}$

(a) $\frac{1}{a^2 + 5a + 6}$ (b) $\frac{a+2}{a^2 + 5a + 6}$
 (c) 0 (d) $\frac{(a+2)^2}{a^2 + 5a + 6}$

3. $\left(\frac{x+1}{x^2-1} - \frac{2}{x}\right)$ expressed as a rational expression is

(a) $\frac{x^2-2}{x(x^2-1)}$ (b) $\frac{-(x-2)}{x(x-1)}$
 (c) $\frac{2(x+1)}{(x^3-1)}$ (d) $\frac{1}{x(x+1)}$

4. The expression $\frac{(x-1)(x-2)(x^2-9x+14)}{(x-7)(x^2-3x+2)}$ in lowest terms is

(a) $\frac{1}{x-7}$ (b) $(x-7)$ (c) $(x-2)$ (d) $\frac{1}{x-2}$

5. $\frac{a-c}{(a-b)(x-a)} + \frac{b-c}{(b-a)(x-b)}$ is equal to

(a) $\frac{b-a}{(x-b)(x-a)}$ (b) $\frac{x-c}{(x-a)(x-b)}$
 (c) $\frac{b-a}{(x-b)(x-c)}$ (d) None of these

6. The sum of the rational expression $\frac{x-3}{x^2+1}$ and its reciprocal is

(a) $\frac{x^3-3x^2+x-3}{x^4+3x^2-6x+10}$ (b) $\frac{x^4+3x^2-6x+10}{x^3-3x^2+x-3}$
 (c) $\frac{x^4-3x^2+6x+10}{x^3-3x^2+x-3}$ (d) $\frac{(x-1)^3}{(x+3)^2}$

7. The value of

$\frac{a}{(a-b)(a-c)} + \frac{b}{(b-c)(b-a)} + \frac{c}{(c-a)(c-b)}$
 (a) 2a (b) 0 (c) 2b (d) b-c

8. If $A = \frac{x-1}{x+1}$ and $B = \frac{x+1}{x-1}$, then $(A+B)^2$ is

(a) $\frac{4x^4+8x^2-4}{x^4-2x^2+1}$ (b) $\frac{4x^4+8x^2+4}{x^4-2x^2+1}$
 (c) $\frac{4x^4+8x^2+4}{x^4+2x+1}$ (d) None of these

9. The value of

$\frac{1}{(1-a)(1-b)} + \frac{a^2}{(1-a)(b-a)} - \frac{b^2}{(b-1)(a-b)}$ is
 (a) $1+a$ (b) $1-a^2$
 (c) $1-b^2$ (d) None of these

10. The value of $\frac{\left[\frac{x}{y} - \frac{y}{x}\right]\left[\frac{y}{z} - \frac{z}{y}\right]\left[\frac{z}{x} - \frac{x}{z}\right]}{\left[\frac{1}{x^2} - \frac{1}{y^2}\right]\left[\frac{1}{y^2} - \frac{1}{z^2}\right]\left[\frac{1}{z^2} - \frac{1}{x^2}\right]}$ is

(a) $x^2y^2z^2$ (b) $-x^2y^2z^2$
 (c) 1 (d) None of these

11. What is the simplified form of

$\left(\frac{x^2-3x+2}{x^3-8}\right) \div \left(\frac{x^2-9}{x^2+7x+12}\right) \times \left(\frac{x^3+2x^2+4x}{x^2+3x-4}\right)$?
 (a) $\frac{x}{x-3}$ (b) $\frac{x-2}{x-3}$
 (c) $\frac{x}{x+3}$ (d) $\frac{x+3}{x+4}$

12. If $pq + qr + rp = 0$, then what is the value of

$\frac{p^2}{p^2-qr} + \frac{q^2}{q^2-rp} + \frac{r^2}{r^2-pq}$?
 (a) 0 (b) 1
 (c) -1 (d) 3

13. If $x + y + z = 0$, then what is the value of

$\frac{1}{x^2+y^2-z^2} + \frac{1}{y^2+z^2-x^2} + \frac{1}{z^2+x^2-y^2}$?
 (a) $\frac{1}{x^2+y^2+z^2}$ (b) 1
 (c) -1 (d) 0

14. If $a = \frac{1+x}{2-x}$, then what is $\frac{1}{a+1} + \frac{2a+1}{a^2-1}$ equal to?

(a) $\frac{(1+x)(2+x)}{2x-1}$ (b) $\frac{(1-x)(2-x)}{x-2}$
 (c) $\frac{(1+x)(2-x)}{2x-1}$ (d) $\frac{(1-x)(2-x)}{2x+1}$

15. If $\frac{1}{x+1} + \frac{2}{y+2} + \frac{1009}{z+1009} = 1$, then what is the value of $\frac{x}{x+1} + \frac{y}{y+2} + \frac{z}{z+1009}$?
- (a) 0 (b) 2 (c) 3 (d) 4

16. What is $\frac{x^2 - 3x + 2}{x^2 - 5x + 6} \div \frac{x^2 - 5x + 4}{x^2 - 7x + 12}$ equal to?
- (a) $\frac{x+3}{x-3}$ (b) 1 (c) $\frac{x+1}{x+3}$ (d) 2

17. If $x + y + z = 0$, then what is $\frac{xyz}{(x+y)(y+z)(z+x)}$ equal to (where, $x \neq -y, y \neq -z, z \neq -x$)?
- (a) -1 (b) 1
(c) $xy + yz + zx$ (d) None of these

PREVIOUS YEARS' QUESTIONS

18. What is $\frac{(x^2 + y^2)(x - y) - (x - y)^3}{x^2y - xy^2}$ equal to? ☑ 2013 II
- (a) 1 (b) 2 (c) 4 (d) -2
19. What is $\frac{1}{a-b} - \frac{1}{a+b} - \frac{2b}{a^2+b^2} - \frac{4b^3}{a^4+b^4} - \frac{8b^7}{a^8-b^8}$ equal to? ☑ 2014 II
- (a) $a+b$ (b) $a-b$ (c) 1 (d) 0
20. If $a^2 - by - cz = 0$, $ax - b^2 + cz = 0$ and $ax + by - c^2 = 0$, then the value of $\frac{x}{a+x} + \frac{y}{b+y} + \frac{z}{c+z}$ will be ☑ 2016 I
- (a) $a+b+c$ (b) 3 (c) 1 (d) 0

ANSWERS

1	a	2	a	3	b	4	c	5	b	6	b	7	b	8	b	9	d	10	b
11	a	12	b	13	d	14	c	15	b	16	b	17	a	18	b	19	d	20	c

HINTS AND SOLUTIONS

1. (a) Here, $\frac{x^3 + 3x^2 - 1}{x^2 + \sqrt{x-1}}$ is not a rational expression, since the denominator is not a polynomial.

2. (a) We know $\frac{a+2}{a+3} - \frac{(a+1)}{(a+2)}$

$$= \frac{(a+2)^2 - (a+1)(a+3)}{(a+3)(a+2)}$$

$$= \frac{a^2 + 4a + 4 - (a^2 + 4a + 3)}{a^2 + 5a + 6}$$

$$= \frac{1}{a^2 + 5a + 6}$$

3. (b) Here, $\left(\frac{x+1}{x^2-1} - \frac{2}{x}\right) = \frac{1}{x-1} - \frac{2}{x}$

$$[\because x^2 - 1 = (x+1)(x-1)]$$

$$= \frac{x-2x+2}{x(x-1)} = \frac{-(x-2)}{x(x-1)}$$

4. (c) We know that,

$$\frac{(x-1)(x-2)(x^2-9x+14)}{(x-7)(x^2-3x+2)}$$

$$= \frac{(x-1)(x-2)(x^2-7x-2x+14)}{(x-7)(x^2-2x-x+2)}$$

$$= \frac{(x-1)(x-2)(x-7)(x-2)}{(x-7)(x-2)(x-1)} = x-2$$

5. (b) Here, $\frac{a-c}{(a-b)(x-a)} + \frac{b-c}{(b-a)(x-b)}$

$$= \frac{(a-c)(x-b) - (b-c)(x-a)}{(a-b)(x-a)(x-b)}$$

$$= \frac{ax-ab-xc+bc-(bx-ab-cx+ac)}{(a-b)(x-a)(x-b)}$$

$$= \frac{ax+bc-bx-ac}{(a-b)(x-a)(x-b)}$$

$$= \frac{x(a-b)-c(-b+a)}{(a-b)(x-a)(x-b)}$$

$$= \frac{(x-c)(a-b)}{(a-b)(x-a)(x-b)} = \frac{(x-c)}{(x-a)(x-b)}$$

6. (b) Here, reciprocal of $\frac{x-3}{x^2+1}$ is $\frac{x^2+1}{(x-3)}$

So,

$$\frac{x-3}{x^2+1} + \frac{x^2+1}{x-3} = \frac{(x-3)^2 + (x^2+1)^2}{(x^2+1)(x-3)}$$

$$= \frac{x^2+9-6x+x^4+1+2x^2}{x^3-3x^2+x-3}$$

$$= \frac{x^4+3x^2-6x+10}{x^3-3x^2+x-3}$$

7. (b) We know that,

$$\frac{a}{(a-b)(a-c)} + \frac{b}{(b-c)(b-a)} + \frac{c}{(c-a)(c-b)}$$

$$= \frac{-a}{(a-b)(c-a)} - \frac{b}{(b-c)(a-b)} - \frac{c}{(c-a)(b-c)}$$

$$= \frac{(-a)(b-c) - b(c-a) - c(a-b)}{(a-b)(b-c)(c-a)}$$

$$= \frac{-ab+ac-bc+ab-ac+bc}{(a-b)(b-c)(c-a)}$$

$$= \frac{0}{(a-b)(b-c)(c-a)} = 0$$

8. (b) Here,

$$(A+B)^2 = \left[\left(\frac{x-1}{x+1} \right) + \left(\frac{x+1}{x-1} \right) \right]^2$$

$$= \left[\frac{(x-1)^2 + (x+1)^2}{(x+1)(x-1)} \right]^2 = \left[\frac{2(x^2+1)}{x^2-1} \right]^2$$

$$[\because (a+b)^2 + (a-b)^2 = 2(a^2+b^2)]$$

$$= \frac{4(x^4+2x^2+1)}{x^4-2x^2+1} = \frac{4x^4+8x^2+4}{x^4-2x^2+1}$$

9. (d) Given,

$$\frac{1}{(1-a)(1-b)} + \frac{a^2}{(1-a)(b-a)} - \frac{b^2}{(b-1)(a-b)}$$

$$= \frac{(b-a) + a^2(1-b) - b^2(1-a)}{(1-a)(1-b)(b-a)}$$

$$\begin{aligned}
 &= \frac{b-a+a^2-a^2b-b^2+ab^2}{(1-a)(1-b)(b-a)} \\
 &= \frac{(b-a)+(a^2-b^2)+ab(b-a)}{(1-a)(1-b)(b-a)} \\
 &= \frac{(b-a)-(a+b)(b-a)+ab(b-a)}{(1-a)(1-b)(b-a)} = \frac{1-a-b+ab}{1-b-a+ab} = 1
 \end{aligned}$$

10. (b) Given, $\left[\frac{x}{y} - \frac{y}{x}\right] \left[\frac{y}{z} - \frac{z}{y}\right] \left[\frac{z}{x} - \frac{x}{z}\right]$

$$\begin{aligned}
 &= \left[\frac{1}{x^2} - \frac{1}{y^2}\right] \left[\frac{1}{y^2} - \frac{1}{z^2}\right] \left[\frac{1}{z^2} - \frac{1}{x^2}\right] \\
 &= \left[\frac{x^2-y^2}{x^2y^2}\right] \left[\frac{y^2-z^2}{z^2y^2}\right] \left[\frac{z^2-x^2}{x^2z^2}\right] \\
 &= \frac{(x^2-y^2)(y^2-z^2)(z^2-x^2)}{x^2y^2z^2} \\
 &= -\frac{(x^2-y^2)(y^2-z^2)(z^2-x^2)}{x^4y^4z^4} = -\frac{x^4y^4z^4}{x^2y^2z^2} = -x^2y^2z^2
 \end{aligned}$$

11. (a) Given, $\left(\frac{x^2-3x+2}{x^3-8}\right) \div \left(\frac{x^2-9}{x^2+7x+12}\right) \times \left(\frac{x^3+2x^2+4x}{x^2+3x-4}\right)$

$$\begin{aligned}
 &= \left(\frac{x^2-3x+2}{x^3-8}\right) \times \frac{x^2+7x+12}{x^2-9} \times \left(\frac{x^3+2x^2+4x}{x^2+3x-4}\right) \\
 &= \frac{(x-1)(x-2)}{(x-2)(x^2+4+2x)} \times \frac{(x+4)(x+3)}{(x-3)(x+3)} \times \frac{x(x^2+2x+4)}{(x-1)(x+4)} \\
 &= \frac{x}{x-3} \quad [\because a^3-b^3 = (a-b)(a^2+b^2+ab)]
 \end{aligned}$$

12. (b) Given, $pq+qr+rp=0$

$$\begin{aligned}
 \therefore \frac{p^2}{p^2-qr} + \frac{q^2}{q^2-rp} + \frac{r^2}{r^2-pq} \\
 &= \frac{p^2}{p^2+rp+pq} + \frac{q^2}{q^2+pq+qr} + \frac{r^2}{r^2+qr+rp} \\
 &= \frac{p^2}{p(p+r+q)} + \frac{q^2}{q(p+q+r)} + \frac{r^2}{r(p+q+r)} = \frac{p+q+r}{p+q+r} = 1
 \end{aligned}$$

13. (d) Given, $x+y+z=0 \Rightarrow x+y=-z$
 On squaring both sides, we get $x^2+y^2+2xy=z^2$
 $x^2+y^2-z^2=-2xy$
 Similarly, $y^2+z^2-x^2=-2yz$ and $z^2+x^2-y^2=-2zx$

$$\begin{aligned}
 \therefore \frac{1}{x^2+y^2-z^2} + \frac{1}{y^2+z^2-x^2} + \frac{1}{z^2+x^2-y^2} \\
 &= \frac{1}{-2xy} + \frac{1}{-2yz} + \frac{1}{-2zx} = \frac{1}{2} \left(\frac{z+x+y}{xyz} \right) = 0
 \end{aligned}$$

14. (c) Given, $a = \frac{1+x}{2-x}$ So, $\frac{1}{a+1} + \frac{2a+1}{a^2-1} = \frac{3a}{a^2-1} = \frac{3\left(\frac{1+x}{2-x}\right)}{\left(\frac{1+x}{2-x}\right)^2-1}$

$$\begin{aligned}
 &= \frac{3(1+x)(2-x)}{1+x^2+2x-(4+x^2-4x)} = \frac{3(1+x)(2-x)}{6x-3} = \frac{(1+x)(2-x)}{(2x-1)}
 \end{aligned}$$

15. (b) Given, $\frac{1}{x+1} + \frac{2}{y+2} + \frac{1009}{z+1009} = 1$

$$\begin{aligned}
 \Rightarrow \frac{1}{x+1} - 1 + \frac{2}{y+2} - 1 + \frac{1009}{z+1009} - 1 &= 1-3 \\
 \Rightarrow -\frac{x}{x+1} - \frac{y}{y+2} - \frac{z}{z+1009} &= -2 \\
 \therefore \frac{x}{x+1} + \frac{y}{y+2} + \frac{z}{z+1009} &= 2
 \end{aligned}$$

16. (b) Given, $\frac{x^2-3x+2}{x^2-5x+6} \div \frac{x^2-5x+4}{x^2-7x+12}$

$$\begin{aligned}
 &= \frac{x^2-3x+2}{x^2-5x+6} \times \frac{(x^2-7x+12)}{(x^2-5x+4)} \\
 &= \frac{(x-1)(x-2)}{(x-3)(x-2)} \times \frac{(x-4)(x-3)}{(x-4)(x-1)} = 1
 \end{aligned}$$

17. (a) Given, $x+y+z=0$

$$\begin{aligned}
 \Rightarrow x+y=-z, y+z=-x \text{ and } z+x=-y \\
 \therefore \frac{xyz}{(x+y)(y+z)(z+x)} = \frac{xyz}{(-z)(-x)(-y)} = \frac{xyz}{-xyz} = -1
 \end{aligned}$$

18. (b) Given, $\frac{(x^2+y^2)(x-y)-(x-y)^3}{x^2y-xy^2}$

$$\begin{aligned}
 &= \frac{x^3+xy^2-x^2y-y^3-(x^3-y^3-3x^2y+3xy^2)}{x^2y-xy^2} \\
 &= \frac{x^3+xy^2-x^2y-y^3-x^3+y^3+3x^2y-3xy^2}{x^2y-xy^2} \\
 &= \frac{2x^2y-2xy^2}{x^2y-xy^2} = \frac{2(x^2y-xy^2)}{x^2y-xy^2} = 2
 \end{aligned}$$

19. (d) $\frac{1}{a-b} - \frac{1}{a+b} - \frac{2b}{a^2+b^2} - \frac{4b^3}{a^4+b^4} - \frac{8b^7}{a^8-b^8}$

$$\begin{aligned}
 &= \frac{(a+b)-(a-b)}{(a-b)(a+b)} - \frac{2b}{a^2+b^2} - \frac{4b^3}{a^4+b^4} - \frac{8b^7}{a^8-b^8} \\
 &= \frac{2b}{a^2-b^2} - \frac{2b}{a^2+b^2} - \frac{4b^3}{a^4+b^4} - \frac{8b^7}{a^8-b^8} \\
 &= \frac{2b(a^2+b^2)-2b(a^2-b^2)}{(a^2-b^2)(a^2+b^2)} - \frac{4b^3}{a^4+b^4} - \frac{8b^7}{a^8-b^8} \\
 &= \frac{4b^3}{a^4-b^4} - \frac{4b^3}{a^4+b^4} - \frac{8b^7}{a^8-b^8} \\
 &= \frac{4b^3(a^4+b^4)-4b^3(a^4-b^4)}{(a^4-b^4)(a^4+b^4)} - \frac{8b^7}{a^8-b^8} = 0
 \end{aligned}$$

20. (c) Given, $a^2-by-cz=0$... (i)
 $ax-b^2+cz=0$... (ii)
 and $ax+by-c^2=0$... (iii)

On adding Eqs. (i), (ii) and (iii), we get

$$\begin{aligned}
 x &= \frac{b^2-a^2+c^2}{2a} \Rightarrow \frac{x}{a+x} = \frac{b^2-a^2+c^2}{a^2+b^2+c^2} \\
 \text{Similarly, } \frac{y}{b+y} &= \frac{a^2-b^2+c^2}{a^2+b^2+c^2} \text{ and } \frac{z}{c+z} = \frac{a^2+b^2-c^2}{a^2+b^2+c^2} \\
 \therefore \frac{x}{a+x} + \frac{y}{b+y} + \frac{z}{c+z} &= \frac{(b^2-a^2+c^2)+(a^2-b^2+c^2)+(a^2+b^2-c^2)}{a^2+b^2+c^2} = 1
 \end{aligned}$$

LINEAR EQUATIONS

Generally (3-5) questions have been definitely asked from this chapter in CDS examination. Beside this, the linear equations are also applicable in solving various word problems related to number system, mensuration, time and work, etc.



A linear equation is an equation for a straight line.

So, the equation which has degree 1, i.e., which has linear power of the variables, is called a linear equation.

LINEAR EQUATIONS IN ONE VARIABLE

A linear equation in one variable is an equation which can be written in the form of $ax + b = 0$, or $ax = c$, where a , b , c are real numbers with $a \neq 0$.

e.g. $5x + 8 = 9 - x$, $\frac{2}{3}y + 7 = \frac{y}{2}$ and $t + 3t = 9 - t$ are linear equations in one variable.

Solving a linear Equation in One Variable

1. Simplify both sides of the equation.
2. Use the addition and subtraction properties to get all variables of terms on the LHS and all constant terms on the RHS.
3. Simplify and divide both sides of the equation by the coefficient of the variable.

EXAMPLE 1. Solve $2(x - 3) - (5 - 3x) = 3(x + 1) - 4(2 + x)$.

a. 1

b. -1

c. 0

d. 3

Sol. a. $2(x - 3) - (5 - 3x) = 3(x + 1) - 4(2 + x) \Rightarrow 2x - 6 - 5 + 3x = 3x + 3 - 8 - 4x$
 $\Rightarrow 5x - 11 = -x - 5 \Rightarrow 6x = 6 \Rightarrow x = 1$, Hence, the value of x is 1.

LINEAR EQUATION IN TWO VARIABLES

An equation which can be put in the form $ax + by + c = 0$, where a , b and c are real numbers and a , b both are not zero, is called a linear equation in two variables.

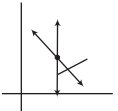
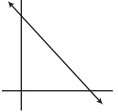
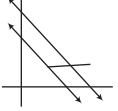
e.g. $2x + 3y = 5$, $\sqrt{2}x + \sqrt{3}y = 0$ and $2a + 3b = 0$ are linear equations in two variables.

Pair of Linear Equations

Two linear equations in the two same variables are called pair of linear equations in two variables. The general form of pair of linear equations in two variables x and y is $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ where, a_1, a_2, b_1, b_2 and c_1, c_2 are all real numbers and $a_1^2 + b_1^2 \neq 0, a_2^2 + b_2^2 \neq 0$.

Consistency Pair of linear Equations

A pair of linear equations which has no solution is called an **inconsistent pair** of linear equations. A pair of linear equations which has atleast one solution is called a **consistent pair** of linear equations. Let us consider two linear equations as $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$. Pair of linear equations in two variables will represent two straight lines, both to be considered together. The table given below represents the three possibilities.

Ratios	Consistency	Graphical Representation	Algebraic interpretation
$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$	Consistent	 Intersecting lines.	Exactly one solution (unique).
$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$	Dependent (consistent)	 Coincident lines.	Infinitely many solutions.
$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$	Inconsistent	 Parallel lines.	No solution.

HOMOGENEOUS PAIR OF LINEAR EQUATIONS

The pair of linear equations $a_1x + b_1y = 0, a_2x + b_2y = 0$ has

1. A unique solution, if $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$ and the solution is $x = 0, y = 0$.
2. An infinite number of solutions, if $\frac{a_1}{a_2} = \frac{b_1}{b_2}$.

This pair of linear equations is always consistent.

EXAMPLE 2. The values of k for which the system $kx + 2y = 5$ and $3x + y = 1$ has

- (i) unique solution (ii) no solution are
- a. $k \neq 6$ and $k = 6$ b. $k \neq 3$ and $k = 3$
- c. $k \neq 2$ and $k = 2$ d. None of these

Sol. a. Given equations are, $kx + 2y - 5 = 0$ and

$$3x + y - 1 = 0$$

On comparing the given equations, with standard form of equations

$$a_1x + b_1y + c_1 = 0 \text{ and } a_2x + b_2y + c_2 = 0$$

$$\text{We have, } a_1 = k, b_1 = 2, c_1 = -5$$

$$\text{and } a_2 = 3, b_2 = 1, c_2 = -1$$

(i) Here, the equations have a unique solution, if

$$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$$

$$\text{i.e. if } \frac{k}{3} \neq \frac{2}{1} \Rightarrow k \neq 6$$

(ii) The equations have no solution, if

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$

$$\therefore \frac{k}{3} = \frac{2}{1} \Rightarrow k = 6$$

EXAMPLE 3. The value of k for which the system of equations $kx - y = 2, 6x - 2y = 3$ has infinitely many solutions is

a. 3

b. 4

c. 12

d. No such value of k exists.

Sol. d. Given equations are,

$$kx - y - 2 = 0 \text{ and } 6x - 2y - 3 = 0$$

On comparing the given equations, with standard form of equations

$$a_1x + b_1y + c_1 = 0 \text{ and } a_2x + b_2y + c_2 = 0$$

$$\text{We have, } a_1 = k, b_1 = -1, c_1 = -2$$

$$\text{and } a_2 = 6, b_2 = -2, c_2 = -3$$

Here, the equations have infinite number of solutions, if

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

$$\text{But, } \frac{a_1}{a_2} = \frac{k}{6}, \frac{b_1}{b_2} = \frac{-1}{-2} = \frac{1}{2} \text{ and } \frac{c_1}{c_2} = \frac{-2}{-3} = \frac{2}{3}$$

$$\text{So, } \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$

Thus, the system has no such value of k , for which the given system has infinitely many solutions.

Algebraic Methods of Solutions

Different algebraic methods for solving pair of linear equations are discussed below in detail.

1. Substitution Method

In this method, we substitute the value of one variable in terms of other variable to solve the pair of linear equations, so this method is called substitution method.

Steps used in this method are given below.

Step I Find the value of one variable x (or y) in terms of other variable i.e. y (or x) from an equation.

EQUATIONS REDUCIBLE TO A PAIR OF LINEAR EQUATIONS

We have several situations when there are two equations that are not linear, but can be reduced to a pair of linear equations. By making suitable substitutions.

EXAMPLE 7. If $\frac{2}{x} + \frac{3}{y} = 13$ and $\frac{5}{x} - \frac{4}{y} = -2$, then

what is the value of $\frac{x}{y}$?

- a. $\frac{2}{3}$ b. $\frac{3}{2}$ c. $\frac{1}{3}$ d. $\frac{1}{2}$

Sol. b. Let $\frac{1}{x} = p$ and $\frac{1}{y} = q$

So, we have, $2p + 3q = 13$... (i)

and $5p - 4q = -2$... (ii)

On multiplying Eq. (i) by 4 and Eq. (ii) by 3 and then adding, we get

$$8p + 15p = 52 - 6 \Rightarrow 23p = 46 \Rightarrow p = 2$$

On putting $p = 2$ in Eq. (i), we get

$$2 \times 2 + 3q = 13 \Rightarrow 3q = 13 - 4 = 9 \Rightarrow q = \frac{9}{3} = 3$$

$$\text{Now, } x = \frac{1}{p} \Rightarrow x = \frac{1}{2} \text{ and } y = \frac{1}{q} \Rightarrow y = \frac{1}{3}$$

Hence, the value of $\frac{x}{y}$ is $\frac{3}{2}$.

EXAMPLE 8. Find the solution of the system of equations $4x + 3y = 18xy$ and $2x - 5y = -4xy$.

- a. $x = 2, y = 3$ b. $x = 3, y = 2$ c. $x = \frac{1}{3}, y = \frac{1}{2}$ d. $x = \frac{1}{2}, y = \frac{1}{3}$

Sol. d. Given equations are, $4x + 3y = 18xy$... (i)

and $2x - 5y = -4xy$... (ii)

On dividing both the equations by xy , we get

$$\frac{4x}{xy} + \frac{3y}{xy} = \frac{18xy}{xy} \text{ and } \frac{2x}{xy} - \frac{5y}{xy} = \frac{-4xy}{xy}$$

$$\text{or } \frac{4}{y} + \frac{3}{x} = 18 \text{ and } \frac{2}{y} - \frac{5}{x} = -4$$

On putting $\frac{1}{y} = A$ and $\frac{1}{x} = B$, we get

$$4A + 3B = 18 \quad \dots \text{(iii)}$$

$$2A - 5B = -4 \quad \dots \text{(iv)}$$

On multiplying Eq. (iv) by 2 and subtracting from Eq. (iii), we get

$$\begin{array}{r} 4A + 3B = 18 \\ 4A - 10B = -8 \\ \hline -7B = 26 \end{array}$$

$$\Rightarrow B = 2$$

On putting, the value of B in Eq. (iii), we get

$$\Rightarrow 4A + 6 = 18 \Rightarrow 4A = 12 \Rightarrow A = 3$$

$$\text{Now } \frac{1}{x} = B = 2 \Rightarrow x = \frac{1}{2} \text{ and } \frac{1}{y} = A = 3 \Rightarrow y = \frac{1}{3}$$

So, $x = \frac{1}{2}$ and $y = \frac{1}{3}$ is the required solution.

EXAMPLE 9. If $\frac{3}{x+y} + \frac{2}{x-y} = 2$ and $\frac{9}{x+y} - \frac{4}{x-y} = 1$,

then what is value of $\frac{x}{y}$?

- a. $\frac{3}{2}$ b. 5 c. $\frac{2}{3}$ d. $\frac{1}{5}$

Sol. b. Given equations are,

$$\frac{3}{x+y} + \frac{2}{x-y} = 2 \quad \dots \text{(i)}$$

$$\text{and } \frac{9}{x+y} - \frac{4}{x-y} = 1 \quad \dots \text{(ii)}$$

$$\text{Let } \frac{1}{x+y} = a \text{ and } \frac{1}{x-y} = b$$

$$\therefore 3a + 2b = 2 \quad \dots \text{(iii)}$$

$$9a - 4b = 1 \quad \dots \text{(iv)}$$

On multiplying Eq. (iii) by 2 and adding with Eq. (iv), we get

$$\Rightarrow 6a + 9a = 5 \Rightarrow 15a = 5$$

$$\therefore a = \frac{1}{3}$$

$$\therefore \frac{1}{x+y} = \frac{1}{3} \Rightarrow x+y = 3 \quad \dots \text{(v)}$$

On putting the value of a in Eq. (iii), we get

$$3 \times \frac{1}{3} + 2b = 2 \Rightarrow b = \frac{1}{2}$$

$$\Rightarrow \frac{1}{x-y} = \frac{1}{2} \Rightarrow x-y = 2 \quad \dots \text{(vi)}$$

On adding Eqs. (v) and (vi), we get, $2x = 5 \Rightarrow x = \frac{5}{2}$

$$\text{From Eq. (v), } x+y = 3 \Rightarrow y = 3 - \frac{5}{2} = \frac{1}{2}$$

$$\therefore \frac{x}{y} = \frac{\frac{5}{2}}{\frac{1}{2}} = 5$$

LINEAR EQUATION IN THREE VARIABLES

An equation in the form of $ax + by + cz = r$, where a, b, c and r are real numbers and a, b and c are not all zeros is called a linear equation in three variables

e.g. $3x + 4y - 7z = 2$, $-2x + y - z = 6$ and $x + y + z = 2$ are all linear equation in three variables.

Use substitution and elimination method to solve the system of three equations in three variables.

EXAMPLE 10. Solve the system $x + y + z = 5$,
 $2x - y + z = 9$ and $x - 2y + 3z = 16$.

a. (2, -1, 4) b. (3, 5, 6) c. (1, -2, 5) d. (6, 1, -3)

Sol. a. Given equations are, $x + y + z = 5$... (i)
 $2x - y + z = 9$... (ii)
 $x - 2y + 3z = 16$... (iii)

First we eliminate y by adding Eqs. (i) and (ii), we get

$$\begin{array}{r} x + y + z = 5 \\ 2x - y + z = 9 \\ \hline 3x + 2z = 14 \end{array} \quad \dots \text{(iv)}$$

Next we eliminate y by multiplying Eq. (i) by 2 and then adding it to Eq. (iii), we get

$$\begin{array}{r} 2x + 2y + 2z = 10 \\ x - 2y + 3z = 16 \\ \hline 3x + 5z = 26 \end{array} \quad \dots \text{(v)}$$

Now subtract Eq. (iv) from Eq. (v), we get

$$3z = 12 \Rightarrow z = 4$$

Substitute $z = 4$ in Eq. (iv), $3x + 2 \times 4 = 14 \Rightarrow 3x = 6 \Rightarrow x = 2$.

Finally put $x = 2$ and $z = 4$ in Eq. (i), we get

$$2 + y + 4 = 5 \Rightarrow y = -1$$

$\therefore (2, -1, 4)$ is the required solution.

APPLICATION OF LINEAR EQUATIONS

The application of linear equations are discussed below.

Problem Based on Ages

If the problem involves finding out the ages of two persons, then take the present age of one person as x and that of the other as y .

Then, ' a ' years ago, age of 1st person was $(x - a)$ years and that of 2nd person was $(y - a)$ years.

After b years, age of 1st person will be $(x + b)$ years and that of 2nd person will be $(y + b)$ years.

Formulate the equations and then solve them.

EXAMPLE 11. Age of X is six times that of Y . After 4 yr, X is 4 times elder to Y . What is the present age of Y ?

a. 4 yr b. 5 yr c. 6 yr d. 7 yr

Sol. c. Let the present age of X and Y be x and y yr, respectively.

Condition I Age of X = Age of $Y \times 6$, $x = 6y$... (i)

After 4 yr, Age of $X = (x + 4)$ yr, Age of $Y = (y + 4)$ yr

Condition II $(x + 4) = (y + 4) \times 4$, $x + 4 = 4y + 16$... (ii)

Put the value of $x = 6y$ in Eq. (ii), we get

$$\Rightarrow 6y + 4 = 4y + 16 \Rightarrow 2y = 12$$

$$\therefore y = 6$$

So, the present age of Y is 6 yr.

Problem Based on Numbers

Let the digit in unit's place be x and that in ten's place be y . Then, the two-digits number is given by $10y + x$. On interchanging the positions of the digits, the digit in unit place becomes y and that in ten's place becomes x , and the number becomes $10x + y$. Formulate the equations and then solve them.

EXAMPLE 12. A number consists of two digits, whose sum is 8. If 18 is added to the number, the digits are reversed. The number is equal to

a. 26 b. 35 c. 53 d. 62

Sol. b. Let the unit's place digit and ten's place digit be x and y , respectively.

$\therefore$ Original number = $10y + x$

Number obtained by reversing the order of digit = $10x + y$

Condition I Sum of digits = 8

$$\Rightarrow x + y = 8 \quad \dots \text{(i)}$$

Condition II $10x + y = 10y + x + 18$

$$\Rightarrow 10x - 10y + y - x = 18 \Rightarrow 9x - 9y = 18$$

$$\Rightarrow x - y = 2 \quad \dots \text{(ii)}$$

On adding Eqs. (i) and (ii), we get

$$x + y + x - y = 8 + 2 \Rightarrow 2x = 10$$

$$\therefore x = 5$$

On putting $x = 5$ in Eq. (i), we get

$$5 + y = 8 \Rightarrow y = 3$$

Hence, Required number = $10y + x = 10 \times 3 + 5 = 35$

Problem Based on Fractions

Let the numerator of the fraction be x and denominator be y , then the fraction is x/y .

Formulate the linear equations on the basis of conditions given and solve them.

EXAMPLE 13. If we add 1 to the numerator and subtract 1 from the denominator, a fraction becomes

1. It also becomes $\frac{1}{2}$, if we only add 1 to the denominator. The fraction is

a. 4/5 b. 3/5 c. 2/5 d. 1/5

Sol. b. Let the numerator and the denominator of the fraction be x and y , respectively.

$$\therefore \text{Fraction} = \frac{x}{y}$$

Condition I When 1 is added to the numerator and 1 is subtract from the denominator, then new fraction = $\frac{x+1}{y-1}$

According to the question,

$$\frac{x+1}{y-1} = 1 \Rightarrow x+1 = y-1 \Rightarrow x-y = -2 \quad \dots \text{(i)}$$

Condition II When 1 is added to denominator, then new fraction = $\frac{x}{y+1}$
 According to the question, $\frac{x}{y+1} = \frac{1}{2} \Rightarrow 2x = y + 1$
 $\Rightarrow 2x - y = 1 \quad \dots(ii)$
 On subtracting Eq. (ii) from Eq. (i), we get
 $-x = -3 \Rightarrow x = 3$
 Put the value of x in Eq. (i), we get
 $3 - y = -2 \Rightarrow y = 5$
 Hence, the fractions is $\frac{x}{y} = \frac{3}{5}$

Problem Based on Distance, Speed and Time

Speed is the distance covered by an object per unit time.

i.e. $\text{Speed} = \frac{\text{Distance}}{\text{Time}}$

If the speed of boat in still water be x km/h and speed of stream be y km/h. Then, the speed of boat downstream = $(x + y)$ km/h and speed of boat upstream = $(x - y)$ km/h

EXAMPLE 14. A motorboat takes 2h to travel a distance of 9 km down the current and it takes 6h to travel the same distance against the current. What is the speed of the boat in still water ?

- a. 3 km/h b. 2 km/h c. 1.5 km/h d. 1 km/h

Sol. a. Let speed of the motorboat in still water be x km/h and speed of the stream be y km/h.
 Then, speed of the motorboat downstream = $(x + y)$ km/h and speed of the boat upstream = $(x - y)$ km/h

Condition I When motorboat goes 9 km downstream, then
 $2 = \frac{9}{x+y} \quad \left[\because \text{Time} = \frac{\text{Distance}}{\text{Speed}} \right]$

$\Rightarrow 2x + 2y = 9 \quad \dots(i)$

Condition II When motorboat goes 9 km upstream, then
 $6 = \frac{9}{x-y} \Rightarrow 6x - 6y = 9 \quad \dots(ii)$

On multiplying Eq. (i) by 3, we get
 $6x + 6y = 27 \quad \dots(iii)$

Now, adding Eqs. (ii) and (iii), we get
 $12x = 36 \Rightarrow x = 3 \text{ km/h}$

Problem Based on Fixed and running Charges

Let the fixed charge of any commodity be ₹ x and rate of running charges be ₹ y , then total cost = $x +$ total running charges.

Formulate the equations and then solve them.

EXAMPLE 15. A railway ticket for a child costs half the full fare but the reservation charge is the same on half tickets as much as on full ticket. One reserved first class ticket for a journey between two stations is ₹ 362, one full and one half reserved first class tickets cost ₹ 554. What is the reservation charge?

- a. ₹ 18 b. ₹ 22 c. ₹ 38 d. ₹ 46

Sol. b. Let full fare and reservation charge be ₹ x and ₹ y , respectively. Then, full ticket = $x + y$

half ticket = $\frac{x}{2} + y$

According to question, $x + y = 362 \quad \dots(i)$

and $(x + y) + \left(\frac{x}{2} + y \right) = 554 = \frac{3}{2}x + 2y = 554 \quad \dots(ii)$

On multiplying Eq. (i) by 2, we get
 $2x + 2y = 724 \quad \dots(iii)$

On subtracting Eq. (ii) from Eq. (iii), we get
 $\frac{x}{2} = 170 \Rightarrow x = 340$

On putting value of x in Eq. (i), we get
 $340 + y = 362 \Rightarrow y = 22$
 Hence, the reservation charge is ₹ 22.

Problems Based on Mensuration

Make use of given formulae to formulate the equation

- Area of rectangle = length $\times$ breadth
 Perimeter of rectangle = $2(\text{length} + \text{Breadth})$
- The sum of angles of a triangle is 180°
- The sum of opposite angles of cyclic quadrilateral is 180°
- In parallelogram, opposite angles are equal.

EXAMPLE 16. The length of a rectangle is 8 cm more than its breadth. If the perimeter of the rectangle of 68 cm, then its length and breadth are

- a. 21 cm, 13 cm b. 14 cm, 23 cm
 c. 19 cm, 20 cm d. 9 cm, 15 cm

Sol. a. Let the length and breadth of a rectangle be x and y , respectively.

Condition I $x = y + 8 \Rightarrow x - y = 8 \quad \dots(i)$

Condition II Perimeter of rectangle = 68 cm
 $\Rightarrow 2[x + y] = 68, \quad x + y = 34 \quad \dots(ii)$

On adding Eq. (i) and (ii), we get
 $2x = 42 \Rightarrow x = 21$

On putting $x = 21$ in Eq. (ii), we get
 $21 + y = 34 \Rightarrow y = 34 - 21 = 13$
 Hence, length of rectangle = 21 cm
 and breadth of rectangle = 13 cm

EXAMPLE 17. The area of a rectangle is decreased by 28 m^2 , if the length is increased by 2 m and the breadth is decreased by 2 m. The area of a rectangle is increased by 33 m^2 , if the length is decreased by 1 m and the breadth is increased by 2 m, then the area of rectangle is

- a. 253 m^2 b. 235 m^2 c. 532 m^2 d. 352 m^2

Sol. a. Let the length and breadth of a rectangle be x and y , respectively.

$\therefore$ Original area of rectangle $= xy \text{ m}^2$

Condition I If length $= (x + 2) \text{ m}$ and breadth $= (y - 2) \text{ m}$

$\therefore$ Area of rectangle $= (x + 2)(y - 2) \text{ m}^2$

According to the question,

$$(x + 2)(y - 2) = xy - 28$$

$$\Rightarrow xy - 2x + 2y - 4 = xy - 28$$

$$\Rightarrow 2x - 2y = 24 \Rightarrow x - y = 12 \quad \dots(i)$$

Condition II If, length $= (x - 1) \text{ m}$ and breadth $= (y + 2) \text{ m}$

$\therefore$ Now the area of rectangle $= (x - 1)(y + 2) \text{ m}^2$

According to the question,

$$(x - 1)(y + 2) = xy + 33$$

$$\Rightarrow xy + 2x - y - 2 = xy + 33$$

$$\Rightarrow 2x - y = 35 \quad \dots(ii)$$

On subtracting Eq. (i) from (ii), we get, $x = 23$

On putting $x = 23$ in Eq. (i), we get,

$$23 - y = 12$$

$$\therefore y = 23 - 12 = 11$$

So, original area of rectangle $= xy = 23 \times 11 = 253 \text{ m}^2$

PRACTICE EXERCISE

1. Find x , if $25x - 19 - [3 - \{4x - 5\}] = 3x - (6x - 5)$.

- (a) $x = 1$ (b) $x = -1$ (c) $x = \frac{1}{2}$ (d) $x = 2$

2. If a number is subtracted from three-fourth of itself, the value so obtained is -130 . Then, what is the number?

- (a) 540 (b) 560 (c) 420 (d) 520

3. Sum of two numbers is 21 and their difference is 11, then the greatest number is

- (a) 5 (b) 16 (c) 9 (d) 10

4. Which of the following equations have $x = 2$ and $y = 1$ as a solution?

I. $2x + 5y = 9$

II. $5x + 3y = 14$

III. $2x + 3y = 7$

IV. $2x - 3y = 1$

Select the correct answer using the codes given below

- (a) I and IV (b) II and III (c) Only I (d) I, III and IV

5. The solution of the pair of equation $\frac{x}{2} + y = 0.8$

and $\frac{7}{x + \frac{y}{2}} = 10$ is

(a) $x = \frac{2}{5}, y = \frac{3}{5}$

(b) $x = \frac{2}{3}, y = 5$

(c) $x = \frac{2}{5}, y = \frac{5}{3}$

(d) $x = \frac{3}{5}, y = \frac{2}{5}$

6. A system of two simultaneous linear equations in two variables has a unique solution if their graphs

- (a) are coincident (b) are parallel
(c) intersect in one point (d) None of these

7. The system of equations $6x + 5y = 11$ and $9x + \frac{15}{2}y = 21$ has

- (a) a unique solution (b) many solution
(c) no solution (d) None of these

8. The sum of two numbers is 2490 and if 6.5% of one number is equal to 8.5% of the other, then numbers are

- (a) 1414, 1076 (b) 1411, 1079
(c) 1412, 1078 (d) None of these

9. Given two linear equations $a_1x + b_1y = c_1$ and $a_2x + b_2y = c_2$, if $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$, then the graph is

- (a) parallel (b) intersection at one point
(c) coincident (d) None of these

10. For what value of k , the following equations will be inconsistent? $4x + 6y = 11$ and $2x + ky = 7$

- (a) $k = -3$ (b) $k = \frac{12}{5}$ (c) $k = 12$ (d) $k = 3$

11. For what value of k , the system of equations has infinitely many solutions $2x - ky = 4$ and $3x + 2y = 6$?

- (a) $\frac{4}{3}$ (b) $-\frac{4}{3}$ (c) $\frac{2}{3}$ (d) $\frac{3}{2}$

12. For what value of α , the system of equation $\alpha x + 3y = \alpha - 3$, $12x + \alpha y = \alpha$ will have a unique solution?

- (a) $\alpha = \pm 6$ (b) $\alpha = 6$ (c) $\alpha \neq \pm 6$ (d) $\alpha = -6$

- 13.** Consider the following statements:
 I. The solution of equation $2x + 3y = 425$ and $3x + 2y = 350$ is a positive integral pair.
 II. for $k = -2$, the equations $x - ky = 2$ and $3x + 6y = -5$ have infinitely many solutions.
 III. $(2, 5)$ and $(-1, 3)$ are solutions of $2x + 5y = 13$
 Select the correct answer using the codes given below:
 (a) Only I (b) Both I and II
 (c) Both II and III (d) All of these
- 14.** A horse and two cows together cost ₹ 680. If a horse costs ₹ 80 more than a cow, then the cost of horse is
 (a) ₹ 170 (b) ₹ 280 (c) ₹ 200 (d) ₹ 220
- 15.** Sunita has 10 paise and 50 paise coins in her purse. If the total number of coins is 17 and their total value is ₹ 4.50, then number of 10 paise coins is
 (a) 9 (b) 7 (c) 10 (d) 5
- 16.** The solution of the equation $\frac{3x - y + 1}{3} = \frac{2x + y + 2}{5} = \frac{3x + 2y + 1}{6}$ is given by which one of the following?
 (a) $x = 2, y = 1$ (b) $x = 1, y = 1$
 (c) $x = -1, y = -1$ (d) $x = 1, y = 2$
- 17.** The line $3x - 5y = -10$ cuts Y-axis at
 (a) $(0, 2)$ (b) $(0, 1)$ (c) $(0, 3)$ (d) $(0, 4)$
- 18.** The equations $px + q = 0$ and $rx + s = 0$ are consistent, if
 (a) $ps = qr$ (b) $ps + qr = 0$ (c) $pq - rs = 0$ (d) $pq + rs = 0$
- 19.** A streamer goes downstream and covers the distance between two ports in 4 h while it covers the same distance upstream in 5 h. If the speed of the stream is 2 km/h, then the speed of the streamer in still water is
 (a) 20 km/h (b) 19 km/h (c) 18 km/h (d) 19.5 km/h
- 20.** The solution of the system of linear equations $0.4x + 0.3y = 1.7$ and $0.7x - 0.2y = 0.8$ is
 (a) $x = 3, y = 2$ (b) $x = 2, y = -3$
 (c) $x = 2, y = 3$ (d) None of these
- 21.** If $\sqrt{2}x - \sqrt{3}y = 0$ and $\sqrt{7}x + \sqrt{2}y = 0$, then find the value of $x + y$.
 (a) 1 (b) 2 (c) 3 (d) 0
- 22.** A man starts his job with a certain monthly salary and earns a fixed increment every year. If his salary was ₹ 1500 after 4 yr of services and ₹ 1800 after 10 yr of service. What was his starting salary?
 (a) ₹ 1300 (b) ₹ 1200 (c) ₹ 50 (d) ₹ 1100
- 23.** If $\frac{44}{x+y} + \frac{30}{x-y} = 10, \frac{55}{x+y} + \frac{40}{x-y} = 13$, then find the value of x and y .
 (a) $x = 2, y = 8$ (b) $x = 8, y = 3$
 (c) $x = 8, y = 2$ (d) $x = 3, y = 8$
- 24.** Consider the following sets of equations
 I. $2x - y = 0$ and $6x - 3y = 0$
 II. $3x - 4y = 0$ and $12x - 20y = 0$ Then,
 (a) Both sets I and II possess unique solution
 (b) Set I possesses unique solution and set II has infinitely many solutions
 (c) Set I possesses infinitely many solutions and set II possess unique solution
 (d) None of the sets I and II possesses a unique solution
- 25.** A fraction becomes $\frac{4}{5}$, if 1 is added to both numerator and denominator. If however 5 is subtracted from both numerator and denominator the fraction becomes $\frac{1}{2}$. Then, the fractions is
 (a) $\frac{7}{9}$ (b) $\frac{9}{7}$ (c) $\frac{3}{5}$ (d) $\frac{4}{3}$
- 26.** For what value of k , the following system of equations $3x + 4y = 6$ and $6x + 8y = k$ represents coincident lines?
 (a) 12 (b) 11 (c) 13 (d) 10
- 27.** The area of a rectangle remains the same if the length is increased by 7 m and the breadth is decreased by 3 m. The area remains unaffected if the length is decreased by 7 m and the breadth is increased by 5 m, then area of rectangle is
 (a) 280 m^2 (b) 320 m^2 (c) 420 m^2 (d) 400 m^2
- 28.** The value of x in the solution of the equation $2^{x+y} = 2^{x-y} = \sqrt{8}$ is
 (a) 0 (b) $\frac{3}{2}$ (c) $\frac{1}{4}$ (d) $\frac{1}{8}$
- 29.** A and B each have a certain number of mangoes. A says to B : "If you give 30 of your mangoes, I will have twice as many as left with you" B replies "If you give me 10, I will have thrice as many as left with you". How many mangoes did A has?
 (a) 41 (b) 62 (c) 34 (d) 32
- 30.** There are two examination rooms A and B. If 10 candidates are sent from room A to room B, the number of students in each room is the same. If 20 candidates are sent from B to A, the number of students in A is double the number of students in B. Then, number of students in room B is
 (a) 40 (b) 100 (c) 80 (d) 60

- 31.** A train started from a station with a certain number of passengers. At the first halt, $\frac{1}{3}$ rd of its passengers got down and 120 passengers got in. At the second halt, half of the passengers got down and 100 persons got in. Then, the train left for its destination with 240 passengers. How many passengers were there in the train when it started ?
 (a) 540 (b) 480 (c) 360 (d) 240
- 32.** The system of equations $x + 2y = 3$ and $3x + 6y = 9$ has
 (a) unique solutions (b) no solution
 (c) infinitely many solutions (d) finite number of solutions
- 33.** The sum of digits of a two-digit number is 8 and the difference between the number and that formed by reversing the digits is 18. What is the difference between the digits of the number?
 (a) 1 (b) 2 (c) 3 (d) 4
- 34.** If $(x, y) = (4, 1)$ is the solution of the pair of linear equations $mx + y = 2x + ny = 5$, then what is $m + n$ equal to?
 (a) -2 (b) -1 (c) 1 (d) 1
- 35.** If $2^a + 3^b = 17$ and $2^{a+2} - 3^{b+1} = 5$, then
 (a) $a=2, b=3$ (b) $a=-2, b=3$
 (c) $a=2, b=-3$ (d) $a=3, b=2$
- 36.** The sum of two numbers is 80. If the larger number exceeds four times the smaller by 5, what is the smaller number ?
 (a) 5 (b) 15 (c) 20 (d) 25
- 37.** If $\frac{2}{x} + \frac{3}{y} = \frac{9}{xy}$ and $\frac{4}{x} + \frac{9}{y} = \frac{21}{xy}$, where $x, y \neq 0$ and $y \neq 0$, then what is the value of $x + y$?
 (a) 2 (b) 3 (c) 4 (d) 8
- 38.** Under what condition do the equation $kx - y = 2$ and $6x - 2y = 3$ have a unique solution?
 (a) $k = 3$ (b) $k \neq 3$ (c) $k = 0$ (d) $k \neq 0$
- 39.** Let there be three simultaneous linear equations in two unknowns, which are non-parallel and non-collinear. What can be the number of solutions (if they do exist)?
 (a) One or infinite (b) Only one
 (c) Exactly two (d) Exactly three
- 40.** A number consists of two digits, whose sum is 10. If 18 is subtracted from the number, digits of the number are reversed. What is the product?
 (a) 15 (b) 18
 (c) 24 (d) 32
- 41.** What is the value of k for which the system of equations $x + 2y - 3 = 0$ and $5x + ky + 7 = 0$ has no solution?
 (a) $-\frac{3}{14}$ (b) $-\frac{14}{3}$ (c) $\frac{1}{10}$ (d) 10
- 42.** What is the solution of the equation $x - y = 0.9$ and $11(x + y)^{-1} = 2$?
 (a) $x = 3.2$ and $y = 2.3$ (b) $x = 1$ and $y = 0.1$
 (c) $x = 2$ and $y = 1.1$ (d) $x = 1.2$ and $y = 0.3$
- 43.** Pooja started her job with certain monthly salary and gets a fixed increment every year. If her salary was ₹ 4200 after 3 yr and ₹ 6800 after 8 yr of service, then what are her initial salary and the annual increment, respectively ?
 (a) ₹ 2640, ₹ 320 (b) ₹ 2460, ₹ 320
 (c) ₹ 2460, ₹ 520 (d) ₹ 2640, ₹ 520
- 44.** A person bought 5 tickets from the station P to a station Q and 10 tickets from the station P to a station R . He paid ₹ 350. If the sum of fare of a ticket from P to Q and a ticket from P to R is ₹ 42, then what is the fare from P to Q ?
 (a) ₹ 12 (b) ₹ 14
 (c) ₹ 16 (d) ₹ 18
- 45.** The Community Relief fund receives a large donation of \$ 2800. The foundation agrees to spend the money on \$ 20 school bags, \$ 25 sweaters, \$ 5 books. They want to buy 200 items and send them to schools in earthquake-hit areas. They must order as many books as school bages and sweaters combined.
 How many of each item should they order?
 (a) (40, 60, 100) (b) (20, 30, 80)
 (c) (50, 100, 60) (d) 40, 80, 25)
- Directions** (Q. Nos. 46-47) *Some money is divided among Rajesh, Sonal and Chetan in such a way that 2 times share of Rajesh, 3 times share of Sonal and 5 times share of Chetan are all equal.*
Now, answer the following questions based on above information.
- 46.** If the sum of 6 times the share with Rajesh and 6 times the share with Sonal is ₹ 150, then find the share of Chetan?
 (a) ₹ 5 (b) ₹ 6 (c) ₹ 10 (d) ₹ 12
- 47.** If the sum of shares of Rajesh, Chetan and Sonal is ₹ 155, then find the share of Rajesh.
 (a) ₹ 75 (b) ₹ 35 (c) ₹ 50 (d) ₹ 25

PREVIOUS YEARS' QUESTIONS

- 48.** The graphs of $ax + by = c$, $dx + ey = f$ will be
 I. Parallel, if the system has no solution.
 II. Coincident, if the system has finite number of solutions.
 III. Intersecting, if the system has only one solutions
 Which of the statement(s) given above is/are correct? **2012 I**
 (a) Both I and II (b) Both II and III
 (c) Both I and III (d) All of these
- 49.** The sum of two numbers is 20 and their product is 75. What is the sum of their reciprocals? **2012 I**
 (a) $\frac{1}{15}$ (b) $\frac{1}{5}$ (c) $\frac{4}{15}$ (d) $\frac{7}{15}$
- 50.** If $3^{x+y} = 81$ and $81^{x-y} = 3$, then what is the value of x ? **2012 I**
 (a) $\frac{17}{16}$ (b) $\frac{17}{8}$ (c) $\frac{17}{4}$ (d) $\frac{15}{4}$
- 51.** If $\frac{a}{b} - \frac{b}{a} = \frac{x}{y}$ and $\frac{a}{b} + \frac{b}{a} = x - y$, then what is the value of x ? **2013 I**
 (a) $\frac{a+b}{a}$ (b) $\frac{a+b}{b}$ (c) $\frac{a-b}{a}$ (d) None of these
- 52.** The system of equations $3x + y - 4 = 0$ and $6x + 2y - 8 = 0$ has **2013 I**
 (a) a unique solution $x = 1$, $y = 1$
 (b) a unique solution $x = 0$, $y = 4$
 (c) no solution
 (d) infinite solutions
- 53.** The sum of two numbers is 7 and the sum of their squares is 25. The product of the two numbers is **2013 I**
 (a) 6 (b) 10 (c) 12 (d) 15

Directions (Q. Nos. 54-55) A number consists of two digits whose sum is 10. If the digits of the number are reversed, then the number is decreased by 36.

- 54.** Which of the following is/are correct?
 I. The number is divisible by a composite number.
 II. The number is a multiple of a prime number.
 Select the correct answer using the codes given below **2013 I**
 (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II
- 55.** What is the products of the two digits? **2013 I**
 (a) 21 (b) 24 (c) 36 (d) 42
- 56.** Ten chairs and six tables together cost ₹ 6200, three chairs and two tables together cost ₹ 1900. The cost of 4 chairs and 5 tables is **2013 I**
 (a) ₹ 3000 (b) ₹ 3300 (c) ₹ 3500 (d) ₹ 3800

- 57.** A bus starts with some passengers. At the first stop, one-fifth of the passengers gets down and 40 passengers get in. At the second stop, half of the passengers gets down and 30 get in. The number of passengers now is 70. The number of passengers with which the bus started was **2013 II**
 (a) 40 (b) 50 (c) 60 (d) 70
- 58.** If $x + y - 7 = 0$ and $3x + y - 13 = 0$, then what is $4x^2 + y^2 + 4xy$ equal to? **2013 II**
 (a) 75 (b) 85 (c) 91 (d) 100
- 59.** If $\frac{x}{2} + \frac{y}{3} = 4$ and $\frac{2}{x} + \frac{3}{y} = 1$, then what is $x + y$ equal to? **2013 II**
 (a) 11 (b) 10 (c) 9 (d) 8
- 60.** If $x + y = 5$, $y + z = 10$ and $z + x = 15$, then which one of the following is correct? **2014 I**
 (a) $z > x > y$ (b) $z > y > x$ (c) $x > y > z$ (d) $x > z > y$
- 61.** The present age of Ravi's father is 4 times Ravi's present age. 5 yr back, Ravi's father was seven times as old as Ravi was at that time. What is the present age of Ravi's father? **2014 I**
 (a) 84 yr (b) 70 yr (c) 40 yr (d) 35 yr
- 62.** The sum of two positive number x and y is 2.5 times their difference. If the product of numbers is 84, then what is the sum of those two numbers? **2014 I**
 (a) 26 (b) 24 (c) 22 (d) 20
- 63.** Two chairs and one table cost ₹ 700 and one chair and Two tables cost ₹ 800. If cost m tables and m chairs is ₹ 30000, then what is m equal to? **2014 I**
 (a) 60 (b) 55 (c) 50 (d) 45
- 64.** A certain number of two digits is three times the sum of its digits. If 45 is added to the number, then the digits will be reversed. What is the sum of the squares of the two digits of the number? **2014 II**
 (a) 41 (b) 45 (c) 53 (d) 64
- 65.** A student was asked to multiply a number by 25. He instead multiplied the number by 52 and got the answer 324 more than the correct answer. The number to be multiplied was **2015 I**
 (a) 12 (b) 15 (c) 25 (d) 32
- 66.** A tin of oil was $\frac{4}{5}$ full. When 6 bottles of oil were taken out from this tin and 4 bottles of oil were poured into it, it was $\frac{3}{4}$ full. Oil of how many bottles can the tin contain? (All bottles are of equal volume) **2015 II**
 (a) 35 (b) 40 (c) 45 (d) 50

- 67.** A number consists of two digits, whose sum is 7. If the digits are reversed, the number is increased by 27. The product of digits of the number is ✎ 2015 II
- (a) 6 (b) 8 (c) 10 (d) 12
- 68.** If $\frac{p}{x} + \frac{q}{y} = m$ and $\frac{q}{x} + \frac{p}{y} = n$, then what is $\frac{x}{y}$ equal to? ✎ 2016 I
- (a) $\frac{np + mq}{mp + nq}$ (b) $\frac{np + mq}{mp - nq}$
 (c) $\frac{np - mq}{mp - nq}$ (d) $\frac{np - mq}{mp + nq}$
- 69.** The value of k , for which the system of equations $3x - ky - 20 = 0$ and $6x - 10y + 40 = 0$ has no solution, is ✎ 2016 I
- (a) 10 (b) 6 (c) 5 (d) 3
- 70.** There are three brothers. The sums of ages of two of them at a time are 4 yr, 6 yr and 8 yr. The age difference between the eldest and the youngest is ✎ 2016 I
- (a) 3 yr (b) 4 yr (c) 5 yr (d) 6 yr
- 71.** The annual incomes of two persons are in the ratio 9 : 7 and their expenses are in the ratio 4 : 3. If each of them saves ₹ 2000 per year, then what is the difference in their annual incomes? ✎ 2016 I
- (a) ₹ 4000 (b) ₹ 4500 (c) ₹ 5000 (d) ₹ 5500
- 72.** Let a two digits number be k times the sum of its digits. If the number formed by interchanging the digits is m times the sum of the digits, then the value of m is ✎ 2016 I
- (a) $9 - k$ (b) $10 - k$ (c) $11 - k$ (d) $k - 1$

ANSWERS

1	a	2	d	3	b	4	d	5	a	6	c	7	c	8	b	9	b	10	d
11	b	12	c	13	a	14	b	15	c	16	b	17	a	18	a	19	c	20	c
21	d	22	a	23	b	24	c	25	a	26	a	27	c	28	b	29	c	30	c
31	d	32	c	33	b	34	a	35	d	36	b	37	c	38	b	39	b	40	c
41	d	42	a	43	d	44	b	45	a	46	b	47	a	48	c	49	c	50	b
51	d	52	d	53	c	54	b	55	a	56	a	57	b	58	d	59	b	60	a
61	c	62	d	63	a	64	c	65	a	66	b	67	c	68	c	69	c	70	b
71	a	72	c																

HINTS AND SOLUTIONS

- 1. (a)** We have, $25x - 19 - [3 - \{4x - 5\}]$
 $= 3x - (6x - 5)$
 $\Rightarrow 25x - 19 - [3 - 4x + 5]$
 $= 3x - 6x + 5$
 $\Rightarrow 25x - 19 + 4x - 8 = -3x + 5$
 $\Rightarrow 29x + 3x = 5 + 27$
 $\Rightarrow 32x = 32 \Rightarrow x = \frac{32}{32} = 1 \Rightarrow x = 1$
 Hence, the value of x is 1.
- 2. (d)** Let the number be x .
 According to the question,
 $\frac{3x}{4} - x = -130 \Rightarrow \frac{3x - 4x}{4} = -130$
 $\Rightarrow \frac{-x}{4} = -130 \Rightarrow x = 520$
 Hence, the number is 520.

- 3. (b)** Let numbers be x and y , respectively.
 Then by given condition,
 $x + y = 21 \quad \dots(i)$
 $x - y = 11 \quad \dots(ii)$
 On adding Eqs. (i) and (ii), we get
 $2x = 32$
 $x = \frac{32}{2} = 16$
 On putting $y = 16$ in Eq. (i), we get
 $x + 16 = 21$
 $\therefore x = 5$
 Hence, the greatest number is 16.
- 4. (d)** Put $x = 2$ and $y = 1$ in each equation
 I. $2x + 5y = 9 \Rightarrow 2(2) + 5(1) = 9$
 $9 = 9$, it is true.
 II. $5x + 3y = 14$
 $\Rightarrow 5(2) + 3(1) = 14$

- $\Rightarrow 13 = 14$, it is false.
- III. $2x + 3y = 7 \Rightarrow 2(2) + 3(1) = 7$
 $7 = 7$, it is true.
- IV. $2x - 3y = 1 \Rightarrow 2(2) - 3(1) = 1$
 $1 = 1$, it is true.
 So, $x = 2$ and $y = 1$
 is a solution of I, III and IV.
- 5. (a)** Given equations are,
 $x + 2y = 16 \quad \dots(i)$
 and $x + \frac{y}{2} = \frac{7}{10} \quad \dots(ii)$
 On multiplying Eq. (i) by 10 and Eq. (ii) by 10, we get
 or $10x + 20y = 160 \quad \dots(iii)$
 and $10x + 5y = 7 \quad \dots(iv)$

On subtracting, Eq. (iv) from Eq. (iii), we get

$$(10x + 20y) - (10x + 5y) = 16 - 7$$

$$\Rightarrow 15y = 9 \Rightarrow y = \frac{9}{15} \Rightarrow y = \frac{3}{5}$$

On putting the value of y in Eq. (i), we get

$$x + 2y = \frac{8}{5}, \quad x + \frac{6}{5} = \frac{8}{5}$$

$$\Rightarrow x = \frac{8}{5} - \frac{6}{5} = \frac{2}{5}$$

6. (c) A pair of linear equations in two variables has a unique solution if their graphs intersect in one point.

7. (c) Here, $\frac{a_1}{a_2} = \frac{6}{9} = \frac{2}{3}, \frac{b_1}{b_2} = \frac{5}{15/2} = \frac{2}{3}$

and $\frac{c_1}{c_2} = \frac{11}{21} \Rightarrow \frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$

So, the system has no solution.

8. (b) Let the numbers be x and $2490 - x$.
6.5% of $x = \frac{65}{100} \times x = \frac{13x}{200}$

$$8.5\% \text{ of } (2490 - x) = \frac{85}{100} (2490 - x)$$

$$= \frac{17}{200} (2490 - x)$$

By given condition, $\frac{13x}{200} = \frac{17(2490 - x)}{200}$

$$\Rightarrow 13x = 17(2490 - x)$$

$$\Rightarrow 13x + 17x = 42330$$

$$\Rightarrow x = \frac{42330}{30} = 1411$$

$$\therefore \text{Second number} = (2490 - x)$$

$$= 2490 - 1411 = 1079$$

9. (b) As this is the case of unique solution $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$, so the graph of the equations

will intersect at one point.

10. (d) Given equations are,

$$4x + 6y = 11 \text{ and } 2x + ky = 7$$

Here, $a_1 = 4, b_1 = 6, c_1 = 11$

and $a_2 = 2, b_2 = k, c_2 = 7$

Here, $\frac{a_1}{a_2} = \frac{4}{2}, \frac{b_1}{b_2} = \frac{6}{k} \text{ and } \frac{c_1}{c_2} = \frac{11}{7}$

Since, system is inconsistent, so

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2} \Rightarrow \frac{4}{2} = \frac{6}{k}$$

$$\Rightarrow 4k = 12 \Rightarrow k = 3$$

Hence, the value of k is 3.

11. (b) Given equations are,

$$2x - ky = 4 \text{ and } 3x + 2y = 6$$

Here, $a_1 = 2, b_1 = -k, c_1 = 4$

and $a_2 = 3, b_2 = 2, c_2 = 6$

Now, $\frac{a_1}{a_2} = \frac{2}{3}, \frac{b_1}{b_2} = \frac{-k}{2} \text{ and } \frac{c_1}{c_2} = \frac{4}{6}$

Since, the system has infinitely many solutions

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}, \quad \frac{2}{3} = \frac{-k}{2} = \frac{2}{3}$$

$$\Rightarrow -3k = 4 \Rightarrow k = \frac{-4}{3}$$

12. (c) Given equations are,

$$\alpha x + 3y = \alpha - 3 \text{ and } 12x + \alpha y = \alpha$$

Here, $a_1 = \alpha, b_1 = 3, c_1 = \alpha - 3$

$$a_2 = 12, b_2 = \alpha, c_2 = \alpha$$

Since, system has unique solution,

So, $\frac{a_1}{a_2} \neq \frac{b_1}{b_2} \Rightarrow \frac{\alpha}{12} \neq \frac{3}{\alpha}$

$$\Rightarrow \alpha^2 \neq 36 \Rightarrow \alpha \neq \pm 6$$

13. (a) I. $2x + 3y = 425, 3x + 2y = 350$

Solving both the equations, we get
 $x = 40$ and $y = 115$.

II. When $k = -2$

$$\frac{a_1}{a_2} = \frac{1}{3}, \frac{b_1}{b_2} = \frac{2}{6} = \frac{1}{3}, \frac{c_1}{c_2} = \frac{-2}{5}$$

$$\therefore \frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$

Hence, the given equations have no solution.

III. (2, 5) is not a solution of
 $2x + 5y = 13$ as

$$2(2) + 5(5) = 4 + 25 = 29 \neq 13.$$

Hence, only I is correct.

14. (b) Let cost of one horse and one cow be ₹ x and ₹ y , respectively.

Condition I $x + 2y = 680$... (i)

Condition II $x = y + 80$

$$\Rightarrow x - y = 80 \text{ ... (ii)}$$

On subtracting Eq. (ii) from Eq. (i), we get

$$(x + 2y) - (x - y) = 680 - 80$$

$$\Rightarrow 3y = 600$$

$$\therefore y = 200$$

On putting $y = 200$ in Eq. (i), we get

$$x + 2 \times 200 = 680$$

$$\Rightarrow x + 400 = 680 \Rightarrow x = 680 - 400$$

$$\therefore x = 280$$

Hence, the cost of one horse is ₹ 280.

15. (c) Let number of 10 paise coins be x and number of 50 paise coins be y .

According to the questions,

Then, $x + y = 17$... (i)

and $10x + 50y = 450$... (ii)

From Eq. (ii), $x + 5y = 45$... (iii)

On subtracting Eq. (i) from Eq. (iii), we get

$$(x + 5y) - (x + y) = 45 - 17$$

$$\Rightarrow 4y = 28 \Rightarrow y = \frac{28}{4} = 7$$

$\therefore$ Number of 10 paise coins

$$= x = 17 - y = 17 - 7 = 10$$

16. (b) Given,

$$\frac{3x - y + 1}{3} = \frac{2x + y + 2}{5} = \frac{3x + 2y + 1}{6}$$

(I) (II) (III)

Taking Ist and IInd terms,

$$5(3x - y + 1) = 3(2x + y + 2)$$

$$\Rightarrow 9x - 5y = 1 \text{ ... (i)}$$

Taking IInd and IIIrd terms

$$6(2x + y + 2) = 5(3x + 2y + 1)$$

$$\Rightarrow 3x + 4y = 7 \text{ ... (ii)}$$

On solving Eqs. (i) and (ii), we get $y = 1$ and $x = 1$

17. (a) Given, $3x - 5y = -10$

At Y-axis, $x = 0$

$$\Rightarrow 0 - 5y = -10 \Rightarrow y = \frac{10}{5} = 2$$

So, the point on Y-axis is (0, 2).

18. (a) Given, $px + q = 0$ and $rx + s = 0$

$$\Rightarrow x = \frac{-q}{p} \text{ and } x = \frac{-s}{r}$$

So, $\frac{-q}{p} = \frac{-s}{r} \Rightarrow ps = qr$

19. (c) Let the speed of the streamer in still water = x km/h

then speed of streamer downstream

$$= (x + 2) \text{ km/h}$$

speed of streamer upstream

$$= (x - 2) \text{ km/h}$$

Distance travelled by streamer

downstream in 4h

$$= 4(x + 2) \text{ km}$$

$$[\because \text{distance} = \text{time} \times \text{speed}]$$

and distance travelled by streamer upstream in 5 h = $5(x - 2)$

So, $4(x + 2) = 5(x - 2)$

$$\Rightarrow 4x + 8 = 5x - 10$$

Hence, $x = 18$ km/h is the speed of streamer in still water.

20. (c) Given, $0.4x + 0.3y = 1.7$ and $0.7x - 0.2y = 0.8$

On multiplying above equation by 10, we get

$$4x + 3y = 17 \text{ ... (i)}$$

and $7x - 2y = 8 \text{ ... (ii)}$

On multiplying Eq. (i) by 2 and Eq. (ii) by 3, we get

$$8x + 6y = 34 \quad \dots(\text{iii})$$

$$\text{and } 21x - 6y = 24 \quad \dots(\text{iv})$$

On adding Eqs. (iii) and (iv), we get

$$(8x + 6y) + (21x - 6y) = 34 + 24$$

$$\Rightarrow 29x = 58$$

$$\therefore x = 2$$

On putting $x = 2$ in Eq. (i), we get

$$8 + 3y = 17 \Rightarrow y = 3$$

Hence, $x = 2$ and $y = 3$ is the required solution.

21. (d) Given, $\sqrt{2}x - \sqrt{3}y = 0$

and $\sqrt{7}x + \sqrt{2}y = 0$

Here, $a_1 = \sqrt{2}$, $b_1 = -\sqrt{3}$

$$a_2 = \sqrt{7} \text{ and } b_2 = \sqrt{2}$$

As the equations are homogeneous equations and also

$$\frac{a_1}{a_2} \neq \frac{b_1}{b_2} \text{ So, equation has one solution,}$$

$$\therefore x = y = 0 \Rightarrow x + y = 0$$

Hence, the value of $x + y$ is zero.

22. (a) Let the starting salary be ₹ x and annual increment be ₹ y .

By given condition, $x + 4y = 1500 \dots(\text{i})$

and $x + 10y = 1800 \dots(\text{ii})$

On subtracting Eq. (i) from Eq. (ii), we get

$$6y = 300 \Rightarrow y = 50$$

On putting $y = 50$ in Eq. (i), we get

$$x + 200 = 1500$$

$$\therefore x = ₹ 1300$$

23. (b) Put $\frac{1}{x+y} = A$ and $\frac{1}{x-y} = B$

$$\Rightarrow 44A + 30B = 10 \dots(\text{i})$$

$$\text{and } 55A + 40B = 13 \dots(\text{ii})$$

On multiplying Eq. (i) by 4 and Eq. (ii) by 3 and subtracting, we get

$$176A + 120B = 40$$

$$165A + 120B = 39$$

$$\frac{11A}{11A} = \frac{1}{1}$$

$$\Rightarrow A = 1/11 \Rightarrow x + y = 11 \dots(\text{iii})$$

On putting the value of A in Eq. (i), we get

$$4 + 30B = 10 \Rightarrow 30B = 6$$

$$\therefore B = \frac{1}{5}, \quad x - y = 5 \dots(\text{iv})$$

On adding Eqs. (iii) and (iv), we get

$$2x = 16$$

$$\therefore x = 8$$

On putting $x = 8$ in Eq. (iii), we get

$$8 + y = 11 \Rightarrow y = 3$$

Hence, $x = 8$ and $y = 3$.

24. (c) Set I $2x - y = 0$ and $6x - 3y = 0$

Here, $a_1 = 2$, $b_1 = -1$

and $a_2 = 6$, $b_2 = -3$

$$\Rightarrow \frac{a_1}{a_2} = \frac{2}{6} = \frac{1}{3} \Rightarrow \frac{b_1}{b_2} = \frac{-1}{-3} = \frac{1}{3}$$

$$\therefore \frac{a_1}{a_2} = \frac{b_1}{b_2}$$

So, system of equations have infinitely many solutions.

Set II $3x - 4y = 0$ and $12x - 20y = 0$

Here, $a_1 = 3$, $b_1 = -4$

and $a_2 = 12$, $b_2 = -20$

$$\therefore \frac{a_1}{a_2} = \frac{3}{12} = \frac{1}{4} \Rightarrow \frac{b_1}{b_2} = \frac{-4}{-20} = \frac{1}{5}$$

Here $\frac{1}{4} \neq \frac{1}{5}$

So, system of equations has unique solution.

25. (a) Let the fraction be $\frac{x}{y}$

Condition I $\frac{x+1}{y+1} = \frac{4}{5}$

$$\Rightarrow 5(x+1) = 4(y+1)$$

$$\Rightarrow 5x - 4y = -1 \dots(\text{i})$$

Condition II $\frac{x-5}{y-5} = \frac{1}{2}$

$$\Rightarrow 2(x-5) = (y-5)$$

$$\Rightarrow 2x - y = 5 \dots(\text{ii})$$

On multiplying Eq. (ii) by 4 and subtracting from Eq. (i), we get

$$5x - 4y = -1$$

$$8x - 4y = 20$$

$$\frac{-}{-} \quad \frac{+}{+} \quad \frac{-}{-}$$

$$-3x = -21$$

$$\therefore x = 7$$

Put the value of x in Eq. (i),

$$\Rightarrow 35 - 4y = -1 \Rightarrow y = 9$$

Hence, the fraction is $\frac{7}{9}$

26. (a) Given equations are,

$$3x + 4y = 6 \dots(\text{i})$$

and $6x + 8y = k \dots(\text{ii})$

Here, $a_1 = 3$, $b_1 = 4$, $c_1 = 6$

$$a_2 = 6$$
, $b_2 = 8$, $c_2 = k$

Since the system of equations represents coincident line

$$\therefore \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} \text{ i.e. } \frac{3}{6} = \frac{4}{8} = \frac{6}{k}$$

$$\Rightarrow 4k = 6 \times 8 \Rightarrow k = \frac{48}{4} = 12$$

Hence, the value of k is 12.

27. (c) Let the length of rectangle be x m and breadth be y m.

$$\therefore \text{Original area of rectangle} = xy$$

Case I Length = $(x + 7)$ m and breadth = $(y - 3)$ m

$$\therefore \text{New area of rectangle} = (x + 7)(y - 3)$$

Given, $xy = xy - 3x + 7y - 21$

$$\Rightarrow 3x - 7y = -21 \dots(\text{i})$$

Case II Length = $(x - 7)$

and Breadth = $(y + 5)$

$$\therefore \text{New area of rectangle} = (x - 7)(y + 5)$$

Given, $xy = xy + 5x - 7y - 35$

$$\Rightarrow 5x - 7y = 35 \dots(\text{ii})$$

On subtracting Eq. (ii) from Eq. (i), we get

$$3x - 7y = -21$$

$$5x - 7y = 35$$

$$\frac{-}{-} \quad \frac{+}{+} \quad \frac{-}{-}$$

$$-2x = -56$$

$$\therefore x = 28 \text{ m}$$

On putting $x = 28$ in Eq. (ii), we get

$$5 \times 28 - 7y = 35 \Rightarrow -7y = 35 - 140$$

$$\therefore y = 15$$

$$\therefore \text{Area of rectangle} = xy = 15 \times 28 = 420 \text{ m}^2$$

28. (b) Given, $2^{x+y} = \sqrt{8}$, $2^{x-y} = \sqrt{8}$

or $2^{x+y} = 2\sqrt{2}$, $2^{x-y} = 2\sqrt{2}$

or $2^{x+y} = 2^{3/2}$, $2^{x-y} = 2^{3/2}$

On comparing both sides, we get

$$x + y = \frac{3}{2} \dots(\text{i})$$

$$\text{and } x - y = \frac{3}{2} \dots(\text{ii})$$

On adding Eqs. (i) and (ii), we get $2x = 3$

$$\Rightarrow x = \frac{3}{2}$$

Hence, the value of x is $\frac{3}{2}$.

29. (c) Let A has x mangoes and B has y mangoes.

Case I $x + 30 = 2(y - 30)$

$$\Rightarrow x + 30 = 2y - 60$$

$$x - 2y = -90 \dots(\text{i})$$

Case II $(y + 10) = 3(x - 10)$

$$y = 3x - 30 - 10$$

$$3x - y = 40 \dots(\text{ii})$$

On solving Eqs. (i) and (ii), we get

$$x = 34, y = 62$$

So, A has 34 mangoes.

30. (c) Let number of students in room A be x and in room B be y .

So, by given condition

$$x - 10 = y + 10 \Rightarrow x - y = 20 \dots (i)$$

$$\text{and } x + 20 = 2(y - 20)$$

$$x - 2y = -60 \dots (ii)$$

On subtracting Eq. (ii) from Eq. (i), we get

$$(x - y) - (x - 2y) = 20 + 60 \Rightarrow y = 80$$

Hence, number of students in room B is 80.

31. (d) Let the number of passengers in the starting be x .

Number of passengers after first halt

$$= \left[x - \frac{x}{3} \right] + 120 = \frac{2}{3}x + 120$$

and number of passengers after second halt

$$= \frac{1}{2} \left[\frac{2}{3}x + 120 \right] + 100$$

But number of passengers after second halt = 240

$$\therefore \frac{1}{2} \left[\frac{2}{3}x + 120 \right] + 100 = 240$$

$$\Rightarrow \frac{2}{3}x + 120 = 280 \Rightarrow \frac{2}{3}x = 160$$

$$\Rightarrow x = 240$$

32. (c) Given system of equations are

$$x + 2y = 3 \text{ and } 3x + 6y = 9$$

$$\text{Here, } \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} = \frac{1}{3}$$

Hence, given system of the equation has infinitely many solutions.

33. (b) Let x be the ten's digit and y be the unit's digit of two-digit number.

$$\text{By given condition, } x + y = 8$$

$$\text{and } (10x + y) - (10y + x) = 18 \dots (i)$$

$$\Rightarrow 9x - 9y = 18 \Rightarrow x - y = 2 \dots (ii)$$

On adding Eqs. (i) and (ii), we get

$$2x = 10 \Rightarrow x = 5$$

On putting $x = 5$ in Eq. (i), we get

$$\Rightarrow y = 8 - 5 = 3 \Rightarrow x = 5 \text{ and } y = 3$$

$\therefore$ Required difference of digits,

$$x - y = 5 - 3 = 2$$

34. (a) Given, $(x, y) = (4, 1)$ and

$$mx + y = 2x + ny = 5$$

On putting $x = 4$ and $y = 1$, we get

$$\therefore m(4) + 1 = 2 \times 4 + n = 5$$

$$\therefore 4m + 1 = 5 \text{ and } 8 + n = 5$$

$$\Rightarrow n = -3 \text{ and } m = 1$$

$$\therefore m + n = 1 - 3 = -2$$

35. (d) Given, $2^a + 3^b = 17$

$$\text{and } 2^{a+2} - 3^{b+1} = 5$$

$$\Rightarrow 2^a \times 2^2 - 3^b \times 3^1 = 5$$

$$\Rightarrow 4 \cdot 2^a - 3 \cdot 3^b = 5$$

$$\text{Let } 2^a = x \text{ and } 3^b = y$$

$$\text{Then, } x + y = 17 \dots (i)$$

$$4x - 3y = 5 \dots (ii)$$

On multiplying Eq. (i) by 3 and adding to Eq. (ii), we get

$$3x + 3y = 51$$

$$4x - 3y = 5$$

$$7x = 56$$

$$\therefore x = 8$$

On putting the value of x in Eq. (i), we get

$$8 + y = 17 \Rightarrow y = 9$$

$$\text{Now, } 2^a = x \Rightarrow 2^a = 8 = (2)^3$$

$$\therefore a = 3 \text{ and } 3^b = y = 9 \Rightarrow 3^b = 3^2$$

$$\therefore b = 2$$

$$\text{Hence, } a = 3 \text{ and } b = 2$$

36. (b) Let smaller number be x .

$$\text{and larger number} = 80 - x$$

$$\text{By given condition, } 80 - x = 4x + 5$$

$$\Rightarrow 5x = 75 \Rightarrow x = 15$$

Hence, the smaller number is 15.

37. (c) Given equations are,

$$\frac{2}{x} + \frac{3}{y} = \frac{9}{xy} \Rightarrow 2y + 3x = 9 \dots (i)$$

$$\text{and } \frac{4}{x} + \frac{9}{y} = \frac{21}{xy} \Rightarrow 4y + 9x = 21 \dots (ii)$$

On solving Eqs. (i) and (ii), we get

$$x = 1 \text{ and } y = 3$$

$$\therefore x + y = 1 + 3 = 4$$

38. (b) Since, the equation $kx - y = 2$ and $6x - 2y = 3$ have a unique solution

$$\therefore \frac{k}{6} \neq \frac{1}{2} \Rightarrow k \neq 3$$

39. (b) Three lines $a_1x + b_1y + c_1 = 0$

$$a_2x + b_2y + c_2 = 0$$

$$\text{and } a_3x + b_3y + c_3 = 0$$

which are non-parallel and non-collinear they have only one solution if they meet in a common point in this case these lines are called concurrent lines.

40. (c) Let the unit's digit be x and ten's digit's be y . Then the two-digit number

$$= 10y + x$$

$$\text{By given condition, } x + y = 10 \dots (i)$$

$$\text{and } 10y + x - 18 = 10x + y$$

$$\Rightarrow 9x - 9y = -18 \Rightarrow x - y = -2$$

On solving Eqs. (i) and (ii), we get

$$x = 4 \text{ and } y = 6$$

$$\therefore \text{Required product} = xy = 4 \times 6 = 24$$

41. (d) Since, given system of equations $x + 2y - 3 = 0$ and $5x + ky + 7 = 0$ has no solution.

$$\text{Then, } \frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$

$$\therefore \frac{1}{5} = \frac{2}{k} \neq \frac{-3}{7} \Rightarrow k = 10$$

42. (a) Given, $x - y = 0.9 \dots (i)$

$$\text{and } 11(x + y)^{-1} = 2 \dots (ii)$$

$$\Rightarrow 2x + 2y = 11$$

On multiplying Eq. (i) by 2 and adding Eqs. (i) and (ii), we get

$$4x = 12.8 \Rightarrow x = 3.2$$

$$\text{From Eq. (i)} \quad y = 3.2 - 0.9 = 2.3$$

43. (d) Let Pooja's initial salary be ₹ x and fixed increment every year be ₹ y .

$$\text{By given condition, } x + 3y = 4200 \dots (i)$$

$$\text{and } x + 8y = 6800 \dots (ii)$$

On subtracting Eq. (ii) from Eq. (i), we get

$$(x + 3y) - (x + 8y) = 4200 - 6800$$

$$\Rightarrow -5y = -2600 \Rightarrow y = ₹520$$

On putting $y = 520$ in Eq. (i), we get

$$x = ₹2640$$

44. (b) Let the fare of ticket from station P to station Q is ₹ x and that from station P to station R is ₹ y .

$$\text{By given condition, } x + y = 42$$

$$\text{and } 5x + 10y = 350$$

On solving Eqs. (i) and (ii), we get

$$x = 14 \text{ and } y = 28$$

Hence, fare from station P to station Q is ₹ 14.

45. (a) Let x represent school bags, y represent sweaters, z represent books. System of equations:

$$x + y + z = 200 \dots (i)$$

$$z = x + y \dots (ii)$$

$$20x + 25y + 5z = 2800 \dots (iii)$$

putting $z = x + y$ in Eq. (i)

$$(x + y) + x + y = 200, 2x + 2y = 200$$

$$x + y = 100, x = 100 - y$$

putting $x = 100 - y$ in Eq. (iii)

$$20(100 - y) + 25y + 5 = 2800$$

$$[2000 - 20y + 25y + 5] = 2800$$

$$2000 - 20y + 25y + 5 = 2800$$

$$5y = 300 \Rightarrow y = 60$$

$$x = 100 - 60 \Rightarrow x = 40$$

$$z = 40 + 60 \Rightarrow z = 100$$

40 school bags, 60 sweaters, 100 books

46. (b) Let the share of Rajesh be R , Sonal be S and Chetan be C , respectively.
According to question, $2R = 3S = 5C$
also, $6R + 6S = 150$
 $\Rightarrow \frac{3}{2} \times 6R + 6S = 150$
 $\Rightarrow 9S + 6S = 150 \Rightarrow S = 10$
 $\therefore C = \frac{3}{5}S = \frac{3}{5} \times 10 = ₹ 6$.

47. (a) $R + C + S = 155$

$$R + \frac{2}{5}R + \frac{2}{3}R = 155$$

$$\frac{15R + 6R + 10R}{15} = 155 \Rightarrow R = ₹ 75$$

48. (c) The graph of $ax + by = c$, $dx + ey = f$ will be coincident, if the system has infinite number of solutions.
So, statement II is false.

Thus, statements I and III are correct.

49. (c) Let the two numbers be x and y , respectively. According to the question,

$$x + y = 20 \quad \dots(i)$$

$$\text{and } xy = 75 \quad \dots(ii)$$

$$\Rightarrow \frac{1}{x} + \frac{1}{y} = \frac{y+x}{xy} = \frac{20}{75} = \frac{4}{15}$$

[from Eqs. (i) and (ii)]

50. (b) Given equations are,

$$3^{x+y} = 81 \Rightarrow 3^{x+y} = 3^4$$

$$\Rightarrow x + y = 4 \quad \dots(i)$$

$$\text{and } 81^{x-y} = 3 \Rightarrow (3^4)^{x-y} = 3^1$$

$$\Rightarrow x - y = \frac{1}{4} \quad \dots(ii)$$

On adding Eqs. (i) and (ii), we get

$$\Rightarrow 2x = \frac{17}{4} \Rightarrow x = \frac{17}{8}$$

51. (d) Given equations are,

$$\frac{a}{b} - \frac{b}{a} = \frac{x}{y} \quad \dots(i)$$

$$\text{and } \frac{a}{b} + \frac{b}{a} = x - y \quad \dots(ii)$$

From Eqs. (i) and (ii),

$$\frac{a}{b} + \frac{b}{a} = x - \frac{x}{\left(\frac{a-b}{b-a}\right)} = \frac{\left(\frac{a-b}{b-a}\right)x - x}{\frac{a-b}{b-a}}$$

$$\Rightarrow \left(\frac{a}{b} + \frac{b}{a}\right)\left(\frac{a-b}{b-a}\right) = x\left(\frac{a-b}{b-a} - 1\right)$$

$$\Rightarrow \left(\frac{a^2}{b^2} - \frac{b^2}{a^2}\right) = x\left(\frac{a^2 - b^2 - ab}{ab}\right)$$

$$\Rightarrow x = \frac{ab}{(a^2 - b^2 - ab)} \times \left(\frac{a^4 - b^4}{a^2 b^2}\right)$$

$$\Rightarrow x = \frac{(a^4 - b^4)}{a^2 - b^2 - ab} \cdot \frac{1}{ab} \\ = \frac{(a-b)(a+b)(a^2 + b^2)}{ab(a^2 - b^2 - ab)}$$

52. (d) Given equations are,

$$3x + y = 4 \quad \dots(i)$$

$$6x + 2y = 8 \quad \dots(ii)$$

$$\text{Here, } a_1 = 3, b_1 = 1 \text{ and } c_1 = 4$$

$$\text{and } a_2 = 6, b_2 = 2 \text{ and } c_2 = 8$$

$$\therefore \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} = \frac{1}{2}$$

So, the system of equations has infinitely many solutions.

53. (c) Let the two numbers be x and y , respectively.

Given, sum of two numbers = 7

$$\Rightarrow x + y = 7 \quad \dots(i)$$

and sum of their squares = 25

$$\Rightarrow x^2 + y^2 = 25 \quad \dots(ii)$$

Now, we have

$$(x + y)^2 = (x^2 + y^2) + 2xy$$

$$\Rightarrow (x + y)^2 - (x^2 + y^2) = 2xy$$

$$\Rightarrow xy = \frac{1}{2}[(x + y)^2 - (x^2 + y^2)]$$

$$= \frac{1}{2}[(7)^2 - (25)]$$

[from Eqs. (i) and (ii)]

$$= \frac{1}{2}(49 - 25) = \frac{1}{2} \times 24 = 12$$

Hence, the product of the two numbers is 12.

Solutions (Q. Nos. 54-55) Let the digit in unit's place be y and that in ten's place be x .

Then, the two-digit number is given by $10x + y$.

Number obtain by reversing the order of digit = $10y + x$

Now, by given condition,

$$x + y = 10 \quad \dots(i)$$

$$\text{and } (x + 10y) + 36 = (y + 10x)$$

$$\Rightarrow -9y + 9x = 36$$

$$\Rightarrow x - y = 4 \quad \dots(ii)$$

On adding Eqs. (i) and (ii), we get

$$2x = 14, x = 7$$

On putting the value of x in Eq. (i), we get

$$7 + y = 10 \Rightarrow y = 3$$

54. (b) The required number is 73.

So, the number is a multiple of a prime number.

55. (a) Required product of two digits

$$= 3 \times 7 = 21$$

56. (a) Let the cost of one chair and one table be ₹ x and ₹ y , respectively.

By given condition,

$$10x + 6y = 6200 \quad \dots(i)$$

$$\text{and } 3x + 2y = 1900$$

$$\Rightarrow 9x + 6y = 5700 \quad \dots(ii)$$

On subtracting Eq. (ii) from Eq. (i), we get

$$(10x + 6y) - (9x + 6y) = 6200 - 5700$$

$$\Rightarrow x = 500$$

$$\therefore x = ₹ 500$$

From Eq. (i), $5000 + 6y = 6200$

$$\Rightarrow 6y = 1200 \Rightarrow y = ₹ 200$$

$\therefore$ Cost of 4 chairs and 5 tables

$$= 4x + 5y = 4 \times 500 + 5 \times 200$$

$$= 2000 + 1000 = ₹ 3000$$

57. (b) Let bus starts with x number of passengers.

After 1st stoppage, number of passengers

$$= x - \frac{x}{5} + 40 = \frac{5x - x + 200}{5}$$

$$= \frac{4x + 200}{5}$$

After 2nd stoppage, number of passengers

$$\frac{4x + 200}{5} - \frac{4x + 200}{5 \times 2} + 30$$

According to question,

$$\Rightarrow \frac{4x + 200}{5} - \frac{4x + 200}{10} + 30 = 70$$

$$\frac{4x + 200}{5} \left(1 - \frac{1}{2}\right) + 30 = 70$$

$$\Rightarrow \frac{4x + 200}{5} \times \frac{1}{2} = 40$$

$$\Rightarrow 4x + 200 = 400 \Rightarrow 4x = 200$$

$$\therefore x = \frac{200}{4} = 50$$

Hence, the number of passengers is 50.

58. (d) We have, $x + y - 7 = 0$

$$\Rightarrow x + y = 7 \quad \dots(i)$$

$$\text{and } 3x + y - 13 = 0$$

$$\Rightarrow 3x + y = 13 \quad \dots(ii)$$

On subtracting Eq. (i) from Eq. (ii), we get

$$3x + y = 13$$

$$x + y = 7$$

$$\underline{\quad \quad \quad}$$

$$2x = 6$$

$$\therefore x = 3$$

On putting the value of x in Eq. (i), we get

$$3 + y = 7 \Rightarrow y = 4$$

$$\text{Now, } 4x^2 + y^2 + 4xy = (2x + y)^2$$

$$= (2 \times 3 + 4)^2$$

$$= (6 + 4)^2$$

$$= 10^2 = 100$$

59. (b) Given, $\frac{x}{2} + \frac{y}{3} = 4 \Rightarrow \frac{3x + 2y}{6} = 4$

$$\Rightarrow 3x + 2y = 24 \quad \dots(i)$$

$$\text{and } \frac{2}{x} + \frac{3}{y} = 1 \Rightarrow \frac{2y + 3x}{xy} = 1$$

$$\Rightarrow 2y + 3x = xy \quad \dots(ii)$$

On comparing Eqs. (i) and (ii), we get $xy = 24$

There are 8 possibilities for x and y , respectively.

$$\begin{array}{ll} 1 \times 24 = 24 & 6 \times 4 = 24 \\ 2 \times 12 = 24, & 8 \times 3 = 24 \\ 3 \times 8 = 24, & 12 \times 2 = 24 \\ 4 \times 6 = 24, & 24 \times 1 = 24 \end{array}$$

On putting all the values of x and y in given equations we get that only $x = 4$ and $y = 6$ satisfy the equations.

$$\therefore x + y = 4 + 6 = 10$$

60. (a) Given equations are,

$$\begin{array}{ll} x + y = 5 & \dots(i) \\ y + z = 10 & \dots(ii) \\ z + x = 15 & \dots(iii) \end{array}$$

from Eq. (i), we have $y = 5 - x$
put the value of x in Eq. (ii),

$$\begin{array}{l} z + 5 - x = 10 \\ \Rightarrow z - x = 5 \quad \dots(iv) \end{array}$$

Add Eq. (iv) and Eq. (iii)

$$\begin{array}{l} (z + x) + (z - x) = 15 + 5 \\ \Rightarrow 2z = 20 \Rightarrow z = 10 \\ \text{from Eq. (iii), we have } 10 + x = 15 \\ \Rightarrow x = 5 \end{array}$$

$$\text{and hence } y = 5 - x = 5 - 5 = 0$$

$\therefore$ correct sequence is $z > x > y$

61. (c) Let present age of Ravi be x yr.

$\therefore$ Present age of Ravi's father = $4x$ yr

Now, 5 yr ago, age of Ravi's father

$$= (4x - 5) \text{ yr}$$

and age of Ravi = $(x - 5)$ yr

According to the question,

$$\begin{array}{l} 4x - 5 = 7(x - 5) \\ \Rightarrow 4x - 5 = 7x - 35 \Rightarrow 3x = 30 \\ \therefore x = 10 \end{array}$$

$\therefore$ Present age of Ravi = $x = 10$ yr

Present age of Ravi's father

$$= 4x = 4 \times 10 = 40 \text{ yr}$$

62. (d) Let the two positive numbers be x and y , respectively. According to the question,

$$\begin{array}{l} (x + y) = 25(x - y) \\ \Rightarrow x + y = 25x - 25y \\ \Rightarrow 35y = 15x \Rightarrow \frac{x}{y} = \frac{7}{3} \text{ or } x = \frac{7}{3}y \end{array}$$

Now, product of numbers $xy = 84$

$$\Rightarrow \frac{7}{3}y \times y = 84 \Rightarrow y^2 = \frac{84 \times 3}{7}$$

$$\Rightarrow y^2 = 12 \times 3$$

$$\therefore y = 6 \Rightarrow x = \frac{7}{3} \times 6 = 14$$

$$\begin{array}{l} \therefore \text{Sum of numbers} = x + y \\ = 14 + 6 = 20 \end{array}$$

63. (a) Let the cost of one chair be ₹ x and that of one table be ₹ y .

$$\text{Then, } 2x + y = 700 \quad \dots(i)$$

$$\text{and } x + 2y = 800 \quad \dots(ii)$$

On multiplying Eq. (ii) by 2 and subtracting from Eq. (i), we get

$$\begin{array}{r} 2x + y = 700 \\ 2x + 4y = 1600 \\ \hline -3y = -900 \Rightarrow y = 300 \end{array}$$

On putting $y = 300$ in Eq. (ii), we get

$$\begin{array}{l} x + 2 \times 300 = 800 \\ x = 800 - 600 = 200 \end{array}$$

Since, m chairs and m tables are to be purchased for ₹ 30000.

$\therefore$ Cost of m table and m chairs = 30000

$$\begin{array}{l} \therefore 200m + 300m = 30000 \\ \Rightarrow m = \frac{30000}{500} \Rightarrow m = 60 \end{array}$$

64. (c) Let the two-digit number be $10x + y$,
Condition I

According to the question,

$$\begin{array}{l} 10x + y = 3(x + y) \\ \Rightarrow 10x + y - 3x - 3y = 0 \\ \Rightarrow 7x - 2y = 0 \quad \dots(i) \end{array}$$

Condition II

$$\begin{array}{l} \Rightarrow 10x + y + 45 - 10y - x = 0 \\ \Rightarrow 9x - 9y + 45 = 0 \\ \Rightarrow x - y = -5 \quad \dots(ii) \end{array}$$

On solving Eqs. (i) and (ii), we get

$$x = 2 \text{ and } y = 7$$

$\therefore$ Sum of the squares of digits

$$= (2)^2 + (7)^2 = 4 + 49 = 53$$

65. (a) Let x be the required number.

$$\begin{array}{l} \therefore 52x - 25x = 324 \\ \Rightarrow 27x = 324 \Rightarrow x = \frac{324}{27} = 12 \end{array}$$

Hence, the required number is 12.

66. (b) Let a tin of oil contain x bottles of oil.

According to the question,

$$\begin{array}{l} \frac{4}{5} - \frac{6}{x} + \frac{4}{x} = \frac{3}{4} \Rightarrow \frac{4}{5} - \frac{3}{4} = \frac{6}{x} - \frac{4}{x} \\ \Rightarrow \frac{1}{20} = \frac{2}{x} \Rightarrow x = 40 \end{array}$$

$\therefore$ A tin contains 40 bottles of oil.

67. (c) Let the number be $10x + y$.

$$\text{We have, } x + y = 7 \quad \dots(i)$$

$$\text{and } 10y + x = 10x + y + 27$$

$$\Rightarrow 9y - 9x = 27 \Rightarrow y - x = 3 \quad \dots(ii)$$

Solving Eqs. (i) and (ii), we get $y = 5$ and $x = 2$

$\therefore$ Number is 25.

$$\text{Product of number} = 2 \times 5 = 10$$

68. (c) Given, $\frac{p}{x} + \frac{q}{y} = m$ and $\frac{q}{x} + \frac{p}{y} = b$

$$\text{Let } \frac{1}{x} = u \text{ and } \frac{1}{y} = v$$

$$\text{Then, } pu + qv = m \quad \dots(i)$$

$$\text{and } qu + pv = n \quad \dots(ii)$$

On solving Eqs. (i) and (ii), we get

$$\begin{array}{l} u = \frac{mp - nq}{p^2 - q^2} \text{ and } v = \frac{mq - np}{q^2 - p^2} \\ \therefore \frac{x}{y} = \frac{1/u}{1/v} = \frac{v}{u} = \frac{-(mq - np)}{mp - nq} \\ = \frac{np - mq}{mp - nq} \end{array}$$

69. (c) Given equations are,

$$3x - ky - 20 = 0 \quad \dots(i)$$

$$6x - 10y + 40 = 0 \quad \dots(ii)$$

since, the given system has no solution

$$\therefore \frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2} \Rightarrow \frac{3}{6} = \frac{k}{10} \Rightarrow k = 5$$

70. (b) Let the ages of three brothers be a, b and c .

$$\text{Then, } a + b = 4, \quad b + c = 6$$

$$\text{and } c + a = 8$$

On solving these three equations, we get

$$a = 3, \quad b = 1 \text{ and } c = 5$$

$\therefore$ Age difference between eldest and youngest = $5 - 1 = 4$ yr

71. (a) Let annual incomes of two persons be $9x$ and $7x$ and expenses be $4y$ and $3y$, respectively.

Then, according to the question,

$$9x - 4y = 2000 \quad \dots(i)$$

$$\text{and } 7x - 3y = 2000 \quad \dots(ii)$$

From Eqs. (i) and (ii), we get

$$9x - 4y = 7x - 3y \Rightarrow y = 2x$$

On putting the value of y in Eq. (i), we get

$$9x - 8x = 2000 \Rightarrow x = ₹ 2000$$

$\therefore$ Difference between their annual incomes = $9x - 7x = 2x = ₹ 4000$

72. (c) Let the unit's place number be y and ten's place number be x .

$$\text{Then, number} = 10x + y$$

Now, after interchanging the digits,

New number = $10y + x$ and sum of digits = $x + y$

According to the question,

$$10x + y = k(x + y) \quad \dots(i)$$

$$\text{and } 10y + x = m(x + y) \quad \dots(ii)$$

On adding Eqs. (i) and (ii), we get

$$11(x + y) = (k + m)(x + y)$$

$$\Rightarrow k + m = 11 \Rightarrow m = 11 - k$$

18

QUADRATIC EQUATIONS AND INEQUALITIES

Regularly (9-10) questions have been asked from this chapter. Questions from this section generally focus on finding roots of quadratic equation and their factorisation.



QUADRATIC EQUATION

A quadratic equation is an equation whose degree is 2, meaning that the highest exponent of variable is 2. The general form of a quadratic equation is $ax^2 + bx + c = 0$, where a , b and c are real numbers and $a \neq 0$.
e.g. $2x^2 + 3x + 5 = 0$, $x^2 + x + 1 = 0$ and $5x^2 + 6x + 8 = 0$ are all quadratic equations.

Roots of a Quadratic Equation

A real number α is said to be a root of the quadratic equation $ax^2 + bx + c = 0$ ($a \neq 0$), if it satisfies the equation i.e. $a\alpha^2 + b\alpha + c = 0$. We can also say that $x = \alpha$ is a solution of the quadratic equation.
e.g. 2 is the root of the quadratic equation, $x^2 - 6x + 8 = 0$ since $(2)^2 - 6(2) + 8 = 0$.

Note A quadratic equation has exactly two roots.

If α be the root of the quadratic equation $ax^2 + bx + c = 0$, then $(x - \alpha)$ is the factor of $ax^2 + bx + c$.

Solutions of Quadratic Equations

Solutions of quadratic equations can be determined by using any of the following two methods

1. By Factorisation

In this method, the middle term of the quadratic equation is splitted and the given quadratic equation is converted into a product of two linear factors. Then, the roots of the equation are obtained by equating each factor equal to zero.

Let the quadratic equation be $ax^2 + bx + c = 0$ and its linear factors are $(px + q)$ and $(rx + s)$, then

$$ax^2 + bx + c = (px + q)(rx + s)$$

$$\text{Now, } ax^2 + bx + c = 0 \Rightarrow (px + q)(rx + s) = 0 \Rightarrow px + q = 0 \text{ or } rx + s = 0 \Rightarrow x = \frac{-q}{p} \text{ or } x = \frac{-s}{r}$$

Thus, $\frac{-q}{p}$ and $\frac{-s}{r}$ are the roots of the equation, $ax^2 + bx + c = 0$.

EXAMPLE 1. Solve the equation $x^2 - 7x + 12 = 0$, and find the value of x .

- a. $-3, -4$ b. $-3, 4$ c. $3, -4$ d. $3, 4$

Sol. d. Given, $x^2 - 7x + 12 = 0$
 $x^2 - 3x - 4x + 12 = 0$ [splitting the middle term]
 $\Rightarrow x(x - 3) - 4(x - 3) = 0$
 $\Rightarrow (x - 3)(x - 4) = 0$
 $\Rightarrow (x - 3) = 0$ or $(x - 4) = 0 \Rightarrow x = 3$ or $x = 4$
 So, $x = 3, 4$ are roots of the given equation.

EXAMPLE 2. Solve the equation $3a^2x^2 + 8abx + 4b^2 = 0$, $a \neq 0$, and find the value of x .

- a. $\frac{2b}{3a}, \frac{2b}{a}$ b. $\frac{-2b}{3a}, \frac{2b}{a}$ c. $\frac{-2b}{3a}, \frac{-2b}{a}$ d. $\frac{2b}{3a}, \frac{-2b}{a}$

Sol. c. The equation is $3a^2x^2 + 8abx + 4b^2 = 0$
 $\Rightarrow 3a^2x^2 + 2abx + 6abx + 4b^2 = 0$
 $\Rightarrow ax(3ax + 2b) + 2b(3ax + 2b) = 0$
 $\therefore (3ax + 2b)(ax + 2b) = 0$
 $\Rightarrow 3ax + 2b = 0$ or $ax + 2b = 0$
 $\therefore x = \frac{-2b}{3a}$ or $\frac{-2b}{a}$

2. By Using the Quadratic Formula

Let the quadratic equation be $ax^2 + bx + c = 0$, where $a \neq 0$. Then, solution of this equation can be found using the formula.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

- If α and β be considered as roots of the quadratic equation, then

$$\alpha = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \text{ and } \beta = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

- The quantity $b^2 - 4ac$ is called the discriminant of the quadratic equation $ax^2 + bx + c = 0$ and is denoted by D . So, $D = b^2 - 4ac$.

EXAMPLE 3. Solve the equation $x^2 - 9x + 18 = 0$ and find the value of x .

- a. $3, 6$ b. $-3, -6$ c. $-3, 6$ d. $3, -6$

Sol. a. We have, $x^2 - 9x + 18 = 0$
 Here, $a = 1$, $b = -9$ and $c = 18$
 $\therefore x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-(-9) \pm \sqrt{(-9)^2 - 4(1)(18)}}{2(1)}$
 $= \frac{9 \pm \sqrt{81 - 72}}{2} = \frac{9 \pm \sqrt{9}}{2} = \frac{9 \pm 3}{2}$
 $\therefore x = \frac{9 + 3}{2}$ or $x = \frac{9 - 3}{2} \Rightarrow x = 6$ or 3
 Hence, the values of x are 6 and 3.

Nature of Roots of Quadratic Equation

Let $D = b^2 - 4ac$ be the discriminant of the quadratic equation, $ax^2 + bx + c = 0$, where $a \neq 0$. Then, the following cases arise

- If $D > 0$, then the two roots are real and distinct.
- If $D = 0$, then the two roots are real and equal and are given by $\alpha = \beta = \frac{-b}{2a}$.
- If $D < 0$, then there are no real roots, i.e. given equation has imaginary roots.

IMPORTANT POINTS

- If $a, b, c \in \mathbb{Q}$ and D is a perfect square, then equation has rational roots.
- If one root of quadratic equation is $p + \sqrt{q}$, then other root will be $p - \sqrt{q}$.

EXAMPLE 4. If the equation $x^2 + 2(1+k)x + k^2 = 0$ has equal roots, then what is the value of k ?

- a. $\frac{1}{2}$ b. $-\frac{1}{2}$
 c. 1 d. -1

Sol. b. Given equation, $x^2 + 2(1+k)x + k^2 = 0$.

If it has equal roots, then $D = 0$

$$\Rightarrow \{2(1+k)\}^2 - 4k^2 = 0$$

$$\Rightarrow 4(1+k^2 + 2k) - 4k^2 = 0$$

$$\Rightarrow 4 + 4k^2 + 8k - 4k^2 = 0$$

$$\Rightarrow 4 + 8k = 0 \Rightarrow k = -\frac{4}{8}$$

$$\therefore k = -\frac{1}{2}$$

Hence, the value of k is $-\frac{1}{2}$.

Roots Under Particular Conditions

Consider the quadratic equation $ax^2 + bx + c = 0$.

- Both roots are positive, if a and b are opposite in sign and a and c are same in sign, i.e. $\frac{-b}{a} > 0$ and $\frac{c}{a} > 0$.
- Both roots are negative, if a, b and c are of same sign, i.e. $\frac{b}{a} > 0$ and $\frac{c}{a} > 0$.
- Roots are of opposite signs, if a and c are of opposite sign, i.e. $\frac{-a}{c} > 0$.
- Roots are equal but opposite in signs, if $b = 0$.
- Roots are reciprocal to each other, if $a = c$.

EXAMPLE 5. What is the least integral value of k for which the equation $x^2 - 2(k-1)x + (2k+1) = 0$ has both the roots positive?

- a. 1 b. $-\frac{1}{2}$ c. 4 d. 0

Sol. a. Both the roots of the equation $ax^2 + bx + c = 0$ are positive, if $-\frac{b}{a} > 0$ and $\frac{c}{a} > 0$

Given equation is $x^2 - 2(k-1)x + (2k+1) = 0$.

Now, $-\frac{b}{a} = \frac{2(k-1)}{1} \Rightarrow 2(k-1) > 0$, if $k > 1$

and $\frac{c}{a} = \frac{2k+1}{1} \Rightarrow (2k+1) > 0$, if $k > -\frac{1}{2}$

$\therefore k > 1$

Hence, the least value k as per given in options is 1.

Relation between Roots and Coefficients

1. Quadratic Equation

Let α, β be the roots of the equation $ax^2 + bx + c = 0$ ($a \neq 0$), then

Sum of the roots $= \alpha + \beta = -\frac{b}{a} = \frac{-\text{Coefficient of } x}{\text{Coefficient of } x^2}$

Product of the roots $= \alpha\beta = \frac{c}{a} = \frac{\text{Constant term}}{\text{Coefficient of } x^2}$

2. Cubic Equation

If α, β, γ are the roots of the cubic equation $ax^3 + bx^2 + cx + d = 0$, $a \neq 0$, then

Sum of roots $= \alpha + \beta + \gamma = -b/a$

Product of roots two at a time $= \alpha\beta + \beta\gamma + \gamma\alpha = c/a$

Product of three roots $= \alpha\beta\gamma = -\frac{d}{a}$

EXAMPLE 6. What are the roots of the quadratic equation $a^2b^2x^2 - (a^2 + b^2)x + 1 = 0$?

- a. $\frac{1}{a^2}, \frac{1}{b^2}$ b. $-\frac{1}{a^2}, -\frac{1}{b^2}$ c. $\frac{1}{a^2}, -\frac{1}{b^2}$ d. $-\frac{1}{a^2}, \frac{1}{b^2}$

Sol. a. Let roots of the equation $a^2b^2x^2 - (a^2 + b^2)x + 1 = 0$ be α and β .

$\therefore$ Sum of roots $= \alpha + \beta = \frac{a^2 + b^2}{a^2b^2}$... (i)

and product of roots $= \alpha\beta = \frac{1}{a^2b^2}$

Now, $\alpha - \beta = \sqrt{(\alpha + \beta)^2 - 4\alpha\beta} = \sqrt{\left(\frac{a^2 + b^2}{a^2b^2}\right)^2 - \frac{4}{a^2b^2}}$

$$\Rightarrow \alpha - \beta = \sqrt{\frac{(a^2 - b^2)^2}{(a^2b^2)^2}} = \frac{a^2 - b^2}{a^2b^2} \quad \dots(ii)$$

On solving Eqs. (i) and (ii), we get

$$\alpha = \frac{1}{b^2} \text{ and } \beta = \frac{1}{a^2}$$

EXAMPLE 7. If one root of the equation $\frac{x^2}{a} + \frac{x}{b} + \frac{1}{c} = 0$ is reciprocal to the other, then which one of the following is correct?

- a. $a = b$ b. $b = c$ c. $ac = 1$ d. $a = c$

Sol. d. Given, $\frac{x^2}{a} + \frac{x}{b} + \frac{1}{c} = 0$... (i)

Now, let one root be α , then the other root is $1/\alpha$.

We know that,

$$\text{Product of roots} = \frac{\text{Constant term}}{\text{Coefficient of } x^2} = \frac{1/c}{1/a}$$

$$\Rightarrow \alpha \cdot \frac{1}{\alpha} = \frac{a}{c} \Rightarrow c = a$$

which is the required relation.

Symmetric Functions of the Roots

Let α and β be the roots of a quadratic equation. An expression in α and β which remains same when α and β are interchanged is known as a symmetric function in α and β . To evaluate the value of symmetric function of the roots, express the given function in terms of $(\alpha + \beta)$ and $\alpha\beta$.

You can use the following results.

- $(\alpha^2 + \beta^2) = [(\alpha + \beta)^2 - 2\alpha\beta]$
- $(\alpha^3 + \beta^3) = [(\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)]$
- $(\alpha - \beta)^2 = [(\alpha + \beta)^2 - 4\alpha\beta]$
- $(\alpha^3 - \beta^3) = [(\alpha - \beta)^3 + 3\alpha\beta(\alpha - \beta)]$

EXAMPLE 8. If α and β are the roots of the equation $x^2 - 6x + 6 = 0$, what is $\alpha^3 + \beta^3 + \alpha^2 + \beta^2 + \alpha + \beta$ equal to?

- a. 150 b. 138 c. 128 d. 124

Sol. b. Here, $\alpha + \beta = \frac{-\text{Coefficient of } x}{\text{Coefficient of } x^2} = 6$

and $\alpha\beta = \frac{\text{Constant term}}{\text{Coefficient of } x^2} = 6$

$$\therefore (\alpha + \beta)^2 = 6^2 \Rightarrow \alpha^2 + \beta^2 + 2\alpha\beta = 36$$

$$\Rightarrow \alpha^2 + \beta^2 = 36 - 2(6) = 24$$

$$\begin{aligned} \therefore (\alpha^3 + \beta^3) + (\alpha^2 + \beta^2) + (\alpha + \beta) &= (\alpha + \beta)(\alpha^2 + \beta^2 - \alpha\beta) + (\alpha^2 + \beta^2) + (\alpha + \beta) \\ &= 6(24 - 6) + (24) + (6) \\ &= 6(18) + 30 = 108 + 30 = 138 \end{aligned}$$

Formation of a Quadratic Equation

If α, β are the roots of a quadratic equation, then the equation is $(x-\alpha)(x-\beta)=0 \Rightarrow x^2-(\alpha+\beta)x+\alpha\beta=0$
i.e. $x^2 - (\text{sum of the roots})x + \text{product of the roots} = 0$

EXAMPLE 9. The quadratic equation whose roots are 3 and -1 , is

- a. $x^2 - 4x + 3 = 0$ b. $x^2 - 2x - 3 = 0$
c. $x^2 + 2x - 3 = 0$ d. $x^2 + 4x + 3 = 0$

Sol. b. Given that, the roots of the quadratic equation are 3 and -1 .

Let $\alpha = 3$ and $\beta = -1$
 $\therefore$ Sum of roots $= \alpha + \beta = 3 - 1 = 2$
and product of roots $= \alpha\beta = (3)(-1) = -3$
So, required quadratic equation is

$$x^2 - (\alpha + \beta)x + \alpha\beta = 0$$

$$\Rightarrow x^2 - (2)x + (-3) = 0$$

$$\Rightarrow x^2 - 2x - 3 = 0$$

Equations Reducible to Quadratic Equations

The equations which at the outset are not quadratic equations, but can be reduced to quadratic equations by suitable substitutions or simplifications are called equations reducible to quadratic equations.

(Rule 1) $ax^{2n} + bx^n + c = 0$, where $n \geq 2$

Use substitution $x^n = y$

$\Rightarrow x^{2n} = y^2$ and equation reduces to $ay^2 + by + c = 0$.

EXAMPLE 10. Solve the quadratic equation $x^4 - 26x^2 + 25 = 0$, and find the value of x .

- a. $\pm 1, \pm 25$ b. $\pm 1, \pm 5$
c. $1, 5$ d. $1, 25$

Sol. b. Put $x^2 = z \Rightarrow x^4 = z^2$

$$x^4 - 26x^2 + 25 = 0 \text{ can be written as}$$

$$z^2 - 26z + 25 = 0 \Rightarrow z^2 - 25z - z + 25 = 0$$

$$\Rightarrow z(z - 25) - 1(z - 25) = 0$$

$$\Rightarrow (z - 1)(z - 25) = 0 \Rightarrow z = 1 \text{ or } z = 25$$

when $z = 1 \Rightarrow x^2 = 1 \Rightarrow x = \pm 1$
when $z = 25 \Rightarrow x^2 = 25 \Rightarrow x = \pm 5$
Hence, the solutions are ± 1 and ± 5 .

(Rule 2) $Px + \frac{Q}{x} = R$

Reduce the given equation to a quadratic equation by multiplying both sides by ' x '. So, $Px^2 + Q = Rx$

or $Px^2 - Rx + Q = 0$

EXAMPLE 11. Solve $2x - \frac{3}{x} = 5$. Then, the value of x is

- a. $1/3, -3$ b. $1/2, -3$ c. $-1/2, 3$ d. $1/2, 4$

Sol. c. Here, $2x - \frac{3}{x} = 5$... (i)

On multiplying Eq. (i) by $x \Rightarrow 2x^2 - 3 = 5x$
 $\Rightarrow 2x^2 - 5x - 3 = 0 \Rightarrow 2x^2 - 6x + x - 3 = 0$
 $\Rightarrow (2x + 1)(x - 3) = 0 \Rightarrow x = -1/2 \text{ or } x = 3$
 Hence, the solution is $x = \frac{-1}{2}, 3$.

(Rule 3) $\sqrt{x+b} + x = c$ or $\sqrt{a-x^2} = bx + c$

- Square both sides to get a quadratic equation without the radical.
- Solve the equation by factorisation or by quadratic formula.

EXAMPLE 12. Solve $\sqrt{2x+9} + x = 13$, then the value of x is

- a. $4, 10$ b. $8, 20$ c. $3, 15$ d. $-4, -10$

Sol. b. Given, $\sqrt{2x+9} + x = 13 \Rightarrow \sqrt{2x+9} = 13 - x$

On squaring both sides, we get $(\sqrt{2x+9})^2 = (13-x)^2$
 $\Rightarrow 2x+9 = 169 + x^2 - 26x$
 $\Rightarrow x^2 - 28x + 160 = 0 \Rightarrow x^2 - 20x - 8x + 160 = 0$
 $\Rightarrow (x-8)(x-20) = 0 \Rightarrow x = 8 \text{ or } x = 20$
 Hence, the solution is $x = 8, 20$.

(Rule 4) $\sqrt{ax+b} \pm \sqrt{cx+d} = e$

or $\sqrt{ax+b} \pm \sqrt{cx+d} \pm \sqrt{ex+f} = 0$

To solve such type of equations, follow the steps given below.

- Square both sides once, so that only one term containing radical is obtained.
- Keep the term containing radical on one side and all other terms on the other side.
- Square again to get, the quadratic equation and solve it.

EXAMPLE 13. Solve the following equation

$\sqrt{4-x} + \sqrt{x+9} = 5$. Then, the value of x is

- a. $1, 5$ b. $-1, 3$ c. $0, -5$ d. $1, 4$

Sol. c. Given, $\sqrt{4-x} + \sqrt{x+9} = 5$

On squaring both sides, we get $(\sqrt{4-x} + \sqrt{x+9})^2 = 25$
 $\Rightarrow (4-x) + (x+9) + 2(\sqrt{4-x}\sqrt{x+9}) = 25$
 $\Rightarrow 2(\sqrt{4-x})(\sqrt{x+9}) = 12 \Rightarrow (\sqrt{4-x})(\sqrt{x+9}) = 6$
 Again, squaring both sides, we get
 $\Rightarrow (4-x)(x+9) = 36 \Rightarrow -x^2 + 36 - 5x = 36$
 $\Rightarrow x^2 + 5x = 0 \Rightarrow x(x+5) = 0 \Rightarrow x = 0 \text{ or } x = -5$
 $\therefore x = 0, -5$ is the required solution.

EXAMPLE 14. Solve $\sqrt{11y-6} + \sqrt{y-1} - \sqrt{4y+5} = 0$.
Then, the value of y is

- a. {5, 6/5} b. {1, 2}
c. {3, 5/2} d. None of these

Sol. a. Given, $\sqrt{11y-6} + \sqrt{y-1} - \sqrt{4y+5} = 0$

$$\sqrt{11y-6} + \sqrt{y-1} = \sqrt{4y+5}$$

On squaring both sides, we get

$$[\sqrt{11y-6} + \sqrt{y-1}]^2 = [\sqrt{4y+5}]^2$$

$$\Rightarrow (11y-6) + (y-1) + 2\sqrt{(11y-6)(y-1)} = (4y+5)$$

$$\Rightarrow 12y-7 + 2\sqrt{11y^2-17y+6} = 4y+5$$

$$\Rightarrow 2\sqrt{11y^2-17y+6} = 12-8y$$

$$\Rightarrow \sqrt{11y^2-17y+6} = 6-4y$$

Again, squaring both sides, we get

$$11y^2-17y+6 = (6-4y)^2$$

$$\Rightarrow 11y^2-17y+6 = 36+16y^2-48y$$

$$\Rightarrow 5y^2-31y+30 = 0$$

Here, $a = 5$, $b = -31$ and $c = 30$

$$\Rightarrow y = \frac{31 \pm \sqrt{(-31)^2 - 4(5)(30)}}{2 \times 5} \left[\because y = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \right]$$

$$\Rightarrow y = \frac{31 \pm \sqrt{361}}{10} \Rightarrow y = \frac{31 \pm 19}{10}$$

$$\Rightarrow y = \frac{31+19}{10} \text{ and } y = \frac{31-19}{10}$$

$$\therefore y = \left\{ 5, \frac{6}{5} \right\} \text{ is the required solution.}$$

(Rule 5) $(x+a)(x+b)(x+c)(x+d)+k=0$, where

$a+b=c+d$ and k may or may not be zero.

To solve such type of equations, put $x^2+(a+b)x=T$.

EXAMPLE 15. Solve the equation $(x+1)(x+2)(x+3)(x+4)-8=0$, and find the number of real roots of this equation.

- a. 1 b. 2 c. 4 d. No real roots

Sol. b. $(x+1)(x+2)(x+3)(x+4)-8=0$

Here, $1+4=2+3$,

$$\text{so, } [(x+1)(x+4)][(x+2)(x+3)]-8=0$$

$$[x^2+5x+4][x^2+5x+6]-8=0$$

Put $x^2+5x=t$, so the equation becomes

$$(t+4)(t+6)-8=0$$

$$\Rightarrow t^2+10t+24-8=0 \Rightarrow t^2+10t+16=0$$

$$\Rightarrow t^2+8t+2t+16=0$$

$$\Rightarrow t(t+8)+2(t+8)=0 \Rightarrow (t+8)(t+2)=0$$

$$\Rightarrow t = -8 \text{ or } -2$$

Now,

$$\begin{array}{l|l} x^2+5x=-8 & x^2+5x=-2 \\ \Rightarrow x^2+5x+8=0 & \Rightarrow x^2+5x+2=0 \\ \Rightarrow x = \frac{-5 \pm \sqrt{25-32}}{2} & \Rightarrow x = \frac{-5 \pm \sqrt{25-8}}{2} \\ \therefore x = \frac{-5 \pm \sqrt{-7}}{2} & \therefore x = \frac{-5 \pm \sqrt{17}}{2} \end{array}$$

Hence, the number of real roots of the given equation is 2.

(Rule 6) (i) $a\left(x^2 + \frac{1}{x^2}\right) + b\left(x + \frac{1}{x}\right) + c = 0$

(ii) $a\left(x^2 + \frac{1}{x^2}\right) + b\left(x - \frac{1}{x}\right) + c = 0$

To solve an equation of the form (i) .

put $\left(x + \frac{1}{x}\right) = y$ and to solve an equation of

the form (ii), put $\left(x - \frac{1}{x}\right) = y$

EXAMPLE 16. Solve $4\left(x^2 + \frac{1}{x^2}\right) - 4\left(x + \frac{1}{x}\right) - 7 = 0$;

$x \neq 0$ and find the real roots of x .

- a. $\{-2, -1/2\}$ b. $\{0, 2\}$
c. $\{2, 1/2\}$ d. None of these

Sol. c. $4\left(x^2 + \frac{1}{x^2}\right) - 4\left(x + \frac{1}{x}\right) - 7 = 0$... (i)

Put $x + \frac{1}{x} = y$

On squaring both sides, we get

$$\left(x + \frac{1}{x}\right)^2 = y^2 \Rightarrow x^2 + \frac{1}{x^2} + 2 = y^2 \Rightarrow x^2 + \frac{1}{x^2} = y^2 - 2$$

$$\Rightarrow 4(y^2 - 2) - 4y - 7 = 0 \quad [\text{from Eq. (i)}]$$

$$\Rightarrow 4y^2 - 8 - 4y - 7 = 0 \Rightarrow 4y^2 - 4y - 15 = 0$$

$$\Rightarrow 4y^2 - 10y + 6y - 15 = 0 \Rightarrow 2y(2y - 5) + 3(2y - 5) = 0$$

$$\Rightarrow (2y - 5)(2y + 3) = 0 \Rightarrow y = \frac{5}{2} \text{ or } -\frac{3}{2}$$

Now, For $y = \frac{5}{2}$ For $y = -\frac{3}{2}$

$$\Rightarrow x + \frac{1}{x} = \frac{5}{2} \quad \Rightarrow x + \frac{1}{x} = -\frac{3}{2}$$

$$\Rightarrow \frac{x^2 + 1}{x} = \frac{5}{2} \quad \Rightarrow \frac{x^2 + 1}{x} = -\frac{3}{2}$$

$$\Rightarrow 2x^2 + 2 = 5x \quad \Rightarrow 2x^2 + 2 = -3x$$

$$\Rightarrow 2x^2 - 5x + 2 = 0 \quad \Rightarrow 2x^2 + 3x + 2 = 0$$

$$\Rightarrow (x-2)(2x-1) = 0 \quad \text{Here, } D = 9 - 16 = -7 < 0$$

$$\therefore x = 2 \text{ or } x = \frac{1}{2} \quad \text{So, there are no real roots.}$$

$$[\because D = b^2 - 4ac]$$

Hence, the values of x are 2 and $\frac{1}{2}$.

Rule 7 $ax^4 + bx^3 + cx^2 + bx + a = 0$

- To solve such type of equations, we divide the equation by x^2 , to obtain $a(x^2 + 1/x^2) + b(x + 1/x) + c = 0$, and, then put $x + \frac{1}{x} = y$

EXAMPLE 17. If $x^4 - 3x^3 - 2x^2 + 3x + 1 = 0$, $x \neq 0$, then the roots of the equation is

- a. $\left\{ \pm 2, \frac{-3 \pm \sqrt{13}}{2} \right\}$ b. $\left\{ \pm 1, \frac{3 \pm \sqrt{13}}{2} \right\}$
 c. $\left\{ \pm 3, \frac{3 \pm \sqrt{11}}{2} \right\}$ d. None of these

Sol. b. $x^4 - 3x^3 - 2x^2 + 3x + 1 = 0$

On dividing throughout by x^2 and rearranging the terms, we get

$$\left(x^2 + \frac{1}{x^2}\right) - 3\left(x - \frac{1}{x}\right) - 2 = 0 \quad \dots(i)$$

Put $x - \frac{1}{x} = y$

On squaring both sides, we get

$$\begin{aligned} x^2 + \frac{1}{x^2} - 2 = y^2 &\Rightarrow x^2 + \frac{1}{x^2} = y^2 + 2 \\ \Rightarrow (y^2 + 2) - 3y - 2 = 0 &\quad [\text{from Eq. (i)}] \\ \Rightarrow y^2 - 3y = 0 &\Rightarrow y(y - 3) = 0 \end{aligned}$$

$$\therefore y = 0 \text{ or } 3$$

For $y = 0$

$$\Rightarrow x - \frac{1}{x} = 0$$

$$\Rightarrow \frac{x^2 - 1}{x} = 0$$

$$\Rightarrow x^2 = 1$$

$$\therefore x = \pm 1$$

For $y = 3$

$$\Rightarrow x - \frac{1}{x} = 3$$

$$\Rightarrow \frac{x^2 - 1}{x} = 3$$

$$\Rightarrow x^2 - 3x - 1 = 0$$

$$\Rightarrow x = \frac{3 \pm \sqrt{9 + 4}}{2}$$

$$\therefore x = \frac{3 \pm \sqrt{13}}{2}$$

So, $x = \left\{ \pm 1, \frac{3 \pm \sqrt{13}}{2} \right\}$ is the required solution.

Note An equation of degree n has n roots.

Word Problems Involving Quadratic Equation

In this section, we will discuss some problems based on practical applications of quadratic equation.

In such type of problems,

- We formulate a quadratic equation in variable x with the help of the given conditions.
- We then solve the quadratic equation to find the answer to the given problem.

- Sometimes it may happen that, out of the roots of the quadratic equation only one satisfies the given condition in the problem. So, the root which does not satisfy the condition of the problem will be rejected.

EXAMPLE 18. A positive number, when increased by 10 equals 200 times its reciprocal. What is number?

- a. 100 b. 10 c. 20 d. 200

Sol. b. Let the positive number be x .

Then, according to the question,

$$\begin{aligned} x + 10 &= \frac{200}{x} \Rightarrow x^2 + 10x = 200 \\ \Rightarrow x^2 + 10x - 200 &= 0 \Rightarrow x^2 + 20x - 10x - 200 = 0 \\ \Rightarrow x(x + 20) - 10(x + 20) &= 0 \Rightarrow (x - 10)(x + 20) = 0 \\ \therefore x &= 10, -20 \\ \text{But } x &\neq -20, \text{ since } x \text{ is a positive number.} \\ \text{So, the required number is } &10. \end{aligned}$$

EXAMPLE 19. Out of group of swans, $7/2$ times the square root of the number are swimming in the pool. While the two remaining are playing outside the pool. What is the total number of swans?

- a. 4 b. 8 c. 12 d. 16

Sol. d. Let total number of swans be x .

Number of swans swimming in the pool $= \frac{7}{2}\sqrt{x}$
 Remaining swans $= 2$

$$\text{By given condition, } \frac{7}{2}\sqrt{x} + 2 = x \Rightarrow \frac{7}{2}\sqrt{x} = x - 2$$

On squaring both sides, we get $\frac{49}{4}x = x^2 + 4 - 4x$

$$\Rightarrow 4x^2 - 65x + 16 = 0, \quad 4x^2 - 64x - x + 16 = 0$$

$$\Rightarrow 4x(x - 16) - 1(x - 16) = 0 \Rightarrow (x - 16)(4x - 1) = 0$$

$$\therefore x = 16 \quad \left[\because x \neq \frac{1}{4} \right]$$

Thus, total number of swans $= 16$

INEQUALITIES

Two real numbers or two algebraic expressions related by the symbols $>$, $<$, $\leq$ or $\geq$ form inequality. Here, the symbols $<$ (less than), $>$ (greater than), $\leq$ (less than or equal) and $\geq$ (greater than or equal) are known as inequality signs.

e.g. $5 < 7$, $x \leq 2$, $x + y \geq 11$ are all inequalities.

Linear Inequalities

A linear inequality is an inequality which involves a linear function i.e. each variable occurs in first degree only and there is no term involving the product of the variables.

e.g. $ax + b \leq 0$, $ax + by + c > 0$, $ax \leq 4$.

Linear inequality in one variable

A linear inequality which has only one variable is called linear inequality in one variable.

e.g. $ax + b < 0$, where $a \neq \left(\frac{-b}{a}\right)$, $4x + 7 \geq 0$

Linear inequality in two variables

A linear inequality which has two variables, is called linear inequality in two variables.

e.g. $3x + 11y \leq 0$, $4x + 3y > 0$

General Rules to Solve Linear Inequalities

- Equal numbers may be added or subtracted from both sides of an inequality without changing the sign of inequality.
e.g. If $a > b$, then for any number c ,
 $a + c > b + c$ or $a - c > b - c$.
- If both sides of an inequality are multiplied or divided by the same positive number, then the sign of inequality remains the same. e.g.
If $a > b$ and $c > 0$, then $\frac{a}{c} > \frac{b}{c}$ and $ac > bc$.
- If both sides of an inequality are multiplied or divided by the same negative number, then the sign of inequality is reversed.
e.g. If $a > b$ and $c < 0$, then $\frac{a}{c} < \frac{b}{c}$ and $ac < bc$.

Graphical Solution of Linear Inequalities in Two Variables

Let the inequality be $ax + by + c \geq 0$.

Step I Consider it as $ax + by + c = 0$ by changing the inequality sign to equality sign, Draw the graph of this equation.

Step II Choose any point not on the line, [usually $(0, 0)$]. Find whether this point satisfy the inequality or not. If it does, then shade the half-plane containing that point. Otherwise shade the other half-plane.

EXAMPLE 20. The solution set of inequality $2x + 3y \geq 6$, $x \geq 0$ and $y \geq 0$ is

- a. $x = 1, y = 1$ b. $x = -1, y = -1$
c. $x = 3, y = 1$ d. None of these

Sol. d. Draw the graph of $2x + 3y = 6$.

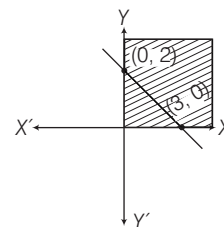
The values of (x, y) satisfying $2x + 3y = 6$ are

x	0	3
y	2	0

Clearly, here $(0, 0)$ does not satisfies the inequality, so we shade the plane which does not contain $(0, 0)$.

Also, since $x \geq 0$ and $y \geq 0$, so the solution set must be in 1st quadrant only.

Hence, the shaded region represents the solution set of the given system of inequalities.



Note (i) If the inequality is strict ($>$ or $<$), graph a dashed line. If the inequality is not strict ($\geq$ or $\leq$), graph a solid line.

(ii) In order to solve a system of two or more linear inequalities with the same variables, graph each inequality on the same set of axes. The intersection of the region of all the inequalities is the required solution set.

EXAMPLE 21. Find the solution set of $-2x + 3y > 6$ and $4x - 6y > 12$.

- a. $x < -1, y > -3$ b. $x < 1, y > 3$
c. $x \geq 3, y \geq 2$ d. No solution

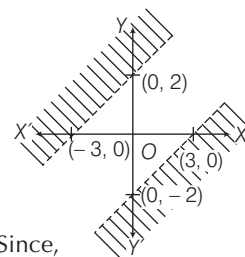
Sol. d. Consider, $-2x + 3y = 6$

x	0	-3
y	2	0

Again, $4x - 6y = 12$

x	0	3
y	-2	0

Now, graph each equation. Since, there is no intersection of these two shaded regions, therefore there is no solution.



Quadratic Inequalities

An equation of the form

$$ax^2 + bx + c \geq 0 \quad \text{or} \quad ax^2 + bx + c \leq 0$$

$$\text{or} \quad ax^2 + bx + c > 0 \quad \text{or} \quad ax^2 + bx + c < 0$$

where, $a \neq 0$ is called a quadratic inequality in one variable x . It is same as quadratic equations except for the inequality sign.

Solution of Quadratic Inequalities

Solving the inequality means finding the values of x that make the inequality true. Solution sets of quadratic inequalities are expressed in the form of intervals. Here are the steps:

- Replace the inequality symbol with an equal sign.
- Solve the quadratic equation by factorisation or by quadratic formula and find the real roots of the equation.
- Plot the two real roots on the numbers line. They divide the line into three sections or three intervals.

- Pick a number from each interval and test it in the original inequality.
- If the inequality holds true for the chosen point, then that interval is the solution of the given quadratic inequality.

Note (i) If the quadratic equation does not have real roots, then the quadratic expression is always positive (or always negative) depending on the sign of a meaning that the solution set will either be empty or the entire real number line.

e.g. $15x^2 - 18x + 7 > 0$ has no real roots as $D = b^2 - 4ac = 64 - 420 < 0$. Since $a > 0$, so the expression is always positive and hence the solution set will be the entire real number line.

(ii) When the inequality has an additional '=' sign ($\geq$ or $\leq$), use closed intervals like $[a, b]$ to indicate that the two end points are also included in the solution set. If the inequality is strict ($>$ or $<$), use open intervals like (a, b) as end points are not included.

EXAMPLE 22. The solution of the inequality

$$x^2 + 4x + 3 \geq 0 \text{ is}$$

- a. $x < -3$ and $x > -1$ b. $x \leq -3$ and $x \geq -1$
c. $x < -3$ and $x \geq -1$ d. $x \leq -3$ or $x > -1$

Sol. b. We have, $x^2 + 4x + 3 \geq 0$

Change the inequality to equality sign and solve

$$x^2 + 4x + 3 = 0 \Rightarrow x^2 + 3x + x + 3 = 0$$

$$\Rightarrow x(x+3) + (x+3) = 0 \Rightarrow (x+1)(x+3) = 0 \Rightarrow x = -1, -3.$$

Place the value of x on the number line to create intervals.

Take $x = -4$, Take $x = -2$, Take $x = 0$

$$\Rightarrow (-4)^2 + 4(-4) + 3 > 0,$$

$$(-2)^2 + 4(-2) + 3 < 0, 0^2 + 4(0) + 3 > 0$$



So, the intervals which satisfy the given equation are

$$(-\infty, -3] \text{ and } [-1, \infty)$$

$$\text{i.e. } x \leq -3 \text{ and } x \geq -1$$

EXAMPLE 23. The real values of x which satisfy $x^2 - 4x + 3 \geq 0$ and $x^2 - 3x - 4 \leq 0$ is

- a. $(-1, 1) \cup (3, 4)$ b. $[-1, 1] \cap [3, 4]$
c. $[-1, 1] \cup [3, 4]$ d. $(-1, 1) \cup [3, 4]$

Sol. c. Given, $x^2 - 4x + 3 \geq 0$

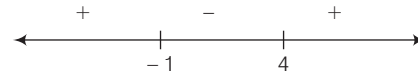
$$\Rightarrow x^2 - 4x + 3 = 0 \Rightarrow (x-1)(x-3) = 0 \Rightarrow x = 1, 3$$



$$\Rightarrow x \in (-\infty, 1] \text{ or } x \in [3, \infty). \Rightarrow x \in (-\infty, 1] \cup [3, \infty).$$

$$\text{Again, } x^2 - 3x - 4 \leq 0 \Rightarrow x^2 - 3x - 4 = 0$$

$$\Rightarrow (x+1)(x-4) = 0 \Rightarrow x = -1, 4$$



$$\Rightarrow x \in [-1, 4]$$

$\therefore$ Solution set is

$$x = (-\infty, 1] \cup [3, \infty) \cap [-1, 4] = [-1, 1] \cup [3, 4]$$

> PRACTICE EXERCISE

1. The quadratic equation has maximum

- (a) one root (b) two roots
(c) four roots (d) three roots

2. The values of x in the equation

$$a^2b^2x^2 - (a^2 + b^2)x + 1 = 0, a \neq 0, b \neq 0 \text{ is}$$

- (a) $1/a^2$ (b) $1/b^2$
(c) $1/a^2, 1/b^2$ (d) None of these

3. The value of ' a ' for which the equation $ax^2 - 2\sqrt{5}x + 4 = 0$ has equal roots is

- (a) $5/4$ (b) $4/5$
(c) $-5/4$ (d) $-5/3$

4. If one root of $3x^2 = 8x + (2k+1)$ is seven times the other, then the value of k is

- (a) $5/3$ (b) $-5/3$
(c) $2/3$ (d) $-3/2$

5. The quadratic equation whose roots are $\frac{4+\sqrt{7}}{2}$

and $\frac{4-\sqrt{7}}{2}$ is

- (a) $4x^2 + 16x + 9 = 0$ (b) $4x^2 - 16x - 9 = 0$
(c) $4x^2 - 16x + 9 = 0$ (d) $4x^2 + 16x - 9 = 0$

6. If α and β are the roots of the equation $x^2 - 8x + p = 0$ and $\alpha^2 + \beta^2 = 40$, then p is equal to

- (a) 12 (b) 10 (c) 9 (d) 11

7. If α and β are the roots of the equation $x^2 - 5x + 6 = 0$, then the value of $\alpha^2 - \beta^2$

- (a) 5 (b) -5 (c) ± 5 (d) ± 4

8. If α, β are the roots of the equation $ax^2 + bx + c = 0$, then an equation whose roots are $1/\alpha$ and $1/\beta$ is

- (a) $bx^2 + ax + c = 0$ (b) $ax^2 - bx + c = 0$
(c) $cx^2 + ax + b = 0$ (d) $cx^2 + bx + a = 0$

- 9.** If α, β are the roots of a quadratic equation such that $\alpha + \beta = 24$ and $\alpha - \beta = 8$, then the equation is
 (a) $x^2 - 24x - 128 = 0$ (b) $x^2 + 24x + 128 = 0$
 (c) $x^2 + 24x - 128 = 0$ (d) None of these
- 10.** Which one of the following is the equation whose roots are respectively three times the roots of the equation $ax^2 + bx + c = 0$?
 (a) $ax^2 + bx + c = 0$ (b) $ax^2 + 3bx + 9c = 0$
 (c) $ax^2 - 3bx + 9c = 0$ (d) $ax^2 + bx + 3c = 0$
- 11.** How many real values of x satisfy the equation $x^{2/3} + x^{1/3} - 2 = 0$?
 (a) only one value (b) two values
 (c) three values (d) No value
- 12.** If α, β are the roots of the quadratic equation $2x^2 - 4x + 1 = 0$. Then, the value of $\frac{1}{\alpha + 2\beta} + \frac{1}{\beta + 2\alpha}$ is equal to
 (a) $\frac{12}{17}$ (b) $\frac{17}{12}$ (c) $\frac{11}{17}$ (d) $\frac{13}{17}$
- 13.** Solve the equation $2^{x+2} + 2^{-x} = 5$ and find the roots of the equation.
 (a) $\{0, 2\}$ (b) $\{-2, 0\}$ (c) $\{0, 1\}$ (d) $\{-1, 0\}$
- 14.** If α and β are roots of the equation $x^2 + p = 0$ where p is a prime, then which equation has the roots $1/\alpha$ and $1/\beta$?
 (a) $\frac{1}{x^2} - \frac{1}{p} = 0$ (b) $px^2 + 1 = 0$ (c) $px^2 - 1 = 0$ (d) $\frac{1}{x^2} + \frac{1}{p} = 0$
- 15.** The roots of the equation $x^2 + px + q = 0$ are 1 and 2. The roots of the equation $qx^2 - px + 1 = 0$ must be
 (a) $\frac{-1}{2}$ and 1 (b) $\frac{1}{2}$ and 1
 (c) $\frac{-1}{2}$ and -1 (d) None of these
- 16.** Which one of the following is the quadratic equation whose roots are reciprocal to the roots of the quadratic equation $2x^2 - 3x - 4 = 0$?
 (a) $3x^2 - 2x - 4 = 0$ (b) $4x^2 + 3x - 2 = 0$
 (c) $3x^2 - 4x - 2 = 0$ (d) $4x^2 - 2x - 3 = 0$
- 17.** The value of x satisfying the equation $\sqrt{x+4} = x - 2$ is
 (a) 0, 5 (b) 0, 4
 (c) 5 (d) None of these
- 18.** If the roots of the quadratic equation $px^2 + qx + r = 0$ are reciprocal to each other, then
 (a) $q = r$ (b) $p = r$
 (c) q divides r (d) p divides q
- 19.** Sum of roots is -1 and sum of their reciprocals is $1/6$, then equation is
 (a) $x^2 - 6x + 1 = 0$ (b) $x^2 - x + 6 = 0$
 (c) $6x^2 + x + 1 = 0$ (d) $x^2 + x - 6 = 0$
- 20.** If sum of the roots of the equation $ax^2 + bx + c = 0$ is equal to the sum of their squares, then which one of the following is correct?
 (a) $a^2 + b^2 = c^2$ (b) $a^2 + b^2 = a + b$
 (c) $2ac = ab + b^2$ (d) $2c + b = 0$
- 21.** If the roots of $x^2 - lx + m = 0$ differ by 1, then
 (a) $l^2 = 4m - 1$ (b) $l^2 = 4m + 2$
 (c) $l = 4m^2 + 1$ (d) $l^2 = 4m + 1$
- 22.** If α, β are the roots of the equation $x^2 - (1 + a^2)x + \frac{1}{2}(1 + a^2 + a^4) = 0$, then $\alpha^2 + \beta^2$ is equal to
 (a) $a^4 + a^2$ (b) a^2
 (c) $(a^2 + a^4)^2$ (d) None of these
- 23.** If the roots of $\frac{1}{x+p} + \frac{1}{x+q} = \frac{1}{r}$ are equal in magnitude and opposite in sign, then product of roots is
 (a) $-\frac{1}{2}(p^2 + q^2)$ (b) $\frac{p^2 + q^2}{2}$
 (c) $\frac{p+q}{2}$ (d) $\frac{1}{2}(p+q)^2$
- 24.** If α, β are the roots of $3x^2 + 2x + 1 = 0$, then the equation whose roots are $\frac{1-\alpha}{1+\alpha}$ and $\frac{1-\beta}{1+\beta}$ is
 (a) $x^2 + 2x + 3 = 0$ (b) $x^2 - 2x + 3 = 0$
 (c) $x^2 + 2x - 3 = 0$ (d) $x^2 - 2x - 3 = 0$
- 25.** If one root of $px^2 + qx + r = 0$ is double of the other root, then which one of the following is correct?
 (a) $2q^2 = 9pr$ (b) $2q^2 = 9p$ (c) $4q^2 = 9r$ (d) $9q^2 = 2pr$
- 26.** If the roots of the equation $x^2 + x + 1 = 0$ are in the ratio $m : n$, then
 (a) $\sqrt{\frac{m}{n}} + \sqrt{\frac{n}{m}} + 1 = 0$ (b) $\sqrt{m} + \sqrt{n} + 1 = 0$
 (c) $\frac{m}{n} + \frac{n}{m} + 1 = 0$ (d) $m + n + 1 = 0$
- 27.** What is one of the value of x in the equation $\sqrt{\frac{x}{1-x}} + \sqrt{\frac{1-x}{x}} = \frac{13}{6}$?
 (a) $\frac{5}{13}$ (b) $\frac{7}{13}$ (c) $\frac{9}{13}$ (d) $\frac{11}{3}$

- 28.** What are the roots of the equation

$$(a + b + x)^{-1} = a^{-1} + b^{-1} + x^{-1} ?$$

- (a) a, b (b) $-a, b$ (c) $a, -b$ (d) $-a, -b$

- 29.** The number of roots of the quadratic equation $8 \sec^2 \phi - 6 \sec \phi + 1 = 0$ is

- (a) n (b) 2 (c) 0 (d) No solution

- 30.** The number of real roots of $3^{2x^2 - 7x + 7} = 9$ is

- (a) 3 (b) 1 (c) 2 (d) 4

- 31.** If $x = \sqrt{2} + 2$, then

- (a) $x^2 + 4x + 2 = 0$ (b) $x^2 - 2x - 2 = 0$
(c) $x^2 - 4x + 2 = 0$ (d) $x^2 - 4x - 2 = 0$

- 32.** For what value of k will the equation $\frac{x^2 - bx}{ax - c} = \frac{k - 1}{k + 1}$ have roots reciprocal to each other?

- (a) $\frac{1+c}{1-c}$ (b) $\frac{c+1}{c-1}$ (c) $a - c$ (d) $\frac{b+c}{c-1}$

- 33.** If ' α ' and ' β ' be the roots of $ax^2 - bx + b = 0$, the value of $\sqrt{\frac{\alpha}{\beta}} + \sqrt{\frac{\beta}{\alpha}}$ is

- (a) a/b (b) $\sqrt{b/a}$ (c) $\sqrt{a/b}$ (d) $-a/b$

- 34.** If the equations $2x^2 - 7x + 3 = 0$ and $4x^2 + ax - 3 = 0$ have a common root, then what is the value of a ?

- (a) -11 or 4 (b) -11 or -4 (c) 11 or -4 (d) 11 or 4

- 35.** If the roots of $x^2 + bx + c = 0$ be α and β and those of $x^2 + px + q = 0$ be $k\alpha$ and $k\beta$, then

- (a) $cb^2 = qp^2$ (b) $qc^2 = b^2p^2$
(c) $qb^2 = cp^2$ (d) None of these

- 36.** If -4 is a root of the equation $x^2 + px - 4 = 0$ and the equation $x^2 + px + q = 0$ has equal roots, then the values of p and q are, respectively

- (a) -3 and $\frac{9}{4}$ (b) 3 and $\frac{9}{4}$ (c) $\frac{9}{4}$ and 3 (d) 4 and 3

- 37.** If α, β are the roots of $2x^2 - 6x + 3 = 0$, then the value of $\left(\frac{\alpha}{\beta} + \frac{\beta}{\alpha}\right) + 3\left(\frac{1}{\alpha} + \frac{1}{\beta}\right) + 2\alpha\beta$ is

- (a) 12 (b) 23 (c) 13 (d) -13

- 38.** If the roots of the equation $x^2 + 2ax + b = 0$, are real and distinct and they differ by at most $2m$, then b lies in the interval

- (a) $(a^2 - m^2, a^2)$ (b) $[a^2 - m^2, a^2)$
(c) $(a^2, a^2 + m^2)$ (d) None of these

- 39.** The solution of the equation

$$(x + 2)(x - 5)(x - 6)(x + 1) = 144 \text{ is}$$

- (a) $\{7, -3\}$ (b) $\{7, -3, 2\}$ (c) $\{7, -3, 2, 1\}$ (d) $\{7, 2\}$

- 40.** The solution of the equation

$$\sqrt{x^2 - 16} - (x - 4) = \sqrt{x^2 - 5x + 4} \text{ is}$$

- (a) $\left\{4, 5, -\frac{13}{3}\right\}$ (b) $\{4, 5\}$ (c) $\{4\}$ (d) $\left\{5, -\frac{13}{3}\right\}$

- 41.** If $\sin \theta$ and $\cos \theta$ are the roots of the equation $ax^2 - bx + c = 0$, then which one of the following is correct?

- (a) $a^2 + b^2 + 2ac = 0$ (b) $a^2 - b^2 + 2ac = 0$
(c) $a^2 + c^2 + 2ab = 0$ (d) $a^2 - b^2 - 2ac = 0$

- 42.** The equation $(1 + n^2)x^2 + 2ncx + (c^2 - a^2) = 0$ will have equal roots, if

- (a) $c^2 = 1 + a^2$ (b) $c^2 = 1 - a^2$
(c) $c^2 = 1 + n^2 + a^2$ (d) $c^2 = (1 + n^2)a^2$

- 43.** What is the condition that the equation $ax^2 + bx + c = 0$, where $a \neq 0$ has both the roots positive?

- (a) a, b and c are of same sign
(b) a and b are of same sign
(c) b and c have the same sign opposite to that of a
(d) a and c have the same sign opposite to that of b

- 44.** The equation whose roots are twice the roots of the equation $x^2 - 2x + 4 = 0$ is

- (a) $x^2 - 2x + 4 = 0$ (b) $x^2 - 2x + 16 = 0$
(c) $x^2 - 4x + 8 = 0$ (d) $x^2 - 4x + 16 = 0$

- 45.** If α and β are the roots of the equation $x^2 + px + q = 0$, then $-\alpha^{-1}, -\beta^{-1}$ are the roots of which one of the following equations?

- (a) $qx^2 - px + 1 = 0$ (b) $q^2 + px + 1 = 0$
(c) $x^2 + px - q = 0$ (d) $x^2 - px + q = 0$

- 46.** If one root of the equation $ax^2 + x - 3 = 0$ is -1 , then what is the other root?

- (a) $\frac{1}{4}$ (b) $\frac{1}{2}$ (c) $\frac{3}{4}$ (d) 1

- 47.** If the equation $(a^2 + b^2)x^2 - 2(ac + bd)x + (c^2 + d^2) = 0$ has equal roots, then which one of the following is correct?

- (a) $ab = cd$ (b) $ad = bc$
(c) $a^2 + c^2 = b^2 + d^2$ (d) $ac = bd$

- 48.** What is the solution of the equation

$$\sqrt{\frac{x}{x+3}} - \sqrt{\frac{x+3}{x}} = -\frac{3}{2} ?$$

- (a) 1 (b) 2 (c) 4 (d) None of these

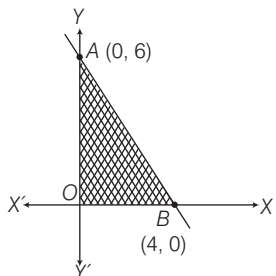
- 49.** What are the roots of the equation

$$4^x - 3 \cdot 2^{x+2} + 32 = 0 ?$$

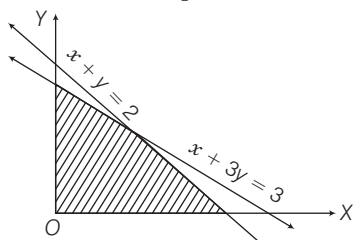
- (a) 1, 2 (b) 3, 4 (c) 2, 3 (d) 1, 3

- 50.** If α and β are the roots of the equation $x^2 - x - 1 = 0$, then what is the value of $(\alpha^4 + \beta^4)$?
 (a) 7 (b) 0
 (c) 2 (d) None of these
- 51.** When the roots of the quadratic equation $ax^2 + bx + c = 0$ are negative and reciprocals of each other, then which one of the following is correct?
 (a) $b = 0$ (b) $c = 0$ (c) $a = c$ (d) $a = -c$
- 52.** If sum as well as product of roots of a quadratic equation is 9, then what is the equation?
 (a) $x^2 + 9x - 18 = 0$ (b) $x^2 - 18x + 9 = 0$
 (c) $x^2 + 9x + 9 = 0$ (d) $x^2 - 9x + 9 = 0$
- 53.** What are the roots of the equation $\log_{10}(x^2 - 6x + 45) = 2$?
 (a) 9, -5 (b) -9, 5 (c) 11, -5 (d) -11, 5
- 54.** The sum of the roots of the equation $\frac{1}{x+a} + \frac{1}{x+b} = \frac{1}{c}$ is zero. What is the product of the roots of the equation?
 (a) $-\frac{(a+b)}{2}$ (b) $\frac{(a+b)}{2}$
 (c) $-\frac{(a^2+b^2)}{2}$ (d) $\frac{(a^2+b^2)}{2}$
- 55.** For what value of k , will the roots of the equation $kx^2 - 5x + 6 = 0$ be in the ratio of 2 : 3?
 (a) 0 (b) 1 (c) -1 (d) 2
- 56.** What is the ratio of sum of squares of roots to the product of the roots of the equation $7x^2 + 12x + 18 = 0$?
 (a) 6 : 1 (b) 1 : 6 (c) -6 : 1 (d) -6 : 7
- 57.** If the roots of the equation $\frac{x(x-1)-(m+1)}{(x-1)(m-1)} = \frac{x}{m}$ are equal, then what is the value of m ?
 (a) 1 (b) $1/2$ (c) 0 (d) $-1/2$
- 58.** If $3^x + 27(3^{-x}) = 12$, then what is the value of x ?
 (a) 1 (b) 2 (c) 1, 2 (d) 0, 1
- 59.** What is the magnitude of difference of the roots of $x^2 - ax + b = 0$?
 (a) $\sqrt{a^2 - 4b}$ (b) $\sqrt{b^2 - 4a}$
 (c) $2\sqrt{a^2 - 4b}$ (d) $\sqrt{b^2 - 4ab}$
- 60.** What can be said about the roots of the equation $x^2 - x - 2 = 0$?
 (a) both of them are integers
 (b) both of them are natural numbers
 (c) the latter of the two is negative
 (d) None of the above
- 61.** An equation equivalent to the quadratic equation $x^2 - 6x + 5 = 0$ is
 (a) $x^2 - 5x + 6 = 0$ (b) $5x^2 - 6x + 1 = 0$
 (c) $|x - 3| = 2$ (d) $6x^2 - 5x + 1 = 0$
- 62.** If the sum of a number and its reciprocal is $\frac{10}{3}$, then the numbers are
 (a) $3, \frac{1}{3}$ (b) $3, -\frac{1}{3}$ (c) $-3, \frac{1}{3}$ (d) $-3, -\frac{1}{3}$
- 63.** Divide 16 into two parts such that the twice of the square of the greater part exceeds the square of the smaller part by 164. Then, the greater part is
 (a) 58 (b) 10 (c) 6 (d) 15
- 64.** The number of straight lines that can connect 'x' points is given by the equation $y = \frac{x(x-1)}{2}$.
 How many points does a figure have if only 15 lines can be drawn connecting them?
 (a) 15 (b) 10 (c) 6 (d) 5
- 65.** The two successive natural numbers whose squares have sum 221 are
 (a) 10 and 11 (b) 11 and 12
 (c) -10 and -11 (d) None of these
- 66.** The two consecutive positive odd integers, the sum of whose squares is 130.
 (a) -7 and -9 (b) 7 and 9
 (c) 7 and 5 (d) 3 and -5
- 67.** The solution set of inequation $2x + 1 \geq 7$ is
 (a) $x \geq 8$ (b) $x \geq 3$
 (c) $x \geq 6$ (d) None of these
- 68.** The values of x satisfying inequation $\frac{x-1}{3} \geq 4$ is
 (a) $x \leq 13$ (b) $x \geq 12$ (c) $x \geq 13$ (d) $x = 13$
- 69.** The values of x satisfying $3x + 2 \leq 5x - (4 - x)$ is
 (a) $x \leq 2$ (b) $x \geq 2$
 (c) $x = 2$ (d) None of these
- 70.** The solution set of x for the inequations $2x + 3 \geq 8$ and $3x + 1 \leq 12$ is
 (a) $\frac{5}{2} < x \leq \frac{11}{3}$ (b) $\frac{5}{2} < x < \frac{11}{3}$
 (c) $\frac{5}{2} \leq x \leq \frac{11}{3}$ (d) $\frac{5}{2} \geq x \geq \frac{11}{3}$
- 71.** The values of x satisfying $\frac{1}{2} \left(\frac{3}{5}x + 4 \right) \geq \frac{1}{3}(x - 6)$ are
 (a) $x \geq 120$ (b) $x \leq 120$ (c) $x \leq 12$ (d) $x \geq 12$
- 72.** $4x^2 - 1 \leq 0$, then the solution set is
 (a) $-\frac{1}{2} \leq x \leq \frac{1}{2}$ (b) $-\frac{1}{2} < x < \frac{1}{2}$
 (c) $-\frac{1}{2} \geq x \geq \frac{1}{2}$ (d) None of these

73. The shaded region, including the boundary in the given graph, is exactly represented by



- (a) $3x + 2y \leq 12, x < 0, y \geq 0$ (b) $3x + 2y \leq 12, x \geq 0, y \geq 0$
 (c) $3x + 2y < 12, x \geq 0, y \geq 0$ (d) $3x + 2y > 12, x \geq 0, y \geq 0$
74. When is the expression $x^2 + 3x - 10$ positive only?
 (a) $x \leq -5$ (b) $x \geq 2$ (c) $-5 < x < 2$ (d) $x < -5$ or $x > 2$
75. The solution set for the quadratic inequation $x^2 - 5x + 6 \geq 0$ is
 (a) $[-\infty, 2] \cup [3, \infty]$ (b) $[-\infty, 2] \cup [3, \infty]$
 (c) $[-\infty, 2] \cup [3, \infty]$ (d) $[2, 3]$
76. All real values of x for which $\frac{x-2}{3x+1} < \frac{x-3}{3x-2}$ are
 (a) $\left[\frac{-1}{3}, \frac{2}{3}\right]$ (b) $\left[\frac{-1}{3}, \frac{2}{3}\right]$ (c) $\left[\frac{-1}{3}, \frac{2}{3}\right]$ (d) R
77. If $x + 2y \leq 3, x > 0$ and $y > 0$, then one of the solution is
 (a) $x = -1, y = 2$ (b) $x = 2, y = 1$
 (c) $x = 1, y = 1$ (d) $x = 0, y = 0$
78. The shaded region in the given figure is the solution set of the inequalities



- (a) $x + y \leq 2, x + 3y \geq 3, x \geq 0, y \geq 0$
 (b) $x + y \geq 2, x + 3y \geq 3, x \geq 0, y \geq 0$
 (c) $x + y \geq 2, x + 3y \leq 3, x \geq 0, y \geq 0$
 (d) $x + y \leq 2, x + 3y \leq 3, x \geq 0, y \geq 0$
79. If $a + b = 2m^2, b + c = 6m, a + c = 2$, where m is a real number and $a \leq b \leq c$, then which one of the following is correct?
 (a) $0 \leq m \leq 1/2$ (b) $-1 \leq m \leq 0$
 (c) $1/3 \leq m \leq 1$ (d) $1 < m \leq 2$
80. If α and β are the roots of the equation $(x^2 - 3x + 2 = 0)$, then which of the following equation has the roots $(\alpha + 1)$ and $(\beta + 1)$?
 I. $x^2 + 5x + 6 = 0$ II. $x^2 - 5x - 6 = 0$

Select the answer using the codes given below.

- (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II
81. Consider the following statements
 I. Every quadratic equation has atleast one real root.
 II. A quadratic equation with integral coefficients has integral roots.
 III. If the coefficient of x^2 and the constant term of a quadratic equation have opposite signs, then the quadratic equation has real roots.

Which of the above statements is/are correct?

- (a) Only I and II (b) Only II and III
 (c) Only III (d) All of these
82. Which of the following are quadratic equations?
 I. $x^2 + \frac{1}{x^2} = 2$ II. $x + \frac{3}{x} = x^2$
 III. $2x^2 - x + 2 = x^2 + 4x - 4$ IV. $x^3 + 6x^2 + 2x - 1 = 0$

Select the correct answer using the codes given below.

- (a) I, III are quadratic (b) II, III and IV are quadratic
 (c) Only III is quadratic (d) None of these
83. Consider the following statements
 I. $x = 1$ is a root of $3x^2 - 2x - 1 = 0$.
 II. $x = -2\sqrt{2}$ is a root of $x^2 + \sqrt{2}x - 4 = 0$.
 III. $x^2 + x + 1 = 0; x = 1, x = -1$.
 IV. $9x^2 - 3x - 2 = 0; x = -\frac{1}{3}, x = \frac{2}{3}$.

Which of the equation(s) given above is/are correct?

- (a) I and II (b) I, II and IV
 (c) III and IV (d) Only IV

Directions (Q. Nos. 84-86) A ball is thrown upwards from a rooftop, 80 m above the ground. It will reach a maximum vertical height and then fall back to the ground. The height of the ball from the ground at time t is h which is given by, $h = -16t^2 + 64t + 80$ Now, answer the following questions based on the above information.

84. What is the height reached by the ball after 1 sec?
 (a) 150 m (b) 128 m
 (c) 64 m (d) None of these
85. What is the maximum height reached by the ball?
 (a) 110 m (b) 132 m
 (c) 144 m (d) cannot be determined
86. How long will it take before hitting the ground?
 (a) 5 sec (b) 6 sec
 (c) 3 sec (d) cannot say

PREVIOUS YEARS' QUESTIONS

- 87.** If the roots of the equation $x^2 - 2ax + a^2 + a - 3 = 0$ are real and less than 3, then which one of the following is correct? **2012 I**
(a) $a < 2$ (b) $2 < a < 3$ (c) $3 < a < 4$ (d) $a > 4$
- 88.** If one of the roots of quadratic equation $7x^2 - 50x + k = 0$ is 7, then what is the value of k ? **2012 I**
(a) 7 (b) 1 (c) 50/7 (d) 7/50
- 89.** Two students A and B solve an equation of the form $x^2 + px + q = 0$. A starts with a wrong value of p and obtains the roots as 2 and 6. B starts with a wrong value of q and gets the roots as 2 and -9 . What are the correct roots of the equation? **2012 II**
(a) 3 and -4 (b) -3 and -4 (c) -3 and 4 (d) 3 and 4
- 90.** If one of the roots of the equation $x^2 - bx + c = 0$ is the square of the other, then which of the following option is correct? **2013 I**
(a) $b^3 = 3bc + c^2 + c$ (b) $c^3 = 3bc + b^2 + b$
(c) $3bc = c^3 + b^2 + b$ (d) $3bc = c^3 + b^3 + b^2$
- 91.** The difference of the roots of the equation $2x^2 - 11x + 5 = 0$ is **2013 I**
(a) 4.5 (b) 4 (c) 3.5 (d) 3
- 92.** The sum of the squares of two numbers is 97 and the squares of their difference is 25. The product of the two numbers is **2013 I**
(a) 45 (b) 36 (c) 54 (d) 63
- 93.** If $x + \frac{1}{x} = 2$, then what is value of $x - \frac{1}{x}$? **2013 I**
(a) 0 (b) 1 (c) 2 (d) -2
- 94.** There are some benches in a classroom having the number of rows 4 more than the number of columns. If each bench is seated with 5 students, there are two seats vacant in a class of 158 students. The number of rows is **2013 I**
(a) 4 (b) 8 (c) 6 (d) 10
- 95.** Which one of the following is factor of $x^2 + \frac{1}{x^2} + 8\left(x + \frac{1}{x}\right) + 14$? **2013 II**
(a) $x + \frac{1}{x} + 1$ (b) $x + \frac{1}{x} + 3$
(c) $x + \frac{1}{x} + 6$ (d) $x + \frac{1}{x} + 7$
- 96.** If α and β are the roots of the equation $x^2 - x - 1 = 0$, then what is $\frac{\alpha^2 + \beta^2}{(\alpha^2 - \beta^2)(\alpha - \beta)}$ equal to? **2013 II**
(a) $\frac{2}{5}$ (b) $\frac{3}{5}$ (c) $\frac{4}{5}$ (d) None of these
- 97.** If $x^2 = 6 + \sqrt{6 + \sqrt{6 + \sqrt{6 + \dots \infty}}}$, then what is one of the values of x equal to? **2013 II**
(a) 6 (b) 5 (c) 4 (d) 3
- 98.** Consider the following statements in respect of the quadratic equation $ax^2 + bx + b = 0$, where $a \neq 0$.
I. The product of the roots is equal to the sum of the roots.
II. The roots of the equation are always unequal and real.
Which of the above statements is/are correct? **2014 I**
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II
- 99.** If the roots of the equation $Ax^2 + Bx + C = 0$ are -1 and 1 , then which one of the following is correct? **2014 I**
(a) A and C are both zero
(b) A and B are both positive
(c) A and C are both negative
(d) A and C are of opposite sign
- 100.** If the roots of the equation $(a^2 - bc)x^2 + 2(b^2 - ac)x + (c^2 - ab) = 0$ are equal, where $b \neq 0$, then which one of the following is correct? **2014 I**
(a) $a + b + c = abc$ (b) $a^2 + b^2 + c^2 = 0$
(c) $a^3 + b^3 + c^3 = 0$ (d) $a^3 + b^3 + c^3 = 3abc$
- 101.** If m and n are the roots of the equation $ax^2 + bx + c = 0$, then the equation whose roots are $(m^2 + 1)/m$ and $(n^2 + 1)/n$ is **2014 I**
(a) $acx^2 + (ab + bc)x + b^2 + (a - c)^2 = 0$
(b) $acx^2 + (ab - bc)x + b^2 + (a - c)^2 = 0$
(c) $acx^2 + (ab - bc)x + b^2 - (a - c)^2 = 0$
(d) $acx^2 + (ab + bc)x + b^2 - (a - c)^2 = 0$
- 102.** In solving a problem, one student makes a mistake in the coefficient of the first degree term and obtains -9 and -1 for the roots. Another student makes a mistake in the constant term of the equation and obtains 8 and 2 for the roots. The correct equation was **2014 I**
(a) $x^2 + 10x + 9 = 0$ (b) $x^2 - 10x + 16 = 0$
(c) $x^2 - 10x + 9 = 0$ (d) None of these
- 103.** If m and n ($m > n$) are the roots of the equation $7(x + 2a)^2 + 3a^2 = 5a(7x + 23a)$, where $a > 0$, then what is $3m - n$ equal to? **2014 II**
(a) $12a$ (b) $14a$ (c) $15a$ (d) $18a$
- 104.** If one of the roots of the equation $px^2 + qx + r = 0$ is three times the other, then which one of the following relations is correct? **2014 II**
(a) $3q^2 = 16pr$ (b) $q^2 = 24pr$
(c) $p = q + r$ (d) $p + q + r = 1$

- (a) $x > 0$ (b) $x < 0$
(c) $\frac{-1-\sqrt{5}}{5} \leq x \leq \frac{-1+\sqrt{5}}{2}$ (d) $x \leq \frac{-1-\sqrt{5}}{2}$ or $x \geq \frac{-1+\sqrt{5}}{2}$

ANSWERS

1	b	2	c	3	a	4	b	5	c	6	a	7	c	8	d	9	d	10	b
11	b	12	a	13	b	14	b	15	c	16	b	17	c	18	b	19	d	20	c
21	d	22	b	23	a	24	b	25	a	26	a	27	c	28	d	29	d	30	c
31	c	32	b	33	b	34	a	35	c	36	b	37	c	38	b	39	b	40	b
41	b	42	d	43	d	44	d	45	a	46	c	47	b	48	a	49	c	50	a
51	c	52	d	53	c	54	c	55	b	56	d	57	d	58	c	59	a	60	a
61	c	62	a	63	b	64	c	65	a	66	b	67	b	68	c	69	b	70	c
71	b	72	a	73	b	74	d	75	a	76	b	77	c	78	d	79	c	80	d
81	c	82	c	83	b	84	b	85	c	86	a	87	a	88	a	89	b	90	a
91	a	92	b	93	a	94	b	95	c	96	b	97	d	98	d	99	d	100	d
101	a	102	c	103	c	104	a	105	b	106	a	107	c	108	c	109	c	110	c
111	a	112	d	113	c	114	a	115	b	116	c	117	a	118	a	119	a	120	d

HINTS AND SOLUTIONS

1. (b) An equation of n degree has n roots.
So, quadratic equation has two roots.

2. (c) $a^2b^2x^2 - a^2x - b^2x + 1 = 0$
 $\Rightarrow a^2x(b^2x - 1) - 1(b^2x - 1) = 0$
 $\Rightarrow (b^2x - 1)(a^2x - 1) = 0$
 So, either $a^2x - 1 = 0 \Rightarrow a^2x = 1$
 or $b^2x - 1 = 0$ or $b^2x = 1$
 $\Rightarrow x = 1/a^2$ or $x = 1/b^2$

3. (a) Given, $ax^2 - 2\sqrt{5}x + 4 = 0$ has equal roots.
 $\therefore$ Discriminant
 $= (-2\sqrt{5})^2 - 4(a)4 = 0$
 $[\because D = B^2 - 4AC]$
 $\Rightarrow 20 - 16a = 0 \Rightarrow a = 5/4$

4. (b) Given, $3x^2 = 8x + (2k + 1)$
 or $3x^2 - 8x - (2k + 1) = 0$
 Let first and second root be α and 7α , respectively.
 $\therefore$ Sum of roots $= \alpha + 7\alpha = \frac{8}{3}$
 $\Rightarrow 8\alpha = \frac{8}{3} \Rightarrow \alpha = \frac{1}{3}$
 So, roots are $\frac{1}{3}$ and $\frac{7}{3}$.
 Also, product of roots $= \frac{1}{3} \times \frac{7}{3} = \frac{7}{9}$
 $\Rightarrow \frac{7}{9} = \frac{-(2k+1)}{3}$
 $\Rightarrow \frac{7}{3} = -(2k+1)$
 $\Rightarrow 7 = -6k - 3 \Rightarrow 10 = -6k$
 $\Rightarrow k = \frac{-5}{3}$

5. (c) Here, sum of roots,

$$S = \frac{4 + \sqrt{7}}{2} + \frac{4 - \sqrt{7}}{2} = 4$$

Product of roots,

$$P = \left(\frac{4 + \sqrt{7}}{2} \right) \left(\frac{4 - \sqrt{7}}{2} \right) = \frac{16 - 7}{4} = \frac{9}{4}$$

The required equation is

$$x^2 - Sx + P = 0$$

$$x^2 - 4x + \frac{9}{4} = 0$$

$$\Rightarrow 4x^2 - 16x + 9 = 0$$

6. (a) Given, $x^2 - 8x + p = 0$
 Sum of roots $\alpha + \beta = 8$ and product of roots $\alpha\beta = p$

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta \quad \dots(i)$$

$$\Rightarrow 40 = (8)^2 - 2(p) [\because \alpha^2 + \beta^2 = 40]$$

$$\Rightarrow 40 - 64 = -2p$$

$$\Rightarrow -24 = -2p \Rightarrow p = 12$$

7. (c) Given, $x^2 - 5x + 6 = 0$

$$\text{Sum of roots} = -\frac{\text{Coefficient of } x}{\text{Coefficient of } x^2}$$

$$\Rightarrow \alpha + \beta = 5$$

$$\text{Product of roots} = \frac{\text{Constant term}}{\text{Coefficient of } x^2}$$

$$\Rightarrow \alpha\beta = 6$$

$$\text{Now, } (\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$$

$$= (5)^2 - 4 \times 6 = 1$$

$$\Rightarrow \alpha - \beta = \pm 1 \text{ and } \alpha + \beta = 5$$

$$\Rightarrow \alpha^2 - \beta^2 = (\alpha + \beta)(\alpha - \beta)$$

$$= 5(\pm 1)$$

$$\Rightarrow \alpha^2 - \beta^2 = \pm 5$$

8. (d) Here, $\alpha + \beta = -b/a$ and $\alpha\beta = c/a$

Now, roots of required equation are

$$\frac{1}{\alpha}, \frac{1}{\beta}$$

$$\Rightarrow S = \frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} = \frac{-b/a}{c/a} = \frac{-b}{c}$$

$$P = \frac{1}{\alpha} \cdot \frac{1}{\beta} = \frac{1}{c/a} = \frac{a}{c}$$

$\therefore$ Required quadratic equation is

$$x^2 - Sx + P = 0$$

$$\Rightarrow x^2 - \left(\frac{-b}{c} \right) x + \frac{a}{c} = 0$$

$$\Rightarrow cx^2 + bx + a = 0$$

9. (d) Given, $\alpha + \beta = 24$ and $\alpha - \beta = 8$

On solving, we get $\alpha = 16$ and $\beta = 8$

$\therefore$ Sum of roots $= \alpha + \beta = 24$

and product of roots $= 16 \times 8 = 128$

So, required equation is

$$x^2 - 24x + 128 = 0.$$

10. (b) Let α and β be the roots of the equation $ax^2 + bx + c = 0$.

$$\text{Then, } \alpha + \beta = \frac{-b}{a} \text{ and } \alpha\beta = \frac{c}{a}$$

$$\text{Now, } 3\alpha + 3\beta = \frac{-3b}{a} \text{ and } 3\alpha \cdot 3\beta = \frac{9c}{a}$$

$\therefore$ Required equation $= x^2 -$

(Sum of roots) x
 + Product of roots

$$\Rightarrow x^2 - \left(\frac{-3b}{a} \right) x + \frac{9c}{a} = 0$$

$$\Rightarrow ax^2 + 3bx + 9c = 0$$

11. (b) Given equation is

$$x^{2/3} + x^{1/3} - 2 = 0$$

$$\Rightarrow (x^{1/3})^2 + x^{1/3} - 2 = 0$$

$$\text{Let } x^{1/3} = x$$

$$\Rightarrow x^2 + x - 2 = 0,$$

It is a quadratic equation in x .

$\therefore$ Discriminant of $x^2 + x - 2 = 0$ is

$$B^2 - 4AC = 1^2 - 4(1)(-2) = 9 > 0$$

Hence, two real values of x satisfy the given equation.

12. (a) Here, $\alpha + \beta = \frac{4}{2} = 2$, $\alpha\beta = \frac{1}{2}$

$$\text{Now, } \frac{1}{\alpha + 2\beta} + \frac{1}{\beta + 2\alpha}$$

$$= \frac{\beta + 2\alpha + \alpha + 2\beta}{(\alpha + 2\beta)(\beta + 2\alpha)}$$

$$= \frac{3\alpha + 3\beta}{\alpha\beta + 2\alpha^2 + 2\beta^2 + 4\alpha\beta}$$

$$= \frac{3(\alpha + \beta)}{2(\alpha + \beta)^2 + \alpha\beta} = \frac{3(2)}{2(2)^2 + \frac{1}{2}} = \frac{12}{17}$$

13. (b) Given equation is $2^{x+2} + 2^{-x} = 5$

$$\Rightarrow 2^x 2^2 + 2^{-x} = 5 \Rightarrow 4 \cdot 2^x + \frac{1}{2^x} = 5$$

$$\text{Put } 2^x = y, 4y + \frac{1}{y} = 5$$

$$\Rightarrow 4y^2 + 1 = 5y \Rightarrow 4y^2 - 5y + 1 = 0$$

$$\Rightarrow (y-1)(4y-1) = 0$$

$$\Rightarrow y-1=0 \text{ or } 4y-1=0$$

$$\Rightarrow y=1 \text{ or } 4y=1$$

$$\Rightarrow y=1, \frac{1}{4}$$

By condition,

$$2^x = y \quad \left| \quad 2^x = y \right.$$

$$\Rightarrow 2^x = 1 \quad \left| \quad \Rightarrow 2^x = \frac{1}{4} \right.$$

$$\Rightarrow 2^x = 2^0 \quad \left| \quad \Rightarrow 2^x = \frac{1}{2^2} = 2^{-2} \right.$$

$$\therefore x=0 \quad \left| \quad \therefore x=-2 \right.$$

Hence, the roots of the equation are 0 and -2.

14. (b) Since, α and β are roots of the equation $x^2 + p = 0$.

$$\therefore \alpha + \beta = 0 \text{ and } \alpha\beta = p$$

$$\text{So, } \frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} = 0$$

$$\text{and } \frac{1}{\alpha} \cdot \frac{1}{\beta} = \frac{1}{\alpha\beta} = \frac{1}{p}$$

$$\therefore \text{ Required equation } x^2 - 0 \cdot x + \frac{1}{p} = 0$$

$$\Rightarrow px^2 + 1 = 0$$

15. (c) The equation is $x^2 + px + q = 0$.

$$\text{Sum of roots} = -p = 1 + 2$$

$$\Rightarrow p = -3$$

$$\text{Product of roots} = q = 1 \times 2 = 2$$

$\therefore$ Equation $qx^2 - px + 1 = 0$ becomes

$$2x^2 - (-3)x + 1 = 0$$

$$\Rightarrow 2x^2 + 3x + 1 = 0$$

$$\Rightarrow (2x+1)(x+1) = 0$$

$$\therefore x = -\frac{1}{2} \text{ or } x = -1$$

16. (b) Given, $2x^2 - 3x - 4 = 0$

For getting a reciprocal roots, we replace

x by $\frac{1}{x}$, we get

$$2\left(\frac{1}{x}\right)^2 - 3\left(\frac{1}{x}\right) - 4 = 0$$

$$\Rightarrow \frac{2}{x^2} - \frac{3}{x} - 4 = 0$$

$$\Rightarrow -4x^2 - 3x + 2 = 0$$

$$\Rightarrow 4x^2 + 3x - 2 = 0$$

17. (c) As, $\sqrt{x+4} = x-2$

On squaring both sides, we get

$$(x+4) = (x-2)^2$$

$$\Rightarrow x+4 = x^2 + 4 - 4x$$

$$\Rightarrow x^2 - 5x = 0 \Rightarrow x = 0, x = 5$$

$$\text{But for } x = 0, \sqrt{0+4} = 0-2$$

$$\sqrt{4} \neq -2$$

So, $x = 5$ is the only solution.

18. (b) Let roots of equation be α and $\frac{1}{\alpha}$.

$\therefore$ Product of roots

$$= \alpha \times \frac{1}{\alpha} = \frac{\text{Constant term}}{\text{Coefficient of } x^2} = \frac{r}{p}$$

$$\Rightarrow 1 = \frac{r}{p} \Rightarrow r = p$$

19. (d) Let roots be α and β , then $\alpha + \beta = -1$

$$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{1}{6} \Rightarrow \frac{\beta + \alpha}{\alpha\beta} = \frac{1}{6}$$

$$\frac{-1}{\alpha\beta} = \frac{1}{6} \Rightarrow \alpha\beta = -6$$

$\therefore$ Required equation is,

$$x^2 - (\alpha + \beta)x + \alpha\beta = 0$$

$$\Rightarrow x^2 - (-1)x + (-6) = 0$$

$$\therefore x^2 + x - 6 = 0$$

20. (c) Let α and β be the roots of the equation

$$ax^2 + bx + c = 0$$

$$\therefore \text{ Sum of roots } (\alpha + \beta) = -\frac{b}{a}$$

$$\text{and product of roots } (\alpha\beta) = \frac{c}{a}$$

By given condition,

$$\alpha + \beta = \alpha^2 + \beta^2$$

$$\Rightarrow \alpha + \beta = (\alpha + \beta)^2 - 2\alpha\beta$$

$$\Rightarrow -\frac{b}{a} = \left(-\frac{b}{a}\right)^2 - 2\left(\frac{c}{a}\right)$$

$$\Rightarrow \frac{-b}{a} = \frac{b^2}{a^2} - \frac{2c}{a} \Rightarrow -ba = b^2 - 2ca$$

$$\Rightarrow 2ac = b^2 + ab$$

21. (d) Here, roots are α and $\alpha + 1$.

$$\therefore \alpha + (\alpha + 1) = l \quad [\text{sum of roots}]$$

$$\Rightarrow 2\alpha = l - 1 \Rightarrow \alpha = \frac{l-1}{2}$$

$$\text{Also, } \alpha(\alpha + 1) = m \text{ or } \alpha^2 + \alpha = m$$

$$\Rightarrow \left(\frac{l-1}{2}\right)^2 + \left(\frac{l-1}{2}\right) = m$$

$$\Rightarrow (l-1)^2 + 2(l-1) = 4m$$

$$\Rightarrow l^2 - 1 = 4m \Rightarrow l^2 = 4m + 1$$

22. (b) Here, $\alpha + \beta = (1 + a^2)$

$$\text{and } \alpha\beta = \frac{1}{2}(a^4 + a^2 + 1)$$

$$\therefore \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$$

$$= (1 + a^2)^2 - (a^4 + a^2 + 1)$$

$$= 1 + a^4 + 2a^2 - a^4 - a^2 - 1$$

$$\therefore \alpha^2 + \beta^2 = a^2$$

23. (a) Here, $\frac{1}{x+p} + \frac{1}{x+q} = \frac{1}{r}$

$$\Rightarrow r(x+p+x+q) = (x+p)(x+q)$$

$$\Rightarrow x^2 + (p+q-2r)x + pq - (p+q)r = 0$$

Let roots be α and $(-\alpha)$, then

$$\alpha + (-\alpha) = 0$$

$$\Rightarrow -(p+q-2r) = 0 \Rightarrow r = \frac{p+q}{2}$$

So, product of roots = $pq - (p+q)r$

$$= pq - (p+q) \cdot \frac{(p+q)}{2}$$

$$= pq - \frac{(p+q)^2}{2} = \frac{-1}{2}(p^2 + q^2)$$

24. (b) Here, $\alpha + \beta = -\frac{2}{3}$ and $\alpha\beta = \frac{1}{3}$

$$\Rightarrow S = \frac{1-\alpha}{1+\alpha} + \frac{1-\beta}{1+\beta}$$

$$= \frac{(1-\alpha)(1+\beta) + (1+\alpha)(1-\beta)}{(1+\alpha)(1+\beta)}$$

$$= \frac{2-2\alpha\beta}{1+(\alpha+\beta)+\alpha\beta} = \frac{2-2 \times \frac{1}{3}}{1+(-2/3)+1/3}$$

$$= \frac{4}{3} \times \frac{3}{2} = 2$$

$$P = \frac{1-\alpha}{1+\alpha} \times \frac{1-\beta}{1+\beta} = \frac{1-(\alpha+\beta)+\alpha\beta}{1+(\alpha+\beta)+\alpha\beta}$$

$$= \frac{1 - (-2/3) + 1/3}{1 + (-2/3) + 1/3} = \frac{2}{2/3} = 3$$

∴ Required equation is

$$x^2 - 5x + P = 0 \text{ or } x^2 - 2x + 3 = 0$$

- 25. (a)** Given, $px^2 + qx + r = 0$

Let the roots be α and β .

By given condition, $\beta = 2\alpha$

$$\text{Product of roots } (\alpha\beta) = \frac{r}{p} = 2\alpha^2$$

$$\Rightarrow \alpha^2 = \frac{r}{2p} \quad \dots(i)$$

$$\text{Sum of roots } (\alpha + \beta) = -\frac{q}{p} = 3\alpha \quad \dots(ii)$$

On squaring Eq. (ii), we get $9\alpha^2 = \frac{q^2}{p^2}$

$$\Rightarrow 9 \left(\frac{r}{2p} \right) = \frac{q^2}{p^2} \quad [\text{from Eq. (i)}]$$

$$\Rightarrow 9rp = 2q^2$$

- 26. (a)** Let roots be α, β , then $\frac{\alpha}{\beta} = \frac{m}{n}$
[given]

$$\alpha + \beta = -1, \alpha\beta = 1$$

$$\sqrt{\frac{m}{n}} + \sqrt{\frac{n}{m}} + 1 = \sqrt{\frac{\alpha}{\beta}} + \sqrt{\frac{\beta}{\alpha}} + 1$$

$$= \frac{\alpha + \beta}{\sqrt{\alpha\beta}} + 1 = \frac{-1}{\sqrt{1}} + 1 = -1 + 1 = 0$$

- 27. (c)** Let $\sqrt{\frac{x}{1-x}} = y \Rightarrow \sqrt{\frac{1-x}{x}} = \frac{1}{y}$
∴ $y + \frac{1}{y} = \frac{13}{6} \Rightarrow (y^2 + 1)6 = 13y$

$$\Rightarrow 6y^2 - 13y + 6 = 0$$

$$\Rightarrow 6y^2 - 9y - 4y + 6 = 0$$

$$\Rightarrow 3y(2y - 3) - 2(2y - 3) = 0$$

$$\Rightarrow (3y - 2)(2y - 3) = 0$$

$$\therefore y = \frac{2}{3} \text{ and } \frac{3}{2}$$

$$\text{When, } y = \frac{2}{3} \Rightarrow \frac{x}{1-x} = \frac{4}{9}$$

$$\Rightarrow 9x = 4 - 4x \Rightarrow x = \frac{4}{13}$$

$$\text{When, } y = \frac{3}{2} \Rightarrow \frac{x}{1-x} = \frac{9}{4}$$

$$\Rightarrow 4x = 9 - 9x \Rightarrow x = \frac{9}{13}$$

- 28. (d)** Given, $\frac{1}{a+b+x} = \frac{1}{a} + \frac{1}{b} + \frac{1}{x}$

$$\therefore \frac{1}{a+b+x} - \frac{1}{x} = \frac{1}{a} + \frac{1}{b}$$

$$\Rightarrow \frac{-(a+b)}{(a+b+x)x} = \frac{(a+b)}{ab}$$

$$\Rightarrow x^2 + (a+b)x + ab = 0$$

$$\Rightarrow (x+a)(x+b) = 0$$

$$\Rightarrow x = -a, -b$$

- 29. (d)** $8 \sec^2 \phi - 6 \sec \phi + 1 = 0$

$$\Rightarrow 8 \sec^2 \phi - 4 \sec \phi - 2 \sec \phi + 1 = 0$$

$$\Rightarrow (4 \sec \phi - 1)(2 \sec \phi - 1) = 0$$

$$\Rightarrow \sec \phi = 1/4 \text{ or } \sec \phi = 1/2$$

$$\text{But } \sec \phi \geq 1 \text{ or } \sec \phi \leq -1$$

Hence, the equation has no solution.

- 30. (c)** Here, $3^{2x^2 - 7x + 7} = 9 = 3^2$

On comparing, we get

$$2x^2 - 7x + 7 = 2$$

$$2x^2 - 7x + 5 = 0$$

$$\text{Here, } D = b^2 - 4ac = 49 - 4(2)(5) = 9$$

Since, $D > 0$, so it has two real roots.

- 31. (c)** Here, $x = \sqrt{2} + 2 \Rightarrow x - 2 = \sqrt{2}$

On squaring both sides, we get

$$(x - 2)^2 = 2 \Rightarrow x^2 - 4x + 4 = 2$$

$$\Rightarrow x^2 - 4x + 2 = 0$$

- 32. (b)** $\frac{x^2 - bx}{ax - c} = \frac{k - 1}{k + 1}$

$$\Rightarrow (x^2 - bx)(k + 1) = (k - 1)(ax - c)$$

$$\Rightarrow x^2k + x^2 - bxc - bx = kax - kc - ax + c$$

$$\Rightarrow (k + 1)x^2 - x(bk + b + ka - a) + kc - c = 0$$

$$\Rightarrow (k + 1)x^2 - x(bk + b + ka - a) + kc - c = 0$$

Since, roots are reciprocal to each other.

So, product = 1

$$\text{i.e. } \frac{kc - c}{(k + 1)} = 1 \Rightarrow kc - c = k + 1$$

$$\Rightarrow k(c - 1) = c + 1 \Rightarrow k = \frac{c + 1}{c - 1}$$

- 33. (b)** Here, $\alpha + \beta = b/a$ and $\alpha\beta = b/a$

$$\text{So, } \sqrt{\frac{\alpha}{\beta}} + \sqrt{\frac{\beta}{\alpha}} = \frac{\alpha + \beta}{\sqrt{\alpha\beta}} = \frac{b/a}{\sqrt{b/a}} = \sqrt{\frac{b}{a}}$$

- 34. (a)** Given, $2x^2 - 7x + 3 = 0$

$$\therefore 2x^2 - 6x - x + 3 = 0$$

$$\Rightarrow 2x(x - 3) - 1(x - 3) = 0$$

$$\Rightarrow (2x - 1)(x - 3) = 0$$

$$\text{When, } x = \frac{1}{2}$$

$$4 \left(\frac{1}{2} \right)^2 + a \left(\frac{1}{2} \right) - 3 = 0$$

$$\Rightarrow 1 + \frac{a}{2} - 3 = 0 \Rightarrow \frac{a}{2} = 2 \Rightarrow a = 4$$

$$\text{When, } x = 3, 4(3)^2 + a(3) - 3 = 0$$

$$\Rightarrow 36 + 3a - 3 = 0 \Rightarrow a = -11$$

$$\therefore a = -11 \text{ or } 4$$

- 35. (c)** Here, for equation $x^2 + bx + c = 0$

$$\alpha + \beta = -b \text{ and } \alpha\beta = c$$

$$\text{and for } x^2 + px + q = 0$$

$$k\alpha + k\beta = -p \text{ and } k^2\alpha\beta = q$$

$$\text{Now, } qb^2 = k^2\alpha\beta(\alpha + \beta)^2$$

$$= k^2(\alpha + \beta)^2\alpha\beta$$

$$= [k(\alpha + \beta)]^2\alpha\beta$$

$$\therefore qb^2 = p^2c$$

- 36. (b)** Since, -4 is a root of

$$x^2 + px - 4 = 0$$

$$(-4)^2 + p(-4) - 4 = 0$$

$$\Rightarrow 16 - 4p - 4 = 0 \Rightarrow p = 3$$

As roots of $x^2 + px + q = 0$ are equal.

$$\therefore D = B^2 - 4AC = 0$$

$$\Rightarrow p^2 - 4q = 0 \text{ or } 3^2 - 4q = 0 \Rightarrow q = \frac{9}{4}$$

$$\text{So, } p = 3, q = \frac{9}{4}$$

- 37. (c)** Here, $\alpha + \beta = 3$ and $\alpha\beta = 3/2$

$$\Rightarrow \left(\frac{\alpha}{\beta} + \frac{\beta}{\alpha} \right) + 3 \left(\frac{1}{\alpha} + \frac{1}{\beta} \right) + 2\alpha\beta$$

$$= \frac{\alpha^2 + \beta^2}{\alpha\beta} + \frac{3(\alpha + \beta)}{\alpha\beta} + 2\alpha\beta$$

$$= \frac{(\alpha + \beta)^2 - 2\alpha\beta}{\alpha\beta} + \frac{3(\alpha + \beta)}{\alpha\beta} + 2\alpha\beta$$

$$= \frac{3^2 - 2 \cdot 3/2}{3/2} + \frac{3 \cdot 3}{3/2} + 2 \left(\frac{3}{2} \right)$$

$$= \frac{6}{3/2} + \frac{9}{3/2} + 3 = 4 + 6 + 3 = 13$$

- 38. (b)** Let α, β be the roots of

$$x^2 + 2ax + b = 0 \quad \dots(i)$$

$$\Rightarrow \alpha + \beta = -2a \text{ and } \alpha\beta = b.$$

By hypothesis, $|\alpha - \beta| \leq 2m$

$$\Rightarrow (\alpha - \beta)^2 \leq 4m^2$$

$$\Rightarrow (\alpha + \beta)^2 - 4\alpha\beta \leq 4m^2$$

$$\Rightarrow 4a^2 - 4b \leq 4m^2$$

$$\Rightarrow a^2 - b \leq m^2 \quad \dots(ii)$$

and discriminant of Eq. (i) is greater than 0.

$$\Rightarrow 4a^2 - 4b > 0 \Rightarrow b < a^2 \quad \dots(iii)$$

From Eqs. (ii) and (iii),

$$b \in [a^2 - m^2, a^2)$$

- 39. (b)** Refer to example 15. [Rule 5]

- 40. (b)** Refer to example 14. [Rule 4]

- 41. (b)** Since, $\sin \theta$ and $\cos \theta$ are the roots of the equation $ax^2 - bx + c = 0$.

$$\therefore \sin \theta + \cos \theta = \frac{b}{a} \quad \dots(i)$$

$$\text{and } \sin \theta \cos \theta = \frac{c}{a}$$

On squaring both sides of Eq. (i), we get

$$\Rightarrow \sin^2 \theta + \cos^2 \theta + 2 \sin \theta \cos \theta = \frac{b^2}{a^2}$$

$$\Rightarrow 1 + 2 \left(\frac{c}{a} \right) = \frac{b^2}{a^2} \Rightarrow 2 \left(\frac{c}{a} \right) = \frac{b^2 - a^2}{a^2}$$

$$\Rightarrow 2ac = b^2 - a^2 \Rightarrow a^2 - b^2 + 2ac = 0$$

- 42.** (d) The equation will have equal roots, if $B^2 - 4AC = 0$
 $\therefore (2nc)^2 - 4(1+n^2)(c^2 - a^2) = 0$
 $\Rightarrow 4n^2 c^2 - 4(c^2 + n^2 c^2 - a^2) = 0$
 $\Rightarrow -n^2 a^2 = 0$
 $\Rightarrow -4c^2 + 4a^2 + 4n^2 a^2 = 0$
 $\Rightarrow c^2 = a^2(1+n^2)$

43. (d) a and c have the same sign opposite to that of b .

44. (d) Let the roots of equation $x^2 - 2x + 4 = 0$ be α and β .
Then, $\alpha + \beta = 2$ and $\alpha\beta = 4$
On taking, $\alpha \rightarrow 2\alpha$ and $\beta \rightarrow 2\beta$
 $2\alpha + 2\beta = 4$ and $2\alpha \cdot 2\beta = 4 \times 4 = 16$
So, the new equation becomes
 $x^2 - 4x + 16 = 0$

45. (a) Since, α and β are the roots of the equation $x^2 + px + q = 0$
 $\therefore \alpha + \beta = -p$ and $\alpha\beta = q$
Now, $-\alpha^{-1} - \beta^{-1} = -\left(\frac{1}{\alpha} + \frac{1}{\beta}\right)$
 $= -\left(\frac{\alpha + \beta}{\alpha\beta}\right) = \frac{p}{q}$
and $\left(-\frac{1}{\alpha}\right)\left(-\frac{1}{\beta}\right) = \frac{1}{\alpha\beta} = \frac{1}{q}$
Hence, required equation is
 $x^2 - (-\alpha^{-1} - \beta^{-1})x + (-\alpha\beta)^{-1} = 0$
 $\Rightarrow x^2 - \frac{p}{q}x + \frac{1}{q} = 0$
 $\Rightarrow qx^2 - px + 1 = 0$

46. (c) Since, one root of the equation $ax^2 + x - 3 = 0$ is -1 .
 $\therefore a(-1)^2 + (-1) - 3 = 0 \Rightarrow a = 4$
 $\therefore 4x^2 + x - 3 = 0$
Let other root of this equation be α .
Now, product of roots = $\frac{c}{a}$
 $\therefore -1 \cdot \alpha = -\frac{3}{4} \Rightarrow \alpha = \frac{3}{4}$

47. (b) Since, the roots of equation $(a^2 + b^2)x^2 - 2(ac + bd)x + (c^2 + d^2) = 0$ are equal.
 $\therefore B^2 = 4AC$
 $\Rightarrow 4(ac + bd)^2 = 4(a^2 + b^2)(c^2 + d^2)$
 $\Rightarrow a^2 c^2 + b^2 d^2 + 2abcd = a^2 c^2 + a^2 d^2 + b^2 c^2 + b^2 d^2$
 $\Rightarrow (ad - bc)^2 = 0$
 $\therefore ad = bc$

48. (a) Given, $\sqrt{\frac{x}{x+3}} - \sqrt{\frac{x+3}{x}} = -\frac{3}{2}$
Let, $y = \sqrt{\frac{x}{x+3}}$, then $\sqrt{\frac{x+3}{x}} = \frac{1}{y}$
 $y - \frac{1}{y} = -\frac{3}{2} \Rightarrow 2y^2 - 2 = -3y$
 $\Rightarrow 2y^2 + 3y - 2 = 0$
 $\Rightarrow 2y(y+2) - 1(y+2) = 0$
 $\Rightarrow (2y-1)(y+2) = 0$
If, $(2y-1) = 0$
 $\Rightarrow y = \frac{1}{2} \Rightarrow y = \sqrt{\frac{x}{x+3}} \Rightarrow \sqrt{\frac{x}{x+3}} = \frac{1}{2}$
On squaring both sides, we get
 $\frac{x}{x+3} = \frac{1}{4}$
 $\Rightarrow 4x = x+3 \Rightarrow x = 1$
or $y+2 = 0 \Rightarrow y = -2$
Since, y cannot be negative.
Hence, $x = 1$ is the required solution.

49. (c) Given, $4^x - 3 \cdot 2^{x+2} + 32 = 0$
 $\Rightarrow 2^{2x} - 3 \cdot 2^x \cdot 2^2 + 32 = 0$
 $\Rightarrow 2^{2x} - 12 \cdot 2^x + 32 = 0$
 $\Rightarrow 2^{2x} - 8 \cdot 2^x - 4 \cdot 2^x + 32 = 0$
 $\Rightarrow (2^x - 8)(2^x - 4) = 0$
Either, $2^x = 8 \Rightarrow x = 3$
or $2^x = 4 \Rightarrow x = 2$

50. (a) Since, α and β are the roots of the equation $x^2 - x - 1 = 0$.
 $\therefore \alpha + \beta = 1$ and $\alpha\beta = -1$
Now, $\alpha^4 + \beta^4 = (\alpha^2 + \beta^2)^2 - 2(\alpha\beta)^2$
 $= [(\alpha + \beta)^2 - 2\alpha\beta]^2 - 2(\alpha\beta)^2$
 $= (1+2)^2 - 2 = 9 - 2 = 7$

51. (c) Let the roots of equation $ax^2 + bx + c = 0$ be $-\alpha$ and $-\frac{1}{\alpha}$.
 $\therefore$ Product of roots = $\frac{c}{a}$
 $(-\alpha)\left(-\frac{1}{\alpha}\right) = \frac{c}{a} \Rightarrow 1 = \frac{c}{a} \Rightarrow c = a$

52. (d) Let the roots of the quadratic equation be α and β .
Then, $\alpha + \beta = 9$ and $\alpha\beta = 9$
Hence, equation is
 $x^2 - (\alpha + \beta)x + (\alpha\beta) = 0$
 $\Rightarrow x^2 - 9x + 9 = 0$

53. (c) Given, $\log_{10}(x^2 - 6x + 45) = 2$
 $\Rightarrow (x^2 - 6x + 45) = 10^2 = 100$
 $\Rightarrow x^2 - 6x - 55 = 0$
 $\Rightarrow x^2 - 11x + 5x - 55 = 0$
 $\Rightarrow x(x-11) + 5(x-11) = 0$
 $\Rightarrow (x+5)(x-11) = 0$
 $\therefore x = 11, -5$

54. (c) Given, $\frac{1}{x+a} + \frac{1}{x+b} = \frac{1}{c}$
 $\Rightarrow \frac{(x+b) + (x+a)}{(x+a)(x+b)} = \frac{1}{c}$
 $\Rightarrow 2cx + (a+b)c = x^2 + (a+b)x + ab$
 $\Rightarrow x^2 + (a+b-2c)x + ab - ac - bc = 0$
Let the roots of above equation be α and β .
Given, $\alpha + \beta = 0$
 $\Rightarrow -(a+b-2c) = 0 \Rightarrow a+b = 2c \dots (i)$
Now, $\alpha\beta = ab - ac - bc$
 $= ab - (a+b)c$
 $= ab - (a+b) \frac{(a+b)}{2}$
[from Eq. (i)]
 $= \frac{2ab - (a^2 + b^2 + 2ab)}{2} = -\frac{(a^2 + b^2)}{2}$

55. (b) Let the roots of the equation $kx^2 - 5x + 6 = 0$ be α and β .
 $\therefore \alpha + \beta = \frac{5}{k}$ and $\alpha\beta = \frac{6}{k}$
Given, $\frac{\alpha}{\beta} = \frac{2}{3} \Rightarrow \alpha = \frac{2}{3}\beta$
 $\therefore \frac{2}{3}\beta + \beta = \frac{5}{k}$ and $\frac{2}{3}\beta^2 = \frac{6}{k}$
 $\Rightarrow \frac{5}{3}\beta = \frac{5}{k}$ and $\beta^2 = \frac{9}{k}$
 $\beta = \frac{3}{k}$ and $\beta^2 = \frac{9}{k} \Rightarrow \frac{9}{k^2} = \frac{9}{k}$
 $\Rightarrow k = 1$ and $k \neq 0$
 $\therefore$ leading coefficient is never zero.

56. (d) Let α and β be the roots of the equation $7x^2 + 12x + 18 = 0$.
 $\therefore \alpha + \beta = \frac{-12}{7}$ and $\alpha\beta = \frac{18}{7}$
 $\Rightarrow \alpha^2 + \beta^2 + 2\alpha\beta = \frac{144}{49}$
 $\Rightarrow \alpha^2 + \beta^2 = \frac{144}{49} - \frac{36}{7} = -\frac{108}{49}$
 $\therefore \frac{\alpha^2 + \beta^2}{\alpha\beta} = \frac{-\frac{108}{49}}{\frac{18}{7}} = -\frac{6}{7} = -6 : 7$

57. (d) Given, $\frac{x(x-1) - (m+1)}{(x-1)(m-1)} = \frac{x}{m}$
 $\Rightarrow m(x^2 - x - m + 1) = x(mx - x - m + 1)$
 $\Rightarrow mx^2 - mx - m(m+1) = mx^2 - x^2 - mx + x$
 $\Rightarrow x^2 - x - m(m+1) = 0$
Let one root be α , then other root is also α .
 $\therefore \alpha + \alpha = 1, \Rightarrow \alpha = \frac{1}{2}$

$$\begin{aligned}\text{Also, } \alpha^2 &= -m(m+1) \\ \Rightarrow \left(\frac{1}{2}\right)^2 &= -m(m+1) \\ \Rightarrow 4m^2 + 4m + 1 &= 0 \\ \Rightarrow (2m+1)^2 &= 0 \\ \therefore m &= -\frac{1}{2}\end{aligned}$$

$$\begin{aligned}58. (c) \text{ Given, } 3^x + 27(3^{-x}) &= 12 \\ \text{Let } 3^x &= y \\ \therefore y + \frac{27}{y} &= 12 \\ \Rightarrow y^2 - 12y + 27 &= 0 \\ \Rightarrow y^2 - 9y - 3y + 27 &= 0 \\ \Rightarrow (y-3)(y-9) &= 0 \Rightarrow y = 3, 9 \\ \Rightarrow 3^x = 3 \text{ or } 3^x = 9 = 3^2 \\ \therefore x = 1 \text{ or } x = 2\end{aligned}$$

$$\begin{aligned}59. (a) \text{ Let the roots of the given equation be } \alpha \text{ and } \beta. \\ \therefore \alpha + \beta = a \text{ and } \alpha\beta = b \\ \text{Now, } |\alpha - \beta| &= \sqrt{(\alpha + \beta)^2 - 4\alpha\beta} \\ &= \sqrt{a^2 - 4b}\end{aligned}$$

$$\begin{aligned}60. (a) x^2 - x - 2 &= 0 \\ (x-2)(x+1) &= 0 \Rightarrow x = 2, -1 \\ \text{So, both the roots are integers.}\end{aligned}$$

$$\begin{aligned}61. (c) x^2 - 6x + 5 &= 0 \Rightarrow (x-5)(x-1) = 0 \\ \Rightarrow x &= 5 \text{ or } 1 \\ \text{Also, } |x-3| &= 2 \Leftrightarrow (x-3) = 2 \\ \text{or } -(x-3) &= 2 \Rightarrow x = 5 \text{ or } x = 1 \\ \therefore x^2 - 6x + 5 &= 0 \\ \text{and } |x-3| &= 2 \text{ are equivalent.}\end{aligned}$$

$$\begin{aligned}62. (a) \text{ Let the number are } x \text{ and } \frac{1}{x}. \\ \text{Then, } x + \frac{1}{x} = \frac{10}{3}, \frac{x^2 + 1}{x} = \frac{10}{3} \\ \Rightarrow 3x^2 - 10x + 3 &= 0 \\ \Rightarrow 3x^2 - 9x - x + 3 &= 0 \\ \Rightarrow 3x(x-3) - 1(x-3) &= 0 \\ \Rightarrow (3x-1)(x-3) &= 0 \\ \therefore x = \frac{1}{3}, x = 3\end{aligned}$$

$$\begin{aligned}63. (b) \text{ Let the smaller part } &= x \text{ and greater part } = 16 - x \\ \text{By given condition,} \\ 2(16-x)^2 - x^2 &= 164 \\ \Rightarrow 2(256 + x^2 - 32x) - x^2 &= 164 \\ \Rightarrow x^2 - 64x + 348 &= 0 \\ \Rightarrow (x-58)(x-6) &= 0 \\ \Rightarrow x = 58, x = 6 \\ \text{Here } x &\neq 58 \\ \therefore x &= 6 \\ \text{and, hence larger part} \\ &= 16 - x = 16 - 6 = 10\end{aligned}$$

$$\begin{aligned}64. (c) \text{ Here, } y = 15, \text{ then } 15 &= \frac{x(x-1)}{2} \\ \Rightarrow x^2 - x - 30 &= 0 \\ \Rightarrow x^2 - 6x + 5x - 30 &= 0 \\ \Rightarrow x(x-6) + 5(x-6) &= 0 \\ \Rightarrow (x+5)(x-6) &= 0 \\ \text{Either } x = 6 \text{ or } x = -5 \\ \text{But } x \neq -5, \text{ so } x &= 6 \\ \text{Thus, figure has 6 points.}\end{aligned}$$

$$\begin{aligned}65. (a) \text{ Let the natural numbers be } x \text{ and } x+1, \text{ respectively.} \\ \text{Then, } x^2 + (x+1)^2 &= 221 \\ \Rightarrow 2x^2 + 2x + 1 &= 221 \\ \Rightarrow 2x^2 + 2x - 220 &= 0 \\ \Rightarrow x^2 + x - 110 &= 0 \\ \Rightarrow (x+11)(x-10) &= 0 \\ \Rightarrow x = -11 \text{ and } x = 10 \\ x \neq -11 \quad [\because \text{numbers are natural}] \\ \text{So, for } x = 10, \text{ next consecutive natural number} \\ &= x + 1 = 10 + 1 = 11\end{aligned}$$

$$\begin{aligned}66. (b) \text{ Let the consecutive positive odd integers be } 2x+1 \text{ and } 2x+3, \\ \text{so } (2x+1)^2 + (2x+3)^2 &= 130 \\ \Rightarrow (4x^2 + 4x + 1) + (4x^2 + 12x + 9) &= 130 \\ \Rightarrow x^2 + 2x - 15 &= 0 \\ \Rightarrow (x+5)(x-3) &= 0 \\ \Rightarrow x = 3, x = -5, \text{ but } x \neq -5 \\ [\because \text{integers are positive.}]\end{aligned}$$

$$\begin{aligned}\therefore \text{Two consecutive integers are} \\ 2x+1 = 7 \text{ and } 2x+3 = 9 \\ \text{i.e. 7 and 9.}\end{aligned}$$

$$\begin{aligned}67. (b) \text{ Here, } 2x+1 \geq 7 \Rightarrow 2x &\geq 7-1 \\ \Rightarrow 2x \geq 6 \Rightarrow x &\geq 3\end{aligned}$$

$$\begin{aligned}68. (c) \text{ Here, } \frac{x-1}{3} \geq 4 \\ \Rightarrow x-1 \geq 12 \Rightarrow x \geq 13\end{aligned}$$

$$\begin{aligned}69. (b) \text{ Here, } 3x+2 \leq 5x-(4-x) \\ \Rightarrow 3x+2 \leq 5x-4+x \\ \Rightarrow 3x+2 \leq 6x-4 \\ \Rightarrow 3x-6x \leq -4-2 \Rightarrow -3x \leq -6 \\ \Rightarrow -x \leq -2 \Rightarrow x \geq 2\end{aligned}$$

$$\begin{aligned}70. (c) \text{ Here, } 2x+3 \geq 8 \Rightarrow 2x &\geq 8-3 \\ \Rightarrow 2x \geq 5 \Rightarrow x &\geq \frac{5}{2}\end{aligned}$$

$$\begin{aligned}\text{Again, } 3x+1 \leq 12 \\ \Rightarrow 3x \leq 11 \Rightarrow x \leq \frac{11}{3} \\ \text{By combining values, we get} \\ \frac{5}{2} \leq x \leq \frac{11}{3}\end{aligned}$$

$$\begin{aligned}71. (b) \frac{1}{2} \left(\frac{3}{5}x + 4 \right) &\geq \frac{1}{3}(x-6) \\ \Rightarrow \frac{3}{10}x + 2 &\geq \frac{1}{3}x - 2 \\ \Rightarrow 9x + 60 &\geq 10x - 60 \\ \Rightarrow -x &\geq -120 \\ [\text{multiplying both sides by } -1] \\ \Rightarrow x &\leq 120\end{aligned}$$

Thus, all real numbers x which are less than or equal to 120 satisfies the inequality.

$$\begin{aligned}72. (a) \text{ We have, } 4x^2 - 1 &\leq 0 \\ \Rightarrow (2x)^2 - 1 &\leq 0 \\ \Rightarrow (2x-1)(2x+1) &\leq 0 \\ \text{So, either } (2x-1) &\geq 0 \\ \text{and } (2x+1) &\leq 0 \quad \dots(i) \\ \text{or } (2x-1) &\leq 0 \\ \text{and } (2x+1) &\geq 0 \quad \dots(ii) \\ \text{From Eq. (i), } 2x &\geq 1 \text{ and } 2x \leq -1 \\ \Rightarrow x &\geq \frac{1}{2} \text{ and } x \leq \frac{-1}{2}\end{aligned}$$

which is not possible.

$$\begin{aligned}\text{From Eq. (ii), } (2x-1) &\leq 0 \\ \text{and } (2x+1) &\geq 0\end{aligned}$$

$$\begin{aligned}2x \leq 1 \text{ and } 2x &\geq -1 \\ x \leq \frac{1}{2} \text{ and } x &\geq \frac{-1}{2}\end{aligned}$$

$$\therefore \frac{-1}{2} \leq x \leq \frac{1}{2} \text{ is the required solution set.}$$

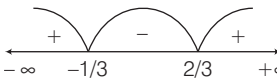
$$\begin{aligned}73. (b) \text{ Here, } 3x+2y = 12 \text{ passes through the points } (0, 6) \text{ and } (4, 0). \text{ So, its graph is line } AB. \text{ Also, } 3x+2y \leq 12 \text{ as the shaded region is below } AB. \\ x = 0 \text{ and } y = 0 \text{ are the Y-axis and X-axis, respectively. } x \geq 0 \text{ and } y \geq 0 \text{ implies the region on the right hand side of Y-axis and region above X-axis, respectively.}\end{aligned}$$

The three portion $x \geq 0, y \geq 0$ and $3x+2y \leq 12$ intersects to give the shaded portion OAB .

$$\begin{aligned}74. (d) \text{ Given, } x^2 + 3x - 10 > 0 \\ \Rightarrow x^2 + 5x - 2x - 10 > 0 \\ \Rightarrow (x+5)(x-2) > 0 \\ \text{Either, } (x+5) > 0 \text{ and } (x-2) > 0 \\ \text{or } (x+5) < 0 \text{ and } (x-2) < 0 \\ \Rightarrow x > 2 \text{ or } x < -5\end{aligned}$$

$$\begin{aligned}75. (a) x^2 - 5x + 6 \geq 0 \\ \Rightarrow (x-2)(x-3) \geq 0 \\ \begin{array}{c} \text{+} \quad \text{---} \quad \text{---} \quad \text{---} \quad \text{+} \\ \text{2} \quad \text{3} \end{array} \\ \Rightarrow x \leq 2 \text{ or } x \geq 3 \\ \Rightarrow x \in [-\infty, 2] \text{ or } [3, \infty] \\ \Rightarrow x \in [-\infty, 2] \cup [3, \infty]\end{aligned}$$

76. (b) $\frac{x-2}{3x+1} < \frac{x-3}{3x-2}$
 $\Leftrightarrow \frac{x-2}{3x+1} - \frac{x-3}{3x-2} < 0$
 $\Leftrightarrow \frac{(x-2)(3x-2) - (x-3)(3x+1)}{(3x+1)(3x-2)} < 0$
 $\Leftrightarrow \frac{7}{(3x+1)(3x-2)} < 0$
 $\Leftrightarrow (3x+1)(3x-2) < 0$
 $\Leftrightarrow 3\left(x + \frac{1}{3}\right)3\left(x - \frac{2}{3}\right) < 0$
 $\Leftrightarrow \left(x + \frac{1}{3}\right)\left(x - \frac{2}{3}\right) < 0$



$\Leftrightarrow -\frac{1}{3} < x < \frac{2}{3}$
 So, $x \in \left[-\frac{1}{3}, \frac{2}{3}\right]$

77. (c) Here, as $x > 0$ and $y > 0$ so both are positive and satisfies $x + 2y \leq 3$.
 $\therefore$ When, $x = 3$ and $y = 1$, we get $x + 2y = 5$.

Clearly, this does not satisfy $x + 2y \leq 3$

When $x = 1$ and $y = 1$,

then, $x + 2y \leq 3$.

So, $x = 1, y = 1$ is one of the solutions.

78. (d) As, $(0, 0)$ satisfies $x + 3y \leq 3$.

And $(0, 0)$ satisfies $x + y \leq 2$.

Clearly, shaded region is the position common to the line $x + 3y = 3$ and below it and that on the line $x + y = 2$ and below it.

So, shaded region is the solution set

$$x + y \leq 2, x + 3y \leq 3, x \geq 0, y \geq 0.$$

79. (c) Given, $a + b = 2m^2$... (i)

$$b + c = 6m \quad \dots (ii)$$

$$\text{and } a + c = 2 \quad \dots (iii)$$

On adding Eqs. (i), (ii) and (iii), we get

$$2(a + b + c) = 2m^2 + 6m + 2$$

$$\Rightarrow a + b + c = m^2 + 3m + 1 \quad \dots (iv)$$

On subtracting Eq. (ii) from Eq. (iv), we get

$$a = m^2 - 3m + 1$$

On subtracting Eq. (iii) from Eq. (iv), we get

$$b = m^2 + 3m - 1$$

On subtracting Eq. (i) from Eq. (iv), we get

$$c = -m^2 + 3m + 1$$

As, $a \leq b$ and $b \leq c$

$$\Rightarrow m^2 - 3m + 1 \leq m^2 + 3m - 1$$

$$\text{and } m^2 + 3m - 1 \leq -m^2 + 3m + 1$$

$$\Rightarrow 6m \geq 2 \text{ and } 2m^2 \leq 2$$

$$\Rightarrow m \geq \frac{1}{3} \text{ and } -1 \leq m \leq 1$$

$$\therefore \frac{1}{3} \leq m \leq 1$$

80. (d) Given equation, $x^2 - 3x + 2 = 0$

$$\Rightarrow x^2 - 2x - x + 2 = 0$$

$$\Rightarrow (x-2)(x-1) = 0 \Rightarrow x = 2, 1$$

Let $\alpha = 1$ and $\beta = 2$

$$\therefore \alpha + 1 = 2 \text{ and } (\beta + 1) = 3.$$

Now, sum of roots = $2 + 3 = 5$

and product of roots = $2 \times 3 = 6$

$\therefore$ Required equation

$$= x^2 - (\text{sum of roots}) + \text{product of roots} = 0$$

$$\Rightarrow x^2 - 5x + 6 = 0$$

Hence, the equation is neither I nor II.

81. (c) I. Every quadratic equation has two roots, which may or may not be real.

II. $x^2 - 4x + 2 = 0$ has integral coefficients but does not have integral roots.

III. Since, discriminant = $b^2 - 4ac$

If a and c have opposite sign, then $b^2 - 4ac \geq 0$

$\therefore$ The quadratic equation has real roots

82. (c) I. $x^2 + \frac{1}{x^2} = 2$

or $x^4 - 2x^2 + 1 = 0$
 is not a quadratic equation.

II. $x + \frac{3}{x} = x^2$

$$\Rightarrow \frac{x^2 + 3}{x} = x^2$$

$$\text{or } x^2 + 3 = x^3$$

$$\Rightarrow x^3 - x^2 - 3 = 0$$

is not a quadratic equation.

III. $2x^2 - x + 2 = x^2 + 4x - 4$

$$\text{or } 2x^2 - x + 2 - x^2 - 4x + 4 = 0$$

$$\text{or } x^2 - 5x + 6 = 0$$

is a quadratic equation.

IV. $x^3 + 6x^2 + 2x - 1 = 0$

is not a quadratic equation.

Hence, only III statement is correct.

83. (b) I. $3x^2 - 2x - 1 = 0$

$$\text{Put } x = 1 \Rightarrow 3(1)^2 - 2(1) - 1 = 0$$

$$\therefore \text{LHS} = \text{RHS}$$

II. $x^2 + \sqrt{2}x - 4 = 0$

$$\text{Put } x = -2\sqrt{2}$$

$$\Rightarrow (-2\sqrt{2})^2 + \sqrt{2}(-2\sqrt{2}) - 4 = 0$$

$$\therefore \text{LHS} = \text{RHS}$$

III. $x^2 + x + 1 = 0; x = 1, x = -1$

Put $x = 1$, we get $1^2 + 1 + 1 = 3 \neq 0$

$\therefore$ LHS $\neq$ RHS

$\therefore x = 1$ is not a solution.

Put $x = -1$, we get

$$(-1)^2 + (-1) + 1 = 1 - 1 + 1 = 1 \neq 0$$

$\therefore$ LHS $\neq$ RHS

$\therefore x = -1$ is not a solution.

IV. $9x^2 - 3x - 2 = 0$

Put $x = -1/3$

$$\Rightarrow 9(-1/3)^2 - 3(-1/3) - 2 = 0$$

$\therefore$ LHS = RHS

Now, put $x = 2/3$

$$\Rightarrow 9(2/3)^2 - 3(2/3) - 2 = 0$$

$\therefore$ LHS = RHS

84. (b) Given, $h = -16t^2 + 64t + 80$

put $t = 1$

$$\Rightarrow h = -16(1)^2 + 64(1) + 80 = 128 \text{ m}$$

85. (c) By rearranging, we get

$$h = -16(t^2 - 4t - 5)$$

$$\Rightarrow h = -16[(t-2)^2 - 9]$$

$$\Rightarrow h = -16(t-2)^2 + 144$$

when the height is maximum, $t = 2$

$\therefore$ maximum height = 144 m

86. (a) When the ball hits the ground $h = 0$

$$\Rightarrow -16t^2 + 64t + 80 = 0$$

$$\Rightarrow t^2 - 4t - 5 = 0$$

$$\Rightarrow t^2 - 5t + t - 5 = 0$$

$$\Rightarrow (t-5)(t+1) = 0 \Rightarrow t = 5, -1$$

Since, the time cannot be negative.

So, $t = 5$ sec

87. (a) Let $f(x) = x^2 - 2ax + a^2 + a - 3$

$$f(3) > 0 \Rightarrow 9 - 6a + a^2 + a - 3 > 0 \dots (i)$$

Also, since roots are real

$$\therefore B^2 - 4AC \geq 0$$

$$\Rightarrow 4a^2 - 4(a^2 + a - 3) \geq 0 \dots (ii)$$

From Eqs. (i) and (ii), we have

$$a^2 - 5a + 6 > 0 \text{ and } a - 3 \leq 0$$

$$\Rightarrow a^2 - 3a - 2a + 6 > 0$$

$$\Rightarrow (a-3)(a-2) > 0$$

$$\Rightarrow a \leq 3 \text{ and } a < 2 \text{ or } a > 3$$

Combining the above inequalities, we have $a < 2$

88. (a) Given, $7x^2 - 50x + k = 0$

Here, $a = 7, b = -50$ and $c = k$

Since, α and β are the roots of the given equation

$$\therefore \alpha + \beta = \frac{-b}{a} \Rightarrow \alpha + \beta = \frac{50}{7}$$

$$\Rightarrow \beta = \frac{1}{7} \quad [\because \alpha = 7, \text{ given}]$$

$$\text{and } \alpha\beta = \frac{c}{a} \Rightarrow 7 \times \frac{1}{7} = \frac{k}{7} \Rightarrow k = 7$$

89. (b) Let α and β be the roots of the quadratic equation

$$x^2 + px + q = 0.$$

Given that, A starts with a wrong value of p and obtains the roots as 2 and 6. But this time q is correct.

i.e. Product of roots

$$= q = \alpha \cdot \beta = 6 \times 2 = 12 \quad \dots(i)$$

and B starts with a wrong value of q and gets the roots as 2 and -9. But this time p is correct.

i.e. Sum of roots

$$= p = \alpha + \beta = -9 + 2 = -7 \quad \dots(ii)$$

We know that,

$$\begin{aligned} (\alpha - \beta)^2 &= (\alpha + \beta)^2 - 4\alpha\beta \\ &= (-7)^2 - 4 \times 12 = 49 - 48 = 1 \end{aligned}$$

[from Eqs. (i) and (ii)]

$$\Rightarrow \alpha - \beta = 1 \quad \dots(iii)$$

From Eqs. (ii) and (iii), $\alpha = -3$ and $\beta = -4$

which are correct roots.

90. (a) Let one root be α , then other roots is α^2

Given equation is $x^2 - bx + c = 0$

$$\therefore \text{Sum of roots} = \alpha + \alpha^2 = -\frac{(-b)}{1}$$

$$\Rightarrow \alpha(\alpha + 1) = b \quad \dots(i)$$

$$\text{and product of roots} = \alpha \cdot \alpha^2 = \frac{c}{1}$$

$$\Rightarrow \alpha^3 = c \Rightarrow \alpha = c^{1/3} \quad \dots(ii)$$

From Eqs. (i) and (ii),

$$c^{1/3} (c^{1/3} + 1) = b \quad \dots(iii)$$

On cubing both sides, we get

$$c(c^{1/3} + 1)^3 = b^3$$

$$\Rightarrow c\{c + 1 + 3c^{1/3}(c^{1/3} + 1)\} = b^3$$

$$\Rightarrow c(c + 1 + 3b) = b^3 \quad [\text{from Eq. (iii)}]$$

$$\Rightarrow b^3 = 3bc + c^2 + c$$

91. (a) Let α and β be the roots of the given quadratic equation $2x^2 - 11x + 5 = 0$.

$$\therefore \alpha + \beta = -\frac{(-11)}{2} = \frac{11}{2} \quad \dots(i)$$

$$\text{and } \alpha \cdot \beta = \frac{5}{2} \quad \dots(ii)$$

Now, $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$

$$\begin{aligned} &= \left(\frac{11}{2}\right)^2 - 4\left(\frac{5}{2}\right) \\ &= \frac{121}{4} - \frac{40}{4} = \frac{81}{4} = \left(\frac{9}{2}\right)^2 \end{aligned}$$

$$\therefore \text{Difference of roots} = (\alpha - \beta) = \frac{9}{2} = 4.5$$

92. (b) Let the two numbers be x and y .

Now, by given condition,

$$x^2 + y^2 = 97 \quad \dots(i)$$

$$\text{and } (x - y)^2 = 25 \quad \dots(ii)$$

$$\Rightarrow (x^2 + y^2) - 2xy = 25$$

$$\Rightarrow 97 - 2xy = 25 \quad [\text{from Eq. (i)}]$$

$$\Rightarrow 2xy = 72$$

$$\therefore xy = 36$$

$$93. (a) \text{ Given that, } x + \frac{1}{x} = 2 \quad \dots(i)$$

On squaring both sides, we get

$$\left(x + \frac{1}{x}\right)^2 = 4 \Rightarrow x^2 + \frac{1}{x^2} + 2 = 4$$

$$\Rightarrow x^2 + \frac{1}{x^2} = 2 \quad \dots(ii)$$

$$\begin{aligned} \text{Now, } \left(x - \frac{1}{x}\right)^2 &= \left(x^2 + \frac{1}{x^2}\right) - 2 \\ &= 2 - 2 = 0 \quad [\text{from Eq. (ii)}] \end{aligned}$$

$$\therefore x - \frac{1}{x} = 0$$

94. (b) Let the number of columns be x .

Then, number of rows = $x + 4$

According to the question,

$$x(x + 4) \times 5 - 2 = 158$$

$$\Rightarrow 5x(x + 4) = 160 \Rightarrow x(x + 4) = 32$$

$$\Rightarrow x^2 + 4x - 32 = 0$$

$$\Rightarrow x^2 + 8x - 4x - 32 = 0$$

$$\Rightarrow x(x + 8) - 4(x + 8) = 0$$

$$\Rightarrow (x + 8)(x - 4) = 0$$

So, $x = 4$ as $x = -8$ is not possible.

$\therefore$ Number of rows

$$= x + 4 = 4 + 4 = 8$$

$$95. (c) \text{ Given, } x^2 + \frac{1}{x^2} + 8\left(x + \frac{1}{x}\right) + 14$$

$$\text{Let } x + \frac{1}{x} = y \quad \dots(i)$$

On squaring both sides, we get

$$\therefore \left(x + \frac{1}{x}\right)^2 = x^2 + \frac{1}{x^2} + 2 = y^2$$

$$\Rightarrow x^2 + \frac{1}{x^2} = y^2 - 2 \quad \dots(ii)$$

Now, on putting values from Eqs. (i) and

(ii) in the given equation, we get

$$y^2 - 2 + 8y + 14 = y^2 + 8y + 12$$

$$= y^2 + 2y + 6y + 12$$

$$= y(y + 2) + 6(y + 2)$$

$$= (y + 6)(y + 2)$$

$$= \left(x + \frac{1}{x} + 6\right)\left(x + \frac{1}{x} + 2\right)$$

$$\text{So, one factor is } \left(x + \frac{1}{x} + 6\right).$$

96. (b) Given, $x^2 - x - 1 = 0$

Here, $a = 1$, $b = -1$ and $c = -1$

$\therefore \alpha$ and β are the roots of the equation

$$x^2 - x - 1 = 0$$

$$\therefore \alpha + \beta = \frac{-b}{a} = 1 \quad \text{and} \quad \alpha\beta = \frac{c}{a} = -1$$

$$\frac{\alpha^2 + \beta^2}{(\alpha^2 - \beta^2)(\alpha - \beta)} = \frac{(\alpha + \beta)^2 - 2\alpha\beta}{(\alpha - \beta)^2(\alpha + \beta)}$$

$$= \frac{(\alpha + \beta)^2 - 2\alpha\beta}{[(\alpha + \beta)^2 - 4\alpha\beta](\alpha + \beta)}$$

$$= \frac{(1)^2 - 2 \times -1}{[(1)^2 - 4 \times -1] \times 1} = \frac{3}{5}$$

97. (d) Given,

$$x^2 = 6 + \sqrt{6 + \sqrt{6 + \sqrt{6 + \dots \infty}}}$$

$$\text{So, } x^2 = 6 + \sqrt{x^2} \Rightarrow x^2 = 6 + x$$

$$\Rightarrow x^2 - x - 6 = 0$$

$$\Rightarrow x^2 + 2x - 3x - 6 = 0$$

$$\Rightarrow x(x + 2) - 3(x + 2) = 0$$

$$\Rightarrow (x - 3)(x + 2) = 0$$

$$\therefore x = 3, -2$$

98. (d) Given, $ax^2 + bx + b = 0$

$$\Rightarrow x^2 + \frac{b}{a}x + \frac{b}{a} = 0$$

$$\therefore \text{Sum of roots, } \alpha + \beta = \frac{-b}{a}$$

$$\text{and products of roots, } \alpha\beta = \frac{b}{a}$$

Hence, product of roots is not equal to the sum of roots,

So, statement I is not correct.

Now, for roots to be real and unequal, $D > 0$

$$\Rightarrow b^2 - 4ac > 0$$

$$\Rightarrow b^2 - 4a(b) > 0 \quad [\because c = b]$$

$$\Rightarrow b^2 - 4ab > 0 \Rightarrow b^2 > 4ab$$

$$\therefore b > 4a$$

So, if $b > 4a$, then roots are unequal and real, so statement II is not always true, it depends on values of a and b .

99. (d) Given, $Ax^2 + Bx + C = 0 \quad \dots(i)$

Since, the given roots are -1 and 1.

$$\therefore \text{Sum of roots} = -1 + 1 = 0$$

$$\text{and product of roots} = 1 \times (-1) = -1$$

$$\Rightarrow \text{quadratic equation is } x^2 -$$

$$(\text{sum of roots})x + \text{product of roots} = 0$$

$$\Rightarrow x^2 - 0 \cdot x - 1 = 0 \Rightarrow x^2 - 1 = 0$$

On comparing with above equation from Eq. (i), we get

$$A = 1 \text{ and } C = -1$$

So, A and C are of opposite sign.

100. (d) Given, $(a^2 - bc)x^2 + 2(b^2 - ac)x + (c^2 - ab) = 0$

Since, the given roots are equal.

$$\therefore D = 0$$

$$\text{i.e. } [2(b^2 - ac)]^2 - 4(a^2 - bc)$$

$$(c^2 - ab) = 0$$

$$\begin{aligned} &\Rightarrow 4(b^4 + a^2c^2 - 2ab^2c) \\ &\quad - 4(a^2c^2 - bc^3 - a^3b + ab^2c) = 0 \\ &\Rightarrow 4b^4 + 4a^2c^2 - 8ab^2c - 4a^2c^2 \\ &\quad + 4bc^3 + 4a^3b - 4ab^2c = 0 \\ &\Rightarrow 4b^4 - 12ab^2c + 4bc^3 + 4a^3b = 0 \\ &\Rightarrow b^3 + c^3 + a^3 - 3abc = 0 \\ &\therefore a^3 + b^3 + c^3 = 3abc \end{aligned}$$

101. (a) Given m and n are the roots of the given equation $ax^2 + bx + c = 0$

$$\therefore \text{Sum of roots} = m + n = -b/a \quad \dots(i)$$

$$\text{and product of roots} = mn = \frac{c}{a} \quad \dots(ii)$$

$$\begin{aligned} \text{Now, } \frac{m^2 + 1}{m} + \frac{n^2 + 1}{n} &= \frac{m^2n + n + mn^2 + m}{mn} \\ &= \frac{mn(m + n) + (m + n)}{mn} \\ &= \frac{(m + n)(mn + 1)}{mn} \\ &= \frac{(m + n)(\frac{c}{a} + 1)}{\frac{c}{a}} = \frac{-b(a + c)}{ac} \end{aligned}$$

[using Eqs. (i) and (ii)]

$$\begin{aligned} \text{and } \frac{m^2 + 1}{m} \times \frac{n^2 + 1}{n} &= \frac{(m^2 + 1)(n^2 + 1)}{mn} \\ &= \frac{m^2n^2 + n^2 + m^2 + 1}{mn} \\ &= \frac{(mn)^2 + (m + n)^2 - 2mn + 1}{mn} \\ &= \frac{\left(\frac{c}{a}\right)^2 + \left(\frac{-b}{a}\right)^2 - 2\left(\frac{c}{a}\right) + 1}{\frac{c}{a}} \\ &= \frac{c^2 + b^2 - 2ac + a^2}{ac} \\ &= \frac{b^2 + (a - c)^2}{ac} \end{aligned}$$

We know that, quadratic equation can be written as

$$x^2 - (\text{Sum of roots})x + \text{Product of roots} = 0$$

$$\Rightarrow x^2 - \left(\frac{-b(a + c)}{ac}\right)x + \left(\frac{b^2 + (a - c)^2}{ac}\right) = 0$$

$$\begin{aligned} &\Rightarrow acx^2 + b(a + c)x + b^2 + (a - c)^2 = 0 \\ &\Rightarrow acx^2 + (ab + bc)x + b^2 + (a - c)^2 = 0 \end{aligned}$$

102. (c) When mistake is done in first degree term, then the roots of the equation are -9 and -1 .

$$\therefore \text{Equation is } (x + 1)(x + 9) = x^2 + 10x + 9 \quad \dots(i)$$

When mistake is done in constant term, then the roots of equation are 8 and 2 .

$$\therefore \text{Equation is } (x - 2)(x - 8) = x^2 - 10x + 16 \quad \dots(ii)$$

From Eqs. (i) and (ii), we get

$$x^2 - 10x + 9 = 0 \text{ is the correct equation.}$$

103. (c) Given equation,

$$\begin{aligned} 7(x + 2a)^2 + 3a^2 &= 5a(7x + 23a) \\ \Rightarrow 7(x^2 + 4a^2 + 4ax) + 3a^2 &= 35ax + 115a^2 \\ \Rightarrow 7x^2 - 7ax - 84a^2 &= 0 \\ \Rightarrow x^2 - ax - 12a^2 &= 0 \\ \Rightarrow (x + 3a)(x - 4a) &= 0 \\ \Rightarrow x = -3a \text{ and } x = 4a \end{aligned}$$

Since, m and n are the roots of the given equation.

$$\text{Let } m = 4a \text{ and } n = -3a (\because m > n)$$

$$\begin{aligned} \therefore 3m - n &= 3(4a) - (-3a) \\ &= 12a + 3a = 15a \end{aligned}$$

104. (a) Given equation is $px^2 + qx + r = 0$.

Let one root of the equation be α .

Then, other root $= 3\alpha$

$$\begin{aligned} \therefore \text{Sum of roots} &= \alpha + 3\alpha = -\frac{q}{p} \\ \Rightarrow 4\alpha &= -\frac{q}{p} \Rightarrow \alpha = -\frac{q}{4p} \quad \dots(i) \end{aligned}$$

$$\text{and product of roots} = (\alpha) \cdot (3\alpha) = \frac{r}{p}$$

$$\begin{aligned} \Rightarrow 3\alpha^2 &= \frac{r}{p} \Rightarrow 3\left(-\frac{q}{4p}\right)^2 = \frac{r}{p} \\ &\quad [\text{from Eq. (i)}] \end{aligned}$$

$$\Rightarrow \frac{3q^2}{16p^2} = \frac{r}{p} \Rightarrow 3q^2p = 16p^2r$$

$$\Rightarrow 3q^2 = 16pr$$

105. (b) I. Given, m and n are the roots of the equation

$$x^2 + ax + b = 0.$$

$$\therefore \text{Sum of roots, } m + n = -a \quad \dots(i)$$

$$\text{and product of roots, } mn = b \quad \dots(ii)$$

Also, given m^2 and n^2 are the roots of the equation

$$x^2 - cx + d = 0.$$

$$\therefore m^2 + n^2 = c \quad \dots(iii)$$

$$\text{and } m^2n^2 = d \quad \dots(iv)$$

On squaring both sides of Eq. (i), we get

$$m^2 + n^2 + 2mn = a^2$$

[from Eqs. (i) and (ii)]

$$\Rightarrow c + 2b = a^2 \Rightarrow c = a^2 - 2b$$

$$\Rightarrow 2b - a^2 = -c$$

Hence, statement I is not correct.

II. From Eq. (ii), $m^2n^2 = b^2$

$$\Rightarrow b^2 = d$$

Hence, statement II is correct.

106. (a) If a is positive and $b^2 - 4ac \leq 0$, then the sign of quadratic polynomial $ax^2 + bx + c$ is always positive.

107. (c) We have, $x + \frac{100}{x} > 50$

$$\text{and } 1 \leq x \leq 100 \Rightarrow x^2 - 50x + 100 > 0$$

$$\begin{aligned} \Rightarrow x &= \frac{50 \pm \sqrt{2500 - 400}}{2} \\ &= \frac{50 \pm 10\sqrt{21}}{2} \end{aligned}$$

$$x < 25 - 5\sqrt{21} \text{ or } x > 25 + 5\sqrt{21}$$

$$\Rightarrow x < 2.087 \text{ or } x > 47.91$$

$$\Rightarrow x = 1 \text{ and } 2 \text{ or } x = 48, 49, \dots, 100$$

$\therefore$ Number of total values of

$$x = 2 + 53 = 55$$

108. (c) We have, $x^2 - 4x - \log_{10} N = 0$

We know that, roots are real, if $D \geq 0$

$$\Rightarrow 16 - 4(-\log_{10} N) \geq 0$$

$$\Rightarrow 16 + 4 \log_{10} N \geq 0$$

$$\Rightarrow \log_{10} N \geq -4$$

$$\Rightarrow N \geq 10^{-4} \Rightarrow N \geq \frac{1}{10000}$$

$$\text{Minimum value of } N = \frac{1}{10000}$$

109. (c) Let $x = \sqrt{4 + \sqrt{4 - \sqrt{4 + \dots}}}$

$$\therefore x = \sqrt{4 + \sqrt{4 - x}}$$

$$\Rightarrow x^2 = 4 + \sqrt{4 - x}$$

$$\Rightarrow (x^2 - 4)^2 = 4 - x$$

$$\Rightarrow (x^2 - 4)^2 + x = 4 \quad \dots(i)$$

$$\text{For option (c), } x = \frac{\sqrt{13} + 1}{2}$$

From Eq. (i),

$$\begin{aligned} \text{LHS} &= \left[\left(\frac{\sqrt{13} + 1}{2} \right)^2 - 4 \right]^2 + \frac{\sqrt{13} + 1}{2} \\ &= \left[\frac{13 + 1 + 2\sqrt{13} - 16}{4} \right]^2 + \frac{\sqrt{13} + 1}{2} \\ &= 4 = \text{RHS} \end{aligned}$$

110. (c) We have, $-2 \leq x \leq 1 \quad \dots(i)$

$$-1 \leq y \leq 2$$

$$\therefore -2 \leq -y \leq 1 \quad \dots(ii)$$

$$\text{and } 3 \leq z \leq 6$$

$$6 \leq 2z \leq 12 \quad \dots(iii)$$

On adding Eqs. (i), (ii) and (iii), we get

$$-2 - 2 + 6 \leq x - y + 2z \leq 1 + 1 + 12$$

$$2 \leq k \leq 14$$

- 111. (a)** Let α and β be the roots of equation $ax^2 + bx + c = 0$.

$$\text{Then, } \alpha + \beta = \frac{-b}{a}$$

$$\text{and } \alpha\beta = \frac{c}{a}$$

$$\text{Now, } \alpha + \beta = \left(\frac{1}{\alpha}\right)^2 + \left(\frac{1}{\beta}\right)^2$$

$$\Rightarrow \alpha + \beta = \frac{\beta^2 + \alpha^2}{(\alpha\beta)^2} = \frac{(\alpha + \beta)^2 - 2\alpha\beta}{(\alpha\beta)^2}$$

$$\Rightarrow \frac{-b}{a} = \frac{b^2/a^2 - 2c/a}{c^2/a^2}$$

$$\Rightarrow ab^2 - 2a^2c = -bc^2$$

$$\Rightarrow ab^2 + bc^2 = 2a^2c$$

- 112. (d)** Here, $a = 1, b = -p, c = q$

$$\text{and } a' = 1, b' = q, c' = -p$$

Applying common root condition,

$$(a'c - ac')^2 = (bc' - b'c)(ab' - a'b)$$

$$\Rightarrow (q + p)^2 = (p^2 - q^2)(q + p)$$

$$\Rightarrow (p + q) = (p - q)(p + q)$$

$$\Rightarrow p - q = 1$$

$$\Rightarrow p - q - 1 = 0$$

- 113. (c)** Let α and α^2 be the roots of equation $x^2 + px + q = 0$.

$$\text{Then, } \alpha \cdot \alpha^2 = q$$

$$\therefore \alpha = (q)^{1/3} \quad \dots(i)$$

$$\text{and } \alpha + \alpha^2 = -p$$

$$\therefore (q)^{1/3} + (q)^{2/3} = -p \quad \dots(ii)$$

On cubing both sides, we get

$$q + q^2 + 3q(q^{1/3} + q^{2/3}) = -p^3$$

$$\Rightarrow q + q^2 + 3q(-p) = -p^3$$

[from Eq. (ii)]

$$\Rightarrow q + q^2 + p^3 = 3pq$$

$$\Rightarrow p^3 + q^2 + q = 3pq$$

- 114. (a)** Consider the line,

$$2x + 6y = 21 \quad \dots(i)$$

The point (0, 0) does not satisfy Eq. (i) but the point (0,0) satisfy the equation $2x + 6y \leq 21$.

Now, consider the line,

$$5x - 2y = 10 \quad \dots(ii)$$

The point (0, 0) does not satisfy

Eq. (ii) but the point (0,0) satisfy the equation $5x - 2y \leq 10$.

- 115. (b)** We have, $A(x) = x^2 + px + q$

$\therefore (x-m)$ and $(x-km)$ are factors of $A(x)$, then m and km are roots of $A(x) = 0$

$$\therefore m + km = -p \Rightarrow m(k+1) = -p$$

$$\Rightarrow m = \frac{-p}{(k+1)} \quad \dots(i)$$

$$\text{and } m \cdot km = q \Rightarrow m^2k = q$$

$$\Rightarrow \frac{p^2}{(k+1)^2} \cdot k = q \quad [\text{from Eq. (i)}]$$

$$\Rightarrow (k+1)^2 q = kp^2$$

- 116. (c)** Given points are $P(5, -1), Q(3, -2)$ and $R(1, 1)$.

If points P, Q and R lie in the solution of inequations $x + y \leq 4$ and $x - y \geq 2$, then points P, Q and R satisfy these inequations.

$$\text{For point } P(5, -1), \quad 5 - 1 \leq 4 \quad [\text{true}]$$

$$5 + 1 \geq 2 \quad [\text{true}]$$

$$\text{For point } Q(3, -2), \quad 3 - 2 \leq 4 \quad [\text{true}]$$

$$3 + 2 \geq 2 \quad [\text{true}]$$

$$\text{and for point } R(1, 1), \quad 1 + 1 \leq 4 \quad [\text{true}]$$

$$1 - 1 \geq 2 \quad [\text{false}]$$

So, the point $R(1, 1)$ does not satisfy the inequations.

Hence, points P and Q lie in the solution of inequations.

- 117. (a)** Given, $x = \frac{\sqrt{a+2b} + \sqrt{a-2b}}{\sqrt{a+2b} - \sqrt{a-2b}}$

By rationalising, we get

$$x = \frac{(\sqrt{a+2b})^2 + (\sqrt{a-2b})^2 + 2\sqrt{(a+2b)(a-2b)}}{(\sqrt{a+2b})^2 - (\sqrt{a-2b})^2}$$

$$\Rightarrow x = \frac{a+2b+a-2b+2\sqrt{a^2-4b^2}}{a+2b-a-2b}$$

$$\Rightarrow x = \frac{2a+2\sqrt{a^2-4b^2}}{4b}$$

$$\Rightarrow x = \frac{a+\sqrt{a^2-4b^2}}{2b}$$

$$\Rightarrow bx^2 - ax + b = 0$$

- 118. (a)** Given, $\sqrt{3x^2 - 7x - 30}$
- $$= (x-5) + \sqrt{2x^2 - 7x - 5}$$

On squaring both sides, we get

$$3x^2 - 7x - 30 = (x-5)^2 + (2x^2 - 7x - 5)$$

$$+ 2(x-5)\sqrt{2x^2 - 7x - 5}$$

$$\Rightarrow 3x^2 - 7x - 30 = x^2 + 25 - 10x + 2x^2 - 7x - 5$$

$$+ (2x-10)\sqrt{2x^2 - 7x - 5}$$

$$\Rightarrow 3x^2 - 7x - 30 - x^2 + 10x + 7x - 2x^2 - 20$$

$$= (2x-10)\sqrt{2x^2 - 7x - 5}$$

$$\Rightarrow (10x - 50)$$

$$= (2x-10)\sqrt{2x^2 - 7x - 5}$$

$$\Rightarrow 10(x-5) = 2(x-5)\sqrt{2x^2 - 7x - 5}$$

$$\Rightarrow \sqrt{2x^2 - 7x - 5} = 5 \quad \dots(i)$$

Again, on squaring both sides of Eq. (i), we get

$$2x^2 - 7x - 5 = 25$$

$$\Rightarrow 2x^2 - 7x - 30 = 0 \quad \dots(ii)$$

If α and β are roots of Eq. (ii), then

$$\alpha\beta = \frac{-30}{2} = -15$$

- 119. (a)** Since, the roots of equation

$$lx^2 + mx + m = 0$$

are in the ratio $p : q$.

Then, we can take roots as pk and qk .

$$\therefore \text{Sum of roots} = pk + qk = -\frac{m}{l}$$

$$\Rightarrow (p+q)k = -\frac{m}{l} \quad \dots(i)$$

and product of roots

$$= (pq)k^2 = \frac{m}{l} \quad \dots(ii)$$

On dividing Eq. (i) by Eq. (ii), we get

$$\frac{(p+q)k}{(pq)k^2} = -1$$

$$\Rightarrow \left(\frac{p+q}{pq}\right) = -k = \frac{m/l}{(p+q)}$$

[using Eq. (i)]

$$\Rightarrow \frac{(p+q)^2}{pq} = \frac{m}{l} \Rightarrow \frac{p+q}{\sqrt{pq}} = \pm \sqrt{\frac{m}{l}}$$

$$\Rightarrow \frac{p+q}{\sqrt{pq}} = -\sqrt{\frac{m}{l}}$$

$$\Rightarrow \sqrt{\frac{p}{q}} + \sqrt{\frac{q}{p}} + \sqrt{\frac{m}{l}} = 0$$

- 120. (d)** We have, $1 + \frac{1}{x} - \frac{1}{x^2} \geq 0$

$$\Rightarrow \frac{x^2 + x - 1}{x^2} \geq 0$$

$$\text{As, } x^2 \geq 0, \text{ then } x^2 + x - 1 \geq 0$$

$$\text{If } x^2 + x - 1 = 0,$$

$$\text{then } x = \frac{-1 \pm \sqrt{5}}{2}$$

$$\therefore \left[x - \left(\frac{-1 + \sqrt{5}}{2} \right) \right] \left[x - \left(\frac{-1 - \sqrt{5}}{2} \right) \right] \geq 0$$

$$\Rightarrow x \leq \frac{-1 - \sqrt{5}}{2} \text{ or } x \geq \frac{-1 + \sqrt{5}}{2}$$

19

SET THEORY

Usually (1-2) questions have been asked from this chapter. Generally questions are asked from these topics more results on operations on set.



SET

A well defined collection of objects, is called a set. The objects in a set are called its members or elements. Sets are usually denoted by the capital letters A, B, C, X, Y and Z etc. And the elements of a set are denoted by small letters a, b, c etc. If x is an element of set A , we can write $x \in A$, which means that ' x belongs to A ' or that x is an element of A .

If x does not belong to A , we can write, $x \notin A$. e.g.

- The collection of vowels in English alphabet is a set A containing five elements namely a, e, i, o and u , where $a \in A$ but $b \notin A$.
- The collection of first four prime numbers is a set A containing the elements 2, 3, 5 and 7, where $3 \in A$ but $1 \notin A$.

Representation of Sets

Sets are generally represented by following two ways:

1. **Roster or Tabular form or listing method** In this method, all the elements of a set are listed, within curly braces $\{ \}$ being separated by commas. e.g.
 - (i) If A is a set of first eight prime numbers, then $A = \{2, 3, 5, 7, 11, 13, 17, 19\}$
 - (ii) If B is a set of squares of first five natural numbers, then $B = \{1, 4, 9, 16, 25\}$
 - (iii) If A is a set of vowels of English alphabets, then $A = \{a, e, i, o, u\}$

Note The order in which the elements are written in a set makes no difference and also the repeated elements are taken only once each.

2. **Set Builder form or Rule method** In this method, instead of listing all the elements of a set, we write the set by some special property or properties satisfied by all its elements and write it as.
 - (i) The set B of all even natural numbers can be written as
 $B = \{x : x \text{ is a natural number and } x = 2n \text{ for } n \in N\}$ or $B = \{x : x \in N, x = 2n, n \in N\}$ and reads it as ' B ' is the set consisting of all elements x such that x has the property of even natural number. The symbol ':' or '1' stands for such that
 - (ii) The set $A = \{3, 5, 7, 9, 11\}$, then it is represented as
 $A = \{x : x = 2n + 1 \text{ where } n \in N, n < 6\}$

- (iii) The set $A = \{0, 1, 4, 9, 16, \dots\}$ can be written as $A = \{x : x = n^2, n \in \mathbb{Z}\}$, where $\mathbb{Z}$ is the set of integers.

Types of Sets

- Empty set** A set which does not contain any element is called an empty set or null set or void set. It is denoted by ϕ or $\{\}$.
e.g. A = set of all odd numbers divisible by 2
and $B = \{x : x \in \mathbb{N} \text{ and } 5 < x < 6\}$
The sets which have atleast one element are called non void or non-empty set.
- Singleton set** A set consisting of a single element is called a singleton set. e.g.
(i) The set $\{5\}$ is a singleton set.
(ii) $\{x | x \in \mathbb{W} \text{ and } x + 6 = 6\} = \{0\}$
which is a singleton set.
- Finite set** A set which consists of a definite number of elements, is called a finite set. Empty set is also a finite set. e.g.
(i) The set $\{1, 2, 3, 4\}$ is a finite set, because it contains a definite number of elements i.e. only 4 elements.
(ii) B = Set of vowels in English Alphabets
 $= \{a, e, i, o, u\}$;
(iii) A = Set of even prime natural numbers.
 $\Rightarrow A = \{2\}$

Note The number of distinct elements contained in a set A is called the cardinal number of A and is denoted as $n(A)$.
If $A = \{a, e, i, o, u\}$, then $n(A) = 5$.

- Infinite set** A set which consists of infinite number of elements is called an infinite set. It is represented by writing a few elements of the set followed by
e.g.
(i) Set of square of natural numbers is an infinite set, because natural numbers are infinite and it can be represented as $\{1, 4, 9, 16, 25, \dots\}$.
(ii) Set of all points in a plane.
- Equal sets** Two sets A and B are said to be equal, if they have exactly the same elements and we write $A = B$. Otherwise, two sets are said to be unequal and we write $A \neq B$.
e.g. Let $A = \{1, 2, 3, 4\}$ and $B = \{4, 3, 1, 2\}$, then $A = B$ because each element of A is in B and *vice-versa*.
- Equivalent sets** Two sets A and B are equivalent, if their cardinal numbers are same i.e. $n(A) = n(B)$ and we write $A \leftrightarrow B$ or $A \sim B$
e.g. Let $A = \{1, 2, 3\}$ and $B = \{4, 5, 6\}$,
then $n(A) = n(B)$
 $A \leftrightarrow B$ or $A \sim B$

- Subsets** Let A and B be two sets. If every element of A is an element of B , then A is called a subset of B .
If A is subset of B , then we can write $A \subseteq B$ which is read as ' A is a subset of B ' or ' A is contained in B '.
(i) Every set is a subset of itself i.e. $A \subseteq A, B \subseteq B$.
(ii) Empty set is a subset of every set i.e. $\phi \subseteq A$.
e.g. Let $A = \{2, 4, 6\}$ and $B = \{6, 4, 2, 8\}$.
Then, $A \subseteq B$ but $B \not\subseteq A$ i.e. A is a subset of B but B is not a subset of A .

Note The total number of subsets of a finite set containing n elements is 2^n .

- Super set** If A is a subset of B , then we say that B is superset of A and we write $B \supseteq A$.
e.g. If $A = \{1, 2, 3, 4\}$ and $B = \{1, 2, 3, 4, 5, 6\}$,
then, $B \supseteq A$.
- Comparability of sets** Two sets A and B are said to be comparable, if either $A \subset B$ or $B \subset A$ or $A = B$, otherwise A and B are said to be incomparable.
e.g. Let $A = \{1, 2, 3\}$, $B = \{1, 2, 4, 6\}$ and $C = \{1, 2, 4\}$
Since, $A \not\subseteq B$ and $B \not\subseteq A$.
So, A and B are incomparable but $C \subset B$ and so B and C are comparable sets.
- Proper subset** If $A \subseteq B$ and $A \neq B$, then A is called a proper subset of B and we write $A \subset B$.
e.g. If $A = \{1, 2, 3, 4, \dots\}$
and $B = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$,
then $A \subset B$.

Note The total number of proper subset of a finite set containing n elements is $(2^n - 1)$.

- Universal set** If there are some sets under consideration, then there happens to be a set which is a superset of each one of the given sets. Such a set is known as the universal set and it is denoted by U .
e.g.
• If $A = \{1, 2, 3\}$, $B = \{2, 3, 4, 5, 7\}$ and $C = \{2, 4, 6, 8\}$, then $U = \{1, 2, 3, 4, 5, 6, 7, 8\}$ is a universal set for A, B and C .
• For the set of all integers, the universal set can be the set of rational numbers or the set of real numbers.
- Power set** The collection of all subsets of a set A , is called power set of A and it is denoted by $P(A)$. In $P(A)$, every element is a set.
e.g. Let $A = \{1, 2, 3\}$
Then, $P(A) = \{\phi, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{2, 3\}, \{3, 1\}, \{1, 2, 3\}\}$

Properties of Power Sets

1. Each element of a power set is a set.
2. If set A has n elements, then $P(A)$ has 2^n elements.
3. If A is an empty set ϕ or $\{\}$, then $P(A)$ has just one element i.e. $P(A) = \{\phi\}$.
4. If $A \subseteq B$, then $P(A) \subseteq P(B)$.

EXAMPLE 1. The set $S = \{x \in N : x + 3 = 3\}$ is a

- a. null set b. singleton set
c. infinite set d. None of these

Sol. a. Given, $S = \{x \in N : x + 3 = 3\}$

$\therefore S = \{\}$
So, S is a null set.

EXAMPLE 2. If $A = \{x : x \text{ is an odd integer}\}$ and $B = \{x : x^2 - 8x + 15 = 0\}$. Then, which one of the following is correct.

☑ 2013 I

- a. $A = B$ b. $A \subseteq B$ c. $B \subseteq A$ d. $A \subseteq B^c$

Sol. c. Given that, $A = \{x : x \text{ is an odd integer}\}$

and $B = \{x : x^2 - 8x + 15 = 0\} = \{x : x^2 - 5x - 3x + 15 = 0\}$
 $= \{x : x(x - 5) - 3(x - 5) = 0\}$
 $= \{x : (x - 5)(x - 3) = 0\} = \{3, 5\}$

Since, B has two odd elements.
 $\therefore B \subseteq A$

EXAMPLE 3. If $A = \{a, b\}$, then power set of A is

- a. $\{\phi, a, b\}$ b. $\{\phi, a, b, A\}$
c. $\{\phi, a, b, ab\}$ d. $\{\phi, \{a\}, \{b\}, A\}$

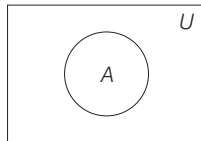
Sol. d. $A = \{a, b\}$

The subsets of A are $\phi, \{a\}, \{b\}, \{a, b\}$.
 $\therefore P(A) = \{\phi, \{a\}, \{b\}, \{a, b\}\} = \{\phi, \{a\}, \{b\}, A\}$

VENN DIAGRAM

‘Venn diagram’ is the diagrammatic representation of various types of sets and operations on sets.

The universal set is usually represented by a rectangular region and other subsets of the universal set are represented by circles inscribed in it. Each inscribed circle represents a set (subset of universal set).



Operations on Sets

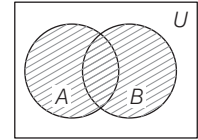
1. **Union of Sets** Let A and B be any two sets. The union of A and B is the set of all the elements of A and all the elements of B , the common elements being taken only once. It is denoted by $A \cup B$ and read as ‘ A union B .’

Thus, $A \cup B = \{x : x \in A \text{ or } x \in B\}$

In the given Venn diagram $A \cup B$ is denoted by the shaded region.

e.g. If $A = \{1, 2, 3, 4\}$

and $B = \{1, 2, 3, 5, 7\}$, then $A \cup B = \{1, 2, 3, 4, 5, 7\}$



Note • $x \in A \cup B \Leftrightarrow x \in A \text{ or } x \in B$

• $x \notin A \cup B \Leftrightarrow x \notin A \text{ and } x \notin B$ • $A \subseteq A \cup B, B \subseteq A \cup B$

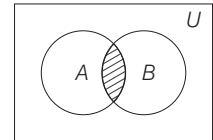
2. **Intersection of sets** Let A and B be any two sets. The intersection of A and B is the set of all those elements which belongs to both A and B . It is denoted by $A \cap B$ and read as A intersection B .

Thus, $A \cap B = \{x : x \in A \text{ and } x \in B\}$

$A \cap B$ is represented by the shaded region in the given Venn diagram

e.g. If $A = \{1, 2, 3, 4\}$ and

$B = \{1, 3, 7, 9\}$, then $A \cap B = \{1, 3\}$.



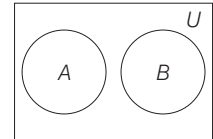
Note • $x \in A \cap B \Leftrightarrow x \in A \text{ and } x \in B$

• $x \notin A \cap B \Leftrightarrow x \notin A \text{ or } x \notin B$ • $A \cap B \subseteq A$ and $A \cap B \subseteq B$

3. **Disjoint sets** Two sets A and B are said to be disjoint, if they have no common element i.e. $A \cap B = \phi$

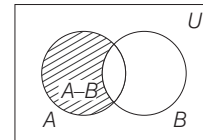
e.g. If $A = \{1, 2, 4\}$, $B = \{3, 5, 6\}$, then $A \cap B = \phi$

So, A and B are disjoint sets.



4. **Difference of sets** Let A and B be two sets. The difference of two sets is the set of all those elements of A which do not belong to B . It is denoted by $A - B$ and read as A minus B .

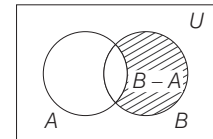
$\therefore A - B = \{x : x \in A \text{ and } x \notin B\}$



The shaded part in the given Venn diagram represents $A - B$.

Similarly, the difference $B - A$ is the set of all those elements of B that do not belong to A .

$\therefore B - A = \{x : x \in B \text{ and } x \notin A\}$



The shaded part in the given Venn diagram represents $B - A$.

e.g.

- If $A = \{2, 4, 6, 8, 10\}$ and $B = \{2, 4, 6, 12, 14\}$, then $A - B = \{8, 10\}$ and $B - A = \{12, 14\}$.
- If $A = \{1, 2, 3\}$ and $B = \{1, 2, 3, 4\}$, then $A - B = \phi$.

Symmetric Difference of Two Sets

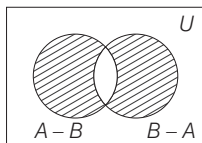
Let A and B be two sets. The symmetric difference of sets A and B is the set $(A - B) \cup (B - A)$ and is denoted by $A \Delta B$ i.e. it is a set consisting of all those members of A which are not in B or all those which are in B but not in A .

Thus, $A \Delta B = \{x : x \in A \text{ but } x \notin B\} \cup \{x : x \in B \text{ but } x \notin A\}$

$$A \Delta B = (A - B) \cup (B - A)$$

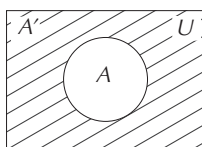
The shaded part in given Venn diagram represents $A \Delta B$.

e.g. If $A = \{1, 2, 3, 4, 5, 6, 7, 8\}$ and $B = \{1, 3, 5, 6, 7, 8, 9\}$,
 $\therefore A - B = \{2, 4\}$ and $B - A = \{9\}$, then $A \Delta B = \{2, 4, 9\}$.



Complement of a Set

Let U be the universal set and A be any set such that $A \subset U$. Then, the complement of A with respect to U is the set of all those elements of U which are not in A . It is denoted by A' or A^C or $U - A$.



Thus, $A' = \{x : x \in U \text{ and } x \notin A\}$

The shaded portion in the above Venn diagram shows the complement of set A . If A is a subset of the universal set U , then its complement A' is also a subset of U . e.g.

- Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, \dots\}$ and $A = \{1, 3, 5, 7, \dots\}$.

Then, the complement of A is

$$A' = U - A = \{2, 4, 6, 8, \dots\}$$

- If $U = \{a, b, c, d, e, f\}$ and $A = \{a, e\}$,
then $A' = \{b, c, d, f\}$.

EXAMPLE 4. If $A = \{0, 1, 2, 3, 4\}$, $B = \{1, 2, 3\}$,
 $C = \{3, 5, 7, 9\}$, then $(A - B) \cap (B - C)$ is

- a. $\{1, 2\}$ b. $\{0, 4\}$ c. $\{0, 1, 2, 4\}$ d. ϕ

Sol. d. Given, $A = \{0, 1, 2, 3, 4\}$, $B = \{1, 2, 3\}$ and $C = \{3, 5, 7, 9\}$

$$\therefore A - B = \{0, 4\} \text{ and } B - C = \{1, 2\}$$

$$\therefore (A - B) \cap (B - C) = \phi$$

Properties of Operations on Sets

- $(A')' = A$
- $A \cup A' = U$
- $A \cap A' = \phi$
- $\phi' = U$
- $U' = \phi$
- $A - B = A \cap B'$
- $B - A = B \cap A'$
- $A - B \subseteq A$
- $A - B \subseteq B$
- $A - \phi = A$ and $A - A = \phi$
- $A \subseteq B \Rightarrow A - B = \phi$
- $A - B = B - A \Leftrightarrow A = B$

Note The sets $A - B$, $A \cap B$ and $B - A$ are mutually disjoint sets i.e. the intersection of any of these two sets is an empty set.

Laws of Algebra of Sets

- Idempotent laws** For any set A , we have
 - $A \cup A = A$
 - $A \cap A = A$
- Identity Laws** For any set A ,
 - $A \cup \phi = A$
 - $A \cap U = A$
 i.e. ϕ and U are identity elements for union and intersection, respectively.
- Commutative laws** For any two sets A and B , we have
 - $A \cup B = B \cup A$
 - $A \cap B = B \cap A$
- Associative laws** If A , B and C are any three sets, then
 - $(A \cup B) \cup C = A \cup (B \cup C)$
 - $A \cap (B \cap C) = (A \cap B) \cap C$
- Distributive laws** If A , B and C are any three sets, then
 - $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
 - $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
- De-Morgan's laws** If A and B are any two sets, then
 - $(A \cup B)' = A' \cap B'$
 - $(A \cap B)' = A' \cup B'$

More Results on Operations on Sets

- If A , B and C are any three sets, then
 - $A - (B \cap C) = (A - B) \cup (A - C)$
 - $A - (B \cup C) = (A - B) \cap (A - C)$
 - $A \cap (B - C) = (A \cap B) - (A \cap C)$
 - $A \cap (B \Delta C) = (A \cap B) \Delta (A \cap C)$
 - $(A - B) \cup (B - A) = (A \cup B) - (A \cap B)$
- If A , B and C are finite sets and U be the finite universal set, then
 - $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
 - $n(A \cup B) = n(A) + n(B) \Leftrightarrow A, B$ are disjoint sets.
 - $n(A - B) = n(A) - n(A \cap B)$
 i.e. $n(A - B) + n(A \cap B) = n(A)$
 - Number of elements which belong to exactly one of A or B
 $= n(A \Delta B) = n(A) + n(B) - 2n(A \cap B)$
 - If A , B and C are finite sets, then
 $n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$
 - $n(A' \cup B') = n((A \cap B)') = n(U) - n(A \cap B)$
 - $n(A' \cap B') = n((A \cup B)') = n(U) - n(A \cup B)$

EXAMPLE 5. If a set A contains 60 elements and another set B contains 70 elements and there are 50 elements in common, then how many elements does $A \cup B$ contain?

- a. 130 b. 100 c. 80 d. 70

Sol. c. Here, $n(A) = 60$, $n(B) = 70$ and $n(A \cap B) = 50$

$$\begin{aligned} \text{Now, } n(A \cup B) &= n(A) + n(B) - n(A \cap B) \\ &= 60 + 70 - 50 = 130 - 50 = 80 \end{aligned}$$

EXAMPLE 6. A group of 40 people who speaks either english or Hindi, out of these 12 speak both English and Hindi and 22 speak Hindi. How many people speak only english not Hindi?

- a. 30 b. 10 c. 18 d. 28

Sol. c. Let E be the set of students who speak English and H be the set of students who speak Hindi.

$$\text{Given, } n(E \cup H) = 40, \quad n(H) = 22 \text{ and } n(E \cap H) = 12$$

$$\begin{aligned} \therefore n(E \cup H) &= n(E) + n(H) - n(E \cap H) \\ 40 &= n(E) + 22 - 12 \Rightarrow n(E) = 30 \end{aligned}$$

$$\begin{aligned} \text{Number of people who speak English only} \\ &= n(E) - n(E \cap H) = 30 - 12 = 18 \end{aligned}$$

Ordered Pairs

An ordered pair consists of two objects or elements in a given fixed order.

If A and B are two sets and $a \in A$, $b \in B$, then the ordered pair of elements a and b is denoted by (a, b) .

The natural numbers and their squares can be represented by ordered pair in the following way.

$$(1, 1), (2, 4), (3, 9), (4, 16), \dots$$

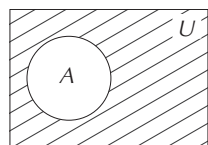
Two ordered pairs (a, b) and (c, d) will be equal, if and only if $a = c$ and $b = d$.

Note $\{a, b\} = \{b, a\}$ but $(a, b) \neq (b, a)$.

PRACTICE EXERCISE

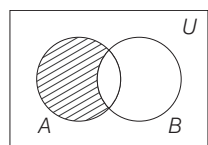
- If $A = \{5, 6, 7\}$ and $B = \{7, 8, 9\}$, then $A \cup B$ is equal to
(a) $\{5, 6, 7, 8, 9\}$ (b) $\{7, 8, 9\}$ (c) $\{5, 6, 7\}$ (d) ϕ
- Given that $A = \{2, 6, 8, 9\}$, $B = \{7, 8, 9, 12\}$, then $B - A$ is equal to
(a) $\{7, 8, 9, 12\}$ (b) $\{7, 12\}$
(c) $\{2, 6, 7, 8, 9, 12\}$ (d) $\{2, 6, 8, 9, 12\}$
- If U is the universal set of all natural numbers and $A = \{1, 2, 3, 4, 5\}$, then compute $A \cap U$.
(a) $\{1, 2, 3, 4\}$ (b) ϕ (c) $\{1, 2, 3, 4, 5\}$ (d) U
- The set $\{2, 4, 16, 256, \dots\}$ can be represented as which one of the following?
(a) $\{x \in N \mid x = 2^{2^n}, n \in N\}$
(b) $\{x \in N \mid x = 2^{2^n}, n = 0, 1, 2, \dots\}$
(c) $\{x \in N \mid x = 2^{4^n}, n = 0, 1, 2, \dots\}$
(d) $\{x \in N \mid x = 2^{2^n}, n = 0, 1, 2, \dots\}$
- If P and Q are any two sets and $P \subset Q$, then
(a) $P \cap Q = \phi$ (b) $P' \cap Q = P$ (c) $P \cap Q = P$ (d) $P \cap Q = Q$
- Which one of the following is a true statement?
(a) $(A - B) \cap (B - A) = \phi$ (b) $(A - B) \cap (B - A) = A$
(c) $(A - B) \cap (B - A) = U$ (d) $(A - B) \cap (B - A) = B$
- If P and Q are any two sets, then $P \cup Q = P \cap Q$, if
(a) P is the empty set (b) Q is the empty set
(c) Both P and Q are empty sets
(d) P and Q are non-empty sets
- The number of non-empty proper subsets of set $A = \{2, 5, 7, 10\}$ is
(a) 16 (b) 15 (c) 14 (d) 8
- Which of the following statements is false for the sets A, B and C , where
 $A = \{x \mid x \text{ is letter of the word 'BOWL'}\}$
 $B = \{x \mid x \text{ is a letter of the word 'ELBOW'}\}$
 $C = \{x \mid x \text{ is a letter of the word 'BELLOW'}\}$
(a) $A \subset B$ (b) $B \supset C$
(c) $B \neq C$ (d) B is a proper subset of C
- If A and B are two sets, then $A \cap (A \cup B)$ equals
(a) A (b) B
(c) ϕ (d) None of these
- The smallest set B such that
 $B \cup \{1, 2\} = \{1, 2, 3, 5, 9\}$ is
(a) $\{3, 5, 9\}$ (b) $\{3, 5, 8\}$
(c) $\{1, 2, 3\}$ (d) None of these
- If $U = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$, $P = \{0, 1, 2, 3\}$,
 $Q = \{2, 3, 4, 5\}$, and $R = \{4, 5, 6\}$, then $Q' \cap (P \cup R)$ is equal to
(a) $\{0, 1, 6\}$ (b) $\{0, 1, 2, 3, 4, 5\}$
(c) $\{6, 7, 8, 9\}$ (d) $\{2, 3\}$
- If P is a non-empty set, then $(P')'$ is equal to
(a) ϕ (b) U (c) $U - P$ (d) P

14. The shaded region in the adjoining diagram is



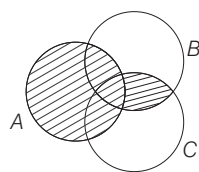
- (a) $A \cup A'$ (b) U (c) A' (d) $A \cap A'$

15. The shaded region in the adjoining diagram represents



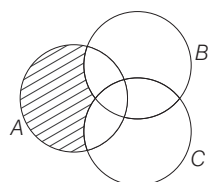
- (a) $(A \cap B)'$ (b) $A \cup B$ (c) $A - B$ (d) $A \cap B$

16. The shaded portion in the following Venn diagram represents



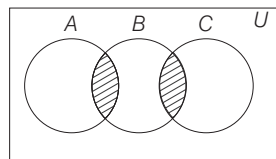
- (a) $A \cup (B \cap C)$ (b) $A \cup (B \cup C)$
(c) $(A \cap B) \cap C$ (d) $(A \cap B) \cup C$

17. The shaded region in the adjoining diagram is



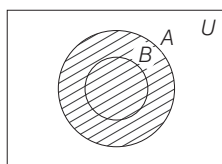
- (a) $A \cap (B - C)$ (b) $A - (B \cup C)$
(c) $A \cap (B \cup C)$ (d) $A \cup (B \cap C)$

18. In the Venn diagram below, shaded portion represents

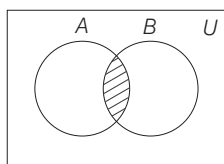


- (a) $A \cup B \cup C$ (b) $A \cap B \cap C$
(c) $(A \cap B) \cup (B \cap C)$ (d) $A \cup B \cap C$

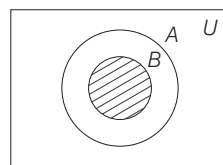
19. The Venn diagram for $A \cup B$ when $B \subset A$ is



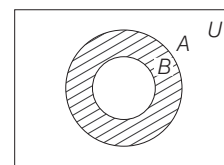
(a)



(b)



(c)



(d)

20. Which one of the following is a correct statement?

- (a) $\phi \in \phi$ (b) $\phi \notin P(\phi)$ (c) $\phi = P(\phi)$ (d) $\phi \in P(\phi)$

21. Let $A = \left\{ 3, \pi, \sqrt{2}, \frac{2}{7}, -5, 3 + \sqrt{7} \right\}$. The subset of A containing all the elements from it which are irrational numbers is

- (a) $\left\{ 3, \frac{2}{7}, -5 \right\}$ (b) $\left\{ 3, \pi, \frac{2}{7}, -5, 3 + \sqrt{7} \right\}$
(c) $\{ \pi, \sqrt{2}, 3 + \sqrt{7} \}$ (d) $\{ 3, -5 \}$

22. If $U = \{x : x \in N\}$, $A = \{x : x \text{ is an odd number}\}$, then A' is equal to

- (a) $\{x : x \text{ is an even number}\}$ (b) $\{x : x \text{ is an odd number}\}$
(c) $\{x : x \text{ is a natural number}\}$ (d) $\{x : x \text{ is an integer}\}$

23. If two sets are disjoint, then their intersection is

- (a) null set (b) singleton
(c) a infinite set (d) None of these

24. If A and B are two non-empty sets, then $A - (A - B)$ equals

- (a) B (b) $A - B$ (c) $A \cap B$ (d) $A' \cap B'$

25. Let two sets A and B have $2n$ and $4n$ elements respectively, where n is a natural number. What can be the minimum number of elements in $A \cup B$?

- (a) $2n$ (b) $3n$ (c) $4n$ (d) $6n$

26. If $P = \{x : x^2 - 3x + 2 = 0\}$, $Q = \{x : x^2 + 4x - 12 = 0\}$, then $P - Q$ is

- (a) $\{1, 2\}$ (b) $\{2\}$ (c) $\{1\}$ (d) $\{4, 3\}$

27. If $A = \{(2^{2n} - 3n - 1) | n \in N\}$ and

$B = \{9(n - 1) | n \in N\}$, then which one of the following is correct?

- (a) $A \subset B$ (b) $A \subset A$ (c) $A = B$
(d) Neither A is a subset of B nor B is a subset of A

28. If A, B and C are any three sets, then

- (a) $A - (B \cup C) = (A - B) \cup (A - C)$
(b) $A - (B \cup C) = (A - B) - (A - C)$
(c) $A - (B \cup C) = (A - B) \cap (A - C)$
(d) $A - (B \cup C) = (A \cup B) - (A \cup C)$

29. If P and Q are two sets such that

$n(P) = m$, $n(Q) = n$ and $n(P \cap Q) = p$, then $n(P \cup Q)$ is equal to

- (a) $m + n$ (b) $m + n + p$ (c) $m + n - p$ (d) $m - n - p$

30. A and B are two sets such that $n(A) = 17$, $n(B) = 23$, $n(A \cup B) = 38$. Then, $n(A \cap B)$ is

- (a) 40 (b) 78 (c) 2 (d) None of these

- 31.** Let $A = \{x : x^2 - 6x + 8 = 0\}$ and $B = \{x : 2x^2 + 3x - 2 = 0\}$. Then, which one of the following is correct?
 (a) $A \subseteq B$ (b) $B \subseteq A$
 (c) Neither $A \subseteq B$ nor $B \subseteq A$ (d) $A = B$
- 32.** If $A = \{1, 2, 3, 4\}$, $B = \{2, 3, 4, 5\}$, $C = \{1, 3, 4, 5, 6, 7\}$, then $A \cap (B \cup C)$ is equal to
 (a) $\{1, 2, 3, 4, 5, 6, 7\}$ (b) $\{1, 2, 3, 4\}$
 (c) $\{1, 2, 3, 4, 5\}$ (d) $\emptyset$
- 33.** Let the set A and B be given by $A = \{1, 2, 3, 4\}$ and $B = \{2, 4, 6, 8, 10\}$ and the universal set $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, then $(A \cup B)'$ is
 (a) $\{2, 4\}$ (b) U
 (c) $\{1, 3, 5, 6, 7, 8, 9, 10\}$ (d) $\{5, 7, 9\}$
- 34.** What is $\{(A \cup B)' \cap A\} - (A - B)$ equal to?
 (a) $\emptyset$ (b) A (c) B (d) B'
- 35.** Which one of the following is a correct statement?
 I. $\{a\} \in \{\{a\}, \{b\}, \{c\}\}$ II. $\{a\} \subseteq \{\{a\}, b, c\}$
 III. $\{a, b\} \subseteq \{\{a\}, b, c\}$ IV. $a \subseteq \{\{a\}, b, c\}$
 (a) Only I (b) Only II (c) III and IV (d) Only IV
- 36.** Which one of the following is an infinite set?
 (a) $\{x : x \text{ is a whole number less than or equal to } 1000\}$
 (b) $\{x : x \text{ is a natural number less than } 1000\}$
 (c) $\{x : x \text{ is a positive integer less than or equal to } 1000\}$
 (d) $\{x : x \text{ is an integer and less than } 1000\}$
- 37.** Which one of the following is not correct in respect of the sets A and B ?
 (a) If $A \subseteq B$, then $B \cup A = B$
 (b) If $A \subseteq B$, then $A \cap (A - B) = \emptyset$
 (c) If $A \subseteq B$, then $B \cap A = A$
 (d) If $A \cap B = \emptyset$, then either $A = \emptyset$ or $B = \emptyset$
- 38.** If $R \in \overleftrightarrow{PQ}, S \notin \overleftrightarrow{PQ}$ and S does not lie on $\overleftrightarrow{PQ}$ extended, then
 (a) $\overleftrightarrow{PQ} \cap \overleftrightarrow{RS} = \emptyset$ (b) $\overleftrightarrow{PQ} \cap \overleftrightarrow{RS} = \{R\}$
 (c) $\overleftrightarrow{PQ} \cap \overleftrightarrow{RS} = \{S\}$ (d) $\overleftrightarrow{PQ} \cap \overleftrightarrow{RS} = \{P\}$
- 39.** In a group of 1000 people, there are 750 people who can speak Hindi and 400 who can speak English. How many can speak Hindi only?
 (a) 600 (b) 150 (c) 300 (d) 500
- 40.** In a committee 50 people speak French, 20 speak Spanish and 10 speak both Spanish and French. How many speak atleast one of these two languages?
 (a) 60 (b) 50 (c) 30 (d) 70
- 41.** Every student in a class of 42 students, studies atleast one of the subjects, Mathematics, English and Commerce, 14 students study Mathematics, 20 Commerce and 24 English. 3 students study Mathematics and Commerce, 2 English and Commerce and there is no student who studies all the three subjects. The number of students who study Mathematics but not Commerce is
 (a) 4 (b) 3 (c) 12 (d) 11
- 42.** In an examination, 52% candidates failed in English and 42% failed in Mathematics. If 17% candidates failed in both English and Mathematics, what percentage of candidates passed in both the subjects?
 (a) 18% (b) 21% (c) 23% (d) 25%
- 43.** State which of the following statements about sets is/are true?
 I. Every subset of a finite set is finite.
 II. $\emptyset$ is a subset of $\{0\}$.
 Select the correct answer using the codes given below
 (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II
- 44.** Which of the following is/are examples of empty set?
 I. $A = \{x : x + 3 = 3, x \in I\}$
 II. $B = \{x : x \text{ is a positive even integer and prime}\}$
 III. $C = \{x : x^2 = 16, x \text{ is odd integer}\}$
 Select the correct answer using the codes given below
 (a) Only I (b) I and III (c) Only III (d) I, II and III
- 45.** Consider the following statements :
 I. $A' \cup B = (A \cap B)'$ II. $(\phi)'\cup\phi$
 III. $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
 Which of the statement(s) given above is/are correct?
 (a) Only II (b) I and III (c) Only III (d) II and III
- 46.** Let $P =$ Set of all integral multiples of 3
 $Q =$ Set of all integral multiples of 4
 $R =$ Set of all integral multiples of 6
 Consider the following relations
 I. $P \cup Q = R$ II. $P \subset R$ III. $R \subset (P \cup Q)$
 Which of the relation(s) given above is/are correct?
 (a) Only I (b) Only II (c) Only III (d) II and III
- 47.** Consider the following statements:
 I. Set of all points of a given line is a infinite set.
 II. The set of all birds in a zoo is a infinite set.
 III. Good books in a school library is a set.
 Which of the statement(s) given above is/are correct?
 (a) Only I (b) Only II (c) Only III (d) All of these
- 48.** Given that the set $A = \{0, 1, 2, 3\}$, which of the following statements about A are true?
 I. A is a finite set.
 II. A is a subset of the set of integers.
 III. $\{1, 2\}$ is a proper subset of A .
 IV. A is the null set.
 Select the correct answer using the codes given below
 (a) I, II and III (b) I and IV
 (c) I and III (d) All of these

- 49.** State which of the sets given below are infinite set?
 I. Set of all concentric circles.
 II. $\{x : x \text{ is a multiple of } 2, x \text{ is an integer}\}$
 III. The set of lines which are parallel to X -axis.
 IV. The set of positive integers greater than 100.

Select the correct answer using the codes given below

- (a) I and II (b) II and III
 (c) Only I (d) All of these
- 50.** Which of the following sets are equivalent?
 I. $A = \{1, 2, 3, 4, 5\}, B = \{7, 8, 9, 10, 11\}$
 II. $A = \{x, y, z\}, B = \{p, q\}$
 III. $P = \{2, 4, 6, 8\}, R = \{a, b, c, d\}$
 IV. $A = \{36, 39, 42, 45\}, B = \{42, 39, 45, 36\}$

Select the correct answer using the codes given below

- (a) I, II and IV (b) II, III and IV
 (c) I, III and IV (d) None of these

Directions (Q. Nos. 51-52) Answer the questions based on the following information.

In a survey of 250 students, it was found that 150 play cricket, 100 play basketball and 120 play football, further, 30 of them play both basketball and football, 50 play both cricket and basketball and 60 play both cricket and football.

- 51.** The maximum number of students who play all the three sports
 (a) 20 (b) 10 (c) 15 (d) None of these
- 52.** If 5 students play none of the three sports then numbers of students who play at least two sports
 (a) 100 (b) 110 (c) 120 (d) 130

PREVIOUS YEARS' QUESTIONS

- 53.** If $A = \{x : x \text{ is an even natural number}\}$, $B = \{x : x \text{ is a natural number and multiple of } 5\}$ and $C = \{x : x \text{ is a natural number and multiple of } 10\}$, then what is the value of $A \cap (B \cup C)$? **2012 I**
 (a) $\{10, 20, 30, \dots\}$ (b) $\{5, 10, 15, 20, \dots\}$
 (c) $\{2, 4, 6, \dots\}$ (d) $\{20, 40, 60, \dots\}$
- 54.** Which one of the following is a null set? **2012 II**
 (a) $A = \{x \text{ is a real number} : x > 1 \text{ and } x < 1\}$
 (b) $B = \{x : x + 3 = 3\}$
 (c) $C = \{\emptyset\}$
 (d) $D = \{x \text{ is a real number} : x \geq 1 \text{ and } x \leq 1\}$
- 55.** Let $x \in \{2, 3, 4\}$ and $y \in \{4, 6, 9, 10\}$. If A be the set of all order pairs (x, y) such that x is a factor of y . Then, how many elements does the set A contain? **2012 II**
 (a) 12 (b) 10 (c) 7 (d) 6

- 56.** Consider the following in respect of the sets A and B . **2013 I**

I. $(A \cap B) \subseteq A$ II. $(A \cap B) \subseteq B$ III. $A \subseteq (A \cup B)$

Which of the statement(s) given above is/are correct?

- (a) I and II (b) II and III (c) I and III (d) All of these

- 57.** In a school there are 30 teachers who teach Mathematics or Physics. Of these teachers, 20 teach Mathematics and 15 teach Physics, 5 teach both Mathematics and Physics. The number of teachers teaching only Mathematics is **2013 I**
 (a) 5 (b) 10 (c) 15 (d) 20

- 58.** In a class of 110 students, x students take both Mathematics and Statistics, $x + 20$ students take Mathematics and $x + 30$ students take Statistics. There are no students who take neither Mathematics nor Statistics. What is x equal to? **2013 II**
 (a) 15 (b) 20 (c) 25 (d) 30

- 59.** If A is a non-empty subset of a set E , then what is $E \cup (A \cap \phi) - (A - \phi)$ equal to? **2014 I**
 (a) A (b) Complement of A
 (c) ϕ (d) E

- 60.** If A and B are any two non-empty subsets of a set E , then what is $A \cup (A \cap B)$ equal to? **2014 I**
 (a) $A \cap B$ (b) $A \cup B$ (c) A (d) B

- 61.** Out of 105 students taking an examination English and Mathematics, 80 students pass in English, 75 students pass in Mathematics, 10 students fail in both the subjects. How many students pass in only one subject? **2014 I**
 (a) 26 (b) 30 (c) 35 (d) 45

- 62.** Let A denotes the set of quadrilaterals having two diagonals equal and bisecting each other. Let B denotes the set of quadrilaterals having diagonals bisecting each other at 90° . Then, $A \cap B$ denotes **2015 II**

- (a) the set of parallelograms (b) the set of rhombuses
 (c) the set of squares (d) the set of rectangles

- 63.** Let S be a set of first fourteen natural numbers. The possible number of pairs (a, b) , where $a, b \in S$ and $a \neq b$ such that ab leaves remainder 1 when divided by 15, is **2016 I**
 (a) 3 (b) 5 (c) 6 (d) None of these

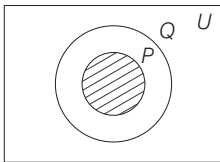
- 64.** In a gathering of 100 people, 70 of them can speak Hindi, 60 can speak English and 30 can speak French. Further, 30 of them can speak both Hindi and English, 20 can speak both Hindi and French. If x is the number of people who can speak both English and French, then which one of the following is correct? (Assume that everyone can speak atleast one of the three languages) **2016 I**
 (a) $10 < x \leq 30$ (b) $0 \leq x < 8$ (c) $x = 9$ (d) $x = 8$

ANSWERS

1	a	2	b	3	c	4	b	5	c	6	a	7	c	8	c	9	a	10	a
11	a	12	a	13	d	14	c	15	c	16	a	17	b	18	c	19	a	20	d
21	c	22	a	23	a	24	c	25	c	26	c	27	a	28	c	29	c	30	c
31	c	32	b	33	d	34	a	35	a	36	d	37	d	38	b	39	a	40	a
41	d	42	c	43	c	44	c	45	c	46	c	47	a	48	a	49	d	50	c
51	a	52	b	53	a	54	a	55	d	56	d	57	c	58	b	59	b	60	c
61	c	62	c	63	d	64	a												

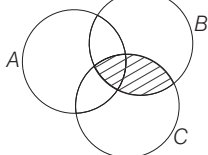
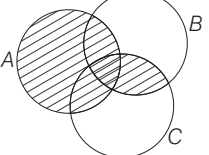
HINTS AND SOLUTIONS

- (a) $A \cup B = \{5, 6, 7\} \cup \{7, 8, 9\}$
 $= \{5, 6, 7, 8, 9\}$
- (b) $B - A = \{7, 8, 9, 12\} - \{2, 6, 8, 9\}$
 $= \{7, 12\}$
- (c) Given, $A = \{1, 2, 3, 4, 5\}$ and $A \subseteq U$.
 $\Rightarrow A \cap U = \{1, 2, 3, 4, 5\} \cap \{U\}$
 $= \{1, 2, 3, 4, 5\}$
- (b) Let $A = \{2, 4, 16, 256, \dots\}$
for $n = 0$, $2^{2^0} = 2^1 = 2$
for $n = 1$, $2^{2^1} = 2^2 = 4$
for $n = 2$, $2^{2^2} = 2^4 = 16$
Thus,
 $A = \{x \in N \mid x = 2^{2^n}, n = 0, 1, 2, \dots\}$
- (c) $\because P \subset Q$



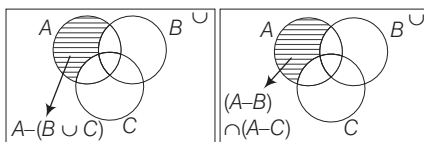
So, $P \cap Q = P$

- (a) Let $x \in A - B \Rightarrow x \in A$ and $x \notin B$
 $\Rightarrow x \notin B - A$
Also, $x \in B - A \Rightarrow x \in B$
and $x \notin A \Rightarrow x \notin A - B$
So, $(A - B) \cap (B - A) = \phi$
- (c) Only if P and Q both are empty set or P and Q have same elements, then $P \cup Q = P \cap Q$.
- (c) Number of elements in $A = 4$
 $\therefore$ Total number of non-empty
Proper subsets of $A = 2^n - 2$
 $= 2^4 - 2 = 14$

- (a) Given, $A = \{B, O, W, L\}$
 $B = \{B, O, W, L, E\}$
 $C = \{B, O, W, L, E\}$
 $\therefore A \subset B$ and $B = C$
- (a) Clearly, $A \cap (A \cup B) = A$
- (a) $B \cup \{1, 2\} = \{1, 2, 3, 5, 9\}$
 $\Rightarrow B = \{3, 5, 9\}$ is the required smallest set
- (a) Here, $(P \cup R) = \{0, 1, 2, 3, 4, 5, 6\}$
 $Q' = U - Q = \{0, 1, 6, 7, 8, 9\}$
So, $Q' \cap (P \cup R) = \{0, 1, 6\}$
- (d) Let $U = \{1, 2, 3, 4, 5, 6\}$ and
 $P = \{1, 3, 4\}$
Then, $P' = U - P = \{2, 5, 6\}$
and $(P')' = U - P' = \{1, 3, 4\} = P$
- (c) The shaded region is $U - A = A'$.
- (c) The shaded region represents the elements in A and not in B , so it represents $A - B$.
- (a) We have,
 $(B \cap C)$  $A \cup (B \cap C)$ 
- (b) The shaded region represents the elements in A and not in B or C .
 $\therefore A - (B \cup C)$
- (c) Clearly, $(A \cap B) \cup (B \cap C)$
- (a) Clearly, (a) is correct.
- (d) Since, $P(\phi) = \{\phi\}$
Hence, $\phi \in P(\phi)$

- (c) As, $\pi, \sqrt{2}, 3 + \sqrt{7}$ are irrational numbers.
So, required set is $\{\pi, \sqrt{2}, 3 + \sqrt{7}\}$.
- (a) Here, $U = \{1, 2, 3, 4, 5, \dots\}$
and $A = \{1, 3, 5, 7, 9, 11, \dots\}$
So, $A' = U - A = \{2, 4, 6, 8, 10, \dots\}$
 $\therefore A' = \{x : x \text{ is an even number}\}$
- (a) Disjoint sets have no common element.
e.g. $A = \{1, 2, 3\}, B = \{4, 5, 6\}$
 $\Rightarrow A \cap B = \phi$
- (c) $A - (A - B) = A - (A \cap B')$
 $= A \cap (A \cap B')' = A \cap (A' \cup B)$
 $= (A \cap A') \cup (A \cap B)$
 $= \phi \cup (A \cap B) = A \cap B$
- (c) $\because n(A \cap B) = 2n$
 $\therefore n(A \cup B) = n(A) + n(B) - n(A \cap B)$
 $= 2n + 4n - 2n = 4n$
Hence, minimum number of elements in $A \cup B$ is $4n$.
- (c) Here, $P = \{x : x^2 - 3x + 2 = 0\}$
 $\Rightarrow P = \{x : (x - 1)(x - 2) = 0\}$
 $\Rightarrow P = \{x : x = 1, x = 2\}$
 $\therefore P = \{1, 2\}$ and
 $Q = \{x : x^2 + 4x - 12 = 0\}$
 $\Rightarrow Q = \{x : (x + 6)(x - 2) = 0\}$
 $\Rightarrow Q = \{-6, 2\}$
So, $P - Q = \{1, 2\} - \{-6, 2\} = \{1\}$
- (a) Given, $A = \{(2^{2n} - 3n - 1) \mid n \in N\}$
 $= \{0, 9, 54, 243, \dots\}$
and $B = \{9(n - 1) \mid n \in N\}$
 $= \{0, 9, 18, 27, \dots\}$
From above, it is clear that $A \subset B$.

28. (c)



From the above two figures, shaded portion of

$$A - (B \cup C) = (A - B) \cap (A - C)$$

$$29. (c) n(P \cup Q) = n(P) + n(Q) - n(P \cap Q)$$

$$\therefore n(P \cup Q) = m + n - p$$

$$30. (c) n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

$$\therefore n(A \cap B) = n(A) + n(B) - n(A \cup B) = 17 + 23 - 38 = 2$$

$$31. (c) \text{ Given, } A = \{x : x^2 - 6x + 8 = 0\}$$

$$\Rightarrow A = \{x : (x - 4)(x - 2)\}$$

$$\therefore A = \{4, 2\}$$

$$\text{And } B = \{x : 2x^2 + 3x - 2 = 0\}$$

$$\Rightarrow B = \{x : (2x - 1)(x + 2)\}$$

$$\therefore B = \left\{\frac{1}{2}, -2\right\}$$

Hence, neither $A \subseteq B$ nor $B \subseteq A$.

$$32. (b) (B \cup C) = \{1, 2, 3, 4, 5, 6, 7\}$$

$$\therefore A \cap (B \cup C) = \{1, 2, 3, 4\} \cap \{1, 2, 3, 4, 5, 6, 7\} = \{1, 2, 3, 4\}$$

$$33. (d) (A \cup B) = \{1, 2, 3, 4, 6, 8, 10\}$$

$$(A \cup B)' = U - (A \cup B) = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\} - \{1, 2, 3, 4, 6, 8, 10\} = \{5, 7, 9\}$$

$$34. (a) \{(A \cup B)' \cap A\} - (A - B) = \{(U - (A \cup B)) \cap A\} - (A - B) = \{(U \cap A) - \{(A \cup B) \cap A\}\} - (A - B) = \{A - A\} - (A - B) = \phi - (A - B) = \phi$$

$$35. (a) \{a\} \text{ is an element of } \{\{a\}, \{b\}, \{c\}\}, \text{ subsets of } \{\{a\}, \{b\}, \{c\}\} \text{ are } \phi, \{\{a\}, \{b\}, \{c\}\}, \{\{a\}, \{b\}\}, \{\{a\}, \{c\}\}, \{\{b\}, \{c\}\}, \{\{a\}, \{b\}, \{c\}\}$$

Hence, only statement I is correct.

$$36. (d) \{x : x \text{ is an integer and less than } 1000\} = \{\dots, 998, 999\}$$

i.e. $x \in (-\infty, 1000)$ is an infinite set.

$$37. (d) \text{ If } A \cap B = \phi, \text{ then it is not necessary that either.}$$

$$A = \phi \text{ or } B = \phi.$$

$$38. (b) \text{ Since } R \in \overleftrightarrow{PQ} \text{ i.e. } R \text{ belongs to } \overleftrightarrow{PQ}.$$

Again $S, \overleftrightarrow{PQ}$ and S does not lie on $\overleftrightarrow{PQ}$ extended

$$\therefore \overleftrightarrow{PQ} \cap \overleftrightarrow{RS} = \{R\}$$

where R is point of intersection of the straight lines PQ and RS .

$$39. (a) \text{ Here, } H = \text{People, who can speak Hindi}$$

$E = \text{People, who can speak English}$

$$\text{Given, } n(H \cup E) = 1000, n(H) = 750,$$

$$n(E) = 400$$

$$n(H \cup E) = n(H) + n(E) - n(H \cap E)$$

$$1000 = 750 + 400 - n(H \cap E)$$

$$\Rightarrow n(H \cap E) = 1150 - 1000 = 150$$

Number of people who can speak Hindi only

$$= n(H) - n(H \cap E)$$

$$= 750 - 150 = 600$$

$$40. (a) \text{ Let } S \text{ be the set of people who speak spanish and } F \text{ be the set of people who speak French}$$

$$\therefore n(S) = 20, n(F) = 50, n(S \cap F) = 10$$

$$\Rightarrow n(S \cup F) = n(S) + n(F) - n(S \cap F)$$

$$= 20 + 50 - 10 = 60$$

$$41. (d) \text{ Here, } M = \text{Students who study Mathematics}$$

$E = \text{Students who study English}$

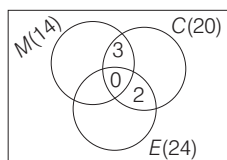
$C = \text{Students who study Commerce}$

$$\therefore n(M \cup E \cup C) = 42$$

$$\text{Also, } n(M) = 14, n(E) = 24, n(C) = 20,$$

$$n(M \cap C) = 3, n(E \cap C) = 2$$

$$\text{and } n(M \cap E \cap C) = 0$$



From the Venn diagram, it is clear that number of students who study Mathematics but not Commerce

$$= 14 - 3 = 11$$

$$42. (c) \text{ Let number of failed candidates in Mathematics and English be } M \text{ and } E, \text{ respectively.}$$

$$\text{Given, candidates failed in English, } n(E) = 52\%$$

$$\text{Candidates failed in Mathematics, } n(M) = 42\%$$

$$\text{Candidates failed in both English and Mathematics,}$$

$$n(E \cap M) = 17\%$$

$$\therefore \text{Total candidates, failed}$$

$$n(E \cup M) = n(E) + n(M) - n(E \cap M)$$

$$= 52 + 42 - 17 = 77\%$$

$$\therefore \text{Total candidates passed} = (100 - 77) = 23\%$$

$$43. (c) \text{ Clearly, both I and II are true.}$$

$$44. (c) \text{ I. } A = \{0\} \quad \text{II. } B = \{2\}$$

$$\text{III. } C = \{ \}; \pm 4 \text{ is not an odd integer}$$

Here, only III is empty set.

$$45. (c) \text{ By distributive law in sets}$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

So, only III is correct.

$$46. (c) \text{ Here, } P = \{\dots - 6, -3, 0, 3, 6, \dots\},$$

$$Q = \{\dots - 8, -4, 0, 4, 8, \dots\} \text{ and}$$

$$R = \{\dots - 12, -6, 0, 6, 12, \dots\}$$

$$\text{I. } P \cup Q = \{\dots - 8, -6, -4, -3, 0, 3, 4, 6, 8, \dots\} \neq R$$

$$\text{II. } P \not\subset R \text{ as } 3 \in P \text{ but } 3 \notin R.$$

$$\text{III. } R \subset (P \cup Q) \text{ is true.}$$

Hence, only statement III is correct.

$$47. (a) \text{ I. There are infinite points lie on a line segment so, it is an infinite set.}$$

$$\text{II. Number of birds in zoo are countable so, it is a finite set.}$$

$$\text{III. It is not a well-defined set.}$$

Hence, only I is correct.

$$48. (a) \text{ I. } A \text{ is a finite set.}$$

$$\text{II. As all elements of } A \text{ are integers, so } A \text{ is subset of integers.}$$

$$\text{III. } \{1, 2\} \text{ is a proper subset of } A.$$

$$\text{IV. } A \neq \phi.$$

So, I, II and III are true.

$$49. (d) \text{ All sets are infinite set.}$$

$$50. (c) \text{ Equivalent sets have same cardinal numbers. Here, cardinal numbers of I, III, IV sets are same.}$$

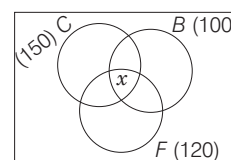
Solutions (Q. No. 51-52) Let C , B and F denotes the number of students who play cricket, basketball and football respectively.

$$\text{Given, } n(C) = 150, n(B) = 100,$$

$$n(F) = 120, n(B \cap F) = 30,$$

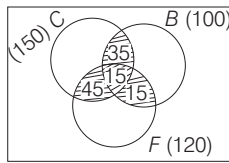
$$n(C \cap B) = 50 \text{ and } n(C \cap F) = 60$$

Let x be the number of students who play all the three sports



51. (a) $n(C \cup B \cup F) = n(C) + n(B) + n(F) - n(C \cap B) - n(C \cap F) - n(B \cap F) + n(C \cap B \cap F)$
 $\Rightarrow n(C \cup B \cup F) = 150 + 100 + 120 - 60 - 50 - 30 + x$
 $\Rightarrow n(C \cup B \cup F) = 230 + x$
 Since, maximum value of $n(C \cup B \cup F)$ is 250
 Hence, maximum value of $x = 250 - 230 = 20$

52. (b) Here, $n(C \cup B \cup F) = 250 - 5 = 245$
 $\therefore 245 = 230 + x \Rightarrow x = 15$



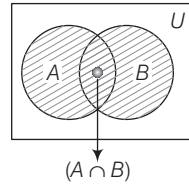
Number of students who play atleast two sports = $15 + 35 + 45 + 15 = 110$

53. (a) Given, $A = \{2, 4, 6, 8, 10, 12, \dots\}$
 $B = \{5, 10, 15, 20, 25, 30, \dots\}$
 and $C = \{10, 20, 30, 40, 50, 60, \dots\}$
 Now,
 $(B \cup C) = \{5, 10, 15, 20, 25, 30, \dots\}$
 $\therefore A \cap (B \cup C) = \{2, 4, 6, 8, 10, 12, \dots\} \cap \{5, 10, 15, 20, 25, 30, \dots\}$
 $= \{10, 20, 30, \dots\}$

54. (a) From option (a), $A = \{x \text{ is a real number} : x > 1 \text{ and } x < 1\}$.
 Since, there is no such element which is greater than 1 and less than 1.
 So, A is a null set.
 From option (b), $B = \{x : x + 3 = 3\} = \{0\}$ = Singleton set
 From option (c), $C = \{\emptyset\}$ = Singleton set
 From option (d), $D = \{x \text{ is a real number} : x \geq 1 \text{ and } x \leq 1\}$
 $= \{1\}$ = Singleton set

55. (d) Given that, $x \in \{2, 3, 4\}$
 and $y \in \{4, 6, 9, 10\}$
 and also $A = (x, y)$, such that x is a factor of y .
 $\therefore A = \{(2, 4), (2, 6), (2, 10), (3, 6), (3, 9), (4, 4)\}$
 Hence, A contain 6 elements.

56. (d) From figure,



- I. $(A \cap B) \subseteq A$ [true]
 II. $(A \cap B) \subseteq B$ [true]
 III. $A \subseteq (A \cup B)$ [true]

So, all three statements are correct.
 Shaded region = $(A \cup B)$

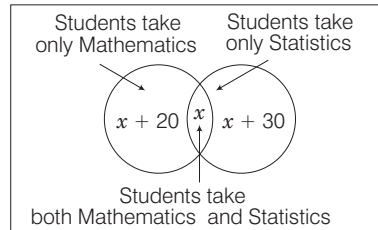
57. (c) Let M and P be the teachers who teach Mathematics and Physics, respectively.

Given,
 $n(M \cup P) = 30$, $n(M) = 20$, $n(P) = 15$
 and $n(M \cap P) = 5$

$\therefore$ Number of teachers teaching only Mathematics is

$$n(\text{only } M) = n(M) - n(M \cap P) = 20 - 5 = 15$$

58. (b) Venn diagram of given conditions is as shown in figure given below



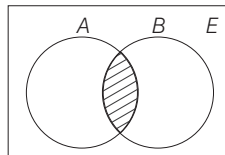
Total students = Students those take only Mathematics + Student those take only Statistics + Student those take both Mathematics and Statistics

$$\therefore 110 = x + 20 + x + x + 30$$

$$\Rightarrow 110 = 3x + 50 \Rightarrow 3x = 60$$

$$\therefore x = 20$$

59. (b) $E \cup \{(A \cap \phi) - (A - \phi)\}$
 $= E \cup \phi - A = E - A = A'$
 $[\because (A \cap \phi) = \phi \text{ and } (A - \phi) = A]$
60. (c) Since, A and B are non-empty subsets of E .



$$\therefore A \cup (A \cap B) = A \cup (\text{Shaded portion}) = A$$

If A and B are two non-empty disjoint sets.

then, $A \cap B = \phi$

$$\therefore A \cup (A \cap B) = A \cup \phi = A$$

61. (c) Let E and M represent the students pass in English and Mathematics.

Given, $n(U) = 105$, $n(E) = 80$,

$n(M) = 75$, and $n(E \cap M) = 10$

$$\text{Now, } n(E \cup M) = n(U) - n(E \cap M) = 105 - 10 = 95$$

Also, $n(E \cup M) = n(E) + n(M)$

$$- n(E \cap M)$$

$$\Rightarrow 95 = 80 + 75 - n(E \cap M)$$

$$\Rightarrow n(E \cap M) = 60$$

The number of students who pass in only one subject

$$= n(E) + n(M) - 2n(E \cap M)$$

$$= 80 + 75 - 2 \times 60$$

$$= 155 - 120 = 35$$

62. (c) A = diagonal equal and bisecting each other.

A is square or rectangle. and B diagonal bisecting each other at 90° .

So, $A \cap B$ = the set of squares.

63. (d) The possible set of pairs (a, b) such that ab leaves remainder 1 when divided by 15 are $(2, 8)$, $(8, 2)$, $(7, 13)$ and $(13, 7)$.

$$\therefore \text{Number of possible set of pairs} = 4$$

64. (a) Let A, B and C be the number of people who can speak Hindi, English and French.

Then, $n(A) = 70$, $n(B) = 60$,

$$n(C) = 30, n(A \cap B) = 30,$$

$$n(A \cap C) = 20, \text{ and } n(B \cap C) = x$$

$$n(A \cup B \cup C) = n(A) + n(B) + n(C)$$

$$- n(A \cap B) - n(B \cap C)$$

$$- n(A \cap C) + n(A \cap B \cap C)$$

$$\Rightarrow 100 = 70 + 60 + 30 - 30 - 20$$

$$- x + n(A \cap B \cap C)$$

$$\Rightarrow x = 10 + n(A \cap B \cap C)$$

Also, $n(A \cap B \cap C)$ can't be more than $n(A \cap C)$

$$\Rightarrow 10 \leq x \leq 30$$

MEASUREMENTS OF ANGLES AND TRIGONOMETRIC RATIOS

Generally (13-14) questions have been asked from this chapter. Generally questions asked from this chapter are based on the trigonometric identities and formulae. So, detailed study of this chapter will help you to score in exam as good number of questions have been asked from this chapter.



Trigonometry is the branch of Mathematics which deals with the measurements of sides and angles of triangles and the problems based on these angles.

ANGLE

An angle is considered as the figure obtained by rotating a given ray about its endpoint. The original ray is called the initial side and the ray into which the initial side rotates is called terminal side. In figure, OA is initial side and OB is the terminal side of $\angle AOB$. The point O is called its vertex.

QUADRANTS

Let XOX' and YOY' be two lines at right angles in the plane of paper. These two perpendicular lines divide the plane of the paper into four equal parts, these four parts are known as four quadrants and lines XOX' and YOY' are known as X -axis and Y -axis, respectively.

The parts XOY , YOX' , $X'OY'$ and $Y'OX$ are known as 1st, 2nd, 3rd and 4th quadrant, respectively.

Relation between Angles, Radius and Arc Length

If l is the length of a circle of radius r . And this arc subtends an angle θ radians at the centre of the circle, then $l = r\theta \Rightarrow \theta = \frac{l}{r}$

Relation between Degree and Radian

A circle subtends an angle, at the centre whose radian measure is 2π and its degree measure is 360° . It follows that

$$\begin{aligned} 2\pi \text{ radian} &= 360^\circ \\ \pi \text{ radian} &= 180^\circ \\ \therefore 1 \text{ radian} &= \frac{180}{\pi} \text{ degree} = 57^\circ 16' 22'' \text{ (approx)} \\ 1 \text{ degree} &= \frac{\pi}{180} \text{ radian} = 0.01746 \text{ rad (approx)} \end{aligned}$$

Note 1 radian is written as 1^c .

EXAMPLE 1. The radian measure corresponding to $-22^\circ 30'$ is

a. $\frac{\pi^c}{8}$ b. $-\frac{\pi^c}{8}$ c. $\frac{\pi^c}{4}$ d. $-\frac{\pi^c}{4}$

Sol. b. As, $180^\circ = \pi$

$$\therefore -22^\circ 30' = -\left(22 + \frac{1}{2}\right)^\circ = -\left(\frac{45}{2}\right)^\circ = -\left(\frac{45}{2} \times \frac{\pi}{180}\right)^c = -\frac{\pi^c}{8}$$

IMPORTANT POINTS

- The angle between two consecutive digits in a clock is $30^\circ = \left(\frac{\pi}{6} \text{ radian}\right)$.
- The hour hand rotates through an angle of 30° in one hour i.e. $\left(\frac{1}{2}\right)^\circ$ in one minute.
- The minute hand rotates through an angle of 6° in one minute.

EXAMPLE 2. If an arc 24π cm long of a circle subtends an angle of 72° at its centre, then the radius of the circle is

a. 30 cm b. 40 cm c. 50 cm d. 60 cm

Sol. d. Given, length of arc (l) = (24π) cm

$$\text{Angle } (\theta) = 72^\circ = \left(72 \times \frac{\pi}{180}\right)^c = \left(\frac{2\pi}{5}\right)^c$$

Let r be the radius of circle

$$\text{Since, } \theta = \frac{l}{r} \Rightarrow r = \frac{l}{\theta} = \frac{24\pi}{\frac{2\pi}{5}} = 60 \text{ cm}$$

EXAMPLE 3. Consider the following statements:

- The angular measure in radian of a circular arc of fixed length subtending at its centre decreases, if the radius of the arc increases.
- 1800° is equal to 5π radian.

Which of the statements given above is/are correct? **2012 I**

- a. Only I b. Only II
c. Both I and II d. Neither I nor II

Sol. a. I. We know that,

$$\text{Radius} = \frac{\text{arc}}{\text{angle}} \quad (\text{given, arc length is constant})$$

$$\therefore \text{Radius} \propto \frac{1}{\text{angle}}$$

So, angular measure in radian decreases, if the radius of the arc increases.

$$\text{II. } 1800^\circ \times \frac{\pi}{180^\circ} = 10\pi$$

Hence, only statement I is correct.

EXAMPLE 4. The angle between the hour hand and the minute hand of a clock at half-past three is

a. 54° b. 63° c. 75° d. 85°

Sol. c. Angle traced by hour hand in 1 hr = 30°

$$\text{Angle traced by hour hand in } 3\frac{1}{2} \text{ hr} = \left(30 \times \frac{7}{2}\right)^\circ = 105^\circ$$

$$\text{Angle traced by minute hand in 1 min} = 6^\circ$$

$$\text{Angle traced by minute hand in 30 min} = 6 \times 30^\circ = 180^\circ$$

$$\text{Angle between two hands} = 180^\circ - 105^\circ = 75^\circ$$

TRIGONOMETRIC RATIOS

The ratio of the sides of a right angled triangle with respect to its angle are called trigonometric ratios.

The side opposite to the right angle is called the hypotenuse. Relative to the angle θ , the side opposite to angle θ is called the perpendicular side and the remaining side is called base.

- For angle θ , perpendicular = BC , base = AB and hypotenuse = AC
- For angle α , perpendicular = AB , base = BC and hypotenuse = AC

Trigonometric ratios of θ in right angled $\triangle ABC$ are defined below

$$\sin \theta = \frac{\text{Perpendicular}}{\text{Hypotenuse}} = \frac{BC}{AC} = \frac{P}{H}$$

$$\cos \theta = \frac{\text{Base}}{\text{Hypotenuse}} = \frac{AB}{AC} = \frac{B}{H}$$

$$\tan \theta = \frac{\text{Perpendicular}}{\text{Base}} = \frac{BC}{AB} = \frac{P}{B}$$

$$\sec \theta = \frac{\text{Hypotenuse}}{\text{Base}} = \frac{AC}{AB} = \frac{H}{B}$$

$$\cot \theta = \frac{\text{Base}}{\text{Perpendicular}} = \frac{AB}{BC} = \frac{B}{P}$$

$$\operatorname{cosec} \theta = \frac{\text{Hypotenuse}}{\text{Perpendicular}} = \frac{AC}{BC} = \frac{H}{P}$$

Relation Between T-ratios

$$(i) \sin \theta = \frac{1}{\operatorname{cosec} \theta} \quad (ii) \cos \theta = \frac{1}{\sec \theta} \quad (iii) \tan \theta = \frac{1}{\cot \theta}$$

$$(iv) \tan \theta = \frac{\sin \theta}{\cos \theta} \quad (v) \cot \theta = \frac{\cos \theta}{\sin \theta}$$

EXAMPLE 5. If $\sin \theta = \frac{8}{17}$, then the other trigonometric ratios $\cos \theta$, $\tan \theta$, $\operatorname{cosec} \theta$, $\sec \theta$, $\cot \theta$ are

- a. $\frac{15}{17}, \frac{8}{15}, \frac{17}{8}, \frac{17}{15}, \frac{15}{8}$ b. $\frac{15}{17}, \frac{8}{15}, \frac{17}{8}, \frac{-17}{15}, \frac{8}{15}$
 c. $\frac{17}{15}, \frac{15}{8}, \frac{8}{17}, \frac{15}{17}, \frac{8}{15}$ d. None of these

Sol. a. As, $\sin \theta = \frac{8}{17} = \frac{\text{perpendicular}}{\text{hypotenuse}}$

In right angled $\triangle PQR$,

$$\therefore PQ^2 + QR^2 = PR^2 \Rightarrow PQ^2 = PR^2 - QR^2$$

$$\Rightarrow PQ = \sqrt{PR^2 - QR^2} = \sqrt{17^2 - 8^2}$$

$$= \sqrt{289 - 64} = \sqrt{225} = 15$$

$$\therefore \cos \theta = \frac{PQ}{PR} = \frac{15}{17}, \quad \tan \theta = \frac{RQ}{PQ} = \frac{8}{15}$$

$$\operatorname{cosec} \theta = \frac{PR}{RQ} = \frac{17}{8}, \quad \sec \theta = \frac{PR}{PQ} = \frac{17}{15}$$

$$\text{and } \cot \theta = \frac{PQ}{RQ} = \frac{15}{8}$$

EXAMPLE 6. If $\cot A = \frac{1}{(\sqrt{2}-1)}$, then the value of $\sin A \cos A$ is

- a. $\frac{1}{4}$ b. $\frac{1}{2}$ c. $\frac{1}{\sqrt{2}}$ d. $\frac{\sqrt{2}}{4}$

Sol. d. $\therefore \cot A = \frac{1}{\sqrt{2}-1} = \frac{\text{base}}{\text{perpendicular}}$

$$\Rightarrow \tan A = \frac{\sqrt{2}-1}{1} \left(\because \tan A = \frac{1}{\cot A} \right)$$

In right angled $\triangle ABC$,

$$\therefore AC^2 = AB^2 + BC^2$$

$$\Rightarrow AC = \sqrt{AB^2 + BC^2}$$

$$= \sqrt{1^2 + (\sqrt{2}-1)^2} = \sqrt{1+2+1-2\sqrt{2}} = \sqrt{4-2\sqrt{2}}$$

$$\therefore \sin A = \frac{BC}{AC} = \frac{\sqrt{2}-1}{\sqrt{4-2\sqrt{2}}} \quad \text{and} \quad \cos A = \frac{AB}{AC} = \frac{1}{\sqrt{4-2\sqrt{2}}}$$

$$\therefore \sin A \cos A = \frac{\sqrt{2}-1}{\sqrt{4-2\sqrt{2}}} \cdot \frac{1}{\sqrt{4-2\sqrt{2}}} = \frac{\sqrt{2}-1}{4-2\sqrt{2}}$$

$$= \frac{\sqrt{2}-1}{2\sqrt{2}(\sqrt{2}-1)} = \frac{1}{2\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{\sqrt{2}}{4}$$

TRIGONOMETRIC IDENTITIES

A statement of equality involving trigonometric ratios of an angle is called a trigonometric identity. It is valid for all values of the angles. The three important identities are

- $\sin^2 \theta + \cos^2 \theta = 1$
- $1 + \tan^2 \theta = \sec^2 \theta$
- $1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$

Note $\sin^2 A + \cos^2 B$ cannot be equal to 1 because the angles are different.

EXAMPLE 7. What is the value of $\frac{\tan A - \sin A}{\sin^3 A}$? **2013 I**

- a. $\frac{\sec A}{1 - \cos A}$ b. $\frac{\sec A}{1 + \cos^2 A}$
 c. $\frac{\sec A}{1 + \cos A}$ d. None of these

Sol. c. $\frac{\tan A - \sin A}{\sin^3 A} = \frac{\frac{\sin A}{\cos A} - \sin A}{\sin^3 A} = \frac{(1 - \cos A)}{\cos A \sin^2 A} \times \frac{(1 + \cos A)}{(1 + \cos A)}$

$$= \frac{1 - \cos^2 A}{\cos A \sin^2 A (1 + \cos A)} = \frac{\sin^2 A}{\cos A \sin^2 A (1 + \cos A)}$$

$$= \frac{1}{\cos A} \cdot \frac{1}{1 + \cos A} = \frac{\sec A}{1 + \cos A}$$

Values of Trigonometric Ratios for Some Specific Angles

Angles	0°	30°	45°	60°	90°
$\sin \theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\tan \theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	∞
$\cot \theta$	∞	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0
$\sec \theta$	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	∞
$\operatorname{cosec} \theta$	∞	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1

EXAMPLE 8. What is the value of $\sin^3 60^\circ \cot 30^\circ - 2 \sec^2 45^\circ + 3 \cos 60^\circ \tan^2 45^\circ - \tan^2 60^\circ$?

- a. $\frac{35}{8}$ b. $\frac{-35}{8}$ c. $\frac{-11}{8}$ d. $\frac{11}{8}$

Sol. b. $\sin^3 60^\circ \cot 30^\circ - 2 \sec^2 45^\circ + 3 \cos 60^\circ \tan^2 45^\circ - \tan^2 60^\circ$

$$= \left(\frac{\sqrt{3}}{2} \right)^3 \times \sqrt{3} - 2 \times (\sqrt{2})^2 + 3 \times \frac{1}{2} \times (1)^2 - (\sqrt{3})^2$$

$$= \frac{9}{8} - 4 + \frac{3}{2} - 3 = \frac{9 - 32 + 12 - 24}{8} = \frac{21 - 56}{8} = \frac{-35}{8}$$

Sign of trigonometric ratio in different quadrants

II Quadrant $(90^\circ < \theta < 180^\circ)$ $\sin \theta$ and cosec θ are positive. All other trigonometric ratios are negative.	I Quadrant $(0 < \theta < 90^\circ)$ All trigonometric ratios are positive.
III Quadrant $(180^\circ < \theta < 270^\circ)$ $\tan \theta$ and cot θ are positive. All other trigonometric ratios are negative.	IV Quadrant $(270^\circ < \theta < 360^\circ)$ $\cos \theta$ and sec θ are positive. All other trigonometric ratios are negative.

EXAMPLE 9. If $0 < \theta < \frac{\pi}{4}$, then what is $\sqrt{1 - 2\sin\theta\cos\theta}$

equal to?

2014 II

- a. $\cos\theta - \sin\theta$ b. $\sin\theta - \cos\theta$
 c. $\pm(\cos\theta - \sin\theta)$ d. $\cos\theta\sin\theta$

Sol. a. Given, $0 < \theta < \frac{\pi}{4}$, then $\sqrt{1 - 2\sin\theta\cos\theta}$

$$\begin{aligned}
 &= \sqrt{\sin^2\theta + \cos^2\theta - 2\sin\theta\cos\theta} \quad [\because \sin^2\theta + \cos^2\theta = 1] \\
 &= \sqrt{(\cos\theta - \sin\theta)^2} \\
 &= \cos\theta - \sin\theta \quad \left[\because 0 < \theta < \frac{\pi}{4}, \cos\theta > \sin\theta, \text{ so we take } (\cos\theta - \sin\theta)^2 \right]
 \end{aligned}$$

Domain and Range of Trigonometrical Functions

Trigonometric Ratio	Domain	Range
$\sin \theta$	R	$[-1, 1]$
$\cos \theta$	R	$[-1, 1]$
$\tan \theta$	$R \sim \left\{ (2n+1)\frac{\pi}{2}, n \in I \right\}$	$(-\infty, \infty) = R$
$\cot \theta$	$R \sim \{n\pi, n \in I\}$	$(-\infty, \infty) = R$
$\sec \theta$	$R \sim \left\{ (2n+1)\frac{\pi}{2}, n \in I \right\}$	$(-\infty, -1] \cup [1, \infty)$
cosec θ	$R \sim \{n\pi, n \in I\}$	$(-\infty, -1] \cup [1, \infty)$

The behaviour of the trigonometric function in different quadrants are defined in following table.

Trigonometric Ratio	Ist Quadrant (θ increase from 0 to $\pi/2$)	IInd Quadrant (θ increase from $\pi/2$ to π)	III Quadrant (θ increase from π to $3\pi/2$)	IVth Quadrant (θ increase from $\frac{3\pi}{2}$ to 2π)
$\sin \theta$	Increases from 0 to 1	Decreases from 1 to 0	Decreases from 0 to -1	Increases from -1 to 0
$\cos \theta$	Decreases from 1 to 0	Decreases from 0 to -1	Increases from -1 to 0	Increases from 0 to 1
$\tan \theta$	Increases from 0 to ∞	Increases from $-\infty$ to 0	Increases from 0 to ∞	Increases from $-\infty$ to 0
$\cot \theta$	Decreases from ∞ to 0	Decreases from 0 to $-\infty$	Decreases from ∞ to 0	Decreases from 0 to $-\infty$
$\sec \theta$	Increases from 1 to ∞	Increases from $-\infty$ to -1	Decreases from -1 to $-\infty$	Decreases from ∞ to 1
cosec θ	Decreases from ∞ to 1	Increases from 1 to ∞	Increases from $-\infty$ to -1	Decreases from -1 to $-\infty$

EXAMPLE 10. Which one of the following statements is true in respect of the expression $\sin 31^\circ + \sin 32^\circ$?

- a. Its value is 0 b. Its value is 1
 c. Its value is less than 1 d. Its value is greater than 1

Sol. d. We know that, $\sin 30^\circ = \frac{1}{2}$

Value of $\sin$ increases 0° to 90°

$$\therefore \sin 31^\circ > \sin 30^\circ \text{ and } \sin 32^\circ > \sin 30^\circ$$

$$\Rightarrow \sin 31^\circ > \frac{1}{2} \text{ and } \sin 30^\circ > \frac{1}{2}$$

Adding both

$$\therefore \sin 31^\circ + \sin 31^\circ \Rightarrow \frac{1}{2} + \frac{1}{2}$$

$$\Rightarrow \sin 31^\circ + \sin 32^\circ > 1$$

IMPORTANT POINTS

- $|\sin \theta| \leq 1$
- $|\cos \theta| \leq 1$
- $\sec \theta \geq 1$
- $\sec \theta \leq -1$
- $\csc \theta \geq 1$
- $\csc \theta \leq -1$
- $\tan \theta$ and $\cot \theta$ can take any value.

TRIGONOMETRIC RATIOS OF ALLIED ANGLES

Two angles are said to be allied when their sum or difference is either zero or a multiple of 90° .

The angles like $-\theta, 90^\circ \pm \theta, 180^\circ \pm \theta, 360^\circ \pm \theta$ etc., are angles allied to the angle θ , if θ is measured in degrees.

Angle	$\sin \theta$	$\cos \theta$	$\tan \theta$	$\cot \theta$	$\sec \theta$	cosec θ
$-\theta$	$-\sin \theta$	$\cos \theta$	$-\tan \theta$	$-\cot \theta$	$\sec \theta$	$-\csc \theta$
$90^\circ - \theta$	$\cos \theta$	$\sin \theta$	$\cot \theta$	$\tan \theta$	cosec θ	$\sec \theta$
$90^\circ + \theta$	$\cos \theta$	$-\sin \theta$	$-\cot \theta$	$-\tan \theta$	$-\csc \theta$	$\sec \theta$
$180^\circ - \theta$	$\sin \theta$	$-\cos \theta$	$-\tan \theta$	$-\cot \theta$	$-\sec \theta$	cosec θ
$180^\circ + \theta$	$-\sin \theta$	$-\cos \theta$	$\tan \theta$	$\cot \theta$	$-\sec \theta$	$-\csc \theta$
$270^\circ - \theta$	$-\cos \theta$	$-\sin \theta$	$\cot \theta$	$\tan \theta$	$-\csc \theta$	$-\sec \theta$
$270^\circ + \theta$	$-\cos \theta$	$\sin \theta$	$-\cot \theta$	$-\tan \theta$	cosec θ	$-\sec \theta$
$360^\circ - \theta$	$-\sin \theta$	$\cos \theta$	$-\tan \theta$	$-\cot \theta$	$\sec \theta$	$-\csc \theta$
$360^\circ + \theta$	$\sin \theta$	$\cos \theta$	$\tan \theta$	$\cot \theta$	$\sec \theta$	cosec θ

EXAMPLE 11. What is the value of $\sec(90^\circ - \theta) \sin \theta \sec 45^\circ$?

- a. 1 b. $\frac{\sqrt{3}}{2}$ c. $\sqrt{2}$ d. $\sqrt{3}$

Sol. c. Given, $\sec(90^\circ - \theta) \sin \theta \sec 45^\circ = \operatorname{cosec} \theta \times \sin \theta \times \sqrt{2}$
 $= \frac{1}{\sin \theta} \times \sin \theta \times \sqrt{2} = \sqrt{2}$

EXAMPLE 12. The value of $\frac{\sin 135^\circ - \cos 120^\circ}{\sin 135^\circ + \cos 120^\circ}$ is

- a. $2 + 3\sqrt{2}$ b. $2 - 3\sqrt{2}$ c. $3 - 2\sqrt{2}$ d. $3 + 2\sqrt{2}$

Sol. d. $\sin 135^\circ = \sin(180^\circ - 45^\circ) = \sin 45^\circ = \frac{1}{\sqrt{2}}$
 $\cos 120^\circ = \cos(180^\circ - 60^\circ) = -\cos 60^\circ = -\frac{1}{2}$
 $\therefore \frac{\sin 135^\circ - \cos 120^\circ}{\sin 135^\circ + \cos 120^\circ} = \frac{\frac{1}{\sqrt{2}} - \left(-\frac{1}{2}\right)}{\frac{1}{\sqrt{2}} + \left(-\frac{1}{2}\right)} = \frac{\frac{\sqrt{2} + 1}{2}}{\frac{\sqrt{2} - 1}{2}} \times \frac{\sqrt{2} + 1}{\sqrt{2} + 1}$
 $= \frac{(\sqrt{2} + 1)^2}{2 - 1} = 2 + 1 + 2\sqrt{2} = 3 + 2\sqrt{2}$

Sum, Difference and Product Formulae

- $\sin(A + B) = \sin A \cos B + \cos A \sin B$
- $\sin(A - B) = \sin A \cos B - \cos A \sin B$
- $\cos(A + B) = \cos A \cos B - \sin A \sin B$
- $\cos(A - B) = \cos A \cos B + \sin A \sin B$
- $\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$
- $\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$
- $\cot(A + B) = \frac{\cot A \cot B - 1}{\cot A + \cot B}$
- $\cot(A - B) = \frac{\cot A \cot B + 1}{\cot B - \cot A}$
- $\sin(A + B) \sin(A - B) = \sin^2 A - \sin^2 B$
 $= \cos^2 B - \cos^2 A$
- $\cos(A + B) \cos(A - B) = \cos^2 A - \sin^2 B = \cos^2 B - \sin^2 A$
- $2 \sin A \cos B = \sin(A + B) + \sin(A - B)$
- $2 \cos A \sin B = \sin(A + B) - \sin(A - B)$
- $2 \cos A \cos B = \cos(A + B) + \cos(A - B)$
- $2 \sin A \sin B = \cos(A - B) - \cos(A + B)$
- $\sin A + \sin B = 2 \sin \frac{(A + B)}{2} \cdot \cos \frac{(A - B)}{2}$

- $\sin A - \sin B = 2 \sin \frac{(A - B)}{2} \cdot \cos \frac{(A + B)}{2}$
- $\cos A + \cos B = 2 \cos \frac{(A + B)}{2} \cdot \cos \frac{(A - B)}{2}$
- $\cos A - \cos B = -2 \sin \frac{(A + B)}{2} \cdot \sin \frac{(A - B)}{2}$
 $= 2 \sin \frac{(A + B)}{2} \cdot \sin \frac{(B - A)}{2}$
- $\tan A + \tan B = \frac{\sin(A + B)}{\cos A \cos B}$
- $\tan A - \tan B = \frac{\sin(A - B)}{\cos A \cos B}$
- $\cot A + \cot B = \frac{\sin(A + B)}{\sin A \sin B}$
- $\cot A - \cot B = \frac{\sin(B - A)}{\sin A \sin B}$

EXAMPLE 13. If $A - B = \frac{\pi}{3}$, then the value of $\cos A \cos B + \sin A \sin B$ is

- a. $1/2$ b. 1
c. $3/2$ d. None of these

Sol. a. $\cos A \cos B + \sin A \sin B = \cos(A - B)$

$$= \cos\left(\frac{\pi}{3}\right) = \frac{1}{2} \quad \left(\because A - B = \frac{\pi}{3}\right)$$

EXAMPLE 14. The value of $\frac{\sqrt{3} \cos 23^\circ - \sin 23^\circ}{2}$ is

- a. $\cos 7^\circ$ b. $\sin 53^\circ$ c. $\cos 53^\circ$ d. $\sin 7^\circ$

Sol. c. $\frac{\sqrt{3} \cos 23^\circ - \sin 23^\circ}{2} = \frac{\sqrt{3}}{2} \cos 23^\circ - \frac{1}{2} \sin 23^\circ$
 $= \cos 30^\circ \cos 23^\circ - \sin 30^\circ \sin 23^\circ$
 $= \cos(30^\circ + 23^\circ) = \cos 53^\circ$
 $[\because \cos(A + B) = \cos A \cos B - \sin A \sin B]$

EXAMPLE 15. The value of $\frac{\tan 47^\circ + \tan 43^\circ}{1 - \tan 47^\circ \tan 43^\circ}$ is

- a. 1 b. ∞
c. 0 d. -1

Sol. b. $\therefore \frac{\tan A + \tan B}{1 - \tan A \tan B} = \tan(A + B) \Rightarrow \frac{\tan 47^\circ + \tan 43^\circ}{1 - \tan 47^\circ \tan 43^\circ}$
 $= \tan(47^\circ + 43^\circ) = \tan 90^\circ = \infty$

EXAMPLE 16. The value of $\sin 20^\circ \sin 40^\circ \sin 60^\circ \sin 80^\circ$ is

- a. $\frac{1}{2}$ b. $\frac{3}{4}$ c. $\frac{1}{8}$ d. $\frac{3}{16}$

Sol. d. $\sin 20^\circ \sin 40^\circ \sin 80^\circ \sin 60^\circ$

$$\begin{aligned}
 &= \sin 60^\circ [\sin 80^\circ \sin 40^\circ] \sin 20^\circ \\
 &= \frac{\sqrt{3}}{2} [\sin 80^\circ \sin 40^\circ] \sin 20^\circ \\
 &= \frac{\sqrt{3}}{2} \times \frac{1}{2} [2 \sin 80^\circ \sin 40^\circ] \sin 20^\circ \\
 &= \frac{\sqrt{3}}{4} [\cos(80^\circ - 40^\circ) - \cos(80^\circ + 40^\circ)] \sin 20^\circ \\
 &= \frac{\sqrt{3}}{4} [\cos 40^\circ - \cos 120^\circ] \sin 20^\circ \\
 &= \frac{\sqrt{3}}{4} \left[\cos 40^\circ - \left(-\frac{1}{2}\right) \right] \sin 20^\circ \quad \left(\because \cos 120^\circ = -\frac{1}{2} \right) \\
 &= \frac{\sqrt{3}}{4} \left[\cos 40^\circ + \frac{1}{2} \right] \sin 20^\circ \\
 &= \frac{\sqrt{3}}{8} [\sin(40^\circ + 20^\circ) - \sin(40^\circ - 20^\circ) + \sin 20^\circ] \\
 &= \frac{\sqrt{3}}{8} [\sin 60^\circ - \sin 20^\circ + \sin 20^\circ] \\
 &= \frac{\sqrt{3}}{8} \times \sin 60^\circ = \frac{\sqrt{3}}{8} \times \frac{\sqrt{3}}{2} = \frac{3}{16}
 \end{aligned}$$

EXAMPLE 17. $\frac{\cos 4x + \cos 3x + \cos 2x}{\sin 4x + \sin 3x + \sin 2x}$ is equal to

- a. $\cot 3x$ b. $\tan 3x$ c. $\cot x$ d. $\cot 2x$

Sol. a. $\frac{\cos 4x + \cos 3x + \cos 2x}{\sin 4x + \sin 3x + \sin 2x} = \frac{(\cos 4x + \cos 2x) + \cos 3x}{(\sin 4x + \sin 2x) + \sin 3x}$

$$\begin{aligned}
 &= \frac{2 \cos \left(\frac{4x+2x}{2} \right) \cos \left(\frac{4x-2x}{2} \right) + \cos 3x}{2 \sin \left(\frac{4x+2x}{2} \right) \cos \left(\frac{4x-2x}{2} \right) + \sin 3x} \\
 &= \frac{2 \cos 3x \cdot \cos x + \cos 3x}{2 \sin 3x \cdot \cos x + \sin 3x} = \frac{\cos 3x (2 \cos x + 1)}{\sin 3x (2 \cos x + 1)} = \cot 3x
 \end{aligned}$$

Trigonometric Ratios of Multiple Angles

- $\sin 2\theta = 2 \sin \theta \cos \theta = \frac{2 \tan \theta}{1 + \tan^2 \theta}$
- $\cos 2\theta = \cos^2 \theta - \sin^2 \theta = 2 \cos^2 \theta - 1$
 $= 1 - 2 \sin^2 \theta = \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta}$
- $\frac{1 + \cos 2\theta}{2} = \cos^2 \theta$ and $\frac{1 - \cos 2\theta}{2} = \sin^2 \theta$
- $\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$ • $\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta$
- $\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$ • $\tan 3\theta = \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta}$

EXAMPLE 18. The value of $3 \sin 15^\circ - 4 \sin^3 15^\circ$ is

- a. 1 b. $\frac{1}{2}$ c. $\frac{1}{\sqrt{2}}$ d. 0

Sol. c. As, $3 \sin \theta - 4 \sin^3 \theta = \sin 3\theta$

$$\text{So, } 3 \sin 15^\circ - 4 \sin^3 15^\circ = \sin 3(15^\circ) = \sin 45^\circ = \frac{1}{\sqrt{2}}$$

Trigonometric Ratios of Submultiple Angles

- $\sin \theta = 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}$
- $\cos \theta = \cos^2 \frac{\theta}{2} - \sin^2 \frac{\theta}{2} = 2 \cos^2 \frac{\theta}{2} - 1 = 1 - 2 \sin^2 \frac{\theta}{2}$
- $\frac{1 - \cos \theta}{\sin \theta} = \tan \frac{\theta}{2}$ • $\frac{1 + \cos \theta}{\sin \theta} = \cot \frac{\theta}{2}$
- $\frac{1 - \cos \theta}{1 + \cos \theta} = \tan^2 \frac{\theta}{2}$ • $\frac{1 + \cos \theta}{1 - \cos \theta} = \cot^2 \frac{\theta}{2}$
- $\sin \theta = \frac{2 \tan \frac{\theta}{2}}{1 + \tan^2 \frac{\theta}{2}}$ • $\cos \theta = \frac{1 - \tan^2 \frac{\theta}{2}}{1 + \tan^2 \frac{\theta}{2}}$

Maximum and Minimum Values of Trigonometrical Functions

We know that, $-1 \leq \sin \theta \leq 1$, $-1 \leq \cos \theta \leq 1$

Maximum and minimum value of a trigonometrical function of the form $a \sin \theta \pm b \cos \theta$ are $\sqrt{a^2 + b^2}$

and $-\sqrt{a^2 + b^2}$, respectively.

EXAMPLE 19. The maximum value and minimum value of $8 \sin \theta \cos \theta + 4 \cos 2\theta$ is

- a. 4 and -4 b. 16 and -16
c. $4\sqrt{2}$ and $-4\sqrt{2}$ d. None of these

Sol. c. Here, $8 \sin \theta \cos \theta + 4 \cos 2\theta = 4 \sin 2\theta + 4 \cos 2\theta$

$$\text{So, maximum value} = \sqrt{4^2 + 4^2} = 4\sqrt{2}$$

$$\text{and minimum value} = -\sqrt{4^2 + 4^2} = -4\sqrt{2}$$

EXAMPLE 20. Solve $\cos \theta + \sin \theta = \sqrt{2}$, then the value of θ is

- a. $\pi/3$ b. $\pi/4$ c. $\pi/6$ d. $\pi/2$

Sol. b. $\cos \theta + \sin \theta = \sqrt{2}$, $\frac{\cos \theta}{\sqrt{2}} + \frac{\sin \theta}{\sqrt{2}} = 1$
 $(\because \text{dividing throughout by } \sqrt{2})$
 $\Rightarrow \cos \theta \cos 45^\circ + \sin \theta \sin 45^\circ = 1$
 $\cos(\theta - 45^\circ) = 1 = \cos 0^\circ$
 $\therefore \theta - 45^\circ = 0 \Rightarrow \theta = 45^\circ = \pi/4$

EXAMPLE 21. The value of ϕ for maximum value of $\sin 3\phi + \cos 3\phi$ is

- a. $\sqrt{2}$ b. 90° c. 1 d. 15°

Sol. d. $\sin 3\phi + \cos 3\phi = \sqrt{2} \left(\frac{1}{\sqrt{2}} \cos 3\phi + \frac{1}{\sqrt{2}} \sin 3\phi \right)$
 $= \sqrt{2} (\sin 45^\circ \cos 3\phi + \cos 45^\circ \sin 3\phi)$
 $= \sqrt{2} \sin (45^\circ + 3\phi)$

$\therefore$ The maximum value occurs when

$$\sin (45^\circ + 3\phi) = 1 = \sin 90^\circ$$

$$\therefore 45^\circ + 3\phi = 90^\circ \Rightarrow \phi = 15^\circ$$

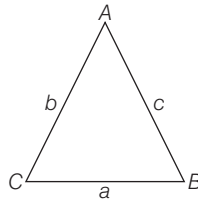
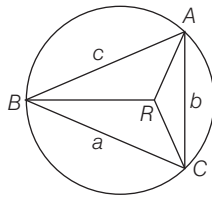
Properties of Triangle

1. **Circumcircle of a triangle** The circle passing through the vertices of a $\triangle ABC$ is called circumcircle. Its radius R is called the circumradius and its centre is known as circumcentre.

$$\text{Here, } R = \frac{a}{2\sin A} = \frac{b}{2\sin B} = \frac{c}{2\sin C}$$

2. **Sine rule** In a $\triangle ABC$, if a, b, c be the three sides opposite to the angles A, B, C respectively, then

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$



3. **Cosine rule** In a $\triangle ABC$, if a, b, c be the sides opposite to angles A, B and C respectively, then

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}, \cos B = \frac{c^2 + a^2 - b^2}{2ac} \text{ and}$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

EXAMPLE 22. A triangle ABC is inscribed in a circle. If sum of the squares of sides of the triangle is equal to twice the square of the diameter, then $\sin^2 A + \sin^2 B + \sin^2 C$ is equal to

- a. 2 b. 3 c. 4 d. None of these

Sol. a. Let the radius of inscribed circle be R .

$$\therefore R = \frac{a}{2\sin A} = \frac{b}{2\sin B} = \frac{c}{2\sin C}$$

$$\Rightarrow \sin A = \frac{a}{2R}, \sin B = \frac{b}{2R} \text{ and } \sin C = \frac{c}{2R}$$

$$\sin^2 A + \sin^2 B + \sin^2 C = \frac{a^2 + b^2 + c^2}{4R^2}$$

$$\Rightarrow \sin^2 A = \frac{a^2}{4R^2}, \sin^2 B = \frac{b^2}{4R^2} \text{ and } \sin^2 C = \frac{c^2}{4R^2}$$

According to the question,

$$a^2 + b^2 + c^2 = 2 \times 2R^2 = 8R^2$$

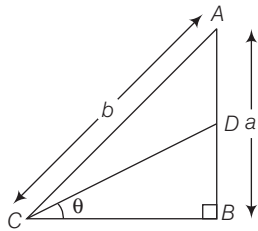
$$\therefore \sin^2 A + \sin^2 B + \sin^2 C = \frac{8R^2}{4R^2} = 2$$

> PRACTICE EXERCISE

- A unit radian is approximately equal to
 (a) $57^\circ 17' 43''$ (b) $57^\circ 16' 22''$
 (c) $57^\circ 17' 47''$ (d) $57^\circ 17' 49''$
- Find the radian measure corresponding to the degree $-47^\circ 30'$.
 (a) $-\frac{19\pi}{72}$ rad (b) $\frac{17\pi}{72}$ rad
 (c) $\frac{13\pi}{72}$ rad (d) $-\frac{15\pi}{72}$ rad
- The length of a pendulum is 60 cm. The angle through which it swings when its tip describes an arc of length 16.5 cm is
 (a) $15^\circ 30'$ (b) $15^\circ 45'$ (c) $16^\circ 15'$ (d) $16^\circ 45'$
- A railway train is moving on a circular curve of radius 1500 m at a speed of 90 km/h. Through what angle has it turned in 11 seconds?
 (a) 12° (b) $16^\circ 30'$ (c) $10^\circ 30'$ (d) $11^\circ 40'$
- The angle between the minute hand and the hour hand of a clock when the time is 7:35 pm is
 (a) $16^\circ 45'$ (b) $17^\circ 30'$ (c) $18^\circ 15'$ (d) $19^\circ 30'$
- When do the hands of a clock coincide between 5 pm and 6 pm?
 (a) 5:30 pm (b) 5:27:16 pm
 (c) 5:32:16 pm (d) 5:28:56 pm
- If $\sin A = \frac{1}{3}$, then $\cos A \cdot \operatorname{cosec} A + \tan A \cdot \sec A$ is equal to
 (a) $\frac{16\sqrt{2} + 3}{8}$ (b) $\frac{4\sqrt{2} + 3}{8}$
 (c) $\frac{\sqrt{3} + 2}{4}$ (d) $\frac{\sqrt{3} - 1}{8}$
- The value of $\cos 15^\circ - \sin 15^\circ$ is equal to
 (a) $\frac{1}{3}$ (b) $\frac{1}{\sqrt{2}}$ (c) $\frac{1}{2}$ (d) $\frac{\sqrt{3}}{2}$

9. If $3 \tan \theta = 4$, then $\sqrt{\frac{1 - \sin \theta}{1 + \sin \theta}}$ is equal to
 (a) $\frac{1}{2}$ (b) $\frac{2}{3}$ (c) $\frac{1}{3}$ (d) None of these
10. If $\tan A = 1$ and $\tan B = \sqrt{3}$, then $\cos A \cdot \cos B - \sin A \cdot \sin B$ is equal to
 (a) $\frac{1 + \sqrt{3}}{2\sqrt{2}}$ (b) $\frac{1 - \sqrt{3}}{2\sqrt{2}}$ (c) $\frac{2\sqrt{2}}{3}$ (d) 1
11. The value of $\operatorname{cosec}^2 \theta - 2 + \sin^2 \theta$ is always
 (a) less than zero (b) non-negative
 (c) zero (d) 1
12. What is the value of $\sin^2 15^\circ + \sin^2 20^\circ + \sin^2 25^\circ + \dots + \sin^2 75^\circ$?
 (a) 0 (b) $\frac{13}{2}$ (c) 6 (d) $\frac{11}{2}$
13. What is the value of $\frac{5 \sin 75^\circ \sin 77^\circ + 2 \cos 13^\circ \cos 15^\circ}{\cos 15^\circ \sin 77^\circ} - \frac{7 \sin 81^\circ}{\cos 9^\circ}$?
 (a) -1 (b) 0 (c) 1 (d) 2
14. What is $\log(\tan 1^\circ) + \log(\tan 2^\circ) + \log(\tan 3^\circ) + \dots + \log(\tan 89^\circ)$ equal to?
 (a) 0 (b) 1 (c) 2 (d) -1
15. If $\tan \theta + \frac{1}{\tan \theta} = 2$, then the value of $\tan^2 \theta + \frac{1}{\tan^2 \theta}$ is equal to
 (a) 6 (b) 4 (c) 2 (d) 3
16. If $\sec \theta = \frac{13}{5}$, then $\frac{2 \sin \theta - 3 \cos \theta}{4 \sin \theta - 9 \cos \theta}$ is equal to
 (a) $1/3$ (b) -3 (c) 3 (d) $\sqrt{3}$
17. $\sin(45^\circ + A) - \cos(45^\circ - A)$ is equal to
 (a) $1/\sqrt{2}$ (b) 1 (c) 0 (d) $\sqrt{2}$
18. What is $\sqrt{\frac{1 + \sin \theta}{1 - \sin \theta}}$ equal to?
 (a) $\sec \theta - \tan \theta$ (b) $\sec \theta + \tan \theta$
 (c) $\operatorname{cosec} \theta + \cot \theta$ (d) $\operatorname{cosec} \theta - \cot \theta$
19. $\sin(n+1)A \sin(n-1)A + \cos(n+1)A \cos(n-1)A$ is equal to
 (a) $\sin 2A$ (b) $\cos 2A$ (c) $\tan 2A$ (d) $\cot 2A$
20. The value of $\frac{\cos 11^\circ + \sin 11^\circ}{\cos 11^\circ - \sin 11^\circ}$ is
 (a) $\tan 56^\circ$ (b) $\tan 34^\circ$ (c) $\cot 56^\circ$ (d) $\tan 11^\circ$
21. If $\sin \theta = \frac{a^2 - b^2}{a^2 + b^2}$, then $\tan \theta$ is equal to
 (a) $\frac{a^2 + b^2}{2ab}$ (b) $\frac{2ab}{a^2 - b^2}$ (c) $\frac{a^2 - b^2}{2ab}$ (d) $\frac{ab}{a^2 + b^2}$
22. If $\cos \theta = 0.96$, then $\frac{1}{\sin \theta} + \frac{1}{\tan \theta}$ is equal to
 (a) 0.98 (b) 3 (c) 4 (d) 7
23. If $0 < x < 45^\circ$ and $45^\circ < y < 90^\circ$, then which one of the following is correct?
 (a) $\sin x = \sin y$ (b) $\sin x < \sin y$
 (c) $\sin x > \sin y$ (d) $\sin x \leq \sin y$
24. For what value of θ is $(\sin \theta + \operatorname{cosec} \theta) = 2.5$, where $0 < \theta \leq 90^\circ$?
 (a) 30° (b) 45° (c) 60° (d) 90°
25. If $\frac{\cos x}{\cos y} = n$ and $\frac{\sin x}{\sin y} = m$, then $(m^2 - n^2) \sin^2 y$ is equal to
 (a) $1 - n^2$ (b) $1 + n^2$ (c) m^2 (d) n^2
26. If $p = \tan^2 x + \cot^2 x$, then which one of the following is correct?
 (a) $p \leq 2$ (b) $p \geq 2$ (c) $p < 2$ (d) $p > 2$
27. The difference of the two angles in degree measure is 1 and their sum in circular measure is also 1. What are the angles in circular measure?
 (a) $\left(\frac{1}{2} - \frac{\pi}{360}\right), \left(\frac{1}{2} + \frac{\pi}{360}\right)$ (b) $\left(\frac{1}{2} - \frac{90}{\pi}\right), \left(\frac{1}{2} + \frac{90}{\pi}\right)$
 (c) $\left(\frac{1}{2} - \frac{\pi}{180}\right), \left(\frac{1}{2} + \frac{\pi}{180}\right)$ (d) None of these
28. What is the value of $[(1 - \sin^2 \theta) \sec^2 \theta + \tan^2 \theta] (\cos^2 \theta + 1)$ when $0 < \theta < 90^\circ$?
 (a) 2 (b) > 2 (c) ≥ 2 (d) < 2
29. If $7 \cos^2 \theta + 3 \sin^2 \theta = 4$ and $0 < \theta < \frac{\pi}{2}$, then what is the value of $\tan \theta$?
 (a) $\sqrt{7}$ (b) $\frac{7}{3}$ (c) 3 (d) $\sqrt{3}$
30. If $\cos 1^\circ = p$ and $\cos 89^\circ = q$, then which one of the following is correct?
 (a) p is close to 0 and q is close to 1
 (b) $p < q$
 (c) $p = q$
 (d) p is close to 1 and q is close to 0
31. If $0 \leq \theta < \frac{\pi}{2}$ and $p = \sec^2 \theta$, then which one of the following is correct?
 (a) $p < 1$ (b) $p = 1$ (c) $p > 1$ (d) $p \geq 1$
32. What is the value of $\cos 1^\circ \cos 2^\circ \cos 3^\circ \dots \cos 90^\circ$?
 (a) $\frac{1}{2}$ (b) 0 (c) 1 (d) 2
33. If $\tan A = \frac{1 - \cos B}{\sin B}$, then what is $\frac{2 \tan A}{1 - \tan^2 A}$ equal to?
 (a) $\frac{\tan B}{2}$ (b) $2 \tan B$ (c) $\tan B$ (d) $4 \tan B$

- 34.** What is $\cot 15^\circ \cot 20^\circ \cot 70^\circ \cot 75^\circ$ equal to?
 (a) -1 (b) 0 (c) 1 (d) 2
- 35.** If $\frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta} = 2$ with $0 < \theta < 90^\circ$, then what is θ equal to?
 (a) 30° (b) 45° (c) 60° (d) 75°
- 36.** In a right $\triangle ABC$, right angled at B , the ratio of AB to AC is $1 : \sqrt{2}$, then $\frac{2 \tan A}{1 - \tan^2 A}$ is equal to
 (a) 2 (b) 1 (c) 3 (d) undefined
- 37.** In figure, $AD = DB$, $\angle B = 90^\circ$, then $\cos \theta$ is equal to



- (a) $\frac{2\sqrt{b^2 - a^2}}{\sqrt{4b^2 - 3a^2}}$ (b) $\frac{a}{\sqrt{4b^2 - 3a^2}}$
 (c) $\frac{\sqrt{4b^2 - 3a^2}}{2}$ (d) None of these
- 38.** If $0^\circ \leq x \leq 90^\circ$ and $\sin x + \sqrt{3} \cos x = 1$, then what is the value of x ?
 (a) 30° (b) 45° (c) 60° (d) 90°
- 39.** If $\tan A = \sqrt{2} - 1$, then the value of $\operatorname{cosec} A \cdot \sec A$ is equal to
 (a) $\frac{1}{\sqrt{2}}$ (b) $\frac{2}{\sqrt{2}}$ (c) $2\sqrt{2}$ (d) $\frac{\sqrt{3}}{2}$
- 40.** $\tan \theta = 3$ and θ lies in third quadrant, then the value of $\sin \theta$ is
 (a) $\frac{1}{\sqrt{10}}$ (b) $-\frac{1}{\sqrt{10}}$ (c) $\frac{-3}{\sqrt{10}}$ (d) $\frac{3}{\sqrt{10}}$
- 41.** The value of $(\sin 20^\circ \cos 70^\circ + \cos 20^\circ \sin 70^\circ)$ is
 (a) 1 (b) 0 (c) -1 (d) $\frac{1}{2}$
- 42.** If ϕ and θ are supplementary angles, then
 (a) $\sin \theta = \sin \phi$ (b) $\cos \theta = \cos \phi$
 (c) $\tan \theta = \tan \phi$ (d) $\sec \phi = \operatorname{cosec} \theta$
- 43.** If $x + y = 90^\circ$, then what is the value of $\sqrt{\cos x \operatorname{cosec} y - \cos x \sin y}$?
 (a) $\cos x$ (b) $\sin x$ (c) $\sqrt{\cos x}$ (d) $\sqrt{\sin x}$
- 44.** The value of $\sqrt{2 + \sqrt{2 + 2 \cos 4\theta}}$ is
 (a) $2 \cos^2 \theta$ (b) $2 \cos 2\theta$ (c) $2 \cos \theta$ (d) $2 \cos \frac{\theta}{2}$
- 45.** If $\tan A = \frac{5}{6}$ and $\tan B = \frac{1}{11}$, then $A + B$ is equal to
 (a) $\frac{\pi}{2}$ (b) $\frac{\pi}{4}$ (c) $\frac{3\pi}{2}$ (d) π
- 46.** The value of $\frac{1}{\sqrt{2}}(\cos A - \sin A)$ is
 (a) $\cos\left(\frac{\pi}{3} + A\right)$ (b) $\cos\left(\frac{\pi}{2} + A\right)$
 (c) $\cos\left(\frac{\pi}{4} + A\right)$ (d) $\sin\left(\frac{\pi}{4} + A\right)$
- 47.** $\sin \frac{\pi}{4} \cos \frac{\pi}{12} - \cos \frac{\pi}{4} \sin \frac{\pi}{12}$ is equal to
 (a) $\frac{1}{\sqrt{2}}$ (b) $\sqrt{3}$ (c) $\frac{\sqrt{3}}{2}$ (d) $\frac{1}{2}$
- 48.** If $\tan \theta = \frac{x \sin \phi}{1 - x \cos \phi}$ and $\tan \phi = \frac{y \sin \theta}{1 - y \cos \theta}$, then $\frac{x}{y}$ is equal to
 (a) $\frac{\sin \phi}{\sin \theta}$ (b) $\frac{\sin \phi}{\cos \theta}$ (c) $\frac{\sin \theta}{\sin \phi}$ (d) $\frac{\sin \theta}{1 - \cos \phi}$
- 49.** If $\sin(\theta + \phi) = 2 \sin(\theta - \phi)$, then
 (a) $\cot \phi = 3 \tan \theta$ (b) $\tan \theta = 3 \tan \phi$
 (c) $\sin \theta = 3 \sin \phi$ (d) $\sin \phi = \sin 2\theta$
- 50.** If $\sin(\theta + \alpha) = \cos(\theta + \alpha)$, then $\tan \theta$ is equal to
 (a) $\frac{1 - \tan \alpha}{1 + \tan \alpha}$ (b) $\frac{\tan \alpha + 1}{\tan \alpha - 1}$ (c) $\frac{1 - \cot \alpha}{1 + \cot \alpha}$ (d) $\frac{\sin 2(\theta + \alpha)}{\cos(\theta + \alpha)}$
- 51.** $\cos^2 \frac{\theta - \phi}{2} - \sin^2 \frac{\theta + \phi}{2}$ is equal to
 (a) $\cos \phi \sin \theta$ (b) $\cos 2\theta \sin \phi$
 (c) $\cos \theta \cos \phi$ (d) $\sin \theta \sin \phi$
- 52.** If $x = r \sin \theta \cos \phi$, $y = r \sin \theta \sin \phi$, $z = r \cos \theta$, then $x^2 + y^2 + z^2$ is equal to
 (a) $r^2 \cos^2 \phi$ (b) $r^2 \sin^2 \theta + r^2 \cos^2 \phi$
 (c) r^2 (d) $\frac{1}{r^2}$
- 53.** If $\cot \theta = \frac{7}{24}$ and $\pi < \theta < \frac{3\pi}{2}$, then the value of $\cos \theta - \sin \theta$ is
 (a) $\frac{19}{25}$ (b) $\frac{18}{25}$
 (c) $\frac{17}{25}$ (d) $\frac{18}{25}$
- 54.** If $\alpha + \beta = \frac{\pi}{2}$ and $\sin \alpha = \frac{1}{2}$, then β is
 (a) 30° (b) 60° (c) 45° (d) $22 \frac{1^\circ}{2}$
- 55.** If $\sin x - \cos x = 0$, then what is the value of $\sin^4 x + \cos^4 x$?
 (a) 1 (b) $\frac{3}{4}$ (c) $\frac{1}{2}$ (d) $\frac{1}{4}$

56. If $3 \cos x = 5 \sin x$, then the value of $\frac{5 \sin x - 2 \sec^3 x + 2 \cos x}{5 \sin x + 2 \sec^3 x - 2 \cos x}$ is
 (a) $\frac{361}{2397}$ (b) $\frac{271}{979}$ (c) $\frac{541}{979}$ (d) $\frac{127}{979}$
57. $\sqrt{\frac{1 + \cos x}{1 - \cos x}}$ is equal to
 (a) $\sec x + \tan x$ (b) $\operatorname{cosec} x + \cot x$
 (c) $\sec x - \tan x$ (d) $\operatorname{cosec} x - \cot x$
58. $\cos^4 x - \sin^4 x$ is equal to
 (a) $2 \cos^2 x - 1$ (b) $2 \sin^2 x - 1$
 (c) $\sin^2 \theta - \cos^2 \theta$ (d) None of these
59. $\frac{\sin \theta}{1 + \cos \theta}$ is equal to
 (a) $\frac{1 - \cos \theta}{\sin \theta}$ (b) $\frac{\sin \theta}{\cos \theta}$ (c) $\frac{\cos \theta - 1}{\sin \theta}$ (d) $\frac{1 + \sin \theta}{\cos \theta}$
60. If $0 \leq \theta \leq 90^\circ$, then $\left(\frac{5 \cos \theta - 4}{3 - 5 \sin \theta} - \frac{3 + 5 \sin \theta}{4 + 5 \cos \theta} \right)$ is equal to
 (a) 0 (b) 1 (c) $\frac{1}{4}$ (d) $\frac{1}{2}$
61. If $\cos(\alpha + \beta) = \frac{4}{5}$ and $\sin(\alpha - \beta) = \frac{5}{13}$, α, β lies between 0 and $\frac{\pi}{4}$, then the value of $\tan 2\alpha$ is
 (a) $\frac{56}{33}$ (b) $\frac{56}{23}$ (c) $\frac{43}{33}$ (d) $\frac{34}{33}$
62. $\frac{\tan(45^\circ + x)}{\tan(45^\circ - x)}$ is equal to
 (a) $\left(\frac{1 + \tan x}{1 - \tan x} \right)^2$ (b) $\left(\frac{1 - \tan x}{1 + \tan x} \right)^2$
 (c) $\frac{2 \sin^2 \frac{x}{2}}{1 + \cos^2 \frac{x}{2}}$ (d) None of these
63. What is the simplest value of $\frac{(1 - \sin A \cos A)(\sin^2 A - \cos^2 A)}{\cos A (\sec A - \operatorname{cosec} A)(\sin^3 A + \cos^3 A)}$?
 (a) $\sin A$ (b) $\cos A$ (c) $\sec A$ (d) $\operatorname{cosec} A$
64. If $A + B = 45^\circ$, then $(1 + \tan A)(1 + \tan B)$ is equal to
 (a) 1 (b) -1 (c) 2 (d) -2
65. $\left(\frac{\cos 2B - \cos 2A}{\sin 2B + \sin 2A} \right)$ is equal to
 (a) $\tan(A - B)$ (b) $\tan(A + B)$ (c) $\cot(A - B)$ (d) $\cot(A + B)$
66. $\sec^2 x + \tan^2 x = 7$, the value of x is
 (a) 15° (b) 30° (c) 45° (d) 60°
67. $\cos 3\theta + \sin 3\theta$ is maximum, when θ is
 (a) 15° (b) 30° (c) 45° (d) 60°
68. If $\sin^2 x + \sin^2 y + \sin^2 z = (\sin x + \sin y + \sin z)^2$, then which of the following expressions must necessarily vanish?
 (a) $\tan x + \tan y + \tan z$ (b) $\cos x + \cos y + \cos z$
 (c) $\frac{1}{\sin x} + \frac{1}{\sin y} + \frac{1}{\sin z}$ (d) $\frac{1}{\cos x} + \frac{1}{\cos y} + \frac{1}{\cos z}$
69. If $\sin x + \sin^2 x = 1$, then the value of $\cos^2 x + \cos^4 x$ is equal to
 (a) -1 (b) 2 (c) -2 (d) 1
70. $\cos 35^\circ + \cos 85^\circ + \cos 155^\circ$ is
 (a) 0 (b) $\frac{1}{\sqrt{3}}$ (c) $\frac{1}{\sqrt{2}}$ (d) $\cos 275^\circ$
71. $\sin 12^\circ \sin 24^\circ \sin 48^\circ \sin 84^\circ$ is
 (a) $\frac{1}{16}$ (b) $\frac{3}{64}$ (c) $\frac{3}{15}$ (d) 0
72. If $\tan \theta = \frac{p}{q}$, then the value of $p \cos 2\theta + q \sin 2\theta$ is
 (a) p (b) q
 (c) $\frac{q(3q^2 - p^2)}{p^2 + q^2}$ (d) $\frac{p(3q^2 - p^2)}{p^2 + q^2}$
73. If $x = y \cos \frac{2\pi}{3} = z \cos \frac{4\pi}{3}$, then $xy + yz + zx$ is equal to
 (a) 1 (b) -1 (c) 0 (d) 2
74. If $\sin \theta + \cos \theta = m$ and $\sec \theta + \operatorname{cosec} \theta = n$, then $n(m^2 - 1)$ is equal to
 (a) $\frac{1}{2m}$ (b) m (c) $2m$ (d) $\frac{m}{2}$
75. If $\operatorname{cosec} \theta - \sin \theta = a^3$, $\sec \theta - \cos \theta = b^3$, then $a^2 b^2 (a^2 + b^2)$ is equal to
 (a) -1 (b) 1 (c) 2 (d) -2
76. The maximum value and minimum value of $(1 + \cos 2x)$ are
 (a) -1 and 1 (b) 1 and 2 (c) $\frac{-1}{2}$ and $\frac{1}{2}$ (d) 0 and 2
77. If $\tan \theta + \tan \phi = a$ and $\cot \theta + \cot \phi = b$, then $\cot(\theta + \phi)$ is equal to
 (a) $\frac{1}{a} + \frac{1}{b}$ (b) $\frac{1}{a} - \frac{1}{b}$
 (c) $a - b$ (d) $a + b$
78. $\tan 75^\circ - \tan 30^\circ - \tan 75^\circ \tan 30^\circ$ is equal to
 (a) -1 (b) 1 (c) 0 (d) 2
79. The value of $\sin^2(15^\circ + A) - \sin^2(15^\circ - A)$ is equal to
 (a) $\frac{1}{2} \cos 2A$ (b) $\frac{1}{2} \sin 2A$ (c) $\frac{1}{2} \tan 2A$ (d) $\cot 2A$

- 80.** The value of $\frac{\sin 38^\circ - \cos 68^\circ}{\cos 68^\circ + \sin 38^\circ}$ is equal to
 (a) $\sqrt{3} \tan 8^\circ$ (b) $\sqrt{3} \cot 8^\circ$
 (c) $\sqrt{3} \sin 38^\circ$ (d) $\sqrt{3} \sin 8^\circ$
- 81.** $2 \cos x - \cos 3x - \cos 5x$ is equal to
 (a) $8 \cos^3 x \sin^2 x$ (b) $12 \cos^2 x \sin^3 x$
 (c) $16 \cos^3 x \sin^2 x$ (d) $32 \cos^3 x \sin x$
- 82.** $\cos 4x$ is equal to
 (a) $1 + 2 \sin^2 2x$ (b) $2 \cos^2 2x$
 (c) $1 - 8 \sin^2 x \cos^2 x$ (d) $1 + 8 \sin^2 x \cos^2 x$
- 83.** $2 \sin A \cos^3 A - 2 \sin^3 A \cos A$ is equal to
 (a) $\frac{1}{2} \sin 4A$ (b) $\frac{1}{2} \cos 4A$ (c) $\frac{1}{2} \tan 4A$ (d) $\frac{1}{2} \cot 4A$
- 84.** If $a \cos \theta - b \sin \theta = c$, then what is the value of $a \sin \theta + b \cos \theta$?
 (a) $\pm \sqrt{a^2 + b^2 + c^2}$ (b) $\pm \sqrt{a^2 - b^2 + c^2}$
 (c) $\pm \sqrt{a^2 + b^2 - c^2}$ (d) $\pm \sqrt{a^2 - b^2 - c^2}$
- 85.** If $\cos x + \cos^2 x = 1$, then the value of $\sin^8 x + 2 \sin^6 x + \sin^4 x$ is
 (a) 0 (b) -1 (c) 2 (d) 1
- 86.** If $m = \operatorname{cosec} x - \sin x$ and $n = \sec x - \cos x$, then $\tan x$ is equal to
 (a) $\left(\frac{n}{m}\right)^{2/3}$ (b) $\left(\frac{n}{m}\right)^{1/3}$
 (c) $\left(\frac{n}{m}\right)$ (d) $\left(\frac{n}{m}\right)^2$
- 87.** If $2 \cos \theta = x + \frac{1}{x}$, then $2 \cos 3\theta$ is equal to
 (a) $x^3 + \frac{1}{x^3}$ (b) $x^2 + \frac{1}{x^2}$ (c) $\frac{x^2}{1+x^2}$ (d) None of these
- 88.** If $\cos \theta \geq \frac{1}{2}$ in the first quadrant, then which one of the following is correct?
 (a) $\theta \leq \frac{\pi}{3}$ (b) $\theta \geq \frac{\pi}{3}$ (c) $\theta \leq \frac{\pi}{6}$ (d) $\theta \geq \frac{\pi}{6}$
- 89.** If $\sin A = \frac{3}{5}$ and $\cos B = \frac{12}{13}$, then the value of $\frac{\tan A - \tan B}{1 + \tan A \tan B}$ is equal to
 (a) $\frac{23}{16}$ (b) $\frac{16}{63}$ (c) $\frac{1}{63}$ (d) $\frac{13}{63}$
- 90.** If $B + C = 60^\circ$, then which of the following is correct statement?
 (a) $\sin(120^\circ - B) = \sin(120^\circ - C)$
 (b) $\sin(120^\circ + B) = \sin(180^\circ + C)$
 (c) $\cos(120^\circ - B) = \sin(120^\circ + C)$
 (d) $\tan(B) = \tan(120^\circ + C)$
- 91.** What is the value of x in the equation $x \frac{\operatorname{cosec}^2 30^\circ \sec^2 45^\circ}{8 \cos^2 45^\circ \sin^2 60^\circ} = \tan^2 60^\circ - \tan^2 30^\circ$?
 (a) $x = 1$ (b) $x = 2$ (c) $x = \frac{1}{2}$ (d) $x = \frac{3}{2}$
- 92.** Under which one of the following conditions is the trigonometrical identity $\sin x/(1 + \cos x) = (1 - \cos x)/\sin x$ true?
 (a) x is not a multiple of 360°
 (b) x is not an odd multiple of 180°
 (c) x is not a multiple of 180°
 (d) None of the above
- 93.** If $\cos x = k \cos(x - 2y)$, then $\tan(x - y) \tan y$ is equal to
 (a) $\frac{1+k}{1-k}$ (b) $\frac{1-k}{1+k}$ (c) $\frac{2k}{k+1}$ (d) $\frac{k-1}{2k+1}$
- 94.** The value of $\frac{\sin(x+y) - 2 \sin x + \sin(x-y)}{\cos(x+y) - 2 \cos x + \cos(x-y)}$ is
 (a) $\cot x$ (b) $\tan x$ (c) $\sin x$ (d) $\operatorname{cosec} x$
- 95.** If $\sin x + \sin y = a$ and $\cos x + \cos y = b$, then $\tan \frac{x+y}{2}$ is
 (a) $\frac{4}{a^2 + b^2}$ (b) $\frac{b}{a}$ (c) $\frac{a}{b}$ (d) $\frac{4}{a^2 - b^2}$
- 96.** The largest hand of a clock is 42 cm long, then the distance covered by the extremity in 20 min is
 (a) 88 cm (b) 80 cm (c) 82 cm (d) 84 cm
- 97.** If $0^\circ < x < 90^\circ$ and $2 \sin x + 15 \cos^2 x = 7$, then what is the value of $\tan x$?
 (a) $\frac{2}{3}$ (b) $\frac{3}{2}$ (c) $\frac{3}{4}$ (d) $\frac{4}{3}$
- 98.** If ABC is a triangle and $A + B + C = 180^\circ$, then $\tan A + \tan B + \tan C$ is
 (a) $\tan A \tan B \tan C$ (b) $\tan \frac{A+B}{2} + \tan \frac{B+C}{2}$
 (c) $\frac{\tan A + \tan B}{\tan C}$ (d) $\cot A \cot B \cot C$
- 99.** If $x = a \sec \theta \cos \phi$, $y = b \sec \theta \sin \phi$ and $z = c \tan \theta$, then the value of $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2}$ is
 (a) 9 (b) 0 (c) 1 (d) 4
- 100.** Which one of the following statements is correct?
 (a) The squares of the tangents of the angles 30° , 45° , 60° are in G.P.
 (b) The squares of the sines of the angles 30° , 45° , 60° are in G.P.
 (c) The squares of the secants of the angles 30° , 45° , 60° are in A.P.
 (d) The squares of the tangents of the angles 30° , 45° , 60° are in A.P.

- 101.** If A, B, C and D are the successive angles of a cyclic quadrilateral, then what is $\cos A + \cos B + \cos C + \cos D$ are equal to?
 (a) 4 (b) 2 (c) 1 (d) 0
- 102.** Which one of the following is correct?
 (a) There is only one θ with $0^\circ < \theta < 90^\circ$ such that $\sin \theta = a$, where a is a real number
 (b) There is more than one θ with $0^\circ < \theta < 90^\circ$ such that $\sin \theta = a$, where a is a real number
 (c) There is no θ with $0^\circ < \theta < 90^\circ$ such that $\sin \theta = a$, where a is a real number
 (d) There are exactly two θ 's with $0^\circ < \theta < 90^\circ$ such that $\sin \theta = a$, where a is a real number
- 103.** If $\sin(B + C - A) = \cos(C + A - B) = \tan(A + B - C) = 1$, then the angles A, B, C which are positive acute angles are, respectively
 (a) $45^\circ, 80^\circ, 105^\circ$ (b) $22\frac{1^\circ}{2}, 67\frac{1^\circ}{2}, 45^\circ$
 (c) $20^\circ, 70^\circ, 90^\circ$ (d) $30^\circ, 60^\circ, 90^\circ$
- 104.** If A, B, C and D be the angles of a cyclic quadrilateral taken in order, then $\cos(180^\circ + A) + \cos(180^\circ + B) + \cos(180^\circ + C) + \cos(180^\circ + D)$ is equal to
 (a) $2(\cos A + \cos B)$ (b) $2(\cos A + \cos D)$
 (c) 0 (d) $4 \cos A$
- 105.** If ABC is a right angled triangle at C and having u units, v units and w units as the lengths of its sides opposite to be vertices A, B and C respectively, then what is $\tan A + \tan B$ equal to?
 (a) $\frac{u^2}{vw}$ (b) 1 (c) $u + v$ (d) $\frac{w^2}{uv}$
- 106.** The Earth takes 24 h to rotate about its own axis. Through what angle will it turn in 4 h and 12 min?
 (a) 63° (b) 64° (c) 65° (d) 70°
- 107.** Assume the Earth to be a sphere of radius R . What is the radius of the circle of latitude $40^\circ S$?
 (a) $R \cos 40^\circ$ (b) $R \sin 80^\circ$
 (c) $R \sin 40^\circ$ (d) $R \tan 40^\circ$
- 108.** If $\sin 3\theta = \cos(\theta - 2^\circ)$, where 3θ and $(\theta - 2^\circ)$ are acute angles, then what is the value of θ ?
 (a) 22° (b) 23°
 (c) 24° (d) 25°
- 109.** Consider the following statements:
 I. $\frac{\cos^2 \theta - \sin^2 \theta}{\cos^2 \theta + \sin^2 \theta} = \cos^2 \theta (1 + \tan \theta) (1 - \tan \theta)$
 II. $\frac{1 + \sin \theta}{1 - \sin \theta} = (\tan \theta + \sec \theta)^2$
 Which of the statement(s) given above is/are correct?
 (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II
- 110.** If $A = \frac{\pi}{6}$ and $B = \frac{\pi}{3}$, then which of the following is/are correct?
 I. $\sin A + \sin B = \cos A + \cos B$
 II. $\tan A + \tan B = \cot A + \cot B$
 Select the correct answer using the codes given below
 (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II
- 111.** For any quadrilateral $ABCD$ which of the following statements are true?
 I. $\sin(A + B) + \sin(C + D) = 0$
 II. $\cos(A + B) + \cos(C + D) = 0$
 Select the correct answer using the codes given below
 (a) Only I (b) Only II
 (c) Both I and II (d) None of these
- 112.** Consider the following statements
 I. 1° in radian measure is less than 0.03 radians.
 II. 1 radian in degree measure is greater than 45° .
 Which of the above statement is/are correct?
 (a) Only II (b) Only I
 (c) Neither I nor II (d) Both I of II
- 113.** Consider the following statements
 I. $\sin \theta \cdot \sin(60^\circ + \theta) \cdot \sin(60^\circ - \theta) = \frac{1}{4} \sin 3\theta$
 II. $\cos \theta \cdot \sin(30^\circ + \theta) \cdot \sin(30^\circ - \theta) = \frac{1}{4} \cos 3\theta$
 III. $\sin \theta \cdot \cos(30^\circ + \theta) \cdot \cos(30^\circ - \theta) = \frac{1}{4} \sin 3\theta$
 Which of the above statement is/are correct?
 (a) Only I (b) Only II
 (c) Only III (d) All of the above
- 114.** If ABC is a right angled triangle, then which of the following statements are correct?
 I. $\sin(A + B) = \sin C$
 II. $\sin\left(\frac{A + B}{2}\right) = \cos \frac{C}{2}$
 III. $\tan \frac{(A + B - C)}{2} = \cot C$
 IV. $\tan \frac{(A - B - C)}{2} = -\cot A$
 Select the correct answer using the codes given below
 (a) I and II (b) I, II and III
 (c) I, II and IV (d) All of these

Directions. Q. Nos. (115-117) Let $\sin(A+B)=1$ and $\sin(A-B)=\frac{1}{2}$, where $A, B \in \left[0, \frac{\pi}{2}\right]$.

115. What is the value of A ?

- (a) $\frac{\pi}{6}$ (b) $\frac{\pi}{3}$ (c) $\frac{\pi}{4}$ (d) $\frac{\pi}{8}$

116. What is the value of $\tan(A+2B)\tan(2A+B)$?

- (a) -1 (b) 0 (c) 1 (d) 2

117. What is the value of $\sin^2 A - \sin^2 B$?

- (a) 0 (b) $\frac{1}{2}$ (c) 1 (d) 2

Directions. Q. Nos. (118-120) ABC is an obtuse triangle and $\tan A, \tan B$ are the roots of the equation $3x^2 - 2\sqrt{3}x + 1 = 0$.

118. The measure of angle C is

- (a) $\pi/3$ (b) $\pi/2$ (c) $\frac{2\pi}{3}$ (d) $\frac{5\pi}{6}$

119. Find the value of $\sin C + \cos C$.

- (a) $\frac{2}{\sqrt{3}+1}$ (b) $\frac{\sqrt{3}+1}{2}$ (c) $\frac{\sqrt{3}-1}{2}$ (d) $\sqrt{3} + \frac{1}{2}$

120. If $2\sin B = \sqrt{\frac{1+\tan^2 C}{\operatorname{cosec}^2 C}}$, then the value of B is

- (a) $\frac{\pi}{2}$ (b) $\frac{\pi}{3}$ (c) $\frac{2\pi}{3}$ (d) $\frac{\pi}{6}$

PREVIOUS YEARS' QUESTIONS

121. The expression $\sin^2 x + \cos^2 x - 1 = 0$ is satisfied by how many values of x ? ☑ 2012 I

- (a) Only one value of x (b) Two values of x
(c) Infinite values of x (d) No value of x

122. If $3\sin x + 5\cos x = 5$, then what is the value of $(3\cos x - 5\sin x)$? ☑ 2012 I

- (a) 0 (b) 2 (c) 3 (d) 5

123. If $p = a\sin x + b\cos x$ and $q = a\cos x - b\sin x$, then what is the value of $p^2 + q^2$? ☑ 2012 I

- (a) $a+b$ (b) ab (c) $a^2 + b^2$ (d) $a^2 - b^2$

124. If α and β are complementary angles, then what

is $\sqrt{\operatorname{cosec} \alpha \cdot \operatorname{cosec} \beta} \left(\frac{\sin \alpha}{\sin \beta} + \frac{\cos \alpha}{\cos \beta} \right)^{-\frac{1}{2}}$ equal to? ☑ 2012 I

- (a) 0 (b) 1
(c) 2 (d) None of these

125. Consider the following statements

I. $\frac{\cot 30^\circ + 1}{\cot 30^\circ - 1} = 2(\cos 30^\circ + 1)$

II. $2\sin 45^\circ \cos 45^\circ - \tan 45^\circ \cot 45^\circ = 0$

Which of the above statement(s) is/are correct?

- (a) Only I (b) Only II ☑ 2012 I
(c) Both I and II (d) Neither I nor II

126. Consider the following statements

I. $\sin^2 1^\circ + \cos^2 1^\circ = 1$

II. $\sec^2 33^\circ - \cot^2 57^\circ = \operatorname{cosec}^2 37^\circ - \tan^2 53^\circ$

Which of the above statement(s) given above is/are correct? ☑ 2012 I

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

127. Consider the following statements:

I. There is only one value of x in the first quadrant that satisfies $\sin x + \cos x = 2$.

II. There is only one value of x in the first quadrant that satisfies $\sin x - \cos x = 0$.

Which of the above statement(s) given above is/are correct? ☑ 2012 I

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

128. If $\sin \theta \cos \theta = \sqrt{3}/4$, then $\sin^4 \theta + \cos^4 \theta$ is equal to ☑ 2012 II

- (a) $7/8$ (b) $5/8$ (c) $3/8$ (d) $1/8$

129. If the angle θ is in the first quadrant and $\tan \theta = 3$, then what is the value of $(\sin \theta + \cos \theta)$? ☑ 2012 II

- (a) $\frac{1}{\sqrt{10}}$ (b) $\frac{2}{\sqrt{10}}$ (c) $\frac{3}{\sqrt{10}}$ (d) $\frac{4}{\sqrt{10}}$

130. If $0^\circ < \theta < 90^\circ$, then all the trigonometric ratios can be obtained when ☑ 2012 II

- (a) only $\sin \theta$ is given
(b) only $\cos \theta$ is given
(c) only $\tan \theta$ is given
(d) any one of the six ratios is given

131. What is the value of $\sin A \cos A \tan A + \cos A \sin A \cot A$? ☑ 2012 II

- (a) $\sin^2 A + \cos^2 A$ (b) $\sin^2 A + \tan^2 A$
(c) $\sin^2 A + \cot^2 A$ (d) $\operatorname{cosec}^2 A + \cot^2 A$

132. What is the value of $\frac{\sin \theta}{1 + \cos \theta} + \frac{1 + \cos \theta}{\sin \theta}$? ☑ 2012 II

- (a) $2 \operatorname{cosec} \theta$ (b) $2 \sec \theta$
(c) $\sec \theta$ (d) $\operatorname{cosec} \theta$

133. What is the value of $\sec^2 D - \tan^2 D$? ☑ 2013 I

- (a) $1/2$ (b) $2/3$
(c) 1 (d) None of these

134. If $\cos A + \cos^2 A = 1$, then what is the value of $2(\sin^2 A + \sin^4 A)$? ☑ 2013 I

- (a) 4 (b) 2 (c) 1 (d) $1/2$

- 135.** $(1 - \tan A)^2 + (1 + \tan A)^2 + (1 - \cot A)^2 + (1 + \cot A)^2$ is equal to ☑ 2013 I

(a) $\sin^2 A \cos^2 A$ (b) $\sec^2 A \operatorname{cosec}^2 A$
(c) $2 \sec^2 A \operatorname{cosec}^2 A$ (d) None of these

- 136.** If $\alpha^2 = \frac{1 + 2 \sin \theta \cos \theta}{1 - 2 \sin \theta \cos \theta}$, then what is the value of $\frac{\alpha + 1}{\alpha - 1}$? ☑ 2013 I

(a) $\sec \theta$ (b) 1 (c) 0 (d) $\tan \theta$

- 137.** If $\sin \theta = \frac{x^2 - y^2}{x^2 + y^2}$, then which one of the following is correct? ☑ 2013 I

(a) $\cos \theta = \frac{2xy}{x^2 - y^2}$ (b) $\cos \theta = \frac{2xy}{x^2 + y^2}$
(c) $\cos \theta = \frac{x - y}{x^2 + y^2}$ (d) $\cos \theta = \frac{xy(x - y)}{x^2 + y^2}$

- 138.** Consider the following statements for $0 \leq \theta \leq 90^\circ$:
I. The value of $\sin \theta + \cos \theta$ is always greater than 1.
II. The value of $\tan \theta + \cot \theta$ is always greater than 1.
Which of the statement(s) given above is/are correct?
(a) Only I (b) Only II ☑ 2013 I
(c) Both I and II (d) Neither I nor II

Directions (Q. Nos. 139-141) Read the following information carefully to answer the questions that follow.

The angles A, B, C and D of a quadrilateral ABCD are in the ratio 1 : 2 : 4 : 5.

- 139.** What is the value of $\cos(A + B)$? ☑ 2013 I

(a) 0 (b) $1/2$
(c) 1 (d) None of these

- 140.** What is the value of $\operatorname{cosec}(C - D + B)$? ☑ 2013 I

(a) 1 (b) 2 (c) 3 (d) 4

- 141.** Consider the following statements:
I. ABCD is a cyclic quadrilateral.

II. $\sin(B - A) = \cos(D - C)$

Which of the statement(s) given above is/are correct? ☑ 2013 I

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

- 142.** If $\sin \theta + \cos \theta = \sqrt{3}$, then what is $\tan \theta + \cot \theta$ equal to? ☑ 2013 II

(a) 1 (b) $\sqrt{2}$ (c) 2 (d) $\sqrt{3}$

- 143.** What is $\operatorname{cosec}(75^\circ + \theta) - \sec(15^\circ - \theta)$ equal to?

☑ 2013 II

(a) 0 (b) 1 (c) $2 \sin \theta$ (d) $2 \cos \theta$

- 144.** If $5 \sin \theta + 12 \cos \theta = 13$, then what is $5 \cos \theta - 12 \sin \theta$ equal to? ☑ 2013 II

(a) -2 (b) -1 (c) 0 (d) 1

- 145.** If $\sin \theta - \cos \theta = 0$, then what is $\sin^4 \theta + \cos^4 \theta$ equal to? ☑ 2013 II

(a) 1 (b) $\frac{3}{4}$ (c) $\frac{1}{2}$ (d) $\frac{1}{4}$

- 146.** What is $\frac{\cos^2(45^\circ + \theta) + \cos^2(45^\circ - \theta)}{\tan(60^\circ + \theta) \tan(30^\circ - \theta)}$ equal to? ☑ 2013 II

(a) -1 (b) 0 (c) 1 (d) 2

- 147.** If $\tan \theta + \sec \theta = m$, then what is $\sec \theta$ equal to?

☑ 2013 II

(a) $\frac{m^2 - 1}{2m}$ (b) $\frac{m^2 + 1}{2m}$ (c) $\frac{m + 1}{m}$ (d) $\frac{m^2 + 1}{m}$

- 148.** If α, β and γ are acute angles such that $\sin \alpha = \frac{\sqrt{3}}{2}$, $\cos \beta = \frac{\sqrt{3}}{2}$ and $\tan \gamma = 1$, then what is $\alpha + \beta + \gamma$ equal to? ☑ 2013 II

(a) 105° (b) 120°
(c) 135° (d) 150°

- 149.** What is $\frac{(1 + \sec \theta - \tan \theta) \cos \theta}{(1 + \sec \theta + \tan \theta)(1 - \sin \theta)}$ equal to? ☑ 2013 II

(a) 1 (b) 2 (c) $\tan \theta$ (d) $\cot \theta$

- 150.** If $\triangle ABC$ is right angled at C, then what is $\cos(A + B) + \sin(A + B)$ equal to? ☑ 2013 II

(a) 0 (b) $\frac{1}{2}$ (c) 1 (d) 2

- 151.** Consider the following statements:

I. $\tan \theta$ increases faster than $\sin \theta$ as θ increases.

II. The value of $\sin \theta + \cos \theta$ is always greater than 1.

Which of the statement(s) given above is/are correct? ☑ 2013 II

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

- 152.** The value of $\cos 25^\circ - \sin 25^\circ$ is ☑ 2014 I

(a) positive but less than 1
(b) positive but greater than 1
(c) negative
(d) 0

- 153.** In a right angled $\triangle ABC$, right angle at B, if $\cos A = \frac{4}{5}$, then what is $\sin C$ is equal to? ☑ 2014 I

(a) $\frac{3}{5}$ (b) $\frac{4}{5}$ (c) $\frac{3}{4}$ (d) $\frac{2}{5}$

- 154.** If α and β are complementary angles, then what is $\sqrt{\cos\alpha \operatorname{cosec}\beta - \cos\alpha \sin\beta}$ equal to? $\checkmark$ 2014 I
 (a) $\sec\beta$ (b) $\cos\alpha$ (c) $\sin\alpha$ (d) $-\tan\beta$
- 155.** If $\sec\theta + \tan\theta = 2$, then what is the value of $\sec\theta$? $\checkmark$ 2014 I
 (a) $\frac{3}{2}$ (b) $\sqrt{2}$ (c) $\frac{5}{2}$ (d) $\frac{5}{4}$
- 156.** What is $\operatorname{cosec}(75^\circ + \theta) - \sec(15^\circ - \theta) - \tan(55^\circ + \theta) + \cot(35^\circ - \theta)$ equal to? $\checkmark$ 2014 I
 (a) -1 (b) 0 (c) 1 (d) $\frac{3}{2}$
- 157.** What is $\sin 25^\circ \sin 35^\circ \sec 65^\circ \sec 55^\circ$ equal to? $\checkmark$ 2014 I
 (a) -1 (b) 0 (c) $\frac{1}{2}$ (d) 1
- 158.** If $2 \cot\theta = 3$, then what is $\frac{2 \cos\theta - \sin\theta}{2 \cos\theta + \sin\theta}$ equal to? $\checkmark$ 2014 I
 (a) $\frac{2}{3}$ (b) $\frac{1}{3}$ (c) $\frac{1}{2}$ (d) $\frac{3}{4}$
- 159.** If $\sin\theta \cos\theta = \frac{1}{2}$, then what is $\sin^6\theta + \cos^6\theta$ equal to? $\checkmark$ 2014 I
 (a) 1 (b) 2 (c) 3 (d) $\frac{1}{4}$
- 160.** If $\cos x + \sec x = 2$, then what $\cos^n x + \sec^n x$ equal to, where n is a positive integer? $\checkmark$ 2014 I
 (a) 2 (b) 2^{n-2} (c) 2^{n-1} (d) 2^n
- 161.** If $\sin\theta + 2 \cos\theta = 1$, where $0 < \theta < \pi/2$, then what is $2 \sin\theta - \cos\theta$ equal to? $\checkmark$ 2014 I
 (a) -1 (b) $1/2$ (c) 2 (d) 1
- 162.** If $\tan 8\theta = \cot 2\theta$, where $0 < 8\theta < \frac{\pi}{2}$, then what is the value of $\tan 5\theta$? $\checkmark$ 2014 I
 (a) $\frac{1}{\sqrt{3}}$ (b) 1 (c) $\sqrt{3}$ (d) 0
- 163.** If $\sin(A+B) = 1$, where $0 < B < 45^\circ$, then what is $\cos(A-B)$ equal to? $\checkmark$ 2014 I
 (a) $\sin 2B$ (b) $\sin B$ (c) $\cos 2B$ (d) $\cos B$
- 164.** At what point of time after 3 O'clock, hour hand and the minute hand of a clock occur at right angles for the first time? $\checkmark$ 2014 I
 (a) 9 O'clock (b) $4 \text{ h } 37\frac{1}{6} \text{ min}$
 (c) $3 \text{ h } 30\frac{8}{11} \text{ min}$ (d) $3 \text{ h } 32\frac{8}{11} \text{ min}$
- 165.** If $\tan\theta + \cot\theta = 2$, then what is $\sin\theta + \cos\theta$ equal to? $\checkmark$ 2014 II
 (a) $\frac{1}{2}$ (b) $\frac{1}{\sqrt{3}}$ (c) $\sqrt{2}$ (d) 1
- 166.** What is $\frac{\sec x}{\cot x + \tan x}$ equal to? $\checkmark$ 2014 II
 (a) $\sin x$ (b) $\cos x$ (c) $\tan x$ (d) $\cot x$
- 167.** What is $(1 + \cot x - \operatorname{cosec} x)(1 + \tan x + \sec x)$ equal to? $\checkmark$ 2014 II
 (a) 1 (b) 2 (c) $\sin x$ (d) $\cos x$
- 168.** What is $(\operatorname{cosec} x - \sin x)(\sec x - \cos x)(\tan x + \cot x)$ equal to? $\checkmark$ 2014 II
 (a) $\sin x + \cos x$ (b) $\sin x - \cos x$
 (c) 2 (d) 1
- 169.** What is $\frac{\sin x - \cos x + 1}{\sin x + \cos x - 1}$ equal to? $\checkmark$ 2014 II
 (a) $\frac{\sin x - 1}{\cos x}$ (b) $\frac{\sin x + 1}{\cos x}$ (c) $\frac{\sin x - 1}{\cos x + 1}$ (d) $\frac{\sin x + 1}{\cos x + 1}$
- 170.** What is $(\sin^2 x - \cos^2 x)(1 - \sin^2 x \cos^2 x)$ equal to? $\checkmark$ 2014 II
 (a) $\sin^4 x - \cos^4 x$ (b) $\sin^6 x - \cos^6 x$
 (c) $\cos^8 x - \sin^8 x$ (d) $\sin^8 x - \cos^8 x$
- 171.** What is $(\sin x \cos y + \cos x \sin y)(\sin x \cos y - \cos x \sin y)$ equal to? $\checkmark$ 2014 II
 (a) $\cos^2 x - \cos^2 y$ (b) $\cos^2 x - \sin^2 y$
 (c) $\sin^2 x - \cos^2 y$ (d) $\sin^2 x - \sin^2 y$
- 172.** If $A + B + C = 180^\circ$, then $\cot A \cot B + \cot B \cot C + \cot C \cot A$ is equal to $\checkmark$ 2014 II
 (a) -1 (b) 2 (c) π (d) 1
- 173.** If $\sin x + \operatorname{cosec} x = 2$, then what is $\sin^9 x + \operatorname{cosec}^9 x$ equal to? $\checkmark$ 2014 II
 (a) 2 (b) 18 (c) 512 (d) 1024
- 174.** If $\sin x + \cos x = p$ and $\sin^3 x + \cos^3 x = q$, then what is $p^3 - 3p$ equal to? $\checkmark$ 2014 II
 (a) 0 (b) $-2q$ (c) $2q$ (d) $4q$
- 175.** Consider the following statements $\checkmark$ 2014 II
 I. $\sin 1^\circ > \sin 1$ II. $\cos 1^\circ < \cos 1$
 Which of the statement(s) given above is/are correct?
 (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II
- 176.** The value of $\operatorname{cosec}^2 67^\circ + \sec^2 57^\circ - \cot^2 33^\circ - \tan^2 23^\circ$ is $\checkmark$ 2015 I
 (a) $2\sqrt{2}$ (b) 2 (c) $\sqrt{2}$ (d) 0
- 177.** If $\tan(A+B) = \sqrt{3}$ and $\tan A = 1$, then $\tan(A-B)$ is equal to $\checkmark$ 2015 I
 (a) 0 (b) 1 (c) $\frac{1}{\sqrt{3}}$ (d) $\sqrt{2}$
- 178.** If $\tan A + \cot A = 4$, then $\tan^4 A + \cot^4 A$ is equal to $\checkmark$ 2015 I
 (a) 110 (b) 191 (c) 80 (d) 194

- 179.** ABC is a triangle right angled at B and $AB:BC = 3:4$. What is $\sin A + \sin B + \sin C$ equal to? ☑ 2015 I
 (a) 2 (b) $\frac{11}{5}$ (c) $\frac{12}{5}$ (d) 3
- 180.** If $\sin x + \cos x = C$, then $\sin^6 x + \cos^6 x$ is equal to ☑ 2015 I
 (a) $\frac{1+6C^2-3C^4}{16}$ (b) $\frac{1+6C^2-3C^4}{4}$
 (c) $\frac{1+6C^2+3C^4}{16}$ (d) $\frac{1+6C^2+3C^4}{4}$
- 181.** If $\frac{3-\tan^2 A}{1-3\tan^2 A} = k$, where k is a real number, then $\operatorname{cosec} A(3 \sin A - 4 \sin^3 A)$ is equal to ☑ 2015 I
 (a) $\frac{2k}{k-1}$ (b) $\frac{2k}{k-1}$, where $\frac{1}{3} \leq k \leq 3$
 (c) $\frac{2k}{k-1}$, where $k < \frac{1}{3}$ or $k > 3$ (d) $\frac{2k}{k+1}$
- 182.** If $p = \sqrt{\frac{1-\sin x}{1+\sin x}}$, $q = \frac{1-\sin x}{\cos x}$, $r = \frac{\cos x}{1+\sin x}$, then which of the following is/are correct? ☑ 2015 I
 I. $p = q = r$ II. $p^2 = qr$
 Select the correct answer using the codes given below
 (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II
- 183.** Consider the following identity
 I. $\frac{\cos A}{1-\tan A} + \frac{\sin A}{1-\cot A} = \sin A + \cos A$
 II. $(1-\sin A - \cos A)^2 = 2(1-\sin A)(1+\cos A)$
 Which of the above identity/identities is/are correct?
 (a) Only I (b) Only II ☑ 2015 I
 (c) Both I and II (d) Neither I nor II
- 184.** If a and b are positive, then the relation $\sin \theta = \frac{2a+3b}{3b}$ is ☑ 2015 II
 (a) not possible (b) possible, if $a = b$
 (c) possible, if $a > b$ (d) possible, if $a < b$
- 185.** The minimum value of $\cos^2 x + \cos^2 y - \cos^2 z$ is ☑ 2015 II
 (a) -1 (b) 0
 (c) 2 (d) 2
- 186.** If $\tan \theta + \sec \theta = 2$, then $\tan \theta$ is equal to ☑ 2015 II
 (a) $\frac{3}{4}$ (b) $\frac{5}{4}$ (c) $\frac{3}{2}$ (d) $\frac{5}{2}$
- 187.** $\frac{\cos \theta}{1-\sin \theta}$ is equal to $\left(\text{where, } \theta \neq \frac{\pi}{2} \right)$ ☑ 2015 II
 (a) $\frac{\tan \theta - 1}{\tan \theta + 1}$ (b) $\frac{1+\sin \theta}{\cos \theta}$ (c) $\frac{\tan \theta + 1}{\tan \theta - 1}$ (d) $\frac{1+\cos \theta}{\sin \theta}$
- 188.** If $\tan(x+40)^\circ \tan(x+20)^\circ \tan(3x)^\circ \tan(70-x)^\circ \tan(50-x)^\circ = 1$, then the value of x is equal to ☑ 2015 II
 (a) 30 (b) 20 (c) 15 (d) 10
- 189.** The value of $32 \cot^2 \left(\frac{\pi}{4} \right) - 8 \sec^2 \left(\frac{\pi}{3} \right) + 8 \cos^3 \left(\frac{\pi}{6} \right)$ is equal to ☑ 2015 II
 (a) $\sqrt{3}$ (b) $2\sqrt{3}$ (c) 3 (d) $3\sqrt{3}$
- 190.** If $x = a \cos \theta$ and $y = b \cot \theta$, then $(ax^{-1} - by^{-1})(ax^{-1} + by^{-1})$ is equal to ☑ 2015 II
 (a) 0 (b) 1 (c) $\tan^2 \theta$ (d) $\sin^2 \theta$
- 191.** If θ is an acute angle and $\sin \theta \cos \theta = 2 \cos^3 \theta - 1.5 \cos \theta$, then what is $\sin \theta$ equal to? ☑ 2015 II
 (a) $\frac{\sqrt{5}-1}{4}$ (b) $\frac{1-\sqrt{5}}{4}$ (c) $\frac{\sqrt{5}+1}{4}$ (d) $-\frac{\sqrt{5}+1}{4}$
- 192.** A clock is started at noon by 10 minutes past 5, through what angle, the hour hand moves? ☑ 2015 II
 (a) 160° (b) 145° (c) 150° (d) 155°
- 193.** Consider the following statements ☑ 2015 II
 I. $\sin 66^\circ$ is less than $\cos 66^\circ$.
 II. $\sin 26^\circ$ is less than $\cos 26^\circ$.
 Which of the above statement(s) is/are correct?
 (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II
- 194.** Consider the following statements.
 I. $\frac{1+\tan^2 \theta}{1+\cot^2 \theta} = \left(\frac{1-\tan \theta}{1-\cot \theta} \right)^2$ is true for all $0 < \theta < \frac{\pi}{2}$, $\theta \neq \frac{\pi}{4}$
 II. $\cot \theta = \frac{1}{\tan \theta}$ is true for $\theta = 45^\circ$ only.
 Which of the above statements is/are correct? ☑ 2015 II
 (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II
- 195.** In a $\triangle ABC$, if $A - B = \frac{\pi}{2}$, then $C + 2B$ is equal to ☑ 2016 I
 (a) $\frac{2\pi}{3}$ (b) $\frac{3\pi}{4}$ (c) π (d) $\frac{\pi}{2}$
- 196.** $\left(\frac{\sin 35^\circ}{\cos 55^\circ} \right)^2 - \left(\frac{\cos 55^\circ}{\sin 35^\circ} \right)^2 + 2 \sin 30^\circ$ is equal to ☑ 2016 I
 (a) -1 (b) 0 (c) 1 (d) 2
- 197.** If $\tan \theta + \cot \theta = \frac{4}{\sqrt{3}}$, where $0 < \theta < \frac{\pi}{2}$, then $\sin \theta + \cos \theta$ is equal to ☑ 2016 I
 (a) 1 (b) $\frac{\sqrt{3}-1}{2}$ (c) $\frac{\sqrt{3}+1}{2}$ (d) $\sqrt{2}$

HINTS AND SOLUTIONS

1. (b) We know that, π radian = 180°
 $\Rightarrow 1 \text{ radian} = \frac{180^\circ}{\pi} = \frac{180^\circ}{22} \times 7$
 $= \frac{630^\circ}{11} = 57\frac{3^\circ}{11} = 57^\circ + \frac{3 \times 60}{11} \text{ min}$
 $= 57^\circ + 16' + \frac{4}{11} \text{ min}$
 $= 57^\circ + 16' + \frac{4}{11} \times 60 \text{ s}$
 $= 57^\circ + 16' + 21.8''$
 $= 57^\circ 16' 21.8'' = 57^\circ 16' 22''$

2. (a) $-47^\circ 30' = -\left(47 + \frac{30}{60}\right)^\circ$
 $= -\left(47 + \frac{1}{2}\right)^\circ \quad \left[\because 1' = \left(\frac{1}{60}\right)^\circ\right]$
 $= -\left(\frac{94+1}{2}\right)^\circ = \frac{-95^\circ}{2}$
 $= \frac{-95}{2} \times \frac{\pi}{180} = \frac{-19\pi}{72} \text{ rad}$

3. (b) Here, $r = 60 \text{ cm}$, $l = 16.5 \text{ cm}$
 $\therefore \theta = \frac{l}{r} = \left(\frac{16.5}{60}\right)^\circ$
 $= \left(\frac{16.5}{60} \times \frac{180}{\pi}\right)^\circ = \left(\frac{16.5}{60} \times \frac{180}{22} \times 7\right)^\circ$
 $= \left(\frac{63}{4}\right)^\circ = 15^\circ 45'$

4. (c) Speed of train = 90 km/h
 $= \left(90 \times \frac{5}{18}\right) \text{ m/sec} = 25 \text{ m/sec}$
 Distance moved in 11 sec
 $= (25 \times 11) \text{ m} = 275 \text{ m}$
 $\therefore l = 275 \text{ m}$, $r = 1500 \text{ m}$
 $\therefore \theta = \frac{l}{r} \Rightarrow \theta = \left(\frac{275}{1500}\right)^\circ$
 $= \left(\frac{275}{1500} \times \frac{180}{\pi}\right)^\circ = \left(\frac{275}{1500} \times \frac{180}{22} \times 7\right)^\circ$
 $= \left(\frac{21}{2}\right)^\circ = 10^\circ 30'$

5. (b) Angle traced by hour hand in 1 hr = 30°
 Angle traced by hour hand in $\frac{91}{12}$ hrs = $\left(30 \times \frac{91}{12}\right)^\circ = 227^\circ 30'$
 Angle traced by minute hand in 1 min = 6°
 Angle traced by minute hand in 35 min = $(6 \times 35)^\circ = 210^\circ$
 $\therefore$ Required angle = $227^\circ 30' - 210^\circ$
 $= 17^\circ 30'$

6. (b) Let the hands of a clock coincide after ' t ' min.

Angle traced by hour hand in

$$t \text{ min} = \frac{1}{2}t$$

Angle traced by minute hand in

$$t \text{ min} = 6t$$

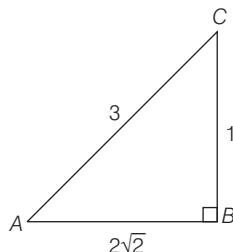
Angle between hour hand and minute hand at 5 pm = $5 \times 30 = 150^\circ$

$$\therefore 6t - \frac{1}{2}t = 150^\circ \Rightarrow t = \frac{150 \times 2}{11} = \frac{300}{11}$$

$$= 27 + \frac{3}{11} = 27 \text{ min } 16 \text{ sec}$$

Therefore, required time = 5:27:16 pm

7. (a) Given, $\sin A = \frac{1}{3}$



In right angled $\triangle ABC$,

$$AB^2 = AC^2 - BC^2 = 3^2 - 1^2 = 8$$

$$\Rightarrow AB = 2\sqrt{2}$$

$$\therefore \cos A = \frac{AB}{AC} = \frac{2\sqrt{2}}{3}$$

$$\text{and } \sec A = \frac{AC}{AB} = \frac{3}{2\sqrt{2}}$$

$$\tan A = \frac{BC}{AB} = \frac{1}{2\sqrt{2}}$$

$$\text{and } \operatorname{cosec} A = \frac{AC}{BC} = 3$$

$$\therefore \cos A \cdot \operatorname{cosec} A + \tan A \cdot \sec A$$

$$= \frac{2\sqrt{2}}{3} \cdot \frac{3}{1} + \frac{1}{2\sqrt{2}} \cdot \frac{3}{2\sqrt{2}}$$

$$= 2\sqrt{2} + \frac{3}{8} = \frac{16\sqrt{2} + 3}{8}$$

8. (b) $\cos 15^\circ - \sin 15^\circ = \cos 15^\circ - \sin (90^\circ - 75^\circ)$

$$= \cos 15^\circ - \cos 75^\circ$$

$$= 2 \sin \frac{15^\circ + 75^\circ}{2} \cdot \sin \frac{75^\circ - 15^\circ}{2}$$

$$\left(\because \cos C - \cos D = 2 \sin \frac{C+D}{2} \sin \frac{D-C}{2} \right)$$

$$= 2 \sin 45^\circ \cdot \sin 30^\circ = 2 \cdot \frac{1}{\sqrt{2}} \cdot \frac{1}{2} = \frac{1}{\sqrt{2}}$$

9. (c) Given, $3 \tan \theta = 4 \Rightarrow \tan \theta = \frac{4}{3}$

$$\sqrt{\frac{1-\sin \theta}{1+\sin \theta}} = \sqrt{\frac{1-\sin \theta}{1+\sin \theta}} \times \frac{1-\sin \theta}{1-\sin \theta}$$

$$= \frac{1-\sin \theta}{\sqrt{1-\sin^2 \theta}}$$

$$= \frac{1-\sin \theta}{\cos \theta} = \frac{1}{\cos \theta} - \frac{\sin \theta}{\cos \theta}$$

$$= \sec \theta - \tan \theta$$

$$= \sqrt{1+\tan^2 \theta} - \tan \theta$$

$$= \sqrt{1+\left(\frac{4}{3}\right)^2} - \frac{4}{3}$$

$$= \sqrt{1+\frac{16}{9}} - \frac{4}{3} = \frac{5}{3} - \frac{4}{3} = \frac{1}{3}$$

10. (b) Given, $\tan A = 1 = \tan 45^\circ$

$$\Rightarrow A = 45^\circ \text{ and } \tan B = \sqrt{3} = \tan 60^\circ$$

$$\therefore B = 60^\circ$$

Now, $\cos A \cos B - \sin A \sin B$

$$= \cos 45^\circ \cos 60^\circ - \sin 45^\circ \sin 60^\circ$$

$$= \frac{1}{\sqrt{2}} \times \frac{1}{2} - \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} = \frac{1-\sqrt{3}}{2\sqrt{2}}$$

11. (b) $\operatorname{cosec}^2 \theta - 2 + \sin^2 \theta$

$$= (\sin \theta - \operatorname{cosec} \theta)^2$$

Hence, it is always non-negative.

12. (b) $\sin^2 15^\circ + \sin^2 20^\circ + \sin^2 25^\circ$

$$+ \dots + \sin^2 75^\circ$$

$$= (\sin^2 15^\circ + \sin^2 75^\circ)$$

$$+ (\sin^2 20^\circ + \sin^2 70^\circ) + \dots + \sin^2 45^\circ$$

$$= [\sin^2 15^\circ + \sin^2 (90^\circ - 15^\circ)]$$

$$+ [\sin^2 20^\circ + \sin^2 (90^\circ - 20^\circ)] + \dots + \sin^2 45^\circ$$

$$= (\sin^2 15^\circ + \cos^2 15^\circ)$$

$$+ (\sin^2 20^\circ + \cos^2 20^\circ) + \dots + 1/2$$

$$= 6 + 1/2 = 13/2$$

$$(\because \sin^2 \theta + \cos^2 \theta = 1)$$

13. (b) $\frac{5 \sin 75^\circ \sin 77^\circ + 2 \cos 13^\circ \cos 15^\circ}{\cos 15^\circ \sin 77^\circ}$

$$- \frac{7 \sin 81^\circ}{\cos 9^\circ}$$

$$= \frac{5 \cos 15^\circ \sin 77^\circ + 2 \sin 77^\circ \cos 15^\circ}{\cos 15^\circ \sin 77^\circ}$$

$$- \frac{7 \cos 9^\circ}{\cos 9^\circ}$$

$$= \frac{7 \cos 15^\circ \cdot \sin 77^\circ}{\cos 15^\circ \cdot \sin 77^\circ} - \frac{7 \cos 9^\circ}{\cos 9^\circ}$$

$$= 7 - 7 = 0$$

$$\begin{aligned}
 14. (a) & \log(\tan 1^\circ) + \log(\tan 2^\circ) + \dots + \log(\tan 89^\circ) \\
 &= \log(\tan 1^\circ \tan 2^\circ \dots \tan 45^\circ \dots \tan 88^\circ \tan 89^\circ) \\
 &= \log[(\tan 1^\circ \cot 1^\circ)(\tan 2^\circ \cot 2^\circ) \dots \tan 45^\circ] \\
 &= \log(1^\circ \cdot 1^\circ \dots 1^\circ) = 0 \quad [\because \tan 89^\circ = \tan(90^\circ - 1^\circ) = \cot 1^\circ]
 \end{aligned}$$

$$\begin{aligned}
 15. (c) & \because \text{Given, } \tan \theta + \frac{1}{\tan \theta} = 2 \\
 & \text{On squaring both side, we get } \left(\tan \theta + \frac{1}{\tan \theta} \right)^2 = 2^2 \\
 & \Rightarrow \tan^2 \theta + \frac{1}{\tan^2 \theta} + 2 = 4 \Rightarrow \tan^2 \theta + \frac{1}{\tan^2 \theta} = 2
 \end{aligned}$$

$$\begin{aligned}
 16. (c) & \text{Given, } \sec \theta = \frac{13}{5} \\
 \therefore \tan \theta &= \sqrt{\sec^2 \theta - 1} = \sqrt{\frac{169}{25} - 1} = \sqrt{\frac{144}{25}} = \frac{12}{5} \\
 \therefore \frac{2 \sin \theta - 3 \cos \theta}{4 \sin \theta - 9 \cos \theta} &= \frac{\frac{2 \sin \theta}{\cos \theta} - 3}{4 \frac{\sin \theta}{\cos \theta} - 9} \\
 & \quad \text{(dividing numerator and denominator by } \cos \theta) \\
 &= \frac{2 \tan \theta - 3}{4 \tan \theta - 9} = \frac{2 \times \frac{12}{5} - 3}{4 \times \frac{12}{5} - 9} = \frac{9/5}{3/5} = \frac{9}{3} = 3
 \end{aligned}$$

$$\begin{aligned}
 17. (c) & \sin(45^\circ + A) - \cos(45^\circ - A) \\
 &= \sin(45^\circ + A) - \cos[90^\circ - (45^\circ + A)] = \sin(45^\circ + A) - \sin(45^\circ + A) \\
 &= 0
 \end{aligned}$$

$$\begin{aligned}
 18. (b) & \sqrt{\frac{1 + \sin \theta}{1 - \sin \theta}} = \sqrt{\frac{(1 + \sin \theta)(1 + \sin \theta)}{(1 - \sin \theta)(1 + \sin \theta)}} \\
 &= \sqrt{\frac{(1 + \sin \theta)^2}{1 - \sin^2 \theta}} = \sqrt{\frac{(1 + \sin \theta)^2}{\cos^2 \theta}} = \frac{1 + \sin \theta}{\cos \theta} = \frac{1}{\cos \theta} + \frac{\sin \theta}{\cos \theta} \\
 &= \sec \theta + \tan \theta
 \end{aligned}$$

$$\begin{aligned}
 19. (b) & \sin(n+1)A \sin(n-1)A + \cos(n+1)A \cos(n-1)A \\
 &= \cos[(n+1)A - (n-1)A] \\
 & \quad [\because \cos A \cos B + \sin A \sin B = \cos(A - B)] \\
 &= \cos[nA + A - nA + A] = \cos 2A
 \end{aligned}$$

$$\begin{aligned}
 20. (a) & \frac{\cos 11^\circ + \sin 11^\circ}{\cos 11^\circ - \sin 11^\circ} \\
 & \text{Divide numerator and denominator by } \cos 11^\circ \\
 &= \frac{1 + \tan 11^\circ}{1 - \tan 11^\circ} = \frac{\tan 45^\circ + \tan 11^\circ}{1 - \tan 45^\circ \tan 11^\circ} = \tan(45^\circ + 11^\circ) = \tan 56^\circ
 \end{aligned}$$

$$\begin{aligned}
 21. (c) & \tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{\sin \theta}{\sqrt{1 - \sin^2 \theta}} = \frac{\frac{a^2 - b^2}{a^2 + b^2}}{\sqrt{1 - \left(\frac{a^2 - b^2}{a^2 + b^2}\right)^2}} \\
 &= \frac{\frac{a^2 - b^2}{a^2 + b^2}}{\frac{\sqrt{(a^2 + b^2)^2 - (a^2 - b^2)^2}}{a^2 + b^2}} = \frac{a^2 - b^2}{\sqrt{4a^2 b^2}} = \frac{a^2 - b^2}{2ab}
 \end{aligned}$$

$$\begin{aligned}
 22. (d) & \text{Given, } \cos \theta = 0.96, \sin \theta = \sqrt{1 - \cos^2 \theta} = \sqrt{1 - (0.96)^2} \\
 &= \sqrt{1 - 0.9216} = 0.28, \tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{0.28}{0.96} = \frac{7}{24} \\
 & \frac{1}{\sin \theta} + \frac{1}{\tan \theta} = \frac{100}{28} + \frac{24}{7} = \frac{196}{28} = 7
 \end{aligned}$$

$$23. (b) \text{ As we know, } \sin x \text{ is increasing from } 0 \text{ to } 90^\circ.$$

$$\therefore \sin y > \sin x$$

$$24. (a) \text{ Given, } (\sin \theta + \operatorname{cosec} \theta) = 2.5$$

$$\begin{aligned}
 \Rightarrow & \left(\sin \theta + \frac{1}{\sin \theta} \right) = \frac{5}{2} \Rightarrow 2 \sin^2 \theta - 5 \sin \theta + 2 = 0 \\
 \Rightarrow & 2 \sin^2 \theta - 4 \sin \theta - \sin \theta + 2 = 0 \\
 \Rightarrow & 2 \sin \theta (\sin \theta - 2) - 1(\sin \theta - 2) = 0 \\
 \Rightarrow & (2 \sin \theta - 1)(\sin \theta - 2) = 0 \Rightarrow \sin \theta = \frac{1}{2} \quad (\because \sin \theta \neq 2)
 \end{aligned}$$

$$\therefore \theta = 30^\circ$$

$$25. (a) \text{ Given, } \frac{\cos x}{\cos y} = n \text{ and } \frac{\sin x}{\sin y} = m \quad \dots(i)$$

$$\begin{aligned}
 \text{Now, } (m^2 - n^2) \sin^2 y &= \left(\frac{\sin^2 x}{\sin^2 y} - \frac{\cos^2 x}{\cos^2 y} \right) \sin^2 y \\
 &= \frac{(1 - \cos^2 x) \cos^2 y - \cos^2 x (1 - \cos^2 y)}{\cos^2 y} \\
 &= \frac{\cos^2 y - \cos^2 x}{\cos^2 y} = 1 - n^2 \quad [\text{from Eq. (i)}]
 \end{aligned}$$

$$26. (b) \text{ Given, } p = \tan^2 x + \cot^2 x = \tan^2 x + \frac{1}{\tan^2 x}$$

$$\therefore \text{AM} \geq \text{GM} \Rightarrow \tan^2 x + \frac{1}{\tan^2 x} \geq 2 \left(\tan^2 x \cdot \frac{1}{\tan^2 x} \right)^{1/2}$$

$$\Rightarrow \tan^2 x + \frac{1}{\tan^2 x} \geq 2 \Rightarrow p \geq 2$$

$$27. (a) \text{ Let angles in circular measure are } A \text{ and } B, \text{ then degree measures will be } A \left(\frac{180}{\pi} \right) \text{ and } B \left(\frac{180}{\pi} \right).$$

$$\text{By given condition, } A + B = 1 \quad \dots(i)$$

$$A \left(\frac{180}{\pi} \right) - B \left(\frac{180}{\pi} \right) = 1 \quad \dots(ii)$$

$$\text{On multiplying Eq. (i) by } \frac{180}{\pi} \text{ and adding to Eq. (ii), we get}$$

$$2A \left(\frac{180}{\pi} \right) = \frac{180}{\pi} + 1 \Rightarrow A = \frac{1}{2} \times \frac{11}{180} \left[\frac{180}{\pi} + 1 \right] = \frac{1}{2} + \frac{\pi}{360}$$

$$\therefore B = 1 - A = 1 - \left(\frac{1}{2} + \frac{\pi}{360} \right) = \frac{1}{2} - \frac{\pi}{360}$$

$$\begin{aligned}
 28. (b) & [(1 - \sin^2 \theta) \sec^2 \theta + \tan^2 \theta] (\cos^2 \theta + 1) \\
 &= [\cos^2 \theta \cdot \sec^2 \theta + \tan^2 \theta] (\cos^2 \theta + 1) \quad (\because \sin^2 \theta + \cos^2 \theta = 1) \\
 &= (1 + \tan^2 \theta) (\cos^2 \theta + 1) = \sec^2 \theta (\cos^2 \theta + 1) \\
 &= \sec^2 \theta \cdot \cos^2 \theta + \sec^2 \theta = 1 + \sec^2 \theta > 2 \quad (\because \sec^2 \theta - \tan^2 \theta = 1) \\
 & \quad (\because \sec^2 \theta > 1 \text{ for } 0 < \theta < 90^\circ)
 \end{aligned}$$

$$\begin{aligned}
 29. (d) & \text{Given, } 7 \cos^2 \theta + 3 \sin^2 \theta = 4 \Rightarrow 7 \cos^2 \theta + 3(1 - \cos^2 \theta) = 4 \\
 \Rightarrow & 3 - 4 \cos^2 \theta = 4 \Rightarrow \cos^2 \theta = 1/4
 \end{aligned}$$

$$\text{Now, } \tan^2 \theta = \sec^2 \theta - 1 = \frac{1}{\cos^2 \theta} - 1 \Rightarrow \tan^2 \theta = 4 - 1 = 3$$

$$\therefore \tan \theta = \sqrt{3} \quad (\because 0 < \theta < \pi/2)$$

$$30. (d) \text{ We know that, the value of } \cos \theta \text{ is decreasing in the interval } 0 \leq \theta \leq 90^\circ$$

$$\therefore \cos 1^\circ > \cos 89^\circ \Rightarrow p > q$$

Also, $\cos 1^\circ$ is close to 1 and $\cos 89^\circ$ is close to 0.

Hence, option (d) is correct.

31. (d) We know in the interval $\theta \in (0, \pi/2)$, $\sec^2 \theta$ is increasing from 1 to ∞ .

$$\therefore p \geq 1$$

32. (b) $\because \cos 90^\circ = 0$

$$\therefore \cos 1^\circ \cos 2^\circ \cos 3^\circ \dots \cos 90^\circ = 0 \quad (\because \cos 90^\circ = 0)$$

33. (c) Given, $\tan A = \frac{1 - \cos B}{\sin B}$

$$\begin{aligned} \therefore \frac{2 \tan A}{1 - \tan^2 A} &= \frac{2 \cdot \frac{(1 - \cos B)}{\sin B}}{1 - \left(\frac{(1 - \cos B)}{\sin B} \right)^2} \\ &= \frac{2(1 - \cos B) \sin B}{\sin^2 B - 1 - \cos^2 B + 2 \cos B} = \frac{2(1 - \cos B) \sin B}{-2 \cos^2 B + 2 \cos B} \\ &= \frac{2 \sin B (1 - \cos B)}{2 \cos B (1 - \cos B)} = \frac{\sin B}{\cos B} = \tan B \end{aligned}$$

34. (c) $\cot 15^\circ \cot 20^\circ \cot 70^\circ \cot 75^\circ$

$$\begin{aligned} &= \tan(90^\circ - 15^\circ) \tan(90^\circ - 20^\circ) \cot 70^\circ \cot 75^\circ \\ &= \tan 75^\circ \tan 70^\circ \frac{1}{\tan 70^\circ} \cdot \frac{1}{\tan 75^\circ} = 1 \end{aligned}$$

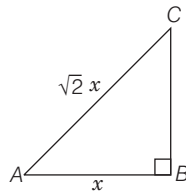
35. (b) Given, $\frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta} = 2$

$$\therefore \sin^2 \theta + \cos^2 \theta = 2 \sin \theta \cos \theta$$

$$\Rightarrow \sin 2\theta = 1 = \sin 90^\circ \Rightarrow 2\theta = 90^\circ \Rightarrow \theta = 45^\circ$$

36. (d) Given, in right angled $\triangle ABC$, $\angle B = 90^\circ$ and $AB : AC = 1 : \sqrt{2}$

$$\text{Let } AB = x, AC = \sqrt{2}x$$



$$\Rightarrow AB^2 + BC^2 = AC^2 \quad (\text{using pythagoras theorem})$$

$$\Rightarrow x^2 + BC^2 = (\sqrt{2}x)^2$$

$$\Rightarrow BC^2 = 2x^2 - x^2 \Rightarrow BC = x$$

$$\therefore \tan A = \frac{BC}{AB} = \frac{x}{x} = 1$$

$$\therefore \frac{2 \tan A}{1 - \tan^2 A} = \frac{2(1)}{1 - 1^2} = \frac{2}{0} = \infty \quad (\text{undefined})$$

37. (a) Given, $AD = DB = \frac{a}{2}$ and $\angle B = 90^\circ$

$$\text{In right angled } \triangle ABC, BC^2 + AB^2 = AC^2$$

$$\Rightarrow BC^2 + a^2 = b^2 \Rightarrow BC = \sqrt{b^2 - a^2}$$

$$\text{In right angled } \triangle BCD, CD^2 = BC^2 + BD^2$$

$$CD^2 = (\sqrt{b^2 - a^2})^2 + \left(\frac{a}{2}\right)^2 = \left(b^2 - a^2 + \frac{a^2}{4}\right)$$

$$\Rightarrow CD^2 = \frac{4b^2 - 3a^2}{4} \Rightarrow CD = \frac{\sqrt{4b^2 - 3a^2}}{2}$$

$$\therefore \cos \theta = \frac{BC}{CD} = \frac{\sqrt{b^2 - a^2}}{\frac{\sqrt{4b^2 - 3a^2}}{2}} = \frac{2\sqrt{b^2 - a^2}}{\sqrt{4b^2 - 3a^2}}$$

38. (d) Given, $\sin x + \sqrt{3} \cos x = 1$

$$\text{On dividing both sides by 2, we get } \frac{1}{2} \sin x + \frac{\sqrt{3}}{2} \cos x = \frac{1}{2}$$

$$\Rightarrow \sin x \sin 30^\circ + \cos x \cos 30^\circ = 1/2$$

$$\Rightarrow \cos(x - 30^\circ) = \cos 60^\circ \Rightarrow x - 30^\circ = 60^\circ$$

$$\therefore x = 90^\circ$$

39. (c) $\operatorname{cosec} A \cdot \sec A = \frac{\sqrt{1 + \tan^2 A}}{\tan A} \cdot \sqrt{1 + \tan^2 A}$

$$= \frac{1 + \tan^2 A}{\tan A} = \frac{1 + (\sqrt{2} - 1)^2}{\sqrt{2} - 1} \quad (\because \tan A = \sqrt{2} - 1)$$

$$= \frac{1 + 2 + 1 - 2\sqrt{2}}{\sqrt{2} - 1} = \frac{4 - 2\sqrt{2}}{(\sqrt{2} - 1)} = \frac{2\sqrt{2}(\sqrt{2} - 1)}{(\sqrt{2} - 1)} = 2\sqrt{2}$$

40. (c) Given, $\tan \theta = \frac{3}{1} \Rightarrow \sin \theta = 3 \cos \theta \Rightarrow \sin \theta = 3\sqrt{1 - \sin^2 \theta}$

$$\text{Squaring both sides, we get } \sin^2 \theta = 9(1 - \sin^2 \theta)$$

$$\Rightarrow 10 \sin^2 \theta = 9 \Rightarrow \sin \theta = \sqrt{\frac{9}{10}} = -\frac{3}{\sqrt{10}}$$

($\because \theta$ lies in IIIrd quadrant, so $\sin \theta$ will be negative)

41. (a) $\sin 20^\circ \cos 70^\circ + \cos 20^\circ \sin 70^\circ$

$$= \sin(20^\circ + 70^\circ) = \sin 90^\circ = 1$$

$$[\because \sin A \cos B + \cos A \sin B = \sin(A + B)]$$

42. (a) $\phi + \theta = 180^\circ \Rightarrow \theta = 180^\circ - \phi \Rightarrow \sin \theta = \sin(180^\circ - \phi)$

$$\sin \theta = \sin \phi$$

43. (b) $\sqrt{\cos x \operatorname{cosec} y - \cos x \sin y}$

$$= \sqrt{\cos x \cdot \operatorname{cosec}(90^\circ - x) - \cos x \cdot \sin(90^\circ - x)}$$

($\because x + y = 90^\circ$, given)

$$= \sqrt{\cos x \cdot \sec x - \cos^2 x}$$

$$= \sqrt{1 - \cos^2 x} = \sqrt{\sin^2 x} = \sin x$$

44. (c) $\sqrt{2 + \sqrt{2 + 2 \cos 4\theta}} = \sqrt{2 + \sqrt{2(1 + \cos 4\theta)}}$

$$= \sqrt{2 + \sqrt{2(2 \cos^2 2\theta)}} = \sqrt{2 + \sqrt{4 \cos^2 2\theta}} = \sqrt{2 + 2 \cos 2\theta}$$

$$= \sqrt{2(1 + \cos 2\theta)} = \sqrt{2(2 \cos^2 \theta)} = \sqrt{4 \cos^2 \theta} = 2 \cos \theta$$

45. (b) Given, $\tan A = \frac{5}{6}$ and $\tan B = \frac{1}{11}$

$$\therefore \tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B} = \frac{\frac{5}{6} + \frac{1}{11}}{1 - \frac{5}{6} \cdot \frac{1}{11}} = \frac{\frac{55 + 6}{66}}{\frac{66 - 5}{66}} = \frac{61}{61} = 1$$

$$\Rightarrow \tan(A + B) = 1 \Rightarrow \tan(A + B) = \tan \frac{\pi}{4} \Rightarrow A + B = \frac{\pi}{4}$$

46. (c) $\frac{1}{\sqrt{2}} (\cos A - \sin A) = \frac{\cos A}{\sqrt{2}} - \frac{\sin A}{\sqrt{2}}$

$$= \cos \frac{\pi}{4} \cos A - \sin \frac{\pi}{4} \sin A = \cos \left(\frac{\pi}{4} + A \right)$$

$$[\because \cos A \cos B - \sin A \sin B = \cos(A + B)]$$

47. (d) $\sin \frac{\pi}{4} \cdot \cos \frac{\pi}{12} - \cos \frac{\pi}{4} \cdot \sin \frac{\pi}{12} = \sin \left(\frac{\pi}{4} - \frac{\pi}{12} \right)$

$$[\because \sin A \cos B - \cos A \sin B = \sin(A - B)]$$

$$= \sin \left(\frac{\pi}{6} \right) = \frac{1}{2}$$

48. (c) Given, $\tan \theta = \frac{x \sin \phi}{1 - x \cos \phi} \Rightarrow x \sin \phi = \tan \theta - x \cos \phi \tan \theta$

$$\Rightarrow x = \frac{\tan \theta}{\sin \phi + \cos \phi \tan \theta} = \frac{\sin \theta}{\cos \theta \sin \phi + \cos \phi \sin \theta} = \frac{\sin \theta}{\sin(\theta + \phi)}$$

Similarly, $y = \frac{\sin \phi}{\sin(\theta + \phi)}$, Hence, $\frac{x}{y} = \frac{\sin \theta}{\sin \phi}$

49. (b) Given, $\sin(\theta + \phi) = 2 \sin(\theta - \phi)$

$$\Rightarrow \sin \theta \cos \phi + \cos \theta \sin \phi = 2(\sin \theta \cos \phi - \cos \theta \sin \phi)$$

$$\Rightarrow 3 \cos \theta \sin \phi = \sin \theta \cos \phi \Rightarrow \frac{3 \sin \phi}{\cos \phi} = \frac{\sin \theta}{\cos \theta}$$

$$\Rightarrow 3 \tan \phi = \tan \theta \text{ or } \tan \theta = 3 \tan \phi$$

50. (a) Given, $\sin(\theta + \alpha) = \cos(\theta + \alpha) \Rightarrow \frac{\sin(\theta + \alpha)}{\cos(\theta + \alpha)} = 1$

$$\Rightarrow \tan(\theta + \alpha) = 1 \Rightarrow \frac{\tan \theta + \tan \alpha}{1 - \tan \theta \tan \alpha} = 1$$

$$\Rightarrow \tan \theta + \tan \alpha = 1 - \tan \theta \tan \alpha$$

$$\Rightarrow \tan \theta(1 + \tan \alpha) = 1 - \tan \alpha \Rightarrow \tan \theta = \frac{1 - \tan \alpha}{1 + \tan \alpha}$$

51. (c) $\cos^2 \frac{\theta - \phi}{2} - \sin^2 \frac{\theta + \phi}{2}$

$$= \cos \left(\frac{\theta - \phi}{2} + \frac{\theta + \phi}{2} \right) \cdot \cos \left(\frac{\theta - \phi}{2} - \frac{\theta + \phi}{2} \right)$$

$$[\because \cos^2 A - \sin^2 B = \cos(A + B) \cos(A - B)]$$

$$= \cos \theta \cdot \cos(-\phi) = \cos \theta \cdot \cos \phi$$

52. (c) $x^2 + y^2 + z^2 = r^2 \sin^2 \theta \cos^2 \phi + r^2 \sin^2 \theta \sin^2 \phi + r^2 \cos^2 \theta$

$$= r^2 \sin^2 \theta (\cos^2 \phi + \sin^2 \phi) + r^2 \cos^2 \theta$$

$$= r^2 \sin^2 \theta + r^2 \cos^2 \theta = r^2 (\sin^2 \theta + \cos^2 \theta) = r^2$$

53. (c) Given, $\cot \theta = \frac{7}{24}$, $\operatorname{cosec}^2 \theta = 1 + \cot^2 \theta = 1 + \frac{49}{576} = \frac{625}{576}$

$$\therefore \operatorname{cosec} \theta = \pm \frac{25}{24}$$

$$\Rightarrow \sin \theta = \pm \frac{24}{25}, \cos^2 \theta = 1 - \sin^2 \theta = 1 - \frac{576}{625} = \frac{49}{625}$$

$$\therefore \cos \theta = \pm \frac{7}{25}, \text{ As } \pi < \theta < \frac{3\pi}{2}$$

$$\therefore \sin \theta \text{ and } \cos \theta \text{ both are negative.}$$

$$\therefore \sin \theta = \frac{-24}{25}, \cos \theta = \frac{-7}{25}$$

$$\therefore \cos \theta - \sin \theta = \frac{-7}{25} + \frac{24}{25} = \frac{17}{25}$$

54. (b) Given, $\sin \alpha = \frac{1}{2} = \sin 30^\circ \Rightarrow \alpha = 30^\circ$

Also, $\beta = \frac{\pi}{2} - \alpha = 90^\circ - 30^\circ = 60^\circ$

55. (c) Given, $\sin x - \cos x = 0 \Rightarrow \sin x = \cos x \Rightarrow \tan x = 1$

$$\Rightarrow \tan x = \tan \frac{\pi}{4} \Rightarrow x = \frac{\pi}{4}$$

$$\therefore \sin^4 x + \cos^4 x = \sin^4 \frac{\pi}{4} + \cos^4 \frac{\pi}{4} = \left(\frac{1}{\sqrt{2}} \right)^4 + \left(\frac{1}{\sqrt{2}} \right)^4$$

$$= \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

56. (b) Given, $3 \cos x = 5 \sin x \Rightarrow \frac{\sin x}{\cos x} = \frac{3}{5} \Rightarrow \tan x = \frac{3}{5}$

$$\frac{5 \sin x - 2 \sec^3 x + 2 \cos x}{5 \sin x + 2 \sec^3 x - 2 \cos x} = \frac{5 \tan x - 2 \sec^4 x + 2}{5 \tan x + 2 \sec^4 x - 2}$$

(dividing numerator and denominator by $\cos x$)

$$= \frac{5 \tan x - 2(1 + \tan^2 x)^2 + 2}{5 \tan x + 2(1 + \tan^2 x)^2 - 2}$$

$$= \frac{5 \times \frac{3}{5} - 2 \left(1 + \frac{9}{25} \right)^2 + 2}{5 \times \frac{3}{5} + 2 \left(1 + \frac{9}{25} \right)^2 - 2} = \frac{3 + 2 - 2 \times \frac{1156}{625}}{3 - 2 + 2 \times \frac{1156}{625}}$$

$$= \frac{3125 - 2312}{625 + 2312} = \frac{813}{2937} = \frac{271}{979}$$

57. (b) $\sqrt{\frac{1 + \cos x}{1 - \cos x}} = \sqrt{\frac{(1 + \cos x)}{(1 - \cos x)} \times \frac{1 + \cos x}{1 + \cos x}}$

$$= \frac{1 + \cos x}{\sin x} = \frac{1}{\sin x} + \frac{\cos x}{\sin x} = \operatorname{cosec} x + \cot x$$

58. (a) $\cos^4 x - \sin^4 x = (\cos^2 x - \sin^2 x)(\sin^2 x + \cos^2 x)$

$$= [(\cos^2 x) - (1 - \cos^2 x)] = 2 \cos^2 x - 1 \quad (\because \sin^2 x + \cos^2 x = 1)$$

59. (a) $\frac{\sin \theta}{1 + \cos \theta} = \frac{\sin \theta}{1 + \cos \theta} \times \frac{1 - \cos \theta}{1 - \cos \theta} = \frac{\sin \theta(1 - \cos \theta)}{\sin^2 \theta} = \frac{1 - \cos \theta}{\sin \theta}$

60. (a) $\frac{5 \cos \theta - 4}{3 - 5 \sin \theta} - \frac{3 + 5 \sin \theta}{4 + 5 \cos \theta}$

$$= \frac{(5 \cos \theta - 4)(4 + 5 \cos \theta) - (3 + 5 \sin \theta)(3 - 5 \sin \theta)}{(3 - 5 \sin \theta)(4 + 5 \cos \theta)}$$

$$= \frac{(25 \cos^2 \theta - 16) - (9 - 25 \sin^2 \theta)}{(3 - 5 \sin \theta)(4 + 5 \cos \theta)}$$

$$= \frac{25(\sin^2 \theta + \cos^2 \theta) - 25}{(3 - 5 \sin \theta)(4 + 5 \cos \theta)} = \frac{25 - 25}{(3 - 5 \sin \theta)(4 + 5 \cos \theta)} = 0$$

61. (a) $\sin(\alpha + \beta) = \sqrt{1 - \cos^2(\alpha + \beta)} = \sqrt{1 - \frac{16}{25}} = \sqrt{\frac{9}{25}} = \frac{3}{5}$

$$\cos(\alpha - \beta) = \sqrt{1 - \sin^2(\alpha - \beta)} = \sqrt{1 - \frac{25}{169}} = \sqrt{\frac{144}{169}} = \frac{12}{13}$$

$$\therefore \tan(\alpha + \beta) = \frac{\sin(\alpha + \beta)}{\cos(\alpha + \beta)} = \left(\frac{3}{5} \times \frac{5}{4} \right) = \frac{3}{4}$$

$$\tan(\alpha - \beta) = \frac{\sin(\alpha - \beta)}{\cos(\alpha - \beta)} = \frac{5}{13} \times \frac{13}{12} = \frac{5}{12}$$

$$\therefore \tan(2\alpha) = \tan[(\alpha + \beta) + (\alpha - \beta)]$$

$$= \frac{\tan(\alpha + \beta) + \tan(\alpha - \beta)}{1 - \tan(\alpha + \beta) \cdot \tan(\alpha - \beta)} = \frac{\frac{3}{4} + \frac{5}{12}}{1 - \frac{3}{4} \times \frac{5}{12}} = \frac{56}{33}$$

62. (a) $\tan(45^\circ + x) = \frac{\tan 45^\circ + \tan x}{1 - \tan 45^\circ \tan x} = \frac{1 + \tan x}{1 - \tan x}$

$$\tan(45^\circ - x) = \frac{\tan 45^\circ - \tan x}{1 + \tan 45^\circ \tan x} = \frac{1 - \tan x}{1 + \tan x}$$

$$\therefore \frac{\tan(45^\circ + x)}{\tan(45^\circ - x)} = \frac{\frac{1 + \tan x}{1 - \tan x}}{\frac{1 - \tan x}{1 + \tan x}} = \left[\frac{1 + \tan x}{1 - \tan x} \right]^2$$

63. (a)
$$\frac{(1 - \sin A \cos A)(\sin^2 A - \cos^2 A)}{\cos A (\sec A - \operatorname{cosec} A)(\sin^3 A + \cos^3 A)}$$

$$= \frac{(1 - \sin A \cos A)(\sin^2 A - \cos^2 A)}{\left[\cos A \left(\frac{1}{\cos A} - \frac{1}{\sin A} \right) (\sin A + \cos A) \right] \frac{(\sin^2 A + \cos^2 A - \sin A \cos A)}{[(1 - \sin A \cos A)(\sin A \cdot \cos A) \times (\sin A + \cos A)(\sin A - \cos A)]}}$$

$$= \frac{(1 - \sin A \cos A)(\sin^2 A - \cos^2 A)}{[\cos A (\sin A - \cos A)(\sin A + \cos A)]} = \sin A$$
64. (c)
$$(1 + \tan A)(1 + \tan B)$$

$$= (1 + \tan A)[1 + \tan(45^\circ - A)]$$

$$= (1 + \tan A) \left[1 + \frac{\tan 45^\circ - \tan A}{1 + \tan 45^\circ \tan A} \right] \quad (\because A + B = 45^\circ)$$

$$= (1 + \tan A) \left[1 + \frac{1 - \tan A}{1 + \tan A} \right]$$

$$= (1 + \tan A) \left[\frac{1 + \tan A + 1 - \tan A}{1 + \tan A} \right]$$

$$= (1 + \tan A) \left[\frac{2}{1 + \tan A} \right] = 2$$
65. (a) $\because \cos A - \cos B = 2 \sin \frac{(A+B)}{2} \cdot \sin \frac{(B-A)}{2}$
 and $\sin A + \sin B = 2 \sin \frac{(A+B)}{2} \cdot \cos \frac{(A-B)}{2}$
 $\therefore \frac{\cos 2B - \cos 2A}{\sin 2B + \sin 2A} = \frac{2 \sin(A+B) \sin(A-B)}{2 \sin(A+B) \cos(A-B)} = \tan(A-B)$
66. (d) $\sec^2 x + \tan^2 x = 7 \quad (\because \sec^2 x = 1 + \tan^2 x)$
 $1 + \tan^2 x + \tan^2 x = 7 \Rightarrow 1 + 2 \tan^2 x = 7$
 $\Rightarrow 2 \tan^2 x = 6 \Rightarrow \tan x = \sqrt{3} = \tan 60^\circ$
 $\therefore x = 60^\circ$
67. (a) $\cos 3\theta + \sin 3\theta = \sqrt{2} \left(\frac{1}{\sqrt{2}} \cos 3\theta + \frac{1}{\sqrt{2}} \sin 3\theta \right)$
 $= \sqrt{2} (\sin 45^\circ \cos 3\theta + \cos 45^\circ \sin 3\theta) = \sqrt{2} [\sin(45^\circ + 3\theta)]$
 $\cos 3\theta + \sin 3\theta$ is maximum when $\sin(45^\circ + 3\theta)$ is maximum
 The maximum value of $\sin(45^\circ + 3\theta) = 1 = \sin 90^\circ$
 $\Rightarrow 45^\circ + 3\theta = 90^\circ \Rightarrow \theta = 15^\circ$
68. (c) $\sin^2 x + \sin^2 y + \sin^2 z = (\sin x + \sin y + \sin z)^2$
 $\Rightarrow \sin x \sin y + \sin y \sin z + \sin z \sin x = 0$
 $[\because (a+b+c)^2 = a^2 + b^2 + c^2 + 2(ab+bc+ca)]$
 On dividing both sides by $\sin x \sin y \sin z$
 $\Rightarrow \frac{1}{\sin x} + \frac{1}{\sin y} + \frac{1}{\sin z} = 0$
69. (d) Given, $\sin x + \sin^2 x = 1 \quad \dots(i)$
 $\Rightarrow \sin x = 1 - \sin^2 x = \cos^2 x \quad (\because \sin^2 x + \cos^2 x = 1)$
 $\therefore \cos^2 x + \cos^4 x = (\sin x) + \sin^2 x = 1 \quad [\text{from Eq. (i)}]$
70. (a) $(\cos 35^\circ + \cos 85^\circ) + \cos 155^\circ$
 $= 2 \cos 60^\circ \cos 25^\circ + \cos(180^\circ - 25^\circ)$
 $= 2 \times \frac{1}{2} \cos 25^\circ - \cos 25^\circ = \cos 25^\circ - \cos 25^\circ = 0$

71. (a) Refer to example 16.

72. (d) Given, $\tan \theta = \frac{p}{q}$,

$$p \cos 2\theta + q \sin 2\theta = p \left(\frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} \right) + q \left(\frac{2 \tan \theta}{1 + \tan^2 \theta} \right)$$

$$= p \left(\frac{1 - \frac{p^2}{q^2}}{1 + \frac{p^2}{q^2}} \right) + q \left(\frac{\frac{2p}{q}}{1 + \frac{p^2}{q^2}} \right) = p \left(\frac{q^2 - p^2}{q^2 + p^2} \right) + \left(\frac{2pq^2}{q^2 + p^2} \right)$$

$$= \frac{p(q^2 - p^2 + 2q^2)}{p^2 + q^2} = \frac{p(3q^2 - p^2)}{p^2 + q^2}$$

73. (c) Given, $x = y \cos \frac{2\pi}{3} = z \cos \frac{4\pi}{3} \Rightarrow x = \frac{-y}{2} = \frac{-z}{2} = k \quad (\text{let})$

$$x = k, y = -2k, z = -2k$$

$$\Rightarrow xy + yz + zx = k(-2k) + (-2k)(-2k) + (-2k)k$$

$$= -2k^2 + 4k^2 - 2k^2 = 0$$

74. (c) Given, $\sin \theta + \cos \theta = m \quad \dots(i)$

$$\text{and } \sec \theta + \operatorname{cosec} \theta = n \Rightarrow \frac{1}{\cos \theta} + \frac{1}{\sin \theta} = n$$

$$\Rightarrow \frac{\sin \theta + \cos \theta}{\sin \theta \cos \theta} = n$$

$$\Rightarrow \frac{m}{\sin \theta \cos \theta} = n \Rightarrow \sin \theta \cos \theta = \frac{m}{n} \quad [\text{from Eq. (i)}]$$

On squaring Eq. (i), we get $\sin^2 \theta + \cos^2 \theta + 2 \sin \theta \cos \theta = m^2$

$$1 + 2 \frac{m}{n} = m^2 \Rightarrow \frac{2m}{n} = m^2 - 1 \quad [\because \sin^2 \theta + \cos^2 \theta = 1]$$

$$\Rightarrow 2m = (m^2 - 1)n$$

75. (b) Given, $\operatorname{cosec} \theta - \sin \theta = a^3 \Rightarrow \frac{1}{\sin \theta} - \sin \theta = a^3$

$$\Rightarrow \frac{1 - \sin^2 \theta}{\sin \theta} = a^3 \Rightarrow \frac{\cos^2 \theta}{\sin \theta} = a^3 \Rightarrow \cos^2 \theta = a^3 \sin \theta \quad \dots(i)$$

$$\text{and } \sec \theta - \cos \theta = b^3 \Rightarrow \frac{1}{\cos \theta} - \cos \theta = b^3 \Rightarrow \frac{1 - \cos^2 \theta}{\cos \theta} = b^3$$

$$\Rightarrow \frac{\sin^2 \theta}{\cos \theta} = b^3 \Rightarrow \sin^2 \theta = b^3 \cos \theta \quad \dots(ii)$$

On putting $\cos \theta = \frac{\sin^2 \theta}{b^3}$ in Eq. (i), we get

$$\frac{\sin^4 \theta}{b^6} = a^3 \sin \theta \Rightarrow \sin^3 \theta = a^3 b^6$$

$$\therefore \sin \theta = ab^2 \quad \dots(iii)$$

$$\text{Similarly, } \cos \theta = a^2 b \quad \dots(iv)$$

On squaring and adding Eqs. (iii) and (iv), we get

$$1 = a^2 b^4 + a^4 b^2 \Rightarrow 1 = a^2 b^2 (a^2 + b^2)$$

76. (d) As, we know that $\cos \theta \geq -1$ and $\cos \theta \leq 1$

$$\therefore \cos 2x \geq -1 \Rightarrow 1 + \cos 2x \geq 0$$

$$\cos 2x \leq 1 \Rightarrow 1 + \cos 2x \leq 2$$

So, minimum value of $1 + \cos 2x$ is 0 and maximum value of $1 + \cos 2x$ is 2.

$$\begin{aligned}
 77. (b) \cot(\theta + \phi) &= \frac{1}{\tan(\theta + \phi)} = \frac{1 - \tan\theta \tan\phi}{\tan\theta + \tan\phi} \\
 &= \frac{1}{\tan\theta + \tan\phi} - \frac{\tan\theta \tan\phi}{\tan\theta + \tan\phi} \\
 &= \frac{1}{\tan\theta + \tan\phi} - \frac{1}{\left(\frac{1}{\tan\theta} + \frac{1}{\tan\phi}\right)} \\
 &= \frac{1}{\tan\theta + \tan\phi} - \frac{1}{\cot\theta + \cot\phi} = \frac{1}{a} - \frac{1}{b} \\
 &\quad [\because \tan\theta + \tan\phi = a, \text{ or } \cot\theta + \cot\phi = b]
 \end{aligned}$$

$$\begin{aligned}
 78. (b) \because \tan 75^\circ &= \tan(45^\circ + 30^\circ) \\
 \Rightarrow \tan 75^\circ &= \frac{\tan 45^\circ + \tan 30^\circ}{1 - \tan 45^\circ \tan 30^\circ} \Rightarrow \tan 75^\circ = \frac{1 + \tan 30^\circ}{1 - \tan 30^\circ} \\
 \Rightarrow \tan 75^\circ - \tan 75^\circ \cdot \tan 30^\circ &= 1 + \tan 30^\circ \\
 \Rightarrow \tan 75^\circ - \tan 30^\circ - \tan 75^\circ \cdot \tan 30^\circ &= 1
 \end{aligned}$$

$$\begin{aligned}
 79. (b) \sin^2(15^\circ + A) - \sin^2(15^\circ - A) &= \sin(15^\circ + A + 15^\circ - A) \\
 &\quad \cdot \sin(15^\circ + A - 15^\circ + A) \\
 &\quad [\because \sin^2 A - \sin^2 B = \sin(A + B) \cdot \sin(A - B)] \\
 &= \sin 30^\circ \sin(2A) = \frac{1}{2} \sin 2A
 \end{aligned}$$

$$\begin{aligned}
 80. (a) \frac{\sin 38^\circ - \cos 68^\circ}{\cos 68^\circ + \sin 38^\circ} &= \frac{\cos(90^\circ - 38^\circ) - \cos 68^\circ}{\cos 68^\circ + \sin(90^\circ - 52^\circ)} \\
 &= \frac{\cos 52^\circ - \cos 68^\circ}{\cos 68^\circ + \cos 52^\circ} = \frac{2 \sin\left(\frac{52^\circ + 68^\circ}{2}\right) \sin\left(\frac{68^\circ - 52^\circ}{2}\right)}{2 \cos\left(\frac{52^\circ + 68^\circ}{2}\right) \cos\left(\frac{68^\circ - 52^\circ}{2}\right)} \\
 &= \frac{\sin 60^\circ \cdot \sin 8^\circ}{\cos 60^\circ \cdot \cos 8^\circ} = \tan 60^\circ \cdot \tan 8^\circ = \sqrt{3} \tan 8^\circ
 \end{aligned}$$

$$\begin{aligned}
 81. (c) 2 \cos x - (\cos 3x + \cos 5x) \\
 = 2 \cos x - 2 \cos 4x \cdot \cos x &= 2 \cos x(1 - \cos 4x) \\
 = 2 \cos x \cdot 2 \sin^2 2x &= 4 \cos x(2 \sin x \cos x)^2 = 16 \cos^3 x \sin^2 x
 \end{aligned}$$

$$\begin{aligned}
 82. (c) \cos 4x &= 1 - 2 \sin^2 2x = 1 - 2(2 \sin x \cos x)^2 \\
 &= 1 - 2(4 \sin^2 x \cos^2 x) = 1 - 8 \sin^2 x \cos^2 x
 \end{aligned}$$

$$\begin{aligned}
 83. (a) 2 \sin A \cos^3 A - 2 \sin^3 A \cos A \\
 = 2 \sin A \cos A (\cos^2 A - \sin^2 A) \\
 = \sin 2A (\cos 2A) = \frac{2 \sin 2A \cos 2A}{2} = \frac{1}{2} \sin 4A
 \end{aligned}$$

$$\begin{aligned}
 84. (c) \text{ Given, } a \cos \theta - b \sin \theta &= c \\
 \text{On squaring both sides, we get} \\
 a^2 \cos^2 \theta + b^2 \sin^2 \theta - 2ab \cos \theta \sin \theta &= c^2 \\
 \Rightarrow a^2 (1 - \sin^2 \theta) + b^2 (1 - \cos^2 \theta) - 2ab \sin \theta \cos \theta &= c^2 \\
 \Rightarrow a^2 + b^2 - c^2 = a^2 \sin^2 \theta + b^2 \cos^2 \theta + 2ab \sin \theta \cos \theta \\
 \Rightarrow (a \sin \theta + b \cos \theta)^2 &= a^2 + b^2 - c^2 \\
 \Rightarrow a \sin \theta + b \cos \theta &= \pm \sqrt{a^2 + b^2 - c^2}
 \end{aligned}$$

$$\begin{aligned}
 85. (d) \text{ Given, } \cos x + \cos^2 x &= 1 \Rightarrow \cos x = 1 - \cos^2 x = \sin^2 x \quad \dots(i) \\
 \sin^8 x + 2 \sin^6 x + \sin^4 x &= (\cos x)^4 + 2 \cos^3 x + \cos^2 x \\
 &= (\cos^2 x + \cos x)^2 \quad [\text{from Eq. (i)}] \\
 &= (1)^2 = 1
 \end{aligned}$$

$$\begin{aligned}
 86. (b) \frac{n}{m} &= \frac{\sec x - \cos x}{\operatorname{cosec} x - \sin x} = \frac{\frac{1 - \cos^2 x}{\cos x}}{\frac{1 - \sin^2 x}{\sin x}} \\
 &= \frac{\sin^2 x}{\cos x} \cdot \frac{\sin x}{\cos^2 x} = \left(\frac{\sin x}{\cos x}\right)^3 \\
 \Rightarrow (\tan x)^3 &= \left(\frac{n}{m}\right) \Rightarrow \tan x = \left(\frac{n}{m}\right)^{1/3}
 \end{aligned}$$

$$87. (a) 2 \cos \theta = x + \frac{1}{x} \quad \dots(i)$$

$$\text{On cubing both sides, we get } 8 \cos^3 \theta = \left(x + \frac{1}{x}\right)^3$$

$$8 \cos^3 \theta = x^3 + \frac{1}{x^3} + 3x \cdot \frac{1}{x} \left(x + \frac{1}{x}\right)$$

$$8 \cos^3 \theta = x^3 + \frac{1}{x^3} + 3(2 \cos \theta) \quad [\text{from Eq. (i)}]$$

$$\Rightarrow 8 \cos^3 \theta - 6 \cos \theta = x^3 + \frac{1}{x^3}$$

$$\Rightarrow 2[4 \cos^3 \theta - 3 \cos \theta] = x^3 + \frac{1}{x^3} \Rightarrow 2[\cos 3\theta] = x^3 + \frac{1}{x^3}$$

$$88. (a) \text{ We know that, the value of } \cos \theta \text{ decreases in the interval } 0 \leq \theta \leq \pi/2$$

$$\because \cos \theta \geq \frac{1}{2} \Rightarrow \cos \theta \geq \cos \frac{\pi}{3} \Rightarrow \theta \leq \frac{\pi}{3}$$

$$89. (b) \text{ Given, } \sin A = \frac{3}{5} \text{ and } \cos B = \frac{12}{13}$$

$$\therefore \tan A = \frac{\sin A}{\cos A} = \frac{\sin A}{\sqrt{1 - \sin^2 A}} = \frac{3/5}{\sqrt{1 - (3/5)^2}} = \frac{3/5}{4/5} = \frac{3}{4}$$

$$\text{and } \tan B = \frac{\sin B}{\cos B} = \frac{\sqrt{1 - \cos^2 B}}{\cos B} = \frac{\sqrt{1 - \frac{144}{169}}}{12/13} = \frac{5/13}{12/13} = \frac{5}{12}$$

$$\begin{aligned}
 \frac{\tan A - \tan B}{1 + \tan A \tan B} &= \frac{\frac{3}{4} - \frac{5}{12}}{1 + \frac{3}{4} \cdot \frac{5}{12}} = \frac{\frac{9 - 5}{12}}{\frac{48 + 15}{48}} \\
 &= \frac{\frac{4}{12}}{\frac{63}{48}} = \frac{1}{3} \times \frac{48}{63} = \frac{16}{63}
 \end{aligned}$$

$$90. (a) \text{ Given, } B + C = 60^\circ \Rightarrow B = 60^\circ - C$$

$$\begin{aligned}
 \sin(120^\circ - B) &= \sin[120^\circ - (60^\circ - C)] = \sin(60^\circ + C) \\
 &= \sin[180^\circ - (60^\circ + C)] = \sin(120^\circ - C)
 \end{aligned}$$

$$91. (a) \text{ Given, } \frac{x \cdot \operatorname{cosec}^2 30^\circ \sec^2 45^\circ}{8 \cos^2 45^\circ \sin^2 60^\circ} = \tan^2 60^\circ - \tan^2 30^\circ$$

$$\begin{aligned}
 \Rightarrow \frac{x \times (2)^2 \times (\sqrt{2})^2}{8 \times \left(\frac{1}{\sqrt{2}}\right)^2 \times \left(\frac{\sqrt{3}}{2}\right)^2} &= (\sqrt{3})^2 - \left(\frac{1}{\sqrt{3}}\right)^2 \\
 \Rightarrow \frac{x \times 4 \times 2}{8 \times \frac{1}{2} \times \frac{3}{4}} &= 3 - \frac{1}{3} \Rightarrow \frac{8x}{3} = \frac{8}{3} \Rightarrow x = 1
 \end{aligned}$$

$$92. (c) \frac{\sin x}{1 + \cos x} = \frac{1 - \cos x}{\sin x}$$

The above identity is possible for all values of x except multiples of 180° . Since, for $x = 180^\circ$, $\sin x = 0$ and $1 + \cos x = 0$.

93. (b) Given, $\cos x = k \cos(x - 2y) \Rightarrow \frac{1}{k} = \frac{\cos(x - 2y)}{\cos x}$

Apply componendo and dividendo theorem

$$\frac{1-k}{1+k} = \frac{\cos(x-2y) - \cos x}{\cos(x-2y) + \cos x} = \frac{2 \sin \frac{x-2y+x}{2} \sin \frac{x-x+2y}{2}}{2 \cos \frac{x-2y+x}{2} \cos \frac{x-2y-x}{2}}$$

$$= \frac{2 \sin(x-y) \sin y}{2 \cos(x-y) \cos y} = \tan(x-y) \tan y$$

94. (b) $\frac{\sin(x+y) - 2 \sin x + \sin(x-y)}{\cos(x+y) - 2 \cos x + \cos(x-y)}$

$$= \frac{\sin(x+y) + \sin(x-y) - 2 \sin x}{\cos(x+y) + \cos(x-y) - 2 \cos x}$$

$$= \frac{2 \sin \frac{x+y+x-y}{2} \cdot \cos \frac{x+y-x+y}{2} - 2 \sin x}{2 \cos \frac{(x+y+x-y)}{2} \cdot \cos \frac{(x+y-x+y)}{2} - 2 \cos x}$$

$$= \frac{2 \sin x \cos y - 2 \sin x}{2 \cos x \cos y - 2 \cos x} = \frac{2 \sin x (\cos y - 1)}{2 \cos x (\cos y - 1)} = \frac{\sin x}{\cos x} = \tan x$$

95. (c) Given, $\sin x + \sin y = a \Rightarrow 2 \sin \frac{x+y}{2} \cos \frac{x-y}{2} = a \dots(i)$

and $\cos x + \cos y = b \Rightarrow 2 \cos \frac{x+y}{2} \cdot \cos \frac{(x-y)}{2} = b \dots(ii)$

On dividing Eq. (i) from Eq. (ii), we get

$$\therefore \frac{2 \sin \frac{x+y}{2} \cos \frac{x-y}{2}}{2 \cos \frac{x+y}{2} \cos \frac{(x-y)}{2}} = \frac{a}{b} \Rightarrow \tan \frac{x+y}{2} = \frac{a}{b}$$

96. (a) Here, r = length of largest hand of clock

l = distance covered by largest hand of clock

Angle traced in 60 min = 360°

Angle traced in 20 min = $\frac{360^\circ}{60} \times 20 = 120^\circ \Rightarrow \theta = \frac{2\pi}{3}$

As, $\theta = \frac{l}{r} \Rightarrow l = r\theta, \Rightarrow l = 42 \times \frac{2\pi}{3} = \frac{42 \times 2 \times 22}{3 \times 7} = 88 \text{ cm}$

97. (d) Given, $2 \sin x + 15 \cos^2 x = 7 \Rightarrow 2 \sin x + 15 - 15 \sin^2 x = 7$

$$\Rightarrow 15 \sin^2 x - 2 \sin x - 8 = 0$$

$$\Rightarrow 15 \sin^2 x - 12 \sin x + 10 \sin x - 8 = 0$$

$$\Rightarrow 3 \sin x (5 \sin x - 4) + 2 (5 \sin x - 4) = 0$$

$$\Rightarrow (3 \sin x + 2) (5 \sin x - 4) = 0$$

$$\Rightarrow \sin x = \frac{4}{5} \left(\because \sin x \neq -\frac{2}{3}, 0^\circ < x < 90^\circ \right)$$

$$\therefore \cos x = \sqrt{1 - \sin^2 x} = \sqrt{1 - \left(\frac{4}{5}\right)^2} = \sqrt{\left(\frac{3}{5}\right)^2} = \frac{3}{5}$$

$$\therefore \tan x = \frac{\sin x}{\cos x} = \frac{4/5}{3/5} = \frac{4}{3}$$

98. (a) In $\triangle ABC$, $A + B + C = 180^\circ$

$$\therefore A + B = 180^\circ - C \Rightarrow \tan(A + B) = \tan(180^\circ - C)$$

$$\Rightarrow \frac{\tan A + \tan B}{1 - \tan A \tan B} = -\tan C$$

$$\Rightarrow \tan A + \tan B = -\tan C + \tan A \tan B \tan C$$

$$\Rightarrow \tan A + \tan B + \tan C = \tan A \tan B \tan C$$

99. (c) Given, $x = a \sec \theta \cos \phi$, $y = b \sec \theta \sin \phi$ and $z = c \tan \theta$,

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = \frac{a^2 \sec^2 \theta \cos^2 \phi}{a^2} + \frac{b^2 \sec^2 \theta \sin^2 \phi}{b^2} - \frac{c^2 \tan^2 \theta}{c^2}$$

$$= \sec^2 \theta [\cos^2 \phi + \sin^2 \phi] - \tan^2 \theta = \sec^2 \theta - \tan^2 \theta = 1$$

100. (a) The squares of the tangents of the angles 30° , 45° and 60° are in G.P.

$$\Rightarrow \tan^2 30^\circ, \tan^2 45^\circ, \tan^2 60^\circ \text{ are in G.P.} \Rightarrow \left(\frac{1}{\sqrt{3}}\right)^2, (1)^2, (\sqrt{3})^2$$

are in G.P. $\Rightarrow \frac{1}{3}, 1, 3$ are in G.P.

which is true as $1^2 = \frac{1}{3} \times 3 \Rightarrow 1 = 1$

101. (d) We know that, in a cyclic quadrilateral sum of opposite angle is 180° .

$$\therefore A + C = 180^\circ \dots(i)$$

$$\text{and } B + D = 180^\circ \dots(ii)$$

$$\therefore \cos A + \cos B + \cos C + \cos D$$

$$= \cos A + \cos B + \cos(180^\circ - A) + \cos(180^\circ - B)$$

$$= \cos A + \cos B - \cos A - \cos B = 0$$

$$[\because \cos(180^\circ - \theta) = -\cos \theta]$$

102. (a) We know that, for $0^\circ < \theta < 90^\circ$, there exist only one θ such that $\sin \theta = a$.

103. (b) $\therefore \sin(B + C - A) = 1 = \sin 90^\circ \Rightarrow B + C - A = 90^\circ \dots(i)$

$$\therefore \cos(C + A - B) = 1 = \cos 0^\circ$$

$$\therefore C + A - B = 0^\circ \dots(ii)$$

and $\tan(A + B - C) = 1 = \tan 45^\circ$

$$\therefore A + B - C = 45^\circ \dots(iii)$$

On adding Eqs. (i), (ii) and (iii), we get

$$A + B + C = 135^\circ \dots(iv)$$

On subtracting Eqs. (i), (ii), (iii) from Eq. (iv), we get

$$2A = 45^\circ, 2B = 135^\circ, 2C = 90^\circ$$

$$A = 22 \frac{1}{2}^\circ, B = 67 \frac{1}{2}^\circ, C = 45^\circ$$

104. (c) $\cos(180^\circ + A) + \cos(180^\circ + B)$

$$+ \cos(180^\circ + C) + \cos(180^\circ + D)$$

$$= -\cos A - \cos B - \cos C - \cos D$$

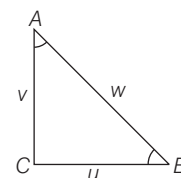
$$= -\cos A - \cos B - \cos(180^\circ - A) - \cos(180^\circ - B)$$

$$(\because A + C = B + D = 180^\circ)$$

$$= -\cos A - \cos B + \cos A + \cos B = 0$$

105. (d) In $\triangle ABC$,

$$\tan A = \frac{BC}{AC} = \frac{u}{v} \text{ and } \tan B = \frac{v}{u}$$



Also, $u^2 + v^2 = w^2$ (by pythagoras theorem) $\dots(i)$

$$\therefore \tan A + \tan B = \frac{u}{v} + \frac{v}{u} = \frac{u^2 + v^2}{uv} = \frac{w^2}{uv} \quad [\text{from Eq. (i)}]$$

106. (a) $\therefore$ In 24 h, Earth rotate about its own axis = 360°

In 1 h Earth rotate about its own axis

$$= \frac{360^\circ}{24} = 15^\circ$$

In 4 h Earth rotate about its own axis

$$= 15^\circ \times 4 = 60^\circ$$

Since, in 60 min Earth rotate about its own axis = 15°

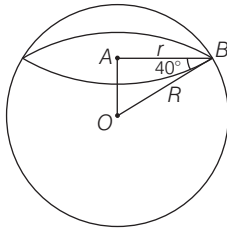
In 12 min Earth rotate about its own

$$\text{axis} = \frac{15^\circ \times 12}{60^\circ} = 3^\circ$$

$\therefore$ In 4 h 12 min Earth rotate about its own axis

$$= 60^\circ + 3^\circ = 63^\circ$$

107. (a) In right angled $\triangle OAB$,



$$\cos 40^\circ = \frac{AB}{OB} \Rightarrow \cos 40^\circ = \frac{r}{R}$$

$$\Rightarrow r = R \cos 40^\circ$$

So, the radius of the circle of latitude 40°S is $R \cos 40^\circ$.

108. (b) Given, $\sin 3\theta = \cos(\theta - 2^\circ)$

$$\Rightarrow \sin 3\theta = \sin[90^\circ - (\theta - 2^\circ)]$$

$$\Rightarrow 3\theta = 90^\circ - \theta + 2^\circ$$

$$\Rightarrow 4\theta = 92^\circ \Rightarrow \theta = \frac{92^\circ}{4} = 23^\circ$$

109. (c) I. RHS = $\cos^2 \theta (1 + \tan \theta)(1 - \tan \theta)$

$$= \cos^2 \theta (1 - \tan^2 \theta)$$

$$= \cos^2 \theta \left(\frac{\cos^2 \theta - \sin^2 \theta}{\cos^2 \theta} \right)$$

$$= \frac{\cos^2 \theta - \sin^2 \theta}{\cos^2 \theta + \sin^2 \theta} = \text{LHS}$$

$$\text{II. LHS} = \frac{1 + \sin \theta}{1 - \sin \theta} = \frac{(1 + \sin \theta)^2}{1 - \sin^2 \theta}$$

$$= \left(\frac{1 + \sin \theta}{\cos \theta} \right)^2$$

$$= (\sec \theta + \tan \theta)^2$$

Hence, both statements are correct.

110. (c) Given, $A = \frac{\pi}{6}$ and $B = \frac{\pi}{3}$

$$\text{I. LHS } \sin A + \sin B = \sin \frac{\pi}{6} + \sin \frac{\pi}{3}$$

$$= \frac{1}{2} + \frac{\sqrt{3}}{2} = \frac{1 + \sqrt{3}}{2}$$

$$\text{RHS } \cos A + \cos B = \cos \frac{\pi}{6} + \cos \frac{\pi}{3}$$

$$= \frac{\sqrt{3}}{2} + \frac{1}{2} = \frac{\sqrt{3} + 1}{2}$$

$$\Rightarrow \sin A + \sin B = \cos A + \cos B$$

$$\text{II. LHS } \tan A + \tan B = \tan \frac{\pi}{6} + \tan \frac{\pi}{3}$$

$$= \frac{1}{\sqrt{3}} + \sqrt{3} = \frac{4}{\sqrt{3}}$$

$$\text{RHS } \cot A + \cot B = \cot \frac{\pi}{6} + \cot \frac{\pi}{3}$$

$$= \sqrt{3} + \frac{1}{\sqrt{3}} = \frac{4}{\sqrt{3}}$$

$$\Rightarrow \tan A + \tan B = \cot A + \cot B$$

Hence, both statements are true.

111. (a) $\therefore A + B + C + D = 360^\circ$

$$\therefore A + B = 360^\circ - (C + D)$$

$$\therefore \sin(A + B) = \sin[360^\circ - (C + D)]$$

$$= -\sin(C + D)$$

$$\therefore \sin(A + B) + \sin(C + D) = 0$$

$$\text{Also, } \cos(A + B) = \cos[360^\circ - (C + D)]$$

$$\cos(A + B) = \cos(C + D)$$

Hence, only statement I is correct.

112. (d) I. $1^\circ = \frac{\pi}{180} \text{ radian} = \frac{3.14}{180} = 0.017$

$$= 0.02 \text{ radians (approx)}$$

which is less than 0.03 radians.

$$\text{II. } 1 \text{ radian} = \frac{180}{\pi} \text{ degree}$$

$$= \frac{180}{3.14} = 57.32 \text{ degree}$$

which is greater than 45° .

Hence, both statements are true.

113. (d) I. $\sin \theta \cdot \sin(60^\circ + \theta) \cdot \sin(60^\circ - \theta)$

$$= \sin \theta [\sin^2 60^\circ - \sin^2 \theta]$$

$$= \sin \theta \left[\frac{3}{4} - \sin^2 \theta \right]$$

$$= \frac{1}{4} [3 \sin \theta - 4 \sin^3 \theta] = \frac{1}{4} \sin 3\theta$$

$$\text{II. } \cos \theta \cdot \sin(30^\circ + \theta) \cdot \sin(30^\circ - \theta)$$

$$= \cos \theta [\sin^2 30^\circ - \sin^2 \theta]$$

$$= \cos \theta \left[\frac{1}{4} - (1 - \cos^2 \theta) \right]$$

$$= \frac{1}{4} [4 \cos^3 \theta - 3 \cos \theta] = \frac{1}{4} \cos 3\theta$$

$$\text{III. } \sin \theta \cdot \cos(30^\circ + \theta) \cdot \cos(30^\circ - \theta)$$

$$= \sin \theta [\cos^2 30^\circ - \sin^2 \theta]$$

$$= \sin \theta \left[\frac{3}{4} - \sin^2 \theta \right] = \frac{1}{4} \sin 3\theta$$

Hence, all three statements are correct.

114. (d) In right angled $\triangle ABC$,

$$A + B + C = 180^\circ$$

$$\text{I. } \sin(A + B) = \sin(180^\circ - C) = \sin C$$

$$\text{II. } \sin \left(\frac{A + B}{2} \right) = \sin \left(\frac{180^\circ - C}{2} \right)$$

$$= \sin \left(90^\circ - \frac{C}{2} \right) = \cos \frac{C}{2}$$

$$\text{III. } \tan \left(\frac{A + B - C}{2} \right)$$

$$= \tan \frac{(180^\circ - C) - C}{2}$$

$$= \tan \frac{(180^\circ - 2C)}{2}$$

$$= \tan(90^\circ - C) = \cot C$$

$$\text{IV. } \tan \frac{(A - B - C)}{2} = \tan \frac{A - (B + C)}{2}$$

$$= \tan \frac{A - (180^\circ - A)}{2}$$

$$= \tan \frac{A - (90^\circ - A)}{2}$$

$$= -\tan(90^\circ - A) = -\cot A$$

Hence, all options are correct.

115. (b) Given, $\sin(A + B) = 1$

$$\text{and } \sin(A - B) = 1/2$$

$$\therefore A + B = \frac{\pi}{2} \quad \dots(i)$$

$$\therefore A - B = \frac{\pi}{6} \quad \dots(ii)$$

On adding Eqs. (i) and (ii), we get

$$2A = \frac{2\pi}{3} \Rightarrow A = \frac{\pi}{3} \text{ and } B = \frac{\pi}{6}$$

116. (c) Now, $\tan(A + 2B) \cdot \tan(2A + B)$

$$= \tan \left(\frac{\pi}{3} + \frac{\pi}{3} \right) \cdot \tan \left(\frac{2\pi}{3} + \frac{\pi}{6} \right)$$

$$= \tan \left(\frac{2\pi}{3} \right) \cdot \tan \left(\frac{5\pi}{6} \right)$$

$$= \tan \left(\frac{\pi}{2} + \frac{\pi}{6} \right) \cdot \tan \left(\frac{\pi}{2} + \frac{\pi}{3} \right)$$

$$= \left[-\cot \frac{\pi}{6} \right] \left[-\cot \frac{\pi}{3} \right] = (-\sqrt{3}) \left(\frac{-1}{\sqrt{3}} \right) = 1$$

117. (b) Now, $\sin^2 A - \sin^2 B$

$$= \sin^2 \left(\frac{\pi}{3} \right) - \sin^2 \left(\frac{\pi}{6} \right)$$

$$= \left(\frac{\sqrt{3}}{2} \right)^2 - \left(\frac{1}{2} \right)^2 = \frac{3}{4} - \frac{1}{4} = \frac{2}{4} = \frac{1}{2}$$

118. (c) Since, $\tan A, \tan B$ are the roots of equation $3x^2 - 2\sqrt{3}x + 1 = 0$

$$\therefore \tan A + \tan B = \frac{2\sqrt{3}}{3}$$

$$\text{and } \tan A \cdot \tan B = 1/3$$

$$\therefore \tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \cdot \tan B}$$

$$= \frac{\frac{2\sqrt{3}}{3}}{1 - 1/3} = \sqrt{3} = \tan \frac{\pi}{3}$$

$$\therefore A + B = \pi/3$$

$$\text{Now, } A + B + C = \pi$$

$$\Rightarrow C = \pi - (A + B) = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$$

$$\begin{aligned}
 119. (c) \text{ Now, } \sin C + \cos C &= \sin(\pi - (A + B)) + \cos(\pi - (A + B)) \\
 &[\text{as } A + B + C = \pi, C = \pi - (A + B)] \\
 &= \sin(A + B) - \cos(A + B) \\
 &= \sin(\pi/3) - \cos(\pi/3) \\
 &= \frac{\sqrt{3}}{2} - \frac{1}{2} = \frac{\sqrt{3} - 1}{2}
 \end{aligned}$$

$$\begin{aligned}
 120. (b) \ 2 \sin B &= \sqrt{\frac{1 + \tan^2 C}{\operatorname{cosec}^2 C}} \\
 &= \sqrt{\frac{1 + \tan^2(2\pi/3)}{\operatorname{cosec}^2(2\pi/3)}} \\
 &= \sqrt{\frac{1 + \tan^2(\pi - \pi/3)}{\operatorname{cosec}^2(\pi - \pi/3)}} \\
 2 \sin B &= \sqrt{\frac{1 + \tan^2 \pi/3}{\operatorname{cosec}^2 \pi/3}} \\
 &= \sqrt{\frac{1 + 3}{4/3}} = \sqrt{\frac{4}{4} \times 3} = \sqrt{3} \\
 \sin B &= \frac{\sqrt{3}}{2} = \sin \frac{\pi}{3}, B = \frac{\pi}{3}
 \end{aligned}$$

$$\begin{aligned}
 121. (c) \text{ Given that, } \sin^2 x + \cos^2 x - 1 &= 0 \\
 \Rightarrow \sin^2 x + \cos^2 x &= 1
 \end{aligned}$$

which is an identity of trigonometric ratio and always true for every real value of x .
So, the equation has infinite solutions.

$$\begin{aligned}
 122. (c) \text{ Given that, } 3 \sin x + 5 \cos x &= 5 \\
 \text{On squaring both sides, we get} \\
 9 \sin^2 x + 25 \cos^2 x + 30 \sin x \cos x &= 25 \\
 \Rightarrow 9(1 - \cos^2 x) + 25(1 - \sin^2 x) &+ 30 \sin x \cos x = 25 \\
 \Rightarrow 9 + 25 - \{9 \cos^2 x + 25 \sin^2 x &- 30 \sin x \cos x\} = 25 \\
 \Rightarrow 9 = (3 \cos x - 5 \sin x)^2 & \\
 \Rightarrow 3 \cos x - 5 \sin x &= 3
 \end{aligned}$$

$$\begin{aligned}
 123. (c) \text{ Given, } p &= a \sin x + b \cos x \quad \text{and} \\
 q &= a \cos x - b \sin x \\
 \Rightarrow p^2 &= a^2 \sin^2 x + b^2 \cos^2 x \\
 &\quad + 2ab \sin x \cos x \\
 \text{and } q^2 &= a^2 \cos^2 x + b^2 \sin^2 x \\
 &\quad - 2ab \sin x \cos x \\
 \therefore p^2 + q^2 &= a^2 (\sin^2 x + \cos^2 x) + b^2 (\cos^2 x + \sin^2 x) \\
 &= a^2 + b^2
 \end{aligned}$$

$$\begin{aligned}
 124. (b) \text{ Given, } \alpha + \beta &= 90^\circ \quad \dots(i) \\
 \therefore \sqrt{(\operatorname{cosec} \alpha \cdot \operatorname{cosec} \beta)} &\left(\frac{\sin \alpha}{\sin \beta} + \frac{\cos \alpha}{\cos \beta} \right)^{-1/2} \\
 &= \frac{1}{(\sin \alpha \sin \beta)^{1/2}} \\
 &\quad \left(\frac{\sin \alpha \cos \beta + \cos \alpha \sin \beta}{\sin \beta \cos \beta} \right)^{-1/2} \\
 &= \frac{1}{(\sin \alpha \sin \beta)^{1/2}} \left\{ \frac{\sin(\alpha + \beta)}{\sin \beta \cos \beta} \right\}^{-1/2}
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{1}{(\sin \alpha \sin \beta)^{1/2}} \left\{ \frac{\sin 90^\circ}{\cos(90^\circ - \alpha) \sin \beta} \right\}^{-1/2} \\
 &\quad [\text{from Eq. (i)}]
 \end{aligned}$$

$$= \frac{1}{(\sin \alpha \sin \beta)^{1/2}} \times (\sin \alpha \sin \beta)^{1/2} = 1$$

$$125. (c) \text{ I. } \frac{\cot 30^\circ + 1}{\cot 30^\circ - 1} = 2(\cos 30^\circ + 1)$$

$$\Rightarrow \frac{\sqrt{3} + 1}{\sqrt{3} - 1} = 2 \left(\frac{\sqrt{3}}{2} + 1 \right)$$

$$\Rightarrow \frac{\sqrt{3} + 1}{\sqrt{3} - 1} \times \frac{\sqrt{3} + 1}{\sqrt{3} + 1} = 2 \left(\frac{\sqrt{3} + 2}{2} \right)$$

$$\Rightarrow \frac{3 + 1 + 2\sqrt{3}}{3 - 1} = \sqrt{3} + 2$$

$$\Rightarrow \frac{4 + 2\sqrt{3}}{2} = \sqrt{3} + 2$$

$$\Rightarrow \frac{2(2 + \sqrt{3})}{2} = \sqrt{3} + 2$$

$$\Rightarrow \sqrt{3} + 2 = \sqrt{3} + 2$$

So, it is true.

$$\text{II. } 2 \sin 45^\circ \cos 45^\circ - \tan 45^\circ \cot 45^\circ = 0$$

$$\Rightarrow 2 \times \left(\frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} \right) - 1 \times 1 = 0$$

$$\text{or } 2 \times \frac{1}{2} - 1 \times 1 = 0$$

$$\Rightarrow 1 - 1 = 0, \text{ It is true.}$$

Hence, both statements I and II are true.

$$\begin{aligned}
 126. (c) \text{ I. We know that, } \sin^2 \theta + \cos^2 \theta &= 1 \\
 \therefore \sin^2 1^\circ + \cos^2 1^\circ &= 1
 \end{aligned}$$

II. We have

$$\begin{aligned}
 \sec^2 33^\circ - \cot^2 57^\circ &= \operatorname{cosec}^2 37^\circ - \tan^2 53^\circ \\
 \text{LHS} &= \sec^2 33^\circ - \cot^2 57^\circ
 \end{aligned}$$

$$\begin{aligned}
 &= \sec^2(90^\circ - 57^\circ) - \cot^2 57^\circ \\
 &= \operatorname{cosec}^2 57^\circ - \cot^2 57^\circ = 1
 \end{aligned}$$

$$\begin{aligned}
 \text{RHS} &= \operatorname{cosec}^2 37^\circ - \tan^2 53^\circ \\
 &= \operatorname{cosec}^2 37^\circ - \tan^2(90^\circ - 53^\circ) \\
 &= \operatorname{cosec}^2 37^\circ - \cot^2 37^\circ = 1
 \end{aligned}$$

$$\begin{aligned}
 \text{LHS} &= \text{RHS} \\
 \text{Hence, both statements are correct.}
 \end{aligned}$$

$$\begin{aligned}
 127. (b) \text{ I. Given that, } \sin x + \cos x &= 2 \\
 \Rightarrow (\sin x + \cos x)^2 &= 4 \\
 \Rightarrow (\sin^2 x + \cos^2 x) + 2 \sin x \cos x &= 4 \\
 \Rightarrow 1 + \sin 2x &= 4 \Rightarrow \sin 2x = 3
 \end{aligned}$$

As maximum value of $\sin \theta$ is 1.

Therefore, no value of x satisfy above equation. Thus, statement I is false.

$$\begin{aligned}
 \text{II. } \sin x - \cos x &= 0 \\
 \Rightarrow \tan x &= 1 = \tan \pi/4
 \end{aligned}$$

It x lies in first quadrant, then $\tan x = \tan \pi/4 \Rightarrow x = \pi/4$

Thus, statement II is true.

$$128. (b) \text{ Given that, } \sin \theta \cdot \cos \theta = \frac{\sqrt{3}}{4} \quad \dots(i)$$

$$\begin{aligned}
 \therefore \sin^4 \theta + \cos^4 \theta &= (\sin^2 \theta + \cos^2 \theta)^2 - 2 \sin^2 \theta \cdot \cos^2 \theta \\
 &= (1)^2 - 2 (\sin \theta \cdot \cos \theta)^2 \\
 &= 1 - 2 \left(\frac{\sqrt{3}}{4} \right)^2 = 1 - 2 \cdot \frac{3}{16} = 1 - \frac{3}{8} = \frac{5}{8}
 \end{aligned}$$

$$129. (d) \text{ Given that, } \theta \text{ lies in first quadrant and } \tan \theta = 3$$

$$\therefore \tan^2 \theta = 9$$

On adding both sides by 1, we get

$$\Rightarrow 1 + \tan^2 \theta = 10$$

$$\Rightarrow \sec^2 \theta = 10 \Rightarrow \sec \theta = \sqrt{10}$$

(since, θ lies in first quadrant)

$$\Rightarrow \cos \theta = \frac{1}{\sqrt{10}} \quad \dots(ii)$$

$$\therefore \sin^2 \theta = 1 - \cos^2 \theta = 1 - \frac{1}{10} = \frac{9}{10}$$

$$\Rightarrow \sin \theta = \frac{3}{\sqrt{10}} \quad \dots(iii)$$

$$\text{Now, } \sin \theta + \cos \theta = \frac{3}{\sqrt{10}} + \frac{1}{\sqrt{10}} = \frac{4}{\sqrt{10}}$$

$$130. (d) \text{ If } 0^\circ < \theta < 90^\circ, \text{ then all the trigonometric ratios can be obtained when any one of the six ratios is given.}$$

$$\begin{aligned}
 131. (a) \sin A \cdot \cos A \cdot \tan A + \cos A \cdot \sin A \cdot \cot A &= \sin A \cdot \cos A \cdot \frac{\sin A}{\cos A} + \cos A \cdot \sin A \cdot \frac{\cos A}{\sin A} \\
 &= \sin^2 A + \cos^2 A
 \end{aligned}$$

$$\begin{aligned}
 132. (a) \frac{\sin \theta}{1 + \cos \theta} + \frac{1 + \cos \theta}{\sin \theta} &= \frac{\sin^2 \theta + (1 + \cos \theta)^2}{\sin \theta (1 + \cos \theta)} \\
 &= \frac{\sin^2 \theta + 1 + \cos^2 \theta + 2 \cos \theta}{\sin \theta (1 + \cos \theta)} \\
 &= \frac{\sin^2 \theta + \cos^2 \theta + 1 + 2 \cos \theta}{\sin \theta (1 + \cos \theta)} \\
 &= \frac{1 + 1 + 2 \cos \theta}{\sin \theta (1 + \cos \theta)} \\
 &= \frac{2(1 + \cos \theta)}{\sin \theta (1 + \cos \theta)} = 2 \operatorname{cosec} \theta
 \end{aligned}$$

$$\begin{aligned}
 133. (c) \sec^2 D - \tan^2 D &= 1 \\
 \text{Since, it is a trigonometric identity.}
 \end{aligned}$$

$$\begin{aligned}
 134. (b) \text{ Given, } \cos A + \cos^2 A &= 1 \\
 \Rightarrow \cos A &= 1 - \cos^2 A = \sin^2 A \quad \dots(i) \\
 (\because \sin^2 \theta + \cos^2 \theta &= 1)
 \end{aligned}$$

$$\begin{aligned}
 \text{Now, } 2(\sin^2 A + \sin^4 A) &= 2(\sin^2 A + \cos^2 A) \quad [\text{from Eq. (i)}] \\
 &= 2 \cdot (1) = 2
 \end{aligned}$$

$$\begin{aligned}
 135. (c) & (1 - \tan A)^2 + (1 + \tan A)^2 + (1 - \cot A)^2 + (1 + \cot A)^2 \\
 &= 2[1 + \tan^2 A] + 2[1 + \cot^2 A] \\
 & \quad [\because (a-b)^2 + (a+b)^2 = 2(a^2 + b^2)] \\
 &= 2 \sec^2 A + 2 \operatorname{cosec}^2 A = 2 \left(\frac{1}{\cos^2 A} + \frac{1}{\sin^2 A} \right) \\
 &= 2 \left(\frac{\sin^2 A + \cos^2 A}{\sin^2 A \cdot \cos^2 A} \right) = \frac{2 \cdot (1)}{\sin^2 A \cdot \cos^2 A} = 2 \sec^2 A \cdot \operatorname{cosec}^2 A
 \end{aligned}$$

$$\begin{aligned}
 136. (d) \text{ Given, } a^2 &= \frac{1 + 2 \sin \theta \cdot \cos \theta}{1 - 2 \sin \theta \cdot \cos \theta} \\
 \Rightarrow a^2 &= \frac{(\sin^2 \theta + \cos^2 \theta) + 2 \sin \theta \cdot \cos \theta}{(\sin^2 \theta + \cos^2 \theta) - 2 \sin \theta \cdot \cos \theta} \\
 \Rightarrow a^2 &= \frac{(\sin \theta + \cos \theta)^2}{(\sin \theta - \cos \theta)^2} \Rightarrow \frac{a}{1} = \frac{\sin \theta + \cos \theta}{\sin \theta - \cos \theta} \\
 \Rightarrow \frac{a+1}{a-1} &= \frac{(\sin \theta + \cos \theta) + (\sin \theta - \cos \theta)}{(\sin \theta + \cos \theta) - (\sin \theta - \cos \theta)} \\
 & \quad (\text{applying componendo dividendo formula}) \\
 \Rightarrow \frac{a+1}{a-1} &= \frac{2 \sin \theta}{2 \cos \theta} = \tan \theta
 \end{aligned}$$

$$\begin{aligned}
 137. (b) \text{ Given, } \sin \theta &= \frac{x^2 - y^2}{x^2 + y^2} \Rightarrow \cos^2 \theta = 1 - \sin^2 \theta = 1 - \left(\frac{x^2 - y^2}{x^2 + y^2} \right)^2 \\
 &= \frac{(x^2 + y^2)^2 - (x^2 - y^2)^2}{(x^2 + y^2)^2} = \frac{4x^2 y^2}{(x^2 + y^2)^2} = \left(\frac{2xy}{x^2 + y^2} \right)^2 \\
 \therefore \cos \theta &= 2xy/x^2 + y^2
 \end{aligned}$$

$$\begin{aligned}
 138. (b) \text{ Let } f(\theta) &= \sin \theta + \cos \theta \\
 \text{Maximum and minimum value of } a \cos \theta + b \sin \theta &\text{ is} \\
 -\sqrt{a^2 + b^2} \leq a \cos \theta + b \sin \theta &\leq \sqrt{a^2 + b^2} \\
 \therefore -\sqrt{1+1} \leq \cos \theta + \sin \theta &\leq \sqrt{1+1} \\
 \Rightarrow -\sqrt{2} \leq \cos \theta + \sin \theta &\leq \sqrt{2} \\
 \text{for } 0 \leq \theta \leq 90^\circ, \quad 1 \leq \cos \theta + \sin \theta &\leq \sqrt{2} \\
 \text{Hence, statement I is false.}
 \end{aligned}$$

$$\text{And let } g(\theta) = \tan \theta + \cot \theta = \tan \theta + \frac{1}{\tan \theta} \quad (\because \text{AM} \geq \text{GM})$$

$$\Rightarrow \frac{\tan \theta + \frac{1}{\tan \theta}}{2} \geq \left(\tan \theta \cdot \frac{1}{\tan \theta} \right)^{\frac{1}{2}}$$

$$\Rightarrow (\tan \theta + \cot \theta) \geq 2$$

So, $(\tan \theta + \cot \theta)$ is always greater than 1.

Hence, statement II is true.

Solutions (Q. Nos. 139-141) Let the angle of A, B, C and D of a quadrilateral be $x, 2x, 4x$ and $5x$.

$$\therefore x + 2x + 4x + 5x = 360^\circ$$

($\because$ sum of all interior in a quadrilateral is 360°)

$$\Rightarrow 12x = 360^\circ, x = 30^\circ$$

$$\therefore \angle A = x = 30^\circ, \angle B = 2x = 60^\circ$$

$$\angle C = 4x = 120^\circ, \angle D = 5x = 150^\circ$$

$$139. (a) \text{ Now, } \cos(A + B) = \cos(30^\circ + 60^\circ) = \cos 90^\circ = 0$$

$$\begin{aligned}
 140. (b) \text{ Now, } \operatorname{cosec}(C - D + B) &= \operatorname{cosec}(120^\circ - 150^\circ + 60^\circ) \\
 &= \operatorname{cosec}(30^\circ) = \operatorname{cosec} 30^\circ = 2
 \end{aligned}$$

141. (d) If $ABCD$ is a cyclic quadrilateral, then sum of opposite angles should be 180° but here

$$\angle A + \angle C = 30^\circ + 120^\circ = 150^\circ \neq 180^\circ$$

$$\text{and } \angle B + \angle D = 60^\circ + 150^\circ = 210^\circ \neq 180^\circ$$

So, statement I is incorrect.

$$\text{Now, } \sin(B - A) = \cos(D - C)$$

$$\Rightarrow \sin(60^\circ - 30^\circ) = \cos(150^\circ - 120^\circ)$$

$$\Rightarrow \sin 30^\circ = \cos 30^\circ \Rightarrow \frac{1}{2} \neq \frac{\sqrt{3}}{2}$$

So, statement II is also incorrect.

$$142. (a) \text{ Given, } \sin \theta + \cos \theta = \sqrt{3}$$

On squaring both sides, we get $(\sin \theta + \cos \theta)^2 = (\sqrt{3})^2$

$$\Rightarrow \sin^2 \theta + \cos^2 \theta + 2 \sin \theta \cos \theta = 3 \Rightarrow 1 + 2 \sin \theta \cos \theta = 3$$

$$\Rightarrow \sin \theta \cos \theta = \frac{3-1}{2} = 1 \quad \dots(i)$$

$$\text{Now, } \tan \theta + \cot \theta = \frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta} = \frac{\sin^2 \theta + \cos^2 \theta}{\sin \theta \cos \theta} = \frac{1}{\sin \theta \cos \theta}$$

$$\text{From Eq. (i), } \tan \theta + \cot \theta = \frac{1}{1} = 1$$

$$\begin{aligned}
 143. (a) \operatorname{cosec}(75^\circ + \theta) - \sec(15^\circ - \theta) &= \operatorname{cosec}(75^\circ + \theta) \\
 &\quad - \sec[90^\circ - (75^\circ + \theta)] \\
 &= \operatorname{cosec}(75^\circ + \theta) - \operatorname{cosec}(75^\circ + \theta) = 0
 \end{aligned}$$

$$144. (c) \text{ Given, } 5 \sin \theta + 12 \cos \theta = 13$$

On squaring both sides, we get

$$25 \sin^2 \theta + 144 \cos^2 \theta + 120 \sin \theta \cos \theta = 169$$

$$\Rightarrow 25(1 - \cos^2 \theta) + 144(1 - \sin^2 \theta) + 120 \sin \theta \cos \theta = 169$$

$$\Rightarrow 25 - 25 \cos^2 \theta + 144 - 144 \sin^2 \theta + 120 \sin \theta \cos \theta = 169$$

$$\Rightarrow 25 \cos^2 \theta + 144 \sin^2 \theta - 120 \sin \theta \cos \theta = 169 - 169$$

$$\Rightarrow (5 \cos \theta - 12 \sin \theta)^2 = 0 \Rightarrow 5 \cos \theta - 12 \sin \theta = 0$$

$$145. (c) \text{ Given, } \sin \theta - \cos \theta = 0$$

$$\therefore \sin \theta = \cos \theta$$

Since, $\sin \theta$ and $\cos \theta$ are equal for $\theta = 45^\circ$.

$$\therefore \sin^4 \theta + \cos^4 \theta = (\sin 45^\circ)^4 + (\cos 45^\circ)^4$$

$$= \left(\frac{1}{\sqrt{2}} \right)^4 + \left(\frac{1}{\sqrt{2}} \right)^4 = \frac{1}{4} + \frac{1}{4} = \frac{1+1}{4} = \frac{2}{4} = \frac{1}{2}$$

$$\begin{aligned}
 146. (c) & \frac{\cos^2(45^\circ + \theta) + \cos^2(45^\circ - \theta)}{\tan(60^\circ + \theta) \tan(30^\circ - \theta)} \\
 &= \frac{\cos^2(45^\circ + \theta) + \cos^2[90^\circ - (45^\circ + \theta)]}{\tan(60^\circ + \theta) \cdot \tan[90^\circ - (60^\circ + \theta)]} \\
 &= \frac{\cos^2(45^\circ + \theta) + \sin^2(45^\circ + \theta)}{\tan(60^\circ + \theta) \cdot \cot(60^\circ + \theta)} = \frac{1}{1} = 1
 \end{aligned}$$

$$147. (b) \text{ Given, } \tan \theta + \sec \theta = m \Rightarrow \tan \theta = m - \sec \theta$$

On squaring both sides, we get

$$\Rightarrow \tan^2 \theta = m^2 + \sec^2 \theta - 2m \sec \theta$$

$$\Rightarrow \tan^2 \theta = m^2 + 1 + \tan^2 \theta - 2m \sec \theta$$

$$\Rightarrow 0 = m^2 + 1 - 2m \sec \theta \Rightarrow \sec \theta = \frac{m^2 + 1}{2m}$$

148. (c) $\because \sin \alpha = \frac{\sqrt{3}}{2} \Rightarrow \sin \alpha = \sin 60^\circ$

$\Rightarrow \alpha = 60^\circ \quad \left(\because \sin 60^\circ = \frac{\sqrt{3}}{2} \right)$

Now, $\cos \beta = \frac{\sqrt{3}}{2} \Rightarrow \beta = 30^\circ \quad \left(\because \cos 30^\circ = \frac{\sqrt{3}}{2} \right)$

and $\tan \gamma = 1 \Rightarrow \gamma = 45^\circ \quad (\because \tan 45^\circ = 1)$

$\therefore \alpha + \beta + \gamma = 60^\circ + 30^\circ + 45^\circ = 135^\circ$

149. (a)
$$\frac{(1 + \sec \theta - \tan \theta) \cos \theta}{(1 + \sec \theta + \tan \theta)(1 - \sin \theta)} = \frac{\left(1 + \frac{1}{\cos \theta} - \frac{\sin \theta}{\cos \theta}\right) \cos \theta}{\left(1 + \frac{1}{\cos \theta} + \frac{\sin \theta}{\cos \theta}\right)(1 - \sin \theta)}$$
$$= \frac{\left(\frac{\cos \theta + 1 - \sin \theta}{\cos \theta}\right) \cos \theta}{\frac{(\cos \theta + 1 + \sin \theta)(1 - \sin \theta)}{\cos \theta}}$$
$$= \frac{\cos \theta + 1 - \sin \theta}{\cos \theta + 1 + \sin \theta - \sin \theta \cos \theta - \sin \theta - \sin^2 \theta}$$
$$= \frac{\cos \theta}{\cos \theta + 1 - \sin \theta} \quad (\because 1 - \sin^2 \theta = \cos^2 \theta)$$
$$= \frac{\cos \theta}{\cos \theta + \cos^2 \theta - \sin \theta \cos \theta} = \frac{\cos \theta + 1 - \sin \theta}{\cos \theta (\cos \theta + 1 - \sin \theta)} = 1$$

150. (c) In right angled $\triangle ABC$, if $\angle C$ is 90° , then

$\angle A + \angle B = 180^\circ - 90^\circ = 90^\circ$

Now, $\cos(A + B) + \sin(A + B) = \cos 90^\circ + \sin 90^\circ = 0 + 1 = 1$

151. (a) Only statement I is correct as $\tan \theta$ increases faster than $\sin \theta$ as θ increases while statement II is wrong as the value of $\sin \theta + \cos \theta$ is not always greater than 1. It may also be equal to 1.

152. (a) Since, value of $\cos \theta$ decreases, from 0° to 90° and at 45° it is equal to the value of $\sin \theta$.

Similarly, value of $\sin \theta$ increases from 0 to 90° and at 45° it is equal to the value of $\cos \theta$.

For $0^\circ < \theta < 45^\circ$, $\cos \theta > \sin \theta$

So, value of $\cos 25^\circ - \sin 25^\circ$ is always positive but less than 1.

153. (b) In right angled $\triangle ABC$, $\angle B = 90^\circ$

$\angle C = 180^\circ - (\angle B + \angle A) = 180^\circ - 90^\circ - \angle A = 90^\circ - \angle A$

$\therefore \sin C = \sin(90^\circ - A) = \cos A = 4/5$

154. (c) Since, α and β are complementary angle.

$\therefore \alpha = 90^\circ - \beta$

Now, $\sqrt{\cos \alpha \operatorname{cosec} \beta - \cos \alpha \sin \beta} = \sqrt{\frac{\cos \alpha}{\sin \beta} - \cos \alpha \sin \beta}$
$$= \sqrt{\frac{\cos \alpha}{\cos(90^\circ - \beta)} - \cos \alpha \cos(90^\circ - \beta)}$$
$$= \sqrt{\frac{\cos \alpha}{\cos \alpha} - \cos \alpha \cdot \cos \alpha} = \sqrt{1 - \cos^2 \alpha} = \sqrt{\sin^2 \alpha} = \sin \alpha$$

155. (d) Given, $\sec \theta + \tan \theta = 2$... (i)

By trigonometric identity, $\sec^2 \theta - \tan^2 \theta = 1$

$\Rightarrow (\sec \theta + \tan \theta)(\sec \theta - \tan \theta) = 1 \Rightarrow \sec \theta - \tan \theta = 1/2$... (ii)

On adding Eqs. (i) and (ii), we get

$\Rightarrow 2 \sec \theta = \frac{1}{2} + 2 \Rightarrow \sec \theta = \frac{5}{4}$

156. (b) $\operatorname{cosec}(75^\circ + \theta) - \sec(15^\circ - \theta) - \tan(55^\circ + \theta) + \cot(35^\circ - \theta)$
$$= \operatorname{cosec}(75^\circ + \theta) - \sec[90^\circ - (75^\circ + \theta)]$$
$$= \operatorname{cosec}(75^\circ + \theta) - \sec(15^\circ + \theta) + \cot[90^\circ - (55^\circ + \theta)]$$
$$= \operatorname{cosec}(75^\circ + \theta) - \operatorname{cosec}(75^\circ + \theta) - \tan(55^\circ + \theta) + \tan(55^\circ + \theta) = 0$$

157. (d) $\sin 25^\circ \sin 35^\circ \sec 65^\circ \sec 55^\circ$
$$= \sin 25^\circ \cdot \sin 35^\circ \cdot \frac{1}{\cos 65^\circ} \cdot \frac{1}{\cos 55^\circ}$$
$$= \sin 25^\circ \cdot \sin 35^\circ \cdot \frac{1}{\cos(90^\circ - 25^\circ)} \cdot \frac{1}{\cos(90^\circ - 35^\circ)}$$
$$= \sin 25^\circ \cdot \sin 35^\circ \cdot \frac{1}{\sin 25^\circ} \cdot \frac{1}{\sin 35^\circ} = 1$$

158. (c) Given, $2 \cot \theta = 3 \Rightarrow \cot \theta = \frac{3}{2}$
$$\therefore \frac{2 \cos \theta - \sin \theta}{2 \cos \theta + \sin \theta} = \frac{2 \cot \theta - 1}{2 \cot \theta + 1} = \frac{2 \times \frac{3}{2} - 1}{2 \times \frac{3}{2} + 1} = \frac{3 - 1}{3 + 1} = \frac{2}{4} = \frac{1}{2}$$

159. (d) $\sin^6 \theta + \cos^6 \theta = (\sin^2 \theta)^3 + (\cos^2 \theta)^3$
$$= (\sin^2 \theta + \cos^2 \theta)(\sin^4 \theta + \cos^4 \theta - \sin^2 \theta \cos^2 \theta)$$
$$= (\sin^2 \theta + \cos^2 \theta)^2 - 2 \sin^2 \theta \cos^2 \theta - \sin^2 \theta \cos^2 \theta$$
$$= (1 - 3 \sin^2 \theta \cos^2 \theta) = 1 - 3 \times \frac{1}{4} = 1 - \frac{3}{4} = \frac{1}{4}$$

160. (a) Given, $\cos x + \sec x = 2$... (i)

$\Rightarrow \cos x + \frac{1}{\cos x} = 2 \Rightarrow \cos^2 x + 1 = 2 \cos x$

$\Rightarrow \cos^2 x - 2 \cos x + 1 = 0 \Rightarrow (\cos x - 1)^2 = 0 \Rightarrow \cos x = 1$

$\therefore \sec x = \frac{1}{\cos x} = 1$

$\therefore \cos^n x + \sec^n x = (1)^n + (1)^n = 1 + 1 = 2$

161. (c) Given, $\sin \theta + 2 \cos \theta = 1$

On squaring both sides, we get $(\sin \theta + 2 \cos \theta)^2 = 1$

$\Rightarrow \sin^2 \theta + 4 \cos^2 \theta + 4 \sin \theta \cos \theta = 1$

$\Rightarrow (1 - \cos^2 \theta) + 4(1 - \sin^2 \theta) + 4 \sin \theta \cos \theta = 1$

$\Rightarrow -(\cos^2 \theta + 4 \sin^2 \theta) + 4 \sin \theta \cos \theta = 1 - 5$

$\Rightarrow \cos^2 \theta + 4 \sin^2 \theta - 4 \sin \theta \cos \theta = 4$

$\Rightarrow (2 \sin \theta - \cos \theta)^2 = 4 \Rightarrow 2 \sin \theta - \cos \theta = 2$

162. (b) Given, $\tan 8\theta = \cot 2\theta$

$\Rightarrow \tan 8\theta = \tan(90^\circ - 2\theta) \Rightarrow 8\theta = 90^\circ - 2\theta \Rightarrow \theta = 9^\circ$

$\therefore \tan 5\theta \Rightarrow \tan 45^\circ = 1$

163. (a) $\because \sin(A + B) = 1 = \sin 90^\circ \Rightarrow (A + B) = 90^\circ$

$\Rightarrow A = 90^\circ - B$

Now, $\cos(A - B) = \cos A \cos B + \sin A \sin B$

$= \cos(90^\circ - B) \cos B + \sin(90^\circ - B) \sin B$

$= \sin B \cos B + \cos B \sin B = 2 \sin B \cos B = \sin 2B$

164. (d) Clock will make right angle at $(5n + 15) \times \frac{12}{11}$ min past n .

Here, $n = 3$

$\therefore (5 \times 3 + 15) \times \frac{12}{11}$ min past 3 = $30 \times \frac{12}{11}$ min past 3

$= 32 \frac{8}{11}$ min past 3 i.e. 3 h $32 \frac{8}{11}$ min

165. (c) Given, $\tan \theta + \cot \theta = 2$

$$\Rightarrow \frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta} = 2 \Rightarrow \frac{\sin^2 \theta + \cos^2 \theta}{\sin \theta \cos \theta} = 2$$

$$\Rightarrow \frac{1}{\sin \theta \cos \theta} = 2 \Rightarrow \sin \theta \cos \theta = \frac{1}{2} \quad \dots(i)$$

$$\text{Now, } (\sin \theta + \cos \theta)^2 = \sin^2 \theta + \cos^2 \theta + 2 \sin \theta \cos \theta$$

$$= 1 + 2 \times \frac{1}{2} = 1 + 1 \quad [\text{from Eq. (i)}]$$

$$\Rightarrow (\sin \theta + \cos \theta)^2 = 2 \Rightarrow \sin \theta + \cos \theta = \sqrt{2}$$

166. (a) $\frac{\sec x}{\cot x + \tan x} = \frac{1/\cos x}{\frac{\cos x}{\sin x} + \frac{\sin x}{\cos x}}$

$$= \frac{1/\cos x}{\frac{\cos^2 x + \sin^2 x}{\sin x \cos x}} = \frac{1}{\cos x} \times \frac{\sin x \cos x}{1} = \sin x$$

167. (b) $(1 + \cot x - \operatorname{cosec} x)(1 + \tan x + \sec x)$

$$= (1 + \cot x - \operatorname{cosec} x) \left(1 + \frac{1}{\cot x} + \sec x \right)$$

$$= \frac{(1 + \cot x - \operatorname{cosec} x)(1 + \cot x + \operatorname{cosec} x)}{\cot x}$$

$$= \frac{(1 + \cot x)^2 - (\operatorname{cosec} x)^2}{\cot x}$$

$$= \frac{1^2 + \cot^2 x + 2 \cot x - \operatorname{cosec}^2 x}{\cot x}$$

$$= \frac{1 + 2 \cot x - (\operatorname{cosec}^2 x - \cot^2 x)}{\cot x} = \frac{1 + 2 \cot x - 1}{\cot x} = 2$$

168. (d) $(\operatorname{cosec} x - \sin x)(\sec x - \cos x)(\tan x + \cot x)$

$$= \left(\frac{1}{\sin x} - \sin x \right) \left(\frac{1}{\cos x} - \cos x \right) \left(\frac{\sin x}{\cos x} + \frac{\cos x}{\sin x} \right)$$

$$= \frac{(1 - \sin^2 x)(1 - \cos^2 x)(\sin^2 x + \cos^2 x)}{\sin x \cos x \cdot \sin x \cos x} = \frac{\cos^2 x \sin^2 x \times 1}{\sin^2 x \cos^2 x} = 1$$

169. (b) $\frac{\sin x - \cos x + 1}{\sin x + \cos x - 1} = \frac{(\sin x - \cos x) + 1}{(\sin x + \cos x) - 1} \times \frac{(\sin x + \cos x) + 1}{(\sin x + \cos x) + 1}$

$$= \frac{(\sin x - \cos x + 1)(\sin x + \cos x + 1)}{(\sin x + \cos x)^2 - 1}$$

$$[\because (a-b)(a+b) = a^2 - b^2]$$

$$\frac{\sin^2 x + \sin x \cos x + \sin x - \cos x \sin x - \cos^2 x}{- \cos x + \sin x + \cos x + 1}$$

$$= \frac{\sin^2 x + \cos^2 x + 2 \sin x \cos x - 1}{1 + 2 \sin x \cos x - 1}$$

$$= \frac{\sin^2 x + 2 \sin x \cos x - (1 - \sin^2 x) + 1}{2 \sin x \cos x} \quad [\because \cos^2 x = 1 - \sin^2 x]$$

$$= \frac{\sin^2 x + 2 \sin x \cos x - 1 + \sin^2 x + 1}{2 \sin x \cos x}$$

$$= \frac{2 \sin^2 x + 2 \sin x \cos x}{2 \sin x \cos x} = \frac{2 \sin x (\sin x + 1)}{2 \sin x \cos x} = \frac{\sin x + 1}{\cos x}$$

170. (b) $(\sin^2 x - \cos^2 x)(1 - \sin^2 x \cos^2 x)$

$$= (\sin^2 x - \cos^2 x)[(\sin^2 x + \cos^2 x)^2 - \sin^2 x \cos^2 x]$$

$$(\because \sin^2 x + \cos^2 x = 1)$$

$$= (\sin^2 x - \cos^2 x)[(\sin^4 x + \cos^4 x + 2 \sin^2 x \cos^2 x) - \sin^2 x \cos^2 x]$$

$$= (\sin^2 x - \cos^2 x)(\sin^4 x + \cos^4 x + \sin^2 x \cos^2 x)$$

$$= (\sin^2 x)^3 - (\cos^2 x)^3 \quad [\because (a-b)(a^2 + b^2 + ab) = a^3 - b^3]$$

$$= \sin^6 x - \cos^6 x$$

171. (d) $(\sin x \cdot \cos y + \cos x \cdot \sin y)(\sin x \cdot \cos y - \cos x \cdot \sin y)$

$$= \sin(x+y) \cdot \sin(x-y) = \sin^2 x - \sin^2 y$$

$$[\because \sin^2 A - \sin^2 B = \sin(A+B)\sin(A-B)]$$

172. (d) In $\triangle ABC$, $A + B + C = 180^\circ \Rightarrow A + B = 180^\circ - C$

$$\tan(A+B) = \tan(180^\circ - C) = -\tan C$$

$$\Rightarrow \frac{\tan A + \tan B}{1 - \tan A \tan B} = -\tan C$$

$$\tan A + \tan B + \tan C = \tan A \tan B \tan C$$

On dividing both sides by $\tan A \tan B \tan C$

$$\frac{1}{\tan B \tan C} + \frac{1}{\tan A \tan C} + \frac{1}{\tan A \tan B} = 1$$

$$\Rightarrow \cot B \cot C + \cot A \cot C + \cot A \cot B = 1$$

173. (a) $\sin x + \operatorname{cosec} x = 2 \Rightarrow \sin x + \frac{1}{\sin x} = 2 \Rightarrow \sin^2 x + 1 = 2 \sin x$

$$\Rightarrow \sin^2 x - 2 \sin x + 1 = 0 \Rightarrow (\sin x - 1)^2 = 0 \Rightarrow \sin x = 1$$

$$\operatorname{cosec} x = \frac{1}{\sin x} = 1$$

$$\therefore \sin^9 x + \operatorname{cosec}^9 x = (1)^9 + (1)^9 = 1 + 1 = 2$$

174. (b) Given, $\sin x + \cos x = p$... (i)

and $\sin^3 x + \cos^3 x = q$... (ii)

On cubing Eq. (i) on both sides, we get

$$\sin^3 x + \cos^3 x + 3 \sin x \cos x (\sin x + \cos x) = p^3$$

$$\Rightarrow q + 3 \sin x \cos x (p) = p^3 \quad [\text{from Eqs. (i) and (ii)}] \dots (iii)$$

On squaring Eq. (i) on both sides, we get

$$\sin^2 x + \cos^2 x + 2 \sin x \cos x = p^2$$

$$\Rightarrow \sin x \cos x = \frac{p^2 - 1}{2} \quad (\because \sin^2 x + \cos^2 x = 1)$$

On putting this value in Eq. (iii), we get $q + \frac{3(p^2 - 1)p}{2} = p^3$

$$\Rightarrow 2q + 3p^3 - 3p = 2p^3 \Rightarrow p^3 - 3p = -2q$$

175. (d) Since, 1 radian $= 57^\circ 16' 22''$

So, $\sin 1^\circ < \sin 1$ and $\cos 1^\circ > \cos 1$

Hence, neither statement I nor II is correct.

176. (b) $\operatorname{cosec}^2 67^\circ + \sec^2 57^\circ - \cot^2 33^\circ - \tan^2 23^\circ$

$$= \operatorname{cosec}^2(90^\circ - 23^\circ) + \sec^2(90^\circ - 33^\circ) - \cot^2 33^\circ - \tan^2 23^\circ$$

$$= \sec^2 23^\circ + \operatorname{cosec}^2 33^\circ - \cot^2 33^\circ - \tan^2 23^\circ$$

$$= 1 + \tan^2 23^\circ + 1 + \cot^2 33^\circ - \cot^2 33^\circ - \tan^2 23^\circ = 2$$

$$(\because 1 + \tan^2 q = \sec^2 q \text{ and } 1 + \cot^2 q = \operatorname{cosec}^2 q)$$

177. (c) Given, $\tan(A+B) = \sqrt{3} = \tan 60^\circ$

$$\therefore A + B = 60^\circ \quad \dots(i)$$

and $\tan A = 1 \Rightarrow \tan A = \tan 45^\circ$

$$\therefore A = 45^\circ$$

From Eq. (i), $A + B = 60^\circ \Rightarrow 45^\circ + B = 60^\circ \Rightarrow B = 15^\circ$

Now, $\tan(A-B) = \tan(45^\circ - 15^\circ) = \tan 30^\circ = 1/\sqrt{3}$

Hence, the value of $\tan(A-B)$ is $1/\sqrt{3}$.

178. (d) Given, $\tan A + \cot A = 4$

On squaring both sides, we get

$$\begin{aligned} (\tan A + \cot A)^2 &= (4)^2 \\ \Rightarrow \tan^2 A + \cot^2 A + 2 \tan A \cot A &= 16 \\ \Rightarrow \tan^2 A + \cot^2 A + 2 &= 16 \\ \Rightarrow \tan^2 A + \cot^2 A &= 14 \end{aligned}$$

Again, squaring both sides, we get

$$\begin{aligned} (\tan^2 A + \cot^2 A)^2 &= (14)^2 \\ \Rightarrow \tan^4 A + \cot^4 A + 2 \tan^2 A \cot^2 A &= 196 \\ \Rightarrow \tan^4 A + \cot^4 A + 2 &= 196 \\ \Rightarrow \tan^4 A + \cot^4 A &= 194 \end{aligned}$$

179. (c) In right angled $\triangle ABC$, $AB:BC = 3:4$

$$\text{or } \frac{AB}{BC} = \frac{3}{4}$$

Now, in $\triangle ABC$

$$AC^2 = AB^2 + BC^2$$

$$\begin{aligned} &= 3^2 + 4^2 \\ &= 9 + 16 = 25 \end{aligned}$$

$$\Rightarrow AC = 5 \Rightarrow \sin A = \frac{BC}{AC} = \frac{4}{5}$$

$$\sin B = 90^\circ = \sin 90^\circ = 1, \sin C = \frac{AB}{AC} = \frac{3}{5}$$

$$\begin{aligned} \text{Now, } \sin A + \sin B + \sin C &= \frac{4}{5} + 1 + \frac{3}{5} \\ &= \frac{4+5+3}{5} = \frac{12}{5} \end{aligned}$$

180. (b) Given, $\sin x + \cos x = C$

On squaring both sides, we get

$$\begin{aligned} (\sin x + \cos x)^2 &= C^2 \\ \Rightarrow \sin^2 x + \cos^2 x + 2 \sin x \cos x &= C^2 \\ \Rightarrow 1 + 2 \sin x \cos x &= C^2 \\ \Rightarrow \sin x \cos x &= \frac{C^2 - 1}{2} \quad \dots(i) \end{aligned}$$

Now, $\sin^6 x + \cos^6 x = (\sin^2 x)^3$

$$\begin{aligned} &+ (\cos^2 x)^3 \\ &= (\sin^2 x + \cos^2 x)[\sin^4 x + \cos^4 x \\ &\quad - \sin^2 x \cos^2 x] \end{aligned}$$

$$[\because a^3 + b^3 = (a+b)(a^2 + b^2 - ab)]$$

$$= 1[(\sin^2 x + \cos^2 x)^2 - 3 \sin^2 x \cos^2 x]$$

$$= (1 - 3 \sin^2 x \cos^2 x)$$

$$\therefore \sin^6 x + \cos^6 x = 1 - 3 \sin^2 x \cos^2 x$$

$$= 1 - 3 \left(\frac{C^2 - 1}{2} \right)^2 \quad [\text{from Eq. (i)}]$$

$$= 1 - 3 \left(\frac{C^4 + 1 - 2C^2}{4} \right)$$

$$= \frac{4 - 3C^4 - 3 + 6C^2}{4} = \frac{1 + 6C^2 - 3C^4}{4}$$

181. (c) Given, $\frac{3 - \tan^2 A}{1 - 3 \tan^2 A} = k$

$$\Rightarrow 3 - \tan^2 A = k(1 - 3 \tan^2 A)$$

$$\Rightarrow 3 - \tan^2 A = k - 3k \tan^2 A$$

$$\Rightarrow 3k \tan^2 A - \tan^2 A = k - 3$$

$$\Rightarrow \tan^2 A(3k - 1) = k - 3$$

$$\Rightarrow \tan^2 A = \frac{k - 3}{3k - 1}$$

$$\text{where } \frac{k - 3}{3k - 1} > 0 \Rightarrow k < \frac{1}{3} \text{ or } k > 3$$

$$\text{Now, } \operatorname{cosec} A (3 \sin A - 4 \sin^3 A)$$

$$= \operatorname{cosec} A \times \sin A (3 - 4 \sin^2 A)$$

$$= \frac{1}{\sin A} \times \sin A (3 - 4 \sin^2 A)$$

$$\left(\because \operatorname{cosec} A = \frac{1}{\sin A} \right)$$

$$= 3 - 4 \sin^2 A = 3 - 4 \left[\frac{\tan^2 A}{1 + \tan^2 A} \right]$$

$$\begin{aligned} &= \frac{3 - \tan^2 A}{1 + \tan^2 A} = \frac{3 - \frac{k - 3}{3k - 1}}{1 + \frac{k - 3}{3k - 1}} \\ &= \frac{9k - 3 - k + 3}{3k - 1 + k - 3} \end{aligned}$$

$$= \frac{8k}{4k - 4} = \frac{2k}{k - 1}$$

$$\text{where } k < \frac{1}{3} \text{ or } k > 3$$

182. (c) Given, $p = \sqrt{\frac{1 - \sin x}{1 + \sin x}}$

$$p = \sqrt{\frac{(1 - \sin x)(1 - \sin x)}{(1 + \sin x)(1 - \sin x)}}$$

$$= \frac{1 - \sin x}{\sqrt{1 - \sin^2 x}} = \frac{1 - \sin x}{\cos x}$$

$$r = \frac{\cos x}{1 + \sin x} \times \frac{(1 - \sin x)}{(1 - \sin x)} = \frac{\cos(1 - \sin x)}{1 - \sin^2 x}$$

$$r = \frac{\cos x (1 - \sin x)}{\cos^2 x} = \frac{1 - \sin x}{\cos x}$$

$$\Rightarrow p = q = r$$

$$\begin{aligned} \text{Also, } q \times r &= \frac{1 - \sin x}{\cos x} \times \frac{\cos x}{1 + \sin x} \\ &= \frac{1 - \sin x}{1 + \sin x} = p^2 \end{aligned}$$

Hence, both statements are correct.

183. (a) I. LHS = $\frac{\cos A}{1 - \tan A} + \frac{\sin A}{1 - \cot A}$

$$= \frac{\cos A}{1 - \frac{\sin A}{\cos A}} + \frac{\sin A}{1 - \frac{\cos A}{\sin A}}$$

$$= \frac{\cos^2 A}{\cos A - \sin A} + \frac{\sin^2 A}{\sin A - \cos A}$$

$$= \frac{\cos^2 A - \sin^2 A}{(\cos A - \sin A)}$$

$$= \frac{(\cos A - \sin A)(\cos A + \sin A)}{(\cos A - \sin A)}$$

$$= (\cos A + \sin A) = \text{RHS}$$

$$\text{II. } [(1 - \sin A) - \cos A]^2$$

$$= (1 - \sin A)^2 + \cos^2 A - 2 \cos A(1 - \sin A)$$

$$= 1 + \sin^2 A - 2 \sin A + \cos^2 A - 2 \cos A(1 - \sin A)$$

$$= 1 + \sin^2 A + \cos^2 A - 2 \sin A - 2 \cos A(1 - \sin A)$$

$$= 2 - 2 \sin A - 2 \cos A(1 - \sin A)$$

$$= 2(1 - \sin A) - 2 \cos A(1 - \sin A)$$

$$= 2(1 - \sin A)(1 - \cos A) \neq \text{RHS}$$

Hence, only statement I is correct.

184. (a) We have, $\sin \theta = \frac{2a + 3b}{3b} = 1 + \frac{2a}{3b}$

Since, $\sin \theta$ is always smaller or equal to 1 but $1 + \frac{2a}{3b} > 1$.

Hence, it is not possible.

185. (a) Since, $0 \leq \cos^2 x \leq 1$

$$\therefore -1 \leq \cos^2 x + \cos^2 y - \cos^2 z \leq 2$$

$\therefore$ Minimum value of the given expression is -1 .

186. (a) We have, $\tan \theta + \sec \theta = 2$

$$\Rightarrow \sec \theta = 2 - \tan \theta$$

On squaring both sides, we get

$$\sec^2 \theta = 4 + \tan^2 \theta - 4 \tan \theta$$

$$\Rightarrow 1 + \tan^2 \theta = 4 + \tan^2 \theta - 4 \tan \theta$$

$$\Rightarrow 4 \tan \theta = 3 \Rightarrow \tan \theta = \frac{3}{4}$$

187. (b) We have, $\frac{\cos \theta}{1 - \sin \theta} \times \frac{1 + \sin \theta}{1 + \sin \theta}$

$$= \frac{\cos \theta(1 + \sin \theta)}{1 - \sin^2 \theta} = \frac{\cos \theta(1 + \sin \theta)}{\cos^2 \theta}$$

$$= \frac{1 + \sin \theta}{\cos \theta}$$

188. (c) We have,

$$\tan(x + 40)^\circ \tan(x + 20)^\circ \tan(3x)^\circ$$

$$\tan(70 - x)^\circ \tan(50 - x)^\circ = 1$$

$$[\because \cot(90^\circ - \theta) = \tan \theta]$$

$$\Rightarrow \tan(x + 40)^\circ \tan(x + 20)^\circ \tan(3x)^\circ$$

$$\cot(90^\circ - 70^\circ + x)^\circ$$

$$\cot(90^\circ - 50^\circ + x)^\circ = 1$$

$$\Rightarrow \tan(x + 40)^\circ \tan(x + 20)^\circ \tan(3x)^\circ$$

$$\cot(20 + x)^\circ \cot(40 + x)^\circ = 1$$

$$\Rightarrow \tan(3x)^\circ = 1 \Rightarrow \tan(3x)^\circ = \tan 45^\circ$$

$$\Rightarrow 3x = 45 \Rightarrow x = 15$$

189. (d) We have, $32 \cot^2 \frac{\pi}{4} - 8 \sec^2 \left(\frac{\pi}{3} \right) + 8 \cos^3 \left(\frac{\pi}{6} \right)$

$$= 32 \cdot (1) - 8 \cdot (2)^2 + 8 \cdot \left(\frac{\sqrt{3}}{2} \right)^3$$

$$= 32 - 8 \cdot (4) + 8 \cdot \frac{3\sqrt{3}}{8}$$

$$= 32 - 32 + 3\sqrt{3} = 3\sqrt{3}$$

190. (b) We have, $x = a \cos \theta$ and $y = b \cot \theta$

$$\therefore \left(\frac{a}{x} - \frac{b}{y} \right) \left(\frac{a}{x} + \frac{b}{y} \right) = \frac{a^2}{x^2} - \frac{b^2}{y^2}$$

$$= \sec^2 \theta - \tan^2 \theta = 1$$

191. (a) We have, $\sin \theta \cos \theta = 2 \cos^3 \theta - \frac{3}{2} \cos \theta$

$$\Rightarrow 2 \sin \theta = 4 \cos^2 \theta - 3$$

$$\Rightarrow 2 \sin \theta = 4 - 4 \sin^2 \theta - 3$$

$$\Rightarrow 4 \sin^2 \theta + 2 \sin \theta - 1 = 0,$$

$$\Rightarrow \sin \theta = \frac{-2 \pm \sqrt{4 + 16}}{8} = \frac{-2 \pm 2\sqrt{5}}{8}$$

$$\Rightarrow \sin \theta = \frac{-1 \pm \sqrt{5}}{4}$$

Since, θ is acute angle

$$\therefore \sin \theta > 0$$

$$\Rightarrow \sin \theta = \frac{\sqrt{5} - 1}{4}$$

192. (d) Hours moved by hour hand = 5 hrs
10 min

$$= 5 + \frac{10}{60} = \frac{31}{6} \text{ hrs}$$

Angle traced by hour hand in 1 hr = 30°

$$\therefore \text{Angle traced by hour hand is}$$

$$\left(\frac{31}{6} \right) \text{ hrs} = 30 \times \frac{31}{6} = 155^\circ$$

193. (b) I. If $45^\circ < \theta < 90^\circ$, then $\sin \theta > \cos \theta$

$$\therefore \sin 66^\circ > \cos 66^\circ$$

II. When $0^\circ < \theta < 45^\circ$, $\cos \theta > \sin \theta$

$$\therefore \cos 26^\circ > \sin 26^\circ$$

or $\sin 26^\circ < \cos 26^\circ$.

Hence only II is correct.

194. (a) I. LHS = $\frac{1 + \tan^2 \theta}{1 + \cot^2 \theta} = \frac{\sec^2 \theta}{\csc^2 \theta}$

$$= \tan^2 \theta$$

$$\text{RHS} = \left(\frac{1 - \tan \theta}{1 - \cot \theta} \right)^2 = \left[\frac{\tan \theta [\cot \theta - 1]}{1 - \cot \theta} \right]^2$$

$$= (-\tan \theta)^2 = \tan^2 \theta$$

$$\therefore \text{LHS} = \text{RHS}$$

$$\text{II. } \cot \theta = \frac{1}{\tan \theta} \Rightarrow \tan \theta \cdot \cot \theta = 1$$

which true for all value of θ .

So, statement II is false.

Hence, only statement I is correct.

195. (d) We have, $A + B + C = \pi$

$$\Rightarrow \frac{\pi}{2} + B + B + C = \pi [A = B + \pi/2]$$

$$\Rightarrow 2B + C = \pi - \frac{\pi}{2} = \frac{\pi}{2}$$

$$196. (c) \left(\frac{\sin 35^\circ}{\cos 55^\circ} \right)^2 - \left(\frac{\cos 55^\circ}{\sin 35^\circ} \right)^2 + 2 \sin 30^\circ$$

$$= \left[\frac{\sin(90^\circ - 55^\circ)}{\cos 55^\circ} \right]^2$$

$$- \left[\frac{\cos(90^\circ - 35^\circ)}{\sin 35^\circ} \right]^2 + 2 \times \frac{1}{2}$$

$$= \left(\frac{\cos 55^\circ}{\cos 55^\circ} \right)^2 - \left(\frac{\sin 35^\circ}{\sin 35^\circ} \right)^2 + 1$$

$$= 1^2 - 1^2 + 1 = 1$$

197. (c) We have, $\tan \theta + \cot \theta = \frac{4}{\sqrt{3}}$

$$\Rightarrow \frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta} = \frac{4}{\sqrt{3}}$$

$$\Rightarrow \frac{\sin^2 \theta + \cos^2 \theta}{\cos \theta \cdot \sin \theta} = \frac{4}{\sqrt{3}}$$

$$\Rightarrow \frac{1}{\sin 2\theta} = \frac{4}{\sqrt{3}}$$

$$\Rightarrow \sin 2\theta = \frac{\sqrt{3}}{2}$$

$$\Rightarrow 2\theta = \frac{\pi}{3} \Rightarrow \theta = \frac{\pi}{6}$$

$$\therefore \sin \theta + \cos \theta = \sin \frac{\pi}{6} + \cos \frac{\pi}{6}$$

$$= \frac{1}{2} + \frac{\sqrt{3}}{2} = \frac{\sqrt{3} + 1}{2}$$

198. (b) We have,

$$p = \cot \theta + \tan \theta = \csc \theta \cdot \sec \theta$$

$$q = \sec \theta - \cos \theta = \sin^2 \theta \cdot \sec \theta$$

$$\therefore (p^2 q)^{2/3} - (q^2 p)^{2/3}$$

$$= (\csc^2 \theta \cdot \sec^2 \theta \cdot \sin^2 \theta \cdot \sec \theta)^{2/3}$$

$$- (\sin^4 \theta \cdot \sec^2 \theta \cdot \csc \theta \cdot \sec \theta)^{2/3}$$

$$= (\sec^3 \theta)^{2/3} - (\sin^3 \theta \cdot \sec^3 \theta)^{2/3}$$

$$= \sec^2 \theta - \sin^2 \theta \cdot \sec^2 \theta$$

$$= \sec^2 \theta (1 - \sin^2 \theta) = 1$$

199. (d) Given, $\frac{x}{a} - \frac{y}{b} \tan \theta = 1$... (i)

$$\text{and } \frac{x}{a} \tan \theta + \frac{y}{b} = 1 \quad \dots \text{(ii)}$$

On solving Eqs. (i) and (ii), we get

$$\frac{x}{a} = \frac{1 + \tan \theta}{\sec^2 \theta} \text{ and } \frac{y}{b} = \frac{1 - \tan \theta}{\sec^2 \theta}$$

$$\therefore \frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{(1 + \tan \theta)^2 + (1 - \tan \theta)^2}{\sec^4 \theta}$$

$$= \frac{2 \sec^2 \theta}{\sec^4 \theta}$$

$$= 2 \cos^2 \theta$$

200. (c) $\therefore 3 - \tan^2 \theta = \alpha (1 - 3 \tan^2 \theta)$

$$\Rightarrow (3\alpha - 1) \tan^2 \theta = \alpha - 3$$

$$\Rightarrow \tan^2 \theta = \frac{\alpha - 3}{3\alpha - 1}$$

$$\text{As, } \tan^2 \theta \geq 0, \text{ then } \frac{\alpha - 3}{3\alpha - 1} \geq 0 \Rightarrow \alpha \geq 3$$

$$\text{or, } \alpha < \frac{1}{3} \Rightarrow \alpha \in \left(-\infty, \frac{1}{3} \right) \cup [3, \infty)$$

201. (a) To exchange the position, both hands has to cover 360° together.

$$\text{Angle traced by hour hand in 1 min} = (1/2)^\circ$$

Angle traced by minute hand in 1 min = 6°

Let the required time be t min. Then,

$$6t + \frac{1}{2}t = 360^\circ \Rightarrow t = \frac{360 \times 2}{13}$$

$$= \frac{720}{13} \text{ min} = 55.38 \text{ min}$$

202. (c) I. $\sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}} = \sqrt{\frac{1 - \cos \theta}{1 + \cos \theta} \times \frac{1 - \cos \theta}{1 - \cos \theta}}$

$$= \sqrt{\frac{(1 - \cos \theta)^2}{1 - \cos^2 \theta}} = \sqrt{\frac{(1 - \cos \theta)^2}{\sin^2 \theta}}$$

$$= \frac{1 - \cos \theta}{\sin \theta} = \csc \theta - \cot \theta$$

Hence, statement I is true.

$$\text{II. } \sqrt{\frac{1 + \cos \theta}{1 - \cos \theta}} = \sqrt{\frac{1 + \cos \theta}{1 - \cos \theta} \times \frac{1 + \cos \theta}{1 + \cos \theta}}$$

$$= \sqrt{\frac{(1 + \cos \theta)^2}{\sin^2 \theta}} = \frac{1 + \cos \theta}{\sin \theta}$$

$$= \csc \theta + \cot \theta$$

Hence, statement II is true.

203. (b) $\therefore \frac{\cos^2 \theta - 3 \cos \theta + 2}{\sin^2 \theta} = 1$

$$\Rightarrow \cos^2 \theta - 3 \cos \theta + 1 + 1 - \sin^2 \theta = 0$$

$$\Rightarrow \cos^2 \theta - 3 \cos \theta + 1 + \cos^2 \theta = 0$$

$$\Rightarrow 2 \cos^2 \theta - 3 \cos \theta + 1 = 0$$

$$\Rightarrow (2 \cos \theta - 1)(\cos \theta - 1) = 0$$

$$\Rightarrow \cos \theta = \frac{1}{2} \text{ or, } \cos \theta = 1$$

$$\Rightarrow \theta = \frac{\pi}{3} [\because 0 < \theta < \pi/2]$$

Hence, only II is correct.

204. (d) I. As, $\cos x$ lies between -1 and 1 , then $\cos x = 2^{m+1}$ does not exist for positive value of m .

II. The given relation $m \geq m + n$ is true for $m, n \in N$ and $m > 1$ and $n > 1$.

Thus, statement II is not true.

Hence, neither I nor II is correct

HEIGHT AND DISTANCE

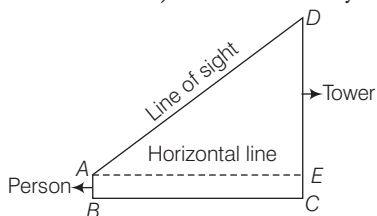
Regularly (1-2) questions have been asked from this chapter. Generally this is the simplest application of trigonometry related to our real world and a little practice can get you command over this section.



It is an important application of trigonometry which helps us to find the height of any object and distance of that object from any point which are not directly measurable. If the angle of elevation/depression from a point is known.

Line of sight

A line of sight is the line drawn from the eye of an observer to the point, where the object is viewed by the observer.



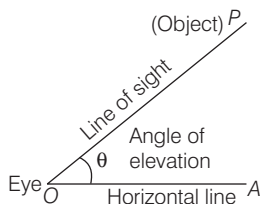
Horizontal Line

The line of sight which is parallel to ground level is known as horizontal line.

ANGLE OF ELEVATION

Angle of elevation is defined as an angle formed by the line of sight with the horizontal line when an object is viewed in the upward direction.

Let P be the position of the object above the horizontal line OA and O be the eye of the observer, then $\angle AOP$ is called angle of elevation.



IMPORTANT POINTS

- If the observer moves towards the perpendicular line (tower/building), then angle of elevation increases and if the observer moves away from the perpendicular line (tower/building), then the angle of elevation decreases.
- If the angle of elevation of Sun above a tower decreases, then the length of shadow of a tower increases.
- If height of tower is doubled and the distance between the observer and foot of the tower is also doubled, then the angle of elevation remains same.

EXAMPLE 1. The angle of elevation of the top of a tower of height $100\sqrt{3}$ m from a point at a distance of 100 m from the foot of the tower on a horizontal plane is

- a. 30° b. 60° c. 75° d. 105°

Sol. b. Let ϕ be the angle of elevation.

Given, $AC = \text{Height of tower}$
 $= 100\sqrt{3}$ m

and $AB = 100$ m

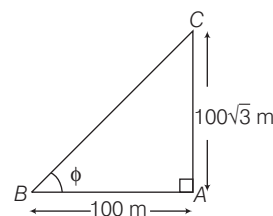
In right angled $\triangle ABC$,

$$\tan \phi = \frac{AC}{AB} = \frac{100\sqrt{3}}{100} = \sqrt{3}$$

$$\Rightarrow \tan \phi = \tan 60^\circ$$

$$\Rightarrow \phi = 60^\circ$$

Hence, angle of elevation is 60° .



EXAMPLE 2. The angle of elevation of a tower at a point is 45° . After going 40 m towards the foot of the tower, the angle of elevation of the tower becomes 60° . Then, the height of the tower is

- a. $20(\sqrt{3} + 1)$ m b. $\frac{20\sqrt{3}}{\sqrt{3} + 1}$ m
c. $\frac{40\sqrt{3}}{\sqrt{3} + 1}$ m d. $\frac{40\sqrt{3}}{\sqrt{3} - 1}$ m

Sol. d. Let $PQ (= h)$ be the height of the tower. Let S and R be the points where the angles subtended are 45° and 60° , respectively.

In right angled ΔPQR ,

$$\tan 60^\circ = \frac{PQ}{RQ}$$

$$\Rightarrow \sqrt{3} = \frac{h}{x} \Rightarrow x = \frac{h}{\sqrt{3}} \quad \dots(i)$$

In right angled ΔPQS , $\tan 45^\circ = \frac{PQ}{SQ} = \frac{PQ}{SR + RQ} = \frac{h}{40 + x}$

$$\Rightarrow h = 40 + x \Rightarrow h = 40 + \frac{h}{\sqrt{3}} \quad [\text{from Eq. (i)}]$$

$$\Rightarrow h \left(1 - \frac{1}{\sqrt{3}}\right) = 40 \Rightarrow h \left(\frac{\sqrt{3} - 1}{\sqrt{3}}\right) = 40$$

$$\therefore h = \frac{40\sqrt{3}}{\sqrt{3} - 1} \text{ m}$$

EXAMPLE 3. The angles of elevation of the top of a tower from two points at distances ' a ' and ' b ' ($a > b$) from the base and in the same straight line with it are complementary. Then, the height of the tower is

- a. $\frac{1}{\sqrt{ab}}$ b. $\sqrt{ab}$ c. $\frac{a}{b}$ d. ab

Sol. b. Let $AD = h$ be the height of the tower and the angle of elevation of the top of tower at B and C be $(90^\circ - \theta)$ and θ , respectively.

In right angled ΔADC ,

$$\tan \theta = \frac{AD}{CD} = \frac{h}{b} \quad \left[\because \tan \theta = \frac{P}{B} \right]$$

$$h = b \tan \theta \quad \dots(i)$$

In right angled ΔADB , $\tan (90^\circ - \theta) = \frac{AD}{BD} \Rightarrow \cot \theta = \frac{h}{a}$
[$\because \tan (90^\circ - \theta) = \cot \theta$]

$$\Rightarrow \frac{h}{a} = \cot \theta \Rightarrow \frac{a}{h} = \tan \theta \quad \dots(ii)$$

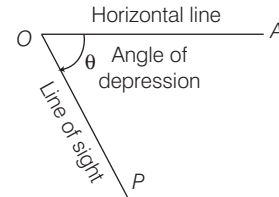
From Eqs. (i) and (ii), we get

$$h = b \left(\frac{a}{h}\right) \Rightarrow h^2 = ab$$

$$\therefore h = \sqrt{ab}$$

ANGLE OF DEPRESSION

Angle of depression is defined as an angle formed by the line of sight with the horizontal line when an object is viewed in the downward direction.



Let P be the position of the object below the horizontal line OA and O be the eye of the observer, then $\angle AOP$ is called angle of depression.

IMPORTANT POINTS

- The angle of depression of a point P as seen from a point O is numerically equal to the angle of elevation of O as seen from P .
- Angle of elevation and angle of depression are always acute angle.
- Unless mentioned, the height of the observer is not considered.

EXAMPLE 4. The angle of depression of two ships from the top of a light house are 45° and 30° towards East. If the ships are 200 m apart, the height of the light house is

- a. $\frac{200}{\sqrt{3} - 1}$ m b. $\frac{200}{\sqrt{3} + 1}$ m c. $\frac{100}{\sqrt{3} - 1}$ m d. $\frac{100}{\sqrt{3} + 1}$ m

Sol. a. Let $AB (= h)$ be the height of the light house and D, C be the ships.

Given, $DC = 200$ m

$$\angle ADC = \angle XAD = 30^\circ$$

$$\text{and } \angle ACB = \angle XAC = 45^\circ$$

Now, in ΔACB ,

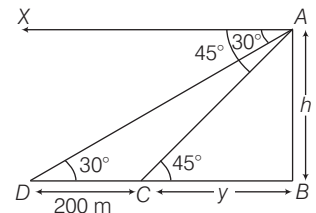
$$\tan 45^\circ = \frac{AB}{CB} \Rightarrow 1 = \frac{h}{y} \Rightarrow h = y \quad \dots(i)$$

$$\text{and in } \Delta ADB, \tan 30^\circ = \frac{AB}{BD} \Rightarrow \frac{1}{\sqrt{3}} = \frac{h}{200 + y}$$

$$\Rightarrow 200 + y = \sqrt{3} h \Rightarrow 200 + h = \sqrt{3} h \quad [\text{from Eq. (i)}]$$

$$\Rightarrow 200 = \sqrt{3} h - h \Rightarrow 200 = h(\sqrt{3} - 1)$$

$$\therefore h = \left(\frac{200}{\sqrt{3} - 1}\right) \text{ m}$$



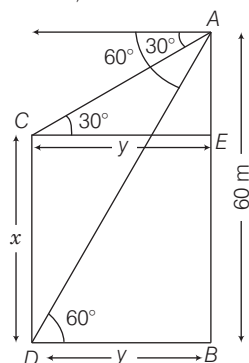
EXAMPLE 5. From the top of a building 60 m high the angles of depression of the top and bottom of a tower are observed to be 30° and 60° . Then, the height of the tower is

- a. 10 m b. 20 m c. 30 m d. 40 m

Sol. d. Let x be the height of the tower.

Here, $AB = 60$ m and $BD = y$ m

In right angled $\triangle ABD$,



$$\frac{AB}{BD} = \tan 60^\circ \Rightarrow \frac{60}{y} = \sqrt{3}$$

$$\Rightarrow y = \frac{60}{\sqrt{3}} \quad \dots(i)$$

In right angled $\triangle AEC$,

$$\frac{AE}{EC} = \tan 30^\circ \Rightarrow \frac{AE}{BD} = \frac{1}{\sqrt{3}} \quad [\because CE = DB]$$

$$\Rightarrow \frac{AE}{y} = \frac{1}{\sqrt{3}} \Rightarrow \frac{AE}{60} = \frac{1}{\sqrt{3}} \quad [\text{from Eq. (i)}]$$

$$\Rightarrow AE = \frac{1}{\sqrt{3}} \times \frac{60}{\sqrt{3}} = 20 \text{ m} \quad [\because \sqrt{3} \times \sqrt{3} = 3]$$

$$\therefore x = AB - AE = 60 - 20 = 40 \text{ m}$$

Hence, the height of tower is 40 m

> PRACTICE EXERCISE

- The length of shadow of a tree is 16 m when the angle of elevation of the Sun is 60° . What is the height of the tree?
(a) 8 m (b) 16 m (c) $16\sqrt{3}$ m (d) $\sqrt{3}$ m
- From a light house the angles of depression of two ships on opposite sides of the light house are observed to 30° and 45° . If the height of light house is h , then what is the distance between the ships?
(a) $(\sqrt{3} + 1)h$ (b) $(\sqrt{3} - 1)h$ (c) $\sqrt{3}h$ (d) $\left(1 + \frac{1}{\sqrt{3}}\right)h$
- The angle of elevation of the top of a tower from the bottom of a building is twice that from its top. What is the height of the building, if the height of the tower is 75 m and the angle of elevation of the top of the tower from the bottom of the building is 60° ?
(a) 25 m (b) 37.5 m (c) 50 m (d) 60 m
- A radio transmitter antenna of height 100 m stands at the top of a tall building. At a point on the ground, the angle of elevation of bottom of the antenna is 45° and that of top of antenna is 60° . What is the height of the building?
(a) 100 m (b) 50 m (c) $50(\sqrt{3} + 1)$ m (d) $25\sqrt{3}$ m
- The shadow of a tower is 15 m when the Sun's altitude is 30° . What is the length of the shadow when the Sun's altitude is 60° ?
(a) 3 m (b) 4 m (c) 5 m (d) 6 m
- The angle of elevation of a moon when the length of the shadow of a pole is equal to its height is
(a) 60° (b) 45° (c) 90° (d) 30°
- An aeroplane flying horizontally 1 km above the ground is observed at an elevation of 60° . After 10 s, its elevation is observed to be 30° , the uniform speed of the aeroplane in km/h is
(a) $\frac{240}{\sqrt{2}}$ (b) $240\sqrt{3}$
(c) 240 (d) $260\sqrt{3}$
- A person of height 2m wants to get a fruit which is on a pole of height $(10/3)$ m. If he stands at a distance of $(4/\sqrt{3})$ m from the foot of the pole, then the angle at which he should throw the stone, so that it hits the fruit is
(a) 60° (b) 45° (c) 90° (d) 30°
- A vertical tower stands on a horizontal plane and is surmounted by a vertical flagstaff of height h . At a point on the plane, the angle of elevation of the bottom of the flagstaff is α and that of the top of the flagstaff is β . Then, the height of the tower is
(a) $\frac{h \tan \beta}{\tan \alpha - \tan \beta}$ (b) $\frac{h \tan \alpha}{\tan \alpha + \tan \beta}$
(c) $\frac{h \tan \beta}{\tan \alpha + \tan \beta}$ (d) $\frac{h \tan \alpha}{\tan \beta - \tan \alpha}$
- From a window (h m high above the ground) of a house in a street, the angle of elevation and depression of the top and the foot of another house on the opposite side of the street are θ and ϕ , respectively. Then, the height of the opposite house is
(a) $h \tan \theta \cot \phi$ (b) $h [\tan \theta \cot \phi + 1]$
(c) $h [\cot \theta \tan \phi + 1]$ (d) $h \cot \theta \tan \phi$

- 11.** The length of the shadow of a person s cm tall when the angle of elevation of the Sun is α is p cm. It is q cm when the angle of elevation of the Sun is β . Which one of the following is correct when $\beta = 3\alpha$?
- (a) $p - q = s \left(\frac{\tan \alpha - \tan 3\alpha}{\tan 3\alpha \tan \alpha} \right)$ (b) $p - q = s \left(\frac{\tan 3\alpha - \tan \alpha}{3 \tan 3\alpha \tan \alpha} \right)$
 (c) $p - q = s \left(\frac{\tan 3\alpha - \tan \alpha}{\tan 3\alpha \tan \alpha} \right)$ (d) $p - q = s \left(\frac{\tan 2\alpha}{\tan 3\alpha \tan \alpha} \right)$
- 12.** Two posts are ' K ' m apart and the height of one is double that of the other. If from the middle point of the line joining their feet, an observer finds the angular elevations of their tops to be complementary, then the height (in metre) of the shorter post is
- (a) $\sqrt{3} K$ (b) $\frac{K}{2\sqrt{2}}$ (c) $2 K$ (d) $\frac{K}{\sqrt{2}}$
- 13.** A flagstaff 5 m high stands on a building of 25 m height. At an observer at a height of 30 m, the flagstaff and the building subtend equal angles. The distance of the observer from the top of the flagstaff is
- (a) $\frac{3}{2}$ m (b) $3\sqrt{\frac{5}{2}}$ m (c) $\frac{5\sqrt{3}}{2}$ m (d) $5\sqrt{\frac{3}{2}}$ m
- 14.** A man is watching from the top of a tower a boat speeding away from the tower. The boat makes an angle of depression of 45° with the man's eye when at a distance of 60 m from the bottom of tower. After 5 s, the angle of depression becomes 30° . What is the approximate speed of the boat assuming that it is running in still water?
- (a) 31.5 km/h (b) 36.5 km/h (c) 38.5 km/h (d) 40.5 km/h
- 15.** Suppose the angle of elevation of the top of a tree at a point E due East of the tree is 60° and that at a point F due West of the tree is 30° . If the distance between the points E and F is 160 ft, then what is the height of the tree?
- (a) $40\sqrt{3}$ ft (b) 60 ft (c) $\frac{40}{\sqrt{3}}$ ft (d) 23 ft
- 16.** A telegraph post gets broken at a point against a storm and its top touches the ground at a distance 20 m from the base of the post making an angle 30° with the ground. What is the height of the post?
- (a) $\frac{40}{\sqrt{3}}$ m (b) $20\sqrt{3}$ m (c) $40\sqrt{3}$ m (d) 30 m
- 17.** If the angle of elevation of a cloud from a point h m above a lake is β and the angle of depression of its reflection in the lake is α , then the height of the cloud is
- (a) $\frac{h \operatorname{cosec}(\alpha - \beta)}{\operatorname{cosec}(\alpha - \beta)}$ (b) $h \operatorname{cosec}(\alpha - \beta) \sin(\alpha - \beta)$
 (c) $h \sin(\alpha + \beta) \operatorname{cosec}(\alpha - \beta)$ (d) $\frac{h \operatorname{cosec}(\alpha + \beta)}{\sin(\alpha - \beta)}$
- 18.** The angle of elevation of the top of an incomplete vertical pillar at a horizontal distance of 100 m from its base is 45° . If the angle of elevation of the top of complete pillar at the same point is to be 60° , then the height of the incomplete pillar is to be increased by
- (a) $50\sqrt{2}$ m (b) 100 m
 (c) $100(\sqrt{3} - 1)$ m (d) $100(\sqrt{3} + 1)$ m
- 19.** A man on the top of a vertical observation tower observes a car moving at a uniform speed coming directly towards it. If it takes 12 min for the angle of depression to change from 30° to 45° , how soon after this will the car reach the observation tower?
- (a) 14 min 35 s (b) 15 min 49 s
 (c) 16 min 23 s (d) 18 min 5 s
- 20.** You are stationed at a radar base and you observe an unidentified plane at an altitude $h = 1000$ m flying towards your radar base at an angle of elevation $= 30^\circ$. After exactly one minute, your radar sweep reveals that the plane is now at an angle of elevation $= 60^\circ$ maintaining the same altitude. What is the speed (in m/s) of the plane?
- (a) 15.58 m/s (b) 19.25 m/s
 (c) 18 m/s (d) 11.25 m/s
- 21.** The angle of elevation of the top of the tower from a point on the ground is $\sin^{-1}(3/5)$. If the point of observation is 20 m away from the foot of the tower, what is the height of the tower?
- (a) 9 m (b) 18 m (c) 15 m (d) 12 m
- 22.** A balloon of radius r makes an angle α at the eye of an observer and the angle of elevation of its centre is β . The height of its centre from the ground level is given by
- (a) $r \sin \beta \operatorname{cosec} \alpha / 2$ (b) $r \operatorname{cosec} \alpha / 2 \sin \alpha$
 (c) $r \operatorname{cosec} \alpha \sin \beta$ (d) None of these
- Directions** (Q. Nos. 23-25) The height of a tower is h and the angle of elevation of the top of the tower is α . When observer on moving a distance $h/2$ towards the tower, the angle of elevation becomes β .
- 23.** What is the value of $\cot \alpha - \cot \beta$?
- (a) $1/2$ (b) $2/3$ (c) 1 (d) 2
- 24.** If $\alpha = 30^\circ$, then find the value of $\cot \beta$.
- (a) $\sqrt{3} - 1/2$ (b) $\frac{2}{2\sqrt{3} - 2}$
 (c) $\frac{1}{1 + \sqrt{3}}$ (d) $\frac{1}{1 - \sqrt{3}}$
- 25.** If $\tan \beta = 4$, then distance between observer and tower when angle of elevation is β
- (a) h (b) $h/4$ (c) $\frac{3h}{4}$ (d) $\frac{h}{3}$

PREVIOUS YEARS' QUESTIONS

- 26.** What is the angle of elevation of the Sun when the shadow of a pole is $\sqrt{3}$ times the length of the pole? ☑ 2012 I
 (a) 30° (b) 45°
 (c) 60° (d) None of these
- 27.** From the top of a cliff 200 m high, the angles of depression of the top and bottom of a tower are observed to be 30° and 45° , respectively. What is the height of the tower? ☑ 2012 I
 (a) 400 m (b) $400\sqrt{3}$ m
 (c) $400/\sqrt{3}$ m (d) None of these
- 28.** The angles of elevation of the top of a tower from two points which are at distances of 10 m and 5 m from the base of the tower and in the same straight line with it are complementary. The height of the tower is ☑ 2012 II
 (a) 5 m (b) 15 m (c) $\sqrt{50}$ m (d) $\sqrt{75}$ m
- 29.** A ladder 20 m long is placed against a wall, so that the foot of the ladder is 10 m from the wall. The angle of inclination of the ladder to the horizontal will be ☑ 2012 II
 (a) 30° (b) 45°
 (c) 60° (d) 75°

Directions (Q. Nos. 30-33) Read the following information carefully to answer the questions that follow.

As seen from the top and bottom of a building of height h m, the angles of elevation of the top of a tower of

height $\frac{(3 + \sqrt{3})h}{2}$ m are α and β , respectively. ☑ 2013 I

- 30.** If $\beta = 30^\circ$, then what is the value of $\tan \alpha$?
 (a) $1/2$ (b) $1/3$
 (c) $1/4$ (d) None of these
- 31.** If $\alpha = 30^\circ$, then what is the value of $\tan \beta$?
 (a) 1 (b) $1/2$
 (c) $1/3$ (d) None of these
- 32.** If $\alpha = 30^\circ$ and $h = 30$ m, then what is the distance between the base of the building and the base of the tower?
 (a) $(15 + 15\sqrt{3})$ m (b) $(30 + 15\sqrt{3})$ m
 (c) $(45 + 15\sqrt{3})$ m (d) None of these
- 33.** If $\beta = 30^\circ$ and if θ is the angle of depression of the foot of the tower as seen from the top of the building, then what is the value of $\tan \theta$?
 (a) $\frac{(3 - \sqrt{3})}{3\sqrt{3}}$ (b) $\frac{(3 + \sqrt{3})}{3\sqrt{3}}$
 (c) $\frac{(2 - \sqrt{3})}{3\sqrt{3}}$ (d) None of these

- 34.** What is the angle of elevation of the Sun, when the shadow of a pole of height x m is $\frac{x}{\sqrt{3}}$ m? ☑ 2013 II
 (a) 30° (b) 45° (c) 60° (d) 75°
- 35.** A spherical balloon of radius r subtends angle 60° at the eye of an observer. If the angle of elevation of its centre is 60° and h is the height of the centre of the balloon, then which one of the following is correct? ☑ 2013 II
 (a) $h = r$ (b) $h = \sqrt{2}r$
 (c) $h = \sqrt{3}r$ (d) $h = 2r$
- 36.** The angle of elevation of the top of a tower 30 m high from the foot of another tower in the same plane is 60° and the angle of elevation of the top of the second tower from the foot of the first tower is 30° . The distance between the two towers is m times the height of the shorter tower. What is m equal to? ☑ 2014 I
 (a) $\sqrt{2}$ (b) $\sqrt{3}$ (c) $\frac{1}{2}$ (d) $\frac{1}{3}$
- 37.** The shadow of a tower standing on a level plane is found to be 50 m longer when the Sun's elevation is 30° , then when it is 60° . What is the height of the tower? ☑ 2014 I
 (a) 25 m (b) $25\sqrt{3}$ m (c) $\frac{25}{\sqrt{3}}$ m (d) 30 m
- 38.** From a certain point on a straight road, a person observe a tower in the West direction at a distance of 200 m. He walks some distance along the road and finds that the same tower is 300 m South of him. What is the shortest distance of the tower from the road? ☑ 2014 II
 (a) $\frac{300}{\sqrt{13}}$ m (b) $\frac{500}{\sqrt{13}}$ m (c) $\frac{600}{\sqrt{13}}$ m (d) $\frac{900}{\sqrt{13}}$ m
- 39.** If from the top of a post a string twice the length of the post is stretched tight to a point on the ground, then what angle will the string make with the post? ☑ 2014 II
 (a) $\frac{\pi}{6}$ (b) $\frac{\pi}{4}$ (c) $\frac{5\pi}{12}$ (d) $\frac{\pi}{3}$
- 40.** The angle of elevation of a cloud from a point 200 m above a lake is 30° and the angle of depression of its reflection in the lake is 60° . The height of the cloud is ☑ 2015 I
 (a) 200 m (b) 300 m (c) 400 m (d) 600 m
- 41.** From the top of a tower, the angles of depression of two objects P and Q (situated on the ground on the same side of the tower) separated at a distance of $100(3 - \sqrt{3})$ m are 45° and 60° , respectively. The height of the tower is ☑ 2015 I
 (a) 200 m (b) 250 m
 (c) 300 m (d) None of these

- 42.** The angles of elevation of the top of a tower from two points P and Q at distance m^2 and n^2 respectively, from the base and in the same straight line with it are complementary. The height of the tower is **2015 I**

(a) $(mn)^{1/2}$ (b) $mn^{1/2}$
(c) $m^{1/2}n$ (d) mn

- 43.** A pole is standing erect on the ground which is horizontal. The tip of the pole is tied tight with a rope of length $\sqrt{12}$ m to a point on the ground. If the rope is making 30° with the horizontal, then the height of the pole is **2015 II**

(a) $2\sqrt{3}$ m (b) $3\sqrt{2}$ m
(c) 3 m (d) $\sqrt{3}$ m

- 44.** An aeroplane flying at a height of 3000 m passes vertically above another aeroplane at an instant, when the angles of elevation of the two planes from some point on the ground are 60° and 45° , respectively. Then, the vertical distance between the two planes is **2015 II**

(a) $1000(\sqrt{3}-1)$ m (b) $1000\sqrt{3}$ m
(c) $1000(3-\sqrt{3})$ m (d) $3000\sqrt{3}$ m

- 45.** Two observers are stationed due North of a tower (of height x m) at a distance y m from each other. The angles of elevation of the tower observed by them are 30° and 45° , respectively. Then, x/y is equal to **2016 I**

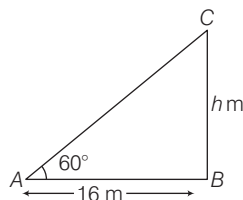
(a) $\frac{\sqrt{2}-1}{2}$ (b) $\frac{\sqrt{3}-1}{2}$ (c) $\frac{\sqrt{3}+1}{2}$ (d) 1

ANSWERS

1	c	2	a	3	c	4	c	5	c	6	b	7	b	8	d	9	d	10	b
11	c	12	b	13	d	14	a	15	a	16	b	17	c	18	c	19	c	20	b
21	c	22	a	23	a	24	a	25	b	26	a	27	d	28	c	29	c	30	b
31	a	32	c	33	a	34	c	35	c	36	b	37	b	38	c	39	d	40	c
41	c	42	d	43	d	44	c	45	c										

HINTS AND SOLUTIONS

- 1.** (c) Let the height of the tree be h m.
 $\therefore BC = h$ m and $AB = 16$ m



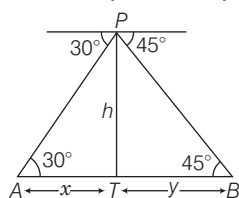
In right angled $\triangle ABC$,

$$\tan 60^\circ = \frac{BC}{AB} = \frac{h}{16} \Rightarrow \sqrt{3} = \frac{h}{16}$$

$$\Rightarrow h = 16\sqrt{3} \text{ m}$$

Hence, the height of the tree is $16\sqrt{3}$ m.

- 2.** (a) In right angled $\triangle PBT$, $\tan 45^\circ = \frac{h}{y} \Rightarrow 1 = \frac{h}{y}$



$$\therefore y = h \quad \dots(i)$$

and in right angled $\triangle PTA$,

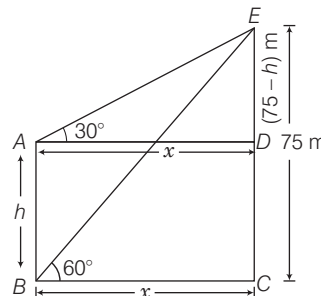
$$\tan 30^\circ = \frac{h}{x} \Rightarrow x = \sqrt{3}h \quad \dots(ii)$$

$\therefore$ Required distance, $AB = x + y$

$$x + y = \sqrt{3}h + h = h(\sqrt{3} + 1) \text{ m}$$

- 3.** (c) Let height of the building be h m and distance between building and tower be x m.

$\therefore AB = h$ m and $BC = x$ m



In right angled $\triangle ADE$,

$$\tan 30^\circ = \frac{ED}{AD} \Rightarrow \frac{1}{\sqrt{3}} = \frac{75-h}{x}$$

$$\Rightarrow x = 75\sqrt{3} - h\sqrt{3} \quad \dots(i)$$

and in right angled $\triangle BCE$,

$$\tan 60^\circ = \frac{CE}{BC} \Rightarrow \sqrt{3} = \frac{75}{x}$$

$$\Rightarrow x\sqrt{3} = 75$$

$$\Rightarrow (75\sqrt{3} - h\sqrt{3})\sqrt{3} = 75 \text{ [from Eq. (i)]}$$

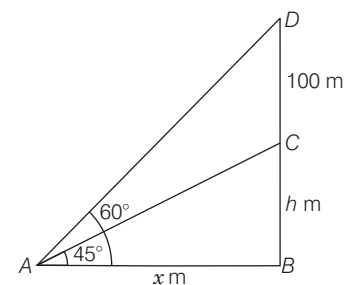
$$\Rightarrow 75 \times 3 - 3h = 75$$

$$\Rightarrow 3h = 75 \times 3 - 75 \Rightarrow h = \frac{75 \times 2}{3}$$

$$\therefore h = 50 \text{ m}$$

Hence, the height of the building is 50 m.

- 4.** (c) Let BC be a building of height h m and CD be a height of antenna. and let distance between A and $B = x$ m



In right angled $\triangle ABC$,

$$\tan 45^\circ = \frac{BC}{AB} = \frac{b}{x} \Rightarrow x = b \quad \dots(i)$$

and in right angled $\triangle ABD$,

$$\tan 60^\circ = \frac{DB}{BA} = \frac{DC + CB}{AB} = \frac{100 + b}{x}$$

$$\Rightarrow \sqrt{3} = \frac{100 + b}{b} \quad [\text{from Eq. (i)}]$$

$$\Rightarrow \sqrt{3}b = 100 + b$$

$$\Rightarrow (\sqrt{3} - 1)b = 100 \Rightarrow b = \frac{100}{\sqrt{3} - 1}$$

$$\Rightarrow b = \frac{100}{(\sqrt{3} - 1)} \times \frac{(\sqrt{3} + 1)}{(\sqrt{3} + 1)}$$

$$= 50(\sqrt{3} + 1) \text{ m}$$

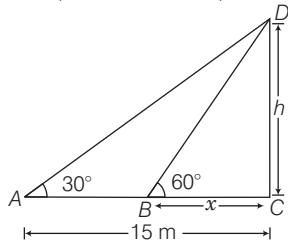
Hence, the height of the building is $50(\sqrt{3} + 1)$ m.

5. (c) Let the height of the tower be b m and length of the shadow (BC) be x m.

In right angled $\triangle ACD$,

$$\tan 30^\circ = \frac{CD}{AC}$$

$$\Rightarrow \frac{1}{\sqrt{3}} = \frac{b}{15} \Rightarrow b = \frac{15}{\sqrt{3}} \text{ m} \quad \dots(i)$$



and in right angled $\triangle BCD$,

$$\tan 60^\circ = \frac{CD}{BC} \Rightarrow \sqrt{3} = \frac{b}{x} \Rightarrow \frac{b}{\sqrt{3}} = x$$

$$\therefore x = \frac{15}{3} = 5 \text{ m}$$

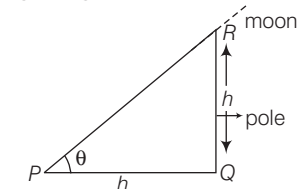
[from Eq. (i)]

Hence, the length of shadow is 5 m.

When sun's altitude is 60° .

6. (b) Let the height of pole be b m, then $PQ = RQ = b$

In right angled $\triangle PRQ$,



$$\tan \theta = \frac{RQ}{PQ} = \frac{b}{h} = 1$$

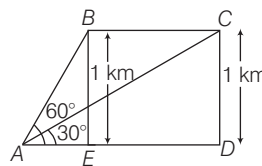
$$\Rightarrow \tan \theta = 1 = \tan 45^\circ \therefore \theta = 45^\circ$$

Hence, the angle of elevation of moon is 45° .

7. (b) Let A be the observer. When the aeroplane was at point B the angle of elevation was 60° and after 10 s when it was at point C the angle of elevation was 30° .

Here, in the figure,

$$BE = CD = 1 \text{ km} = 1000 \text{ m}$$



In right angled $\triangle BAE$,

$$\tan 60^\circ = \frac{BE}{AE} = \frac{1000}{AE}$$

$$\Rightarrow AE = 1000 \cot 60^\circ = \frac{1000}{\sqrt{3}} \text{ m} \quad \dots(i)$$

In right angled $\triangle ACD$,

$$\tan 30^\circ = \frac{DC}{AD} = \frac{1000}{AD}$$

$$\Rightarrow AD = 1000 \cot 30^\circ = 1000\sqrt{3}$$

$$\therefore ED = AD - AE = 1000\sqrt{3} - \frac{1000}{\sqrt{3}} = \frac{2000}{\sqrt{3}} \text{ m}$$

So, distance travelled by plane in 10 s, then

$$BC = DE = \frac{2000}{\sqrt{3}} \text{ m}$$

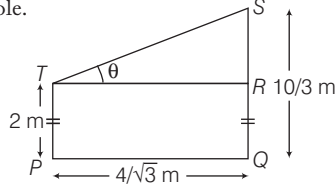
$$\therefore \text{Speed of aeroplane} = \frac{\text{Distance}}{\text{Time}}$$

$$= \frac{\frac{2000}{\sqrt{3}}}{10} \text{ m/s} = \frac{2000 \times 60 \times 60}{10\sqrt{3} \times 1000} \text{ km/h}$$

$$= 240\sqrt{3} \text{ km/h}$$

Hence, the uniform speed of the aeroplane is $240\sqrt{3}$ km/h.

8. (d) Let TP be the man and SQ be the pole.



Let $\angle STR = \theta$

Now, $SR = SQ - RQ$

$$= \frac{10}{3} - 2 = \frac{4}{3} \text{ m} [\because RQ = TP]$$

In right angled $\triangle STR$,

$$\tan \theta = \frac{SR}{TR} = \frac{SR}{PQ} \quad [\because TR = PQ]$$

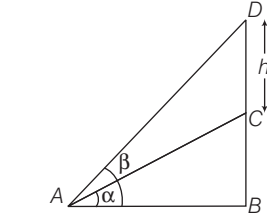
$$= \frac{4}{3} \times \frac{\sqrt{3}}{4} = \frac{1}{\sqrt{3}} = \tan 30^\circ$$

$$\therefore \theta = 30^\circ$$

9. (d) Let BC be the tower and CD be the flagstaff.

$$\therefore \angle BAC = \alpha \text{ and } \angle BAD = \beta$$

In right angled $\triangle ABC$,



$$\tan \alpha = \frac{BC}{AB} \quad \dots(i)$$

and in right angled $\triangle ABD$,

$$\frac{BD}{AB} = \frac{BC + b}{AB} = \tan \beta \quad \dots(ii)$$

On dividing Eq. (ii) by Eq. (i), we get

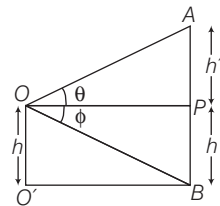
$$\frac{BC + b}{BC} = \frac{\tan \beta}{\tan \alpha}$$

$$\Rightarrow (BC + b) \tan \alpha = BC \tan \beta$$

$$\Rightarrow BC (\tan \beta - \tan \alpha) = b \tan \alpha$$

$$\therefore BC = \frac{b \tan \alpha}{\tan \beta - \tan \alpha}$$

10. (b) Let O be the window and AB be the house on the opposite side of the street.



Let $AP = b'$

and $BP = OO' = b$

In right angled $\triangle AOP$,

$$\tan \theta = \frac{AP}{OP} = \frac{b'}{OP} \quad \dots(i)$$

and in right angled $\triangle BOP$,

$$\tan \phi = \frac{BP}{OP} = \frac{b}{OP} \quad \dots(ii)$$

$$\therefore \frac{b'}{b} = \frac{\tan \theta}{\tan \phi} \quad [\text{by Eq. (i) } \div \text{ Eq. (ii)}]$$

$$\Rightarrow b' = b \tan \theta \cot \phi$$

$\therefore$ Height of the house

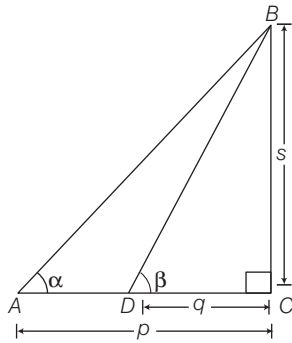
$$AB = AP + PB = b + b'$$

$$= b \tan \theta \cot \phi + b$$

$$= b [\tan \theta \cot \phi + 1]$$

11. (c) In right angled $\triangle BAC$,

$$\tan \alpha = \frac{BC}{AC} = \frac{s}{p} \Rightarrow p = \frac{s}{\tan \alpha} \quad \dots(i)$$



In right angled $\triangle BDC$,

$$\tan \beta = \frac{BC}{DC} = \frac{s}{q} \Rightarrow q = \frac{s}{\tan 3\alpha} \quad \dots(ii)$$

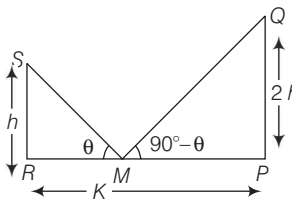
[$\because \beta = 3\alpha$, given]

On subtracting Eq. (ii) from Eq. (i), we get

$$p - q = \frac{s}{\tan \alpha} - \frac{s}{\tan 3\alpha}$$

$$= s \left(\frac{\tan 3\alpha - \tan \alpha}{\tan \alpha \tan 3\alpha} \right)$$

12. (b) Let PQ and RS be the two posts such that $PQ = 2RS$ [given]



Given, M is the mid-point of RP .

$$\therefore RM = PM = \frac{K}{2} \quad [\because RP = K \text{ given}]$$

Let $\angle RMS = \theta$

$$\therefore \angle QMP = 90^\circ - \theta$$

Let $RS = h$, then $PQ = 2h$

Now, in right angled $\triangle PMQ$,

$$\tan (90^\circ - \theta) = \frac{PQ}{MP}$$

$$\Rightarrow \cot \theta = \frac{2h}{K/2} = \frac{4h}{K}$$

$$\text{or } \cot \theta = \frac{4h}{K} \quad \dots(i)$$

In right angled $\triangle SRM$,

$$\tan \theta = \frac{SR}{RM} = \frac{h}{K/2}$$

$$\Rightarrow \frac{2h}{K} = \tan \theta \quad \dots(ii)$$

On multiplying Eq. (i) by Eq. (ii), we get

$$\Rightarrow \frac{4h}{K} \times \frac{2h}{K} = 1$$

$$\Rightarrow 8h^2 = K^2 \Rightarrow h^2 = \frac{K^2}{8}$$

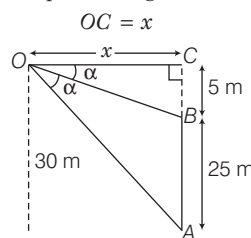
$$\therefore h = \frac{K}{2\sqrt{2}} \text{ m}$$

Hence, the height of shorter post is

$$\frac{K}{2\sqrt{2}} \text{ m.}$$

13. (d) Let O be an observer at a height 30 m. Let x be the distance of observer from the top of the flagstaff.

i.e.



Here, AB be the tower of the flag staff = 25 m and BC be the flag staff = 5 m

In right angled $\triangle OBC$,

$$\tan \alpha = \frac{BC}{OC} = \frac{5}{x}$$

$$\Rightarrow x = 5 \cot \alpha \quad \dots(i)$$

In right angled $\triangle OCA$,

$$\tan 2\alpha = \frac{AC}{OC} = \frac{30}{x} \quad [\because AC = AB + BC]$$

$$\therefore \tan 2\alpha = \frac{30}{5 \cot \alpha} \quad [\text{from Eq. (i)}]$$

$$\Rightarrow \tan 2\alpha = 6 \tan \alpha$$

$$\Rightarrow \frac{2 \tan \alpha}{1 - \tan^2 \alpha} = 6 \tan \alpha$$

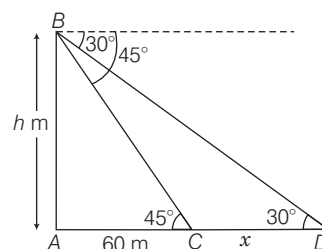
$$\left[\because \tan 2\alpha = \frac{2 \tan \alpha}{1 - \tan^2 \alpha} \right]$$

$$\Rightarrow 3 - 3 \tan^2 \alpha = 1 \Rightarrow \tan^2 \alpha = \frac{2}{3}$$

$$\Rightarrow \tan \alpha = \sqrt{\frac{2}{3}} \text{ or } \cot \alpha = \sqrt{3/2}$$

$$\text{or } x = 5 \sqrt{\frac{3}{2}} \text{ m } [\text{from Eq. (i)}]$$

14. (a) Let height of tower AB be h m and distance between C and D be x m.



In right angled $\triangle ACB$, $\tan 45^\circ = \frac{AB}{AC}$

$$\Rightarrow 1 = \frac{h}{60} \Rightarrow h = 60 \text{ m} \quad \dots(i)$$

Now, in right angled $\triangle ADB$,

$$\tan 30^\circ = \frac{AB}{AD} = \frac{AB}{AC + CD} = \frac{60}{60 + x}$$

[from Eq. (i)]

$$\Rightarrow \frac{1}{\sqrt{3}} = \frac{60}{60 + x} \Rightarrow 60 + x = 60\sqrt{3}$$

$$\Rightarrow x = 60(\sqrt{3} - 1) = 60(1.73 - 1)$$

$$= 60 \times 0.73 = 43.8 \text{ m}$$

Now, given time = 5 s

We know that, Speed = $\frac{\text{Distance}}{\text{Time}}$

$$\therefore \text{Speed of boat} = \frac{43.8}{5} \times \frac{18}{5} = \frac{788.4}{25}$$

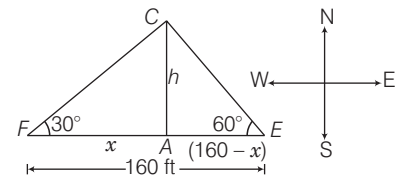
$$= 31.5 \text{ km/h}$$

15. (a) Let $AC = h$ ft = Height of a tree and x ft = Distance between A and F then $AE = (160 - x)$ ft

In right angled $\triangle AFC$,

$$\tan 30^\circ = \frac{AC}{AF} = \frac{h}{x} \Rightarrow \frac{1}{\sqrt{3}} = \frac{h}{x}$$

$$\Rightarrow x = \sqrt{3}h \quad \dots(i)$$



and in right angled $\triangle AEC$,

$$\tan 60^\circ = \frac{AC}{AE} = \frac{h}{160 - x}$$

$$\Rightarrow \sqrt{3} = \frac{h}{160 - x} \quad [\because \tan 60^\circ = \sqrt{3}]$$

$$\Rightarrow \sqrt{3}(160 - x) = h$$

$$\Rightarrow \sqrt{3}(160 - \sqrt{3}h) = h \quad [\text{from Eq. (i)}]$$

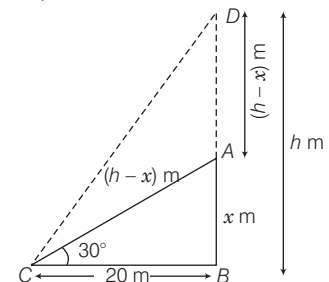
$$\Rightarrow 160\sqrt{3} - 3h = h \Rightarrow 4h = 160\sqrt{3}$$

$$\therefore h = 40\sqrt{3} \text{ ft}$$

Hence, the height of the tree is $40\sqrt{3}$ ft.

16. (b) Let the height of the post be h m, and $AB = x$ m.

Given, $BC = 20$ m



In right angled $\triangle ABC$,

$$\tan 30^\circ = \frac{AB}{BC} \Rightarrow \frac{1}{\sqrt{3}} = \frac{x}{20}$$

$$\Rightarrow x = \frac{20}{\sqrt{3}} \text{ m} \quad \dots(i)$$

$$\text{and } \cos 30^\circ = \frac{BC}{AC} = \frac{20}{b-x}$$

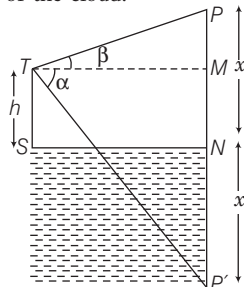
$$\Rightarrow \frac{\sqrt{3}}{2} = \frac{20}{b-x} \Rightarrow b-x = \frac{40}{\sqrt{3}}$$

From Eq. (i), putting the value of x ,

$$b = \frac{40}{\sqrt{3}} + \frac{20}{\sqrt{3}} = \frac{60}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = 20\sqrt{3} \text{ m}$$

Hence, the height of the post is $20\sqrt{3}$ m.

17. (c) Let P be the cloud and P' its image in the lake. Let T be the point 'h' m above the surface of the lake and let x be the height of the cloud.



$$\therefore ST = h$$

$$\text{Also, } NP = NP' = x \quad [\text{say}]$$

$$\text{Then, } PM = x - h$$

$$P'M = x + h$$

In right angled $\triangle PTM$,

$$\frac{PM}{TM} = \tan \beta$$

$$\therefore x - h = TM \tan \beta \quad \dots(i)$$

In right angled $\triangle P'TM$,

$$\frac{P'M}{TM} = \tan \alpha$$

$$x + h = TM \tan \alpha \quad \dots(ii)$$

From Eqs. (i) and (ii), we get

$$\frac{x-h}{x+h} = \frac{\tan \beta}{\tan \alpha}$$

Using componendo and dividendo rule, we get

$$\frac{x-h+x+h}{x-h-x-h} = \frac{\tan \beta + \tan \alpha}{\tan \beta - \tan \alpha}$$

$$\Rightarrow \frac{2x}{-2h} = \frac{\tan \beta + \tan \alpha}{\tan \beta - \tan \alpha}$$

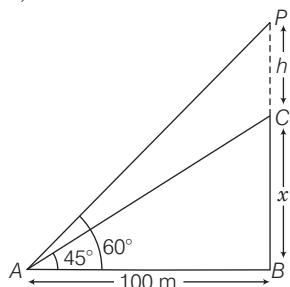
$$\Rightarrow \frac{x}{h} = \frac{\tan \beta + \tan \alpha}{\tan \alpha - \tan \beta}$$

$$\begin{aligned} \Rightarrow \frac{x}{h} &= \frac{\sin \beta \cos \alpha + \cos \beta \sin \alpha}{\sin \alpha \cos \beta - \cos \alpha \sin \beta} \\ &= \frac{\sin(\alpha + \beta)}{\sin(\alpha - \beta)} \end{aligned}$$

$$\therefore x = h \sin(\alpha + \beta) \operatorname{cosec}(\alpha - \beta)$$

18. (c) Let the height of the incomplete pillar be x m and the increasing height be $PC = h$ m.

Given, $AB = 100$ m



In right angled $\triangle ABC$,

$$\tan 45^\circ = \frac{BC}{AB} = \frac{x}{100} \Rightarrow x = 100 \text{ m} \dots(i)$$

and in right angled $\triangle APB$,

$$\tan 60^\circ = \frac{PB}{AB} = \frac{PC + BC}{AB} = \frac{x+h}{100}$$

$$\Rightarrow x + h = 100\sqrt{3}$$

$$\Rightarrow h = 100\sqrt{3} - x = 100\sqrt{3} - 100$$

[from Eq. (i)]

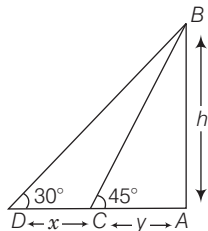
$$\therefore h = 100(\sqrt{3} - 1) \text{ m}$$

Hence, the height of the incomplete pillar is $100(\sqrt{3} - 1)$ m.

19. (c) Let AB be the tower and C, D be the two positions of the car.

Then, $\angle ACB = 45^\circ$, $\angle ADB = 30^\circ$.

Let $AB = h$, $CD = x$ and $AC = y$.



In right angled $\triangle ABC$, we get

$$\frac{AB}{AC} = \tan 45^\circ = 1 \Rightarrow \frac{h}{y} = 1$$

$$\Rightarrow y = h \quad \dots(i)$$

In right angled $\triangle ABD$, we get

$$\frac{AB}{AD} = \tan 30^\circ = \frac{1}{\sqrt{3}} \Rightarrow \frac{h}{x+y} = \frac{1}{\sqrt{3}}$$

$$\Rightarrow x + y = \sqrt{3}h \quad \dots(ii)$$

$$\therefore x = \sqrt{3}h - h = h(\sqrt{3} - 1)$$

[by Eqs. (i) and (ii)]

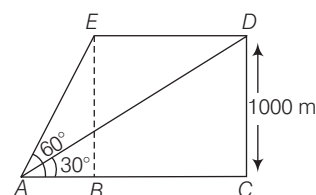
Now, $h(\sqrt{3} - 1)$ is covered in 12 min.

So, h will be covered in

$$\left[\frac{12}{h(\sqrt{3} - 1)} \times h \right] = \frac{12}{(\sqrt{3} - 1)} \text{ min.}$$

$$= \left(\frac{1200}{73} \right) \text{ min} \approx 16 \text{ min } 23 \text{ s}$$

20. (b) Let the radar base is at point A . The plane is at point D in the first sweep and at point E in the second sweep. The distance it covers in the one minute interval is DE .



From the figure,

In right angled $\triangle ADC$, we get

$$\tan 30^\circ = \frac{DC}{AC} = \frac{1000}{AC} \Rightarrow AC = \frac{1000}{\tan 30^\circ}$$

Similarly, in $\triangle EAB$, we get

$$\tan 60^\circ = \frac{EB}{AB} = \frac{1000}{AB} \Rightarrow AB = \frac{1000}{\tan 60^\circ}$$

Total distance covered by plane in 1 min, then

$$DE = AC - AB$$

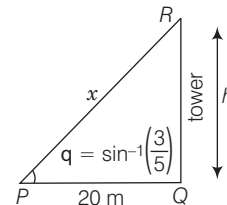
$$\begin{aligned} DE &= \frac{1000}{\tan 30^\circ} - \frac{1000}{\tan 60^\circ} \\ &= 1000\sqrt{3} - \frac{1000}{\sqrt{3}} = 1154.70 \text{ m} \end{aligned}$$

The speed of the plane is given by

$$s = \text{distance covered/time taken}$$

$$= DE / 60 = 19.25 \text{ m/s.}$$

21. (c)



Let P be the point of observation and QR be the tower.

Given that, $\theta = \sin^{-1}\left(\frac{3}{5}\right)$, $PQ = 20$ m.

Let the height of the tower, $QR = h$ and $PR = x$

From the right angled $\triangle PQR$,

$$\sin \theta = \frac{QR}{PR} \Rightarrow \sin \left[\sin^{-1}\left(\frac{3}{5}\right) \right] = h/x$$

$$\Rightarrow \frac{3}{5} = \frac{h}{x} \Rightarrow x = \frac{5h}{3} \quad \dots(i)$$

From pythagoras theorem, we get,

$$PQ^2 + QR^2 = PR^2$$

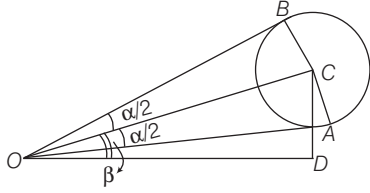
$$\Rightarrow 20^2 + h^2 = x^2$$

$$\Rightarrow 20^2 + h^2 = \left(\frac{5h}{3}\right)^2 \quad [\text{using Eq. (i)}]$$

$$\Rightarrow 20^2 + h^2 = \frac{25h^2}{9} \Rightarrow \frac{16h^2}{9} = 20^2$$

$$\therefore h = \frac{3 \times 20}{4} = 15 \text{ m}$$

22. (a) Let O be the position of the man's eye and C be the centre of the balloon.



$$\text{Now, } \angle BOC = \angle COA = \frac{\alpha}{2}$$

$$\text{Given, } CA = BC = r$$

$$\text{In right angled } \triangle COD, \sin \beta = \frac{CD}{OC}$$

$$\therefore CD = OC \sin \beta \quad \dots(i)$$

$$\text{In right angled } \triangle COA,$$

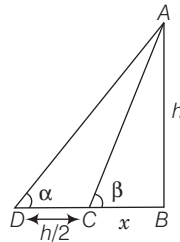
$$\sin \frac{\alpha}{2} = \frac{CA}{OC}$$

$$\therefore OC = \frac{r}{\sin \frac{\alpha}{2}} = r \operatorname{cosec} \frac{\alpha}{2}$$

$$\text{From Eq. (i), } CD = r \operatorname{cosec} \frac{\alpha}{2} \sin \beta$$

Hence, the height of centre of the balloon is $r \sin \beta \operatorname{cosec} \frac{\alpha}{2}$.

23. (a) Let AB be the tower of height h m, BC be x m and DC be $h/2$ m.



In right angled $\triangle ABD$,

$$\tan \alpha = \frac{h}{x + h/2} \Rightarrow \cot \alpha = \frac{x + h/2}{h} \quad \dots(i)$$

Now, in right angled $\triangle ABC$,

$$\tan \beta = \frac{h}{x} \Rightarrow \cot \beta = \frac{x}{h} \quad \dots(ii)$$

Subtracting eq. (ii) from eq. (i), we get

$$\cot \alpha - \cot \beta = \frac{x + h/2}{h} - \frac{x}{h} = \frac{h/2}{h} = \frac{1}{2}$$

24. (a) We have, $\cot \alpha - \cot \beta = 1/2$
 $\Rightarrow \cot 30^\circ - \cot \beta = 1/2$
 $\Rightarrow \cot \beta = \cot 30^\circ - 1/2$
 $= \sqrt{3} - 1/2$

25. (b) $\tan \beta = 4 \Rightarrow \cot \beta = 1/4$

we have, $\cot \alpha - \cot \beta = 1/2$

$$\Rightarrow \cot \alpha = \cot \beta + 1/2 = 1/4 + 1/2 = \frac{3}{4}$$

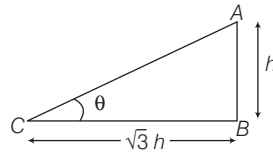
$$\text{In } \triangle ABD, \cot \alpha = \frac{x + h/2}{h}$$

$$\therefore \frac{3}{4} = \frac{x + h/2}{h}$$

$$\Rightarrow 3h = 4x + 2h \Rightarrow x = \frac{h}{4}$$

Hence, required distance is $\frac{h}{4}$

26. (a) Let θ be the angle of elevation of Sun and height of the pole be h m.



In right angled $\triangle ABC$,

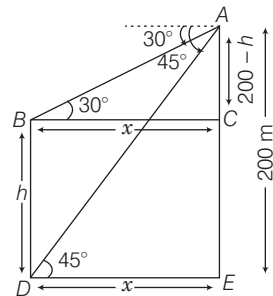
$$\tan \theta = \frac{AB}{BC} \Rightarrow \tan \theta = \frac{h}{\sqrt{3}h}$$

$$\Rightarrow \tan \theta = \frac{1}{\sqrt{3}} = \tan 30^\circ$$

$$\therefore \theta = 30^\circ$$

Hence, the angle of elevation of Sun is 30° .

27. (d) Let AE be the height of the cliff and h m (BD) be the height of the tower.



In right angled $\triangle ABC$,

$$\tan 30^\circ = \frac{AC}{BC} = \frac{200 - h}{x}$$

$$\Rightarrow \frac{1}{\sqrt{3}} = \frac{200 - h}{x}$$

$$\Rightarrow x = (200 - h) \sqrt{3} \quad \dots(i)$$

and in right angled $\triangle ADE$,

$$\tan 45^\circ = \frac{AE}{DE} = \frac{200}{x}$$

$$\Rightarrow 1 = \frac{200}{x} \Rightarrow x = 200 \text{ m}$$

$$\text{From Eq. (i), } 200 = (200 - h) \sqrt{3}$$

$$\Rightarrow 200 = 200\sqrt{3} - h\sqrt{3}$$

$$\Rightarrow h\sqrt{3} = 200(\sqrt{3} - 1)$$

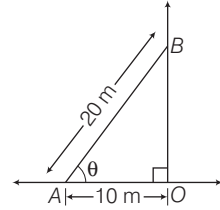
$$\therefore h = 200 \left(\frac{\sqrt{3} - 1}{\sqrt{3}} \right) \text{ m}$$

Hence, the height of tower is

$$\frac{200(\sqrt{3} - 1)}{\sqrt{3}} \text{ m.}$$

28. (c) Refer to example 3.

29. (c) Let θ be the inclination of the ladder to the horizontal.



Now, in right angled $\triangle AOB$,

$$\cos \theta = \frac{AO}{AB} = \frac{10}{20} = \frac{1}{2}$$

$$\Rightarrow \cos \theta = \cos 60^\circ$$

$$\therefore \theta = 60^\circ$$

Hence, the angle of inclination of the ladder is 60° .

Solutions (Q. Nos. 30-33)

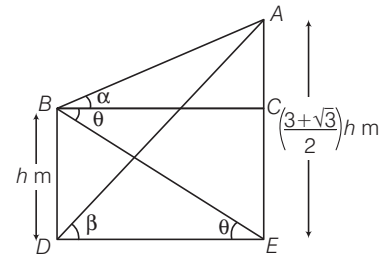
30. (b) Given, $\beta = 30^\circ$

In right angled $\triangle ADE$, $\tan \beta = \frac{AE}{DE}$

$$\tan 30^\circ = \frac{AE}{DE} \Rightarrow \frac{1}{\sqrt{3}} = \frac{AE}{DE}$$

$$\Rightarrow DE = \sqrt{3} AE = \sqrt{3} \left(\frac{3 + \sqrt{3}}{2} \right) h$$

$$\Rightarrow BC = DE = \frac{3}{2} (1 + \sqrt{3}) h \quad \dots(i)$$



$$[\because BC = DE]$$

Now, in right angled $\triangle ABC$,

$$\Rightarrow \tan \alpha = \frac{AC}{BC}$$

$$\Rightarrow BC \tan \alpha = (AE - CE)$$

$$= (AE - BD) [\because BD = CE]$$

$$\Rightarrow BC \tan \alpha = \left(\frac{3 + \sqrt{3}}{2} \right) h - h$$

$$= h \left(\frac{3 + \sqrt{3} - 2}{2} \right)$$

$$\Rightarrow \frac{3}{2} (1 + \sqrt{3}) h \tan \alpha = \left(\frac{1 + \sqrt{3}}{2} \right) h$$

[from Eq. (i)]

$$\therefore \tan \alpha = \frac{1}{3}$$

31. (a) Given, $\alpha = 30^\circ$

In right angled $\triangle ABC$,

$$\tan \alpha = \tan 30^\circ = \frac{AC}{BC}, \frac{1}{\sqrt{3}} = \frac{AC}{BC}$$

$$\begin{aligned} \Rightarrow BC &= \sqrt{3} AC = \sqrt{3} (AE - CE) \\ &= \sqrt{3} (AE - BD) [\because BD = CE] \\ &= \sqrt{3} \left(\frac{3 + \sqrt{3}}{2} - 1 \right) h \\ &= \frac{\sqrt{3}}{2} (1 + \sqrt{3}) h \quad \dots(ii) \end{aligned}$$

Now, in right angled $\triangle ADE$,

$$\begin{aligned} \tan \beta &= \frac{AE}{DE} \\ \Rightarrow \tan \beta &= \frac{AE}{BC} [\because DE = BC] \\ &= \frac{\left(\frac{3 + \sqrt{3}}{2} \right) h}{\frac{\sqrt{3}}{2} (1 + \sqrt{3}) h} = \frac{\sqrt{3}(1 + \sqrt{3})}{\sqrt{3}(1 + \sqrt{3})} \\ \therefore \tan \beta &= 1 \end{aligned}$$

32. (c) Given, $\alpha = 30^\circ$ and $h = 30$ m

In right angled $\triangle ABC$,

$$\begin{aligned} \tan \alpha &= \tan 30^\circ = \frac{1}{\sqrt{3}} = \frac{AC}{BC} \\ \Rightarrow \frac{BC}{\sqrt{3}} &= (AE - CE) = (AE - BD) \\ \Rightarrow BC &= \sqrt{3} \left(\frac{3 + \sqrt{3}}{2} - 1 \right) h \\ \Rightarrow BC &= \sqrt{3} \left(\frac{1 + \sqrt{3}}{2} \right) \cdot 30 \\ &= (\sqrt{3} + 3) \cdot 15 \\ \therefore DE = BC &= (45 + 15\sqrt{3}) \text{ m} \\ &[\because DE = BC] \end{aligned}$$

33. (a) Given, $\beta = 30^\circ$

In right angled $\triangle ADE$, $\tan \beta = \frac{AE}{DE}$

$$\begin{aligned} \Rightarrow \tan 30^\circ &= \frac{\left(\frac{3 + \sqrt{3}}{2} \right) h}{DE} \\ \Rightarrow \frac{1}{\sqrt{3}} &= \frac{\sqrt{3} \left(\frac{1 + \sqrt{3}}{2} \right) h}{DE} \\ \Rightarrow DE &= \frac{3}{2} (1 + \sqrt{3}) h \quad \dots(i) \end{aligned}$$

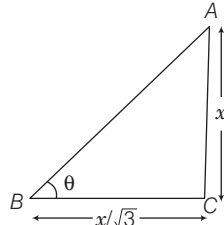
In right angled $\triangle BDE$,

$$\begin{aligned} \tan \theta &= \frac{BD}{DE} = \frac{h}{DE} \\ \Rightarrow \tan \theta &= \frac{h}{\frac{3}{2} (1 + \sqrt{3}) h} [\text{from Eq. (i)}] \\ \Rightarrow \tan \theta &= \frac{2}{3} \frac{(\sqrt{3} - 1)}{(\sqrt{3} + 1)(\sqrt{3} - 1)} \\ \Rightarrow \tan \theta &= \frac{2(\sqrt{3} - 1)}{3 \cdot 2} \end{aligned}$$

$$\Rightarrow \tan \theta = \frac{(\sqrt{3} - 1)}{3} \cdot \frac{\sqrt{3}}{\sqrt{3}}$$

$$\therefore \tan \theta = \frac{(3 - \sqrt{3})}{3\sqrt{3}}$$

34. (c) Let θ be the angle of elevation,

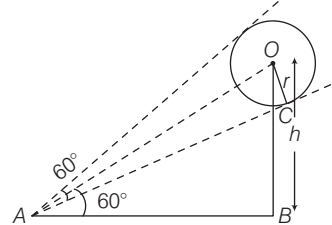


In right angled $\triangle ABC$,

$$\tan \theta = \frac{AC}{BC} = \frac{x}{x/\sqrt{3}} = \frac{\sqrt{3}x}{x} = \sqrt{3}$$

$$\begin{aligned} \text{Here, } \tan \theta &= \sqrt{3} = \tan 60^\circ \\ \therefore \theta &= 60^\circ [\because \tan 60^\circ = \sqrt{3}] \end{aligned}$$

35. (c) Given, radius of circle $(OC) = r$



In right angled $\triangle ABO$,

$$\sin 60^\circ = \frac{OB}{AO} \Rightarrow AO = \frac{OB}{\sin 60^\circ} \quad \dots(i)$$

Now, in right angled $\triangle AOC$,

$$\sin \frac{60^\circ}{2} = \frac{OC}{AO} \Rightarrow AO = \frac{OC}{\sin 30^\circ} \quad \dots(ii)$$

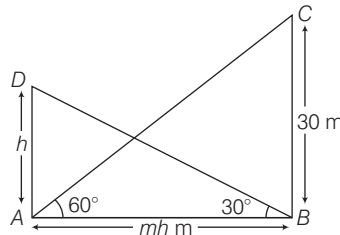
From Eqs. (i) and (ii),

$$\frac{OB}{\sin 60^\circ} = \frac{OC}{\sin 30^\circ} \Rightarrow \frac{h}{\frac{\sqrt{3}}{2}} = \frac{r}{\frac{1}{2}}$$

$$\therefore h = \sqrt{3} r$$

36. (b) Let h m be the height of shorter tower and the distance between the two towers is mh m.

Given, $\angle ABD = 30^\circ$ and $\angle BAC = 60^\circ$



In right angled $\triangle ABD$,

$$\tan 30^\circ = \frac{h}{mh} \Rightarrow \frac{1}{\sqrt{3}} = \frac{1}{m}$$

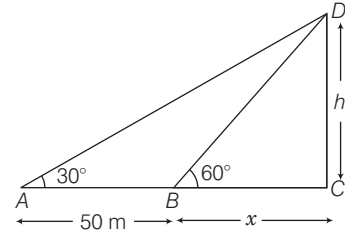
$$\Rightarrow m = \sqrt{3}$$

Hence, the value of m is $\sqrt{3}$.

37. (b) Let h m be the height of the tower and BC be x m.

In right angled $\triangle BCD$,

$$\begin{aligned} \tan 60^\circ &= \frac{DC}{BC} = \frac{h}{x} \\ \Rightarrow \sqrt{3} &= \frac{h}{x} \Rightarrow h = x\sqrt{3} \quad \dots(i) \end{aligned}$$



Now, in right angled $\triangle ACD$,

$$\begin{aligned} \tan 30^\circ &= \frac{DC}{AC} \\ &= \frac{DC}{AB + BC} = \frac{h}{50 + x} \end{aligned}$$

$$\Rightarrow \frac{1}{\sqrt{3}} = \frac{x\sqrt{3}}{50 + x} \quad [\text{from Eq. (i)}]$$

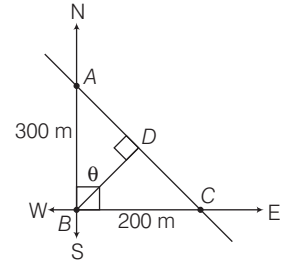
$$\Rightarrow 50 + x = 3x \Rightarrow x = 25 \text{ m}$$

$$\therefore h = 25\sqrt{3} \text{ m}$$

Hence, the height of tower is $25\sqrt{3}$ m.

38. (c) Let person be at point C and observes a tower in West direction at B .

$$\therefore BC = 200 \text{ m}$$



He walks some distance and reach at A . Now, he observe tower in South direction at B .

$$\therefore AB = 300 \text{ m}$$

Let BD be the shortest distance of tower from the road, which is a perpendicular distance.

If $\angle ABD = \theta$, then

$$\angle CBD = 90^\circ - \theta$$

[$\because$ angle between S and $W = 90^\circ$]

In right angled $\triangle ADB$,

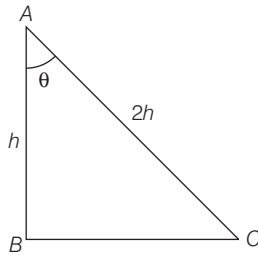
$$\cos \theta = \frac{BD}{AB} \Rightarrow \cos \theta = \frac{BD}{300} \quad \dots(i)$$

In right angled $\triangle CDB$,

$$\begin{aligned} \cos (90^\circ - \theta) &= \frac{BD}{BC} \\ \Rightarrow \sin \theta &= \frac{BD}{200} \quad \dots(ii) \end{aligned}$$

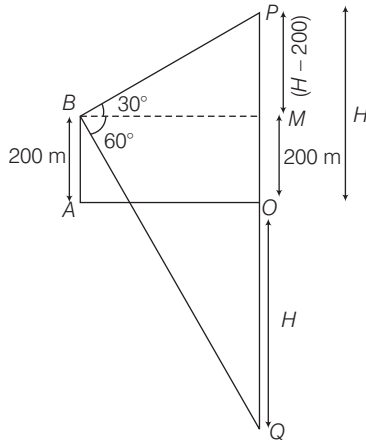
We know that, $\cos^2 \theta + \sin^2 \theta = 1$
 $\Rightarrow \left(\frac{BD}{300}\right)^2 + \left(\frac{BD}{200}\right)^2 = 1$
 [from Eqs. (i) and (ii)]
 $\Rightarrow BD^2 \left[\frac{40000 + 90000}{90000 \times 40000} \right] = 1$
 $\Rightarrow BD^2 \left[\frac{130000}{3600000000} \right] = 1$
 $\Rightarrow BD^2 = \frac{360000}{13}$
 $\therefore BD = \sqrt{\frac{360000}{13}} = \frac{600}{\sqrt{13}} \text{ m}$

39. (d) Let AB be the height of the post, AC be the string and the angle made by string with the post be θ .



Now, $\cos \theta = \frac{AB}{AC} = \frac{h}{2h} \Rightarrow \frac{1}{2} = \cos 60^\circ$
 $\therefore \theta = \frac{\pi}{3}$

40. (c) Let P be the cloud at a height H m above the level of the water in the lake and Q be its image in the water.

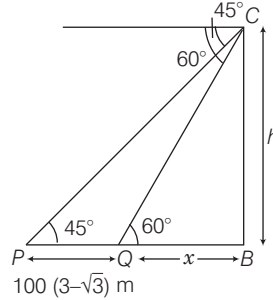


$\therefore OQ = OP = H$
 Given, $\angle PBM = 30^\circ$ and $\angle MBQ = 60^\circ$
 In right angled $\triangle PBM$,
 $\tan 30^\circ = \frac{PM}{BM} = \frac{H - 200}{BM}$
 $\Rightarrow \frac{1}{\sqrt{3}} = \frac{H - 200}{BM}$
 $\Rightarrow BM = \sqrt{3}(H - 200) \dots(i)$

In right angled $\triangle QMB$,
 $\tan 60^\circ = \frac{MQ}{BM} = \frac{H + 200}{BM}$
 $\Rightarrow \sqrt{3} = \frac{H + 200}{\sqrt{3}(H - 200)} \text{ [from Eq. (i)]}$
 $\Rightarrow H + 200 = 3(H - 200)$
 $\Rightarrow H + 200 = 3H - 600$
 $\Rightarrow 2H = 800$
 $\therefore H = 400 \text{ m}$

Hence, the height of the cloud is 400 m.

41. (c) Let $BC = h$ m be height of tower. Let P and Q be the points, where the angle subtended are 45° and 60° , respectively.

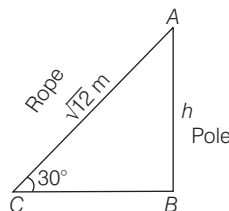


In right angled $\triangle BCQ$,
 $\tan 60^\circ = \frac{BC}{BQ} \Rightarrow \sqrt{3} = \frac{h}{x}$
 $\Rightarrow x = \frac{h}{\sqrt{3}} \dots(i)$

In right angled $\triangle PBC$,
 $\tan 45^\circ = \frac{BC}{PB} = \frac{BC}{PQ + QB}$
 $\Rightarrow 1 = \frac{h}{100(3 - \sqrt{3}) + x}$
 $\Rightarrow 100(3 - \sqrt{3}) + x = h$
 $\Rightarrow 100(3 - \sqrt{3}) + \frac{h}{\sqrt{3}} = h \text{ [from Eq. (i)]}$
 $\Rightarrow h - \frac{h}{\sqrt{3}} = 100(3 - \sqrt{3})$
 $\Rightarrow \frac{h(\sqrt{3} - 1)}{\sqrt{3}} = 100\sqrt{3}(\sqrt{3} - 1)$
 $\Rightarrow h = 100\sqrt{3} \times \sqrt{3}$
 $\therefore h = 300 \text{ m}$

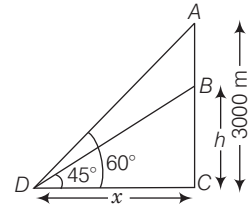
42. (d) Refer to example 3.

43. (d) AB is a pole and AC is rope.



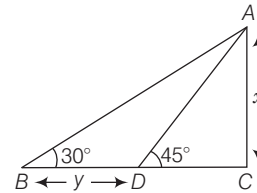
In $\triangle ABC$, $\sin 30^\circ = \frac{AB}{AC} = \frac{h}{\sqrt{12}}$
 $\Rightarrow \frac{h}{\sqrt{12}} = \frac{1}{2}$
 $\therefore h = \frac{\sqrt{12}}{2} = \frac{2\sqrt{3}}{2} = \sqrt{3} \text{ m}$

44. (c) Let A and B be the position of two planes and D be a point.



In $\triangle BCD$, $\tan 45^\circ = \frac{h}{x}$
 $\Rightarrow h = x$
 In $\triangle ACD$, $\tan 60^\circ = \sqrt{3} = \frac{3000}{x}$
 $\therefore x = \frac{3000}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = 1000\sqrt{3} \text{ m}$
 $\therefore AB = 3000 - h$
 $= 3000 - 1000\sqrt{3}$
 $= 1000(3 - \sqrt{3}) \text{ m}$

45. (c) In right angled $\triangle ADC$, $\tan 45^\circ = \frac{AC}{DC}$
 $\Rightarrow DC = \frac{AC}{\tan 45^\circ} = x \dots(i)$



In right angled $\triangle BCA$, we get
 $\tan 30^\circ = \frac{AC}{BC} \Rightarrow \frac{x}{DC + BD}$
 $\Rightarrow \frac{1}{\sqrt{3}} = \frac{x}{y + x} \Rightarrow y + x = \sqrt{3}x$
 $\Rightarrow y = x(\sqrt{3} - 1)$
 $\Rightarrow \frac{x}{y} = \frac{1}{(\sqrt{3} - 1)}$
 $\Rightarrow \frac{x}{y} = \frac{1 \times (\sqrt{3} + 1)}{(\sqrt{3} - 1)(\sqrt{3} + 1)}$
 $\Rightarrow \frac{x}{y} = \frac{(\sqrt{3} + 1)}{(\sqrt{3})^2 - (1)^2}$
 $\therefore \frac{x}{y} = \frac{(\sqrt{3} + 1)}{2}$

22

LINES AND ANGLES

Usually (2-3) questions have been asked from this chapter. And this is the easiest topic of geometry and so candidate can easily score marks in examinations.



In this chapter, we will study about the properties of the angles which formed, when two lines intersect each other and when a line intersect two or more parallel lines at distinct points.

Basic Terms and Definitions

Point

The figure of which length, breadth and height cannot be measured is called a point. It is infinitesimal.

Line Segment

The straight path between two points P and Q is called a line segment.

This can be represented as $\overline{PQ}$.

- P and Q are called the end points of the line segment.
- The line segment has a definite length.
- Distance between P and Q is called the length of the line segment PQ .

Line

A line segment extended endlessly on both sides, is

called a line. It is denoted by $\overleftrightarrow{PQ}$

or $\overleftrightarrow{QP}$.

- A line is a set of infinite number of points.
- A line has no end points and no definite length.

Ray

A ray extends indefinitely in one direction. This is exhibited by an arrow i.e. $\overrightarrow{PQ}$.

- P is called the initial point of the ray. The ray has no definite length.

Collinear Points

Three or more points are said to be collinear if a single straight line passes through them. Here, A , B and C are collinear.

Non-collinear Points

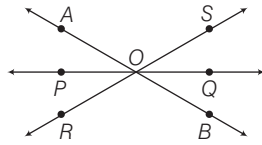
Three or more points not lying on a single straight line are called non-collinear points. Here, A , B and C are non-collinear points.

Intersecting Lines

Two lines having a common point are called intersecting lines. This common point is the point of intersection i.e. O .

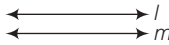
Concurrent Lines

Three or more lines intersecting at the same point are said to be concurrent. This common point is the point of concurrence i.e. O .



Non Intersecting lines/Parallel Lines

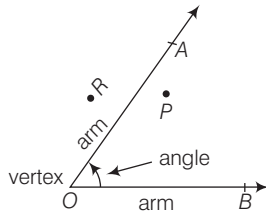
If two lines lie in the same plane and do not intersect when produced on either side, then such lines are said to be parallel to each other.



If l and m are two parallel lines, we write $l \parallel m$ and read it as l is parallel to m .

ANGLES

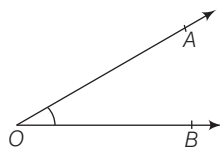
An angle is formed by two rays with a common initial point. Let 'O' is the initial point, then O is called the vertex.



Where, P is a point in the interior of $\angle AOB$ and R is a point in the exterior of $\angle AOB$.

Types of Angle

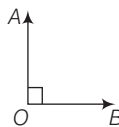
1. **Acute angle** An angle whose measure is more than 0° , but less than 90° , is called an acute angle.



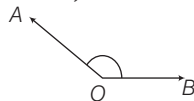
In the given figure, $0^\circ < \angle AOB < 90^\circ$

2. **Right angle** An angle whose measure is 90° , is called a right angle.

In the given figure, $\angle AOB = 90^\circ$.



3. **Obtuse angle** An angle whose measure is more than 90° but less than 180° , is called an obtuse angle.



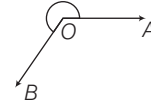
In the given figure, $90^\circ < \angle AOB < 180^\circ$.

4. **Straight angle** An angle whose measure is 180° , is called a straight angle.

In the figure, $\angle AOB = 180^\circ$



5. **Reflex angle** An angle whose measure is more than 180° , but less than 360° , is called a reflex angle.



In the given figure, $180^\circ < \angle AOB < 360^\circ$

6. **Complete angle** An angle whose measure is 360° , is called a complete angle. In the given figure, $\angle AOA$ is a complete angle.

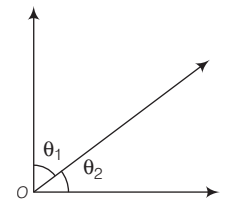


Pairs of Angle

1. Complementary Angles

Two angles are said to be complementary, if the sum of their measure is 90° . Thus, $\angle \theta_1$ and $\angle \theta_2$ are complementary, if $\angle \theta_1 + \angle \theta_2 = 90^\circ$

- Complementary angles are complement of each other.
- Complement of x is $(90^\circ - x)$.



EXAMPLE 1. The measure of an angle which is 28° more than its complement is

- a. 23° b. 59° c. 77° d. None of these

Sol. b. Let the measure of the required angle be x° . Then, measure of its complement $= 90^\circ - x$

$$\therefore x - (90^\circ - x) = 28^\circ \Leftrightarrow 2x = 118^\circ$$

$$\Leftrightarrow x = 59^\circ$$

Hence, the measure of the required angle is 59° .

EXAMPLE 2. The measure of the complement of an angle of $48^\circ 36' 24''$ is

- a. $41^\circ 23' 36''$ b. $42^\circ 23' 36''$ c. $41^\circ 24' 36''$ d. $42^\circ 24' 36''$

Sol. a. As, $90^\circ = 89^\circ 59' 60''$

$$\therefore \text{Complement of an angle of } (48^\circ 36' 24'')$$

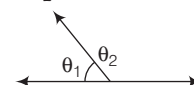
$$= \text{Angle of } [90^\circ - 48^\circ 36' 24'']$$

$$= \text{Angle of } [89^\circ 59' 60'' - 48^\circ 36' 24'']$$

$$= \text{Angle of } (41^\circ 23' 36'')$$

2. Supplementary Angles

Two angles are said to be supplementary, if the sum of their measures is 180° . Thus, $\angle \theta_1$ and $\angle \theta_2$ are supplementary if $\theta_1 + \theta_2 = 180^\circ$



- Supplementary angles are supplement of each other.
- Supplement of x is $(180^\circ - x)$.

EXAMPLE 3. The measure of an angle, which is 32° less than its supplement is

- a. 31° b. 64° c. 74° d. 148°

Sol. c. Let the measure of the required angle be x . Then, measure of its supplement $= (180^\circ - x)$

$$\therefore (180^\circ - x) - x = 32^\circ \Leftrightarrow 2x = 148^\circ \Leftrightarrow x = 74^\circ$$

EXAMPLE 4. Two supplementary angles are in the ratio 3 : 2. Then, the measurement of the smaller angle is

- a. 36° b. 72° c. 108° d. 112°

Sol. b. Let the supplementary angles be $3x$ and $2x$, respectively. Then, according to the definition of supplementary angle.

$$3x + 2x = 180^\circ$$

$$\Rightarrow 5x = 180^\circ \Rightarrow x = 36^\circ$$

$$\therefore \text{Angles will be } 3x = 3 \times 36 = 108^\circ$$

$$\text{and } 2x = 2 \times 36 = 72^\circ.$$

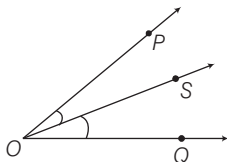
Thus, the smaller angle is 72° .

Bisector of an Angle

A ray which divides the angle into two equal parts, is called the bisector of an angle.

If a ray OS is the bisector of $\angle POQ$, then $\angle POS = \angle QOS$.

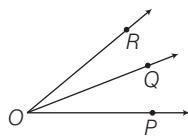
$$\bullet \angle POS = \angle QOS = \frac{1}{2} \angle POQ$$



Adjacent Angles

Two angles are said to be adjacent, if

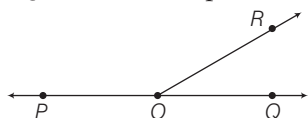
- they have a common vertex.
- they have a common arm
- their non-common arms are on either side of the common arm.



Here, $\angle POQ$ and $\angle ROQ$ are adjacent angles.

Linear Pairs of Angles

If the non-common arms of two adjacent angles form a line, then these angles are called linear pair of angles. $\angle ROP$ and $\angle ROQ$ form a linear pair of angles.

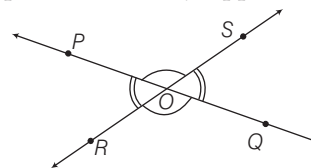


$$\therefore \angle POQ = \angle ROP + \angle ROQ = 180^\circ$$

Vertically Opposite Angles

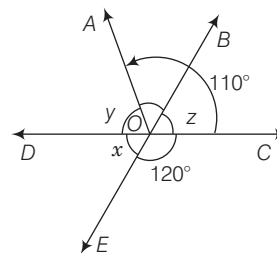
Two angles are called a pair of vertically opposite angles, if their arms form two pairs of opposite rays.

If two lines PQ and RS intersect at a point O , then the pair of $\angle POR$ and $\angle QOS$ or pair of $\angle POS$ and $\angle ROQ$ is said to be a pair of vertically opposite angles.



- Vertically opposite angles are always equal i.e. $\angle POS = \angle ROQ$ and $\angle POR = \angle SOQ$.
- Sum of the angles around a point is 360° .

EXAMPLE 5. In the given figure if DOC is a straight ray OB is bisector of $\angle AOC$, where $\angle AOC = 110^\circ$ and $\angle COE = 120^\circ$, then the sum of $\angle x$, $\angle y$ and $\angle z$ is



- a. 160° b. 115° c. 170° d. 180°

Sol. d. Since, DOC is a straight line

$$\angle AOC + \angle DOA = 180^\circ \quad [\text{Linear pair}]$$

$$110^\circ + \angle DOA = 180^\circ \quad [\because \angle AOC = 110^\circ]$$

$$\angle DOA = 70^\circ$$

$$\angle DOA = \angle y = 70^\circ$$

Also, ray OB is angle bisector of $\angle AOC$

$$\angle z = \frac{\angle AOC}{2} = \frac{110^\circ}{2} = 55^\circ$$

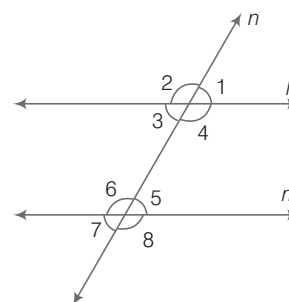
$\therefore$ Line BE and DC intersect each other at O .

$$\angle x = \angle z = 55^\circ \quad [\because \text{vertically opposite angle}]$$

$$\therefore \angle x + \angle y + \angle z = 55^\circ + 70^\circ + 55^\circ = 180^\circ$$

Some Other Angles

Let l and m are parallel lines and n is the transversal which cuts these parallel lines. The different angles formed are as follows.



- Corresponding angles** Corresponding angle pairs are
 $\angle 1$ and $\angle 5$, $\angle 2$ and $\angle 6$, $\angle 4$ and $\angle 8$, $\angle 3$ and $\angle 7$ and all the corresponding pair are equal
i.e., $\angle 1 = \angle 5$, $\angle 2 = \angle 6$, $\angle 4 = \angle 8$ and $\angle 3 = \angle 7$.
- Vertically opposite angles** Vertically opposite angles pairs are
 $\angle 1$ and $\angle 3$, $\angle 4$ and $\angle 2$, $\angle 8$ and $\angle 6$, $\angle 5$ and $\angle 7$ and all vertically opposite angles are equal
i.e., $\angle 1 = \angle 3$, $\angle 4 = \angle 2$, $\angle 8 = \angle 6$ and $\angle 5 = \angle 7$.
- Alternate angles** Alternate angles pairs are
 $\angle 3$ and $\angle 5$, $\angle 4$ and $\angle 6$ and they are equal
i.e., $\angle 3 = \angle 5$ and $\angle 4 = \angle 6$.
 - The sum of interior angles on the same side of transversal is equal to 180° .
i.e., $\angle 3 + \angle 6 = 180^\circ$ and $\angle 4 + \angle 5 = 180^\circ$
 - The sum of exterior angles on the same side of transversal is equal to 180° .
i.e., $\angle 2 + \angle 7 = 180^\circ$ and $\angle 1 + \angle 8 = 180^\circ$

EXAMPLE 6. In the given figure, $PQ \parallel RS$. The value of x is

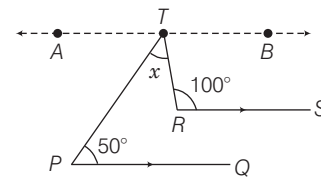
- a. 50° b. 80° c. 75° d. 65°

Sol. a. Draw $AB \parallel PQ$

$$\therefore \angle ATP = \angle TPQ = 50^\circ \quad [\text{alternate angles}]$$

$$\text{and } \angle BTR + \angle TRS = 180^\circ$$

[Angles on the same side of transversal]



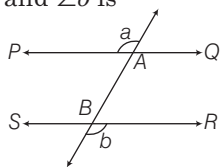
$$\Rightarrow \angle BTR = 80^\circ$$

$$\text{and } \angle ATP + x + \angle BTR = 180^\circ \quad [\because ATB \text{ is a straight line}]$$

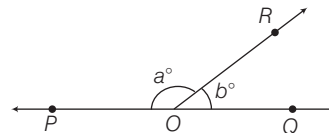
$$\therefore x = 180^\circ - (50^\circ + 80^\circ) = 50^\circ$$

PRACTICE EXERCISE

- Three lines intersect each other in pairs. What is the total number of angles so formed?
 (a) 3 (b) 6 (c) 9 (d) 12
- An angle is 14° more than its complement. Then, its measure is
 (a) 166° (b) 86° (c) 76° (d) 52°
- The measure of an angle is twice the measure of its supplementary angle. So, its measure is
 (a) 120° (b) 60° (c) 100° (d) 90°
- What is the least number of straight lines for a bounded plane figure?
 (a) 1 (b) 2 (c) 3 (d) 4
- The supplement of $\frac{3}{5}$ of a right angle is
 (a) 122° (b) 126° (c) 130° (d) 132°
- In the given figure, if $PQ \parallel SR$, then the relation between $\angle a$ and $\angle b$ is
- If $\angle 1 = (5x - 20)^\circ$ and $\angle 7 = (2x + 10)^\circ$, then $\angle 7$ is
 (a) 38° (b) 10°
 (c) 30° (d) 70°
- The measure of complementary angle of $12^\circ 25' 40''$ is
 (a) $77^\circ 34' 20''$ (b) $77^\circ 36' 20''$ (c) $77^\circ 24' 20''$ (d) $77^\circ 34'$
- $\angle POR$ and $\angle QOR$ form a linear pair and if $a - b = 80^\circ$, then the values of a and b are, respectively

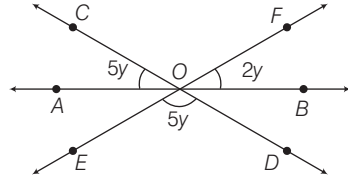


- (a) $\angle a \neq \angle b$ (b) $\angle a < \angle b$ (c) $\angle a = \angle b$ (d) $\angle a > \angle b$



- (a) $95^\circ, 85^\circ$ (b) $108^\circ, 72^\circ$
 (c) $130^\circ, 50^\circ$ (d) $105^\circ, 75^\circ$

10. In the given figure, the value of y is

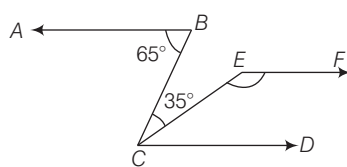


- (a) 25° (b) 35° (c) 15° (d) 40°

11. AB and CD are two parallel lines. PQ cuts AB and CD at E and F , respectively. EL is the bisector of $\angle FEB$. If $\angle LEB = 35^\circ$; then $\angle CFQ$ will be

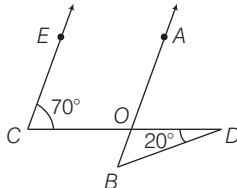
- (a) 110° (b) 85° (c) 70° (d) 95°

12. AB and CD are two parallel lines. The points B and C are joined such that $\angle ABC = 65^\circ$. A line CE is drawn making angle of 35° with the line CB , EF is drawn parallel to AB , as show in figure, then $\angle CEF$ is equal to



- (a) 160° (b) 155° (c) 150° (d) 145°

13. In the given figure, if $EC \parallel AB$, $\angle ECD = 70^\circ$, $\angle BDO = 20^\circ$, then $\angle OBD$ is equal to

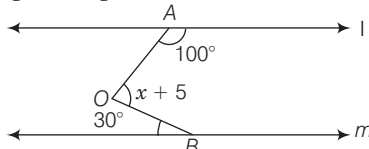


- (a) 70° (b) 60° (c) 50° (d) 20°

14. Two parallel lines AB and CD are intersected by a transversal line EF at M and N , respectively. The lines MP and NP are the bisectors of the interior angles BMN and DNM on the same side of the transversal. Then, $\angle MPN$ is equal to

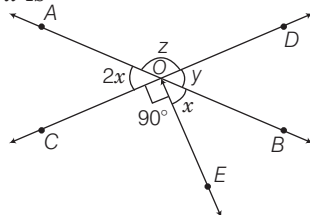
- (a) 90° (b) 45° (c) 135° (d) 60°

15. In the given figure if $l \parallel m$, then the value of x is



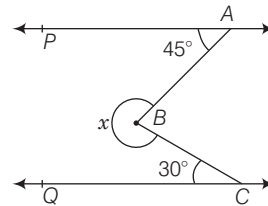
- (a) 105° (b) 100° (c) 110° (d) 115°

16. In the given figure, if $\angle COE = 90^\circ$, then the value of x is



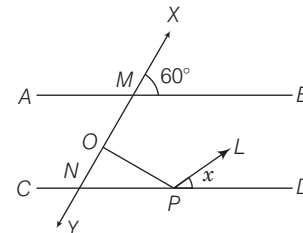
- (a) 120° (b) 60° (c) 45° (d) 30°

17. The value of x in the given figure is



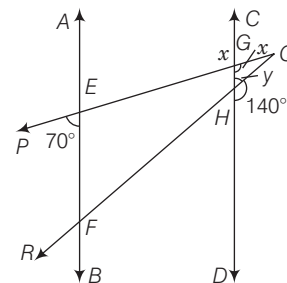
- (a) 75° (b) 185°
(c) 285° (d) 245°

18. In the given figure, $AB \parallel CD$, $\angle DPL = \frac{1}{2} \angle NPO$, and $OP \perp XY$ the value of x° is



- (a) 30° (b) 40° (c) 15° (d) 25°

19. In the given figure, $AB \parallel CD$ and they cut PQ and QR at E, F and G, H respectively. Then find the value of $x + y$



- (a) 120° (b) 130°
(c) 150° (d) 132°

20. Two parallel lines are cut by a transversal, then which of the following are true?

- I. Pair of alternate interior angles are congruent.
II. Pair of corresponding angles are congruent.
III. Pair of interior angles on the same side of the transversal are supplementary.

Select the correct answer using the codes given below

- (a) I, II and III are true (b) I and III are true
(c) I and II are true (d) II and III are true

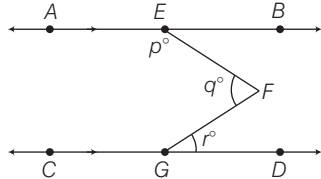
21. Consider the following statements related to three lines L_1, L_2 and L_3 in the same plane.

- I. If L_2 and L_3 are both parallel to L_1 , then they are parallel to each other.
II. If L_2 and L_3 are both perpendicular to L_1 then they are parallel to each other.
III. If there is acute angle between L_1 and L_3 , then L_2 is parallel to L_3 .

Which of the statement(s) given above is/are correct?

- (a) I and II (b) II and III
(c) All of these (d) None of these

- 22.** In the given figure, $AB \parallel CD$, then which one of the following is true?



- (a) $p + q - r = 180^\circ$ (b) $p + q + r = 180^\circ$
(c) $p - q + r = 180^\circ$ (d) $p + q - 2r = 180^\circ$

- 23.** LM is a straight line and O is a point on LM . Line ON is drawn not coinciding with OL or OM . If $\angle MON$ is one-third of $\angle LON$, then what is $\angle MON$ equal to?

- (a) 45° (b) 60°
(c) 75° (d) 80°

- 24.** Consider the following statements :

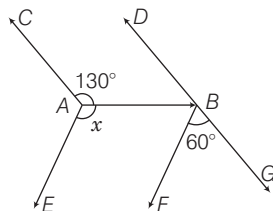
Two lines intersected by a transversal are parallel, if

- I. The pairs of corresponding angles are equal.
II. The interior angles on the same side of the transversal are supplementary.

Which of the statement(s) given above is/are correct?

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

- 25.** In the given figure $AC \parallel BD$ and $AE \parallel BF$. What is the value of $\angle x$?



- (a) 130° (b) 110°
(c) 70° (d) 50°

- 26.** The necessary conditions for the line l and m to be parallel when these lines are intersected by a transversal line n is that

- I. Interior angles on the same side are equal.
II. Corresponding angles are equal.
III. Vertically opposite angles are equal.
IV. Alternate interior angles are equal.

Select the correct answer using the codes given below

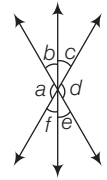
- (a) II and III (b) I, II and III (c) II and IV (d) I, II and IV

- 27.** In the given figure, which of the following statements must be true?

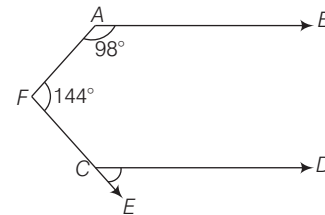
- I. $a + b = d + c$ II. $a + c + e = 180^\circ$
III. $b + f = c + e$ IV. $a + b + c = d + e + f$

Select the correct answer using the codes given below

- (a) Only I (b) I, II and III
(c) II, III and IV (d) All of these



- 28.** In the given figure, AB is parallel to CD . If $\angle BAF = 98^\circ$ and $\angle AFC = 144^\circ$, then $\angle ECD$ is equal to?



- (a) 62° (b) 64°
(c) 82° (d) 84°

- 29.** Consider the following statements :

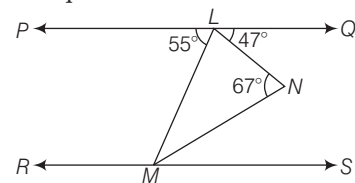
If two parallel lines are intersected by a transversal, then

- I. each pair of corresponding angles are equal.
II. each pair of alternate angle are unequal.

Which of the statement(s) given above is/are correct?

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

- 30.** In the given figure, PQ is parallel to RS . Then $\angle NMS$ is equal to

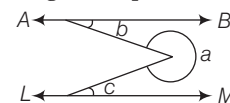


- (a) 20° (b) 23° (c) 27° (d) 47°

- 31.** The line segments AB and CD intersect at O . OF is the internal bisector of obtuse $\angle BOC$ and OE is the internal bisector of acute $\angle AOC$. If $\angle BOC = 130^\circ$, then what is the measure of $\angle FOE$?

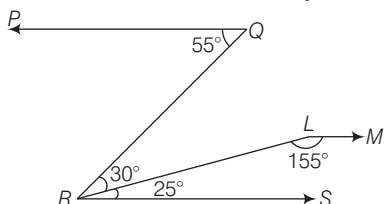
- (a) 90° (b) 110° (c) 115° (d) 120°

- 32.** In the given figure below, AB is parallel to LM . What is the angle a equal to?



- (a) $\pi + b + c$ (b) $2\pi - b + c$
(c) $2\pi - b - c$ (d) $2\pi + b - c$

33. In the given figure, PQ is parallel to RS . What is the angle between the lines PQ and LM ?



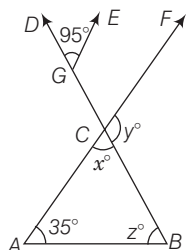
- (a) 175° (b) 177° (c) 179° (d) 180°

34. The length of a line segment AB is 2 units. It is divided into two parts at the point C such that $AC^2 = AB \times CB$. What is the length of CB ?

- (a) $3 + \sqrt{5}$ units (b) $3 - \sqrt{5}$ unit
(c) $2 - \sqrt{5}$ unit (d) $\sqrt{3}$ units

Directions (Q.Nos 35-36) Read the information and answer the questions.'

In the given figure, the lines CB and AC of a triangle ABC are extended to D and F , respectively and $CF \parallel GE$.



35. The sum of the value of x , y and z .
(a) 250° (b) 180° (c) 230° (d) 110°
36. The value of the angle which equals one-fifth of the supplement of z ?
(a) 50° (b) 26° (c) 10° (d) 130°

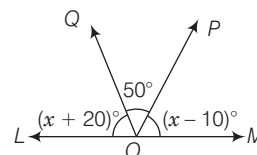
PREVIOUS YEARS' QUESTIONS

37. AB is a straight line. C is a point whose distance from AB is 3 cm. What is the number of points which are at a distance of 1 cm from AB and 5 cm from C ? **2012 I**
(a) 1 (b) 2 (c) 3 (d) 4
38. The ratio of two complementary angle is 1 : 5. What is the difference between the two angles? **2012 I**
(a) 60° (b) 90°
(c) 120° (d) Cannot be determined
39. Consider the following statements :
If two straight lines intersect, then
I. vertically opposite angles are equal.
II. vertically opposite angles are supplementary.
III. adjacent angles are complementary.

Which of the statement(s) given above is/are correct? **2012 I**

- (a) Only III (b) Only I (c) I and III (d) II and III

40. In the given figure, LOM is a straight line. What is the value of x° ? **2012 II**



- (a) 45° (b) 60° (c) 70° (d) 80°

41. If the arms of one angle are respectively parallel to the arms of another angle, then the two angles are **2013 I**

- (a) Neither equal nor supplementary
(b) not equal but supplementary
(c) equal but not supplementary
(d) Either equal or supplementary

42. The complement angle of 80° is **2015 I**

- (a) $\frac{18}{\pi}$ radian (b) $\frac{5\pi}{9}$ radian
(c) $\frac{\pi}{18}$ radian (d) $\frac{9}{5\pi}$ radian

43. Let OA, OB, OC and OD be rays in the anti-clockwise direction? Such that $\angle AOB = \angle COD = 100^\circ$, $\angle BOC = 82^\circ$ and $\angle AOD = 78^\circ$.

Consider the following statements

I. AOC and BOD are lines.

II. $\angle BOC$ and $\angle AOD$ are supplementary. **2015 I**

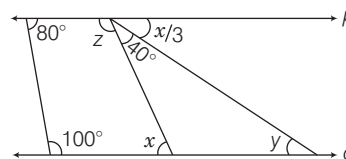
Which of the statement(s) given above is/are correct?

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

44. If a transversal intersects four parallel straight lines, then the number of distinct values of the angles formed will be **2016 I**

- (a) 2 (b) 4 (c) 8 (d) 16

- 45.



In the figure given above, p and q are parallel lines. What are the values of the angles x , y and z ? **2016 I**

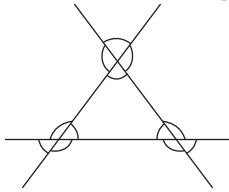
- (a) $x = 80^\circ$, $y = 40^\circ$, $z = 100^\circ$
(b) $x = 80^\circ$, $y = 50^\circ$, $z = 105^\circ$
(c) $x = 70^\circ$, $y = 40^\circ$, $z = 110^\circ$
(d) $x = 60^\circ$, $y = 20^\circ$, $z = 120^\circ$

ANSWERS

1	d	2	d	3	a	4	c	5	b	6	c	7	c	8	a	9	c	10	c
11	a	12	c	13	c	14	a	15	a	16	d	17	c	18	c	19	c	20	a
21	a	22	a	23	a	24	c	25	b	26	c	27	c	28	a	29	a	30	a
31	a	32	c	33	d	34	b	35	c	36	b	37	d	38	a	39	b	40	b
41	b	42	c	43	d	44	a	45	d										

HINTS AND SOLUTIONS

1. (d) We know that, when two lines intersect each other, it makes 4 angles. Since, the total number of pairs = 3



∴ Total number of angles = $3 \times 4 = 12$

2. (d) Let angle be x and its complement be $90^\circ - x$.

According to the question,

$$x = (90^\circ - x) + 14$$

$$\Rightarrow 2x = 104^\circ$$

$$\Rightarrow x = \frac{104^\circ}{2} = 52^\circ$$

3. (a) Let angle be x and its supplementary = $180^\circ - x$

$$\text{Then, } x = 2(180^\circ - x)$$

$$\Rightarrow 3x = 360^\circ$$

$$\Rightarrow x = 120^\circ$$

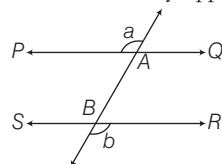
4. (c) The least number of straight lines for a bounded plane figure is 3.

5. (b) Given, angle = $\frac{3}{5}$ of right angle
 $= \frac{3}{5} \times 90^\circ = 3 \times 18^\circ = 54^\circ$

Supplement of $54^\circ = (180^\circ - 54^\circ)$

= An angle of measure 126°

6. (c) We have, $\angle QAB = \angle a$
 [vertically opposite angles]



and $\angle QAB = \angle b$

$$\Rightarrow \angle a = \angle b \quad [\text{corresponding angles}]$$

7. (c) Here, $\angle 1 = \angle 5$
 [corresponding angles]

$$\angle 7 = \angle 5 \quad (\text{vertical opposite})$$

$$\Rightarrow \angle 1 = \angle 7 \quad (\text{angle})$$

$$\text{So, } 5x - 20^\circ = 2x + 10^\circ$$

$$3x = 30^\circ \Rightarrow x = 10^\circ$$

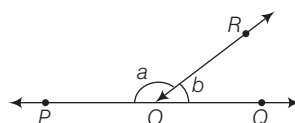
Hence,

$$\angle 7 = 2x + 10^\circ = 2(10^\circ) + 10^\circ = 30^\circ$$

8. (a) Complementary angle of $12^\circ 25' 40''$
 $= 90^\circ - 12^\circ 25' 40''$
 $= 89^\circ 59' 60'' - 12^\circ 25' 40''$
 $= (89 - 12)^\circ + (59' - 25') + (60'' - 40'')$
 $= 77^\circ + 34' + 20'' = 77^\circ 34' 20''$

9. (c) Since, $\angle POR$ and $\angle QOR$ is a linear pair.

$$\therefore \angle POR + \angle ROQ = 180^\circ$$



$$\Rightarrow a + b = 180^\circ \quad \dots (i)$$

$$\text{and } a - b = 80^\circ \quad \dots (ii)$$

On solving Eqs. (i) and (ii), we get

$$2a = 260^\circ \Rightarrow a = 130^\circ$$

$$\text{and } b = 180^\circ - 130^\circ = 50^\circ$$

$$\text{So, } a = 130^\circ, b = 50^\circ$$

10. (c) Since, AOB is a straight line.

$$\therefore \angle AOB = 180^\circ$$

$$\angle AOC + \angle COF + \angle FOB = 180^\circ$$

[straight line]

$$5y + 5y + 2y = 180^\circ$$

$$[\because \angle COF = \angle DOE, \text{ vertically opposite angles}]$$

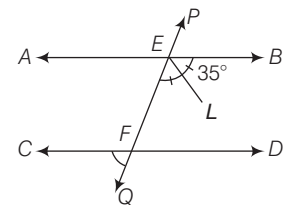
$$\Rightarrow 12y = 180^\circ$$

$$\Rightarrow y = \frac{180^\circ}{12} = 15^\circ$$

Hence, the value of y is 15° .

11. (a) Given, $\angle LEB = 35^\circ$

$$\angle FEB = 2 \times \angle LEB = 2 \times 35^\circ = 70^\circ$$



$$\Rightarrow \angle AEB = \angle AEF + \angle BEF = 180^\circ$$

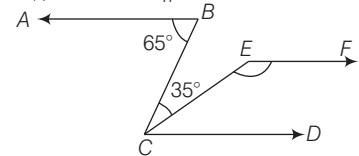
[straight line]

$$\Rightarrow \angle AEF = 180^\circ - 70^\circ = 110^\circ$$

$$\Rightarrow \angle CFQ = \angle AEF = 110^\circ$$

[corresponding angles]

12. (c) Since, $AB \parallel CD$



$$\therefore \angle ABC = \angle BCD = 65^\circ$$

[alternate angles]

$$\angle ECD = 65^\circ - \angle BCE$$

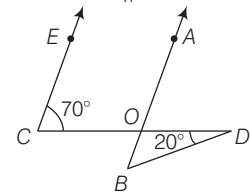
$$= 65^\circ - 35^\circ = 30^\circ$$

$$\angle CEF + \angle ECD = 180^\circ$$

[Angles on the same side of transversal]

$$\Rightarrow \angle CEF = 180^\circ - 30^\circ = 150^\circ$$

13. (c) Since, $EC \parallel AB$



$$\angle AOD = \angle ECO$$

[∵ corresponding angles]

$$\Rightarrow \angle AOD = 70^\circ$$

$$\text{So, } \angle BOD = 180^\circ - 70^\circ$$

$$= 110^\circ \quad [\text{linear point}]$$

Now, in $\triangle BOD$

$$\angle OBD + \angle BOD + \angle ODB = 180^\circ$$

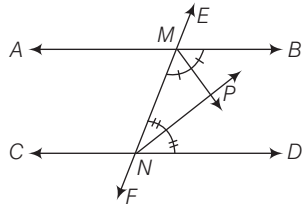
$$\Rightarrow \angle OBD = 180^\circ - (110 + 20)^\circ = 50^\circ$$

14. (a) Given, $\angle PMN = \frac{1}{2} \angle BMN$

$$\text{and } \angle PNM = \frac{1}{2} \angle DNM$$

$$\text{As, } \angle BMN + \angle DNM = 180^\circ$$

[angles on the same side of transversal]



$$\begin{aligned} \therefore \text{In } \triangle PMN, \angle PMN + \angle PNM &= 90^\circ \\ \Rightarrow \angle MPN &= 180^\circ - (\angle PMN + \angle PNM) \\ &\quad [\text{angle sum property}] \\ \therefore \angle MPN &= 180^\circ - 90^\circ = 90^\circ \end{aligned}$$

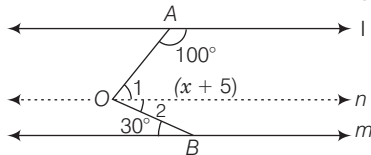
15. (a) Draw a line n passing through O and parallel to l and m

$$\text{Since, } l \parallel n, \angle 1 + 100^\circ = 180^\circ$$

[Sum of the interior angles on the same side of the transversal]

$$\angle 1 = 80^\circ$$

$$\text{Since, } n \parallel m, \angle 2 = 30^\circ \text{ [alternate angles]}$$



$$\begin{aligned} \text{Now, } \angle AOB &= \angle 1 + \angle 2 \\ &= (80^\circ + 30^\circ) = 110^\circ \end{aligned}$$

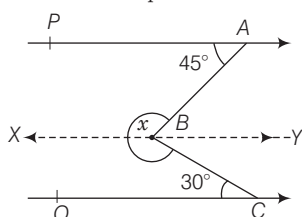
$$\begin{aligned} \text{But, } \angle AOB &= (x + 5)^\circ = 110^\circ \\ x &= (110 - 5)^\circ = 105^\circ \end{aligned}$$

16. (d) Here, $\angle BOD = \angle AOC$
 $\therefore 2x = y$ [vertically opposite angles]
 Now, COB is a straight line.

$$\begin{aligned} \angle COD &= 180^\circ \\ \Rightarrow \angle COE + \angle EOB + \angle BOD &= 180^\circ \\ \Rightarrow 90^\circ + x + 2x &= 180^\circ \\ \Rightarrow 3x &= 90^\circ \Rightarrow x = 30^\circ \end{aligned}$$

Hence, the value of x is 30° .

17. (c) From figure, $PA \parallel QC$
 Draw a line XY parallel to PA and QC .



$$\begin{aligned} \therefore PA \parallel XY, \\ \Rightarrow \angle AXY &= \angle PAB = 45^\circ \\ &\quad [\text{alternate angles}] \end{aligned}$$

$$\begin{aligned} \text{and } \therefore XY \parallel QC, \angle QCB &= \angle CBY = 30^\circ \\ &\quad [\text{alternate angles}] \end{aligned}$$

$$\begin{aligned} \Rightarrow \angle ABC &= \angle AXY + \angle CBY \\ \Rightarrow \angle ABC &= 45^\circ + 30^\circ = 75^\circ \\ \therefore x &= 360^\circ - \angle ABC = 360^\circ - 75^\circ \\ \therefore x &= 285^\circ \end{aligned}$$

18. (c) $\therefore AB \parallel CD, \angle ONP = \angle XMB = 60^\circ$
 [corresponding angles]
 $\angle OPN = 90^\circ - \angle ONP$
 $= 90^\circ - 60^\circ \Rightarrow \angle OPN = 30^\circ$
 But, $\angle DPL = \frac{1}{2} \angle NPO = \frac{1}{2} 30^\circ$
 $\Rightarrow \angle DPL = 15^\circ \Rightarrow x = 15^\circ$

19. (c) Since, $AB \parallel CD$ and PQ is transversal
 $\angle PEF = \angle EGH$ [corresponding angles]
 $\Rightarrow \angle EGH = 70^\circ$ [$\because \angle PEF = 70^\circ$]
 Now, $\angle EGH + \angle HGQ = 180^\circ$
 [linear pair]
 $\Rightarrow \angle HGQ = 180^\circ - 70^\circ = 110^\circ$
 Also, $\angle DHQ + \angle GHQ = 180^\circ$
 [linear pair]
 $\angle GHQ = 180^\circ - 140^\circ = 40^\circ$
 $\therefore x + y = 110^\circ + 40^\circ = 150^\circ$

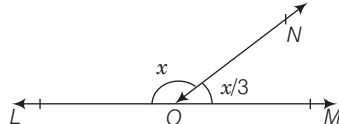
20. (a) All the three statements are true.

21. (a) Only statements I and II are true.

22. (a) Draw $FH \parallel AB \parallel CD$
 (sum of interior angles)
 $\therefore \angle 1 + p = 180^\circ \dots (i)$
-
- $$\begin{aligned} \angle 2 + 180^\circ - r &= 180^\circ \dots (ii) \\ &\quad (\text{sum of interior angle}) \end{aligned}$$

$$\begin{aligned} \text{On adding Eqs. (i) and (ii), we get} \\ \angle 1 + \angle 2 + p + 180^\circ - r &= 360^\circ \\ \Rightarrow p + q - r &= 180^\circ [\because \angle 1 + \angle 2 = q^\circ] \end{aligned}$$

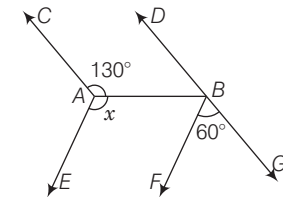
23. (a) Given that, $\angle MON = \frac{1}{3} \angle LON$
 Let $\angle LON = x$, then, $\angle MON = \frac{x}{3}$



$$\begin{aligned} \text{We know that,} \\ \angle LON + \angle MON &= 180^\circ \text{ [linear pair]} \\ \Rightarrow x + \frac{x}{3} &= 180^\circ \Rightarrow x = \frac{180^\circ \times 3}{4} = 135^\circ \\ \text{Thus, } \angle MON &= \frac{x}{3} = \frac{135^\circ}{3} = 45^\circ \end{aligned}$$

24. (c) Both the statements I and II are correct.

25. (b) Since, $AC \parallel BD$



$$\begin{aligned} \therefore \angle DBA &= 180^\circ - 130^\circ = 50^\circ \\ &\quad [\text{interior angle}] \\ &\quad [\because \angle CAB = 130^\circ] \end{aligned}$$

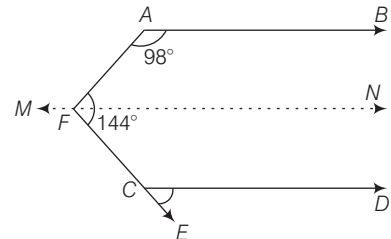
DBG is straight line.

$$\begin{aligned} \therefore \angle DBA + \angle ABF + \angle FBG &= 180^\circ \\ &\quad [\text{linear pairs}] \\ \Rightarrow 50^\circ + \angle ABF + 60^\circ &= 180^\circ \\ \Rightarrow \angle ABF &= 70^\circ \\ \text{Since, } AE \parallel BF \\ \therefore x &= 180^\circ - \angle ABF \\ &= 180^\circ - 70^\circ = 110^\circ \end{aligned}$$

26. (c) When two lines are parallel intersected by a transversal then corresponding as well as alternate interior angles are equal.
 Hence, the statement II and IV are correct.

27. (c) We have, $a = d, b = e$ and $c = f$
 [vertically opposite angles]
 $\text{and } a + b + c = d + e + f = 180^\circ$
 II. $a + b + c = 180^\circ \Rightarrow a + c + e = 180^\circ$
 III. $b + f = c + e$
 IV. $a + b + c = d + e + f$
 Hence, statements II, III and IV are true.

28. (a) Draw a line MN parallel to AB and CD .



$$\begin{aligned} \Rightarrow \angle AFN &= 180^\circ - 98^\circ = 82^\circ \\ &\quad (\text{sum of interior angles}) \\ \angle CFN &= 144^\circ - 82^\circ = 62^\circ \\ \text{and } \angle ECD &= \angle CFN = 62^\circ \\ &\quad [\text{corresponding angles}] \end{aligned}$$

29. (a) If two parallel lines are intersected by a transversal, then each pair of corresponding angles and of alternate angles are equal.
 Therefore, statement I is correct.

30. (a) Since, $PQ \parallel RS$

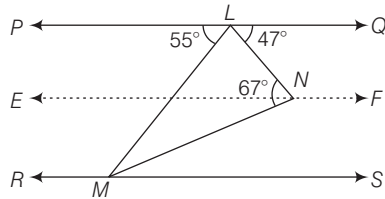
$$\therefore \angle PLM = \angle LMS = 55^\circ$$

[alternate angle]

Draw a line EF which is parallel to PQ .

$$\text{Then, } \angle QLN = \angle LNE = 47^\circ$$

$$\therefore \angle ENL + \angle MNE = 67^\circ$$



$$\Rightarrow 47^\circ + \angle MNE = 67^\circ$$

$$\Rightarrow \angle MNE = 67^\circ - 47^\circ$$

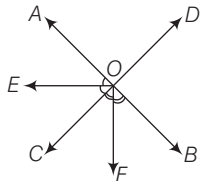
$$\Rightarrow \angle MNE = 20^\circ$$

Similarly, $EF \parallel RS$, then

$$\angle ENM = \angle NMS = 20^\circ$$

[alternate angle]

31. (a) Given, $\angle BOC = 130^\circ$



$\therefore AOB$ is a straight line.

$$\therefore \angle BOC + \angle AOC = 180^\circ \text{ [linear pair]}$$

$$\Rightarrow 130^\circ + \angle AOC = 180^\circ$$

$$\Rightarrow \angle FOC + \angle FOC = 130^\circ$$

$$\Rightarrow \angle AOC = 50^\circ$$

$$\text{Now, } \angle BOC = 130^\circ$$

$$\Rightarrow \angle BOF + \angle FOC = 130^\circ$$

[$\therefore OF$ is bisector of $\angle BOC$]

$$\Rightarrow \angle FOC = 65^\circ$$

$$\text{Now, } \angle AOC = 50^\circ$$

$$\Rightarrow \angle AOE + \angle EOC = 50^\circ$$

$$\Rightarrow \angle EOC + \angle EOC = 50^\circ$$

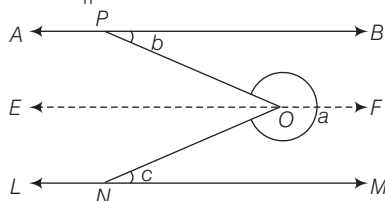
$$\Rightarrow \angle EOC = 25^\circ$$

[$\therefore OE$ is bisector of $\angle AOC$]

$$\therefore \angle EOF = \angle EOC + \angle FOC$$

$$= 25^\circ + 65^\circ = 90^\circ$$

32. (c) Draw a line parallel to AB i.e. $EF \parallel AB$.



$$\therefore \angle EOP = \angle OPB = b \text{ [alternate angle]}$$

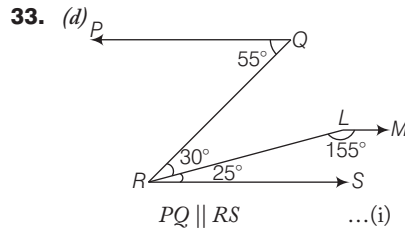
$$\text{and } \angle EON = \angle ONM = c$$

[alternate angle]

$$\Rightarrow \angle PON = b + c$$

$$\therefore \angle PON + a = 2\pi$$

$$\therefore a = 2\pi - \angle PON = 2\pi - b - c$$



$$\text{Since, } \angle PQR = \angle QRS = 30^\circ + 25^\circ = 55^\circ$$

[alternate angle]

$$PQ \parallel RS \quad \dots(i)$$

$$\text{and } \angle SRL + \angle RLM = 180^\circ$$

$$\Rightarrow RS \parallel LM \quad \dots(ii)$$

From Eqs. (i) and (ii), we get

$$PQ \parallel LM$$

So, the angle between the lines PQ and LM is 180° .

34. (b) Given, $AC^2 = AB \times CB$

$$\Rightarrow x^2 = 2 \times (2 - x)$$

$$A \quad \xrightarrow{x} \quad C \quad \xrightarrow{(2-x)} \quad B$$

$$\xrightarrow{2}$$

$$\Rightarrow x^2 = 4 - 2x$$

$$\Rightarrow x^2 + 2x - 4 = 0$$

$$\Rightarrow x = \frac{-2 \pm \sqrt{4 + 16}}{2 \times 1}$$

$$\Rightarrow x = -1 \pm \sqrt{5}$$

$$\text{Now, } BC = 2 - (-1 + \sqrt{5}) = 3 - \sqrt{5}$$

$$(\text{neglect } 3 + \sqrt{5} \because 3 + \sqrt{5} > 2)$$

35. (c) We have, $\angle DGE = \angle GCF = 95^\circ$
[corresponding angles]

$$\text{Also, } \angle GCF + \angle FCB = 180^\circ$$

[linear pair]

$$\Rightarrow 95^\circ + y = 180^\circ$$

$$\Rightarrow y = 180^\circ - 95^\circ = 85^\circ$$

$$\text{Again, } \angle ACB + y = 180^\circ$$

$$\Rightarrow x = 180^\circ - 85^\circ = 95^\circ$$

$$\text{Now, In } \triangle ABC \angle A + \angle B + \angle C = 180^\circ$$

$$\Rightarrow 35^\circ + z + 95^\circ = 180^\circ$$

$$\Rightarrow z + 130^\circ = 180^\circ$$

$$\Rightarrow z = 180^\circ - 130^\circ = 50^\circ$$

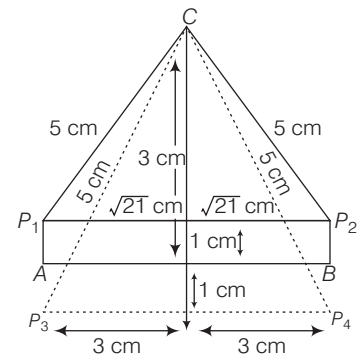
$$\therefore x + y + z = 95^\circ + 85^\circ + 50^\circ = 230^\circ$$

36. (b) Supplement of $z = 130^\circ$

$$\therefore \frac{1}{5} \text{th of } 130^\circ = \frac{130^\circ}{5} = 26^\circ$$

37. (d) $\therefore$ Required number of points

$$= 4(P_1, P_2, P_3, P_4)$$



38. (a) Given that, $\frac{\alpha}{\beta} = \frac{1}{5}$

$$\text{Let } \alpha = k \text{ and } \beta = 5k$$

Sum of two complementary angles

$$= 90^\circ$$

$$\Rightarrow \alpha + \beta = 90^\circ,$$

$$\alpha = 90^\circ - \beta$$

$$\Rightarrow k = 90^\circ - 5k$$

$$\Rightarrow k = 15^\circ$$

$$\therefore \alpha = 15^\circ \text{ and } \beta = 75^\circ$$

$$\therefore \text{Difference between angles}$$

$$= 75^\circ - 15^\circ = 60^\circ$$

39. (b) Here, AB and CD are two lines.



If two straight lines intersect, then vertically opposite angles are equal.

40. (b) From the given figure,

$$\angle LOQ + \angle QOP + \angle POM = 180^\circ$$

[straight line]

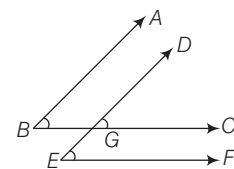
$$\therefore (x^\circ + 20^\circ) + 50^\circ + (x^\circ - 10^\circ) = 180^\circ$$

$$\Rightarrow 2x^\circ + 60^\circ = 180^\circ$$

$$\Rightarrow 2x^\circ = 120^\circ$$

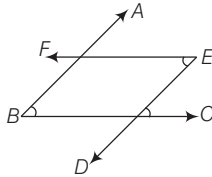
$$\therefore x^\circ = 60^\circ$$

41. (b) Case I When both pairs of arms are parallel same sense.



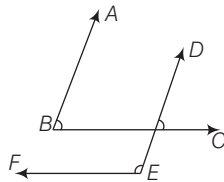
$$\text{Here, } \angle ABC = \angle DEF$$

Case II When both pairs of arms are parallel in opposite sense.



Here, $\angle ABC = \angle DEF$

Case III When one pair of arms is parallel in same direction and other pairs are parallel in opposite direction.



Here, $\angle ABC + \angle DEF = 180^\circ$

So, the two angles are not equal but supplementary.

- 42.** (c) Let the angle be θ .

$$\therefore \theta = 80^\circ$$

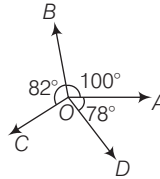
Complement angle

$$= 90^\circ - \theta = 90^\circ - 80^\circ = 10^\circ$$

[$\because$ sum of two complement angles is 90°]

$$= 10 \times \frac{\pi}{180^\circ} = \frac{\pi}{18} \text{ radian.}$$

- 43.** (d)



Given, $\angle AOB = \angle COD = 100^\circ$,

$$\angle BOC = 82^\circ$$

and $\angle AOD = 78^\circ$

If, AOC is a straight line, then

$$\angle AOB + \angle BOC = 180^\circ$$

$$\Rightarrow 100^\circ + 82^\circ = 180^\circ$$

$$\Rightarrow 182^\circ \neq 180^\circ$$

If, BOD is a straight line, then

$$\angle BOA + \angle AOD = 180^\circ$$

$$\Rightarrow 100^\circ + 78^\circ = 180^\circ$$

$$\Rightarrow 178^\circ \neq 180^\circ$$

If, $\angle BOC$ and $\angle AOD$

are supplementary angles, then

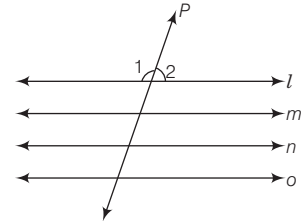
$$\angle BOC + \angle AOD = 180^\circ$$

$$\Rightarrow 82^\circ + 78^\circ = 180^\circ$$

$$\Rightarrow 160^\circ \neq 180^\circ$$

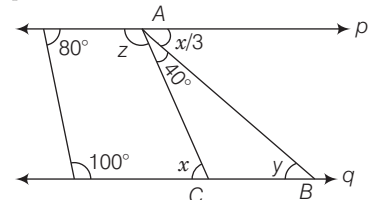
Hence, statements I and II are incorrect.

- 44.** (a) If, line P intersect four parallel lines l, m, n and o , then 16 angles will be formed.



As these lines are parallel, hence distinct angle will be $\angle 1$ and $\angle 2$.

- 45.** (d) In the given figure, lines p and q are parallel.



$$\therefore x = 40^\circ + \frac{x}{3} \text{ [alternate angles]}$$

$$\Rightarrow \frac{2x}{3} = 40^\circ \Rightarrow x = 60^\circ$$

$$y = x/3 \text{ [alternate angle]}$$

$$y = \frac{60^\circ}{3} = 20^\circ, z + 40^\circ + \frac{x}{3} = 180^\circ$$

$$\Rightarrow z + 20^\circ = 180^\circ - 40^\circ$$

$$\Rightarrow z = 120^\circ$$

23

TRIANGLES

Generally (8-10) questions have been asked from this chapter. Generally questions are asked from the topic of pythagoras theorem, similarity of triangles and mid-point theorem.

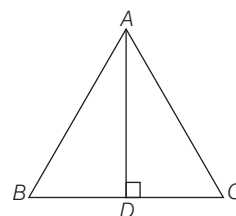


TRIANGLE

A plane (closed) figure bounded by three line segments is called a triangle. It is denoted by Δ .

ΔABC has

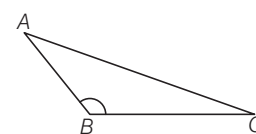
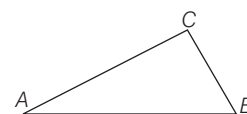
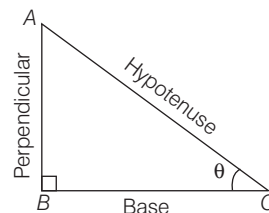
- three vertices, namely A , B and C .
- three sides, namely AB , BC and CA .
- three angles, namely $\angle A$, $\angle B$ and $\angle C$.
- Sum of three angles of a triangle is 180° . i.e., $\angle A + \angle B + \angle C = 180^\circ$
- Area of a triangle $= \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times BC \times AD$



Classification of Triangle

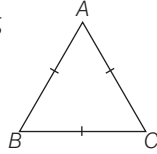
1. On the Basis of Angles

- Right Angled Triangle** A triangle in which one of the angle measures 90° is called a right angled triangle. The side opposite to the right angle is called its hypotenuse and the remaining two sides are called as perpendicular and base depending upon conditions. Here, ΔABC is a right angled triangle in which $\angle B = 90^\circ$ and AC is hypotenuse.
- Acute Angled Triangle** A triangle in which every angle measure more than 0° but less than 90° is called an acute angled triangle. Here, ΔABC is an acute angled triangle.
- Obtuse Angled Triangle** A triangle in which one of the angle measures more than 90° but less than 180° is called an obtuse angled triangle. Here, ΔABC is an obtuse angled triangle and $\angle ABC$ is the obtuse angle.

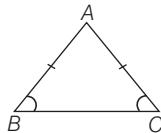


2. On the Basis of Sides

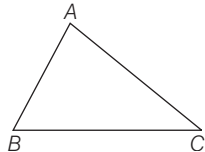
- (i) **Equilateral Triangle** A triangle having all sides equal is called an equilateral triangle. Here $\triangle ABC$, is an equilateral triangle in which $AB = BC = AC$. All angles are equal and are of measure 60° .



- (ii) **Isosceles Triangle** A triangle in which two sides are equal, is called an isosceles triangle. Here, $\triangle ABC$ is an isosceles triangle as $AB = AC$.
- Angle opposite to equal sides are equal.
i.e. $\angle B = \angle C$



- (iii) **Scalene Triangle** A triangle in which all the sides are of different lengths is called a scalene triangle. $\triangle ABC$ is a scalene triangle as $AB \neq BC \neq AC$.

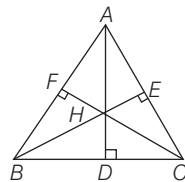


Note The sum of the lengths of three sides of a triangle is called its perimeter.
So, perimeter of $\triangle ABC = AB + BC + AC$

Some Term Related to a Triangle

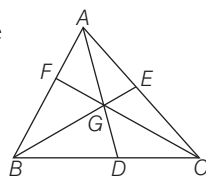
Altitudes and Orthocentre The altitude of a triangle is a line segment perpendicular drawn from vertex to the side opposite to it. The side on which the perpendicular is being drawn is called its base.

Here, In $\triangle ABC$, AD , BE and CF are altitudes drawn on BC , AC and AB , respectively.



- Altitudes of a triangle are concurrent.
- The point of intersection of all the three altitudes of a triangle is called its orthocentre.

Medians and Centroid A line segment joining the mid-point of a side to the opposite vertex is called median. Here, In $\triangle ABC$, AD , BE and CF are medians.

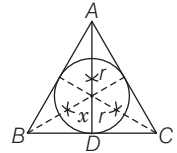


- The medians of a triangle are concurrent.
- The point of intersection of all the three medians of a triangle is called its centroid. It is denoted by G .

- Centroid divides the median in the ratio $2 : 1$.
- A median bisects the area of the triangle i.e. $\text{Ar}(\triangle ABD) = \text{Ar}(\triangle ADC) = \frac{1}{2} \text{Ar}(\triangle ABC)$, etc.

Incentre and Angle Bisector The point of intersection of all the three angle bisectors of a triangle is called its incentre. It is denoted by I .

- The circle with centre I and touches all the sides is called incircle and radius of this circle is called inradius denoted by ' r '.

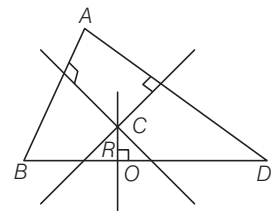


- Inradius $= \frac{1}{3} \times \text{Height} = \frac{\text{Side}}{2\sqrt{3}}$

$ID \rightarrow$ Inradius, $AD \rightarrow$ Height

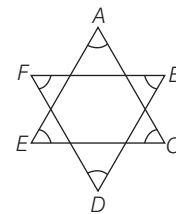
Perpendicular Bisector and Circumcentre The point of intersection of the perpendicular bisectors of the sides of a triangle is called its circumcentre. It is denoted by C .

The circle with centre C and passing through vertices A , B and C is called circumcircle. Radius of circumcircle is called circumradius denoted by R .



Circumradius $(CO) = \frac{2}{3} \times \text{Height} = \frac{\text{Side}}{3}$

EXAMPLE 1. In the diagram given below what is the sum of all the angles $\angle A$, $\angle B$, $\angle C$, $\angle D$, $\angle E$ and $\angle F$?



- a. 120° b. 180°
c. 290° d. 360°

Sol. d. Since, sum of the angles of a triangle is 180° .

$$\text{In } \triangle AEC, \angle A + \angle C + \angle E = 180^\circ \quad \dots(i)$$

$$\text{and In } \triangle BDF, \angle B + \angle D + \angle F = 180^\circ \quad \dots(ii)$$

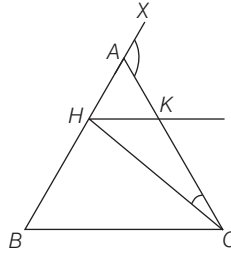
On adding the Eqs. (i) and (ii), we get

$$\angle A + \angle B + \angle C + \angle D + \angle E + \angle F = 180^\circ + 180^\circ = 360^\circ$$

EXAMPLE 2. ABC is an isosceles triangle in which $AB = AC$, $CH \perp BC$ and HK is parallel to BC . If the exterior $\angle CAX = 137^\circ$, then what is the measure of $\angle HCK$?

- a. $68\frac{1}{2}^\circ$ b. 43°
c. $25\frac{1}{2}^\circ$ d. 137°

Sol. $\therefore \angle CAX = 137^\circ$



$$\therefore \angle ABC = \frac{1}{2}(137^\circ) = 68\frac{1}{2}^\circ$$

$$\therefore \text{Again, } BC = CH \text{ and } \angle ABC = 68\frac{1}{2}^\circ$$

$$\text{Therefore, } \angle CHB = 68\frac{1}{2}^\circ, \text{ Therefore, } \angle HCB = 43^\circ$$

$$\text{Hence, } \angle HCK = 68\frac{1}{2}^\circ - 43^\circ = 25\frac{1}{2}^\circ$$

Some Important Results of a Triangle

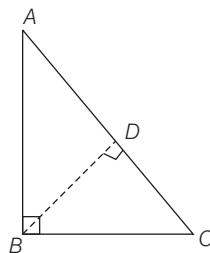
- If two angles of a triangle are equal, then the sides opposite to them are also equal.
- In a triangle, the side opposite to a greater angle is the longest side.
- The sum of all the three interior angles of a triangle is 180° .
- If a side of a triangle is produced, then the exterior angle so formed is equal to the sum of the two interior opposite angles.
- An exterior angle of a triangle is greater than either of the interior opposite angles.
- A triangle must have at least two acute angles.
- If a perpendicular is drawn from the vertex of a right angled triangle to the hypotenuse, then the triangles on both sides of the perpendicular are similar to the original triangle and also to each other.
- In a right angle $\triangle ABC$, $\angle B = 90^\circ$ and AC is hypotenuse. The perpendicular BD is dropped on hypotenuse AC from right angle vertex B , then

$$(i) \quad BD = \frac{AB \times BC}{AC}$$

$$(ii) \quad AD = \frac{AB^2}{AC}$$

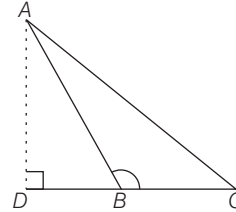
$$(iii) \quad CD = \frac{BC^2}{AC}$$

$$(iv) \quad \frac{1}{BD^2} = \frac{1}{AB^2} + \frac{1}{BC^2}$$

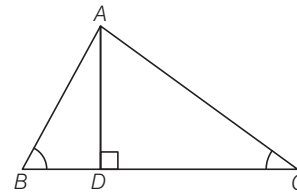


- The area of the equilateral triangle described on the side of a square is half the area of the equilateral triangle described on its diagonal.

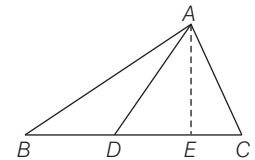
- In $\triangle ABC$, if $\angle B > 90^\circ$ and AD is perpendicular dropped on BC , then $AC^2 = AB^2 + BC^2 + 2BC \cdot BD$



- In $\triangle ABC$, if $\angle B < 90^\circ$ and $AD \perp BC$, then $AC^2 = AB^2 + BC^2 - 2BC \cdot BD$



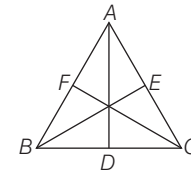
- In any triangle, the sum of the square of any two sides is equal to twice the square of half of the third side together with twice the square of the median which bisect the third side.



Here, AD is median, so

$$AB^2 + AC^2 = 2AD^2 + 2\left(\frac{1}{2}BC\right)^2$$

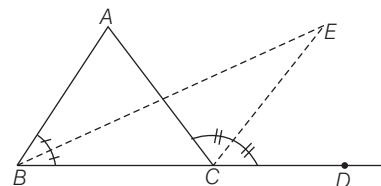
- In a $\triangle ABC$, three times the sum of the squares of the sides of a triangle is equal to four times the sum of the squares of the medians of the triangle.



So, in triangle, if AD, BE, FC are the medians, then

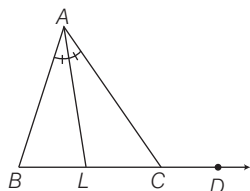
$$3(AB^2 + BC^2 + AC^2) = 4(AD^2 + BE^2 + CF^2)$$

- In an equilateral $\triangle ABC$, the side BC is trisected at D . Then, $9AD^2 = 7AC^2$
- When the bisector of one internal base angle and the bisector of external angle meet at a point then the formed angle is equal to one half of the vertical angle.

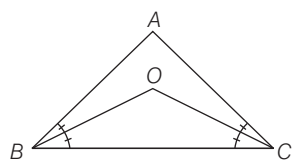


$$\text{Here, } \angle BEC = \frac{1}{2} \angle BAC$$

- The side BC of $\triangle ABC$ is produced to D . The bisector of $\angle A$ meets BC in L . Then $\angle ABC + \angle ACD = 2\angle ALC$.

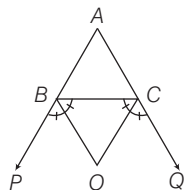


- In a $\triangle ABC$ the bisector of $\angle B$ and $\angle C$ intersect each other at a point O .



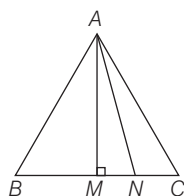
$$\text{then, } \angle BOC = 90^\circ + \frac{1}{2} \angle A$$

- In a $\triangle ABC$, the side AB and AC are produced to P and Q , respectively. The bisectors of $\angle PBC$ and $\angle QCB$ intersect at a point O .



$$\text{Then, } \angle BOC = 90^\circ - \frac{1}{2} \angle A$$

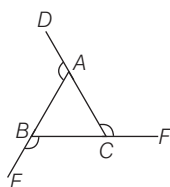
- In $\triangle ABC$, $\angle B > \angle C$. If AN is the bisector of $\angle BAC$ and $AM \perp BC$, then



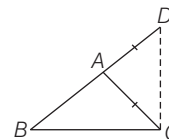
$$\angle MAN = \frac{1}{2} (\angle B - \angle C)$$

- The bisectors of the base angles of a triangle can never enclose a right angle.
- A triangle is an isosceles if and only if any two altitudes are equal.
- If the three sides of a triangle be produced in order, then the sum of all the exterior angles so formed is 360° .

$$\text{So, } \angle DAB + \angle EBC + \angle ACF = 360^\circ$$



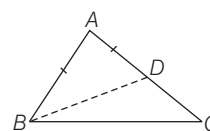
- Sum of any two sides of a triangle is greater than its third side.



Here, in $\triangle ABC$, $AB + AC > BC$

similarly, $AB + BC > AC$, $BC + AC > AB$

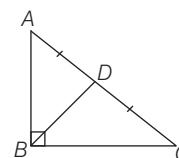
- The difference between any two sides of a triangle is less than its third side.



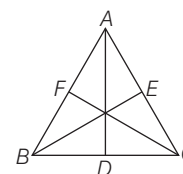
Here, in $\triangle ABC$, $AC - AB < BC$

Similarly, $BC - AC < AB$, $BC - AB < AC$

- If the bisector of the vertical angle of a triangle bisects the base, then that triangle is an isosceles.
- If the altitude from one vertex of a triangle bisects the opposite side, then the triangle is an isosceles.
- The perpendiculars drawn from the vertices of equal angles of an isosceles triangle to the opposite sides are equal.
- Medians of equilateral triangle are equal.
- If D is the mid-point of the hypotenuse AC of a right angled $\triangle ABC$. Then, $BD = \frac{1}{2} AC$

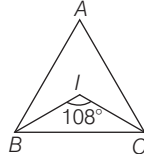


- The sum of any two sides of a triangle is greater than twice the median drawn to the third side.
- Perimeter of a triangle is greater than the sum of its three medians.



$$\text{So, } AB + BC + AC > AD + BE + CF.$$

EXAMPLE 3. In the given figure, I is the incentre of $\triangle ABC$. What is the measure of angle A .



- a. 54° b. 18° c. 36° d. None of these

Sol. c. Here, I is the incentre of the $\triangle ABC$.

$\therefore BI$ and CI are the bisectors of $\angle B$ and $\angle C$, then

we know that, $\angle BIC = 90^\circ + \frac{1}{2} \angle A$

$$\Rightarrow 108^\circ = 90^\circ + \frac{1}{2} \angle A$$

$$\Rightarrow \frac{1}{2} \angle A = 108^\circ - 90^\circ = 18^\circ$$

$$\therefore \angle A = 36^\circ$$

EXAMPLE 4. In a triangle ABC , $\angle BAC = 90^\circ$ and AD is perpendicular to BC . If $AD = 6$ cm and $BD = 4$ cm, then the length of BC is

- a. 8 cm b. 10 cm c. 9 cm d. 13 cm

Sol. d. In $\triangle ABC$, $\angle BAC = 90^\circ$

$$\therefore AB = \sqrt{AD^2 + BD^2} \quad [\text{By pythagoras theorem}]$$

$$= \sqrt{36 + 16} = \sqrt{52}$$

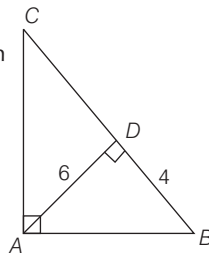
We know that, in a right angled triangle the perpendicular is drawn on hypotenuse from right angle vertex, then

$$BD = \frac{AB^2}{BC}$$

$$AB^2 = BD \times BC$$

$$\Rightarrow (\sqrt{52})^2 = BC \times 4$$

$$\Rightarrow BC = \frac{52}{4} = 13 \text{ cm}$$



Congruent Figures

The geometrical figures having the same shape and size are known as congruent figures.

e.g. Two circles of the same radii and two squares of the same sides are congruent.

Congruent Triangles

Two triangles are said to be congruent, if both are exactly of same size i.e. all angles and sides are equal to corresponding angles and sides of other.

- Every triangle is congruent to itself $\triangle ABC \cong \triangle ABC$
- If $\triangle ABC \cong \triangle DEF$, then $\triangle DEF \cong \triangle ABC$
- If $\triangle ABC \cong \triangle DEF$ and $\triangle DEF \cong \triangle PQR$, then $\triangle ABC \cong \triangle PQR$

In congruent triangles, corresponding parts are equal and we write in short 'CPCT', i.e. corresponding part of congruent triangles.

Criteria for Congruence of Triangles

Theorem 1 Two triangles are congruent if two sides and the included angle of one triangle are equal to the corresponding sides and the included angle of other triangle. (SAS criteria)

Theorem 2 Two triangles are congruent if two angles and the included side of one triangle are equal to the corresponding two angles and the included side of the other triangles. (ASA criteria)

Theorem 3 Two triangles are congruent if three sides of one triangle are equal to the corresponding three sides of the other triangle (SSS).

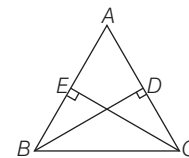
Theorem 4 Two triangles are congruent if the hypotenuse and other side of one triangle are equal to the hypotenuse and the corresponding side of the other triangle (RHS).

Theorem 5 Two triangles are congruent if two angles and a side other than the included side of one triangle are equal to the corresponding two angles and a side other than the included side of other triangle. (AAS Criteria)

EXAMPLE 5. In a $\triangle ABC$, the altitudes BD and CE are equal and $\angle A = 36^\circ$. What is the value of the $\angle B$?

- a. 72° b. 84° c. 18° d. 36°

Sol. a. For the $\triangle BDC$ and $\triangle BEC$,

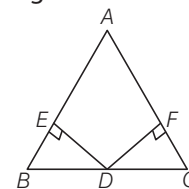


$$BD = EC, BC = BC \text{ and } \angle BEC = \angle BDC = 90^\circ$$

Thus, $\triangle BEC \cong \triangle BDC$ [by RHS rule]

$$\therefore \angle B = \angle C = \frac{180^\circ - 36^\circ}{2} = 72^\circ \text{ each}$$

EXAMPLE 6. In the given figure, D is the midpoint of BC , $DE \perp AB$ and $DF \perp AC$ such that $DE = DF$, then which of the following is correct?



- a. $AB = AC$ b. $AC = BC$
c. $AB = BC$ d. None of these

Sol. a. In right angled $\triangle BED$ and right angled $\triangle CFD$

$$\begin{aligned} DE &= DF && \text{(given)} \\ \text{hypotenuse } BD &= \text{hypotenuse } CD && [\because D \text{ is the mid-point of } BC] \\ \therefore \triangle BED &\cong \triangle CFD && [\text{by RHS congruency}] \\ \Rightarrow \angle B &= \angle C \Rightarrow AC = AB && [\text{sides opposite to equal angles}] \end{aligned}$$

Similar Figures

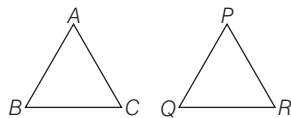
The geometrical figures having the same shape, but different sizes are known as similar figures.

- The congruent figures are always similar, but two similar figures need not be congruent.
e.g. Any two circles are similar. Any two rectangles are similar.

Similar Triangles

Two triangles are said to be similar to each other, if

- their corresponding sides are proportional.
- their corresponding angles are equal.



Here, $\triangle ABC$ and $\triangle PQR$ are similar triangles.

$$\begin{aligned} \therefore \quad \angle A &= \angle P, \angle B = \angle Q, \angle C = \angle R \\ \text{and} \quad \frac{AB}{PQ} &= \frac{BC}{QR} = \frac{AC}{PR} \end{aligned}$$

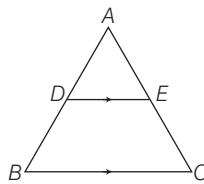
Then, $\triangle ABC \sim \triangle PQR$

where, symbol $\sim$ is read as, 'is similar to'.

Some Results on Similar Triangles

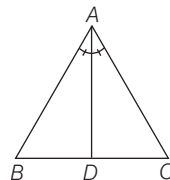
Theorem 1 If a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct point, then it divides these sides in the same ratio. It is also called basic proportionality theorem.

$$\begin{aligned} \text{Here, } DE \parallel BC, \text{ then } \frac{AD}{DB} &= \frac{AE}{EC} \\ \text{or } \frac{AD}{AB} &= \frac{AE}{AC} \\ \text{or } \frac{AB}{DB} &= \frac{AC}{EC} \end{aligned}$$



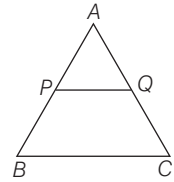
Theorem 2 The internal bisector of an angle of a triangle divides the opposite sides internally in the ratio of the sides containing the angle.

$$\begin{aligned} \text{Here, } AD \text{ is internal bisector of } \angle A, \text{ then} \\ \frac{AB}{AC} &= \frac{BD}{DC} \end{aligned}$$

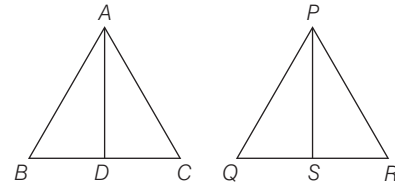


Theorem 3 The line joining the mid-points of any two sides of a triangle is parallel to the third side and is half of the third side.

Here, P and Q are mid-point of AB and AC . So, $PQ = \frac{1}{2} BC$.



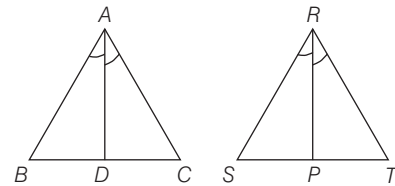
Theorem 4 If two triangles are equiangular, then the ratio of their corresponding sides is the same as the ratio of the corresponding altitudes.



Here, $\triangle ABC \sim \triangle PQR$ and AD and PS are altitude on BC and QR , respectively, then

$$\frac{BC}{QR} = \frac{AD}{PS}$$

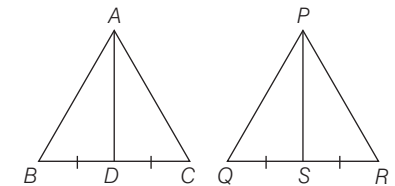
Theorem 5 If two triangles are equiangular, then the ratio of the corresponding sides is the same as the ratio of the corresponding angle bisector segments.



Here, $\triangle ABC$ and $\triangle RST$ are equiangular/similar and AD , RP are the angle bisectors of $\angle A$ and $\angle R$, then

$$\frac{BC}{ST} = \frac{AD}{RP}$$

Theorem 6 If two triangles are equiangular, then the ratio of the corresponding sides is the same as the ratio of the corresponding medians.



Here, $\triangle ABC$ and $\triangle PQR$ are equiangular and AD , PS are the medians, then

$$\frac{BC}{QR} = \frac{AD}{PS}$$

Note: If two triangles are similar, then ratio of their corresponding sides is same as ratio of their perimeters.

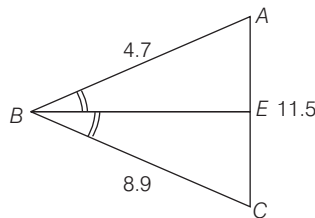
Criteria for Similarity of Two Triangles

1. **AAA similarity** If the corresponding angles of two triangles are equal then their corresponding sides are proportional and hence the two triangles are similar.

Corollary AA similarity If two angles of one triangle are equal to the corresponding two angles of another triangle, then the two triangles are similar.

2. **SSS similarity** If the corresponding sides of two triangles are proportional, then their corresponding angles are equal, and hence the two triangles are similar.
3. **SAS similarity** If one angle of a triangle is equal to the corresponding angle of the other triangle and the sides including these angles are proportional, then the two triangles are similar.
4. **RHS similarity** If hypotenuse and one side of a right triangle are proportional to the hypotenuse and corresponding side of other right triangle then the two triangles are similar.

EXAMPLE 7. If $AB = 4.7$, $BC = 8.9$, $CA = 11.5$, then EA is



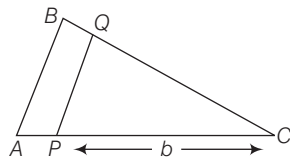
- a. 6.07 b. 3.97
c. 2.37 d. None of these

Sol. b. Since, BE is the bisector of $\angle ABC$.

$$\therefore \frac{AB}{BC} = \frac{AE}{EC} \Rightarrow \frac{4.7}{8.9} = \frac{AE}{11.5 - AE} \quad [\because EC = AC - AE]$$

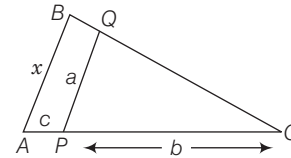
$$\Rightarrow AE = 3.97$$

EXAMPLE 8. In the given triangle, AB is parallel to PQ . $AP = c$, $PC = b$, $PQ = a$, $AB = x$. What is the value of x ?



- a. $a + \frac{ab}{c}$ b. $a + \frac{bc}{a}$
c. $b + \frac{ca}{b}$ d. $a + \frac{ac}{b}$

Sol. d. In $\triangle ABC$ and $\triangle PQC$.



$$AB \parallel PQ$$

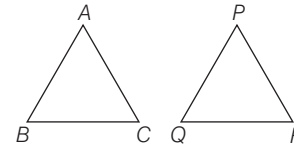
$$\therefore \triangle ABC \sim \triangle PQC$$

$$\therefore \frac{PC}{AC} = \frac{PQ}{AB} \Rightarrow \frac{b}{c+b} = \frac{a}{x}$$

$$\therefore x = \frac{a(c+b)}{b} = \frac{ac}{b} + a$$

Areas of Similar Triangles

Theorem 1 The ratio of areas of two similar triangles is equal to the square of the ratio of their corresponding sides.



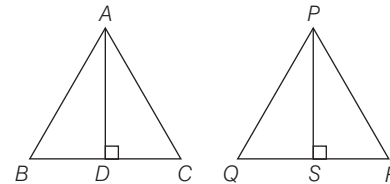
Here, $\triangle ABC \sim \triangle PQR$

$$\frac{\text{Area}(\triangle ABC)}{\text{Area}(\triangle PQR)} = \frac{AB^2}{PQ^2} = \frac{BC^2}{QR^2} = \frac{AC^2}{PR^2}$$

Theorem 2 The areas of two similar triangles is equal to the ratio of the squares of corresponding altitudes.

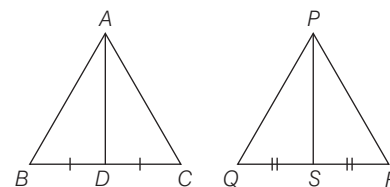
Here,

$$\triangle ABC \sim \triangle PQR$$



$$\text{Then, } \frac{\text{Area}(\triangle ABC)}{\text{Area}(\triangle PQR)} = \frac{AD^2}{PS^2}$$

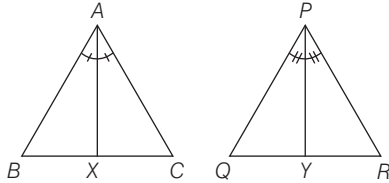
Theorem 3 The areas of two similar triangles is equal to the ratio of the squares of the corresponding medians.



Here, $\triangle ABC \sim \triangle PQR$,

$$\text{then } \frac{\text{Area}(\triangle ABC)}{\text{Area}(\triangle PQR)} = \frac{AD^2}{PS^2}$$

Theorem 4 The areas of two similar triangles is equal to the ratio of squares of the corresponding angle bisector segments.



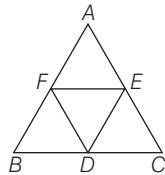
Here, $\Delta ABC \sim \Delta PQR$

So, here $\frac{\text{Area}(\Delta ABC)}{\text{Area}(\Delta PQR)} = \frac{AX^2}{PY^2}$

Theorem 5 If the areas of two similar triangles are equal, then the triangles are congruent.

or
Equal and similar triangles are congruent.

Theorem 6 The line segments joining the mid-points of the sides of a triangle form four triangles, each of which is similar to the original triangle.



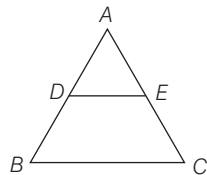
Here, D, E and F are mid-point of BC, AC and AB. Then, ΔAFE , ΔFBD , ΔEDC and ΔDEF is similar to ΔABC .

Here, also

$$\frac{\text{Area}(\Delta DEF)}{\text{Area}(\Delta ABC)} = \frac{DE^2}{AB^2} = \frac{\left(\frac{1}{2}AB\right)^2}{AB^2} = \frac{1}{4}$$

So, area (ΔDEF) : area (ΔABC) = 1 : 4

EXAMPLE 9. In the figure given below, BC is parallel to DE and $DE : BC = 3 : 5$. What is the ratio of area of the ΔABC to that of ΔADE ?



- a. 3 : 1 b. 5 : 3 c. 9 : 2 d. 25 : 9

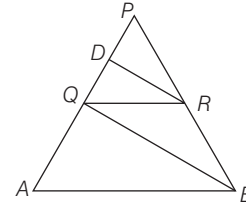
Sol. d. $\therefore$ Given, $DE : BC = 3 : 5$

Since, $DE \parallel BC \Rightarrow \angle ADE = \angle ABC$
and $\angle AED = \angle ACB$

$\therefore \Delta ABC \sim \Delta ADE$ [by AA similarity]

$$\therefore \frac{\text{Area of } \Delta ABC}{\text{Area of } \Delta ADE} = \left(\frac{BC}{DE}\right)^2 = \frac{25}{9} \text{ or, } 25 : 9$$

EXAMPLE 10. In a given figure, QR is parallel to AB and DR is parallel to QB. What is the number of distinct pairs of similar triangles?



- a. 1 b. 2 c. 3 d. 4

Sol. c. Since, QR is parallel to AB.

$$\therefore \Delta PQR \sim \Delta PAB$$

Also, DR is parallel to QB, $\Delta PQB \sim \Delta PDR$

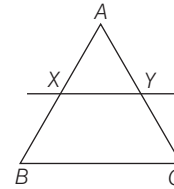
Again, $DR \parallel QB$ and $QR \parallel AB$,

$$\therefore \Delta DQR \sim \Delta QAB$$

EXAMPLE 11. In a triangle, a line XY is drawn parallel to BC meeting AB in X and AC in Y. The area of the ΔABC is 2 times the area of the ΔAXY . In what ratio X divides AB?

- a. $1 : \sqrt{2}$ b. $\sqrt{2} : 1$ c. $(\sqrt{2} - 1) : 1$ d. $1 : (\sqrt{2} - 1)$

Sol. d. $\therefore \frac{\text{Area}(\Delta ABC)}{\text{Area}(\Delta AXY)} = \frac{AB^2}{AX^2}$



$$\Rightarrow \frac{2 \text{ Area}(\Delta AXY)}{\text{Area}(\Delta AXY)} = \frac{AB^2}{AX^2} \Rightarrow \frac{2}{1} = \frac{AB^2}{AX^2}, \frac{AB}{AX} = \sqrt{2}$$

$$\Rightarrow AB = \sqrt{2} AX \Rightarrow AX + BX = \sqrt{2} AX$$

$$\Rightarrow BX = AX(\sqrt{2} - 1)$$

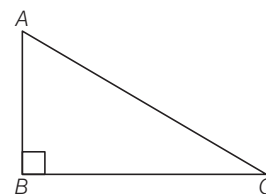
$$\therefore \frac{AX}{BX} = \frac{1}{\sqrt{2} - 1}$$

So, X divides AB in $1 : (\sqrt{2} - 1)$.

PYTHAGORAS THEOREM

In a right angled triangle, the square of the hypotenuse is equal to the sum of the square of the other two sides.

i.e. In ΔABC , if $\angle B = 90^\circ$, then $AB^2 + BC^2 = AC^2$



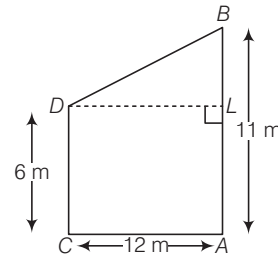
Converse of Pythagoras theorem

In a triangle, if the square of one side is equal to the sum of square of the other two sides then angle opposite to the first side is a right angle i.e. In $\triangle ABC$, if $AC^2 = AB^2 + BC^2$, then $\angle B = 90^\circ$.

EXAMPLE 12. Two poles of heights 6m and 11m stand vertically on a plane ground at a distance of 12m from each other. The distance between their tops is

- a. 9 m b. 10 m
c. 13 m d. 15 m

Sol. c. Let AB and CD be two poles such that $AB = 11\text{m}$, $CD = 6\text{ m}$ and $AC = 12\text{ m}$.



Draw $DL \perp AB$

Then, $BL = AB - AL = AB - CD = (11 - 6)\text{m} = 5\text{m}$

and $DL = AC = 12\text{m}$

In right angled $\triangle BDL$,

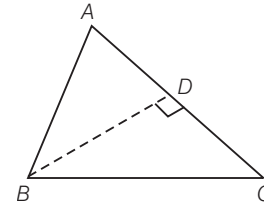
$$BD^2 = DL^2 + BL^2 = 5^2 + 12^2 = 169$$

$$\Rightarrow BD = \sqrt{169} = 13\text{m}$$

> PRACTICE EXERCISE

- If the bisector of an angle of a triangle bisects the opposite side, then the triangle is
(a) equilateral (b) isosceles
(c) scalene (d) right angled triangle
- The line segments joining the mid-points of the sides of a triangle form four triangles each of which is
(a) similar to the original triangle
(b) congruent to the original triangle
(c) an equilateral triangle (d) an isosceles triangle
- In a $\triangle ABC$, BD and CE are perpendicular on AC and AB , respectively. If $BD = CE$, then the $\triangle ABC$ is
(a) equilateral (b) isosceles (c) right angled (d) scalene
- If the length of hypotenuse of a right angled triangle is 5 cm and its area is 6 cm^2 , then the length of the remaining sides are
(a) 1 cm and 3 cm (b) 3 cm and 2 cm
(c) 3 cm and 4 cm (d) 4 cm and 2 cm
- $\triangle ABC$ is such that $AB = 3\text{ cm}$, $BC = 2\text{ cm}$ and $AC = 2.5\text{ cm}$. $\triangle DEF$ is similar to $\triangle ABC$. If $EF = 4\text{ cm}$, then the perimeter of $\triangle DEF$ is
(a) 5 cm (b) 7.5 cm (c) 15 cm (d) 18 cm
- A soldier goes to a warfield and runs in the following manner. From the starting point, he goes West 25 m, then due North 60 m, then due East 80 m, and finally due South 12 cm. The distance between the starting point and the finishing point is
(a) 177 m (b) 103 m (c) 83 m (d) 73 m

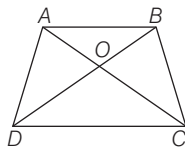
- Let ABC be an isosceles triangle in which $AB = AC$ and $BD \perp AC$. Then, $BD^2 - CD^2$ is equal to



- (a) $2DC \cdot AD$ (b) $2AD \cdot BC$ (c) $3DC \cdot AD$ (d) $\frac{1}{2} AD \cdot DC$

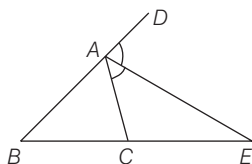
- D and E are the points on the sides AB and AC respectively of a $\triangle ABC$ and $AD = 8\text{ cm}$, $DB = 12\text{ cm}$, $AE = 6\text{ cm}$ and $EC = 9\text{ cm}$, then BC is equal to
(a) $\frac{2}{5} DE$ (b) $\frac{5}{2} DE$ (c) $\frac{3}{2} DE$ (d) $\frac{2}{3} DE$
- A vertical stick 15m long casts a shadow 12m long on the ground. At the same time, a tower casts a shadow 50m long on the ground. The height of the tower is
(a) 60 m (b) 62 m (c) 62.5 m (d) 63 m
- The areas of two similar triangles are 81 cm^2 and 49 cm^2 , respectively. The ratio of their corresponding heights is
(a) 9 : 7 (b) 7 : 9 (c) 6 : 5 (d) 81 : 49
- If D and E are points on the sides AB and AC , respectively of a $\triangle ABC$ such that $DE \parallel BC$. If $AD = x$, $DB = x - 2$, $AE = x + 2$ and $EC = x - 1$. The value of x is
(a) 2.5 (b) 2 (c) 3 (d) 4

- 12.** In the adjoining figure, $ABCD$ is a trapezium in which $AB \parallel CD$ and its diagonals intersect at O . If $AO = (3x - 1)$, $OC = (5x - 3)$, $BO = (2x + 1)$ and $OD = (6x - 5)$, then x is equal to



- (a) 1 (b) 2 (c) 3 (d) 4

- 13.** In the adjoining figure, AE is the bisector of exterior $\angle CAD$ meeting BC produced in E . If $AB = 10$ cm, $AC = 6$ cm and $BC = 12$ cm, then CE is equal to



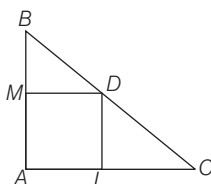
- (a) 6 cm (b) 12 cm (c) 18 cm (d) 20 cm

- 14.** If D, E and F are respectively the mid-points of sides BC, AC and AB of a $\triangle ABC$. If $EF = 3$ cm, $FD = 4$ cm and $AB = 10$ cm, then DE, BC and CA , respectively will be equal to

- (a) 6, 8 and 20 cm (b) $\frac{10}{3}$, 9 and 12 cm
(c) 4, 6 and 8 cm (d) 5, 6 and 8 cm

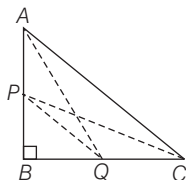
- 15.** In $\triangle PQR$, $\angle Q = 3a$, $\angle P = a$, $\angle R = b$ and $3b - 5a = 30$, then the triangle is
- (a) scalene (b) isosceles
(c) equilateral (d) right angled

- 16.** In $\triangle ABC$ show in the figure $\angle A = 90^\circ$. Let D be a point on BC such that $BD : DC = 1 : 3$. If DM and DL , respectively are perpendicular on AB and AC , then DM and LC are in the ratio of



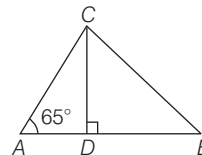
- (a) 1 : 3 (b) 1 : 2 (c) 1 : 1 (d) 4 : 1

- 17.** In a right angled $\triangle ABC$, right angled at B , if P and Q are points on the sides AB and AC respectively, then



- (a) $AQ^2 + CP^2 = 2(AC^2 + PQ^2)$
(b) $2(AQ^2 + CP^2) = AC^2 + PQ^2$
(c) $AQ^2 + CP^2 = AC^2 + PQ^2$
(d) $AQ + CP = \frac{1}{2}(AC + PQ)$

- 18.** In $\triangle ABC$, $\angle C = 90^\circ$ and $CD \perp AB$, also $\angle A = 65^\circ$, then $\angle CBA$ is equal to

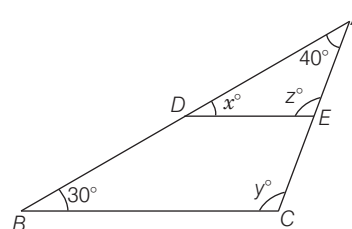


- (a) 25° (b) 35° (c) 65° (d) 40°

- 19.** The angles of a triangle are in the ratio 2 : 3 : 4. The angles of triangle are, respectively

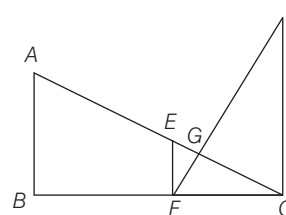
- (a) $30^\circ, 60^\circ, 90^\circ$ (b) $40^\circ, 60^\circ, 80^\circ$
(c) $60^\circ, 40^\circ, 80^\circ$ (d) $20^\circ, 60^\circ, 80^\circ$

- 20.** In figure, D and E are points on sides AB, AC of $\triangle ABC$ such that $DE \parallel BC$. If $\angle B = 30^\circ$ and $\angle A = 40^\circ$, then x, y and z are, respectively



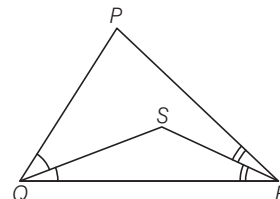
- (a) $30^\circ, 110^\circ, 110^\circ$ (b) $30^\circ, 105^\circ, 105^\circ$
(c) $30^\circ, 85^\circ, 85^\circ$ (d) $30^\circ, 95^\circ, 95^\circ$

- 21.** In figure, AB, EF and CD are parallel lines. Given that $GE = 5$ cm, $GC = 10$ cm and $DC = 18$ cm, then EF is equal to



- (a) 11 cm (b) 5 cm (c) 6 cm (d) 9 cm

- 22.** In the given figure, $PQ > PR$. QS and RS are the bisectors of $\angle Q$ and $\angle R$ respectively, then which of the following is correct?



- (a) $SQ < SR$ (b) $SQ = SR$
(c) $SQ > SR$ (d) None of these

- 23.** In a $\triangle ABC$, $AB = AC$ and $60^\circ < A < 90^\circ$. If $AB = c$, $AC = b$ and $BC = a$, then which of the following is correct?

- (a) $b < a < 2b$ (b) $c/3 < a < 3c$
(c) $b < a < b\sqrt{3}$ (d) $c < a < c\sqrt{2}$

24. In $\triangle ABC$, $AD \perp BC$, then

- (a) $AB^2 - BD^2 = AC^2 - CD^2$ (b) $AB^2 + BD^2 = AC^2 - CD^2$
 (c) $AB^2 + BD^2 = AC^2 + CD^2$ (d) $AB^2 + AC^2 = BD^2 + DC^2$

25. $\triangle ABC$ is a right angled at C and P is the length of the perpendicular from C to AB . If $BC = a$, $AC = b$, $AB = c$, then

- (a) $\frac{a}{b} = \frac{p}{c}$ (b) $pc = ab$ (c) $\frac{1}{a} + \frac{1}{b} = \frac{1}{ab}$ (d) None of these

26. ABC is a triangle and the perpendicular drawn from A meets BC in D . If $AD^2 = BD \cdot DC$, then which one of the following is correct?

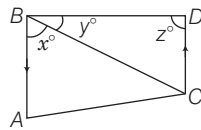
- (a) ABC must be an obtuse angled triangle
 (b) ABC must be an acute angled triangle
 (c) Either $\angle B \geq 45^\circ$ or $\angle C \geq 45^\circ$ (d) $BC^2 = AB^2 + AC^2$

27. If Δ is the area of a right angled triangle and b is one of the sides containing the right angle, then what is the length of the altitude on the hypotenuse?

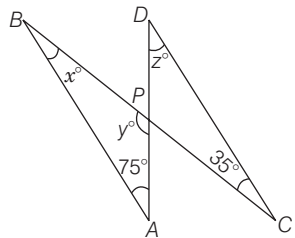
- (a) $\frac{2 \Delta b}{\sqrt{b^4 + 4\Delta^2}}$ (b) $\frac{2 \Delta^2 b}{\sqrt{b^4 + 4\Delta^2}}$
 (c) $\frac{2 \Delta b^2}{\sqrt{b^4 + 4\Delta^2}}$ (d) $\frac{2 \Delta^2 b^2}{\sqrt{b^4 + \Delta^2}}$

28. In figure, $AB \parallel CD$. If $x = \frac{4}{3}y$ and $y = \frac{3}{8}z$, then the value of x is

- (a) 48° (b) 96° (c) 36° (d) None of these

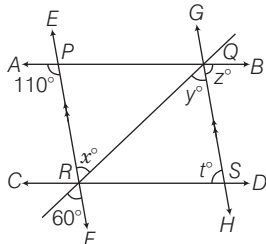


29. In the given figure, $AB \parallel CD$, then the values of x, y and z are, respectively



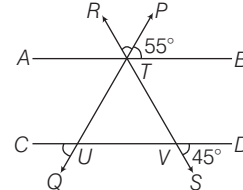
- (a) $75^\circ, 35^\circ, 80^\circ$ (b) $70^\circ, 35^\circ, 60^\circ$
 (c) $35^\circ, 70^\circ, 75^\circ$ (d) $70^\circ, 35^\circ, 80^\circ$

30. In the given figure, $AB \parallel CD$, and $EF \parallel GH$. The values of x, y, z and t are respectively



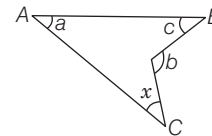
- (a) $60^\circ, 75^\circ, 75^\circ, 60^\circ$ (b) $50^\circ, 75^\circ, 75^\circ, 65^\circ$
 (c) $60^\circ, 70^\circ, 60^\circ, 70^\circ$ (d) $60^\circ, 60^\circ, 70^\circ, 70^\circ$

31. In the given figure, $AB \parallel CD$, $\angle PTB = 55^\circ$ and $\angle DVS = 45^\circ$, then what is the sum of the measures of $\angle CUQ$ and $\angle RTP$?



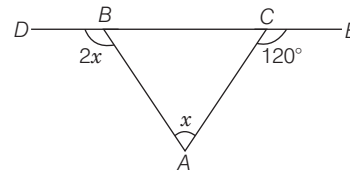
- (a) 180° (b) 135° (c) 110° (d) 100°

32. What is the value of x in the figure given below?



- (a) $b - a - c$ (b) $b - a + c$ (c) $b + a - c$ (d) $\pi - (a + b + c)$

33. In the given figure, what is the value of x ?

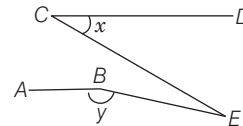


- (a) 30° (b) 40° (c) 45° (d) 60°

34. PQR is a triangle right angled at Q . If X and Y are the mid-points of the sides PQ and QR respectively, then which one of the following is not correct?

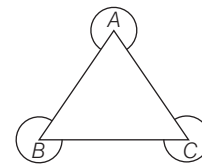
- (a) $RX^2 + PY^2 = 5XY^2$ (b) $RX^2 + PY^2 = XY^2 + PR^2$
 (c) $4(RX^2 + PY^2) = 5PR^2$ (d) $RX^2 + PY^2 = 3(PQ^2 + QR^2)$

35. In the given figure, AB is parallel to CD . If $\angle DCE = x$ and $\angle ABE = y$, then what is $\angle CEB$ equal to?



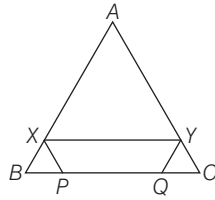
- (a) $y - x$ (b) $(x + y)/2$ (c) $x + y - (\pi/2)$ (d) $x + y - \pi$

36. In the figure given below, what is the sum of the angles formed around A, B, C except the angles of the $\triangle ABC$?

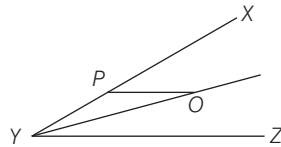


- (a) 360° (b) 720° (c) 900° (d) 1000°

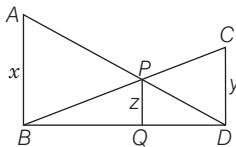
- 37.** In the given figure, ABC is an equilateral triangle of side length 30 cm. XY is parallel to BC , XP is parallel to AC and YQ is parallel to AB . If $(XY + XP + YQ)$ is 40 cm, then what is PQ equal to?



- (a) 5 cm (b) 12 cm
(c) 15 cm (d) None of these
- 38.** O is any point on the bisector of the acute angle $\angle XYZ$. The line OP is parallel to ZY . Then, $\triangle YPO$ is
- (a) scalene
(b) isosceles but not right angled
(c) equilateral
(d) right angled and isosceles
- 39.** The median BD of the $\triangle ABC$ meets AC at D . If $BD = \frac{1}{2} AC$, then which one of the following is correct?
- (a) $\angle ACB = 1$ right angle (b) $\angle BAC = 1$ right angle
(c) $\angle ABC = 1$ right angle (d) None of these
- 40.** ABC is a triangle, X is a point outside the $\triangle ABC$ such that $CD = CX$, where D is the point of intersection of BC and AX and $\angle BAX = \angle XAC$. Which one of the following is correct?
- (a) $\triangle ABD$ and $\triangle ACX$ are similar (b) $\angle ABD < \angle ACD$
(c) $AC = CX$ (d) $\angle ADB > \angle DXC$

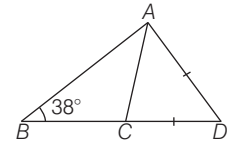


- 41.** In the figure given, $\angle ABD = \angle PQD = \angle CDQ = \frac{\pi}{2}$. If $AB = x$, $PQ = z$ and $CD = y$, then which one of the following is correct?



- (a) $\frac{1}{x} + \frac{1}{y} = \frac{1}{z}$
(b) $\frac{1}{x} + \frac{1}{z} = \frac{1}{y}$
(c) $\frac{1}{z} + \frac{1}{y} = \frac{1}{x}$
(d) $\frac{1}{x} + \frac{1}{y} = \frac{2}{z}$
- 42.** $\triangle PQR$ is right angled at Q , $PR = 5$ cm and $QR = 4$ cm. If the lengths of sides of another $\triangle ABC$ are 3 cm, 4 cm and 5 cm, then which one of the following is correct?
- (a) Area of $\triangle PQR$ is double that of $\triangle ABC$
(b) Area of $\triangle ABC$ is double that of $\triangle PQR$
(c) $\angle B = \frac{\angle Q}{2}$
(d) Both triangles are congruent

- 43.** In the figure, $\angle B = 38^\circ$, $AC = BC$ and $AD = CD$. What is the value $\angle D$?



- (a) 26° (b) 28°
(c) 38° (d) 52°

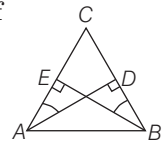
- 44.** Consider the following statements in respect of any triangle.

- I. The three medians of a triangle divide it into six triangles of equal area.
II. The perimeter of a triangle is greater than the sum of the lengths of its three medians.

Which of the statement(s) given above is/are correct?

- (a) Only I (b) Only II (c) Both I and II (d) Neither I nor II

- 45.** Consider the following in respect of the given figure :



- I. $\triangle DAC \sim \triangle EBC$
II. $CA/CB = CD/CE$
III. $AD/BE = CD/CE$

Which of the statement(s) given above is/are correct?

- (a) II and III (b) I and II (c) I and III (d) All of these

- 46.** Consider the following statements :

- I. Let PQR be a triangle in which $PQ = 3$ cm, $QR = 4$ cm and $RP = 5$ cm. If D is a point in the plane of the $\triangle PQR$ such that D is either outside it or inside it, then

$$DP + DQ + DR > 6 \text{ cm}$$

- II. PQR is a right angled triangle.

Which one of the following is correct in respect of the above two statements?

- (a) Both statements I and II are individually true and statement II is the correct explanation of statement I
(b) Both statements I and II are individually true but statement II is not the correct explanation of statement I
(c) Statement I is true and statement II is false
(d) Statement I is false and statement II is true

- 47.** I. Let LMN be a triangle. Let P, Q be the mid-points of the sides LM, LN , respectively. If $PQ^2 = MP^2 + NQ^2$, then LMN is a right angled triangle at L .

- II. If in a $\triangle ABC$, $AB^2 > BC^2 + CA^2$, then $\angle ACB$ is obtuse.

Which of the following is correct in the light of the above statements?

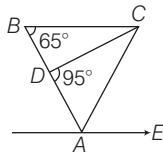
- (a) Both statements I and II are individually true and statement II is the correct explanation of statement I
(b) Both statements I and II are individually true but statement II is not the correct explanation of statement I
(c) Statement I is true and statement II is false
(d) Statement I is false and statement II is true

PREVIOUS YEARS' QUESTIONS

48. In a $\triangle ABC$, $\frac{1}{2}\angle A + \frac{1}{3}\angle C + \frac{1}{2}\angle B = 80^\circ$, then what is the value of $\angle C$? ☑ 2012 II
 (a) 35° (b) 40° (c) 60° (d) 70°

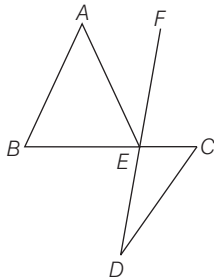
49. In a $\triangle ABC$, side AB is extended beyond B , side BC beyond C and side CA beyond A . What is the sum of the three exterior angles? ☑ 2012 II
 (a) 270° (b) 305° (c) 360° (d) 540°

50. In the given figure, ABC is a triangle, BC is parallel to AE . If $BC = AC$, then what is the value of $\angle CAE$? ☑ 2012 II



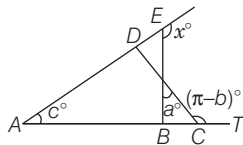
- (a) 20° (b) 30° (c) 40° (d) 50°

51. In the given figure, AB is parallel to CD . $\angle ABC = 65^\circ$, $\angle CDE = 15^\circ$ and $AB = AE$. What is the value of $\angle AEF$? ☑ 2012 II



- (a) 30° (b) 35° (c) 40° (d) 45°

52. The angles x° , a° , c° and $(\pi - b)^\circ$ are indicated in the figure given below : ☑ 2012 II



Which one of the following is correct? ☑ 2012 II

- (a) $x^\circ = a^\circ + c^\circ - b^\circ$ (b) $x^\circ = b^\circ - a^\circ - c^\circ$
 (c) $x^\circ = a^\circ + b^\circ + c^\circ$ (d) $x^\circ = a^\circ - b^\circ - c^\circ$

53. Consider the following statements:

- I. If G is the centroid of $\triangle ABC$, then $GA = GB = GC$.
 II. If H is the orthocentre of $\triangle ABC$, then $HA = HB = HC$.

Which of the statement(s) given above is are correct? ☑ 2013 II

- (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II

54. I. Let the side DE of a $\triangle DEF$ be divided at S , so that $\frac{DS}{SE} = \frac{1}{\sqrt{2}}$. If a line through S parallel to EF meets DF at T , then the area of $\triangle DEF$ is twice the area of the $\triangle DST$.

- II. The areas of the similar triangles are proportional to the squares of the corresponding sides.

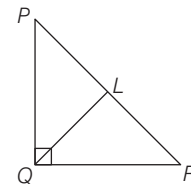
Which one of the following is correct in respect of the above statements? ☑ 2013 II

- (a) Both statements I and II are true and statement II is the correct explanation of statement I
 (b) Both statements I and II are true but statement II is not the correct explanation of statement I
 (c) Statement I is true but statement II is false
 (d) Statement II is true but statement I is false

55. Let ABC be a triangle with $AB = 3$ cm and $AC = 5$ cm. If AD is a median drawn from the vertex A to the side BC , then which one of the following is correct? ☑ 2013 II

- (a) AD is always greater than 4 cm but less than 5 cm
 (b) AD is always greater than 5 cm
 (c) AD is always less than 4 cm
 (d) None of the above

56. In the figure given above, $\angle PQR = 90^\circ$ and QL is a median, $PQ = 5$ cm and $QR = 12$ cm. Then, QL is equal to ☑ 2013 II



- (a) 5 cm (b) 5.5 cm (c) 6 cm (d) 6.5 cm

57. In a $\triangle ABC$, $\angle BCA = 90^\circ$ and CD is perpendicular to AB . If $AD = 4$ cm and $BD = 9$ cm, then the value of DC will be ☑ 2013 II

- (a) $\sqrt{18}$ cm (b) $\sqrt{20}$ cm (c) $\sqrt{65}$ cm (d) 6 cm

58. ABC is a triangle, where $BC = 2AB$, $\angle C = 30^\circ$ and $\angle A = 90^\circ$. The magnitude of the side AC is ☑ 2013 II

- (a) $\frac{2BC}{3}$ (b) $\frac{3BC}{4}$ (c) $\frac{BC}{\sqrt{3}}$ (d) $\frac{\sqrt{3} BC}{2}$

59. Let ABC be an equilateral triangle. If the side BC is produced to the point D so that $BC = 2CD$, then AD^2 is equal to ☑ 2013 II

- (a) $3CD^2$ (b) $4CD^2$ (c) $5CD^2$ (d) $7CD^2$

60. ABC is a right angled triangle such that $AB = a - b$, $BC = a$ and $CA = a + b$. D is a point on BC such that $BD = AB$. The ratio of $BD : DC$ for any value of a and b is given by ☑ 2013 II

- (a) 3 : 2 (b) 4 : 3 (c) 5 : 4 (d) 3 : 1

- 61.** The side BC of a $\triangle ABC$ is produced to D , bisectors of the $\angle ABC$ and $\angle ACD$ meet at P . If $\angle BPC = x^\circ$ and $\angle BAC = y^\circ$, then which one of the following option is correct? **2013 II**
 (a) $x^\circ = y^\circ$ (b) $x^\circ + y^\circ = 90^\circ$
 (c) $x^\circ + y^\circ = 180^\circ$ (d) $2x^\circ = y^\circ$
- 62.** The side AC of a $\triangle ABC$ is produced to D such that $BC = CD$. If $\angle ACB$ is 70° , then what is $\angle ADB$ equal to? **2013 II**
 (a) 35° (b) 45° (c) 70° (d) 110°
- 63.** The heights of two trees are x and y , where $x > y$. The tops of the trees are at a distance z apart. If s is the shortest distance between the trees, then what is s^2 equal to? **2013 II**
 (a) $x^2 + y^2 - z^2 - 2xy$ (b) $x^2 + y^2 - z^2$
 (c) $x^2 + y^2 + z^2 - 2xy$ (d) $z^2 - x^2 - y^2 + 2xy$
- 64.** In a $\triangle ABC$, $\angle B = 90^\circ$ and $\angle C = 2\angle A$, then what is AB^2 equal to? **2013 II**
 (a) $2BC^2$ (b) $3BC^2$ (c) $4BC^2$ (d) $5BC^2$
- 65.** PQR is an equilateral triangle. O is the point of intersection of altitudes PL , QM and RN . If $OP = 8$ cm, then what is the perimeter of the $\triangle PQR$? **2013 II**
 (a) $8\sqrt{3}$ cm (b) $12\sqrt{3}$ cm (c) $16\sqrt{3}$ cm (d) $24\sqrt{3}$ cm
- 66.** $\triangle DEF$ is formed by joining the mid-points of the sides of $\triangle ABC$. Similarly, a $\triangle PQR$ is formed by joining the mid-points of the sides of the $\triangle DEF$. If the sides of the $\triangle PQR$ are of lengths 1, 2 and 3 units, what is the perimeter of the $\triangle ABC$? **2013 II**
 (a) 18 units (b) 24 units
 (c) 48 units (d) Cannot be determined
- 67.** E is the mid-point of the median AD of a $\triangle ABC$. If BE produced meets the side AC at F , then CF is equal to **2013 II**
 (a) $AC/3$ (b) $2AC/3$ (c) $AC/2$ (d) None of these
- 68.** If AD is the internal angular of $\triangle ABC$ with $AB = 3$ cm and $AC = 1$ cm, then what is $BD : BC$ equal to? **2014 I**
 (a) 1 : 3 (b) 1 : 4 (c) 2 : 3 (d) 3 : 4
- 69.** In a $\triangle ABC$, AD is perpendicular to BC and BE is perpendicular to AC . Which of the following is correct? **2014 I**
 (a) $CE \times CB = CA \times CD$ (b) $CE \times CA = CD \times CB$
 (c) $AD \times BD = AE \times BE$ (d) $AB \times AC = AD \times BE$
- 70.** The sides of a triangle are in geometric progression with common ratio $r < 1$. If the triangle is a right angled triangle, the square of common ratio is given by **2014 I**
 (a) $\frac{\sqrt{5} + 1}{2}$ (b) $\frac{\sqrt{5} - 1}{2}$ (c) $\frac{\sqrt{3} + 1}{2}$ (d) $\frac{\sqrt{3} - 1}{2}$
- 71.** If triangles ABC and DEF are similar such that $2AB = DE$ and $BC = 8$ cm, then what is EF equal to? **2014 I**
 (a) 16 cm (b) 12 cm (c) 10 cm (d) 8 cm
- 72.** The sides of a right angled triangle are equal to three consecutive numbers expressed in centimeters. What can be the area of such a triangle? **2014 I**
 (a) 6 cm^2 (b) 8 cm^2 (c) 10 cm^2 (d) 12 cm^2
- 73.** The three sides of a triangle are 15, 25, x units. Which one of the following is correct? **2014 I**
 (a) $10 < x < 40$ (b) $10 \leq x \leq 40$
 (c) $10 \leq x < 40$ (d) $10 < x \leq 40$
- 74.** In a $\triangle ABC$, if $\angle B = 2$, $\angle C = 2\angle A$. Then, what is the ratio of AC to AB ? **2014 II**
 (a) $\sqrt{2} : 1$ (b) $\sqrt{3} : 1$ (c) 1 : 1 (d) $1 : \sqrt{2}$
- 75.** Three straight lines are drawn through the three vertices of a $\triangle ABC$, the line through each vertex being parallel to the opposite side. The $\triangle DEF$ is bounded by these parallel lines. **2014 II**
 Consider the following statements in respect of the $\triangle DEF$.
 I. Each side of $\triangle DEF$ is double the side of $\triangle ABC$ to which it is parallel.
 II. Area of $\triangle DEF$ is four times the area of $\triangle ABC$.
 Which of the statement(s) given above is/are correct?
 (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II
- 76.** In a $\triangle ABC$, AD is the median through A and E is the mid-point of AD and BE produced meets AC at F . Then, AF is equal to **2014 II**
 (a) $AC/5$ (b) $AC/4$ (c) $AC/3$ (d) $AC/2$
- 77.** The point O is equidistant from the three sides of a $\triangle ABC$. Consider the following statements: **2015 II**
 I. $\angle OAC + \angle OCB + \angle OBA = 90^\circ$
 II. $\angle BOC = 2\angle BAC$
 III. The perpendiculars drawn from any point on OA to AB and AC are always equal
 Which of the above statements are correct?
 (a) I and II (b) II and III (c) I and III (d) All of these
- 78.** An equilateral $\triangle BOC$ is drawn inside a square $ABCD$. If $\angle AOD = 2\theta$, what is $\tan \theta$ equal to? **2015 II**
 (a) $2 - \sqrt{3}$ (b) $1 + \sqrt{2}$ (c) $4 - \sqrt{3}$ (d) $2 + \sqrt{3}$
- 79.** In a $\triangle PQR$, point X is on PQ and point Y is on PR such that $XP = 1.5$ units, $XQ = 6$ units, $PY = 2$ units and $YR = 8$ units. Which of the following are correct?
 I. $QR = 5XY$ II. QR is parallel to XY .
 III. $\triangle PYX$ is similar to $\triangle PRQ$.

Select the correct answer using the codes given below.

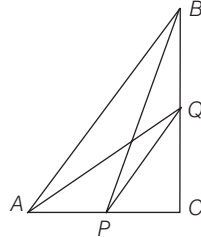
☑ 2016 I

- (a) I and II (b) II and III
(c) I and III (d) All of these

80. ABC is a triangle right angled at C as shown below. Which one of the following is correct?

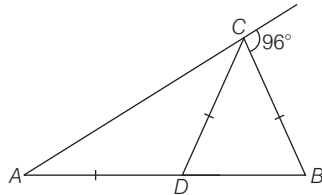
☑ 2016 I

- (a) $AQ^2 + AB^2 = BP^2 + PQ^2$
(b) $AQ^2 + PQ^2 = AB^2 + BP^2$
(c) $AQ^2 + BP^2 = AB^2 + PQ^2$
(d) $AQ^2 + AP^2 = BK^2 + KQ^2$



81. In the given figure, $AD = CD = BC$. What is the value of $\angle CDB$?

☑ 2016 I



- (a) 32° (b) 64°
(c) 78°
(d) Cannot be determined due to insufficient data

82. ABC is an equilateral triangle and X, Y and Z are the points on BC, CA and AB respectively, such that $BX = CY = AZ$. Which of the following is/are correct?

☑ 2016 I

- I. XYZ is an equilateral triangle.
II. $\triangle XYZ$ is similar to $\triangle ABC$.

Select the correct answer using the codes given below.

- (a) Only I (b) Only II (c) Both I and II (d) Neither I nor II

83. Let ABC and $A'B'C'$ be two triangles in which $AB > A'B', BC > B'C'$ and $CA > C'A'$. Let D, E and F be the mid-points of the sides BC, CA and AB , respectively. Let D', E' and F' be the mid-points of the sides $B'C', C'A'$ and $A'B'$, respectively.

Consider the following statements

☑ 2016 I

- I. $AD > A'D', BE > B'E'$ and $CF > C'F'$ are always true.

$$\text{II. } \frac{AB^2 + BC^2 + CA^2}{AD^2 + BE^2 + CF^2} = \frac{A'B'^2 + B'C'^2 + C'A'^2}{A'D'^2 + B'E'^2 + C'F'^2}$$

Which one of the following is correct in respect of the above statements?

- (a) Both statement I and statement II are true and statement II is the correct explanation of statement I
(b) Both statement I and statement II are true but statement II is not the correct explanation of statement I
(c) Statement I is true but statement II is false
(d) Statement I is false but statement II is true

84. ABC and DEF are similar triangles. If the ratio of side AB to side DE is $(\sqrt{2} + 1) : \sqrt{3}$, then the ratio of area of $\triangle ABC$ to that of $\triangle DEF$ is

☑ 2016 I

- (a) $(3 - 2\sqrt{2}) : 3$ (b) $(9 - 6\sqrt{2}) : 2$
(c) $1 : (9 - 6\sqrt{2})$ (d) $(3 + 2\sqrt{2}) : 2$

85. Let $\triangle ABC$ and $\triangle DEF$ be such that $\angle ABC = \angle DEF$, $\angle ACB = \angle DFE$ and $\angle BAC = \angle EDF$. Let L be the mid-point of BC and M be the mid-point of EF .

Consider the following statements

☑ 2016 I

- I. $\triangle ABL$ and $\triangle DEM$ are similar.
II. $\triangle ALC$ is congruent to $\triangle DMF$ even, if $AC \neq DF$.

Which one of the following is correct in respect of the above statements?

- (a) Both statement I and statement II are true and statement II is the correct explanation of statement I
(b) Both statement I and statement II are true but statement II is not the correct explanation of statement I
(c) Statement I is true but statement II is false
(d) Statement I is false but statement II is true

86. ABC is a triangle in which D is the mid-point of BC and E is the mid-point of AD .

- I. The area of $\triangle ABC$ is equal to four times the area of $\triangle BED$.

- II. The area of $\triangle ADC$ is twice the area of $\triangle BED$.

Select the correct answer using the codes given below.

☑ 2016 I

- (a) Only I (b) Only II (c) Both I and II (d) Neither I nor II

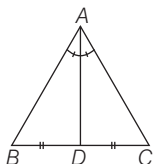
ANSWERS

1	b	2	a	3	b	4	c	5	c	6	d	7	a	8	b	9	c	10	a
11	d	12	b	13	c	14	d	15	d	16	a	17	c	18	a	19	b	20	a
21	d	22	c	23	d	24	a	25	b	26	d	27	a	28	a	29	c	30	d
31	b	32	a	33	d	34	d	35	d	36	c	37	d	38	b	39	c	40	a
41	a	42	d	43	b	44	c	45	d	46	b	47	b	48	c	49	c	50	d
51	b	52	c	53	d	54	a	55	c	56	d	57	d	58	d	59	d	60	d
61	d	62	a	63	d	64	b	65	d	66	b	67	b	68	d	69	b	70	b
71	a	72	a	73	a	74	a	75	c	76	c	77	c	78	d	79	d	80	c
81	b	82	c	83	b	84	c	85	c	86	c								

HINTS AND SOLUTIONS

1. (b) Let ABC is any triangle and AD is the bisector of angle A .

Also, $BD = DC$ (given)



$\therefore AD$ is the internal bisector of $\angle A$.

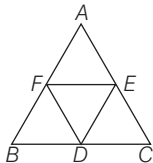
$$\therefore \frac{AB}{AC} = \frac{BD}{DC}$$

$$\Rightarrow \frac{AB}{AC} = \frac{BD}{DC} = 1 \quad [\because BD = DC \text{ given}]$$

$$\Rightarrow AB = AC$$

Hence, triangle is an isosceles triangle.

2. (a) The line segments joining the mid-points of the sides of a triangle form four triangles each of which is similar to the original triangle.



Here, $\triangle BDF \sim \triangle ABC$

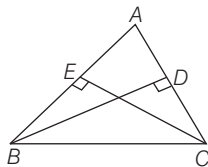
Also, $\triangle DEC, \triangle DEF$

$\triangle AFE \sim \triangle ABC$

3. (b) As $BD = EC$,

$$\angle AEC = \angle BDA = 90^\circ \text{ each}$$

$[\because BD \perp AC \text{ and } CE \perp AB]$



Also, $\angle A = \angle A$ [common]

$\therefore \triangle BDA \cong \triangle AEC$ [by AAS congruency]

$$\Rightarrow AB = AC \quad [\text{by cpct}]$$

So, triangle is an isosceles triangle.

4. (c) Let the other side be b and p .

$$\therefore \frac{1}{2} b \times p = 6 \Rightarrow b \times p = 12$$

$$\Rightarrow b = \frac{12}{p}$$

Also, by pythagoras theorem

$$b^2 = b^2 + p^2$$

$$5^2 = \left(\frac{12}{p}\right)^2 + p^2$$

$$\Rightarrow 25 = \frac{144}{p^2} + p^2$$

$$25p^2 = 144 + p^4$$

$$\Rightarrow p^4 - 25p^2 + 144 = 0$$

$$\Rightarrow p^4 - 16p^2 - 9p^2 + 144 = 0$$

$$\Rightarrow p^2(p^2 - 16) - 9(p^2 - 16) = 0$$

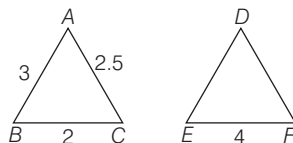
$$\Rightarrow (p^2 - 9)(p^2 - 16) = 0$$

$$\Rightarrow p = 3 \text{ or } p = 4$$

Hence, other sides are 3 cm and 4 cm.

5. (c) As $\triangle ABC \sim \triangle DEF$

$$\Rightarrow \frac{AB}{DE} = \frac{AC}{DF} = \frac{BC}{EF}$$



$$\therefore \frac{BC}{EF} = \frac{2}{4} = \frac{1}{2}$$

$$\therefore DE = 2AB = 2 \times 3 = 6 \text{ cm}$$

$$\text{and } DF = 2 \times AC = 2 \times 2.5 = 5 \text{ cm}$$

$$\therefore \text{Perimeter of } \triangle DEF = (6 + 5 + 4) = 15 \text{ cm}$$

Shortcut Method

Perimeter of $\triangle ABC$

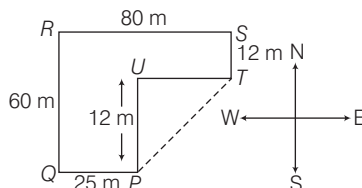
Perimeter of $\triangle DEF$

= Ratio of corresponding sides

$$\therefore \frac{(3 + 2 + 2.5)}{\text{Perimeter of } \triangle DEF} = \frac{1}{2}$$

$$\therefore \text{Perimeter of } \triangle DEF = 2(7.5) = 15 \text{ cm.}$$

6. (d) Let P be the starting point of his run, then PT is the distance between the starting and the finishing point.



$$\therefore PU = RQ - ST = 60 - 12 = 48 \text{ m}$$

$$\text{and } UT = RS - QP = 80 - 25 = 55 \text{ m}$$

$$\therefore \text{In } \triangle PUT, PT^2 = (PU)^2 + (TU)^2$$

$$\therefore PT = \sqrt{(48)^2 + (55)^2} = \sqrt{2304 + 3025}$$

$$= \sqrt{5329} = 73 \text{ m}$$

7. (a) As ADB is a right angled triangle.

$$\text{So, } AB^2 = AD^2 + BD^2$$

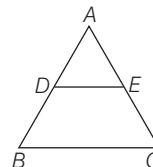
$$\Rightarrow AC^2 = AD^2 + BD^2 \quad [\because AB = AC]$$

$$\Rightarrow (AD + DC)^2 = AD^2 + BD^2$$

$$\Rightarrow AD^2 + DC^2 + 2AD \cdot DC = AD^2 + BD^2$$

$$\Rightarrow BD^2 - CD^2 = 2DC \cdot AD$$

8. (b) As in $\triangle ADE$ and $\triangle ABC$,
 $\frac{AD}{AB} = \frac{8}{20} = \frac{2}{5}$ and $\frac{AE}{AC} = \frac{6}{15} = \frac{2}{5}$



$$\therefore \frac{AD}{AB} = \frac{AE}{AC}$$

and $\angle A = \angle A$ [common]

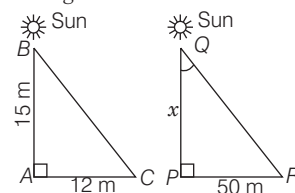
$\therefore \triangle ADE \sim \triangle ABC$

[by SAS similarity]

$$\therefore \frac{DE}{BC} = \frac{AD}{AB} \Rightarrow \frac{DE}{BC} = \frac{2}{5}$$

$$\Rightarrow BC = \frac{5}{2} DE.$$

9. (c) Let AB be a vertical stick and AC be its shadow. Also, let PQ be a tower having shadow PR .



$$\angle A = \angle P$$

$$\angle B = \angle Q$$

$[\because \text{Sun with tower and stick forms same angle}]$

As, $\triangle ABC \sim \triangle PQR$

[$\therefore$ By AA similarity]

$$\therefore \frac{AB}{PQ} = \frac{AC}{PR} \Rightarrow \frac{15}{x} = \frac{12}{50}$$

$$\Rightarrow x = \frac{15 \times 50}{12} = 62.5 \text{ m}$$

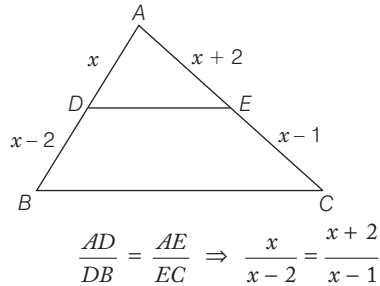
Hence, the height of the tower is 62.5 m.

10. (a) Let the ratio of their corresponding height be $h_1 : h_2$.

But the ratio of the areas of two similar triangles is equal to the ratio of the squares of their corresponding heights.

$$\therefore \frac{h_1^2}{h_2^2} = \frac{81}{49} \Rightarrow h_1 : h_2 = 9 : 7$$

11. (d) As $DE \parallel BC$, by basic proportionality theorem,



$$\begin{aligned} \Rightarrow x(x-1) &= (x+2)(x-2) \\ \Rightarrow x^2 - x &= x^2 - 2x + 2x - 4 \\ \Rightarrow x &= 4 \end{aligned}$$

Hence, the value of x is 4.

12. (b) As $AB \parallel CD$ and the diagonals of a trapezium divide each other proportionally.

So, $\frac{AO}{OC} = \frac{BO}{OD}$

$$\Rightarrow \frac{3x-1}{5x-3} = \frac{2x+1}{6x-5}$$

$$\begin{aligned} (3x-1)(6x-5) &= (5x-3)(2x+1) \\ \Rightarrow 18x^2 - 15x - 6x + 5 &= 10x^2 + 5x - 6x - 3 \\ \Rightarrow 8x^2 - 20x + 8 &= 0 \\ \Rightarrow 4x^2 - 10x + 4 &= 0 \\ \Rightarrow 2(2x^2 - 5x + 2) &= 0 \\ \Rightarrow 2(2x^2 - 4x - x + 2) &= 0 \\ \Rightarrow 2[2x(x-2) - 1(x-2)] &= 0 \\ \Rightarrow 2(2x-1)(x-2) &= 0 \\ \text{But as } x = \frac{1}{2} \text{ will make } OC \text{ negative.} \\ \therefore x &= 2 \end{aligned}$$

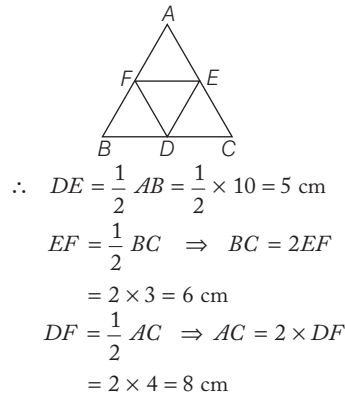
13. (c) Exterior angle bisector theorem The exterior bisector of an angle of a triangle divides the opposite side externally in the ratio of the sides containing the angle.

$$\therefore \frac{BE}{CE} = \frac{AB}{AC} \text{ as } AE \text{ is an exterior angle bisector.}$$

Let $CE = x$, $BE = BC + EC = 12 + x$

$$\begin{aligned} \Rightarrow \frac{12+x}{x} &= \frac{10}{6} \\ \Rightarrow (12+x)6 &= 10x \\ \Rightarrow 72 + 6x &= 10x \Rightarrow 4x = 72 \\ \Rightarrow x &= 18 \text{ cm} \end{aligned}$$

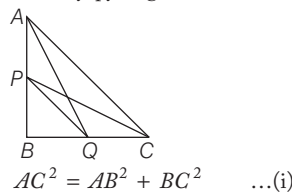
14. (d) As the line joining the mid-points of any two sides of a triangle is parallel to the third side and is half of the third side.



15. (d) In $\triangle PQR$, $\angle P + \angle Q + \angle R = 180^\circ$
- $$\begin{aligned} \Rightarrow a + 3a + b &= 180^\circ \\ \Rightarrow 4a + b &= 180^\circ \quad \dots(i) \\ \text{Given, } -5a + 3b &= 30^\circ \quad \dots(ii) \end{aligned}$$
- Solving Eqs. (i) and (ii), we get $a = 30^\circ$ and $b = 60^\circ$
- $$\therefore \angle P = 30^\circ, \angle Q = 90^\circ \text{ and } \angle R = 60^\circ$$
- So, $\triangle PQR$ is right angled triangle.

16. (a) Consider $\triangle BMD$ and $\triangle DLC$,
As $\angle BMD = \angle DLC = 90^\circ$ [each]
Also, $\angle BDM = \angle DCL$ corresponding angle
 $\therefore \triangle BMD \sim \triangle DLC$ [by AA similarity]
 $\therefore \frac{BD}{DC} = \frac{DM}{LC} = \frac{BM}{DL}$
 $\Rightarrow \frac{BD}{DC} = \frac{DM}{LC} = \frac{1}{3}$
 $\therefore DM : LC = 1 : 3$

17. (c) In $\triangle ABC$, by pythagoras theorem,



And in $\triangle PBQ$,

$$PQ^2 = PB^2 + BQ^2 \quad \dots(ii)$$

Adding Eqs. (i) and (ii),

$$AC^2 + PQ^2 = (AB^2 + BC^2) + PB^2 + BQ^2$$

$$= (AB^2 + BQ^2) + (PB^2 + BC^2)$$

$$AC^2 + PQ^2 = AQ^2 + CP^2$$

18. (a) In $\triangle BCA$, $\angle CAB = 65^\circ$ and $\angle ACB = 90^\circ$ [$\because \angle C = 90^\circ$ given]
 $\angle CBA = 180^\circ - (\angle BCA + \angle CAB)$
[by angle sum property of a triangle]
 $= 180^\circ - (65^\circ + 90^\circ)$
 $= 180^\circ - 155^\circ = 25^\circ$

19. (b) Let the angles of a triangle be $2x, 3x, 4x$, then
- $$2x + 3x + 4x = 180^\circ$$
- [by angle sum property of a triangle]
- $$9x = 180^\circ \Rightarrow x = 20^\circ$$

So, angles are $2x = 40^\circ$,
 $3x = 60^\circ$, $4x = 80^\circ$.

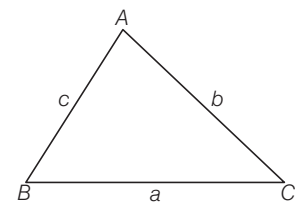
20. (a) In $\triangle ABC$, $\angle A + \angle B + y = 180^\circ$
 $y = 180^\circ - (40 + 30)^\circ = 110^\circ$
 $\angle ADE = \angle ABC$ [corresponding angle]
 $x = 30^\circ$
Similarly, $y = z = 110^\circ$

21. (d) In $\triangle GEF$ and $\triangle GCD$, we have
 $\angle EFG = \angle GDC$ [alternate angles]
 $\angle EGF = \angle CGD$
[vertically opposite angles]
 $\triangle GEF \sim \triangle GCD$ [by AA similarity]

$$\begin{aligned} \therefore \frac{GE}{CG} &= \frac{EF}{DC} \\ \Rightarrow \frac{5}{10} &= \frac{EF}{18} \\ \Rightarrow EF &= \frac{5 \times 18}{10} = 9 \text{ cm} \end{aligned}$$

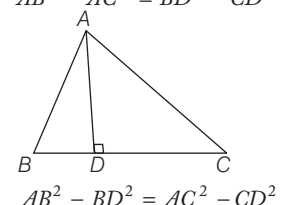
22. (c) Given, In $\triangle PQR$, $PQ > PR$
 $\Rightarrow \angle PRQ > \angle PQR$
[angle opp to larger side is greater]
 $\Rightarrow \frac{1}{2} \angle PRQ > \frac{1}{2} \angle PQR$
 $\Rightarrow \angle SRQ > \angle SQR \Rightarrow SQ > SR$
[side opp to greater angle is larger]

23. (d) Given, $AB = AC$

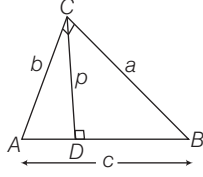


$$\begin{aligned} \Rightarrow \angle ABC &= \angle ACB \\ \text{If } \angle A &= 60^\circ, \text{ then} \\ \triangle ABC &\text{ is an equilateral triangle } a = b = c \\ \text{If } \angle A &= 90^\circ, \text{ then} \\ \triangle ABC &\text{ is right angled triangle} \\ a^2 &= b^2 + c^2 = c^2 + c^2 = 2c^2 [\because b = c] \\ \Rightarrow a &= \sqrt{2} c \\ \therefore \text{ If } 60^\circ < A < 90^\circ, \text{ then } c < a < c\sqrt{2} \end{aligned}$$

24. (a) Here, in $\triangle ABC$
 $AB^2 = AD^2 + BD^2 \quad \dots(i)$
In right angled $\triangle ADC$, we have
 $AC^2 = AD^2 + CD^2 \quad \dots(ii)$
Subtracting Eq. (ii) from Eq. (i),



25. (b) Since, c is the base and p is the altitude of $\triangle ABC$.

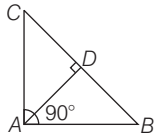


Here, area of $\triangle ABC = \frac{1}{2} pc$... (i)

Also, area of $\triangle ABC = \frac{1}{2} ab$... (ii)

From Eqs. (i) and (ii), we get
 $\frac{1}{2} pc = \frac{1}{2} ab \Rightarrow pc = ab$

26. (d) Hence, $AD^2 = BD \cdot DC$



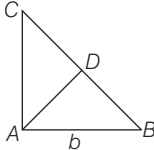
$\Rightarrow \frac{AD}{BD} = \frac{DC}{AD}$

$\therefore \triangle ADB \sim \triangle CDA \Rightarrow \angle BAD = \angle ACD$

Now, $\angle CAB = \angle CAD + \angle BAD$
 $= \angle CAD + \angle ACD = 90^\circ$

So, $\triangle ABC$ must be right angled triangle.
Hence, $BC^2 = AC^2 + AB^2$

27. (a) In $\triangle ABC$,



Area of $\triangle ABC = \frac{1}{2} \times \text{Base} \times \text{Altitude}$

$\Delta = \frac{1}{2} b \times AC, AC = \frac{2\Delta}{b}$

In $\triangle ABC$, Using pythagoras theorem,

$AC^2 + AB^2 = BC^2$
 $\Rightarrow BC = \sqrt{\frac{4\Delta^2}{b^2} + b^2}$

Again in $\triangle ABC$, area of $\triangle ABC$

$\Delta = \frac{1}{2} \times BC \times AD$
 $\Rightarrow AD = \frac{2\Delta}{BC} = \frac{2\Delta b}{\sqrt{4\Delta^2 + b^4}}$

28. (a) As, $AB \parallel CD$ and BC cuts them.
 $\therefore \angle BCD = \angle ABC = x$ [alternate $\angle$ s]

$\therefore$ In $\triangle BCD$, $x + y + z = 180^\circ$
 $\frac{4}{3}y + y + z = 180^\circ$

$\Rightarrow \frac{7y}{3} + z = 180^\circ$

$\frac{7}{3} \left(\frac{3}{8}z \right) + z = 180^\circ \Rightarrow \frac{7}{8}z + z = 180^\circ$
 $\left[\because y = \frac{3}{8}z \right]$

$\frac{15z}{8} = 180^\circ \Rightarrow z = 96^\circ$

So, $y = \frac{3}{8} \times 96^\circ = 36^\circ$

and $x = \frac{4}{3} \times 36^\circ = 48^\circ$

29. (c) We have, $x = 35^\circ$ [alternate angles]
 $z = 75^\circ$ [alternate angles]

In $\triangle ABP$, $x + y + 75^\circ = 180^\circ$

$y = 180^\circ - (75^\circ + 35^\circ) \Rightarrow y = 70^\circ$

Hence the value of

$x = 35^\circ, y = 70^\circ, z = 75^\circ$

30. (d) $x^\circ = 60^\circ$ [vertically opposite angles]
 $y = 60^\circ$ [corresponding angles]

$\angle PRS = 110^\circ$ [alternate angles]

$\angle QRS + x^\circ = 110^\circ$ [Alternate angles]

$\angle QRS = 110^\circ - 60^\circ = 50^\circ$

$\therefore t = 180^\circ - (y + \angle QRS)$
 $= 180^\circ - (60^\circ + 50^\circ)$

$t = 70^\circ$

Also, $t = z$ [alternate angles]

$\therefore z = 70^\circ$

31. (b) Since, $\angle PTB = 55^\circ$ [given]
Then, $\angle TUV = 55^\circ$

[corresponding angle]

Also, $\angle TUV = \angle CUQ = 55^\circ$

[vertically opposite angle]

Also given, $\angle DVS = 45^\circ$

Then, $\angle UVT = 45^\circ$

[vertically opposite angle]

In $\triangle UTV$,

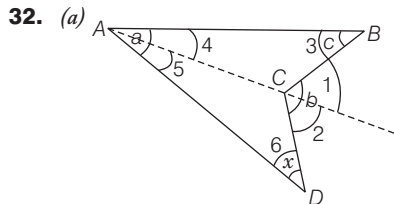
$\angle T = 180^\circ - (55^\circ + 45^\circ) = 80^\circ$

[angle sum property of a triangle]

$\Rightarrow \angle T = \angle PTR = 80^\circ$

[vertically opposite angle]

$\therefore \angle CUQ + \angle RTP = 55^\circ + 80^\circ = 135^\circ$



$\angle 1 = \angle 3 + \angle 4, \angle 2 = \angle 5 + \angle 6$

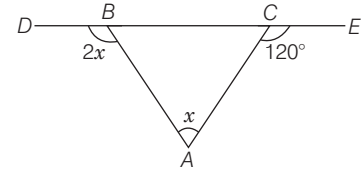
$\Rightarrow \angle 1 + \angle 2 = \angle 3 + \angle 4 + \angle 5 + \angle 6$

[since, exterior angle is equal to sum of two opposite interior angles]

$\Rightarrow b = c + a + x$

$\therefore x = b - c - a$

33. (d) $\because \angle ABC = 180^\circ - \angle DBA = 180^\circ - 2x$



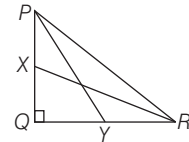
and $\angle ACB = 180^\circ - \angle ACE$

$= 180^\circ - 120^\circ = 60^\circ$

In $\triangle ABC$, $\angle ABC + \angle ACB + \angle BAC = 180^\circ$

$\Rightarrow 180^\circ - 2x + 60^\circ + x = 180^\circ \Rightarrow x = 60^\circ$

34. (d) In $\triangle PQY$, by pythagoras theorem,
 $PY^2 = PQ^2 + QY^2$



$\Rightarrow PY^2 = PQ^2 + \left(\frac{QR}{2} \right)^2$

$\left[QY = YR = \frac{QR}{2} \right] \dots (i)$

and in $\triangle XQR$, $RX^2 = QX^2 + QR^2$

$\Rightarrow RX^2 = \left(\frac{PQ}{2} \right)^2 + QR^2$

$\left[\because PX = XQ = \frac{PQ}{2} \right] \dots (ii)$

On adding Eqs. (i) and (ii), we get

$PY^2 + RX^2 = \frac{5PQ^2}{4} + \frac{5QR^2}{4}$

$\Rightarrow PY^2 + RX^2 = 5/4(PQ^2 + QR^2)$

$\Rightarrow 4(PY^2 + RX^2) = 5(PR^2)$

In $\triangle XQR$, by pythagoras theorem,

$XR^2 = XQ^2 + QR^2 \dots (iii)$

and In $\triangle PQY$, by pythagoras theorem,

$PY^2 = PQ^2 + QY^2 \dots (iv)$

On adding Eqs. (iii) and (iv),

$XR^2 + PY^2 = XQ^2 + QR^2$

$+ PQ^2 + QY^2$

$\Rightarrow XR^2 + PY^2 = XQ^2 + QY^2$

$+ QR^2 + PQ^2$

$\Rightarrow XR^2 + PY^2 = XY^2 + PR^2$

and again,

$XR^2 + PY^2 = XY^2 + PR^2$

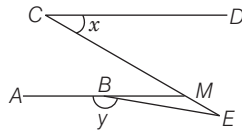
$= XY^2 + (2XY)^2$

$XR^2 + PY^2 = XY^2 + 4XY^2$

$\Rightarrow XR^2 + PY^2 = 5XY^2$

Thus, option (d) is incorrect.

35. (d) We produced line AB to meet line CE at M .



Since, $AM \parallel CD$.

$$\angle DCM = \angle BMC = x$$

[alternate angles]

CME is a straight line

$$\therefore \angle BME = \pi - x$$

Also, ABM is a straight line.

$$\therefore \angle EBM = \pi - y$$

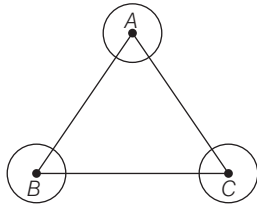
Now, in $\triangle BEM$

$$\angle EBM + \angle BME + \angle BEM = \pi$$

$$\Rightarrow \pi - y + \pi - x + \angle E = \pi$$

$$\Rightarrow \angle E = x + y - \pi$$

36. (c) $\therefore \angle A = 360^\circ - \text{Ext } \angle A$

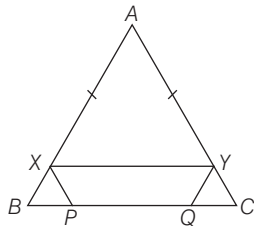


Similarly, $\angle B = 360^\circ - \text{Ext } \angle B$ and $\angle C = 360^\circ - \text{Ext } \angle C$

We know that, the sum of all the angles of a triangle is 180° .

$$\begin{aligned} \therefore \angle A + \angle B + \angle C &= 180^\circ \\ \Rightarrow 360^\circ - \text{Ext } \angle A + 360^\circ - \text{Ext } \angle B &+ 360^\circ - \text{Ext } \angle C = 180^\circ \\ \Rightarrow \text{Ext } \angle A + \text{Ext } \angle B + \text{Ext } \angle C &= 1080^\circ - 180^\circ = 900^\circ \end{aligned}$$

37. (d) Since, $XP \parallel AC$ and $YQ \parallel AB$



$$\therefore \angle XBP = \angle YQC \text{ and } \angle XPB = \angle YCQ$$

Also, $XY \parallel BC$

$\therefore XYQB$ and $XYCP$ are parallelogram

$$\Rightarrow XB = YQ \text{ and } XP = YC$$

So, $\triangle XBP$ and $\triangle YCQ$ are congruent triangles.

Now, $XY \parallel BC$

$$\therefore \frac{AX}{AB} = \frac{XY}{BC} \Rightarrow AX = XY$$

$$[\because AB = BC = 30 \text{ cm}]$$

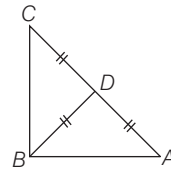
$$\begin{aligned} \text{Also, } XY + XP + YQ &= 40 \text{ [given]} \\ \Rightarrow AX + XB + YQ &= 40 \\ &[\because XY = AX, XP = XB] \\ \Rightarrow AB + YQ &= 40 \\ \Rightarrow YQ &= 40 - 30 = 10 \text{ cm} \\ \therefore YQ &= XP = 10 \text{ cm} \\ \therefore BP &= CQ = 10 \text{ cm} \\ \therefore PQ &= 30 - BP - CQ \\ &= 30 - 10 - 10 = 10 \text{ cm} \end{aligned}$$

38. (b) As, $OP \parallel YZ$
 $\Rightarrow \angle POY = \angle OYZ$ [alternate angles]
 $\Rightarrow \angle PYO = \angle POY$
 [since OY is angle bisector of $\angle Y$]
 $\therefore PY = PO$
 As $\angle XYZ$ is an acute angle.
 $\therefore \frac{1}{2} \angle XYZ < 45^\circ$
 $\therefore \angle POY = \angle PYO < 45^\circ$
 $\therefore \angle YPO > 90^\circ$
 Hence, $\triangle PYO$ is an isosceles triangle but not a right angled triangle.

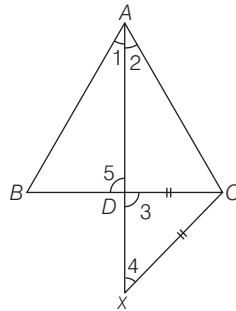
39. (c) Here, we see that

$$CD = BD = DA$$

This is possible only when ABC is right angled triangle.



40. (a) In $\triangle DCX$, $CD = CX$ [given]
 $\angle 3 = \angle 4$
 [opposite angle of same sides]
 But $\angle 3 = \angle 5$, So, $\angle 4 = \angle 5$
 In $\triangle ABD$ and $\triangle ACX$,
 $\angle 1 = \angle 2$ [given]



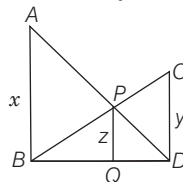
$$\begin{aligned} \angle 4 &= \angle 5 \\ \therefore \triangle ABD &\sim \triangle ACX \end{aligned}$$

[by AA similarity]

41. (a) Since, $AB \perp BD$ and $PQ \perp BD$

$$\Rightarrow AB \parallel PQ$$

$$\text{So, } \triangle ABD \sim \triangle PQD$$



$$\text{Then, } \frac{x}{z} = \frac{BD}{QD} \quad \dots(i)$$

Since $CD \parallel PQ$

So, $\triangle BCD \sim \triangle BPQ$,

$$\therefore \frac{z}{y} = \frac{BQ}{BD}$$

$$\Rightarrow \frac{z}{y} = \frac{BD - QD}{BD}$$

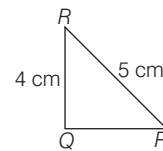
$$\Rightarrow \frac{z}{y} = 1 - \frac{QD}{BD}$$

$$\Rightarrow \frac{z}{y} = 1 - \frac{z}{x}$$

$$\Rightarrow \frac{z}{y} = 1 - \frac{z}{x} \quad [\text{from Eq. (i)}]$$

$$\Rightarrow \frac{z}{x} + \frac{z}{y} = 1 \Rightarrow \frac{1}{x} + \frac{1}{y} = \frac{1}{z}$$

42. (d) In right angle $\triangle PQR$, by pythagoras theorem,



$$QP^2 = (5)^2 - (4)^2 = 9 \Rightarrow QP = 3 \text{ cm}$$

In second $\triangle ABC$ whose sides are 3 cm, 4 cm and 5 cm. So, the sides of both triangle are same, hence they are congruent.

43. (b) Given, $AC = BC$

$$\therefore \angle ABC = \angle BAC$$

[angles of an isosceles triangle]

$$\Rightarrow \angle BAC = \angle ABC = 38^\circ$$

In $\triangle ABC$,

$$\angle ACB = 180^\circ - (\angle BAC + \angle ABC)$$

[by angle sum property of a triangle]

$$= 180^\circ - (38^\circ + 38^\circ)$$

$$= 180^\circ - 76^\circ = 104^\circ$$

$$\text{In } \triangle ACD, \angle ACD = 180^\circ - 104^\circ = 76^\circ$$

[by linear pair]

$$\text{and } \angle CAD = \angle ACD = 76^\circ$$

[$\because CD = AD$]

$$\therefore \angle ADC = 180^\circ - (\angle CAD + \angle ACD)$$

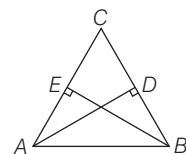
$$= 180^\circ - (76^\circ + 76^\circ) = 28^\circ$$

44. (c) I. It is true that the three medians of a triangle divide it into six triangles of equal area.

II. It is also true that, the perimeter of a triangle is greater than the sum of the lengths of its three medians.

Hence, I and II are correct.

45. (d) In $\triangle CAD$ and $\triangle CBE$,



$\angle C = \angle C$ [common]
 $\angle CEB = \angle ADC$ [each 90°]
 $\therefore \triangle CAD \sim \triangle CBE$ [by AA similarity]

Sides will be in same proportion.

$$\frac{CA}{CB} = \frac{CD}{CE} \text{ and } \frac{AD}{BE} = \frac{CD}{CE}$$

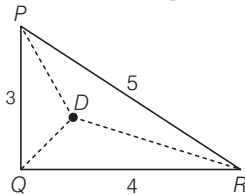
Hence, all three statements are correct.

46. (b) Given, $PQ = 3$ cm, $QR = 4$ cm and $RP = 5$ cm

Here, $RP^2 = PQ^2 + QR^2$

So, PQR is a right angled triangle.

Let D is an interior point in $\triangle PQR$



- In $\triangle DPQ$, $DP + DQ > 3$... (i)
 similarly, $DQ + DR > 4$... (ii)
 and $DP + DR > 5$... (iii)

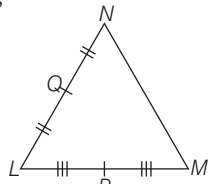
adding (i), (ii) and (iii), we get

$$2(DP + DQ + DR) > 12$$

$$\Rightarrow DP + DQ + DR > 6$$

Hence, both statements I and II are individually true and statement II is not correct explanation of statement I.

47. (b) Given,



- I. $PQ^2 = MP^2 + NQ^2$
 $\Rightarrow PQ^2 = LP^2 + LQ^2$
 [$\because LP = MP$ and $NQ = LQ$]
 $\Rightarrow \angle QLP = 90^\circ$
 It means, $\triangle NLM$ is a right angled triangle.

- II. It also true that if in a $\triangle ABC$, $AB^2 > BC^2 + CA^2$, then $\angle ACB$ is obtuse.

Hence, both statements are individually true but statement II is not the correct explanation of statement I.

48. (c) Given that,

$$\frac{1}{2}\angle A + \frac{1}{3}\angle C + \frac{1}{2}\angle B = 80^\circ$$

$$\Rightarrow 3\angle A + 2\angle C + 3\angle B = 480^\circ$$

$$\Rightarrow 3(\angle A + \angle B) + 2\angle C = 480^\circ \dots (i)$$

Also, in $\triangle ABC$,

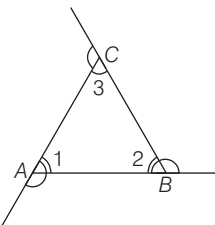
$$\angle A + \angle B + \angle C = 180^\circ$$

$$\Rightarrow 3(\angle A + \angle B) + 2\angle C = 540^\circ \dots (ii)$$

On subtracting Eq. (i) from Eq. (ii), we get

$$\angle C = 60^\circ$$

49. (c)



By Exterior angle theorem,

$$\text{ext. } \angle A = \angle 2 + \angle 3$$

$$\text{ext. } \angle B = \angle 1 + \angle 3$$

$$\text{ext. } \angle C = \angle 1 + \angle 2$$

Sum of the three exterior angles

$$= (\angle 1 + \angle 2) + (\angle 2 + \angle 3) + (\angle 3 + \angle 1)$$

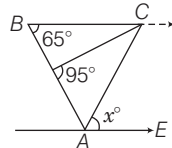
$$= 2(\angle 1 + \angle 2 + \angle 3)$$

$$= 2 \times 180^\circ = 360^\circ$$

$$[\because \angle 1 + \angle 2 + \angle 3 = 180^\circ]$$

50. (d) Given that, $BC \parallel AE$

$$\angle CBA + \angle EAB = 180^\circ$$



$$\Rightarrow \angle EAB = 180^\circ - 65^\circ = 115^\circ$$

$$\therefore BC = AC$$

Hence, $\triangle ABC$ is an isosceles triangle.

$$\Rightarrow \angle CBA = \angle CAB = 65^\circ$$

$$\text{Now, } \angle EAB = \angle EAC + \angle CAB$$

$$\Rightarrow 115^\circ = x + 65^\circ \Rightarrow x = 50^\circ$$

51. (b) Given that, $\angle ABC = 65^\circ$

$$\text{and } \angle CDE = 15^\circ$$

Since, $AB \parallel CD$

$$\angle DCB = \angle ABC = 65^\circ$$

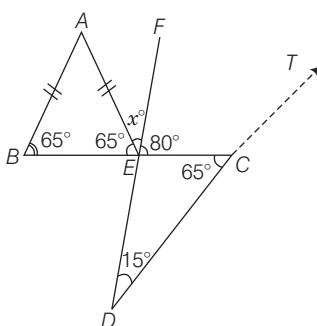
[alternate angles]

$$\text{Also, } \angle DCE = \angle DCB \text{ [same angle]}$$

$$\text{Now, } \angle FEC = \angle CDE + \angle DCE$$

[exterior angle]

$$15^\circ + 65^\circ = 80^\circ$$



$$\therefore \angle DEC + \angle FEC = 180^\circ \text{ [linear pair]}$$

Given that, $AB = AE$

$$\therefore \angle ABE = \angle AEB = 65^\circ$$

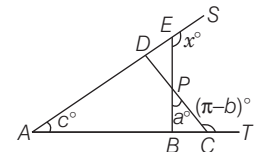
$$\therefore \angle AEB + \angle AEF + \angle FEC = 180^\circ$$

[straight line]

$$\Rightarrow 65^\circ + x^\circ + 80^\circ = 180^\circ$$

$$\therefore x^\circ = 180^\circ - 145^\circ = 35^\circ$$

52. (c) $\therefore \angle PCT + \angle PCB = \pi$ [linear pair]



$$\angle PCB = \pi - (\pi - b^\circ) = b^\circ \dots (i)$$

In $\triangle BPC$, $\angle PCB + \angle BPC + \angle PBC = \pi$

[by angle sum property of a triangle]

$$\Rightarrow \angle PBC = \pi - \angle PCB - \angle BPC$$

$$= \pi - b^\circ - a^\circ \dots (ii)$$

$$\therefore \angle ABE + \angle EBC = \pi \text{ [linear pair]}$$

$$\Rightarrow \angle ABE = \pi - \angle PBC$$

$$[\because \angle PBC = \angle EBC]$$

$$= \pi - (\pi - b^\circ - a^\circ) = a^\circ + b^\circ \dots (iii)$$

Now, in $\triangle ABE$

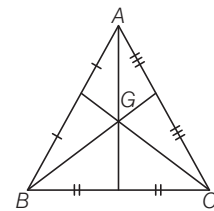
Sum of two interior angles = Exterior angle

$$\angle EAB + \angle ABE = \angle BES$$

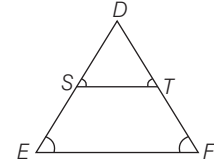
$$c^\circ + b^\circ + a^\circ = x^\circ$$

$$\therefore x^\circ = a^\circ + b^\circ + c^\circ$$

53. (d) $GA = GB = GC$ is true only and only for equilateral triangle and here it is not given that ABC is an equilateral triangle. So, statement I is not correct. Similarly, statement II will also hold only for equilateral triangle.



54. (a) Given, $\frac{DS}{DE} = \frac{1}{\sqrt{2}}$



$$\therefore DE = \sqrt{2} \times DS$$

Since, ST is parallel to EF .

$$\therefore \angle S = \angle E \text{ and } \angle T = \angle F$$

So, $\triangle DTS$ and $\triangle DEF$ are similar.

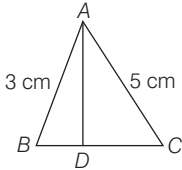
Ratio of areas = Ratio of squares of corresponding sides.

$$\Rightarrow \frac{\Delta DST}{\Delta DEF} = \left(\frac{1}{\sqrt{2}}\right)^2 = \frac{1}{2}$$

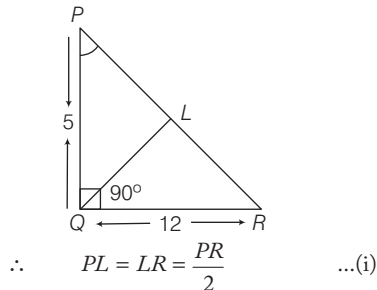
$$\Rightarrow \Delta DEF = 2 \Delta DST$$

Hence, both statements I and II are true and statement II is the correct explanation of statement I.

55. (c) We know that, The sum of any two sides of a triangle is greater than twice the median drawn to the third side.
i.e. $(AB + AC) > 2AD$
 $\Rightarrow (3 + 5) > 2AD \Rightarrow AD < 4$
Hence, AD is always less than 4 cm.



56. (d) Given that, $PQ = 5$ cm, $QR = 12$ cm and QL is a median.



$$\begin{aligned} \therefore PL = LR = \frac{PR}{2} \quad \dots(i) \\ \text{In } \Delta PQR, (PR)^2 = (PQ)^2 + (QR)^2 \\ \text{[by pythagoras theorem]} \\ (PR)^2 = (5)^2 + (12)^2 \\ (PR)^2 = 25 + 144 \Rightarrow 169 = (13)^2 \\ \Rightarrow PR^2 = (13)^2 \Rightarrow PR = 13 \end{aligned}$$

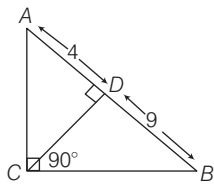
Now, by mid-point theorem, if L is the mid-point of the hypotenuse PR of a right angled ΔPQR , then

$$QL = \frac{1}{2} PR = \frac{1}{2} (13) = 6.5 \text{ cm}$$

57. (d) In ΔABC and ΔACD ,
 $\therefore \frac{AC}{AB} = \frac{4}{AC}$
 $\therefore AC^2 = 4 \times 13 = 52$... (i)
[$\because AB = 4 + 9 = 13$]

$$\text{In } \Delta ABC \text{ and } \Delta BCD, \frac{BC}{AB} = \frac{9}{BC}$$

$$\begin{aligned} \Rightarrow BC^2 = 9 \times 13 = 117 \quad \dots(ii) \\ \text{Now, } \frac{1}{CD^2} = \frac{1}{AC^2} + \frac{1}{BC^2} = \frac{1}{52} + \frac{1}{117} \end{aligned}$$



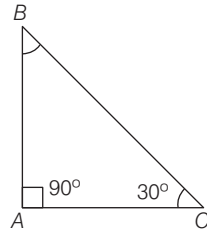
$$= \frac{9 + 4}{13 \times 4 \times 9} \Rightarrow \frac{1}{CD^2} = \frac{1}{36}$$

$$\therefore CD = 6 \text{ cm}$$

Shortcut Method

$$\begin{aligned} CD^2 &= AD \cdot DB \\ CD &= \sqrt{AD \cdot DB} = \sqrt{4 \times 9} \\ &= \sqrt{36} = 6 \text{ cm} \end{aligned}$$

58. (d) Given that, $\angle A = 90^\circ$ and $\angle C = 30^\circ$



$\therefore ABC$ is a right angled triangle.

Also, given that $BC = 2AB$

$$\Rightarrow AB = \frac{BC}{2} \quad \dots(i)$$

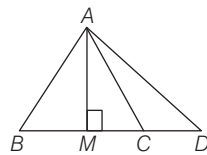
By pythagoras theorem,

$$\begin{aligned} BC^2 &= AC^2 + AB^2 \\ \Rightarrow (2AB)^2 &= AC^2 + AB^2 \\ \Rightarrow AC^2 &= 4AB^2 - AB^2 = 3AB^2 \\ \Rightarrow AC &= \sqrt{3} \cdot AB = \frac{\sqrt{3}}{2} \cdot (2AB) \\ \therefore AC &= \frac{\sqrt{3}}{2} \cdot BC \quad \text{[from Eq.(i)]} \end{aligned}$$

59. (d) Draw $AM \perp BC$

Let $AB = BC = AC = x$

Then, $BM = MC = \frac{x}{2}$



$$\text{Also, } CD = \frac{x}{2} \quad [\because BC = 2CD]$$

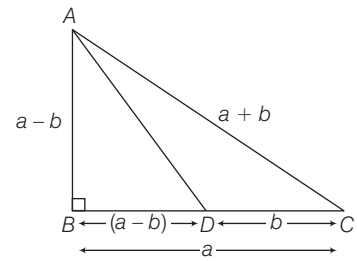
Now, in ΔAMC , by pythagoras theorem,

$$\begin{aligned} AM^2 &= AC^2 - MC^2 \\ &= x^2 - \frac{x^2}{4} = \frac{3x^2}{4} \end{aligned}$$

In ΔAMD , by pythagoras theorem,

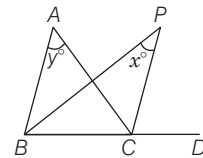
$$\begin{aligned} AD^2 &= AM^2 + MD^2 \\ &= \frac{3x^2}{4} + x^2 = \frac{7x^2}{4} \\ &= 7 \left(\frac{x^2}{4} \right) = 7CD^2 \quad \left[\because CD = \frac{x}{2} \right] \end{aligned}$$

60. (d) In right angle ΔABC ,
By pythagoras theorem,
 $(a + b)^2 = (a - b)^2 + a^2$



$$\begin{aligned} \Rightarrow a^2 + b^2 + 2ab &= a^2 + b^2 - 2ab + a^2 \\ \Rightarrow 4ab &= a^2 \Rightarrow 4b = a \\ \text{Now, } \frac{BD}{DC} &= \frac{a-b}{b} \\ &= \frac{4b-b}{b} = \frac{3b}{b} = \frac{3}{1} \\ \text{or } 3 : 1 \end{aligned}$$

61. (d) Given that, $\angle BPC = x^\circ$
and $\angle BAC = y^\circ$



Since, BP and CP are the angle bisectors of $\angle ABC$ and $\angle ACD$, respectively.

$$\therefore \angle ABC = 2\angle PBC \quad \dots(i)$$

$$\text{and } \angle ACD = 2\angle PCD \quad \dots(ii)$$

since sum of two interior angles is equal to exterior angle

$$\therefore \angle PCD = x^\circ + \angle PBC \quad \dots(iii)$$

$$\text{and } \angle ACD = y^\circ + \angle ABC$$

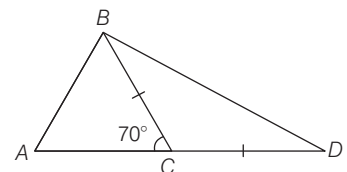
$$\Rightarrow 2\angle PCD = y^\circ + 2\angle PBC$$

[using (i) and (ii)]

$$\Rightarrow 2[x^\circ + \angle PBC] = y^\circ + 2\angle PBC$$

$$\Rightarrow 2x^\circ = y^\circ$$

62. (a) $\therefore \angle ACB + \angle BCD = 180^\circ$ [linear pair]
 $\angle BCD = 180^\circ - 70^\circ = 110^\circ$



In ΔBCD , $BC = CD$

$$\Rightarrow \angle CBD = \angle CDB \quad \dots(i)$$

[angles opposite to equal side]

Also, $\angle BCD + \angle CBD + \angle CDB = 180^\circ$

[by angle sum property of a triangle]

$$2\angle CDB = 180^\circ - \angle BCD$$

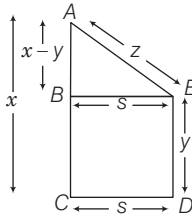
$$= 180^\circ - 110^\circ = 70^\circ$$

$$\therefore \angle CDB = \angle ADB = \frac{70^\circ}{2} = 35^\circ$$

63. (d) Now, applying pythagoras theorem in $\triangle ABE$,

$$AE^2 = AB^2 + BE^2$$

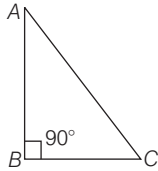
$$\Rightarrow z^2 = (x-y)^2 + s^2$$



$$\Rightarrow z^2 = x^2 + y^2 - 2xy + s^2$$

$$\therefore s^2 = z^2 - x^2 - y^2 + 2xy$$

64. (b) Given, $\angle C = 2\angle A$ and $\angle B = 90^\circ$
 $\therefore \angle A + \angle C = 90^\circ$



$$\Rightarrow \angle A + 2\angle A = 90^\circ \quad [\because \angle C = 2\angle A]$$

$$\Rightarrow 3\angle A = 90^\circ \Rightarrow \angle A = 30^\circ$$

So, $\angle C = 90^\circ - 30^\circ = 60^\circ$

Now, if the angles of a triangle are of measure 30° , 60° and 90° .

Then, side opposite to 30° i.e.

$$BC = \frac{1}{2} \times \text{hypotenuse}$$

$$\Rightarrow BC = \frac{1}{2} AC \Rightarrow AC = 2BC$$

In right angled $\triangle ABC$,

by pythagoras theorem

$$AC^2 = AB^2 + BC^2$$

$$\Rightarrow 4BC^2 = AB^2 + BC^2 \quad [\because AC = 2BC]$$

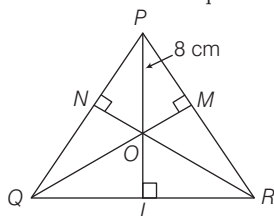
$$\Rightarrow AB^2 = 4BC^2 - BC^2 = 3BC^2$$

65. (d) Since, PQR is an equilateral triangle. Then, PL is also the median of $\triangle PQR$. Similarly, RN and QM are also the median of $\triangle PQR$ and O is the centroid.

$$\text{So, } \frac{PO}{OL} = \frac{2}{1} \Rightarrow OL = \frac{PO}{2} = \frac{8}{2} = 4 \text{ cm}$$

$$\text{Now, altitude of } \triangle PQR = \frac{\sqrt{3}a}{2}$$

[where, a = length of side of an equilateral $\triangle PQR$]



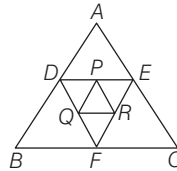
$$PO + OL = \frac{\sqrt{3}a}{2}, \quad 8 + 4 = \frac{\sqrt{3}a}{2}$$

$$a = \frac{12 \times 2}{\sqrt{3}} = \frac{24}{\sqrt{3}} \text{ cm}$$

$$\therefore \text{Perimeter of } \triangle PQR = 3a$$

$$= \frac{3 \times 24}{\sqrt{3}} = 24\sqrt{3} \text{ cm}$$

66. (b) Perimeter of $\triangle PQR = 1+2+3=6$ units



$\therefore P, Q, R$ are the mid-points of DE, DF and FE of $\triangle DEF$.

$$\therefore 2PQ = FE$$

[by mid-point theorem]

Similarly, $DF = 2PR$ and $DE = 2QR$

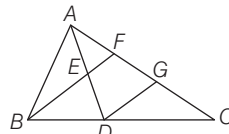
$$\therefore \text{Perimeter of } \triangle DEF = 2 \times 6 = 12 \text{ units}$$

$$\text{Similarly, perimeter of } \triangle ABC$$

$$= 2 \times \text{perimeter of } \triangle DEF$$

$$= 2 \times 12 = 24 \text{ units}$$

67. (b) Draw a line segment DG parallel to BF . Then, in $\triangle ADG$,



$$BF \parallel DG$$

$$\therefore EF \parallel DG$$

and $AE = ED$

[since, E is mid-point of AD]

$$\therefore AF = FG \quad \dots(i)$$

Similarly, in $\triangle BCF$

$$DG \parallel BF \text{ and } BD = DC$$

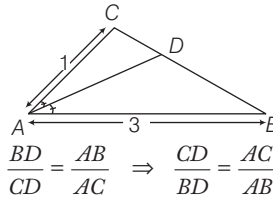
$$\therefore FG = GC \quad \dots(ii)$$

From Eqs. (i) and (ii), we get

$$CF = \frac{2}{3} AC$$

68. (d) In $\triangle ABC$, AD is the internal angle bisector of $\angle A$

Using property of internal angle bisector.



$$\frac{BD}{CD} = \frac{AB}{AC} \Rightarrow \frac{CD}{BD} = \frac{AC}{AB}$$

On adding both sides, we get

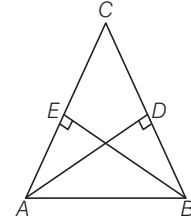
$$\Rightarrow \frac{CD}{BD} + 1 = \frac{AC}{AB} + 1$$

$$\Rightarrow \frac{CD + BD}{BD} = \frac{AC + AB}{AB}$$

$$\Rightarrow \frac{BC}{BD} = \frac{3+1}{3} \Rightarrow \frac{BD}{BC} = \frac{3}{4}$$

$$\therefore BD : BC = 3 : 4$$

69. (b) In $\triangle ADC$ and $\triangle BEC$



$$\angle BEC = \angle ADC = 90^\circ$$

$$\angle ACD = \angle BCE \quad [\text{common}]$$

$\therefore \triangle ADC \sim \triangle BEC$ [by AA similarity]

$$\frac{AC}{BC} = \frac{CD}{CE}$$

$$\Rightarrow AC \times CE = BC \times CD$$

$$\Rightarrow CE \times CA = CD \times CB$$

70. (b) Let the sides of triangle be $\frac{a}{r}, a, ar$ and since $r < 1$.

$$\therefore \frac{a}{r} > a > ar$$

Now, triangle is right angled.

Using pythagoras theorem,

$$\left(\frac{a}{r}\right)^2 = (a)^2 + (ar)^2$$

$$\Rightarrow \frac{a^2}{r^2} = a^2 + a^2 r^2$$

$$\Rightarrow \frac{a^2}{r^2} = a^2(1 + r^2) \Rightarrow r^2 + r^4 = 1$$

$$\text{Put } r^2 = x,$$

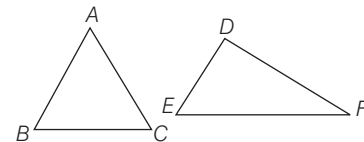
$$\Rightarrow x^2 + x - 1 = 0$$

$$x = \frac{-1 \pm \sqrt{1 - 4(-1)}}{2} = \frac{-1 \pm \sqrt{5}}{2}$$

Since, sides of triangle cannot be negative

$$\therefore r^2 = \frac{\sqrt{5} - 1}{2}$$

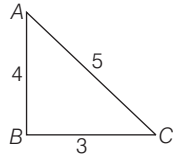
71. (a) $\triangle ABC \sim \triangle DEF$



$$\therefore \frac{AB}{DE} = \frac{BC}{EF} \Rightarrow \frac{1}{2} = \frac{8}{EF}$$

$$EF = 16 \text{ cm}$$

72. (a) Since, the triangle is right angled. So, all the three consecutive sides must satisfy pythagoras theorem.

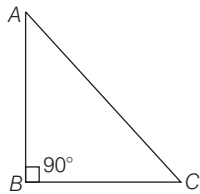


Hence, 3, 4 and 5 are the sides of triangle which satisfy pythagoras theorem.

$$\therefore \text{Area of triangle} = \frac{1}{2} \times 4 \times 3 = 6 \text{ cm}^2$$

73. (a) In a triangle, Sum of two sides is always greater than the third side
i.e. $x < 40$... (i)
Difference of two sides is always less than third side
i.e. $10 < x$... (ii)
From Eqs. (i) and (ii), $10 < x < 40$

74. (a) Given, in $\triangle ABC$, $\angle B = 2\angle C = 2\angle A$
We know that, sum of all angles of a triangle = 180°
 $\Rightarrow \angle A + \angle B + \angle C = 180^\circ$
 $\Rightarrow \angle A + 2\angle A + \angle A = 180^\circ$
 $[\because 2\angle C = 2\angle A = \angle B]$
 $\Rightarrow 4\angle A = 180^\circ$
 $\Rightarrow \angle A = \frac{180^\circ}{4} = 45^\circ$



$$\therefore \angle B = 90^\circ \text{ and } \angle C = 45^\circ$$

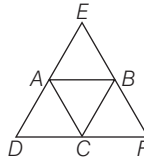
Thus, $\triangle ABC$ is a right angled triangle, right angle at B and $AB = BC$.

In $\triangle ABC$, by pythagoras theorem,

$$\begin{aligned} AB^2 + BC^2 &= AC^2 \\ \Rightarrow AB^2 + AB^2 &= AC^2 & [\because AB = BC] \\ \Rightarrow 2AB^2 &= AC^2 \\ \Rightarrow \sqrt{2}AB &= AC \\ &[\text{taking square root on both sides}] \\ \Rightarrow \frac{AC}{AB} &= \frac{\sqrt{2}}{1} \\ \therefore AC : AB &= \sqrt{2} : 1 \end{aligned}$$

75. (c) I. On drawing the three straight lines through the three vertices of $\triangle ABC$, we get the following figure
Here, $AB \parallel DF$, $BC \parallel DE$ and $AC \parallel EF$

Obviously, A, B and C are the mid-points of DE, EF and DF, respectively.



By mid-point theorem,

$$BC = \frac{1}{2} DE \text{ or } DE = 2BC$$

Similarly, $DF = 2AB$
and $EF = 2AC$

Hence, statement I is correct.

II. Also, area of

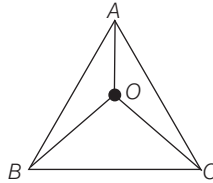
$$\triangle ABC = \frac{1}{4} \text{ area of } \triangle DEF$$

or area of $\triangle DEF = 4$ area of $\triangle ABC$

Hence, statement II is also correct.

76. (c) See solved answer of 67.

77. (c) Given, O is equidistant from AB, BC and AC
 $\therefore$ O is the incentre of $\triangle ABC$



Join OA, OB and OC.

In $\triangle ABC$, OA, OB and OC are angle bisectors.

$$\therefore \angle OAB = \angle OAC = \frac{1}{2} \angle BAC$$

$$\therefore \angle OBA = \angle OBC = \frac{1}{2} \angle ABC$$

$$\text{and } \angle OCB = \angle OCA = \frac{1}{2} \angle ACB$$

Now, in $\triangle ABC$,

$$\angle ABC + \angle BAC + \angle ACB = 180^\circ$$

$[\because \text{angle sum property}]$

$$\therefore \angle OAC + \angle OCB + \angle OBA = 90^\circ$$

[option]

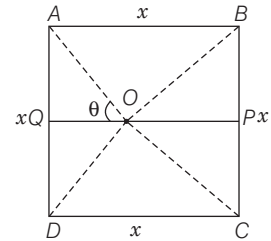
II. This statement is false as

$$\angle BOC = 90^\circ + \frac{1}{2} \angle BAC$$

III. The statement perpendiculars drawn from any point on OA to AB and AC are always equal so it is true, because O is equidistant from AB and AC.

Hence, the statements I and III are correct.

78. (d)



Let side of square ABCD be x and drawn equilateral $\triangle BOC$ inside ABCD.

Such that, $BO = OC = BC = x$

$$\text{and } BP = \frac{BC}{2} = \frac{x}{2}$$

$$\begin{aligned} \therefore PO &= \sqrt{x^2 - \left(\frac{x}{2}\right)^2} = \sqrt{x^2 - \frac{x^2}{4}} \\ &= \sqrt{\frac{3x^2}{4}} = \frac{\sqrt{3}}{2}x \end{aligned}$$

$$\text{and } OQ = AB - PO = x - \frac{\sqrt{3}}{2}x$$

$$= x \left(\frac{2 - \sqrt{3}}{2} \right)$$

Since, $\angle AOD = 90^\circ$

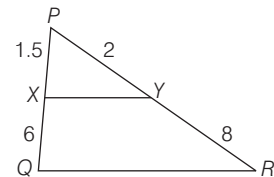
$$\Rightarrow \angle AOQ = \theta$$

In $\triangle AOQ$,

$$\begin{aligned} \tan \theta &= \frac{AQ}{OQ} = \frac{AD}{OQ} = \frac{\frac{x}{2}}{x \left(\frac{2 - \sqrt{3}}{2} \right)} \\ &= \frac{x}{2} \times \frac{2}{x(2 - \sqrt{3})} \\ &= \frac{1}{2 - \sqrt{3}} \times \frac{2 + \sqrt{3}}{2 + \sqrt{3}} = \frac{2 + \sqrt{3}}{4 - 3} \\ &[\because (a + b)(a - b) = a^2 - b^2] \\ &= 2 + \sqrt{3} \end{aligned}$$

79. (d) In $\triangle PQR$,

$$\frac{PX}{XQ} = \frac{1.5}{6} = \frac{1}{4} \text{ and } \frac{PY}{YR} = \frac{2}{8} = \frac{1}{4}$$



$$\text{So, } \frac{PX}{XQ} = \frac{PY}{YR}$$

$$\text{I. } \frac{PX}{XY} = \frac{PQ}{QR} \Rightarrow \frac{1.5}{XY} = \frac{7.5}{QR}$$

$$\Rightarrow QR = 5XY$$

$$\text{II. Also, } \frac{PX}{PQ} = \frac{PY}{PR} = \frac{1}{5}$$

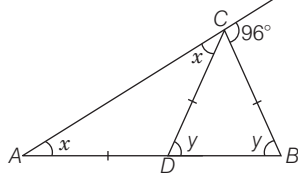
$$\Rightarrow QR \text{ is parallel of } XY.$$

III. $\triangle PYX$ is similar to $\triangle PRQ$.

Hence, all statements are correct.

80. (c) See solved answer 17.

81. (b) Let $\angle CDB = y$ and $\angle CAD = x$
As, $AD = CD$



$$\Rightarrow \angle ACD = \angle CAD = x \text{ and } CD = CB$$

$$\Rightarrow \angle CBD = \angle CDB = y$$

From exterior angle property,

$$\angle CDB = \angle CAD + \angle ACD$$

$$\Rightarrow y = x + x \Rightarrow y = 2x$$

$$\text{Now, } \angle ACD + \angle DCB + 96^\circ = 180^\circ$$

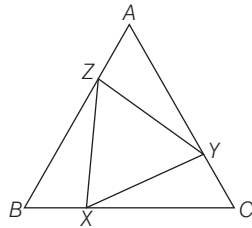
$$\Rightarrow x + 180^\circ - 2y + 96^\circ = 180^\circ$$

$$\Rightarrow \frac{y}{2} - 2y + 96^\circ = 0 \Rightarrow \frac{3}{2}y = 96^\circ$$

$$\therefore y = 64^\circ$$

82. (c) In an equilateral $\triangle ABC$,

$$AB = BC = CA \text{ and } \angle B = \angle C = \angle A$$



Given that, $BX = CY = AZ$

Now, in $\triangle XYC$, $\triangle ZYA$ and $\triangle XZB$,

$$BX = CY = AZ \quad \dots(i)$$

$$\Rightarrow (BC - XC) = (AC - AY)$$

$$= AB - BZ$$

$$\Rightarrow XC = AY = BZ \quad \dots(ii)$$

$$\text{and } \angle B = \angle C = \angle A \quad \dots(iii)$$

From Eqs. (i), (ii) and (iii), we get $\triangle XYC$, $\triangle ZYA$ and $\triangle XZB$ are congruent triangles.

$$\therefore XY = YZ = XZ$$

So, $\triangle XYZ$ is an equilateral triangle and $\triangle XYZ$ is similar to $\triangle ABC$.

So, both statements are correct.

83. (b) I. This statement is always true.

II. In any triangle three times the sum of squares of the sides of a triangle is equal to four times the sum of squares of medians

$$\text{i. e. } 3(AB^2 + BC^2 + CA^2) = 4(AD^2 + BE^2 + CF^2)$$

$$\text{Similarly } 3(A'B'^2 + B'C'^2 + C'A'^2)$$

$$= 4(A'D'^2 + B'E'^2 + C'F'^2)$$

$$\therefore \frac{AB^2 + BC^2 + CA^2}{AD^2 + BE^2 + CF^2} = \frac{A'B'^2 + B'C'^2 + C'A'^2}{A'D'^2 + B'E'^2 + C'F'^2} = \frac{4}{3}$$

Hence, both statements are correct but statement II is not correct explanation of statement I.

$$\begin{aligned} 84. (c) \frac{\text{Area of } \triangle ABC}{\text{Area of } \triangle DEF} &= \frac{AB^2}{DE^2} \\ &= \frac{(\sqrt{2} + 1)^2}{(\sqrt{3})^2} = \frac{3 + 2\sqrt{2}}{3} \\ &= \frac{(3 + 2\sqrt{2})(3 - 2\sqrt{2})}{3(3 - 2\sqrt{2})} \\ &= \frac{1}{9 - 6\sqrt{2}} \end{aligned}$$

Hence, the required ratio is

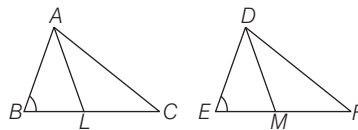
$$1 : (9 - 6\sqrt{2}).$$

85. (c) Given that, $\angle ABC = \angle DEF$,

$$\angle ACB = \angle DFE$$

and

$$\angle BAC = \angle EDF$$



So, $\triangle ABC$ and $\triangle DEF$ are similar.

$$\therefore \frac{AB}{DE} = \frac{BC}{EF} = \frac{AC}{DF}$$

Now, L is the mid-point of BC , then

$$BL = \frac{1}{2}BC$$

Also, M is the mid-point of EF , then

$$EM = \frac{1}{2}EF \Rightarrow \frac{AB}{DE} = \frac{2BL}{2EM} = \frac{BL}{EM}$$

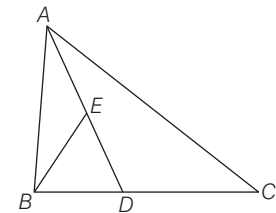
and $\angle ABL = \angle DEM$

$\therefore \triangle ABL$ is similar to $\triangle DEM$.

[By SAS Similarity]

Hence, statement I is true but statement II is false.

86. (c) In $\triangle ABC$, AD is the median and median AD bisects the area of $\triangle ABC$.
Area of $\triangle ABD = \frac{1}{2} \times \text{Area of } \triangle ABC$



Since, E is the mid-point of AD . Then, BE is the median of $\triangle ABD$. So, BE bisects area of $\triangle ABD$.

$$\begin{aligned} \text{I. Area of } \triangle BED &= \frac{1}{2} \times \text{Area of } \triangle ABD \\ &= \frac{1}{2} \left[\frac{1}{2} \times \text{Area of } \triangle ABC \right] \\ &= \frac{1}{4} \times \text{Area of } \triangle ABC \end{aligned}$$

Hence, statement I is correct.

II. Area of $\triangle ADC = \text{Area of } \triangle ABD$

$$= 2 \times \text{Area of } \triangle BED$$

So, statement II is also correct.

Hence, the statements I and II are correct.

24

QUADRILATERAL AND POLYGON

Regularly (2-3) questions have been asked from this chapter. Generally, questions from this chapter are tricky and mostly statement based. So, a clear concept of all the properties related to quadrilateral is necessary to do well.



QUADRILATERAL

A figure enclosed by four sides is called a quadrilateral. A quadrilateral has four angles and sum of these angles is equal to 360° .

Various types of quadrilateral are discussed below.

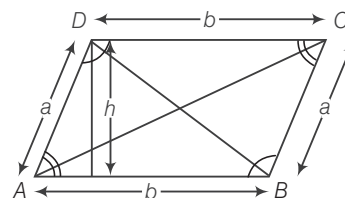
Parallelogram

A quadrilateral, in which opposite sides are parallel is called a parallelogram.

- (i) Area = Base $\times$ Height = $b \times h$ (ii) Perimeter = $2(a + b)$

Properties of Parallelogram

- (i) Diagonals of a parallelogram bisect each other.
- (ii) Each diagonal of a parallelogram divides it into two congruent triangles.
- (iii) Sum of any two adjacent angles is 180° .
- (iv) A parallelogram inscribed in a circle is a rectangle.
- (v) A parallelogram circumscribed about a circle is a rhombus.
- (vi) Two parallelograms have equal areas if they are on the same base and between the same parallel lines.
- (vii) Lines joining the midpoints of the sides of parallelogram is a parallelogram.
- (viii) The opposite angles of parallelogram are equal. ($\angle A = \angle C$ and $\angle B = \angle D$) (from above figure)
- (ix) The sum of the squares of the four sides is equal to the sum of squares of diagonal.
$$AC^2 + BD^2 = AB^2 + BC^2 + CD^2 + AD^2$$



EXAMPLE 1. In a parallelogram $ABCD$, the bisectors of $\angle A$ and $\angle B$ meet at O . Then, the value of $\angle AOB$ is

- a. 55° b. 75° c. 90° d. 120°

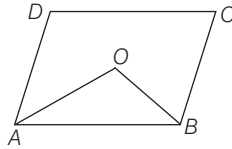
Sol. c. As, $ABCD$ is a parallelogram.

$\therefore \angle A + \angle B = 180^\circ$ [cointerior angle]

$$\frac{1}{2} \angle A + \frac{1}{2} \angle B = 90^\circ$$

$$\Rightarrow \angle OAB + \angle OBA = 90^\circ$$

$$\text{In } \triangle AOB, \angle AOB = 180^\circ - (\angle OAB + \angle OBA) = 180^\circ - 90^\circ = 90^\circ$$



EXAMPLE 2. In the adjoining figure, $ABCD$ is a parallelogram and E, F are the centroids of $\triangle ABD$ and $\triangle BCD$, respectively, then the length of EF is

- a. AE b. OB c. $\frac{1}{3} AE$ d. $\frac{1}{3} FC$

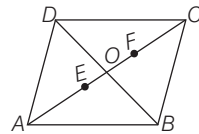
Sol. a. As E is the centroid of $\triangle ABD$ and AO is one of its medians.

$$\Rightarrow OA : EO = 3 : 1 \Rightarrow EO = \frac{1}{3} OA$$

$$\text{Similarly, } FO = \frac{1}{3} OC$$

$$\therefore EO + OF = \frac{1}{3} OA + \frac{1}{3} OC = \frac{1}{3} AC = AE$$

$$\therefore EF = AE$$

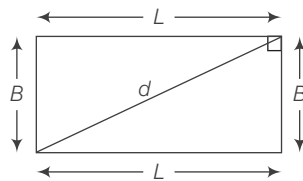


Rectangle

It is a parallelogram with opposite sides equal and each angle is equal to 90° .

(i) Area = Length $\times$ Breadth
 $= L \times B$

(ii) Perimeter = $2(L + B)$ (iii) Diagonal (d) = $\sqrt{L^2 + B^2}$

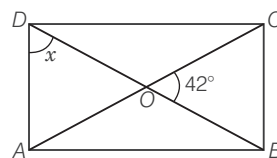


Properties of Rectangle

- The diagonals of a rectangle are of equal lengths and they bisect each other.
- All rectangles are parallelograms but reverse is not true.
- Diagonal of a rectangle inscribed in a circle is equal to the diameter of the circle.
- The figure formed by joining the mid-points of the sides of a rectangle is a rhombus.
- A rectangle and a parallelogram have equal area if they are on the same base and between same parallel lines.

EXAMPLE 3. In adjoining figure, if $ABCD$ is a rectangle and diagonals AC and BD intersect each other at O . Find the value of x

- a. 75° b. 70° c. 69° d. 63°



Sol. c. Since, diagonals of rectangle bisect each other

$$\therefore OD = OA \Rightarrow \angle ODA = \angle OAD$$

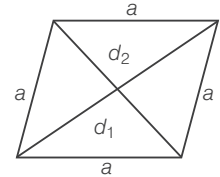
$$\text{Also, } \angle DOA = \angle COB = 42^\circ$$

$$\text{In } \triangle DOA, \angle DOA + \angle ODA + \angle OAD = 180^\circ$$

$$\Rightarrow 42^\circ + x + x = 180^\circ \Rightarrow 2x = 138^\circ \Rightarrow x = 69^\circ$$

Rhombus

It is a parallelogram with all 4 sides equal. The opposite angles in a rhombus are equal but they are not right angle.



(i) Area = $\frac{1}{2} \times d_1 \times d_2$

(ii) Perimeter = $4a$

(iii) Side (a) = $\frac{1}{2} \sqrt{d_1^2 + d_2^2}$

(iv) $4a^2 = d_1^2 + d_2^2$

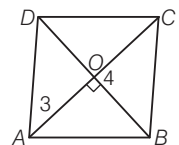
where, a = side, d_1 and d_2 are diagonals.

Properties of Rhombus

- The diagonals of a rhombus bisect each other at right angles (90°). But they are not necessarily equal.
- Diagonals bisect the vertex angles.
- The figure formed by joining the mid-points of the sides of rhombus is a rectangle.
- All rhombus are parallelogram but reverse is not true.
- A rhombus may or may not be a square but all squares are rhombus.

EXAMPLE 4. In a rhombus $ABCD$, diagonals intersect each other at O . If $AO = 3$ cm and $OB = 4$ cm, then find the perimeter of $ABCD$.

- a. 14 cm b. 30 cm c. 20 cm d. 28 cm



Sol. c. The diagonals of a rhombus are perpendicular bisectors. Which means they form right angles at the point of their intersection.

$\therefore$ In $\triangle AOB$, by pythagoras theorem,

$$AO^2 + OB^2 = AB^2 \Rightarrow AB = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25}$$

$$\therefore AB = 5 \text{ cm}$$

Hence, perimeter of rhombus

$$ABCD = AB + BC + CD + DA = 5 + 5 + 5 + 5 = 20 \text{ cm}$$

Square

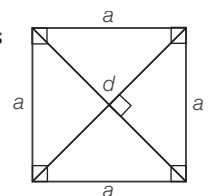
It is a parallelogram with all four sides equal and each angle is equal to 90° .

(i) Area = (side) $^2 = a^2$ or $\frac{1}{2} d^2$

(ii) Perimeter = $4 \times \text{side} = 4a$

(iii) Diagonal (d) = $a\sqrt{2}$

where, a = side, d = diagonal

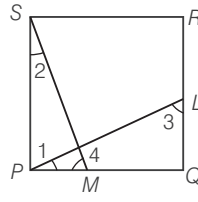


Properties of Square

- Diagonal of a square are equal and bisect each other at right angles (90°).
- All square are rhombus but converse is not true.
- Diagonal of square is the diameter of the circumscribing circle that circumscribes the square.
- Side of a circumscribed square is equal to the diameter of the inscribed circle.
- The figure formed by joining the mid-points of the sides of square is a square.

EXAMPLE 5. PQRS is a square, P is joined to a point L on QR and S is joined to a point M on PQ. If $PL = SM$, then

- $\angle POM > 90^\circ$
- $\angle POM = 90^\circ$
- $\angle POM < 90^\circ$
- None of these

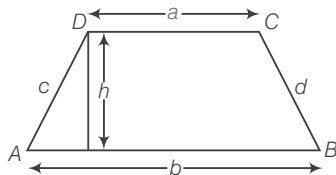


Sol. b. In $\triangle SPM$ and $\triangle PQL$, $\angle SPM = \angle PQL = 90^\circ$

$$\begin{aligned}
 SP &= PQ \text{ and } SM = PL \Rightarrow \triangle SPM \cong \triangle PQL \quad [\text{By AAS}] \\
 \Rightarrow \angle 1 &= \angle 2 \text{ and } \angle 3 = \angle 4 \\
 \text{In } \triangle SPM, \quad \angle 2 + 90^\circ + \angle 4 &= 180^\circ \Rightarrow \angle 2 + \angle 4 = 90^\circ \\
 \text{In } \triangle OPM, \quad \angle 1 + \angle 4 + \angle POM &= 180^\circ \\
 \Rightarrow \angle 2 + \angle 4 + \angle POM &= 180^\circ \\
 \Rightarrow 90^\circ + \angle POM &= 180^\circ \quad [\because \angle 2 + \angle 4 = 90^\circ] \\
 \Rightarrow \angle POM &= 90^\circ
 \end{aligned}$$

Trapezium

It is a quadrilateral with any one pair of opposite sides parallel.

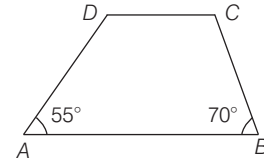


- Area = $\frac{1}{2}$ (Sum of the parallel sides $\times$ Height)
 $= \frac{1}{2} (a + b)h$ where, a and b are parallel sides and h is the height or perpendicular distance between a and b .
- Perimeter = $AB + BC + CD + AD$

Properties

- If the non-parallel sides are equal then the diagonals will also be equal to each other.
- Diagonals intersect each other proportionally in the ratio of lengths of parallel sides.
- If a trapezium is inscribed in a circle, then it is an isosceles trapezium with equal oblique sides.
- The line joining the mid-points of non-parallel sides is half the sum of parallel side and is called median.
i.e. Median = $\frac{1}{2} \times$ sum of parallel sides.

EXAMPLE 6. In the adjoining figure, ABCD is a trapezium in which $AB \parallel DC$. If $\angle A = 55^\circ$ and $\angle B = 70^\circ$, then the value of $\angle C$ and $\angle D$ is



- 75° and 85°
- 90° and 120°
- 110° and 125°
- 115° and 120°

Sol. c. As, $AB \parallel CD$

$$\text{So, } \angle A + \angle D = 180^\circ \Rightarrow \angle D = 180^\circ - 55^\circ = 125^\circ$$

$$\text{Also, } \angle B + \angle C = 180^\circ \Rightarrow \angle C = 180^\circ - \angle B$$

$$\Rightarrow \angle C = 180^\circ - 70^\circ = 110^\circ$$

$$\text{So, } \angle C = 110^\circ \text{ and } \angle D = 125^\circ$$

POLYGON

A polygon is a closed, plane figure bounded by 'n' straight lines ($n \geq 3$). Each of the line segment forming the polygon is called its sides.

A polygon may be a triangle, quadrilateral, pentagon etc. Polygons are classified according to the number of sides as given below:

Number of sides	Name
3	Triangle
4	Quadrilateral
5	Pentagon
6	Hexagon
7	Heptagon
8	Octagon
9	Nonagon
10	Decagon

Regular Polygon

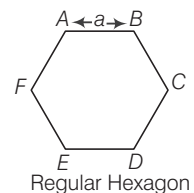
In regular polygons, all sides and all interior angles are equal. If each side of a regular polygon of n sides = a , then

$$(i) \text{ Area of regular pentagon} = 5a^2 \frac{\sqrt{3}}{4}$$

$$(ii) \text{ Area of regular hexagon} = 6a^2 \frac{\sqrt{3}}{4}$$

$$(iii) \text{ Area of regular octagon} = 2(\sqrt{2} + 1)a^2$$

$$(iv) \text{ Each exterior angle} = \frac{360^\circ}{n}$$



Regular Hexagon

(v) Each interior angle $= 180^\circ - \text{Exterior angle}$

$$= \frac{(n-2) \times 180}{n}$$

(vi) Number of diagonals $= \left\{ \frac{n(n-1)}{2} - n \right\}$

(vii) Sum of all interior angles $= (n-2) \times 180^\circ$.

EXAMPLE 7. A polygon has 35 diagonals. Then, the number of sides of that polygon is

- a. 7 b. 10 c. 11 d. 12

Sol. b. Let number of sides be n , then

$$\begin{aligned} \frac{n(n-1)}{2} - n &= 35 \Rightarrow \frac{n^2 - n - 2n}{2} = 35 \\ \Rightarrow n^2 - 3n - 70 &= 0 \Rightarrow n^2 - 10n + 7n - 70 = 0 \\ \Rightarrow n(n-10) + 7(n-10) &= 0 \Rightarrow (n-10)(n+7) = 0 \\ \Rightarrow n &= 10 \text{ and } n \neq -7 \end{aligned}$$

Hence, the number of sides of the polygon is 10.

EXAMPLE 8. The angles of a hexagon are x° , $(x-5)^\circ$, $(x-5)^\circ$, $(2x-5)^\circ$, $(2x-5)^\circ$ and $(2x+20)^\circ$. Then, the value of x is

- a. 60° b. 75° c. 80° d. 90°

Sol. c. Sum of interior angles of a hexagon $= (n-2) \times 180^\circ$

$$= 4 \times 180^\circ = 720^\circ$$

$$\begin{aligned} \therefore x^\circ + (x-5)^\circ + (x-5)^\circ + (2x-5)^\circ \\ + (2x-5)^\circ + (2x+20)^\circ &= 720^\circ \\ \therefore 9x &= 720^\circ = \frac{720^\circ}{9} = 80^\circ \end{aligned}$$

Hence, the value of x is 80° .

EXAMPLE 9. The difference between the interior and exterior angles of a regular polygon is 60° . Then, how many sides are there in that polygon?

- a. 5 b. 6 c. 7 d. 8

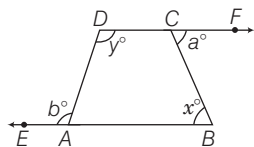
Sol. b. Here, (interior angle) $-$ (exterior angle) $= 60^\circ$

$$\begin{aligned} \Rightarrow \frac{(n-2) \times 180}{n} - \frac{360}{n} &= 60 \\ \Rightarrow \frac{1}{n} [(n-2) \times 180 - 360] &= 60 \\ \Rightarrow \frac{1}{n} [180n - 360 - 360] &= 60 \Rightarrow \frac{1}{n} [180n - 720] = 60 \\ \Rightarrow 180n - 720 &= 60n \Rightarrow 180n - 60n = 720 \\ \Rightarrow 120n &= 720 \Rightarrow n = 6 \end{aligned}$$

Therefore, the polygon contains 6 sides.

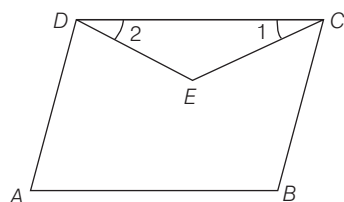
> PRACTICE EXERCISE

1. The sides BA and DC of quadrilateral $ABCD$ are produced as shown in figure. Then, which of the following statement is correct?



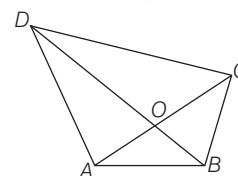
- (a) $2x^\circ + y^\circ = a^\circ + b^\circ$ (b) $x^\circ + \frac{1}{2}y^\circ = \frac{a^\circ + b^\circ}{2}$
(c) $x^\circ + y^\circ = a^\circ + b^\circ$ (d) $x^\circ + a^\circ = y^\circ + b^\circ$

2. In the quadrilateral $ABCD$, the line segments bisecting $\angle C$ and $\angle D$ meet at E . Then, the correct statement is



- (a) $\angle A + \angle B = \angle CED$ (b) $\angle A + \angle B = 2 \angle CED$
(c) $\angle A + \angle B = 3 \angle CED$ (d) None of these

3. If $ABCD$ is a quadrilateral whose diagonals AC and BD intersect at O , then



- (a) $(AB + BC + CD + DA) < (AC + BD)$
(b) $(AB + BC + CD + DA) > 2(AC + BD)$
(c) $(AB + BC + CD + DA) > (AC + BD)$
(d) $(AB + BC + CD + DA) = 2(AC + BD)$

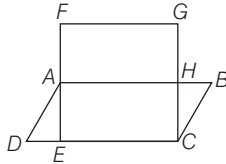
4. If area of a parallelogram with sides p and q is R and that of a rectangle with sides p and q is S , then

- (a) $R > S$ (b) $R < S$
(c) $R = S$ (d) None of these

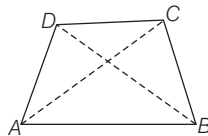
5. Two parallelogram stand on equal bases and between the same parallel. The ratio of their areas is

- (a) 1:2 (b) 2:1 (c) 1:3 (d) 1:1

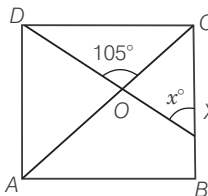
6. If $ABCD$ is a rhombus, then
 (a) $AC^2 + BD^2 = 4AB^2$ (b) $AC^2 + BD^2 = AB^2$
 (c) $AC^2 + BD^2 = 2AB^2$ (d) $2(AC^2 + BD^2) = 3AB^2$
7. A point O in the interior of a rectangle $ABCD$ is joined with each of the vertices A, B, C and D . Then,
 (a) $OB + OD = OC + OA$ (b) $OB^2 + OA^2 = OC^2 + OD^2$
 (c) $OB \cdot OD = OC \cdot OA$ (d) $OB^2 + OD^2 = OC^2 + OA^2$
8. In a trapezium $ABCD$, if $AB \parallel CD$, then $AC^2 + BD^2$ is equal to
 (a) $BC^2 + AD^2 + 2AB \cdot CD$ (b) $AB^2 + CD^2 + 2AD \cdot BC$
 (c) $AB^2 + CD^2 + 2AB \cdot CD$ (d) $BC^2 + AD^2 + 2BC \cdot AD$
9. $ABCD$ is a trapezium in which $AB \parallel CD$ and $AB = 2CD$. Its diagonals intersect each other at O , then the ratio of the areas of the $\triangle AOB$ and $\triangle COD$ is
 (a) 1:2 (b) 2:1 (c) 4:1 (d) 1:4
10. In the figure, $ABCD$ is a parallelogram with $AD = a$ units, $DC = 2a$ units and $DE : EC = 1 : 2$. $CEFG$ is a rectangle with $FE = 3AE$. What is the ratio of the area of a parallelogram and the rectangle?



11. In the adjoining figure, $ABCD$ is a quadrilateral in which AB is the longest side and CD is the shortest side, then

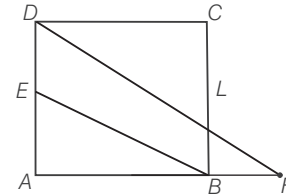


- (a) $\angle C > \angle A$ and $\angle D > \angle B$ (b) $\angle C > \angle A$ and $\angle B > \angle D$
 (c) $\angle C < \angle A$ and $\angle D < \angle B$ (d) $\angle C < \angle A$ and $\angle D = \angle B$
12. In the given figure, $ABCD$ is a square. A line segment DX cuts the side BC at X and the diagonal AC at O such that $\angle COD = 105^\circ$ and $\angle OXC = x$. The value of x is



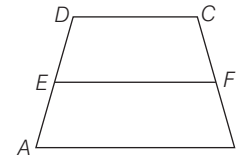
- (a) 40° (b) 60° (c) 80° (d) 85°

13. In the given figure, $ABCD$ is a parallelogram and E is the mid-point of AD . A line through D , drawn parallel to EB , meets AB produced at F and BC at L . Then,



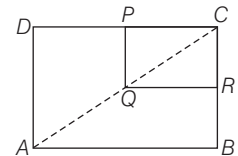
- (a) $AF = 2DC$ (b) $2AF = 3DC$
 (c) $2DF = 3DL$ (d) $AF + DF = 3DC + DL$

14. Let $ABCD$ be a trapezium in which $AB \parallel DC$ and let E be the mid-point of AD . Let F be a point on BC such that $EF \parallel AB$. Then, consider the following statements.



- (a) F is the mid-point of BC (b) $EF = \frac{1}{2}(AB + DC)$
 (c) Both (a) and (b) (d) None of these

15. In the figure given below, $ABCD$ and $PQRC$ are rectangles, where Q is the mid-point of AC , then

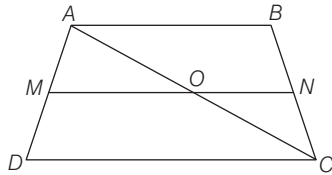


- (a) $DP = PC$ (b) $\frac{1}{3}PR = \frac{3}{2}DB$
 (c) $DP = PQ$ (d) $DP = \frac{1}{2}AC$

16. How many diagonals are there in a octagon?
 (a) 8 (b) 16 (c) 18 (d) 20
17. The angles of a pentagon are in the ratio 1:2:3:5:9, the largest angle is
 (a) 81° (b) 135° (c) 243° (d) 249°
18. The ratio of an interior angle to the exterior angle of a regular polygon is 5 : 1. The number of sides of polygon is
 (a) 10 (b) 11 (c) 12 (d) 14
19. The measure of each angle of a regular hexagon is
 (a) 90° (b) 120° (c) 105° (d) 135°
20. Each interior angle of a regular polygon is 150° . The number of sides of the polygon is
 (a) 4 (b) 8 (c) 12 (d) 16

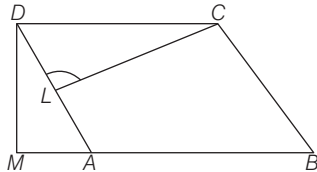
- 21.** Let $LMNP$ be a parallelogram and NR be perpendicular to LP . If the area of the parallelogram is six times the area of $\triangle RNP$ and $RP = 6$ cm what is LR equal to?
 (a) 15 cm (b) 12 cm (c) 9 cm (d) 8 cm

- 22.** $ABCD$ is a trapezium in which $AB \parallel CD$. M and N are the mid-points of AD and BC , respectively. If $AB = 12$ cm and $MN = 14$ cm find CD .



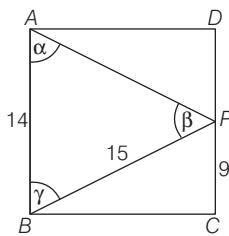
- (a) 2 cm (b) 5 cm (c) 12 cm (d) 16 cm

- 23.** $ABCD$ is a parallelogram. CL is perpendicular to AD and DM is perpendicular to BA produced. If $CD = 16$ units, $DM = 12$ units and $CL = 15$ units then $AD = ?$



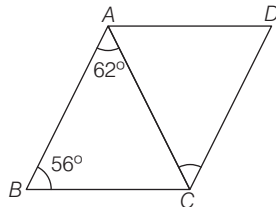
- (a) 12.8 units (b) 13.6 units (c) 11.1 units (d) 12.4 units

- 24.** $ABCD$ is a rectangle. $PC = 9$ cm, $BP = 15$ cm, $AB = 14$ cm. Then, the angles of $\triangle APB$ are such that



- (a) $\alpha > \beta > \gamma$ (b) $\alpha > \gamma > \beta$ (c) $\beta > \gamma > \alpha$ (d) $\alpha < \beta < \gamma$

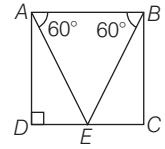
- 25.** $ABCD$ is a rhombus with $\angle ABC = 56^\circ$ and $\angle BAC = 62^\circ$, then $\angle ACD$ is equal to



- (a) 90° (b) 60° (c) 56° (d) 62°

- 26.** One angle of a pentagon is 140° . If the remaining angles are in the ratio $1 : 2 : 3 : 4$, then the size of the greatest angle is
 (a) 150° (b) 180° (c) 160° (d) 170°

- 27.** In the given figure $ABCD$ is a quadrilateral with AB parallel to DC and AD parallel to BC , $\angle ADC$ is a right angle. If the perimeter of the $\triangle ABE$ is 6 units, what is the area of the quadrilateral?

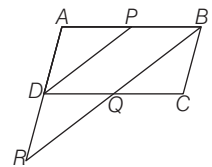


- (a) $2\sqrt{3}$ sq units (b) 4 sq units
 (c) 3 sq units (d) $4\sqrt{3}$ sq units

- 28.** A square and a rhombus have the same base and the rhombus is inclined at 30° . What is the ratio of area of the square to the area of the rhombus?

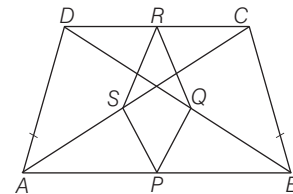
- (a) $\sqrt{2} : 1$ (b) $2 : 1$ (c) $1 : 1$ (d) $2 : \sqrt{3}$

- 29.** P is the mid-point of side AB of parallelogram $ABCD$. A line through B parallel to PD meets DC at Q and AD produced at R . Then BR is equal to



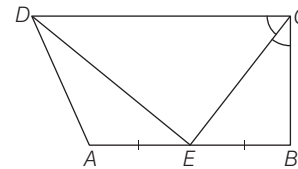
- (a) BQ (b) $\frac{1}{2}$ (c) $2BQ$ (d) None of these

- 30.** $ABCD$ is a trapezium in which $AB \parallel DC$ and $AD = BC$. If P, Q, R, S be respectively the mid-point of BA, BD, CD and CA , then $PQRS$ is a



- (a) rhombus (b) rectangle
 (c) parallelogram (d) square

- 31.** $ABCD$ is a parallelogram, E is the mid-point of AB and CE bisects $\angle BCD$. Then $\angle DEC$ is



- (a) 60° (b) 90° (c) 100° (d) 120°

- 32.** $ABCD$ is a square, P, Q, R and S are points on the sides AB, BC, CD and DA , respectively such that $AP = BQ = CR = DS$. What is the value of $\angle SPQ$?

- (a) 30° (b) 45° (c) 60° (d) 90°

- 33.** The middle points of the parallel sides AB and CD of a parallelogram $ABCD$ are P and Q , respectively. If AQ and CP divide the diagonal BD into three parts BX, XY and YD , then which one of the following is correct?

- (a) $BX \neq XY \neq YD$ (b) $BX = YD \neq XY$
 (c) $BX = XY = YD$ (d) $XY = 2BX$

- 34.** A parallelogram and a rectangle stand on the same base and on the same side of the base with the same height. If l_1, l_2 be the perimeters of the parallelogram and the rectangle respectively, then which one of the following is correct?

(a) $l_1 < l_2$ (b) $l_1 = l_2$
(c) $l_1 > l_2$ but $l_1 \neq 2l_2$ (d) $l_1 = 2l_2$

- 35.** Two similar parallelograms have corresponding sides in the ratio $1:k$. What is the ratio of their areas?

(a) $1:3k^2$ (b) $1:4k^2$ (c) $1:k^2$ (d) $1:2k^2$

- 36.** Let $WXYZ$ be a square. Let P, Q and R be the mid-points of WX, XY and ZW , respectively and K, L be the mid-points of PQ and PR , respectively. What is the value of

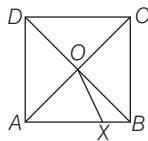
$\frac{\text{area of } \triangle PKL}{\text{area of square } WXYZ}$?

(a) $\frac{1}{32}$ (b) $\frac{1}{16}$
(c) $\frac{1}{8}$ (d) $\frac{1}{64}$

- 37.** ABC is a triangle in which $AB = AC$. Let BC be produced to D . From a point E on the line AC let EF be a straight line such that EF is parallel to AB . Consider the quadrilateral $ECDF$ thus formed. If $\angle ABC = 65^\circ$ and $\angle EFD = 80^\circ$, then what is the value of $\angle FDC$?

(a) 43° (b) 41°
(c) 37° (d) 35°

- 38.** In the given figure, $ABCD$ is a square in which $AO = AX$. What is $\angle XOB$?

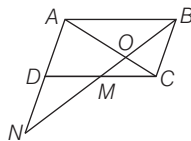


(a) 22.5° (b) 25°
(c) 30° (d) 45°

- 39.** Two sides of a parallelogram are 10 cm and 15 cm. If the altitude corresponding to the side of length 15 cm is 5 cm, then what is the altitude to the side of length 10 cm?

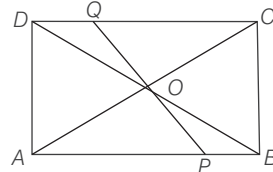
(a) 5 cm (b) 7.5 cm
(c) 10 cm (d) 15 cm

- 40.** In the given figure, M is the mid-point of the side CD of the parallelogram $ABCD$. What is $ON : OB$?



(a) $3:2$ (b) $2:1$ (c) $3:1$ (d) $5:2$

- 41.** $ABCD$ is a parallelogram whose diagonals AC and BD intersect at O . A line segment PQ through O meets AB at P and DC at Q .



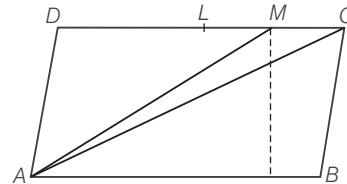
(a) $\text{ar}(\text{quadrilateral } APQD) = \frac{1}{2} \text{ar}(\text{gm } ABCD)$

(b) $\text{ar}(\text{quadrilateral } APQD) = \frac{1}{4} \text{ar}(\text{gm } ABCD)$

(c) $\text{ar}(\text{quadrilateral } APQD) = \frac{1}{3} \text{ar}(\text{gm } ABCD)$

(d) None of the above

- 42.** In the figure, $ABCD$ is a $\parallel\text{gm}$, L is the mid-point of DC and M is the mid-point of LC . If $\text{ar}(\triangle AMC) = 4 \text{ cm}^2$, then $\text{ar}(\parallel\text{gm } ABCD)$ is



(a) 32 cm^2 (b) 16 cm^2
(c) 64 cm^2 (d) Data insufficient

- 43.** Consider the following statements.

I. All squares are not parallelograms.

II. All parallelograms are trapezium.

III. All squares are rhombuses and also rectangles.

IV. All rhombuses are parallelograms.

Which of the above statements are correct?

(a) I and II (b) III and IV (c) I and III (d) II and IV

- 44.** Consider the following statements.

I. If the side of a rhombus is 10 cm. Its diagonals should have values 16 cm and 12 cm.

II. The diagonals of a rhombus cut at right angles.

III. The diagonals do not bisect each other.

Which of the above statements are true?

(a) I and II (b) II and III (c) I and III (d) I, II and III

- 45.** Consider the following statements in respect of a quadrilateral

I. The line segments joining the mid-points of the two pairs of opposite sides bisect each other at the point of intersection.

II. The area of the quadrilateral formed by joining the mid-points of the four adjacent sides is half of the total area of the quadrilateral.

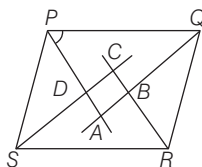
Which of the statement(s) given above is/are correct?

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

PREVIOUS YEARS' QUESTIONS

Directions (Q.Nos. 46-47) Let $ABCD$ be a quadrilateral, the diagonals AC and BD meet at O and the perpendicular drawn from A to CD , meet CD at E . Further, $AO : OC = BO : OD$, $AB = 30$ cm, $CD = 40$ cm and the area of the quadrilateral $ABCD$ is 1050 cm^2 .

- 46.** What is the value of BE ? ☑ 2012 I
 (a) 30 cm (b) $30\sqrt{2}$ cm
 (c) $30\sqrt{3}$ cm (d) None of these
- 47.** What is the area of the $\triangle ADC$? ☑ 2012 I
 (a) 300 cm^2 (b) 450 cm^2
 (c) 600 cm^2 (d) None of these
- 48.** $ABCD$ is a rhombus with diagonals AC and BD . Then, which one among the following is correct? ☑ 2012 I
 (a) AC and BD bisect each other but not necessarily perpendicular to each other
 (b) AC and BD are perpendicular to each other but not necessarily bisect each other
 (c) AC and BD bisect each other and perpendicular to each other
 (d) AC and BD neither bisect each other nor perpendicular to each other
- 49.** The sides of a parallelogram are 12 cm and 8 cm long and one of the diagonals is 10 cm long. If d is the length of other diagonal, then which one of the following is correct? ☑ 2012 I
 (a) $d < 8$ cm (b) $8 \text{ cm} < d < 10$ cm
 (c) $10 \text{ cm} < d < 12$ cm (d) $d > 12$ cm
- 50.** Let X be any point within a square $ABCD$. On AX a square $AXYZ$ is described such that D is within it. Which one of the following is correct? ☑ 2012 II
 (a) $AX = DZ$ (b) $\angle ADZ = \angle BAX$
 (c) $AD = DZ$ (d) $BX = DZ$
- 51.** $ABCD$ is a parallelogram. If the bisectors of the $\angle A$ and $\angle C$ meet the diagonal BD at points P and Q respectively, then which one of the following is correct? ☑ 2012 II
 (a) $PCQA$ is a straight line (b) $\triangle APQ$ is similar to $\triangle PCQ$
 (c) $AP = CP$ (d) $AP = AQ$
- 52.** The locus of a point in rhombus $ABCD$ which is equidistant from A and C is ☑ 2012 II
 (a) a fixed point on diagonal BD
 (b) diagonal BD (c) diagonal AC
 (d) None of the above
- 53.** In the figure given below, $PQRS$ is a parallelogram. If AP , AQ , CR and CS are bisectors of $\angle P$, $\angle Q$, $\angle R$ and $\angle S$ respectively, then $ABCD$ is a ☑ 2013 I
 (a) square (b) rhombus
 (c) rectangle (d) None of these



- 54.** $ABCD$ is a trapezium with parallel sides $AB = 2$ cm and $DC = 3$ cm. E and F are the mid-points of the non-parallel sides. The ratio of area of $ABFE$ to area of $EFCD$ is ☑ 2013 I
 (a) $9 : 10$ (b) $8 : 9$ (c) $9 : 11$ (d) $11 : 9$
- 55.** If the diagonals of a rhombus are 4.8 cm and 1.4 cm, then what is the perimeter of the rhombus? ☑ 2013 II
 (a) 5 cm (b) 10 cm (c) 12 cm (d) 20 cm
- 56.** $ABCD$ is a quadrilateral such that $BC = BA$ and $CD > AD$. Which one of the following is correct? ☑ 2013 II
 (a) $\angle BAD = \angle BCD$ (b) $\angle BAD < \angle BCD$
 (c) $\angle BAD > \angle BCD$ (d) $2\angle BAD = \angle BCD$
- 57.** Let $ABCD$ be a parallelogram. Let P , Q , R and S be the mid-points of sides AB , BC , CD and DA , respectively. Consider the following statements.
 I. Area of triangle $APS <$ Area of triangle DSR , if $BD < AC$.
 II. Area of triangle $ABC = 4$ (Area of triangle BPQ).
 Select the correct answer using the codes given below ☑ 2014 I
 (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II
- 58.** In a trapezium, the two non-parallel sides are equal in length, each being of 5 cm. The parallel sides are at a distance of 3 cm apart. If the smaller side of the parallel sides is of length 2 cm, then the sum of the diagonals of the trapezium is ☑ 2014 I
 (a) $10\sqrt{5}$ cm (b) $6\sqrt{5}$ cm (c) $5\sqrt{5}$ cm (d) $3\sqrt{5}$ cm
- 59.** Two light rods $AB = a + b$, $CD = a - b$ symmetrically lying on a horizontal plane. They are kept intact by two strings AC and BD . The perpendicular distance between rods is a . The length of AC is given by ☑ 2014 I
 (a) a (b) b (c) $\sqrt{a^2 - b^2}$ (d) $\sqrt{a^2 + b^2}$
- 60.** Let $ABCD$ be a parallelogram. Let X and Y be the mid-points of the sides BC and AD , respectively. Let M and N be the mid-points of the sides AB and CD , respectively. Consider the following statements.
 I. The straight line MX cannot be parallel to YN .
 II. The straight lines AC , BD , XY and MN meet at a point.
 Which of the statement(s) given above is/are correct? ☑ 2014 II
 (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II
- 61.** If each interior angle of a regular polygon is 135° , then the number of diagonals of the polygon is equal to ☑ 2015 I
 (a) 54 (b) 48 (c) 20 (d) 18

- 62.** If a star figure is formed by elongating the sides of a regular pentagon, then the measure of each angle at the angular points of the star figure is $\checkmark$ 2015 I
(a) 36° (b) 35° (c) 32° (d) 30°
- 63.** If X is any point within a square $ABCD$ and on AX a square $AXYZ$ is described, which of the following is/are correct?
I. $BX = DZ$ or $BZ = DX$
II. $\angle ABX = \angle ADZ$ or $\angle ADX = \angle ABZ$
Select the correct answer using the code given below. $\checkmark$ 2015 II
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II
- 64.** In the adjoining figure, $ABCD$ is a parallelogram. P is a point on BC such that $PB:PC = 1:2$. DP and AB when both produced meet at Q . If area of $\triangle BPQ$ is 20 sq units, the area of $\triangle DCP$ is $\checkmark$ 2015 I
(a) 20 sq units (b) 30 sq units
(c) 40 sq units (d) None of these
- 65.** $ABCD$ is a parallelogram with AB and AD as adjacent sides. If $\angle A = 60^\circ$ and $AB = 2AD$, then the diagonal BD will be equal to $\checkmark$ 2015 II
(a) $\sqrt{2}AD$ (b) $\sqrt{3}AD$ (c) $2AD$ (d) $3AD$
- 66.** If each interior angle of a regular polygon is 140° then number of vertices of the polygon is equal to $\checkmark$ 2016 I
(a) 10 (b) 9 (c) 8 (d) 7
- 67.** Consider the following statements.
I. There exists a regular polygon whose exterior angle is 70° .
II. Let $n \geq 5$, then the exterior angle of any regular polygon of n sides is acute.
Which of the above statements is/are correct? $\checkmark$ 2016 I
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II
- 68.** Consider the following statements.
I. If $n \geq 3$ and $m \geq 3$ are distinct positive integers, then the sum of the exterior angles of a regular polygon of m sides is different from the sum of the exterior angles of a regular polygon of n sides.
II. Let m, n be integers such that $m > n \geq 3$. Then, the sum of the interior angles of a regular polygon of m sides is greater than the sum of the interior angles of a regular polygon of n sides and their sum is $(m+n)\pi/2$.
Which of the above statements is/are correct? $\checkmark$ 2016 I
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

ANSWERS

1	c	2	b	3	c	4	b	5	d	6	a	7	d	8	a	9	c	10	b
11	a	12	b	13	a	14	b	15	a	16	d	17	c	18	c	19	b	20	c
21	b	22	d	23	a	24	a	25	d	26	c	27	a	28	b	29	c	30	a
31	b	32	d	33	c	34	c	35	c	36	b	37	d	38	a	39	b	40	b
41	a	42	a	43	b	44	a	45	c	46	b	47	c	48	c	49	d	50	d
51	b	52	a	53	c	54	c	55	b	56	c	57	b	58	b	59	d	60	b
61	c	62	a	63	c	64	d	65	b	66	b	67	b	68	d				

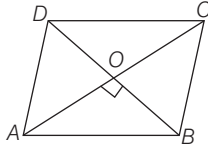
HINTS AND SOLUTIONS

- 1.** (c) As $\angle A + b^\circ = 180^\circ$ [linear pair]
 $\Rightarrow \angle A = 180^\circ - b$
 Also, $\angle C + a^\circ = 180^\circ$ [linear pair]
 $\Rightarrow \angle C = 180^\circ - a^\circ$
 But $\angle A + \angle B + \angle C + \angle D = 360^\circ$
 [by angle sum property of a quadrilateral]
 $\Rightarrow (180^\circ - b^\circ) + x^\circ + (180^\circ - a^\circ) + y^\circ = 360^\circ$
 $\Rightarrow x^\circ + y^\circ = a^\circ + b^\circ$
- 2.** (b) $\because \angle 1 = \frac{1}{2} \angle C$ and $\angle 2 = \frac{1}{2} \angle D$
 In $\triangle DEC$, $\angle 1 + \angle 2 + \angle CED = 180^\circ$
 [angle sum property of a triangle]
 $\therefore \angle CED = 180^\circ - (\angle 1 + \angle 2)$
 In quadrilateral $ABCD$,
 $\angle A + \angle B + \angle C + \angle D = 360^\circ$
 $\angle A + \angle B + 2(\angle 1 + \angle 2) = 360^\circ$
 $\angle A + \angle B = 360^\circ - 2(\angle 1 + \angle 2)$
 $\angle A + \angle B = 2[180^\circ - (\angle 1 + \angle 2)]$
 $\angle A + \angle B = 2 \angle CED$
- 3.** (c) In $\triangle ABC$, $\triangle ACD$, $\triangle BCD$ and $\triangle ABD$
 $AB + BC > AC$, $CD + DA > AC$
 $BC + CD > BD$, $DA + AB > BD$
 [Sum of any two sides is greater than the third side.]
 adding above inequalities,
 $2(AB + BC + CD + DA) > 2(AC + BD)$
 $(AB + BC + CD + DA) > (AC + BD)$
- 4.** (b) Let h be the height of the parallelogram. Then, clearly $h < q$
 $\therefore R = p \times h < p \times q = S$
 $R < S$

5. (d) Two parallelograms on the same base and between the same parallels are equal in area. So, the ratio of their areas is 1 : 1.

6. (a) As $ABCD$ is a rhombus.

So, $AO = OC = \frac{1}{2} AC$



$$BO = OD = \frac{1}{2} BD, \angle AOB = 90^\circ$$

$$\therefore AB^2 = OA^2 + OB^2$$

$$AB^2 = \frac{AC^2}{4} + \frac{BD^2}{4}$$

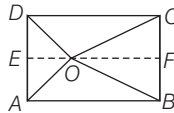
$$\Rightarrow 4AB^2 = AC^2 + BD^2$$

7. (d) Draw $EF \parallel AB$.

In right angled $\triangle OEA$ and $\triangle OFC$,

$$OA^2 = OE^2 + AE^2 \quad \dots (i)$$

and $OC^2 = OF^2 + CF^2 \quad \dots (ii)$



On adding Eqs. (i) and (ii), we get

$$\therefore OA^2 + OC^2 = OE^2 + AE^2 + OF^2 + CF^2 \quad \dots (iii)$$

In right angled $\triangle DEO$ and $\triangle BOF$,

$$OD^2 = OE^2 + DE^2 \quad \dots (iv)$$

$$OB^2 = OF^2 + BF^2 \quad \dots (v)$$

On adding Eqs. (iv) and (v), we get

$$\Rightarrow \text{Area of } \parallel \text{gm } ABCD$$

$$\Rightarrow OD^2 + OB^2 = OE^2 + OF^2 + DE^2 + BF^2 \quad \dots (vi)$$

As $FB = EA$ and $DE = CF$

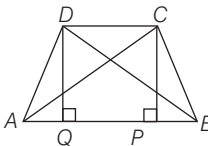
$$\therefore OD^2 + OB^2 = OE^2 + OF^2 + AE^2 + CF^2 \quad \dots (vii)$$

Here, from Eqs. (iii) and (vii), we get

$$OA^2 + OC^2 = OD^2 + OB^2$$

8. (a) Draw $DQ \perp AB$ and $CP \perp AB$.

In $\triangle ABD$, $\angle A$ is acute.



$$\text{So, } BD^2 = AD^2 + AB^2 - 2AB \cdot AQ \quad \dots (i)$$

In $\triangle ABC$, $\angle B$ is acute.

$$\text{So, } AC^2 = BC^2 + AB^2 - 2AB \cdot BP \quad \dots (ii)$$

Adding Eqs. (i) and (ii), we get

$$\therefore AC^2 + BD^2 = (BC^2 + AD^2)$$

$$+ 2AB(AB - BP - AQ)$$

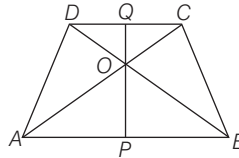
$$= (BC^2 + AD^2) + 2AB \cdot PQ$$

$$[\because PQ = AB - AQ - BP]$$

$$= BC^2 + AD^2 + 2AB \cdot CD$$

$$[\because PQ = DC]$$

9. (c) Let PQ be the perpendicular distance between the two parallel sides AB and CD .
[$\because PO = 2OQ$]



$$\text{So, } \frac{\text{Area of } \triangle AOB}{\text{Area of } \triangle COD} = \frac{\frac{1}{2} AB \times OP}{\frac{1}{2} CD \times OQ}$$

$$= \frac{2CD \times 2OQ}{CD \times OQ} = \frac{4}{1} \text{ or } 4 : 1$$

10. (b) Area of parallelogram $ABCD$
 $= 2 (\text{Area of } \triangle ADE)$

$$+ (\text{Area of rectangle } AECH)$$

$$= 2 \times \frac{1}{2} DE \times AE + AE \times EC$$

$$= DE \times AE + AE \times EC$$

$$= AE(DE + EC) = AE \times [DC]$$

$$\therefore \frac{DE}{EC} = \frac{1}{2} \Rightarrow \frac{DE}{EC} + 1 = \frac{1}{2} + 1$$

$$\Rightarrow \frac{DE + EC}{EC} = \frac{3}{2} \Rightarrow \frac{DC}{EC} = \frac{3}{2}$$

$$\text{Area of parallelogram } ABCD = \frac{FE}{3} \times \frac{3EC}{2} = \frac{FE \times EC}{2}$$

$$[\because FE = 3AE]$$

$$= \frac{\text{Area of rectangle } EFGC}{2}$$

$\therefore$ Required ratio = 1 : 2

11. (a) Join AC and BD .

As $AB > BC$

$$\Rightarrow \angle ACB > \angle BAC \quad \dots (i)$$

Also, $AD > DC$

$$\Rightarrow \angle ACD > \angle CAD \quad \dots (ii)$$

On adding Eqs. (i) and (ii), we get

$$\therefore \angle ACB + \angle ACD > \angle BAC + \angle CAD$$

$$\Rightarrow \angle C > \angle A, \text{ similarly } \angle D > \angle B$$

12. (b) $\angle COD = 105^\circ$

$$\therefore \angle BOA = 105^\circ$$

$$[\text{vertically opposite angles}]$$

$$\therefore \angle AOD = \angle COX = \lambda$$

$$\Rightarrow 105^\circ + 105^\circ + x + x = 360^\circ$$

$$\Rightarrow 2x = 150^\circ$$

$$\Rightarrow x = 75^\circ$$

$$\text{Also, } \angle OCX = \frac{1}{2} \times \angle C = \frac{1}{2} \times 90^\circ = 45^\circ$$

In $\triangle OCX$,

$$\angle OCX + \angle COX + \angle OXC = 180^\circ$$

[by angle sum of property of a triangle]

$$\Rightarrow 45^\circ + 75^\circ + \angle OXC = 180^\circ$$

$$\Rightarrow \angle OXC = 180^\circ - 120^\circ = 60^\circ$$

$$\Rightarrow x = 60^\circ$$

13. (a) $EB \parallel DL$ and $ED \parallel BL$

$\Rightarrow EBLD$ is a parallelogram.

$$\therefore BL = ED = \frac{1}{2} AD = \frac{1}{2} BC = LC$$

Also, in $\triangle DCL$ and $\triangle FBL$,

$$LC = BL, \angle DLC = \angle FLB$$

$\therefore$ [vertically opposite angles]

and $\angle CDL = \angle BFL$ [alternate angles]

$\therefore \triangle DCL \cong \triangle FBL$ [By AAS rule]

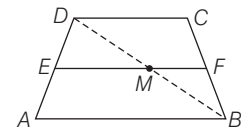
$$\therefore DC = BF \text{ and } DL = FL$$

$$\therefore BF = DC = AB$$

$$\Rightarrow 2AB = 2DC$$

$$\Rightarrow AF = 2DC \quad [\because BF = AB]$$

14. (b) Join BD , cutting EF at M .



So, M is mid-point of BD .

[$\because E$ is mid-point of AD and $EM \parallel AB$]

$$\therefore EM = \frac{1}{2} AB \quad \dots (i)$$

Similarly, F is mid-point of BC in $\triangle BCD$.

$$\therefore FM = \frac{1}{2} DC \quad \dots (ii)$$

$$\therefore EF = EM + MF = \frac{1}{2} (AB + DC)$$

[from Eqs. (i) and (ii)]

15. (a) $\angle CRQ = \angle CBA = 90^\circ \Rightarrow QR \parallel AB$

In $\triangle ABC$, Q is the mid-point of AC and $QR \parallel AB$.

So, R is mid-point of BC .

Similarly, P is the mid-point of DC .

$$\therefore DP = PC$$

16. (d) Number of diagonals of a polygon of

$$8 \text{ sides} = \frac{n(n-1)}{2} - n$$

$$= \frac{8(8-1)}{2} - 8 = 20$$

17. (c) Sum of all angles of a pentagon

$$= (5-2) \times 180^\circ = 3 \times 180^\circ = 540^\circ$$

Let the angle be $x, 2x, 3x, 5x$ and $9x$.

$$\therefore x + 2x + 3x + 5x + 9x = 540^\circ$$

$$20x = 540^\circ \Rightarrow x = 27^\circ$$

$$\therefore \text{Largest angle} = 9x = 9 \times 27^\circ = 243^\circ$$

18. (c) By condition,

$$\frac{\text{Interior angle of a regular polygon}}{\text{Exterior angle of a regular polygon}} = \frac{5}{1}$$

$$\Rightarrow \frac{\frac{(n-2)}{n} \times 180^\circ}{\frac{360^\circ}{n}} = \frac{5}{1} \Rightarrow \frac{(n-2)}{2} = \frac{5}{1}$$

$$\Rightarrow n - 2 = 10 \Rightarrow n = 12$$

19. (b) Here, $n = 6$
 Sum of interior angles of a hexagon
 $= (n-2)180^\circ = (6-2) \times 180^\circ$
 $= 4 \times 180^\circ = 720^\circ$
 Number of angles = 6
 $\therefore$ Each angle of a regular hexagon
 $= \frac{720^\circ}{6} = 120^\circ$

20. (c) Each interior angle of a regular polygon

$$= \frac{(n-2) \times 180^\circ}{n}$$

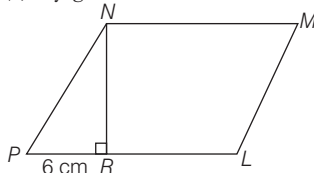
$$\therefore \frac{(n-2) \times 180^\circ}{n} = 150^\circ \quad (\text{given})$$

$$\Rightarrow \frac{(n-2)180}{n} = 150$$

$$\Rightarrow 30n = 360 \Rightarrow n = \frac{360}{30}$$

$$\Rightarrow n = 12$$

21. (b) By given condition,



Area of parallelogram $LMNP$
 $= 6 \times \text{Area of } \triangle NRP$

$$\therefore NR \times PL = 6 \times \frac{1}{2} \times NR \times PR$$

$$\Rightarrow PL = 3PR$$

$$\Rightarrow PR + RL = 3PR \quad [\because PL = PR + RL]$$

$$\Rightarrow RL = 2PR = 2 \times 6 = 12 \text{ cm}$$

22. (d) $ABCD$ is a trapezium in which $AB \parallel DC$ and M, N are the mid-points of AD and BC .

Hence, $MN \parallel AB$ and $MN \parallel DC$.

In $\triangle ACB$, ON passes through the mid-point N of BC and $ON \parallel AB$

$$\therefore ON = \frac{1}{2}AB = \frac{1}{2} \times 12 \text{ cm} = 6 \text{ cm}$$

$$\text{And, } MO = MN - ON = (14 - 6) \text{ cm} = 8 \text{ cm}$$

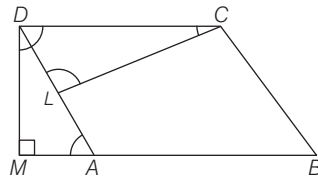
Again, MO passes through the mid-point M of AD and $MO \parallel DC$

$$\therefore MO = \frac{1}{2}DC = \frac{1}{2}CD$$

Hence, $CD = 2(MO) = 2(8) = 16 \text{ cm}$

23. (a) $\triangle ADM \sim \triangle DCL$

Since, $\angle DMA = \angle CLD = 90^\circ$ and $\angle CDL = \angle DAM$ as they are alternate angles



$$(\angle D / \angle C) = (\angle M / \angle L)$$

$$\Rightarrow \frac{AD}{CD} = \frac{DM}{CL} = \frac{12 \times 16}{15}$$

$$= 12.8 \text{ units}$$

24. (a) Evidently $DP = 14 - 9 = 5 \text{ cm}$

From $\triangle BPC$,

$$BC^2 = 15^2 - 9^2 = 12^2$$

$$\Rightarrow BC = 12 \text{ cm}$$

$$\text{From } \triangle APD, AP^2 = AD^2 + DP^2$$

$$= 12^2 + 5^2 = 169$$

$$\Rightarrow AP = 13 \text{ cm}$$

In $\triangle ABP$, $AP < AB < BP$.

Therefore, $\gamma < \beta < \alpha$ (i.e.) $\alpha > \beta > \gamma$

25. (d) Since, $AB = BC$

$$\therefore \angle BAC = \angle BCA = 62^\circ$$

Also as $AB \parallel CD$ and AC transversal

So, $\angle BAC = 62^\circ = \angle ACD$

(alternate interior angles)

$$\therefore \angle ACD = 62^\circ$$

26. (c) One angle of the pentagon is 140° .
 Since, the remaining angles are in the ratio $1 : 2 : 3 : 4$.

$\therefore$ Let the remaining angles be x° , $(2x)^\circ$, $(3x)^\circ$ and $(4x)^\circ$.

But the sum of interior angles of a pentagon

$$(2 \times 5 - 4) \times 90 = 6 \times 90^\circ = 540^\circ$$

$$\therefore 140 + x + 2x + 3x + 4x = 540$$

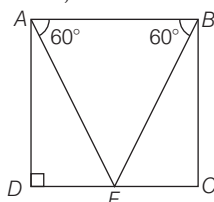
$$\Rightarrow 10x = 400 \Rightarrow x = 40$$

$\therefore$ The angles of the pentagon are 140° , 40° , 80° , 120° and 160° .

Hence, the size of the greatest angle
 $= 160^\circ$

27. (a) Given that, $AB \parallel DC$ and $AD \parallel BC$

In $\triangle ABE$,



$$\angle EAB = \angle ABE = 60^\circ$$

$$\Rightarrow \angle AEB = 60^\circ$$

So, $\triangle ABE$ is an equilateral triangle.

Now, perimeter of $\triangle ABE = 6$

$$\Rightarrow AB + BE + EA = 6$$

$$\Rightarrow AB = 2 \text{ units}$$

Now, In $\triangle ADE$ and $\triangle BCE$,

$$\angle ADE = \angle BCE \quad [\text{each } 90^\circ]$$

$$\angle DAC = \angle CBE \quad [\text{each } 30^\circ]$$

and $AE = BE$

$$\therefore \triangle ADE \cong \triangle BCE \quad [\text{By AAS}]$$

$$\Rightarrow DE = CE \quad [\text{By CPCT}]$$

In $\triangle ADE$ by pythagoras theorem,

$$AE^2 = AD^2 + ED^2$$

$$\Rightarrow 4 = AD^2 + 1$$

$$(\because E \text{ is mid-point of } CD)$$

$$\Rightarrow AD = \sqrt{3} \text{ units}$$

Hence, area of quadrilateral

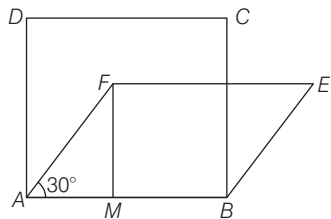
$$ABCD = AB \times AD$$

$$= 2 \times \sqrt{3} = 2\sqrt{3} \text{ sq units}$$

28. (b) $ABCD$ is square and $ABEF$ is a rhombus.

$$\frac{FM}{AF} = \sin 30^\circ = \frac{1}{2}$$

$$\therefore FM = \frac{AF}{2}, AF = AB$$



$$\text{Area of square} = a^2 \quad (AB = AD = a)$$

$$\text{Area of rhombus} = \frac{a \times a}{2} \quad \left(FM = \frac{a}{2} \right)$$

(Area of rhombus = base $\times$ height)

$$\therefore \frac{\text{Area of square}}{\text{Area of rhombus}} = \frac{2}{1}$$

29. (c) In $\triangle ARB$, P is the mid-point of AB and $PD \parallel BR$.

$$\Rightarrow D \text{ is the mid-point of } AR.$$

$\therefore ABCD$ is a parallelogram.

$$\Rightarrow DC \parallel AB \Rightarrow DQ \parallel AB$$

Thus, in $\triangle ARB$, D is the mid-point of AR and $DQ \parallel AB$.

$$\therefore Q \text{ is the mid-point of } RB \Rightarrow BR = 2BQ$$

30. (a) In $\triangle BDC$, Q and R are the mid-points of BD and CD respectively.

$$\therefore QR \parallel BC \text{ and } QR = \frac{1}{2}BC$$

$$\text{Similarly, } PS \parallel BC \text{ and } PS = \frac{1}{2}BC$$

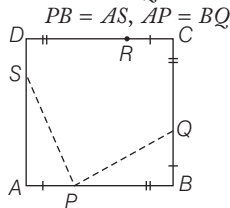
$$\therefore PS \parallel QR \text{ and } PS = QR$$

$$\left[\text{each equal to } \frac{1}{2}BC \right]$$

Similarly, $PQ \parallel SR$ and $PQ = SR$
 [each equal to $1/2 AD$]
 $\therefore PS = QR = SR = PQ$ [$\because AD = BC$]
 Hence, $PQRS$ is a rhombus.

31. (b) $AB \parallel DC$ and EC is a transversal
 $\Rightarrow \angle BEC = \angle ECD$
 $\Rightarrow \angle BEC = \angle ECB$ [$\because \angle ECD = \angle ECB$]
 $\Rightarrow EB = BC \Rightarrow AE = AD$
 Now, $AE = AD \Rightarrow \angle ADE = \angle AED$
 $\Rightarrow \angle ADE = \angle EDC$
 [$\because$ alternate Int. angles]
 $\therefore DE$ bisects $\angle ADC$
 Again, $\angle ADC + \angle BCD = 180^\circ$
 [Co. Int. angles]
 $\Rightarrow \frac{1}{2} \angle ADC + \frac{1}{2} \angle BCD = 90^\circ$
 $\Rightarrow \angle EDC + \angle DCE = 90^\circ$
 But, $\angle EDC + \angle DEC + \angle DCE = 180^\circ$
 [$\because$ sum of the $\angle$ s of a Δ is 180°]
 $\therefore \angle DEC = 180^\circ - 90^\circ$
 $\therefore \angle DEC = 90^\circ$

32. (d) In ΔAPS and ΔBPQ , [given]



and $\angle A = \angle B = 90^\circ$
 $\therefore \Delta APS \cong \Delta BPQ$ (by SAS rule)
 $\therefore SP = PQ, \angle SPA = \angle BQP$
 and $\angle ASP = \angle BPQ$
 $\angle SPQ = 180^\circ - \angle APS - \angle BPQ$
 $= 180^\circ - (\angle APS + \angle ASP)$
 $= 180^\circ - 90^\circ = 90^\circ$

33. (c) As $ABCD$ is a || gm

$\therefore AB \parallel DC$ and $AB = DC$

and $\frac{1}{2} AB = \frac{1}{2} DC$

$\Rightarrow AP = QC$

$\therefore APCQ$ is a || gm

$\Rightarrow AQ \parallel PC$

In $\Delta BAY, XP \parallel AY$

and P is the mid-point of AB

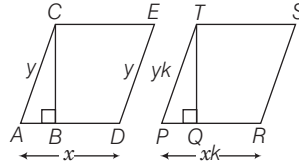
$\therefore BX = YX$

Similarly, in $\Delta DXC, DY = YX$

$\therefore BX = XY = DY$

34. (c) If a parallelogram and a rectangle stand on the same base and on the same side of the base with the same height, then perimeter of parallelogram is greater than perimeter of rectangle.
 $\therefore l_1 > l_2$

35. (c) Let the sides of two parallelograms are x, y and xk, yk , respectively.



Since, sides of two parallelogram are in the ratio $1 : k$.

$\therefore \Delta ABC \sim \Delta PQT$

$\therefore \frac{AC}{PT} = \frac{BC}{QT}$

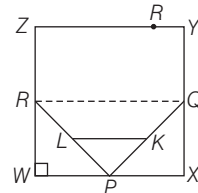
$\Rightarrow \frac{BC}{QT} = \frac{y}{yk} = \frac{1}{k}$

Let $BC = z$ and $QT = zk$

$\therefore$ Ratio of areas of two similar

parallelograms $= \frac{x \times z}{xk \times zk} = \frac{1}{k^2}$ or $1 : k^2$

36. (b) Area of $(\Delta PRQ) = \frac{1}{2}$ Area of $(WXYZ)$



$= \frac{1}{2} \left[\frac{1}{2} \text{Area of } (WXYZ) \right]$

$= \frac{1}{4} \text{Area of } (WXYZ)$... (i)

$\frac{\text{Area of } (\Delta PRQ)}{\text{Area of } (\Delta PLK)} = \frac{RP^2}{LP^2}$

(by properties of similar triangle)

$\Rightarrow \frac{\text{Area of } (\Delta PRQ)}{\text{Area of } (\Delta PLK)} = \frac{(2LP)^2}{LP^2}$

[$\because L$ and K are the mid-point of PR and PQ]

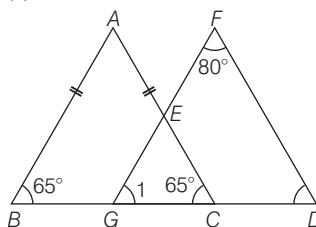
$\Rightarrow \text{Area of } (\Delta PRQ) = 4 \text{Area of } (\Delta PLK)$

$\Rightarrow \frac{1}{4} \text{Area of } (WXYZ)$

$= 4 \text{Area of } (\Delta PLK)$ [from Eq. (i)]

$\Rightarrow \frac{\text{Area of } (\Delta PLK)}{\text{Area of } (WXYZ)} = \frac{1}{16}$

37. (d) Here, $\angle B = \angle C = 65^\circ$ [$\because AB = AC$]



$\angle 1 = \angle B = 65^\circ$ (corresponding angles)

In ΔFGD ,

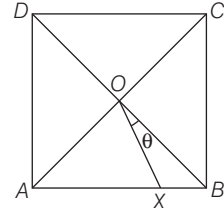
$\angle 1 + \angle F + \angle D = 180^\circ$

(by angle sum of property of a triangle)

$\Rightarrow 65^\circ + 80^\circ + \angle D = 180^\circ$

$\Rightarrow \angle D = 35^\circ$

38. (a) Let $\angle XO B = \theta$



In ΔOXB ,

$\angle XO B + \angle OBX + \angle OXB = 180^\circ$

(by angle sum of property of a triangle)

$\Rightarrow \theta + 45^\circ + \angle OXB = 180^\circ$

$\Rightarrow \angle OXB = 180^\circ - 45^\circ - \theta = 135^\circ - \theta$

Here, $\angle OXA + \angle OXB = 180^\circ$

(linear pair)

$\Rightarrow \angle OXA + 135^\circ - \theta = 180^\circ$

$\Rightarrow \angle OXA = 45^\circ + \theta$

In ΔOXA , $AO = AX$ (given)

$\therefore \angle OXA = \angle AOX = 45^\circ + \theta$

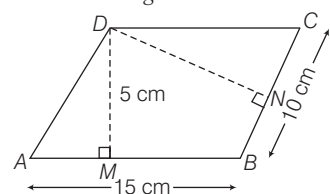
Since, $\angle AOX + \angle XO B = 90^\circ$

$\Rightarrow 45^\circ + \theta + \theta = 90^\circ$

$\Rightarrow 2\theta = 45^\circ \Rightarrow \theta = 22.5^\circ$

39. (b) Area of parallelogram

$= \text{Base} \times \text{height} = 15 \times 5 = 75 \text{ cm}^2$



Area of parallelogram $= \text{Base} \times \text{Height}$

$= 10 \times DN$

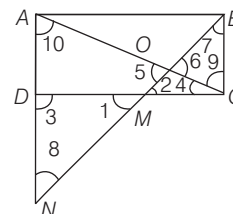
$\therefore 10 \times DN = 75$

$\Rightarrow DN = \frac{75}{10} = 7.5 \text{ cm}$

40. (b) In ΔDMN and ΔBMC ,

$DM = MC$

(given)



$\angle 1 = \angle 2$ (vertically opposite angle)

$\angle 3 = \angle 4 + \angle 9$

(alternate interior angle)

$\Delta DMN \cong \Delta BMC$ (as ASA rule)

$$DN = BC = AD \quad (\text{by CPCT})$$

$$\text{So, } AN = 2BC \Rightarrow \frac{AN}{BC} = \frac{2}{1} \dots(i)$$

In $\triangle OAN$ and $\triangle OBC$,

$$\angle 5 = \angle 6 \quad (\text{vertically opposite angle})$$

$$\angle 7 = \angle 8 \quad (\text{alternate interior angle})$$

$$\angle 9 = \angle 10 \quad (\text{altenate angle})$$

$\therefore \triangle OAN \sim \triangle OBC$ (by AAA similarity)

So, the sides will be in same ratio

$$\frac{AN}{BC} = \frac{ON}{OB} \Rightarrow \frac{2}{1} = \frac{ON}{OB} \text{ or } 2 : 1$$

[from Eq. (i)]

41. (a) As $ABCD$ is a parallelogram.

$$\therefore AB \parallel CD \text{ and } AB = CD$$

Draw $QR \perp AB$

In $\triangle DOQ$ and $\triangle POB$,

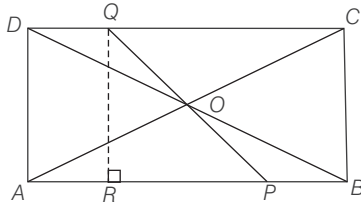
$$\angle ODQ = \angle OPB$$

$$\angle OQD = \angle OBP$$

and $OD = OB$

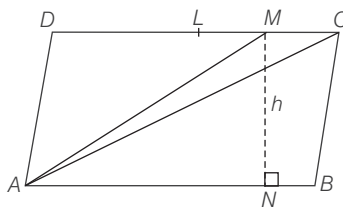
$$\therefore \triangle OQD \cong \triangle OPB \quad [\text{By AAS}]$$

$$\Rightarrow PB = DQ$$



$$\begin{aligned} \text{Now, ar (} \parallel \text{gm } ABCD) &= QR \times AB \\ &= QR \times (AP + PB) = QR \times (AP + DQ) \\ &= 2 \times \text{ar (quadrilateral } APDQ) \end{aligned}$$

42. (a) Draw $MN \perp AB$ and let $MN = h$



$$\begin{aligned} \text{Now, ar (trapezium } ABCM) &= \text{ar } (\triangle AMC) + \text{ar } (\triangle ABC) \\ &\Rightarrow \frac{1}{2}(MC + AB) \times h = 4 + \frac{1}{2} \text{ar} \\ &\quad (\text{parallelogram } ABCD) \\ &\Rightarrow \frac{1}{2} \left(\frac{AB}{4} + AB \right) \times h = 4 + \frac{1}{2} \times AB \times h \\ &\quad \left[\because MC = \frac{AB}{4} \right] \end{aligned}$$

$$\Rightarrow \frac{AB \times h}{8} = 4 \Rightarrow AB \times h = 32$$

$$\Rightarrow \text{ar (} \parallel \text{gm } ABCD) = 32 \text{ cm}^2$$

43. (b) I. All squares are parallelograms.

II. No parallelogram is a trapezium.

III. All squares are rhombuses and also rectangles.

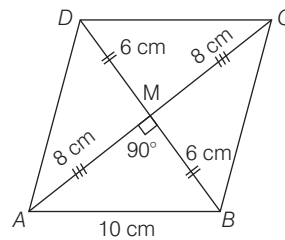
IV. All rhombuses are parallelograms.

So, statements (III) and (IV) are correct.

44. (a) In a rhombus $ABCD$, if AC and BD are two diagonals then

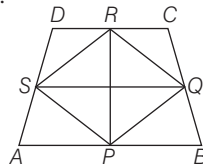
$$\begin{aligned} AB^2 + BC^2 + CD^2 + AD^2 &= AC^2 + BD^2 \end{aligned}$$

$$\begin{aligned} \Rightarrow (10)^2 + (10)^2 + (10)^2 + (10)^2 &= (16)^2 + (12)^2 \\ \Rightarrow 400 &= 400 \end{aligned}$$



Hence, both I and II are true but III is false.

45. (c) $PQRS$ can be shown parallelogram, so the diagonal PR and SQ bisect each other.



II. Area ($\triangle RSQ$)

$$= \frac{1}{2} \text{Area (} \triangle SQCD) \dots(i)$$

$$\text{and area } \triangle (PSQ) = \frac{1}{2} \text{Area (} \triangle PQAS) \dots(ii)$$

On adding Eqs. (i) and (ii), we get

$$\text{Area (} \triangle RSQ) + \text{Area (} \triangle PSQ) = \frac{1}{2}$$

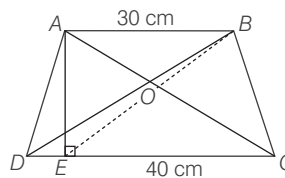
$$[\text{Area (} \triangle SQCD) + \text{Area (} \triangle PQAS)]$$

$$\Rightarrow \text{Area (} \triangle PQRS) = \frac{1}{2} \text{Area (} \triangle ABCD)$$

Hence, both statements are true.

Solutions (Q.Nos. 46-47)

Given, $AO : OC = BO : OD$
and $AB = 30 \text{ cm}$ and $CD = 40 \text{ cm}$



$$\therefore \triangle AOB \sim \triangle COD$$

$$\Rightarrow \frac{OC}{OA} = \frac{AB}{CD} = \frac{30}{40} = \frac{3}{4}$$

$$\therefore \angle OAB = \angle OCD$$

$$\text{and } \angle OBA = \angle ODC$$

It means $DC \parallel AB$.

So, it is a trapezium.

Area of trapezium

$$ABCD = \frac{1}{2} (AB + CD) \times AE$$

$$\Rightarrow 1050 = \frac{1}{2} (30 + 40) \times AE$$

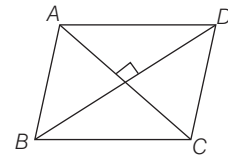
$$\Rightarrow AE = 30 \text{ cm}$$

$$\begin{aligned} 46. (b) \text{ In right } \triangle EAB, EB &= \sqrt{AE^2 + AB^2} \\ &= \sqrt{30^2 + 30^2} = 30\sqrt{2} \text{ cm} \end{aligned}$$

$$\begin{aligned} 47. (c) \text{ Area of } \triangle ADC &= \frac{1}{2} \times CD \times AE \\ &= \frac{1}{2} \times 40 \times 30 = 600 \text{ cm}^2 \end{aligned}$$

48. (c) $ABCD$ is a rhombus.

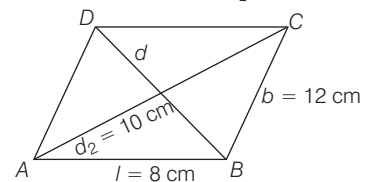
$$\therefore AB = BC = CD = DA$$



and diagonals bisect each other at right angles.

49. (d) In parallelogram,

$$d^2 + d_2^2 = 2(l^2 + b^2)$$



$$\therefore d^2 + (10)^2 = 2(64 + 144)$$

$$\Rightarrow d^2 = 2 \times 208 - 100$$

$$\Rightarrow d^2 = 416 - 100 = 316$$

$$\Rightarrow d = \sqrt{316} \Rightarrow d = 17.76 \text{ cm}$$

$$\therefore d > 12$$

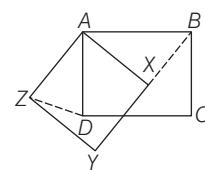
50. (d) In $\triangle ABX$ and $\triangle ADZ$,

$AB = AD$ (side of a square $ABCD$)

and $AX = AZ$ (side of a square $AXYZ$)

Let $\angle BAX = \theta$

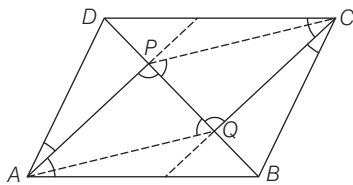
$$\therefore \angle XAD = 90^\circ - \theta$$



Also, $AXYZ$ is a square.

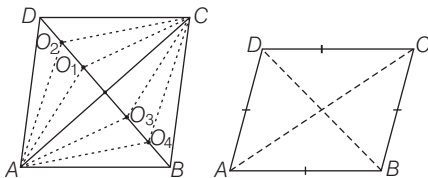
$\therefore \angle ZAX = 90^\circ$
 $\Rightarrow \angle ZAD + \angle XAD = 90^\circ$
 $\Rightarrow \angle ZAD = 90^\circ - (90^\circ - \theta) = \theta$
 i.e. $\angle BAX = \angle ZAD$
 $\therefore \triangle ABX \cong \triangle ADZ$
 $\therefore BX = DZ$ (by CPCT)

- 51. (b)** Since, line segment AP and CQ bisects the $\angle A$ and $\angle C$, respectively.
 Then, $AP \parallel CQ$
 Now, in $\triangle APQ$ and $\triangle CQP$,
 $\therefore AP \parallel CQ$



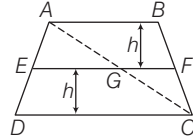
$\therefore \angle APQ = \angle PQC$ (alternate angle)
 $PQ = PQ$ (common)
 Also, $PC \parallel AQ$
 $\therefore \angle CPQ = \angle PQA$ (alternate angle)
 $\therefore \triangle APQ \cong \triangle CQP$ (by ASA)
 Hence, $\triangle APQ$ is similar to $\triangle PCQ$.

- 52. (a)** We know that, in a rhombus $ABCD$ diagonals bisect each other at point O which means that the distance of O from four vertices of rhombus i.e., A, B, C and D are equal. Even, if we take any fixed point on diagonal BD of rhombus $ABCD$ and join with vertices A and C , we get a point which is equidistant from A and C . (by property of congruent triangle). Hence, the locus of a point in rhombus $ABCD$ which is equidistant from A and C is a fixed point on diagonal BD .



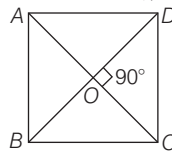
- 53. (c)** In parallelogram $PQRS$
 $\angle P + \angle S = 180^\circ$
 $\Rightarrow \frac{1}{2} \angle P + \frac{1}{2} \angle S = 90^\circ$
 $\Rightarrow \angle DPS + \angle DSP = 90^\circ$
 In $\triangle DPS$,
 $\angle SDP = 180^\circ - (\angle DPS + \angle DSP) = 90^\circ$
 $\therefore \angle ADC = \angle SDP = 90^\circ$
 (vertically opposite $\angle S$)
 Similarly,
 $\angle DAB = \angle ABC = \angle BCD = 90^\circ$
 Hence, $ABCD$ is a rectangle.

- 54. (c)** Join AC . In $\triangle ACD$, $EG \parallel DC$ and E and G are mid-points of AD and AC , respectively.
 In $\triangle ADC$, $EG \parallel DC$
 $\therefore EG = \frac{1}{2} DC = \frac{3}{2}$



Similarly, in $\triangle ABC$
 $GF = \frac{1}{2} AB = 1$
 $EF = EG + GF = 1 + 1 = 2$
 $\therefore$ Area of trapezium = $\frac{1}{2} \times (\text{sum of parallel sides} \times \text{height})$
 Now, required ratio = $\frac{\text{Area of } ABFE}{\text{Area of } EFCD}$
 $= \frac{\frac{1}{2} \left(2 + \frac{5}{2} \right) \times h}{\frac{1}{2} \left(3 + \frac{5}{2} \right) \times h} = \frac{9}{11}$ or $9 : 11$

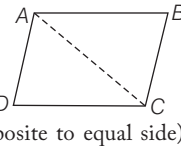
- 55. (b)** Here, $OD = \frac{BD}{2} = \frac{4.8}{2} = 2.4$
 $\Rightarrow OC = \frac{AC}{2} = \frac{1.4}{2} = 0.7$



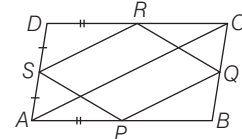
Since, in rhombus diagonal bisect at 90° .
 Then, in $\triangle ODC$, $OD^2 + OC^2 = CD^2$
 $\Rightarrow CD = \sqrt{OD^2 + OC^2}$
 $= \sqrt{(2.4)^2 + (0.7)^2}$
 $CD = \sqrt{6.25} \Rightarrow CD = 2.5 = \frac{5}{2}$

$\therefore$ Perimeter of rhombus
 $= 4a = 4 \times \frac{5}{2} = 10$ cm

- 56. (c)** Join AC .
 Now, in $\triangle ABC$
 $\therefore AB = BC$ (given)
 $\therefore \angle BAC = \angle BCA$
 ... (i)
 (angles opposite to equal side)
 In $\triangle ADC$,
 $\therefore CD > AD$
 $\therefore \angle DAC > \angle DCA$... (ii)
 (since in a triangle, angle opposite to greater side is bigger than the angle opposite to smaller side)
 On adding Eqs. (i) and (ii), we get
 $\angle BAD > \angle BCD$

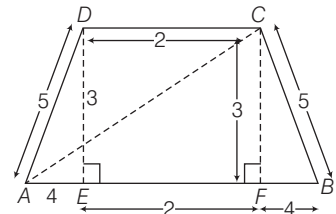


- 57. (b)** In $\triangle APS$ and $\triangle DSR$
 $AP = DR$ [$\because P$ and R are mid points]
 $AS = DS$ [$\because S$ is the mid point]
 and $SP = SR$ [$\because P$ or RS us a]
 $\therefore \triangle APS \cong \triangle DRS$ [By SSS]
 $\Rightarrow$ Area of $\triangle APS$ = Area of $\triangle DSR$



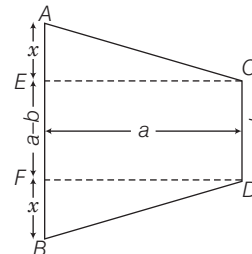
$\therefore AS = SD$ and $AP = DR$
 In $\triangle ABC$ and $\triangle PBQ$
 $\angle ABC = \angle PBQ$ [common]
 $\angle BAC = \angle BPQ$ [$\because PQ \parallel AC$]
 $\therefore \triangle ABC \sim \triangle PBQ$
 $\frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle PBQ)} = \frac{AB^2}{PB^2} \Rightarrow \left[\frac{2PB}{PB} \right]^2$
 $\therefore \text{ar}(\triangle ABC) = 4 \text{ar}(\triangle BPQ)$
 Hence, only statement II is correct.

- 58. (b)**



In $\triangle BCF$,
 By pythagoras theorem,
 $(5)^2 = (3)^2 + (BF)^2 \Rightarrow BF = 4$ cm
 $\therefore AB = 2 + 4 + 4 = 10$ cm
 Now in $\triangle ACF$, $AC^2 = CF^2 + FA^2$
 $\Rightarrow AC^2 = 3^2 + 6^2$, $AC = \sqrt{45}$ cm
 Similarly, $BD = \sqrt{45}$ cm
 $\therefore$ Sum of diagonal = $2 \times \sqrt{45}$
 $= 2 \times 3\sqrt{5} = 6\sqrt{5}$ cm

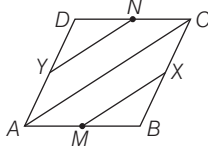
- 59. (d)** Since, they are symmetrically lying on horizontal plane.



$\therefore AC = BD \therefore AE = BF = x$ (say)
 Now, $AB = (a - b) + 2x$
 i.e. $a + b = a - b + 2x$
 $\Rightarrow 2b = 2x$
 $\therefore x = b$

Now, in $\triangle ACE$, $x^2 + a^2 = AC^2$
 $\Rightarrow AC^2 = b^2 + a^2$
 $\therefore AC = \sqrt{b^2 + a^2}$

60. (b) Given, $ABCD$ is a parallelogram. X and Y are mid-points of BC and AD , respectively. M and N are the mid-points of AB and CD , respectively.



Now, join AC .

In $\triangle ABC$, M and X are mid-points of AB and BC .

$$\therefore MX \parallel AC \text{ and } MX = \frac{1}{2} AC \quad \dots(i)$$

In $\triangle ADC$, Y and N are mid-points of AD and CD .

$$\therefore YN \parallel AC \text{ and } YN = \frac{1}{2} AC \quad \dots(ii)$$

From Eqs. (i) and (ii), we get

$$MX \parallel YN$$

So, statement I is not correct.

II Obviously, straight lines AC, BD, XY and MN meet at a point. So, statement II is correct.

61. (c) Given, each interior angle of a regular polygon = 135°

$\therefore$ Each exterior angle

$$= 180^\circ - \text{interior angle} \\ = 180^\circ - 135^\circ = 45^\circ$$

$$\therefore \text{Number of sides in the polygon } (n) \\ = \frac{360^\circ}{\text{Each exterior angle}} = \frac{360^\circ}{45^\circ} = 8$$

$$\therefore \text{Number of diagonals of a polygon} \\ = \frac{n(n-1)}{2} - n = \frac{8(8-1)}{2} - 8 \\ = \frac{8 \times 7}{2} - 8 \\ = 28 - 8 = 20$$

Hence, the number of diagonals of a polygon is 20.

62. (a) $\therefore$ Each interior angle of a regular polygon of n side

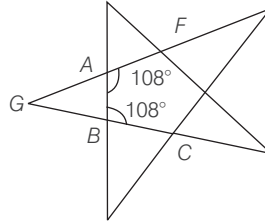
$$= \frac{(n-2) \times 180^\circ}{n}$$

Here, $n = 5$

$\therefore$ Each interior angle of the pentagon

$$= \frac{(5-2) \times 180^\circ}{5} = \frac{3 \times 180^\circ}{5} \\ = 3 \times 36^\circ = 108^\circ$$

Since, GAF is a straight line.



$$\therefore \angle GAF = 180^\circ$$

$$\Rightarrow \angle GAB + \angle BAF = 180^\circ$$

$$\Rightarrow \angle GAB + 108^\circ = 180^\circ$$

$$\Rightarrow \angle GAB = 180^\circ - 108^\circ = 72^\circ$$

Similarly,

$$\angle GBA = 180^\circ - 108^\circ = 72^\circ$$

In $\triangle ABG$,

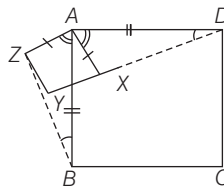
$$\angle GAB + \angle ABG + \angle BGA = 180^\circ$$

$$\Rightarrow 72^\circ + 72^\circ + \angle BGA = 180^\circ$$

$$\Rightarrow \angle BGA = 180^\circ - 144^\circ, \angle BGA = 36^\circ$$

63. (c) Here, $\angle DAB = \angle XAZ = 90^\circ$

$$\Rightarrow \angle DAB - \angle XAB = \angle XAZ - \angle XAB$$



$$\Rightarrow \angle DAX = \angle BAZ$$

Now, in $\triangle DAX$ and $\triangle BAZ$,

$$AD = AB \quad (\text{sides of squares})$$

$$\Rightarrow \angle DAX = \angle BAZ \quad (\text{proved earlier})$$

$$\Rightarrow AX = AZ \quad (\text{sides of square})$$

$$\Rightarrow \triangle DAX \cong \triangle BAZ \quad (\text{SAS criterion})$$

$$\Rightarrow BZ = DX \quad (\text{by CPCT})$$

$$\text{and } \angle ABZ = \angle ADX \quad (\text{by CPCT})$$

Hence, both statements are correct.

64. (d) Given as $\frac{PB}{PC} = \frac{1}{2} \quad \dots(i)$

$$BQ \parallel DC \quad (\because AB \parallel DC)$$

In $\triangle BPQ$ and $\triangle DPC$

$$\angle BPQ = \angle DPC$$

($\because$ vertically opposite angles)

$$\Rightarrow \angle QBP = \angle PCD$$

($\because$ alternate interior angle)

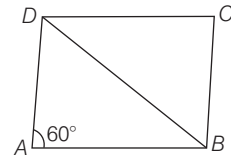
$$\Rightarrow \triangle BPQ \sim \triangle DPC \quad [\text{by AA similarity}]$$

We know that area of two similar triangles are in the ratio of the square of the corresponding sides

$$\frac{\text{ar}(\triangle BPQ)}{\text{ar}(\triangle DPC)} = \left(\frac{PB}{PC}\right)^2 \\ \Rightarrow \frac{20}{\text{ar}(\triangle DPC)} = \left(\frac{1}{2}\right)^2 \quad [\text{from Eq. (i)}]$$

$$\therefore \text{ar}(\triangle DPC) = 80 \text{ sq units}$$

65. (b)



Let $AD = x$, then $AB = 2x$

using cosine rule in $\triangle ABD$, we have

$$\cos A = \frac{AD^2 + AB^2 - BD^2}{2 \cdot AD \cdot AB} \\ \Rightarrow \cos 60^\circ = \frac{x^2 + (2x)^2 - BD^2}{2 \cdot x \cdot (2x)}$$

$$\Rightarrow \frac{1}{2} = \frac{5x^2 - BD^2}{4x^2}$$

$$\Rightarrow 2x^2 = 5x^2 - BD^2$$

$$\Rightarrow BD^2 = 3x^2$$

$$\Rightarrow BD = \sqrt{3}x = \sqrt{3}AD.$$

66. (b) Given, each interior angle = 140°

Then, each exterior angle

$$= (180^\circ - 140^\circ) = 40^\circ$$

$\therefore$ Number of sides

$$= \frac{360^\circ}{\text{Each exterior angle}} = \frac{360^\circ}{40^\circ} = 9$$

Hence, the number of vertices of polygon is 9.

67. (b) Exterior angle of regular polygon

$$= \frac{360^\circ}{\text{Number of sides}}$$

I. If exterior angle is 70°

Then, number of sides

$$= \frac{360^\circ}{70^\circ} = \frac{36}{7} = \text{A rational number}$$

So, statement I is incorrect.

II. If number of sides ≥ 5 , then

$$\text{Exterior angle} = \frac{360^\circ}{5} = 72^\circ$$

For number of sides ≥ 5 , there is always an acute exterior angle.

So, statement II is correct.

68. (d) I. In regular polygon,

$$\text{Sum of exterior angles} = 360^\circ$$

[which is always constant]

So, statements I is incorrect.

II. Sum of interior angles of m sides of regular polygon = $(m-2) \times 180^\circ$

Sum of interior angles of m sides and n sides of regular polygon

$$= (m-2) \times 180^\circ + (n-2) \times 180^\circ$$

$$= (m+n-4) \times 180^\circ$$

So, statement II is incorrect.

Hence, neither I nor II is correct.

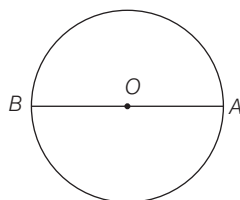
CIRCLE

Usually (3-6) questions have been asked from this chapter. Generally questions are asked from the topics related to theorem of circle, tangent to a circle and locus.



A circle is a curved figure consisting of all points in a plane that are at a fixed distance from a fixed point. The fixed point is called centre and the fixed distance is called radius of the circle.

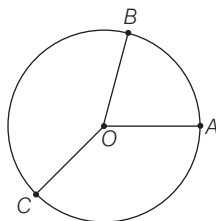
- Circles are simple closed curves which divide the plane into two regions : an interior and an exterior.
- In the given figure, O is the centre of circle, OA is the radius of the circle and AB is the diameter of the circle.



RADIUS

Radius is the fixed distance between the centre of the circle and the points lying on the circle.

Here, in the given figure OA , OB and OC all are having fixed distance, so they are known as radius of the circle. It is denoted by r .

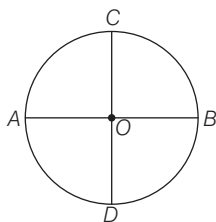


Diameter

Diameter is any line segment that passes through the centre of the circle and whose end points lie on the circle. The diameter of a circle is twice the radius. Here, AB and CD are the diameters of the circle.

$$AB = CD = 2(OA) \\ = 2(OB) = 2(OC) = 2(OD)$$

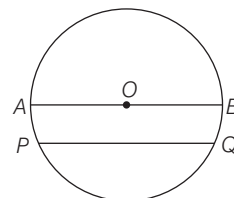
A circle can have an infinite number of diameters.



Chord

A chord is a line segment whose end points lie on the circle. A diameter is the longest chord in a circle.

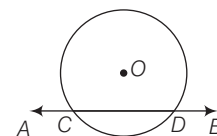
Here, in the given figure PQ is a chord and AB is the diameter of the circle which is the longest chord of that circle.



Secant

A line which intersects the circle at two points is called secant of the circle.

Here, in the figure AB is a secant cutting the circle at two distinct points as C and D .

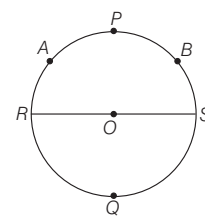


Arc

Any part of a circle between two points is called an arc of the circle.

Any two points, say A and B of a circle divide it into two parts called arcs. The smaller arc APB is called minor arc and larger arc AQB is called major arc.

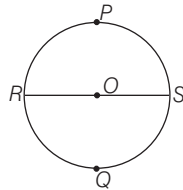
It is usually denoted by $\widehat{APB}$ and $\widehat{AQB}$.



Semi Circle

A diameter divides the circle into two equal parts. Each of the two is called a semi-circle.

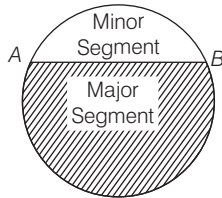
In the above figure, $RPSR$ and $RQSR$ are semi-circles.



Segment

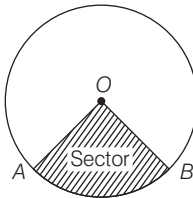
The area enclosed by an arc and its corresponding chord is called a segment of the circle.

- The segment containing the minor arc is called minor segment.
- The segment containing the major arc is called major segment.



Sector

The area enclosed by any two radii and the arc determined by the end points of the radii is called a sector of the circle.



Circumference

The length of the complete circle is called its circumference.

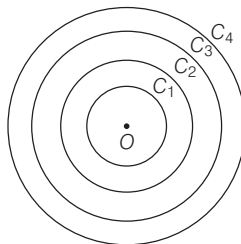
$$\begin{aligned} \text{Circumference} &= 2 \times \pi \times \text{Radius of the circle} \\ &\text{or} \\ &= \pi \times \text{Diameter of the circle} \end{aligned}$$

Concentric Circles

Two or more circles having the same centre are called concentric circles.

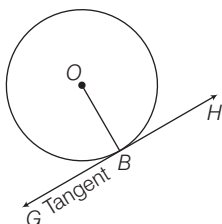
Here, in the given figure C_1, C_2, C_3 and C_4 are known as concentric circles as they have a common centre 'O'.

An infinite number of circles can be drawn with same centre.



Tangent

A tangent to a circle is a straight line that touches the circle at a single point.

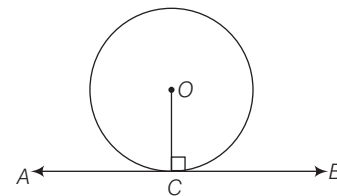


Here, in the given figure GBH is a tangent touching the circle at a single point B .

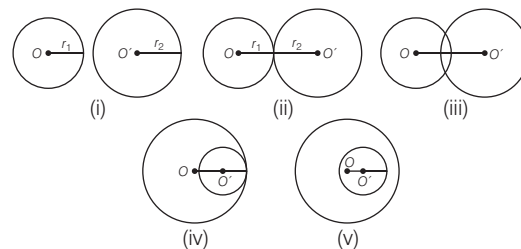
Length of tangent The distance between external point P from which tangent is drawn and the point of contact of the tangent is called length of tangent.

Properties of Tangent To a Circle

- Only two tangents can be drawn from a point outside the circle.
- Only one tangent can be drawn through a point lying on the circle.
- No tangent can be drawn from a point lying inside the circle.
- A tangent at any point of a circle is perpendicular to the radius through the point of contact.



Relative Position of two Circles



Two circles will

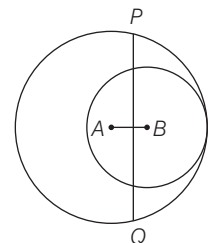
- be disjoint, when $OO' > r_1 + r_2$.
- be touching externally, when $OO' = r_1 + r_2$.
- be intersecting when $OO' < r_1 + r_2$.
- be touching internally, when $OO' = |r_2 - r_1|$
- one of the circle will lie inside the other, when $OO' < |r_2 - r_1|$.

EXAMPLE 1. Two circles with centres A and B touch each other internally, as shown in the figure given below. Their radii are 5 and 3 units, respectively. Perpendicular bisector of AB meets the bigger circle in P and Q . What is the length of PQ ?

- a. $2\sqrt{6}$ b. $\sqrt{34}$ c. $4\sqrt{6}$ d. $6\sqrt{2}$

Sol. c. Let PQ intersect AB at O . Join AP and AQ . Produce AB to intersect the circle at M .

Here, $AP = AM = 5$ units, $BM = 3$ units



$$\therefore AB = AM - BM = 5 - 3 = 2 \text{ units}$$

PQ bisects AB perpendicularly.

$$\therefore AO = OB = 1 \text{ unit}$$

Now, in right $\triangle AOP$,

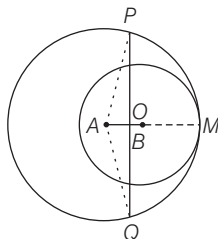
$$(PA)^2 = (OA)^2 + (OP)^2$$

$$\Rightarrow (5)^2 = (1)^2 + (OP)^2$$

$$\Rightarrow 24 = (OP)^2 \Rightarrow OP = 2\sqrt{6} \text{ units}$$

$$\therefore PQ = 2OP = 2 \times 2\sqrt{6} = 4\sqrt{6} \text{ units}$$

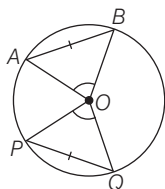
Hence, the length of PQ is $4\sqrt{6}$ units.



Important Theorems of Circles

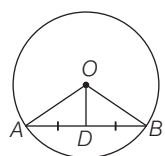
Theorem 1 If two arcs of a circle are congruent, then the corresponding chords are equal.

Theorem 2 Equal arcs (or chords) subtend equal angle at the centre. Here, if $AB = PQ$, then $\angle AOB = \angle POQ$.



Theorem 3 The perpendicular from the centre of a circle to a chord bisects the chord.

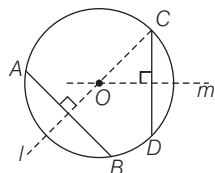
Here, if $OD \perp AB$, then $AD = DB$.



Theorem 4 The line joining the centre to the mid-point of a chord is perpendicular to the chord.

Here, if $AD = DB$, then $\angle ADO = \angle ODB = 90^\circ$.

Theorem 5 The perpendicular bisectors of two chords of a circle intersect at its centre.

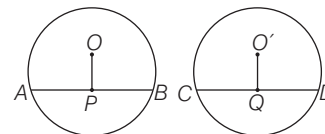


Here, AB, CD are the chords and l, m are perpendicular bisector of AB and CD . So, l and m meet at centre 'O'.

Theorem 6 There is one and only one circle passing through three non-collinear points.

- An infinite number of circles can be drawn to pass through a single point.
- An infinite number of circles can be drawn to pass through two given points.
- A unique circle can be drawn to pass through three given non-collinear points.

Theorem 7 Equal chords of congruent circles are equidistant from the corresponding centres.

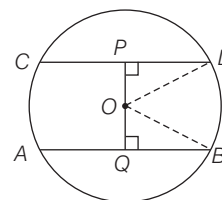


Here, if $AB = CD$, then $OP = O'Q$.

Conversely if $OP = O'Q$, then $AB = CD$.

Theorem 8 Equal chords of a circle are equidistant from the centre.

Here, if AB and CD are equal chords of circle, then $OP = OQ$.



Conversely, if $OP = OQ$, then also $AB = CD$

i.e. chords at equal distance from the centre are equal.

EXAMPLE 2. AB and CD are two parallel chords on the opposite sides of the centre of the circle. If $AB = 10$ cm, $CD = 24$ cm and the radius of the circle is 13 cm, then what is the distance between the chords?

- a. 10 cm b. 17 cm
c. 24 cm d. None of these

Sol. b. From O draw $OL \perp AB$ and $OM \perp CD$. Join OA and OC .

$$AL = \frac{1}{2} AB = \frac{10}{2} \text{ cm} = 5 \text{ cm},$$

$$OA = 13 \text{ cm}$$

In right angled $\triangle OLA$, by pythagoras theorem,

$$OL^2 = OA^2 - AL^2 = (13)^2 - (5)^2$$

$$= (169 - 25) = 144$$

$$\Rightarrow OL = \sqrt{144} = 12 \text{ cm}$$

$$\text{Now, } CM = \frac{1}{2} \times CD = 12 \text{ cm}$$

$$\text{and } OC = 13 \text{ cm}$$

In $\triangle OMC$,

$$OM^2 = OC^2 - CM^2$$

$$= (13)^2 - (12)^2$$

$$= (169 - 144) = 25$$

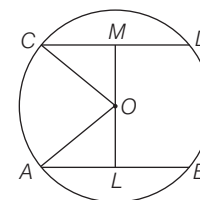
$$\Rightarrow OM = \sqrt{25} = 5 \text{ cm}$$

$$\therefore ML = OM + OL$$

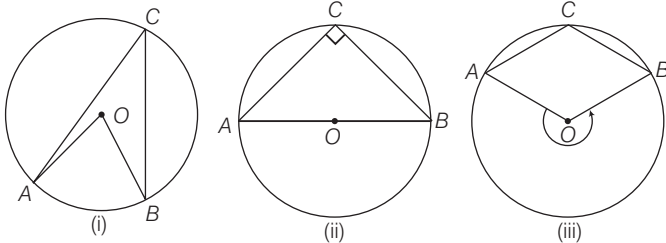
$$= (5 + 12) \text{ cm}$$

$$= 17 \text{ cm}$$

[by pythagoras theorem]

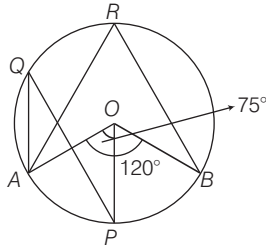


Theorem 9 The angle subtended by an arc of a circle at the centre is double the angle subtended by it at any point on the circumference of circle. Here, three case arises.



In all the three cases $\angle AOB = 2\angle ACB$.

EXAMPLE 3 In the figure given below, if $\angle AOP = 75^\circ$ and $\angle AOB = 120^\circ$, then what is the value of $\angle AQP$?



- a. 45° b. 37.5° c. 30° d. 22.5°

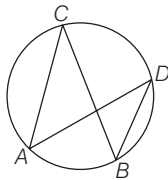
Sol. b. We know that the angle subtended by an arc of a circle at the centre is double the angle subtended by it at any point on the circumference of a circle.

$$2\angle AQP = \angle AOP$$

$$\angle AQP = \frac{1}{2} \times \angle AOP = \frac{75^\circ}{2} = 37.5^\circ$$

Theorem 10 The angle in a semi-circle is a right angle.

Theorem 11 Angles in the same segment of a circle are equal.

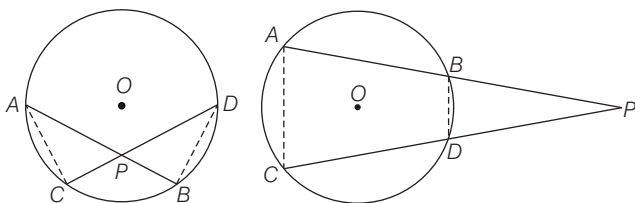


Here, $\angle ACB = \angle ADB$

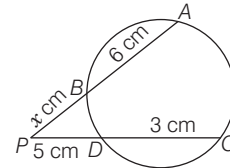
Theorem 12 If two chords AB and CD of a circle intersect inside or outside the circle when produced at a point P

then,

$$AP \times PB = DP \times PC$$



EXAMPLE 4. In the given figure, chords AB and CD of a circle intersect externally at P . If $AB = 6$ cm, $CD = 3$ cm and $PD = 5$ cm, then the measurement of PB is



- a. 2.5 cm b. 4 cm c. 10 cm d. None of these

Sol. b. Let the measurement of PB be x .

$$PA \times PB = PC \times PD \Rightarrow (x + 6) \times x = 8 \times 5$$

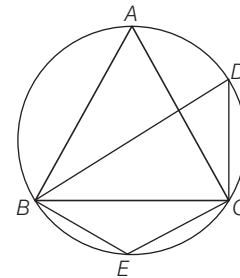
$$\Rightarrow x^2 + 6x - 40 = 0 \Rightarrow x^2 + 10x - 4x - 40 = 0$$

$$\Rightarrow x(x + 10) - 4(x + 10) = 0 \Rightarrow (x + 10)(x - 4) = 0$$

$$\Rightarrow x = 4, \text{ as } x \neq -10 \quad [\text{side cannot be negative}]$$

$$\therefore PB = 4 \text{ cm}$$

EXAMPLE 5. In the adjoining figure, $\triangle ABC$ is an isosceles triangle with $AB = AC$ and $\angle ABC = 50^\circ$. Then, the measure of $\angle BDC$ is



- a. 80° b. 100° c. 40° d. 160°

Sol. a. Given, $AB = AC \Rightarrow \angle ACB = \angle ABC = 50^\circ$

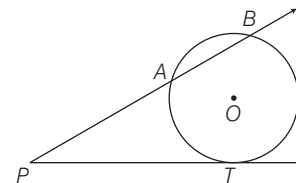
$$\text{In } \triangle ABC, \angle BAC = 180^\circ - (50^\circ + 50^\circ) = 80^\circ$$

[by angle sum property of a triangle]

$$\therefore \angle BDC = \angle BAC = 80^\circ$$

[angles lying in the same segment of the circle]

Theorem 13 Let PT be a line tangent to the circle at point T . From external point P , draw a secant which intersect the circle at two point A and B , then $PA \times PB = PT^2$

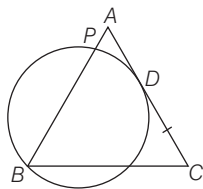


EXAMPLE 6. In a $\triangle ABC$, $AB = AC$. A circle through B touches AC at D and intersects AB at P . If D is the mid-point of AC , then which one of the following is correct?

- a. $AB = 2AP$ b. $AB = 3AP$ c. $AB = 4AP$ d. $2AB = 5AP$

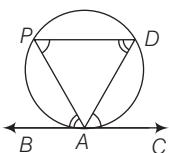
Sol. c. By using theorem,

$$\begin{aligned} AB \times AP &= AD^2 \\ &= \left(\frac{AC}{2}\right)^2 = \frac{1}{4}(AC)^2 \\ \Rightarrow AB \times AP &= \frac{1}{4}(AB)^2 \\ [\because AC &= AB \text{ given}] \\ \Rightarrow AB &= 4AP \end{aligned}$$

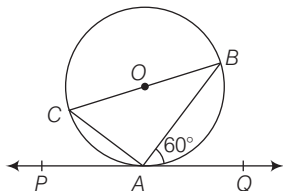


Theorem 14 Alternate Segment

Theorem In the given figure B, if BAC is a tangent to a circle at point A and if AD is any chord, then $\angle DAC = \angle APD$ and $\angle PAB = \angle PDA$.



EXAMPLE 7. In the given figure, PAQ is the tangent. BC is the diameter of the circle. If $\angle BAQ = 60^\circ$, then $\angle ABC$ is



- a. 25° b. 30° c. 45° d. 50°

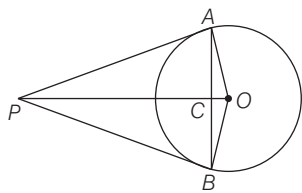
Sol. b. Since BC is the diameter of the circle.

$$\begin{aligned} \therefore \angle BAC &= 90^\circ & [\because \text{angle in a semi-circle is } 90^\circ] \\ \angle ACB &= 60^\circ & [\because \text{alternate angle theorem}] \end{aligned}$$

$$\begin{aligned} \text{Now, } \angle ACB + \angle BAC + \angle ABC &= 180^\circ \\ \Rightarrow \angle ABC + 90^\circ + 60^\circ &= 180^\circ \\ \Rightarrow \angle ABC &= 180^\circ - (90^\circ + 60^\circ) = 30^\circ \end{aligned}$$

- Theorem 15** If two tangents are drawn to a circle from an external point, then they subtend equal angles at the centre. These two tangents are equally inclined to the line segment joining the centre to that point. In the given figure,

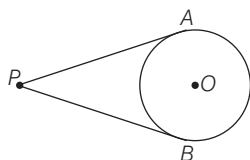
$$\begin{aligned} \angle AOP &= \angle BOP & \text{and} \\ \angle APO &= \angle BPO \end{aligned}$$



- Theorem 16** Line joining centre and external point is perpendicular bisector of line joining point of contacts of tangents to the circle. In the above figure, $OP \perp AB$ and $AC = BC$

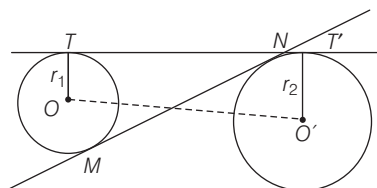
- Theorem 17** The length of two tangents drawn from an external point to a circle are equal.

Here, $PA = PB$



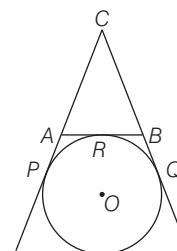
- Theorem 18** Tangents drawn at the end points of a diameter of a circle are always parallel.

- Length of the direct common tangent is $TT' = \sqrt{(OO')^2 - (r_1 - r_2)^2}$ and length of transverse common tangent is $MN = \sqrt{(OO')^2 - (r_1 + r_2)^2}$



EXAMPLE 8. In the given figure, CP and CQ are tangent to a circle with centre O . ARB is another tangent touching the circle at R . If $CP = 11$ cm, $BC = 7$ cm length of BR is

- a. 1 cm b. 2 cm
c. 4 cm d. 3 cm



Sol. c. Since, the length of two tangents drawn from external point are equal.

$$\therefore CQ = CP = 11 \text{ cm and } BQ = BR$$

$$\begin{aligned} \text{Now, } BQ &= CQ - BC \\ &= (11 - 7) \text{ cm} = 4 \text{ cm} \end{aligned}$$

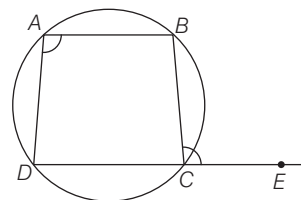
$$\therefore BR = BQ = 4 \text{ cm}$$

Cyclic Quadrilateral

A quadrilateral whose all vertices lie on a circle is called a cyclic quadrilateral.

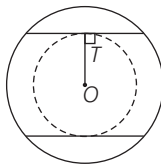
Properties of Cyclic Quadrilateral

- The sum of either pair of opposite angles of a cyclic quadrilateral is 180° i.e. the opposite angles of a cyclic quadrilateral are supplementary.
- If the sum of a pair of opposite angles of a quadrilateral is 180° , then quadrilateral is cyclic.
- The exterior angle, formed by producing a side of a cyclic quadrilateral is equal to the interior opposite angle.

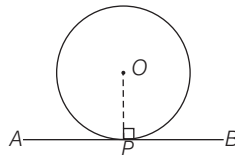


$$\text{Here, } \angle BAD = \angle BCE$$

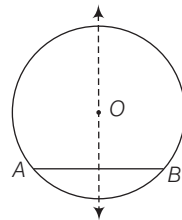
4. The locus of the mid-points of equal chords is a circle with the same centre as the given circle and radius equal to the distance from the centre of the given circle to the given chord.



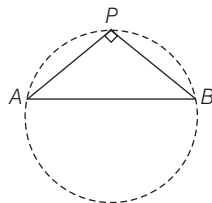
5. The locus of the centre of circle touching a given line at a given point is a line perpendicular to the given line through the given point.



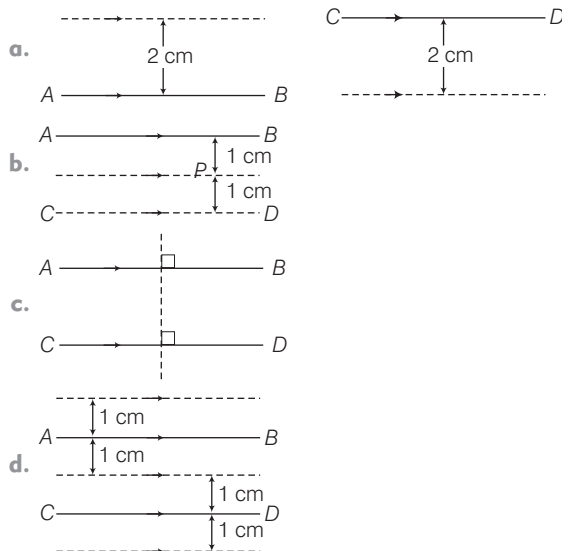
6. The locus of the centre of all circles passing through two given points is the perpendicular bisector of the line segment joining the two points.



7. If A, B are fixed points, then the locus of a point P such that $\angle APB = 90^\circ$ is the circle with AB as diameter.



EXAMPLE 10. Given two parallel straight lines AB and CD which are 2 cm apart which of the following dotted lines may represent the locus of a point P such that it is equidistant from AB and CD ?

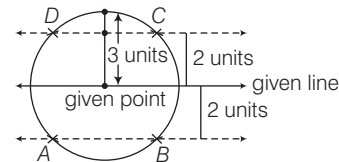


Sol. b.

EXAMPLE 11. What is the number of points in a plane two units from a given line and three units from a given point of the line?

- a. 1 b. 2 c. 3 d. 4

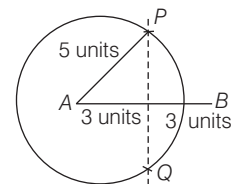
Sol. d. The answer will be 4 points. The dotted lines are first locus condition and the circle is the second locus condition. These two loci intersect at points marked with A, B, C and D .



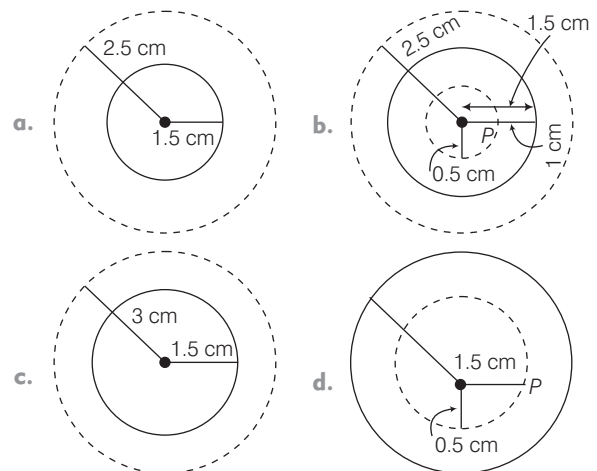
EXAMPLE 12. Two points A and B are 6 units apart. How many points are there that are equidistant from both A and B and also 5 units from A ?

- a. 0 b. 1 c. 2 d. 3

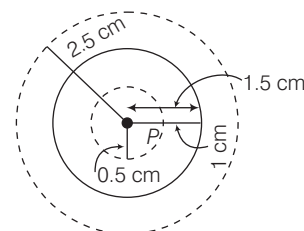
Sol. c. The answer will be 2 points. The dotted line is first locus condition and the circle is the second locus condition. These two loci intersect at two points marked with P and Q .



EXAMPLE 13. Given a circle with radius 1.5 cm. Which of the following dotted lines may represent the locus of a point P such that P is at a distance of 1 cm from the nearest point on the circle?



Sol. b.

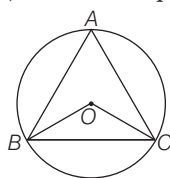


PRACTICE EXERCISE

1. In a circle with centre O and radius 5 cm, AB is a chord of length 8 cm. If $OM \perp AB$, then what is the length of OM ?

(a) 4 cm (b) 5 cm
(c) 3 cm (d) None of these

2. An equilateral $\triangle ABC$ is inscribed in a circle with centre O . Then, $\angle BOC$ is equal to



(a) 120° (b) 75° (c) 180° (d) 60°

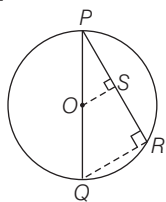
3. A square $ABCD$ is inscribed in a circle with centre O . Then, the angle subtended by each side of the square at the centre O is

(a) 120° (b) 180° (c) 45° (d) 90°

4. The points of concurrency of the perpendicular bisectors of the sides of a triangle is called its

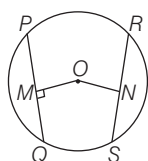
(a) incentre (b) circumcentre
(c) excentre (d) orthocentre

5. In the given figure, PQ is the diameter of a circle with centre at O . OS is perpendicular to PR . Then, OS is equal to



(a) $\frac{1}{4}QR$ (b) $\frac{1}{3}QR$ (c) $\frac{1}{2}QR$ (d) QR

6. In the given figure, OM and ON are the perpendiculars drawn on the chords PQ and RS . If $OM = ON = 6$ cm. Then,



(a) $PQ \geq RS$ (b) $PQ < RS$ (c) $PQ \leq RS$ (d) $PQ = RS$

7. The locus of the mid-points of all radii of a circle is a

(a) circle (b) parallelogram
(c) rhombus (d) square

8. The perpendicular bisectors of the sides of a triangle pass through the

(a) different point (b) more than 2 points
(c) same point (d) None of these

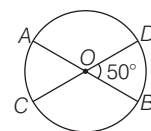
9. If a point P moves such that the sum of the squares of its distance from two fixed points A and B is a constant, then the locus of P is

(a) a circle
(b) a straight line
(c) an arbitrary curve
(d) the perpendicular bisector of AB

10. The locus of points equidistant from two fixed points is a straight line which

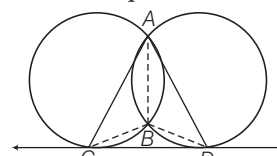
(a) is at right angles to the line joining the two fixed points
(b) bisects the line joining the two fixed points
(c) is the perpendicular bisector of the line joining the two fixed points
(d) None of the above

11. Diameter AB and CD of a circle intersect at O . If $m\angle BOD = 50^\circ$, then $m\angle AOD$ is



(a) 50° (b) 180° (c) 130° (d) 310°

12. CD is a direct common tangent to two circles intersecting each other at A and B . Then, $\angle CAD + \angle CBD$ is equal to?



(a) 180° (b) 90° (c) 360° (d) 120°

13. In a circle of radius 17 cm, two parallel chords are drawn on opposite sides of a diameter. The distance between the chords is 23 cm. If the length of one chord is 16 cm, then the length of the other chord is

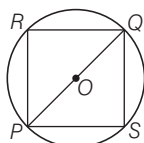
(a) 34 cm (b) 15 cm (c) 23 cm (d) 30 cm

14. If AB is a chord of a circle, P and Q are the two points on the circle different from A and B , then

(a) the angles subtended at P and Q by AB are always equal
(b) the sum of the angles subtended by AB at P and Q is always equal to two right angles
(c) the angles subtended by AB at P and Q are either equal or supplementary
(d) None of the above

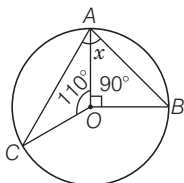
- 15.** The locus of the centre of circles which passes through two given points is
 (a) perpendicular to the line joining the given points at one of those points
 (b) perpendicular bisector of the line joining the given points
 (c) parallel to the line joining the given points
 (d) None of the above

- 16.** In the adjoining figure, POQ is the diameter of the circle, R and S are any two points on the circles. Then,



- (a) $\angle PRQ > \angle PSQ$ (b) $\angle PRQ < \angle PSQ$
 (c) $\angle PRQ = \angle PSQ$ (d) $\angle PRQ = \frac{1}{2} \angle PSQ$

- 17.** If O is the centre of the circle, then the value of x in the adjoining figure is

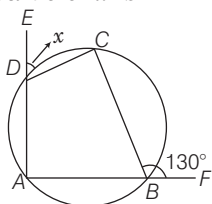


- (a) 80° (b) 70° (c) 60° (d) 50°

- 18.** If S is a circle with centre C and P be a movable point outside S , then the locus of P such that the tangents from P to S are of constant length is

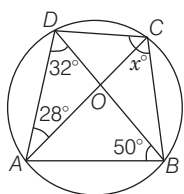
- (a) the straight line CP
 (b) the circle through P with centre at C
 (c) a circle intersecting S
 (d) a circle touching S

- 19.** In the given figure A, B, C, D are the concyclic points. The value of x is



- (a) 50° (b) 60° (c) 70° (d) 90°

- 20.** If O is the centre of the circle, then x is



- (a) 72° (b) 62° (c) 82° (d) 52°

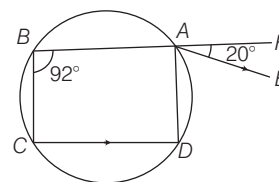
- 21.** Two circles touch each other internally. Their radii are 2 cm and 3 cm. The biggest chord of the outer circle which is outside the inner circle is of length

- (a) $2\sqrt{2}$ cm (b) $3\sqrt{2}$ cm (c) $2\sqrt{3}$ cm (d) $4\sqrt{2}$ cm

- 22.** If two circles are such that the centre of one lies on the circumference of the other, then the ratio of the common chord of the two circles to the radius of any one of the circles is

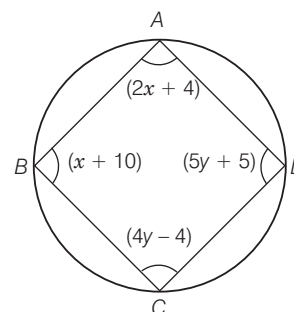
- (a) $2 : 1$ (b) $\sqrt{3} : 1$ (c) $\sqrt{5} : 1$ (d) $4 : 1$

- 23.** In the given figure, $ABCD$ is a cyclic quadrilateral. AE is drawn parallel to CD and BA is produced. If $\angle ABC = 92^\circ$ and $\angle FAE = 20^\circ$, then $\angle BCD$ is equal to



- (a) 88° (b) 98° (c) 108° (d) 72°

- 24.** The values of $x + y$ in the figure is equal to

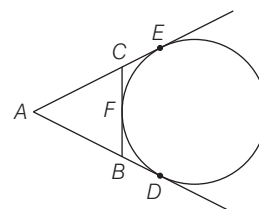


- (a) 90° (b) 85° (c) 75° (d) 65°

- 25.** Let A and B be two points. What is the locus of the point P such that $\angle APB = 90^\circ$?

- (a) The line AB itself
 (b) The point P itself
 (c) The circumference of the circle with AB as diameter
 (d) The line perpendicular to AB and bisecting AB

- 26.** In the adjoining figure AD, AE and BC are tangent to the circle at D, E, F respectively, then



- (a) $AD = AB + BC + AC$ (b) $2AD = AB + BC + AC$
 (c) $AD = \frac{1}{4}(AB + BC + AC)$ (d) $3AD = AB + BC + AC$

27. S_1 and S_2 are two circles on a plane with radii 4 cm, and 2 cm, respectively and the distance between their centres is 3 cm. Which one of the following statements is true?

(a) S_2 lies entirely within the circle S_1
 (b) S_1 and S_2 touch each other internally
 (c) S_1 and S_2 touch each other externally
 (d) S_1 and S_2 intersect in two distinct points

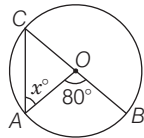
28. ACB is a tangent to a circle at C , CD and CE are chords such that $\angle ACE > \angle ACD$. If $\angle ACD = \angle BCE = 50^\circ$, then

(a) $CD = CE$
 (b) ED is not parallel to AB
 (c) ED passes through the centre of the circle
 (d) $\triangle CDE$ is a right angled triangle

29. A bicycle is running straight towards North. What is the locus of the centre of the front wheel of the bicycle whose diameter is d ?

(a) A line parallel to the path of the wheel of the bicycle at a height d cm
 (b) A line parallel to the path of the wheel of the bicycle at a height $d/2$ cm
 (c) A circle of radius $d/2$ cm
 (d) A circle of radius d cm

30. If 'O' is the centre of circle, then x is equal to



(a) 80° (b) 60° (c) 40° (d) 20°

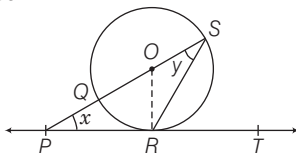
31. If two equal circles touch each other externally, the common tangent divides the line of centres in the ratio

(a) 1 : 1 (b) 2 : 1 (c) 1 : 2 (d) 3 : 2

32. With the vertices of a $\triangle ABC$ as centre three circles are described, each touching the other two circle externally. If the sides of the triangle are 9 cm, 7 cm and 6 cm. Then, the radius of the circle are

(a) 4, 5, 2 (b) 4, 5, 6
 (c) 3, 2, 3 (d) All equal to 3 cm

33. In the given figure PT touches the circle with centre O at R . Diameter SQ when produced meet PT at P . If $\angle SPR = x$ and $\angle QSR = y$, then $x + 2y$ is equal to

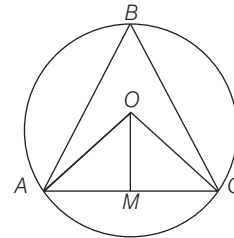


(a) 180° (b) 90°
 (c) 135° (d) None of these

34. What is the locus of centres of circles which touch a given line at a given point?

(a) A line perpendicular to the given line, passing through the given point
 (b) A line parallel to the given line
 (c) A circle tangent to the given line at the given point
 (d) A closed curve other than a circle

35. In the given figure, O is the centre of the circle, $OA = 3$ cm, $AC = 3$ cm and OM is perpendicular to AC . What is $\angle ABC$ equal to?



(a) 60° (b) 45°
 (c) 30° (d) None of these

36. The distance between the centres of two circles having radii 4.5 cm and 3.5 cm, respectively is 10 cm. What is the length of the transverse common tangent of these circles?

(a) 8 cm (b) 7 cm
 (c) 6 cm (d) None of these

37. ABC is an equilateral triangle inscribed in a circle with $AB = 5$ cm. Let the bisector of the angle, A meet BC in X and the circle in Y . What is the value of $AX \cdot AY$?

(a) 16 cm^2 (b) 20 cm^2 (c) 25 cm^2 (d) 30 cm^2

38. Two unequal circles are touching each other externally at P , APB and CPD are two secants cutting the circles at A, B, C and D . Which one of the following is correct?

(a) $ACBD$ is parallelogram (b) $ACBD$ is a trapezium
 (c) $ACBD$ is a rhombus (d) None of these

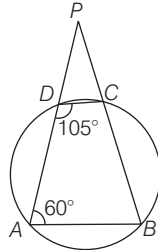
39. $ABCD$ is a quadrilateral, the sides of which touch a circle. Which one of the following is correct?

(a) $AB + AD = CB + CD$ (b) $AB : CD = AD : BC$
 (c) $AB + CD = AD + BC$ (d) $AB : AD = CB : CD$

40. Let PAB be a secant to a circle intersecting at points A and B and PC is a tangent. Which one of the following is correct?

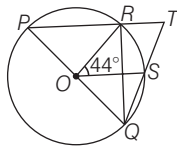
(a) The area of rectangle with PA, PB as sides is equal to the area of square with PC as sides
 (b) The area of rectangle with PA, PC as sides is equal to the area of square with PB as sides
 (c) The area of rectangular with PC, PB as sides is equal to the area of square with PA as side
 (d) The perimeter of rectangle with PA, PB as sides is equal to the perimeter of square with PC as side

41. In the given figure, if $\angle BAD = 60^\circ$, $\angle ADC = 105^\circ$, then what is $\angle DPC$ equal to?



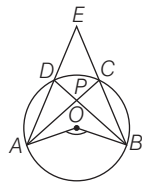
- (a) 40° (b) 45° (c) 50° (d) 60°

42. In the given figure, PQ is a diameter of the circle whose centre is at O . If $\angle ROS = 44^\circ$ and OR is a bisector of $\angle PRQ$, then what is the value of $\angle RTS$?



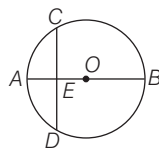
- (a) 46° (b) 64°
(c) 69° (d) None of these

43. In the given figure, O is the centre of the circle. AC and BD intersect at P . If $\angle AOB = 100^\circ$ and $\angle DAP = 30^\circ$, what is $\angle APB$?



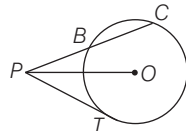
- (a) 77° (b) 80° (c) 85° (d) 90°

44. In the given figure, AB is a diameter of a circle and CD is perpendicular to AB , if $AB = 10$ cm and $AE = 2$ cm, then what is the length of ED ?



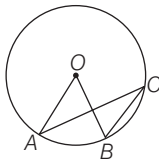
- (a) 5 cm (b) 4 cm (c) $\sqrt{10}$ cm (d) $\sqrt{20}$ cm

45. In the given figure, PT is a tangent to a circle of radius 6 cm. If P is at a distance of 10 cm from the centre O and $PB = 5$ cm, then what is the length of the chord BC ?



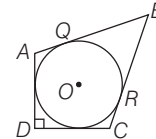
- (a) 7.8 cm (b) 8 cm (c) 8.4 cm (d) 9 cm

46. In the given figure, $\angle AOB = 46^\circ$, AC and OB intersect each other at right angles. What is the measure of $\angle OBC$? (O is the centre of the circle.)



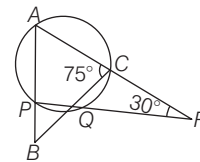
- (a) 44° (b) 46° (c) 67° (d) 78.5°

47. In the given figure, a circle is inscribed in a quadrilateral $ABCD$. Given that, $BC = 38$ cm, $QB = 27$ cm, $DC = 25$ cm and AD is perpendicular to DC . What is the radius of the circle?



- (a) 11 cm (b) 14 cm (c) 15 cm (d) 16 cm

48. In the figure given below, what is $\angle CBA$?



- (a) 30° (b) 45° (c) 50° (d) 60°

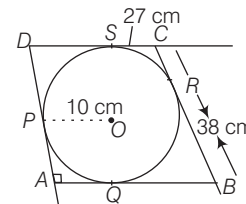
49. PQ is a common chord of two circles. APB is a secant line joining points A and B on the two circles. Two tangents AC and BC are drawn. If $\angle ACB = 45^\circ$, then what is $\angle AQB$ equal to?

- (a) 75° (b) 90° (c) 120° (d) 135°

50. $ABCD$ is concyclic quadrilateral. The tangents at A and C intersect each other at P . If $\angle ABC = 100^\circ$, then what is $\angle APC$ equal to?

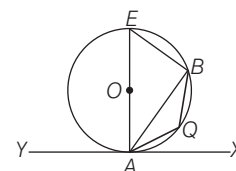
- (a) 10° (b) 20° (c) 30° (d) 40°

51. In the given figure, quad. $ABCD$ is circumscribed, touching the circle at P, Q, R and S such that $\angle DAB = 90^\circ$. If $CS = 27$ cm and $CB = 38$ cm and radius of the circle is 10 cm, then $AB = ?$



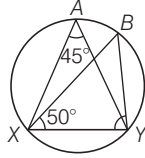
- (a) 28 cm (b) 21 cm (c) 19 cm (d) 17 cm

52. In the figure given below, YAX is a tangent to the circle with centre O . If $\angle BAX = 70^\circ$ and $\angle BAQ = 40^\circ$, then what is $\angle ABQ$ equal to?



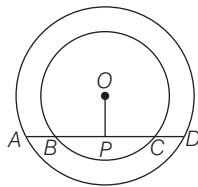
- (a) 20° (b) 30°
(c) 35° (d) 40°

53. In the given figure, what is $\angle BYX$ equal to?



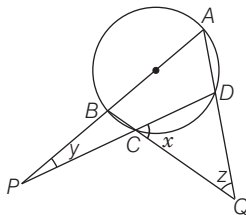
- (a) 85° (b) 50° (c) 45° (d) 90°

54. In the given figure, AD is a straight line, OP perpendicular to AD and O is the centre of both circles. If $OA = 20$ cm, $OB = 15$ cm and $OP = 12$ cm, what is AB equal to?



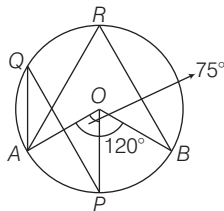
- (a) 7 cm (b) 8 cm (c) 10 cm (d) 12 cm

55. In the given figure, if $\frac{x}{3} = \frac{y}{4} = \frac{z}{5}$, where $\angle DCQ = x$, $\angle BPC = y$ and $\angle DQC = z$, then what are the values of x, y and z , respectively?



- (a) $33^\circ, 44^\circ$ and 55° (b) $36^\circ, 48^\circ$ and 60°
(c) $39^\circ, 52^\circ$ and 65° (d) $42^\circ, 56^\circ$ and 70°

56. In the given figure, if $\angle AOP = 75^\circ$ and $\angle AOB = 120^\circ$, then what is $\angle AQP$?



- (a) 45° (b) 37.5° (c) 30° (d) 22.5°

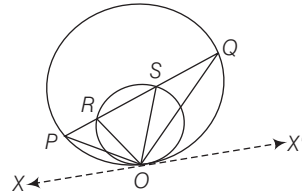
57. A circle of radius 5 cm has chord RS at a distance of 3 units from it. Chord PQ intersects with chord RS at T such that $TS = 1/3$ of RT . Find minimum value of PQ .

- (a) $6\sqrt{3}$ (b) $4\sqrt{3}$ (c) $8\sqrt{3}$ (d) $2\sqrt{3}$

58. Two mutually perpendicular chords AB and CD intersect at P . $AP = 4$ cm, $PB = 6$ cm, $CP = 3$ cm. Find radius of the circle.

- (a) $\frac{5\sqrt{5}}{2}$ cm (b) $\frac{3\sqrt{5}}{4}$ cm (c) $\frac{6\sqrt{5}}{2}$ cm (d) $\frac{2\sqrt{5}}{3}$ cm

- 59.



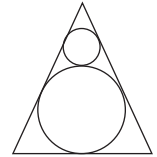
In the above figure which of the following holds good?

- (a) $\angle SOQ = \angle ROP$ (b) $2\angle ROP = \angle SOR$
(c) $\angle POR = \angle ASO$ (d) $\angle QOX' = \angle SOR + \angle ROP$

60. Semi-circle C_1 is drawn with a line segment PQ as its diameter with centre at R . Semicircles C_2 and C_3 are drawn with PR and QR as diameters respectively, both C_2 and C_3 lying inside C_1 . A full circle C_4 is drawn in such a way that it is tangent to all the three semicircles C_1, C_2 and C_3 , C_4 lies inside C_1 and outside both C_2 and C_3 . The radius of C_4 is

- (a) $\frac{1}{3}PQ$ (b) $\frac{1}{6}PQ$ (c) $\frac{1}{\sqrt{2}}PQ$ (d) $\frac{1}{4}PQ$

61. Two circles are placed in an equilateral triangle as shown in the figure. What is the ratio of the area of the smaller circle to that of the equilateral triangle?



- (a) $\pi : 36\sqrt{3}$ (b) $\pi : 18\sqrt{3}$ (c) $\pi : 27\sqrt{3}$ (d) $\pi : 42\sqrt{3}$

62. Consider the following statements :

- I. The opposite angles of a cyclic quadrilateral are supplementary.
II. Angle subtended by an arc at the centre is double the angle subtended by it at any point on the remaining part of the circle.

Which one of the following is/are correct in respect of the above statements?

- (a) Statement I $\Rightarrow$ Statement II
(b) Statement II $\Rightarrow$ Statement I
(c) Statement I $\Leftrightarrow$ Statement II
(d) Neither Statement I $\Rightarrow$ Statement II nor Statement II $\Rightarrow$ Statement I

63. Consider the following statements :

- I. Let P be a point on a straight line L . Let Q, R and S be the points on the same plane containing the line L such that PQ, PR and PS are perpendicular to L . Then, there exists no triangle with vertices Q, R, S .
II. Let C be a circle passing through three distinct points D, E and F such that the tangent at D to the circle C is parallel to EF . Then, DEF is an isosceles triangle.

Which of the statement(s) given above is/are correct?

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

- 64.** The incircle of a $\triangle ABC$ touches the sides AB, BC and AC at the points P, Q, R , respectively, then which of the following statement is/are correct?

I. $AP + BQ + CR = PB + QC + RA$
 II. $AP + BQ + CR = \frac{1}{2}$ (perimeter of $\triangle ABC$)

III. $AP + BQ + CR = 3(AB + BC + CA)$

Select the correct answer using the codes given below

- (a) I, II and III are correct (b) Only I is correct
 (c) II and III are correct (d) I and II are correct

- 65.** Points X and Y are two different points on a circle. Point M is located so that line segment XM and line segment YM have equal length. Which of the following could be true?

I. M is the centre of the circle.
 II. M is on arc XY III. M is outside of the circle.
 (a) Only 1 (b) Only II (c) I and II (d) I, II and III

- 66.** Which of the following statement(s) are correct?

I. A circle is the locus of points equidistant from a fixed point.
 II. The locus of centre of circle rolling on the circumference of fixed circle is a circle concentric with fixed circle.
 III. The locus of the points which are equidistant from three distinct points on a line, is a line parallel to the given line.
 IV. If the bisectors of $\angle B$ and $\angle C$ of a quadrilateral $ABCD$ intersect in P , then P is equidistant from AB and CD .

Select the correct answer using the codes given below

- (a) All are correct (b) Only IV
 (c) Both I and II (d) I, II and IV

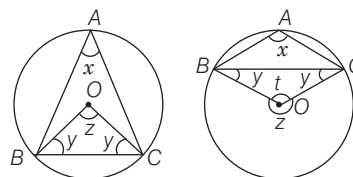
- 67.** A square and a circle intersect at more than one point. Consider the following statements:

I If atleast two of the intersection points are on the vertices of the square then the square is cyclic.
 II If there are exactly four intersection points then the square has less area than the circle.
 III If there are exactly four intersection point then the diameter of circle is equal to the diagonal of square.
 IV If atleast two intersection points are on the vertices of square. Then it may be possible to have the sides of square equal to the diameter of circle.

Which of the statement (s) given above is/are correct ?

- (a) Only I (b) Only IV
 (c) Both I and II (d) II, III and IV

Directions (Q. Nos. 68-69) BC is a chord with centre O . A is a point on an arc BC as shown in the figure below.



- 68.** What is the value of $\angle BAC + \angle OBC$ if A is the point on the major arc?

(a) 110° (b) 70° (c) 90° (d) 60°

- 69.** What is the value of $\angle BAC - \angle OBC$ if A is the point on the minor arc?

(a) 90° (b) 180° (c) 130° (d) 50°

PREVIOUS YEARS' QUESTIONS

- 70.** A circular ring with centre O is kept in the vertical position by two weightless thin strings TP and TQ attached to the ring at P and Q . The line OT meets the ring at E where as a tangential string at E meets TP and TQ at A and B , respectively. If the radius of the ring is 5 cm and $OT = 13$ cm, then what is the length of AB ? **2012 I**

(a) $10/3$ cm (b) $20/3$ cm (c) 10 cm (d) $40/3$ cm

- 71.** The locus of the mid-points of all equal chords in a circle is **2012 I**

(a) The circumference of the circle concentric with the given circle and having radius equal to the chords
 (b) The circumference of the circle concentric with the given circle and having radius equal to the distance of the chords from the centre
 (c) The circumference of the circle concentric with the given circle and having radius equal to half of the radius of the given circle
 (d) The circumference of the circle concentric with the given circle and having radius equal to half of the distance of the chords from the centre

- 72.** Consider the following statements

I. The locus of points which are equidistant from two parallel lines is a line parallel to both of them and drawn mid way between them.
 II. The perpendicular distances of any point on this locus line from two original parallel lines are equal. Further, no point outside this locus line has this property.

Which of the statement(s) given above is/are correct? **2012 II**

- (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II

- 73.** Consider a circle with centre at O and radius r . Points A and B lie on its circumference and a point M lies outside of it such that M, A and O lie on the same straight line. Then, the ratio of MA to MB is **2013 I**

(a) equal to 1 (b) equal to r
 (c) greater than 1 (d) less than 1

74. Consider the following statements :

- I. The perpendicular bisector of a chord of a circle does not pass through the centre of the circle.
 II. The angle in a semi-circle is a right angle.

Which of the statement(s) given above is/are correct? **2013 II**

- (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II

75. The diameter of a circle with centre at C is 50 cm. CP is a radial segment of the circle. AB is a chord perpendicular to CP and passes through P . CP produced intersects the circle at D . If $DP = 18$ cm, then what is the length of AB ?

2013 II

- (a) 24 cm (b) 32 cm
 (c) 40 cm (d) 48 cm

76. Consider the following statements in respect of two chords XY and ZT of a circle intersecting at P .

I. $PX \cdot PY = PZ \cdot PT$

II. PXZ and PTY are similar triangles.

Which of the statement(s) given above is/are correct? **2013 II**

- (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II

77. In a $\triangle ABC$, $AB = BC = CA$. The ratio of the radius of the circumcircle to that of the incircle is

- (a) 2 : 1 (b) 3 : 1 **2014 I**
 (c) 3 : 2 (d) None of these

78. AB and CD are two chords of a circle meeting externally at P . Then, which of the following is/are correct?

I. $PA \times PD = PC \times PB$

II. $\triangle PAC$ and $\triangle PDB$ are similar.

Select the correct answer using the codes given below **2014 I**

- (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II

79. For a triangle, the radius of the circumcircle is double the radius of the inscribed circle, then which one of the following is correct? **2014 II**

- (a) The triangle is a right angled
 (b) The triangle is an isosceles
 (c) The triangle is an equilateral
 (d) None of the above

80. If the chord of an arc of a circle is of length x , the height of the arc is y and the radius of the circle is z . Then, which one of the following is correct? **2014 II**

- (a) $y(2z - y) = x^2$ (b) $y(2z - y) = 4x^2$
 (c) $2y(2z - y) = x^2$ (d) $4y(2z - y) = x^2$

81. If the angle between the radii of a circle is 130° , then the angle between the tangents at the ends of the radii is **2015 I**

- (a) 90° (b) 70° (c) 50° (d) 40°

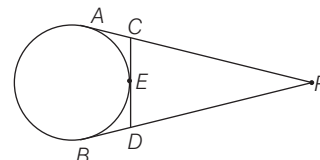
82. Out of two concentric circles, the diameter of the outer circle is 26 cm and the chord MN of length 24 cm is tangent to the inner circle. The radius of the inner circle is **2015 I**

- (a) 5 cm (b) 6 cm (c) 8 cm (d) 10 cm

83. AD is the diameter of a circle and AB is a chord. If $AD = 34$ cm, $AB = 30$ cm, then the distance of AB from the centre of the circle is **2015 I**

- (a) 17 cm (b) 15 cm (c) 13 cm (d) 8 cm

84. From an external point P , tangents PA and PB are drawn to the circle as shown in the above figure.



CD is the tangent to the circle at E . If $AP = 16$ cm, then the perimeter of the $\triangle PCD$ is equal to **2015 II**

- (a) 24 cm (b) 28 cm (c) 30 cm (d) 32 cm

85. Chord CD intersects the diameter AB of a circle at right angle at a point P in the ratio 1:2. If diameter of circle is d , then CD is equal to **2015 II**

- (a) $\frac{\sqrt{2}d}{3}$ (b) $\frac{2d}{3}$ (c) $\frac{2\sqrt{2}d}{3}$ (d) $\frac{2\sqrt{3}d}{3}$

86. The diameter of a wheel that makes 452 revolutions to move 2 km and 26 dm is equal to **2015 II**

- (a) $1\frac{9}{22}$ m (b) $1\frac{13}{22}$ m (c) $2\frac{5}{11}$ m (d) $2\frac{7}{11}$ m

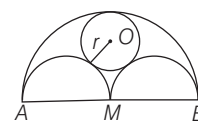
87. A boy is cycling such that the wheels of the cycle are making 140 revolutions per minute. If the radius of the wheel is 30 cm, then the speed of the cycle is **2015 II**

- (a) 15.5 km/h (b) 15.84 km/h (c) 16 km/h (d) 16.36 km/h

88. The two adjacent sides of a cyclic quadrilateral are 2 cm and 5 cm and the angle between them is 60° . If the third side is 3 cm, then the fourth side is of length **2015 II**

- (a) 2 cm (b) 3 cm (c) 4 cm (d) 5 cm

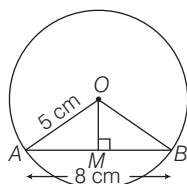
89. AB is a line segment of length $2a$, with M as mid-point. Semi-circles are drawn on one side with AM , MB and AB as diameter as shown in the above figure. A circle with centre O and radius r is drawn such that this circle touches all the three semi-circles. The value of r , is **2015 II**



- (a) $\frac{2a}{3}$ (b) $\frac{a}{2}$ (c) $\frac{a}{3}$ (d) $\frac{a}{4}$

HINTS AND SOLUTIONS

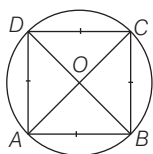
1. (c) Given, $OA = 5$ cm, $AM = \frac{1}{2} \times AB = \frac{8}{2}$
 $\Rightarrow AM = 4$ cm



In right angled $\triangle OMA$,
 $OM^2 = OA^2 - AM^2$
 [by pythagoras theorem]
 $= 5^2 - 4^2 = 25 - 16 = 25 - 16 = 9$
 $\Rightarrow OM = 3$ cm

2. (a) $\therefore \angle BOC = 2\angle A = 2 \times 60^\circ$
 (Angle subtended by an arc at the center is double the angle subtended by it at any point on the circumference of circle)
 $\Rightarrow \angle BOC = 120^\circ$

3. (d) A square has four equal side.
 $\therefore$ Each side subtends the same angle at the centre O.
 Let angle subtended be x° .

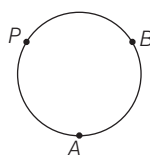


$$\text{So, } 4x^\circ = 360^\circ$$

$$x^\circ = \frac{360^\circ}{4} = 90^\circ$$

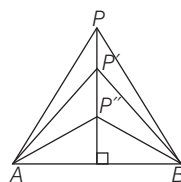
4. (b) Clearly, the point of concurrency of the perpendicular bisectors of the sides of a triangle is known as circumcentre.
5. (c) As, O is mid-point of diameter PQ and $\angle PRQ = 90^\circ$ (angle in semi-circle)
 So, $OS \parallel QR$ and $OS \perp PR$
 $\therefore \frac{PO}{PQ} = \frac{OS}{QR}$
 $\Rightarrow \frac{OS}{QR} = \frac{1}{2} \Rightarrow OS = \frac{1}{2} QR$
6. (d) $PQ = RS$ as chords equidistant from the centre are equal.
7. (a) The locus of the mid-points of all radii of a circle is a circle.
8. (c) The perpendicular bisectors of the sides of a triangle passes through the same point. The perpendicular bisectors are concurrent and point is called the circumcentre.

9. (a) Let P be the moving point.
 Then, $PA^2 + PB^2 = \text{constant}$



So, the locus of P is a circle.

10. (c) The locus of points equidistant from fixed points is a straight line which is the perpendicular bisector of the line joining the two fixed points.
 Here, A, B are fixed point and P is locus.

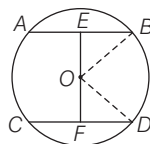


11. (c) Given that, $\Rightarrow \angle BOD = 50^\circ$
 $\Rightarrow \angle AOD = 180^\circ - \angle BOD = 130^\circ$
 [linear pair]

12. (a) Here, $\angle CAB = \angle CBD$
 [angles in alternate segments]
 and $\angle DAB = \angle CDB$
 [angles in alternate segments]
 $\Rightarrow \angle CAD = \angle CAB + \angle DAB$
 $= \angle CBD + \angle CDB$
 On adding $\angle CBD$ both sides, we get
 $\angle CAD + \angle CBD$
 $= \angle CBD + \angle CDB + \angle CBD$
 $= 180^\circ$ [angles of a triangle]

13. (d) Here, $BE = \frac{1}{2} AB = \frac{16}{2} \text{ cm} = 8 \text{ cm}$

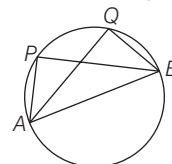
$OB = OD = 17 \text{ cm}$ (radii)



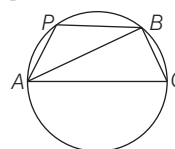
In right angled $\triangle OEB$,
 $OE = \sqrt{OB^2 - BE^2}$
 $= \sqrt{17^2 - 8^2} = \sqrt{289 - 64}$
 $= \sqrt{225} = 15 \text{ cm}$
 $\therefore OF = EF - OE = (23 - 15) = 8 \text{ cm}$
 In right angled $\triangle OFD$,
 $FD = \sqrt{OD^2 - OF^2}$
 $= \sqrt{17^2 - 8^2} = \sqrt{289 - 64}$
 $= \sqrt{225} = 15 \text{ cm}$
 $\therefore CD = 2FD = 30 \text{ cm}$

14. (c) There are two possibilities

Case I When P and Q are on the same side of AB. In this case $\angle APB = \angle AQB$ (angle in the same segment).



Case II When P and Q are on the opposite of AB. In this case PAQB is a cyclic quadrilateral.



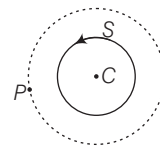
So, $\angle APB + \angle AQB = 180^\circ$

15. (b) Clearly, the locus of the centre of circles which passes through two given point is perpendicular bisector of the line joining the given points.

16. (c) As POQ is the diameter of the circle, therefore, $\angle PRQ$ and $\angle PSQ$ are two semicircles and $\angle PRQ = \angle PSQ$.
 $\angle PRQ = \angle PSQ = 90^\circ$
 [angles in semicircle]

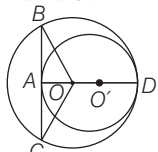
17. (a) $\angle COB = 360^\circ - (110^\circ + 90^\circ) = 160^\circ$
 $\therefore x = \angle CAB = \frac{1}{2} \angle COB$
 $= \frac{1}{2} \times 160^\circ = 80^\circ$

18. (b) If S is a circle with centre C and P be a movable point outside S, then the locus of P such that the tangent from P to S are of constant length is the circle through P with centre at C.

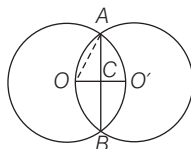


19. (a) $\angle CBF = \angle CDA$
 (angles in alternate segments)
 $\Rightarrow \angle CDA = 130^\circ$
 $\angle CDA + x = 180^\circ$ [linear pair]
 $x = 180^\circ - 130^\circ = 50^\circ$
 Hence, the value of x is 50° .
20. (c) $\angle ACB = \angle ADB = 32^\circ$
 [angle in same segment]
 $\angle ACD = \angle ABD = 50^\circ$
 $\therefore x = \angle BCD = \angle ACB + \angle ACD$
 $= 32^\circ + 50^\circ = 82^\circ$

21. (d) Here, $OB = OD = 3$ cm and
 $O'D = O'A = 2$ cm
 $OO' = OD - O'D = 3 - 2 = 1$ cm
 $\therefore OA = O'A - OO' = 2 - 1 = 1$ cm
 In $\triangle OAB$,
 $AB = \sqrt{OB^2 - OA^2}$
 [by pythagoras theorem]
 $= \sqrt{3^2 - 1^2} = \sqrt{9 - 1}$
 $= \sqrt{8}$ cm
 $\therefore$ Required length $= BC = 2AB$
 $= 2\sqrt{8}$ cm $= 4\sqrt{2}$ cm



22. (b) Here, let O, O' be the centres of the circles.
 As, the centre of each lies on the circumference of the other, therefore the two circles will have the same radius. Let the radius be r .
 $\therefore OC = O'C = \frac{r}{2}$



In right angled $\triangle OCA$, by pythagoras theorem

$$\begin{aligned} \therefore AC &= \sqrt{OA^2 - OC^2} \\ &= \sqrt{r^2 - \frac{r^2}{4}} = \frac{\sqrt{3}}{2} r \\ \Rightarrow AB &= 2AC = \sqrt{3} r \\ \therefore \frac{\text{Common chord}}{\text{Radius}} &= \frac{AB}{OA} = \frac{\sqrt{3}r}{r} \end{aligned}$$

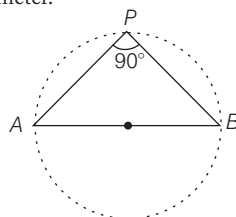
Hence, ratio of common chord to the radius is $\sqrt{3} : 1$.

23. (c) $\therefore \angle B + \angle D = 180^\circ$
 (sum of two opposite angle in cyclic quadrilateral)
 $\Rightarrow \angle D = 180^\circ - \angle B$
 $= 180^\circ - 92^\circ = 88^\circ$
 $\Rightarrow \angle DAE = \angle D = 88^\circ$ [alternate angle]
 $\Rightarrow \angle FAD = 88^\circ + 20^\circ = 108^\circ$
 $\Rightarrow \angle BCD = \angle FAD = 108^\circ$
 [angles in alternate segments]
 $\Rightarrow \angle BCD = 108^\circ$
24. (d) As, $\angle B + \angle D = 180^\circ$
 and $\angle A + \angle C = 180^\circ$
 [sum of two opposite angle in a cyclic quadrilateral]
 $x + 10 + 5y + 5 = 180^\circ$
 $\Rightarrow x + 5y = 165^\circ \dots(i)$
 $\therefore \angle A + \angle C = 180^\circ$
 $\Rightarrow 2x + 4 + 4y - 4 = 180^\circ$
 $\Rightarrow 2x + 4y = 180^\circ \dots(ii)$

On solving, Eqs. (i) and (ii), we get x and y are 40° and 25° , respectively.

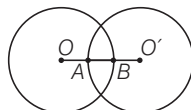
$$\therefore x + y = 40^\circ + 25^\circ = 65^\circ$$

25. (c) The locus of a point is the circumference of the circle with AB as diameter.



26. (b) As, the tangents drawn to a circle from a point outside it are equal. We have,
 $AD = AE$, $BD = BF$ and $CE = CF$
 $\Rightarrow AD = AB + BD = AB + BF$
 and $AD = AC + EC = AC + CF$
 $\therefore 2AD = AB + AC + (BF + CF)$
 $2AD = AB + AC + BC$

27. (d) $OB = 4$ cm



$$O'A = 2 \text{ cm}, OO' = 3 \text{ cm}$$

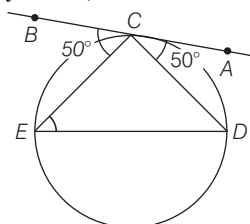
$$\text{As, } OO' \neq OB + O'A$$

So, circle does not touch each other externally.

$$\text{Also, } OO' \neq OB - O'A$$

So, circle does not touch internally, hence they intersect each other at two distinct points.

28. (a) Join ED , then



$$\angle DEC = \angle ACD = 50^\circ$$

[angle in alternate segment]

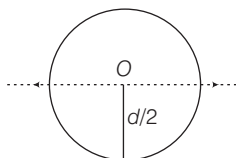
$$\angle EDC = \angle BCE = 50^\circ$$

[angle in alternate segment]

$$\therefore \angle DEC = \angle EDC$$

$$\text{So, } CD = CE$$

29. (b)



The locus of the centre of the front wheel of the cycle is a line parallel to the path of the wheel of the bicycle at a height $d/2$ cm.

30. (c) $\therefore \angle OAC = \angle OCA$
 [angle opposite to equal sides]
 $\therefore 2\angle OAC = 80^\circ$
 [external angle of $\triangle OAC$]
 $\angle OAC = \frac{80^\circ}{2} = 40^\circ \Rightarrow x = 40^\circ$

Hence, the value of x is 40° .

31. (a) Since, the direct common tangents of two circles divides the line joining their centres externally in the ratio of their radii. Here, both the circles being of equal radii. Hence, their ratio is $1 : 1$.

32. (a) Let $AB = 9$ cm, $BC = 7$ cm,

$$AC = 6$$
 cm

Let x, y and z be radii of circles with centre A, B, C .

$$x + y = 9 \dots(i)$$

$$y + z = 7 \dots(ii)$$

$$\text{and } z + x = 6 \dots(iii)$$

On adding Eqs. (i), (ii) and (iii), we get

$$2(x + y + z) = 22$$

$$\Rightarrow (x + y + z) = 11 \dots(iv)$$

On putting $x + y = 9$ in Eq. (iv), we get

$$9 + z = 11 \Rightarrow z = 2 \text{ cm}$$

On putting value of z in Eq. (ii), we get

$$y + 2 = 7 \Rightarrow y = 5$$

Again, putting value of y in Eq. (i), we get

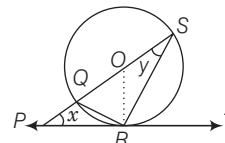
$$x + 5 = 9 \Rightarrow x = 4 \text{ cm}$$

So, radii are 4 cm, 5 cm and 2 cm.

33. (b) $\angle QRS = 90^\circ$ [angle in semi-circle]
 and $\angle QRP = \angle QSR = y^\circ$

[angle in alternate segments]

$$\text{Also, } \angle PRS = 90 + y^\circ$$



In $\triangle PRS$,

$$\angle SRP + \angle RPS + \angle PSR = 180^\circ$$

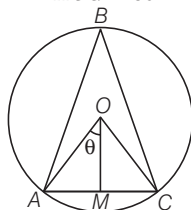
[$\therefore$ sum of all angles of triangle is 180°]

$$\Rightarrow (90 + y^\circ) + x^\circ + y^\circ = 180^\circ$$

$$\Rightarrow x + 2y^\circ = 90^\circ$$

34. (a) A line perpendicular to the given line, passing through the given point is the required locus.

35. (c) Given, $OA = 3$ cm and $AC = 3$ cm and $OA = OC = 3$ cm (radius of a circle)
In $\triangle AOC$, all sides are equal.
 $\therefore \triangle AOC$, is an equilateral triangle
 $\Rightarrow \angle AOC = 60^\circ$



Hence, $\angle ABC = \frac{1}{2} \angle AOC$

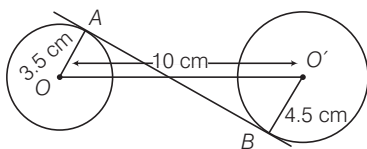
[Angle subtended by an arc at the centre is double the angle subtended by it at any point on the circle]
 $= \frac{60^\circ}{2} = 30^\circ$

36. (c) From figure, length of the transverse common tangent of these circles

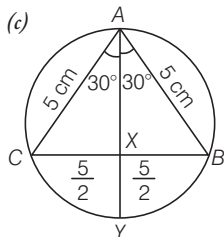
$$= \sqrt{(\text{distance between the centres of circles})^2 - (\text{sum of radius})^2}$$

$$= \sqrt{10^2 - (4.5 + 3.5)^2} = \sqrt{10^2 - 8^2}$$

$$= \sqrt{(100 - 64)} = \sqrt{36} = 6 \text{ cm}$$



37. (c)



In equilateral triangle, angle bisector and median are same

$$\therefore CX = BX = \frac{BC}{2} = \frac{5}{2} \text{ cm}$$

$$\text{In } \triangle CXA, \cos 30^\circ = \frac{AX}{AC} \Rightarrow \frac{\sqrt{3}}{2} = \frac{AX}{5}$$

$$\Rightarrow AX = \frac{\sqrt{3}}{2} \times 5 = \frac{5\sqrt{3}}{2} \text{ cm}$$

AY and BC are the chord of circle.

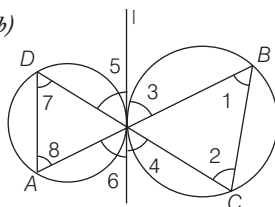
$$\therefore AX \cdot XY = BX \cdot XC$$

$$\Rightarrow \frac{5\sqrt{3}}{2} \cdot XY = \frac{5}{2} \cdot \frac{5}{2} \Rightarrow XY = \frac{5}{2\sqrt{3}}$$

$$\therefore AY \cdot AX = \left(\frac{5\sqrt{3}}{2} + \frac{5}{2\sqrt{3}} \right) \times \frac{5\sqrt{3}}{2}$$

$$= 25 \text{ cm}^2$$

38. (b)



l is the common tangent of two circles

Then, $\angle 1 = \angle 4$ and $\angle 5 = \angle 8$

[angle in alternate segment]

But $\angle 4 = \angle 5$ [opposite angles]

$$\therefore \angle 1 = \angle 8$$

Similarly $\angle 2 = \angle 7$

But these are alternate angles

$\Rightarrow AD \parallel BC \Rightarrow ACBD$ is a trapezium.

39. (c) We know that, two tangents drawn from an external point to a circle are equal in length.

$$\therefore AP = AS \dots (i)$$

$$BP = BQ \dots (ii)$$

$$CR = CQ \dots (iii)$$

$$DR = DS \dots (iv)$$

On adding Eqs. (i), (ii), (iii) and (iv), we get

$$(AP + BP) + (CR + DR)$$

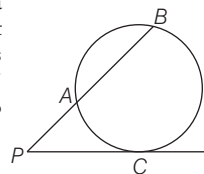
$$= (AS + DS) + (BQ + CQ)$$

$$\Rightarrow AB + CD = AD + BC$$

40. (a) If a secant to a circle intersect circle at points A and B and PC is a tangent to circle, then

$$PC^2 = PA \times PB$$

which is equivalent to saying that area of rectangle with PA and PB as sides is equal to the area of square with PC as sides.



41. (b) Given, $\angle BAD = 60^\circ$

and $\angle ADC = 105^\circ$

As ABCD is a cyclic quadrilateral

$$\therefore \angle DCP = \angle BAD = 60^\circ$$

and $\angle ADC + \angle CDP$

$$= 180^\circ \text{ [linear line]}$$

$$\Rightarrow 105^\circ + \angle CDP = 180^\circ$$

$$\Rightarrow \angle CDP = 75^\circ$$

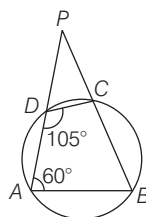
In $\triangle CPD$,

$$\angle DCP + \angle CDP + \angle DPC = 180^\circ$$

[by angle sum property of a triangle]

$$\Rightarrow 60^\circ + 75^\circ + \angle DPC = 180^\circ$$

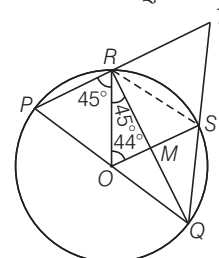
$$\Rightarrow \angle DPC = 180^\circ - 135^\circ = 45^\circ$$



42. (d) $\angle PRQ = 90^\circ$ as angle in a semi-circle is 90° .

Since, OR is a bisector of $\angle PRQ$.

$$\therefore \angle PRO = \angle ORQ = 45^\circ$$



Also, $OP = OR$ [radius]

$$\therefore \angle OPR = 45^\circ$$

in $\triangle ORS$, $OR = OS$

$$\Rightarrow \angle ORS = \angle OSR$$

in $\triangle ORS$, by angle sum property of a triangle,

$$\angle ORS + \angle OSR + \angle ROS = 180^\circ$$

$$2\angle ORS = 180^\circ - 44^\circ$$

$$2\angle ORS = 136^\circ$$

$$\therefore \angle ORS = \angle OSR = \frac{136^\circ}{2} = 68^\circ$$

$$\therefore \angle MRS = 68^\circ - 45^\circ = 23^\circ$$

$$\Rightarrow \angle PRS = 90^\circ + 23^\circ = 113^\circ$$

By properties of cyclic quadrilateral,

$$\angle PRS + \angle PQS = 180^\circ$$

$$\Rightarrow \angle PQS = 180^\circ - 113^\circ = 67^\circ$$

in $\triangle PTQ$,

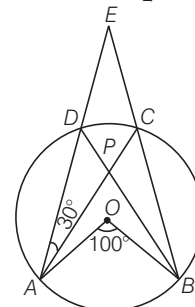
$$\angle QPT + \angle PQT + \angle PTQ = 180^\circ$$

[by angle sum property of a triangle]

$$\Rightarrow \angle PTQ = 180^\circ - 45^\circ - 67^\circ = 68^\circ$$

$$\therefore \angle RTS = \angle PTQ = 68^\circ$$

43. (b) Since, $\angle ADB = \frac{1}{2} \angle AOB = 50^\circ$



In $\triangle DPA$, by angle sum property of a triangle

$$\angle DAP + \angle ADP + \angle DPA = 180^\circ$$

$$\Rightarrow 30^\circ + 50^\circ + \angle DPA = 180^\circ$$

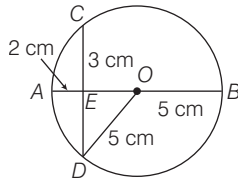
$$\Rightarrow \angle DPA = 100^\circ$$

Also, DPB is a straight line,

$$\therefore \angle DPA + \angle APB = 180^\circ$$

$$\Rightarrow \angle APB = 180^\circ - 100^\circ = 80^\circ$$

44. (b) In $\triangle OED$,



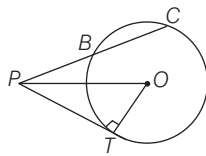
[by pythagoras theorem]

$$\begin{aligned} (OD)^2 &= (DE)^2 + (EO)^2 \\ \Rightarrow (5)^2 &= (DE)^2 + (3)^2 \\ \Rightarrow (DE)^2 &= 25 - 9 = 16 \\ \Rightarrow DE &= 4 \text{ cm} \end{aligned}$$

45. (a) Given, $PO = 10$ cm, $OT = 6$ cm and $PB = 5$ cm

In right angled $\triangle OTP$,

$$(OP)^2 = (PT)^2 + (OT)^2$$

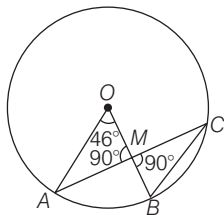


$$\begin{aligned} \Rightarrow (10)^2 &= (PT)^2 + 6^2 \\ \Rightarrow PT^2 &= 100 - 36 \Rightarrow PT^2 = 64 \\ \Rightarrow PT &= \sqrt{64} \\ \therefore PT &= 8 \text{ cm} \end{aligned}$$

By using theorem of circle,

$$\begin{aligned} (PT)^2 &= PB \times PC \\ \Rightarrow 8^2 &= 5 \times (BC + PB) \\ \Rightarrow 64 &= 5(BC + 5) \Rightarrow 5BC = 39 \\ \therefore BC &= 7.8 \text{ cm} \end{aligned}$$

46. (c) Given, $\angle AOB = 46^\circ$,



$$\therefore \angle ACB = \frac{1}{2} \angle AOB = \frac{1}{2} \times 46^\circ = 23^\circ$$

[angle subtended on the circumference is half of the angle subtend on centre.]

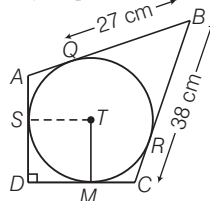
In $\triangle MCB$,

$$\angle C + \angle B + \angle BMC = 180^\circ$$

[by angle sum property of a triangle]

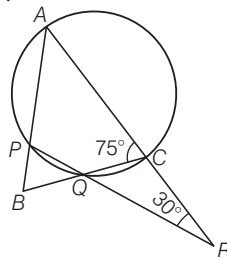
$$\begin{aligned} \Rightarrow 23^\circ + \angle B + 90^\circ &= 180^\circ \\ \Rightarrow \angle B &= 67^\circ \end{aligned}$$

47. (b) We know that, the tangents drawn to the circle from a point outside the circle are always equal.



$$\begin{aligned} \therefore BQ &= BR = 27 \text{ cm} \\ \Rightarrow RC &= BC - BR \\ &= 38 - 27 = 11 \text{ cm} \\ \therefore RC &= CM = 11 \text{ cm} \\ \text{Now, } DM &= 25 - 11 = 14 \text{ cm} \\ \therefore ST &= DM = 14 \text{ cm} \end{aligned}$$

48. (d) We know that, the sum of opposite angles in a cyclic quadrilateral is always 180° .



$$\begin{aligned} \angle ACQ + \angle APQ &= 180^\circ \\ \Rightarrow 75^\circ + \angle APQ &= 180^\circ \\ \Rightarrow \angle APQ &= 180^\circ - 75^\circ = 105^\circ \\ \text{Since, } \angle APQ + \angle BPQ &= 180^\circ \end{aligned}$$

[linear pair]

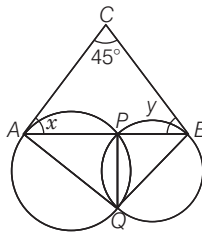
$$\begin{aligned} \therefore 105^\circ + \angle BPQ &= 180^\circ \\ \Rightarrow \angle BPQ &= 180^\circ - 105^\circ = 75^\circ \\ \text{Since, } \angle ACQ \text{ is an exterior angle of } \triangle RCQ. \end{aligned}$$

$$\begin{aligned} \therefore \angle ACQ &= \angle CRQ + \angle CQR \\ \Rightarrow 75^\circ &= 30^\circ + \angle CQR \\ \Rightarrow \angle CQR &= 45^\circ = \angle PQB \end{aligned}$$

[vertically opposite angles]

$$\begin{aligned} \text{In } \triangle BPQ, \angle B + \angle P + \angle Q &= 180^\circ \\ \text{[by angle sum property of a triangle]} \\ \Rightarrow \angle B + 75^\circ + 45^\circ &= 180^\circ \Rightarrow \angle B = 60^\circ \end{aligned}$$

49. (d)



$$\begin{aligned} \text{In } \triangle CAB, 45^\circ + \angle x + \angle y &= 180^\circ \\ \text{[by angle sum property of triangle]} \end{aligned}$$

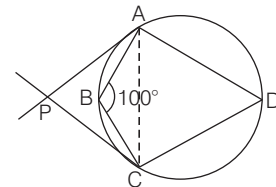
$$\Rightarrow \angle x + \angle y = 180^\circ - 45^\circ = 135^\circ \dots (i)$$

$$\angle x = \angle AQP \text{ and } \angle y = \angle BQP$$

[angles in alternate segment]

$$\begin{aligned} \therefore \angle AQB &= \angle AQP + \angle BQP \\ &= \angle x + \angle y = 135^\circ \end{aligned}$$

50. (b) We know that, the sum of opposite angles of a cyclic quadrilateral is always 180° .



$$\begin{aligned} \therefore \angle B + \angle D &= 180^\circ \\ \Rightarrow 100 + \angle D &= 180^\circ \Rightarrow \angle D = 80^\circ \\ \therefore \angle ACP &= \angle PAC = 80^\circ \end{aligned}$$

[by theorem of alternate interior segment]

In $\triangle PAC$,

$$\angle P + \angle PAC + \angle PCA = 180^\circ$$

[by angle sum property of a triangle]

$$\begin{aligned} \Rightarrow \angle P + 80^\circ + 80^\circ &= 180^\circ \\ \Rightarrow \angle P &= 180^\circ - 160^\circ = 20^\circ \end{aligned}$$

51. (b) Since, the length of tangents from an external point to a circle are equal

$$\therefore CR = CS = 27 \text{ cm}$$

$$\therefore BR = (BC - CR) = 38 - 27 = 11 \text{ cm}$$

$$\Rightarrow BQ = BR = 11 \text{ cm}$$

join OQ, then PAQO is a rectangle.

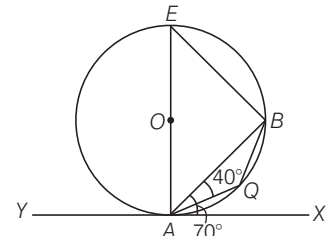
$$\therefore AQ = PQ = 10 \text{ cm}$$

$$\text{So, } AB = (AQ + BQ) = (10 + 11) = 21 \text{ cm}$$

52. (b) Given, $\angle BAX = 70^\circ$

$$\text{and } \therefore \angle BEA = 70^\circ$$

[angles in alternate segment]



Since, AQBE is a cyclic quadrilateral.

$$\therefore \angle AEB + \angle AQB = 180^\circ$$

$$\Rightarrow \angle AQB = 180^\circ - 70^\circ = 110^\circ$$

In $\triangle ABQ$,

$$\angle ABQ + \angle BAQ + \angle AQB = 180^\circ$$

$$\Rightarrow \angle ABQ = 180^\circ - (40^\circ + 110^\circ)$$

$$= 180^\circ - 150^\circ = 30^\circ$$

53. (a) We know that, the angles in the same segment of a circle are equal.

$$\therefore \angle XBY = \angle XAY = 45^\circ$$

In $\triangle BXY$, by angle sum property of a triangle

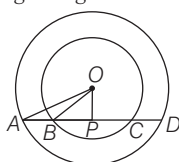
$$\angle BXY + \angle XBY + \angle BYX = 180^\circ$$

$$\Rightarrow 50^\circ + 45^\circ + \angle BYX = 180^\circ$$

$$[\because \angle BXY = 50^\circ]$$

$$\Rightarrow \angle BYX = 180^\circ - 95^\circ = 85^\circ$$

54. (a) In right angled $\triangle OPB$,



[by pythagoras theorem]

$$OB^2 = OP^2 + BP^2$$

$$\Rightarrow (15)^2 = (12)^2 + BP^2$$

$$\Rightarrow 225 - 144 = BP^2$$

$$\Rightarrow (BP)^2 = 81 \Rightarrow BP = 9 \text{ cm}$$

and in right angled $\triangle AOP$, by pythagoras theorem,

$$(OA)^2 = (OP)^2 + (AP)^2$$

$$\Rightarrow (20)^2 = (12)^2 + (AP)^2$$

$$\Rightarrow (AP)^2 = 400 - 144 = 256$$

$$\Rightarrow AP = 16 \text{ cm}$$

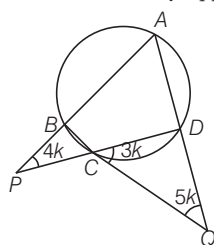
$$\text{Hence, } AB = AP - BP = 16 - 9 = 7 \text{ cm}$$

55. (b) Given, $\frac{x}{3} = \frac{y}{4} = \frac{z}{5} = k$ [say]

$$\therefore x = 3k, y = 4k \text{ and } z = 5k$$

$$\text{Since, } \angle DCQ = \angle BCP = 3k$$

[vertically opposite angle]



In $\triangle DCQ$, by angle sum property of a triangle

$$\angle CDQ = 180^\circ - (3k + 5k) = 180^\circ - 8k$$

By properties of cyclic quadrilateral,

$$\angle CDQ = \angle CBA = 180^\circ - 8k$$

$$\Rightarrow \angle PBC = 8k$$

In $\triangle PBC$, by angle sum property of a triangle

$$\angle P + \angle B + \angle C = 180^\circ$$

$$\therefore 4k + 8k + 3k = 180^\circ$$

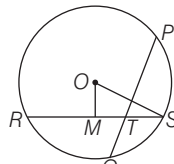
$$\Rightarrow k = \frac{180^\circ}{15} \Rightarrow k = 12^\circ$$

$$\therefore x = 36^\circ, y = 48^\circ, z = 60^\circ$$

56. (b) We know that, angle subtended on the circumference is half of the angle subtended at the centre.

$$\angle AQP = \frac{1}{2} \angle AOP = \frac{1}{2} 75^\circ = 37.5^\circ$$

57. (b)



Given, $OM = 3 \text{ cm}$, $OS = 5 \text{ cm}$.

$$MS = 4 \text{ cm} = RM$$

[Using pythagoras theorem]

$$\Rightarrow RS = 8 \text{ cm}$$

We know that, $TS = 1/3$ of RT

$$\Rightarrow TS = 1/4 \text{ of } RS, \text{ if } RS = 8 \text{ cm}$$

Then, $TS = 2 \text{ cm}$

Also, $RT \times TS = PT \times TQ$

[When there are two intersecting chords, the product of the segments of one chord is equal to the product of the segments of other.]

$$\Rightarrow 6 \times 2 = PT \times TQ$$

$$\Rightarrow PT \times TQ = 12$$

Now, by $AM \geq GM$ inequality,

$$(PT + TQ) / 2 \geq \sqrt{(PT \times TQ)}$$

$$(PT + TQ) / 2 \geq \sqrt{12}$$

$$\Rightarrow PT + TQ \geq 2\sqrt{12} \Rightarrow PQ \geq 2\sqrt{12}$$

$$\text{or } PQ \geq 4\sqrt{3}$$

$$\therefore \text{Minimum } PQ = 4\sqrt{3}$$

58. (a) When 2 chords AB and CD intersect at P then

$$AP \times PB = CP \times PD$$

Hence,

$$4 \times 6 = 3 \times PD, \text{ Thus,}$$

$$PD = 8 \text{ cm}$$

$$\text{Now } AB = AP + PB$$

$$= 10 \text{ cm}$$

$$\text{and } CD = CP + PD$$

$$\text{Thus, } CD = 11 \text{ cm}$$

Consider the circle with center O .

Drop a perpendicular from O to chord AB and CD . This will bisect the chords at X and Y .

$$\therefore AX = \frac{AB}{2} = 5 \text{ cm}$$

$$\text{and } YD = \frac{CD}{2} = \frac{11}{2} \text{ cm}$$

$$\text{Now, } PX = AX - AP = (5 - 4) = 1 \text{ cm}$$

$$OY = PX = 1 \text{ cm}$$

[$\because PXOY$ is a rectangle]

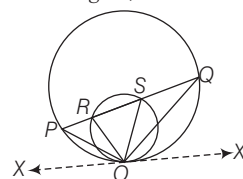
In right triangle OYD

$$OD = \sqrt{OY^2 + YD^2}$$

[by pythagoras theorem]

$$= \sqrt{1^2 + \left(\frac{11}{2}\right)^2} = \frac{5\sqrt{5}}{2} \text{ cm}$$

59. (a) In the figure,



Let $\angle POX = x^\circ$, $\angle ROP = y^\circ$

$$\angle POX = \angle PQO = x$$

$$\text{and } \angle ROX = \angle RSO = x^\circ + y^\circ$$

[angles in alternate segment]

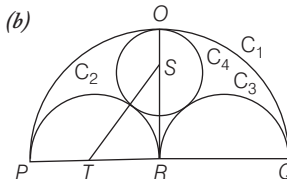
Now, in $\triangle SQO$, $\angle RSO$ is the exterior angle

$$\therefore \angle SQO + \angle SOQ = x^\circ + y^\circ$$

$$\Rightarrow x + \angle SOQ = x^\circ + y^\circ$$

$$\Rightarrow \angle SOQ = y = \angle ROP$$

60. (b)



Assume that the radius of $C_4 = r$ and $PQ = k$

$$\text{Now, } PR = k/2 = RQ = RO$$

$$\Rightarrow RS = (k/2) - r, RT = k/4$$

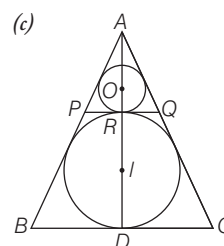
$$ST = (k/4) + r$$

Applying pythagoras theorem in triangle STR .

$$\left(\frac{k}{4} + r\right)^2 = \left(\frac{k}{4}\right)^2 + \left(\frac{k}{2} - r\right)^2$$

$$\Rightarrow r = k/6 = PQ/6$$

61. (c)



In-radius of equilateral triangle of side

$$a = \frac{a}{2\sqrt{3}}$$

$$\text{Diameter of larger circle} = \frac{a}{\sqrt{3}}$$

Let us say common tangent PQ touches the two circle at R , centre of smaller circle is I .

Now, PQ is parallel to BC . AR is perpendicular to PQ . Triangle PQR is also an equilateral triangle and $AORID$ is a straight line.

$$AD = \frac{\sqrt{3}}{2}a, RD = \frac{a}{\sqrt{3}}, AR = \frac{\sqrt{3}}{2}a - \frac{a}{\sqrt{3}}$$

$$AR = \frac{3a - 2a}{2\sqrt{3}} = \frac{a}{2\sqrt{3}}, AR = \frac{1}{3}AD$$

Radius of smaller circle = $1/3$ radius of larger circle

Radius of smaller circle

$$= \frac{1}{3} \cdot \frac{a}{2\sqrt{3}} = \frac{a}{6\sqrt{3}}$$

Area of smaller circle = πr^2

$$\pi \left(\frac{a}{6\sqrt{3}} \right)^2 = \frac{\pi a^2}{108}$$

$$\text{Area of triangle} = \frac{\sqrt{3}}{4} a^2$$

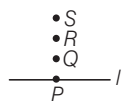
$$\text{Ratio} = \frac{\pi a^2}{108} : \frac{\sqrt{3}}{4} a^2, \pi : 27\sqrt{3}$$

62. (d) I. It is true that the opposite angles of a cyclic quadrilateral are supplementary.

II. It is also true that the angle subtended by an arc at the centre is double the angle subtended by it at any point on the remaining part of the circle.

Hence, both statements are individually true, but neither statements implies to each other.

63. (c) I. It is clear from the figure that points Q, S and R are in a straight line.

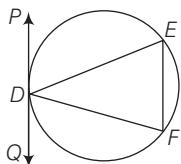


So, statement I is true.

II. Since, PQ is parallel to EF .

$$\therefore \angle PDE = \angle DEF \text{ [alternate angle]}$$

$$\text{Also, } \angle PDE = \angle EFD$$



[angle in the alternate segment of chord ED]

$$\therefore \angle DEF = \angle DFE$$

$$\Rightarrow \triangle DEF \text{ is an isosceles triangle.}$$

So, statement II is also true.

Hence, the statement I and II are correct.

64. (d) As, the tangents drawn from an external point to a circle are equal.

$$\therefore AP = AR \dots (i)$$

$$BQ = BP \dots (ii)$$

$$\text{and } CR = QC \dots (iii)$$

On adding Eqs. (i), (ii) and (iii), we get

$$AP + BQ + CR = BP + QC + RA$$

and perimeter of

$$\triangle ABC = AB + BC + CA$$

$$= (AP + PB) + (BQ + QC) + (CR + RA)$$

$$= (AP + BQ) + (BQ + CR) + (CR + AP)$$

$$= 2(AP + BQ + CR)$$

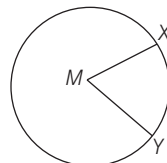
$$\therefore AP + BQ + CR = 1/2$$

[perimeter of $\triangle ABC$]

Hence, the statement I and II are correct.

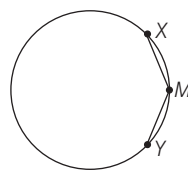
65. (d) I. Consider the possibility that M

could be the centre of the circle. We know this could be true because M being the center point of the circle would make line XM and YM radii of the circle, which would mean that they are equal.



II. The statement presents us with the possibility that

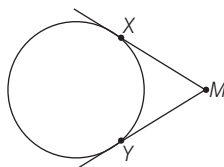
point M lies somewhere on the arc of XY . Well, if point M rested exactly halfway



between X and

Y , then straight lines drawn from X to M and Y to M would certainly be equal.

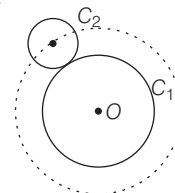
III. The statement presents us with the possibility that M lies somewhere outside of the circle. If two tangents are drawn from point M at X and Y they will be equal.



Hence, the statement I, II and III are correct.

66. (d) I. Circle is a locus of points which are equidistant from a fixed point.

II. Let C_1 be a fixed circle and C_2 be moving circle then locus of centre of C_2 is also a circle.



III. There does not exist any point which is equidistant from three distinct points on a line.

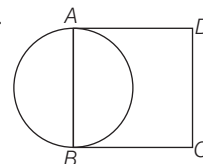
IV. If P lies on bisector of $\angle B$ then P is equidistant from AB and BC also, P lies on bisector of $\angle C$.

$$\Rightarrow P \text{ is equidistant from } BC \text{ and } CD$$

$$\Rightarrow P \text{ is equidistant from } AB \text{ and } CD.$$

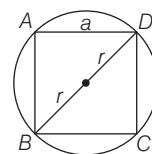
Hence, I, II, and IV are correct statement.

67. (d) I.



So, the square is not cyclic.

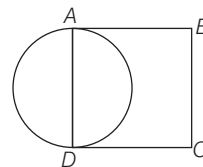
II.



Square has less area than circle.

III. From the above figure it is clear that the diameter of circle is equal to the diagonal of square. $\Rightarrow 2r = a\sqrt{2}$

IV.



So, it is possible that the sides of square is equal to diameter of circle.

Hence statement II, III and IV are correct.

68. (c) We observe that the arc BC makes $\angle BOC = z$ at the centre and $\angle BAC = x$ at a point on the circumference.

$$\therefore z = 2x$$

In $\triangle OBC$, we have

$$\angle OBC + \angle OCB + \angle BOC = 180^\circ$$

$$\Rightarrow y + z + y = 180^\circ$$

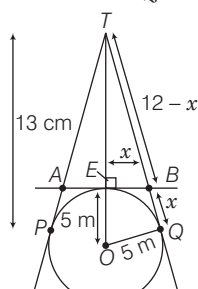
$$\Rightarrow z + 2y = 180^\circ \Rightarrow 2x + 2y = 180^\circ$$

$$\Rightarrow x + y = 90^\circ$$

$$\Rightarrow \angle BAC + \angle OBC = 90^\circ$$

69. (a) In $\triangle OBC$, we have
 $\angle OBC + \angle OCB + \angle BOC = 180^\circ$
 $\Rightarrow y + y + t = 180^\circ \Rightarrow t = 180^\circ - 2y$
 Now, $z = 360^\circ - t \Rightarrow z = 360^\circ - (180^\circ - 2y)$
 $\Rightarrow 2x = 180^\circ + 2y \Rightarrow 2x - 2y = 180^\circ$
 $[\because z = 2x]$
 $\Rightarrow x - y = 90^\circ$
 $\Rightarrow \angle BAC - \angle OBC = 90^\circ$

70. (b) In right angled $\triangle OTQ$,
 $OT^2 = OQ^2 + TQ^2$



$$\Rightarrow (13)^2 = (5)^2 + (TQ)^2$$

$$\Rightarrow TQ^2 = 169 - 25 = 144 \Rightarrow TQ = 12\text{cm}$$

In right angled $\triangle TEB$,

$$TB^2 = EB^2 + TE^2$$

$$\because (EB = BQ)$$

$$\Rightarrow (12 - x)^2 = BQ^2 + TE^2$$

$$\Rightarrow 144 + x^2 - 24x = x^2 + (8)^2$$

$$\Rightarrow 144 + x^2 - 24x = x^2 + 64$$

$$\Rightarrow 24x = 80 \Rightarrow x = \frac{20}{6} = \frac{10}{3}\text{ cm}$$

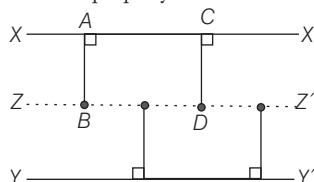
$$\therefore AB = 2EB = 2x = 2 \times \frac{10}{3} = \frac{20}{3}\text{ cm}$$

Hence, the length of AB is $\frac{20}{3}$ cm.

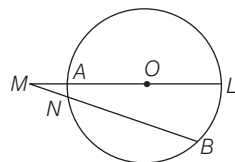
71. (b) The locus obtained is the circumference of the circle concentric with the given circle and having radius equal to the distance of the chords from the centre.

72. (c) Statements I and II are both correct, because the locus of points which are equidistant from two parallel lines is a line parallel to both of them and lie mid way between them.

Also, it is true that the perpendicular distances of any point on this locus line from two original parallel lines are equal. further, no point outside this locus line has this property.



73. (d) Since, secants LA and BN are intersecting at an exterior point M , then



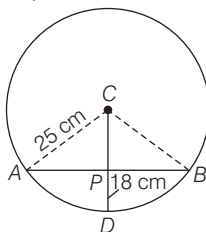
$$LM \times AM = BM \times NM$$

$$\Rightarrow \frac{MA}{MB} = \frac{MN}{LM} < 1$$

74. (b) The perpendicular bisector of the chord of a circle always passes through the centre. So, statement I is wrong.

The angle in a semi-circle is a right angle. So, statement II is correct.

75. (d) Here, $CA = CD = CB = 25$ cm



and $DP = 18$ cm

In right angled $\triangle ACP$,

$$CP = CD - PD = 25 - 18 = 7\text{cm}$$

Now, $AC^2 = CP^2 + AP^2$

[by pythagoras theorem]

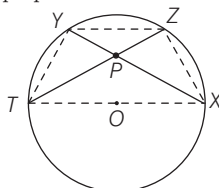
$$\therefore AP = \sqrt{AC^2 - CP^2} = \sqrt{(25)^2 - (7)^2}$$

$$= \sqrt{625 - 49} = \sqrt{576} = 24\text{ cm}$$

Similarly, $PB = 24$ cm

$$\therefore AB = AP + PB = 24 + 24 = 48\text{ cm}$$

76. (c) When two chords of a circle intersect internally, then they are divided in a proportion.



$$\text{i.e. } PX \cdot PY = PZ \cdot PT$$

In $\triangle PXZ$ and $\triangle PTY$, $\angle ZPX = \angle YPT$

[vertically opposite angles]

$$\angle PZX = \angle PYT$$

[angles in same segment]

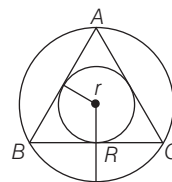
$$\angle PXZ = \angle PTY$$

[angles in same segment]

$$\therefore \triangle PXZ \sim \triangle PTY$$

Hence, the both statements are correct.

77. (a) In $\triangle ABC$, $AB = BC = AC$



Hence, $\triangle ABC$ is equilateral triangle.

Let r be the radius of incircle and R be the radius of circumcircle.

$$\text{Now, radius of incircle, } r = \frac{\text{side}}{2\sqrt{3}} = \frac{AB}{2\sqrt{3}}$$

and radius of circumcircle,

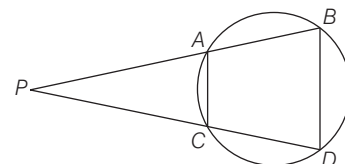
$$R = \frac{\text{side}}{\sqrt{3}} = \frac{AB}{\sqrt{3}}$$

$$\text{So, the required ratio} = \frac{R}{r} = \frac{AB / \sqrt{3}}{AB / 2\sqrt{3}}$$

$$= \frac{2}{1} = 2 : 1$$

78. (b) I. AB and CD are chords when produced meet externally at P .

$$\therefore AP \times BP = CP \times DP$$



II. Since, $ABDC$ is a cyclic quadrilateral

$$\angle PAC = \angle PDB$$

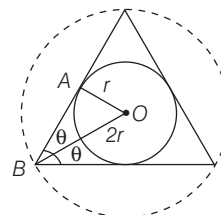
$$\text{and } \angle PCA = \angle PBD$$

Also, $\angle APC = \angle BPD$ [common]

$$\therefore \triangle PAC \sim \triangle PDB$$

So, statement II is true.

79. (c) Let $OA = r$ be the inradius of circle. Then, circumradius = $OB = 2r$



We know that, inradius is the perpendicular distance of centre O from side of triangle and circumradius OB bisect $\angle B$. Again, let $\angle OBA = \theta$

In right angled $\triangle OAB$,

$$\sin \theta = \frac{OA}{OB} = \frac{r}{2r}$$

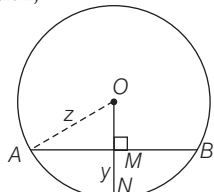
$$\Rightarrow \sin \theta = \frac{1}{2} = \sin 30^\circ \quad \left[\because \sin 30^\circ = \frac{1}{2} \right]$$

$$\therefore \theta = 30^\circ$$

$$\text{Then, } \angle B = 2 \times \theta = 2 \times 30^\circ = 60^\circ$$

Hence, given triangle is an equilateral triangle.

80. (d) Let O be the centre of circle and AB be the chord of an arc. According to the question,



Length of chord $AB = x$, Radius of circle $= OA = z$
and height of arc $MN = y$
Now, $AM = MB = \frac{AB}{2} = \frac{x}{2}$

[Perpendicular from the centre to a chord bisects the chord]
and $OM = ON - MN$
 $= z - y$ [$\because ON = z$ (radius)]

In right angled $\triangle OMA$,
 $OA^2 = OM^2 + AM^2$
[by pythagoras theorem]

$$\Rightarrow z^2 = (z - y)^2 + \left(\frac{x}{2}\right)^2$$

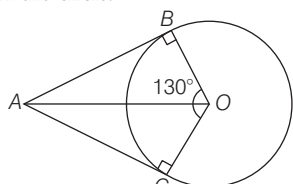
$$\Rightarrow z^2 = z^2 + y^2 - 2yz + \frac{x^2}{4}$$

$$[\because (a - b)^2 = a^2 + b^2 - 2ab]$$

$$\Rightarrow 2yz - y^2 = \frac{x^2}{4} \Rightarrow 4(2yz - y^2) = x^2$$

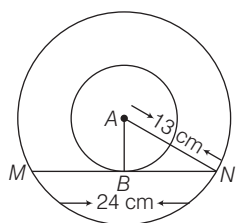
$$\Rightarrow 4y(2z - y) = x^2$$

81. (c) Given, AB and AC are the tangents of the circle.



and $OB = OC$ [radius of circle]
 $\therefore \angle B = \angle C = 90^\circ$
[radius is perpendicular to the tangent]
In quadrilateral $ABOC$,
 $\angle ABO + \angle OCA + \angle BAC + \angle BOC = 360^\circ$
 $\Rightarrow 90^\circ + 90^\circ + \angle BAC + 130^\circ = 360^\circ$
 $\Rightarrow \angle BAC = 180^\circ - 130^\circ = 50^\circ$
Hence, the angle between the tangents at the ends of the radius is 50° .

82. (a) Given, $MN = 24$ cm and
 $AN = \frac{26}{2}$ cm $= 13$ cm



Now, we draw a perpendicular bisector from A to MN , which meets MN at B .

Then, $MB = BN = \frac{MN}{2} = \frac{24}{2} = 12$ cm

[$\because$ perpendicular from centre to the chord bisects the chord]

In right angled $\triangle ABN$,
 $AN^2 = AB^2 + BN^2$
 $\Rightarrow 13^2 = AB^2 + 12^2$
 $\Rightarrow 169 = AB^2 + 144$
 $\Rightarrow AB^2 = 169 - 144 = 25$
 $\Rightarrow AB = 5$ cm

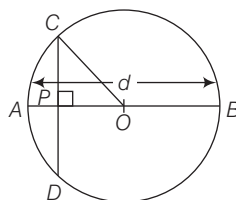
$\therefore$ Radius of the inner circle is 5 cm.

83. (d) Given, $AD = 34$ cm and $AB = 30$ cm
Let C be a point on AB such that $OC \perp AB$

$\Rightarrow OD = OB = 17$ cm
Also, $CB = \frac{AB}{2} = 15$ cm
[Perpendicular from centre to the chord bisects the chord]
 $\Rightarrow OC = \sqrt{OB^2 - CB^2} = \sqrt{17^2 - 15^2}$
 $= \sqrt{289 - 225} = \sqrt{64} = 8$ cm

84. (d) Since, $AP = BP$; $DE = BD$ and $AC = CE$
 $\therefore$ Perimeter of $\triangle PCD$
 $= PC + PD + CD$
 $= PC + PD + CE + DE$
 $= (PC + CE) + (PD + DE)$
 $= (PC + CA) + (PD + BD)$
 $= PA + PB = 2PA = 2 \times 16 = 32$ cm

85. (c) Given, $AB = d$ is diameter of circle and CD intersect AB at P in the ratio 1 : 2.



$\therefore AP = \frac{d}{3}$ and $PB = \frac{2d}{3}$
 $\Rightarrow OA = OC = OB = \frac{d}{2}$
[$\because$ radius of circle $= \frac{1}{2}$ diameter]

In $\triangle OPC$, $OP = OA - AP = \frac{d}{2} - \frac{d}{3} = \frac{d}{6}$
Applying pythagoras theorem in $\triangle OPC$,
 $OC^2 = OP^2 + PC^2$
 $\Rightarrow \left(\frac{d}{2}\right)^2 = \left(\frac{d}{6}\right)^2 + PC^2$
 $\Rightarrow (PC)^2 = \frac{d^2}{4} - \frac{d^2}{36}$
 $= \frac{9d^2 - d^2}{36} = \frac{8d^2}{36}$

$$\Rightarrow PC = \frac{2\sqrt{2}d}{6} = \frac{\sqrt{2}d}{3}$$

$$\therefore CD = 2PC = \frac{2\sqrt{2}d}{3}$$

86. (b) 1 km = 1000 m, 1 dm = 10 m
 $\therefore 2$ km and 26 dm
 $= 2000 + 260 = 2260$ m

Let radius be r m.

Distance covered in revolution $= 2\pi r$

Distance covered in 452 revolution

$$= 452 \times 2\pi r$$

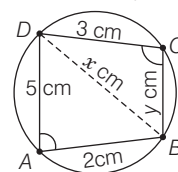
$$\therefore 452 \times 2\pi r = 2260$$

$$\therefore r = \frac{2260 \times 7}{452 \times 2 \times 22} = \frac{35}{44}$$

$$\text{Diameter} = 2 \times \frac{35}{44} = \frac{35}{22} = 1\frac{13}{22} \text{ m}$$

87. (b) Distance covered in 1 revolution
 $= 2\pi (0.3) \text{ m} = 0.6\pi \text{ m}$
 $\therefore$ Distance covered in 140 revolutions
 $= 140 \times 0.6\pi \text{ m}$
 $\Rightarrow$ Speed the cycle
 $= 140 \times 0.6 \times \frac{22}{7} \times 60 \text{ m/h}$
 $= 15840 \text{ m/h} = 15.84 \text{ km/h}$

88. (a) Let $ABCD$ be cyclic quadrilateral.



in which $\angle A = 60^\circ$,

$$AB = 2 \text{ cm}, AD = 5 \text{ cm}$$

Since, $\angle A + \angle C = 180^\circ$

[property of cyclic quadrilateral]

$$\Rightarrow 60^\circ + \angle C = 180^\circ$$

$$\Rightarrow \angle C = 180^\circ - 60^\circ = 120^\circ$$

In $\triangle ABD$, let BD be x cm.

$$\cos 60^\circ = \frac{2^2 + 5^2 - x^2}{2 \times 2 \times 5}$$

[by cosine formula]

$$\Rightarrow \frac{1}{2} = \frac{4 + 25 - x^2}{2 \times 2 \times 5} \Rightarrow 29 - x^2 = 10$$

$$\Rightarrow x^2 = 29 - 10 \Rightarrow x^2 = 19$$

Again, in $\triangle BCD$,

$$\cos 120^\circ = \frac{3^2 + y^2 - x^2}{2 \times 3 \times y}$$

[by cosine formula]

Let length of fourth side of cyclic quadrilateral be y cm.

$$\Rightarrow -\frac{1}{2} = \frac{9 + y^2 - 19}{2 \times 3 \times y} \quad [\because x^2 = 19]$$

$$\Rightarrow -3y = 9 + y^2 - 19$$

$$\begin{aligned}
 &\Rightarrow y^2 + 3y - 10 = 0 \\
 &\Rightarrow y^2 + 5y - 2y - 10 = 0 \\
 &\Rightarrow y(y + 5) - 2(y + 5) = 0 \\
 &\Rightarrow (y + 5)(y - 2) = 0 \\
 &\Rightarrow y + 5 = 0 \text{ or } y - 2 = 0 \\
 &\therefore y = 2 \text{ cm}
 \end{aligned}$$

89. (c) Since, two circles touch each other externally if distance between their centres = Sum of their radii

$$O_1O_2 = r + \frac{a}{2} \text{ and } O_2M = a - r$$

$$O_1M = \frac{a}{2}$$

In right $\triangle O_1MO_2$

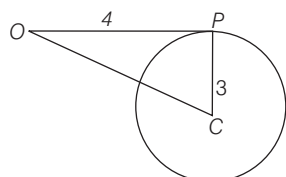
$$(O_1O_2)^2 = (O_1M)^2 + (O_2M)^2$$

$$\Rightarrow \left(r + \frac{a}{2}\right)^2 = \frac{a^2}{4} + (a - r)^2$$

$$\Rightarrow 3r = a$$

$$\therefore r = \frac{a}{3}$$

90. (c) Since, OP is tangent to the circle $OP \perp PC$.



Here, $\triangle OPC$ is right angled triangle with $\angle P = 90^\circ$

$$OC = \sqrt{(3)^2 + (4)^2} = 5 \text{ units}$$

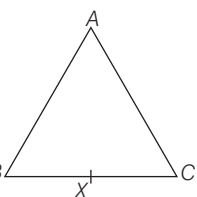
$$\therefore \sin(\angle COP) = \frac{3}{5}$$

91. (d) In $\triangle ABC$, $AB = AC$ i.e. $\triangle ABC$ is isosceles triangle.

If L is the locus of points X inside or on the triangle, such that $BX = CX$. Then, L is a straight line passing through A .

As, AX is bisecting the line BC , then it passes through the centroid.

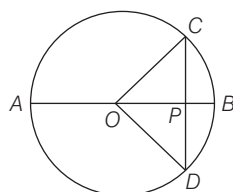
As, $AB = AC$ in $\triangle ABC$, then AX is perpendicular to BC , hence it passes through the orthocentre.



In $\triangle ABC$, $AB = AC$ and $AX \perp BC$. So, AX is angle bisector of $\angle A$, hence it passes through the incentre. Hence, all statements are correct.

92. (b) Given, radius of a circle = 2 unit

$$\therefore CP = PD = \frac{CD}{2} = \frac{2}{2} = 1 \text{ unit}$$

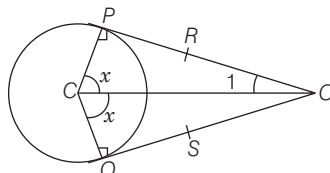


$$\text{Now, } OP = OD^2 - PD^2$$

$$= \sqrt{4 - 1} = \sqrt{3} \text{ units}$$

$$\therefore AP = AO + OP = (2 + \sqrt{3}) \text{ units}$$

93. (c) In the given figure, OP and OQ are tangents to the circle with centre C .



Then, $OP = OQ$ and

$$\angle CPO = \angle CQO = 90^\circ$$

$$\begin{aligned}
 \text{I. We have, } OR \times SQ &= OS \times RP \\
 &\Rightarrow (OP - RP) \times SQ \\
 &= (OQ - SQ) \times PR \\
 &\Rightarrow (OP - RP) \times SQ \\
 &= (OP - SQ) \times PR \\
 &\Rightarrow SQ = PR \Rightarrow OR = OS
 \end{aligned}$$

Hence, statement I is correct.

$$\text{II. } \therefore \triangle OPC \cong \triangle OCQ$$

$$\text{In } \triangle OPC, \quad 90^\circ + x + \angle 1 = 180^\circ$$

$$\Rightarrow x + \angle 1 = 90^\circ$$

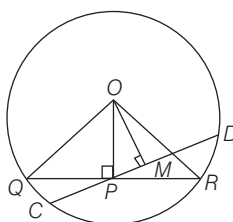
$$\Rightarrow \angle POC + \angle QCO = x + \angle 1 = 90^\circ$$

Hence, statement II is correct.

94. (a) We have, $MP = \sqrt{24}$ cm,

$$OQ = OR = 7 \text{ cm}$$

$$\text{and } QP = PR = \frac{QR}{2} = \frac{2}{2} = 1$$



$$\text{I. In } \triangle OPQ, \quad OP = \sqrt{OQ^2 - QP^2}$$

$$= \sqrt{49 - 1} = \sqrt{48} = 4\sqrt{3} \text{ cm}$$

$$\text{In } \triangle OMP, \quad OP^2 = PM^2 + OM^2$$

$$\Rightarrow OM = \sqrt{48 - 24} = \sqrt{24} \text{ cm}$$

$$\text{So, } OM = PM = \sqrt{24} \text{ cm}$$

$$\therefore \angle OPM = \angle POM = 45^\circ$$

$$\text{Now, } \angle QPD = 90^\circ + \angle OPM$$

$$= 90^\circ + 45^\circ = 135^\circ$$

Hence, statement I is correct.

- II. Now, if $CP = m$ and $PD = n$, then $QP \times PR = CP \times PD$

$$\Rightarrow 1 \times 1 = m \times n \Rightarrow mn = 1$$

Also, $CM = MD$

$$\Rightarrow CP + PM = PD - PM$$

$$\Rightarrow m + \sqrt{24} = n - \sqrt{24}$$

$$\Rightarrow n - m = 2\sqrt{24}$$

$$\therefore m + n = \sqrt{(m - n)^2 + 4mn}$$

$$= \sqrt{96 + 4} = 10$$

So, statement II is also true

$$\text{III. } \frac{\text{ar}(\triangle OPR)}{\text{ar}(\triangle OMP)} = \frac{\frac{1}{2} \times OP \times PR}{\frac{1}{2} \times OM \times PM}$$

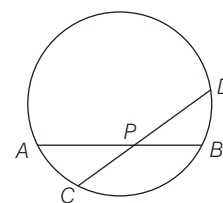
$$= \frac{4\sqrt{3} \times 1}{\sqrt{24} \times \sqrt{24}} = \frac{1}{2\sqrt{3}}$$

So, statement III is not true.

Hence, the statement I, II are correct.

95. (c) From the given circle,

$$AP \times BP = CP \times PD \quad \dots(i)$$



Given that, $A'P' = AP$,

$B'P' = BP$, $C'Q' = CP$ and $D'Q' = DP$

Now,

$$\frac{\text{ar}(\triangle A'P'B')}{\text{ar}(\triangle C'Q'D')} = \frac{\frac{1}{2} \times A'P' \times P'B'}{\frac{1}{2} \times C'Q' \times Q'D'}$$

$$= \frac{AP \times BP}{CP \times DP} = \frac{AP \times BP}{AP \times BP} = 1 \quad [\text{using (i)}]$$

$\therefore \triangle A'P'B'$ and $\triangle C'Q'D'$ are triangles of same area.

So, statement III cannot be false.

Hence, only (c) is correct option.

AREA AND PERIMETER OF PLANE FIGURES

Generally (7-9) questions have been asked from this chapter. Generally, questions are asked from the topics related to the triangles and circles.

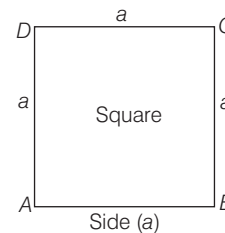


Area of the plane figure is the amount of surface enclosed by its boundary. It is measured in square units. The perimeter of plane figure is total length of the sides enclosing the figure. Unit of Perimeter is same as the unit of sides of a given figure.

SQUARE

Let each side of a square be a unit.

- (i) Perimeter of square = $4(\text{Side}) = 4a$ units
- (ii) Diagonal of square = $\sqrt{2} \times (\text{Side}) = a\sqrt{2}$ units
- (iii) Area of square = Side $\times$ Side = $(a)^2$ sq units = $\frac{1}{2}(\text{Diagonal})^2 = \frac{1}{2}d^2$ sq units



EXAMPLE 1. The perimeter of a square is $2(2x + 4y)$. Then, the area is

- a. $x^2 - 4xy + 4y^2$ b. $x^2 + 4y^2$ c. $x^2 - 4y^2$ d. $x^2 + 4xy + 4y^2$

Sol. d. Here, perimeter of a square = $2(2x + 4y)$

$$\therefore \text{Side of a square} = \frac{\text{Perimeter of a square}}{4} \therefore \text{Side of a square} = \frac{2(2x + 4y)}{4} = x + 2y$$

$$\therefore \text{Area of square} = (\text{Side})^2 \therefore \text{Area of square} = (x + 2y)^2 = x^2 + 4xy + 4y^2$$

EXAMPLE 2. The side of a square exceeds the side of the another square by 4 cm and the sum of the areas of the two squares is 400cm^2 . The dimensions of the squares are

- a. 8 cm and 12 cm b. 6 cm and 10 cm c. 12 cm and 16 cm d. None of these

Sol. c. Let the side of a square be x cm. Then, side of another square = $(x + 4)$ cm.

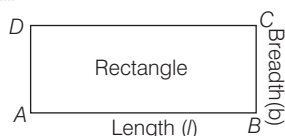
$$\text{Area of first square} = x^2 \text{ cm}^2 \text{ and area of second square} = (x + 4)^2 \text{ cm}^2$$

$$\text{According to the question} \Rightarrow x^2 + (x + 4)^2 = 400 \Rightarrow x^2 + x^2 + 16 + 8x = 400 \Rightarrow 2x^2 + 8x - 384 = 0$$

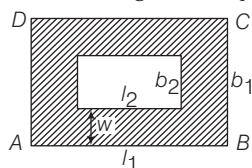
$$\begin{aligned}
 x^2 + 4x - 192 &= 0 \Rightarrow x^2 + 16x - 12x - 192 = 0 \\
 \Rightarrow x(x+16) - 12(x+16) &= 0 \Rightarrow (x-12)(x+16) = 0 \\
 \Rightarrow x-12 &= 0 \text{ or } x+16 = 0 \\
 \therefore x &= 12 \quad [\text{side cannot be negative}] \\
 \text{Hence, side of one square} &= 12 \text{ cm and side of another square} = 12 + 4 = 16 \text{ cm}
 \end{aligned}$$

RECTANGLE

Let l and b be the length and breadth of a rectangle respectively, then



- (i) Area of rectangle = Length $\times$ Breadth = $l \times b$
- (ii) Perimeter of rectangle = $2(\text{Length} + \text{Breadth})$
 $= 2(l + b)$ units
- (iii) Diagonal of rectangle = $\sqrt{(\text{Length})^2 + (\text{Breadth})^2}$
 $= \sqrt{l^2 + b^2}$ units
- (iv) Area of track (Shaded region) = $(l_1 b_1 - l_2 b_2)$ sq units



EXAMPLE 3. The area of a rectangular plot is 180 m^2 . If its length is 18 m. Then, its perimeter is

- a. 28 m b. 56 m c. 360 m d. None of these

Sol. b. Given, area of rectangular plot = 180 m^2

and length of the plot = 18 m

$$\therefore \text{Breadth} = \frac{\text{Area}}{\text{Length}} = \frac{180}{18} = 10 \text{ m}$$

Now, Perimeter of rectangular plot

$$= 2(\text{Length} + \text{Breadth}) = 2(18 + 10) = 56 \text{ m}$$

Hence, the perimeter of rectangular plot is 56 m.

EXAMPLE 4. The area of the floor of a rectangular hall of length 40 m is 960 m^2 . Carpets of size $6 \text{ m} \times 4 \text{ m}$ are available. Then, how many carpets are required to cover the hall?

- a. 20 b. 30 c. 40 d. 45

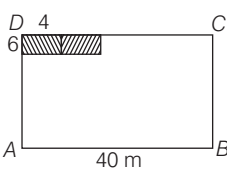
Sol. c. Given, area of the floor = 960 m^2

Area of one carpet = $6 \times 4 = 24 \text{ m}^2$

Number of carpets required

$$= \frac{\text{Area of the floor}}{\text{Area of one carpet}} = \frac{960}{24} = 40$$

Hence, 40 carpets are required to cover the hall.



EXAMPLE 5. A lawn is in the shape of a rectangle of length 60 m and width 40 m. There is a footpath of uniform width 1 m bordering the lawn. The area of the path is

- a. 194 m^2 b. 196 m^2 c. 198 m^2 d. 200 m^2

Sol. b. Given, length of the outer rectangle = 60 m and breadth of the outer rectangle = 40 m

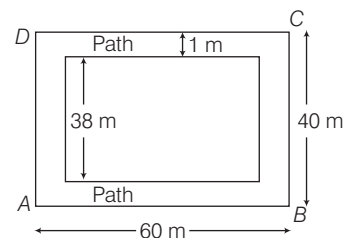
$$\therefore \text{Area of outer rectangle} = 60 \times 40 = 2400 \text{ m}^2$$

$$\therefore \text{Width of path} = 1 \text{ m}$$

$$\therefore \text{Length of the inner rectangle} = 60 - (1 + 1) = 58 \text{ m}$$

$$\text{Breadth of the inner rectangle} = 40 - (1 + 1) = 38 \text{ m}$$

$$\therefore \text{Area of the inner rectangle} = 58 \times 38 = 2204 \text{ m}^2$$



$$\text{Area of the path} = [\text{Area of the outer rectangle}]$$

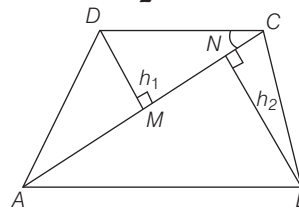
$$- [\text{Area of the inner rectangle}] = (2400 - 2204) = 196 \text{ m}^2$$

Hence, the area of the path is 196 m^2 .

QUADRILATERAL

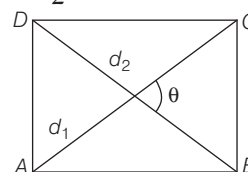
- (i) Let $ABCD$ is a quadrilateral in which $DM = h_1$ and $BN = h_2$ are perpendiculars on diagonal AC from other two vertices B and D , then

$$\begin{aligned}
 \text{Area of the quadrilateral} &= \frac{1}{2} \times \text{Diagonal} \times (h_1 + h_2) \\
 &= \frac{1}{2} \times AC \times (DM + BN) \text{ sq units}
 \end{aligned}$$



- (ii) Let d_1 and d_2 are two diagonals and θ is angle between them, then area of quadrilateral

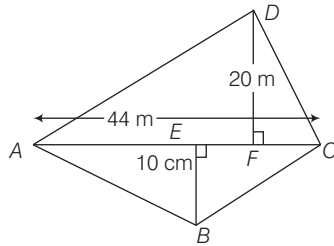
$$= \frac{1}{2} d_1 d_2 \sin \theta$$



EXAMPLE 6. In a quadrilateral $ABCD$, diagonal $AC = 44$ cm and the length of the perpendicular drawn from B and D to AC are 10 cm and 20 cm, respectively. The area of the quadrilateral is

- a. 330 cm^2 b. 440 cm^2 c. 550 cm^2 d. 660 cm^2

Sol. d.



Area of quadrilateral

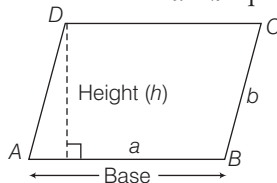
$$= \frac{1}{2} AC(h_1 + h_2) = \frac{1}{2} (44) (20 + 10) = \frac{1}{2} \times 44 \times 30 = 660 \text{ cm}^2$$

Hence, the area of the quadrilateral is 660 cm^2 .

PARALLELOGRAM

Let adjacent sides of a parallelogram are b and a respectively and h is the corresponding altitude (height) of side a , then

- (i) Area of the parallelogram = (Base $\times$ Height)
 $= a \times h$ sq units



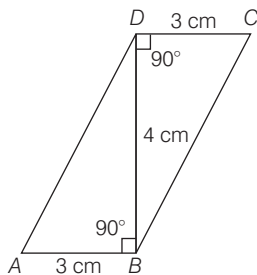
- (ii) Perimeter of a parallelogram = 2(Sum of adjacent side)
 $= 2(a + b)$ units

Note Each diagonal of a parallelogram divides it into two triangles of equal area.

EXAMPLE 7. $ABCD$ is a parallelogram as shown in figure, then its area is

- a. 12 cm^2 b. 14 cm^2 c. 15 cm^2 d. 18 cm^2

Sol. a. Given, base = 3 cm and height = 4 cm



Area of parallelogram,

$$ABCD = \text{Base} \times \text{height} = 3 \times 4 = 12 \text{ cm}^2$$

EXAMPLE 8. In the parallelogram $ABCD$, $AB = 10$ cm. The altitude corresponding to sides AB and AD are respectively 7 cm and 8 cm. Then, AD is

- a. 8.75 cm b. 8.95 cm c. 9 cm d. 9.25 cm

Sol. a. Area of parallelogram $ABCD$

= Base $\times$ Corresponding altitude

$\therefore$ Area of parallelogram

$$ABCD = AB \times DM$$

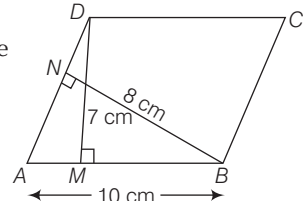
$$= 10 \times 7 = 70 \text{ cm}^2 \quad \dots(i)$$

Also, area of parallelogram

$$ABCD = AD \times BN = 8AD \quad [\because BN = 8 \text{ cm}] \quad \dots(ii)$$

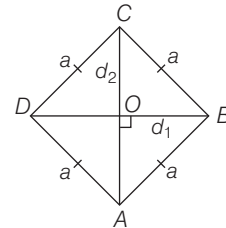
From Eqs. (i) and (ii), $8AD = 70$

$$\therefore AD = \frac{70}{8} = 8.75 \text{ cm}$$



RHOMBUS

Let the length of each sides of a rhombus is a and length of both diagonals are d_1 and d_2 , then



- (i) Area of rhombus = $\frac{1}{2} \times d_1 \times d_2$ sq units

- (ii) Side of rhombus = $\frac{1}{2} \sqrt{d_1^2 + d_2^2}$ units $\Rightarrow 4a^2 = d_1^2 + d_2^2$

- (iii) Perimeter of rhombus = $4 \times (\text{Side})$ units

Note Diagonals of a rhombus bisect each other at right angles.

EXAMPLE 9. If the length of the diagonal of a rhombus is $(a + b)$ and its area is $\frac{a^2 - b^2}{2}$ sq units, then the other diagonal is

- a. $a + b$ b. $a - b$ c. $\frac{a - b}{2}$ d. $\frac{a + b}{2}$

Sol. b. Let second diagonal be d units.

$$\text{Given, area of rhombus} = \frac{a^2 - b^2}{2}$$

$$\therefore \text{Area of rhombus} = \frac{1}{2} \times d_1 \times d_2$$

$$\therefore \frac{1}{2} \times d_1 \times d_2 = \frac{a^2 - b^2}{2} \Rightarrow \frac{1}{2} (a + b) \cdot d = \frac{a^2 - b^2}{2}$$

$$[\because d_1 = a + b]$$

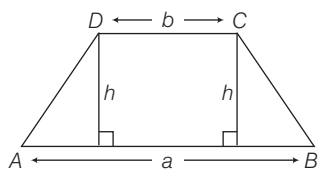
$$\Rightarrow (a + b) d = a^2 - b^2 \Rightarrow d = \frac{(a^2 - b^2)}{a + b} = \frac{(a - b)(a + b)}{(a + b)}$$

$$\therefore d = (a - b) \quad [\because a^2 - b^2 = (a - b)(a + b)]$$

TRAPEZIUM

Let the length of parallel sides of a trapezium are a and b and distance between them is h , then

$$\text{Area of trapezium} = \frac{1}{2} (\text{Sum of parallel sides}) \times (\text{Distance between them})$$



$$= \frac{1}{2} (AB + CD) \times h = \frac{1}{2} (a + b) \times h \text{ sq units}$$

EXAMPLE 10. The difference between two parallel sides of a trapezium is 4 cm and the perpendicular distance between them is 19 cm. Find the lengths of the parallel sides, if the area of the trapezium is 475 cm^2 .

- a. 22 cm and 18 cm b. 25 cm and 21 cm
c. 29 cm and 25 cm d. 27 cm and 23 cm

Sol. d. Let the length of the parallel sides of the trapezium be a cm and b cm, respectively.

Then, according to the question, $(a - b) = 4$... (i)

$$\therefore \text{Area of trapezium} = \frac{1}{2} \times (a + b) \times h$$

$$\frac{1}{2} \times (a + b) \times 19 = 475$$

$$\Rightarrow a + b = \frac{950}{19} = 50$$

$$\therefore a + b = 50 \quad \dots \text{(ii)}$$

On solving Eqs. (i) and (ii), we get $a = 27$ and $b = 23$

Thus, length of parallel sides are 27 cm and 23 cm.

SCALED TRIANGLE

Let the sides of a triangle are a, b, c and h be the corresponding altitude to side a , then

(i) Perimeter of scalene triangle, $2s = (a + b + c)$ units

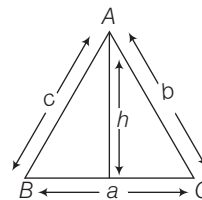
(ii) Semi-perimeter of scalene triangle, $s = \frac{a + b + c}{2}$ units

(iii) Area of triangle = $\sqrt{s(s-a)(s-b)(s-c)}$ sq units

$$\text{where, } s = \frac{a + b + c}{2} \quad [\text{Heron's formula}]$$

$$\text{or area of triangle} = \frac{1}{2} \times a \times h \text{ sq units}$$

(iv) The radius of incircle = $\frac{2(\text{Area})}{\text{Perimeter}}$



(v) The radius of circumcircle = $\frac{abc}{4(\text{Area})}$

EXAMPLE 11. The area of a triangle whose sides are 9 cm, 12 cm and 15 cm is

- a. 45 cm^2 b. 54 cm^2 c. 56 cm^2 d. 64 cm^2

Sol. b. Here $a = 9$ cm, $b = 12$ cm and $c = 15$ cm

$$s = \frac{a + b + c}{2} = \frac{9 + 12 + 15}{2} = 18 \text{ cm}$$

$$\text{Area of a triangle} = \sqrt{s(s-a)(s-b)(s-c)}$$

[Heron's formula]

$$= \sqrt{18(18-9)(18-12)(18-15)}$$

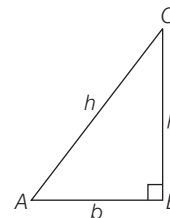
$$= \sqrt{18 \times 9 \times 6 \times 3} = 54 \text{ cm}^2$$

Hence, the area of a triangle is 54 cm^2 .

RIGHT ANGLED TRIANGLE

A figure bounded by three straight lines is called a triangle.

Let perpendicular, base and hypotenuse of a right angled triangle ($\triangle ABC$) be p, b and h , respectively, then



(i) Perimeter of right angled triangle = $AB + BC + CA$
 $= (b + p + h)$ units

(ii) Area of right angled triangle = $\frac{1}{2} \times \text{Base} \times \text{Altitude}$
 $= \frac{1}{2} \times b \times p$ sq units

EXAMPLE 12. The base of triangular field is three times its altitude. If the cost of cultivating the field at 50 per hectare be ₹ 675, then its base and height are

- a. 900 m and 300 m b. 600 m and 300 m
c. 500 m and 200 m d. None of these

Sol. a. Area of the triangular field = $\frac{\text{Total cost}}{\text{Rate}}$

$$= \frac{675}{50} = 13.5 \text{ hec}$$

$$= (13.5 \times 10000) \text{ m}^2 = 135000 \text{ m}^2 \quad \dots \text{(i)}$$

Let altitude of triangular field be x m

and base of field = $3x$ m

Again, area of the field = $\frac{1}{2} \times \text{Base} \times \text{Altitude}$

$$= \frac{1}{2} \times 3x \times x = \frac{3x^2}{2} \quad \dots(ii)$$

From Eqs. (i) and (ii), we get

$$\frac{3x^2}{2} = 135000$$

$$\therefore x^2 = \frac{135000 \times 2}{3} = 90000 \Rightarrow x = 300 \text{ m}$$

So, base = $3x = 3 \times 300 = 900$ m

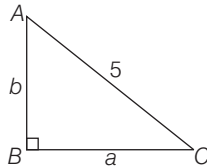
and altitude = $x = 300$ m

EXAMPLE 13. The perimeter of a right triangle is 12 cm. The hypotenuse is 5 cm. The other two sides and area of the triangle are

- a. 3, 4 and 6 cm^2 b. 4, 3 and 12 cm^2
c. 6, 2 and 6 cm^2 d. None of these

Sol. a. Let other two sides of a right triangle be a cm and b cm such that $a > b$.

Given, perimeter of a right angled triangle = 12 cm



$$\therefore a + b + 5 = 12 \Rightarrow a + b = 7 \text{ cm} \quad \dots(i)$$

Also, by pythagoras theorem,

$$a^2 + b^2 = 25$$

Also, $(a + b)^2 + (a - b)^2 = 2(a^2 + b^2)$

$$(a - b)^2 = 2(a^2 + b^2) - (a + b)^2$$

$$\Rightarrow (a - b)^2 = 2(25) - (7)^2 = 50 - 49 = 1$$

$$\therefore a - b = 1 \quad [\because a > b] \dots(ii)$$

On solving Eqs. (i) and (ii), we get

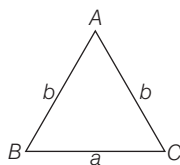
$$a = 4 \text{ cm and } b = 3 \text{ cm}$$

$$\therefore \text{Area of triangle} = \frac{1}{2} \times \text{Base} \times \text{Altitudes}$$

$$\Rightarrow \frac{1}{2} \times 3 \times 4 = 6 \text{ cm}^2$$

ISOSCELES TRIANGLE

Let sides of an isosceles triangle are a , b and b , then



(i) Perimeter of isosceles triangle
= $a + b + b = a + 2b$ units

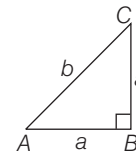
(ii) Area of isosceles triangle

$$= (s - b)(\sqrt{s(s - a)}) = \frac{1}{4} a \sqrt{4b^2 - a^2} \text{ sq units}$$

where, a = Base and b = Equal sides and
 s = semi-perimeter

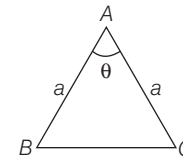
(iii) Area of a right isosceles triangle, in which equal sides from a right angle is given by

$$\text{Area} = \frac{1}{2} a^2 \text{ sq units}$$



(iv) If θ is the angle between two equal sides of isosceles triangle, then

$$\text{Area of triangle} = \frac{1}{2} a^2 \sin \theta$$



EXAMPLE 14. The perimeter of an isosceles triangle is 32 cm while equal sides together measure 20 cm. Then, area of an isosceles triangle is

- a. 48 cm^2 b. 84 cm^2 c. 44 cm^2 d. 41 cm^2

Sol. a. Let third side of an isosceles triangle be a cm.

Given, $b = 10$ cm

$$\therefore \text{Perimeter of isosceles triangle} = a + b + b = a + 2b$$

$$\Rightarrow 32 = a + 2 \times 10 = a + 20 \Rightarrow 32 - 20 = a$$

$$\therefore a = 12 \text{ cm}$$

Now, area of an isosceles triangle

$$= \frac{1}{4} a \sqrt{4b^2 - a^2} = \frac{1}{4} \times 12 \sqrt{4 \times 10^2 - 12^2}$$

$$= 3\sqrt{400 - 144} = 3 \times 16 = 48 \text{ cm}^2$$

EQUILATERAL TRIANGLE

Let a be the side of an equilateral triangle, then

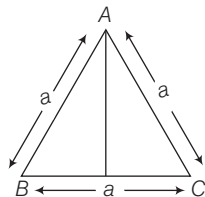
(i) Height (altitude) of an equilateral triangle = $\frac{\sqrt{3}}{2} a$

(ii) Area of an equilateral triangle = $\frac{\sqrt{3}}{4} a^2$

(iii) Perimeter of an equilateral triangle = $3 \times \text{Side} = 3a$

(iv) The radius of incircle = $\frac{a}{2\sqrt{3}}$

- (v) The radius of circumcircle = $\frac{a}{\sqrt{3}}$



EXAMPLE 15. The perimeter of an equilateral triangle whose area is $4\sqrt{3}$ cm² is

- a. 4 cm b. 3 cm c. 12 cm d. 7 cm

Sol. c. Area of an equilateral triangle = $\frac{\sqrt{3}}{4} (\text{Side})^2$

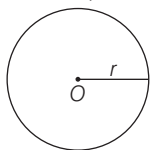
$$\Rightarrow \frac{\sqrt{3}}{4} (\text{Side})^2 = 4\sqrt{3}, (\text{Side})^2 = 16$$

$$\therefore \text{Side} = 4 \text{ cm}$$

$$\text{Hence, the perimeter of an equilateral triangle} \\ = 3 \times \text{Side} = 3 \times 4 = 12 \text{ cm}$$

CIRCLE

Let the radius of a circle be r , then



- (i) Circumference of circle = $2\pi r = \pi D$ units
[$\because D$ is diameter, $D = 2r$]
(ii) Area of circle = πr^2 sq units
(iii) Distance covered by a wheel in one revolution
= Circumference of the wheel

EXAMPLE 16. The circumference of a circle whose area is 24.64 m² is

- a. 17.2 m b. 17.4 m c. 17.6 m d. 18.0 m

Sol. c. Let the radius of the circle be r m.

$$\therefore \pi r^2 = 24.64$$

$$\Rightarrow \frac{22}{7} r^2 = 24.64 \Rightarrow r^2 = \frac{7 \times 24.64}{22}$$

$$\Rightarrow r = \sqrt{\frac{7 \times 24.64}{22}} = 2.8$$

$$\text{Thus, circumference} = 2\pi r = 2 \times \frac{22}{7} \times 2.8 = 17.6 \text{ m}$$

EXAMPLE 17. The diameter of the driving wheel of a car is 140 cm. Then, in order to keep the speed of 66 km/h, how many revolutions per minute must the wheel make?

- a. 250 b. 275 c. 290 d. 295

Sol. a. Given, radius of the wheel = 70 cm = 0.7 m

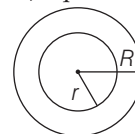
$$\text{Circumference of the wheel} = 2\pi r = \left(2 \times \frac{22}{7} \times 0.7\right) = 4.4 \text{ m}$$

$$\text{Distance to be covered in 1 min} = \frac{66 \times 1000}{60} = 1100 \text{ m}$$

$$\therefore \text{Number of revolutions per minute} = \frac{1100}{4.4} = 250$$

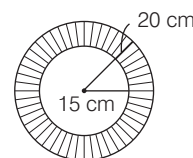
Circular Ring

If ' R ' and ' r ' be outer and inner radii of a ring, then the area of ring = $\pi(R^2 - r^2)$ sq units



EXAMPLE 18. The area of a ring whose outer and inner radii are respectively 20 cm and 15 cm is

- a. 440 cm² b. 550 cm² c. 565 cm² d. 675 cm²



Sol. b. Given, radius of outer circle (R) = 20 cm

Radius of inner circle (r) = 15 cm

$$\therefore \text{Area of the ring} = [\text{Area of outer circle} \\ - \text{Area of inner circle}]$$

$$= (\pi R^2 - \pi r^2) = \pi(R^2 - r^2) = \frac{22}{7}(20^2 - 15^2)$$

$$= \frac{22}{7}(20 + 15)(20 - 15) = \frac{22}{7} \times 35 \times 5 = 550 \text{ cm}^2$$

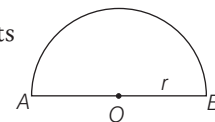
Semi-circle

A diameter divides a circle into two equal parts. Each of these two arcs is called semi-circle. If r is the radius of a circle, then

(i) Area of semi-circle = $\frac{1}{2} \pi r^2$ sq units

(ii) Perimeter of semi-circle

$$= (\pi r + 2r) = (\pi r + D) \text{ units}$$



EXAMPLE 19. If the perimeter of a semi-circular protractor is 36 cm, then its diameter is

- a. 6 cm b. 7 cm c. 7.5 cm d. 14 cm

Sol. d. Let the radius of the protractor be r cm, then perimeter

$$= (\pi r + 2r) = (\pi + 2)r = \frac{36}{7}r \Rightarrow 36 = \frac{36r}{7}$$

$$\therefore r = 7 \text{ cm}$$

$$\text{Hence, diameter of the protractor} = 2r = 2 \times 7 = 14 \text{ cm}$$

Quadrant of a Circle

If r is the radius of a circle, then

$$\begin{aligned} \text{(i) The perimeter of the quadrant} \\ &= \frac{1}{4} (\text{Circumference of a circle}) + 2r \\ &= \frac{1}{4} \times 2\pi r + 2r = \left(\frac{\pi}{2} + 2\right)r \text{ units} \end{aligned}$$

$$\text{(ii) Area of the quadrant} = \frac{1}{4} (\text{Area of circle}) = \frac{\pi r^2}{4} \text{ sq units}$$

If two diameters are perpendicular to each other, then they divide the circle into four quadrants.

Area of Sector

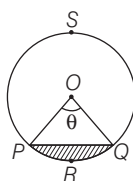
If θ be the angle at the centre of a circle of radius r , then

$$\text{(i) Length of the arc } \overline{PQ} = \frac{2\pi r \theta}{360^\circ} = \frac{\pi r \theta}{180^\circ}$$

$$\text{(ii) Area of sector } OPRQO = \frac{\pi r^2 \theta}{360^\circ}$$

$$\begin{aligned} \text{(iii) Area of segment } PRQP \\ &= (\text{Area of sector } OPRQO) - \text{Area of } \triangle OPQ \\ &= \frac{\pi r^2 \theta}{360^\circ} - \frac{1}{2} r^2 \sin \theta \end{aligned}$$

$$\text{(iv) Area of major segment } QPSQ = (\text{Area of circle}) - (\text{Area of minor segment } PRQP)$$



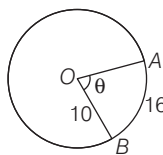
EXAMPLE 20. The arc AB of the circle with centre at O and radius 10 cm has length 16 cm. What is the area of the sector bounded by the radii OA , OB and the arc AB ?

- a. $40\pi \text{ cm}^2$ b. 40 cm^2 c. 80 cm^2 d. $20\pi \text{ cm}^2$

Sol. c. Arc length $= 2\pi r \cdot \frac{\theta}{360^\circ}$

$$\Rightarrow 16 = 2\pi r \cdot \frac{\theta}{360^\circ} \Rightarrow \frac{\theta}{360^\circ} = \frac{16}{2\pi r}$$

$$\begin{aligned} \text{Now, area of sector } OABO &= \pi r^2 \cdot \frac{\theta}{360^\circ} \\ &= \pi r^2 \cdot \frac{16}{2\pi r} = 8r = 8 \times 10 = 80 \text{ cm}^2 \end{aligned}$$



REGULAR POLYGON

A polygon that has all sides equal and all interior angles equal are regular polygon.

Let, n = number of sides, a = length of side
 r = radius of inscribed circle (or inradius)
 R = radius of circumcircle

$$\begin{aligned} \text{Area of polygon} &= \frac{1}{4} na^2 \cot\left(\frac{180^\circ}{n}\right) \\ &= nr^2 \tan\left(\frac{180^\circ}{n}\right) = \frac{1}{2} nR^2 \sin\left(\frac{360^\circ}{n}\right) \end{aligned}$$

$$\text{Perimeter of polygon} = n \times a = 2nr \sin\left(\frac{\pi}{n}\right)$$

$$\text{Inradius of polygon, } r = \frac{1}{2} a \cot\left(\frac{180^\circ}{n}\right)$$

$$\text{circumradius of polygon, } R = \frac{1}{2} a \operatorname{cosec}\left(\frac{180^\circ}{n}\right)$$

$$\text{(i) Area of regular pentagon} = \frac{1}{4} \sqrt{5(5+2\sqrt{5})} a^2 \text{ sq units}$$

$$\text{(ii) Area of regular hexagon} = \frac{3\sqrt{3}a^2}{2} \text{ sq units}$$

$$\text{(iii) Area of regular octagon} = 2(\sqrt{2}+1)a^2 \text{ sq units}$$

EXAMPLE 21. If the area of a regular hexagon is $96\sqrt{3} \text{ cm}^2$, then its perimeter is

- a. 36 cm b. 48 cm c. 54 cm d. 64 cm

Sol. b. Given that, area of a regular hexagon $= 96\sqrt{3}$
 $\Rightarrow \frac{3}{2} \sqrt{3} (\text{Side})^2 = 96\sqrt{3} \Rightarrow (\text{Side})^2 = 64 = (8)^2$

$$\therefore \text{Side} = 8 \text{ cm}$$

$$\begin{aligned} \text{Hence, perimeter of a regular hexagon} \\ &= 6 \times (\text{Side}) = 6 \times 8 = 48 \text{ cm} \end{aligned}$$

Some Useful Results

- Angle inscribed by minute hand in 60 min $= 360^\circ$
- Angle inscribed by hour hand in 12 h $= 360^\circ$
- Angle inscribed by minute hand in 1 min $= 6^\circ$
- If the length of a square/rectangle is increased/decreased by $x\%$ and the breadth is increased/decreased by $y\%$, the net effect on the area is given by

$$\text{Net effect} = \left[\pm x \pm y + \frac{(\pm x)(\pm y)}{100} \right] \%$$

- If the length of a square/rectangle is increased/decreased by $x\%$ and the breadth is increased/decreased by $y\%$ the net effect on the area is given by

$$\text{Net effect} = \left[\pm x \pm y + \frac{(\pm x)(\pm y)}{100} \right] \%$$

Note (+) Increase
 (–) Decrease

- If the side of a square/rectangle/triangle is doubled the area is increased by 300%, i.e. the area becomes four times of itself.
- If the radius of a circle is decreased by $x\%$, then, area is decreased by $\left(-2x + \frac{x^2}{100}\right)\%$. And, if radius is increased

$$\text{by } x\%, \text{ then area is increased by } \left(2x + \frac{x^2}{100}\right)\%.$$

- If a room of dimensions $(l \times b)$ m is to be paved with square tiles, then

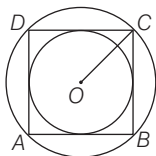
(i) the side of the largest square tile = HCF of l and b

(ii) the least number of tiles required

$$= \frac{l \times b}{(\text{HCF of } l \text{ and } b)^2}$$

- Area of a square inscribed in a circle of radius r is $2r^2$ and the side of a square inscribed in a circle of radius r is $\sqrt{2}r$.
- The area of the largest triangle inscribed in a semi-circle of radius r is r^2 .

EXAMPLE 22. The ratio of the areas of the incircle and circumcircle of a square are



- a. 1 : 1 b. 2 : 1 c. 1 : 2 d. 3 : 1

Sol. c. Let side $AB = BC = CD = AD = x$

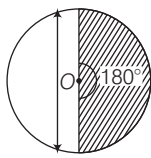
$$\therefore \text{Diagonal of square} = \sqrt{2}x$$

$$\therefore \text{Radius of circumcircle} = \frac{\sqrt{2}x}{2} = \frac{x}{\sqrt{2}}$$

$$\therefore \text{Radius of incircle} = \frac{x}{2}$$

$$\therefore \text{Required ratio} = \left(\frac{\pi x^2}{4} : \frac{\pi x^2}{2} \right) = 2 : 4 = 1 : 2$$

EXAMPLE 23. The minute hand of a clock is 14 cm long. The area of the face of the clock inscribed by the minute hand in 30 min is



- a. 308 cm^2 b. 312 cm^2 c. 412 cm^2 d. 416 cm^2

Sol. a. Angle inscribed by the minute hand in 60 min = 360°

Angle inscribed by the minute hand in 30 min

$$= \frac{360^\circ}{60} \times 30 = 180^\circ$$

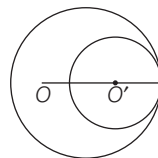
So, $\theta = 180^\circ$ and $r = 14$ cm.

Required area swept by minute hand in 30 min

$$= \frac{\theta}{360^\circ} \times \pi r^2 = \frac{180^\circ}{360^\circ} \times \frac{22}{7} \times 14 \times 14$$

$$= \frac{1}{2} \times \frac{22}{7} \times 14 \times 14 = 308 \text{ cm}^2$$

EXAMPLE 24. Two circles touch internally. The sum of their areas is $116\pi \text{ cm}^2$ and distance between their centres is 6 cm. Then, the radii of the circles are



- a. 4 cm and 9 cm
b. 5 cm and 10 cm
c. 4 cm and 8 cm
d. 4 cm and 10 cm

Sol. d. Let the radius of outer circle be R and radius of inner circle = r cm

$$\therefore \text{Given, } \pi R^2 + \pi r^2 = 116\pi, \quad R^2 + r^2 = 116$$

If O and O' be the centre of these circles, then

$$OO' = (R - r)$$

Also, $(R - r) = 6$ [given]

$$\text{So, } (R + r)^2 + (R - r)^2 = 2(R^2 + r^2)$$

$$\Rightarrow (R + r)^2 = 2(116) - 36 = 196$$

$$\Rightarrow R + r = \sqrt{196}$$

$$\text{So } R + r = 14 \quad \dots(i)$$

$$R - r = 6 \quad \dots(ii)$$

On solving Eqs. (i) and (ii), we get

$$r = 4 \text{ cm and } R = 10 \text{ cm}$$

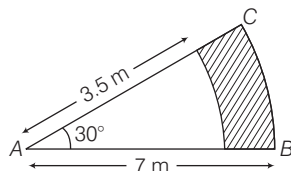
> PRACTICE EXERCISE

1. The diagonal of a square field measures 50 m. The area of square field is
(a) 1250 m^2 (b) 1200 m^2 (c) 1205 m^2 (d) 1025 m^2
2. The length of a rectangle is 2 cm more than its breadth. The perimeter is 48 cm. The area of the rectangle (in cm^2) is
(a) 96 (b) 128 (c) 143 (d) 144
3. In a circle of radius 42 cm, an arc subtends an angle of 72° at the centre. The length of the arc is
(a) 52.8 cm (b) 53.8 cm (c) 72.8 cm (d) 79.8 cm
4. An isosceles right angle triangle has area 200 cm^2 . The length of its hypotenuse is
(a) $15\sqrt{2}\text{ cm}$ (b) $\frac{10}{\sqrt{2}}\text{ cm}$ (c) $10\sqrt{2}\text{ cm}$ (d) $20\sqrt{2}\text{ cm}$
5. With in a rectangular garden 10 m wide and 20 m long, we wish to pave a walk around the borders of uniform width so as to leave an area of 96 m^2 for flowers. The width of the walk is
(a) 1 m (b) 2 m (c) 2.5 m (d) 2.56 m
6. The least number of square slabs that can be fitted in a room 10.5 m long and 3 m wide, is
(a) 12 (b) 13 (c) 14 (d) 15
7. If the side of a square be increased by 50%, the percent increase in area is
(a) 50 (b) 100 (c) 125 (d) 150
8. The area of the largest circle that can be drawn inside a square of side 14 cm in length, is
(a) 84 cm^2 (b) 96 cm^2 (c) 104 cm^2 (d) 154 cm^2
9. If the radius of a circle is decreased by 50%, its area will decrease by
(a) 25% (b) 50% (c) 75% (d) 100%
10. The area of the circle whose circumference is equal to the perimeter of a square of side 11 cm is
(a) 154 cm^2 (b) 144 cm^2 (c) 134 cm^2 (d) 124 cm^2
11. A wire is in the form of a circle of radius 42 cm. It is bent into a square. The side of the square is
(a) 33 cm (b) 66 cm (c) 78 cm (d) 112 cm
12. A horse is tied to a pole with 28 m long string. The area which the horse can graze is equal to
(a) 246 m^2 (b) 2404 m^2 (c) 2464 m^2 (d) 2164 m^2
13. The area of ring is 418 cm^2 . If the radius of the smaller circle is 6 cm. The radius of the bigger circle is
(a) 18 cm (b) 16 cm (c) 13 cm (d) 10 cm
14. The length of a rectangle is increased by 60%. By what per cent would the width have to be decreased to maintain the same area?
(a) $37\frac{1}{2}\%$ (b) 60% (c) 75% (d) 120%
15. The perimeter of a rectangular field is 240 m and the ratio between the length and breadth is 5 : 3. The area of the field is
(a) 33750 m^2 (b) 3375 m^2 (c) 3500 m^2 (d) 3950 m^2
16. The inner circumference of a circular park is 440 m. The track is 14 m wide. The diameter of the outer circle of the track is
(a) 168 m (b) 169 m (c) 144 m (d) 108 m
17. If the length and breadth of a rectangular plot are increased by 50% and 20% respectively, then the new area is how many times the original area?
(a) $\frac{5}{9}$ (b) 10 (c) $\frac{9}{5}$ (d) None of these
18. The area of an isosceles triangle, each of whose equal sides is 13 cm and whose base is 24 cm is
(a) 60 cm^2 (b) 55 cm^2 (c) 50 cm^2 (d) 40 cm^2
19. The difference between the sides at right angles in a right angled triangle is 14 cm. The area of the triangle is 120 cm^2 . The perimeter of the triangle
(a) 68 cm (b) 64 cm (c) 60 cm (d) 58 cm
20. In a four sided field, the length of the longer diagonal is 128 m. The lengths of the perpendicular from the opposite vertices upon this diagonal are 22.7 m and 17.3 m. The area of the field is
(a) 2246 m^2 (b) 2460 m^2 (c) 2540 m^2 (d) 2560 m^2
21. The adjacent sides of a parallelogram are 36 cm and 27 cm in length. If the distance between the shorter sides is 12 cm. The distance between the longer sides is
(a) 9 cm (b) 10 cm (c) 11 cm (d) 12 cm
22. The area of the quadrilateral whose sides measures are 9 cm, 40 cm, 28 cm and 15 cm and in which the angle between the first two sides is a right angle is
(a) 206 cm^2 (b) 306 cm^2 (c) 356 cm^2 (d) 380 cm^2
23. A bicycle wheel makes 5000 revolutions in moving 11 km. The diameter of the wheel is
(a) 50 cm (b) 60 cm (c) 70 cm (d) 80 cm

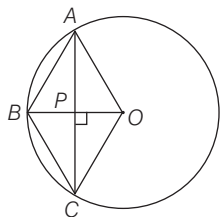
- 24.** Two circles touch externally. The sum of their areas is $130\pi \text{ cm}^2$ and the distance between their centres is 14 cm. The radii of the circles are
 (a) 11 cm, 3 cm (b) 10 cm, 4 cm
 (c) 9 cm, 5 cm (d) 8 cm, 6 cm

- 25.** The minute hand of a clock is 12 cm long. The area of the face of the clock described by the minute hand in 35 min.
 (a) 284 cm^2 (b) 294 cm^2 (c) 274 cm^2 (d) 264 cm^2

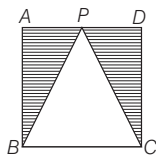
- 26.** In the given figure, sectors of two concentric circles of radii 7 cm and 3.5 cm are shown. The area of the shaded region



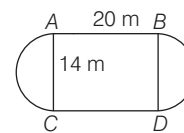
- (a) $\frac{77}{4} \text{ cm}^2$ (b) $\frac{77}{8} \text{ cm}^2$
 (c) $\frac{77}{2} \text{ cm}^2$ (d) None of these
- 27.** Of the two square fields, the area of one is 1 hectare, while the other one is broader by 2%. The difference in their areas is
 (a) 104 m^2 (b) 200 m^2 (c) 204 m^2 (d) 404 m^2
- 28.** If the diameter of the circle is increased by 100%, its area is increased by
 (a) 100% (b) 200% (c) 300% (d) 400%
- 29.** If the two parallel sides of a trapezium are 15 cm and 25 cm respectively and the distance between them is 7 cm, then the area of the trapezium is
 (a) 105 cm^2 (b) 125 cm^2
 (c) 140 cm^2 (d) None of these
- 30.** The diagonals of a rhombus are 24 cm and 10 cm, respectively. The perimeter of the rhombus is
 (a) 50 cm (b) 52 cm (c) 60 cm (d) 68 cm
- 31.** In the given figure $OABC$ is a rhombus whose three vertices A, B, C lie on the circle of radius 10 cm. The area of rhombus is



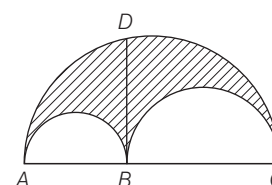
- (a) $50\sqrt{3} \text{ cm}^2$ (b) $100\sqrt{3} \text{ cm}^2$ (c) $75\sqrt{3} \text{ cm}^2$ (d) $125\sqrt{3} \text{ cm}^2$
- 32.** In the adjoining figure, $ABCD$ is a rectangle such that $AB = 2AD = a$. P is the mid-point of AD . The area of the shaded region is
 (a) $\frac{1}{3}a^2$ (b) $\frac{1}{2}a^2$ (c) $\frac{1}{4}a^2$ (d) a^2



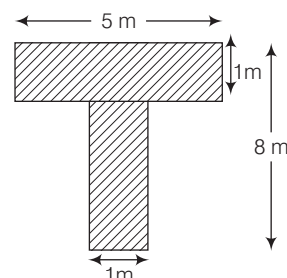
- 33.** A garden is in the form of a rectangle with semi-circular ends on the either side as shown in the diagram below. The length and breadth of the rectangle are 20 m and 14 m, respectively. The cost of levelling the plot at ₹ 25
 (a) ₹ 10850 (b) ₹ 5425 (c) ₹ 8510 (d) ₹ 4255



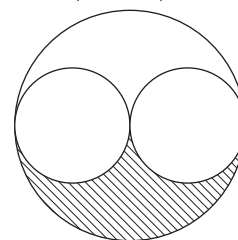
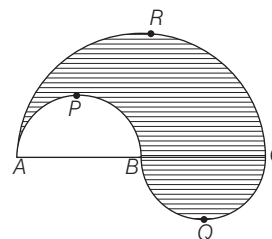
- 34.** Consider the adjoining figure, let $AB = 4 \text{ cm}$, $BC = 14 \text{ cm}$, then the area (shaded) bounded by three semi-circles as shown in the adjoining figure in cm^2 , is π times



- (a) 48 (b) 24 (c) 14 (d) 12
- 35.** Area of shaded portion as shown in the figure is



- (a) 10 m^2 (b) 12 m^2 (c) 14 m^2 (d) 16 m^2
- 36.** The boundary of the shaded region in the adjoining diagram consists of three semi-circular arc, the smaller two being equal. If the diameter of the large one is 10 cm, then the length of the boundary is
 (a) 31 cm (b) $10\pi \text{ cm}$ (c) $20\pi \text{ cm}$ (d) $19\pi \text{ cm}$
- 37.** In the adjoining figure, the larger circle with radius 4 cm is touched internally by two smaller circles which also touch each other externally at the centre O of the larger circle. The area of shaded region is (in cm^2)



- (a) π (b) 2π (c) 3π (d) 4π

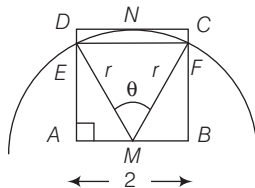
- 38.** The lengths of two sides of a right angled triangle which contain the right angle are a and b , respectively. Three squares are drawn on the three sides of the triangle on the outer side. What is the total area of the triangle and the three squares?

(a) $2(a^2 + b^2) + ab$ (b) $2(a^2 + b^2) + 2.5 ab$
(c) $2(a^2 + b^2) + 0.5 ab$ (d) $25(a^2 + b^2)$

- 39.** A grassy field has the shape of an equilateral triangle of side 6 m. A horse is tied to one of its vertices with a rope of length 4.2 m. The percentage of the total area of the field which is available for grazing is best approximated by

(a) 50% (b) 55% (c) 59% (d) 62%

- 40.** From the mid-point of the side of a square of length 2 units, a circle of radius 2 units is circumscribed. The area of intersection of the square and the circle is



(a) $\frac{4}{3}\pi + 3\sqrt{3}$ (b) $\pi + \sqrt{3}$ (c) $\frac{2}{3}\pi + \sqrt{3}$ (d) $\frac{2}{3}\pi + 2\sqrt{3}$

- 41.** In a circle with radius 2 cm, on its chord $2\sqrt{2}$ cm long taken as a diameter, another circle is constructed. The area of part of that circle, which is outside the greater circle is

(a) 1 cm^2 (b) 2 cm^2 (c) 6 cm^2 (d) 10 cm^2

- 42.** If a rectangular park has length l , width w , area a and perimeter p , which equation describing this park must be true?

I. $wp - 2w^2 = 2a$ II. $2a - 2w^2 - wp = 0$

III. $p^2 - 8a = 4(l^2 + w^2)$

Select the correct answer using the codes given below

(a) I and II (b) II and III
(c) I and III (d) All of these

- 43.** A regular polygon with number of sides 12 is inscribed in a circle of area $9\pi \text{ cm}^2$. If r is the radius of a circle inscribed in a polygon, then consider the following statements.

I. Radius of circumcircle is 3 cm.

II. Radius of incircle is $3\cos 15^\circ$.

III. Ratio of area of circumcircle to area of polygon is $\pi : 3$.

Which of the following statement(s) are correct?

(a) I and II (b) II and III
(c) I and III (d) All of these

- 44.** What is the area of rectangle R ?

I. The length of rectangle R is twice the width.

II. The area of rectangle R is twice the perimeter.

(a) Statement I alone is sufficient but statement II alone is not sufficient.

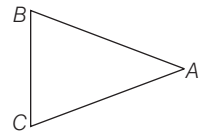
(b) Statement II alone is sufficient but statement I alone is not sufficient.

(c) Both statements together are sufficient.

(d) Neither statement I nor II is sufficient.

Directions (Q. Nos. 45-46) Answer the questions based on the following information.

A cow is tethered at point A by a rope. Neither the rope nor the cow is allowed to enter the triangle ABC . $\angle BAC = 30^\circ$, $AB = AC = 10 \text{ m}$.



- 45.** What is the area that can be grazed by the cow if the length of the rope is 8 m?

(a) $134\pi \frac{1}{3} \text{ m}^2$ (b) $121\pi \text{ m}^2$ (c) $132\pi \text{ m}^2$ (d) $\frac{176\pi}{3} \text{ m}^2$

- 46.** What is the area that can be grazed by the cow if the length of the rope is 12 m?

(a) $133\pi \frac{1}{6} \text{ m}^2$ (b) $121\pi \text{ m}^2$ (c) $132\pi \text{ m}^2$ (d) $\frac{176\pi}{3} \text{ m}^2$

PREVIOUS YEARS' QUESTIONS

- 47.** If the outer and inner diameters of a stone parapet around a well are 112 cm and 70 cm, respectively. Then, what is the area of the parapet? ☑ 2012 I

(a) 264 cm^2 (b) 3003 cm^2
(c) 6006 cm^2 (d) 24024 cm^2

- 48.** The area of a rectangle, whose one side is a is $2a^2$. What is the area of a square having one of the diagonals of the rectangle as side? ☑ 2012 I

(a) $2a^2$ (b) $3a^2$ (c) $4a^2$ (d) $5a^2$

- 49.** If the altitude of an equilateral triangle is $\sqrt{3} \text{ cm}$, then what is its perimeter? ☑ 2012 I

(a) 3 cm (b) $3\sqrt{3} \text{ cm}$ (c) 6 cm (d) $6\sqrt{3} \text{ cm}$

- 50.** If the area of a rectangle whose length is 5 more than twice its width is 75 sq units. What is the perimeter of the rectangle? ☑ 2012 I

(a) 40 units (b) 30 units
(c) 24 units (d) 20 units

- 51.** A square, a circle and an equilateral triangle have same perimeter.

Consider the following statements

I. The area of square is greater than the area of the triangle.

II. The area of circle is less than the area of triangle.

Which of the statement(s) given above is/are correct? **☑ 2012 I**

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

- 52.** If the area of a circle is equal to the area of a square with side $2\sqrt{\pi}$ units, then what is the diameter of the circle? **☑ 2012 I**
(a) 1 unit (b) 2 units (c) 4 units (d) 8 units

- 53.** In the $\triangle ABC$, the base BC is trisected at D and E . The line through D , parallel to AB , meets AC at F and the line through E parallel to AC meets AB at G . If EG and DF intersect at H , then what is the ratio of the sum of the area of parallelogram $AGHF$ and the area of the $\triangle DHE$ to the area of the $\triangle ABC$? **☑ 2012 II**
(a) $1/2$ (b) $1/3$ (c) $1/4$ (d) $1/6$

- 54.** If the area of a circle inscribed in an equilateral triangle is 154 cm^2 , then what is the perimeter of the triangle? **☑ 2012 II**
(a) 21 cm (b) $42\sqrt{3}$ cm (c) $21\sqrt{3}$ cm (d) 42 cm

- 55.** If the circumferences of two circles are in the ratio $2 : 3$, then what is the ratio of their areas? **☑ 2012 II**
(a) $2 : 3$ (b) $4 : 9$ (c) $1 : 3$ (d) $8 : 27$

- 56.** If the circumference of a circle is equal to the perimeter of square, then which one of the following is correct? **☑ 2012 II**
(a) Area of circle = Area of square
(b) Area of circle $\geq$ Area of square
(c) Area of circle $>$ Area of square
(d) Area of circle $<$ Area of square

- 57.** The area enclosed between the circumferences of two concentric circles is $16\pi \text{ cm}^2$ and their radii are in the ratio $5 : 3$. What is the area of the outer circle? **☑ 2012 II**
(a) $9\pi \text{ cm}^2$ (b) $16\pi \text{ cm}^2$ (c) $25\pi \text{ cm}^2$ (d) $36\pi \text{ cm}^2$

- 58.** The perimeter of a rectangle is 82 m and its area is 400 m^2 . What is the breadth of the rectangle? **☑ 2012 II**
(a) 18 m (b) 16 m (c) 14 m (d) 12 m

- 59.** Consider the following statements
I. Area of a segment of a circle is less than area of its corresponding sector.
II. Distance travelled by a circular wheel of diameter $2d$ cm in one revolution is greater than $6d$ cm.

Which of the above statement(s) is/are correct?

- (a) Only I (b) Only II **☑ 2012 II**
(c) Both I and II (d) Neither I nor II

- 60.** The minute hand of a watch is 2.5 cm long. The distance its extreme end transverses in 40 min is
(a) $10\pi/3$ cm (b) $3\pi/10$ cm **☑ 2013 I**
(c) $10/3$ cm (d) 10 cm

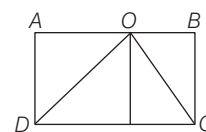
- 61.** The short and long hands of a clock are 4 cm and 6 cm long, respectively. Then, the ratio of distances travelled by tips of short hand in 2 days and long hand in 3 days is **☑ 2013 I**
(a) $4 : 9$ (b) $2 : 9$ (c) $2 : 3$ (d) $1 : 27$

- 62.** The area of a square inscribed in a circle of radius 8 cm is **☑ 2013 I**
(a) 32 cm^2 (b) 64 cm^2 (c) 128 cm^2 (d) 256 cm^2

- 63.** A circular water fountain 6.6 m in diameter is surrounded by a path of width 1.5 m. The area of this path (in m^2) is **☑ 2013 I**
(a) 13.62π (b) 13.15π
(c) 12.15π (d) None of these

- 64.** If an isosceles right angled triangle has area 1 sq unit, then what is its perimeter? **☑ 2013 I**
(a) 3 units (b) $2\sqrt{2} + 1$ units
(c) $(\sqrt{2} + 1)$ units (d) $2(\sqrt{2} + 1)$ units

- 65.** In the figure given below, the area of rectangle $ABCD$ is 100 cm^2 , O is any point on AB and $CD = 20$ cm. Then, the area of $\triangle COD$ is **☑ 2013 I**
(a) 40 cm^2 (b) 45 cm^2 (c) 50 cm^2 (d) 80 cm^2



- 66.** One side of a parallelogram is 8.06 cm and its perpendicular distance from opposite side is 2.08 cm. What is the approximate area of the parallelogram? **☑ 2013 II**
(a) 12.56 cm^2 (b) 14.56 cm^2 (c) 16.76 cm^2 (d) 22.56 cm^2

- 67.** The perimeter of a rectangle having area equal to 144 cm^2 and sides in the ratio $4 : 9$ is **☑ 2013 II**
(a) 52 cm (b) 56 cm (c) 60 cm (d) 64 cm

- 68.** What is the area between a square of side 10 cm and two inverted semi-circular, cross-section each of radius 5 cm inscribed in the square?
(a) 17.5 cm^2 (b) 18.5 cm^2 **☑ 2013 II**
(c) 20.5 cm^2 (d) 21.5 cm^2

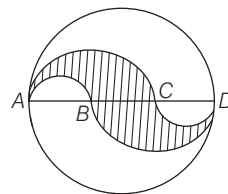
- 69.** How many 200 mm lengths can be cut from 10 m of ribbon? **☑ 2013 II**
(a) 50 (b) 40 (c) 30 (d) 20

- 70.** The area of an isosceles $\triangle ABC$ with $AB = AC$ and altitude $AD = 3$ cm is 12 cm^2 . What is its perimeter? **☑ 2013 II**
(a) 18 cm (b) 16 cm (c) 14 cm (d) 12 cm

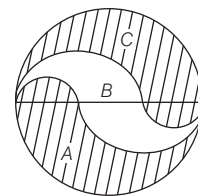
- 71.** If the area of an equilateral triangle is x and its perimeter is y , then which one of the following is correct? **☑ 2013 II**
(a) $y^4 = 432x^2$ (b) $y^4 = 216x^2$
(c) $y^2 = 432x^2$ (d) None of these

- 72.** What is the area of a circle whose area is equal to that of a triangle with sides 7 cm, 24 cm and 25 cm? **2013 II**
 (a) 80 cm^2 (b) 84 cm^2 (c) 88 cm^2 (d) 90 cm^2
- 73.** If AB and CD are two diameters of a circle of radius r and they are mutually perpendicular, then what is the ratio of the area of the circle to the area of the ΔACD ? **2014 I**
 (a) $\frac{\pi}{2}$ (b) π (c) $\frac{\pi}{4}$ (d) 2π
- 74.** A rectangle of maximum area is drawn inside a circle of diameter 5 cm. What is the maximum area of such a rectangle? **2014 I**
 (a) 25 cm^2 (b) 12.5 cm^2
 (c) 12 cm^2 (d) None of these
- 75.** What is the area of the larger segment of a circle formed by a chord of length 5 cm subtending an angle of 90° at the centre? **2014 I**
 (a) $\frac{25}{4}\left(\frac{\pi}{2} + 1\right)\text{ cm}^2$ (b) $\frac{25}{4}\left(\frac{\pi}{2} - 1\right)\text{ cm}^2$
 (c) $\frac{25}{4}\left(\frac{3\pi}{2} + 1\right)\text{ cm}^2$ (d) None of these
- 76.** Consider an equilateral triangle of side of unit length. A new equilateral triangle is formed by joining the mid-points of one, then a third equilateral triangle is formed by joining the mid-points of second. The process is continued. The perimeter of all triangles, thus formed is **2014 I**
 (a) 2 units (b) 3 units (c) 6 units (d) Infinity
- 77.** ABC is triangle right angled at A , $AB = 6\text{ cm}$ and $AC = 8\text{ cm}$. Semi-circles drawn (outside the triangle) on AB , AC and BC as diameters which enclose areas x , y and z square units, respectively. What is $x + y - z$ equal to? **2014 I**
 (a) 48 cm^2 (b) 32 cm^2
 (c) 0 (d) None of these
- 78.** A square is inscribed in a circle of diameter $2a$ and another square is circumscribing circle. The difference between the areas of outer and inner squares is **2014 I**
 (a) a^2 (b) $2a^2$ (c) $3a^2$ (d) $4a^2$
- 79.** The area of a sector of a circle of radius 36 cm is $72\pi\text{ cm}^2$ the length of the corresponding arc of the sector is **2014 I**
 (a) $\pi\text{ cm}$ (b) $3\pi\text{ cm}$ (c) $3\pi\text{ cm}$ (d) $4\pi\text{ cm}$
- 80.** What is the total area of three equilateral triangles inscribed in a semi-circle of radius 2 cm? **2014 I**
 (a) 12 cm^2 (b) $\frac{3\sqrt{3}}{4}\text{ cm}^2$ (c) $\frac{9\sqrt{3}}{4}\text{ cm}^2$ (d) $3\sqrt{3}\text{ cm}^2$
- 81.** How many circular plates of diameter d be taken out of a square plate of side $2d$ with minimum loss of material? **2014 I**
 (a) 8 (b) 6 (c) 4 (d) 2
- 82.** The sides of a triangle are in the ratio $\frac{1}{2} : \frac{1}{3} : \frac{1}{4}$. If its perimeter is 52 cm, then what is the length of the smallest side? **2014 II**
 (a) 9 cm (b) 10 cm (c) 11 cm (d) 12 cm
- 83.** The sides of a triangle are 25 cm, 39 cm and 56 cm. The perpendicular from the opposite vertex on the side of 56 cm is **2014 II**
 (a) 10 cm (b) 12 cm (c) 15 cm (d) 16 cm
- 84.** The area of the largest triangle that can be inscribed in a semi-circle of radius r is **2015 I**
 (a) r^2 (b) $2r^2$ (c) $3r^2$ (d) $4r^2$
- 85.** The area of a rhombus with side 13 cm and one diagonal 10 cm will be **2015 I**
 (a) 140 cm^2 (b) 130 cm^2 (c) 120 cm^2 (d) 110 cm^2
- 86.** Four equal-sized maximum circular plates are cut off from a square paper sheet of area 784 cm^2 . The circumference of each plate is **2015 I**
 (a) 11 cm (b) 22 cm (c) 33 cm (d) 44 cm
- 87.** From a circular piece of cardboard of radius 3 cm, two sectors of 40° each have been cut off. The area of the remaining portion is **2015 I**
 (a) 11 cm^2 (b) 22 cm^2 (c) 33 cm^2 (d) 44 cm^2
- 88.** Three equal circles each of diameter d are drawn on a plane in such a way that each circle touches the other two circles. A big circle is drawn in such a manner that it touches each of the small circles internally. The area of the big circle is **2015 I**
 (a) πd^2 (b) $\pi d^2(2 - \sqrt{3})^2$
 (c) $\frac{\pi d^2(\sqrt{3} + 1)^2}{2}$ (d) $\frac{\pi d^2(\sqrt{3} + 2)^2}{12}$
- 89.** The angles of a triangle are in the ratio 4 : 1 : 1. Then, the ratio of the largest side to the perimeter is **2015 I**
 (a) $\frac{2}{3}$ (b) $\frac{1}{2 + \sqrt{3}}$ (c) $\frac{\sqrt{3}}{2 + \sqrt{3}}$ (d) $\frac{2}{1 + \sqrt{3}}$
- 90.** Let a , b , c be the sides of a right triangle, where c is the hypotenuse. The radius of the circle which touches the sides of the triangle is **2015 I**
 (a) $(a + b - c)/2$ (b) $(a + b + c)/2$
 (c) $(a + 2b + 2c)/2$ (d) $(2a + 2b - c)/2$
- 91.** The ratio of the outer and inner perimeters of a circular path is 23 : 22. If the path is 5 m wide, the diameter of the inner circle is **2015 I**
 (a) 55 m (b) 110 m (c) 220 m (d) 230 m

- 92.** The sides of a triangular field are 41 m, 40 m and 9 m. The number of rose beds that can be prepared in the field if each rose bed, on an average, needs 900 cm^2 space, is **2015 I**
(a) 2000 (b) 1800 (c) 900 (d) 800
- 93.** The diameter of a wheel that makes 452 revolutions to move 2 km and 26 dm is equal to **2015 II**
(a) $1\frac{9}{22}$ m (b) $1\frac{13}{22}$ m (c) $1\frac{5}{11}$ m (d) $1\frac{7}{11}$ m
- 94.** From a rectangular sheet of sides 18 cm and 14 cm, a semi-circular portion with smaller side as diameter is taken out. Then, the area of the remaining sheet will be **2015 II**
(a) 98 cm^2 (b) 100 cm^2 (c) 108 cm^2 (d) 175 cm^2
- 95.** A square and an equilateral triangle have equal perimeter. If the diagonal of the square is $12\sqrt{2} \text{ cm}$, then the area of the triangle is **2015 II**
(a) $24\sqrt{2} \text{ cm}^2$ (b) $24\sqrt{3} \text{ cm}^2$ (c) $48\sqrt{3} \text{ cm}^2$ (d) $64\sqrt{3} \text{ cm}^2$
- 96.** ABCD is a square. If the sides AB and CD are increased by 30%, sides BC and AD are increased by 20%, then the area of the resulting rectangle exceeds the area of the square by **2015 II**
(a) 50% (b) 52% (c) 54% (d) 56%
- 97.** The area of a trapezium is 336 cm^2 . If its parallel sides are in the ratio 5 : 7 and the perpendicular distance between them is 14 cm, then the smaller of the parallel sides is **2015 II**
(a) 20 cm (b) 22 cm (c) 24 cm (d) 26 cm
- 98.** The circumference of a circle is 100 cm. The side of the square inscribed in the circle is **2015 II**
(a) $50\sqrt{2} \text{ cm}$ (b) $\frac{100}{\pi} \text{ cm}$ (c) $\frac{50\sqrt{2}}{\pi} \text{ cm}$ (d) $\frac{100\sqrt{2}}{\pi} \text{ cm}$
- 99.** There are 437 fruit plants in an orchard planted in rows. The distance between any two adjacent rows is 2 m and the distance between any two adjacent plants is 2 m. Each row has the same number of plants. There is 1 m clearance on all sides of the orchard. What is the cost of fencing the area at the rate of ₹ 100 per metre? **2015 II**
(a) ₹ 15600 (b) ₹ 16800 (c) ₹ 18200
(d) More information is required
- 100.** A square is inscribed in a right triangle with legs x and y and has common right angle with the triangle. The perimeter of the square is given by **2015 II**
(a) $\frac{2xy}{x+y}$ (b) $\frac{4xy}{x+y}$ (c) $\frac{2xy}{\sqrt{x^2+y^2}}$ (d) $\frac{4xy}{\sqrt{x^2+y^2}}$
- 101.** A circle of radius r is inscribed in a regular polygon with n sides (the circle touches all sides of the polygon). If the perimeter of the polygon is p then the area of the polygon is **2015 II**
(a) $(p+n)r$ (b) $(2p-n)r$ (c) $\frac{pr}{2}$ (d) None of these
- 102.** A circular path is made from two concentric circular rings in such a way that the smaller ring when allowed to roll over the circumference of the bigger ring, it takes three full revolutions. If the area of the pathway is equal to n times the area of the smaller ring, then n is equal to **2016 I**
(a) 4 (b) 6 (c) 8 (d) 10
- 103.** The number of rounds that a wheel of diameter $7/11 \text{ m}$ will make in traversing 4 km will be **2016 I**
(a) 500 (b) 1000 (c) 1700 (d) 2000
- 104.** The base of an isosceles triangle is 300 units and each of its equal sides is 170 units. Then, the area of the triangle is **2016 I**
(a) 9600 sq units (b) 10000 sq units
(c) 12000 sq units (d) None of these
- 105.** Four equal disc are placed such that each one touches two others. If the area of empty space enclosed by them is $150/847 \text{ cm}^2$, then the radius of each disc is equal to **2016 I**
(a) $7/6 \text{ cm}$ (b) $5/6 \text{ cm}$ (c) $1/2 \text{ cm}$ (d) $5/11 \text{ cm}$
- 106.** AD is the diameter of a circle with area 707 m^2 and $AB = BC = CD$ as shown in the figure above. All curves inside the circle are semicircles with their diameters on AD. What is the cost of levelling the shaded region at the rate of ₹ 63 per m^2 ? **2016 I**
(a) ₹ 29700 (b) ₹ 22400 (c) ₹ 14847 (d) None of these



- 107.** A rhombus is formed by joining mid-points of the sides of a rectangle in the suitable order. If the area of the rhombus is 2 sq units, then the area of the rectangle is **2016 I**
(a) $2\sqrt{2}$ sq units (b) 4 sq units
(c) $4\sqrt{2}$ sq units (d) 8 sq units



- 108.** A circle of 3 m radius is divided into three areas by semi-circles of radii 1 m and 2 m as shown in the figure above. The ratio of the three areas A, B and C will be **2016 I**
(a) 2 : 3 : 2 (b) 1 : 1 : 1 (c) 4 : 3 : 4 (d) 1 : 2 : 1

ANSWERS

1	a	2	c	3	a	4	d	5	b	6	c	7	c	8	d	9	c	10	a
11	b	12	c	13	c	14	a	15	b	16	a	17	c	18	a	19	c	20	d
21	a	22	b	23	c	24	a	25	d	26	b	27	d	28	c	29	c	30	b
31	a	32	c	33	a	34	c	35	b	36	b	37	d	38	c	39	c	40	c
41	b	42	c	43	d	44	c	45	d	46	c	47	c	48	d	49	c	50	a
51	a	52	c	53	b	54	b	55	b	56	c	57	c	58	b	59	c	60	a
61	d	62	c	63	c	64	d	65	c	66	c	67	a	68	d	69	a	70	a
71	a	72	b	73	b	74	c	75	c	76	c	77	c	78	b	79	d	80	d
81	c	82	d	83	c	84	a	85	c	86	d	87	b	88	d	89	c	90	a
91	c	92	a	93	b	94	d	95	d	96	d	97	a	98	c	99	b	100	b
101	c	102	c	103	d	104	c	105	d	106	c	107	b	108	b				

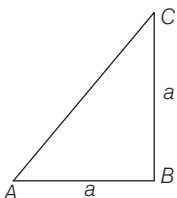
HINTS AND SOLUTIONS

1. (a) Area of square = $\frac{1}{2} \times (\text{Diagonal})^2$
 $= \frac{1}{2} \times 50 \times 50$
 $= 1250 \text{ m}^2$

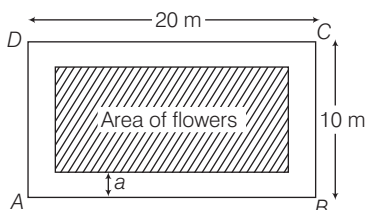
2. (c) Let length of rectangle be x cm.
 and breadth of rectangle = $(x - 2)$ cm
 $\therefore$ Perimeter of rectangle = $2(l + b)$
 So, $2[x + (x - 2)] = 48$
 $\Rightarrow 4x - 4 = 48$
 $\Rightarrow x = \frac{52}{4} = 13$
 $\therefore$ Length = 13 cm and breadth = 11 cm
 Hence, area of rectangle
 $= l \times b = 13 \times 11$
 $= 143 \text{ cm}^2$

3. (a) Length of the arc = $\frac{2\pi r\theta}{360^\circ}$
 $= \frac{2 \times 22 \times 42 \times 72}{7 \times 360} = 52.8 \text{ cm}$

4. (d) Area of an isosceles right triangle with side
 $'a' = \frac{1}{2} a^2 = 200 \text{ cm}^2$
 $\therefore$ Side = $a = 20 \text{ cm}$
 Hence, hypotenuse
 $= \sqrt{a^2 + a^2}$
 $= \sqrt{2}a = 20\sqrt{2} \text{ cm}$



5. (b) Let the width of the walk be ' a ' m.



Given, length of outer rectangle = 20 m
 and breadth of outer rectangle = 10 m
 $\therefore$ Width of walk = a m
 $\therefore$ Length of inner rectangle
 $= (20 - a - a) = (20 - 2a)$ m
 and breadth of inner rectangle
 $= (10 - a - a) = (10 - 2a)$ m
 Now, area of inner rectangle
 $= (20 - 2a)(10 - 2a)$
 $\therefore (20 - 2a)(10 - 2a) = 96$ [given]
 $\Rightarrow 4a^2 - 60a + 104 = 0$
 $\Rightarrow a^2 - 15a + 26 = 0$
 $\Rightarrow (a - 13)(a - 2) = 0$
 But $a \neq 13$, so $a = 2$ m

6. (c) Side of the greatest square tile
 $= \text{GCM of the length and breadth of the room}$
 $= \text{GCM of } 10.5 \text{ and } 3 \text{ is } 1.5 \text{ m}$
 Area of room = $10.5 \times 3 \text{ m}^2$
 $\therefore$ Number of tiles needed
 $= \frac{l \times b}{(\text{H.C.F of } l \text{ \& } b)^2}$
 $= \frac{10.5 \times 3}{2.25} = 14 \text{ tiles}$

7. (c) Let side of a square be a . Area of square = a^2

$\therefore$ The side of a square be increased by 50%, then new side = $a + \frac{a}{2} = \frac{3a}{2}$

New area = $\frac{9a^2}{4}$

Increase in area = $\frac{9a^2}{4} - a^2 = \frac{5a^2}{4}$

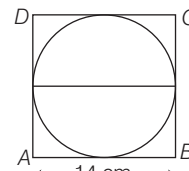
$\therefore$ Percent increase in area = $\frac{5a^2}{4a^2} \times 100$
 $= 125\%$

Shortcut Method

Side of square is increased by 50%.

$\therefore$ Net effect on area
 $= \left[50 + 50 + \frac{50 \times 50}{100} \right] \% = 125\%$

8. (d) Given, diameter of circle = Side of square = 14 cm
 $\therefore r = 7 \text{ cm}$



Area of circle = $\pi r^2 = \frac{22}{7} \times 7 \times 7$
 $= 154 \text{ cm}^2$

9. (c) Let original radius of circle be r .
 $\therefore$ Area of circle = πr^2
 $\therefore$ Radius of a circle is decreased by 50%, then
 New radius = $\frac{r}{2}$

$$\text{New area of circle} = \frac{\pi r^2}{4}$$

$\therefore$ Decrease in area

$$= \pi r^2 - \frac{\pi r^2}{4} = \frac{3\pi r^2}{4}$$

Decrease percent in area

$$= \left(\frac{3}{4} \pi r^2 \times \frac{1}{\pi r^2} \times 100 \right) \% = 75\%$$

Short trick Area is decreased by

$$= \left(-2x + \frac{x^2}{100} \right) = \left(-2 \times 50 + \frac{(50)^2}{100} \right)$$

$$= \left(-100 + \frac{2500}{100} \right) = -75\%$$

Area is decreased by 75%.

10. (a) Perimeter of square

$$= 4 \times 11 = 44 \text{ cm}$$

$$2\pi r = 44 \text{ cm} \quad [\text{by condition}]$$

$$\therefore r = \frac{44}{2\pi} = 7 \text{ cm}$$

Thus, area of circle

$$= \pi r^2 = \frac{22}{7} \times 7 \times 7 = 154 \text{ cm}^2$$

11. (b) Circumference of circle

$$= 2\pi r = 2 \times \frac{22}{7} \times 42 = 264 \text{ cm}$$

$\therefore$ Length of wire = 264 cm

Wire is bent into a square.

$\therefore$ Perimeter of square = 264 cm

$$\Rightarrow 4 \times \text{Sides of square} = 264$$

$$\therefore \text{Side of square} = \frac{264}{4} = 66 \text{ cm}$$

12. (c) Given, length of string = Radius of circle = 28 m

Area over which the horse can graze

$$= \pi r^2 = \frac{22}{7} \times 28 \times 28 = 2464 \text{ m}^2$$

13. (c) $\therefore$ Area of ring = $\pi(R^2 - r^2)$

Here, $r = 6 \text{ cm}$, $R = ?$

$$\text{Area of ring} = 418 = \frac{22}{7} (R^2 - 6^2)$$

[given]

$$\Rightarrow R^2 - 36 = \frac{418 \times 7}{22}$$

$$\Rightarrow R^2 = 133 + 36 = 169$$

$$\Rightarrow R = \sqrt{169} = 13 \text{ cm}$$

14. (a) Let the length and breadth of rectangle be l and b , respectively.

$\therefore$ Area of rectangle = $l \times b$ sq units

$$\therefore \text{New length} = l + \frac{60}{100} \times l = \frac{8l}{5}$$

New breadth = a

$$\text{Then, } lb = \frac{8l}{5} \times a \Rightarrow a = \frac{5b}{8}$$

[by condition]

Decrease percent

$$= \left[\left(b - \frac{5}{8}b \right) \times \frac{1}{b} \times 100 \right] \%$$

$$= 37 \frac{1}{2} \%$$

15. (b) Let the length and breadth of rectangular field be $5x$ and $3x$, respectively.

$$l = 5x \text{ and } b = 3x$$

$$\text{Perimeter of rectangular field} = 2(l + b)$$

$$2(5x + 3x) = 240$$

$$\Rightarrow 8x = 120 \Rightarrow x = 15$$

$$\therefore l = 5x = 5 \times 15 = 75 \text{ m}$$

$$\text{and } b = 3x = 3 \times 15 = 45 \text{ m}$$

$$\text{Now, area of rectangle} = l \times b$$

$$= 75 \times 45 = 3375 \text{ m}^2$$

16. (a) Given, inner circumference

$$= 2\pi r = 440 \text{ m}$$

$$\Rightarrow r = \frac{440}{2 \times 22} \times 7 = 70 \text{ m}$$

Width of track = 14 m

$$\therefore \text{Radius of outer circle} = (70 + 14) \text{ m}$$

$$= 84 \text{ m}$$

$\therefore$ Diameter of outer circle

$$= 2 \times 84 = 168 \text{ m}$$

17. (c) Let length and breadth of rectangular be x and y , respectively.

The original area = xy

New length = $x + x \times 50\%$

$$= x + \frac{x}{2} = \frac{3x}{2}$$

New breadth = $y + y \times 20\%$

$$= y + \frac{y}{5} = \frac{6y}{5}$$

$$\text{New area} = \frac{3x}{2} \times \frac{6y}{5} = \frac{9}{5} xy$$

$$= \frac{9}{5} (\text{original area})$$

Short trick Area is increased by

$$= 50 + 20 + \frac{50 \times 20}{100} = 80\%$$

$$\text{New Area} = A + \frac{80}{100} \times A = \frac{9}{5} A$$

18. (a) Let each equal side = $a = 13 \text{ cm}$ and base $b = 24 \text{ cm}$

$\therefore$ Area of the isosceles triangle

$$= \frac{1}{4} b \cdot \sqrt{4a^2 - b^2}$$

$$= \left[\frac{1}{4} \times 24 \times \sqrt{4 \times 169 - 24 \times 24} \right]$$

$$= 60 \text{ cm}^2$$

19. (c) Let the sides containing right angle be $x \text{ cm}$ and $(x - 14) \text{ cm}$.

Now, area of right triangle

$$= \left[\frac{1}{2} x \times (x - 14) \right] \text{ cm}^2$$

But area = 120 cm^2 [given]

$$\Rightarrow \frac{1}{2} x(x - 14) = 120$$

$$\Rightarrow x^2 - 14x - 240 = 0$$

$$\Rightarrow (x - 24)(x + 10) = 0$$

$$\Rightarrow x \neq -10 \therefore x = 24$$

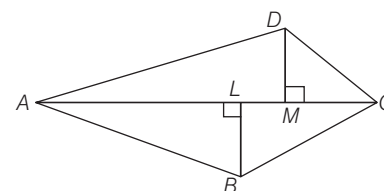
Other side = $24 - 14 = 10 \text{ cm}$

$$\text{Hypotenuse} = \sqrt{24^2 + 10^2} = \sqrt{676} \text{ cm}$$

$$= 26 \text{ cm}$$

$$\therefore \text{Perimeter} = (24 + 10 + 26) = 60 \text{ cm}$$

20. (d) Here, $AC = 128 \text{ m}$, $BL = 22.7 \text{ m}$, $DM = 17.3 \text{ m}$



$\therefore$ Area of the field

$$= \frac{1}{2} [AC(BL + DM)]$$

$$= \frac{1}{2} \times 128 (22.7 + 17.3)$$

$$= 64 \times 40 = 2560 \text{ m}^2$$

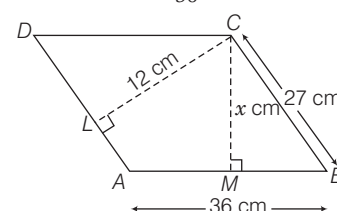
21. (a) Let distance between the longer sides be $x \text{ cm}$.

Area of parallelogram

$$= AD \times LC = MC \times AB$$

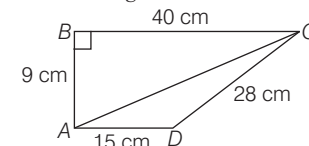
$$\therefore 36 \times x = 27 \times 12$$

$$\Rightarrow x = \frac{27 \times 12}{36} = 9$$



$\therefore$ Distance between the longer sides = 9 cm

22. (b) Applying pythagoras theorem in $\triangle ABC$, we get



$$9^2 + 40^2 = AC^2$$

$$\Rightarrow AC = \sqrt{1681} = 41 \text{ cm}$$

$$\text{Area of } \triangle ABC = \frac{1}{2} \times 9 \times 40 \text{ cm}^2$$

$$= 180 \text{ cm}^2$$

In $\triangle ACD$,

$$s = \frac{41 + 28 + 15}{2} = \frac{84}{2} = 42 \text{ cm}$$

$$\begin{aligned}\Delta &= \sqrt{s(s-a)(s-b)(s-c)} \\ &= \sqrt{42(42-41)(42-28)(42-15)} \\ &= \sqrt{42 \times 1 \times 14 \times 27} \\ &= 14 \times 3 \times 3 = 126 \text{ cm}^2\end{aligned}$$

$\therefore$ Area of quadrilateral = Area of $\triangle ABC$ + Area of $\triangle ADC$
 $= 180 + 126 = 306 \text{ cm}^2$

23. (c) Distance covered in one revolution
 $= \frac{11 \times 1000 \times 100}{5000} = 220 \text{ cm}$

$\therefore$ The circumference of the wheel
 $= 220 \text{ cm}$

Let the diameter be 'D'.

Then, $\pi D = 220 \Rightarrow \frac{22}{7} \times D = 220$

$\therefore D = \frac{220 \times 7}{22} = 70 \text{ cm}$

24. (a) Let radius of given circles be x cm and $(14 - x)$ cm.

$\therefore$ Sum of areas of circle
 $= [\pi x^2 + \pi (14 - x)^2]$
 $130\pi = \pi x^2 + \pi (14 - x)^2$
 [by condition]

$\Rightarrow 130 = 2x^2 - 28x + 196$

$\Rightarrow x^2 - 14x + 33 = 0$

$\Rightarrow (x - 11)(x - 3) = 0$

$\Rightarrow x = 11 \text{ or } x = 3$

So, the radii of circles are 11 cm and 3 cm.

25. (d) Angle inscribed by minute hand in 60 min = 360° .

Angle inscribed in 35 min = $\frac{360^\circ}{60} \times 35$
 $= 210^\circ$

given, $r = 12 \text{ cm}$

$\therefore$ Area swept by the minute-hand in 35 min

= Area of sector with $r = 12 \text{ cm}$ and $\theta = 210^\circ$

$= \frac{22}{7} \times \left(12 \times 12 \times \frac{210}{360} \right) \text{ cm}^2 = 264 \text{ cm}^2$

26. (b) Area of the shaded region
 $= (\text{Area of sector with } r = 7 \text{ cm}, \theta = 30^\circ)$
 $- (\text{Area of sector with } r = 3.5 \text{ cm}, \theta = 30^\circ)$

$= \left[\left(\frac{22}{7} \times 7 \times 7 \times \frac{30}{360} \right) - \left(\frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \times \frac{30}{360} \right) \right] \text{ cm}^2$

$= \left(\frac{77}{6} - \frac{77}{24} \right) \text{ cm}^2 = \frac{77}{8} \text{ cm}^2$

27. (d) Area of field = 1 hect = 10000 m^2

$\therefore$ side = $\sqrt{10000} \text{ m} = 100 \text{ m}$

$\therefore$ Side of other field = 102% of 100
 $= 102$

$\therefore$ Area of the field
 $= 102 \times 102 = 10404 \text{ m}^2$

Thus, difference of area

$= (10404 - 10000) \text{ m}^2$
 $= 404 \text{ m}^2$

28. (c) Let diameter of the circle be $2r$.

$\therefore$ Area of the circle = πr^2

Diameter is increased by 100%

New diameter = $4r$

and new area = $4\pi r^2$

$\therefore$ Increase in area = $4\pi r^2 - \pi r^2 = 3\pi r^2$

$\therefore$ Increase percentage in area

$= \left(\frac{3\pi r^2}{\pi r^2} \times 100 \right) \% = 300\%$

Short trick Area is increased by

$= \left(2x + \frac{x^2}{100} \right) \% = \left(2 \times 100 + \frac{100^2}{100} \right) \%$
 $= (200 + 100) \% = 300\%$

29. (c) Area of trapezium = $1/2$

(Sum of parallel sides)

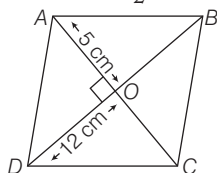
$\times$ Distance between them

$= 1/2 (25 + 15) \times 7 = 140 \text{ cm}^2$

30. (b) Since, diagonal of rhombus bisect each other and perpendicular to each other.

$\therefore AO = \frac{10}{2} \text{ cm} = 5 \text{ cm},$

$DO = \frac{24}{2} \text{ cm} = 12 \text{ cm}$



$\therefore AD = \sqrt{OA^2 + OD^2} = \sqrt{5^2 + 12^2}$
 $= \sqrt{169} = 13 \text{ cm}$

$\therefore$ Perimeter = $4 \times 13 = 52 \text{ cm}$

31. (a) Here, $OA = OB = OC = 10 \text{ cm}$

Also, $AC = 2CP$

and $CP = \sqrt{OC^2 - OP^2} = \sqrt{10^2 - 5^2}$
 $= \sqrt{75} = 5\sqrt{3} \text{ cm}$

As, $AC = 2PC$

$\therefore AC = 10\sqrt{3} \text{ cm}$

$\therefore$ Area of the rhombus $OABC$

$= \frac{1}{2} \times d_1 \times d_2 = \frac{1}{2} \times OB \times AC$

$= \frac{1}{2} \times 10 \times 10\sqrt{3} = 50\sqrt{3} \text{ cm}^2$

32. (c) Area of shaded region

= Area of $\triangle ABP$ + Area of $\triangle PDC$

$= \frac{1}{2} \times AB \times AP + \frac{1}{2} DC \times PD$

$= \frac{1}{2} \times AB \times AP + \frac{1}{2} AB \times PD$

[$\because AB = DC$]

$= \frac{1}{2} AB (AP + PD)$

$= \frac{1}{2} AB \times AD$ [$\because AD = (AP + PD)$]

$= \frac{1}{2} AB \times \frac{AB}{2} = \frac{1}{4} a^2$

[$\because AB = a$ and $AD = \frac{AB}{2}$]

33. (a) Area of figure = Area of rectangle

+ 2 (Area of semi-circular ends)

$= 20 \times 14 + 2 \times \frac{1}{2} \times \frac{22}{7} \times 7 \times 7$

$= 280 + 154 = 434 \text{ m}^2$

$\therefore$ Cost of levelling = ₹ (434×25)
 $= ₹ 10850$

34. (c) Length of

$AC = AB + BC = 4 + 14 = 18 \text{ cm}$

The area (shaded) bounded by three semi-circle = Area of semi-circle with AC as diameter - [(Area of semi-circle with diameter AB + Area of semi-circle with diameter BC)]

$= \frac{1}{2} \pi (9)^2 - \left[\frac{1}{2} \pi (7)^2 + \frac{1}{2} \pi (2)^2 \right]$

$= \pi \frac{81}{2} - \frac{1}{2} [49\pi + 4\pi]$

$= \frac{\pi(81)}{2} - \frac{1}{2} [53\pi]$

$= \frac{\pi}{2} (81 - 53) = \frac{\pi(28)}{2} = 14\pi$

$\therefore$ Area of shaded region

= 14 times π

35. (b) Area of shaded region = Area of

horizontal rectangle

+ Area of vertical rectangle

$= 5 \times 1 + (8 - 1) \times 1 = 5 + 7 = 12 \text{ m}^2$

36. (b) Length of boundary = Length of arc ($APB + BQC + ARC$)

$= \pi \left(\frac{AB}{2} \right) + \pi \left(\frac{BC}{2} \right) + \pi \left(\frac{AC}{2} \right)$

$= \pi \left(\frac{5}{2} \right) + \pi \left(\frac{5}{2} \right) + \pi (5)$

$= \frac{20\pi}{2} = 10\pi \text{ cm}$

37. (d) Radius of smaller circle = 2 cm

$\therefore$ Area of shaded region

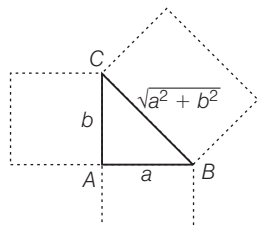
$= \frac{1}{2} [\text{Area of larger circle} - 2(\text{Area of smaller circle})]$

$= \frac{1}{2} [\pi 4^2 - 2(\pi 2^2)]$

$= \frac{1}{2} [16\pi - 8\pi]$

$= \frac{1}{2} (8\pi) = 4\pi \text{ cm}^2$

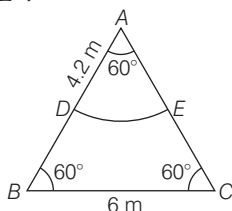
38. (c)

 $\therefore$ Total area

$$= a^2 + b^2 + (\sqrt{a^2 + b^2})^2 + \frac{1}{2}ab$$

$$= 2(a^2 + b^2) + 0.5ab$$

39. (c) Suppose, a horse is tied at vertex A. Then, area available grazing field is ADE.



Now, area of sector $ADE = \frac{\pi r^2 \theta}{360^\circ}$

$$= \frac{22 \times (4.2)^2 \times 60^\circ}{7 \times 360^\circ} = 9.24 \text{ m}^2$$

and area of equilateral

$$\Delta ABC = \frac{\sqrt{3}}{4} (\text{Side})^2 = \frac{\sqrt{3}}{4} \times (6)^2$$

$$= 15.57$$

$$\therefore \text{Required percentage} = \frac{9.24}{15.57} \times 100$$

$$= 59.34\% = 59\% \text{ (approx)}$$

40. (c) Given, radius of circumcircle,
 $r = 2$ unitsside of square, $AB = 2$ unitsSince, $EF = ME = MF = 2$ units $\therefore MEF$ is an equilateral triangleIn ΔAEM , by pythagoras theorem

$$AE^2 = EM^2 - AM^2 = (2)^2 - (1)^2$$

$$\therefore AE = \sqrt{3} \text{ units}$$

Area of intersection region square and circle = Area of sector $MENF$ + Area of ΔAEM + Area of ΔFMB

$$= \frac{\theta}{360^\circ} \times \pi r^2 + \frac{1}{2} \times AE \times AM$$

$$+ \frac{1}{2} \times MB \times FB$$

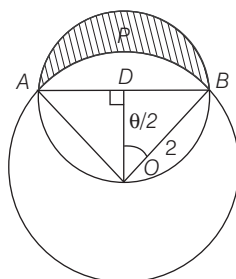
$$= \frac{60^\circ}{360^\circ} \times \pi \times (2)^2 + \frac{1}{2} \times \sqrt{3} \times 1$$

$$+ \frac{1}{2} \times \sqrt{3} \times 1 [\because FB = EA = \sqrt{3}]$$

$$= \frac{4\pi}{6} + \sqrt{3} = \frac{2}{3}\pi + \sqrt{3}$$

41. (b) Given, radius, $r = 2$ cm
length of chord, $AB = 2\sqrt{2}$ cm

$$\text{In } \Delta ODB, \sin \frac{\theta}{2} = \frac{BD}{OB}$$



$$\Rightarrow \sin \frac{\theta}{2} = \frac{2\sqrt{2}}{2} = \frac{1}{\sqrt{2}} \Rightarrow \frac{\theta}{2} = 45^\circ$$

$$\therefore \theta = 90^\circ$$

Area of segment $APBA$ = Area of sector
– Area of ΔOAB

$$= \frac{1}{4} \cdot \pi r^2 - \frac{1}{2} \times OA \times OB$$

$$= \frac{1}{4} \times \pi \times (2)^2 - \frac{1}{2} \times 2 \times 2$$

$$= (\pi - 2) \text{ cm}^2$$

Area of shaded region = Area of
semicircle with AB as diameter – Area
of segment $APBA$

$$= \frac{1}{2} \times \pi \times (BD)^2 - (\pi - 2)$$

$$= \frac{1}{2} \times \pi \times (\sqrt{2})^2 - (\pi - 2) = 2 \text{ cm}^2$$

42. (c) We have, $a = l \times w$... (i)
and $p = 2(l + w)$... (ii)
Putting the value of l from Eq. (i) in (ii),
we get

$$wp - 2w^2 = 2a$$

$$\text{Now, } p^2 - 8a = [2(l + w)]^2 - 8lw$$

$$= 4(l^2 + w^2 + 2lw) - 8lw$$

$$= 4l^2 + 4w^2 + 8lw - 8lw$$

$$= 4(l^2 + w^2)$$

Hence, the statement I and III are correct.

43. (d) I. Let R be the radius of circumcircle

$$\text{Area of circumcircle} = \pi R^2$$

$$\Rightarrow 9\pi = \pi R^2 \Rightarrow R = 3 \text{ cm}$$

II. $\frac{\text{Inradius of polygon}}{\text{circumradius of polygon}} = \frac{r}{R}$

$$= \frac{\frac{1}{2} a \cot \left(\frac{180^\circ}{n} \right)}{\frac{1}{2} a \operatorname{cosec} \left(\frac{180^\circ}{n} \right)} = \cos \left(\frac{180^\circ}{n} \right)$$

$$\Rightarrow r = R \cos \left(\frac{180^\circ}{12} \right) = 3 \cos 15^\circ$$

$$\text{III. Area of polygon} = \frac{1}{2} n R^2 \sin \left(\frac{360^\circ}{n} \right)$$

$$= \frac{1}{2} \times 12 \times R^2 \sin \left(\frac{360^\circ}{12} \right)$$

$$= 6R^2 \sin 30^\circ = 3R^2$$

$$\therefore \frac{\text{Area of circumcircle}}{\text{Area of polygon}} = \frac{\pi R^2}{3R^2} = \frac{\pi}{3}$$

or $\pi : 3$

Hence, all three statements are correct.

44. (c) $\therefore$ From statement Ilength = $2 \times$ width

$$\therefore \text{Area of rectangle} = 2 \times \text{width} \times \text{width}$$

$$= 2 (\text{width})^2$$

From statement II,

$$\therefore \text{Area of rectangle} = 2 \times \text{Perimeter}$$

$$= 2 \times 2(\text{length} + \text{width})$$

$$= 4 (\text{length} + \text{width})$$

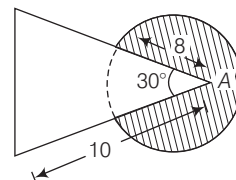
From I and II,

$$2 (\text{width})^2 = 4 (2 \text{ width} + \text{width})$$

$$2 (\text{width})^2 = 12 \text{ width}$$

$$\Rightarrow \text{width} = 6 \text{ units.}$$

So, both statements together are sufficient.

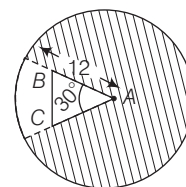
45. (d) The length of the rope is 8 m, then the cow will be able to graze an area equal to the area of the circle with radius = 8 m, subtracting from that the area of the sector of the same circle with angle 30° ,

$$\text{which is equal to } \pi(8)^2 - \frac{\theta}{360} \pi(8)^2$$

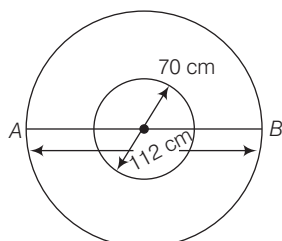
$$= \pi \times 64 - \frac{30}{360} \pi \times 64 = \frac{176\pi}{3} \text{ m}^2$$

46. (c) Since, the length of the rope is more than that of sides AB and AC , $\therefore$ Required Area =(area of the circle with radius 12)
– (area of the sector of the same circle
with angle 30°)

$$= \pi(12)^2 - \frac{30}{360} \pi(12)^2 = 132\pi \text{ m}^2$$

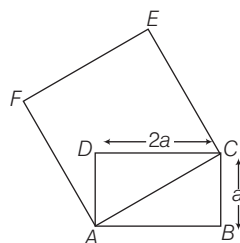


47. (c) $\therefore$ Outer diameter = 112 cm
and inner diameter = 70 cm
 $\therefore$ Required area = $\frac{1}{4}\pi(112^2 - 70^2)$



$$\begin{aligned} &= \frac{1}{4}(12544 - 4900)\pi \\ &= \frac{1}{4} \times 7644 \times \frac{22}{7} \\ &= \frac{1}{4} \times 24024 = 6006 \text{ cm}^2 \end{aligned}$$

48. (d) Given that, Area of rectangle = $2a^2$
 $= l \times b$
 $\Rightarrow l \times b = 2a^2 = l \times a \Rightarrow l = 2a$

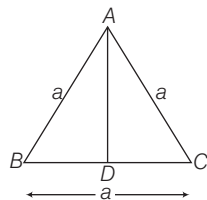


Now, in $\triangle ACD$, by pythagoras theorem

$$\begin{aligned} AC^2 &= AD^2 + CD^2 \\ &= a^2 + 4a^2 = 5a^2 \end{aligned}$$

$\therefore$ Side of square, $AC = a\sqrt{5}$ unit
Hence, area of the square = $(a\sqrt{5})^2$
 $= 5a^2$ sq units

49. (c)



Height of equilateral triangle (AD)
 $= \frac{\sqrt{3}}{2} \times \text{Side} \Rightarrow \sqrt{3} = \frac{\sqrt{3} \times \text{Side}}{2}$

$$\Rightarrow 2\sqrt{3} = \sqrt{3} \times \text{Side}$$

$$\therefore \text{Side} = \frac{2\sqrt{3}}{\sqrt{3}} = 2 \text{ cm}$$

$\therefore$ Perimeter of an equilateral triangle
 $= 3a = 3 \times 2 = 6 \text{ cm}$

50. (a) Let the width of the rectangle be x unit.
 $\therefore$ Length = $(2x + 5)$ unit
According to the question,

$$\begin{aligned} \text{Area} &= x(2x + 5) \Rightarrow 75 = 2x^2 + 5x \\ \Rightarrow 2x^2 + 5x - 75 &= 0 \\ \Rightarrow 2x^2 + 15x - 10x - 75 &= 0 \\ \Rightarrow x(2x + 15) - 5(2x + 15) &= 0 \\ \Rightarrow (2x + 15)(x - 5) &= 0 \\ \therefore x = 5 \text{ and } \frac{-15}{2} \end{aligned}$$

Since, width cannot be negative.

$\therefore$ Width = 5 units

and length = $2 \times 5 + 5 = 15$ units

$\therefore$ Perimeter of the rectangle
 $= 2(15 + 5) = 40$ units

51. (a) Let the radius of circle is r and the side of a square is a , then by given condition,

$$2\pi r = 4a \Rightarrow a = \frac{\pi r}{2}$$

$$\begin{aligned} \therefore \text{Area of square} &= \left(\frac{\pi r}{2}\right)^2 = \frac{\pi^2 r^2}{4} \\ &= \frac{(3.14)^2 r^2}{4} = \frac{9.86 r^2}{4} = 2.46 r^2 \end{aligned}$$

and area of circle = $\pi r^2 = 3.14 r^2$

Again, let the side of equilateral triangle be x .

Then, by given condition, $3x = 2\pi r$

$$\Rightarrow x = \frac{2\pi r}{3}$$

$$\begin{aligned} \therefore \text{Area of equilateral triangle} &= \frac{\sqrt{3}}{4} x^2 \\ &= \frac{\sqrt{3}}{4} \times \frac{4\pi^2 r^2}{9} = \frac{\pi^2}{3\sqrt{3}} r^2 = 189 r^2 \end{aligned}$$

Hence, Area of circle > Area of square
> Area of equilateral triangle

Hence, only I statements is correct.

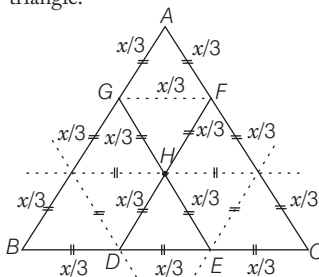
52. (c) Given that, Area of the circle = Area of the square = $(\text{Side})^2$

$$\therefore \pi r^2 = (2\sqrt{\pi})^2 \Rightarrow \pi r^2 = 4\pi$$

$$\Rightarrow r^2 = \frac{4\pi}{\pi} = 4 \therefore r = \sqrt{4} = 2 \text{ units}$$

$$\therefore \text{Diameter of circle (d)} = 2 \cdot r = 2 \cdot 2 = 4 \text{ units}$$

53. (b) Here, $\triangle ABC$ forms an equilateral triangle.



where, AGHF form a rhombus and $\triangle HDE$ is also an equilateral triangle.

$\therefore$ Area of rhombus = (Area of $\triangle AGF$
+ Area of $\triangle GFH$)

$$= \frac{\sqrt{3}}{4} \left(\frac{x}{3}\right)^2 + \frac{\sqrt{3}}{4} \left(\frac{x}{3}\right)^2 = 2 \cdot \frac{\sqrt{3}}{4} \left(\frac{x}{3}\right)^2$$

$$\text{Now, area of } \triangle HDE = \frac{\sqrt{3}}{4} \left(\frac{x}{3}\right)^2$$

$$\text{and area of } \triangle ABC = \frac{\sqrt{3}}{4} x^2$$

By given condition,

Area of rhombus AGHF

+ Area of $\triangle HDE$

Area of $\triangle ABC$

$$= \frac{2 \cdot \frac{\sqrt{3}}{4} \left(\frac{x}{3}\right)^2 + \frac{\sqrt{3}}{4} \left(\frac{x}{3}\right)^2}{\frac{\sqrt{3}}{4} x^2}$$

$$= \frac{3 \cdot \frac{\sqrt{3}}{4} \left(\frac{x}{3}\right)^2}{\frac{\sqrt{3}}{4} x^2} = \frac{3}{9} = \frac{1}{3}$$

54. (b) We know that, the radius of a circle inscribed in an equilateral triangle = $a / 2\sqrt{3}$ where, a be the length of the side of an equilateral triangle.

Given that, area of a circle inscribed in an equilateral triangle = 154 cm^2

$$\therefore \pi \left(\frac{a}{2\sqrt{3}}\right)^2 = 154$$

$$\Rightarrow \left(\frac{a}{2\sqrt{3}}\right)^2 = \frac{154 \times 7}{22} = 7 \times 7 = (7)^2$$

$$\Rightarrow \frac{a}{2\sqrt{3}} = 7 \Rightarrow a = 14\sqrt{3} \text{ cm}$$

$$\therefore \text{Perimeter of an equilateral triangle} = 3a = 3(14\sqrt{3}) = 42\sqrt{3} \text{ cm}$$

55. (b) Let the radii of two circles be r_1 and r_2 , respectively.

$$\text{Given, } \frac{\text{Circumference of Ist circle}}{\text{Circumference of IInd circle}} = \frac{2}{3}$$

$$\Rightarrow \frac{2\pi r_1}{2\pi r_2} = \frac{2}{3} \Rightarrow \frac{r_1}{r_2} = \frac{2}{3} \Rightarrow \left(\frac{r_1}{r_2}\right)^2 = \frac{4}{9}$$

$$\therefore \frac{\text{Area of Ist circle}}{\text{Area of IInd circle}} = \frac{\pi r_1^2}{\pi r_2^2} = \left(\frac{r_1}{r_2}\right)^2 = \frac{4}{9} \dots(i)$$

$$\text{or } 4 : 9$$

56. (c) Let the radius of a circle be r and a be the length of the side of a square.

Given, circumference of a circle

= Perimeter of a square

$$\Rightarrow 2\pi r = 4a \Rightarrow a = \frac{\pi}{2} r = 1.57r$$

Now, area of the circle (A_c)

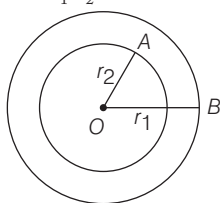
$$= \pi r^2 = 3.14 r^2$$

and area of the square (A_s)

$$= a^2 = 2.4649 r^2$$

$\therefore$ Area of circle > Area of square

57. (c) Given that, ratio of their radii = 5 : 3
i.e. $r_1 : r_2 = 5 : 3$



$$\Rightarrow \frac{r_1}{r_2} = \frac{5}{3} \quad \dots(i)$$

Let, $r_1 = 5x$ and $r_2 = 3x$

Also, given that area enclosed between the circumferences of two concentric circles = $16\pi \text{ cm}^2$

$$\therefore \pi(r_1^2 - r_2^2) = 16\pi$$

$$\Rightarrow (5x)^2 - (3x)^2 = 16$$

$$\Rightarrow 25x^2 - 9x^2 = 16$$

$$\Rightarrow 16x^2 = 16 \Rightarrow x^2 = 1$$

$$\therefore x = 1$$

$$\therefore r_1 = 5 \text{ and } r_2 = 3$$

$$\therefore \text{Area of the outer circle} = \pi r_1^2 = \pi(5)^2 = 25\pi \text{ cm}^2$$

58. (b) Given that, perimeter of a rectangle = 82 m

$$\therefore 2(\text{Length} + \text{Breadth}) = 82 \text{ m}$$

$$\Rightarrow \text{Length} + \text{Breadth} = 41 \text{ m}$$

$$\Rightarrow l + b = 41 \text{ m} \quad \dots(i)$$

Also, its area = 400 m^2

$$\Rightarrow l \cdot b = 400 \quad \dots(ii)$$

$$\begin{aligned} \text{Now, } (l-b)^2 &= (l+b)^2 - 4lb \\ &= (41)^2 - 4(400) \\ &= 1681 - 1600 = 81 \end{aligned}$$

$$\therefore l - b = 9 \quad \dots(iii)$$

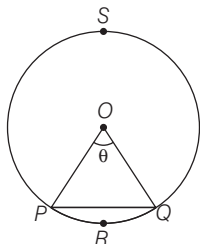
On solving Eqs. (i) and (iii), we get

$$2l = 50 \Rightarrow l = 25 \text{ m and } b = 16 \text{ m}$$

$\therefore$ Required breadth (b) = 16 m

59. (c) I. We know that, area of segment (PRQP)

$$\begin{aligned} &= \text{Area of sector (OPRQO)} \\ &\quad - \text{Area of } \triangle OPQ \\ &= \frac{\pi r^2 \theta}{360} - \frac{1}{2} r^2 \sin \theta \end{aligned}$$



So, the area of a segment of a circle is always less than area of its corresponding sector.

- II. Distance travelled by a circular wheel of diameter $2d$ cm in one revolution = $2\pi \frac{(2d)}{2} = 2 \times 3.14 \times d = 6.28d$

which is greater than $6d$ cm.

Hence, both statements are correct.

60. (a) $\therefore$ Angle described in 60 min by minute hand of a clock = 360°
and angle described in 40 min by minute hand of a clock = $\frac{360^\circ}{60} \times 40 = 240^\circ$

$$\therefore \text{Required distance} = \frac{2\pi(2.5) \times 240^\circ}{360^\circ} = \frac{10\pi}{3} \text{ cm}$$

61. (d) Given that, length of hour hand = 4 cm

and length of minute hand = 6 cm

$\therefore$ Hour hand rotating in 1 day = $2 \times 360^\circ = 720^\circ$

$\therefore$ Hour hand rotating in 2 days = $2 \times 720^\circ = 1440^\circ$

Similarly,

Minute hand rotating in 1 day = $24 \times 360^\circ$

$\therefore$ Minute hand rotating in 3 days = $72 \times 360^\circ$

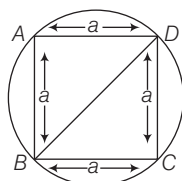
$\therefore$ Distance travelled by hour hand = $\frac{2\pi(4) \times 1440^\circ}{360^\circ} = 32\pi$

and distance travelled by minute hand = $\frac{2\pi(6) \times 72 \times 360^\circ}{360} = 6 \times 144\pi \text{ cm}$

$\therefore$ Required ratio = $32\pi : 6 \times 144\pi = 1 : 27$

62. (c) Let side of a square be a cm.

Given that, radius of a circle = 8 cm and diameter of a circle = 16 cm



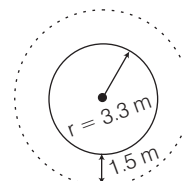
$\therefore$ Length of a diagonal of a square = Diameter of a circle

$$\Rightarrow a\sqrt{2} = 16$$

$$\therefore a = 8\sqrt{2} \text{ cm}$$

$$\therefore \text{Area of square } ABCD = a^2 = (8\sqrt{2})^2 = 64 \times 2 = 128 \text{ cm}^2$$

63. (c) Area of the path = Area of circular water [fountain + path] - Area of circular water fountain



$$\begin{aligned} &= \pi(3.3 + 1.5)^2 - \pi(3.3)^2 \\ &= [(4.8)^2 - (3.3)^2] \pi \\ &= (23.04 - 10.89) \pi \\ &= 12.15 \pi \text{ m}^2 \end{aligned}$$

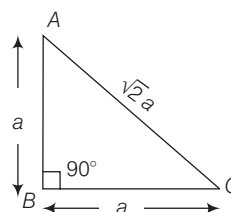
64. (d) Let the equal sides of isosceles right triangle be a cm.

Then $AB = BC = a$

In $\triangle ABC$, by pythagoras theorem

$$AC^2 = AB^2 + BC^2 = 2AB^2$$

$$\Rightarrow AC^2 = 2a^2 \therefore AC = \sqrt{2}a$$

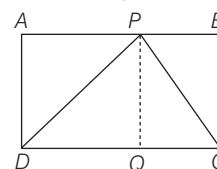


$$\text{Area of } \triangle ABC = \frac{1}{2} \times a \times a$$

$$\Rightarrow 1 = \frac{1}{2} a^2 \Rightarrow a = \sqrt{2}$$

$$\begin{aligned} \therefore \text{Perimeter of } \triangle ABC &= 2a + a\sqrt{2} \\ &= 2\sqrt{2} + \sqrt{2} \cdot \sqrt{2} \\ &= 2(\sqrt{2} + 1) \text{ units} \end{aligned}$$

65. (c) Given that, $CD = 20$ cm and area of rectangle $ABCD = 100 \text{ cm}^2$



$$\Rightarrow AD \times CD = 100 \text{ cm}^2$$

$$\Rightarrow AD \times 20 = 100$$

$$\therefore AD = 5 \text{ cm}$$

$$\begin{aligned} \therefore \text{Area of } \triangle PDC &= \frac{1}{2} \times PQ \times CD \\ &= \frac{1}{2} \times 5 \times 20 = 5 \times 10 = 50 \text{ cm}^2 \end{aligned}$$

$$[\because AD = PQ]$$

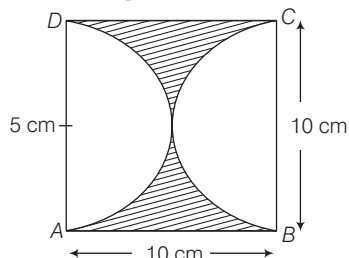
66. (c) Area of parallelogram

$$= \text{Base} \times \text{Height}$$

$$= 8.06 \times 2.08 = 16.76 \text{ cm}^2$$

67. (a) Let $b = 4x$ and $l = 9x$
 $\therefore$ Area of rectangle $= l \times b$
 $144 = 4x \times 9x$
 $\Rightarrow x^2 = \frac{144}{36} \Rightarrow x^2 = 4 \therefore x = 2$
 Now, $b = 4 \times 2 = 8$ cm and
 $l = 9 \times 2 = 18$ cm
 $\therefore$ Perimeter of rectangle $= 2(b + l)$
 $= 2(8 + 18) = 2 \times 26 = 52$ cm

68. (d) Area between square and semi-circles
 $=$ Area of square $- 2$ (Area of semi-circle)



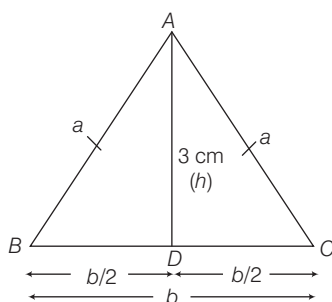
$$= (10)^2 - 2 \times \frac{1}{2} \times \frac{22}{7} \times (5)^2 = 100 - 78.5$$

$$= 21.5 \text{ cm}^2$$

Hence, the area between the square and semi-circular cross-section is 21.5 cm^2 .

69. (a) $1 \text{ m} = 1000 \text{ mm}$
 $\therefore 10 \text{ m} = 10000 \text{ mm}$
 Number of 200 mm lengths that can be cut from 10 m of ribbon
 $= \frac{10000}{200} = 50$

70. (a) Area of the $\triangle ABC = \frac{1}{2} \times b \times h$



$$\Rightarrow 12 = \frac{1}{2} \times b \times 3$$

$$\therefore b = \frac{12 \times 2}{3} = 8 \text{ cm}$$

$$\text{Here, } BD = CD = \frac{b}{2} = \frac{8}{2} = 4 \text{ cm}$$

In right angled $\triangle ABD$, by pythagoras theorem,

$$AB = \sqrt{BD^2 + AD^2}$$

$$\Rightarrow a = \sqrt{4^2 + 3^2} = \sqrt{16 + 9}$$

$$= \sqrt{25} = 5 \text{ cm}$$

Now, perimeter of an isosceles triangle

$$= 2a + b = 2 \times 5 + 8$$

$$= 10 + 8 = 18 \text{ cm}$$

71. (a) Area of equilateral triangle

$$= \frac{\sqrt{3}a^2}{4} = x \quad \dots(i)$$

$$\text{and perimeter} = 3a = y \Rightarrow a = \frac{y}{3} \dots(ii)$$

Now, putting the value of a from Eq. (ii) in Eq. (i), we get

$$\frac{\sqrt{3} \left(\frac{y}{3} \right)^2}{4} = x \Rightarrow x = \frac{\sqrt{3} \times y^2}{9 \times 4}$$

$$\Rightarrow x = \frac{y^2}{3\sqrt{3} \times 4} \Rightarrow x = \frac{y^2}{12\sqrt{3}}$$

$$\Rightarrow 12\sqrt{3} x = y^2$$

On squaring both sides, we get

$$y^4 = 432x^2$$

72. (b) Here, $a = 7$ cm, $b = 24$ cm

$$\text{and } c = 25 \text{ cm}$$

Semi-perimeter of triangle

$$s = \frac{a + b + c}{2} = \frac{7 + 24 + 25}{2} = \frac{56}{2}$$

$$= 28 \text{ cm}$$

According to the question,

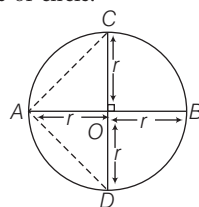
Area of circle = Area of triangle

$$= \sqrt{s(s-a)(s-b)(s-c)}$$

$$= \sqrt{28(28-7)(28-24)(28-25)}$$

$$= \sqrt{28 \times 21 \times 4 \times 3} = \sqrt{7056} = 84 \text{ cm}^2$$

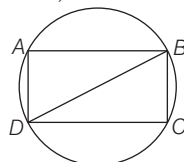
73. (b) Diameters of a circle intersect at the centre of circle.



$$\text{Required ratio} = \frac{\text{Area of circle}}{\text{Area of } \triangle ACD}$$

$$= \frac{\pi r^2}{\frac{1}{2} \times 2r \times r} = \pi$$

74. (c) Area of rectangle inscribed in a circle is maximum, when



Diameter of the circle = Diagonal of the rectangle

Now, let x and y be the length and breadth of the rectangle, respectively.

Now in $\triangle ABD$, by pythagoras theorem

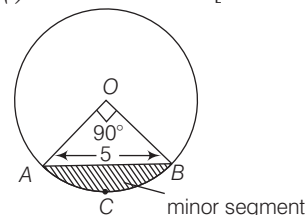
$$AB^2 + AD^2 = (5)^2 \Rightarrow x^2 + y^2 = 25$$

Since, they form pythagorean triplet,

$$\therefore x = 4 \text{ and } y = 3$$

$$\text{So, area of rectangle} = 3 \times 4 = 12 \text{ cm}^2$$

75. (c) Let $AO = OB = r$ [radius of circle]



In $\triangle AOB$, by pythagoras theorem,

$$AB^2 = OA^2 + OB^2$$

$$\Rightarrow (5)^2 = r^2 + r^2 \quad [\because AB = 5 \text{ cm}]$$

$$\therefore r^2 = \frac{25}{2} \text{ cm}$$

Now, area of sector

$$ACBOA = \frac{\theta}{360^\circ} \times \pi r^2$$

$$= \frac{90^\circ}{360^\circ} \times \pi \times \frac{25}{2} = \frac{25\pi}{8} \text{ cm}^2$$

Now, area of minor segment $ACBA$

$=$ Area of sector $ACBOA$ $-$ Area of triangle

$$= \frac{25\pi}{8} - \frac{r^2}{2} = \frac{25\pi}{8} - \frac{25}{4}$$

$$= \left(\frac{25\pi - 50}{8} \right) \text{ cm}^2$$

Area of major segment $=$ Area of circle $-$ Area of minor segment

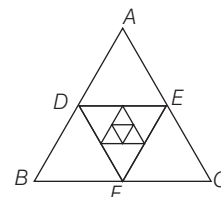
$$= \pi r^2 - \left(\frac{25\pi - 50}{8} \right)$$

$$= \frac{25\pi}{2} - \frac{(25\pi - 50)}{8}$$

$$= \frac{100\pi - 25\pi + 50}{8} = \frac{75\pi + 50}{8}$$

$$= \frac{25}{8} (3\pi + 2) = \frac{25}{4} \left(\frac{3\pi}{2} + 1 \right) \text{ cm}^2$$

76. (c)



Perimeters of triangles

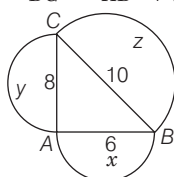
$$= 3 + 3 \times \frac{1}{2} + 3 \times \frac{1}{4} + 3 \times \frac{1}{8} + \dots$$

$$= 3 \left[1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots \right] = 3 \times \frac{1}{1 - \frac{1}{2}}$$

$$= 6 \text{ units}$$

77. (c) In $\triangle ABC$, by pythagoras theorem,

$$BC^2 = AB^2 + AC^2$$



$$= 6^2 + 8^2 = 36 + 64 = 100 \text{ cm}$$

$$\therefore BC = 10 \text{ cm}$$

$$\text{Radius of semi-circle } x = \frac{6}{2} = 3$$

$$\text{Now, area of semi-circle} = x = \frac{\pi(3)^2}{2} = \frac{9\pi}{2} \text{ cm}^2$$

$$\text{Radius of semi-circle, } y = \frac{8}{2} \text{ cm}$$

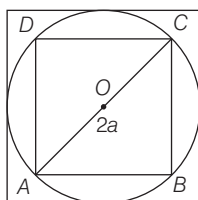
$$\text{Area of semi-circle} = y = \frac{16\pi}{2} \text{ cm}^2$$

$$\text{Radius of semi-circular, } z = \frac{10}{2} = 5 \text{ cm}^2$$

$$\text{Area of semi-circle} = z = \frac{25\pi}{2} \text{ cm}^2$$

$$\text{Now, } x + y - z = \frac{9\pi}{2} + \frac{16\pi}{2} - \frac{25\pi}{2} = 0$$

78. (b)



For inscribed circle, Diameter of circle = Diagonal of square = $2a$

Now, in $\triangle ABC$ by pythagoras theorem,

$$AB^2 + BC^2 = AC^2 \Rightarrow 2AB^2 = 4a^2$$

$$AB^2 = 2a^2 \Rightarrow AB = \sqrt{2}a$$

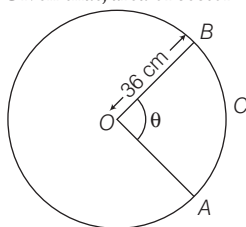
$$\therefore \text{Area of inner square} = AB^2 = (\sqrt{2}a)^2 = 2a^2$$

For circumscribed square, Diameter of a circle = Side of square

$$\therefore \text{Area of circumscribed square} = (2a)^2 = 4a^2$$

$$\therefore \text{Difference between areas of outer and inner squares} = 4a^2 - 2a^2 = 2a^2$$

79. (d) Given that, area of sector = $72\pi \text{ cm}^2$



$$\Rightarrow \frac{\pi r^2 \theta}{360^\circ} = 72\pi$$

$$\Rightarrow \theta = \frac{72 \times 360}{36 \times 36} = 20^\circ$$

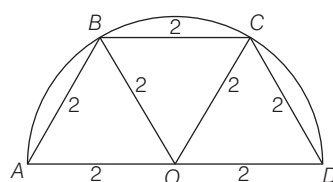
$$\text{Now, length of arc } ACB = \frac{\pi r \theta}{180^\circ} = \frac{\pi \times 36 \times 20}{180} = 4\pi \text{ cm}$$

80. (d) Since, $\triangle AOB$, $\triangle BOC$ and $\triangle COD$ are equilateral.

$$\therefore \text{Side of an equilateral triangle} = \text{radius of semi-circle} = 2 \text{ cm}$$

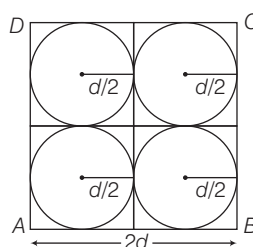
Now, total area of three equilateral triangles

$$= 3 \times \frac{\sqrt{3}}{4} (\text{Side})^2$$



$$= 3 \times \frac{\sqrt{3}}{4} \times 4 = 3\sqrt{3} \text{ cm}^2$$

81. (c)



From the figure it is clear that, 4 circular plates of diameter d can be made from.

square plate of side $2d$ with minimum loss of material.

82. (d) Given, sides of a triangle are in the ratio = $\frac{1}{2} : \frac{1}{3} : \frac{1}{4}$ i.e. $6 : 4 : 3$

Let the sides of a triangle be $6x$, $4x$ and $3x$, respectively.

We know that,

Perimeter of a triangle = Sum of all sides of a triangle

$$\Rightarrow 52 = 6x + 4x + 3x$$

$$\Rightarrow 52 = 13x$$

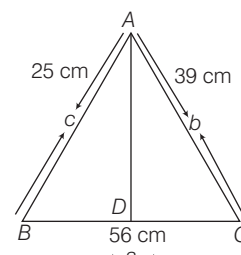
$$\therefore x = \frac{52}{13} = 4$$

$$\therefore \text{Smallest side of a triangle} = 3x = 3 \times 4 = 12 \text{ cm}$$

83. (c) Let the three sides of a triangle be a , b and c , respectively.

$$\therefore a = 56 \text{ cm, } b = 39 \text{ cm}$$

$$\text{and } c = 25 \text{ cm}$$



Perimeter of triangle, $2s = a + b + c$

$$\Rightarrow 2s = 25 + 39 + 56$$

$$\Rightarrow s = 60 \text{ cm}$$

Now, area of triangle

$$= \sqrt{s(s-a)(s-b)(s-c)}$$

$$= \sqrt{60(60-25)(60-39)(60-56)}$$

$$= \sqrt{60 \times 35 \times 21 \times 4} = 420 \text{ cm}^2 \quad \dots(i)$$

$$\text{Again area of triangle} = \frac{1}{2} \times \text{Base}$$

$$\times \text{Altitude}$$

$$= \frac{1}{2} \times BC \times AD \quad \dots(ii)$$

From Eqs. (i) and (ii), we get

$$\frac{1}{2} \times BC \times AD = 420$$

$$\Rightarrow \frac{1}{2} \times 56 \times AD = 420$$

$$\Rightarrow AD = \frac{420}{28} = 15 \text{ cm}$$

Hence, the length of AD is 15 cm.

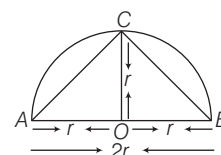
84. (a) Let the r be the radius of semi-circle.

In $\triangle ABC$,

$$AO = OB = OC = r \text{ [radius of circle]}$$

$\therefore$ Area of largest triangle

$$= \frac{1}{2} \times \text{Base} \times \text{Height}$$



$$= \frac{1}{2} \times AB \times OC = \frac{1}{2} \times 2r \times r$$

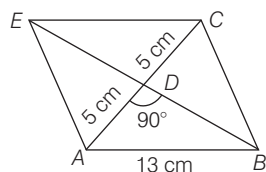
$$[\because AB = OA + OB]$$

$$= r^2$$

Hence, the area of the largest triangle is r^2 .

85. (c) Given, a side of rhombus (a) = 13 cm and a diagonal of rhombus (d_1) = 10 cm

Now, in right angled $\triangle ABD$,



$$\begin{aligned} AB^2 &= AD^2 + BD^2 \\ \Rightarrow 13^2 &= 5^2 + BD^2 \\ \Rightarrow BD^2 &= 169 - 25 = 144 \\ \therefore BD &= 12 \text{ cm} \\ \therefore EB &= 2 \times BD = 2 \times 12 = 24 \text{ cm} \\ \text{Length of second diagonal } (d_2) &= 24 \text{ cm} \end{aligned}$$

$$\begin{aligned} \text{Now, area of rhombus} &= \frac{1}{2} \times d_1 \times d_2 \\ &= \frac{1}{2} \times 10 \times 24 = 120 \text{ cm}^2 \end{aligned}$$

Shortcut Method

Here, $a = 13$ cm, $d_1 = 10$ cm

We know that,

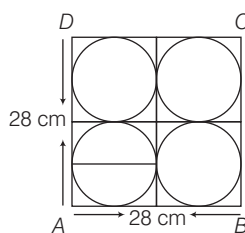
$$\begin{aligned} 4a^2 &= d_1^2 + d_2^2 \\ \Rightarrow 4 \times 13^2 &= 10^2 + d_2^2 \\ \Rightarrow 4 \times 169 &= 100 + d_2^2 \\ \Rightarrow 676 - 100 &= d_2^2 \\ \Rightarrow d_2 &= \sqrt{576} = 24 \text{ cm} \end{aligned}$$

$$\begin{aligned} \text{Now, area of rhombus} &= \frac{1}{2} d_1 d_2 \\ &= \frac{1}{2} \times 10 \times 24 = 120 \text{ cm}^2 \end{aligned}$$

86. (d) Let the side of a square paper sheet be a cm.

Given, area of square paper sheet = 784 cm²

$$\Rightarrow a^2 = 784 \Rightarrow a = 28 \text{ cm}$$



$$\therefore \text{Diameter of one circle} = \frac{28}{2} = 14 \text{ cm}$$

$$\text{and radius of one circle} = \frac{14}{2} = 7 \text{ cm}$$

$$\begin{aligned} \therefore \text{Circumference of each plate} &= 2\pi r \\ &= 2\pi \times 7 = 2 \times \frac{22}{7} \times 7 = 44 \text{ cm} \end{aligned}$$

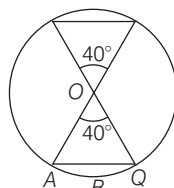
Hence, the circumference of each plate is 44 cm.

87. (b) Given, radius of circle, $r = 3$ cm

Area of remaining portion

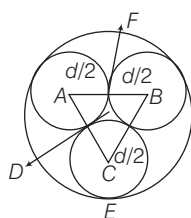
= Area of circle - 2 (Area of sector OABQO).

$$\begin{aligned} &= \pi r^2 - \frac{2\pi r^2 \theta}{360^\circ} \\ &= \pi r^2 \left[1 - \frac{2\theta}{360^\circ} \right] \\ &= \pi r^2 \left[1 - \frac{2 \times 40}{360^\circ} \right] \\ &= \frac{22}{7} \times 9 \left[1 - \frac{8}{36} \right] \\ &= \frac{22}{7} \times 9 \left[1 - \frac{2}{9} \right] = \frac{22}{7} \times \frac{7}{9} \times 9 = 22 \text{ cm}^2 \end{aligned}$$



88. (d) Given, diameter of each circle = d

$\therefore$ Radius of each circle = $d/2$



$$CE = AF = BF = d/2$$

Here, $\triangle ABC$ is an equilateral triangle.

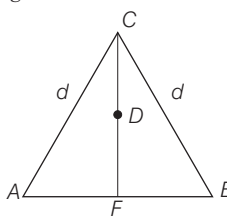
$\therefore$ Altitude of equilateral triangle

$$= \frac{\sqrt{3}}{2} (\text{Side}) = \frac{\sqrt{3}}{2} (AB) = \frac{\sqrt{3}}{2} d$$

$$[\because AB = AF + FB = d]$$

Let D be the centroid of $\triangle ABC$

$\therefore$ Ratio of centroid of equilateral triangle = 2 : 1



$$\therefore \text{Length of } DC = \frac{2}{3} \times \frac{\sqrt{3}}{2} d = \frac{\sqrt{3}}{3} d$$

Now, radius of big circle,

$$\begin{aligned} R = DE &= DC + CE = \frac{\sqrt{3}}{3} d + \frac{d}{2} \\ &= d \left[\frac{\sqrt{3}}{3} + \frac{1}{2} \right] = \frac{d \times \sqrt{3}(2 + \sqrt{3})}{6} \end{aligned}$$

$\therefore$ Area of big circle = πR^2

$$\begin{aligned} &= \pi \left[d \times \frac{\sqrt{3}(2 + \sqrt{3})}{6} \right]^2 \\ &= \pi \times \left[\frac{d^2 \times 3(2 + \sqrt{3})^2}{36} \right] \end{aligned}$$

$$= \frac{\pi d^2 (2 + \sqrt{3})^2}{12}$$

Hence, the area of big circle is

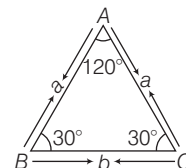
$$= \frac{\pi d^2 (2 + \sqrt{3})^2}{12}$$

89. (c) Let the angles of a triangle be $4x$, x and x , respectively.

$$\begin{aligned} \text{i.e. } \angle A &= 4x, \\ \angle B &= x \text{ and} \\ \angle C &= x \end{aligned}$$

$\therefore$ Sum of all angles of a triangle = 180°

$$\therefore \angle A + \angle B + \angle C = 180^\circ$$



$$\Rightarrow 4x + x + x = 180^\circ \Rightarrow 6x = 180^\circ$$

$$\Rightarrow x = 30^\circ$$

$$\therefore \angle A = 4x = 30 \times 4 = 120^\circ,$$

$$\angle B = x = 30^\circ$$

$$\text{and } \angle C = x = 30^\circ$$

So, it is clear that, given triangle is isosceles triangle. Let the sides isosceles triangle be a , a and b , respectively.

$\therefore$ Perimeter of triangle

$$= a + a + b = 2a + b$$

$$\text{Using sine rule, } \frac{b}{\sin 120^\circ} = \frac{a}{\sin 30^\circ}$$

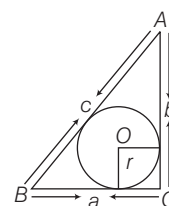
$$\Rightarrow \frac{b}{\sqrt{3}/2} = \frac{a}{1/2} \Rightarrow \frac{2b}{\sqrt{3}} = 2a \Rightarrow a = \frac{b}{\sqrt{3}}$$

$$\text{Now, } \frac{b}{2a + b} = \frac{b}{2 \times \frac{b}{\sqrt{3}} + b}$$

$$= \frac{b}{\frac{2b + \sqrt{3}b}{\sqrt{3}}} = \frac{\sqrt{3}}{2 + \sqrt{3}}$$

90. (a) Given, a , b and c be the sides of a right triangle, where c is the hypotenuse.

$$\therefore c^2 = a^2 + b^2 \quad \dots(i)$$



and perimeter of triangle $2s = a + b + c$

$$\therefore s = \frac{a + b + c}{2}$$

$$\text{Area of triangle} = \frac{1}{2} \times BC \times AC$$

$$\Rightarrow \Delta = \frac{1}{2} ab \quad \dots(ii)$$

From Eq. (i), $c^2 = a^2 + b^2$

$$\Rightarrow c^2 = (a + b)^2 - 2ab$$

$$\Rightarrow 2ab = (a+b)^2 - c^2$$

$$\Rightarrow ab = \frac{(a+b)^2 - c^2}{2}$$

Now, area of triangle = $\frac{ab}{2}$ [from Eq. (ii)]

$$\Delta = \frac{[(a+b)^2 - c^2]}{2}, \Delta = \frac{[(a+b)^2 - c^2]}{4}$$

$$\therefore \text{Radius of circle, } r = \frac{\Delta}{\frac{1}{2}(a+b+c)} = \frac{(a+b-c)}{4 \times \left(\frac{a+b+c}{2}\right)} = \frac{(a+b-c)}{2}$$

Hence, the radius of circle is $\frac{(a+b-c)}{2}$

91. (c) Let the radius of outer circle and inner circle be R and r , respectively.

Perimeter of outer circle = $2\pi R$

and perimeter of inner circle = $2\pi r$

$$\therefore \frac{2\pi R}{2\pi r} = \frac{23}{22}$$

$$\Rightarrow \frac{R}{r} = \frac{23}{22}$$

From the figure, $R = 5 + r$

$$\therefore \frac{5+r}{r} = \frac{23}{22} \Rightarrow 110 + 22r = 23r$$

$$\Rightarrow r = 110 \text{ m}$$

So, diameter of the inner circle is 220 m.

92. (a) Let the sides of a triangular field be a , b and c , respectively.

$$\therefore a = 41 \text{ m}, b = 40 \text{ m and } c = 9 \text{ m}$$

Perimeter of triangular field = $a + b + c$

$$\Rightarrow 2s = a + b + c = (41 + 40 + 9)$$

$$\Rightarrow 2s = 90 \Rightarrow s = 45 \text{ m}$$

$\therefore$ Area of triangular field

$$= \frac{\sqrt{s(s-a)(s-b)(s-c)}}{1}$$

$$= \frac{\sqrt{45(45-41)(45-40)(45-9)}}{1}$$

$$= \frac{\sqrt{45 \times 4 \times 5 \times 36}}{1}$$

$$= \frac{\sqrt{9 \times 5 \times 5 \times 4 \times 9 \times 4}}{1}$$

$$= 9 \times 5 \times 4 = 180 \text{ m}^2$$

Given, area of each rose bed = 900 cm^2

$$= \frac{900}{100 \times 100} \text{ m}^2$$

$\therefore$ Number of rose beds

$$= \frac{\text{Area of triangular field}}{\text{Area of one rose bed}}$$

$$= \frac{180 \times 100 \times 100}{900} = 2000$$

93. (b) 1 km = 1000 m 1 dm = 10 m

$$\therefore 2 \text{ km and } 26 \text{ dm}$$

$$= 2000 + 260 = 2260 \text{ m}$$

Let radius be r m.

Distance covered in revolution = $2\pi r$

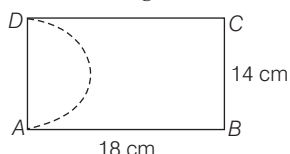
$$\text{Distance covered in 452 revolution} = 452 \times 2\pi r$$

$$\therefore 452 \times 2\pi r = 2260$$

$$r = \frac{2260 \times 7}{452 \times 2 \times 22} = \frac{35}{44}$$

$$\text{Diameter} = 2 \times \frac{35}{44} = \frac{35}{22} = 1\frac{13}{22} \text{ m}$$

94. (d) Area of remaining portion = Area of rectangle - Area of semi-circle



$$= 18 \times 14 - \frac{1}{2} \times \frac{22}{7} \times 7 \times 7$$

$$= 252 - 77 = 175 \text{ cm}^2$$

95. (d) Let length of side of square and equilateral triangle be x and y , respectively.

$$\text{Then, } 4x = 3y \quad \dots(i)$$

Given, diagonal of square = $12\sqrt{2}$

$$\Rightarrow x\sqrt{2} = 12\sqrt{2}$$

$$\therefore x = 12 \text{ cm}$$

$$\therefore y = \frac{4x}{3} = \frac{4}{3} \times 12$$

$$= 16 \text{ cm}$$

$$\text{So, area of triangle} = \frac{\sqrt{3}}{4} \times 16 \times 16$$

$$= 64\sqrt{3} \text{ cm}^2$$

96. (d) Let the side of square be x .

Area of square = x^2

Now, increased length

$$= x \left(1 + \frac{30}{100} \right) = \frac{13x}{10}$$

and increased breadth

$$= x \left(1 + \frac{20}{100} \right) = \frac{12x}{10}$$

Now, new area of rectangle

$$= \frac{13x}{10} \times \frac{12x}{10} = \frac{156x^2}{100}$$

Increased in area

$$= \frac{156x^2}{100} - x^2 = \frac{56x^2}{100}$$

Percentage increased in area

$$= \frac{56x^2}{100x^2} \times 100$$

$$= 56\%$$

Shortcut Method

Here, $x = 30\%$, $y = 20\%$

$$\therefore \text{Net effect} = \left[x + y + \frac{xy}{100} \right] \%$$

$$= \left[30 + 20 + \frac{30 \times 20}{100} \right] = 56\%$$

Hence, area is increased by 56%.

97. (a) Area of trapezium

$$= \frac{1}{2} (\text{Sum of parallel side}) \times \text{Distance}$$

between them

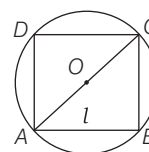
$$\Rightarrow \frac{1}{2} (5x + 7x) \times 14 = 336$$

$$\Rightarrow 12x \times 7 = 336 \Rightarrow x = 4$$

The length of smaller side

$$= 4 \times 5 = 20 \text{ cm}$$

98. (c) We have, $2\pi r = 100$



$$\therefore 2\pi r = \frac{100}{\pi} \text{ cm}$$

Let the side of square be l cm.

$$\therefore l\sqrt{2} = \frac{100}{\pi} \Rightarrow l = \frac{50\sqrt{2}}{\pi} \text{ cm}$$

99. (b) There are 437 fruits plants.

$$\therefore 437 = 19 \times 23$$

$\therefore$ There are 19 rows and 23 trees or there are 23 rows and 19 trees.

Given, the distance between any two adjacent plants is 2 m and the distance between any two adjacent rows is 2 m.

$\therefore$ Length of orchard

$$= [1 + 22(2) + 1] = 46 \text{ m}$$

Breadth of orchard

$$= [1 + 18(2) + 1] = 38 \text{ m}$$

$\therefore$ Perimeter of orchard

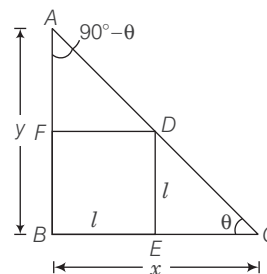
$$= [(46 \times 2) + (38 \times 2)] = 168 \text{ m}$$

$\therefore$ Cost of fencing at the rate of

$$= ₹ 100 \text{ per metre}$$

$$= 168 \times 100 = ₹ 16800$$

100. (b) Let length of the side of square be l .



$$\text{In } \triangle DEC, \tan \theta = \frac{DE}{EC} = \frac{l}{x-l} \dots(i)$$

$$\text{In } \triangle AFD, \tan (90^\circ - \theta) = \cot \theta = \frac{FD}{AF}$$

$$= \frac{l}{y-l} \dots(ii)$$

On multiplying Eqs. (i) and (ii), we get

$$1 = \frac{l^2}{(x-l)(y-l)}$$

$$\Rightarrow xy - xl - yl + l^2 = l^2$$

$$\therefore l(x+y) = xy \Rightarrow l = \frac{xy}{x+y}$$

$$\therefore \text{Perimeter of square} = 4l = \frac{4xy}{x+y}$$

101. (c) Let a be the side of regular polygon then, inradius, $r = \frac{a}{2 \tan\left(\frac{\pi}{n}\right)}$

$$\Rightarrow a = 2r \tan\left(\frac{\pi}{n}\right)$$

$\therefore$ Perimeter of polygon,

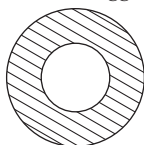
$$p = na = 2nr \tan\left(\frac{\pi}{n}\right)$$

$$\Rightarrow \tan\left(\frac{\pi}{n}\right) = \frac{p}{2nr} \quad \dots(i)$$

$$\text{Area of polygon} = nr^2 \tan\left(\frac{\pi}{n}\right)$$

$$= nr^2 \times \frac{p}{2nr} = \frac{pr}{2} \quad [\text{from Eq. (i)}]$$

102. (c) Let the radius of smaller ring be r_1 and the radius of bigger ring be r .



Since, the smaller ring takes three full revolutions to roll over the circumference of bigger ring.

$$\therefore 2\pi r = 3(2\pi r_1) \Rightarrow r = 3r_1$$

$$\text{Area of pathway} = \pi r^2 - \pi r_1^2 = n(\pi r_1^2)$$

$$\Rightarrow \pi r^2 = (n+1)\pi r_1^2$$

$$\Rightarrow r^2 = (n+1)r_1^2 \Rightarrow n = 9 - 1 = 8$$

103. (d) Number of rounds that a wheel of diameter D will make in traversing a distance $= n(\pi D)$

Given that, distance = 4 km

$$\Rightarrow n(\pi D) = 4000 \text{ m}$$

$$\Rightarrow n\left(\pi \times \frac{7}{11}\right) = 4000$$

$$\Rightarrow n\left(\frac{22}{7} \times \frac{7}{11}\right) = 4000$$

$$\Rightarrow n = 2000$$

Hence, the number of rounds is 2000.

104. (c) Here, base = 300 units and side = 170 units

We know that,

Area of an isosceles triangle

$$= \frac{\text{Base}}{4} \sqrt{4(\text{Side})^2 - (\text{Base})^2}$$

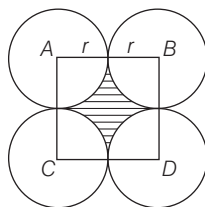
$$= \frac{300}{4} \sqrt{4(170)^2 - (300)^2}$$

$$= 75 \sqrt{4 \times 28900 - 90000}$$

$$= 75 \sqrt{115600 - 90000} = 75 \sqrt{25600}$$

$$= 75 \times 160 = 12000 \text{ sq units}$$

105. (d) Let the radius of the circle be r . Then, side of square $ABCD = 2r$



$\therefore$ Area of empty space = Area of square - 4 [Area of quadrant]

$$\Rightarrow \frac{150}{847} = (2r)^2 - 4 \left[\frac{1}{4} \times \pi r^2 \right]$$

$$\Rightarrow \frac{150}{847} = \frac{6}{7} r^2$$

$$\Rightarrow r^2 = \frac{25}{121} \quad \left[\because \pi = \frac{22}{7} \right]$$

$$\Rightarrow r = \frac{5}{11} \text{ cm}$$

106. (c) Area of circle with diameter (AD)

$$= \frac{\pi(AD)^2}{4}$$

$$\text{So, } \frac{\pi(AD)^2}{4} = 707$$

$$\Rightarrow AD = \sqrt{\frac{707}{\pi}} \times 4 = 2\sqrt{\frac{707}{\pi}} \text{ m}$$

$$\text{As, } AB = BC = CD = \frac{AD}{3} = \frac{2}{3} \sqrt{\frac{707}{\pi}} \text{ m}$$

and $AC = 2AB = BD$

$$= 2CD = \frac{4}{3} \sqrt{\frac{707}{\pi}} \text{ m}$$

$\therefore$ Area of shaded portion = 2

[Area of semi-circle with AC as diameter - Area of semi-circle with AB as diameter]

$$= \frac{2\pi(AC)^2}{8} - \frac{2\pi(AB)^2}{8}$$

$$= \frac{2\pi}{8} \left[\frac{16}{9} \cdot \frac{707}{\pi} - \frac{4}{9} \cdot \frac{707}{\pi} \right] = \frac{1}{3} \times 707 \text{ m}^2$$

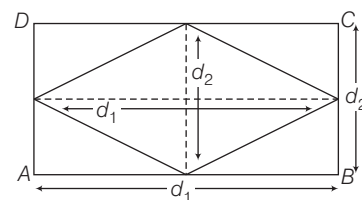
$\therefore$ Cost of levelling the shaded portion

$$= \frac{707}{3} \times 63 = ₹ 14847$$

107. (b) We know that, Area of rhombus

$$= \frac{1}{2} \times \text{product of diagonals}$$

$$= \frac{1}{2} \times d_1 \times d_2$$



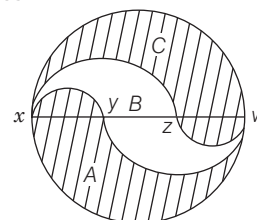
From the above figure,

$$\text{Area of rectangle} = d_1 \times d_2$$

$$= 2 (\text{Area of rhombus})$$

$$= 2 \times 2 = 4 \text{ sq units}$$

108. (b) In the given question, A radii of 3 m is divided in such a way that the radii of smaller semi-circle is 1 m and radii of bigger semi-circle is 2 m.



Area of shaded portion A

= Area of semi-circle of radii 3 m - Area of semi-circle of radii 2 m + Area of semi-circle of radii 1 m

$$= \frac{1}{2} \pi (3)^2 - \frac{1}{2} \pi (2)^2 + \frac{1}{2} \pi (1)^2$$

$$= \frac{1}{2} \pi (9 - 4 + 1) = 3\pi$$

Area of portion B

= 2 [Area of semi-circle of radii 2 m - Area of semi-circle of radii 1 m]

$$= 2 \left[\frac{1}{2} \pi (2)^2 - \frac{1}{2} \pi (1)^2 \right] = 3\pi$$

Similarly, area of shaded portion C

= Area of portion A

Hence, the ratio of areas A, B and C is 1 : 1 : 1.

27

SURFACE AREA AND VOLUME OF SOLIDS

Regularly (10-12) questions have been asked from this chapter. Questions from this chapter are direct formula based, so one must be through with all the formulae and their application. It is easy to score section in the exam.



In this chapter, we will study how to find the surface area and volume of solid figures, like parallelepiped, cube, cuboid, cylinder, cone, frustum of cone, sphere and hemisphere and the combination of solid figures.

SOLID FIGURES

The objects which occupy space (i.e. they have three dimensions) are called solid figures. The solid figures can be derived from the plane figures.

A solid figure has surface area and volume.

Surface Area of Solid Figure

Surface area of a solid body is the area of all of its surfaces together. It is always measured in square units. Surface area is also referred to as the total surface area. The area of four walls or the area excluding top and bottom is called curved surface area or lateral surface area.

Volume of Solid Figure

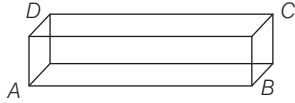
The measure of space occupied by a solid is called its volume. It is always measured in cubic units.

Cuboid

A figure which is surrounded by six rectangular surfaces, is called cuboid. It is also called rectangular parallelepiped.

- A cuboid has 8 corners, 12 edges, 6 faces and 4 diagonals.

- Volume of cuboid = Area of base $\times$ height = $l \times b \times h$
where, l = length, b = breadth and h = height



- Total surface area of cuboid = $2(lb + bh + lh)$
- Diagonal of the cuboid = $\sqrt{l^2 + b^2 + h^2}$
- Lateral surface area or Area of four walls = $2(l + b)h$.

EXAMPLE 1. The volume of a cuboid is 880 cm^3 , the area of its base is 88 cm^2 . Then, its height is

- a. 10 cm b. 12 cm c. 14 cm d. 16 cm

Sol. a. Here, volume of a cuboid = 880 cm^3

Area of the base = 88 cm^2

$$\text{Height} = \frac{\text{Volume of the cuboid}}{\text{Area of the base}} = \frac{880}{88} = 10 \text{ cm}$$

EXAMPLE 2. The length of the longest rod that can be placed in a room 12 m long, 9 m broad and 8 m high is

- a. 15 m b. 16 m
c. 17 m d. 17.5 m

Sol. c. Since, the length of the longest rod is the diagonal of the room

$$\begin{aligned} &= \sqrt{l^2 + b^2 + h^2} = \sqrt{(12)^2 + (9)^2 + (8)^2} \\ &= \sqrt{144 + 81 + 64} = \sqrt{289} = 17 \text{ m} \end{aligned}$$

EXAMPLE 3. The areas of three adjacent faces of a cuboid are x , y and z . If its volume is V , then which of the following is true?

- a. $V = x^3 y^2 z^2$ b. $V^2 = xyz$
c. $V = \sqrt[3]{xyz}$ d. $V = \frac{x^2 y}{z}$

Sol. b. Let the dimensions of a cuboid be l , b and h respectively, then volume of cuboid, $V = lbh$

Also, $x = lb$, $y = bh$ and $z = hl$

$\therefore V = lbh$

On squaring both sides, we get

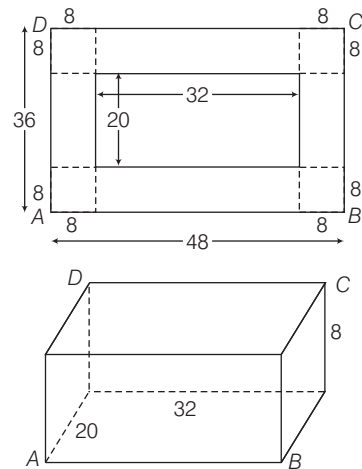
$$V^2 = l^2 b^2 h^2 = (lb) \cdot (bh) \cdot (hl) = xyz$$

EXAMPLE 4. A metallic sheet is of rectangular shape with dimensions $48 \text{ cm} \times 36 \text{ cm}$ from each of its corners a square of 8 cm is cut-off. An open box is made of the remaining sheet, what is the volume of the box?

- a. 13824 cm^3 b. 1728 cm^3
c. 5120 cm^3 d. None of these

Sol. c. Given length of metallic sheet = 48 cm

Breadth of metallic sheet = 36 cm



When square of side 8 cm is cut-off from each corner and the flaps turned up, we get an open box whose

$$\text{Length} = 48 - (8 + 8) = 32 \text{ cm}$$

$$\text{Breadth} = 36 - (8 + 8) = 20 \text{ cm}$$

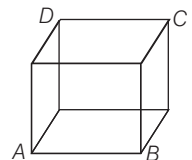
and height = 8 cm

$$\therefore \text{Volume of the box} = l \times b \times h = 32 \times 20 \times 8 = 5120 \text{ cm}^3$$

Cube

A cuboid whose length, breadth and height are same is called a cube.

- A cube has 6 surfaces, 12 edges, 8 corners and 4 diagonals.
- Cube is a special case of cuboid which has 6 square faces.
- Volume of a cube = $(\text{side})^3$
- Total surface area of a cube = $6 \times (\text{side})^2$
- Diagonal of a cube = $\sqrt{3} \times \text{edge}$
- Lateral surface area or area of four walls = $4 \times (\text{side})^2$



EXAMPLE 5. If the surface area of a cube is 96 cm^2 , then, its volume is

- a. 16 cm^3 b. 64 cm^3
c. 12 cm^3 d. 32 cm^3

Sol. b. Surface area of cube = $6 \times a^2$, where a = Side of a cube

$$\Rightarrow 6a^2 = 96 \Rightarrow a^2 = 16 \Rightarrow a = 4 \text{ cm}$$

$$\therefore \text{Volume of the cube} = a^3 = 4^3 = 64 \text{ cm}^3$$

EXAMPLE 6. The surface area and the length of the diagonal of the cube, if the volume of a cube is 2197 cm^3 , are

- a. 1012 cm^2 and 21.516 cm b. 1024 cm^2 and 24.516 cm
c. 1014 cm^2 and 22.516 cm d. None of these

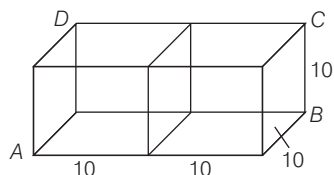
Sol. c. Volume of cube = (side)³ = 2197 cm³
 $\Rightarrow$ Side of cube = $\sqrt[3]{\text{volume of a cube}} = 13$ cm
 $\Rightarrow$ Surface area of cube
 $= 6 (\text{side})^2 = 6 (13)^2 = 6 \times 169 = 1014$ cm²
 and length of the diagonal of the cube = $\sqrt{3} \times \text{side}$
 $= \sqrt{3} \times 13 = 1.732 \times 13 = 22.516$ cm

EXAMPLE 7. How many 6 m cubes can be cut from a cuboid measuring 36 m $\times$ 15 m $\times$ 8 m?

- a. 10 b. 15
 c. 19 d. 20

Sol. d. Volume of given cuboid = $(36 \times 15 \times 8)$ m³
 Volume of the cube to be cut = $(6 \times 6 \times 6)$ m³
 $\therefore$ Number of cubes that can be cut from the cuboid
 $= \frac{\text{Volume of the cuboid}}{\text{Volume of the cube}} = \frac{36 \times 15 \times 8}{6 \times 6 \times 6} = 20$

EXAMPLE 8. Two cubes each of 10 cm edge are joined end-to-end. Then, the surface area of the resulting cuboid is



- a. 100 cm² b. 1000 cm²
 c. 2000 cm² d. None of these

Sol. b. Length of the resulting cuboid = $10 + 10 = 20$ cm
 Breadth of resulting cuboid = 10 cm
 Height of resulting cuboid = 10 cm
 $\therefore$ Surface area of resulting cuboid = $2(lb + bh + hl)$
 $= 2[20 \times 10 + 10 \times 10 + 20 \times 10]$
 $= 2[200 + 100 + 200] = 2 \times 500 = 1000$ cm²

IMPORTANT FORMULAE

- Volume of prism = Area of base $\times$ height
- Lateral surface area of prism = Perimeter of base $\times$ height
- Volume of pyramid = $1/3 \times$ Area of base $\times$ height
- Lateral surface area = $1/2 \times$ Perimeter of base $\times$ slant height

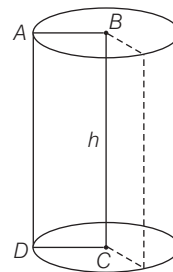
Right Circular Cylinder

A right circular cylinder is a solid figure obtained by revolving the rectangle, say $ABCD$ about its one side, say BC . Let base radius of right circular cylinder be r and its height be h . Then,

(i) Volume of the cylinder
 $= (\text{Area of base}) \times \text{Height}$
 $= \pi r^2 h$ cu units

(ii) Curved surface area = Circumference of the base $\times$ Height
 $= 2\pi r h$ sq units

(iii) Total surface area = Curved surface + Area of two ends
 $= 2\pi r h + 2\pi r^2 = 2\pi r (h + r)$ sq units



EXAMPLE 9. The volume of a cylinder is 448π cm³ and height 7 cm. Then, its lateral surface area and total surface area are

- a. 349 cm² and 753.286 cm²
 b. 352 cm² and 754.286 cm²
 c. 353 cm² and 755.286 cm²
 d. None of the above

Sol. b. Given, volume of the cylinder = 448π cm³
 and height of the cylinder = 6 cm

Let radius be r , then $\pi r^2 h = 448 \pi$

$$\therefore r^2 = \frac{448 \pi}{h \pi} = \frac{448}{7} = 64 \Rightarrow r = 8 \text{ cm}$$

$$\therefore \text{Lateral surface area of the cylinder} = 2\pi r h$$

$$= 2 \times \frac{22}{7} \times 8 \times 7 = 352 \text{ cm}^2$$

$$\text{Total surface area of the cylinder} = 2\pi r (h + r)$$

$$= 2 \times \frac{22}{7} \times 8 (7 + 8)$$

$$= 2 \times \frac{22}{7} \times 8 \times 15 = \frac{5280}{7} \approx 754.286 \text{ cm}^2$$

EXAMPLE 10. The radii of two cylinders are in the ratio 2 : 3 and their heights are in the ratio 5 : 3. Then, the ratio of their volumes is

- a. 4 : 9 b. 16 : 25 c. 20 : 27 d. None of these

Sol. c. For first cylinder,

Let radius = r_1 , height = h_1 , volume = V_1

For second cylinder,

Let radius = r_2 , height = h_2 and volume = V_2

$$\text{Then, } \frac{r_1}{r_2} = \frac{2}{3} \text{ and } \frac{h_1}{h_2} = \frac{5}{3} \Rightarrow r_1 = \frac{2r_2}{3} \text{ and } h_1 = \frac{5h_2}{3}$$

Required ratio of their volumes be $V_1 : V_2$.

$$\Rightarrow \pi r_1^2 h_1 : \pi r_2^2 h_2 \Rightarrow r_1^2 h_1 : r_2^2 h_2$$

$$\Rightarrow \frac{4}{9} r_2^2 \frac{5h_2}{3} : r_2^2 h_2 \Rightarrow \frac{20}{27} : 1 \Rightarrow 20 : 27$$

EXAMPLE 11. A cylindrical bucket of diameter 28 cm and height 12 cm is full of water. The water is emptied into a rectangular tub of length 66 cm and breadth 28 cm. The height to which water rises in the tub is

- a. 2 cm b. 4 cm c. 6 cm d. 8 cm

Sol. b. Volume of water in the bucket

$$= \pi r^2 h = \frac{22}{7} \times 14 \times 14 \times 12 = 7392 \text{ cm}^3$$

Let h be the height to which water rises in the tub.

$$\therefore \text{Volume of water in the tub} = 66 \times 28 \times h \text{ cm}^3$$

According to the question,

$$66 \times 28 \times h = 7392 \Rightarrow h = \frac{7392}{66 \times 28} = 4 \text{ cm}$$

Hence, the height to which water rises in the tub is 4 cm.

Hollow Cylinder

A hollow cylinder is one which is empty from inside and has some difference between the internal and external radius.

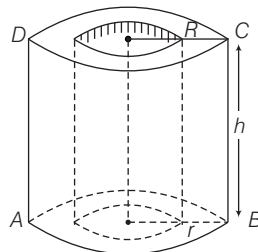
Let R and r be the external and internal radii of the hollow cylinder and h be its height.

Then,

$$(i) \text{ Volume of hollow cylinder} = \pi(R^2 - r^2)h \text{ cu units}$$

$$(ii) \text{ Curved surface area} = 2\pi R h + 2\pi r h = 2\pi(R + r)h \text{ sq units}$$

$$(iii) \text{ Total surface area} = 2\pi r h + 2\pi R h + 2\pi(R^2 - r^2) \text{ sq units} \\ = 2\pi(R + r)(h + R - r) \text{ sq units}$$



EXAMPLE 12. A hollow cylindrical tube, open at both ends is made of iron 1 cm thick. The volume of iron used in making the tube, if the external diameter is 12 cm and the length of tube is 70 cm, is

- a. 2420 cm³ b. 2520 cm³
c. 2720 cm³ d. 2900 cm³

Sol. a. Here, external radius (R_1) = 6 cm

and internal radius (R_2) = 6 - 1 = 5 cm

Height of hollow cylinder (h) = 70 cm

$\therefore$ Volume of iron used in making tube

= External volume - Internal volume

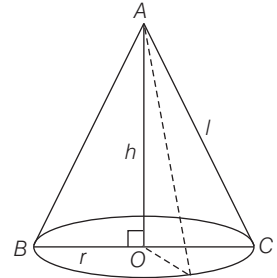
$$= \pi R^2 h - \pi r^2 h = \pi h(R^2 - r^2)$$

$$= \frac{22}{7} \times 70 \times (36 - 25)$$

$$= 220 \times 11 = 2420 \text{ cm}^3$$

Right Circular Cone

A right circular cone is a solid, generated by the revolution of a right angled triangle about one of its sides containing the right angle. Let height of a right circular cone be h , slant height be l and its radius be r . Then,



(i) The slant height of the cone,

$$l = AC = \sqrt{r^2 + h^2} \text{ units}$$

(ii) Volume of cone = $\frac{1}{3} \pi r^2 h$ cu units

(iii) Curved surface area of cone = $\pi r l$ sq units

(iv) Total surface area of a cone = Curved surface area + Base area = $\pi r(l + r)$ sq units

EXAMPLE 13. How many metres of cloth 5 m wide will be required to make a conical tent, the radius of whose base is 7 m and whose height is 24 m?

- a. 100 m b. 105 m c. 109 m d. 110 m

Sol. d. Given radius of base (r) = 7 m

and vertical height of tent (h) = 24 m

Slant height of the tent (l)

$$= \sqrt{r^2 + h^2} = \sqrt{24^2 + 7^2} = \sqrt{625} = 25 \text{ m}$$

$$\text{Now, curved surface area} = \pi r l = \frac{22}{7} \times 7 \times 25 = 550 \text{ m}^2$$

$\therefore$ Width of cloth = 5 m

$\therefore$ Length required to make conical tent

$$= \frac{550}{5} = 110 \text{ m}$$

EXAMPLE 14. The radius and the height of a right circular cone are in the ratio 5:12. If its volume is 314 m³, the slant height and the radius are

- a. 12 m, 5 m b. 13 m, 4 m c. 1 m, 4 m d. 13 m, 5 m

Sol. d. Let radius of cone be $5x$ and the height of the cone be $12x$.

$$\therefore \text{Volume of cone} = 314 = \frac{1}{3} \pi r^2 h$$

$$\Rightarrow \frac{1}{3} \times 3.14 \times (5x)^2 \times (12x) = 314 \Rightarrow 314x^3 = 314$$

$$\Rightarrow x^3 = 1 \Rightarrow x = 1$$

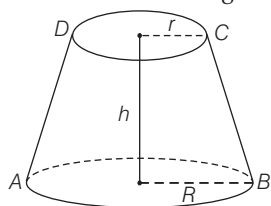
$\therefore$ Radius = 5 m and height = 12 m

$$\therefore \text{Slant height} = \sqrt{r^2 + h^2} = \sqrt{25 + 144} = \sqrt{169} = 13 \text{ m}$$

Hence, the slant height and the radius are 5 m and 13 m.

Frustum of a Cone

If a right circular cone is cut-off by a plane parallel to the base of the cone, then the portion of the cone between the cutting plane and base of the cone is called the frustum of the cone. Let R, r be the radii of base and top ($R > r$) of the frustum of a cone. Let height of frustum of a cone be h and slant height be l , then



- (i) Slant height of frustum of right circular cone

$$= BC = \sqrt{h^2 + (R-r)^2} \text{ units}$$

- (ii) Volume of frustum of right circular cone

$$= \frac{\pi h}{3} [R^2 + r^2 + Rr] \text{ cu units}$$

- (iii) Curved (lateral) surface area of frustum of right circular cone = $\pi l (R+r)$ sq units

- (iv) Total surface area of frustum of right circular cone
= Area of base + Area of top + Lateral surface area
= $\pi [R^2 + r^2 + l(R+r)]$ sq units

- (v) Total surface area of bucket = $\pi [(R+r)l + r^2]$ sq unit
[$\because$ it is open at the bigger end.]

EXAMPLE 15. The radii of the ends of a bucket of height 24 cm are 15 cm and 5 cm. Then, its capacity is

- a. 8000 cm³ b. 8100 cm³
c. 8171.43 cm³ d. 8200.43 cm³

Sol. c. Given, $h = 24$ cm, $R = 15$ cm and $r = 5$ cm

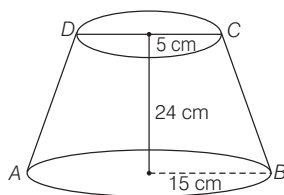
Capacity of bucket = Volume of frustum of a cone

$$= \frac{\pi h}{3} [R^2 + r^2 + Rr]$$

$$= \frac{22}{7} \times \frac{24}{3} [(15)^2 + 5^2 + 15 \times 5]$$

$$= \frac{22}{7} \times 8 [225 + 25 + 75]$$

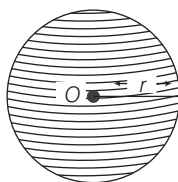
$$= \frac{176}{7} (325) = 8171.43 \text{ cm}^3$$



Sphere

The set of all points in space, which are equidistant from a fixed point, is called a sphere. Let radius of sphere be r , then

- (i) Volume of sphere = $\frac{4}{3} \pi r^3$ cu units



- (ii) Surface area of sphere = $4\pi r^2$ sq units

- (iii) Volume of a hollow sphere = $\frac{4}{3} \pi (R^3 - r^3)$ cu units

where, r = Inner radius and R = Outer radius

EXAMPLE 16. Given that the volume of a metal sphere is 38808 cm³. Then, its radius and its surface area are

- a. 7 cm and 616 cm² b. 21 cm and 5544 cm²
c. 14 cm and 2464 cm² d. None of these

Sol. b. Given, volume of the metal sphere = 38808 cm³

$$\text{But volume of sphere} = \frac{4}{3} \pi r^3$$

$$\therefore \frac{4}{3} \pi r^3 = 38808 \Rightarrow r^3 = 38808 \times \frac{3}{4} \times \frac{7}{22} = 9261$$

$$\Rightarrow r = (9261)^{1/3} = 21 \text{ cm}$$

$$\begin{aligned} \therefore \text{Surface area of the sphere} &= 4\pi r^2 \\ &= 4 \times \frac{22}{7} \times 21 \times 21 \\ &= 5544 \text{ cm}^2 \end{aligned}$$

EXAMPLE 17. How many balls each of radius 2 cm can be made by melting a big ball whose radius is 8 cm.

- a. 4 balls b. 16 balls c. 64 balls d. 128 balls

Sol. c. Given radius of big ball (R) = 8 cm

$$\therefore \text{Volume of big ball} = \frac{4}{3} \pi R^3 = \frac{4}{3} \pi (8)^3$$

$$\text{Radius of small ball } (r) = 2 \text{ cm}$$

$$\therefore \text{Volume of small balls} = \frac{4}{3} \pi (2)^3$$

$$\therefore \text{Required number of balls} = \frac{\text{Volume of big ball}}{\text{Volume of small ball}}$$

$$= \frac{\frac{4}{3} \pi (8)^3}{\frac{4}{3} \pi (2)^3} = \frac{8 \times 8 \times 8}{2 \times 2 \times 2} = 64 \text{ balls}$$

EXAMPLE 18. A copper sphere of diameter 18 cm is drawn into a wire of diameter 4 mm. Then, the length of the wire is

- a. 243 m b. 343 m c. 443 m d. 972 m

Sol. a. Let the length of wire be h cm.

$$\text{Volume of sphere} = \text{Volume of wire} \quad [\text{by condition}]$$

$$\Rightarrow \left(\frac{4}{3} \pi \times 9 \times 9 \times 9 \right) = \left(\pi \times \frac{2}{10} \times \frac{2}{10} \times h \right)$$

$$\begin{aligned} \Rightarrow \frac{h}{25} &= 972 \Rightarrow h = (972 \times 25) \text{ cm} \\ &= \frac{972 \times 25}{100} = 243 \text{ m} \end{aligned}$$

Hence, the length of the wire is 243 m.

Hemisphere

A plane passing through the centre cuts the sphere in two equal parts, each part is called a hemisphere.

Let the radius of hemisphere be r , then



- (i) Volume of hemisphere = $\frac{2}{3}\pi r^3$ cu units
- (ii) Curved surface area of hemisphere = $2\pi r^2$ sq units
- (iii) Total surface area = $2\pi r^2 + \pi r^2 = 3\pi r^2$ sq units
- (iv) Volume of hemispherical shell = $\frac{2}{3}\pi(R^3 - r^3)$

where, R = outer radius, r = inner radius

EXAMPLE 19. The volumes of two hemispheres are in the ratio 8:27. What is the ratio of their radii?

- a. 2 : 3 b. 3 : 2 c. 1 : 2 d. 2 : 1

Sol. a. Let volumes be V_1 and V_2 .

$$\therefore V_1 : V_2 = 8 : 27 \Rightarrow \frac{2}{3}\pi r_1^3 : \frac{2}{3}\pi r_2^3 = 8 : 27$$

$$\Rightarrow r_1^3 : r_2^3 = 8 : 27 \Rightarrow r_1 : r_2 = 2 : 3$$

IMPORTANT FORMULAE

- If the side of a cube is increased by $x\%$, then its volume is increased by $\left[\left(1 + \frac{x}{100}\right)^3 - 1\right] \times 100\%$.
- If the length, breadth and height of cuboid are made x , y and z times respectively, then its volume is increased by $(xyz - 1) \times 100\%$.
- If the length, breadth and height of a cuboid are increased by $x\%$, $y\%$ and $z\%$ respectively, then its volume is increased by $\left[x + y + z + \frac{xy + yz + zx}{100} + \frac{xyz}{(100)^2}\right] \%$.
- If the sides and diagonal of a cuboid are given, then the total surface area in terms of diagonal and sides is given by
Total surface area = $(\text{Sum of the sides})^2 - (\text{Diagonal})^2$.
- If the side of a cube is increased by $x\%$, then the surface area is increased by $\left(2x + \frac{x^2}{100}\right) \%$.

> PRACTICE EXERCISE

1. Three equal cubes are placed adjacently in a row. The ratio of total surface area of the new cuboid to that of the sum of the surface areas of the three cubes is
(a) 3 : 1 (b) 6 : 5 (c) 7 : 9 (d) 6 : 7
2. If side of a cube is increased by 12%, by how much percent does its volume increase?
(a) 35.2% (b) 40.5% (c) 45.0% (d) 42.4%
3. The curved surface area of a cylinder is 1320 cm^2 and its base has diameter 21 cm, then the height of the cylinder is
(a) 10 cm (b) 20 cm (c) 22 cm (d) 25 cm
4. A cylindrical vessel can hold 154 g of water. If the radius of its base is 3.5 cm and 1 cm^3 of water weights 1 g. The depth of the water is
(a) 2 cm (b) 3 cm
(c) 4 cm (d) 5 cm
5. The curved surface area of a cylindrical pillar is 264 m^2 and its volume is 924 m^3 . The diameter of the pillar is
(a) 3 m (b) 6 m
(c) 7 m (d) 14 m
6. How many metres of cloth 50 m wide will be required to make a conical tent, the radius of whose base is 7 m and whose height is 24 m?
(a) 9 m (b) 11 m (c) 12 m (d) 13 m
7. The radius and height of a right circular cone are in the ratio of 5 : 12 and its volume is 2512 cm^3 . The slant height of the cone is
(a) 24 cm (b) 25 cm (c) 26 cm (d) 27 cm
8. If the height of a cone is doubled, then its volume is increased by
(a) 100% (b) 200% (c) 300% (d) 400%
9. If the surface areas of two spheres are in the ratio of 4 : 25, then the ratio of their volumes is
(a) 2 : 25 (b) 4 : 75 (c) 8 : 125 (d) 16 : 125
10. A cone and a cylinder are of the same height. Their radii of the bases are in ratio of 2 : 1. The ratio of their volumes is
(a) 2 : 1 (b) 3 : 2 (c) 4 : 3 (d) 1 : 3
11. If the height and diameter of a right circular cylinder are 32 cm and 6 cm respectively, then the radius of the sphere whose volume is equal to the volume of the cylinder is
(a) 3 cm (b) 4 cm (c) 6 cm (d) 8 cm

- 12.** Two solid spheres of gold having diameters 3 cm and 4 cm are molten and then cast into one big sphere of gold. If the radius of this sphere is x , then what is the value of x^3 ?
 (a) 125 cm^3 (b) 15.625 cm^3
 (c) 11.375 cm^3 (d) 9.875 cm^3
- 13.** The total surface area of a cone, whose slant height is equal to the radius R of its base, is S . If A is the area of a circle of radius $2R$, then which one of the following is correct?
 (a) $A = S$ (b) $A = 2S$ (c) $A = S/2$ (d) $A = 4S$
- 14.** A cylinder having base of circumference 60 cm is rolling without sliding at a rate of 5 rounds per second. How much distance will the cylinder roll in 5 s?
 (a) 15 m (b) 1.5 m (c) 30 m (d) 3 m
- 15.** 27 drops of water form a big drop of water. If the radius of each smaller drop is 0.2 cm, then what is the radius of the bigger drop?
 (a) 0.4 cm (b) 0.6 cm (c) 0.8 cm (d) 1.0 cm
- 16.** A circus tent is made of canvas and is in the form of a right circular cylinder and a right circular cone above it, the diameter and height of the cylindrical part of the tent are 126 m and 5 m, respectively. The total height of the tent is 21 m. Then, the cost of the canvas used for tent at the rate of ₹ 12 per m^2
 (a) ₹ 14850 (b) ₹ 168200 (c) ₹ 178200 (d) ₹ 112000
- 17.** A solid sphere of radius 6 cm is melted into a hollow cylinder of uniform thickness. If the external radius of the base of the cylinder is 5 cm and its height is 32 cm. The uniform thickness of the cylinder is
 (a) 1.5 cm (b) 3 cm (c) 1.2 cm (d) 1 cm
- 18.** Given a solid cylinder of radius 10 cm and length 1000 cm a cylindrical hole is made into it to obtain a cylindrical shell of uniform thickness and having volume equal to one-fourth of the original volume of the original cylinder. The thickness of the cylindrical shell is
 (a) $5(\sqrt{5} - 2) \text{ cm}$ (b) $7(2 - \sqrt{3}) \text{ cm}$
 (c) 10 cm (d) $5\sqrt{2} \text{ cm}$
- 19.** A tent is of the shape of right circular cylinder upto a height of 3 m and then becomes a right circular cone with a maximum height of 13.5 m above the ground. The cost of painting the inner side of the tent at the rate of ₹ 2 per m^2 , if the radius of the base is 14 m is
 (a) ₹ 2048 (b) ₹ 2068 (c) ₹ 2008 (d) ₹ 2088
- 20.** A container is in the form of a right circular cylinder surmounted by a hemisphere of the same radius 15 cm as the cylinder. If the volume of the container is $32400 \pi \text{ cm}^3$, then the height h of the container satisfies which one of the following?
 (a) $135 \text{ cm} < h < 150 \text{ cm}$ (b) $140 \text{ cm} < h < 147 \text{ cm}$
 (c) $145 \text{ cm} < h < 148 \text{ cm}$ (d) $139 \text{ cm} < h < 145 \text{ cm}$
- 21.** A conical flask of base radius r and height h is full of milk. The milk is now poured into a cylindrical flask of radius $2r$. What is the height to which the milk will rise in the flask?
 (a) $\frac{h}{3}$ (b) $\frac{h}{6}$ (c) $\frac{h}{9}$ (d) $\frac{h}{12}$
- 22.** The radius and height of a right circular cone are in the ratio 3 : 4 and its volume is $96 \pi \text{ cm}^3$. What is the lateral surface area?
 (a) $24 \pi \text{ cm}^2$ (b) $36 \pi \text{ cm}^2$ (c) $48 \pi \text{ cm}^2$ (d) $60 \pi \text{ cm}^2$
- 23.** What is the number of wax balls, each of radius 1 cm, that can be molded out of a sphere of radius 8 cm?
 (a) 256 (b) 512 (c) 768 (d) 1024
- 24.** 10 cylindrical pillars of a building have to be painted. The diameter of each pillar is 70 cm and the height is 4 m. What is the cost of painting at the rate of ₹ 5 per m^2 ?
 (a) ₹ 400 (b) ₹ 440 (c) ₹ 480 (d) ₹ 500
- 25.** The radii of the circular ends of a bucket of height 40 cm are of lengths 35 cm and 14 cm. What is the volume of the bucket?
 (a) 60060 cm^3 (b) 70040 cm^3
 (c) 80080 cm^3 (d) 80160 cm^3
- 26.** If S is the total surface area of a cube and V is its volume, then which one of the following is correct?
 (a) $V^3 = 216 S^2$ (b) $S^3 = 216 V^2$
 (c) $S^3 = 6 V^2$ (d) $S^2 = 36 V^3$
- 27.** A cylindrical tank 7 m in diameter, contains water to a depth of 4 m. What is the total area of wetted surface?
 (a) 110.5 m^2 (b) 126.5 m^2
 (c) 131.5 m^2 (d) 136.5 m^2
- 28.** The radii of two cylinders are in the ratio 2 : 3 and their curved surface areas are in the ratio 5 : 3. What is the ratio of their volumes?
 (a) 20 : 27 (b) 10 : 9 (c) 9 : 10 (d) 27 : 20
- 29.** A cylindrical vessel of height 10 cm has base radius 60 cm. If d is the diameter of a spherical vessel of equal volume, then what is the value of d ?
 (a) 30 cm (b) 60 cm (c) 90 cm (d) 120 cm
- 30.** A hollow sphere of internal and external diameters 4 cm and 8 cm, respectively is melted into a cone of base diameter 8 cm. The height of the cone is
 (a) 11 cm (b) 12 cm (c) 14 cm (d) 16 cm

- 31.** If the diameter of a sphere is doubled, how does its surface area change?
 (a) It increases two times (b) It increases three times
 (c) It increases four times (d) It increases eight times
- 32.** A cistern 6 m long and 4 m wide contains water to a depth of 1.25 m. What is the area of wetted surface?
 (a) 40 m^2 (b) 45 m^2 (c) 49 m^2 (d) 73 m^2
- 33.** The outer and inner diameters of a circular pipe are 6 cm and 4 cm, respectively. If its length is 10 cm, then what is the total surface area in cm^2 ?
 (a) 35π (b) 110π
 (c) 150π (d) None of these
- 34.** A toy is in the form of a cone mounted on a hemisphere such that the diameter of the base of the cone is equal to that of the hemisphere. If the diameter of the base of the cone is 6 cm and its height is 4 cm, what is the surface area in cm^2 of the toy? (Take $\pi = 3.14$)
 (a) 93.62 (b) 103.62 (c) 113.62 (d) 115.50
- 35.** A solid cylinder of height 9 m has its curved surface area equal to one-third of the total surface area. What is the radius of the base?
 (a) 9 m (b) 18 m (c) 27 m (d) 30 m
- 36.** The volume of a sphere is 8 times that of another sphere. What is the ratio of their surface areas?
 (a) 8:1 (b) 4:1 (c) 2:1 (d) 4:3
- 37.** The dimensions of a field are 12 m $\times$ 10 m. A pit 5 m long, 4 m wide and 2 m deep is dug in one corner of the field and the Earth removed has been evenly spread over the remaining area of the field. The level of the field is raised by
 (a) 30 cm (b) 35 cm (c) 38 cm (d) 40 cm
- 38.** A cube of 9 cm edge is immersed completely in a rectangular vessel containing water. If the dimensions of base are 15 cm and 12 cm. Then, the rise in water level in the vessel is
 (a) 4.05 cm (b) 4 cm (c) 3.5 cm (d) 3 cm
- 39.** From a solid cube of edge 3 m, a solid of largest sphere is curved out. What is the volume of solid left?
 (a) $(27 - 2.25\pi) \text{ m}^3$ (b) $(27 - 4.5\pi) \text{ m}^3$
 (c) $2.25\pi \text{ m}^3$ (d) $4.5\pi \text{ m}^3$
- 40.** A rectangular tank is $80 \times 40 \text{ cm}^3$. Water flows into it through a pipe of cross-section area 40 cm^2 at the speed of 10 km/h. The rise in the level of water in the tank in $\frac{1}{2}$ h is
 (a) $\frac{3}{2}$ cm (b) $\frac{4}{3}$ cm (c) $\frac{5}{8}$ cm (d) 6 cm
- 41.** A measuring jar of internal diameter 10 cm is partially filled with water. Four equal spherical balls of diameter 2 cm each are dropped in it and they sink down in the water completely. The change in the level of water in the jar is
 (a) $\frac{16}{65}$ cm (b) $\frac{15}{16}$ cm (c) $\frac{16}{75}$ cm (d) None of these
- 42.** The height of a cone is 30 cm. A small cone is cut off at the top by a plane parallel to the base. If its volume be $\frac{1}{27}$ of the volume of the given cone, then the height above the base where the section is made, is
 (a) 12 cm (b) 15 cm (c) 20 cm (d) 22 cm
- 43.** From a wooden cylindrical block, whose diameter is equal to its height, a sphere of maximum possible volume is carved out. What is the ratio of the volume of the utilised wood to that of the wasted wood?
 (a) 2:1 (b) 1:2 (c) 2:3 (d) 3:2
- 44.** The base diameter of a right circular cylinder is 3 cm. There is a section making an angle of 30° with the cross section. What is its area?
 (a) $\frac{9\pi}{4} \text{ cm}^2$ (b) $\frac{3\sqrt{3}\pi}{2} \text{ cm}^2$ (c) $\frac{9\pi}{8} \text{ cm}^2$ (d) $\frac{9\sqrt{3}\pi}{8} \text{ cm}^2$
- 45.** A cone is inscribed in a hemisphere such that their bases are common. If C is the volume of the cone and H that of the hemisphere, then what is the value of $C:H$?
 (a) 1:2 (b) 2:3 (c) 3:4 (d) 4:5
- 46.** If the diameter of a wire is decreased by 10%, by how much per cent (approximately) will the length be increased to keep the volume constant?
 (a) 5% (b) 17% (c) 20% (d) 23%
- 47.** The diameter of a solid metallic right circular cylinder is equal to its height. After cutting out the largest possible solid sphere S from this cylinder, the remaining material is recast to form a solid sphere S_1 . What is the ratio of the radius of sphere S to that of sphere S_1 ?
 (a) $1:2^{\frac{1}{3}}$ (b) $2^{\frac{1}{3}}:1$ (c) $2^{\frac{1}{3}}:3^{\frac{1}{3}}$ (d) $3^{\frac{1}{3}}:2^{\frac{1}{2}}$
- 48.** A square has its side equal to the radius of a sphere. The square revolves round a side to generate a surface of total area S . If A is the surface area of the sphere, then which one of the following is correct?
 (a) $A = 3S$ (b) $A = 2S$ (c) $A = S$ (d) $A < S$
- 49.** A swimming pool is 24 m long and 15 m broad. When x number of men dive into the pool, the height of the water rises by 1 cm. If the average amount of water displaced by one man is 0.1 m^3 , then what is the value of x ?
 (a) 36 (b) 72 (c) 108 (d) 360

- 50.** Water is distributed to a town of 50000 inhabitants from a rectangular reservoir consisting of 3 equal compartments. Each compartment has length and breadth 200 m, 100 m, respectively and 12 m depth of water in the beginning. The allowance is 20 L per head per day. For how many days will the supply of water hold out?
(a) 240 days (b) 720 days (c) 800 days (d) 900 days
- 51.** A right circular cylinder and a right circular cone have equal bases and equal volumes. But the lateral surface area of the right circular cone is $15/8$ times the lateral surface area of the right circular cylinder. What is the ratio of radius to height of the cylinder?
(a) 3 : 4 (b) 9 : 4 (c) 15 : 8 (d) 8 : 15
- 52.** The volume of a cuboid whose sides are in the ratio of 1 : 2 : 4 is same as that of a cube. What is the ratio of length of diagonal of cuboid to that of cube?
(a) $\sqrt{1.25}$ (b) $\sqrt{1.75}$ (c) $\sqrt{2}$ (d) $\sqrt{3.5}$
- 53.** A field is 125 m long and 15 m wide. A tank $10\text{ m} \times 7.5\text{ m} \times 6\text{ m}$ was dug in it and the Earth, thus dug out was spread equally on the remaining field. The level of the field thus raised is equal to which one of the following?
(a) 15 cm (b) 20 cm (c) 25 cm (d) 30 cm
- 54.** If C_1 is a right circular cone with base radius r_1 cm and height h_1 cm and C_2 is a right circular cylinder with base radius r_2 cm and height h_2 cm and if $r_1 : r_2 = 1 : n$ (where n is a positive integer) and their volumes are equal, then which one of the following is correct?
(a) $h_1 = 3nh_2$ (b) $h_1 = 3n^2h_2$
(c) $h_1 = 3h_2$ (d) $h_1 = n^2h_2$
- 55.** A right circular cone is cut by a plane parallel to its base in such a way that the slant heights of the original and the smaller cone thus obtained are in the ratio 2 : 1. If V_1 and V_2 are respectively the volumes of the original cone and of the new cone, then what is $V_1 : V_2$?
(a) 2 : 1 (b) 3 : 1 (c) 4 : 1 (d) 8 : 1
- 56.** The surface area of a sphere is 616 cm^2 . If its radius is changed so that the area gets reduced by 75%, then the radius becomes
(a) 1.6 cm (b) 2.3 cm (c) 2.5 cm (d) 3.5 cm
- 57.** A sphere is inscribed in a cubical box such that the sphere is tangent to all six faces of the box. What is the ratio of the volume of the cubical box to the volume of sphere?
(a) 6π (b) 36π (c) $\frac{4\pi}{3}$ (d) $\frac{6}{\pi}$
- 58.** From a solid cylinder of height 4 cm and radius 3 cm, a conical cavity of height 4 cm and of base radius 3 cm is hollowed out. What is the total surface area of the remaining solid?
(a) $15\pi\text{ cm}^2$ (b) $22\pi\text{ cm}^2$ (c) $33\pi\text{ cm}^2$ (d) $48\pi\text{ cm}^2$
- 59.** The curved surface of a cylinder is 1000 cm^2 . A wire of diameter 5 mm is wound around it, so as to cover it completely. What is the length of the wire used?
(a) 22 m (b) 20 m
(c) 18 m (d) None of these
- 60.** In order to fix an electric pole along a roadside, a pit with dimensions $50\text{ cm} \times 50\text{ cm}$ is dug with the help of a spade. The pit is prepared by removing Earth by 250 strokes of spade. If one stroke of spade removes 500 cm^3 of Earth, then what is the depth of the pit?
(a) 2 m (b) 1 m (c) 0.75 m (d) 0.5 m
- 61.** A figure is formed by revolving a rectangular sheet of dimensions $7\text{ cm} \times 4\text{ cm}$ about its length. What is the volume of the figure thus formed?
(a) 352 cm^3 (b) 296 cm^3
(c) 176 cm^3 (d) 616 cm^3
- 62.** The diagonals of the three faces of a cuboid are x, y, z , respectively. What is the volume of the cuboid?
(a) $\frac{xyz}{2\sqrt{2}}$
(b) $\frac{\sqrt{(y^2 + z^2)(z^2 + x^2)(x^2 + y^2)}}{2\sqrt{2}}$
(c) $\frac{\sqrt{(y^2 + z^2 - x^2)(z^2 + x^2 - y^2)(x^2 + y^2 - z^2)}}{2\sqrt{2}}$
(d) None of the above
- 63.** The volume of a cone is equal to that of a sphere. If the diameter of base of cone is equal to the diameter of the sphere, what is the ratio of height of cone to the diameter of the sphere?
(a) 2 : 1 (b) 1 : 2 (c) 3 : 1 (d) 4 : 1
- 64.** The length, breadth and height of a rectangular parallelopiped are in ratio 6 : 3 : 1. If the surface area of a cube is equal to the surface area of this parallelopiped, then what is the ratio of the volume of the cube to the volume of the parallelopiped?
(a) 1 : 1 (b) 5 : 4 (c) 7 : 5 (d) 3 : 2
- 65.** A hollow hemisphere is made of a sheet of a metal 1 cm thick. If the outer radius is 5 cm. What is the weight of the hemisphere (1 cm^3 of the metal weight 9 g)?
(a) $54\pi\text{ g}$ (b) $366\pi\text{ g}$ (c) $122\pi\text{ g}$ (d) $108\pi\text{ g}$

- 66.** Half of a large cylindrical tank open at the top is filled with water and identical heavy spherical balls are to be dropped into the tank without spilling water out. If the radius and the height of the tank are equal and each is four times the radius of a ball, what is the maximum number of balls that can be dropped?

(a) 12 (b) 24 (c) 36 (d) 48

- 67.** A cylinder is circumscribed about a hemisphere and a cone is inscribed in the cylinder so as to have its vertex at the centre of one end, and the other end as its base. The volume of the cylinder, hemisphere and the cone are respectively in the ratio

(a) 2 : 3 : 2 (b) 3 : 2 : 1 (c) 3 : 1 : 2 (d) 1 : 2 : 3

- 68.** A conical cavity is drilled in a circular cylinder of height 15 cm and base radius 8 cm. The height and the base radius of the cone are also same. Then the whole surface area of the remaining solid is

(a) $440 \pi \text{ cm}^2$ (b) $240 \pi \text{ cm}^2$ (c) $640 \pi \text{ cm}^2$ (d) $960 \pi \text{ cm}^2$

- 69.** Smaller lead shots are to be prepared by using the material of a spherical lead shot of radius 1 cm. Same possibilities are listed in the statements given below

- I. The material is just sufficient to prepare 8 shots each of radius 0.5 cm.
 II. A shot of radius 0.75 cm and a second shot of radius 0.8 cm can be prepared from the available material.

Which of the above statement is/are correct?

(a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II

- 70.** Consider the following statements:

- I. The volume of a cuboid is the product of the lengths of its coterminal edges.
 II. The surface area of a cuboid is twice the sum of the products of lengths of its coterminal edges taken two at a time.

Which of the above statement(s) is/are correct?

(a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II

- 71.** Consider the following statements :

The length of a side of a cube is 1 cm. Which of the following can be the distance between any two vertices?

I. 1 cm II. $\sqrt{2}$ cm III. $\sqrt{3}$ cm

Select the correct answer using the code given below.

(a) Only I (b) Only II (c) Only III (d) All of these

- 72.** Consider the following statements.

- I. When eight drops of water combine to form a single drop, the surface area of all the eight drops is greater than the surface area of big drop.

- II. Square of volume of a spherical body is directly proportional to cube of its surface area.

- III. A sphere occupies the biggest space but has the smallest surface area.

Which of the above statement(s) is/are correct?

(a) I and II (b) I and III (c) II and III (d) All of these

- 73.** Consider the following statements:

- I. The curved surface area of a right circular cone of base radius r and height h is given by

$$\pi r(\sqrt{h^2 + r^2}).$$

- II. The right circular cone of base radius r and height h when cut opened along the slant height, forms a rectangle of length πr and breadth $\sqrt{h^2 + r^2}$.

- III. If a right circular cone is cut off by a plane parallel to the base of the cone, then its curved surface area is $\frac{\pi h}{3} [R^2 + r^2 + Rr]$.

- IV. If a right triangle is being revolved about one of its edges. The surface generated by the hypotenuse is the lateral surface of the right circular cone.

Which of the above statement(s) are correct?

(a) I and II (b) I, II and III (c) I, III and IV (d) All of these

Directions (Q. Nos. 74-76) Read the following information carefully and answer the given questions that follow.

A cone of height 10 cm and radius 5 cm is divided into two parts by drawing a plane through the mid-point of its axis, parallel to its base.

- 74.** What is the ratio of the volume of the original cone to the volume of the frustum left?

(a) $\frac{4}{3}$ (b) $\frac{7}{8}$ (c) $\frac{8}{7}$ (d) $\frac{9}{8}$

- 75.** What is the area of the top circle of the frustum in square centimetre?

(a) 25π (b) $\frac{5\pi}{4}$ (c) $\frac{25\pi}{4}$ (d) None of these

- 76.** What is the ratio of the curved surface area of the original cone and the curved surface area of the frustum?

(a) 3 : 1 (b) 3 : 2 (c) 4 : 1 (d) 4 : 3

Directions (Q. Nos. 77-79) Read the following information carefully and answer the given questions that follow.

Let C be a right circular cone. It is given that the two ends of a frustum of C are of radii 3 cm and 6 cm and the height of the frustum is 9 cm.

- 77.** What is the slant height of the given frustum?

(a) $3\sqrt{10}$ cm (b) $6\sqrt{10}$ cm (c) 12 cm (d) 15 cm

- 78.** What is the height of the cone?

(a) 9 cm (b) 12 cm (c) 13.5 cm (d) 18 cm

79. What is the total surface area of the given frustum?

(a) $9\pi(2\sqrt{10} + 5)\text{cm}^2$ (b) $9\pi(3\sqrt{10} + 5)\text{cm}^2$
 (c) $9\pi(3\sqrt{10} + 4)\text{cm}^2$ (d) $27\pi(\sqrt{10} + 1)\text{cm}^2$

PREVIOUS YEARS' QUESTIONS

80. What will be the cost to plaster the inner surface of a well 14 m deep and 4 m in diameter at the rate of ₹ 25 per m^2 ? **2012 I**

(a) ₹ 4000 (b) ₹ 4200 (c) ₹ 4400 (d) ₹ 5400

81. What is the length of the uniform wire of diameter 0.4 cm that can be drawn from a solid sphere of radius 9 cm? **2012 I**

(a) 243 m (b) 240 m (c) 60.75 m (d) 60 m

82. The total surface area of a cube is 150 cm^2 . What is its volume? **2012 I**

(a) 64 cm^3 (b) 81 cm^3 (c) 125 cm^3 (d) 160 cm^3

83. If the volume of a cube is 729 cm^3 , then what is the length of its diagonal? **2012 I**

(a) $9\sqrt{2}\text{ cm}$ (b) $9\sqrt{3}\text{ cm}$ (c) 18 cm (d) $18\sqrt{3}\text{ cm}$

84. The curved surface area of a right circular cone of radius 14 cm is 440 cm^2 . What is the slant height of the cone? **2012 I**

(a) 10 cm (b) 11 cm (c) 12 cm (d) 13 cm

85. A large solid metallic cylinder whose radius and height are equal to each other is to be melted and 48 identical solid balls are to be recast from the liquid metal, so formed. What is the ratio of the radius of a ball to the radius of the cylinder?

(a) 1 : 16 (b) 1 : 12 **2012 I**
 (c) 1 : 8 (d) 1 : 4

86. What are the dimensions (length, breadth and height, respectively) of a cuboid with volume 720 cm^3 , surface area 484 cm^2 and the area of the base 72 cm^2 ? **2012 I**

(a) 9, 8 and 10 cm (b) 12, 6 and 10 cm
 (c) 18, 4 and 10 cm (d) 30, 2 and 12 cm

87. If the surface area of a sphere is 616 cm^2 , then what is its volume? **2012 I**

(a) $4312/3\text{ cm}^3$ (b) $4102/3\text{ cm}^3$
 (c) 1257 cm^3 (d) 1023 cm^3

Directions (Q.Nos. 88-89) Read the following information carefully and answer the given questions that follow.

The areas of the ends of a frustum of a cone are P and Q , where $P < Q$ and H is its thickness.

88. What is the difference in radii of the ends of the frustum? **2012 II**

(a) $\frac{\sqrt{Q} - \sqrt{P}}{\sqrt{\pi}}$ (b) $\frac{\sqrt{Q} - \sqrt{P}}{\pi}$
 (c) $\sqrt{Q} - \sqrt{P}$ (d) None of these

89. What is the volume of the frustum? **2012 II**

(a) $3H(P + Q + \sqrt{PQ})$ (b) $H(P + Q + \sqrt{PQ})$
 (c) $H(P + Q + \sqrt{PQ})/3$ (d) $H(P + Q - \sqrt{PQ})/3$

90. Let the largest possible right circular cone and largest possible sphere be fitted into two cubes of same length. If C and S denote the volume of cone and volume of sphere, respectively. Then, which one of the following is correct? **2012 II**

(a) $C = 2S$ (b) $S = 2C$ (c) $C = S$ (d) $C = 3S$

91. 10 circular plates each of thickness 3 cm, are placed one above the other and a hemisphere of radius 6 cm is placed on the top just to cover the cylindrical solid. What is the volume of the solid so formed? **2012 II**

(a) $264\pi\text{ cm}^3$ (b) $252\pi\text{ cm}^3$
 (c) $236\pi\text{ cm}^3$ (d) None of these

92. A right circular metal cone (solid) is 8 cm high and the radius is 2 cm. It is melted and recast into a sphere. What is the radius of the sphere? **2012 II**

(a) 2 cm (b) 3 cm (c) 4 cm (d) 5 cm

93. The volume of a cube is numerically equal to sum of its edges. What is the total surface area?

(a) 12 sq units (b) 36 sq units **2012 II**
 (c) 72 sq units (d) 144 sq units

94. The diameter of base of a right circular cone is 7 cm and slant height is 10 cm, then what is its lateral surface area? **2012 II**

(a) 110 cm^2 (b) 100 cm^2 (c) 70 cm^2 (d) 49 cm^2

95. What is the height of a solid cylinder of radius 5 cm and total surface area is 660 cm^2 ? **2012 II**

(a) 10 cm (b) 12 cm (c) 15 cm (d) 16 cm

96. If the ratio of the diameters of two spheres is 3 : 5, then what is the ratio of their surface areas? **2012 II**

(a) 9 : 25 (b) 9 : 10 (c) 3 : 5 (d) 27 : 125

97. What is the volume of the largest sphere that can be curved out of a cube of edge 3 cm? **2012 II**

(a) $9\pi\text{ cm}^3$ (b) $6\pi\text{ cm}^3$ (c) $4.5\pi\text{ cm}^3$ (d) $3\pi\text{ cm}^3$

98. What is the quantity of cloth required to roll up to form a right circular tent whose base is of radius 12 m and height 5 m? **2013 I**

(a) $40\pi\text{ m}^2$ (b) $60\pi\text{ m}^2$ (c) $78\pi\text{ m}^2$ (d) $156\pi\text{ m}^2$

99. The volume of the material of a hemispherical shell with outer and inner radii 9 cm and 7 cm, respectively is approximately **2013 I**

(a) 808 cm^3 (b) 800 cm^3 (c) 816 cm^3 (d) 824 cm^3

100. The ratio of surface area to diameter of a sphere whose volume is $36\pi\text{ cm}^3$, is **2013 I**

(a) 3π (b) 6π (c) 6 (d) None of these

- 101.** A cylindrical tube open at both ends is made of metal. The internal diameter of the tube is 6 cm and length of the tube is 10 cm. If the thickness of the metal used is 1 cm, then the outer curved surface area of the tube is **✓ 2013 I**
 (a) $140\pi \text{ cm}^2$ (b) $146.5\pi \text{ cm}^2$
 (c) $70\pi \text{ cm}^2$ (d) None of these
- 102.** From a solid wooden right circular cylinder, a right circular cone whose radius and height are same as the radius and height of the cylinder, respectively is carved out. What is the ratio of the volume of the utilised wood to that of the wasted wood? **✓ 2013 I**
 (a) 1 : 2 (b) 2 : 1 (c) 2 : 3 (d) 1 : 3
- 103.** If the heights and the areas of the base of a right circular cone and a pyramid with square base are the same, then they have **✓ 2013 I**
 (a) same volume and same surface area
 (b) same surface area but different volumes
 (c) same volume but different surface areas
 (d) different volumes and different surface areas
- 104.** The volume of a right circular cone of height 3 cm and slant height 5 cm is **✓ 2013 I**
 (a) 49.3 cm^3 (b) 50.3 cm^3
 (c) 52 cm^3 (d) 53 cm^3
- 105.** A bucket is of a height 25 cm. Its top and bottom radii are 20 cm and 10 cm, respectively. Its capacity (in L) is **✓ 2013 I**
 (a) $17.5\pi/3$ (b) 17.5π (c) 20π (d) 25π
- 106.** The height of a cylinder is 15 cm. The lateral surface area is 660 cm^2 . Its volume is **✓ 2013 II**
 (a) 1155 cm^3 (b) 1215 cm^3 (c) 1230 cm^3 (d) 2310 cm^3
- 107.** The diameter of the Moon is approximately one-fourth of the diameter of the Earth. What is the ratio (approximate) of their volumes? **✓ 2013 II**
 (a) 1 : 16 (b) 1 : 64 (c) 1 : 4 (d) 1 : 128
- 108.** A conical cap has the base diameter 24 cm and height 16 cm. What is the cost of painting the surface of the cap at the rate of 70 paise per cm^2 ? **✓ 2013 II**
 (a) ₹ 520 (b) ₹ 524 (c) ₹ 528 (d) ₹ 532
- 109.** What is the whole surface area of a cone of base radius 7 cm and height 24 cm? **✓ 2013 II**
 (a) 654 cm^2 (b) 704 cm^2 (c) 724 cm^2 (d) 964 cm^2
- 110.** A tent is in the form of a right circular cylinder surmounted by a cone. The diameter of the cylinder is 24 m. The height of the cylindrical portion is 11 m, while the vertex of the cone is 16 m above the ground. What is the area of the curved surface for conical portion? **✓ 2013 II**
 (a) $3434/9 \text{ m}^2$ (b) $3431/8 \text{ m}^2$
 (c) $3432/7 \text{ m}^2$ (d) $3234/7 \text{ m}^2$
- 111.** If x is the curved surface area and y is the volume of a right circular cylinder, then which one of the following is correct? **✓ 2014 I**
 (a) Only the ratio of the height to radius of the cylinder is independent of x
 (b) Only the ratio of height to radius of the cylinder is independent of y
 (c) Either (a) or (b)
 (d) Neither (a) nor (b)
- 112.** A cylinder is surmounted by a cone at one end, a hemisphere at the other end. The common radius is 3.5 cm, the height of the cylinder is 6.5 cm and the total height of the structure is 12.8 cm. The volume V of the structure lies between **✓ 2014 I**
 (a) 370 cm^3 and 380 cm^3 (b) 380 cm^3 and 390 cm^3
 (c) 390 cm^3 and 400 cm^3 (d) None of these
- 113.** The diameter of the base of a cone is 6 cm and its altitude is 4 cm. What is the approximate curved surface area of the cone? **✓ 2014 I**
 (a) 45 cm^2 (b) 47 cm^2 (c) 49 cm^2 (d) 51 cm^2
- 114.** A drainage tile is a cylindrical shell 21 cm long. The inside and outside diameters are 4.5 cm and 5.1 cm, respectively. What is the volume of the clay required for the tile? **✓ 2014 I**
 (a) $6.96\pi \text{ cm}^3$ (b) $6.76\pi \text{ cm}^3$
 (c) $5.76\pi \text{ cm}^3$ (d) None of these
- 115.** A cube has each edge 2 cm and a cuboid is 1 cm long, 2 cm wide and 3 cm high. The paint in a certain container is sufficient to paint an area equal to 54 cm^2 . **✓ 2014 I**
 Which one of the following is correct?
 (a) Both cube and cuboid can be painted
 (b) Only cube can be painted
 (c) Only cuboid can be painted
 (d) Neither cube nor cuboid can be painted
- 116.** A cone of radius r cm and height h cm is divided into two parts by drawing a plane through the middle point of its height and parallel to the base. What is the ratio of the volume of the original cone to the volume of the smaller cone? **✓ 2014 I**
 (a) 4 : 1 (b) 8 : 1 (c) 2 : 1 (d) 6 : 1
- 117.** If 64 identical small spheres are made out of big sphere of diameter 8 cm, then what is surface area of each small sphere? **✓ 2014 I**
 (a) $\pi \text{ cm}^2$ (b) $2\pi \text{ cm}^2$ (c) $4\pi \text{ cm}^2$ (d) $8\pi \text{ cm}^2$
- 118.** The dimensions of a field are 15 m by 12 m. A pit 8 m long, 2.5 m wide and 2 m deep is dug in one corner of the field and the earth removed is evenly spread over the remaining area of the field. The level of the field is raised by **✓ 2014 I**
 (a) 15 cm (b) 20 cm (c) 25 cm (d) $\frac{200}{9} \text{ cm}$

- 119.** What is the diameter of the largest circle lying on the surface of a sphere of surface area 616 cm^2 ?

☑ 2014 I

(a) 14 cm (b) 10.5 cm (c) 7 cm (d) 3.5 cm

- 120.** The volume of a hollow cube is $216x^3$. What is the surface area of the largest sphere which can be enclosed in it?

☑ 2014 I

(a) $18\pi x^2$ (b) $27\pi x^2$ (c) $36\pi x^2$ (d) $72\pi x^2$

Directions (Q. Nos. 121-122) Read the following information carefully and answer the given questions that follow.

A right angled triangle having hypotenuse 25 cm and legs in the ratio 3 : 4 is made to revolve about its hypotenuse. ($\pi = 3.14$)

- 121.** What is the volume of the double cone so formed?

☑ 2014 II

(a) 3124 cm^3 (b) 3424 cm^3 (c) 3768 cm^3 (d) 3924 cm^3

- 122.** What is the surface area of the double cone so formed?

☑ 2014 II

(a) 1101.2 cm^2 (b) 1111.4 cm^2 (c) 1310.4 cm^2 (d) 1318.8 cm^2

- 123.** If the side of a cube is increased by 100%, then by what percentage is the surface area of the cube increased?

☑ 2014 II

(a) 150% (b) 200% (c) 300% (d) 400%

- 124.** A cylinder circumscribes a sphere. What is the ratio of volume of the sphere to that of the cylinder?

☑ 2014 II

(a) 2 : 3 (b) 1 : 2 (c) 3 : 4 (d) 3 : 2

Directions (Q. Nos. 125-126) Read the following information carefully and answer the given questions that follow.

A toy is in the form of a cone mounted on the hemisphere with the same radius. The diameter of the base of the conical portion is 12 cm and its height is 8 cm.

- 125.** What is the total surface area of the toy?

☑ 2014 II

(a) $132\pi \text{ cm}^2$ (b) $112\pi \text{ cm}^2$ (c) $96\pi \text{ cm}^2$ (d) $66\pi \text{ cm}^2$

- 126.** What is the volume of the toy?

☑ 2014 II

(a) $180\pi \text{ cm}^3$ (b) $240\pi \text{ cm}^3$ (c) $300\pi \text{ cm}^3$ (d) $320\pi \text{ cm}^3$

- 127.** The areas of the three adjacent faces of a cuboidal box are x , $4x$ and $9x$ sq unit. What is the volume of the box?

☑ 2014 II

(a) $6x^2$ cu units (b) $6x^{3/2}$ cu units
(c) $3x^{3/2}$ cu units (d) $2x^{3/2}$ cu units

- 128.** What is the number of pairs of perpendicular planes in a cuboid?

☑ 2014 II

(a) 4 (b) 8
(c) 12 (d) None of these

- 129.** Consider the following statements in respect of four spheres A, B, C and D having respective radii 6, 8, 10 and 12 cm.

I. The surface area of sphere C is equal to the sum of surface areas of spheres A and B .

II. The volume of sphere D is equal to the sum of volumes of spheres A, B and C .

☑ 2014 II

Which of the above statement(s) is/are correct?

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

- 130.** The diameter of a metallic sphere is 6 cm. The sphere is melted and drawn into a wire of uniform circular cross-section. If the length of the wire is 36 m, then what is its radius?

☑ 2014 II

(a) 0.1 cm (b) 0.01 cm (c) 0.001 cm (d) 1.0 cm

- 131.** What is the maximum distance between two points of a cube of side 2 cm?

☑ 2014 II

(a) $\sqrt{3}$ cm (b) $2\sqrt{3}$ cm (c) $4\sqrt{3}$ cm (d) $2\sqrt{2}$ cm

- 132.** The total outer surface area of a right circular cone of height 24 cm with a hemisphere of radius 7 cm upon its base is

☑ 2015 I

(a) $327\pi \text{ cm}^2$ (b) $307\pi \text{ cm}^2$
(c) $293\pi \text{ cm}^2$ (d) $273\pi \text{ cm}^2$

- 133.** A sphere and a cube have same surface area. The ratio of square of their volumes is

☑ 2015 I

(a) 6 : π (b) 5 : π (c) 3 : 5 (d) 1 : 1

- 134.** The radius of a sphere is equal to the radius of the base of a right circular cone, and the volume of the sphere is double the volume of the cone. The ratio of the height of the cone to the radius of its base is

☑ 2015 I

(a) 2 : 1 (b) 1 : 2 (c) 2 : 3 (d) 3 : 2

- 135.** If the radius of a sphere is increased by 10%, then the volume will be increased by

☑ 2015 I

(a) 33.1% (b) 30% (c) 50% (d) 10%

- 136.** If three metallic spheres of radii 6 cm, 8 cm and 10 cm are melted to form a single sphere, then the diameter of the new sphere will be

☑ 2015 I

(a) 12 cm (b) 24 cm (c) 30 cm (d) 36 cm

- 137.** If the height of a right circular cone is increased by 200% and the radius of the base is reduced by 50%, then the volume of the cone

☑ 2015 I

(a) remains unaltered (b) decreases by 25%
(c) increases by 25% (d) increases by 50%

- 138.** A rectangular paper of 44 cm long and 6 cm wide is rolled to form a cylinder of height equal to width of the paper. The radius of the base of the cylinder so rolled is

☑ 2015 I

(a) 3.5 cm (b) 5 cm (c) 7 cm (d) 14 cm

- 139.** The diagonals of three faces of a cuboid are 13, $\sqrt{281}$ and 20 linear units. Then, the total surface area of the cuboid is **✎ 2015 I**
 (a) 650 sq units (b) 658 sq units
 (c) 664 sq units (d) 672 sq units
- 140.** A cylindrical vessel of radius 4 cm contains water. A solid sphere of radius 3 cm is lowered into the water until it is completely immersed. The water level in the vessel will rise by **✎ 2015 I**
 (a) 1.5 cm (b) 2 cm (c) 2.25 cm (d) 4.5 cm
- 141.** A rectangular block of wood having dimensions $3\text{m} \times 2\text{m} \times 1.75\text{m}$ has to be painted on all its faces. The layer of paint must be 0.1 mm thick. Paint comes in cubical boxes having their edges equal to 10 cm. The minimum number of boxes of paint to be purchased is **✎ 2015 I**
 (a) 5 (b) 4 (c) 3 (d) 2
- 142.** If the surface area of a cube is 13254 cm^2 , then the length of its diagonal is **✎ 2015 II**
 (a) $44\sqrt{2}\text{ cm}$ (b) $44\sqrt{3}\text{ cm}$ (c) $47\sqrt{2}\text{ cm}$ (d) $47\sqrt{3}\text{ cm}$
- 143.** A water tank, open at the top, is hemispherical at the bottom and cylindrical above it. The radius is 12 m and the capacity is $3312\pi\text{ m}^3$. The ratio of the surface areas of the spherical and cylindrical portions, is **✎ 2015 II**
 (a) 3 : 5 (b) 4 : 5 (c) 1 : 1 (d) 6 : 5
- 144.** A large water tank has the shape of a cube. If 128 m^3 of water is pumped out, the water level goes down by 2m. Then, the maximum capacity of the tank is **✎ 2015 II**
 (a) 512 m^3 (b) 480 m^3 (c) 324 m^3 (d) 256 m^3
- 145.** The areas of three mutually perpendicular faces of a cuboid are x, y, z . If V is the volume, then xyz is equal to **✎ 2015 II**
 (a) V (b) V^2 (c) $2V$ (d) $2V^2$
- 146.** How many right angled triangles can be formed by joining the vertices of a cuboid? **✎ 2015 II**
 (a) 24 (b) 28 (c) 32 (d) None of these
- 147.** Three rectangles R_1, R_2 and R_3 have the same area. Their lengths x_1, x_2 and x_3 respectively are such that $x_1 < x_2 < x_3$. If V_1, V_2 and V_3 are the volumes of the cylinders formed from the rectangles R_1, R_2 and R_3 respectively by joining the parallel sides along the breadth, then which one of the following is correct? **✎ 2015 II**
 (a) $V_3 < V_2 < V_1$ (b) $V_1 < V_3 < V_2$
 (c) $V_1 < V_2 < V_3$ (d) $V_3 < V_1 < V_2$
- 148.** Let V be the volume of an inverted cone with vertex at origin and the axis of the cone is along positive Y-axis. The cone is filled with water up to half of its height. The volume of water is **✎ 2015 II**
 (a) $\frac{V}{8}$ (b) $\frac{V}{6}$ (c) $\frac{V}{3}$ (d) $\frac{V}{2}$
- 149.** 30 metallic cylinders of same size are melted and cast in the form of cones having the same radius and height as those of the cylinders. Consider the following statements
 I. A maximum of 90 cones will be obtained.
 II. The curved surface of the cylinder can be flattened in the shape of a rectangle but the curved surface of the cone when flattened has the shape of triangle.
 Which one of the following statement(s) is/are correct in respect of the above? **✎ 2015 II**
 (a) Both statement I and statement II are correct and statement II is the correct explanation of statement I
 (b) Both statement I and statement II are correct and statement II is not the correct explanation of statement I
 (c) statement I is correct but statement II is not correct
 (d) statement I is not correct but statement II is correct
- 150.** Water is filled in a container in such a manner that its volume doubles every 5 min. If it takes 30 min for the container to be full, in how much time will it be one-fourth full? **✎ 2015 II**
 (a) 7.5 min (b) 15 min (c) 20 min (d) 17.5 min
- 151.** Consider the following statements:
 I. If the height of a cylinder is doubled, the area of the curved surface is doubled.
 II. If the radius of a hemispherical solid is doubled, its total surface area becomes fourfold.
 Which of the statement(s) given above is/are correct? **✎ 2015 II**
 (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II
- 152.** From the solid gold in the form of a cube of side length 1 cm, spherical solid balls each having the surface area $\pi^{1/3}\text{ cm}^2$ are to be made. Assuming that there is no loss of the material in process of making the balls, the maximum number of balls made will be **✎ 2015 II**
 (a) 3 (b) 4 (c) 6 (d) 9
- 153.** A hollow cylindrical drum has internal diameter of 30 cm and a height of 1 m. What is the maximum number of cylindrical boxes of diameter 10 cm and height 10 cm each that can be packed in the drum? **✎ 2015 II**
 (a) 60 (b) 70 (c) 80 (d) 90
- 154.** A pipe with square cross-section is supplying water to a cistern which was initially empty. The area of cross-section is 4 cm^2 and the nozzle velocity of water is 40 m/s . The dimensions of the cistern $10\text{ m} \times 8\text{ m} \times 6\text{ m}$. Then, the cistern will be full in **✎ 2015 II**
 (a) 9.5 h (b) 9 h
 (c) 8 h and 20 min (d) 8 h

ANSWERS

1	c	2	b	3	b	4	c	5	d	6	b	7	c	8	a	9	c	10	c
11	c	12	c	13	b	14	a	15	b	16	c	17	d	18	a	19	b	20	a
21	d	22	d	23	b	24	b	25	c	26	b	27	b	28	b	29	b	30	c
31	c	32	c	33	b	34	b	35	b	36	b	37	d	38	a	39	b	40	c
41	c	42	c	43	a	44	b	45	a	46	d	47	b	48	c	49	a	50	b
51	b	52	b	53	c	54	b	55	d	56	d	57	d	58	d	59	b	60	d
61	a	62	c	63	a	64	d	65	b	66	b	67	b	68	a	69	c	70	c
71	d	72	d	73	c	74	c	75	c	76	d	77	a	78	d	79	b	80	c
81	a	82	c	83	b	84	a	85	d	86	a	87	a	88	a	89	c	90	b
91	d	92	a	93	c	94	a	95	d	96	a	97	c	98	d	99	a	100	b
101	d	102	a	103	c	104	b	105	a	106	d	107	b	108	c	109	b	110	c
111	d	112	a	113	b	114	d	115	a	116	b	117	c	118	c	119	a	120	c
121	c	122	d	123	c	124	a	125	a	126	b	127	b	128	c	129	c	130	a
131	b	132	d	133	a	134	a	135	a	136	b	137	b	138	c	139	c	140	c
141	c	142	d	143	b	144	a	145	b	146	a	147	a	148	a	149	b	150	c
151	c	152	c	153	d	154	c												

HINTS AND SOLUTIONS

1. (c) Let 'a' be the edge of three equal cubes. Surface area of a single cube
 $= 6 (\text{Side})^2 = 6a^2$

$\therefore$ Sum of the surface areas of the three cubes is $= 3(6a^2) = 18a^2$

Length of resulting cuboid $= 3a$

Breadth of resulting cuboid $= a$

Height of resulting cuboid $= a$

$\therefore$ Total surface area of cuboid

$$= 2(lb + bh + hl)$$

$$= 2(3a^2 + a^2 + 3a^2) = 14a^2$$

$$\text{Required ratio} = \frac{14a^2}{18a^2} = \frac{7}{9} \text{ i.e. } 7:9$$

2. (b) Here, $x = 12$,

According to the formula, percentage increase in volume

$$= \left[\left(1 + \frac{x}{100} \right)^3 - 1 \right] \times 100\%$$

$$= \left[\left(1 + \frac{12}{100} \right)^3 - 1 \right] \times 100\%$$

$$= \left[\left(1 + \frac{3}{25} \right)^3 - 1 \right] \times 100\%$$

$$= \left[\left(\frac{28}{25} \right)^3 - 1 \right] \times 100\%$$

$$= \left[\frac{21952}{15625} - 1 \right] \times 100\% = [1.405 - 1] \times 100\%$$

$$= 0.405 \times 100\% = 40.5\%$$

3. (b) Given, diameter of the base of the cylinder $= 21$ cm

$$\therefore \text{Radius} = \frac{21}{2} \text{ cm}$$

$$\text{As, curved surface area} = 2\pi rh = 1320$$

$$\Rightarrow 2 \times \frac{22}{7} \times \frac{21}{2} \times h = 1320$$

$$\therefore h = \frac{1320}{22 \times 3} = 20 \text{ cm}$$

4. (c) 1 cm^3 of water weighs 1 g.

$$\therefore \text{Volume of } 154 \text{ g of water} = 154 \text{ cm}^3$$

$$\therefore \text{Volume of cylinder} = \pi r^2 h = 154 \text{ cm}^3$$

$$\Rightarrow \frac{22}{7} (3.5)^2 \times h = 154$$

$$\Rightarrow \frac{77h}{2} = 154$$

$$\Rightarrow h = \frac{154 \times 2}{77} = 4 \text{ cm}$$

5. (d) Here, $2\pi rh = 264 \text{ m}^2$

$$\text{and } \pi r^2 h = 924 \text{ m}^3 \quad [\text{given}]$$

$$\therefore \frac{\pi r^2 h}{2\pi rh} = \frac{924}{264}$$

$$\Rightarrow \frac{r}{2} = \frac{7}{2} \Rightarrow r = 7 \text{ m}$$

$$\therefore \text{Diameter of pillar}$$

$$= r \times 2 = 7 \times 2 = 14 \text{ m}$$

6. (b) Given, $r = 7$ m, $h = 24$ m

Slant height of cone,

$$= \sqrt{r^2 + h^2} = \sqrt{24^2 + 7^2}$$

$$= \sqrt{576 + 49} = \sqrt{625} = 25 \text{ m}$$

Curved surface area $= \pi rl$

$$= \frac{22}{7} \times 7 \times 25 = 550 \text{ m}^2$$

Width of cloth $= 50$ m

$$\text{Length required} = \frac{550}{50} = 11 \text{ m}$$

7. (c) Let radius and height of a right circular cone be $5x$ and $12x$ respectively.

$$\therefore r = 5x \text{ and } h = 12x$$

$$\text{Volume of cone} = \frac{1}{3} \pi r^2 h = 2512$$

$$\Rightarrow \frac{1}{3} \times 3.14 \times (5x)^2 \times 12x = 2512$$

$$\Rightarrow x^3 = \frac{2512}{314} = 8$$

$$\Rightarrow x = 2$$

$$\therefore \text{Radius} = 10 \text{ cm, height} = 24 \text{ cm}$$

$\therefore$ Slant height,

$$l = \sqrt{r^2 + h^2} = \sqrt{10^2 + 24^2}$$

$$= \sqrt{676} = 26 \text{ cm}$$

8. (a) Let r and H be the radius and height of the cone respectively.

$$\therefore \text{Original volume} = \frac{1}{3} \pi r^2 h = V$$

New radius = r and new height = $2h$

$$\text{New volume} = \frac{1}{3} \pi r^2 \times 2h = 2V$$

$$\therefore \text{Increase in volume} = 2V - V = V$$

$$\begin{aligned} \text{Increase percentage of volume} \\ = \left(\frac{V}{V} \times 100 \right) \% = 100\% \end{aligned}$$

9. (c) Ratio of surface area

$$= \frac{4\pi r^2}{4\pi R^2} = \left(\frac{4}{25} \right) = \left(\frac{2}{5} \right)^2 \Rightarrow \frac{r}{R} = \frac{2}{5}$$

Now, ratio of volumes

$$\begin{aligned} \frac{\frac{4}{3} \pi r^3}{\frac{4}{3} \pi R^3} &= \left(\frac{r}{R} \right)^3 = \left(\frac{2}{5} \right)^3 = \frac{8}{125} \\ &= 8 : 125 \end{aligned}$$

10. (c) Let radius of cylinder = x and radius of cone = $2x$

Height of each = h

$$\begin{aligned} \therefore \text{Required ratio} &= \frac{\text{Volume of cone}}{\text{Volume of cylinder}} \\ &= \frac{\frac{1}{3} \pi 4x^2 h}{\pi x^2 h} = \frac{4}{3} = 4 : 3 \end{aligned}$$

11. (c) Volume of sphere = Volume of cylinder

$$\therefore \frac{4}{3} \pi R^3 = \pi \times 3 \times 3 \times 32$$

$$\Rightarrow R^3 = 3 \times 3 \times 3 \times 8 \Rightarrow R = 6 \text{ cm}$$

12. (c) Volume of first sphere = $\frac{4}{3} \pi \left(\frac{3}{2} \right)^3$

$$\text{and volume of second sphere} = \frac{4}{3} \pi (2)^3$$

$\therefore$ Volume of big sphere

$$= \frac{4}{3} \pi \left\{ \left(\frac{3}{2} \right)^3 + (2)^3 \right\} [\text{by condition}]$$

$$\Rightarrow \frac{4}{3} \pi x^3 = \frac{4}{3} \pi \left\{ \frac{27}{8} + 8 \right\}$$

$$x^3 = 91/8 = 11.375 \text{ cm}^3$$

13. (b) Given, $l = R$

$\therefore$ Total surface area of cone,

$$S = \pi R(R + l) = \pi R(R + R) = 2\pi R^2$$

$$\therefore \text{Area of circle, } A = \pi (2R)^2 = 4\pi R^2$$

$$\Rightarrow A = 2S$$

14. (a) In one round, distance covered by cylinder = 60 cm

In one second, distance covered by cylinder = $60 \times 5 = 300$ cm

In five seconds, distance covered by cylinder = $300 \times 5 = 1500$ cm = 15 m

15. (b) Let the radius of big drop and small drop be R and r respectively.

By given condition,

$27 \times \text{Volume of smaller drops} = \text{Volume of bigger drop}$

$$\therefore 27 \times \frac{4}{3} \pi r^3 = \frac{4}{3} \pi R^3$$

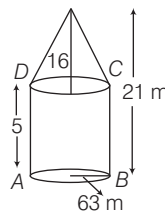
$$\Rightarrow 27 \times (0.2)^3 = R^3 \quad [\because r = 0.2 \text{ cm}]$$

$$\Rightarrow (3 \times 0.2)^3 = R^3 \Rightarrow R = 0.6 \text{ cm}$$

16. (c) Common radius = $\frac{126}{2} = 63$ m

Curved surface area of cylindrical part

$$= 2\pi rh$$



$$= 2 \times \frac{22}{7} \times 63 \times 5 = 1980 \text{ m}^2$$

Height of cone = $21 - 5 = 16$ m

Curved surface area of conical part = πrl

$$= \frac{22}{7} \times 63 \times \sqrt{(63)^2 + (16)^2}$$

$$= \frac{22}{7} \times 63 \times \sqrt{3969 + 256}$$

$$= \frac{22}{7} \times 63 \times 65 = 12870 \text{ m}^2$$

$\therefore$ Total surface area

$$= 1980 + 12870 = 14850 \text{ m}^2$$

$\therefore$ Total cost of canvas used

$$= 14850 \times 12 = ₹ 178200$$

17. (d) Given, radius of cylinder = 6 cm

Volume of sphere

$$= \frac{4}{3} \pi r^3 = \frac{4}{3} \times \frac{22}{7} \times 6 \times 6 \times 6$$

$$= \frac{88 \times 2 \times 36}{7} \text{ cm}^3$$

Let r be its internal radius, R be its external radius, then material used to cast the cylinder

$$= \pi h (R^2 - r^2) = \frac{22}{7} \times 32 (25 - r^2)$$

$$\text{Hence, } \frac{22}{7} \times 32 \times (25 - r^2)$$

$$= \frac{88 \times 2 \times 36}{7}$$

$$\Rightarrow (25 - r^2) = \frac{88 \times 2 \times 36 \times 7}{22 \times 32 \times 7} = 9$$

$$\Rightarrow r^2 = 25 - 9 = 16 \Rightarrow r = 4 \text{ m}$$

Thickness of cylinder

$$= R - r = 5 - 4 = 1 \text{ cm}$$

18. (c) Volume of original cylinder

$$= [\pi \times (10)^2 \times 1000] \text{ cm}^3$$

Volume of shell

$$= \frac{1}{4} \times \pi \times 100000 = 25000 \pi \text{ cm}^3$$

Let r be the radius of shell, then

$$\pi r^2 h = 25000 \pi$$

$$\Rightarrow r^2 = \frac{25000 \pi}{\pi \times 1000} = 25$$

$$\Rightarrow r = 5 \text{ cm}$$

Now, thickness = $2r = 2 \times 5 = 10$ cm

$$\Rightarrow R^2 = 100 + 25 = 125$$

$$\Rightarrow R = \sqrt{125}$$

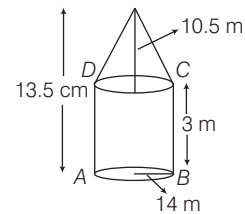
$\therefore$ Thickness of cylindrical shell

$$= R - 10 = \sqrt{125} - 10$$

$$= 5\sqrt{5} - 5 \times 2 = 5(\sqrt{5} - 2) \text{ cm}$$

19. (b) Curved surface area of cylinder

$$= 2 \times \frac{22}{7} \times 14 \times 3 = 264 \text{ m}^2$$



Height of cone = $13.5 - 3 = 10.5$ m

Slant height of cone

$$= \sqrt{(10.5)^2 + (14)^2} = \sqrt{110.25 + 196}$$

$$= \sqrt{306.25} = 17.5 \text{ m}$$

Curved surface area of cone = πrl

$$= \frac{22}{7} \times 14 \times 17.5 = 770 \text{ m}^2$$

Total area to be painted

$$= 264 + 770 = 1034 \text{ m}^2$$

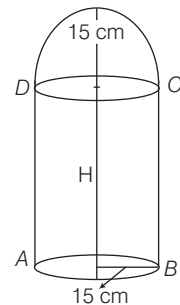
$$\therefore \text{Cost of painting} = 1034 \times 2 = ₹ 2068$$

20. (a) Let height of the cylinder be H .

By given condition,

Volume of hemisphere + Volume of cylinder = Volume of container

$$\therefore \frac{2}{3} \pi r^3 + \pi r^2 H = 32400 \pi$$



$$\Rightarrow \frac{2}{3} \pi \times 3375 + \pi \times 225 H = 32400 \pi$$

$$\Rightarrow 2\pi \times 1125 + \pi \times 225 H = 32400 \pi$$

On dividing both sides by 225π , we get

$$\Rightarrow 10 + H = 144 \Rightarrow H = 134 \text{ cm}$$

$\therefore$ Height of container

$$= 15 + 134 = 149 \text{ cm}$$

- 21. (d)** Since, milk is in a conical flask whose base radius and height are r and h , respectively.

$$\therefore \text{Volume of milk} = \frac{1}{3} \pi r^2 h$$

Now, this milk is poured into a cylindrical flask whose base radius is $2r$. Let H be the height of cylindrical flask.

$\therefore$ Volume of milk

= Volume of cylindrical flask

$$\Rightarrow \frac{1}{3} \pi r^2 h = \pi (2r)^2 H$$

$$\Rightarrow \frac{1}{3} \pi r^2 h = 4\pi r^2 H \Rightarrow H = \frac{h}{12}$$

- 22. (d)** Let radius and height of a cone be r and h .

$$\therefore \frac{r}{h} = \frac{3}{4} \Rightarrow h = \frac{4r}{3} \quad \dots(i)$$

$$\text{Volume of cone} = \frac{1}{3} \pi r^2 h$$

$$\therefore 96\pi = \frac{1}{3} \pi \times r^2 \times \frac{4r}{3}$$

$$\Rightarrow r^3 = \frac{96 \times 3 \times 3}{4} = 216 \quad [\text{from Eq. (i)}]$$

$$\Rightarrow r = 6 \text{ cm and } h = 8 \text{ cm}$$

$\therefore$ Lateral surface area = πrl

$$= \pi r \sqrt{r^2 + h^2} \quad [\because l^2 = r^2 + h^2]$$

$$= \pi \times 6 \sqrt{36 + 64} = 60\pi \text{ cm}^2$$

- 23. (b)** $\therefore$ Volume of wax balls

$$= \frac{4}{3} \pi (1)^3 = \frac{4}{3} \pi \text{ cm}^3$$

$$\text{and volume of sphere} = \frac{4}{3} \pi (8)^3 \text{ cm}^3$$

$$\therefore \text{Required number of balls} = \frac{\frac{4}{3} \pi (8)^3}{\frac{4}{3} \pi} = 512$$

- 24. (b)** Given, $r = \frac{70}{2} \text{ cm} = 0.35 \text{ m}$

and $h = 4 \text{ m}$

$\therefore$ Surface area of 10 cylinders

$$= 10 (2\pi rh)$$

$$= 10 \left(2 \times \frac{22}{7} \times 0.35 \times 4 \right) = 88 \text{ m}^2$$

$\therefore$ Total cost of 10 pillars painting at the rate of ₹ 5 per m^2

$$= 88 \times 5 = ₹ 440$$

- 25. (c)** Given,

$$R = 35 \text{ cm}, r = 14 \text{ cm and } h = 40 \text{ cm}$$

$\therefore$ Volume of the bucket

$$= \frac{\pi h}{3} (R^2 + r^2 + Rr)$$

$$= \frac{22 \times 40}{7 \times 3} (35^2 + 14^2 + 35 \times 14)$$

$$= \frac{880}{21} (1225 + 196 + 490)$$

$$= \frac{880}{21} \times 1911 = 80080 \text{ cm}^3$$

- 26. (b)** Let side of a cube be a unit.

$\therefore$ Volume of cube, $V = a^3$

and total surface area of cube, $S = 6(a)^2$

$$S^3 = 6^3 (a^6) = 216 V^2$$

- 27. (b)** Total area of wetted surface

= Curved surface area of cylinder + Area of base of cylinder

$$= 2\pi rh + \pi r^2$$

$$= \pi [2 \times 3.5 \times 4 + (3.5)^2]$$

$$= \frac{22}{7} (28 + 12.25) = \frac{22}{7} \times 40.25$$

$$= 126.5 \text{ m}^2$$

- 28. (b)** Let height h_1 , radius r_1 , area S_1 and volume V_1 of first cylinder.

Similarly, height h_2 , radius r_2 , area S_2 and volume V_2 of second cylinder.

$$\text{By given condition, } \frac{r_1}{r_2} = \frac{2}{3} \quad \dots(i)$$

$$\therefore \frac{S_1}{S_2} = \frac{5}{3} \Rightarrow \frac{2\pi r_1 h_1}{2\pi r_2 h_2} = \frac{5}{3}$$

$$\Rightarrow \frac{h_1}{h_2} \times \frac{2}{3} = \frac{5}{3} \Rightarrow \frac{h_1}{h_2} = \frac{5}{2} \quad \dots(ii)$$

$$\therefore \frac{V_1}{V_2} = \frac{\pi r_1^2 h_1}{\pi r_2^2 h_2} = \left(\frac{r_1}{r_2} \right)^2 \left(\frac{h_1}{h_2} \right) = \left(\frac{2}{3} \right)^2 \left(\frac{5}{2} \right) = \frac{10}{9} \text{ or, } 10 : 9$$

- 29. (b)** By given condition,

Volume of cylindrical vessel = Volume of sphere

$$\therefore \pi r^2 h = \frac{4}{3} \pi R^3$$

$$\Rightarrow (60)^2 \times 10 = \frac{4}{3} \left(\frac{d}{2} \right)^3$$

$$\Rightarrow (60)^2 \times 10 = \frac{4}{3} \times \frac{d^3}{8}$$

$$\Rightarrow d^3 = (60)^2 \times (60)$$

$$\therefore d = 60 \text{ cm}$$

- 30. (c)** Given, inner radius of hollow sphere

$$(r) = \frac{4}{2} = 2 \text{ cm}$$

Outer radius of hollow sphere

$$(R) = \frac{8}{2} = 4 \text{ cm}$$

Volume of metal of the sphere

$$= \frac{4}{3} \pi R^3 - \frac{4}{3} \pi r^3 = \frac{4}{3} \pi (4^3 - 2^3)$$

$$= \frac{4}{3} \pi \times 56 \text{ cm}^3$$

$$\text{Radius of base of cone } (x) = \frac{8}{2} = 4 \text{ cm}$$

$$\therefore \frac{1}{3} \pi x^2 h = \frac{4}{3} \pi \times 56 \quad [\text{by condition}]$$

$$\Rightarrow h = \frac{\frac{4}{3} \times 56 \times 3}{16} = 14 \text{ cm}$$

Hence, the height of cone is 14 cm.

- 31. (c)** Surface area of sphere, $S_1 = 4\pi r^2$

If radius is $2r$, then surface area of sphere,

$$S_2 = 4\pi (2r)^2 = 16\pi r^2$$

$$\therefore S_2 = 4S_1$$

Hence, it increases four times.

- 32. (c)** Given, $l = 6 \text{ m}$, $b = 4 \text{ m}$

and $h = 1.25 \text{ m}$

Area of wetted surface

$$= 2(l \times b + b \times h) + l \times b$$

$$= 2(7.5 \times 5) + 24$$

$$= 25 + 24 = 49 \text{ m}^2$$

- 33. (b)** Given, $R = 3 \text{ cm}$, $r = 2 \text{ cm}$, $h = 10 \text{ cm}$

Total surface area

$$= 2\pi (R + r)(R + h - r)$$

$$= 2\pi (3 + 2)(3 + 10 - 2)$$

$$= 110\pi \text{ cm}^2$$

- 34. (b)** For conical part, $r = \frac{6}{2} = 3 \text{ cm}$,

$$h = 4 \text{ cm}, l = \sqrt{h^2 + r^2} = 5 \text{ cm}$$

Surface area of conical part = πrl

$$= 3.14 \times 3 \times 5 = 47.1 \text{ cm}^2$$

For hemispherical part, $r = \frac{6}{2} = 3 \text{ cm}$

Surface area of hemispherical part

$$= 2\pi r^2 = 2 \times 3.14 \times 3 \times 3$$

$$= 56.52 \text{ cm}^2$$

$$\therefore \text{Surface area of toy} = 47.1 + 56.52 = 103.62 \text{ cm}^2$$

- 35. (b)** Let r and h be the radius and height of the cylinder. Here, $h = 9 \text{ m}$

By given condition,

$$2\pi r \times h = \frac{2\pi r}{3} (h + r)$$

$$\Rightarrow 9 = \frac{1}{3} (9 + r) \Rightarrow 27 = 9 + r$$

$$\therefore r = 18 \text{ m}$$

36. (b) Let r_1 and r_2 be the radii of two spheres.

$$\therefore \frac{\frac{4}{3}\pi r_1^3}{\frac{4}{3}\pi r_2^3} = \frac{8}{1} \Rightarrow \frac{r_1}{r_2} = \frac{2}{1}$$

$$\therefore \text{Ratio of their surface areas} = \frac{4\pi r_1^2}{4\pi r_2^2} = \left(\frac{2}{1}\right)^2 = 4:1$$

37. (d) Area of the field = Length $\times$ Breadth
 $= 12 \times 10 = 120 \text{ m}^2$

$$\text{Area of the pit's surface} = 5 \times 4 = 20 \text{ m}^2$$

$$\text{Area on which the Earth is to be spread} = 120 - 20 = 100 \text{ m}^2$$

Volume of Earth dug out

$$= 5 \times 4 \times 2 = 40 \text{ m}^3$$

$$\therefore \text{Level of field raised} = \frac{40}{100} = \frac{2}{5} \text{ m}$$

$$= \frac{2}{5} \times 100 = 40 \text{ cm}$$

38. (a) Edge of cube = 9 cm

For rectangular vessel,

Length = 15 cm and breadth = 12 cm

Let the rise in the water level is x cm.

Volume of cube = $9^3 = 9 \times 9 \times 9 \text{ cm}^3$

Volume of water risen in the vessel
 $= 15 \times 12 \times x \text{ cm}^3$

$$\Rightarrow 15 \times 12 \times x = 9 \times 9 \times 9$$

$$\therefore x = \frac{9 \times 9 \times 9}{15 \times 12} = \frac{81}{20} = 4.05 \text{ cm}$$

39. (b) The maximum diameter of a sphere in a cube is of 3 m.

$$\therefore \text{Volume of sphere, } V_1 = \frac{4}{3} \pi (1.5)^3$$

$$= 4.5 \pi \text{ m}^3$$

Volume of cube, $V_2 = (3)^3 = 27 \text{ m}^3$

$$\text{and Volume of solid left} = V_2 - V_1$$

$$= (27 - 4.5 \pi) \text{ m}^3$$

40. (c) Length of water flowing in 1h = 10 km

$$\therefore \text{Length of the water flowing in } \frac{1}{2} \text{ h}$$

$$= 5 \text{ km} = 5000 \text{ m}$$

Area of pipe

$$= 40 \text{ cm}^2 = \frac{40}{10000} \text{ m}^2 = \frac{1}{250} \text{ m}^2$$

$$\therefore \text{Volume of water flowing in } \frac{1}{2} \text{ h}$$

$$= 5000 \times \frac{1}{250} = 20 \text{ m}^3$$

Area of bottom of tank

$$= 80 \times 40 = 3200 \text{ m}^2$$

$\therefore$ Depth of water

$$= \frac{\text{Volume}}{\text{Area}} = \frac{20}{3200} = \frac{1}{160} \text{ m}$$

$$= \frac{1}{160} \times 100 \text{ cm} = \frac{5}{8} \text{ cm}$$

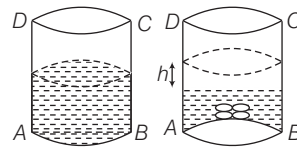
41. (c) Given radius of balls = 1 cm

$$\text{Volume of each ball} = \frac{4}{3} \times \pi \times 1 \times 1 \times 1$$

$$= \frac{4}{3} \pi \text{ cm}^3$$

$\therefore$ Volume of 4 balls

$$= 4 \times \frac{4}{3} \pi = \frac{16}{3} \pi \text{ cm}^3$$



Volume of water increased

= Volume of balls

$$\Rightarrow \text{Area of base} \times \text{height} = \frac{16}{3} \pi$$

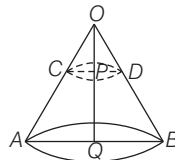
$$\Rightarrow \pi \times 5 \times 5 \times h = \frac{16}{3} \pi$$

$$\therefore h = \frac{16}{3 \times 25} = \frac{16}{75} \text{ cm}$$

42. (c) Given, height of the cone = 30 cm

$\therefore$ Volume of original cone (AOB)

$$= \frac{1}{3} \pi R^2 H$$



$$= \frac{1}{3} \times \pi R^2 \times 30 = 10 \pi R^2 \text{ cm}^3$$

Volume of small cone OCD

$$= \frac{1}{27} \text{ of volume cone AOB}$$

$$\Rightarrow \frac{1}{3} \pi r^2 h = \frac{1}{27} \times 10 \pi R^2$$

$$\Rightarrow h = \frac{10 \pi R^2}{27} \times \frac{3}{\pi r^2} \Rightarrow h = \frac{10}{9} \left(\frac{R}{r}\right)^2$$

$$\therefore \frac{R}{r} = \frac{30}{h} \quad [\because \triangle OQB \sim \triangle OPD]$$

$$\therefore h = \frac{10}{9} \left(\frac{30}{h}\right)^2 = \frac{10}{9} \times \frac{900}{h^2}$$

$$\Rightarrow h^3 = 10 \times 100$$

$$\Rightarrow h = \sqrt[3]{1000} = 10 \text{ cm}$$

$\therefore$ Required height

$$= H - h = 30 - 10 = 20 \text{ cm}$$

43. (a) Let r be the radius of cylindrical block, then height will be $2r$.

$$\text{Volume of block} = \pi(r^2)(2r) = 2\pi r^3$$

A sphere of maximum possible volume is curved out whose radius will be r .

$$\therefore \text{Volume of sphere} = \frac{4}{3} \pi r^3$$

$$\therefore \text{Volume of utilised wood} = \frac{4}{3} \pi r^3$$

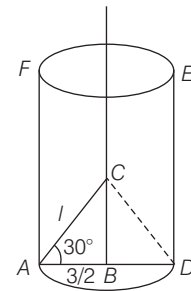
and volume of wasted wood

$$= 2\pi r^3 - \frac{4}{3} \pi r^3 = \frac{6\pi r^3 - 4\pi r^3}{3} = \frac{2\pi r^3}{3}$$

$$\therefore \text{Required ratio} = \frac{4}{3} \pi r^3 : \frac{2}{3} \pi r^3 = 2:1$$

44. (b) Let the slant height of cone = l cm

In $\triangle ABC$,



$$\cos 30^\circ = \frac{AB}{AC} = \frac{3/2}{l} \quad [\because AC = l \text{ cm}]$$

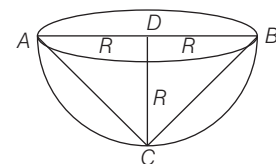
$$\Rightarrow l = \frac{3/2}{\sqrt{3}/2} = \sqrt{3} \text{ cm} \quad [\because \cos 30^\circ = \frac{\sqrt{3}}{2}]$$

$\therefore$ Area of cone $ACD = \pi r l$

$$= \pi \times \frac{3}{2} \times \sqrt{3} = \frac{3\sqrt{3}\pi}{2} \text{ cm}^2$$

45. (a) Volume of cone, $C = \frac{1}{3} \pi R^2 H$

$$= \frac{1}{3} \pi R^3 \quad [\because H = R]$$



$$\text{Volume of hemisphere, } H = \frac{2}{3} \pi R^3$$

$$\therefore C : H = \frac{1}{3} \pi R^3 : \frac{2}{3} \pi R^3 = 1:2$$

46. (d) Volume of wire = $\pi r^2 h$

Now, new radius of the wire

$$= \frac{r \times 90}{100} = \frac{9r}{10}$$

Let new length of the wire be L_2 .

$$\therefore \text{Volume of new wire} = \pi \left(\frac{9r}{10}\right)^2 \times L_2$$

$$= \frac{81}{100} \pi r^2 L_2$$

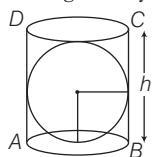
$$\text{By given condition, } \pi r^2 h = \frac{81}{100} \pi r^2 L_2$$

$$\Rightarrow L = \frac{100}{81} h$$

$$\text{Increases in length} = \frac{100}{81} b - b = \frac{19}{81} b$$

$$\text{Percentage increase} = \frac{19/81b}{b} \times 100\% = 23.45\% = 23\% \text{ (approx)}$$

47. (b) Let the height of cylinder be h .



$$\text{Then, radius of cylinder} = \frac{b}{2}$$

$$\text{Also, radius of the sphere, } r = \frac{b}{2}$$

$$\therefore \text{Volume of cylinder} = \pi \left(\frac{b}{2}\right)^2 h = \frac{\pi b^3}{4}$$

and volume of sphere.

$$S = \frac{4}{3} \pi r^3 = \frac{4}{3} \pi \left(\frac{b}{2}\right)^3 = \frac{\pi b^3}{6}$$

$$\therefore \text{Volume of remaining material} = \frac{\pi b^3}{4} - \frac{\pi b^3}{6} = \frac{\pi b^3}{12}$$

$\therefore$ Volume of sphere in remaining material,

$$S_1 = \frac{4}{3} \pi r_1^3$$

$$\text{By given condition, } \frac{4}{3} \pi r_1^3 = \frac{\pi b^3}{12}$$

$$\Rightarrow r_1^3 = \frac{b^3}{16} \Rightarrow r_1 = \frac{b}{2 \cdot 2^{1/3}}$$

$$\text{Required ratio} = \frac{r}{r_1} = \frac{b/2}{b/2 \cdot 2^{1/3}} = 2^{1/3}$$

$$\Rightarrow r : r_1 = 2^{1/3} : 1$$

48. (c) Let side of square be x

$\therefore$ Radius of sphere = x

Surface area of sphere, $A = 4\pi x^2$

Since, square revolves round a side to generate a cylinder whose height and radius are x and x , respectively.

$$\therefore S = 2\pi x(x+x) = 4\pi x^2$$

So, it is clear that, $A = S$.

49. (a) Given, length of pool = 24 m

and breadth of pool = 15 m

Rise in height of water

$$= 1 \text{ cm} = 0.01 \text{ m}$$

Volume of water displaced by x men

$$= 24 \times 15 \times 0.01 = 3.6 \text{ m}^3$$

But volume of water displaced by x men

$$= 0.1 x \text{ m}^3$$

$$\therefore 0.1 x = 3.6 \Rightarrow x = \frac{3.6}{0.1} = 36$$

50. (b) Given, length of one compartment = 200 m

and breadth of one compartment

$$= 100 \text{ m}$$

and height of water in one compartment

$$= 12 \text{ m}$$

Volume of one compartment

$$= 200 \times 100 \times 12$$

$$= 240000 \text{ m}^3$$

$\therefore$ Volume of 3 compartments

$$= 3 \times 240000 = 720000 \text{ m}^3$$

$$= 720000000 \text{ L}$$

$$[\because 1 \text{ m}^3 = 1000 \text{ L}]$$

Total requirement of water 50000

inhabitants in 1 day

$$= 50000 \times 20 = 1000000 \text{ L}$$

$\therefore$ Required number of days

$$= \frac{720000000}{1000000} = 720 \text{ days}$$

51. (b) Let r and h be the radius and height of the cone and r and H be the radius and height of the cylinder.

$$\therefore \text{Volume of cone} = \frac{1}{3} \pi r^2 h$$

and volume of cylinder = $\pi r^2 H$

$$\text{By given condition, } \frac{1}{3} \pi r^2 h = \pi r^2 H$$

$$\Rightarrow h = 3H \dots(i)$$

Lateral surface area of cone = $\pi r l$

and lateral surface area of cylinder

$$= 2\pi r H$$

$$\text{By given condition, } \pi r l = \frac{15}{8} \times 2\pi r H$$

$$\Rightarrow l = \frac{15}{4} H \Rightarrow l^2 = \frac{225}{16} H^2$$

$$\Rightarrow r^2 + b^2 = \frac{225}{16} H^2 [\because r^2 + b^2 = l^2]$$

$$\Rightarrow r^2 + 9H^2 = \frac{225}{16} H^2 \text{ [from Eq. (i)]}$$

$$\Rightarrow r^2 = \frac{81}{16} H^2 \Rightarrow \frac{r^2}{H^2} = \frac{81}{16}$$

$$\Rightarrow \frac{r}{H} = \frac{9}{4}$$

$$\therefore r : H = 9 : 4$$

52. (b) Let the sides of the cuboid be x , $2x$ and $4x$ and the side of the cube is y .

$$\therefore \text{Volume of cuboid} = x \times 2x \times 4x = 8x^3$$

and volume of cube = y^3

By given condition,

Volume of cuboid = Volume of cube

$$\therefore 8x^3 = y^3 \Rightarrow y = 2x \dots(i)$$

$\therefore$ Diagonal of cuboid

$$= \sqrt{x^2 + 4x^2 + 16x^2} = \sqrt{21}x$$

and diagonal of cube = $y\sqrt{3} = 2x\sqrt{3}$

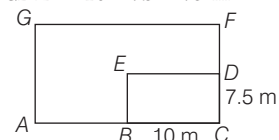
[from Eq. (i)]

Hence, the required ratio

$$= \frac{\sqrt{21}x}{2x\sqrt{3}} = \sqrt{\frac{21}{4 \times 3}} = \sqrt{1.75}$$

53. (c) Area of tank,

$$BCDE = 10 \times 7.5 = 75 \text{ m}^2$$



Area of remaining field

$$ABEDFGA = 125 \times 15 - 75 = 1800 \text{ m}^2$$

$$\text{Volume of Earth dug} = 10 \times 7.5 \times 6 = 450 \text{ m}^3$$

By given condition, $1800 \times b = 450$

$$\Rightarrow b = \frac{1}{4} \text{ m} = \frac{1}{4} \times 100 \text{ cm} = 25 \text{ cm}$$

54. (b) Let $r_1 = k$ and $r_2 = nk$

Since, $V_1 = V_2$

$$\therefore \frac{1}{3} \pi r_1^2 h_1 = \frac{1}{3} \pi r_2^2 h_2$$

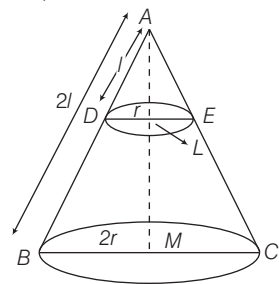
$$\Rightarrow \frac{1}{3} \pi k^2 \times h_1 = \frac{1}{3} \pi n^2 k^2 h_2$$

$$\Rightarrow h_1 = 3n^2 h_2$$

55. (d) Let r be the radius, h the height and l the slant height of smaller cone

$$\text{Given, } \frac{\text{slant height of original cone}}{\text{slant height of smaller cone}} = \frac{2}{1}$$

Since, $\triangle ABM$ and $\triangle ADL$ are similar.



Therefore, radius and height of original cone will be $2r$ and $2h$ respectively.

$$\therefore \frac{V_1}{V_2} = \frac{\frac{1}{3} \pi (2r)^2 \times 2h}{\frac{1}{3} \pi r^2 h} = \frac{8r^2 h}{r^2 h} = 8$$

$$\therefore V_1 : V_2 = 8 : 1$$

56. (d) By given condition,

Surface area of new sphere = 25% of 616

$$4\pi r^2 = 154$$

$$\Rightarrow r^2 = \frac{154}{22 \times 4} = \left(\frac{7}{2}\right)^2 \Rightarrow r = 3.5 \text{ cm}$$

57. (d) Let side of a cube be 'a' unit.

Then, radius of sphere is $\frac{a}{2}$ unit.

$$\therefore \frac{\text{Volume of cube}}{\text{Volume of sphere}} = \frac{(a)^3}{\frac{4\pi}{3} \left(\frac{a}{2}\right)^3} = \frac{6}{\pi}$$

58. (d) Total surface area = Curved surface area of cylinder + Curved surface area of cone + Top surface area of cylinder
 $= 2\pi rh + \pi rl + \pi r^2 = \pi [2rh + r^2 + rl]$
 $= \pi [2 \times 3 \times 4 + 3^2 + 3\sqrt{3^2 + 4^2}]$
 $= 48\pi \text{ cm}^2$

59. (b) Curved surface of a cylinder
 $= 1000 \text{ cm}^2, 2\pi rh = 1000 \dots(i)$

Length of wire used in one round

$$= \text{Perimeter of cylinder's base} = 2\pi r$$

$\therefore$ Number of rounds

$$= \frac{\text{Height of cylinder}}{\text{Diameter of wire}} = \frac{h}{0.5}$$

$\therefore$ Required length of wire

$$= 2\pi r \times \frac{h}{0.5} = \frac{2\pi rh}{0.5}$$

$$= \frac{1000}{0.5} = 2000 \text{ cm or } 20 \text{ m}$$

60. (d) Let h be the depth of the pit.

$\therefore$ Volume of Earth dug

$$= 500 \times 250 \text{ cm}^3 = 125000 \text{ cm}^3$$

But volume of pit = $50 \times 50 \times h$

$$\therefore h = \frac{125000}{50 \times 50} = 50 \text{ cm} = 0.5 \text{ m}$$

61. (a) Since, sheet is revolved about its length, therefore a cylinder is formed with
 $h = 7 \text{ cm}$ and $r = 4 \text{ cm}$

$\therefore$ Volume of the figure, thus formed

$$= \pi r^2 h = \frac{22}{7} \times 4 \times 4 \times 7 = 352 \text{ cm}^3$$

62. (c) Let l, b and h be the sides of cuboid.

then, $l^2 + b^2 = x^2 \dots(i)$

$$b^2 + h^2 = y^2 \dots(ii)$$

and $h^2 + l^2 = z^2 \dots(iii)$

Adding Eq. (i), (ii) and (iii), we get

$$2(l^2 + b^2 + h^2) = x^2 + y^2 + z^2$$

$$\Rightarrow l^2 + b^2 + h^2 = \frac{1}{2}(x^2 + y^2 + z^2)$$

$\dots(iv)$

From Eqs. (i), (ii), (iii) and (iv)

$$b = \sqrt{\frac{y^2 + z^2 - x^2}{2}}, l = \sqrt{\frac{z^2 + x^2 - y^2}{2}}$$

and

$$h = \sqrt{\frac{x^2 + y^2 - z^2}{2}}$$

Hence, volume of cuboid = lbh

$$= \sqrt{\frac{(y^2 + z^2 - x^2)(z^2 + x^2 - y^2)(x^2 + y^2 - z^2)}{2 \times 2 \times 2}}$$

$$= \frac{1}{2\sqrt{2}} \sqrt{(y^2 + z^2 - x^2)(z^2 + x^2 - y^2)(x^2 + y^2 - z^2)}$$

63. (a) Let the radius of cone and sphere be r and height of cone be h_1 .

By given condition,

Volume of cone = Volume of sphere

$$\therefore \frac{1}{3} \pi r^2 h_1 = \frac{4}{3} \pi (r)^3 \Rightarrow \frac{h_1}{2r} = \frac{2}{1}$$

or, $2 : 1$

64. (d) Let the length, breadth and height of a rectangular parallelepiped be $6x, 3x$ and x , respectively.

Also, let the side of a cube be a .

By given condition,

Surface area of a cube = Surface area of rectangular parallelepiped

$$\Rightarrow 6(a)^2 = 2(6x \times 3x + 3x \times x + x \times 6x)$$

$$\Rightarrow 6a^2 = 2(18x^2 + 3x^2 + 6x^2)$$

$$\Rightarrow 6a^2 = 54x^2$$

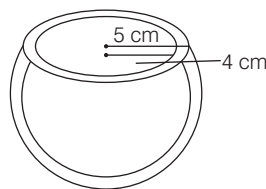
$$\therefore a = 3x$$

Now,

$$\frac{\text{Volume of cube}}{\text{Volume of rectangular parallelepiped}} = \frac{a^3}{6x \times 3x \times x} = \frac{(3x)^3}{18x^3} = \frac{27x^3}{18x^3} = \frac{3}{2}$$

or, $3 : 2$

65. (b) Given thickness of a metal = 1 cm



$\therefore$ Inner radius

= outer radius - thickness

$$= 5 - 1 = 4 \text{ cm}$$

$$\text{Volume of hemisphere} = \frac{2}{3} \pi (R^3 - r^3)$$

$$= \frac{2}{3} \pi (5^3 - 4^3) = \frac{2}{3} \pi (125 - 64)$$

$$= \frac{2}{3} \pi \times 61 \text{ cm}^3$$

$\therefore$ Weight of 1 cm³ of metal = 9 g

$\therefore$ Weight of all metal

$$= \frac{2}{3} \pi \times 61 \times 9 \text{ g} = 366 \pi \text{ g}$$

66. (b) Let the radius of ball be r .

$\therefore$ Radius of base of cylinder = $4r$

and height of cylinder = $4r$

$$\therefore \text{Volume of spherical ball} = \frac{4}{3} \pi r^3$$

and volume of water

$$= \frac{1}{2} \pi (4r)^2 (4r) = 32 \pi r^3$$

Also, volume of remaining portion of cylinder = $32 \pi r^3$

Let number of spherical balls be n .

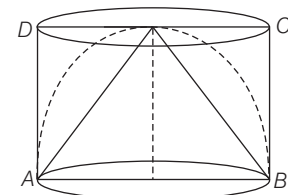
$$\therefore 32 \pi r^3 = n \times \frac{4}{3} \pi r^3$$

$$\Rightarrow n = 8 \times 3 = 24$$

67. (b) From the information given in the question and the figure it is clear that

Radius of the hemisphere = radius of cone = height of cone = height of cylinder. Let it be r .

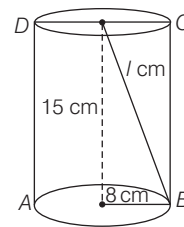
Then, ratio of volume of cylinder, hemisphere and cone.



$$= \pi r^3 : \frac{2}{3} \pi r^3 : \frac{1}{3} \pi r^3$$

$$= 1 : \frac{2}{3} : \frac{1}{3} = 3 : 2 : 1$$

68. (a) $l = \sqrt{15^2 + 8^2} = 17 \text{ cm}$



Total surface area of the remaining solid

$$= 2\pi rh + \pi r^2 + \pi rl$$

$$= 2\pi \times 8 \times 15 + \pi(8)^2 + \pi \times 8 \times 17$$

$$= 240\pi + 64\pi + 136\pi$$

$$= 440 \pi \text{ cm}^2$$

69. (c) Volume of spherical lead shot

$$= \frac{4}{3} \pi (1)^3 = \frac{4}{3} \pi \text{ cm}^3$$

I. Volume of 8 shots

$$= \frac{4}{3} \pi (0.5)^3 \times 8$$

$$= \frac{4}{3} \pi \text{ cm}^3$$

II. Volume of both shots

$$\begin{aligned}
 &= \frac{4}{3} \pi (0.75)^3 + \frac{4}{3} \pi (0.8)^3 \\
 &= \frac{4}{3} \pi \left[\left(\frac{3}{4}\right)^3 + \left(\frac{4}{5}\right)^3 \right] \\
 &= \frac{4}{3} \pi \left(\frac{27}{64} + \frac{64}{125} \right) \\
 &= \frac{4}{3} \pi \left(\frac{3375 + 4096}{8000} \right) \\
 &= \frac{4}{3} \pi \left(\frac{7471}{8000} \right) = \frac{4}{3} \pi (0.93) \text{ cm}^3
 \end{aligned}$$

Hence, statement I, II are true.

70. (c) I. Coterminous edges are those who have same boundaries and for a cuboid, these may be considered as length, breadth and height.

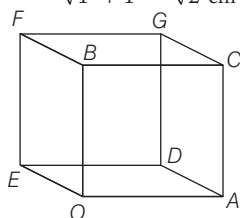
So, given statement is true

II. The surface area of a cuboid is twice the sum of the products of lengths of its coterminous edges taken two at a time.

Hence, both statements I and II are correct.

71. (d) I. The distance between vertices B and C is 1 cm.

II. The distance between A and B is $\sqrt{1^2 + 1^2} = \sqrt{2}$ cm



III. The distance between vertices B

and D is $\sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$ cm

Hence, the statements I and II, III are correct.

72. (d) I. It is true that the surface area of all the eight drops is greater than the surface area of big drop.

II. Let radius of spherical body be 'a'.

$$\therefore \text{Volume of sphere, } V = \frac{4}{3} \pi a^3$$

$$\therefore \text{Surface area of sphere, } S = 4 \pi a^2$$

$$\Rightarrow V^2 = \frac{16}{9} \pi^2 a^6 \propto S^3$$

III. It is true that sphere occupies biggest space but has smallest surface area. Hence, all statements are true.

73. (c) I. It is a correct statement.

II. The length may be πr but the breadth is always less than

$\sqrt{b^2 + r^2}$ of formed rectangle,

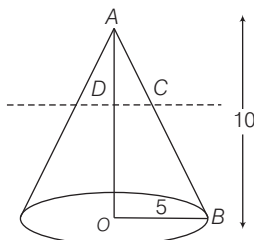
So, statement II is false.

Statement III and IV are correct.

Solution. (74-76)

As cone is divided into two parts by a plane through mid-point of its axis, therefore it will bisect the slant height.

$\triangle AOB \sim \triangle ADC$



$$\therefore \frac{DC}{OB} = \frac{AD}{AO} = \frac{1}{2} \Rightarrow DC = \frac{OB}{2}$$

$\therefore$ Radius of smaller cone will be half of original cone

$$\begin{aligned}
 74. (c) \text{ The volume of the original cone} \\
 &= \frac{1}{3} \times \pi \times (5)^2 \times 10 = \frac{250\pi}{3}
 \end{aligned}$$

The volume of the frustum.

$$\begin{aligned}
 &= \frac{250\pi}{3} - \frac{1}{3} \times \pi \left(\frac{5}{2}\right)^2 \times 5 \\
 &= \frac{250\pi}{3} - \frac{125\pi}{12} = \frac{875\pi}{12}
 \end{aligned}$$

Thus, dividing the two volumes, we get $\frac{8}{7}$

$$\begin{aligned}
 75. (c) \text{ Area of the top circle of the frustum} \\
 &= \pi \left(\frac{5}{2}\right)^2 = \frac{25\pi}{4}
 \end{aligned}$$

76. (d) Let l be slant height of original cone. Curved surface area of cone

$$= \pi \times 5 \times l = 5\pi l$$

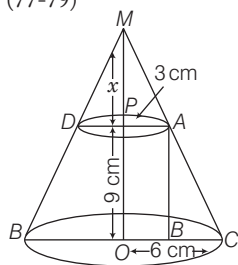
Curved surface area of frustum

$$= 5\pi l - \pi \times \frac{5}{2} \times \frac{l}{2} = \frac{15}{4} \pi l$$

Thus, the ratio of the two will be $\frac{4}{3}$

or 4 : 3.

Solution. (77-79)



Here, $r_1 = 3$ cm, $r_2 = 6$ cm and $h = 9$ cm

$$\begin{aligned}
 77. (a) \text{ In } \triangle ABC, AC^2 &= AB^2 + BC^2 \\
 AC^2 &= 9^2 + (6-3)^2 = 81 + 9 = 90 \\
 \therefore AC &= 3\sqrt{10} \text{ cm}
 \end{aligned}$$

78. (d) By using properties of similar triangle in $\triangle MPA$ and $\triangle MOC$.

$$\frac{3}{6} = \frac{x}{x+9} \Rightarrow 3x + 27 = 6x$$

$$\Rightarrow 27 = 3x$$

$$\Rightarrow x = 9 \text{ cm}$$

$$\begin{aligned}
 \text{Hence, height of cone} &= 9 + 9 \\
 &= 18 \text{ cm}
 \end{aligned}$$

79. (b) Total surface area of the frustum

$$= \pi[R+r]l + \pi R^2$$

$$\text{where, } l = \sqrt{h^2 + (R-r)^2}$$

$$= \pi[(6+3)\sqrt{81+9} + 9 + 36]$$

$$= \pi(9\sqrt{90} + 45)$$

$$= 9\pi(3\sqrt{10} + 5) \text{ cm}^2$$

80. (c) Curved surface area of the well

$$= 2\pi rh = 2 \times \frac{22}{7} \times 2 \times 14 = 176 \text{ m}^2$$

$\therefore$ Expense of getting per square metre plastered = ₹ 25

$$\therefore \text{Expense of } 176 \text{ m}^2 = 176 \times 25 = ₹ 4400$$

81. (a) Given that,

Radius of sphere (r) = 9 cm = 0.09 m

and diameter of wire (d) = 0.4 cm

$$\Rightarrow R = 0.2 \text{ cm} = 0.002 \text{ m}$$

Given condition,

Volume of sphere = Volume of wire

$$\Rightarrow \frac{4}{3} \pi r^3 = \pi R^2 h \Rightarrow h = \frac{4}{3} \cdot \frac{r^3}{R^2}$$

$$= \frac{4}{3} \times \frac{0.09 \times 0.09 \times 0.09}{0.002 \times 0.002} = 81 \times = 243 \text{ m}$$

82. (c) Total surface area of cube

$$= 6 \times (\text{Side})^2$$

$$\therefore 150 = 6 \times (\text{Side})^2$$

$$\Rightarrow \text{Side}^2 = \frac{150}{6} = 25$$

$$\therefore \text{Side} = \sqrt{25} = 5 \text{ cm}$$

$$\begin{aligned}
 \therefore \text{Volume of cube} &= (\text{Side})^3 \\
 &= 5 \times 5 \times 5 = 125 \text{ cm}^3
 \end{aligned}$$

83. (b) Volume of cube = $(\text{Side})^3$

$$\therefore 729 = a^3 \Rightarrow a = 9 \text{ cm}$$

$$\begin{aligned}
 \therefore \text{Diagonal of cube} &= \text{Side} \times \sqrt{3} \\
 &= 9 \times \sqrt{3} = 9\sqrt{3} \text{ cm}
 \end{aligned}$$

84. (a) Curved surface area of right circular cone = $\pi r l$

$$\therefore 440 = \frac{22}{7} \times 14 \times l \Rightarrow l = \frac{440 \times 7}{22 \times 14}$$

$$= 10 \text{ cm}$$

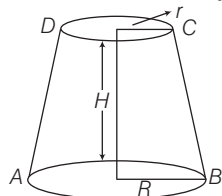
85. (d) Volume of cylinder = $\pi r_1^2 h$
 Volume of sphere = $\frac{4}{3} \pi r_2^3$
 Number of spheres = 48
 $\therefore \frac{\text{Volume of cylinder}}{\text{Volume of sphere}} = \frac{\pi r_1^2 h}{\frac{4}{3} \pi r_2^3}$
 $\Rightarrow \frac{\pi r_1^2 h}{\frac{4}{3} \pi r_2^3} = 48 \Rightarrow \frac{\pi r_1^3}{\frac{4}{3} \pi r_2^3} = 48$
 $\Rightarrow \frac{3}{4} \left(\frac{r_1}{r_2} \right)^3 = 48 \Rightarrow \left(\frac{r_1}{r_2} \right)^3 = \frac{48 \times 4}{3}$
 $\Rightarrow \frac{r_1}{r_2} = (64)^{1/3} \Rightarrow \frac{r_1}{r_2} = \frac{4}{1} \Rightarrow \frac{r_2}{r_1} = \frac{1}{4}$

Hence, the ratio of radius of ball to cylinder is 1:4.

86. (a) Volume of the cuboid = 720 cm^3
 Area of base = $lb = 72$
 Height of the cuboid
 $= \frac{\text{Volume of the cuboid}}{\text{Base area of the cuboid}}$
 $= \frac{720}{72} = 10 \text{ cm}$
 Surface area of the cuboid
 $= 2(lb + bh + hl)$
 $\Rightarrow 484 = 2(72 + b \times 10 + l \times 10)$
 $\Rightarrow 20(l + b) = 340 \Rightarrow l + b = 17$
 So, it is obvious that length, breadth and height of the cuboid is 9 cm, 8 cm and 10 cm.

87. (a) Surface area of the sphere
 $= 4\pi r^2$
 $\Rightarrow 616 = 4\pi r^2 \Rightarrow \pi r^2 = \frac{616}{4} = 154$
 $\Rightarrow r^2 = \frac{154 \times 7}{22} = 49$
 $\therefore r = \sqrt{49} = 7 \text{ cm}$
 $\therefore \text{Volume of the sphere} = \frac{4}{3} \pi r^3$
 $= \frac{4}{3} \times \frac{22}{7} \times 7 \times 7 \times 7 = \frac{4312}{3} \text{ cm}^3$

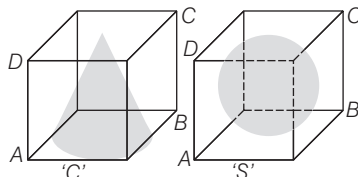
88. (a) Given that,
 Area of first end = $P = \pi r^2$
 and area of second end = $Q = \pi R^2$



Given, $P < Q \Rightarrow r = \sqrt{\frac{P}{\pi}}$ and $R = \sqrt{\frac{Q}{\pi}}$
 $\therefore$ Difference in radii of the ends of the frustum = $R - r$
 $= \sqrt{\frac{Q}{\pi}} - \sqrt{\frac{P}{\pi}} = \frac{\sqrt{Q} - \sqrt{P}}{\sqrt{\pi}}$

89. (c) We know that,
 Volume of frustum
 $= \frac{\pi H}{3} (R^2 + r^2 + Rr)$
 $= \frac{\pi H}{3} \left\{ \left(\sqrt{\frac{Q}{\pi}} \right)^2 + \left(\sqrt{\frac{P}{\pi}} \right)^2 + \sqrt{\frac{Q}{\pi}} \cdot \sqrt{\frac{P}{\pi}} \right\}$
 $= \frac{\pi H}{3} \left\{ \frac{Q}{\pi} + \frac{P}{\pi} + \frac{\sqrt{PQ}}{\pi} \right\}$
 $= \frac{H}{3} (P + Q + \sqrt{PQ})$

90. (b) Let the side of both cubes be a ,
 then the height and radius of a cone,
 $h = a$ and $r = \frac{a}{2}$



and radius of sphere (R) = $\frac{a}{2}$

$\therefore$ Volume of cone (C) = $\frac{1}{3} \pi r^2 h$
 $= \frac{1}{3} \pi \left(\frac{a}{2} \right)^2 a \Rightarrow C = \frac{\pi a^3}{12} \dots (i)$
 and volume of sphere (S) = $\frac{4}{3} \pi R^3$
 $= \frac{4}{3} \pi \left(\frac{a}{2} \right)^3 = \frac{\pi a^3}{6} \dots (ii)$

From Eqs. (i) and (ii), $S = 2C$

91. (d) If 10 circular plates each of thickness 3 cm, are placed one above the other, then it forms a cylinder with height ($3 \times 10 = 30$ cm) and a hemisphere of radius 6 cm is placed on the top just to cover the cylinder that means its radius is also 6 cm.

$\therefore$ Radius of hemi-sphere (R) = 6 cm
 Radius of cylinder (r) = 6 cm
 and height of cylinder (h) = 30 cm
 $\therefore$ Volume of the solid = Volume of cylinder + Volume of hemisphere
 $= \pi r^2 h + \frac{2}{3} \pi R^3$
 $= \pi (6)^2 \times 30 + \frac{2}{3} \pi (6)^3$
 $= \pi \times 36 \times 30 + \frac{2}{3} \pi \times 216$

$= 1080\pi + 2\pi \times 72$
 $= 1080\pi + 144\pi = 1224\pi \text{ cm}^3$

92. (a) Given that, the height and radius of a right circular metal cone (solid) are 8 cm and 2 cm, respectively.

i.e. $h = 8$ cm and $r = 2$ cm

Let the radius of the sphere be R .

Then, by condition, $\frac{1}{3} \pi r^2 h = \frac{4}{3} \pi R^3$
 $\Rightarrow 4 \times 8 = 4R^3 \Rightarrow R^3 = (2)^3$
 $\therefore R = 2$

Hence, radius of the sphere = 2 cm

93. (c) Let the edge of a square be x . Then, its volume is x^3 and sum of its edges is $12x$.
 Now, by condition, $x^3 = 12x$

$\Rightarrow x(x^2 - 12) = 0$

$\Rightarrow x^2 = 12 \quad [\because x \neq 0]$

$\therefore$ Its total surface area = $6x^2$
 $= 6(12) = 72 \text{ sq units}$

94. (a) Given that,
 Diameter of a right circular cone = 7 cm
 $\therefore$ Radius of a right circular cone = $\frac{7}{2}$ cm

and slant height of a right circular cone (l) = 10 cm

$\therefore$ Lateral surface area of a cone = $\pi r l$
 $= \frac{22}{7} \times \frac{7}{2} \times 10 = 11 \times 10 = 110 \text{ cm}^2$

95. (d) Let the height and radius of solid cylinder be h and r cm, respectively.

Given that, $r = 5$ cm

and total surface area = 660 cm^2

$\Rightarrow 2\pi r h + 2\pi r^2 = 660$

$\Rightarrow 2\pi r(h + r) = 660$

$\Rightarrow (h + 5) = \frac{330}{5\pi} = \frac{330}{5} \times \frac{7}{22}$

$\Rightarrow h = \frac{66 \times 7}{22} - 5 = 21 - 5 = 16$

$\therefore$ Required height = 16 cm

96. (a) Let the diameters of two spheres be

d_1 and d_2 , respectively.

$\therefore d_1 : d_2 = 3 : 5$

$\therefore$ Ratio of their surface areas

$= \frac{4\pi r_1^2}{4\pi r_2^2} = \frac{(2r_1)^2}{(2r_2)^2} = \frac{d_1^2}{d_2^2}$

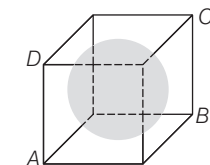
$= \left(\frac{d_1}{d_2} \right)^2 = \left(\frac{3}{5} \right)^2 = \frac{9}{25} = 9 : 25$

97. (c) From figure, it is clear that

Diameter of a sphere = Side of the cube
 $= 3 \text{ cm}$

$\therefore$ Radius = $\frac{3}{2} \text{ cm}$

∴ Volume of the largest sphere



$$= \frac{4}{3} \pi (\text{Radius})^3 = \frac{4}{3} \pi \left(\frac{3}{2}\right)^3$$

$$= \frac{4}{3} \pi \cdot \frac{27}{8} = \frac{9}{2} \pi = 4.5 \pi \text{ cm}^3$$

98. (d) Given that, radius of a right circular cone (r) = 12 m

Height of a right circular cone (h) = 5 m

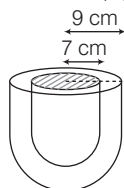
∴ Lateral height of cone (l) = $\sqrt{r^2 + h^2}$

$$= \sqrt{144 + 25} = \sqrt{169} = 13 \text{ m}$$

∴ Required quantity of cloth to roll up to form a right circular tent = $\pi r l$

$$= \pi \times (12) \times (13) = 156 \pi \text{ m}^2$$

99. (a) Given that, Outer radius of hemispherical shell (R) = 9 cm



and inner radius of hemispherical shell (r) = 7 cm

∴ Volume of a hemispherical shell

$$= \frac{2}{3} \pi (R^3 - r^3)$$

$$= \frac{2}{3} \times \frac{22}{7} \times (729 - 343)$$

$$= \frac{2}{3} \times \frac{22}{7} \times 386 = \frac{16984}{21} = 808.76$$

$$\approx 808 \text{ cm}^3 \text{ (approx)}$$

100. (b) Let r be the radius of the sphere.

Given that, volume of sphere = 36π

$$\Rightarrow \frac{4}{3} \pi r^3 = 36 \pi \Rightarrow r^3 = 27 = (3)^3$$

$$\therefore r = 3 \text{ cm}$$

∴ Diameter of sphere

$$= 2r = 2(3) = 6 \text{ cm}$$

and surface area of sphere = $4\pi r^2$

$$= 4\pi (3)^2 = 36 \pi \text{ cm}^2$$

∴ Required ratio

$$= \frac{\text{Surface area of sphere}}{\text{Diameter of sphere}} = \frac{36 \pi}{6} = 6 \pi$$

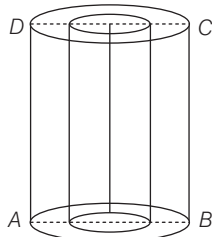
101. (d) Given that,

Internal diameter of the tube = 6 cm

∴ Internal radius (r) = 3 cm

Height of the tube (h) = 10 cm

Thickness of the metal = 1 cm



∴ Outer radius (R)

$$= \text{Thickness of the metal} + \text{Internal radius}$$

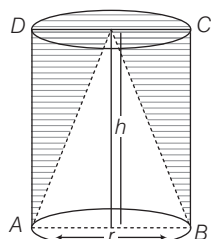
$$= 1 + 3 = 4 \text{ cm}$$

∴ Outer curved surface area

$$= 2\pi r h = 2 \times \pi \times 4 \times 10 = 80 \pi$$

102. (a) Let the height and radius of right circular cylinder be h and r , respectively.

Then, volume of cylinder = $\pi r^2 h$



$$\text{Volume of circular cone} = \frac{1}{3} \pi r^2 h$$

∴ Required ratio

$$= \frac{\text{Volume of utilised wood}}{\text{Volume of wasted wood}}$$

$$= \frac{\text{Volume of right circular cone}}{\text{Volume of right circular cylinder} - \text{Volume of right circular cone}}$$

$$= \frac{\frac{1}{3} \pi r^2 h}{\pi r^2 h - \frac{1}{3} \pi r^2 h} = \frac{\frac{1}{3} \pi r^2 h}{\frac{2}{3} \pi r^2 h} = \frac{1}{2} = 1:2$$

103. (c) Since, volume of cone and pyramid = $\frac{1}{3} \times \text{Base area} \times \text{Height}$

Therefore, they have same volume but their surface areas are not same as nothing can be said about their slant heights.

104. (b) Given that, height of cone (h) = 3 cm and slant height of cone (l) = 5 cm

Let r be the radius of cone. Then,

$$l = \sqrt{r^2 + h^2} = 5 \Rightarrow r^2 + h^2 = 25$$

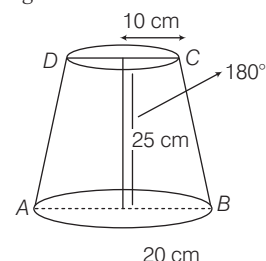
$$\Rightarrow r^2 = 25 - 9 = 16 \Rightarrow r = 4 \text{ cm}$$

$$\therefore \text{Volume of cone} = \frac{1}{3} \pi r^2 h$$

$$= \frac{1}{3} \times \frac{22}{7} \times 16 \times 3 = \frac{352}{7} = 50.3 \text{ cm}^3$$

105. (a) Given that,

Height of bucket = 25 cm



Radius of top (R) = 20 cm

and radius of bottom (r) = 10 cm

∴ Capacity of bucket

$$= \frac{\pi}{3} h (R^2 + r^2 + Rr)$$

$$= \frac{\pi}{3} \times 25 (400 + 100 + 200) \text{ cm}^3$$

$$= \frac{\pi}{3} \times 25 \times 700 \text{ cm}^3$$

$$= \frac{\pi}{3} \times \frac{175 \times 100}{1000} \text{ L} = \frac{17.5 \pi}{3} \text{ L}$$

106. (d) Given that, $h = 15$ cm

Let r be the radius of cylinder.

Lateral surface area of cylinder = $2\pi r h$

$$\Rightarrow 2\pi r h = 660 \quad [\text{given}]$$

$$\Rightarrow \pi r h = 330$$

$$\Rightarrow \frac{22}{7} \times r \times 15 = 330 \Rightarrow \frac{22}{7} \times r = 22$$

$$\therefore r = 7 \text{ cm}$$

∴ Volume of cylinder = $\pi r^2 h$

$$= \frac{22}{7} \times 49 \times 15$$

$$= 22 \times 7 \times 15 = 2310 \text{ cm}^3$$

107. (b) Given that, the diameter of Moon is approximately one-fourth of the diameter of Earth.

If radius of Moon = r

Then, radius of Earth = $4r$

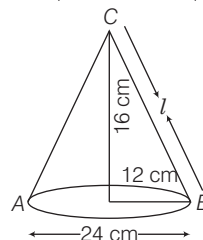
$$\therefore \frac{\text{Volume of Moon}}{\text{Volume of Earth}} = \frac{\frac{4}{3} \pi r^3}{\frac{4}{3} \pi (4r)^3}$$

$$= \frac{r^3}{64 r^3} = \frac{1}{64} = 1:64$$

108. (c) Slant height, $l = \sqrt{b^2 + r^2}$

$$= \sqrt{16^2 + 12^2}$$

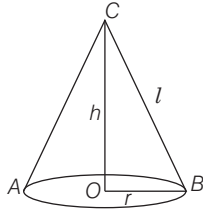
$$= \sqrt{256 + 144} = \sqrt{400} = 20 \text{ cm}$$



$$\begin{aligned}\text{Curved surface area} &= \pi r l \\ &= \frac{22}{7} \times 12 \times 20 \text{ cm}^2\end{aligned}$$

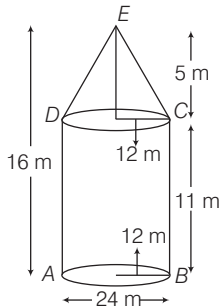
$$\begin{aligned}\text{Cost of painting the surface of the cap} &= \frac{22}{7} \times 12 \times 20 \times 0.70 = ₹ 528\end{aligned}$$

109. (b) Slant height, $l = \sqrt{h^2 + r^2}$
 $= \sqrt{(24)^2 + (7)^2} = \sqrt{576 + 49}$
 $= \sqrt{625} = 25$



$$\begin{aligned}\text{Total surface area} &= \pi r (l + r) \\ &= \frac{22}{7} \times 7 (25 + 7) \\ &= \frac{22}{7} \times 7 \times 32 = 704 \text{ cm}^2\end{aligned}$$

110. (c) Radius of cone, $r = \frac{24}{2} = 12 \text{ cm}$
 Slant height of the cone, $l = \sqrt{5^2 + 12^2}$
 $= \sqrt{25 + 144} = \sqrt{169} = 13 \text{ m}$



$$\begin{aligned}\text{Curved surface area for conical portion} &= \pi r l = \frac{22}{7} \times 12 \times 13 = \frac{3432}{7} \text{ m}^2\end{aligned}$$

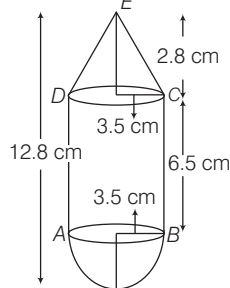
111. (d) Curved surface area of cylinder
 $= 2\pi r h = x$
 Volume of cylinder $= \pi r^2 h = y$
 $\Rightarrow \frac{2\pi r h}{\pi r^2 h} = \frac{x}{y} \Rightarrow r = \frac{2y}{x}$
 Also, $h = \frac{x}{2\pi r}$

$$\begin{aligned}\therefore \text{Required ratio} &= \frac{h}{r} = \frac{\frac{x}{2\pi r}}{\frac{2y}{x}} \\ &= \frac{x}{2\pi \cdot \frac{2y}{x}} \times \frac{x}{2y} = \frac{x^3}{8\pi y^2}\end{aligned}$$

So, the ratio is not independent of x or y .

112. (a) Let common radius be $r \text{ cm}$.
 Then, height of cylinder $= h$

and height of cone $= h'$
 $\therefore$ Volume of the complete structure
 $= \frac{1}{3} \pi r^2 h' + \pi r^2 h + \frac{2}{3} \pi r^3$
 $= \pi r^2 \left(\frac{h'}{3} + h + \frac{2}{3} r \right)$
 $= \pi (3.5)^2 \left(\frac{2.8}{3} + 6.5 + \frac{2}{3} \times 3.5 \right)$
 $= \pi \times 3.5 \times 3.5 \times 9.76 = 375.86 \text{ cm}^3$



Hence, volume (V) of the structure lies between 370 cm^3 and 380 cm^3 .

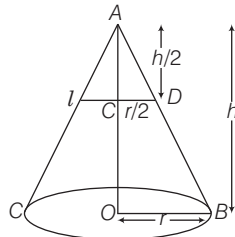
113. (b) Given, radius of cone $= \frac{6}{2} = 3 \text{ cm}$
 and height of cone $= 4 \text{ cm}$
 Now, curved surface area $= \pi r l$
 where, $l = \sqrt{r^2 + h^2} = \sqrt{3^2 + 4^2} = 5 \text{ cm}$
 $\therefore$ Curved surface area $= \pi \times 3 \times 5 = 15\pi$
 $= \frac{15 \times 22}{7} \approx 47 \text{ cm}^2$

114. (d) Volume of clay required
 $= \pi \left[\left(\frac{5.1}{2} \right)^2 - \left(\frac{4.5}{2} \right)^2 \right] \times 21$
 $= \pi [(2.55)^2 - (2.25)^2] \times 21$
 $= \pi (0.3 \times 4.8) \times 21 = 30.24 \pi \text{ cm}^2$

115. (a) Surface area of cube which can be painted $= 6 (\text{Side})^2 = 6(2)^2 = 24 \text{ cm}^2$
 Now, surface area of cuboid which can be painted

$$\begin{aligned}&= 2(lb + bh + lh) \\ &= 2(2 + 6 + 3) = 22 \text{ cm}^2 \\ \text{Total surface area of both cube and cuboid} &= 22 + 24 = 46 \text{ cm}^2 < 54 \text{ cm}^2 \\ \text{Hence, both cube and cuboid can be painted.}\end{aligned}$$

116. (b) Let the cone is divided into two parts by a line l .



In $\triangle AOB$ and $\triangle ACD$, $\triangle AOB \sim \triangle ACD$

By basic proportionality theorem,

$$CD = \frac{r}{2}, \text{ since } AC = \frac{h}{2}$$

$$\begin{aligned}\text{Now, ratio} &= \frac{\text{Volume of original cone}}{\text{Volume of smaller cone}} \\ &= \frac{\frac{1}{3} \pi r^2 h}{\frac{1}{3} \pi \left(\frac{r}{2} \right)^2 \left(\frac{h}{2} \right)} = \frac{8}{1}\end{aligned}$$

$\therefore$ Required ratio $= 8 : 1$

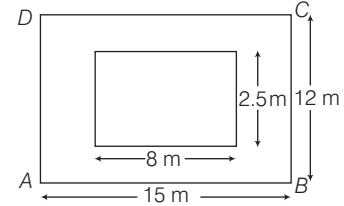
117. (c) Volume of each small sphere
 $= \frac{\text{Volume of bigger sphere}}{\text{Number of small spheres}} = \frac{\frac{4}{3} \pi (4)^3}{64}$
 $= \frac{4}{3} \times \frac{\pi \times 4 \times 4 \times 4}{64} = \frac{4}{3} \pi \text{ cm}^3$

Let radius of small sphere be r' .

$$\therefore \frac{4}{3} \pi r'^3 = \frac{4}{3} \pi \Rightarrow r' = 1 \text{ cm}$$

Now, surface area of small sphere
 $= 4\pi r'^2 = 4\pi \text{ cm}^2$

118. (c) Volume of Earth dug out
 $= 8 \times 2.5 \times 2 = 40 \text{ m}^3$



Area where Earth is spread $\times$ Field level raised

$$\begin{aligned}&= \text{Volume of Earth dug out} \\ \Rightarrow [(12 \times 15) - (8 \times 2.5)] \times h &= 40 \\ \therefore h &= \frac{40}{180 - 20} = \frac{40}{160} = \frac{1}{4} \text{ m} \\ &= \frac{100}{4} \text{ cm} = 25 \text{ cm}\end{aligned}$$

119. (a) Given, surface area of sphere
 $= 616 \text{ cm}^2$

$$\Rightarrow 4\pi r^2 = 616 \Rightarrow r^2 = \frac{616 \times 7}{4 \times 22}$$

$$\Rightarrow r^2 = 7 \times 7 \therefore r = 7 \text{ cm}$$

$\therefore$ Diameter of the largest circle lying on sphere

$$= 2 \times r = 14 \text{ cm}$$

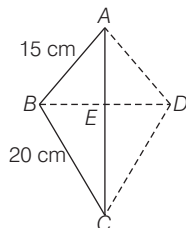
120. (c) Given, volume of cube $= 216x^3$
 $(\text{Side})^3 = 216x^3 \Rightarrow \text{Side} = 6x$

Since, sphere is enclosed in hollow cube.

$\therefore$ Diameter of sphere $= 6x$

$$\begin{aligned}\text{Now, surface area of sphere} &= 4\pi r^2 \\ &= 4\pi \left(\frac{6x}{2} \right)^2 = 36\pi x^2\end{aligned}$$

121. (c) Let ABC be a right angled triangle.
Then, hypotenuse, $AC = 25$ cm



Let $AB = 3x$ and $BC = 4x$
Thus, $AC^2 = AB^2 + BC^2$
[by pythagoras theorem]
 $\Rightarrow (25)^2 = (3x)^2 + (4x)^2$
 $\Rightarrow (25)^2 = 9x^2 + 16x^2$
 $\Rightarrow 25 = x^2 \Rightarrow x = 5$
 $\therefore AB = 15$ cm and $BC = 20$ cm
Now, $\triangle ABC$ revolves about AC , so it forms two cones ABD and BCD .
Since, $\triangle AEB$ and $\triangle ABC$ are similar.
 $\therefore \frac{BE}{BC} = \frac{AB}{AC} \Rightarrow \frac{BE}{20} = \frac{15}{25}$
 $\Rightarrow BE = \frac{15 \times 20}{25} = 12$ cm

So, radius of the base of cone
 $= BE = 12$ cm
In right angled $\triangle AEB$,
 $AE = \sqrt{(AB)^2 - (BE)^2}$
 $= \sqrt{(15)^2 - (12)^2}$
 $= \sqrt{225 - 144} = \sqrt{81} = 9$ cm
So, height of cone $ABD = AE = 9$ cm
 $\therefore$ Height of cone $BCD = AC - AE$
 $= 25 - 9 = 16$ cm
Now, volume of cone $ABD = \frac{1}{3} \pi r^2 h$
 $= \frac{1}{3} \pi (12)^2 \times 9 = 432\pi$ cm³

and volume of cone
 $BCD = \frac{1}{3} \pi (12)^2 \times 16 = 768\pi$ cm³
 $\therefore$ Required volume of double cone
 $= 432\pi + 768\pi = 1200\pi$
 $= 1200 \times 3.14$ [$\because \pi = 3.14$]
 $= 3768$ cm³

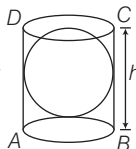
122. (d) $\therefore$ Surface area of cone $ABD = \pi rl$
 $= \pi \times 12 \times 15 = 180\pi$ cm²
and surface area of cone
 $BCD = \pi \times 12 \times 20 = 240\pi$ cm²
 $\therefore$ Required surface area of double cone
 $= 180\pi + 240\pi = 420\pi$
 $= 420 \times 3.14 = 1318.8$ cm²

123. (c) Let side of a cube be x .
Then, surface area of cube $= 6x^2$
If the side of cube is increased by 100%.

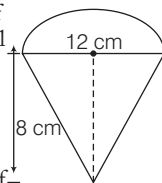
Then, new side of cube $= x + 100\%$ of x
 $= x + x = 2x$
 $\therefore$ New surface area of cube $= 6(2x)^2$
 $= 6 \times 4x^2 = 24x^2$
Now, increase percentage in surface area
 $= \frac{24x^2 - 6x^2}{6x^2} \times 100$
 $= \frac{18}{6} \times 100 = 300\%$

124. (a) Let radius of the sphere be r .

Since, cylinder circumscribes a sphere.
 $\therefore$ Radius of the base of cylinder $= r$ and height of cylinder $= 2r$
 $=$ Diameter of sphere
Now, volume of sphere $= \frac{4}{3} \pi r^3$
and volume of cylinder
 $= \pi r^2 h = \pi r^2 (2r) = 2\pi r^3$
 $\therefore$ Required ratio $= \frac{\frac{4}{3} \pi r^3}{2\pi r^3} = \frac{4}{3 \times 2}$
 $= \frac{2}{3} = 2 : 3$



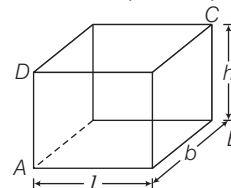
125. (a) Given, diameter of the base of the conical portion $= 12$ cm
 $\therefore$ Radius of conical portion $= 6$ cm
Radius of hemisphere $= 6$ cm and height of conical portion $= 8$ cm
 $\therefore$ Slant height of conical portion
 $= \sqrt{6^2 + 8^2} = \sqrt{36 + 64}$
 $= \sqrt{100} = 10$ cm
[$\because l = \sqrt{r^2 + h^2}$]



Now, total surface area of the toy
 $=$ Curved surface area of conical portion
 $+ \text{Curved surface area of hemisphere}$
 $= \pi rl + 2\pi r^2 = \pi (rl + 2r^2)$
 $= \pi (6 \times 10 + 2 \times 6 \times 6)$
 $= \pi (60 + 72) = 132\pi$ cm²

126. (b) Volume of the toy
 $=$ Volume of conical portion
 $+ \text{Volume of hemisphere}$
 $= \frac{1}{3} \pi r^2 h + \frac{2}{3} \pi r^3$
 $= \pi \left(\frac{1}{3} r^2 h + \frac{2}{3} r^3 \right)$
 $= \pi \left[\frac{1}{3} \times (6)^2 \times 8 + \frac{2}{3} \times (6)^3 \right]$
 $= \pi [96 + 144] = 240\pi$ cm³

127. (b) Let length, breadth and height of a cuboidal box be l , b and h , respectively.



Given, areas of the three adjacent faces are x , $4x$ and $9x$ sq units.

Now, $lb = x$
[$\because$ area of rectangular face $=$ length $\times$ breadth]

Similarly, $bh = 4x$ and $lh = 9x$

Now, $(lb) \cdot (bh) \cdot (lh) = (x) \cdot (4x) \cdot (9x)$

$$\Rightarrow (lbh)^2 = 36x^3 \Rightarrow lbh = \sqrt{36x^3}$$

$$\therefore lbh = 6x^{3/2}$$

Hence, volume of cuboidal box
 $= lbh = 6x^{3/2}$ cu units

128. (c) In a cuboid, 4 perpendicular face pairs in bottom surface, 4 perpendicular face pairs in top surface and 4 perpendicular face pairs in vertical surface.

Hence, total perpendicular pairs are 12.

129. (c) I. Surface area of sphere A
 $= 4\pi(6)^2 = 144\pi$ cm²
Surface area of sphere $B = 4\pi(8)^2$
 $= 256\pi$ cm²
Surface area of sphere $C = 4\pi(10)^2$
 $= 400\pi$ cm²

Now, sum of surface area of spheres A and B

$$= 144\pi + 256\pi = 400\pi$$
 cm²
 $= \text{Surface area of sphere } C$

Hence, statement I is correct.

- II. $\therefore$ Volume of sphere D
 $= \frac{4}{3} \pi (12)^3 = 2304\pi$ cm³

$$\text{Volume of sphere } A = \frac{4}{3} \pi (6)^3 \text{ cm}^3$$

$$\text{Volume of sphere } B = \frac{4}{3} \pi (8)^3 \text{ cm}^3$$

and volume of sphere

$$C = \frac{4}{3} \pi (10)^3 \text{ cm}^3$$

Now, sum of volumes of spheres A , B and C

$$= \frac{4}{3} \pi [6^3 + 8^3 + 10^3]$$

$$= \frac{4}{3} \pi [216 + 512 + 1000]$$

$$= 2304\pi \text{ cm}^3 = \text{Volume of sphere } D$$

Hence, statement II is also correct.

- 130. (a)** Given, diameter of a sphere, $d = 6$ cm
 $\therefore$ Radius of a sphere, $r = \frac{d}{2} = \frac{6}{2} = 3$ cm

Let the radius of wire be R cm.

Also, given the length of wire, $H = 36$ m
 $= 3600$ cm

According to the question,

Volume of sphere = Volume of wire

$$\Rightarrow \frac{4}{3}\pi r^3 = \pi R^2 H$$

$$\Rightarrow \frac{4}{3} \times (3)^3 = R^2 \times 3600$$

$$\Rightarrow R^2 = \frac{4 \times 3^2}{3600} = \frac{(6)^2}{(60)^2}$$

$$\therefore R = \frac{6}{60} = \frac{1}{10} = 0.1 \text{ cm}$$

- 131. (b)** Given, side of a cube = 2 cm
 $\therefore$ Maximum distance between two points of a cube
 $=$ Length of diagonal
 $= \sqrt{3} \times \text{Side} = 2\sqrt{3}$ cm

- 132. (d)** Let the slant height of cone be l cm.

Given, $h = 24$ cm and $r = 7$ cm

$$\therefore l^2 = h^2 + r^2$$

$$\Rightarrow l^2 = 24^2 + 7^2 = 576 + 49$$

$$\Rightarrow l^2 = 625 \Rightarrow l = 25 \text{ cm}$$

Total surface area = Curved surface area of cone + curved surface area of hemisphere

$$= \pi r l + 2\pi r^2 = \pi r (l + 2r)$$

$$= \pi \times 7(25 + 2 \times 7) = 7\pi[25 + 14]$$

$$= 7\pi \times 39 = 273\pi \text{ cm}^2$$

- 133. (a)** Let the radius of the sphere and side of cube be r and a respectively.

$\therefore$ Total surface area of sphere

$$(S_1) = 4\pi r^2 \text{ sq units}$$

and total surface area of cube

$$(S_2) = 6a^2 \text{ sq units}$$

Now, according to the question,

$$S_1 = S_2 \Rightarrow 4\pi r^2 = 6a^2 \Rightarrow \frac{r^2}{a^2} = \frac{6}{4\pi} \quad \dots(i)$$

$$\therefore \text{Volume of sphere } (V_1) = \frac{4}{3}\pi r^3 \text{ cu units}$$

and volume of cube $(V_2) = a^3$ cu units

Now,

$$\left(\frac{V_1}{V_2}\right)^2 = \left(\frac{\frac{4}{3}\pi r^3}{a^3}\right)^2 = \frac{16}{9}\pi^2 \left(\frac{r^2}{a^2}\right)^3$$

$$= \frac{16}{9}\pi^2 \left(\frac{6}{4\pi}\right)^3 \quad [\text{from Eq. (i)}]$$

$$= \frac{16}{9}\pi^2 \times \frac{6 \times 6 \times 6}{4 \times 4 \times 4 \times \pi^3}$$

$$= \frac{6}{\pi} = 6 : \pi$$

Hence, the ratio of square of their volume is $6 : \pi$.

- 134. (a)** Let radius of sphere and cone be r .

$$\therefore \text{Volume of sphere } V_1 = \frac{4}{3}\pi r^3$$

$$\text{and volume of cone } V_2 = \frac{1}{3}\pi r^2 h$$

Now, according to the question,

$$V_1 = 2V_2 \Rightarrow \frac{4}{3}\pi r^3 = 2 \times \frac{1}{3}\pi r^2 h$$

$$\Rightarrow 2r^3 = r^2 h \Rightarrow 2r = h \Rightarrow \frac{h}{r} = \frac{2}{1}$$

$$\therefore h : r = 2 : 1$$

- 135. (a)** Let the radius of sphere be r .

$$\therefore \text{Volume of sphere} = \frac{4}{3}\pi r^3$$

According to the question,

If the radius of a sphere is increased by 10%.

Then, new radius $r' = r + r \times 10\%$

$$= r + \frac{r}{10} = \frac{11r}{10}$$

$$\therefore \text{New volume of sphere} = \frac{4}{3}\pi r'^3$$

$$= \frac{4\pi}{3} \left(\frac{11r}{10}\right)^3 = \frac{4}{3}\pi \times \frac{1331r^3}{1000}$$

Increased volume

$$= \frac{4}{3}\pi \times \frac{1331r^3}{1000} - \frac{4}{3}\pi r^3 = \frac{4}{3}\pi r^3 \times \frac{331}{1000}$$

Increased percentage

$$= \frac{\frac{4}{3}\pi r^3 \times \frac{331}{1000}}{\frac{4}{3}\pi r^3} \times 100\% = 33.1\%$$

- 136. (b)** Since, three metallic spheres are melted to form a single sphere.

So, the sum of volume of three spherical solid sphere.

$$= \frac{4}{3}\pi(6)^3 + \frac{4}{3}\pi(8)^3 + \frac{4}{3}\pi(10)^3$$

$$= \frac{4}{3}\pi[6^3 + 8^3 + 10^3]$$

Let r be the radius of new sphere

$$\therefore \frac{4}{3}\pi r^3 = \frac{4}{3}\pi(6^3 + 8^3 + 10^3)$$

$$\Rightarrow r^3 = 6^3 + 8^3 + 10^3$$

$$\Rightarrow r^3 = 216 + 512 + 1000$$

$$\Rightarrow r^3 = 1728$$

$$\Rightarrow r^3 = 12 \times 12 \times 12$$

$$\Rightarrow r = 12 \text{ cm}$$

$\therefore$ Diameter of the new sphere

$$= 2r = 2 \times 12 = 24 \text{ cm}$$

- 137. (b)** Let the radius and height of the cone be r and h , respectively.

$\therefore$ Initial volume of cone

$$(V) = \frac{1}{3}\pi r^2 h \quad \dots(i)$$

new height, $H = h + h \times 200\%$

$$= h + \frac{200h}{100} = 3h$$

and new radius, $R = r - r \times 50\%$

$$= r - \frac{50r}{100} = \frac{r}{2}$$

$$\therefore \text{New volume of cone } (V_2) = \frac{1}{3}\pi R^2 H$$

$$= \frac{1}{3}\pi \left(\frac{r}{2}\right)^2 \times 3h = \frac{1}{3}\pi \times \frac{r^2}{4} \times 3h$$

$$= \frac{1}{3}\pi r^2 h \times \frac{3}{4} = \frac{3V}{4} \quad [\text{from Eq. (i)}]$$

$$\text{Decrease in volume} = V - \frac{3V}{4} = \frac{V}{4}$$

$\therefore$ Decrease percentage in volume

$$= \frac{\text{Decrease in Volume}}{\text{Initial Volume}} \times 100\%$$

$$= \frac{V/4}{V} \times 100\% = \frac{V}{4V} \times 100\% = 25\%$$

- 138. (c)** Given a rectangular paper of 44 cm long and 6 cm wide is rolled to form a cylinder of height equal to width of the paper.

$\therefore$ Circumference of the base of cylinder
 $= 44$

$$\text{i.e. } 2\pi r = 44 \Rightarrow r = \frac{44}{2\pi} = \frac{44 \times 7}{2 \times 22}$$

$$\therefore r = 7 \text{ cm}$$

Hence, the radius of the base of the cylinder is 7 cm.

- 139. (c)** Let the dimensions of cuboid be l , b and h respectively.

For first face

$$l^2 + b^2 = 13^2 \quad [\because l^2 + b^2 = d^2]$$

$$\Rightarrow l^2 + b^2 = 169 \quad \dots(i)$$

For second face

$$b^2 + h^2 = (\sqrt{281})^2$$

$$\Rightarrow b^2 + h^2 = 281 \quad \dots(ii)$$

For third face

$$h^2 + l^2 = 20^2 \Rightarrow h^2 + l^2 = 400 \quad \dots(iii)$$

On adding Eqs. (i), (ii) and (iii), we get

$$2(l^2 + b^2 + h^2) = 850$$

$$\Rightarrow l^2 + b^2 + h^2 = 425 \quad \dots(iv)$$

Subtracting Eq. (i) from Eq. (iv), we get

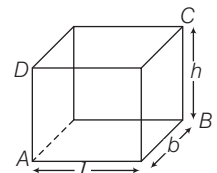
$$b^2 = 256 \Rightarrow b = 16 \text{ units}$$

$$\text{similarly, } b^2 = 25 \Rightarrow b = 5 \text{ units}$$

$$\text{and } l^2 = 144 \Rightarrow l = 12 \text{ units}$$

$\therefore$ Total surface area of cuboid

$$= 2(lb + bh + lh)$$



$$\begin{aligned}
 &= 2(12 \times 5 + 5 \times 16 + 16 \times 12) \\
 &= 2(60 + 80 + 192) \\
 &= 2 \times 332 = 664 \text{ sq units}
 \end{aligned}$$

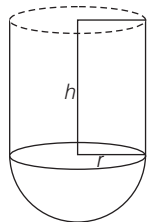
Hence, the total surface area of cuboid is 664 sq units.

- 140. (c)** Given, radius of vessel, $(r) = 4$ cm and radius of sphere $(R) = 3$ cm
Let height of vessel be h cm, then volume of vessel = $\pi r^2 h = \pi(4^2)h = 16\pi h$
and volume of sphere = $\frac{4}{3}\pi R^3 = 36\pi$
Let H be the rise in water level
 $\therefore$ Volume of water displaced
= volume of sphere
 $\Rightarrow 16\pi H = 36\pi \Rightarrow H = \frac{36}{16} = 2.25$ cm

- 141. (c)** Volume of paint required
= $2 \times (3 \times 2 \times 0.0001) +$
 $2 \times (2 \times 1.75 \times 0.0001) +$
 $2 \times (3 \times 1.75 \times 0.0001)$
= $0.0012 + 0.0007 + 0.00105 = 0.00295 \text{ m}^3$
Volume of cubical box = 10^3 .
= $1000 \text{ cm}^3 = 0.001 \text{ m}^3$
 $\therefore$ Boxes required = $\frac{0.00295}{0.001} \approx 3$.

- 142. (d)** Let the sides of a cube be l cm.
Then, $6l^2 = 13254$
 $\Rightarrow l^2 = \frac{13254}{6} = 2209 \Rightarrow l = 47$
 $\therefore$ Length of diagonal = $l\sqrt{3} = 47\sqrt{3}$ cm

- 143. (b)** Let the height of cylindrical portion be h .
Then, $\frac{2}{3}\pi r^3 + \pi r^2 h = 3312\pi$
 $\Rightarrow \pi r^2 \left[\frac{2}{3}r + h \right] = 3312\pi$
 $\Rightarrow 12 \times 12 \left[\frac{2}{3} \times 12 + h \right] = 3312$
 $\Rightarrow 8 + h = 23 \Rightarrow h = 15$ m



$$\begin{aligned}
 \therefore \frac{\text{Surface area of hemisphere}}{\text{Surface area of cylinder}} &= \frac{2\pi r^2}{2\pi rh} \\
 &= \frac{r}{h} = \frac{12}{15} = \frac{4}{5} \\
 \therefore r : h &= 4 : 5
 \end{aligned}$$

- 144. (a)** Let the length of water tank be lm .
volume of tank = $l \times l \times l = l^3$

volume of water tank after taking out water

$$\begin{aligned}
 &= l \times l \times (l - 2) = l^2(l - 2) \\
 \therefore l^3 - 128 &= l^2(l - 2) \\
 \Rightarrow 2l^2 &= 128 \Rightarrow l^2 = 64 \Rightarrow l = 8 \text{ m} \\
 \text{So, volume of cubical tank} &= 8^3 = 512 \text{ m}^3.
 \end{aligned}$$

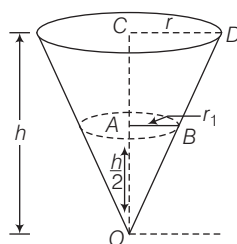
- 145. (b)** Let the sides of the cuboid be l, b and h respectively.
Volume of cuboid,
 $V = l \times b \times h \dots(i)$
Given, $x = lb, y = bh$ and $z = lh$
 $\therefore xyz = lb \times bh \times lh$
 $= (lbh)^2 = V^2$ [using Eq. (i)]

- 146. (a)** On each face, we can draw four right angled triangles by choosing one vertex to serve as the right angled and two adjacent vertices. Since there are six faces, you can form $6 \times 4 = 24$ triangles.

- 147. (a)** Let the area of each rectangle be A .
Then, breadth of R_1, R_2 and R_3 are $\frac{A}{x_1}, \frac{A}{x_2}$ and $\frac{A}{x_3}$ respectively
Cylinder is formed by joining parallel side of breadth.
Here, length becomes height and $2\pi r = \text{breadth} = \frac{A}{x} \Rightarrow r = \frac{A}{2\pi x}$
 $\therefore V = \pi r^2 h = \pi \frac{A^2}{4\pi^2 x^2} \times x = \frac{A^2}{4\pi x}$
Now, V_1, V_2 and V_3 are $\frac{A^2}{4\pi x_1}, \frac{A^2}{4\pi x_2}$ and $\frac{A^2}{4\pi x_3}$ respectively
 $\therefore x_1 < x_2 < x_3 \Rightarrow \frac{1}{x_1} > \frac{1}{x_2} > \frac{1}{x_3}$
 $\Rightarrow \frac{A^2}{4\pi x_1} > \frac{A^2}{4\pi x_2} > \frac{A^2}{4\pi x_3}$
 $\therefore V_3 < V_2 < V_1$

- 148. (a)** Since, $\triangle OAB$ and $\triangle OCD$ are similar
 $\therefore \frac{r}{r_1} = \frac{h}{\frac{h}{2}} = \frac{2}{1}$

$$\text{We have, } V = \frac{1}{3}\pi r^2 h$$



$$\Rightarrow r^2 = \frac{3V}{\pi h} \Rightarrow r_1^2 = \frac{r^2}{4} \Rightarrow r_1 = \frac{r}{2}$$

$$\begin{aligned}
 \text{Hence, volume of water} &= \frac{1}{3}\pi r_1^2 \frac{h}{2} \\
 &= \frac{1}{6}\pi b \frac{r^2}{4} = \frac{1}{24}\pi b \frac{3V}{\pi b} = \frac{3V}{24} = \frac{V}{8}
 \end{aligned}$$

- 149. (b)** Let radius be r and height be h of the cylinder.

$$\text{Then, } 30 \cdot \pi r^2 h = \frac{1}{3}\pi r^2 h \times n$$

$$\Rightarrow n = 30 \times 3 = 90 \text{ cones}$$

Statement I is correct. Statement II is also correct. But statement II is not correct explanation of statement of I.

- 150. (c)** At 30 min, the container was full
 $\therefore$ At 5 min before i.e. in 25 min, the container was half filled and at 20 min, container was one-fourth full.

- 151. (c)** I. Area of curved surface = $2\pi rh$
 $\therefore 2\pi r(2h) = 2 \cdot 2\pi rh$

$\therefore$ Curved surface area is also doubled.

It is correct.

II. Total surface area of hemisphere = $3\pi r^2$

$$\therefore 3\pi(2r)^2 = 3\pi 4r^2 = 4 \cdot 3\pi r^2 = 4 \text{ times the area}$$

It is also correct.

- 152. (c)** Let n spherical ball be made.

$$\text{Then, } 4\pi r^2 = \pi \frac{1}{3}$$

$$\Rightarrow r^2 = \frac{\pi \frac{1}{3}}{4\pi} = \frac{1}{4}\pi^{-\frac{2}{3}} \Rightarrow r^2 = \left(\frac{1}{2}\pi^{-\frac{1}{3}}\right)^2$$

$$\Rightarrow r = \frac{1}{2}\pi^{-\frac{1}{3}}$$

$$\text{Given, } n \cdot \frac{4}{3}\pi \left(\frac{1}{2}\pi^{-\frac{1}{3}}\right)^3 = 1 \times 1 \times 1$$

$$\Rightarrow n \cdot \frac{4}{3}\pi \times \frac{1}{8} \times \frac{1}{\pi} = 1 \Rightarrow \frac{n}{6} = 1$$

$$\therefore n = 6$$

- 153. (d)** Let n cylindrical boxes be packed.

$$\Rightarrow n \times \pi \times 5 \times 5 \times 10$$

$$= \pi \times 15 \times 15 \times 100$$

$$\therefore n = 90 \text{ boxes}$$

- 154. (c)** We have, area of cross-section of pipe

$$= 4 \text{ cm}^2 = 4 \times 10^{-4} \text{ m}^2$$

and $4 \times 10^{-4} \times 40 \text{ m}^3$ water flows in 1 s.

Let cistern fill in t s.

$$\Rightarrow 4 \times 10^{-4} \times 40 \times t = 10 \times 8 \times 6$$

$$\Rightarrow t = 30000 \text{ s} = \frac{30000}{60 \times 60} \text{ h}$$

$$= \frac{50}{6} \text{ h} = 8 \text{ h and } 20 \text{ min}$$

28

STATISTICS

Usually (10-13) questions have been asked from this chapter. This chapter is very important from examination point of view and generally questions are asked from the graphical representation of data, measures of central tendency.



Statistics is the branch of Mathematics which deals with the collection, analysis and interpretation of numerical data. In this chapter, we shall study measures of central tendency i.e. mean, median and mode of ungrouped data and grouped data. Concept of cumulative frequency, the cumulative frequency distribution how to draw cumulative frequency curves (ogive) and graphical representation of data will also be discussed.

Collection of Data

Collection of data is the first step in statistics towards achieving the goal or conclusion.

On the basis of collection, data are of two types

1. **Primary data** The data collected actually in the process of investigation by the investigator is called primary data. It is original and first hand information.
2. **Secondary data** The data collected by someone and used by any other person known as secondary data.

Presentation of Data

Raw or Ungrouped data When the data presented is random and is not prepared according to some order, it is known as raw or ungrouped data. It does not give us a clear picture of the class.

Grouped data When the data is arranged in any manner like ascending or descending order etc., it is called grouped data. It can also be presented in the form of a table called frequency distribution table.

Class Intervals Class intervals are the groups in which all the observations are divided. Each class is bounded by two figures (numbers) which are called class limits. The figure on the left side of a class, is called its lower limit and that on the right side of a class, is called its upper limit.

Class mark It is the mid-point of the class interval.
i.e. $\text{Class mark} = \frac{\text{Lower class limit} + \text{Upper class limit}}{2}$

Range or a class size Difference between the upper limit and the lower limit of a class is called its class size.

$\text{Range} = \text{Upper limit} - \text{Lower limit}$

e.g. Range of the observations 4, 7, 8, 10, 12
 $= 12 - 4 = 8$

Frequency of an observation The number of times an observation occurs is called its frequency.

Frequency distribution

The tabular arrangement of data, showing the frequency of each observation is called a frequency distribution. It is a method of presenting the data in a summarized form. Frequency distribution is also known as frequency table.

There are two types of frequency distribution which are as follows:

1. Discrete frequency distribution

A frequency distribution is called a discrete frequency distribution, if data are presented in a way such that exact measurements of the units are clearly shown.

Marks	Number of students (frequency)
40	7
60	3
80	3
100	2
Total	15

2. Continuous frequency distribution

Exclusive form A frequency distribution in which upper limit of each class is excluded and lower limit is included, is called an exclusive form.

e.g. In the class 0-10 of marks obtained by students, a student who has obtained 10 marks is not included in this class. It will be counted in the next class i.e 10-20.

Inclusive method In this method, the classes are so formed that the upper limit of a class is included in that class. The following example illustrates the method. In class 1000-1099, we include workers having wages between ₹ 1000 and ₹ 1099. If the income of a worker is exactly ₹ 1100, then it will be included in the next class 1100-1199.

Exclusive method		Inclusive method	
Wages (in ₹)	Number of workers	Wages (in ₹)	Number of works
1000-1100	125	1000-1099	125
1100-1200	150	1100-1199	150
1200-1300	200	1200-1299	200
1300-1400	250	1300-1399	250
1400-1500	175	1400-1499	175
1500-1600	100	1500-1599	100
Total	1000	Total	1000

It is clear from the above example that both the inclusive and exclusive methods give us the same class frequency although the class intervals are apparently different in the two cases.

In the above example on inclusive method, the difference between the lower limit of a class and upper limit of the preceding class is 1.

Therefore, we subtract $\frac{1}{2}$ from the lower limit and add $\frac{1}{2}$

to upper limit of each class to make it continuous.

The adjusted classes would be as follows:

Wages (in ₹)	Number of workers
999.5-1099.5	125
1099.5-1199.5	150
1199.5-1299.5	200
1299.5-1399.5	250
1399.5-1499.5	175
1499.5-1599.5	100

Cumulative frequency

If the frequency of first class interval is added to the frequency of second class and this sum is added to third class and so on, then frequencies so obtained are known as cumulative frequency.

EXAMPLE 1. Consider the table given below :

Marks	Number of students (frequency)	Cumulative frequency
0-10	13	13
10-20	7	20
20-30	5	25
30-40	4	29
40-50	1	30
50-60	7	37
60-70	3	40
70-80	4	44
80-90	5	49
90-100	1	50
Total	50	

Then, find the value of the following

(i) frequency of class 10-20. (ii) class size.

(iii) mid value of 60-70. (iv) total frequencies.

a. 10, 65, 50, 60

b. 7, 10, 65, 60

c. 50, 65, 10, 10

d. 7, 10, 65, 50

Sol. d. (i) Here, frequency of class 10-20 is 7.

(ii) Class size = Upper limit – Lower limit = $30 - 20 = 10$

(iii) Mid value = $\frac{\text{Upper limit} + \text{Lower limit}}{2} = \frac{60 + 70}{2} = 65$

(iv) Total frequencies = 50

EXAMPLE 2. The class mark of the interval 12.5-17.5 is

a. 5

b. 12.5

c. 15

d. 17.5

Sol. c. Class mark = $\frac{\text{Lower limit} + \text{Upper limit}}{2}$

$$= \frac{12.5 + 17.5}{2} = \frac{30}{2} = 15$$

EXAMPLE 3. The class marks of a distribution are 54, 64, 74, 84, 94 and 104. Then, the class size is

- a. 5 b. 10 c. 54 d. 104

Sol. b. Since, class size is the difference between the class marks of two adjacent classes.

$$\therefore \text{Class size} = 64 - 54 = 10$$

MEASURES OF CENTRAL TENDENCY

An average or central value of a statistical series is the value of the variable which describes the characteristic of the entire distribution. The following are the five measure of central tendency

- (i) Arithmetic mean or mean (ii) Geometric mean
(iii) Harmonic mean (iv) Median
(v) Mode

Out of these measures of central tendency. Arithmetic mean, median and mode are sometimes known as measures of location.

Arithmetic Mean

The mean (or average) of number of observations is the sum of the values of all the observations divided by the total number of observations.

$$\text{Mean} = \frac{\text{Sum of observations}}{\text{Number of observation}}$$

Properties of Arithmetic Mean

1. If every observation is increased by a constant, then the mean of the observations, so obtained also increases by the same constant.
2. If every observation is decreased by a constant, then the mean of the observation, so obtained also decreases by the same constant.
3. If each observation is multiplied by a constant, then the mean of the resulting observations can be obtained by multiplying the mean by the same constant.
4. If each observation is divided by a constant, then the mean of the resulting observation can be obtained by dividing the mean by the same constant.
5. The average of natural numbers from 1 to n is $\frac{(n+1)}{2}$.
6. The average of odd numbers from 1 to n is $\left[\frac{\text{Last odd number} + 1}{2} \right]$ and the average of even numbers from 1 to n is $\left[\frac{\text{Last even number} + 2}{2} \right]$.

1. Arithmetic mean of ungrouped or individual observations

If $x_1, x_2, x_3, \dots, x_n$ are n observations, then

$$(i) \text{ Mean } (\bar{x}) = \frac{x_1 + x_2 + \dots + x_n}{n} = \frac{1}{n} \sum_{i=1}^n x_i$$

It is also called direct method.

$$(ii) \text{ Mean } (\bar{x}) = A + \frac{1}{n} \sum_{i=1}^n d_i$$

where, A = Assumed mean and $d_i = x_i - A$.

It is also called shortcut method.

EXAMPLE 4. If the heights of 5 persons are 144 cm, 152 cm, 150 cm, 158 cm and 155 cm, respectively. Find the mean height.

- a. 152.5 cm b. 150 cm c. 149.8 cm d. 151.8 cm

Sol. d. Mean height = $\frac{\text{Sum of the heights}}{\text{Number of persons}}$

$$= \frac{144 + 152 + 150 + 158 + 155}{5} = \frac{759}{5} = 151.8 \text{ cm}$$

EXAMPLE 5. If the average of 6, 8, 5, 7, x and 4 is 7. Then, the value of x is

- a. 10 b. 11 c. 12 d. 15

Sol. c. $\therefore$ Sum of observation = $6 + 8 + 5 + 7 + x + 4 = 30 + x$
 and number of observation = 6

$$\therefore \text{Average} = \frac{30 + x}{6} \Rightarrow 7 = \frac{30 + x}{6}$$

$$\Rightarrow 42 = 30 + x \Rightarrow x = 42 - 30 = 12$$

Hence, the value of x is 12.

EXAMPLE 6. The average of 27 observations is 35. If 5 is added to each observation, what will be the new mean?

- a. 10 b. 20 c. 30 d. 40

Sol. d. Given, $\bar{x} = 35$ and $n = 27$

$$\text{Sum of observation} = n\bar{x} = 27 \times 35 = 945$$

$$\text{and new total of observation} = 945 + 27 \times 5 = 1080$$

$$\therefore \text{New mean} = \frac{1080}{27} = 40$$

Shortcut Method

$$\text{New mean} = \text{Previous mean} + \text{Number added to each term} \\ = 35 + 5 = 40$$

EXAMPLE 7. The mean of 53 observations is 18. If each observation is multiplied by 3. What will be the new mean?

- a. 18 b. 36 c. 53 d. 54

Sol. d. Given, $\bar{x} = 18$ and $n = 53$

$$\text{So, sum of observation} = n\bar{x} = 53 \times 18 = 954$$

$$\text{New total of observation} = 954 \times 3 = 2862$$

$$\therefore \text{New mean} = \frac{2862}{53} = 54$$

Shortcut Method

New mean = Previous mean $\times$ Constant multiplied to each term $= 18 \times 3 = 54$

2. **Mean of grouped or continuous observations** If $x_1, x_2, x_3, \dots, x_n$ are n observations whose corresponding frequencies are $f_1, f_2, f_3, \dots, f_n$, then

(i) Direct Method

$$(a) \text{ Mean, } \bar{x} = \frac{x_1 f_1 + x_2 f_2 + \dots + x_n f_n}{f_1 + f_2 + \dots + f_n}$$

$$= \frac{\sum_{i=1}^n f_i x_i}{\sum_{i=1}^n f_i} = \frac{\sum f_i x_i}{\sum f_i}$$

$$(b) \text{ Mean, } \bar{x} = A + \frac{\sum_{i=1}^n f_i d_i}{\sum_{i=1}^n f_i} = A + \frac{\sum f_i d_i}{\sum f_i}$$

where, A = Assumed mean and $d_i = x_i - A$.

$$(ii) \text{ Step Deviation Method } (\bar{x}) = A + \left(\frac{\sum f_i u_i}{\sum f_i} \right) \times h$$

where, h = Difference between two consecutive class marks.

EXAMPLE 8. The arithmetic mean of the marks from the following table is

Marks	0-10	10-20	20-30	30-40	40-50	50-60
Number of students	12	18	27	20	17	6

a. 20 **b.** 28 **c.** 2800 **d.** 100

Sol. **b.** We have,

Marks	Class mark, (x)	f	fx
0-10	5	12	60
10-20	15	18	270
20-30	25	27	675
30-40	35	20	700
40-50	45	17	765
50-60	55	6	330
Total		$\Sigma f = 100$	$\Sigma fx = 2800$

$$\therefore \bar{x} = \frac{\Sigma fx}{\Sigma f} = \frac{2800}{100} = 28$$

EXAMPLE 9. Find the mean wage from the data given below :

Wages (in ₹)	800	820	860	900	920	980	1000
Number of workers	7	14	19	25	20	10	5

a. 890 **b.** 890.5 **c.** 891.2 **d.** 100

Sol. **c.** Let the assumed mean be $A = 900$.
The given data can be written as under :

Wages (in ₹) (x_i)	Number workers (f_i)	$d = x_i - A$	$f_i d_i$
800	7	-100	-700
820	14	-80	-1120
860	19	-40	-760
900	25	0	0
920	20	20	400
980	10	80	800
1000	5	100	500
Total	$\Sigma f_i = 100$		$\Sigma f_i d_i = -880$

Here, $A = 900$

$$\therefore \text{Mean} = A + \frac{\Sigma f_i d_i}{\Sigma f_i} = 900 + \frac{-880}{100} = 900 - 8.80 = 891.2$$

$$\therefore \text{Mean wage} = ₹891.2$$

EXAMPLE 10. Find the mean of the following data.

x	15	25	35	45	55	65
f	4	28	15	20	17	16

a. 44.2 **b.** 42.4 **c.** 40.6 **d.** 41.6

Sol. **d.** Here, the values of class marks x_i are given, so let $A = 35$ and class width, h = difference between two consecutive class marks = 10, then

x_i	f_i	$u_i = \frac{x_i - A}{h}$	$f_i u_i$
15	4	$\frac{15 - 35}{10} = \frac{-20}{10} = -2$	$4 \times (-2) = -8$
25	28	$\frac{25 - 35}{10} = \frac{-10}{10} = -1$	$28 \times (-1) = -28$
35 = (A)	15	$\frac{35 - 35}{10} = \frac{0}{10} = 0$	$15 \times 0 = 0$
45	20	$\frac{45 - 35}{10} = \frac{10}{10} = 1$	$20 \times 1 = 20$
55	17	$\frac{55 - 35}{10} = \frac{20}{10} = 2$	$17 \times 2 = 34$
65	16	$\frac{65 - 35}{10} = \frac{30}{10} = 3$	$16 \times 3 = 48$
Total	$\Sigma f_i = 100$		$\Sigma f_i u_i = 66$

Thus, we have $\Sigma f_i u_i = 66$, $\Sigma f_i = 100$, $h = 10$

$$\therefore \bar{x} = A + h \left(\frac{\Sigma f_i u_i}{\Sigma f_i} \right) = 35 + 10 \times \frac{66}{100}$$

$$= 35 + \frac{660}{100} = 35 + 6.6 \Rightarrow \bar{x} = 41.6$$

Weighted Arithmetic Mean

If corresponding weight of $x_1, x_2, \dots, x_n$ are $w_1, w_2, \dots, w_n$ respectively, then

Weighted arithmetic mean

$$= \frac{w_1 x_1 + w_2 x_2 + \dots + w_n x_n}{w_1 + w_2 + \dots + w_n} = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i}$$

EXAMPLE 11. The weighted arithmetic mean of the first n natural numbers, the weights being the corresponding numbers, is

- a. $\frac{n+1}{2}$ b. $\frac{n+2}{2}$ c. $\frac{2n+1}{3}$ d. None of these

Sol. c. First n natural numbers are $1, 2, 3, \dots, n$; whose corresponding weights are $1, 2, 3, \dots, n$, respectively.

$$\begin{aligned} \therefore \text{Weight arithmetic mean} &= \frac{1 \times 1 + 2 \times 2 + \dots + n \times n}{1 + 2 + \dots + n} \\ &= \frac{1^2 + 2^2 + \dots + n^2}{1 + 2 + \dots + n} = \frac{n(n+1)(2n+1)}{6n(n+1)} = \frac{2n+1}{3} \end{aligned}$$

Combined Arithmetic Mean

If two sets of observations are given, then the combined mean of both the sets can be calculated by the following formula.

$$\bar{x} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2}$$

where, $\bar{x}$ = Mean of sets of observations

$\bar{x}_1$ = Mean of first set of observations

n_1 = Number of observations in first set

$\bar{x}_2$ = Mean of second set of observations

n_2 = Number of observations in second set

EXAMPLE 12. The average salary of male employees in a firm was ₹ 5200 and that of females was ₹ 4200. The mean salary of all the employees was ₹ 5000. The percentage of male and female employees are, respectively.

- a. 80 and 20 b. 20 and 80 c. 60 and 40 d. 52 and 48

Sol. a. Let $x_1 = 5200$, $x_2 = 4200$ and $\bar{x} = 5000$

$$\text{We know that, } \bar{x} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2}$$

$$\Rightarrow 5000(n_1 + n_2) = 5200n_1 + 4200n_2 \Rightarrow \frac{n_1}{n_2} = \frac{4}{1}$$

$\therefore$ The percentage of male employees in the firm

$$= \frac{4}{4+1} \times 100 = 80$$

and the percentage of female employees in the firm

$$= \frac{1}{4+1} \times 100 = 20$$

Geometric Mean

If a , G and b are in GP, then the geometric mean between a and b is, $G = \sqrt{ab}$.

EXAMPLE 13. What is the geometrical mean of the variate which takes values 210, 201, 202, 20, 12, 10, 2, 1 and 0?

- a. 10 b. 9 c. 8 d. 0

Sol. d. The given variates are 210, 201, 202, 20, 12, 10, 2, 1 and 0.

$$\begin{aligned} \therefore \text{Geometric mean of given variates} \\ = \sqrt[9]{210 \times 201 \times 202 \times 20 \times 10 \times 2 \times 1 \times 0} = \sqrt[9]{0} = 0 \end{aligned}$$

Harmonic Mean

The harmonic mean of n positive numbers $a_1, a_2, \dots, a_n$ is

$$\frac{1}{H} = \frac{1}{n} \left(\frac{1}{a_1} + \frac{1}{a_2} + \frac{1}{a_3} + \dots + \frac{1}{a_n} \right)$$

where, H denotes the harmonic mean.

Median

After arranging the given data in ascending or descending order of magnitude, the value of the middle most observation is called the median of the data.

1. **Median of an individual series** If number of observations is n . Then, arrange the observations in ascending or descending order.

(i) If n is an odd number, then

$$\text{Median} = \text{Value of the } \left(\frac{n+1}{2} \right) \text{th observation}$$

(ii) If n is an even number, then

$$\text{Value of the } \left(\frac{n}{2} \right) \text{th observation}$$

$$+ \text{Value of } \left(\frac{n}{2} + 1 \right) \text{th observation}$$

$$\text{Median} = \frac{\quad}{2}$$

2. **Median of a discrete frequency series** First arrange the data in ascending or descending order and find cumulative frequency. Now, find $\frac{N}{2}$, where $N = \Sigma f_i$.

See the cumulative frequency just greater than $\frac{N}{2}$.

The corresponding value of x is median.

3. **Median of a continuous series** In this case, the class corresponding to the cumulative frequency just greater than $N/2$ is called the median class and the value of median is obtained by the following formula.

$$\text{Median} = l + \left(\frac{\frac{N}{2} - c}{f} \right) \times h$$

where, l = Lower limit of median class

f = Frequency of median class
 h = Size of median class
 c = Cumulative frequency of class before median class

EXAMPLE 14. From the data given, the median of the average deposit balance of saving for the branch during March 1982 is

Average deposit balance (in ₹)	Number of deposit
0 – 100	26
100-200	68
200-300	145
300-400	242
400-500	188
500-600	65
600-700	16

a. 356 b. 300 c. 56.2 d. 356.2

Sol. d.

Average deposit balance (in ₹)	f	Cumulative frequency (cf)
Less than 100	26	26
100-200	68	94
200-300	145	239
300-400	242	481
400-500	188	669
500-600	65	734
600-700	16	750

$$\frac{N}{2} = \frac{750}{2} = 375$$

The frequency just greater than 375 is 481.

∴ Median class is 300-400.

$$\text{Median} = l + \frac{\left(\frac{N}{2} - c\right)}{f} \times h = 300 + \frac{375 - 239}{242} \times 100$$

$$= 300 + 56.2 = 356.2$$

Mode

The data value which occurs maximum number of times in a data distribution is called mode.

- Mode of individual series** The value which is repeated maximum number of times is called mode of the series.

EXAMPLE 15. Find the mode for the following series 2.5, 2.3, 2.2, 2.2, 2.2, 2.4, 2.7, 2.7, 2.5, 2.3, 2.2, 2.6 and 2.2.

a. 2.2 b. 2.3
c. 2.7 d. 2.6

Sol. a. Arranging the data in the form of a frequency table, we have

Value	Frequency
2.2	4
2.3	2
2.4	1
2.5	2
2.6	1
2.7	2

We see that, the value 2.2 has the maximum frequency 4. So, the mode for the given series is 2.2.

- Mode of a discrete frequency series** In this case, mode is the value of the variate corresponding to the maximum frequency.

EXAMPLE 16. Compute the modal value for the following frequency distribution.

x	95	105	115	125	135	145	155	165	175
f	4	2	18	22	21	19	10	3	2

a. 115 b. 125 c. 22 d. 120

Sol. b. From the given table, it is clear that 125 has the highest frequency i.e. 22. Hence, modal value of the given frequency distribution is 125.

- Mode of a continuous series** The class which has maximum frequency is called modal class or group. The mode is given by the formula,

$$\text{Mode} = l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h$$

where, l = Lower limit of modal class

h = Size of class interval

f_1 = Frequency of modal class

f_0 = Frequency of the class preceding the modal class

f_2 = Frequency of the class succeeding the modal class

EXAMPLE 17. The mode of the following distribution is

Class interval	Frequency
0-10	5
10-20	8
20-30	7
30-40	12
40-50	28
50-60	20
60-70	10
70-80	10

a. 46 b. 6.66
c. 46.67 d. None of these

Sol. c. Here, maximum frequency is 28. Thus, the class 40-50 is the modal class. Here, $f_1 = 28, f_0 = 12, f_2 = 20, l = 40$ and $h = 10$

$$\begin{aligned}\therefore \text{Mode} &= l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) \times h = 40 + \frac{10(28 - 12)}{(2 \times 28 - 12 - 20)} \\ &= 40 + \frac{10 \times 16}{24} = 40 + 6.666 = 46.67 \text{ (approx)}\end{aligned}$$

Relation between Mean, Median and Mode

The empirical relationship between the three measures of central tendency is given by

$$\text{Mode} = 3 (\text{Median}) - 2 (\text{Mean})$$

In case, mean = median = mode, then distribution is said to be symmetric distribution.

EXAMPLE 18. If in a frequency distribution, the mean and median are 20 and 21 respectively, then its mode is approximate by

- a. 24 b. 23 c. 25 d. None of these

Sol. b. Here, mean = 20 and median = 21

$$\text{Mode} = 3 \times \text{Median} - 2 \times \text{Mean} = 63 - 40 = 23$$

Some Useful Formulae

If $x_1, x_2, \dots, x_n$ are n observations with their mean M , then deviation d_i is given by $d_i = |x_i - M|$.

- Mean deviation for individual series is given by $\frac{\sum |d_i|}{n}$.
- Mean deviation for discrete series is given by $\frac{\sum f_i |d_i|}{\sum f_i}$.
- Standard deviation (σ) for ungrouped data is given by $\sigma = \sqrt{\frac{\sum |d_i|^2}{n}}$.
- Standard deviation for grouped data is given by $\sigma = \sqrt{\frac{\sum f_i |d_i|^2}{n}}$.
- Variance = σ^2
- Coefficient of variation = $\frac{\text{Standard deviation}}{\text{Mean}} \times 100$
- Coefficient of mean deviation = $\frac{\text{Mean deviation}}{\text{Mean}} \times 100$

EXAMPLE 19. The mean deviation of the data, 3, 5, 6, 7, 8, 10, 11, 14 is

- a. 4 b. 3.25 c. 2.75 d. 2.4

Sol. c. Mean = $\frac{3+5+6+7+8+10+11+14}{8} = \frac{64}{8} = 8$

$$\begin{aligned}\therefore \sum \delta &= |3-8| + |5-8| + |6-8| + |7-8| + |8-8| \\ &\quad + |10-8| + |11-8| + |14-8| = 22\end{aligned}$$

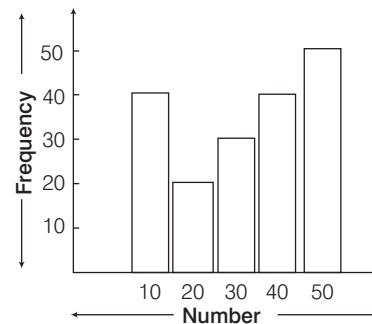
$$\therefore \text{Mean deviation} = \frac{\sum \delta}{n} = \frac{22}{8} = 2.75$$

Graphical Representation of Data

There are many ways of picturing a frequency distribution of continuous type which are discussed below

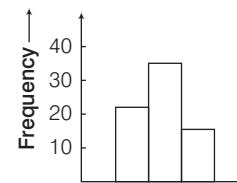
1. Bar Graph

In a bar graph, bars of uniform width are drawn with various heights. The heights of a bar represents the frequency of observation.



2. Histogram

A histogram is the graphical representation of a frequency distribution in the form of the rectangles with class intervals as bases, and the corresponding frequencies as height. There being no gap between any two consecutive rectangles.



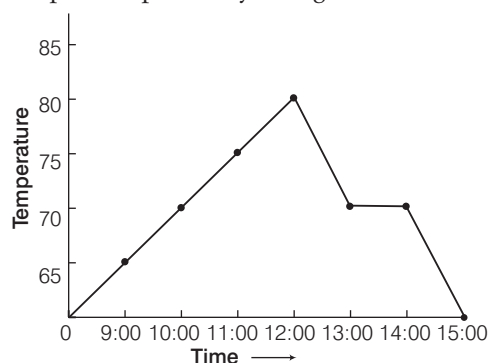
POINTS TO BE REMEMBERED

- A histogram consists of a set of adjacent rectangles, whose bases are equal to class sizes and height are equal to class frequencies.
- The total area of the histogram = Sum of areas of all rectangles. If the class intervals are of same size (width), then the area of histogram = NK , where K = Size of a class and N = Sum of frequencies of all classes.
- If the frequency distribution is discontinuous (inclusive), change it to continuous (exclusive) and then construct a histogram.

3. Frequency polygon

A frequency polygon can be drawn by joining the mid-points of the respective tops of the rectangles of a histogram in case of equal class intervals.

A frequency polygon for a grouped data can also be drawn independently by plotting the mid-points of the all classes along X -axis and frequencies along Y -axis and joining the plotted points by straight line.



4. Ogive (Cumulative Frequency Curve)

When we plot the upper or lower class limits along X -axis and cumulative frequencies along Y -axis, And on joining them we get a curve called an ogive.

We have two types of ogive curves

(a) less than ogive (i.e. the rising curve)

It is the graph between upper limits and cumulative frequencies of a distribution.

(b) more than ogive (i.e. a falling curve)

It is the graph between lower limits and cumulative frequencies of a distribution.

- Note**
- If we have both ogive (i.e. less than type and more than type), then these two ogives intersect each other at a point.
 - From this point, draw a perpendicular on X -axis, the point at which it cuts the X -axis gives the median. i.e. the x -coordinate of intersection point gives the median.

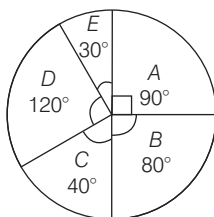
Pie-Diagram (or Pie chart)

A type of graph in which a circle is divided into sectors such that each sector represents a proportion of the whole.

Thus, in a pie chart

- Data is represented by sectors of a circle.
- Each part of data makes a certain central angle.
- Sum of all the angles of sector is 360° .

- Central angle = $\left(\frac{\text{Frequency} \times 360}{\text{Total frequency}} \right)^\circ$

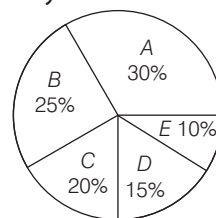


EXAMPLE 20. In statistics, a suitable graph for representing the partitioning of total into subpart is

- an ogive
- a pictograph
- a histogram
- a pie chart

Sol. d. In statistics, a suitable graph for representing the partitioning of total into sub parts is a pie chart.

EXAMPLE 21. The following diagram show the expenditure of a family on various items A, B, C, D and E .



Study the diagram carefully and answer the following questions.

(i) The angle of pie diagram showing the expenditure incurred on item A is

- 30°
- 35°
- 108°
- None of these

(ii) Which two expenditures together will form an angle of 90° at the centre of pie diagram?

- B and C
- C and A
- D and E
- None of these

(iii) If the income of the family is ₹ 3000 per month, then expenditure of item C will be

- ₹ 400
- ₹ 500
- ₹ 600
- ₹ 800

Sol. (i) c. Angle for $100\% = 360^\circ$

$$\therefore \text{Angle for expenditure of } A = 30\% \text{ of } 360^\circ \\ = \frac{30}{100} \times 360 = 108^\circ$$

(ii) c. $\because 100\% = 360^\circ$

$$\therefore \text{An angle of } 90^\circ \text{ at the centre of pie diagram in percentage is } = \frac{90 \times 100}{360} = 25\%$$

Expenditure of items D and E makes upto $(15 + 10) = 25\%$.

So, expenditure of D and E together will form an angle of 90° at the centre of pie diagram.

(iii) c. Expenditure on item C

$$= 20\% \text{ of } ₹ 3000 = \frac{20}{100} \times 3000 = ₹ 600$$

> PRACTICE EXERCISE

1. Frequency polygon can be drawn after drawing
 - (a) ogive
 - (b) bar chart
 - (c) histogram
 - (d) None of these
2. An ogive is used to determine
 - (a) mean
 - (b) median
 - (c) GM
 - (d) HM
3. The mid-value of a class interval is 42. If the class size is 10, then the upper and lower limits of the class are
 - (a) 37.5 and 47.5
 - (b) 47 and 37
 - (c) 37 and 47
 - (d) 47.5 and 37.5
4. The actual lower class limits of the following classes 10-19, 20-29, 30-39 and 40-49 are
 - (a) 9.5, 19, 29 and 39.5
 - (b) 10, 20, 30 and 40
 - (c) 9.5, 19.5, 29.5 and 39.5
 - (d) 18.5, 28.5, 38.5 and 48.5
5. If the mean of five observations $x, x + 2, x + 4, x + 6$ and $x + 8$ is 11, then the mean of first three observations is
 - (a) 9
 - (b) 11
 - (c) 13
 - (d) None of these
6. The combined mean of three groups is 12 and the combined mean of first two groups is 3. If the first, second and third groups have 2, 3 and 5 items respectively, then mean of third group is
 - (a) 10
 - (b) 21
 - (c) 12
 - (d) 18
7. 10 is the mean of a set of 7 observations and 5 is the mean of a set of 3 observations. The mean of the combined set is given by
 - (a) 15
 - (b) 10
 - (c) 8.5
 - (d) 7.5
8. In a class of 50 students, 10 have failed and their average marks are 28. The total marks obtained by the entire class are 2800. The average marks of those who have passed are
 - (a) 43
 - (b) 53
 - (c) 63
 - (d) 70
9. Which of the following statements about the median is true?
 - (a) It is not affected by extreme values
 - (b) It can be found even, if some items are not known
 - (c) It is useful when the data cannot be measured quantitatively
 - (d) All of the above
10. The middle item of the series arranged in ascending or descending order is called
 - (a) mean
 - (b) median
 - (c) mode
 - (d) standard deviation
11. The following pie chart shows the marks obtained by a student in an examination who scored 540 marks in all. The subject in which the student scored 108 marks is

Mathematics 90°
English 63°
Social Science 72°
Science 75°
Hindi 60°

 - (a) Science
 - (b) Hindi
 - (c) English
 - (d) Social Science
12. A distribution consists of three components with frequencies 45, 40 and 15 having their means 2, 2.5 and 2 respectively. The mean of the combined distribution is
 - (a) 2.1
 - (b) 2.2
 - (c) 2.3
 - (d) 2.4
13. If the values $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \dots, \frac{1}{n}$ occur at frequencies 1, 2, 3, 4, 5, 6, ..., n respectively, in a frequency distribution, then the mean is
 - (a) 1
 - (b) n
 - (c) $\frac{1}{n}$
 - (d) $\frac{2}{n+1}$
14. If the geometric mean of three observations 40, 50 and x is 10, then the value of x is
 - (a) $\frac{1}{2}$
 - (b) 4
 - (c) 6
 - (d) 2
15. Suppose, X is some statistical variable with mean μ . Let $x_1, x_2, \dots, x_n$ be its deviations from mean with the respective frequencies $f_1, f_2, \dots, f_n$. What is the value of the sum $x_1 f_1 + x_2 f_2 + \dots + x_n f_n$?
 - (a) 0
 - (b) 1
 - (c) μ
 - (d) $\mu + 1$
16. Which one of the following statements is not correct with reference to a histogram?
 - (a) Frequency curve is obtained by joining the mid-points of the top of the adjacent rectangles with smooth curves
 - (b) Histogram is drawn for continuous data
 - (c) The height of the bar is proportional to the frequency of that class
 - (d) Mode of the distribution can be obtained from the histogram

17.

X	0	1	2	3	4
Frequency	4	f	9	g	4

The table above gives the frequency distribution of a discrete variable X with two missing frequencies. If the total frequency is 25 and the arithmetic mean of X is 2, then what is the value of the missing frequency f ?

- (a) 4 (b) 5 (c) 6 (d) 7

- 18.** The yield of paddy per plot of one acre were obtained from a number of plots from two different districts in a state and are summarized in the following table.

Yield of paddy per plot in quintals	Number of plots in district A	Number of plots in district B
38.0-41.0	25	14
41.0-44.0	36	29
44.0-47.0	59	35
47.0-50.0	30	54
50.0-53.0	25	41

Which one of the following statements is correct?

- (a) The mode for district A is higher than the mode for district B
 (b) The mode for district B is higher than the mode for district A
 (c) Both the distributions are symmetric
 (d) Both the distributions have the same mean
- 19.** Average score of 50 students in a class is 44. Later on it was found that the score 23 was incorrectly recorded as 73. The correct average score is
- (a) 42 (b) 43 (c) 45 (d) 46
- 20.** In the "less than" type of ogive the cumulative frequency is plotted against
- (a) the lower limit of the concerned class interval
 (b) the upper limit of the concerned class interval
 (c) the mid value of the concerned class interval
 (d) any value of the concerned class interval
- 21.** A, B and C are three sets of values of x
 A : 2, 3, 7, 1, 3, 2, 3
 B : 7, 5, 9, 12, 5, 3, 8
 C : 4, 4, 11, 7, 2, 3, 4
- Select the correct statement from among the following
- (a) Mean of A is equal to mode of C
 (b) Mean of C is equal to median of B
 (c) Median of B is equal to mode of A
 (d) Median, mean and mode of A are same
- 22.** A student obtains 75%, 80% and 85% marks in three subjects. If the marks of any other subject are added, then their average cannot be less than
- (a) 60% (b) 65% (c) 70% (d) 80%

- 23.** If the mean and median of a set of numbers are 8.9 and 9 respectively, then the mode will be
- (a) 7.2 (b) 8.2 (c) 9.2 (d) 10.2

- 24.** If every number of a finite set is increased by any number k , then the measure of central tendency should also increase by k . Which one of the following measures of central tendency does not have this property?

- (a) Arithmetic mean
 (b) Median
 (c) Mid-range i.e. the arithmetic mean of the largest and the smallest numbers
 (d) Geometric mean

- 25.** If the median of the distribution (arranged in ascending order) 1, 3, 5, 7, 9, x , 15, 17, 19, 21 is 10, what is the value of x ?

- (a) 11 (b) 13 (c) $9 < x < 15$ (d) $9 \leq x \leq 15$

- 26.** If the width of each of the ten classes in a frequency distribution is 2.5 and the lower class boundary of the lowest class is 5.1, then the upper class boundary of the highest class is

- (a) 30.1 (b) 30 (c) 31.1 (d) 27.56

- 27.** The average value of the median of 2, 8, 3, 7, 4, 6, 7 and the mode of 2, 9, 3, 4, 9, 6, 9 is

- (a) 9 (b) 8 (c) 7.5 (d) 6

- 28.** When 10 is subtracted from each of the given observations, the mean is reduced to 60%. If 5 is added to all the given observations, the mean will be

- (a) 25 (b) 30 (c) 60 (d) 65

- 29.** A data has highest value 120 and the lowest value 71. A frequency distribution in descending order with seven classes is to be constructed.

The limits of the second class interval shall be

- (a) 77 and 78 (b) 78 and 85
 (c) 85 and 113 (d) 113 and 120

- 30.** If mean of y and $\frac{1}{y}$ is M , then what is the mean of y^3 and $\frac{1}{y^3}$?

- (a) $\frac{M(M^2 - 3)}{3}$ (b) M^3
 (c) $M^3 - 3$ (d) $M(4M^2 - 3)$

- 31.** For the following frequency distribution

Class interval	0-5	5-10	10-15	15-20	20-25	25-30
Frequency	10	15	30	80	40	20

If m is the value of mode, then which one of the following is correct?

- (a) $5 < m < 10$ (b) $10 < m < 15$
 (c) $15 < m < 20$ (d) $20 < m < 25$

- 32.** Square diagrams are drawn to represent the following data

Country	Pakistan	India	Myanmar	China
Labour Production (in ₹)	36	81	25	100

Using the scale $1 \text{ cm}^2 = ₹ 25$ what is the length of the representative square for India?

- (a) 1.8 cm (b) 1.2 cm (c) 1 cm (d) 2 cm
- 33.** The standard deviation of 7, 9, 11, 13, 15 is
(a) 2.4 (b) 2.5 (c) 2.7 (d) 2.82
- 34.** The total number of cellphones sold for Motorola, Samsung and Sony was 45664. The number of cellphones sold for these companies were in the ratio 3:5:8 respectively. If these data were shown on a pie chart, calculate the angle represented by the number of cellphones sold by Motorola.
(a) 75° (b) 67.5° (c) 70° (d) 74.5°
- 35.** If the population figures are given for each State of India, then the data can be classified as
(a) qualitative (b) quantitative
(c) chronological (d) geographical
- 36.** Which one of the following represents statistical data?
(a) The names of all owners of shops located in a shopping complex
(b) A list giving the names of all States of India
(c) A list of all European countries and their respective capital cities
(d) The volume of a rainfall in certain geographical area, recorded every month for 24 consecutive months
- 37.** Prime numbers are the numbers which comes in the table of 1 and itself only.
I. The mean of first seven prime numbers is greater than their median.
II. Mean is always greater than median.
Select the correct option from the options given below
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II
- 38.** Consider the following statements in respect of a histogram
I. The histogram consists of vertical rectangular bars with a common base such that there is no gap between consecutive bars.
II. The height of the rectangle is determined by the frequency of the class it represents.
Which of the statements given above is/are correct?
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

- 39.** Consider the following statements in respect of histogram

- I. Histogram is an equivalent graphical representation of the frequency distribution.
II. Histogram is suitable for continuous random variables, where the total frequency of an interval is evenly distributed over the interval.

Which of the statements given above is/are correct?

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

- 40.** Examples of data are given below

- I. Information on households collected by an investigator by door-to-door visits.
II. Data on the percentage of literates, sex-wise, for the different districts of a state collected from records of the census of India.
III. General information about families, collected by telephonic interviews.

Which one of the following in respect of the above is correct?

- (a) I and II are primary data (b) I and III are primary data
(c) II and III are primary data (d) I, II and III are primary data

- 41.** Consider the following types of data

- I. Marks of students who appeared for a test of 100 marks.
II. Collar sizes of 200 shirts sold in a week.
III. Monthly incomes of 250 employees of a factory.

For which of the above data, mode is a suitable measure of central tendency?

- (a) I and II (b) Only II (c) I and III (d) All of these

- 42.** The cumulative frequency curve of a frequency distribution with 6 classes and total frequency 60 is a straight line. Consider the following statements

- I. The first and the last classes have a frequency of 10 each.
II. Both the middle classes have a total frequency of 30.
III. The frequency distribution does not have a mode.

Which of the statements given above are correct?

- (a) I and II (b) I and III (c) II and III (d) All of these

- 43.** State which of the following variables are discrete?

- I. Number of children in a family.
II. Wages of workers.
III. The ages of students.
IV. Weights of a set of a students.

Select the correct answer using the codes given below

- (a) I and II (b) I, II and III
(c) All of these (d) None of these

Directions (Q. Nos. 44-45) The number of goals scored by a soccer team in a season were as follows:

Number of goals	Number of matches
0	3
1	6
2	2
3	x
4	5

44. If the mode is 1, the largest possible value of x is
(a) 2 (b) 5 (c) 3 (d) 6
45. If the median is 2, the smallest possible value of x is
(a) 2 (b) 5 (c) 3 (d) 6

Directions (Q. Nos. 46-47) The itemwise expenditure of a non-government organisation for the year 2008-2009 is given below.

Item	Expenditure (in ₹ lakh)
Salary of employees	6
Social welfare activities	7
Office contingency	3
Vehicle maintenance	4
Rent and hire charges	2.5
Miscellaneous expenses	1.5

The above data are represented by a pie diagram.

46. What is the sectorial angle of the largest sector?
(a) 120° (b) 105° (c) 90° (d) 85°
47. What is the difference in the sectorial angles of the largest and the smallest sectors?
(a) 90° (b) 85° (c) 82.5° (d) 77.5°

Directions (Q. Nos. 48-49) The following table gives the frequency distribution of life length in hours of 100 electric bulbs having median life 20 h.

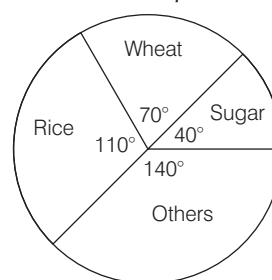
Life of bulbs (in h)	Number of bulbs
8-13	7
13-18	x
18-23	40
23-28	y
28-33	10
33-38	2

48. What is the missing frequency of x ?
(a) 31 (b) 27 (c) 24 (d) 14
49. What is the missing frequency of y ?
(a) 27 (b) 24 (c) 14 (d) 11

Directions (Q. Nos. 50-52) The arithmetic mean, geometric mean and median of 6 positive numbers a, a, b, b, c, c where $a < b < c$ are $\frac{7}{3}, 2, 2$ respectively.

50. What is the sum of the squares of all the six numbers?
(a) 40 (b) 42 (c) 45 (d) 48
51. What is the value of c ?
(a) 1 (b) 2 (c) 3 (d) 4
52. What is the mode?
(a) 1 (b) 2
(c) 1, 2 and 4 (d) None of these

Directions (Q. Nos. 53-55) Study the following pie diagram and answer the questions.



53. The percentage of wheat production is
(a) 35% (b) 70% (c) 19.44% (d) 20%
54. If the sugar production is 9000 kg, then the wheat production is
(a) 20000 kg (b) 24000 kg
(c) 15750 kg (d) None of these
55. If the total production is 180000 kg, the difference in sugar and wheat production is
(a) 10000 (b) 15000 (c) 20000 (d) None of these

Directions (Q. Nos. 56-58) Consider the following frequency distribution

Class interval	1-4	4-7	7-10	10-13	13-16	16-19
Frequency	6	30	40	16	4	4

56. What is the median of the data?
(a) 9.4 (b) 7.68 (c) 8.05 (d) 8.32
57. What is modal size of the given data?
(a) 8.14 (b) 7.88 (c) 7.62 (d) 8.48
58. The value of (Mean + Mode - 2 Median) is equal to
(a) 0 (b) 0.2 (c) 0.4 (d) 0.1

PREVIOUS YEARS' QUESTIONS

59. Let the observations at hand be arranged in increasing order. Which one of the following measures will not be affected when the smallest and the largest observations are removed?

- (a) Mean (b) Median
(c) Mode (d) Standard deviation

2012 I

- 60.** Consider the following statements :
- I. The data collected by the investigator to be used by himself are called primary data.
- II. The data obtained from government agencies are called secondary data.

Which of the statement(s) given above is/are correct? **☑ 2012 I**

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

- 61.** Which one among the following statements is correct? **☑ 2012 I**

- (a) Simple bar diagrams are those diagrams which show two characteristics of the data
(b) In pie diagrams, all the items are converted into angles
(c) A bar diagram is one in which data are shown in terms of bars
(d) Bar diagrams present data through length and breadth

- 62.** Consider the following distribution :

Value of the variable	1	2	3	4	5
Frequency	3	f	6	5	3

For what value of f , is the arithmetic mean of the above distribution 3.1? **☑ 2012 I**

- (a) 2 (b) 3 (c) 4 (d) 5

- 63.** Which of the following pair(s) is/are correctly matched?

- I. Weight of a person : Continuous variable
II. Educational qualification of the person : Attribute
(a) Only I (b) Only II **☑ 2012 II**
(c) Both I and II (d) Neither I nor II

- 64.** The mean of 100 values is 45. If 15 is added to each of the first forty values and 5 is subtracted from each of the remaining sixty values, then the new mean becomes **☑ 2012 II**

- (a) 45 (b) 48 (c) 51 (d) 55

- 65.** Which one of the following relations for the numbers 10, 7, 8, 5, 6, 8, 5, 8 and 6 is correct?

- (a) Mean = Median (b) Mean = Mode **☑ 2012 II**
(c) Mean > Median (d) Mean > Mode

- 66.** In histogram the width of the bars is proportional to **☑ 2012 II**

- (a) Frequency (b) Number of classes
(c) Class interval (d) None of these

- 67.** Which one of the following statements is correct? **☑ 2013 I**

- (a) A frequency polygon is obtained by connecting the corner points of the rectangles in a histogram
(b) A frequency polygon is obtained by connecting the mid-points of the tops of the rectangles in a histogram
(c) A frequency polygon is obtained by connecting the corner points of the class intervals in a histogram
(d) None of the above

- 68.** Consider the following statements :

- I. A frequency distribution condenses the data and reveals its important features.
II. A frequency distribution is an equivalent representation of original data.

Which of the statement(s) given above is/are correct? **☑ 2013 I**

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

Directions (Q. Nos. 69-71) Read the following information carefully to answer the questions that follow. **☑ 2013 I**

In a frequency distribution having class intervals 0-10, 10-20, 20-30 and 30-40 the respective frequencies are x , $x+8$, $x-2$ and $x-4$ and the arithmetic mean of the distribution is 17.8.

- 69.** The value of x is
(a) 3 (b) 6 (c) 8 (d) 12

- 70.** The median lies in which one of the following class intervals?

- (a) 0-10 (b) 10-20 (c) 20-30 (d) 30-40

- 71.** The modal class is

- (a) 0-10 (b) 10-20 (c) 20-30 (d) 30-40

- 72.** The mean of 7 observations is 7. If each observation is increased by 2, then the new mean is **☑ 2013 I**

- (a) 12 (b) 10 (c) 9 (d) 8

- 73.** There are 45 male and 15 female employees in an office. If the mean salary of the 60 employees is ₹ 4800 and the mean salary of the male employees is ₹ 5000, then the mean salary of the female employees is **☑ 2013 I**

- (a) ₹ 4200 (b) ₹ 4500 (c) ₹ 5600 (d) ₹ 6000

Directions (Q. Nos. 74-75) Read the following information carefully and answer the questions given below. **☑ 2013 II**

The median of the following distribution is 14.4 and the total frequency is 20.

Class interval	0-6	6-12	12-18	18-24	24-30
Frequency	4	x	5	y	1

- 74.** What is x equal to?

- (a) 4 (b) 5 (c) 6 (d) 7

- 75.** What is the relation between x and y ?

- (a) $2x = 3y$ (b) $3x = 2y$ (c) $x = y$ (d) $2x = y$

- 76.** If m is the mean of p, q, r, s, t, u and v , then what is $(p-m) + (q-m) + (r-m) + (s-m) + (t-m) + (u-m) + (v-m)$ equal to? **☑ 2013 II**

- (a) 0 (b) s
(c) $\frac{(p+v)}{2}$ (d) None of these

- 77.** The average of u, v, w, x, y and z is 10. What is the average of $u + 10, v + 20, w + 30, x + 40, y + 50$ and $z + 60$? ☑ 2013 II

(a) 30 (b) 35 (c) 40 (d) 45

- 78.** The mean of the following distribution is 18.

Class interval	Frequency
11-13	3
13-15	6
15-17	9
17-19	13
19-21	f
21-23	5
23-25	4

What is the value of f ? ☑ 2014 II

(a) 8 (b) 9 (c) 10 (d) 11

- 79.** Consider the following statements pertaining to a frequency polygon of a frequency distribution of a continuous variable having seven class intervals of equal width.

- I. The original frequency distribution can be reconstructed from the frequency polygon.
 II. The frequency polygon touches the X-axis in its extreme right and extreme left.

Which of the statement(s) given above is/are correct? ☑ 2014 II

(a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II

- 80.** Consider the following in respect of variate which takes values 2, 2, 2, 2, 7, 7, 7 and 7.

- I. The median is equal to mean.
 II. The mode is both 2 and 7.

Which of the statement(s) given above is/are correct? ☑ 2014 II

(a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II

- 81.** Which of the following items of information is a good example of statistical data? ☑ 2014 II

- (a) A table of logarithms of numbers
 (b) A list of names of 120 students of a class
 (c) A list of annual incomes of the members of a club
 (d) Holiday list of the offices of Government of India in the year 2013

- 82.** The following table gives 'less than' type frequency distribution of income per day.

Income (in ₹) less than	Number of persons
1500	100
1250	80
1000	70
750	55
500	32
250	12

What is the modal class?

☑ 2014 II

- (a) 250-500 (b) 500-750
 (c) 750-1000 (d) None of these

- 83.** The geometric mean of $(x_1, x_2, x_3, \dots, x_n)$ is X and the geometric mean of $(y_1, y_2, y_3, \dots, y_n)$ is Y .

Which of the following statement is/are correct?

I. The geometric mean of

$(x_1 y_1, x_2 y_2, x_3 y_3, \dots, x_n y_n)$ is XY .

II. The geometric mean of $\left(\frac{x_1}{y_1}, \frac{x_2}{y_2}, \frac{x_3}{y_3}, \dots, \frac{x_n}{y_n}\right)$ is $\frac{X}{Y}$.

Select the correct answer using the codes given below

- (a) Only I (b) Only II ☑ 2014 II
 (c) Both I and II (d) Neither I nor II

- 84.** Consider the following statements in respect of a discrete set of numbers.

- I. The arithmetic mean uses all the data and is always uniquely defined.
 II. The median uses only one or two numbers from the data and may not be unique.

Which of the statement(s) given above is/are correct? ☑ 2014 II

- (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II

- 85.** If x_i 's are the mid-points of the class intervals of grouped data, f_i 's are the corresponding frequencies and $\bar{x}$ is the mean, then what is $\sum f_i(x_i - \bar{x})$ equal to? ☑ 2014 II

- (a) 0 (b) -1 (c) 1 (d) 2

- 86.** When we take class intervals on the X-axis and corresponding frequencies on the Y-axis and draw rectangles with the areas proportional to the frequencies of the respective class intervals, the graph so obtained is called ☑ 2014 II

- (a) bar diagram (b) frequency curve
 (c) ogive (d) None of these

- 87.** Consider the following data:

- I. Number of complaints lodged due to road accidents in a state within a year for 5 consecutive years.

- II. Budgetary allocation of the total available funds to the various items of expenditure.

Which of the above data is/are suitable for representation of a pie diagram? ☑ 2014 II

- (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II

- 88.** Consider the following statements related to cumulative frequency polygon of a frequency distribution, the frequencies being cumulated from the lower end of the range

- I. The cumulative frequency polygon gives an equivalent representation of frequency distribution table.
- II. The cumulative frequency polygon is a closed polygon with one horizontal and one vertical side. The other sides have non-negative slope.

Which of the statement(s) given above is/are correct? **2014 II**

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II
- 89.** The class which has maximum frequency is known as **2014 II**
(a) median class (b) mean class
(c) modal class (d) None of these
- 90.** Ten observations 6, 14, 15, 17, $x + 1$, $2x - 13$, 30, 32, 34 and 43 are written in ascending order. The median of the data is 24. What is the value of x ? **2014 II**
(a) 15 (b) 18 (c) 20 (d) 24

Directions (Q. Nos. 91-94) Consider the following frequency distribution. **2015 I**

Class	Frequency
0-10	4
10-20	5
20-30	7
30-40	10
40-50	12
50-60	8
60-70	4

- 91.** What is the mean of the distribution?
(a) 37.2 (b) 38.1 (c) 39.2 (d) 40.1
- 92.** What is the median class?
(a) 20-30 (b) 30-40
(c) 40-50 (d) 50-60
- 93.** What is the median of the distribution?
(a) 37 (b) 38 (c) 39 (d) 40
- 94.** What is the mode of the distribution?
(a) 38.33 (b) 40.66
(c) 42.66 (d) 43.33
- 95.** There are five parties A, B, C, D and E in an election. Out of total 100000 votes cast, 36000 were cast to party A, 24000 to party B, 18000 to party C, 7000 to party D and rest to party E. What angle will be allocated for party E in the pie chart? **2015 I**
(a) 15° (b) 54° (c) 60° (d) 72°

- 96.** If a variable takes discrete values $a + 4$, $a - 3.5$, $a - 2.5$, $a - 3$, $a - 2$, $a + 0.5$, $a + 5$ and $a - 0.5$, where $a > 0$, then the median of the data set is **2015 II**

(a) $a - 2.5$ (b) $a - 1.25$ (c) $a - 1.5$ (d) $a - 0.75$

- 97.** If each of n numbers $x_i = i$ ($i = 1, 2, 3, \dots, n$) is replaced by $(i + 1)x_i$, then the new mean is **2015 II**

(a) $\frac{n+3}{2}$ (b) $\frac{n(n+1)}{2}$
(c) $\frac{(n+1)(n+2)}{3n}$ (d) $\frac{(n+1)(n+2)}{3}$

- 98.** The weighted arithmetic mean of first 10 natural numbers whose weights are equal to the corresponding numbers is equal to **2015 II**

(a) 7 (b) 14 (c) 35 (d) 38.5

- 99.** The election result in which six parties contested was depicted by a pie chart. Party A had an angle 135° on this pie chart. If it secured 21960 votes, then how many valid votes in total were cast? **2016 I**

(a) 51240 (b) 58560 (c) 78320 (d) 87840

- 100.** The mean and median of 5 observations are 9 and 8, respectively. If 1 is subtracted from each observation, then the new mean and the new median will respectively be **2016 I**

(a) 8 and 7 (b) 9 and 7
(c) 8 and 9 (d) Cannot be determined due to insufficient data

- 101.** The age distribution of 40 children are as follows

Age (in years)	5-6	6-7	7-8	8-9	9-10	10-11
Number of children	4	7	9	12	6	2

Consider the following statements in respect of the above frequency distribution

- I. The median of the age distribution is 7 yr.
II. 70% of the children are in the age group 6-9 yr.
III. The modal age of the children is 8 yr.

Which of the above statements are correct?

2016 I
(a) I and II (b) II and III (c) I and III (d) I, II and III

- 102.** Suppose $x_i = \lambda^i$ for $0 \leq i \leq 10$, where $\lambda > 1$. Which one of the following is correct? **2016 I**

(a) AM < Median (b) GM < Median
(c) GM = Median (d) AM = Median

- 103.** Suppose $x_i = \frac{1}{i}$ for $i = 1, 2, 3, \dots, 11$. Which one of the following is not correct? **2016 I**

(a) AM > $1/6$ (b) GM > $1/6$
(c) HM > $1/6$ (d) Median = HM

ANSWERS

1	c	2	b	3	b	4	c	5	a	6	b	7	c	8	c	9	d	10	b
11	d	12	b	13	d	14	a	15	a	16	c	17	a	18	b	19	b	20	b
21	d	22	a	23	c	24	d	25	a	26	a	27	c	28	b	29	b	30	d
31	c	32	a	33	d	34	b	35	b	36	d	37	c	38	c	39	c	40	b
41	b	42	b	43	b	44	b	45	c	46	b	47	c	48	b	49	c	50	b
51	d	52	d	53	c	54	c	55	b	56	c	57	b	58	d	59	b	60	c
61	b	62	b	63	a	64	b	65	a	66	c	67	b	68	c	69	d	70	b
71	b	72	c	73	a	74	a	75	b	76	a	77	d	78	a	79	a	80	c
81	c	82	b	83	c	84	c	85	a	86	d	87	c	88	a	89	c	90	c
91	a	92	b	93	c	94	d	95	b	96	b	97	d	98	d	99	b	100	a
101	b	102	c	103	d														

HINTS AND SOLUTIONS

- (c) Frequency polygon can be drawn by joining the mid-points of the respective tops in histogram.
- (b) An ogive is used to determine the median.
- (b) Let the lower limit be x . Then, the upper limit of class interval = $x + 10$
 $\therefore$ mid-value = 42
 $\therefore \frac{x + (x + 10)}{2} = 42 \Rightarrow 2x + 10 = 84$
 $\Rightarrow 2x = 74 \Rightarrow x = 37$
 $\therefore$ Lower limit = 37
Upper limit = $37 + 10 = 47$
- (c) Lower class limits are obtained by subtracting 0.5 from the lower limit, so clearly 9.5, 19.5, 29.5 and 39.5 are the actual lower class limits.
- (a) Here, sum of 5 observations
 $= x + (x + 2) + (x + 4) + (x + 6) + (x + 8)$
 $= 5x + 20$
and total number of observation = 5
 $\therefore$ Mean of total observations
 $= \frac{\text{Sum of 5 observation}}{\text{Total number of observation}}$
 $= \frac{5x + 20}{5} = 11 \Rightarrow 5x + 20 = 55$
 $\Rightarrow 5x = 55 - 20 = 35 \Rightarrow x = 7$
 $\therefore$ Mean of first three
 $= \frac{x + (x + 2) + (x + 4)}{3} = \frac{3x + 6}{3}$
 $= \frac{3 \times 7 + 6}{3} = \frac{27}{3} = 9 \quad (\because x = 7)$

Hence, the mean of first three observation is 9.

- (b) Total sum of items
 $= 2 \times 12 + 3 \times 12 + 5 \times 12$
 $= 24 + 36 + 60 = 120$
Total sum of items of first two group
 $= (2 + 3) \times 3 = 15$
Total sum of 5 items of group third
 $= 120 - 15 = 105$
 $\therefore$ Mean of third group = $\frac{105}{5} = 21$
Hence, the mean of third group is 21.
- (c) Given, $n_1 = 7, \bar{x}_1 = 10$
and $n_2 = 3, \bar{x}_2 = 5$
 $\therefore$ Combined mean = $\frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2}$
 $= \frac{7 \times 10 + 3 \times 5}{7 + 3} = \frac{85}{10} = 8.5$
Hence, the mean of combined set is 8.5.
- (c) Given, total number of students = 50
Total number of students passed
 $= 50 - 10 = 40$
Total marks obtained by failed students
 $= 28 \times 10 = 280$
 $\therefore$ Total marks of 40 students
 $= 2800 - 280 = 2520$
 $\therefore$ Required average of passed students
 $= \frac{2520}{40} = 63$
- (d) Clearly, all the statements are true about median.

- (b) Median is the middle item of the series arranged in ascending or descending order.

- (d) Central angle for 540 marks = 360° .
 $\therefore$ Central angle for 108 marks
 $= \frac{108}{540} \times 360^\circ = 72^\circ$

Here, the student scored 108 marks in Social Science.

- (b) Given, $n_1 = 45, \bar{x}_1 = 2, n_2 = 40$
 $\bar{x}_2 = 2.5$ and $n_3 = 15, \bar{x}_3 = 2$.
 $\therefore$ Required mean
 $= \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2 + n_3 \bar{x}_3}{n_1 + n_2 + n_3}$
 $= \frac{45 \times 2 + 40 \times 2.5 + 15 \times 2}{45 + 40 + 15}$
 $= \frac{90 + 100 + 30}{45 + 40 + 15} = \frac{220}{100} = 2.2$

Hence, the mean of the combined distribution is 2.2.

- (d) Here,
 $\Sigma f_i = (1 + 2 + 3 + \dots + n) = \frac{n(n+1)}{2}$
 $\Sigma f_i \times x_i = \left(1 \times 1 + 2 \times \frac{1}{2} + 3 \times \frac{1}{3} + \dots + n \times \frac{1}{n} \right)$
 $= (1 + 1 + 1 + \dots + n \text{ times}) = n$
 $\therefore \text{Mean} = \frac{\Sigma f_i x_i}{\Sigma f_i} = \frac{n}{\frac{1}{2} n(n+1)} = \frac{2}{n+1}$

14. (a) Geometric mean of 40, 50 and $x = (40 \times 50 \times x)^{1/3}$
 $\Rightarrow (40 \times 50 \times x)^{1/3} = 10$ (given)
 $\Rightarrow 40 \times 50 \times x = 10^3 \Rightarrow x = \frac{1000}{40 \times 50} = \frac{1}{2}$
Hence, the value of x is $\frac{1}{2}$.

15. (a) We know that the sum of deviation from their mean is zero.
 $\therefore x_1 f_1 + x_2 f_2 + \dots + x_n f_n = 0$

16. (c) The height of the bar is not proportional to the frequency of the class.

17. (a)

x	f	$f \times x$
0	4	0
1	f	f
2	9	18
3	g	$3g$
4	4	16
Total	$\Sigma f = 17 + f + g$	$\Sigma fx = 34 + f + 3g$

$$\therefore \text{Mean} = \frac{\Sigma fx}{\Sigma f} = \frac{34 + f + 3g}{17 + f + g} \Rightarrow 2 = \frac{34 + f + 3g}{17 + f + g} \quad (\because \text{mean} = 2)$$

$$\Rightarrow 34 + 2f + 2g = 34 + f + 3g \Rightarrow f - g = 0 \quad \dots(i)$$

$$\text{Given, } \Sigma f = 25$$

$$\therefore 17 + f + g = 25 \Rightarrow f + g = 25 - 17$$

$$\Rightarrow f + g = 8 \quad \dots(ii)$$

$$\text{On adding Eqs. (i) and (ii), we get } 2f = 8 \Rightarrow f = 4$$

$$\text{On putting } f = 4 \text{ Eq. (i), we get } 4 - g = 0 \Rightarrow g = 4$$

18. (b) For district A, as the class 44.0-47.0 has maximum frequency, so it is the modal class.

$$\therefore l = 44, f_1 = 59, f_0 = 36, f_2 = 30, \text{ and } h = 3$$

$$\therefore \text{Mode} = l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 44 + \frac{59 - 36}{2 \times 59 - 36 - 30} \times 3$$

$$= 44 + \frac{23 \times 3}{52} = 44 + 1.33 = 45.33$$

For district B, as the class 47.0-50.0 has the maximum frequency, so it is the modal class.

$$\therefore l = 47, f_1 = 54, f_0 = 35, f_2 = 41 \text{ and } h = 3$$

$$\therefore \text{Mode} = l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 47 + \frac{54 - 35}{2 \times 54 - 35 - 41} \times 3$$

$$= 47 + \frac{19 \times 3}{32} = 47 + 1.78 = 48.78$$

Thus, the mode for district B is higher than the mode for district A.

19. (b) Given, average score of 50 students = 44

$$\text{Total score} = 44 \times 50 = 2200$$

$$\text{Correct score of 50 students} = (2200 - 73 + 23) = 2150$$

$$\therefore \text{Correct average score} = \frac{2150}{50} = 43$$

20. (b) In the less than type of ogive, the cumulative frequency is plotted against the upper limit of the concerned class interval.

21. (d) Mean of A = $\frac{2+3+7+1+3+2+3}{7} = \frac{21}{7} = 3$

Mode of A = 3, having maximum frequency.

Arrange data in ascending order, we have 1, 2, 2, 3, 3, 3, 7.

and median of A = $\left(\frac{7+1}{2}\right)$ th value = 4th

value = 3

Here, mean, median and mode are equal.

22. (a) Total marks in 3 subjects = $(75 + 80 + 85) = 240$ out of 300.
In any other subject the marks are atleast 0.

$$\text{Average of 4 subjects} = \frac{240}{4} = 60\%$$

$\therefore$ Average cannot be less than 60%.

23. (c) Given, mean = 8.9 and median = 9

$$\therefore \text{Mode} = (3 \times \text{Median}) - (2 \times \text{Mean})$$

$$= (3 \times 9 - 2 \times 8.9) = 27 - 17.8 = 9.2$$

24. (d) If every number of a finite set is increased by any number k , then measure of central tendency should also increase by k .

But geometric mean does not have this property.

25. (a) Given distribution is 1, 3, 5, 7, 9, x , 15, 17, 19, 21.
Number of terms = 10 (even)

$$\therefore \text{Median} = \frac{\text{Value of } \frac{10}{2} \text{th term} + \text{Value of } \left(\frac{10}{2} + 1\right) \text{th term}}{2}$$

$$10 = \frac{\text{Value of 5th term} + \text{Value of 6th term}}{2}$$

$$\Rightarrow 10 = \frac{9 + x}{2} \Rightarrow 20 = 9 + x \Rightarrow x = 11$$

26. (a) Required class boundary = Lower class boundary of lowest class + Width of class $\times$ Number of class
 $= 5.1 + 2.5 \times 10 = 30.1$

27. (c) Arranging the observation in ascending order 2, 3, 4, 6, 7, 7, 8

$\therefore$ Number of items = 7

$$\therefore \text{Median} = \left(\frac{n+1}{2}\right) \text{th value} = \left(\frac{7+1}{2}\right) = 4 \text{th value} = 6$$

So, median = 6 and mode of 2, 9, 3, 4, 9, 6, 9 is 9.

$$\therefore \text{Average of median and mode} = \frac{1}{2}(6 + 9) = 7.5$$

28. (b) Let the mean of observations $x_1, x_2, x_3, \dots, x_n$ be $\bar{x}$.

On subtracting 10 from each, the mean would be $(\bar{x} - 10)$.

$$(\bar{x} - 10) = 60\% \text{ of } \bar{x}$$

$$\Rightarrow (\bar{x} - 10) = \frac{60}{100} \bar{x} = \frac{3}{5} \bar{x} \Rightarrow \left(1 - \frac{3}{5}\right) \bar{x} = 10$$

$$\Rightarrow \frac{2}{5} \bar{x} = 10 \Rightarrow \bar{x} = \frac{5 \times 10}{2} = 25$$

Now, 5 is added to all the given observations, then new mean becomes $= \bar{x} + 5 = 25 + 5 = 30$

29. (b) Range of the data = $120 - 71 = 49$

$$\therefore \text{Class size} = \frac{\text{Range}}{\text{Number of classes}} = \frac{49}{7} = 7$$

The class are 71-78, 78-85, 85-92, 92-97 etc.

So, limits of second class interval are 78 and 85.

30. (d) Here, mean, $M = \frac{y + \frac{1}{y}}{2} \dots (i)$

New mean

$$= \frac{y^3 + \frac{1}{y^3}}{2} = \frac{\left(y + \frac{1}{y}\right)^3 - 3\left(y + \frac{1}{y}\right)}{2}$$

$$= \frac{\left(y + \frac{1}{y}\right) \left[\left(y + \frac{1}{y}\right)^2 - 3\right]}{2} \text{ [from Eq. (i)]}$$

$$= M[(2M)^2 - 3] = M(4M^2 - 3)$$

31. (c) Since, maximum frequency is 80, hence mode will be between 15 to 20.

32. (a) ₹ 25 = 1 cm² $\Rightarrow$ ₹ 1 = $\frac{1}{25}$ cm²

$\therefore$ ₹ 81 = $\frac{81}{25}$ cm² = Area of square

Side of square = $\sqrt{\frac{81}{25}} = \frac{9}{5} = 1.8$ cm

33. (d) Given, $n = 5$

Mean = $\frac{7 + 9 + 11 + 13 + 15}{5} = \frac{55}{5} = 11$

$\Sigma |d_i|^2 = |7 - 11|^2 + |9 - 11|^2 + |11 - 11|^2$
 $+ |13 - 11|^2 + |15 - 11|^2 = 40$

$\therefore \sigma = \sqrt{\frac{\Sigma d_i^2}{n}} = \sqrt{\frac{40}{5}} = 2\sqrt{2} = 2 \times 1.41$
 $= 2.82$

34. (b) Number of cellphones sold by Motorola = $\frac{3}{16} \times 45664$

Central angle of Motorola

$$= \left(\frac{\frac{3}{16} \times 45664}{45664} \times 360^\circ \right) = \left(\frac{3}{16} \times 360^\circ \right)$$

$$= 67.5^\circ$$

35. (b) If the population figures are given for each State of India, then data can be classified as quantitative.

36. (d) The volume of rainfall in certain geographical area, recorded every month for 24 consecutive months.

37. (c) I. First seven prime numbers are 2, 3, 5, 7, 11, 13, 17.

Mean = $\frac{2 + 3 + 5 + 7 + 11 + 13 + 17}{7}$
 $= \frac{58}{7} = 8.28$

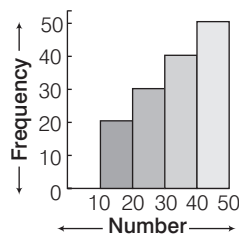
Median = Value of $\left(\frac{7+1}{2}\right)$ th term

= Value of 4th term = 7

Since, prime numbers are always increasing, therefore here mean is always greater than median.

Hence, both statements are correct.

38. (c) Since, the histogram consists of vertical rectangular bars with a common base such that there is no gap between consecutive bars and also the height of the rectangle is determined by the frequency of the class it represents. Hence, both statements are correct in respect of a histogram.



39. (c) We know that, histogram is an equivalent graphical representation of the frequency distribution and is suitable for continuous random variables where the total frequency of an interval is evenly distributed over the interval.

Hence, both the given statements are correct.

40. (b) Primary data is original research that is obtained through first hand investigation. So, statements I and III are primary data.

41. (b) Collar sizes of 200 shirts sold in a week, here mode is a suitable measure of central tendency.

42. (b) Since, the frequency in a straight line, so we take all classes have equal frequency, i.e. 10.

I. It is true that first and last class have 10 frequency.

II. Both the middle classes have frequency, $10 + 10 = 20$

III. Since, all have equal frequency, so we cannot determine the mode.

43. (b) A discrete frequency distribution is such a distribution in which data are presented in a way that exact measurements of the units are clearly shown. Clearly weights of a set of a students is continuous, while other three are discrete.

44. (b) Since, modal size of given data is 1 and its corresponding frequency is 6.

So, x cannot be equal to or more than 6.

Hence, the maximum value of x is 5.

45. (c)

Number of goals	Frequency	CF
0	3	3
1	6	9
2	2	11
3	x	$11 + x$
4	5	$16 + x$
Total	$N = 16 + x$	

Since, median is 2 and the corresponding CF is 11.

So, $\frac{N}{2}$ must be greater than 9.

$\Rightarrow \frac{16 + x}{2} > 9 \Rightarrow x > (18 - 16) \Rightarrow x > 2$

So, smallest possible value of $x = 3$.

46. (b) Here, total expenditure
 $= 6 + 7 + 3 + 4 + 25 + 1.5$
 $= ₹ 24$ lakh

Sectorial angle of largest sector
 $= \frac{360^\circ}{24} \times 7 = 105^\circ$

47. (c) Difference in the sectorial angles of the largest and the smallest sectors

$= 105^\circ - \frac{360^\circ}{24} \times 1.5$
 $= 105^\circ - 22.5^\circ = 82.5^\circ$

48. (b) Here, $l = 18$, $\frac{N}{2} = 50$, $c = 7 + x$,

$f = 40$ and $h = 5$

$\therefore$ Median = $l + \frac{\left(\frac{N}{2} - c\right)}{f} \times b$
 $= 18 + \frac{(50 - (7 + x))}{40} \times 5$

$\therefore 20 = 18 + \frac{5}{40} (43 - x)$

[$\because$ median = 20]

$\Rightarrow 800 = 720 + 215 - 5x$

$\Rightarrow 5x = 935 - 800$

$\Rightarrow 5x = 135 \Rightarrow x = \frac{135}{5} \Rightarrow x = 27$

Hence, the value of x is 27.

49. (c) We have,

$7 + x + 40 + y + 10 + 2 = 100$

$x = 27$

$\Rightarrow 86 + y = 100 \Rightarrow y = 100 - 86$

$\Rightarrow y = 14$

Hence, the value of y is 14.

Solutions (Q. Nos. 50-52) Since, $a < b < c$

Therefore, series is in increasing order i.e. a, a, b, b, c, c

$\therefore$ Median

$$= \frac{\left(\frac{6}{2}\right)\text{th term} + \left(\frac{6}{2} + 1\right)\text{th term}}{2}$$

$$= \frac{3\text{rd term} + 4\text{th term}}{2}$$

$\Rightarrow 2 = \frac{b + b}{2} \Rightarrow b = 2$

Also, arithmetic mean

$$= \frac{a + a + b + b + c + c}{6}$$

$\Rightarrow \frac{7}{3} = \frac{a + b + c}{3} \Rightarrow a + c + 2 = 7$

$$\Rightarrow a + c = 7 - 2 = 5 \quad \dots(i)$$

and geometric mean $= (a^2 \times b^2 \times c^2)^{1/6}$

$$\Rightarrow 2 = (abc)^{1/3} \Rightarrow abc = 8 \Rightarrow ac = \frac{8}{b} = 4 \quad [\because b = 2] \dots(ii)$$

$\therefore$ From Eqs. (i) and (ii), we get, $a = 1$, $c = 4$ and $b = 2$

50. (b) Required sum $= 2(a)^2 + 2(b)^2 + 2(c)^2$
 $= 2(1)^2 + 2(2)^2 + 2(4)^2 = 2 + 8 + 32 = 42$

51. (d) The value of c is 4.

52. (d) $\therefore$ Mode = 3 (median) - 2 (mean)
 $= 3(2) - 2\left(\frac{7}{3}\right) = 6 - \frac{14}{3} = \frac{18-14}{3} = \frac{4}{3}$

53. (c) Percentage of wheat production $= \frac{70^\circ}{360^\circ} \times 100\% = 19.44\%$

Hence, the percentage of wheat production is 19.44%.

54. (c) If $40^\circ = 9000 \text{ kg} \Rightarrow 70^\circ = \frac{9000 \times 70}{40} = 15750 \text{ kg}$

Hence, the wheat production is 15750 kg.

55. (b) $360^\circ = 180000 \text{ kg}$

Difference in Sugar and wheat angle $= (70^\circ - 40^\circ) = 30^\circ$

$\therefore$ Required production $= \frac{180000 \times 30}{360} = 15000 \text{ kg}$

Hence, the difference in sugar and wheat production is 15000 kg.

Solutions (Q. Nos. 56-58)

Class interval	f _i	cf	x _i	f _i x _i
1-4	6	6	2.5	15.0
4-7	30	36	5.5	165.0
7-10	40	76	8.5	340.0
10-13	16	92	11.5	184.0
13-16	4	96	14.5	58.0
16-19	4	100	17.5	70.0
Total	N = 100			832.0

56. (c) We have, $\frac{N}{2} = \frac{100}{2} = 50$, so cf just greater than 50 is 76 which lies in class 7-10.

$\therefore$ Median class = 7-10, $l = 7$, $f = 40$, $c = 36$, $h = 3$

$$\text{Median} = l + \left(\frac{\frac{N}{2} - c}{f} \right) \times h = 7 + \left(\frac{50 - 36}{40} \right) \times 3 = 7 + \frac{42}{40} = 8.05$$

57. (b) The maximum frequency is 40, which lies in the class = 7-10

$\therefore$ Modal class = 7-10 $\Rightarrow l = 7$, $f_1 = 40$, $f_0 = 30$, $f_2 = 16$, $h = 3$

$$\text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) \times h$$

$$= 7 + \left(\frac{40 - 30}{2 \times 40 - 30 - 16} \right) \times 3 = 7 + \frac{30}{34} = 7 + 0.88 = 7.88$$

58. (d) Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{832}{100} = 8.32$

$\therefore$ Mean + Mode - 2 Median $= 8.32 + 7.88 - 2 \times 8.05$
 $= 16.2 - 16.1 = 0.1$

59. (b) In an increasing order arrangements of observations, the median will not be affected when the smallest and the largest observations are removed.

60. (c) Primary data are those data which is collected by investigator himself, while secondary data are those which is collected by other persons.

Hence, both statements are correct.

61. (b) In pie diagrams, all the items are converted into angles.

62. (b) Given that, the arithmetic mean of the above distribution = 3.1

$$\Rightarrow \frac{\sum_{i=1}^5 x_i f_i}{\sum_{i=1}^5 f_i} = 3.1 \Rightarrow \frac{x_1 f_1 + x_2 f_2 + x_3 f_3 + x_4 f_4 + x_5 f_5}{f_1 + f_2 + f_3 + f_4 + f_5} = 3.1$$

$$\Rightarrow \frac{(1)(3) + (2)(f) + (3)(6) + (4)(5) + (5)(3)}{3 + f + 6 + 5 + 3} = 3.1$$

$$\Rightarrow \frac{3 + 2f + 18 + 20 + 15}{17 + f} = 3.1 \Rightarrow 2f + 56 = 3.1f + 52.7$$

$$\Rightarrow 1.1f = 3.3 \Rightarrow f = 3$$

Hence, the value of f is 3.

63. (a) The weight of a person is continuous variable because the weight of the persons can be divide into different but certain range of intervals. But educational qualification of the person does not show the attributes. Hence, statement I is true and statement II is false.

64. (b) Let 100 value be $x_1, x_2, x_3, \dots, x_{100}$

$$\therefore \text{Mean} = \frac{\text{Sum of values}}{\text{Total number of values}}, 45 = \frac{x_1 + x_2 + x_3 + \dots + x_{100}}{100}$$

$$\Rightarrow x_1 + x_2 + x_3 + \dots + x_{100} = 4500 \quad \dots(i)$$

Now, according to the question

$\therefore$ Sum of new value

$$= (x_1 + 15 + x_2 + 15 + \dots + x_{40} + 5) + x_{41} - 5 + x_{42} - 5 + \dots + x_{100} - 5$$

$$\therefore \text{New mean} = \frac{\text{Sum of new value}}{\text{Number of values}}$$

$$\frac{(x_1 + x_2 + x_3 + \dots + x_{40}) + 40 \times 15 + (x_{41} + x_{42} + \dots + x_{100}) - 5 \times 60}{100}$$

$$= \frac{(x_1 + x_2 + \dots + x_{100}) + 600 - 300}{100} = \frac{4500 + 300}{100} \quad [\text{from Eq. (i)}]$$

$$= \frac{4800}{100} = 48$$

Hence, the new mean is 48.

65. (a) Given numbers are 10, 7, 8, 5, 6, 8, 5, 8 and 6.

First we arrange these numbers in ascending order.

$$5, 5, 6, 6, 7, 8, 8, 8, 10$$

Total term, $n = 9$ (odd)

Now,

(i) Mean $= \frac{5 + 5 + 6 + 6 + 7 + 8 + 8 + 8 + 10}{9} = \frac{63}{9} = 7$

(ii) Median $= \left(\frac{n+1}{2} \right)^{\text{th}} \text{ term} = \left(\frac{9+1}{2} \right)^{\text{th}} \text{ term} = 5^{\text{th}} \text{ term} = 7$

(iii) Mode = 8 (higher frequency term)

$\therefore$ Mean = Median

66. (c) In histogram the width of the bars is proportional to class interval.

67. (b) A frequency polygon can be drawn joining the mid-points of the respective tops of the rectangle of a histogram in the case of equal class intervals.

68. (c) A frequency distribution is a comprehensive way to classify raw data of a quantitative variable. It shows how different values of a variable are distributed in different classes along with their corresponding class frequencies. It is an equivalent representation of original data.

69. (d)

CI	x	f	$f \times x$
0-10	5	x	$5x$
10-20	15	$x+8$	$15(x+8)$
20-30	25	$x-2$	$25(x-2)$
30-40	35	$x-4$	$35(x-4)$
Total		$\Sigma f = 4x+2$	$\Sigma fx = 80x-70$

$$\therefore \text{Mean} = \frac{\Sigma fx}{\Sigma f} \Rightarrow 17.8 = \frac{80x-70}{4x+2}$$

$$\Rightarrow 17.8(4x+2) = 80x-70 \Rightarrow 71.2x+35.6 = 80x-70$$

$$\Rightarrow 8.8x = 105.6 \Rightarrow x = \frac{105.6}{8.8} = 12$$

70. (b)

Class interval (ci)	0-10	10-20	20-30	30-40
Frequencies (f)	12	20	10	8
Cumulative frequency (cf)	12	32	42	50

Here, $N = 50 \Rightarrow \frac{N}{2} = 25$

The frequency just greater than 25 is 32.

So, median class is 10-20.

71. (b) Here, modal class is 10-20, because it has maximum frequency i.e. 20.

72. (c) Given that, mean of 7 observations = 7

$$\Rightarrow \text{New mean} = \text{Previous mean} + \frac{\text{Number added to each term}}{7}$$

$$= 7 + 2 = 9$$

73. (a) Given that, number of male employees (M) = 45

Number of female employees (F) = 15

Mean salary of male employees ($\bar{x}_M$) = ₹ 5000

Total number of employees = $(M + F) = 45 + 15 = 60$

Mean salary of total employees ($\bar{x}_{MF}$) = ₹ 4800

Let mean salary of female employees be $\bar{x}_F$.

$$\bar{x}_{MF} = \frac{M\bar{x}_M + F\bar{x}_F}{(M+F)} \Rightarrow 4800 = \frac{45 \times 5000 + 15 \times \bar{x}_F}{60}$$

$$\Rightarrow 4800 \times 60 - 45 \times 5000 = 15 \times \bar{x}_F$$

$$\therefore \bar{x}_F = \frac{4800 \times 60 - 45 \times 5000}{15} = \frac{300(16 \times 4 - 50)}{15}$$

$$= 300 \times 14 = ₹ 4200$$

Hence, the mean salary of female employees is ₹ 4200.

74. (a) Median = $l + \left(\frac{\frac{n}{2} - cf}{f} \right) \times h$

$$\Rightarrow 14.4 = 12 + \left(\frac{\frac{20}{2} - (4+x)}{5} \right) \times 6 = 12 + \frac{10-4-x}{5} \times 6$$

$$\Rightarrow 14.4 - 12 = \frac{6-x}{5} \times 6 \Rightarrow 2.4 = \frac{36-6x}{5}$$

$$\Rightarrow 12 = 36 - 6x \Rightarrow 6x = 24 \Rightarrow x = 4$$

75. (b) We have, $4 + x + 5 + y + 1 = 20 \Rightarrow x + y = 10$

$$\Rightarrow 4 + y = 10 \quad (\because x = 4) \Rightarrow y = 6$$

$$\text{Now, } \frac{x}{y} = \frac{4}{6} = \frac{2}{3} \Rightarrow 3x = 2y$$

76. (a) $\therefore \text{Mean} = \frac{\text{Sum of the observations}}{\text{Number of observations}}$

$$\Rightarrow \frac{p+q+r+s+t+u+v}{7} = m$$

$$\Rightarrow p+q+r+s+t+u+v = 7m \quad \dots(i)$$

$$\therefore (p-m) + (q-m) + (r-m) + (s-m) + (t-m)$$

$$+ (u-m) + (v-m)$$

$$= (p+q+r+s+t+u+v) - 7m \quad [\text{from Eq. (i)}]$$

$$= 7m - 7m = 0$$

77. (d) $\therefore \text{Average} = \frac{\text{Sum of the observations}}{\text{Number of observations}}$

$$\Rightarrow \frac{u+v+w+x+y+z}{6} = 10 \quad (\because \text{average} = 10)$$

$$\Rightarrow u+v+w+x+y+z = 10 \times 6 = 60 \quad \dots(ii)$$

Now, sum of new observations

$$u+10+v+20+w+30+x+40+y+50+z+60$$

$$= (u+v+w+x+y+z) + 10+20+30+40+50+60$$

$$= 60 + 10 + 20 + 30 + 40 + 50 + 60 = 270 \quad [\text{from Eq. (i)}]$$

Now, average of $u+10, v+20, w+30, x+40, y+50$

$$\text{and } z+60 = \frac{270}{6} = 45$$

78. (a)

Class interval	x_i	f_i	$x_i f_i$
11-13	12	3	36
13-15	14	6	84
15-17	16	9	144
17-19	18	13	234
19-21	20	f	$20f$
21-23	22	5	110
23-25	24	4	96
Total		$\Sigma f_i = 40 + f$	$\Sigma x_i f_i = 704 + 20f$

Here, $\Sigma f_i = 40 + f$ and $\Sigma x_i f_i = 704 + 20f$ and mean, $\bar{x} = 18$

$$\therefore \text{Mean, } \bar{x} = \frac{\Sigma x_i f_i}{\Sigma f_i} \Rightarrow 18 = \frac{704 + 20f}{40 + f}$$

$$\Rightarrow 704 + 20f = 720 + 18f$$

$$\Rightarrow 20f - 18f = 720 - 704 \Rightarrow 2f = 16 \Rightarrow f = 8$$

Hence, the value of f is 8.

79. (a) Frequency polygon is formed by joining the mid-points of histogram. The original frequency distribution can be reconstructed from frequency polygon. Frequency polygon does not touch the X -axis in its extreme right and extreme left.

Hence, only statement I is true.

80. (c) I. Mean of all observations = $\frac{2 \times 4 + 7 \times 4}{8} = \frac{8 + 28}{8} = 4.5$

For median, first we arrange the given observations in ascending order = 2, 2, 2, 2, 7, 7, 7, 7, $n = 8$ (even)

$$\therefore \text{Median} = \frac{\frac{n}{2} \text{th term} + \left(\frac{n}{2} + 1\right) \text{th term}}{2}$$

$$= \frac{\frac{8}{2} \text{th term} + \left(\frac{8}{2} + 1\right) \text{th term}}{2} = \frac{4\text{th} + 5\text{th}}{2} = \frac{2 + 7}{2} = 4.5$$

II. Mode is both 2 and 7, since frequency of occurrence is same i.e. maximum frequency.

81. (c) A list of annual incomes of the members of a club is a good example of statistical data.

82. (b)

Income less than	Class interval	Number of persons	Frequency
1500	1250-1500	100	20
1250	1000-1250	80	10
1000	750-1000	70	15
750	500-750	55	23
500	250-500	32	20
250	0-250	12	12

Here, maximum frequency is 23.

So, the modal class is 500-750.

83. (c) Geometric mean of $(x_1, x_2, x_3, \dots, x_n) = (x_1 \cdot x_2 \cdot \dots \cdot x_n)^{1/n} = X$
and geometric mean of $(y_1, y_2, y_3, \dots, y_n)$
 $= (y_1 \cdot y_2 \cdot \dots \cdot y_n)^{1/n} = Y$
 $\therefore$ Geometric mean of $(x_1 y_1, x_2 y_2, \dots, x_n y_n)$
 $= (x_1 y_1 \cdot x_2 y_2 \cdot \dots \cdot x_n y_n)^{1/n}$
 $= (x_1 \cdot x_2 \cdot \dots \cdot x_n)^{1/n} \times (y_1 \cdot y_2 \cdot \dots \cdot y_n)^{1/n} = XY$
 Geometric mean of $\left(\frac{x_1}{y_1}, \frac{x_2}{y_2}, \dots, \frac{x_n}{y_n}\right) = \left(\frac{x_1}{y_1} \cdot \frac{x_2}{y_2} \cdot \dots \cdot \frac{x_n}{y_n}\right)^{1/n}$
 $= \frac{(x_1 \cdot x_2 \cdot \dots \cdot x_n)^{1/n}}{(y_1 \cdot y_2 \cdot \dots \cdot y_n)^{1/n}} = \frac{X}{Y}$

84. (c) Arithmetic mean uses all the data and is always uniquely defined. Median uses only one or two numbers from the data and may not be unique. $\left(\frac{n+1}{2}\right)$ th term for odd n and half the sum of $\left(\frac{n}{2} + 1\right)$ th and $\frac{n}{2}$ th terms for even n .
85. (a) If x_i 's are the mid-points of the class intervals of grouped data, f_i 's are the corresponding frequencies and $\bar{x}$ is the mean, then $\sum f_i (x_i - \bar{x}) = 0$
86. (d) When we take class intervals on the X-axis and corresponding frequencies on the Y-axis and draw rectangles with the areas proportional to the frequencies of the respective class intervals, the graph so obtained is called histogram.
87. (c) Both statements I and II are suitable for representation of a pie diagram.
88. (a) Here, statement I is correct but statement II is not correct.
89. (c) The class which has maximum frequency is known as modal class.
90. (c) Given observations in ascending order are 6, 14, 15, 17, $x + 1$, $2x - 13$, 30, 32, 34 and 43.

Here, $n = 10$

(even)

$$\therefore \text{Median} = \frac{\text{Value of } \left(\frac{n}{2}\right) \text{th term} + \text{Value of } \left(\frac{n}{2} + 1\right) \text{th term}}{2}$$

$$= \frac{\text{Value of } \left(\frac{10}{2}\right) \text{th term} + \text{Value of } \left(\frac{10}{2} + 1\right) \text{th term}}{2}$$

$$= \frac{\text{Value of 5th term} + \text{Value of 6th term}}{2}$$

$$= \frac{x + 1 + 2x - 13}{2} = \frac{3x - 12}{2}$$

But given, median = 24 $\Rightarrow \frac{3x - 12}{2} = 24$

$\Rightarrow 3x - 12 = 24 \times 2 = 48 \Rightarrow 3x = 48 + 12$

$\Rightarrow 3x = 60 \Rightarrow x = 20$

Hence, the value of x is 20.

Solutions (Q. Nos. 91-94)

Class interval	Frequency (f)	Class mark (x)	f(x)	cf
0-10	4	5	20	4
10-20	5	15	75	9
20-30	7	25	175	16
30-40	10	35	350	26
40-50	12	45	540	38
50-60	8	55	440	46
60-70	4	65	260	50
Total	50		1860	

91. (a) Mean = $\frac{\sum fx}{\sum f} = \frac{1860}{50} = 37.2$

Hence, the value of mean is 37.2.

92. (b) Here, $N = 50$, Now, $\frac{N}{2} = \frac{50}{2} = 25$

which lies in the cumulative frequency corresponding class interval for cf 26 is 30-40.

93. (c) From the table, $l = 30$, $f = 10$, $C = 16$ and $h = 10$

$$\therefore \text{Median} = l + \frac{\left(\frac{N}{2} - C\right)}{f} \times h$$

$$= 30 + \frac{(25 - 16)}{10} \times 10 = 30 + 9 = 39$$

94. (d) Modal class of the given data is 40-50, because it has largest frequency among the given classes of the data i.e. 12.

Here, $l = 40$, $f_1 = 12$, $f_0 = 10$, $f_2 = 8$ and $h = 10$

$$\therefore \text{Mode} = l + \left\{ \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right\} \times h = 40 + \left\{ \frac{12 - 10}{2 \times 12 - 10 - 8} \right\} \times 10$$

$$= 40 + \frac{2 \times 10}{24 - 18} = 40 + \frac{20}{6} = 40 + 3.33 = 43.33$$

Hence, the mode of given data is 43.33.

95. (b) Given, total number of votes = 100000

Party E get votes = $100000 - (36000 + 24000 + 18000 + 7000)$
 $= 100000 - 85000 = 15000$

$\therefore$ Angle allocated for party E = $\frac{15000}{100000} \times 360^\circ = 54^\circ$

96. (b) We have, discrete values $a + 4, a - 3.5, a - 2.5, a - 3, a - 2, a + 0.5, a + 5$ and $a - 0.5$
Now, ascending order is $a - 3.5, a - 3, a - 2.5, a - 2, a - 0.5, a + 0.5, a + 4, a + 5$.
 $\therefore$ Median = $\frac{a - 2 + a - 0.5}{2} = \frac{2a - 2.5}{2} = a - 1.25$

97. (d) We have, $x_i = i$
Mean = $\frac{2x_1 + 3x_2 + 4x_3 + \dots + (n+1)x_n}{n}$
 $= \frac{2 \times 1 + 3 \times 2 + 4 \times 3 + \dots + (n+1) \times n}{n}$
 $= \frac{\Sigma n(n+1)}{n} = \frac{\Sigma n^2 + \Sigma n}{n} = \frac{\frac{n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2}}{n}$
 $= \frac{n(n+1)}{2n} \left[\frac{2n+1}{3} + 1 \right] = \frac{(n+1)(n+2)}{3}$

98. (d) Weighted AM = $\frac{1 \times 1 + 2 \times 2 + 3 \times 3 + \dots + 10 \times 10}{10}$
 $= \frac{10 \times 11 \times 21}{6 \times 10} = \frac{21 \times 11}{6} = \frac{77}{2} = 38.5$

99. (b) Let the total votes be x .
Then, central angle of party A = $\frac{360^\circ}{x} \times 21960$
 $\Rightarrow 135^\circ = \frac{360^\circ}{x} \times 21960 \Rightarrow x = \frac{360^\circ \times 21960}{135^\circ} = 58560$

100. (a) $\therefore \frac{\text{Sum of 5 observations}}{\text{Number of observations}} = 9$
 $\Rightarrow$ Sum of 5 observations = $9 \times 5 = 45$
If 1 is subtracted from each observation, then
New mean of 5 observations = $\frac{\text{Sum of 5 observations} - 5}{5}$
 $= \frac{45 - 5}{5} = 8$

Median of 5 observations = $\left(\frac{5+1}{2}\right)\text{th term} = 3\text{rd term} = 8$

If 1 is subtracted from each observation, then

New median = $8 - 1 = 7$

Hence, the new mean and median are 8 and 7, respectively.

101. (b)
- | Age (in years) | Number of children | Cumulative frequency |
|----------------|--------------------|----------------------|
| 5-6 | 4 | 4 |
| 6-7 | 7 | 11 |
| 7-8 | 9 | 20 |
| 8-9 | 12 | 32 |
| 9-10 | 6 | 38 |
| 10-11 | 2 | 40 |
| Total | $N = 40$ | |

I. Here, $N = 40 \Rightarrow \frac{N}{2} = 20$

Thus, 7-8 is the median class.

$\therefore l = 7, f = 9, C = 11, \frac{N}{2} = 20$ and $h = 1$

$$\begin{aligned} \text{Median} &= l + \frac{\frac{N}{2} - C}{f} \times h \\ &= 7 + \frac{20 - 11}{9} \times 1 \\ &= 7 + 1 = 8 \text{ yr} \end{aligned}$$

Hence, statement I is incorrect.

- II. Total number of children (N) = 40

Number of children in the age group 6-9 = $7 + 9 + 12 = 28$

$\therefore$ Required percentage = $\frac{28}{40} \times 100\% = 70\%$

Hence, statement II is correct.

- III. $\therefore$ Modal group = 8-9,

$\therefore$ Modal age = 8 yr

Since, 9 is not included in this group.

Hence, statement III is correct.

102. (c) We have 11 observations as follow

$$1, \lambda, \lambda^2, \lambda^3, \lambda^4, \lambda^5, \lambda^6, \lambda^7, \lambda^8, \lambda^9, \lambda^{10}$$

Here, number of observations = 11 (odd)

Median = $\left(\frac{11+1}{2}\right)\text{th term} = 6\text{th term} = \lambda^5$... (i)

$\therefore$ AM = $\frac{1 + \lambda + \lambda^2 + \dots + \lambda^{10}}{11} = \left(\frac{\lambda^{11} - 1}{11(\lambda - 1)}\right)$

and GM = $(1 \cdot \lambda \cdot \lambda^2 \cdot \lambda^3 \dots \lambda^{10})^{\frac{1}{11}} = (\lambda^{1+2+3+\dots+10})^{1/11}$
 $= (\lambda)^{55/11} = \lambda^5$... (ii)

From Eqs. (i) and (ii), we get GM = Median

103. (d) We have, 11 observations as follows

$$1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \dots, \frac{1}{11}$$
 (odd)

$\therefore$ Median = $\left(\frac{11+1}{2}\right)\text{th term} = 6\text{th term} = \frac{1}{6}$... (i)

and $\frac{1}{\text{HM}} = \frac{1}{11} \left(\frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots + \frac{1}{11} \right)$

$\Rightarrow \frac{1}{\text{HM}} = \frac{1}{11} (1 + 2 + 3 + \dots + 11)$

$\Rightarrow \frac{1}{\text{HM}} = \frac{66}{11} \Rightarrow \text{HM} = \frac{1}{6}$

$\therefore$ HM = Median [from Eq. (i)]